BIOLOGY

Question Bank

NEET / AIPMT / GUJCET / AFMC / AIIMS...

For Competitive Exams







ગુજરાત માધ્યમિક અને ઉચ્ચતર માધ્યમિક શિક્ષણ બોર્ડના પત્ર-ક્રમાંક
મઉમશબ/સંશોધન/2016/3, તા. 03-09-2016-થી મંજૂર

BIOLOGY

Question Bank

NEET • AIPMT • GUJCET • AFMC • AIIMS...
For Competitive Exams

PLEDGE

India is my country.
All Indians are my brothers and sisters.
I love my country and I am proud of its rich and varied heritage.
I shall always strive to be worthy of it.
I shall respect my parents, teachers and all my elders and treat everyone with courtesy.
I pledge my devotion to my country and its people.
My happiness lies in their well-being and prosperity.

Price : ₹ 00

Gujarat State Board of School Textbooks
'Vidyayan', Sector 10-A, Gandhinagar-382010.

© Gujarat State Board of School Textbooks, Gandhinagar

Copyright of this book is reserved by Gujarat State Board of School Textbooks. No reproduction of this book in whole or in part, in any form is permitted without written permission of the Director, Gujarat State Board of School Textbooks.

Authors

Shri Nitin D. Dave (Convener)
Shri Avinash K. Bhatt
Shri Girish B. Patel
Shri Rajendra S. Patel
Shri Jaysukh B. Harmani
Shri Naresh R. Jhaveri
Shri Sandeep K. Patel
Shri Kalpesh K. Pandya
Shri Mehul S. Patel
Shi Paresh M. Parikh
Shri Mukesh R. Teli
Dr. Prashant R. Pandya
Smt. Sonal T. Bhatia
Smt. Bhargavi G. Vankhede
Shri Nimesh A. Darji

Translators

Shri Nitin D. Dave
Shri Rajendra S. Patel
Shri Paresh M. Parikh
Dr. Kavita Sharma
Shri Mehul S. Patel
Shri Jayshukh B. Harmani
Smt. Bhargavi G. Vankhede
Smt. Jagruti P. Bhatt
Smt. Inbarasi A. Mudaliar
Shri Payal A. Panchal
Smt. Kinjal Y. Modi

Co-ordinator

Dr. Chirag H. Patel
(Subject Coordinator : Physics)

Preparation and Planning

Dr. Kamlesh N. Parmar
(Dy. Director : Academic)

Layout and Planning

Shri Haresh S. Limbachiya
(Dy. Director : Production)

PREFACE

Gujarat State Board of School Textbooks has always made efforts to upgrade the quality of education by providing reference and supplementary material in addition to providing textbooks.

The Academic and Research Committees have prepared best quality reference and supplementary material which is helpful to the students in accordance with the competitive examinations like NEET / GUJCET. In the past also, the Textbook Board has published various reference books for the betterment of education. At school level teaching, there is always a need for extra references. In this context, in order to make the school syllabi more effective, supplementary material is prepared. In this reference, we are happy to publish **Biology Question Bank** for students of **Std. XI and XII.**

If the students, with the help of this question bank practices more, under the guidance of teachers, then the goal of the textbook board is achieved. This question bank is very useful to the Biology teachers and the readers associated with education. We expect that the students may achieve success in the competitive examinations like NEET / GUJCET by using this supplementary material.

The board is thankful to all the experts, authors, lay-out designers and others who have directly or indirectly extended their services in the preparation of Question Bank of Biology.

The board has taken enough care to make this Question Bank of Biology interesting, useful and error-free. However, suggestions from students, teachers, parents and academicians are welcome.

H. N. Chavda
Director
Date : 29-03-2017

Dr. Nitin Pethani
Executive President
Gandhinagar

First Edition: 2017

Published by : H. N. Chavda, Director, on behalf of Gujarat State Board of School Textbooks, Vidyayan, Sector 10-A, Gandhinagar

Printed by:

FUNDAMENTAL DUTIES

It shall be the duty of every citizen of India : *

(a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;

(b) to cherish and follow the noble ideals which inspired our national struggle for freedom;

(c) to uphold and protect the sovereignty, unity and integrity of India;

(d) to defend the country and render national service when called upon to do so;

(e) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities; to renounce practices derogatory to the dignity of women;

(f) to value and preserve the rich heritage of our composite culture;

(g) to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures;

(h) to develop the scientific temper, humanism and the spirit of inquiry and reform;

(i) to safeguard public property and to abjure violence;

(j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement;

(k) to provide opportunities for education by the parent or the guardian, to his child or, as the case may be, ward between the age of six and fourteen years.

* Constitution of India : Section 51-A

INDEX

1	Classification of Living Organisms	1
2	Taxomonical aids	16
3	Classification of Plant Kingdom	28
4	Classification of Animal Kingdom	48
5	Cell structure	68
6	Biomolecules-I (Carbohydrates, Lipids)	87
7	Biomolecules-II (Proteins, Nucleic acids and Enzymes)	108
8	Cell cycle and cell division	131
9	Animal Husbandry and Plant Breeding	149
10	Human Health and Diseases (Immunity Vaccination, Cancer, Aids)	156
11	Human Welfare and Microbes	186
12	Plant Morphology-I (Root, Stem and Leaf)	195
13	Plant Morphology-II (Flower, Fruit, Seed and Family)	216
14	Internal structure of Flowering Plant	238
15	Animal Tissue	260
16	Animal Morphology and Internal Structure - I (Earthworm and Cockroach)	286
17	Animal Morphology and Internal Structure - II (Frog)	311
18	Transport in Plants	338
19	Mineral Nutrition	357
20	Photosynthesis	382
21	Respiration	410
22	Digestion and Absorption	443
23	Breathing and gases exchange	462
24	Body fluid and circulation	482
25	Excretory substances and it Elimination	509
26	Locomotion and Movement	534
27	Organisms and Population	550
28	Ecosystem	573
29	Biodiversity and its conservation	595
30	Environmental Issues	620
31	Neural control and co-ordination in Animals	640
32	Chemical co-ordination and control	665
33	Reproduction in Organisms	683
34	Sexual reproduction in flowering plants	711
35	Growth and development in plants	729
36	Human Reproduction	754
37	Reproductive health	788
38	Heredity and Variation	806
39	Molecular basis inheritance	873
40	Evolution	910
41	Biotechnology : Principles and Processes	930
42	Biotechnology and its applications	940
•	Papers of AIPMT	951
•	Papers of AIIMS	1010
•	Papers of GUJCET	1057
•	GUJCET Model paper	1073

About this book...

Dear Students,

All of you will appear for competative exams like GUJCET/NEET, and other such type of test or national level exams. So to provide exam oriented perfect guidance, this question bank is prepared with the help of experienced subject experts by Gujarat State School text-book board.

Salient features of this question bank :

- All points and subjective material for GSEB and GUJCET/NEET type of competative exam syllabus is included.
- Students can revise very thoroughly and speedily, for it every point is discussed with MCQs and its answers
- In all the chapters questions asked in various patterns as well for practical skill and important questions for NEET are included.
- For practice, model question papers, GUJCET, AIPMT and AIIMS question papers are included.
- For all MCQ and competative question paper solutions are included.

Along with NEET syllabus GSEB syllabus is compared chapter wise. We hope that this will be useful and gives proper guidance to all students.

- Authors

Comparison of Syllabus

Sr. No.	GSEB			NEET	
	Standard / Part	Chapter	Name of chapter	Unit	Name of Unit
1.	11 / 1	Chapter 1	Classification of Living Organisms	Unit 1	Diversity in Living world
		Chapter 2	Taxomonical aids		
		Chapter 3	Classification of Plant Kingdom		
		Chapter 4	Classification of Animal Kingdom		
2.	11 / 2	Chapter 1	Plant Morphology-I (Root, Stem and Leaf)	Unit 2	Structural Organization in plants and Animals
		Chapter 2	Plant Morphology-II (Flower, Fruit, Seed and Family)		
		Chapter 3	Internal structure of Flowering Plants		
		Chapter 4	Animal Tissue		
		Chapter 5	Animal Morphology and Internal Structure-I(Earthworm and Cockroach)		
		Chapter 6	Animal Morphology Internal Structure-II (Frog)		

Sr. No.	GSEB			NEET	
	Standard / Part	Chapter	Name of chapter	Unit	Name of Unit
3.	11 / 1	Chapter 5	Cell structure	Unit 3	Cell structure and its functions
		Chapter 6	Biomolecules-I (Carbohydrates, Lipids)		
		Chapter 7	Biomolecules-II (Proteins, Nucleic acids and Enzymes)		
		Chapter 8	Cell cycle and cell division		
4.	12 / 1	Chapter 1	Transport in Plants	Unit 4	Plant Physiology
		Chapter 2	Mineral Nutrition		
		Chapter 3	Photosynthesis		
		Chapter 4	Respiration		
	12/2	Chapter 5	Growth and development in plants		
5.	12/1	Chapter 5	Digestion and Absorption	Unit 5	Animal Physiology
		Chapter 6	Breathing and gases exchange		
		Chapter 7	Body fluid and circulation		
		Chapter 8	Excretory substances and its Elimination		
		Chapter 9	Locomotion and Movement		
	12/2	Chapter 1	Neural control and co-ordination in Animals		
		Chapter 2	Chemical co-ordination and control		
6.	12 / 2	Chapter 3	Reproduction in Organisms	Unit 6	Reproduction
		Chapter 4	Sexual reproduction in flowering plants		
		Chapter 6	Human Reproduction		
		Chapter 7	Reproductive health		
7.	12 / 2	Chapter 8	Heredity and Variation	Unit 7	Genetics and Evolution
		Chapter 9	Molecular basis inheritance		
		Chapter 10	Evolution		
8.	11/1	Chapter 9	Animal Husbandry and Plant Breeding	Unit 8	Biology and Human Welfare
		Chapter 10	Human Health and Diseases		
		Chapter 11	Human Welfare and Microbes		
9.	12 / 2	Chapter 11	Biotechnology : Principles and Processes	Unit 9	Biotechnology and its Significance
		Chapter 12	Biotechnology and its applications		
10.	12 / 1	Chapter 10	Organisms and Population	Unit 10	Ecology and Environment
		Chapter 11	Ecosystem		
		Chapter 12	Biodiversity and its conservation		
		Chapter 13	Environmental Issues		

1 Classification of Living Organisms

Living and non-living the two natural structures are found in the entire world. From which non-living organisms or components are studied by Physics and Chemistry while living organisms are studied by Biology i.e. Life is a property of a living organisms. Many biologists or scientists derived many principles for the origin of life, by that way they derived clear definition of living organism. "Which has proper growth, which shows different types of physiological processes, which develop with the concern of environment and get proper adaptation and they have particular lifespan which reproduce new life that is known as living organism." After that it dies. At the mature age living organism reproduce new offsprings after the death of old living organisms they replace them so that they maintain continuation of life generation to generation of specific species.

(1) Which of the following branches of Science deal with study of non-living components ?

(A) Physics and Chemistry (B) Chemistry and Biology

(C) Biology and Physics (D) Physics and Account

(2) Which of the following characteristic is not true for plant ?

(A) Reproduction (B) Locomotion (C) Growth (D) Adaptation

(3) Living being can produce new living beings similar to itself on reaching maturity is known as....

(A) Regeneration (B) Adaptation (C) Reproduction (D) Differentiation

(4) Organisms incapable for reproduction

(A) Interspecific (B) Mature (C) Reproductive (D) Sterile

(5) Which of the following is the method of reproduction ?

(A) Power of regeneration (B) Biological power

(C) Life power (D) Free energy

(6) By which process, existence of living being is maintained ?

(A) Adaptation (B) Reproduction (C) Development (D) Differentiation

(7) By reproduction existing new living beings...

(A) Die before reproduction (B) Take the position in place of dead ones

(C) Do not reproduce again (D) Do not show adaptation

(8) What is Entropy ?

(A) Anabolism (B) Catabolism (C) Disorder (D) None of the above

(9) Which phylum shows maximum level of regeneration capacity ?

(A) Reptilia (B) Platyhelminthes (C) Porifera (D) Echinodermata

(10) ..... is the major characteristics of the living.

(A) Death (B) Growth (C) Reproduction (D) Adaptation

Answers : (1 -A), (2-B), (3-C), (4-D), (5-A), (6-B), (7-B), (8-C), (9-D), (10-A)

- **Metabolism :**

Metabolism is a complex biochemical reaction. As a result of these reactions degredation or growth take place in living organism. During these processes conversion of energy is required and living organisms get the energy from the food materials.

(11) Which process is essential to perform biological activities ?
(A) Energy (B) Energy transformation
(C) Free energy (D) Food

(12) Where do different types of biochemical processes are constantly carried out ?
(A) Blood plasma (B) Cytoplasm (C) Cell (D) Brain

(13) By which name is complex biochemical processes in each living cell of organisms is known as ?
(A) Growth (B) Metabolism (C) Differentiation (D) Development

(14) In metabolism process, anabolism means...
(A) Degradation process (B) Differentiation process
(C) Creative process (D) Progressive process

(15) In metabolism process, catabolism...
(A) Degradation process (B) Differentiation process
(C) Creative process (D) Progressive process

(16) If the ratio of anabolic process is more than catabolic process ..... occurs.
(A) Differentiation (B) Growth (C) Wear and Tear (D) Degradation

(17) If the ratio of catabolic process is more than anabolic process ..... occurs.
(A) Differentiation (B) Growth (C) Wear and Tear (D) Degradation

(18) In body, which type of processes are protein synthesis and digestion of protein ?
(A) Catabolism, anabolism (B) Catabolism, degradation
(C) Anabolism, catabolism (D) Degradation, anabolism

(19) ..... process of energy takes place during metabolism of a living organism.
(A) Degradation (B) Differentiation (C) Transformation (D) Replication

(20) From where, living organisms get energy required to do biological activities ?
(A) From environment (B) From another organism
(C) From stored energy (D) From their food

Answers : (11-B), (12-B), (13-B), (14-C), (15-A), (16-B), (17-C), (18-C), (19-C), (20-D)

- **Growth, Development, Reaction with Environment, Adaptation, Death :**

To grow in quantity and in number is a character of living organism. With Growth development also occurs. Each and every living organism react with its own environment. In which external factors play important role. So that every living organisms show changes somewhat in their body structure, method of function or behaviour and than adaptation occurs with the environment. It is known as adaptation and afterwards every living organisms dies. Increasing in the entropy specific living organisms dies. Death is one of the meaningful event.

(21) By which process living organisms increases volume of their body ?
(A) Cell-differentiation (B) Cell-division (C) Cell- degradation (D) Cell-growth

(22) In unicellular organisms cell division results in...
(A) Growth of organism (B) Growth of their cell number
(C) Growth of their volume (D) No change in living organism

(23) In which of the following growth takes place throughout their life ?

(A) Plants (B) Porifera (C) Protozoa (D) Vertebrata

(24) By which process, members of the same species produce zygote by mating ?

(A) Differentiation (B) Degradation (C) Fertilization (D) Development

(25) Changes take place in cells on the basis of specific functions is known as .....

(A) Differentiation (B) Degradation (C) Fertilization (D) Development

(26) Due to differentiation in cells, tissues and organs are developed. This is known as .....

(A) Growth (B) Development (C) Tissue formation (D) Organogenesis

(27) What is the form of efficiency of manifestation of feeling ?

(A) Physical (B) Chemical (C) Biological (D) All of the above

(28) Which of the following are external factors for plant reaction ?

(A) Light, water (B) Light, water, temperature

(C) Light, water, temperature, another organisms (D) Option (C) and Pollutants

(29) Living organisms breed with respect of which factor ?

(A) Environmental factor (B) Self choices

(C) Surrounding habitat (D) Self efficiency of breeding

(30) How living organisms do try to show adaptations ?

(A) By body structure (B) By mode of action (C) By behaviour (D) All of the above

(31) Those organisms who develop their characters for sustaining influence on efficiency of breeding are known as .....

(A) More easier (B) More adaptive (C) More dominant (D) None of them

(32) Organisms live in their habitat because...

(A) They get food there. (B) They get protection there.

(C) They are adapted to their habitat. (D) All of the above

(33) Required energy of any system is called .....

(A) Free energy (B) Mechanical energy (C) Chemical energy (D) Stored energy

(34) When do measure of disorder increases in living body ?

(A) During exchange of energy (B) During use of energy

(C) During decomposion of energy (D) During transportation of energy

(35) When entropy increases in living body .....

(A) Amount of free energy decreases (B) Amount of free energy increases

(C) The efficiency decreases (D) (A) and (C) both

(36) Which one of the following balance the cycle of components ?

(A) Adaptation (B) Metabolism (C) Death (D) Variation

(37) What will happen when in all organs entropy increase and reach at the maximum level ?

(A) Adaptation (B) Death (C) Variation (D) Reproduction

(38) Death is meaningful event because .....

(A) The number of living individuals of each species remains limited.

(B) Provide scope for the birth for new creatures.

(C) Components of body further turn to environment.

(D) All of the above

Answers : (21-B), (22-B), (23-A), (24-C), (25-A), (26-D), (27-D), (28-D), (29-A), (30-D), (31-B), (32-C), (33-A), (34-A), (35-D), (36-C), (37-B), (38-D)

- **Efficiency to maintain heredity and variation :**

In all living organisms nucleic acid DNA molecule is responsible for the inheritance properties. Which has complex coded language for internal reactions and properties, which is known as Gene. In one species differences occur in living organisms in own surronding that is called variation. These type of variation occur in specific species characters after the adaptation. Afterwards speciation occurs.

(39) Properties of tissues are based on .....

(A) interaction of their cells. (B) constituents of cells.

(C) function of organelles. (D) arrangement of cells.

(40) Properties of cell are .....

(A) due to constituent of cellular organelles.

(B) result of interaction of cellular activity.

(C) due to function of organelles.

(D) due to constituent of molecular components of organelles.

(41) Which molecule is inherited by the next progeny produced by parents ?

(A) DNA (B) ATP (C) RNA (D) GTP

(42) What is unit of Inheritance ?

(A) Cell (B) Gene (C) Nucleic acid (D) Chromosome

(43) We see around us, we find various living organisms. This property is known as .....

(A) Development (B) Reaction (C) Variation (D) Adaptation

(44) Which process is adopted by the living organisms to their nature to get maximum utilization from it ?

(A) Development (B) Variation (C) Reaction (D) Adaptation

(45) Those organisms who have efficient variations to use environment are known as .....

(A) Successful animals (B) Dominant animals (C) Adaptive animals (D) Active animals

(46) What happens when variations increases in such a way that organisms differ from their original parental characters ?

(A) A new species arise (B) A new genus arise

(C) A new organism arise (D) All of these

(47) What is responsible for formation of new species ?

(A) Adaptation (B) Death (C) Organisation (D) Variation

(48) For which process DNA is responsible in higher classes of organisms ?

(A) For sex determination (B) To maintain legacy

(C) For genetic engineering (D) For organ transplantation

Answers : (39-A), (40-B), (41-A), (42-B), (43-C), (44-D), (45-A), (46-A), (47-D), (48-B)

- **Organisation :**

There are various layers of organisation in living organisms. Which is constructed from atoms to living organisms and living organisms to biosphere.

(49) What will be formed by association of organelles ?
(A) Tissue (B) Organs (C) Cell (D) Organ systems

(50) Organisms of a same species constitute .....
(A) Biomes (B) Social group (C) Ecosystem (D) Population

(51) What is formed by interactions between biotic community and environment ?
(A) Biomes (B) Biosphere (C) Ecosystem (D) Population

(52) Select the correct sequence.
(A) Tissue - Cell - Organ - Organism (B) Organelle - Organ - Tissue - Organism
(C) Molecules - Cell - Organ system - Organism (D) Cell - Organ system - Tissue - Organism

(53) How ecosystem are structurised ?
(A) Interactions between biomes and energy
(B) Interactions between population and species
(C) Interactions between population and environment
(D) Interaction between biotic community and environment

(54) What organises to form a biospheres ?
(A) By gathering of species (B) By gathering of biotic community
(C) By gathering of ecosystems (D) By gathering of populations

Answers : (49-C), (50-D), (51-C), (52-C), (53-D), (54-C)

- **Biodiversity :**

There are various types of living organisms living in biosphere, this is known as biodiversity. Billions of species are present on the earth. Which is examined by biomes - observations.

(55) Animals show diversity in ....._____.
(A) Shape and size (B) Only shape
(C) Life style and structure (D) (A) and (C) both

(56) What is the method adopted by scientists in order to facilitate specific study of living organisms ?
(A) Distribution system (B) Classification system (C) Organisation system (D) (A) and (C) both

(57) When we can observe more and more biodiversity by living organisms ?
(A) By making continuous observation (B) By extending observation fields
(C) By classifying organisms (D) (A) and (B) both

(58) Approximately how many species are identified till today ?
(A) 17 to 18 lacs (B) 27 to 29 lacs (C) 7 to 18 lacs (D) 37 to 40 lacs

(59) What is the number of estimated species in the world ?
(A) 17 lacs (B) 17 lacs to 5 crore (C) 50 lacs (D) 50 lacs to 5 crores

Answers : (55-D), (56-B), (57-D), (58-A), (59-D)

- **Nomenclature :**

When we study about living organisms we identify the living organisms by their local names in Biology. But in one country and their states only one living organisms having different name so that study is troublesome. For this, specific name is required for each living organisms. By the adaptation of rules for living organisms these method is called Nomenclature and perfect description given to specific organisms known as identification. This process is more complex. But it is used for aproved classification method because in classification of living organisms it is classified by the taxon and these organisation is distributed by taxons. So it is known as taxonomy.

(60) What is the method adopted to agreed principals for naming is called ?

(A) Identification (B) Neosystematics (C) Nomenclature (D) Classification

(61) When can identification be possible ?

(A) If description has to be done correctly (B) If naming has to be done correctly

(C) If study has to be done correctly (D) If local names are correct

(62) At primary level, classification is a process in which ..... arrangement is there.

(A) Provision of taxon

(B) Easily observable characters

(C) Organisms are meaningfully classified into groups

(D) Nomenclature of organisms

(63) Classification is a process in which...

(A) Provision of taxon arrangement (B) Easily observable characters are present

(C) Only taxon arrangement (D) (A) and (B) both

(64) Which scientific word is used for proper group for study of living organisms ?

(A) Category (B) Taxon (C) Subclass (D) (A) and (C) both

Answers : (60-C), (61-A), (62-C), (63-D), (64-B)

- **History of classification method :**

- Classification is precise method of logical arrangement of organisms.
- Reference to classification are found in Shshrut Samhita.
- Greek Philosopher Aristotle had proposed classification of organisms.
- Carolus Linnaeus is father of taxonomy. He gave binomial Nomenclature.

(65) Which of the following is reference to classification ?

(A) Manu's writing (B) Sushrut Samihita (C) Ayurveda (D) Yajurveda

(66) Who had not done research in field of classification ?

(A) Bentham & Hooker (B) Aristotle (C) Robert Brown (D) Whittaker

(67) Five kingdom classification was given by .....

(A) Bentham and Hooker (B) Aristotle (C) Linnaeus (D) Whittaker

(68) Which of the following method is known for scientific name of living organisms ?

(A) New systematic classification method (B) Five kingdom classification

(C) Standard classification (D) Binomial nomenclature method

(69) Who developed binomial nomenclature method ?

(A) Bentham and Hooker (B) Carolus Linnaeus (C) Huxley (D) Aristotle

(70) Which greek philosopher proposed classification of organisms ?

(A) Bentham and Hooker (B) Carolus Linnaeus (C) Huxley (D) Aristotle

(71) Who is known as father of taxonomy ?

(A) Aristotle (B) Huxley (C) Whittaker (D) Carolus Linnaeus

(72) Who developed new systematics method ?

(A) Aristotle (B) Sir Julian Huxley (C) Whittaker (D) Carolus Linnaeus

(73) Which one of the following is essential condition to study classification ?

(A) The knowledge of characters of organisms and salient features of groups and taxa.

(B) Expertness to use required instruments.

(C) Detailed study of the field which is to be studied.

(D) Subjective knowledge

(74) Which systematics has developed through compilation of different branches ?

(A) Chemotaxonomy (B) Cytotaxonomy

(C) Numerical taxonomy (D) All of these

Answers : (65-B), (66-C), (67-D), (68-D), (69-B), (70-D), (71-D), (72-B), (73-A), (74-D)

- **Sources to study systematics :**

A scholar of systemics has to take training for field study, obeying their rules. For this type of study specific needed instruments, chemical, etc. are essential during the study. In any situations live in entire fields. Specieman of plants or preservation methods of animal speciemens. Zoological parks, Museum, Botanical gardens, Herbaria, etc. are source of taxomical study. Apart from this speciemens, photograph or slides can be studied.

(75) Which important characters are required in scholar dealing with taxonomy field study ?

(A) Curiosity (B) Concentration (C) Patience (D) All of these

(76) Which equipments should be kept with scholar of taxonomy during study ?

(A) Binocular, Scissors, Pages, Chair (B) Binocular, Camera, Cutter, Forceps, Bags

(C) Binocular, Cutter, Forceps, Umbrella (D) Camera, Pages, Cutter, Big pouches

(77) Forests, mountains, grounds, grassland, rivers, lakes, oceans like places are like ..... source for field study.

(A) Open ecosystem (B) Open books (C) Natural treasure (D) Natural factors

(78) Which plants are breeded in botanical gardens ?

(A) Medicinal plants (B) Economically important

(C) Rare plants (D) All of these

(79) Where dead stuffed bodies, skeleton, fossils of animals are kept ?

(A) Museum (B) Zoo

(C) (A) and (B) both (D) None of the above

Answers : (75-D), (76-B), (77-B), (78-D), (79-A)

- **Principles of Taxonomy :**

The nomenclature and classification is based on definate rules. Scientific names of plants are based as per principles of International Code for Botanical Nomenclature (ICBN), Animal taxonomists have to execute the principle of International Code for Zoological Nomenclature (ICZN).

(80) Principles of which institutes have to execute for scientific names of plants and animals respectively ?

(A) ICBN and ICZN (B) CZN and IABG

(C) ICBN and ICZN (D) WCU and WWF

(81) After what and how the author's name is written ?

(A) Species name in abbreviated form (B) Generic name and capital letters

(C) Generic name and small letters (D) Generic name in abbreviated form

(82) When scientific name is written by hand .....

(A) Each separate word should be underlined by dotted line and if is printed in italic form.

(B) Each separate word should be underlined and it is printed in italic form.

(C) Each separate word should be underlined by dotted line and it is printed in latin word.

(D) Each separate word should be underlined and it is printed in latin form.

Answers : (80-A), (81-A), (82-B)

- **Categories of Classification :**

Classification is not only one step method but it is method of series of sequential steps; in which each taxon shows category. If taxon is part of classified arrangement with all aspects, it is called a taxonomic category. On the basis of characters of different living organisms serially taxons of each living organisms are classified. There is more similarities when we go towards species.

(83) Each taxon in classification shows .....

(A) Taxa (B) Category (C) Sequential steps (D) Class

(84) What is called category given to all living groups at different levels of classification ?

(A) Category (B) Kingdom (C) Taxa (D) Taxonomic category

(85) What is taken as a reference of unit in classification ?

(A) Each class (B) Each kingdom (C) Each taxon (D) Each category

(86) The main group in which all other groups of organisms at different levels of classification are included is known as .....

(A) Kingdom (B) Species (C) Family (D) Class

(87) Select correct sequence for decreasing order of variation .....

(A) Kingdom - Class - Family - Species (B) Class - Kingdom - Species - Family

(C) Species - Family - Class - Kingdom (D) Species - Class - Family - Kingdom

(88) Select correct sequence for increasing order of similarity .....

(A) Family - Genus - Order - Class (B) Class - Order - Family - Genus

(C) Genus - Family - Order - Class (D) Order - Class - Family - Genus

(89) What is called a group of living individuals which have more similarity in most of their characters and are capable of interbreeding and giving rise to fertile offspring are called ?

(A) Genus (B) Family (C) Species (D) Order

(90) What is group of species having a common ancestor called ?

(A) Genus (B) Family (C) Species (D) Order

(91) Extinct human race is known as .....

(A) americana (B) columbidae (C) sapiens (D) erractus

(92) What is called a group of genera ?

(A) Order (B) Class (C) Category (D) Family

(93) Which of the following family is associated with the study of ornithology ?

(A) Blattidae (B) Ranidae (C) Columbidae (D) Megascolecidae

(94) A group of interrelated families constitute .....

(A) Class (B) Phylum (C) Order (D) Divison

(95) A subclass is a group of .....

(A) A group of families (B) A group of series (C) A group of species (D) A group of orders

(96) The categories depends on their common characters...

(A) Genus, species, family (B) Species, family, order

(C) Genus, species, series (D) Series, family, order

(97) Which of the following is included in class oligochaeta ?

(A) Cockroach (B) Frog (C) Earthworm (D) Pigeon

(98) Frog belongs to which order ?

(A) Orthoptera (B) Ophisthopora (C) Anura (D) Inferae

(99) Earthworm belongs to which family ?

(A) Banidae (B) Blattidae (C) Megascolecidiae (D) Asteraceae

(100) The organism from asteraceae family is .....

(A) Frog (B) Cockroach (C) Earthworm (D) Sunflower

(101) Which organism is from Glumiflorae order ?

(A) Cockroach (B) Earthworm (C) Sunflower (D) Maize

Answers : (83-B), (84-C), (85-C), (86-A), (87-A), (88-B), (89-C), (90-A), (91-D), (92-D), (93-C), (94-C), (95-B), (96-A), (97-C), (98-C), (99-C), (100-D), (101-D)

(102) Which statement is not correct for growth ?

(A) Multicellular organisms increase by cell division.

(B) Due to cell division, growth occurs in tissue, organ or body.

(C) In animals growth takes place throughout life span.

(D) After birth living organisms increase volume of their body continuously.

(103) Which statement does not fit with nomenclature ?

(A) Name is given to organisms through rules.

(B) One scientific name is not used for any other living organism.

(C) At the global level each living organism have two scientific names.

(D) Nomenclature of all living organisms is not possible for study.

(104) Which statement is not proper for classification ?

(A) At the primary level living organisms are distributed in specific meaningful groups.

(B) Any living organism is classified on the basis of taxon arrangement.

(C) Some of them are based on characters which can be observed easily.

(D) Name is given by following the rules.

(105) Which statement does not fit with DNA ?

(A) It is unit of adaptation.

(B) It is made up of nucleic acid.

(C) It has mysterious genetic codes to produce required chemicals for processes similar to parents.

(D) It inherits in characters produced by chemicals to perform activities like parents.

(106) Which statement is proper for living organism ?

(A) Each living organism has characteristics like growth, development and reproduction.

(B) Each living organism can grow, can adopt and show locomotion.

(C) Each living organism shows growth, and development but they cannot reproduce.

(D) Each living organism shows locomotion and reproduction capacity but it is not necessary that it grows.

(107) Which statement is related with death ?

(A) During an energy exchange some of the energy is lost in the form of heat.

(B) Maximum entropy occurs on all the levels of organization which stop functioning.

(C) Living organisms have their body organization mechanism or changes in behaviour simultaneously with environment.

(D) When anabolic processes occur more as compared to catabolic processes, then growth is observed.

(108) Which statement is correct for taxon between family and species ?

(A) Reproductive progeny is formed by inter reproduction.

(B) Most related group of genera.

(C) Group of species of common ancestors.

(D) Group of families having inter relationship.

(109) Which of the following characteristics of herbarium is incorrect ?

(A) To collect and care plant specimens.

(B) To collect and care plant books.

(C) To develop medicinal plants, attractive plants as well as rare plants.

(D) To collect plant photographs, slides, maps, etc.

(110) Select the correct statement which is responsible to arise a new species ?

(A) When quantity of variation increases the new organisms differ from their original parental characters.

(B) DNA molecule inherits from parent to the next progeny.

(C) Members of the same species copulate with each other.

(D) Living organisms breed keeping environmental factors in centre.

(111) Which of the following classification of cockroach is correct ?
(A) Vertebrata - Amphibia - Anura - Ranidae - Rana - tigrina
(B) Arthropoda - Insect - Orthoptera - Blattidae - Periplaneta - americana
(C) Arthropoda - Insect - Opisthopora - Periplaneta - Rana - americana
(D) Annelida - Oligochaeta - Blattidae - Periplaneta - americana

(112) Which of the following statement is incorrect ?
(A) Family : A group of genera which are closely related.
(B) Class : A group of series.
(C) Order : A group of interrelated families constitute.
(D) Series : A group of orders.

(113) Which of the following is mismatched pair ?
(A) Classification : Systematic arrangement of living organisms.
(B) Classification : Living organisms are meaningfully classified.
(C) Species name : First letter in capital.
(D) The author's name : Abbreviated form after species name.

(114) Select the correct pair.
(A) Megascolecidae : Anura
(B) Periplaneta : Blattidae
(C) Rana : Orthoptera
(D) Helianthus : Glumiflorae

(115) Select the correct pair :
(A) Zea : Monocotyledons
(B) Sunflower : Gymnosperms
(C) Oligochaeta : Arthropoda
(D) Orthoptera : Vertebrata

(116) Select mismatched pair
(A) Adaptation : Can sustain in their environment
(B) Biospere : Combined biomes of nature
(C) Variation : Different characters within same species among individuals
(D) Taxonomic hierarchy : Aggregation of taxa

Answers : (102-C), (103-A), (104-C), (105-A), (106-C), (107-B), (108-C), (109-C), (110-A), (111-B), (112-B), (113-C), (114-B), (115-A), (116-B)

- **A - Assertion (Statement), R - Reason**

(A) Both A and R are true and R is correct explanation of A.

(B) Both A and R are true and R is not correct explanation of A.

(C) A - true, R - wrong.

(D) A - wrong, R - true.

(117) Statement A : Scientific names of plants are based on the principles of ICBN.
Reason R : ICBN has sketch diagram, photographs and speciman copy for giving names to plants.
(A) (B) (C) (D)

(118) Statement A : Taxon and taxonomic category are same.
Reason R : Taxa aggregate and forms a taxonomic hierarchy.
(A) (B) (C) (D)

(119) Statement A : Taxonomic hierarchy includes seven taxonomic category.

Reason R : To get specific information and specific place of each category they are divided into sub categories.

(A) (B) (C) (D)

(120) Statement A : Death is meaningful event.

Reason R : Components of body further turn to environment due to death and number of living individual of each species remains limited.

(A) (B) (C) (D)

(121) Statement A : If the ratio of anabolic process is more than catabolic process, growth occurs.

Reason R : Growth is an output of metabolism.

(A) (B) (C) (D)

(122) Statement A : Members of the same species can not copulate.

Reason R : Zygote is produced as a result of fertilization.

(A) (B) (C) (D)

(123) Statement A : Energy transformation takes place in metabolism in living organism.

Reason R : Organisms have to perform many biological activities.

(A) (B) (C) (D)

(124) Statement A : Group of genera which are closely related is called family.

Reason R : Blattidae is family which contains pigeons and doves have different genera and species.

(A) (B) (C) (D)

(125) Statement A : Binomial nomenclature method is given by Linneus.

Reason R : Linneus is known as father of taxonomy.

(A) (B) (C) (D)

(126) Statement A : Differentiation and organogenesis takes place during growth.

Reason R : Number of cells increase during growth.

(A) (B) (C) (D)

(127) Statement A : Organisms must be given two names.

Reason R : Species name must be written in small letters.

(A) (B) (C) (D)

(128) Statement A : Each living organism possesses reproduction, growth, development, adaptation and death like abnormal characteristics.

Reason R : For each living organism the conversion of energy is essential.

(A) (B) (C) (D)

(129) Statement A : To define organism concentration is made on specific characters of organisms.

Reason R : Reproduction, growth, development, awarness of environment, adaptation and death are abnormal characters of living organisms.

(A) (B) (C) (D)

(130) Statement A : Different types of biochemical processes are collectively called metabolism.

Reason R : If the ratio of anabolic process is more than catabolic process, growth occurs.

(A) (B) (C) (D)

(131) Statement A : The result of continuous disorders is called entropy.

Reason R : As a result of entropy free energy decreases and the efficiency decreases.

(A) (B) (C) (D)

Answers : (117-A), (118-D), (119-C), (120-A), (121-B), (122-D), (123-A), (124-C), (125-B), (126-D), (127-A), (128-B), (129-B), (130-C), (131-D)

(132) Match the following :

Column - I (Common name)	**Column - II** (Scientic name)	
(a) Earthworm	(i) *Pariplaneta americana*	(A) a - v, b - iv, c - iii, d - ii, e - i
(b) Sunflower	(ii) *Rana tigrina*	(B) a - ii, b - i, c - iv, d - iii, e - v
(c) Frog	(iii) *Helianthus annus*	(C) a - iv, b - iii, c - ii, d - v, e - i
(d) Maize	(iv) *Pheretima posthuma*	(D) a - iii, b - ii, c - v, d - i, e - iv
(e) Cockroach	(v) *Zea mays*	

(133) Match the following :

Column - I (Common name)	**Column - II** (Family)	
(a) Frog	(i) Asteraceae	(A) a - iii, b - iv, c - v, d - i, e - ii
(b) Cockroach	(ii) Poaceae	(B) a - iv, b - v, c - i, d - ii, e - iii
(c) Earthworm	(iii) Ranidae	(C) a - v, b - i, c - ii, d - iii, e - iv
(d) Sunflower	(iv) Blattidae	(D) a - i, b - ii, c - iii, d - iv, e - v
(e) Maize	(v) Megascolecidae	

(134) Match the following :

Column - I (Common name)	**Column - II** (Series)	
(a) Frog	(i) Glumiflorae	(A) a - iii, b - iv, c - v, d - i, e - ii
(b) Cockroach	(ii) Opisthopora	(B) a - iv, b - v, c - i, d - ii, e - iii
(c) Earthworm	(iii) Anura	(C) a - v, b - i, c - ii, d - iii, e - iv
(d) Sunflower	(iv) Orthoptera	(D) a - i, b - ii, c - iii, d - iv, e - v
(e) Maize	(v) Infirae	

(135) Match the following :

Column - I (Taxon)	**Column - II** (Character)
(a) Species	(i) A group of interrelated families
(b) Genus	(ii) A group of genera which are closely related
(c) Series	(iii) Capable of interbreeding and giving rise to fertile offsprings
(d) Family	(iv) Group of orders
(e) Order	(v) A group of species having common ancestor
(A) a - v, b - iii, c - i, d - iv, e - ii	(B) a - i, b - ii, c - iii, d - v, e - iv
(C) a - ii, b - i, c - iii, d - iv, e - v	(D) a - iii, b - v, c - iv, d - ii, e - i

(136) Match the following :

Column - I (Common name)	**Column - II** (Classification method)
(a) Aristotle	(i) Books for identification of plants
(b) Linneus	(ii) New systematics
(c) Bantham and Hooker	(iii) Five kingdom classification
(d) Huxley	(iv) Classification of animals
(e) Whittaker	(v) Binomial nomenclature

(A) a - iv, b - v, c - i, d - ii, e - iii

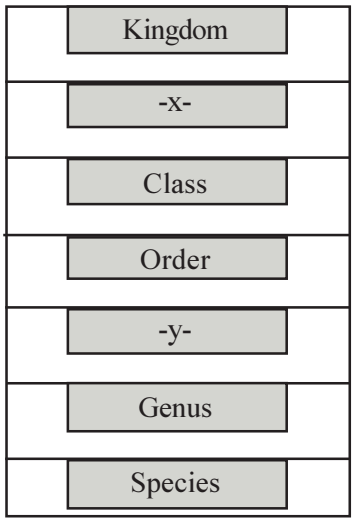
(B) a - v, b - i, c - ii, d - iii, e - iv

(C) a - i, b - ii, c - iii, d - iv, e - v

(D) a - ii, b - iii, c - iv, d - v, e - i

Answers : (132-C), (133-A), (134-A), (135-D), (136-A)

(137) Select proper option for given 'x' and 'y' in given diagram ?



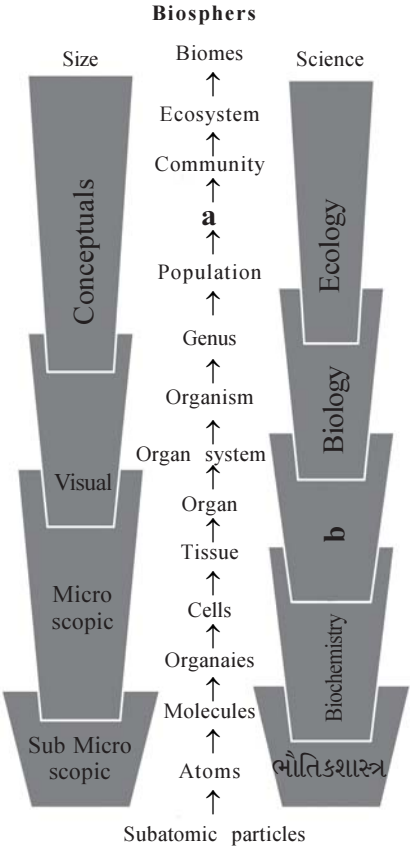
(A) 'x' - Subkingdom, 'y' - order

(B) 'x' - Subkingdom, 'y' - Subgenus

(C) 'x' - Phylum/Divison, 'y' - Family

(D) 'x' - Phylum/Divison, 'y' - Subgenus

(138) Which part is labelled as 'a' and 'b' in given diagram ?



Organisation level of livings

(A) a - Population, b - Biology

(B) a - Community, b - Cytology

(C) a - Species, b - Cytology

(D) a - Community, b - Biology

Answers : (137-C), (138-B)

- **Questions for NEET :**

(139) What is correct for individuals within same species ?

(A) Live in same ecological life style (B) Live in same habitat

(C) Capable of interbreeding (D) Live always in different habitat

(140) Main goal of biological system is .....

(A) Identification and arrangement of organisms based on their cell structure characters.

(B) Classification of organisms based on their morphology.

(C) To put living organism in their taxa and to establish interrelation among them.

(D) Classification of living organisms based on their evolutionary history and establishment of individual development at each level.

(141) Species are considered as .....

(A) Real unit of classification decided by taxonomist.

(B) Basic unit of classification.

(C) Real basic the lowest units of classification.

(D) Artificial concept of human mind which cannot be defined in absolute terms.

(142) The living organisms can be unexceptionally distinguished from the non-living thing on the basis of their ability for .....

(A) Interaction with the environment and progressive evolution

(B) Reproduction

(C) Growth and movement

(D) Responsiveness to touch

(143) What is responsible for creation of new species ?

(A) Inbreeding hybridization (B) Variation

(C) Differentiative reproduction (D) None of these

(144) Which of the following is mismatched pair ?

(A) *Zea mays* - Glumiflorae (B) *Helianthus annus* - Asterales

(C) *Rana tigrina* - Ranidie (D) *Pheritima posthuma* - Mega scolecidae

(145) Which of the following is not included in microscopic level of organisation ?

(A) Organs (B) Tissue (C) Cells (D) Species

(146) Which of the following category includes characters of group in living organisms ?

(A) Family (B) *Taxon* (C) Genus (D) Species

(147) Which of the following is not related ?

(A) Opisthopora (B) Glumiflorae (C) Anura (D) Asteraceae

Answers : (139-C), (140-D), (141-C), (142-B), (143-B), (144-A), (145-D), (146-B), (147-D)

2 Taxonomical Aids

Every living organisms are interrelated in biosphere and it is necessary to know interrelations of these for the classification or taxonomical study. For these study places are important where plant and animal specimens are stored. Measures can also be taken for conservation of endangered and extinction-prone organisms. For these herbarium is most important. In these plant specimens are collected from various places and different types of specific methods applied on them, then they are stored and studied for years. These types of herbarium has many uses.

(1) Which of the following is correct group of diversities in living organisms ?

(A) Structure, function and Behaviour (B) Shape, relation, Habitat

(C) Height, Weight, shape (D) Food, Energy, Function

(2) For what classification of organisms and identity is neccesary ?

(A) In laboratory (B) Field studies

(C) Geographical distribution (D) A and B both

(3) Which branch of science is important for identity and interrelationship ?

(A) Biology (B) Classification (C) Embryology (D) Physiology

(4) Which of the following requires classification studies ?

(A) Agriculture, forestry, industry (B) Biodiversity and bio resources knowledge

(C) To study future of living being (D) All of the above

(5) Which of the following is main source for study of classification ?

(A) Collection of specimens of living organisms (B) Collection of fossils of organisms

(C) Collection of diagrammatic sheets (D) Collection of dead bodies of organisms

(6) What is the use of classification of various geographical area ?

(A) Can prepare maps (B) Can prepare plant groups

(C) Can prepare animal groups (D) B and C both

(7) For which type of organisms conservation programme can be arranged by classification ?

(A) Endangered and extinction-prone organisms (B) Organisms who failed to reproduce

(C) Organisms in very less number (D) All the above

(8) Plant specimens are collected in...

(A) Green house (B) Botanical gardens (C) Plant museum (D) National park

(9) Which is the correct sequence for plant specimen collection ?

(A) Collection - Pressing - mounting - drying - poisoning

(B) Collection - Pressing - drying - poisoning - mounting

(C) Collection - Drying - pressing - poisoning - mounting

(D) Collection - Drying - poisoning - pressing - mounting

(10) Which paper is used to dry plant specimens ?

(A) Parchment paper (B) Blotting paper (C) Filter paper (D) Chromatography paper

(11) What is poisoning ?

(A) Cultivation in special chemical (B) To put naphthalene balls in specimen

(C) To sprinkle special chemical (D) To add special toxic substances

(12) Which treatment is used to preserve dry plant specimens by special chemical ?

(A) Fumigation (B) Pressing (C) Mounting (D) Poisoning

(13) To mount plant specimen on a specific size of sheet is known as...

(A) Cobalt sheet (B) Copper sheet (C) Herbarium sheet (D) Chloride sheet

(14) Which process is done to protect herbarium sheet from fungi, insects and humidity ?

(A) to put in plastic bag (B) to change cupboard

(C) to put Naphthalene balls (D) A and B both

(15) Against which factors fumigation provides protection ?

(A) Fungi, Algae, Humidity (B) Algae, Insects, Humidity

(C) Algae, Fungi, Draught (D) Fungi, Insects, Humidity

(16) Where can we label on Herbarium sheet ?

(A) upper left (B) lower left (C) lower right (D) upper right

(17) What is useful for economical and ethnobotanical information ?

(A) Botanical garden (B) Herbarium (C) Museum (D) All of the above

(18) What provides key for modern classification system ?

(A) Botanical Garden (B) Museum (C) Herbarium (D) None of the above

(19) Where is the museum of Royal Botanical Garden located ?

(A) Kew (B) London (C) Paris (D) Dehradun

(20) Where is the herbarium for forest research institute located ?

(A) England (B) Dehradun (C) Darjeeling (D) Paris

(21) Location of Museum of Natural History ..... .

(A) England (B) Kolkata (C) Dehradun (D) Paris

(22) Which famous museum is located in Kolkata-India ?

(A) Herbarium, Department of Botany (B) Herbarium of forest research

(C) Central National Herbarium (D) Museum of Natural History

(23) Location of herbarium, department of Botany ..... .

(A) France (B) Vadodara (C) England (D) Kolkata

(24) What is helpful to get necessary information for verifying and identifying newly collected plant species ?

(A) Botanical Garden (B) Herbarium (C) Museum (D) None of the above

(25) Where can we study rare plant species, not available in our area ?

(A) Botanical Garden (B) Museum (C) Herbarium (D) None of the above

Answers : (1-A), (2-D), (3-B), (4-D), (5-A), (6-D), (7-A), (8-C), (9-B), (10-B), (11-C), (12-A), (13-C), (14-C), (15-D), (16-C), (17-B), (18-C), (19-A), (20-B), (21-D), (22-C), (23-B), (24-B), (25-C)

- **Botanical gardens and their importance :**

Botanical Garden means scientifically planned collection of trees, shrubs, herbs and climbers and other living plants from different parts of the globe. They are different from public parks and gardens. Various kinds of plants having medicinal and economic importance as well as rare plants are cultivated and maintained in botanical gardens. Thus these gardens play important role in different regions.

(26) What is appropriate for botanical garden ?

(A) Plants from different gardens of the world.

(B) It is different from public park and gardens.

(C) A and B both

(D) None of the above.

(27) Where can we cultivate medicinal, economically important and rare plants ?

(A) Nursery (B) Germplasm bank (C) Herbarium (D) Botanical garden

(28) Which method is used to cultivate rare plants ?

(A) Grafting (B) Fernary (C) Conservatory (D) Green House

(29) Which of the following method is used to grow plants of other places by creating necessary (convinient) environment ?

(A) Green house and Cactus house (B) Fernary and Orchidium

(C) Glass house and Conservatory (D) All of the above

(30) Which is the most beautiful place ?

(A) Museum (B) Sanctuary (C) Zoo (D) Botanical Garden

(31) What is the location of great Banyan tree ?

(A) Indian Botanical Garden (B) Royal Botanical Garden

(C) Lyoid Botanical Garden (D) National Botanical Garden

(32) In which city the great Banyan tree is present ?

(A) Lucknow (B) Darjeeling (C) Shibpur (D) Vaghai

(33) Each plant species is labelled indicating its family, scientific name and location...

(A) to make study easier. (B) self indication is better.

(C) for beautification. (D) for self study.

(34) In which fields, plants present in botanical gardens are used ?

(A) Anatomy, Embryology (B) Plant, biochemistry, cell biology

(C) Physiology and Ecology (D) All of the above

(35) For what reason importance of botanical garden is being increased day-by-day ?

(A) For conservation of endangered organisms (B) More cultivation of plants

(C) For economical production (D) For reproduction related act

(36) Which branch information are provided to public by botanical garden ?

(A) Landscape gardening (B) Fruit culture (C) Anatomy (D) A and B both

(37) When can we provide classification information at place in the botanical garden for research ?

(A) If germplasmbank is present in it. (B) If cloning and hybridization occurs.

(C) If it has aesthetic appeal. (D) If it has museum and library.

(38) Which is the true method for production of new species ?

(A) Grafting tissue culture (B) Cloning, hybridization

(C) Fernery, Horticulture (D) A and B both

(39) What is used for conservation of rare genes ?

(A) Germplasm bank (B) Seed bank (C) Seed storage (D) Gene bank

(40) Which of the following is incorrect for the contribution of botanical garden ?

(A) For conservation of biodiversity and cultural heritage (B) For conservation of environment

(C) For study of botany (D) For conservation of natural resources

(41) What is the number of botanical gardens in IABG ?

(A) 600 (B) 6000 (C) 8000 (D) 800

(42) What is the full form of IABG ?

(A) Indian association of botanical garden (B) Indian agriculture of botanical garden

(C) International association of botanical garden (D) International agriculture of botanical garden

(43) What is the location of Royal Botanical garden ?

(A) England (B) Paris (C) NewYork (D) Kew

(44) What is the location of Lyoid Botanical garden ?

(A) Shibpur (B) Lucknow (C) Darjeeling (D) Vaghai

(45) Where is the Indian botanical garden located ?

(A) Shibpur (B) Lucknow (C) Darjeeling (D) Vaghai

(46) What is the location of National botanical garden ?

(A) Shibpur (B) Lucknow (C) Darjeeling (D) Vaghai

(47) Where is the botanical garden located in Gujarat ?

(A) Vadodara (B) Vaghai (C) Dang (D) Saputara

(48) In which states of India botanical garden is present ?

(A) Uttar Pradesh and West Bengal (B) Gujarat and Madhya Pradesh

(C) Uttar Pradesh and Madhya Pradesh (D) Gujarat and Maharastra

Answers : (26-C), (27-D), (28-C), (29-D), (30-D), (31-A), (32-C), (33-B), (34-D), (35-A), (36-D), (37-D), (38-D), (39-D), (40-A), (41-D), (42-C), (43-D), (44-C), (45-A), (46-B), (47-B), (48-A)

- **Museum :**

Plant, animal and fossil specimens are collected in the museum for study and reference purposes. These type of museum are generally setup in schools and colleges as per requirement. Plant and animal specimens are preserved here and by the various methods animals are preserved in original form.

(49) Which of the following shows speciality of university museum ?

(A) Primary level (B) Rich and less informative

(C) Rich and informative (D) Higher level

(50) Which is the primary level museum ?

(A) School (B) University (C) Private Institute (D) Voluntary Institute

(51) Where can we find ethnogallery ?

(A) Botanical museum (B) Botanical garden (C) Zoo (D) Zoological museum

(52) What is ethno gallery ?

(A) Skeleton gallery (B) Mammalian gallery (C) Fossil gallery (D) Ethano gallery

(53) Which solution is used to preserve specimens in containers or jars ?

(A) Preservative (B) Staining solution (C) Special chemicals (D) Enzymes

(54) Which are different sections of zoological museum ?

(A) Library (B) Laboratory (C) Supervisor's home (D) All the above

(55) Which method is used to preserve larger sized dead bodies of animals ?

(A) Scanning (B) Grafting (C) Stuffing (D) Cloning

(56) What is Stuffing ?

(A) To put animals in chemicals.

(B) To put animals in glass cupboard.

(C) By pressing method.

(D) Stuffing dried plant powder and mixture of preservatives in animals.

(57) Which is the famous museum of India ?

(A) Mumbai (B) Chennai (C) Kolkata (D) Vadodara

(58) What is the location of Natural History Museum ?

(A) Mumbai (B) Chennai (C) Kolkata (D) Vadodara

(59) What is the location of Zoological Survey of India ?

(A) Jodhpur (B) Kolkata (C) Mumbai (D) A and B both

(60) Where can we study Museology ?

(A) Jodhpur (B) Kolkata (C) Vadodara (D) A and B both

(61) Which is famous museum location ?

(A) Jodhpur (B) Kolkata (C) Vadodara (D) A and B both

(62) Which one is the famous museum ?

(A) Natural History Museum (Mumbai) (B) Zoological Survey of India (Jodhpur, Kolkata)

(C) Central National Museum (D) Government Museum (Chennai)

Answers : (49-C), (50-A), (51-D), (52-D), (53-A), (54-D), (55-C), (56-D), (57-B), (58-A), (59-D), (60-C), (61-C), (62-D)

- **Zoological park :**

Zoological park means "an institution where various kinds of living animals are conserved and exhibited in captivity." The zoological parks in India are also sometimes referred as the Zoological gardens. Various objectives of the zoological parks are these. For the improvement that different situation would provide and concentration proper taken on their residences and diet. All the zoological park working under the observation of central Zoo authority. In Gujarat and in our nation so many zoological parks are situated.

(63) What is the another name of zoological park ?

(A) Zoological garden (B) Zoological museum (C) Zoo (D) Zoo habitation

(64) "Ex-situ conservation" – This is main objective of...

(A) Botanical garden (B) Museum (C) Zoological garden (D) Botanical Museum

(65) Which of the following is included in objectives of zoological park ?

(A) Hybridization, Rehabilitation (B) Conservation, animal behaviour

(C) Public awareness, Study, tourism (D) All of the above

(66) Which of the following is not area of zoological park ?

(A) Nocturnal house (B) Skeleton house (C) Insect house (D) None of the abvoe

(67) Which are different sections of zoological parks ?

(A) administration (B) Veterinary section (C) Research section (D) All of the above

(68) Who works under CZA ?

(A) Botanical garden (B) Museum (C) Botanical museum (D) Zoological park

(69) What is the full form of CZA ?

(A) Central zoological authority (B) Central zoo organization

(C) Central zoological organization (D) Central zoo authority

(70) Which of the following manages both private and government organization ?

(A) IABG (B) CZA (C) WWF (D) WCC

(71) Which of the following is not famous park of Gujarat ?

(A) Safari Park (B) Indroda Park (C) Nehru zoological Park (D) Sakkarbaug

(72) Which zoological park is located in Junagadh ?

(A) Sakkarbaug (B) Indroda Park (C) Rani Jijamata Udhyan (D) Nehru Park

(73) Zoological park of Gangtok...

(A) Indroda Park (B) Nehru Park (C) Trivendram Park (D) Himalayan Park

(74) Which zoological park is located at Chennai ?

(A) Nehru Park (B) Arignar Anna (C) Himalayan (D) Rani Jijamata

(75) Where does Rani Jijamata zoo present ?

(A) Ahmedabad (B) Mumbai (C) Hydrabad (D) Chennai

(76) What is the location of Indroda park ?

(A) Ahmedabad (B) Junagadh (C) Sasangir (D) Gandhinagar

(77) Where is National Zoological Park located ?

(A) Mumbai (B) New Delhi (C) Hydrabad (D) Chennai

(78) How can we separate botanical garden from Herbarium ?

(A) to store plants (B) by research

(C) by library (D) by classification

(79) Which is not appropriate ?

(A) Hybridization and Cloning (B) Glass house and Conservatory

(C) Horticulture and Landscape gardening (D) Conservation and hybridization

(80) Which of the following provides facility to students for classification research ?

(A) Botanical garden (B) Herbarium (C) Gene Bank (D) Museum

(81) Whose members' can provide proper understanding of Landscape gardening ?

(A) of Herbarium (B) of gene bank (C) of botanical garden (D) of library

(82) What is developed for conservation of rare genes ?

(A) Gene bank (B) Green house (C) Herbarium (D) Library

(83) Which of the following possess skeleton gallery ?

(A) Zoological garden (B) Zoological museum (C) Botanical garden (D) Herbarium

(84) Which of the following helps to prepare flora and fauna of geographical regions ?

(A) Identification (B) Nomenclature (C) Classification (D) Geographical distribution

(85) In which of the following we can develop Fernery, Conservatory, Orchidium ?

(A) Botanical garden (B) Herbarium (C) Museum (D) Zoological museum

(86) Which of the following helps to measure endangered and extinction prone organisms ?

(A) Classification (B) Geographical isolation (C) Nomenclature (D) Identification

(87) What is done before we put plant specimens in the pigeon-holes of steel or wooden cupboards ?

(A) Pressing (B) Drying

(C) Fixed on proper sized herbarium sheet after pressing and drying

(D) Mounted on definite size of thick sheet

(88) What is related to Paris and Kew ?

(A) Herbarium of forest research, New York Botanical Garden

(B) Royal Botanical Garden, British Museum of Royal Botanical Garden

(C) Museum of Natural History, Royal Botanical Garden

(D) Herbarium of forest research, Lyoid Botanical Garden

(89) How can we separate botanical garden from public park and gardens ?

(A) Plant species are grown for conservation of environment in this type of gardens.

(B) Plant species are grown for natural beauty in this type of gardens.

(C) Various kinds of plants having medicinal and economic value and rare plants are also cultivated and maintained in this garden.

(D) Plant images are protected in this gardens.

(90) It can provide classification interrelationship.

(A) Horticulture (B) Landscape gardening (C) Botanical garden (D) Herbarium

(91) How can we preserve dead bodies of birds and mammals ?

(A) by using preservatives (B) by drying dead bodies

(C) killing and pressing (D) stuffing dried power

(92) Seed bank and Gene banks are protected in it...

(A) Herbarium (B) Botanical garden (C) Museum (D) Zoological park

Answers : (63-A), (64-C), (65-D), (66-B), (67-D), (68-D), (69-D), (70-B), (71-C), (72-A), (73-D), (74-B), (75-B), (76-D), (77-B), (78-C), (79-D), (80-B), (81-C), (82-A), (83-B), (84-C), (85-A), (86-B), (87-C), (88-C), (89-C), (90-C), (91-D), (92-B)

(93) Which of the following is inappropriate pair ?

(A) Hybridization - Tissue culture (B) Aesthetic appeal - the great Banyan tree

(C) Germplasm bank - Gene (D) Public service - Horticulture

(94) Find incorrect pair :

(A) Zoological survey of India - Kolkata (B) National Park - New Delhi

(C) Arignar Anna Park - Karnataka (D) Himalayan Park - Gangtok

(95) Find incorrect pair :

(A) Indian Botanical Garden - Dehradun (B) Lyoid Botanical Garden - Darjeeling

(C) National Botanical Garden - Lucknow (D) Central National Herbarium - Kolkata

(96) Choose two wrong pairs of the following :

(A) Nehru Park - New Delhi (B) Himalayan Park - Hydrabad

(C) Safari Park - Sasangir (D) Rani Jijamata Park - Mumbai

(A) B and C (B) C and D (C) A and B (D) B and D

(97) Which of the following is not correct for zoological park ?

(A) Snake house (B) Nocturnal house (C) Bird house (D) Glass house

(98) Which of the following is not true for Herbarium sheet ?

(A) Specimen is mounted. (B) It is put in sequence in cupboard.

(C) Poisoning is processed. (D) to examine by fumigation

(99) Which of the following is not correct for Herbarium ?

(A) Identification of plant specimens (B) provides facility for taxonomic research

(C) Seed bank can be developed (D) provides key for preparation of modern classification

(100) Which of the following is not correct for museum ?

(A) Maintenance of cultural heritage (B) Facility of Library and Laboratory

(C) Storage of fossils of animals and skeleton (D) Preservation of dead bodies by stuffing method

Answers : (93-A), (94-C), (95-A), (96-C), (97-D), (98-D), (99-C), (100-B)

- **A - Assertion, R - Reason questions :**

Choose the correct option of the following :

(A) A and R both are correct and R is the correct explanation of A.

(B) A and R both are correct but R is not correct explanation of A.

(C) A - True, R - False

(D) A - False, R - True

(101) Assertion A : Herbarium is a store house of plant specimens collected from different localities.
Reason R : There are specific methods to store plant specimens in Herbarium.
(A) (B) (C) (D)

(102) Assertion A : Specific methods like grafting, tissue culture, cloning and hybridization are used in Herbarium.
Reason R : Herbarium provides necessary information for verifying and identifying newly collected plant specimens.
(A) (B) (C) (D)

(103) Assertion A : Different plant species are grown in Botanical garden.
Reason R : Botanical garden provides material for research-work.
(A) (B) (C) (D)

(104) Assertion A : Botanical garden is different from public park and gardens.
Reason R : Plant species are grown for their identification in botanical garden.
(A) (B) (C) (D)

(105) Assertion A : Herbarium provides facilities for taxonomical research to students.
Reason R : Herbarium provides key for the preparation of modern system of classification.
(A) (B) (C) (D)

(106) Assertion A : The purpose of museum is to increase cultured heritage.
Reason R : The purpose of museum is to increase biodiversity.
(A) (B) (C) (D)

(107) Assertion A : A label occurs on lower side of the Herbarium sheet.
Reason R : It shows scientific name, famous name, family, etc...
(A) (B) (C) (D)

Answers : (101-B), (102-D), (103-A), (104-A), (105-B), (106-C), (107-D)

(108) Choose the correct pair :

Column - I	**Column - II**
(a) Museum of Natural History	(i) Vadodara
(b) British museum of Royal botanical garden	(ii) Dehradun
(c) Central National Herbarium	(iii) Kew (England)
(d) Herbarium of forest Research	(iv) Kolkata
(e) Herbarium, Department of Botany, M.S. University	(v) Paris (France)

(A) a-v, b-iii, c-iv, d-ii, e-i
(B) a-ii, b-iii, c-iv, d-v, e-i
(C) a-iii, b-iv, c-v, d-i, e-ii
(D) a-iv, b-v, c-i, d-ii, e-iii

(109) Choose the correct pair :

Column - I (Zoological park)	**Column - II** (Place)	
(a) Safari Park	(i) Trivendram	(A) a-ii, b-iii, c-iv, d-i, e-v
(b) Sakkarbaug	(ii) Gandhinagar	(B) a-iii, b-iv, c-v, d-ii, e-i
(c) Zoological Park	(iii) Sasangir	(C) a-iv, b-v, c-i, d-iii, e-ii
(d) Indroda Park	(iv) Junagadh	(D) a-v, b-i, c-ii, d-iv, e-iii
(e) Trivendram	(v) Vadodara	

(110) Choose the correct pair :

Column - I	**Column - II**	
(a) National Zoological park	(i) Gangtok	(A) a-ii, b-i, c-iii, d-iv, e-v
(b) Rani Jijamata zoo	(ii) Chennai	(B) a-iii, b-ii, c-iv, d-v, e-i
(c) Nehru Zoological park	(iii) Mumbai	(C) a-iv, b-iii, c-v, d-i, e-ii
(d) Himalayan	(iv) New Delhi	(D) a-v, b-iv, c-i, d-ii, e-iii
(e) Arignar Anna	(v) Hydrabad	

(111) Choose the correct pair :

Column - I (Botanical garden)	**Column - II** (Place)	**Column - III** (Area)
(a) Royal botanical garden	(1) Shibpur	(i) Britain
(b) New York botanical garden	(2) Lucknow	(ii) West Bengal
(c) Indian botanical garden	(3) Darjeeling	(iii) U.S.A.
(d) National botanical garden	(4) Vaghai	(iv) Kolkata
(e) Lyoid botanical garden	(5) Kew	(v) Uttar Pradesh
(f) Botanical garden	(6) New York	(vi) Gujarat

(A) (a-5-i) (b-6-iii) (c-1-iv) (d-2-v) (e-3-ii) (f-4-vi)

(B) (a-6-ii) (b-1-iv) (c-2-v) (d-3-vi) (e-4-iii) (f-5-i)

(C) (a-1-iii) (b-2-v) (c-3-vi) (d-4-i) (e-5-iv) (f-6-ii)

(D) (a-2-iv) (b-3-vi) (c-4-i) (d-5-ii) (e-6-v) (f-1-iii)

(112) Choose the correct pair :

Column - I (Museum)	**Column - II** (Place)	
(a) Natural History Museum	(i) Vadodara	(A) a-i, b-iv, c-iii, d-ii
(b) Zoological Survey of India	(ii) Chennai	(B) a-ii, b-i, c-iv, d-iii
(c) Government Museum	(iii) Kolkata	(C) a-iii, b-ii, c-i, d-iv
(d) Study of museology	(iv) Mumbai	(D) a-iv, b-iii, c-ii, d-i

(113) Choose the correct pair :

Column - I	**Column - II**
(a) Promotes ecotourism for employment	(i) Museum
(b) Stuffing method	(ii) Botanical garden
(c) Landscape gardening	(iii) Zoological park
(d) Key to prepare modern classification	(iv) Herbarium
(A) a-iv, b-ii, c-i, d-iii	(B) a-iii, b-i, c-ii, d-iv
(C) a-i, b-iii, c-iv, d-ii	(D) a-ii, b-iv, c-iii, d-i

(114) Choose the correct pair :

Column - I	Column - II	
(a) Fumigation method	(i) Botanical garden	(A) a-iii, b-ii, c-i, d-iv
(b) Conservation	(ii) Herbarium	(B) a-iv, b-iii, c-ii, d-i
(c) Ethanogallery	(iii) Zoological park	(C) a-i, b-iv, c-iii, d-ii
(d) Ex-situ conservation	(iv) Museum	(D) a-ii, b-i, c-iv, d-iii

(115) Choose the correct pair :

Column - I	Column - II
(a) To develop new species	(i) Stuffing
(b) Preservation of dead bodies of birds and mammals	(ii) Preservative
(c) Preservation of specimens in bottles of glass jars	(iii) Germplasm
(d) To obtain improved varities	(iv) Cloning
(A) a-iii, b-iv, c-i, d-ii	(B) a-iv, b-i, c-ii, d-iii
(C) a-i, b-ii, c-iii, d-iv	(D) a-ii, b-iii, c-iv, d-i

(116) Choose the correct pair :

Column - I	Column - II
(a) Herbarium	(i) Ex-situ conservation
(b) Botanical garden	(ii) Key to prepare modern classification
(c) Museum	(iii) Conserve natural resources
(d) Zoological park	(iv) Conserve biodiversity
(A) a-ii, b-iii, c-iv, d-i	(B) a-iii, b-iv, c-i, d-ii
(C) a-iv, b-i, c-ii, d-iii	(D) a-i, b-ii c-iii, d-iv

Answers : (108-A), (109-B), (110-C), (111-A), (112-D), (113-B), (114-D), (115-B), (116-A)

- **Questions for NEET :**

(117) Most famous botanical garden of the world ..... .

(A) Lyoid botanical garden, Darjeeling.
(B) New York botanical garden, U.S.A.
(C) Royal botanical garden, Kew, England.
(D) Royal botanical garden, Sydney, Australia.

(118) Which of the following is important function of botanical garden ?

(A) Tropical plants can be seen in botanical garden
(B) Ex-situ conservation of germplasm bank is done here
(C) It provides natural habitat to wild life
(D) It provides beautiful area for entertainment

(119) Which of the following is an effective method for conservation of biodiversity ?

(A) Development of gene bank
(B) Tissue culture
(C) Creating conserved biosphere
(D) Creating Botanical gardens

(120) What is it known where dry plant specimens are conserved and studied ?

(A) Arboreum
(B) Museum
(C) Vasculum
(D) Herbarium

(121) Which of the following is the largest Herbarium ?

(A) Kew (B) Geneva (C) Berlin (D) Sweden

(122) Where does the conservation of dry plant specimens occur ?

(A) Arboreum (B) Herbarium (C) Botanical garden (D) All of the above

(123) Botanical garden and Zoological garden means...

(A) Storage of native living species (B) Storage of foreign living species

(C) Storage of native and foreign living species (D) Storage of native plants and animals

(124) Which of the following is appropriate for germplasm bank ?

(A) To develop Herbarium sheets. (B) To develop new plant species

(C) To create seed bank (D) All of the above

(125) Choose the correct sequence to fill information on right side of Herbarium ?

(A) Scientific name - Family - Popular name - Location of its collection

(B) Popular name - Family - Location of its collection - Scientific name

(C) Family - Popular name - Scientific name - Location of its collection

(D) Location of its collection - Family - Popular name - Scientific name

(126) Which type of information is given to general public to understand botanical garden ?

(A) Tissue culture (B) Hybridization (C) Cloning (D) Landscape gardening

(127) Which are the fields integrated for research work by using living plant species from botanical gardens ?

(A) Anatomy (B) Cytology (C) Embryology (D) All of the above

Answers : (117-C), (118-B), (119-C), (120-D), (121-A), (122-B), (123-C), (124-C), (125-A), (126-D), (127-D)

Keys – Taxonomical science

- For identification of livings, artificial analytical instrument which is useful according rules.
- Generally every statement exhibits similarity or difference between two.
- Two types of keys - matter given within bracket and generally half-underilined matter, considered.

Botanical garden

- Habitat of living plants, collected for reference.
- Biggest Botanical garden is in Moscow.
- Most famous botanical garden in Kew, botanical garden.

Botanical museum/herberium

- Accumulation of plants on thick paper sheet is generally shown which is done by drying, pressing and preservation.
- Paper size 41 cm × 29 cm
- Biggest Botanical museum is in Kew.

Zoological park

- Method in which animals are protected freely in open area instead of cage. Based on specific proper protection.
- Study of wild animals and habit of their obtaining food can be understood properly.
- *ex-Situ* useful in breeding of nearly endangered species which is for animals.

Museum

- Exhibition of specimens of natural history their storage and preservation place. It is balance of livings showing art and nature together. It provides information of livings from other areas, along with local plants and animals.

3 Classification of Plant Kingdom

- Science is proper and perfect knowledge of everything.
- Biology : The study of living objects.
- Biology is divided into two branches : (1) Botany (2) Zoology
- Botany : This science investigates the internal and external structures of plants through evolutionary and phylogenetical aspect.
- Botanical science represents changes in inernal and external structure of plants based on evolution and speciation. By classification characteristics and functions of plants, cellular structure, Habitat, Adaptations, Nutrition, Interrelations, Reproduction, Life cycle, Importance etc. become more clear.

(1) The study of living objects is called as ..... .

(A) Biology (B) Chemistry (C) Physics (D) B and C

(2) The first formed life is known as ..... .

(A) Fungi (B) Plant kingdom (C) Monera (D) Protista

(3) Which is the main branch of biology ?

(A) Botany (B) Zoology (C) Biochemistry (D) A and B

(4) Which are the two different forms of life of protista ?

(A) Immobile plants (B) Mobile animals (C) Mobile plants (D) A and B

(5) Botany deals with the study of ..... .

(A) External structures of plants (B) Internal structures of plants

(C) Interrelationship of plants (D) A, B, C All

Answers : (1-A), (2-D), (3-D), (4-D), (5-D)

- **Plant kingdom :**

Scientist Eichler has classified the plant kingdom into two main group :

(1) Flowerless or seedless plants-cryptogrames (2) Flowering or seed bearing plant phanerogames

- Nonflowering or seedless plants are called cryptogams.
- Cryptogams are divided into three groups :

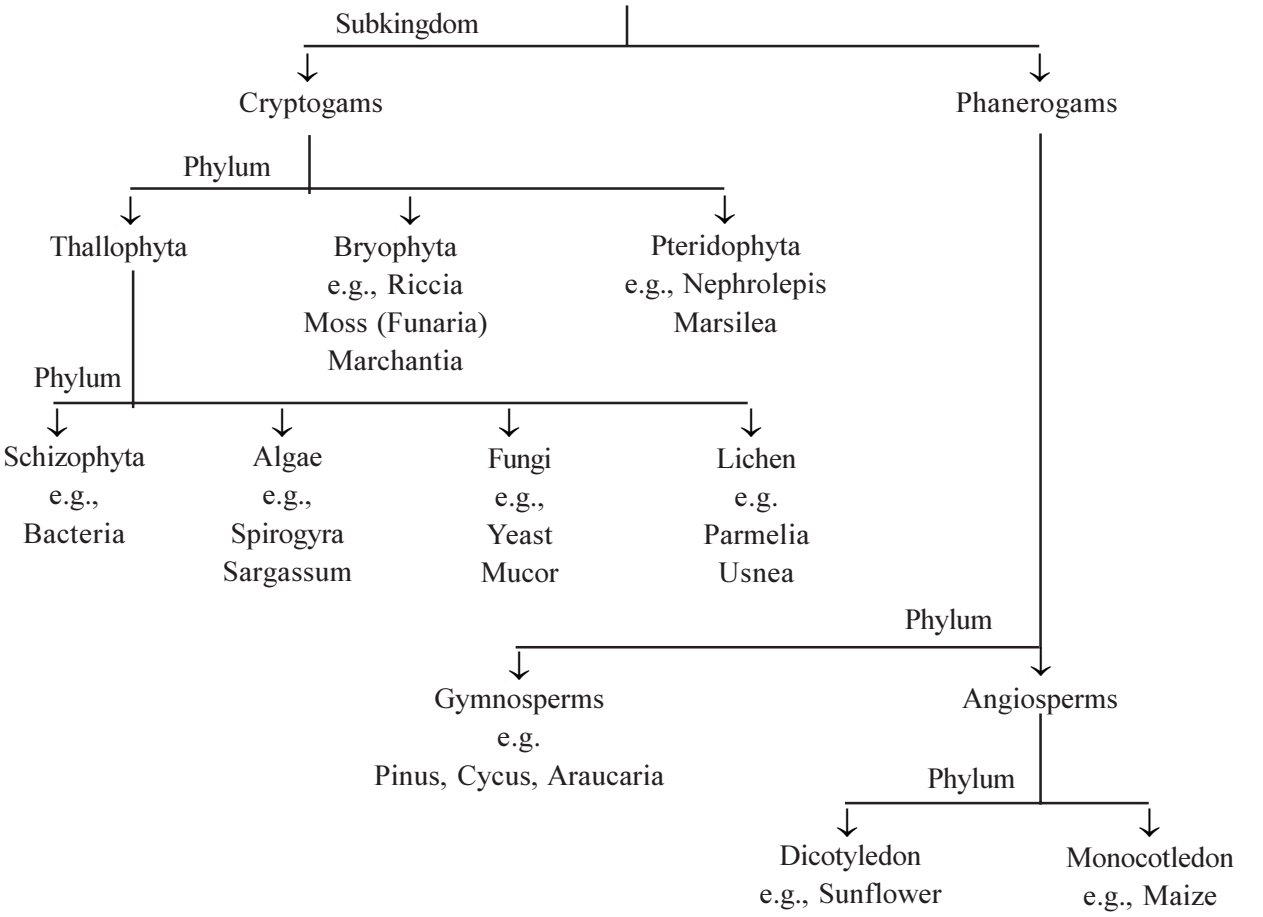
(1) Thallophyta (2) Bryophyta (3) Pteriodophyta

- Phanerogames are divided into two groups :

(1) Gymnosperms (2) Angiosperms

- The bryophyta, pteridophyta and phanerogams are included in embryophyta.

Kingdom - Plant



(6) How many groups of plant kingdom has been classified by Eichler ?

(A) Two (B) Three (C) Four (D) Five

(7) How many divisions are there of Cryptogams ?

(A) Three (B) Two (C) Four (D) Five

(8) Which of the following includes Thallophyta ?

(A) Angiosperms (B) Gymnosperms

(C) Cryptogams (D) Phanerogams

(9) Marchantia and Marsilea are included in which division respectively ?

(A) Bryophyta, Thallophyta (B) Bryophyta, Pteridophyta

(C) Moncot, Dicot (D) Pteridophyta, Bryophyta

(10) What is included in achlorophyllous phylum ?

(A) Lichen (B) Fungi (C) Algae (D) Schizophyta

(11) Which division includes Araucaria ?

(A) Pteridophyta (B) Dicot (C) Angiosperms (D) Gymnosperms

(12) Which of the following is included embryophyta ?

(A) Bryophyta (B) Pteridophyta (C) Phanerogams (D) A, B, C All

Answers : (6-A), (7-A), (8-C), (9-B), (10-B), (11-D), (12-D)

- **Five kingdom classification :**

R. H. Whittaker (1969) gave five kingdom classification on the basis of nucleus, cell structure, Nutrition and major ecological role.

(1) Monera (2) Protista (3) Fungi (4) Plantae (5) Animalia

A Greek Naturalist Theophrastus (370 - 285 B.C.) classified plants into four groups based on their habitat. Theophrastus is regarded as Father of Botany.

A swedish naturalist Linnaeus classified plants into 24 groups on the basis of sexual characters. Linnaaeus (1770 – 1778) is considered as the father of Taxonomy.

(13) Five kingdom classification of plants was given by .....

(A) R. H. Whittaker (B) Carolas Linnaeus (C) Theophrastus (D) Eichler

(14) Who is considered as the father of Botany ?

(A) Whittaker (B) Theophrastus (C) Eichler (D) Linnaeus

(15) Who is considered as the father of Taxonomy ?

(A) Whittaker (B) Linnaeus (C) Eichler (D) Theophrastus

(16) How many groups have been divided by Theophrastus and Linnaeus ?

(A) 24, 4 (B) 4, 24 (C) 5, 4 (D) 2, 3

(17) Which criteria have been used for plant classification by Theophrastus and Linnaeus ?

(A) Habitat (B) Sexual characters (C) Importance (D) A and B

(18) In which kingdom organisms having nucleoid and without membrane bound organelles are included ?

(A) Protista (B) Fungi (C) Monera (D) Plant

(19) Cyanobacteria are included in .....

(A) Fungi (B) Protista (C) Plant (D) Monera

(20) In which kingdom dino flagellates are included ?

(A) Plant kingdom (B) Animal kingdom (C) Protista (D) Monera

(21) Which mode of nutrition occurs in prokaryotes ?

(A) Saprophytic (B) Autotrophic (C) Heterotrophic (D) B and C

(22) In which plant embryo is not formed as a result of sexual reproduction ?

(A) Yeast (B) Mucor

(C) Mushroom and slime mould (D) A, B, C All

(23) Which substance is present in cell wall of fungi ?

(A) Carbohydrates (B) Chitin (C) Lipid (D) Pectin

(24) Which kingdom includes all multicellular, aquatic or terrestrial eukaryotic organisms ?

(A) Monera (B) Plant kingdom (C) Animal kingdom (D) Fungi

(25) In which plant embryo is not formed as a result of sexual reproduction ?

(A) Bryophytes (B) Pteridophytes (C) Algae (D) Angiosperms

(26) Which kingdom is kingdom of Prokaryotes ?

(A) Protista (B) Fungi (C) Plantae (D) Monera

(27) Which fungi is structurally unicellular ?

(A) Bread mould (B) Slime mould (C) Mushroom (D) Yeast

Answers : (13-A), (14-B), (15-B), (16-B), (17-D), (18-C), (19-D), (20-C), (21-D), (22-D), (23-B), (24-B), (25-C), (26-D), (27-D)

- **Three-domains classification :**

The three domains classification system was given by Carl Woese. In this system prokaryotic and eukaryotic organisms are divided into three domains :

(1) Archae domain :

- They are prokaryotic cells without nuclear membrane.
- Cell was does not contain peptidoglycan.
- Archaea live in extreme condition.
- Example : Methanogens, Halophiles, Thermoacidophiles.

(2) Bacteria domain :

- They are prokaryotic cell. Cell wall is made up of peptidoglycan. This kingdom includes most pathogenic prokaryotic organisms.
- Example : Cyanobacteria, Spirochaete, Firmicutes

(3) Eukarya domain :

- They are eukaryotes, cell wall is absent, if present made up of cellulose or fungus cellulose.
- The eukarya domain is divided into four kingdoms :

(1) Protista (2) Fungi (3) Animalia (4) Plantae

---

(28) Who has given three-domain classifications ?

(A) Whittaker (B) Linnaeus (C) Woese (D) Eichler

(29) In which bacteria cell wall is not made up of peptidoglycan ?

(A) Methanogens (B) Halophiles (C) Thermoacidophiles (D) A, B, C All

(30) Which bacteria is responsible for the production of biogas ?

(A) Cyanobacteria (B) Methanogens (C) Spirochaete (D) Halophiles

(31) Which bacteria can live in extreme salty area ?

(A) Spirochaete (B) Halophiles (C) Methanogens (D) Firmicutes

(32) Which bacteria can survive in hot spring ?

(A) Methanogens (B) Firmicutes (C) Thermoacidophiles (D) Halophiles

(33) In which bacteria cell wall is made up of peptidoglycan ?

(A) Cyanobacteria (B) Spirochaete (C) Firmicutes (D) A, B, C All

(34) Which of following is photosynthesizing bacteria ?

(A) Spirochaete (B) Firmicutes (C) Cyanobacteria (D) Halophiles

(35) Which bacteria known as Gram positive and Gram negative bacteria respectively ?

(A) Firmicutes - Spirochaete (B) Halophiles - Methanogens

(C) Cyanobacteria - Spirochaete (D) Spirochaete - Firmicutes

(36) Which are the simple, predominantly unicellular eukaryotic organisms ?

(A) Protista (B) Euglenoids (C) Slime moulds (D) A, B, C All

(37) Which unicellular or multicellular eukaryotes obtain nutrients by absorption ?

(A) Sac - fungi (B) Club - fungi (C) Yeasts and moulds (D) A, B, C All

(38) Which animals obtain nutrients primarily by ingestion ?

(A) Worms (B) Insects (C) Sponges (D) A, B, C All

(39) Which living organisms obtain nutrients by photosynthesis and absorption ?

(A) Bryophytes (B) Pteridophytes (C) Conifers (Gymnosperms) (D) A, B, C All

Answers : (28-C), (29-D), (30-B), (31-B), (32-C), (33-D), (34-C), (35-A), (36-D), (37-D), (38-D), (39-D)

- **Viroids :**
  - Viroids are discoved by Diener. Viroids are smaller than viruses. Viroid consists of a very simple structure and short RNA strand and capsid is absent.
  - Viroids cause potato spindle tuber disease in potato and Alzheimers disease in human being.

(40) Who discovered viroids ?

(A) Invanowsky (B) Diener (C) Whittaker (D) Linnaeus

(41) Which organisms are smaller than viruses ?

(A) Viroids (B) Algae (C) Bacteria (D) Yeast

(42) Which diseases occur by viroids in potato and human respectively ?

(A) Tuber, Alzheimer's (B) Tobacco mosaic disease, Diabetes

(C) Chlorosis, Alzheimer's (D) Alzheimer's, Tuber

(43) Which organisms consists of a very simple structure, short RNA strand and absence of capsid ?

(A) Viruses (B) Viroids (C) Bacteria (D) Fungi

Answers : (40-B), (41-A), (42-A), (43-B)

- **Viruses :**
  - Pasteur gave name viruses, they are causative agents of infectious diseases.
  - Invanowsky discovered Tobacco Mosaic virus.
  - Viruses are utra-microscopic, crystalline, self reproducing and obligate parasite in living cells.
  - Viruses are intermediate between living and non-living things hence they are called living chemical.
  - Virus is living chemical which contains DNA or RNA as nucleoprotein. Surrounding the virus protective protein layer capside is found which is made up of capsomere.
  - Plant Viruses - TMV, BSV
  - Animal Viruses - Polio viruses, Small pox viruses.
  - Viruses that live on bacteria are known as bacteriophages.

(44) Who has given the name virus first ?

(A) Invanowsky (B) Diener (C) Pasteur (D) Woese

(45) Who first time recognized that tobacco mosaic virus are causative organism for tobacco mosaic disease ?

(A) Pasteur (B) Diener (C) Invanowsky (D) Eichler

(46) Who has discovered TMV ?

(A) Diener (B) Invanowsky (C) Pasteur (D) Linnaeus

(47) Which organism can pass through bacteria proof filters ?

(A) Viruses (B) Bacteria (C) Algae (D) Fungi

(48) Which organism behaves as intermediate of living and non-living things ?

(A) Bacteria (B) Viruses (C) Algae (D) Fungi

(49) Which substance is present in Viral Capsid ?

(A) Protein (B) Lipid (C) Carbohydrate (D) Nucleic acid

(50) The organism which engulf bacteria is known as....

(A) Virus (B) Bacteriophages (C) Viroids (D) A and C

Answers : (44-C), (45-C), (46-B), (47-A), (48-B), (49-A), (50-B)

- **Algae :**
  - Algae is the first existing group of plants on the earth. The study of algae is called Algology or Phycology.
  - Professor Iyengar is considered as the father of modern algology in India.
  - Algae is found in fresh water, ocean or moist habitat. The plant body does not possess root, stem and leaves, So it is known as thallus.
  - Algae are unicellular or multicellular, prokaryotic or eukaryotic, filamentous of colonial in form.

- Algae consists chlorophylls and other photosynthetic pigments, therefore algae are autotrophic in nutrition.
- They contain starch as reserve food material.
- Algae reproduce by vegetative, asexual and sexual method.
- Sex organs are naked, after fertilization zygote does not develop into embryo.
- Examples : Nostoc, Chlamydomonas, Spirogyra and volvox.

(51) Which is the first existing group of plants on the earth ?

(A) Fungi (B) Algae (C) Lichen (D) Angiosperm

(52) Who is considered as the father of modern algology in India ?

(A) Linnaeus (B) Eichler (C) Professor Iyengar (D) Woese

(53) Which is the example of Unicellular algae ?

(A) Oscillatoria (B) Spirogyra (C) Nostoc (D) Chlamydomonas

(54) Which plastid is included in photosynthetic pigments ?

(A) Xanthophyll (B) Phycocynin (C) Phycoerythrin (D) A, B, C All

(55) Which types of reproductive method is observed in algae ?

(A) Fragmentation (B) By spores (C) By conjugation (D) A, B, C All

(56) In algae storage of food occur in which form ?

(A) Starch (B) Protein (C) Lipid (D) Nucleic acids

(57) Which plant group possess thallus structure ?

(A) Pteridophyta (B) Angiosperms (C) Algae (D) Gymnosperms

Answers : (51-B), (52-C), (53-D), (54-D), (55-D), (56-A), (57-C)

- **Fungi :**
  - The study of fungi is called Mycology. Fungi show cosmopolitan distribution.
  - The plant body is called mycelium. Cell wall is made up of fungus-cellulose (chitin). Chlorophyll is absent so Fungi cannot prepare its own food material.
  - It is heterotroph or saprophyte. Reserved food is stored as Glycogen and fat droplet form.
  - Fungi reproduce by vegetative, asexual and sexual method.
  - Sexual reproduction may be isogamous, heterogamous or oogamous type.
  - Sexual reproduction take place in three phase : (1) Plasmogamy (2) Karyogamy (3) Meiosis
  - Examples : Yeast, Mucor, Agaricus, Penicillium

(58) The plant body of fungi is called as ..... .

(A) Herbs (B) Mycelium (C) Shrubs (D) Tree

(59) In which form is reserve food material present in fungi ?

(A) Glycogen (B) Animal starch (C) Fat droplets (D) A, C

(60) In fungi how many types of sexual reproduction takes place ?

(A) One (B) Two (C) Three (D) Four

(61) On the basis of nutritions fungi is included in which group ?

(A) Autotrophic (B) Heterotrophic

(C) Saprophytes (D) B and C

(62) Which fungi is used as antibiotics ?

(A) Yeast (B) Mucor (C) Penicillium (D) Mushroom

(63) Which types of spores occurs in fungi ?

(A) Zoospores (B) Aplanospores (C) Pycnidiospors (D) A and B

(64) Which plant is achlorophyllus and unicellular ?

(A) Chlamydomonas (B) Yeast (C) Mould (D) Penicillium

Answers : (58-B), (59-D), (60-C), (61-D), (62-C), (63-D), (64-B)

- **Lichen :**
  - The study of lichen is called lichenology. It is symbiosis of algae and fungi.
  - Lichen discovered by Tulsane.
  - The algal component is called phycobiont and fungal component is called mycobiont.
  - Fungi absorb water and mineral nutrients from environment and provides to algae and algae synthesize food by photosynthesis.
  - Lichens reproduce asexually and sexually. The fruiting body produce sex organ.
  - The fruiting bodies of lichens are called apothecium or perithecium.
  - On the basis of external form lichens are of three types : (1) Crustose lichen (2) Foliose lichen (3) Fruticose lichen
  - Examples : Strigula, Parmelia, Usnea.

(65) Who discoverd lichen ?

(A) Professor Iyengar (B) Tulsane (C) Shiv Ram Kashyap (D) Eichler

(66) In lichen algae and fungi structure respectively called as ..... .

(A) Mycobiont, Phycobiont (B) Apothecium, Perithecium

(C) Phycobiont, Mycobiont (D) Oidiospores, Pycnidiospores

(67) Asexually lichen reproduce by which spores ?

(A) Oidiospores (B) Pycnidiospores (C) Aplanospores (D) A and B

(68) How many types are there of lichen on the basis of external structure ?

(A) One (B) Two (C) Three (D) Four

(69) Which type of shape is found in fruiting body of lichen ?

(A) Cup shaped (B) Flask shaped (C) Round shaped (D) A and B

(70) Which of the following is odd for lichen ?

(A) Crustose lichens (B) Foliose lichens (C) Fruticose lichens (D) Gemmae

(71) The example of lichen is ..... .

(A) Strigula (B) Parmelia (C) Usnea (D) A, B, C All

Answers : (65-B), (66-C), (67-D), (68-C), (69-D), (70-D), (71-D)

- **Bryophytes :**
  - These plants occupy a position between algae and pteridophyta.
  - Professor Shiv Ram Kashyap is considered as the father of Indian bryology.
  - It is autotrophic as it contains chlorophyll.
  - Botanist Rothmelar has divided the bryophytes into three classes : (1) Hepaticopsida (2) Anthocerotopsida (3) Bryopsida.
  - Bryophytes lack vascular tissue, their fertilization occur only in water. After fertilization embryo is not formed.
  - The life cycle of bryophytes has two distinct phase (1) Gametophytic (2) Sporophytic; alternating with each other is called alternation of generation.
  - Example : Riccia, Anthoceros, Marchantia, Funaria.

(72) Which plants occupy position between algae and pteridophyta ?

(A) Algae (B) Bryophytes (C) Angiosperms (D) Gymnosperms

(73) Who classified bryophytes into Hepaticopsida, Anthocerotopsida and Bryopsida ?

(A) Professor Shiv Ram (B) Tulsane (C) Rothmelar (D) Iyengar

(74) In which plant group vascular tissues are absent ?

(A) Bryophyta (B) Pteridophyta (C) Gymnosperms (D) Angiosperms

(75) In which plants fertilization occurs only in water ?

(A) Riccia (B) Funaria (C) Anthoceros (D) A, B, C All

(76) In which plant group gametophytic stage is main and sporophytic phase is subsidiary ?

(A) Pteridophyta (B) Gymnosperms (C) Bryophyta (D) Angiosperms

(77) In which plant group gametophytic plant body is either thalloid or erect ?

(A) Funaria (B) Selaginella (C) Araucaria (D) Morpichh

(78) Which type of reproductive method is observed in anthoceros ?

(A) Fragmentation (B) Gemmae (C) Sexual reproduction (D) A, B, C All

Answers : (72-B), (73-C), (74-A), (75-D), (76-C), (77-A), (78-D)

- **Pteridophytes :**
  - They were the first land plants on the earth. They are usually terrestrial and grow in moist or shady habitat.
  - They consist root, stem and leaves with well developed vascular tissue.
  - Zygote undergoes divisions to form embryo.
  - The sporophylls are of two types : (1) Homophyllous (2) Heterophyllous
  - The pteridophyte shows alternation of generation.
  - Example : Nephrolepis, Equisetum, Selaginella, Rhynia (Fossil)

(79) Which are the first land plants on the earth ?
(A) Pteridophytes (B) Bryophytes (C) Monocot (D) Dicot

(80) In which plant group first vascular tissue is observed ?
(A) Bryophytes (B) Pteridophytes (C) Gymnosperms (D) Angiosperms

(81) From where the spores are produced in pteriophytes ?
(A) Sporangia (B) Microspores (C) Megaspores (D) A, B, C All

(82) Which are the fossil pteriophyte plant ?
(A) Bennettites (B) Rhynia (C) Nephrolepis (D) Selaginella

(83) Which stage is responsible for sexual reproduction in pteridophyta ?
(A) Gametophyte (B) Sporophyte (C) Vegetative (D) A and B

Answers : (79-A), (80-B), (81-A), (82-B), (83-A)

- **Gymnosperms :**
  - Gymnosperms vary in size from small plants to very large gigantic plants.
  - The tallest tree is Sequoia, about 150 meters in height. While *zamia pygmea* is smallest gymnosperm having tuberous stem.
  - Gymnosperm possess two type of leaves : (1) Foliage leaves (2) Scaly leaves
  - They are evergreen, perennial, xerophytic plant which possess vascular tissue.
  - They show alternation of generation.
  - Sporophylls are arranged on central axis in the form of cones. Cones are unisexual and gymnosperms are heterosporous.
  - Example : Cycus, Pinus, Araucaria, Bennettites (Fossil plant), Thuja (Morpichh).

(84) Which plant show air pollination, pre-fertilization and single fertilization ?
(A) Gymnosperms (B) Monocot (C) Angiosperms (D) Dicot

(85) In which plant ovule are orthotropous and lacks the fruit ?
(A) Pinus (B) Cycus (C) Morpichh (D) A, B, C All

(86) In which plant ovules are naked ?
(A) Angiosperm (B) Gymnosperm (C) Monocot (D) Dicot

(87) Which plant is tallest in the world ?

(A) *Zamia pygmea* (B) Agave (C) Sequoia (D) Rafflesia

(88) Which of following plant is smallest gymnosperms having an underground tuberous stem ?

(A) Pinus (B) *Zamia pygmea* (C) *Cycus* (D) Sequoia

(89) Which is the fossil gymnosperm ?

(A) Pinus (B) Bennettites (C) Cycus (D) Morpichh

(90) The characters of gymnosperm is ..... .

(A) Hydrophytes (B) Mesophytes (C) Halophytes (D) Xerophytes

(91) How many types of leaves are there in Gymnosperms ?

(A) One (B) Two (C) Three (D) Four

Answers : (84-A), (85-D), (86-B), (87-C), (88-B), (89-B), (90-D), (91-B)

- **Angiosperm :**
  - Angiosperms, are most dominant and biggest plant group all over world.
  - Plant species of angiosperms vary in size i.e. smallest plant is *Wolffia globosa* which is 2-5 mm in size where as largest plant is *Eucalyptus sp.* with a height of about 90 to 100 meters in Australia.
  - *Rafflesia arnoldi* possesses the largest flower with a weight of about 8 kg and diameter about 1 mt. *Agave sp.* consists largest inflorescence of about 6 mt height.
  - Sporophytic plant body is in the form of herbs. Shrubs, trees, climbers or lianas. Vascular tissue are well developed and ovules are enclosed in the ovary.
  - Flowers are unisexual or bisexual. They include two accessory whorls (calyx and corolla) and two necessary whoris (Androeium and Gynoecium).
  - Pollination takes place through air, insects and birds. Angiosperm shows double and post fertilization.
  - The life cycle shows alternation of generation. After fertilization ovules are transformed into seed and ovary into fruit.

(92) Which group of plants represent maximum species and occupies first position on earth ?

(A) Angiosperm (B) Gymnosperm (C) Bryophyte (D) Pteridophyte

(93) Which characters are observed in angiosperm ?

(A) Hydrophytic (B) Xerophytic (C) Mesophytic (D) A, B, C All

(94) Which is the smallest and largest angiosperm plant ?

(A) *Wolffia globosa, Zamia Pygmea* (B) Eucalyptus, Sequoia

(C) *Wolffia globosa,* Eucalyptus (D) A and B

(95) What is the size of *Wolffia globosa* ?

(A) 2 - 5 mm (B) 1 - 2 mm (C) 5 - 10 mm (D) 3 - 4 mm

(96) What is the height of largest inflorescence of *Agave sp* ?

(A) About 6 meters (B) 5 meters (C) 4 meters (D) 8 meters

(97) What is weight of *Rafflesia arnoldii* ?

(A) 8 Kg (B) 7 Kg (C) 6 Kg (D) 5 Kg

(98) Pollination takes place through which in Angiosperm ?

(A) Air (B) Insects (C) Birds (D) A, B, C All

Answers : (92-A), (93-D), (94-C), (95-A), (96-A), (97-A), (98-D)

- **Bentham and Hooker's classification :**
  - Bentham and Hooker classified the angiosperms into two classes (1) Dicotyledon (2) Monocotyledon.
  - **Dicotyledon :**
    - Embryo possesses two cotyledons. Flowers are pentamerous and leaves show reticulate venation. Example : Sunflower
    - Class dicotyledon is divided into three sub classes.

    (1) Polypetalae : Petals are free in flower. It includes three series :

    (a) Thalamiflorae (b) Disciflorae (c) Calyciflorae

    (2) Gamopetalae : Petals are fused in the flower. It includes three series.

    (a) Inferae (b) Heteromerae (c) Bicarpellatae

    (3) Monochlamydea : Flower possess one whorl of perianth. It has no order but includes 8 series.
  - **Monocotyledon :**
    - Embryo possesses single cotyledon. Leaves show parallel venation. Flower are trimerous. Example : Maize
    - This class does not include any order but it is divided into 7 series. Example : *Alium cepa* local name : Onion

(99) Whose classification is used by most of the well known herbaria of the world ?

(A) Eichler (B) Bentham and Hooker (C) Linnaeus (D) R. H. Whittaker

(100) Thalamiflorae, disciflorae and calyciflorae respectively possess how many orders ?

(A) 6, 4, 5 (B) 3, 3, 4 (C) 5, 4, 6 (D) 6, 5, 4

(101) How many orders are there in inferae, heteromerae and bicarpellatae respectively ?

(A) 4, 3, 2 (B) 3, 3, 4 (C) 6, 4, 5 (D) 6, 5, 4

(102) *Catharanthus roseus* is name of which plant ?

(A) Rose (B) Hibiscus (C) Barmasi (D) Boganvel

(103) Boganvel is included in series ..... .

(A) Polypetalae (B) Gamopetalae (C) Monochlamydeae (D) A and B

(104) Which plant has no order ?

(A) Onion (B) Boganvel (C) Maize (D) A, B, C All

(105) *Hibiscus rosa sinensis* is scientific name of ..... .

(A) Rose (B) Sunflower (C) Shoe-flower (D) Lemon

Answers : (99-B), (100-A), (101-B), (102-C), (103-C), (104-D), (105-C)

- **Plant life cycle and alternation of generation :**
  - The life cycle of plant shows two phases : (1) Gametophytic (2) Sporophytic alternating with each other is known as alternation of generation.
  - The different plant groups shows their alternation of generation into three pattenrs :

    (1) Haplontic life cycle : Example : Volvox and Spirogyra

    (2) Diplontic life cycle : Example : Gymnosperms, Angiosperms and exceptionally fucus alga.

    (3) Haplo-diplontic life cycle : Example : Ectocarpus and other sea weeds.

---

(106) How many patterns of different plant groups show their alternation of generations ?

(A) One (B) Two (C) Three (D) Four

(107) Which plant shows haplontic life cycle ?

(A) Fucus (B) Ectocarpus (C) Volvox (D) Nostoc

(108) Diplontic life cycle occurs in ..... .

(A) Gymnosperm (B) Angiosperm (C) Fucus-algae (D) A, B, C All

(109) Which type of life cycle Spirogyra shows ?

(A) Haplontic (B) Diplontic (C) Haplo-Diplontic (D) Triplontic

(110) In which plants haplo-diplontic life cycle occurs ?

(A) Bryophyte (B) Pteridophyte (C) Ectocarpus (D) A, B, C All

(111) Naked and orthotropous ovules occur in which plant ?

(A) Pinus (B) Maize (C) Sunflower (D) Agave

(112) Which of following is not included in gynoecium ?

(A) Stigma (B) Connective (C) Style (D) Ovary

(113) How many necessary whorls are there in typical flower ?

(A) One (B) Two (C) Three (D) Four

Answers : (106-C), (107-C), (108-D), (109-A), (110-D), (111-A), (112-B), (113-B)

- **True-False (T - F) types questions :**

Select true or false option in following sentences.

(114) (1) Volvox and spirogyra shows haplontic life cycle.

(2) Fucus algae shows diplontic life cycle.

(3) Ectocarpus shows Haplo-diplontic life cycle.

(4) Gametophytic phase is main in pteridophyte.

(5) Sporophytic phase is main in bryophytes.

(A) T, T, T, F, F (B) T, F, T, F, T (C) T, T, F, T, T (D) F, F, T, T, T

(115) (1) Smallest angiospermic plant is *Wolffia.*

(2) The size of Wolffia is 5-10 mm.

(3) The largest angiosperm occurs in Australia.

(4) *Zamia pygmea* having an underground tuberous stem.

(5) Gymnosperm plants have xerophytic characters.

(A) T, T, F, F, T (B) T, F, T, T, T (C) F, T, F, F, T (D) F, T, T, F, T

(116) (1) Pteridophyte possess two type of sporophylls.

(2) Angiosperm shows double fertilization.

(3) In Gymnosperm pollination occurs by insects.

(4) In Bryophytes fertilization occurs by air.

(5) Algae is endospermic plant.

(A) F, T, F, T, F (B) T, F, T, F, T (C) T, T, F, F, F (D) F, F, T, F, T

(117) (1) Bennetties is fossil pteridophytes.

(2) Rhynia is fossil bryophytes.

(3) Bryophytes lack vascular tissue.

(4) Fungal component of lichen is called phycobiont.

(5) In fungus sexual reproduction takes place in two phases.

(A) F, F, T, F, T (B) T, T, T, F, F (C) T, T, F, T, T (D) F, T, F, T, F

(118) (1) Fungi stores food in glycogen form.

(2) Algae stores food in starch form.

(3) Viroids cannot cause disease in plants.

(4) Viruses are discovered by Invanowsky.

(5) Algae is the first existing group of plants on the earth.

(A) T, T, F, T, T (B) F, T, F, T, F (C) F, F, T, F, F (D) T, T, T, F, T

(119) (1) Eichler has classified the plant kingdom into two main groups.

(2) The three domain classification system was given by Woese.

(3) A five kingdom classification system was given by Whittaker.

(4) Shiv Ram Kashyap is considered as the father of algology.

(5) Rothmelar is considered as the father of bryophytes.

(A) F, F, T, F, F (B) T, F, T, F, T (C) T, T, T, F, F (D) T, T, F, F, T

Answers : (114-A), (115-B), (116-C), (117-A), (118-A), (119-C)

- **A - Assertion and R - Reason type questions :**

Which option is correct for given question ?

(A) A and R both are true and R is correct explanation of the A.

(B) A and R both are true and R is not correct explanation of the A.

(C) A is true and R is false.

(D) A is false and R is true.

(120) Assertion A : The first formed life is known as protista.

Reason R : They are of various forms on the basis of their development.

(A) (B) (C) (D)

(121) Assertion A : Cyanobacteria possess rigid cell wall, lacks membrane bound organelles and nucleoprotein as genetic material.

Reason R : Cyanobacteria are included in kingdom monera.

(A) (B) (C) (D)

(122) Assertion A : Methanogens lived in extreme condition.

Reason R : Methanogens are included in Archaea domain.

(A) (B) (C) (D)

(123) Assertion A : Viruses are intermediate of living and non-living.

Reason R : Viruses can pass through bacteria-proof filters.

(A) (B) (C) (D)

(124) Assertion A : Algae possess plastids and various photosynthetic pigments.

Reason R : In algae after fertilization, zygote develops into embryo.

(A) (B) (C) (D)

(125) Assertion A : Lichens is a symbiotic structure of algae and fungi.

Reason R : The fruiting body of lichen is called apothecium or perithecium.

(A) (B) (C) (D)

(126) Assertion A : The bryophyte and pteridophyte plants grow in shady habitat.

Reason R : In bryophyte and pteridophyte after fertilization zygote develops into embryo.

(A) (B) (C) (D)

(127) Assertion A : The height of sequoia is about 150 meters.

Reason R : The agave sp. consists of about 6 meter hight.

(A) (B) (C) (D)

(128) Assertion A : Polypetalae are not classified on the basis of thalamus.

Reason R : Gamopetalae are not classified on the basis of ovary.

(A) (B) (C) (D)

Answers : (120-C), (121-A), (122-A), (123-B), (124-C), (125-B), (126-B), (127-C), (128-D)

(129) Select correct option :

Column - I	Column - II	
(a) Eichler	(i) Five kingdom classification	(A) a - iv, b - iii, c - ii, d - i
(b) Linnaeus	(ii) Father of Botany	(B) a - ii, b - iv, c - i, d - iii
(c) Theophrastus	(iii) Father of classification	(C) a - iv, b - iii, c - i, d - ii
(d) Whittaker	(iv) Plant kingdom classified into two group	(D) a - iii, b - iv, c - i, d - ii

(130) Select correct option :

Column - I	Column - II	
(a) Phycobiont	(i) Haplo-diplontic	(A) a - ii, b - i, c - iv, d - iii
(b) Strigula	(ii) Cup-shaped	(B) a - iii, b - iv, c - ii, d - i
(c) Apothecium	(iii) Algae	(C) a - ii, b - iii, c - iv, d - i
(d) Perithecium	(iv) Lichens	(D) a - iv, b - i, c - ii, d - iii

(131) Select correct option :

Column - I	Column - II	
(a) Wolffia	(i) 1 meter diameter	(A) a - ii, b - iii, c - iv, d - i
(b) Rafflesia	(ii) 150 meter height	(B) a - iii, b - iv, c - i, d - ii
(c) Sequoia	(iii) 6 meter high inflorescence	(C) a - iv, b - iii, c - i, d - ii
(d) Agave	(iv) 2 - 5 mm	(D) a - iv, b - i, c - ii, d - iii

(132) Select correct option :

Column - I	Column - II	
(a) TMV	(i) Human	(A) a - iv, b - i, c - ii, d - iii
(b) Alzheimer's	(ii) Potato	(B) a - iii, b - iv, c - i, d - ii
(c) Spindle tuber	(iii) Animal viruses	(C) a - ii, b - iii, c - iv, d - i
(d) Polio virus	(iv) Tobacco mosaic viruses	(D) a - iv, b - ii, c - i, d - iii

(133) Select correct option :

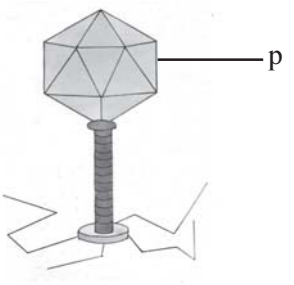
Column - I	Column - II	
(a) Thalamiflorae	(i) 5 order	(A) a - iv, b - iii, c - ii, d - i
(b) Disciflorae	(ii) 3 order	(B) a - ii, b - iii, c - iv, d - i
(c) Heteromerae	(iii) 4 order	(C) a - iii, b - iv, c - ii, d - i
(d) Calyciflorae	(iv) 6 order	(D) a - iv, b - i, c - ii, d - iii

(134) Select correct option :

Column - I	Column - II	
(a) Blue green algae	(i) Spirogyra	(A) a - iv, b - iii, c - ii, d - i
(b) Red algae	(ii) Sargassum	(B) a - ii, b - iii, c - iv, d - i
(c) Brown algae	(iii) Batracospermum	(C) a - iii, b - iv, c - ii, d - i
(d) Green algae	(iv) Nostoc	(D) a - iv, b - i, c - ii, d - iii

Answers : (129-A), (130-B), (131-D), (132-A), (133-A), (134-A)

(135) In given figure 'p' indicates ..... .



(A) Head (B) Capsid

(C) Filaments (D) Tail

(136) In given diagram 'p' indicate which plant ?



(A) Nephrolepis (B) Selaginella

(C) Equisetum (D) Zamia

(137) In given diagram, plant belongs to group ..... .



(A) Bryophytes (B) Pteridophytes

(C) Gymnosperm (D) Angiosperm

(138) In given diagram the storage of food occur in which form ?



(A) Starch (B) Cellulose

(C) Glycogen (D) Chitin

(139) Which plant is shown in given diagram ?



(A) Wolffia (B) Rafflesia

(C) Cycus (D) Pinus

(140) Select the scientific name of plant of given diagram.



(A) *Hibiscus rosa-Sinensis*

(B) *Rosa-indica*

(C) *Helianthus annuus*

(D) *Catharanthus roseus*

Answers : (135-A), (136-B), (137-A), (138-C), (139-B), (140-C)

- **Questions for NEET :**

(141) The locomotion of male gamete of bryophyte and pteridophyte towards female gamete is known as...

(A) Chemotaxis (B) Phototaxis (C) Phototropism (D) Hydrotropism

(142) Which fungi is used as food ?

(A) Mucor (B) Agaricus (C) Penicillium (D) Rizopus

(143) Pteridophyte plant differ from bryophyte on which basis ?

(A) Zoospores (B) Vascular tissue (C) Aplanospores (D) Male reproductive organ

(144) What is the specific character of bryophyte ?

(A) Absence of vascular tissue (B) Absence of Root

(C) To produce spore (D) Alternation of generation

(145) Which types of lifecycle occurs in spirogyra ?

(A) Diplontic (B) Haplo-diplontic (C) Haplontic (D) A and B

(146) Which of following plant possesses seeds but fruits are absent ?

(A) Selaginella (B) Araucaria (C) Maize (D) Sunflower

(147) Who has discovered viruses ?

(A) Diener (B) Pasteur (C) Invanowsky (D) Iyengar

(148) What is bacteriophage ?

(A) Animal virus (B) Virus lived on bacteria

(C) Plant virus (D) Halophiles

(149) Which organism lived in extreme condition ?

(A) Archaebacteria (B) Eubacteria

(C) Cyanobacteria (D) Gram positive bacteria

(150) Which disease is caused by viruses ?

(A) Polio (B) Diphtheria (C) Cold (D) T.B.

(151) The five kingdom classification system given by whittaker is based on which criteria ?

(A) Cell structure (B) Nucleus (C) Types of nutrition (D) A, C

(152) Whittaker put prokaryotes into .....

(A) Protista (B) Plant kingdom (C) Fungi (D) Monera

(153) Who gave word Bryophyta ?

(A) Shiv Ram Kashyap (B) Linnaeus (C) Iyengar (D) Tulsane

(154) Which pigment is present in green algae ?

(A) Xanthophyll (B) Carotenoid (C) Fucoxanthin (D) Chlorophyll a and b

(155) Which plant is included in series calyciflorae ?

(A) Shoe-flower (B) Rose (C) Lemon (D) Maize

Answers : (141-A), (142-B), (143-B), (144-A), (145-C), (146-B), (147-C), (148-B), (149-A), (150-A), (151-D), (152-D), (153-A), (154-D), (155-B)

Classification Theory

Two Kingdom
- Plant kingdom
- Animal kingdom

- Due to organisms like Fungi, Euglena Chlamydomonas, This two kingdom bacteria can not give correct opinion so it is not proper.

Three Kingdom
Pioneer : Hackel(1866)
- Plant kingdom
- Animal kingdom
- Protista (Unicellular living organisms)

Four Kingdom
Pioneer : Copland (1956)
- Protista (Unicellular living organisms)
- Monera
- Plant kingdom
- Animal kingdom

Five Kingdom
Pioneer : R. H. Whittaker (1969)
Classification based on cell structure complexity
- Number of cell (Unicellular or Multi cellular)
- Mode of nutrition

Six Kingdom
Pioneer : Carl Woese (1990)
- (r-RNAsequence
- Eubacteria
- Archaebacteria
- Protista
- Fungi
- Plantae
- Animalea

Monera	**Protista**	**Fungi**	**Plantae**	**Animalia**
Prokaryotic Unicellular organism	Unicellular Eukaryotic organism	Eukaryotic heterotrophic organism	Algae Lichen Bryophyta Pferidophyta Gymnasperms Angiosperms	(Metazoa)

Living World

Viruses / Vivoids
Connecting link between non-living and living as they are inert particles outside the host cell

Prokaryotes
- Archaea
- Bacteria

Domain

Kingdom

Monera
Unicellular organisms
e.g., Streptococcus, E. Coli

Living beings
Organisms that can grow, reproduce and respond

Eukaryotes

Kingdom

Protista
Unicellular organisms except red and green algae.
e.g., Amoeba, Entamoeba

Fungi
Unicellular or multicellular heterotrophic organisms.
e.g., Rhizopus, Aspergillus

Plantae
Photosynthetic Unicellular or multicellular organisms. Cells have cell wall
eg., Chlorella, Cycus, Citrus

Animalia
Heterotrophic, multicellullar organisms, cells lack cell wall. e.g., Hydra, Pheretima, Felis

KINGDOM - PLANTAE

Cryptogamae (Plants without seeds)

Thallophyta
Thalloid, distinct roots, stem and leaves absent

Algae
Unicellular or multicellular, aquatic, green algae is considered as ancestors of land plants. e.g., Fucus, Volvox

Bryophyta
Plant amphibians, Can grow only on damp, watery places as male gametes require water to reach female gamete, protected sex organs, seeds absent, dominant gametophyte. eg., Funaria

Hepaticopsida
Liverworts, thallose or foliose; sporophyte is parasite. e.g., Riccia

Anthocerotopsida
Horn worts, thallose, horn like sporophyte is partial parasite on gametophyte. e.g., Anthoceros

Musci (Bryopsida)
Mosses, foliose, sporophyte partially dependent, sex organs borne on separate branches. e.g., Sphagnum, Sphaerocarpes

Pteridophyta
First true land plants, with vascular system, jacketed sex organs and dominant sporophyle. e.g., Salvinia, Pteris, Nephrolepis

Psilopsida
Most primitive, rootless leaves usually absent.
e.g., Rhynia

Lycopsida
Club moss/ground pine, true root, stem and leaves are present, usually gametophyte symbiotic, non-photosynthetic.
e.g., Lycopodium, Selaginella

Sphenopsida
Horse tail, mostly extinct, have distinct, nodes and internodes. e.g., Equisetum (sole living member)

Pteropsida (Filicopsida)
Ferns, stem, rhizomatous, macrophyllus, gametophyte, independent.
e.g., Adiantum

Phanerogamae (Plants with seeds)

Gymnospermae
Well developed vasculature but vessels and companion cells absent, mostly extinct, ovary absent thus fruit is not formed and seeds remain naked, gametophyte is highly reduced, endosperm is haploid. e.g., Ginkgo, Pinus

Cycadopsida
Comparitively primitive, small plants, female cones absent. e.g., Cycus, Zamia

Coniferopsida
Larger, dominant, gymnosperms, evergreen
e.g., Gingko, Taxus

Gnetopsida
Advanced gymnosperms with vessels in xylem embryo has two Cotyledons, Considered closest to angiosperms. e.g., Gnetum

Angiospermae
Vessels and companion cells are present, ovary present thus seeds remain protected inside fruits till matarity, triploid, endosperm very diversified, found in all habitats. e.g., Oryza, Citrus

Monocotyledoneal
Seeds with single Cotyledon, leaves have paralled venation, trimerous flowers, adventitious roots, lack secondary growth. e.g., Wheat, Rice

Dicotyledoneal
Seeds with two Cotyledon, reticulate venation in leaves, Pentamerous flowers, tap roots, secondary growth present. e.g., Mango, Beans

4 Classification of Animal Kingdom

Classification :

Classification is scientific method in which organisms are arranged in different or same taxa according to similarity and dissimilarity.

- Animal classification is based on their shape, form, size, level of organization, symmetry, coelom, segmentation, etc.
- **Levels of organization :**

  Members of animal kingdom exhibit different levels of organization. Protozoans are unicellular animals. Sponges are multicellular but cells are functions independently. Coelenterata have tissue level of organization. Platyhelminthes have organ level of organization. Organ systems are seen in Annelida, Arthropods, Molluscus, Echinoderms and Chordates.
- Organ systems of all phyla exhibit different structure.

(1) Digestive tract :

- Digestive tract with only one opening is called incomplete digestive tract. e.g., Platyhelminthes.
- Digestive tract has two opening i.e., anterior mouth and posterior anus is called complete digestive tract. e.g. Aschelminthes to chordate.

(2) Circulatory system : Circulatory system is of two types :

(i) Open type of circulatory system example : Arthropoda and Mollusca except - cephalopoda.

(ii) Closed type of circulatory system example : Annelida, cephalopoda and all vertebrata.

---

(1) The classification of Animal is based on which characters ?

(A) Level of organization and symmetry (B) Embryonic layers and coelom.

(C) Notochord and segmentation (D) A, B, C All

(2) The plylum of which animals are structurally and functionally unicellular ?

(A) Coelenterata (B) Protozoa (C) Annelida (D) Porifera

(3) Which animal phylum are multicellular but possesses cellular level of organization ?

(A) Platyhelminthes (B) Arthropoda (C) Porifera (D) Annelida

(4) Which animal phylum have organ level of organization ?

(A) Annelida (B) Arthropoda (C) Platyhelminthes (D) Chordata

(5) Which animal phylum possesses organ system level of organization ?

(A) Annelida and Echinodermata (B) Arthropoda and Mollusca

(C) Chordata (D) A, B, C All

(6) Which animal phylum possesses incomplete digestive tract ?

(A) Aschelminthes (B) Platyhelminthes (C) Chordata (D) Arthropoda

(7) Which animal phylum possesses complete digestive tract ?

(A) Aschelminthes (B) Arthropoda (C) Annelida (D) A, B, C All

(8) Which animal phylum possesses open circulatory system ?

(A) Annelida (B) Amphibia (C) Arthropoda (D) Reptilia

(9) Which Animal phylum possesses close circulatory system ?

(A) Reptilia (B) Annelida (C) Amphibia (D) A, B, C All

(10) In which Animal blood pressure is high and regular ?

(A) Earthworm (B) Centipede (C) Cockroach (D) Peripatus

Answers : (1-D), (2-B), (3-C), (4-C), (5-D), (6-B), (7-D), (8-C), (9-D), (10-A)

- **Symmetry :**
  - According to body organization animal exibit three types of symmetry.
  - Asymmetry example - sponges
  - Bilateral symmetry example - Annelida, Arthropoda
  - Radial symmetry example - Coelenterata and Echinodermata

(11) If the body of an animals divided into two equal left and right part then it is known as ..... .

(A) Asymmetry (B) Bilateral symmetry (C) Radial symmetry (D) A and C

(12) If the body of an animals is divided radially into more than one plane then it is known as ..... .

(A) Bilateral symmetry (B) Asymmetry (C) Radial symmetry (D) A and B

(13) When plane is passing through the central axis does not divide body of animal into equal halves is known as...

(A) Radial symmetry (B) Bilateral symmetry (C) Axial symmetry (D) Asymmetry

Answers : (11-B), (12-C), (13-D)

- **Organization of germinal layers :**
  - On the basis of germinal layer animals are two types : (1) Diploblastic : example : Coelenterata (2) Triploblastic : example : Platyhelminthes to chordate

(14) Which phylum of animal possesses diploblastic organization ?

(A) Annelida (B) Coelenterata (C) Arthropoda (D) Platyhelminthes

(15) In which triploblastic organization is observed for the 1st time ?

(A) Platyhelminthes (B) Annelida (C) Arthropoda (D) Chordata

(16) On the basis of embryonic layer animals are classified into how many groups ?

(A) One (B) Two (C) Three (D) Four

Answers : (14-B), (15-A), (16-B)

- **Coelom :**
  - Coelom is a cavity located between digestive trackt wall and body wall. This cavity is lined by mesoderm. Animal having coelom are called coelomates. e.g., Annelida to chordate.
  - In the animal of phylum aschelminthes mesoderm is present as scattered pouches is called pseudocoelom and animal called pseudocoelomate.
  - Animal do not possess coelom known as acoelomate. Example : Platyhelminthes

Segmentation :

- If animal body is divided into equal internal and external segments it is called metameric segmentation. e.g., Annelida, Arthropoda.

Notochord :

- Notochord is mesodermal in origin, dorsally situated and rod like organ located from head to tail.
- If Notochord is present animals are called chordates. e.g., Pisces to Mammal.
- If notochord is absent, animals are called non-chordates. e.g., Protozoa to Echinodermata.

(17) Which one is originated from mesoderm ?

(A) Coelom (B) Notochord (C) Body cavity (D) A, B, C All

(18) Which phylum respectively included in acoelomate, pseudocoelomate and coelomate ?

(A) Annelida, Aschelminthes, Platyhelminthes (B) Platyhelminthes, Aschelminthes, Annelida

(C) Arthropoda, Mollusca, Aschelminthes (D) Platyhelminthes, Annelida, Aschelminthes

(19) Which animal phylum possesses metameric segmentation ?

(A) Annelida (B) Arthropoda (C) Echinodermata (D) A and B

(20) Which animal phylum is included in Acoelomate ?

(A) Protozoa (B) Platyhelminthes (C) Echinodermata (D) A, B, C All

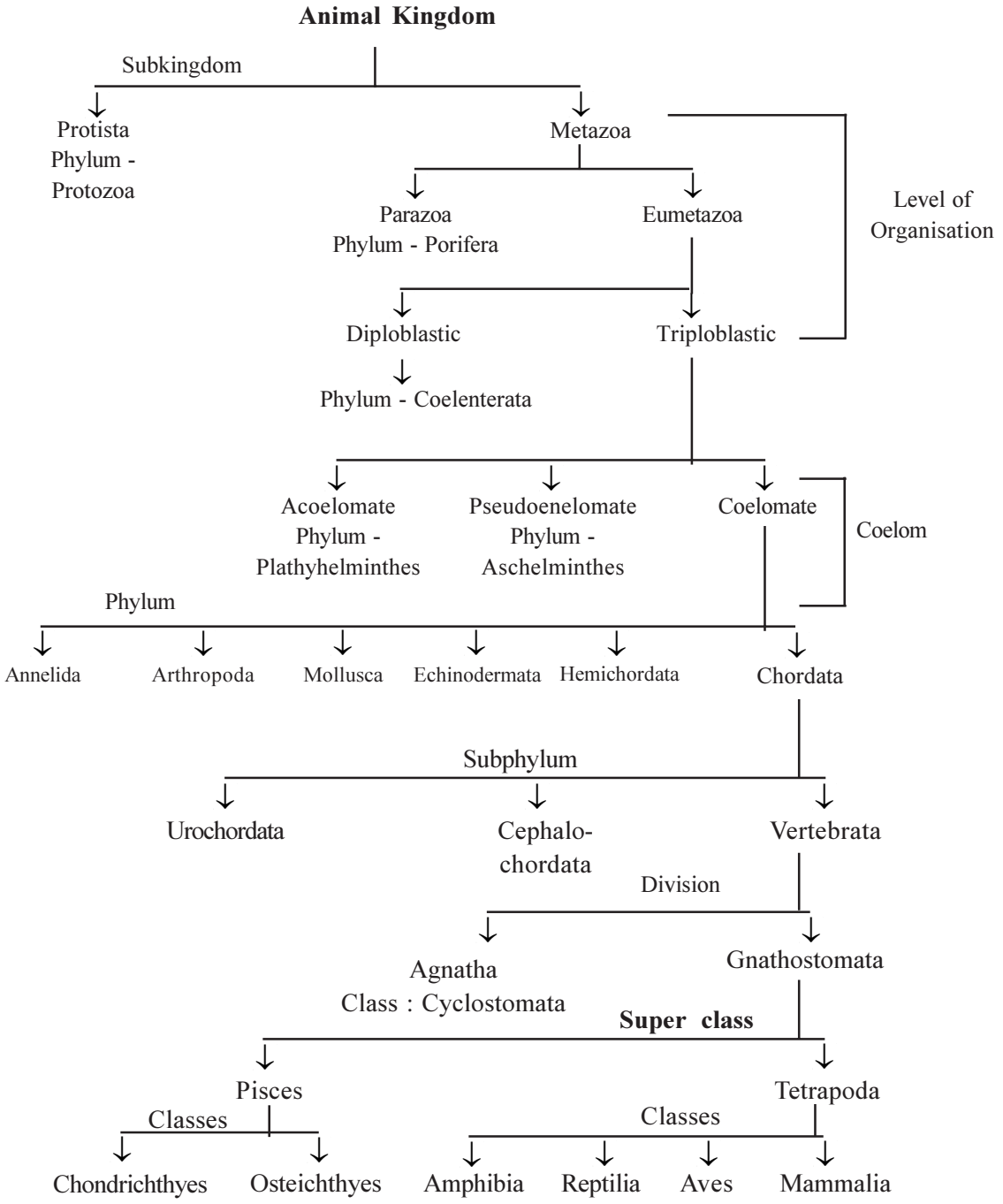
(21) Which animal phylum is included in coelomate ?

(A) Pisces (B) Amphibia (C) Mammalia (D) A, B, C All

(22) In which phylum animal, mesoderm is present as scattered pouches ?

(A) Coelomate (B) Pseudocoelomate (C) Acoelomate (D) Chordate

Answers : (17-D), (18-B), (19-D), (20-D), (21-D), (22-B)



- **Phylum : Protozoa :**
  - Protozoa is the first phylum of animal kingdom.
  - Protozoan animals are microscopic, unicellular, with one or more nuclei. Animal possesses various types of symmetry.
  - Nutrition is generally holozoic, halophytic and some animals are parasitic.
  - Locomotory organelles are pseudopodia, cilia or flagella.
  - Animal of this phylum reproduce asexually or sexually.
  - Example : Amoeba, Euglena, Opalina, Plasmodium

(23) The first phylum of animal kingdom is ..... .

(A) Mammalia (B) Protozoa (C) Arthropoda (D) Annelida

(24) Which animal possesses one or more nuclei ?

(A) Amoeba (B) Euglena (C) Opalina (D) A, B, C All

(25) In the phylum protozoa asexual reproduction occurs by which methods ?

(A) Binary fission (B) Multiple fission (C) Budding (D) A, B, C All

(26) Which locomotory organelles occur in protozoan ?

(A) Pseudopodia (B) Cilia (C) Flagella (D) A, B, C All

(27) Which of following animal reproduce sexually by conjugation ?

(A) Opalina (B) Euglena (C) Plasmodium (D) A, B, C All

(28) Which of following phylum possess all types of symmetry ?

(A) Protozoa (B) Annelida (C) Arthropoda (D) Platyhelminthes

Answers : (23-B), (24-D), (25-D), (26-D), (27-(D), (28-A)

- **Phylum : Porifera :**
  - Multicellular and sessile animals possess body pores and lives solitary or colonial life.
  - All animals are aquatic seen in marine water and some are seen in fresh water.
  - The body of animal is asymmetrical or radial symmetrical structure. It possess many ostia, canals and chambers called spongocoel which open in osculum.
  - Choanocytes line the spongocoel, is it's uniqueness.
  - All sponges are hermophrodite, they reproduce asexually, sexually and show power of regeneration.
  - The development is indirect and during this amphiblastula or parenchymula larva are seen.
  - Example : Lucosolenia, Hyalonema, Spongilla.

(29) Which of animal phylum body possess ostia, canals, chambers and osculum ?

(A) Protozoa (B) Porifera (C) Aschelminthes (D) Platyhelminthes

(30) The endoskeleton of porifera is made up of which substance ?

(A) Spongin Fibre (B) Calcium Carbonate

(C) Different kinds of spicules (D) A and C

(31) Which type of fertilization occur in Hyalonema ?

(A) External fertilization (B) Internal fertilization (C) A and B (D) None of these

(32) Spongocoel is lined by which structure ?

(A) Nematocytes (B) Tentacles (C) Choanocytes (D) Hypostome

(33) Which animals include in phylum porifera ?

(A) Lucosolenia (B) Hyalonema (C) Spongilla (D) A, B, C All

Answers : (29-B), (30-D), (31-B), (32-C), (33-D)

- **Phylum - Coelenterata :**
  - Coelenterates are diploblastic, radially or bilaterally symmetrical possess tissue level of organization.
  - All animals are aquatic. Animal are sedentary or free swimming, solitary or colonial.
  - They have central gastro vascular cavity with single opening hypostome which surrounded by tentacles. Which possess nematocytes. Tentacles helps to capture food as well as for offence and defence.
  - Coelenterate occur in two form polyps and medusae. Some species show polymorphism.
  - They reproduce asexually by budding and fission while sexually by ovum and sperm. Development is indirect and planula larva occurs. Example : Hydra, sea anemone, Jelly fish, coral.

(34) Which is the diploblastic animal phylum ?

(A) Annelida (B) Coelentrata (C) Arthropoda (D) Mollusca

(35) Which type of body organization possess by phylum coelentrata ?

(A) Cellular level (B) Tissue level (C) Organ level (D) Organ system level

(36) The tentacles are related with which functions ?

(A) To capture food (B) For offence (C) For defence (D) A, B, C All

(37) Which type of forms occur in coelenterate respectively for asexual and sexual reproduction ?

(A) Polyps, Medusae (B) Medusae, polyps

(C) Planula, Parenchymula (D) Nematocyte, tentacles

(38) The jelly fish possess which type of body organization ?

(A) Polyps (B) Medusae (C) Polymorphism (D) Colonial

Answers : (34-B), (35-B), (36-D), (37-A), (38-B)

- **Phylum - Platyhelminthes**
  - Platyhelminthes are first triploblastic, bilaterally symmetrical, dorso-ventrally flat acoelomate animals.
  - Animals are endoparasites. They form external hooks or suckers or both are present to attach host.
  - Digestive system is incomplete, branched without anus.
  - Excretory system consists of flame cells and canals.
  - Sexes are not separate - fertilization is internal and development is indirect.
  - Example : Tapeworm, Liver fluke, Planaria.

(39) Which is the first phylum possess triploblastic acoelomate body organization ?

(A) Plathyhelminthes (B) Arthropoda (C) Annelida (D) Mollusca

(40) Which phylum possess organ level body organization ?

(A) Annelida (B) Platyhelminthes (C) Mollusca (D) Arthropoda

(41) Which phylum possess incomplete, branched digestive system without anus ?

(A) Annelida (B) Arthropoda (C) Platyhelminthes (D) Mollusca

(42) Flame cells related with which functions ?

(A) Digestion (B) Respiration (C) Circulation (D) Excretion

(43) Which phylum are endoparasite found in human ?

(A) Arthropoda (B) Annelida (C) Mollusca (D) Platyhelminthes

Answers : (39-A), (40-B), (41-C), (42-D), (43-D)

- **Phylum - Aschelminthes :**
  - Aschelminthes are triploblastic, pseudocoelomate, bilaterally symmetrical with organ system level of organization.
  - Digestive system is complete. Excretion through branched excretory organ.
  - Sexes are separate. Male is smaller than female.
  - Fertilization is internal and development is direct. i.e. Viviparous animal.
  - Example : Ascaris, Wuchereria.

(44) In Animal kingdom, which phylum is included in pseudocoelomate ?

(A) Platyhelminthes (B) Aschelminthes (C) Mollusca (D) Arthropoda

(45) In which phylum the complete digestive system is observed first ?

(A) Aschelminthes (B) Annelida (C) Arthropoda (D) Mollusca

(46) On the basis of sexual dimorphism Wuchereria is which types animal ?

(A) Unisexual (B) Bisexual (C) Dioecious (D) B and C

(47) On the basis of reproduction ascaris includes is .....

(A) Oviparous (B) Viviparous (C) Ovoviviparous (D) Aparous

Answers : (44-B), (45-A), (46-A), (47-B)

- **Phylum - Annelida :**
  - Annelida are triploblastic, Coelomate animal possess bilateral symmetry and metamerically segmentation.
  - Mostly aquatic and some are terrestrial, bourrowing or tubicoulous, free living and occasionally parasite.
  - Locomotory organs are setaes, parapodia.
  - Digestive system is complete. Circulatory system is closed and respiratory pigment haemoglobin is present in blood plasma.
  - Nephridia are present as an excretory and osmoregulatory organ.
  - Nervous system is with paired cerebral ganglia and double ventral nerve cord.
  - Animal are unisexual. e.g., Nereis, Bisexual e.g., Earthworm; Leech - Reproduce by sexual method.
  - Example : Earthworm, Leech, Nereis.

(48) First time true body cavity observed in which phylum ?

(A) Mollusca (B) Annelida (C) Echinodermata (D) Arthropoda

(49) First time closed circulatory system observed in which coelomate invertebrate phylum ?

(A) Annelida (B) Arthropoda (C) Mollusca (D) Echinodermata

(50) Which structure is present as an excretory and osmoregulatory organs in annelida ?

(A) Flame cell (B) Nematocytes (C) Nephridia (D) Kidney

(51) Where does the respiratory pigment haemoglobin present in Leech ?

(A) Red blood cell (B) White blood cell (C) Thrombocyte (D) Blood plasma

(52) Which one is locomotory organ in earthworm ?

(A) Flagella (B) Cilia (C) Parapodia (D) Setaes

(53) Which of the following animal is included in monoecious ?

(A) Earthworm (B) Leech (C) Liver fluke (D) A, B, C All

(54) First time nervous system observed in which phylum ?

(A) Annelida (B) Coelenterata (C) Arthropoda (D) Mollusca

Answers : (48-B), (49-A), (50-C), (51-D), (52-D), (53-D), (54-B)

- **Phylum - Arthropoda :**
  - Arthropoda is largest phylum of Animal kingdom. They are triploblastic, coelomate and have bilateral symmetry.
  - Animals possess jointed appendage in which body is divided into head, thorax and abdomen. Head and thorax fused to form cephalothorax.
  - Sensory organ comprises of simple or compound eye. Chemo and tactile receptors statocysts and auditory organs.
  - Exoskeleton is chitinous and sheds at intervals (ecdysis).
  - Digestive system is complete circulatory system is of open type. Respiration occurs by body surface, gills, trachea and book lungs.
  - Excretory organs are green gland or malpighian tubules.
  - They are unisexual animal and fertilization internal, oviparous or ovoviviparous animal. Development may be director indirect some of them exhibit parthenogenesis.
  - Example : Peripetus, Crab, Scorpion, Prawn, Centipede, Cockroach.

(55) Which phylum shows ecdysis ?

(A) Mollusca (B) Arthropoda (C) Annelida (D) Echinodermata

(56) Which is the excretory organ in arthropoda ?

(A) Green glands (B) Nephridia (C) Flame cell (D) Nematocytes

(57) Which phylum possess open circulatory system ?

(A) Arthropoda (B) Mollusca (C) Amphibia (D) A and B

(58) Which is the respiratory organ in arthropoda ?

(A) Gills (B) book lungs (C) Trachea (D) A, B, C All

(59) In phylum arthropoda exoskeleton is made up of which substance ?

(A) Chitin (B) Cutin (C) Carotene (D) Calcium carbonate

(60) Arthropoda is which type of animal, regarding reproduction ?

(A) Viviparous (B) Ovoviviparous (C) Oviparous (D) B and C

(61) Which is the largest phylum of Animal Kingdom ?

(A) Mollusca (B) Arthropoda (C) Annelida (D) Echinodermata

Answers : (55-B), (56-A), (57-D), (58-D), (59-A), (60-D), (61-B)

- **Phylum - Mollusca :**
  - Animals included in mollusca are triploblastic, coelomate; bilateral symmetry and unsegmented. Soft body is made up of calcium carbonate.
  - Majority animals are aquatic and some are terrestrial.
  - Mantle cavity is located between body wall and mantle. In mantle feather like gills present which act as a respiratory organs.
  - Digestive system is complete and animals have Radula in mouth, which helps in grinding food.
  - Circulatory system is of open type and excretion take place through kidney.
  - They are unisexual or bisexual. Example : Chiton, Pila, Dentalium, Pearl oyster, Sepia, Octopus

(62) Which phylum possess mantle cavity ?

(A) Arthropoda (B) Mollusca (C) Annelida (D) Echinodermata

(63) What is function of radula ?

(A) Intake of food (B) Digestion of food (C) Grinding of food (D) Excretion

(64) Which is the excretory organ in mollusca ?

(A) Nephridia (B) Malpighian tabules (C) Green glands (D) Kidney

(65) In phylum mollusca endoskeleton made up of which substance ?

(A) Calcium carbonate (B) Chitin (C) Cutin (D) Carotene

Answers : (62-B), (63-C), (64-D), (65-A)

- **Phylum - Echinodermata :**
  - They are triploblastic, coelomate, radially symmetrical body with organ system level of organization.
  - Animals are exclusively marine. They possess water vascular system of coelomic origin, including tube feet for locomotary function.
  - Water vascular system also concerned with respiratory and excretory functions.
  - Digestive system is complete.
  - Animals are unisexual, fertilization is external and development is indirect through free swimming larva. They possess regeneration power. Example : Starfish, Seaurchin, Sealily, Sea-cucumber, Brittle star.

(66) Which phylum shows Radial symmetry ?

(A) Echinodermata (B) Annelida (C) Mollusca (D) Arthropoda

(67) Which phylum possess water vascular system ?

(A) Annelida (B) Echinodermata (C) Arthropoda (D) Mollusca

(68) What is function of tube feet ?

(A) Digestion (B) Locomotion (C) Respiration (D) Reproduction

(69) Water vascular system is related with which functions ?

(A) Respiration (B) Circulation (C) Excretion (D) A, B, C All

(70) Which animal possess digestive system complete and digestive duct is straight or coiled ?

(A) Sea-cucumber (B) Brittlestar (C) Sealily (D) A, B, C All

Answers : (66-A), (67-B), (68-B), (69-D), (70-D)

- **Phylum - Hemichordata :**
  - Hemichordata are usually primitive chordates - invertebrate chordate.
  - They are wormlike unsegmented, bilaterally symmetrical, triploblastic, having organ system level of organization. Exclusively marine, solitary or colonial and mostly tubiculous animals.
  - Cylindrical body is divided into probosis, collar and trunk.
  - Digestive tract is complete straight or U-shaped, circulatory system is open type.
  - Excretion take place by single probosis gland which is connected to blood vessels and respiration occurs through gills.
  - Animals are unisexual, fertilization is external and development is direct or indirect through free swimming Tornaria larva. Example : Balanoglossus

(71) Which phylum is included in primitive chordata ?

(A) Echinodermata (B) Hemichordata (C) Arthropoda (D) Mollusca

(72) The excretion occurs in hemichordata by which structure ?

(A) Probosis (B) Flame cells (C) Nephridia (D) Green glands

(73) In which invertebrate, chordate animal closed type of circulatory system is observed ?

(A) Earthworm (B) Pila

(C) Balanoglossus (D) Man

(74) Which type of digestive tract is observed in hemichordata ?

(A) Complete (B) Incomplete

(C) Straight of U-shaped (D) A and C

(75) Which types of larva is observed in hemichordata ?

(A) Tornaria (B) Amphiblastula (C) Parenchymula (D) Planula

(76) The body of balanoglossus divided into which parts ?

(A) Probosis (B) Collar (C) Trunk (D) A, B, C All

Answers : (71-B), (72-A), (73-C), (74-D), (75-A), (76-D)

- **Phylum - Chordata :**
  - Animal possess notochord called chordata. They are triploblastic, coelomate and possess bilateral symmetrical structure.
  - Skeleton is made up of cartilage or bone. Post anal tail present life long or during some stages of life.
  - Digestive system is complete and circulatory system is closed type. They are unisexual animals.
  - Phylum - Chordata is divided into three sub-phylum :
    (1) Urochordata : Notochord is seen during larval stage only and animals are marine. Example : Ascidia, Salpa.
    (2) Cephalochordata : Notochord persists during the whole life span and extended from head to tail. Example : Amphioxus
    (3) Vertebrata : Notochord is modified into vertebral column at the adult stage. Body is divided into head, neck, trunk and tail.

---

(77) The endoskeleton of chordate made up of which substance ?
(A) Cartilage and Bone (B) Cartilage or Bone (C) Hyaline Cartilage (D) Bone

(78) Phylum chordata is divided in to how many sub phylum ?
(A) One (B) Two (C) Three (D) Four

(79) Ascidia included in which sub phylum ?
(A) Non-chordata (B) Urochordata (C) Cephalo chordata (D) Vertebrata

(80) Amphioxus included in which sub phylum ?
(A) Cephalo chordata (B) Vertebrata (C) Non-chordata (D) Urochordata

(81) In vertebrate animal body divided into how many part ?
(A) One (B) Two (C) Three (D) Four

(82) In which phylum nervous system is hollow, single, dorsal and without ganglia ?
(A) Chordata (B) Non-chordata (C) Prochordata (D) Invertebrata

Answers : (77-B), (78-C), (79-B), (80-A), (81-D), (82-A)

- **Class : Cyclostomata**
  - Animals of the class cyclostomata are ectoparasites on fishes during their adult stage. Jaws are absent. Mouth is anterioventral, circular and suctorial.
  - Skin is without scale contain unicellular mucus glands. Internal skeleton is fibrous and cartilaginous.
  - Heart is two chambered, circulatory system is of closed type, respiration occur by gill slits; one pair of kidney as an excretory organ.
  - They are marine but reproduction take place in fresh water. Larva after metamorphosis return to ocean. Example : Lamprey and Hagfish

---

(83) Which animals are included in agnatha, cyclostomata ?
(A) Lamprey and Hag fish (B) Salpa and Ascidia (C) Silver fish and Jelly fish (D) Hag fish, Salpa

(84) In which phylum reproduction take place in fresh water, larva after metamorphosis return to ocean ?
(A) Cephalochordata (B) Cyclostomata (C) Urochordata (D) Vertebrata

(85) In which class skin is without scale contains unicellular mucus glands ?

(A) Amphibia (B) Cyclostomata (C) Reptilia (D) A and B

(86) Which animal possess two chambered heart ?

(A) Shark (B) Lamprey (C) Sea horse (D) A, B, C All

Answers : (83-A), (84-B), (85-D), (86-D)

Class : Chondrichthyes - Osteichthyes

Special character	Chondrichthyes	Osteichthyes
Habitat	Marine	Fresh water and marine
Scales	Placoid	Cycloid or ctenoid
Caudal fin	Heterocercal	Homocercal
Endoskeleton	Cartilage	Bone
Mouth	Anterior-Ventrally	Anterior
Gills	5 to 7 pairs	4 pairs
Heart	Two chambered	Two chambered
Fertilization	Internal	External
	Oviparous or Ovoviviparous	Oviparous
Example	Shark, Ray fish	Sea horse, Labeo, Catla

(87) The exoskeleton of chondrichthyes and osteichthyes made up of which types of scales ?

(A) Placoid (B) Cycloid (C) Epidermal (D) A and B

(88) Which fish possess caudal fin heteroceral ?

(A) Shark (B) Catla (C) Rohu (D) Labeo

(89) In which class fertilization is external and includes oviparous animals ?

(A) Osteichthyes (B) Amphibia (C) Chondrichthyes (D) A and B

(90) In which animals endoskeleton is made up of bones ?

(A) Labeo-catla (B) Shark - sea horse (C) Shark - Ray fish (D) Labeo - Ray fish

(91) In sea horse mouth opens at which side ?

(A) Anterior (B) Posterior (C) Lateral (D) Dorsal

Answers : (87-D), (88-A), (89-D), (90-A), (91-A)

- **Class : Amphibia**
  - Amphibia lives in water and on land, hence they are called Amphibians. They are tetrapod.
  - External skeleton is absent, skin is moist and acts as a respiratory organ.
  - Body is divided into head and trunk. External ear is absent. Internal and middle ears are present.
  - Digestive tract is complete. Which terminates into cloaca. Excretory and reproductive tracts also terminate into cloaca.

- Heart is three chambered. With two auricles and one ventricle.
- They are coldblooded, unisexual animal. Fertilization is external and development is through metamorphosis. Example : Frog, Salamander, Ichthyophis.

(92) Which animal is included in Amphibia, tetrapoda limbless ?

(A) Snake (B) Ichthyophis (C) Salamander (D) Frog

(93) In frog which trackt terminate into cloaca ?

(A) Digestive (B) Excretory (C) Reproductive (D) A, B, C All

(94) Which animal possess three chambered heart ?

(A) Crocodile (B) Catla (C) Salamander (D) Labeo

(95) Which animal class exhibit metamorphosis ?

(A) Amphibia (B) Arthropoda (C) Reptilia (D) A and B

Answers : (92-B), (93-D), (94-C), (95-D)

- **Class : Reptilia**
  - Reptiles represent the first class of vertebrata fully adopted for life on terrestrial habitat.
  - They locomote by crawling. They are carnivores, cold blooded animal.
  - Body divides into head, neck, trunk and tail. External skeleton made up of epidermal scales and skin is dry.
  - Limbs are uniform, short with claws. Limbs are absent in snake.
  - Digestive system is complete, pinna absent, Heart is three chambered (two auricles and one incompletely divided ventricle). In crocodile heart is four chambered.
  - Kidney is present as an excretory organ. Uric acid is released as an excretory substance.
  - They are unisexual, fertilization is internal, oviparous and development is direct in animals. Example : Chameleon, Crocodile, Turtle, Calotes, Wall lizard.

(96) Which is the first class of vertebrata fully adopted for life on terrestrial habitat ?

(A) Aves (B) Amphibia (C) Mammalia (D) Reptilia

(97) Which of the following class animals are poikilothermic ?

(A) Aves (B) Reptilia (C) Amphibia (D) B and C

(98) Which reptilia possess four chambered heart ?

(A) Calotes (B) Crocodile (C) Chameleon (D) Turtle

(99) Which animal releases an excretory substance uric acid ?

(A) Chameleon (B) Turtle (C) Snake (D) A, B, C All

(100) Which animal lacks pinna ?

(A) Kangaroo (B) Bat (C) Turtle (D) Rat

Answers : (96-D), (97-D), (98-B), (99-D), (100-C)

- **Class : Aves**
  - Their body is streamlined and it is divided into head, neck, trunk and tail. Jaw is modified into beak and teeth are absent.
  - Forelimb modifies into wings for flying.
  - Feathers, scales, beak, claws acts as an external skeleton. Bones of internal skeleton are porous and hollow which help for flying.
  - The digestive tract has crop and gizzard. Heart is four chambered and aortic arch turns toward right side.
  - Respiration take place through lungs and air-sacs associated with lungs, help in flying.
  - They are the first vertebrates which are warm-blooded.
  - Animals are unisexual, shows internal fertilization and direct embryonic development. They are oviparous animals.
  - Example : Pigeon, Crow, Peafowl, Ostrich, Penguin

---

(101) Which are first warm-blooded vertebrate animals ?

(A) Ostrich - Pigeon (B) Peacock - Crow (C) Rat - Dolphin (D) A and B

(102) Which structure is absent in digestive system of Aves ?

(A) Teeth (B) Urinary bladder (C) Stomach (D) A, B, C All

(103) What is the function of airsac ?

(A) Flying (B) Swimming (C) Excretion (D) A and B

(104) Which structure is important in class - Aves for flying ?

(A) Forelimbs modified into wings (B) Air sacs

(C) Endoskeleton are porous and hollow (D) A, B, C all

(105) Which bird is flightless ?

(A) Ostrich (B) Emu (C) Peafowl (D) A and B

Answers : (101-D), (102-D), (103-D), (104-D), (105-D)

- **Class : Mammalia**
  - They possess mammary gland so known as mammalia. They are found in all types of habitat.
  - External skeleton includes hairs on body, horns, hoof and nail. Two pair of limbs are present which are used in locomotion. External ear develop as pinna.
  - They possess four types of teeth. i.e. canines, incisors, premolars and molars. Permanent teeth develops after the milk teeth drop.
  - Digestive tract is complete. Heart is four chambered and aortic arch turns toward left.
  - Respiration take place through lungs. They are warm-blooded.
  - The are unisexual, fertilization is internal and shows direct embryonic development, usually viviparous animals. Platypus is oviparous.

    Example : Kangaroo, Rabbit, Rat, Elephant, Dolphin, Whale, Bat.

(106) In which class external ears develop as pinna ?

(A) Amphibia (B) Mammalia (C) Reptilia (D) Aves

(107) Which of the following is correct for Platypus ?

(A) Oviparous (B) Viviparous (C) Ovoviviparous (D) A and B

(108) In which class circulatory system is of closed type and aortic arch turns toward left ?

(A) Reptilia (B) Aves (C) Mammalia (D) Amphibia

(109) How many types of teeth are found in mammalia ?

(A) Two (B) Four (C) Three (D) Five

(110) What is present as external skeleton in mammalia ?

(A) Nail and hoof (B) Hairs (C) Horns (D) A, B, C All

(111) Which mammalian adapted for aerial life ?

(A) Whale (B) Dolphin (C) Platypus (D) Bat

(112) Which class of tetrapode included in homeothermic animals ?

(A) Amphibia - Reptilia (B) Mammalia - Aves

(C) Amphibia - Aves (D) Reptilia - Mammalia

(113) Which animals are included in Mammalia ?

(A) Dolphin (B) Platypus (C) Whale (D) A, B, C All

Answers : (106-B), (107-A), (108-C), (109-B), (110-D), (111-D), (112-B), (113-D)

- **True - False (T - F) types questions :**

Select proper answer from following :

(114) (1) All vertebrates are chordate.

(2) All chordata are not vertebrates.

(3) Aschelminthes are pseudocoelomate.

(4) Phylum Annelida and after it all are coelomate animals.

(A) T, F, T, F (B) T, T, T, T (C) F, T, T, F (D) T, F, T, T

(115) (1) Protozoan reproduce sexually by conjugation, binary fission, multiple fission and budding.

(2) Digestive tract of Aves and Arthropoda possess gizards.

(3) Aves and Reptilia excrete uric acid during excretion.

(4) Ichthyophis and Penguin shows metamorphosis.

(5) Nematocytes is characteristic of reptilia.

(A) T, T, F, T, F (B) F, T, F, T, F (C) F, T, T, F, F (D) T, F, T, F, T

(116) (1) Porifera possess amphiblastula and coelenterata possess planula larva.

(2) Platyhelminthes possess incomplete digestive tract.

(3) Annelida possess haemoglobin in blood plasma.

(4) Arthropoda do not shows ecdysis.

(5) Mollusca possess tube feet.

(A) T, T, T, F, F (B) T, T, F, T, T (C) T, F, T, F, F (D) T, T, F, F, T

(117) (1) In balanoglossus body is divided into probosis, collar and trunk.

(2) Hag fish and Ichthyophis possess skin, without scale.

(3) Chondrichthyes possess placoid scales and osteichthyes possess cycloid scales.

(4) Crocodile and whale possess four chambered heart.

(A) T, T, T, F (B) T, F, T, F (C) F, T, T, F (D) F, F, T, T

(118) (1) In Ascidia notochord is seen during larval stage and in Amphioxus notochord persists during the whole life span.

(2) Vertebrate possess heart two, three, four chambered and located on dorsal side.

(3) Reptila and aves are oviparous while platypus are viviparous.

(4) In amphibia digestive system, Respiratory system and Reproductive system open into cloaca.

(5) Reptila possess epidermal scales and aves lacks scales.

(A) T, T, F, T, F (B) T, F, F, T, F (C) T, F, T, T, F (D) F, T, F, T, F

(119) (1) Kangaroo and ostrich are homeothermic.

(2) Bat possess four types of teeth.

(3) Amphibia lacks external pinna.

(4) Ray-fish possess four pair of gills.

(5) Aves and osteichthyes possess air bladders.

(A) T, F, T, F, T (B) T, T, F, F, T (C) T, T, T, F, T (D) F, T, F, F, T

Answers : (114-B), (115-C), (116-A), (117-A), (118-C), (119-C)

- **A - Assertion, R - Reason type questions :**

Choose the correct answer from given option :

(A) A and R both are true and R is correct explanation of the A.

(B) A and R both are true and R is not correct explanation of the A.

(C) A is true and R is false.

(D) A is false and R is true.

(120) Assertion A : First time closed circulatory system observed in Annelida

Reason R : In earthworm respiratory pigment is present in blood plasma.

(A) (B) (C) (D)

(121) Assertion A : Ascaris are pseudocoelomate.

Reason R : In Ascaris mesoderm is present as scattered pouches.

(A) (B) (C) (D)

(122) Assertion A : Echinodermata possess water vascular system.

Reason R : Water vascular system concerned with locomotion, respiration and excretion.

(A) (B) (C) (D)

(123) Assertion A : Salamander is unisexual animal, shows external fertilization and metamorphosis.

Reason R : Salamander is amphibian animals.

(A) (B) (C) (D)

(124) Assertion A : Aves can easily fly.

Reason R : In Aves feathers, beak, claws act as external skeleton while endoskeleton are porous and hollow and forelimbs modified into wings.

(A) (B) (C) (D)

Answers : (120-A), (121-A), (122-B), (123-A), (124-A)

(125) Select correct option :

Column - I	**Column - II**	
(a) Chondrichthes	(i) Cycloid scales	(A) a - iii, b - i, c - ii
(b) Osteichthyes	(ii) Epidermal scales	(B) a - ii, b - iii, c - i
(c) Reptilia	(iii) Placoid scales	(C) a - iii, b - ii, c - i
		(D) a - ii, b - iii, c - i

(126) Select correct option :

Column - I	**Column - II**	
(a) Frog	(i) Flame cell	(A) a - iv, b - iii, c - i, d - ii
(b) Pisces	(ii) Three chambered heart	(B) a - ii, b - iv, c - i, d - iii
(c) Liverfluke	(iii) Planula larva	(C) a - iv, b - iii, c - ii, d - i
(d) Coelenterata	(iv) Air-bladder	(D) a - ii, b - iii, c - iv, d - i

(127) Select correct option :

Column - I	**Column - II**	
(a) Porifera	(i) Viviparous	(A) a - iii, b - iv, c - ii, d - i
(b) Earthworm	(ii) Choanocytes	(B) a - ii, b - iv, c - iii, d - i
(c) Cockroach	(iii) Nephridia	(C) a - ii, b - iii, c - iv, d - i
(d) Rabbit	(iv) Malpighian tubules	(D) a - ii, b - iii, c - i, d - iv

(128) Select correct option :

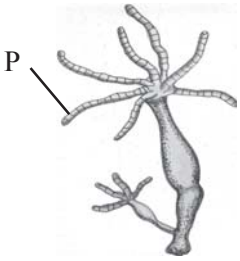
Column - I	**Column - II**	
(a) Ommatidia	(i) Porifera	(A) a - ii, b - iii, c - iv, d - i
(b) Nematocytes	(ii) Arthropoda	(B) a - iii, b - iv, c - ii, d - i
(c) Setae	(iii) Coelenterata	(C) a - iv, b - iii, c - i, d - ii
(d) Osculum	(iv) Nephridia	(D) a - ii, b - iv, c - i, d - iii

(129) Select correct option :

Column - I	**Column - II**	
(a) Triploblastic	(i) Protozoan	(A) a - iv, b - iii, c - ii, d - i
(b) Diploblastic	(ii) Hemichordata	(B) a - iii, b - iv, c - i, d - ii
(c) Contractile vacuole	(iii) Annelida	(C) a - ii, b - i, c - iv, d - iii
(d) Probosis	(iv) Coelenterata	(D) a - iii, b - iv, c - ii, d - i

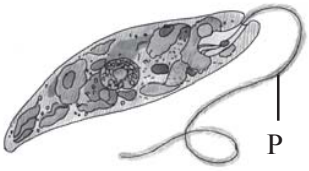
Answers : (125-A), (126-B), (127-C), (128-A), (129-B)

(130) In given figure P indicates...



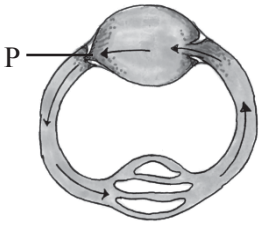
(A) Nemotocytes (B) Tentacles
(C) Hypostome (D) Choanocytes

(131) In given diagram P is related with which functions ?



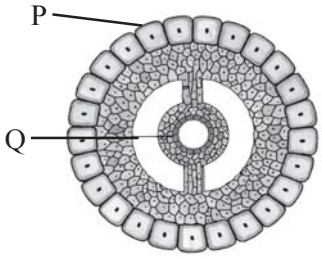
(A) Locomotion (B) Digestion
(C) Circulation (D) Excretion

(132) In given diagram P indicates...



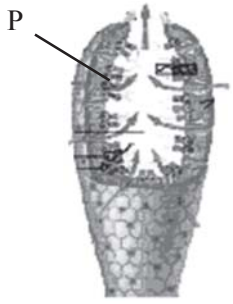
(A) Valve (B) Artery
(C) Ostia (D) Vein

(133) In given diagram P and Q indicates...



(A) Mesoderm, Ectoderm
(B) Ectoderm, Mesoderm
(C) Ectoderm, Endoderm
(D) Endoderm, Ectoderm

(134) In given diagram P indicates...



(A) Spongilla
(B) Choanocytes
(C) Spongocoel
(D) Osculum

Answers : (130-B), (131-A), (132-A), (133-C), (134-B)

- **Questions for NEET :**

(135) Which is the limbless amphibian animal ?
(A) Toad (B) Ichthyophis (C) Frog (D) Salamandar

(136) Which animal is cold blooded ?
(A) Turtle (B) Penguin (C) Whale (D) Kangaroo

(137) The exoskeleton of pisces is made up of ..... .
(A) Bone (B) Cartilage (C) Hair (D) Scales

(138) Undeveloped artery and vein with open circulatory system is characteristic of which phylum ?
(A) Arthropoda (B) Aves (C) Coelenterata (D) Mollusca

(139) Which is common structure in cockroach and earthworm ?

(A) Trachea (B) Closed circulatory system (C) Ventral nervecord (D) Nephridia

(140) In which phylum diploblastic animals are included ?

(A) Platyhelminthes (B) Echinodermata (C) Coelenterata (D) Mollusca

(141) In class aves which organ is modified ?

(A) Fore limb (B) Pelvic girdle (C) Hind limb (D) Pectoral girdle

(142) In which phylum metameric segmentation is observed for the first time ?

(A) Annelida (B) Arthropoda (C) Platyhelminthes (D) Mollusca

(143) In presence or absences of which structure pisces are classified into chondrichthyes and osteichthyes ?

(A) Laterline (B) Middle ear (C) Operculum (D) Mucin

(144) Give the name of excretory organ of prawn.

(A) Flame cell (B) Nephridia (C) Green gland (D) Tube feet

(145) Which of following is oviparous mammalia ?

(A) Dolphin (B) Platypus (C) Penguin (D) Rabbit

(146) Which of the following is not classified as amphibian ?

(A) Ichthyophis (B) Frog (C) Turtle (D) Salamander

(147) Which of following phylum is acoelomate ?

(A) Annelida (B) Arthropoda (C) Echinodermata (D) Platyhelminthes

(148) In which of the following Air bladders is observed ?

(A) Brittle Star (B) Star Fish (C) Chondrichthyes (D) Osteichthyes

(149) Mesoglea occur in between which two germinal layers ?

(A) Ectoderm and Mesoderm (B) Mesoderm and Endoderm

(C) Below Mesoderm (D) Ectoderm and Endoderm

(150) In which animal jaws are absent ?

(A) Rohu and Labeo (B) Lamprey, Hag fish (C) Labeo, Catla (D) Sea horse, Hag fish

Answers : (135-B), (136-A), (137-D), (138-A), (139-C), (140-C), (141-A), (142-A), (143-C), (144-C), (145-B), (146-C), (147-D), (148-D), (149-D), (150-B)

KINGDOM - ANIMALIA (Multicellular, Eukaryotic)

Mesozoa

No tissues, body comprising an external layer of ciliated cells. Minute body, Worm like, Parasites of marine Phylum Mesozoa e.g., Dicyema

Parazoa

No tissues, diploblastic, cellular level of body organization, choanocytes present, called sponges. Phylum Porifera e.g., Leucosolenia, Sycon

Enterozoa or Eumetazoa

Tissue or organ system level of body organization

Division-Radiata

Body radially symmetrical, diploblastic, tissue level of body organization. *Phylum*-Cnidaria = Coelenterata. Cnidoblasts present. e.g., Hydra, Phylum ctenophora e.g., Chenoplana

Division-Bilateria

Body bilaterally symmetrical, triploblastic, organ system level of body organization

Sub-division Acoelomata
No Coelom
Phylum Platyhelminthes
e.g., Fasciota, Taenia

Sub-division Pseudocoelomata
Pseudocoelom
Phylum Nemathelminthes
e.g., Ascaris

Sub-division Eucoelomata
True Coelom

Schizocoelomata
Coelom develops from solid mesoderm
Phylum Annelida e.g., Earthworm
Phylum Arthropoda e.g., Cockroach
Phylum Mollusca e.g., Pila

Enterocoelomata
Coelom develops from enteric pouches
Phylum Echinodermata e.g., Starfish
Phylum Hemichordata e.g., Balanoglossus
Phylum Chordata

Phylum - Chordata

Sub Phylum - Urochordata
e.g., Herdmania

Sub Phylum - Cephalochordata
e.g., Amphioxus

Sub Phylum - Vertebrata

Section-Agnatha
Jawless Vertebrates
Class : Cyclostomata e.g., Petromyzon, Myxine
Class : Ostracodermi e.g., Cephalaspis, Pteraspis

Section-Gnathostomata
Jawed Vertebrates

Superclass - Pisces
True Fishes

Superclass - Tetrapoda
Typically 4 limbs

Class - Placodermi
All fossils have protective bony scales / plates
e.g., *Both riolepis panderi*

Class - Chodrichthyes
Cartilagenous fishes
e.g., Scoliodon

Class - Osteichthyes
Bony fishes e.g., Labeo

Class - Amphibia
e.g., Uraeotyphlus, Salamander, Frog

Class - Reptilia
e.g., Tortoise, Lizards, Snakes

Class - Aves
e.g., Birds

Class - Mammals
e.g., Rat, Man

5 Cell Structure

- All living organisms are composed of cells. Cell is structural and functional unit of life. Organisms are made up of a single cell known as unicellular organisms. Amoeba, Paramoecium, Bacteria, Yeast and Chlamydomonas are their examples. Organisms composed of many cells are know as multicellular organisms.
- Life of the organisms begin as a single cell called zygote. This cell divides repeatedly to produce new cells. This cell differentiates to form tissue, organs and organ systems. As zygote divides by mitotic divisons, the genetic make up of each cell in the body remains the same.
- Any cell of the body is capable of producing new individual, this character of cell is known as totipotency.

(1) The smallest, structural and functional unit of organisms known as ..... .

(A) Tissue (B) Cell (C) Organ (D) Organelles

(2) Which is the unicellular organisms ?

(A) Chlamydomonas (B) Yeast (C) Bacteria (D) A, B, C All

(3) In which type of cell division the genetic material of each cell in the body remains the same ?

(A) Mitosis (B) Amitosis (C) Meiosis (D) A and B

Answers : (1-B), (2 -D), (3-A)

- **Cell and cell theory :**
  - *Robert Hooke (1665)* an English scientist observed a thin slice of cork under his crude microscope and describe that cork composed of small spaces surrounded by firm wall and gave the name cell.
  - *Robert Brown (1831)* discovered the nucleus in the cell.
  - In *1838 Matthias Schleiden* a German Botanist and *Theodore Schwann,* a British zoologist proposed the cell theory.
  - In *1855 Rudolf Virchow* first explained that new cells arise through cell division of preexisting cells.
  - Main points of cell theory :

    (1) All organisms are made up of cells and cellular products.

    (2) Cells is a structural and functional unit of organisms.

    (3) New cells are formed through divisions of preexisting.

(4) Who proposed the cell theory first time ?

(A) Robert Hooke (B) Robert Brown (C) Schleiden - Schwann (D) Robert Koch

(5) Who proposed that new cell arise through cell division of preexisting cells ?

(A) Robert Brown (B) Rudolf Virchow (C) Robert Hooke (D) Schleiden - Schwann

(6) Who gave the name 'Cell' first ?

(A) Robert Brown (B) Virchow (C) Robert Hooke (D) Schleiden - Schwann

(7) Who discovered nucleus in the cell ?

(A) Robert Brown (B) Schleiden - Schwann (C) Robert Hooke (D) Virchow

(8) Matthias Schleiden and Theodore Schwann are respectly scientists of ..... and ..... country.

(A) America, Canada (B) German, India (C) British, German (D) German, British

Answers : (4-C), (5-B), (6-C), (7-A), (8-D)

- **An overview of cell :**
  - A cell is the smallest unit of living organisms. A typical plant cell has a cell wall as its outerside and plasma membrane to its inner side. While a typical animal cell has only plasma membrane.
  - All cells, except bacterial cells contain a nucleus and cytoplasm. The nucleus contains the chromosomes which in contain the genetic material - DNA, it controls the cells activities.
  - Eukaryotic cell possess membrane bound nuclei whereas prokaryotic cell lack a membrane bound nucleus. Eukaryotic cell possess membrance bound organelles whereas prokaryotic cell lack a membrane bound organells. Non-membrane bound organelles-ribosomes are found in prokaryotes, eukaryotes, chloroplasts and in mitochondria. Centrioles found in animal cell, which is associated in cell division.

(9) What is the specificity of plant cell ?

(A) Mitochondria (B) Cell-wall (C) Cytoplasm (D) Ribosomes

(10) Typical Animal cell lacks which structure ..... .

(A) Cell-wall (B) Cytoplasm (C) Mitochondria (D) Ribosomes

(11) Which is the membrane bound organelles ?

(A) Chloroplast (B) Golgi complex (C) Mitochondria (D) A, B, C All

(12) Which is the Non-membrane bound organelles ?

(A) Ribosomes (B) Centrioles (C) A and B (D) None of these

(13) Which one of following is largest animal cell ?

(A) Bacteria (B) Cheek cell (C) Ostrich's egg (D) Mycoplasma

(14) Which is the smallest cell ?

(A) Bacteria (B) Mycoplasma (C) Cheek cell (D) A and C

Answers : (9-B), (10-A), (11-D), (12-C), (13-C), (14-B)

- **Prokaryotic cells :**
  - Prokaryotes are unicellular organisms that lack a nuclear membrane and do not develop or differentiate into multicellular forms. Prokaryotic cells are smaller than eukaryotic cells, however their cell division is very rapid.
  - Example : Bacteria, blue green algae, mycoplasma and pleuro Pneumonia like Organisms (PPLO). Bacteria possess four basic shapes like - Bacillus, coccus, vibrio and spirillum.
  - Prokaryotes lack a nuclear membrane. Many bacterial cells have small circular DNA out side the genomic DNA is called plasmid. Prokaryotes lack the intracellular organelles like

mitochondria, chloroplast, endoplasmic reticulum, golgi bodies, centrioles.

- Prokaryotic cells have three architectural regions : (1) Appendages (2) Cell envelope (3) Cytoplasmic region.
- Appendages called flagellum, which are extended from cytoplasm. Pili or fimbriae arise from the surface of some bacteria. They play important role in conjugation. Flagellum is made up of flagellin protein.
- A cell envelop consisting of a capsule made up of a cell wall and a plasma membrane. Cell envelope demarcated into three layers. The outermost layer is made up of glycocalyx, the second layer is a cell wall and inner to cell wall is third layer called plasma membrane. Capsule serves as a protective layer against attack by phagocytes and by viruses.
- The plasma membrance functions as a selectively permeable layer. Energy production extra cellular polymer production, site of chromosome attachment and extension of plasma membrane into membranous structures.
- Gram named scientist developed two group of Bacteria on the basis of staining procedure. Those bacteria which take, up the gram stain are Gram positive and the other do not are called Gram negative bacteria.
- Prokaryotes possess Ribosome are dense particle of 20 nm in diameter. They are made up of two subunits 50 s and 30 s which combine to form 70 s Prokaryotic ribosomes. Ribosomes are the site of protein synthesis.
- In the cytoplasm a number of inclusion bodies occur called storage granules, which are not bounded by any membrane and lie free in cytoplasm e.g. Phosphate granules, cynnophycean granules and glycogen granules. Gas vacuoles are also reported in blue green and green photosynthetic bacteria.

(15) Which organism is included in Prokaryotes ?

(A) PPLO (B) Bacteria and Mycoplasm

(C) Blue-green algae (D) A, B, C All

(16) How many basic shapes of bacteria are there ?

(A) three (B) four (C) one (D) five

(17) Many bacterial cells have small circular DNA out side the genomic DNA is called as ..... .

(A) Pili (B) Fimbriae (C) Plasmid (D) Flagella

(18) Prokaryotic cells lack architectual region known as ..... .

(A) Appendages (B) Cell envelope (C) Cytoplasmic region (D) Nuclear-membrane

(19) The outermost layer of cell envelop is made up of which substance known as ..... .

(A) Glycocalyx (B) Mannans (C) Pectin (D) Lignin

(20) Which one serve as a protective layer against attack by phagocytes and by viruses ?

(A) Cell wall (B) Capsule (C) Flagella (D) Pili

(21) Which structure is formed in Prokaryotic organism due to extension of plasma membrane ?

(A) Mesosome (B) Vesicles

(C) Tubules and lamellae (D) A, B, C All

(22) Which is important for conjugation in bacteria ?

(A) Pili (B) Fimbriae (C) Flagella (D) A and B

(23) What is the diameter of Prokalyotic ribosome ?

(A) 25 nm (B) 20 nm (C) 30 nm (D) 28 nm

(24) Several ribosomes may attach to a single mRNA and form a chain called as...

(A) Polysomes (B) Polymer (C) Polypeptide (D) Polysaccharide

(25) Which granules are absent in Prokaryotes ?

(A) Cyanophycean granules (B) Phosphate granules

(C) Fat granules (D) Glycogen granules

Answers : (15-D), (16-B), (17-C), (18-D), (19-A), (20-B), (21-D), (22-D), (23-B), (24-A), (25-C)

- **Eukaryotic cells :**
  - Eukaryotes include fungi, animals and plants as well as some unicellular organisms. They possess organized nucleus with nuclear membrane.
  - Plant cell possess cell walls, plastids, large vacuoles which are absent in animal cells. Animal cell possess centriole which is absent in plant cell.

(26) Which cell possess cell wall, Plastids and large vacuoles ?

(A) Plant cell (B) Animal cell (C) Bacteria (D) A and C

(27) Which cell possess centriole ?

(A) Plant cell (B) Blue-green alge (C) Animal cell (D) A and B

Answers : (26-A), (27-C)

- **Plasma membrane :**
  - Cell membrane or plasma membrane forms the outermost envelope of cytoplasm. It is composed of lipids and proteins. Each lipid molecule possesses a polar hydrophilic head on outer surface and a non-polar hydrophobic tail towards the inner surface. Proteins may be peripheral or integral. Peripheral protein are associated with surface while integral protein are partially or totally burried in the membrane.
  - Robertson has proposed a unit membrane concept according to which an uneven protein layer occurs on both the sides of the lipid bilayer.
  - *Singer and Nicolson (1972)* proposed fluid mosaic model which is widely accepted model. According to fluid mosaic model, the plasma membrane is made up of a continuous lipid bilayer and the proteins included unit. This membrane is semi fluid and functionally dynamic.
  - The function of plasma membrance is the transport of the molecules across it. Plasma membrane behaves both as a semipermeable and as a selectively permeable membrane. Transport across the membrane occurs in two main ways - Passive transport and Active transport.

(28) Cell membrane is made up of ..... .

(A) Lipid (B) Protein (C) Carbohydrate (D) A and B

(29) Which protein does not separate easily from cell membrane ?

(A) Integral Protein (B) Peripheral Protein (C) Surface Protein (D) B and C

(30) Who proposed unit membrane concept for plasma membrane ?

(A) Robertson (B) Singer and Nicolson (C) Robert Brown (D) Robert Hook

(31) Who proposed plasma membrane as an unit membrane ?

(A) Singer (B) Nicolson (C) Robert Hook (D) Robertson

(32) Which type of transport occurs by cell membrane ?

(A) Active transport (B) Passive transport (C) Faciliated diffusion (D) A, B, C All

Answers : (28-D), (29-A), (30-A), (31-D), (32-D)

- **Cell wall :**
  - Cell wall is a non-living rigid structure which forms an outer covering for the plasma membrane in plants. It gives the shape to the cell and also protects cell from mechanical damage and infection.
  - Algae cell wall is made up of cellulose, galactans, Mannons and calcium carbonate, while in other plants it is made up of cellulose, hemicellulose, pectins and protein. The fungal cell wall is made up of chitin.
  - The primary cell wall is made up of cellulose, above it a secondary wall is formed which is made up of hemicellulose, lignin and suberin. A middle lamella is made up of calcium pectate which occurs between primary wall of two adjacent cells.

(33) What is the function of plant cell wall ?

(A) To give shape to the cell (B) To protects the cell from mechanical damage and infection

(C) To form an outer covering (D) A, B, C All

(34) Algal cell wall is made up of which substance ?

(A) Mannons and galactans (B) Cellulose

(C) Calcium Carbonate (D) A, B, C All

(35) In plant cell a secondary cell wall formed by deposition of which substance ?

(A) Lignin (B) Suberin (C) Hemi cellulose (D) A, B, C All

(36) In plant, cytoplasm of two neighbouring cells are connected through which structure ?

(A) Middle lamella (B) Plasmodesmata (C) Pectin (D) A and C

Answers : (33-D), (34-D), (35-D), (36-B)

- **Endoplasmic membrane system :**
  - All the membranous organelles in the cells together form an endomembrane system.
  - Endoplasmic reticulum, golgi complex, lysosomes and vacuoles are considered as a membrane system.
  - As the functions of mitochondria, chloroplast and peroxisomes are not co-ordinated with the above components these are not considered as part of endomembrane system.

(37) Which is included in endomembrane system ?

(A) Endoplasmic reticulum (B) Golgi complex (C) Lysosome (D) A, B, C All

(38) Which organelle is not included in endomembrane system ?

(A) Mitochondria (B) Peroxisome (C) Chloroplast (D) A, B, C All

Answers : (37-D), (38-D)

- **Endoplasmic reticulum :**
  - A network of tubular structure spread throughout the cytoplasmic region is called endoplasmic reticulum. The tubules are double walled bag like structure.
  - Endoplasmic reticulum maintain association with plasma membrane and nucleur membrane.
  - The outersurface of endoplasmic reticulum possess ribosomes are called rough endoplasmic reticulum and those do not possess ribosomes on outer surface is called smooth endoplasmic reticulum.
  - Rough endoplasmic reticulum (RER) synthesize protein and smooth endoplasmic reticulum (SER) synthesize lipid.

(39) A network of tubular structure spread throughout the cytoplasmic region is called .....
(A) Lysosomes (B) Endoplasmic reticulum (C) Golgi-complex (D) Ribosomes

(40) How many types of endoplasmic reticulum are there ?
(A) One (B) Two (C) Three (D) Four

(41) In which orgenelle the synthesis of lipid like Steroidal hormones occur ?
(A) RER (B) SER (C) Golgi-complex (D) Ribosomes

Answers : (39-B), (40-B), (41-B)

- **Golgi-apparatus :**
  - First in *1898* an Italian anatomist *Camillo Golgi* observed golgi-apparatus near the nucleus.
  - Golgi-body is formed by a stackpile arrangement of flat membraneous bag like units or cisternae of 0.5 μm to 1.0 μm diameter. Four to eight unit occur in each stack pile. Towards their outer end round vesicles appear.
  - Golgi body is a site for synthesis of Glycolipids and Glycoproteins.

(42) What is the diameter of cisternae in Golgi body ?
(A) 0.5 μm to 1.0 μm (B) 0.1 μm to1.0 μm
(C) 5.0 μm to1.0 μm (D) 0.25 μm to 0.50 μm

(43) Which substance is synthesized by Golgi body ?
(A) Glycolipids (B) Glycoproteins (C) Glycol (D) A and B

Answers : (42-A), (43-D)

- **Lysosomes :**
  - Lysosomes arise as vesicles separated from golgi bodies, which is surrounded by a single layered wall. They are associated with intracellular digestion.
  - Lysosomes possess hydrolases enzyme. They are important in phagocytosis and pinocytosis like processes and also responsible for degradation of worn out cells so they are called 'suicide bags'.

(44) Lysosomes produce from which organelles ?
(A) Golgi body (B) Ribosomes (C) Endoplasmic Reticulum (D) Mitochondria

(45) Lysosomes are not connected with which functions ?
(A) Pinocytosis (B) Respiration (C) Phagocytosis (D) Lysis

(46) Which organells of cell known as suicidal bag ?

(A) Golgi body (B) Chloroplast (C) Lysosomes (D) Mitochondria

(47) Which is the single layered organelle of cell ?

(A) Mitochondria (B) Chloroplast (C) Golgi body (D) Lysosomes

(48) Lysosomes possesses which types of enzyme ?

(A) Lipases (B) Proteases (C) Carbohydrases (D) A, B, C All

Answers : (44-A), (45-B), (46-C), (47-D), (48-D)

- **Vacuoles :**
  - The area in cytoplasm without any cytoplasm is called vacuole. In plant cell a large vacuole present but normally vacuole do not occur in animal cell.
  - Vacuole surrounded by a semipermeable membrane is called tonoplast.
  - In Paramoecium the vacuole is contractile. They create osmotic pressure in cells. Various substance are excreted or stored in them.

(49) In cell the area in cytoplasm without any cytoplasm is known as ..... .

(A) Golgi body (B) Vacuole (C) Ribosomes (D) Lysosomes

(50) What is the nature of Tonoplast ?

(A) Impermeable membrane (B) Permeable membrane

(C) Semi permeable membrane (D) Selective permeable membrane

(51) In which organism contractile vacuoles are present selectively ?

(A) Paramoecium (B) Bacteria (C) Blue-green algae (D) Hydra

(52) What is the function of contractile vacuole ?

(A) Stores substances (B) Excrete the substances

(C) Create osmotic pressure (D) A, B, C All

Answers : (49-B), (50-C), (51-A), (52-D)

- **Mitochondria :**
  - Mitochondria is self replicating organelle that occur in various number, shapes and sizes in the cytoplasm of all eukaryotic cells. The number of mitochondria depends upon the physiological activity of the cells. They are fibrous, cylindrical or granular having diameter of 0.2 – 1.0 μm and length 1.0 – 4.1 μm.
  - Each mitochondrion possesses a double walled envelope. The outer layer is continuous while the inner layer has many foldings are called cristae. Which possesses F_1 particle. F_1 particle is site for oxidative phosphorylation where synthesis of ATP takes place. The area without cristae is known as matrix. Matrix possesses ribosomes, circular DNA. Kreb's cycle occurs in Matrix of mitochondria.
  - Mitochondria is known as power house of cell.

(53) Which is the self replicating organelle of an organism ?

(A) Lysosome (B) Mitochondria (C) Golgi body (D) Centriole

(54) What is the length and diameter of mitochondria ?

(A) 0.2 - 1.0 μm and 1.0 - 4.1 μm (B) 2.0 - 0.1 μm and 0.1 - 1.4 μm

(C) 1.0 - 4.1 μm and 0.2 - 1.0 μm (D) 4 μm and 3 - 5 μm

(55) Which process occurs in matrix and cristae in mitochondria respectively ?

(A) Oxidative phosphorylation and Glycolysis (B) TCA cycle and Glycolysis

(C) Kreb's cycle and oxidative Phosphorylation (D) Glycolysis and Kreb's cycle

(56) Which necessary enzyme are present in cristae F_1 particle of mitochondria ?

(A) Oxidative phosphorylation (B) Glycolysis

(C) Kreb's cycle (D) Light reaction

(57) Which organelles is known as power house of cell ?

(A) Chloroplast (B) Lysosome (C) Ribosomes (D) Mitochondria

Answers : (53-B), (54-C), (55-C), (56-A), (57-D)

- **Plastids :**
  - Plastids are found in all plant cells. Based on the types of pigments plastids are of three types : Chromoplasts, Chloroplasts and leucoplasts.
- **Chromoplasts :**
  - These plastids possess pigments other than chlorophylls. They have pigments like carotene xanthophylls, anthocynin, etc. Various colours in flowers, fruits and seeds are due to their presence.
- **Chloroplasts**
  - Chlorophyll containg pigment is called chloroplasts. They carry out photosynthesis process. Majority of chloroplasts are present in the mesophyll of leaves.
  - Chloroplasts are lens shaped, oval, spherical, discoid or ribbon shaped. They are 5-10 mm in length and 2-4 mm in width. There number varies in various cells. Chlamydomonas possess one and mesophyll have 20 to 40 chloroplast per cells.
  - Chloroplast has a double layer wall. Outer layer is continuous and inner layer forms a highly folded membrane system, which forms a grana. The grana are interconnected by intergranum membrans. The region except the membrane is called stroma.
  - Each granum is made up of a stack of flat lamellar structure called thylacoids.
  - Normally one chloroplast contain 40 to 60 grana each granum possesses 02 to 100 thylakoids. Thylakoids possess chlorophyll and essential materials for synthesis of ATP through photophosphorylation. Stroma contains proteins, (70S) ribosomes, circular DNA and enzyme required in dark reaction.

- **Leucoplast :**
  - They do not contain any pigments. They acts as food storing units those which store starch are called amyloplasts, Elaioplasts store fat or oil and aleuroplasts store proteins.

(58) How many types of plastids are there ?
(A) One (B) Two (C) Three (D) Four

(59) The various colour of flowers, fruits and seeds are due ..... .
(A) Carotene (B) Xanthophylls (C) Anthocyanin (D) A, B, C All

(60) In which part of leaves the chloroplasts are present ?
(A) Epidermis (B) Hypodermis (C) Mesophyll tissue (D) A, B, C All

(61) What is the length of chloroplast ?
(A) 5 - 10 mm (B) 5 - 10 μm (C) 2 - 10 cm (D) 2 - 4 μm

(62) How many number of chloroplasts are present per cell of mesophyll ?
(A) 20 to 40 (B) 30 to 40 (C) 10 to 20 (D) 25 to 35

(63) Where does chloroplast materials essential for synthesis of ATP are present ?
(A) In Stroma (B) In Thylakoid (C) A and B (D) Intergranum membrane

(64) Which reaction occurs in grana and stroma of chloroplasts respectively ?
(A) Dark reaction and Light reaction (B) Light reaction and Dark reaction
(C) Light reaction and Glycolysis (D) Photooxidation and Photophosphorylation

(65) How many thylakoids are present in each granum in chloroplast ?
(A) 02 to 100 (B) 20 to 50 (C) 30 to 40 (D) 40 to 60

(66) What is absent in stroma of chloroplast ?
(A) Protein (B) 70S Ribosomes (C) Circular - DNA (D) 80S Ribosomes

(67) Which substance is absent in leucoplasts ?
(A) Protein (B) Starch (C) Plastids (D) Elaioplasts

(68) What is the shape of chloroplast ?
(A) Lens (B) Oval (C) Spherical (D) A, B, C All

(69) How many chloroplast are present in Chlamydomonas ?
(A) One (B) Two (C) Three (D) Four

Answers :(58-C), (59-D), (60-C), (61-B), (62-A), (63-B), (64-B), (65-A), (66-D), (67-C), (68-D), (69-A)

- **Ribosomes :**
  - Ribosomes are granular structure found freely in the cytoplasm as well as associated with endoplasmic reticulum.
  - In Eukaryotes ribosomes are 80 S types (60 S + 40 S) and In prokaryotes they are 70 S type (50 S + 30 S).
  - In the structure of ribosome ribosomal RNA and Proteins are present. Ribosomes are associated with protein synthesis.

- Association of more than one ribosome with single molecule of mRNA is called polysome or polyribosome.

(70) Which type of Ribosome respectively present in Eukaryotes and Prokaryotes ?

(A) 80 S and 70 S (B) 60 S and 40 S (C) 50 S and 30 S (D) 70 S and 80 S

(71) Which types of RNA present in structure of Ribosomes ?

(A) m - RNA (B) r - RNA (C) t -RNA (D) A, B, C All

(72) Which substances are synthesized by Ribosomes ??

(A) Carbohydrates (B) Lipids (C) Proteins (D) Nucleic acids

Answers : (70-A), (71-B), (72-C)

- **Cytoskeleton :**
  - Cytoskeleton is formed by three kinds of filaments (1) Microfilaments (2) Microtubules (3) Intermediate filaments.
  - Microfilaments are made up of actin like proteins. They may be scattered or arranged in a network or in a parallel fashion. They are associated with cellular form and cellular movement. Amoeboid movement, cyclosis of protoplasm and transport of particulate are due to them.
  - Microtubules are made up of the globular protein - tubulin. They are responsible to maintain cell shape, in cellular movement, cellular transport of materials and for chromosome movement.
  - Intermediate filaments are strong and durable protein fibres. They form a basket and provide support to other filaments and tubules.

(73) How many filaments is/are associated with structure of cytoskeleton ?

(A) One (B) Two (C) Three (D) Four

(74) Microfilaments are made up of which substance ?

(A) Myosin (B) Actin (C) Tubulin (D) Carotene

(75) What is present in microtubule ?

(A) Actin (B) Tubulin (C) Carotene (D) Myosin

(76) Which structure is related with cyclosis of protoplasm ?

(A) Microtubules (B) Intermediate filaments (C) Microfilaments (D) A and B

(77) Which structure is associated with chromosome movement ?

(A) Microtubules (B) Intermediate filaments (C) Microfilaments (D) A and C

(78) Which structure of cytoskeleton forms basket ?

(A) Microtubules (B) Microfilaments (C) Stackpile (D) Intermediate filaments

(79) Which substance is present in cytoskeleton ?

(A) Carbohydrate (B) Protein (C) Lipids (D) Nucleic acid

Answers : (73-C), (74-B), (75-B), (76-C), (77-A), (78-D), (79-B)

- **Cilia and Flagella :**
  - Cilia and flagella are associated with locomotion and movement. They are located on free surface of cells. There is a lot of similarity in their ultrastructure. Both originate in a basal body.
  - Flagella are longer. They are one or two. Euglena possesses flagella.
  - Cilia are short. They are many. Cilia are observed in Paramoecium.
  - Bacteria may be ciliated or flagellated.
  - The axis of cilium and flagellum is called axoneme. It is made up of two central and nine pairs of microtubes along the circumference (9 + 2 arrangement). They are connected with the central microtubules through filaments.

(80) Which are the functions of Cilia and flagella ?

(A) Locomotion (B) Movement (C) Digestion (D) A and B

(81) Which types of arrangement is structure of cilia and flagella ?

(A) 9 + 2 (B) 9 + 0 (C) 2 + 9 (D) 0 + 9

(82) Which organisms possesses cilia ?

(A) Euglena (B) Paramoecium (C) Amoeba (D) Opalina

(83) Which organisms possesses flagella ?

(A) Amoeba (B) Euglena (C) Opalina (D) Paramoecium

(84) From where are the Flagella and cillia originated ?

(A) Centriole (B) Mitochondria (C) Basal body (D) Golgi body

Answers : (80-D), (81-A), (82-B), (83-B), (84-C)

- **Centrosome and Centrioles :**
  - Centrosome containing two cylindrical structure, arranged right angle to each other called centrioles.
  - Each centriole is organized like a wheel. Nine triplets forming angle of about 40° are arranged in peripheral region. Three microtubules are made up of tubulin. Each triplet are interconnected by protein fibres. At the centre there occurs proteinaceous hub. The microtubules of triplets are connected with the hub by filaments. The protoplast surrounding the centrioles is called centrosphere.
  - All animal cells and some algae and fungi have centriole.
  - Centrosome related with formation of bipolar spindle during cell division and in the formation of basal granules, cilia and flagella.

(85) Which of following has Centriole ?

(A) Animal cell (B) Some of algae (C) Some of fungi (D) A, B, C All

(86) Which structure is formed by association with centrosome ?

(A) Basal granules (B) Flagella (C) Cilia (D) A, B, C All

(87) Two cylindrical structure of centrosome arranged right angle to each other then is known as ..... .

(A) Centrosphere (B) Centrioles (C) Golgi-apparatus (D) Lysosome

(88) Which organelles is associated with the formation of the bipolar spindle during cell division ?

(A) Centriole (B) Flagella (C) Cilia (D) Golgi body

(89) At which angle nine triplets of centrioles are arranged in peripheral region ?

(A) 30º (B) 45º (C) 40º (D) 90º

Answers : (85-D), (86-D), (87-B), (88-A), (89-C)

- **Nucleus :**
  - Nucleus is a regulatory centre of various cellular activities. Normally the cells are uninucleate but some cells are binucleate. e.g. zygote and some are multinucleate. e.g. Endosperm while Human RBC and sieve tubes lack nucleus.
  - Nuclear membrane, nucleoplasm, nucleolus and chromatin occur in the structure of nucleus.
  - Nuclear membrane is a double layered structure. Between two layers a space called perinuclear space. The outer layer associated with endoplasmic recticulum. It possesses ribosomes. Nuclearpores occur at various place and they are responsible for transfer of RNA and protein molecules between cytoplasm and nucleoplasm.
  - Nucleus contain nucleoplasm and chromatin. It is a spherical shaped and no membrane surrounds it. They are formed on nucleolar organizer region of some specific chromosomes. Nucleolus is a site for active ribosomal RNA synthesis.

(90) Which of the following is a regulatory centre of various cellular activities in cell ?

(A) Centrioles (B) Nucleus (C) Golgi body (D) Chromosomes

(91) Which cells are anucleated ?

(A) Human RBC (B) Sieve tubes (C) Zygote (D) A and B

(92) Which structure is associated with structure of nucleus ?

(A) Nuclear membrane (B) Nucleolus (C) Chromatin (D) A, B, C All

(93) Which organelle is amembranous ?

(A) Nucleus (B) Chloroplasts (C) Lysosomes (D) Nucleolus

(94) Nuclear organizer of some chromosome synthesized which RNA ?

(A) r - RNA (B) m-RNA (C) t-RNA (D) s-RNA

Answers : (90-B), (91-D), (92-D), (93-D), (94-A)

- **Chromosomes :**
  - Chromatin is made up of DNA, RNA, histone and non-histone kinds of protein.
  - In an interphase of cell the chromosomes are spread as an indistinct network which is called chromatin. In eukaryotic cell during the metaphase of cell division chromosome appears distinct.
  - Each chromosomes has a primary constriction disc shaped structure called Kinetochores. On the position of centromere chromosomes are base of four types :

    (1) Metacentric (2) Sub-metacentric (3) Acrocentric (4) Telocentric

- *Metacentric* : Centromere is located in the centre forming two equal arm of chromosome.
- In *sub-metacentric* chromosome centromere located some distance away from the centre resulting into one shorter and one longer arm of chromosomes.
- In *Acrocentric* chromosome from centromere one arm is short and other is very long, the short arm possesses stalk and satellite.
- In *Telocentric* centromere is located at the end of chromosome.

- **Microbodies :**
  - Membrane bound minute vesicle called microbodies that contain various enzymes are present in both plant and animal cells.

---

(95) Which substance is present in Chromatin ?

(A) DNA and RNA (B) Histone (C) Non Histone (D) A, B, C All

(96) In which stage of cell division chromosomes are visible in distinct shape ?

(A) Prophase (B) Metaphase (C) Anaphase (D) Telophase

(97) Centromere is located at the end of chromosome is known as ..... .

(A) Metacentric (B) Telocentric (C) Acrocentric (D) Sub-metacentric

(98) The chromosomes which possesses stalk and satellite is known as ..... .

(A) Acrocentric (B) Metacentric (C) Telocentric (D) Sub-metacentric

(99) Which structure of plant and animal cell possesses various enzymes ?

(A) Nucleus (B) Microbodies (C) Nucleolus (D) Lysosomes

Answers : (95-D), (96-B), (97-B), (98-A), (99-B)

- **True-False (T - F) type questions :**

Choose the correct option :

(100) (1) Life of organisms begins from zygote.

(2) Any cell of the body is capable of producing a new individual.

(3) New cells are formed through divisions of pre-existing cells.

(4) Mycoplasma is the smallest eukaryotic cell.

(5) Bacteria and all other cell possesses nucleus and other organelles.

(A) T, F, T, T, F (B) T, T, T, F, F (C) T, T, F, T, F (D) F, T, T, F, T,

(101) (1) Prokaryotic cell possesses Mitochondria, Chloroplast, Endoplasmic reticulum and centriole.

(2) Flagella is made up of carbohydrate named flagellin.

(3) Prokaryotes possesses 70 s types and Eukaryotes possesses 80 s type of ribosomes.

(4) Fluid-mosaic model was proposed by Robertson.

(5) Mitochondria, Chloroplast and Peroxisomes are not considered as part of endomembrane system.

(A) F, F, T, F, T (B) T, T, F, F, T (C) T, F, F, T, T (D) F, T, T, T, F

(102) (1) The diameter of cisternae in golgi apparatus is 0.5 μm to 1.0 μm.

(2) Lysosome possesses single layer wall.

(3) One chloroplast has 40-60 Grana.

(4) Aleuroplasts stores starch.

(5) All animal cells possesses centrioles.

(A) F, T, T, F, T (B) T, F, T, F, T (C) T, T, T, F, T (D) T, T, F, T, T

(103) (1) Microfilaments made up of protein.

(2) Microtubules made up of Tubulin.

(3) The arrangement of flagella in Euglena is 9 + 2.

(4) Nucleolus is membrane less structure.

(5) The nucleus is absent in human RBC and sieve tube.

(A) F, T, F, F, T (B) T, F, T, F, T (C) T, F, F, T, T (D) T, T, T, T, T

(104) (1) The chromosomes are four types on the basis of function of centromere.

(2) Chromatin is made up of histone and non-histone kind of proteins.

(3) Acrocentric chromosome possesses satellite.

(4) The Lysosomes known as suicidal bags of cells.

(5) The Matrix of mitochondria and chloroplast possesses ribosomes and circular DNA.

(A) F, T, T, F, T (B) F, T, T, T, T (C) F, F, T, T, T (D) T, F, T, T, F

(105) (1) The endoplasmic reticulum is of two type : (1) RER (2) SER

(2) Fimbriae plays important role in conjugation.

(3) In bacteria outermost layer is made up of glycocalyx.

(4) All living organisms are not made up of cells and their products.

(5) Golgi body is a site for synthesis of glycolipids and glycoproteins.

(A) T, F, T, T, F (B) T, T, F, T, F (C) T, T, T, F, T (D) T, T, T, F, F

Answers : (100-B), (101-A), (102-C), (103-D), (104-B), (105-C)

- **Assertion (A) and Reason (R) type questions :**

  Select right options from following :

  (A) A and R both are true. R is correct explanation of A.

  (B) A and R both are true. R is not correct explanation of A.

  (C) A is true, R is false.

  (D) A is false, R is true.

(106) Assertion A : Cell is the structural and functional unit of living organisms.

Reason R : New cells are formed by cell division of pre existing cells.

(A) (B) (C) (D)

(107) Assertion A : Mitochondria is known as power house of cell.
Reason R : ATP is known as energy currency of the cell.
(A) (B) (C) (D)

(108) Assertion A : Mitochondria is self replicating organelle.
Reason R : Mitochondria possesses circular DNA and ribosomes.
(A) (B) (C) (D)

(109) Assertion A : Cilia and Flagella possesses 9 + 2 type of arrangement.
Reason R : Centriole does not possesse 9 + 0 type arrangement.
(A) (B) (C) (D)

(110) Assertion A : Ribosome is associated with Protein synthesis.
Reason R : Ribosome is non membranous.
(A) (B) (C) (D)

(111) Assertion A : Aleuroplasts stores proteins.
Reason R : Various colours in flowers, fruits and seeds are due to aleuroplasts.
(A) (B) (C) (D)

(112) Assertion A : Mitochondria, Lysosome and chloroplast are parts of endomembrane system.
Reason R : In cell all the membranous organelles combine to form endomembrane system.
(A) (B) (C) (D)

(113) Assertion A : Cytoplasms of two neighbouring cells are connected through plasmodesmata.
Reason R : Cell wall and middle lamella possesses plasmodesmata.
(A) (B) (C) (D)

(114) Assertion A : Blue-green algae is a prokaryotic cell.
Reason R : In prokaryotic cell cell division occurs very rapidly.
(A) (B) (C) (D)

(115) Assertion A : RER possesses ribosomes.
Reason R : Steroidal hormones are synthesized by RER.
(A) (B) (C) (D)

(116) Assertion A : Mitochondria synthesize ATP.
Reason R : F1 particle is present in mitochondria.
(A) (B) (C) (D)

(117) Assertion A : 20 to 40 chloroplasts are present per cell of mesophyll.
Reason R : Each chloroplast has Grana and stroma in their structure.
(A) (B) (C) (D)

Answers : (106-B), (107-B), (108-B), (109-C), (110-B), (111-C), (112-D), (113-B), (114-B), (115-C), (116-A), (117-B)

(118) Select correct option :

Column - I	Column - II	
(a) Microtubules	(i) Glycocalyx	(A) a - iv, b - iii, c - ii, d - i
(b) Microfilaments	(ii) Actin	(B) a - iii, b - ii, c - iv, d - i
(c) Flagella	(iii) Tubulin	(C) a - iii, b - iv, c - ii, d - i
(d) Outer most layer of Bacteria	(iv) Flagellin	(D) a - ii, b - iv, c - i, d - iii

(119) Select correct option :

Column - I	Column - II	
(a) Zygote	(i) Single nucleus	(A) a - iii, b - iv, c - ii, d - i
(b) Human RBC	(ii) Multi nucleate	(B) a - iv, b - iii, c - i, d - ii
(c) Endosperm	(iii) Binucleate	(C) a - ii, b - iv, c - iii, d - i
(d) Typical plant cell	(iv) Absence of nucleus	(D) a - iii, b - iv, c - i, d - ii

(120) Select correct option :

Column - I	Column - II	
(a) Mitochondria	(i) Synthesis of Glycolipids	(A) a - ii, b - i, c - iv, d - iii
(b) Lysosomes	(ii) ATP - formation and storage	(B) a - iii, b - ii, c - i, d - iv
(c) Chloroplast	(iii) Suicidal bag	(C) a - iv, b - iii, c - ii, d - i
(d) Golgi body	(iv) Photosynthesis	(D) a - ii, b - iii, c - iv, d - i

(121) Select correct option :

Column - I	Column - II	
(a) Robert Hooke	(i) Cell theory	(A) a - ii, b - iii, c - iv, d - i
(b) Robert Brown	(ii) Cell	(B) a - iii, b - iv, c - i, d - ii
(c) Camillo Golgi	(iii) Nucleus	(C) a - iv, b - i, c - ii, d - iii
(d) Schleiden and Schwann	(iv) Golgi body	(D) a - ii, b - i, c - iv, d - iii

(122) Select correct option :

Column - I	Column - II	
(a) Chloroplast	(i) Non-membranous structure	(A) a - ii, b - i, c - iv, d - iii
(b) Lysosome	(ii) Create osmotic pressure	(B) a - iii, b - iv, c - ii, d - i
(c) Nucleolus	(iii) Two layered structure	(C) a - iv, b - iii, c - ii, d - i
(d) Contractile Vacuole	(iv) Single layered structure	(D) a - iii, b - iv, c - i, d - ii

Answers : (118-B), (119-A), (120-D), (121-A), (122-D)

(123) Given diagram is known as ..... .

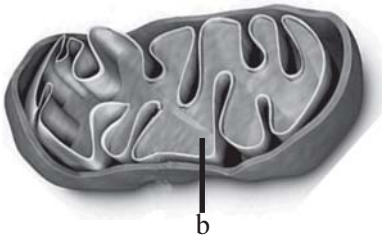
(A) Mitochondria (B) Chloroplast

(C) Golgi body (D) Ribosome

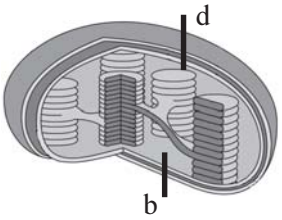
(124) Give the name of b in given diagram.

(A) Cristae (B) Matrix

(C) Mitochondria (D) F_1 Particle



(125) Which process occurs in b in given figure ?



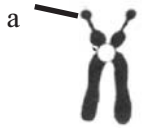
(A) Oxidative phosphorylation

(B) Dark reaction

(C) Light reaction

(D) Phosphorylation

(126) 'a' is significant of which chromosomes ?



(A) Metacentric (B) Sub Metacentric

(C) Acrocentric (D) Telocentric

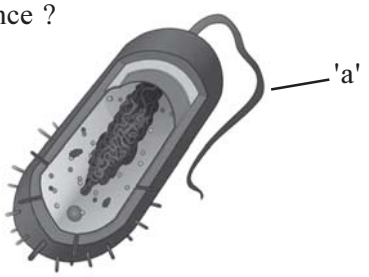
(127) In given diagram, a is made up of which substance ?

(A) Flagellin

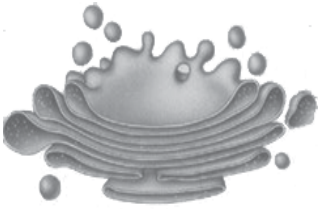
(B) Suberin

(C) Pectin

(D) Glycocalyx



(128) Given diagram is known as ..... .



(A) Golgi body

(B) Mitochondria

(C) Chloroplast

(D) Endoplasmic reticum

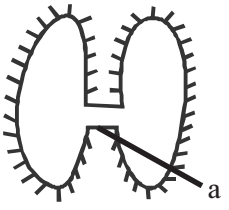
(129) In diagram a – related with which process ?

(A) Conjugation

(B) Protein Synthesis

(C) Digestion

(D) Excretion



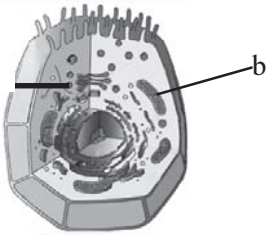
(130) In given diagram b is known as ..... .

(A) Crystals

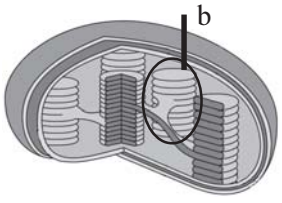
(B) Vacuoles

(C) Nucleus

(D) Mitochondria



(131) In given diagram b known as ..... .



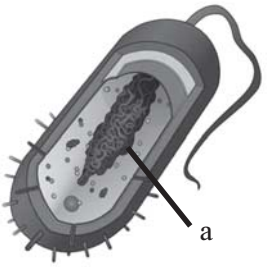
(A) Grana

(B) Stroma

(C) Intergranum membrane

(D) Granum

(132) In given diagram 'a' known as ..... .



(A) DNA (B) RNA

(C) Nucleoid (D) Nucleus

Answers : (123-A), (124-B), (125-B), (126-C), (127-A), (128-A), (129-A), (130-D), (131-A), (132-C)

- **Questions for NEET :**

(133) Which substance is present in higher plant cell wall ?

(A) Lipoprotein (B) Cellulose (C) Peptidoglycan (D) Fungus cellulose

(134) Which substance is synthesized by Granular endoplasmic reticulum (RER) ?

(A) Lipid (B) Protein (C) Carbohydrate (D) Nucleic acid

(135) What is the similarity of prokaryotic and eukaryatic cell ?

(A) Histone (B) Gene (C) Mitosis (D) Endoplasmic reticulum

(136) Which of the following is absent in prokaryotic cell ?

(A) Centrioles (B) Membranous organelles (C) Nucleus (D) A, B, C All

(137) Which are the two sub units of ribosomes in Eukaryotic cell ?

(A) 50 S + 40 S (B) 30 S + 70 S (C) 60 S + 30 S (D) 60 S + 40 S

(138) In which organelle DNA is absent ?

(A) Chloroplasts (B) Mitochondria (C) Ribosomes (D) Cytoplasm

(139) Which organell is related with formation of ATP ?

(A) Mitochondria (B) Golgi-apparatus (C) Ribosomes (D) Chloroplasts

(140) Centrioles related with process known as ..... .

(A) Crossing over (B) Gene transcription (C) Chromosome movement (D) Cell-division

(141) Where does the ribosomes synthesized ?

(A) Golgi-apparatus (B) Nucleolus (C) Cytoplasm (D) Cell-membrane

(142) Aldolase enzyme is related with organelle ?

(A) Chloroplast (B) Ribosomes (C) Golgi-body (D) Lysosome

(143) The fluid mosaic model is related with organelle ..... .

(A) Plasma membrane (B) Nucleus (C) Cytoplasm (D) Cell wall

(144) In higher plants stroma of chloroplast possesses enzyme for ..... .

(A) Dark reaction (B) Light reaction

(C) Phosphorylation (D) A and C

(145) The cristae is related with organelles ..... .

(A) Endoplasmic reticulum (B) Chloroplasts (C) Golgi-apparatus (D) Mitochondria

(146) Where does the peptide synthesis occur in cell ?

(A) Ribosome (B) Chloroplast (C) Mitochondria (D) Lysosome

(147) Which organelles is related with synthesis of new protein and protein transport ?

(A) Chloroplast (B) Endoplasmic reticulum (C) Mitochondria (D) Lysosome

(148) Which is the self replicating organelles ?

(A) Golgi body and Endoplasmic reticulum (B) Chloroplast and Golgi-body

(C) Mitochondria and Chloroplast (D) Lysosome and Mitochondria

(149) Which is false about Mitochondria ?

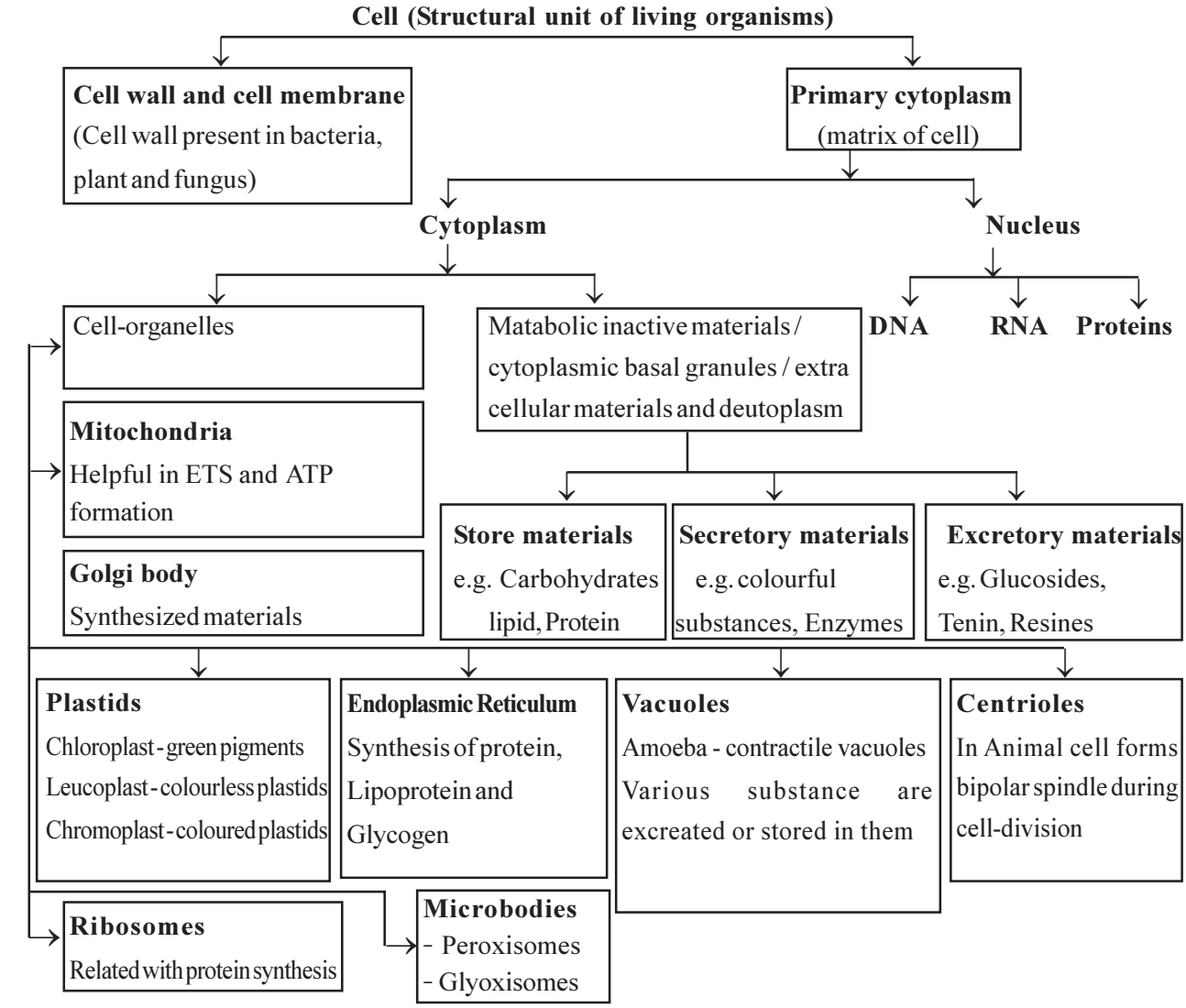
(A) Outermost layer of mitochondria is permeable for all type of molecule.

(B) Outermost layer possesses enzyme for oxidative phosphorylation (ETS).

(C) Outermost layer of mitochondria possesses seive like pores.

(D) Inner most layer possesses many folds.

Answers : (133-B), (134-B), (135-C), (136-D), (137-D), (138-C), (139-A), (140-D), (141-B), (142-D), (143-A), (144-A), (145-D), (146-A), (147-B), (148-C), (149-B)



6 Biomolecules-I (Carbohydrates and Lipids)

- **Cell :** Cell is the structural and functional unit of living organism.
- **Biomolecules :** All the compounds found in living tissues can be called biomolecules.
- **Biochemical reactions :** Metabolic activities carried out by cells are an outcome of various chemical reactions called biochemical reactions.
  - Living organisms obtain a variety of molecules and elements from their environment.
  - With this element living organisms synthesize essential components. For the living body mainly four atoms are important viz. C, H, O, N.
  - These substances are of two types find in living organisms : (1) Inorganic substances (Water, Mineral salts, Mineral elements) (2) Organic substances (Carbohydrates, Lipid, Protein, Enzyme, Hormone)

---

(1) Which are the important organic elements found in living being ?

(A) C, H, Mg, P (B) C, K, Na, N (C) C, H, O, N (D) C, H, N, P

(2) Which of the following is inorganic compound of living ?

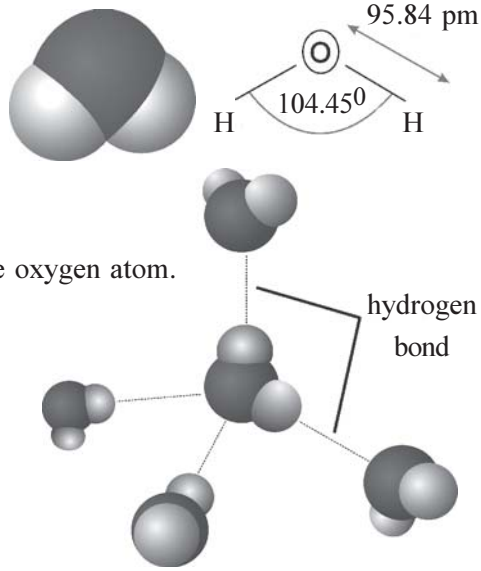
(A) Hormone (B) Minerals (C) Enzyme (D) Nucleotide

(3) Which substance helps in metabolism in living organisms ?

(A) Enzymes (B) Organic molecules (C) Inorganic molecules (D) Minerals

Answers : (1-C), (2-B), (3-A)

- **Water :**
  - In the constitution of living organism 65 % of water occurs
  - In living cell it may be 70 % to 90 %
  - Water usually makes up 55 to 78% of the human body.
  - It has two hydrogen atoms covalently bonded to a single oxygen atom.
  - H - O - H



Properties :

- Colourless, tasteless and odourless liquid.
- Most efficient solvent.

  Most of the chemicals found in organisms are soluble in water, their homogeneity of cytoplasm is maintained. An excellent transport medium for various chemicals.
- O_2 and CO_2 are transported through water.
- Water has polar nature.
- The hydrogen and oxygen units in its constitution bear partial positive and partial negative charges respectively, thus molecules remain bounded with one another.
- Thus normally occurs as liquid.
- Thus it is an efficient solvent also.

- Water has characteristic of high specific and latent heat. Due to these properties any change in the temperature of the surrounding is loss or gain of heat in water does not affect the temperature of water. Since water has a high latent heat, water in lakes and sea does not freeze.
- Cohesive force is very high.

  Thus ascent of sap can maintain.
- It has more density and viscosity.
- The density of water is dependent on the dissolved salt content as well as the temperature of the water.
- The density of water is highest at 4° C similarity, its viscosity is also high.
- This property of water enables planktonic organisms to float and move about freely on the surface of water. They do not encounter any mechanical shock in water.
- Water has high capacity to conduct heat, hence heat is equally distributed in all the parts of the body of an organism.
- Water plays an important role in maintenance of three dimensional form of macromolecules. e.g., Proteins, nucleic acids, etc.
- Water also acts as a reactant and provides H^{+} and OH^{-} for many reactions.

(4) Which is most efficient solvent ?

(A) Water (B) Ethanol (C) Liquid N_2 (D) Above three

(5) Water normally occurs as liquid, because of ..... .

(A) polarity of water (B) Thermal conductive energy of water

(C) latent heat of water (D) cohesive force of water

(6) What is the distance between H and O in water ?

(A) 95.84 meter (B) 10^{-12} meter

(C) 95.84×10^{-12} meter (D) 104.45 picometer

(7) What is the characterstic of latent heat of water ?

(A) Do not encounter mechanical shock in water.

(B) It maintains ascent of sap in plants.

(C) Heat is equally distributed in all parts of body.

(D) Water in lakes does not freeze.

(8) Ascent of sap path is maintained by plant because of ..... .

(A) water high density (B) water high viscosity

(C) high specific heat of water (D) cohesive force of water

(9) Planktonic organisms freely float on the surface of the water because of ....

(A) polarity of water (B) viscosity of water (C) water heat (D) solubility of water

Answers : (4-A), (5-A), (6-C), (7-D), (8-D), (9-B)

- **Minerals :**

Minerals	Importance
(1) Nitrogen (N)	- Essential for the synthesis of proteins and nucleic acids. - It forms a part of vitamins, enzymes.
(2) Phosphorous (P)	- It is an important constituent of nucleic acids, cell membrane, ATP and various enzymes. - In energy exchange processes. - In the vertebrates about 80 % of the total phosphorous is associated with teeth and bones.
(3) Calcium (Ca)	- For strong bones and teeth. - In clotting of blood and contraction of muscles. - The middle lamella between plant cell is make up of calcium pectate. - It determines the permeability of cell membrane.
(4) Sulphur (S)	- It is found as a structural component. e.g., Cysteine, Methionine - It is also structural component of vitamins. e.g., Biotin, Thiamine - It is present in hyaline cartilage, ligaments and bone marrow
(5) Magnesium (Mg)	- It is contituent of chlorophyll, ATP and in synthesis of carbohydrates - Enzymes involved in carbohydrates, lipids and protein metabolism contain Mg.
(6) Iron (Fe)	- Haemoglobin, Myoglobin and Cytochrome compounds contain irons. - Many enzymes concerned with respiration contain iron in their structure
(7) Manganese (Mn)	- It plays important role in respiration and nitrogen metabolism. - In animals it is useful for the growth of bones and in reproduction. - It is a co-factor for the activation of enzymes like phosphotase.
(8) Zinc (Zn)	- It is essential in our body for general growth and reproduction. - It takes part in repair of worn out cells. - Many enzymes are activated in presence of it.
(9) Boron (B)	- It is associated with the transport of sugars in plants. It plays an important role in the production of flowers and fruits. - Cell division and certain other processes.
(10) Copper (Cu)	- It takes part in the synthesis of haemoglobin in animals. Enzyme tyrosinase has copper as structural component. - It takes part in the synthesis of chlorophyll in plants. It is present in haemocyanin a respiratory pigment in certain arthropods.
(11) Molybdenum (Mo)	- It helps in the fixation of nitrogen in plants. - In animals it is a component of intestinal enzymes.
(12) Sodium (Na) and Potassium (K)	- Maintainance of pH, maintainance of osmotic pressure of intra cellular fluids.
(13) Chlorine (Cl)	- It is amongst the main ions in blood; important role in CO_2 transport - It also takes part in the process of digestion of food, Osmo-regulation in blood and maintainance of pH.

(10) Which mineral play important role in energy exchange processes ?

(A) Ca (B) P (C) Mg (D) Fe

(11) Give the true group of sulphur containing vitamines and Amino acid (in sequence).

(A) Cysteine and thiamine (B) Biotine and thiamine

(C) Thiamine and Cysteine (D) Methionine and Biotine

(12) Which mineral is found in hyaline cartilage ?

(A) Sulphur (B) Zinc (C) Sodium (D) Phosphorus

(13) Which mineral is found in middle lamella between plant ?

(A) Phosphorous (B) Sulphur (C) Nitrogen (D) Calcium

(14) What is the importance of Calcium ?

(A) Clotting of blood (B) Contraction of muscles

(C) permeability of cell membrane (D) All of the above

(15) Which mineral occurs in chlorophyll ?

(A) Nitrogen (B) Magnesium (C) Calcium (D) Sulphur

(16) If deficiency of Magnesium in living being ..... process stops.

(A) Metabolism of food (B) Energy exchange

(C) permeability of cell membrane (D) Reproduction

(17) Which is the compound of iron ?

(A) Haemoglobin, biotin, cytochrome (B) Myoglobin, cytochrome, haemoglobin

(C) Cysteine, haemoglobin, tyrosinase (D) Methionin, haemoglobin, myoglobin

(18) Which mineral is important in reproduction ?

(A) Mn (B) Na (C) Cu (D) B

(19) Which process stops in absence of zinc ?

(A) Transport of sugar (B) Nitrogen fixatation (C) Respiration of cell (D) Contraction of muscles

(20) What is importance of boron in plant ?

(A) Transport of sugar (B) Production of flower and fruits

(C) Cell division (D) All of the above

(21) Which mineral is found as respiratory pigment in arthropoda ?

(A) Fe (B) Cu (C) Mo (D) Zn

(22) Which is the main compound of intestinal enzymes ?

(A) Zn (B) Cu (C) Mo (D) Cl

(23) What is the importance of sodium and chlorine (in sequence) ?

(A) Transport of CO_2 and maintenance of osmotic pressure

(B) Maintenance of pH and cell division

(C) Constituent of haemocynin and transport of CO_2

(D) Maintenance of osmotic pressure and water potential of blood

Answers : (10-B), (11-C), (12-A), (13-D), (14-D), (15-B), (16-A), (17-B), (18-A), (19-C), (20-D), (21-B), (22-C), (23-D)

- **Organic substances :**
  - It has main component as carbon.
  - Carbon has 4 valency. $\left(-\overset{|}{\underset{|}{C}}-\right)$
  - Organic biomolecules of two types.

    (1) Simple bio molecules : It has molecular weight less than 1000.

    (2) Macro biomolecules : It has molecular weight greater than 1000. e.g., Protein, Polysaccharides, Fat and Nucleic acid.
  - **Carbohydrates :** $C_n (H_2O)_m$

    They are structurally composed of H and O in ratio 2 : 1 with carbon.
  - It has mainly three types :

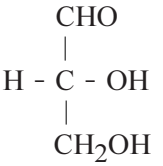
Monosaccharides	Disaccharides	Polysaccharides
• Including only one molecule most simple sugar	• It is made up of two monosaccharides	• Bonding by many unit of monosaccharides
• Value of n and m are similar	• By the glycosidic bond two hexose sugar are joins	• Many glycosidic bonds are present in that its structure
• It has -CHO or $>C = O$ group	• $C_n (H_2O)_{n-1}$ or $C_{12} H_{22} O_{11}$	• $(C_6 H_{10} O_{11})_n$
• It is known as aldose or Ketose sugar		
Properties :	**Properties :**	**Properties :**
• Sweet in taste	• Sweet in taste	• Not sweet in taste
• Soluble in water	• Soluble in water	• Not soluble in water
• Passes simply from the plasma membrane	• Not passes from the plasma membrane	
• Do not hydrolysed	• By an enzyme hydrolysed	

- **Monosaccharides has three types :**

(1) Triose ($C_3H_6O_3$) are two types :

(i) Aldose

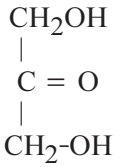
- Glyceraldehyde



- Example : PGAL is formed in dark reaction of photosynthesis

(ii) Ketose

- Dihydroxy acetone

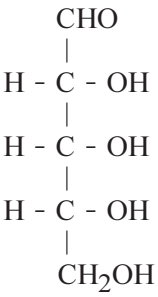


- Example : DHAP is formed in respiration process

(2) Pentose ($C_5H_{10}O_5$) are of two types :

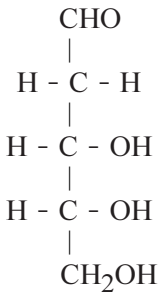
(i) Aldose are two types :

Ribose



- In structure of RNA

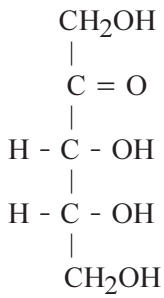
Deoxyribose



- In structure of DNA

(ii) Ketose

Ribulose



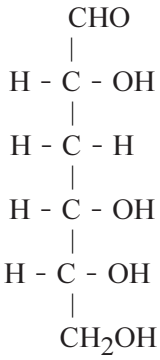
- As a constituent of RuBP

(3) Hexose ($C_6H_{12}O_6$) gives energy. It is of two types :

(i) Aldose is two types :

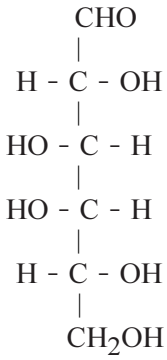
Glucose

- It gets by the digestion of starch



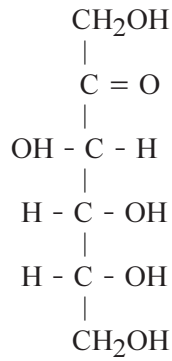
Galactose

- It gets by the digestion of milk



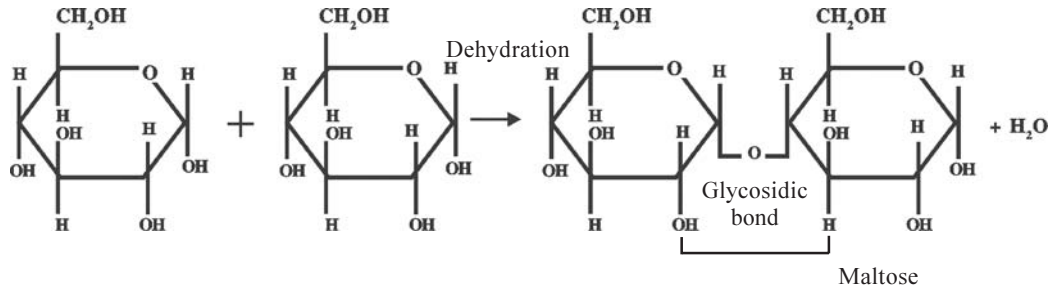
(ii) Ketose

- Fructose is present juices of fruit

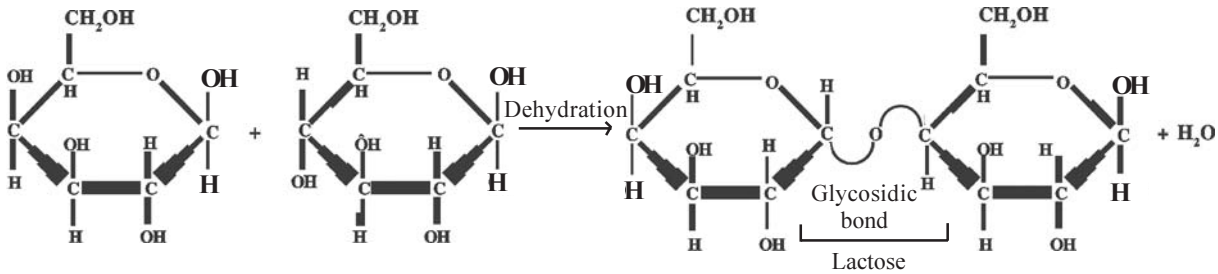


- **Disaccharides :**

Maltose = Glucose + Glucose



Lactose = Galactose + Glucose



- **Polysaccharides :**

Starch :

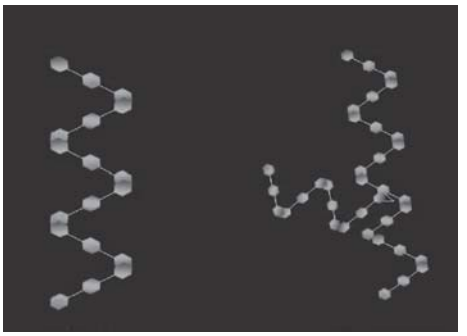
- Amylose is made up of unbranched polysaccharide chains made up of glucose units.
- Branched polysaccharide chains of glucose is amylopectin.
- In plants food is stored as starch.

Glycogen :

- Branched polysaccharide chains of glucose occur in the constitution of glycogen. These chains are branched.
- In animal the food is stored as glycogem.

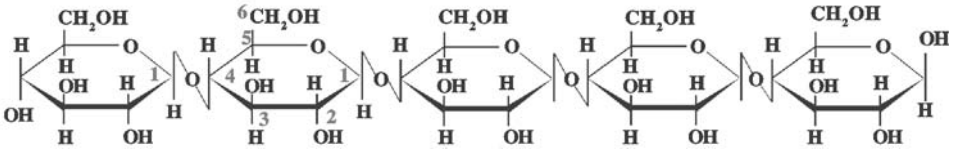
Cellulose :

- Cellulose also made up of polysaccharide chains of glucose and is the structural component of plant cell wall.



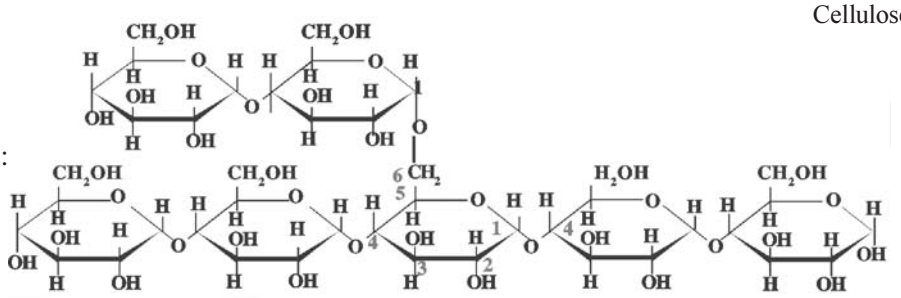
Amylose (Unbranched chain) Amylo pectin (Branched chain)

Amylose :



Cellulose

Amylopectin :



Amylopectin

(24) What are micromolecules ?

(A) Molecular weight less than one thousand Dalton.

(B) Molecular weight more than one thousand Dalton.

(C) Consist of one molecule only.

(D) Consist of more than one molecule.

(25) Which is inappropriate pair ?

(A) DHAP - Triose - respiration

(B) Ribose - Pentose - ATP

(C) Starch - Hexose - Plant

(D) Sucrose - disaccharide - fruit

(26) Which is appropriate pair ?

(A) deoxyribose - ketose pentose sugar

(B) Fructose - ketose hexose sugar

(C) Reibulose - aldose pentose sugar

(D) Galactose - aldose pentose sugar

(27) Which carbohydrate molecule cannot be hydrolysed ?

(A) Sucrose (B) Starch (C) Cellulose (D) Fructose

(28) Which sugar is formed by digestion of glycogen ?

(A) Glucose (B) Maltose (C) Ribose (D) Glyceraldehyde

(29) Which bond is responsible of formation of polysaccharide from disaccharide ?

(A) Easter (B) Peptide (C) Glycosidic (D) Hydrophobic

(30) Which process is done for the formation of glycosidic bond between the sugar ?

(A) Reduction (B) Oxidation (C) Hydrolysis (D) All of the above

(31) Which is the true option of cellulose molecule and structure ?

(A) Disaccharide $(C_6H_{10}O_5)_n$ (B) Disaccharide $C_{12}H_{22}O_{11}$

(C) Polysaccharide $(C_6H_{10}O_5)_n$ (D) Polysaccharide $C_n(H_2O)_{n-1}$

(32) What is amylose ?

(A) Second name of cellulose (B) Unbranched chain of glucose

(C) Branched chain of glucose (D) Second of glucose

(33) Find the inappropriate pair :

(A) Glucose - Fructose (B) Ribose - Ribulose

(C) DHAP - PGAL (D) Maltose - Cellulose

(34) Name the disaccharide present in mammals milk ?

(A) Glucose (B) Fructose (C) Galactose (D) Lactose

(35) Which substance is not made up from hexose ?

(A) Sucrose (B) Starch (C) Glycogen (D) Insulin

Answers : (24-A), (25-C), (26-D), (27-D), (28-A), (29-C), (30-C), (31-C), (32-B), (33-D), (34-D), (35-D)

- **Biological importance of carbohydrates :**
  - They play an important role in metabolism of cell and constitutions of tissues.
  - Ribose and deoxyribose pentose sugars are the structural components of RNA and DNA respectively.
  - It is the main source of energy in living organisms.
  - Cellulose forms the plant cell wall.
  - Starch serves as a stored food in plants.
  - Glycogen served as a reserve food in animals.

(36) Which sugar is found in RNA structure ?

(A) Hexose (B) Triose (C) Pentose (D) Tetrose

(37) Which sugar is found in DNA structure ?

(A) Deoxyribulose (B) Disaccharide

(C) Ribose (D) Deoxyribose

(38) Which sugar is found in cell wall of plant ?

(A) Monosaccharide (B) Polysaccharide (C) Disaccharide (D) Hexoses

(39) Which sugar is found in cell wall structure ?

(A) Starch (B) Cellulose (C) Glycogen (D) Sucrose

(40) What is Glycogen ?

(A) Polysaccharide (B) Food storage of animal

(C) Cell wall of plant (D) Food storage of plant

Answers : (36-C), (37-D), (38-B), (39-B), (40-A)

- **Lipids :**
  - A heterogenous group of compounds related to fatty acids.

  Properties :
  - Insoluble in water but soluble in other solvent like chloroform and benzene. Made up by C, H and O. H numbers are more than O.

  Structure : Alcohol + Fatty acids = Lipid

  (1) **Alchohol :** Two types of alcohol are used :

  (i) **Monohydroxy** (1 - OH group have)

  (ii) **Trihydroxy** (3C and 3-OH group) e.g., : Glycerol

  (2) **Fatty acids :** Two types of fatty acids are used :

  (i) **Saturated fatty acid :**

  - Two successive carbon atoms therein, are linked by a single bond.
  - They are not able to accept hydrogen or halogen atoms.
    - **Short chain fatty acids :** e.g., Butyric acid
    - **Long chain fatty acids :** e.g., Palmitic acid, Stearic acid.

  (ii) **Unsaturated fatty acid :**

  - Two successive carbon atoms at certain places there in are linked by double bond.
  - They are capable of accepting hydrogen or halogen atoms.
    - **Short chain fatty acid :** e.g., Crotonic acid
    - **Long chain fatty acid :** e.g., Oleic acid

---

(41) Which solvent is best for lipid ?

(A) Alcohol, Water, Benzene (B) Alcohol, HCl, Ether

(C) Chloroform, Benzene, Ether (D) Water, Chloroform, Alcohol

(42) Which molecule is not found in lipid ?

(A) C (B) H (C) N (D) O

(43) Which of following is the constituent of glycerol ?

(A) 3C, 3 - OH group (B) 3C, 1 - OH group

(C) 1C, 3 - OH group (D) 1C, 1 - OH group

(44) Which sentence is false for saturated fatty acid ?

(A) They are not able to accept hydrogen.

(B) They help in lipid constituent.

(C) Palmitic acid act like fatty acid.

(D) They linked by successive carbon atoms by double bonds.

(45) Find the appropriate pair :

(A) Butyric acid - saturated fatty acid - long chain

(B) Crotonic acid - saturated fatty acid - short chain

(C) Palmitic acid - saturated fatty acid - long chain

(D) Oleic acid - unsaturated fatty acid - short chain

(46) Which of the following substance is not capable to accept halogen atoms ?

(A) Palmitic acid (B) Crotonic acid (C) Steartic acid (D) All of the above

Answers : (41-C), (42-C), (43-A), (44-D), (45-C), (46-B)

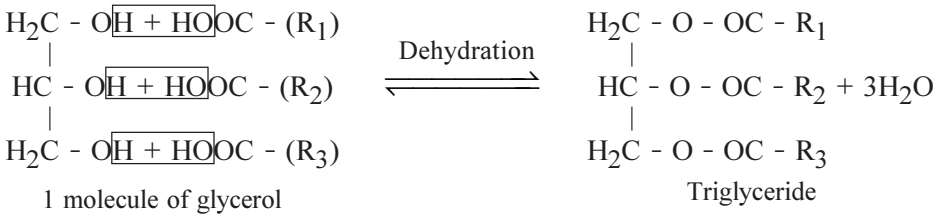
- **Lipids are of three types :** (1) Simple lipids (2) Complex lipid and (3) Steroids.

(1) Simple lipids :

- Alcohol + Fatty acids in their structure.
- They are of two types :

(i) Triglycerides :

- Glycerol + Fatty acids (3 molecules) $\xrightarrow[\text{Dehydration}]{\text{Easter bond}}$ Triglyceride in their structure
- - COOH group of fatty acid and - OH group of glycerol link with H and form easter bond.
- Type : • **Fat** - Saturated fatty acid.
  - e.g., Butter, Ghee, Animal fats
  - **Oil** - Unsaturated fatty acid
  - e.g., Ground nut oil, Coconut oil, oil of til.



(ii) Waxes :

- Monohydroxy alcohol + (1 molecule) of saturated fatty acid in their structure.

(2) Complex lipid :

- Lipid + Non lipid component is present in their structure.
- e.g., Glycolipid (Carbohydrate, Glycogen)

(3) Steroids :

- Only alcohol is present in their structure.
- It is of two types :

Sterol	Steroids
OH group is present	COOH or $\geq C = 0$ group is present
e.g., Cholesterol, Ergosterol	E.g., Cortisone, Progesterone
	(Animal hormone)

(47) Which bond is found in lipid ?

(A) Glycosidic bond (B) Easter bond (C) Phosphodiester bond (D) Hydrogen bond

(48) Which group is present in the formation of easter bond ?

(A) - COOH and - OH (B) $\geq C = 0$ and - OH

(C) $\geq C = 0$ and - COOH (D) - NH_2 and - OH

(49) In which of the following both carbohydrate and lipids are present ?

(A) Phospholipid (B) Waxes (C) Glycolipid (D) Cholesterol

(50) Find in appropriate pair :

(A) Cortisone - Sterol (B) Phospholipid - Phosphate

(C) Oil - Triglyceride (D) Ergesterol - Sterol

(51) When carbohydrate joined with lipid it is called ....

(A) Phospholipid (B) Glycolipid (C) Sterol (D) Simple lipid

(52) What is formed by mixture of fat and oil ?

(A) Alcohol (B) Acid (C) Easter (D) Aldehyde

(53) In which way the waxes and fats differentiate ?

(A) presence of fatty acid (B) Types of lipid

(C) Types of alcohol (D) All of the above

(54) Which is example of animal Hormone ?

(A) Cholesterol (B) Glycolipid (C) Progesterone (D) Ergosterol

Answers : (47-B), (48-A), (49-C), (50-B), (51-B), (52-C), (53-C), (54-C)

- **Biological importance of Lipids :**
  - It liberate a very large amount of energy than carbohydrates.
  - It is stored in body mass as a reserve food.
  - They form insulating layer as follow :

Myelin sheath	Subcutaneous layer	Protective layer
Around the nerve fibre	Subcutaneous loci	Outer surface of the aerial plant organs
Prevent the passage of nerve impulses	Maintain body temperature	It has protective function

- Vitamin A, D, E, K are fat soluble.
- Act as a cotituent of ellular organelle
- For the activation of enzymes : e.g., glucose phosphate
- Significant for the synthesis of steroidal hormone and vitamin D, E.

(55) Which is false for myelin sheath ?

(A) It is around the nerve fibre (B) It prevents the passage of nerve impulss

(C) It is made up of lipid (D) It is made up of non conductive layers

(56) Which lipid forms a protective layer on the outer surface of the aerial plant organs ?

(A) Fats (B) Waxes (C) Phospholipid (D) Ergosterol

(57) Which lipid is found in organelle membrane ?

(A) Glycolipid (B) Lipoprotein (C) Phospholipid (D) Triglyceride

(58) Glucose phosphatase is which type of biomolecule ?

(A) Lipid (B) Enzyme (C) Hormone (D) Carbohydrate

(59) Which vitamines are synthesized by derivatives of lipid ?

(A) Vitamin E, K (B) Vitamin A, D (C) Vitamin D, E (D) Vitamin A, E

Answers : (55-D), (56-B), (57-C), (58-B), (59-C)

- **True - False (T - F) type questions :**

Given sentence are true or false, choose appropriate answer :

(60) (1) Organic molecules include carbohydrate, lipid, protein and minerals.

(2) Living organism receives molecule from food for synthesis of important components.

(3) In water molecule, hydrogen and oxygen joined by cordinate bond.

(4) In the entire body for the bio chemical process water is essential which possess is necessary substance.

(A) T, T, T, F (B) F, F, F, T (C) F, T, T, F (D) F, F, T, T

(61) (1) At $40^{o}C$ temperature the density of water is very high.

(2) Plants absorb N_2 from atmosphere.

(3) Water provides $OH^{+}$and H^{-} ion for chemical reaction.

(4) Respiration and metabolism occur in the presence of Mn in plant.

(A) F, F, F, T (B) T, F, T, T (C) T, F, F, T (D) T, T, T, F

(62) (1) For synthesis of haemoglobin Cu is important in animal.

(2) Main ion of blood is made up of Fe.

(3) Organic molecules is only made up from C, H, O, N and P.

(4) Density of water depends on dissolve salts and lateral heat.

(A) F, T, T, F (B) T, T, T, F (C) T, F, F, F (D) T, F, T, F

(63) (1) Hydrolysis of glyceraldehyde is not possible.

(2) At the end of digetion of milk glucose and sucrose found.

(3) Lactose is not diffusible through the cell membrane.

(4) Chain branches of glucose units are called amylopectin.

(A) F, T, T, F (B) T, F, T, T (C) F, F, T, T (D) T, F, F, T

(64) (1) Lignin is a complex lipid.

(2) Fructose is a ketose sugar found in fruit juice.

(3) In plant food storage is in the form of sucrose.

(4) Fructose is a useful component in respiration process.

(A) F, F, T, T (B) T, F, T, F (C) T, T, F, F (D) F, T, F, F

(65) (1) H is double then O in the composition of lipid.

(2) Waxes is composed of molecule of monohydroxy alcohol not glycerol.

(3) Subcutaneous waxes layer maintains the body temperature.

(4) Fats consist of unsaturated fatty acid.

(A) F, T, T, F (B) F, T, F, F (C) T, T, T, F (D) T, T, T, T

(66) (1) Absence of alcohol in lipid is called complex lipid.

(2) Crotonic acid is not example of unsaturated fatty acid.

(3) In hydrolysis of maltose more than two molecule of monosaccharide is formed.

(4) Lipids liberate a very large amount of energy.

(A) T, T, F, T (B) F, F, F, T (C) F, T, F, T (D) T, F, F, T

(67) (1) Lipid dissolved in water is for storage as reservior food.

(2) In progesterone composition - COOH group is found.

(3) Easter bond derived as C - O - O - C bond.

(4) For the activation of phosphate enzyme, Mn acts as a accessory substance.

(A) T, F, T, T (B) F, F, T, T (C) F, T, T, T (D) T, F, T, F

(68) (1) Main composition of DNA is carbohydrate compound.

(2) Due to high energy value, carbohydrate is main respiratory substance.

(3) Fat, fatty acid related homologous group compounds.

(4) Monosaccharide is classified by number of carbon atoms.

(A) T, T, T, T (B) T, F, T, T (C) F, T, T, T (D) T, F, F, T

Answers : (60-D), (61-A), (62-C), (63-B), (64-D), (65-A), (66-B), (67-C), (68-D)

- **A - Assertion, R - Reason.**

Choose the correct option from the given below :

(A) A and R both are true, R is correct explanation of A.

(B) A and R both are true, R is not correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(69) Assertion A : Water in lakes and sea does not freeze.

Reason R : Water has a high specific heat.

(A) (B) (C) (D)

(70) Assertion A : Water reaches upto even tall trees like sequoia.

Reason R : Water has high thermal conductivity.

(A) (B) (C) (D)

(71) Assertion A : Planktonic organisms do not encounter any mechanical shock in water.

Reason R : Water density and velocity is high.

(A) (B) (C) (D)

(72) Assertion A : Due to the property of cohesive other ions are submerged in water molecule.

Reason R : Due to polarity water act as an active solvent.

(A) (B) (C) (D)

(73) Assertion A : Sugar is transported in plant by Boron.

Reason R : Calcium is in compostion of biotin and Thiamine.

(A) (B) (C) (D)

(74) Assertion A : Sulphur is present in ligaments and hyaline cartilage.

Reason R : Sulphur is in composition of amino acid.

(A) (B) (C) (D)

(75) Assertion A : Calcium determines the permeabilty of cell membrane.

Reason R : Middle layer of plant cell is made up of phospholipid.

(A) (B) (C) (D)

(76) Assertion A : Plant do not get Nitrogen in deficiency of Molybdenum.

Reason R : Molybdenum help in Nitrogen fixation.

(A) (B) (C) (D)

(77) Assertion A : Those which have molecules weighs 2970 Daltons are called macro molecules.

Reason R : Those which have molecular weight less than one thousand Dalton are called micro molecules.

(A) (B) (C) (D)

(78) Assertion A : The general formula of carbohydrate is $C_n(H_2O_n)_m$.

Reason R : Value of m and n is equal in triose sugar.

(A) (B) (C) (D)

(79) Assertion A : Monosaccharide cannot be hydrolysed.

Reason R : Monosaccharide is a simple carbohydrate.

(A) (B) (C) (D)

(80) Assertion A : By adding enzyme and acid hydrolysis of carbohydrate is possible.

Reason R : Monosaccharide passes through membrane very easily.

(A) (B) (C) (D)

(81) Assertion A : Starch is made up of branched glucose.

Reason R : Branched chain of glucose are called amylopectin.

(A) (B) (C) (D)

(82) Assertion A : Butyric acid is a saturated fatty acid.
Reason R : Lipid liberate a double amount of energy than carbohydrate liberate.
(A) (B) (C) (D)

(83) Assertion A : The oil are in liquid at room temperature.
Reason R : All fatty acid of lipid are saturated.
(A) (B) (C) (D)

(84) Assertion A : Complex lipid is not included in any fatty acid.
Reason R : The nomenclature of complex lipids is dependant on the non-lipid component present in it.
(A) (B) (C) (D)

(85) Assertion A : In structure of some molecule some of the loci of respective carbon are linked by double bond.
Reason R : Some of the fatty acid recieveing ability to hydrogen or halogen.
(A) (B) (C) (D)

(86) Assertion A : During the dark reaction of photosynthesis DHAP is produced.
Reason R : DHAP is example of Keto-triose sugar.
(A) (B) (C) (D)

(87) Assertion A : Vitamins D and E are synthesized from the derivatives of polysaccharide.
Reason R : Vitamins A, D, E, K are soluble in lipid.
(A) (B) (C) (D)

(88) Assertion A : Lipid form an insulating layer.
Reason R : Cell wall of plant is made up of cellulose.
(A) (B) (C) (D)

Answers : (69-B), (70-B), (71-A), (72-D), (73-C), (74-B), (75-C), (76-A), (77-A), (78-B), (79-A), (80-D), (81-A), (82-B), (83-B), (84-D), (85-A), (86-D), (87-D), (88-B)

(89) Select the appropriate pair :

Column - I	**Column - II**	
(i) Boron	(p) Maintenance of pH in blood	(A) i - r, ii - s, iii - t, iv - p, v - q
(ii) Magnecium	(q) Intestinal enzyme components	(B) i - r, ii - p, iii - t, iv - q, v - s
(iii) Copper	(r) Process of cell division	(C) i - t, ii - r, iii - s, iv - p, v - q
(iv) Chlorine	(s) Synthesis of ATP	(D) i - s, ii - r, iii - q, iv - p, v - t
(v) Molybdenum	(t) Composition of tyrosinase enzyme	

(90) Select the appropriate pair :

Column - I	**Column - II**	
(i) Phospholipid	(p) Steroid	(A) i - q, ii - r, iii - s, iv - p
(ii) Phosphoglyceraldehyde	(q) Polysaccharide	(B) i - p, ii - s, iii - q, iv - r
(iii) Ergosterol	(r) Complex lipid	(C) i - r, ii - q, iii - p, iv - s
(iv) Lignine	(s) Triose sugar	(D) i - r, ii - s, iii - p, iv - q

(91) Select the appropriate pair :

Column - I	Column - II	
(i) Crotonic acid	(p) Sterone	(A) i - r, ii - q, iii - p, iv - s
(ii) Waxes	(q) Animal fat	(B) i - s, ii - r, iii - p, iv - q
(iii) Cortisone	(r) Monohydroxy alcohol	(C) i - s, ii - p, iii - q, iv - r
(iv) Triglyceride	(s) Unsaturated fatty acid	(D) i - p, ii - r, iii - s, iv - q

(92) Select the appropriate pair :

Column - I	Column - II	
(i) Glycogen	(p) $C_{12}H_{22}O_{11}$	(A) i - q, ii - r, iii - p, iv - s
(ii) Lactose	(q) $C_6H_{12}O_6$	(B) i - q, ii - p, iii - r, iv - s
(iii) Galactose	(r) $(C_6H_{10}O_5)_n$	(C) i - r, ii - p, iii - q, iv - s
(iv) Ribulose	(s) $C_5H_{10}O_5$	(D) i - r, ii - q, iii - p, iv - s

(93) Select the appropriate pair :

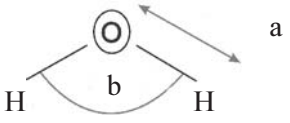
Column - I	Column - II	
(i) Cohesive force	(p) Water temperature	(A) i - q, ii - r, iii - s, iv - p
(ii) Water polarity	(q) Mechanical shock	(B) i - q, ii - p, iii - r, iv - s
(iii) Water density	(r) Ascent of sap in plant	(C) i - r, ii - p, iii - q, iv - s
(iv) Water viscosity	(s) Efficient solvent	(D) i - r, ii - s, iii - p, iv - q

(94) Select the appropriate pair :

Column - I	Column - II
(i) Pass through the cell membrane	(p) Cu^{+2}
(ii) Growth of animal bone	(q) Ca^{+2}
(iii) Structure of chlorophyll	(r) Mn^{+2}
(iv) Synthesis of chlorophyll	(s) Mg^{+2}
(A) i - q, ii - r, iii - s, iv - p	(B) i - q, ii - r, iii - p, iv - s
(C) i - r, ii - q, iii - s, iv - p	(D) i - q, ii - s, iii - r, iv - p

Answers : (89-A), (90-D), (91-B), (92-C), (93-D), (94-A)

(95) What is the measurement of a and b respectively in given diagram ?



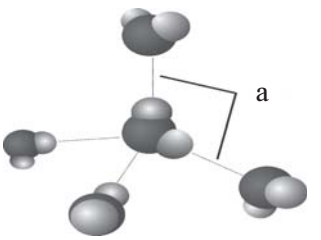
(A) 104.45°, 95.84 pm (B) 95.84 pm, 104.45°

(C) 95.84 nm, 104.45° (D) 104.45°, 95.84 nm

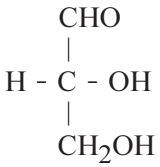
(96) What 'a' indicate in given figure ?

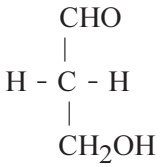
(A) Easter bond (B) Hydrophilic bond

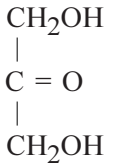
(C) Hydrogen bond (D) Oxygen bond

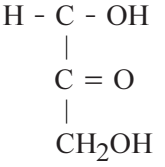


(97) What is constitution of DHAP ?

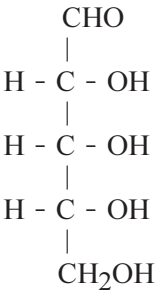
(A) 

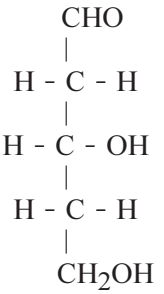
(B) 

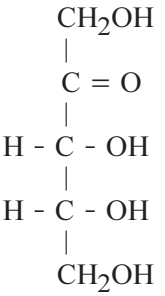
(C) 

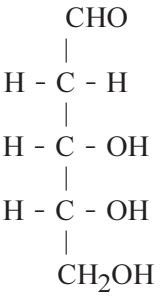
(D) 

(98) Which sugar is found in DNA structure ?

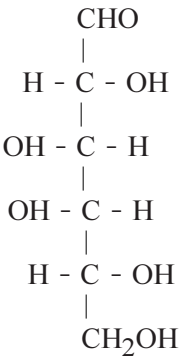
(A) 

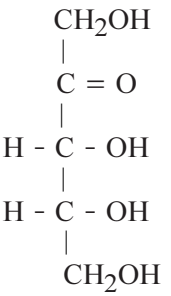
(B) 

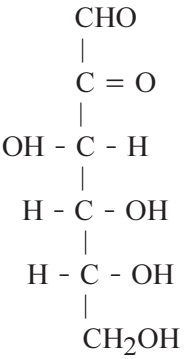
(C) 

(D) 

(99) Which is the true group of a, b, c :

(a) 

(b) 

(c) 

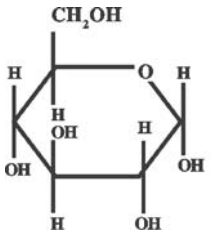
(A) Galactose, Ribulose, Fructose

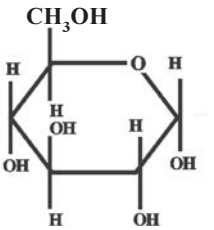
(B) Fructose, Ribose, Galactose

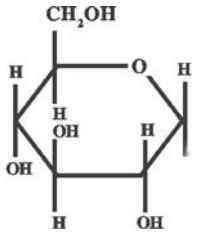
(C) Galactose, Ribose, Fructose

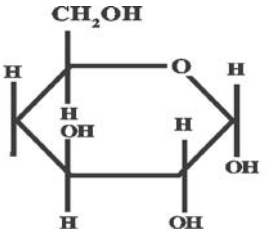
(D) Fructose, Ribulose, Galactose

(100) Which one is structure of glucose ?

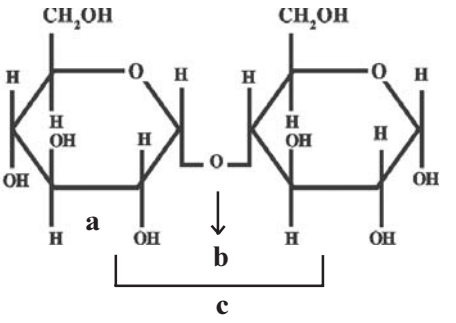
(A) 

(B) 

(C) 

(D) 

(101) Give the name of a, b, c from given structure :



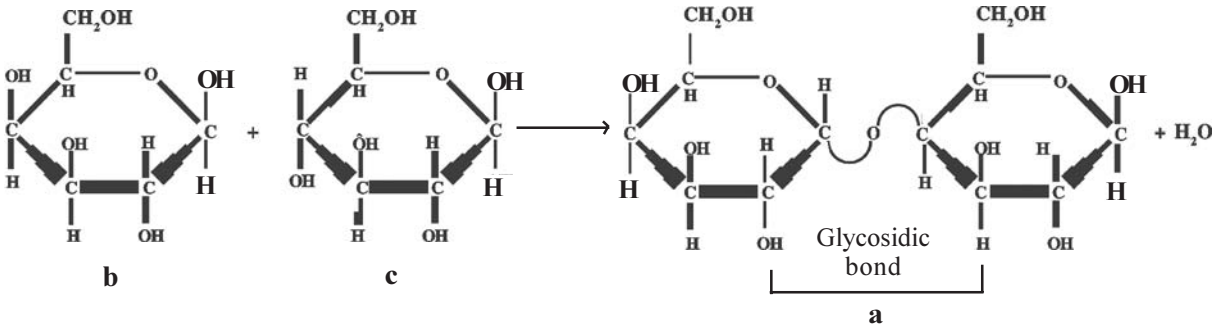
(A) glucose, hydrophilic bond, lactose

(B) galactose, glycosidic bond, lactose

(C) glucose, glycosidic bond, maltose

(D) glucose, glycosidic bond, maltose

(102) Give the name of a, b, c from given structure :



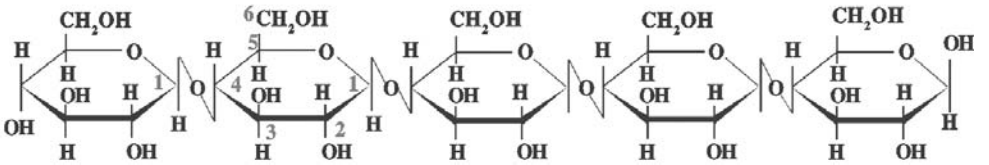
(A) Lactose, galactose, glucose

(B) Lactose, glucose, galactose

(C) Maltose, glucose, galactose

(D) Lactose, galactose, glucose

(103) The following structure is of :



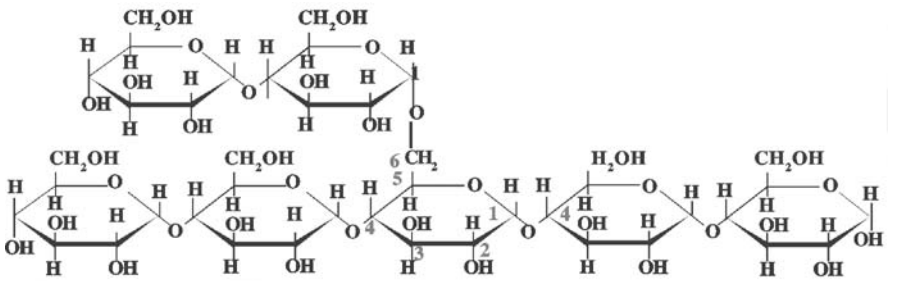
(A) Amylopectin

(B) Amylose

(C) Starch

(D) Cellulose

(104) What indicate this structure ?



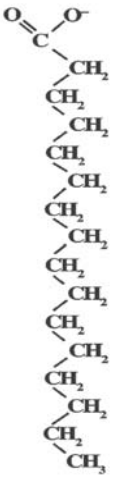
(A) Branched chain of glucose

(B) Unbranched chain of glucose

(C) Polysaccharide chain of glycogen

(D) Polysaccharide chain of cellulose

(105) Which type of fatty acid is seen in given structure ?



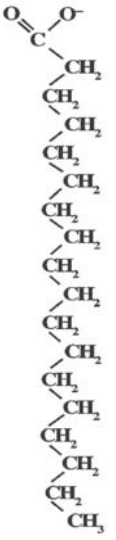
(A) Saturated short chain of fatty acid

(B) Unsaturated long chain of fatty acid

(C) Saturated long chain of fatty acid

(D) Unsaturated short chain of fatty acid

(106) Give the name of fatty acid.



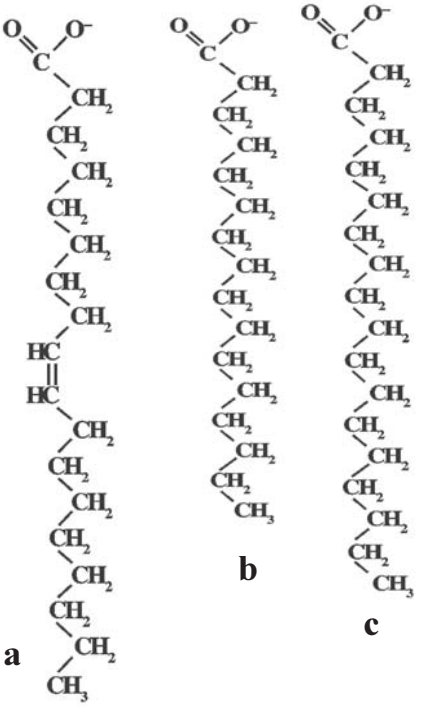
(A) Oleic Acid

(B) Crotonic Acid

(C) Stearic Acid

(D) Palmitic Acid

(107) Which is true structure of a, b, c fatty acid ?



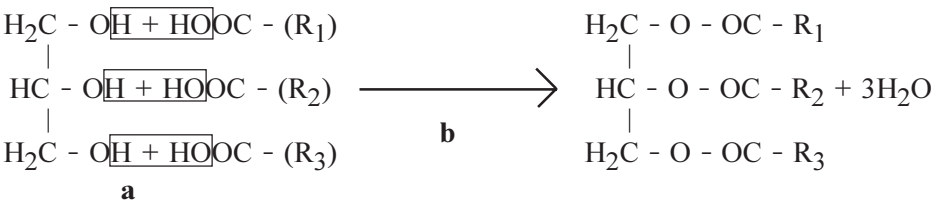
(A) stearic acid, palmitic acid, oleic acid

(B) palmitic acid, stearic acid, oleic acid

(C) oleic acid, palmitic acid, stearic acid

(D) oleic acid, stearic acid, palmitic acid

(108) What indicate a, b ?



(A) Glycerol, dehydration

(B) Fatty acid, dehydration

(C) Triglyceride, dehydration

(D) Mono hydroxy alcohol, dehydration

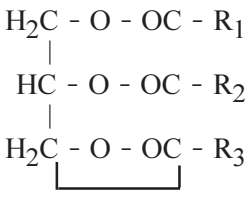
(109) What indicate a ?

(A) Glycosidic bond

(B) Hydrogen bond

(C) Easter bond

(D) Peptide bond



Answers : (95-B), (96-C) , (97-C), (98-D), (99-A), (100-A), (101-C), (102-D), (103-B), (104-A), (105-B), (106-C), (107-C), (108-A), (109-C)

- **Questions for NEET :**

(110) Which one is non reducing sugar ?

(A) Glucose (B) Galactose (C) Sucrose (D) Fructose

(111) In which cell wall Chitin is found ?

(A) Algae (B) Bacteria (C) Fungi (D) Yeast

(112) Which one is reducing sugar ?

(A) Cellulose (B) Starch (C) Sucrose (D) Maltose

(113) In which neutral substance cellulose is found in huge amount ?

(A) Wood (B) Pith of fruit (C) Wheat (D) Cotton

(114) Which one is not disaccharide ?

(A) Maltose (B) Lactose (C) Sucrose (D) Galactose

(115) Cellulose $\xrightarrow{\text{Cellulase}}$?

(A) Glycogen (B) Starch (C) Glucose (D) Fructose

(116) Which are main '6' minerals found in living organism ?

(A) C, H, O, N, S, Mg (B) C, H, O, N, P, Ca

(C) C, H, O, N, P, S (D) C, H, O, N, Mg, Na

(117) By whom biomolecules are produced mostly in earth ?

(A) Some bacteria, algae, green plant (B) Fungi, algae, bacteria

(C) Green plant, Fungi, algae (D) Virus, Bacteria, fungi

(118) Lipid are insoluble in water, because lipid molecule is ..... .

(A) Neutral (B) Hydrophilic (C) Hydrophobic (D) Zwitter ion

[Hint: Hydrophilic = affinity towards water, Hydrophobic = tending to repel to mix with water. Zwitterion = group having +ve and –ve charged groups**]**

(119) Sucrose is made up of ..... .

(A) Glucose + Fructose (B) Glucose + Lactose

(C) Glucose + Galactose (D) Glucose + Maltose

(120) Starch is made up of ..... .

(A) Cellulose (B) Protein (C) Amylopectin (D) Amino acid

(121) What is the simple monomer of carbohydrate ?

(A) Fructose (B) Sucrose (C) Maltose (D) Glucose

(122) Which molecule is basic structure of lipid ?

(A) Glycerol (B) Glutamic acid (C) Galactose (D) Guanine

(123) Thymine works as ..... .

(A) Enzyme (B) Vitamin (C) Protein (D) Hormone

(124) Which one is complex protein of containing carbohydrate ?

(A) Eretroprotein (B) Meteloprotein (C) Glycoprotein (D) Lipoprotein

(125) Which is insoluble polysaccharide in green plant ?

(A) Starch (B) Sucrose (C) Refinose (D) Glycogen

(126) Which chemical bond is present in Amylose ?

(A) α,1, 6- Glycosidic bond (B) $\alpha - 1$- Glycosidic bond

(C) $\beta - 1, 4$ - Glycosidic bond (D) $\beta - 1, 6$ - Glycosidic bond

(127) Take starch solution in test tube add saliva. Keep it in 38° C temperature and pH 2 to 8. After that add I_2. What is your observation ?

(A) Violet colour and sweet test (B) Colourless and sweet test

(C) Violet colour and no test (D) Colourless and no test

(128) Select the appropriate pair :

Column - I	**Column - II**
(i) Triglyceride	(p) Animal Hormone
(ii) Cell membrane lipid	(q) Feather and beak
(iii) Steroid	(r) Phospholipid
(iv) Waxes	(s) Stored lipid in small globule

(A) i - s, ii - r, iii - p, iv - q (B) i - q, ii - r, iii - s, iv - p

(C) i - r, ii - s, iii - p, iv - q (D) i - s, ii - p, iii - q, iv - r

Answers : (110-C), (111-C), (112-D), (113-D), (114-D), (115-C), (116-C), (117-A), (118-C), (119-A), (120-C), (121-D), (122-A), (123-B), (124-C), (125-A), (126-C), (127-B), (128-A)

•

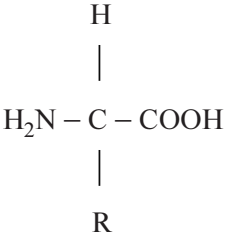
7 Biomolecules - II (Proteins, Nucleic acids and Enzymes)

- **Proteins :** Important compounds of cytoplasm
  - Constitutions of those C, H, O, N and S
  - build up by amino acids → act as a heteropolymer
  - structural component of it is amino acid
  - Collagen : The most abundant protein in animal world
  - The most abundant protein in the whole biosphere
  - Eg., RuBisCO

  (Ribulos biphosphate Carboxylase Oxygenase)

- **Properties :** It is soluble in water, dilute acidic, dilute basic or dilute alcoholic solution.
  - It is destroy or denatured at high temperature, concentrate acid, basic or in alcohol
  - They are destroyed by X-rays or UV-rays
  - Some of the protein not desolved in any solvent
  - Eg., Scleroprotein (Keratin Protein)
  - It is present in hair, feathers, scales, horns, nails claws, etc.

- **Amino acid :** All amino acids possess an amino group ($-NH_2$), a carboxyl group (-COOH), an H and the reamainder part is 'R' group.
  - The carboxyl group (-COOH) is acidic while the amino ($-NH_2$) group is basic.
  - Thus amino acid acts as an electrolyte.



- **Classification :** All the amino acids is similar except for their 'R' group. But the chemical structure of 'R' group (functional group) being different, the properties and types of amino acids are different.
  - The most widely used one is that by Lehninger.
  - Classification is based on the polarity of 'R' group of amino acid.

No.	Kind of amino acids	Examples
1.	Amino acids with nonpolar 'R' group	Alanine, Leucine, Valine, Isoleucine, Methionine, Phenylalanine, Tryptophane, Proline
2.	Amino acids with polar and neutral 'R' group	Aspergine, Cysteine, Serine, Glutamine, Glycine, Threonine, Tyrosine
3.	Amino acids with polar and negetively charged 'R' group	Aspartic acid, Glutamic acid
4.	Amino acids with polar and positively charged 'R' group	Arginine, Histidine, Lysine

(1) Which is the correct group of elements present in the structure of protein ?

(A) C, H, N, O, P (B) C, H, N, O, S (C) C, H, O, N (D) C, H, O, N, P, S

(2) Protein means...

(A) Heteropolymer group of fatty acid (B) Heteropolymer group of Amino acid

(C) Heteropolymer group of Nucleotide (D) Heteropolymer group of monosaccharide

(3) Which is the most abundant protein found in the whole biosphere ?

(A) Scleroprotein (B) Chlorophyll (C) Collagen (D) RuBisCO

(4) Insoluble Protein means ..... .

(A) Collagen (B) Myosin (C) Scleroprotein (D) Actin

(5) Full form of RuBisCO is ..... .

(A) Ribulose Bicarboxylase Oxygenase (B) Ribulose Biphosphate Carboxylase Oxygenase

(C) Ribulose phosphate carboxylase (D) Ribose Biphosphate Carboxylase Oxygenase

(6) Which factor is responsible for protein denaturation ?

(A) X-ray (B) UV-ray (C) Strong Acid (D) All of the above

(7) Which Biomolecule fight against infections organisms ?

(A) Carbohydrate (B) Enzyme (C) Nucleic Acid (D) Protein

(8) Which protein is present in the exoskeleton of an organism ?

(A) Keratin (B) Casein (C) Myosin (D) Melanin

(9) Which is the building block of Protein ?

(A) Peptide (B) Nucleotide (C) Amino Acid (D) Glucose

(10) Which group is correct for the structure of Amino Acids ?

(A) Carboxyl group, Amino group, Hydrogen group, Reactive 'R' group

(B) Ketone group, Amino group, Hydrogen group

(C) Hydroxyl group, Carboxyl group, Amino group

(D) Carboxyl group, Hydrogen group, Hydroxyl group

(11) There are twenty functional Amino Acids present in the structure of protein, because of...

(A) In every Amino Acid numbers of - COOH group differs.

(B) In every Amino Acid propotion of - NH_2 group differs.

(C) Chemical structure of 'R' group is different in Amino Acid.

(D) All of the above

(12) Which of the following indicates the importance of 'R' group ?

(A) Amino Acids are classified on the basis of it (B) Electric charge of an Amino Acid is based upon it

(C) Protein synthesis is based on it (D) Peptide bond is based on it

(13) Which method is widely used for the classification on the basis of 'R' group ?

(A) Electroporesis (B) Reductive method (C) Lehninger (D) Whittekar method

(14) Long chain of protein is synthesized by ..... .

(A) Purine (B) Pyrimidine (C) Amino Acid (D) Sugar

(15) Which Amino Acid possesses polar and negatively charged 'R' group ?

(A) Aspartic Acid (B) Arginine (C) Methionine (D) Threonine

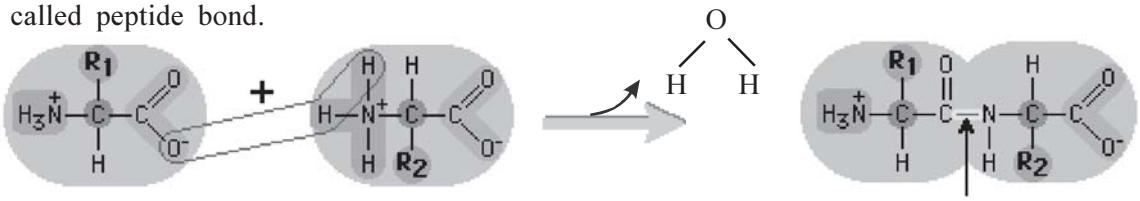
Answers : (1-B), (2-B), (3-D), (4-C), (5-B), (6-D), (7-D), (8-A), (9-C), (10-A), (11-C), (12-A), (13-C), (14-C), (15-A)

- **Protein :** Amino acids $\xrightarrow[\text{Dehydration}]{\text{Peptide bond}}$ Dipeptide

Dipeptide : A dipeptide is formed through the union of two similar or dissimilar amino acid molecules.

Amino acid (1) + Amino acid (2) $\xrightarrow{\text{Peptide bond}}$ Dipeptide

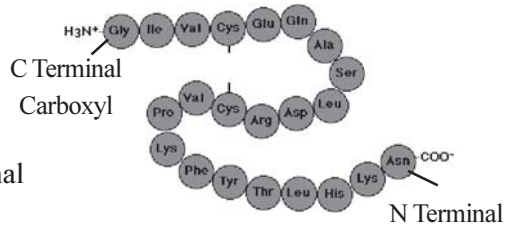
Peptide bond : The bond is formed between the – COOH group of one amino acid and – NH_2 group of another amino acid molecule. A molecule of H_2O is released in this process. Such a bond is called peptide bond.



Amino acid - 1 Amino acid – 2 Peptide bond

Polypeptide bond :

- Many amino acids $\xrightarrow{\text{Peptide bond}}$ Protein
- One or more polypeptide chain → Protein
- Protein exhibit two ends. (1) C-Terminal (2) N-Terminal



Structure of Protein :

(1) Primary structure :

- On the basis of serial and types of amino acid.
- Gene regulation.

(2) Secondary structure :

- Structure is formed by hydrogen bond.
- Spiraly coiled or flat disc like.

(3) Tertiary structure :

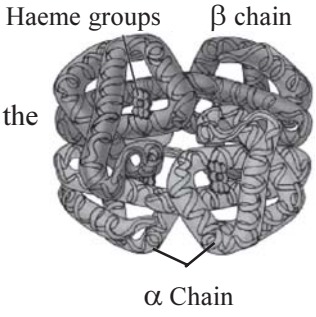
- The three dimensional arrangement on the atoms within a single polypeptide chain.

(4) Quaternary structure :

- A three dimensional form of a whole protein.

Bonds : Disulphide, hydrogen, hydrophobic, ionic bonds are involved in the formation of it.

- Globular or fibrous form.
- E.g., Haemoglobin (Quatenary structure)
- 4 polypeptide chains : 2 – α chain, 2 – β chain and 4 haeme groups.



Importance of Protein :

- Components of the membrane and organelles.
- In structure of an enzyme.
- In the structures of hormones of Pancreas, Pituitary and Parathyroid glands.
- In muscles → Actin and Myosin; In Cilia and flagella → It is responsible for locomotion by globular contractile protein.
- In blood plasma → Immunoglobulin → For the immunity

- In skin → Melanin - responsible for the colour.

Conjugated protein : When proteins are associated with some materials other than amino acids or protein they are known as conjugated protein.

eg., Haemoglobin - For the transportation of respiratory gases and chlorophyll - For the photosynthesis.

(16) The union of two dissimilar amino acid is called...

(A) peptide (B) polypeptide (C) dipeptide (D) protein

(17) Which groups are required for the formation of peptide bond ?

(A) $>$COOH and - NH_2 (B) $>$COOH and - OH

(C) $>$C = O and - NH_2 (D) C = O and - NH_2

(18) Which of the following indicates the peptide bond ?

(A) - C - O - O - C (B) - CO - NH (C) O - C - H (D) C - O - H

(19) Polypeptide means ..... .

(A) Chain of many Amino Acids joined through ester bond

(B) Chain of many Amino Acids joined through peptide bond

(C) Chain of many Amino Acids joined through polypeptide bond

(D) Chain of many Amino Acids joined through glycosidic bond

(20) Polypeptide chain of Amino Acid is ..... .

(A) Enzyme (B) Nucleic Acid (C) Hormone (D) Protein

(21) The primary structure of Protein is due to ..... .

(A) Peptide bond (B) S- S bond (C) Ionic bond (D) Hydrogen bond

(22) The primary structure of protein is controlled by ..... .

(A) Hormone (B) Chromosome (C) Amino Acid (D) Genes

(23) Secondary structure of protein is impossible in the absence of...

(A) S-S bond (B) Hydrogen bond (C) Ionic bond (D) Peptide bond

(24) Because of the presence of Hydrogen bond, which characteristic is considered ?

(A) Chain becomes spiral and complicated (B) Folds formed between the Amino Acid

(C) Formation of strong tubules (D) All of the above

(25) The secondary structure of protein is ..... .

(A) Interactions between the polypeptide chain (B) Three dimensional arrangement of polypeptide chain

(C) Complicated helical structure of polypeptide chain (D) Numbers of Amino Acid in the structure of polypeptide chain

(26) Three dimensional arrangement of one polypeptide chain means...

(A) Quaternary structure of protein (B) Secondary structure of protein

(C) Primary structure of protein (D) Teritary structure of protein

(27) Four polypeptide chains are observed in the structure of haemoglobin molecule under the microscope, so what does it indicates from the following ?

(A) It is conjugated protein (B) It may carry out the exchange of gases

(C) It is the protein with quaterrary structure (D) It is found in the blood of vertebrates

(28) Which bond is necessary to form a quaternary structure of protein ?

(A) Peptide bond (B) Ester bond (C) Phosphodiester bond (D) Disulphide bond

(29) Which bond is unnecessary for the formation of protein ?

(A) Ester bond (B) Hydrophobic bond (C) Peptide bond (D) Ionic bond

(30) Which sentence is wrong for the enzyme ?

(A) Enzymes are made up of protein (B) It is used in the biochemical reactions of a cell

(C) It regulates the different activities of a body (D) It acts as a electrolyte

(31) Which gland hormone is not made up of protein ?

(A) Pancreas (B) Parathyroid (C) Pituitary (D) Adrenal

(32) Which of the following protein does not have the characteristic of contractibility ?

(A) Myosin (B) Globulin (C) Globular (D) Actin

(33) Where does the Actin and Myosin proteins present in the body ?

(A) In a blood (B) In a muscle (C) In a plasma (D) In a skin

(34) Which of the following protein is having the characteristic of immunity ?

(A) Immunoglobulin (B) Albumin (C) Haemoglobin (D) Melanin

(35) Globular protein is found in the ..... .

(A) Muscle (B) Nails (C) Cillia (D) Scales

(36) Meena's skin colour is bright and Seeta's skin colour is dark which protein is responsible for it ?

(A) Melatonin (B) Melanin (C) Haemoglobin (D) Myoglobin

(37) Conjugate protein means ..... .

(A) protein + lipid (B) protein + carbohydrate (C) protein + ion (D) protein + non-protein

(38) Which kind of haemoglobin protein is ?

(A) conjugated protein with quaternary structure (B) conjugated protein with tertiary structure

(C) complex protein with quaternary structure (D) Immuno protein with quaternary structure

(39) Which bond is present in the hormone of parathyroid gland ?

(A) Peptide (B) Glycosidic (C) Ester (D) Phosphodiester

(40) Give one example of conjugate protein with quaternary structure ..... .

(A) Immunoglobulin (B) Myosin (C) Chlorophyll (D) Globular

(41) Which conjugate protein is found in plants ?

(A) Chlorophyll (B) Thylakoid (C) Pectin (D) All of the above

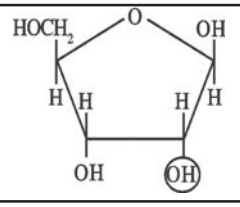
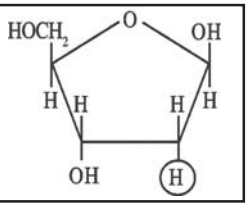
Answers : (16-C), (17-A), (18-B), (19-B), (20-D), (21-A), (22-D), (23-B), (24-D), (25-C), (26-D), (27-C), (28-D), (29-A), (30-C), (31-D), (32-B), (33-B), (34-A), (35-C), (36-B), (37-D), (38-A), (39-A), (40-C), (41-A)

- **Nucleic acid :**
  - Isolation of DNA prepared by - Johannsen Friedrich Miescher.

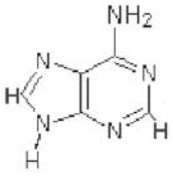
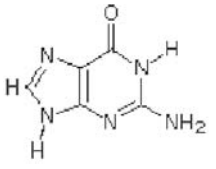
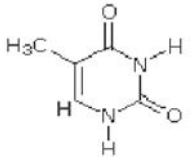
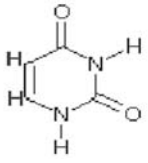
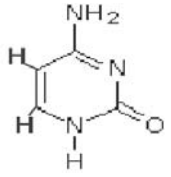
- Found a weakly acidic substance of unknown function in the nuclei of human white blood cells and named this material "nuclein".
- Nuclein = Nucleic acid + Protein, Main component of chromosome in nucleic acid.

Chromosome : Small gene-carrying bodies in the nuclei of complex cells. They are responsible for inheritance in all organisms.

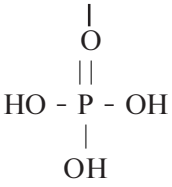
- C, H, O, N and P are in it.
- It is two types (1) DNA and (2) RNA
- Nucleotides = Pentose sugar + Nitrogenous base + Phosphoric acid
- **Pentose sugar :**

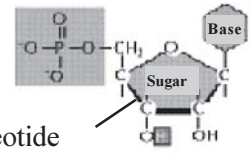
Ribose - In structure of RNA		**Deoxyribose** - In structure of DNA	

- **Nitrogen bases :** **Purine** - Two rings | **Pyrimidine** - Single rings

Adenine	**Guanine**	**Thymine** - Not in structure of RNA	**Uracil** - Not in structure of DNA	**Cytosine**
				

- **Phosphoric acid :**



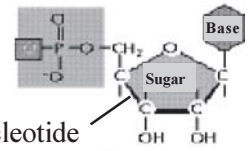


Nucleotide

+

- Nucleoside = Pentose sugar + Nitrogen base
- Nucleotide = Nucleoside + Phosphate
- Dinucleotide = Nucleotide + Nucleotide

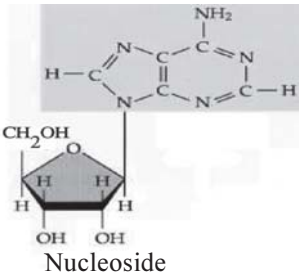
Phosphodiester bond



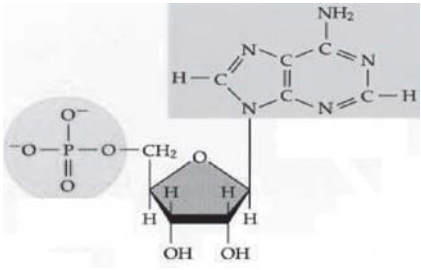
Nucleotide

Phosphodiester bond

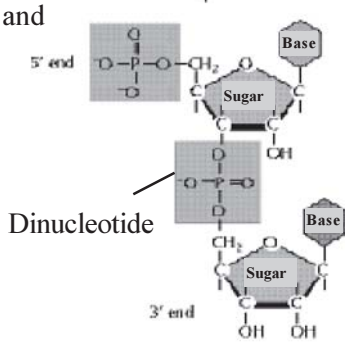
Two sequential nucleotides join through a phosphodiester bond and form a dinucleotide. Such union occurs between third carbon of sugar in one nucleotide and



Nucleoside
(Pentose sugar + N - Base)



Nucleotide



fifth carbon of sugar (Pentose) in the other nucleotide with the phosphate molecule structural bond is known as phosphodiester bond.

(Between the two sequential nucleotides phosphate linked that bond is phosphodiester)

Polynucleotides :

- Many nucleotide units unite, a polynucleotide chain is formed.

- **DNA :** (Deoxy ribose nucleic acid)
  - Watson, Crick, Wilkins gave the model of it.
  - Two polynucleotide chain is comprised.
  - It is anti parallel to each others.
  - Two chains are in a definite manner and spirally twisted. So it like a spiral ladder.
  - The linkages between the two polynucleotide chains are such that a purine base of one chain is linked with a pyrimidine base of the opposite chain.

    A = T and G ≡ C (Uracil is not present in it)
  - Length of complete spiral = 34 $\overset{0}{A}$
  - Distance (width) between the two chain = 20 $\overset{0}{A}$
- **RNA :** (Ribo nucleic acid)
  - It has single polynucleotide chain.
  - Thymine is not present instead of it Uracil is present.
  - Types of RNA :

(1) m - RNA (Messenger RNA) : Transport the genetical information.

(2) t - RNA (Transfer RNA) : It has 75 % nucleotides. It transport of amino acid.

(3) r - RNA (Ribosomal - RNA) : It is find in ribosomes. 80 to 85% of it in total RNA.

---

(42) Nucleic Acids shows the evolution of ..... .

(A) Nuclein (B) Protein (C) Nucleotide (D) Amino Acid

(43) Nuclein ...

(A) Weak Acid + Weak Protein (B) Nucleic Acid

(C) Nucleic Acid + Protein (D) Chromosome

(44) Nucleic Acid means...

(A) Major component of a chromosome (B) An ultrastructure which carry genes in the Nucleus

(C) A weak substance with unknown function (D) All of the above

(45) State the wrong sentence for the chromosome ..... .

(A) It is responsible for the heredity in all the organisms

(B) An ultrastructure which carry genes in the nucleus of a cell

(C) Main component of a Nucleic Acid

(D) All of the above

(46) Nucleic Acid is a Polymer of the following.

(A) Amino Acid (B) Protein (C) Nuclein (D) Nucleotide

(47) Nucleic Acid is related with ..... .

(A) Respiration (B) Photosynthesis (C) Heredity (D) Reproduction

(48) Which of the following molecules are showing different characteristics in different organism ?

(A) Protein (B) Lipid (C) Nucleic Acid (D) Carbohydrate

(49) Which pair is correct of purine nitrogen base ?

(A) Adenine, cytosine (B) Adenine, thymine

(C) Cytosine, thymine (D) Adenine, guanine

(50) Which pair of pyrimidine Nitrogen base is correct of the following ?

(A) Cytosine, thymine, uracil (B) Adenine, guanine, cytosine

(C) Cytosine, thymine (D) Thymine, Uracil

(51) Which element is not there in the Nitrogen base ?

(A) Carbon (B) Phosphorus (C) Hydrogen (D) Nitrogen

(52) Which of the following is an example of nitrogen base consisting two rings ?

(A) Cytosine (B) Thymine (C) Uracil (D) Guanine

(53) Nitrogen base is ..... kind of structure.

(A) Linear (B) Cyclic (C) Helical (D) Flat and sheet

(54) Nucleoside means ..... .

(A) Nucleic acid + Phosphate (B) Nucleotide + Phosphate

(C) Pentose sugar + Nitrogen base (D) Pentose sugar + Nitrogen base + Phosphate

(55) Nucleotide means...

(A) Nucleoside + Phosphate (B) Nitrogen base + Phosphate

(C) Nucleic acid + Protein (D) Protein + Nucleoside

(56) How the DNA and RNA differs from each other ?

(A) Only sugar (B) Sugar and purine

(C) Sugar and Purimidine (D) Sugar and phosphate

(57) Phosphodiester bond is between which two carbon number of nucleotide ?

(A) 3, 5 (B) 3, 3 (C) 2, 5 (D) 1, 5

(58) Which one is required in the structure of Nucleic Acid ?

(A) Ester bond (B) Phosphodiester bond (C) Peptide bond (D) Di-sulphide bond

(59) If a one chain of DNA have the nucleotide sequence like ACGGTTAA then, what should be the opposite chain sequences ?

(A) TGCCAATT (B) GTAACCTT (C) CATTGGCC (D) TACCGGTT

(60) DNA's strands are Antiparallel to each other because of...

(A) Hydrogen bond (B) Di-sulphide bond (C) Phosphodiester bond (D) Glycosidic bond

(61) Which one of the following pair is attached by three weaker Hydrogen bond ?

(A) A,T (B) C,G (C) T,G (D) A,U

(62) Identify pair which is linked through two hydrogen bonds ?

(A) A,T (B) T,A (C) C,G (D) A and B both

(63) What is true for DNA structure ?

(A) A = T and $C \equiv G$ (B) A = C and T = G

(C) A = G and T = C (D) All of the above

(64) Which sequence of nucleotides possesses six pyrimidine nitrogen base ?

(A) GATCATCCT (B) TGCCTAACG

(C) GCUAGACAA (D) UAGCGGUAA

(65) What is the length of one complete spiral ?

(A) 34 $\overset{0}{A}$ (B) 3.4 $\overset{0}{A}$ (C) 20 $\overset{0}{A}$ (D) 2.0 $\overset{0}{A}$

(66) What is the distance between two chains of DNA ?

(A) 30 $\overset{0}{A}$ (B) 3.4 $\overset{0}{A}$ (C) 2.0 $\overset{0}{A}$ (D) 20 $\overset{0}{A}$

(67) DNA is not directly connected in the synthesis of the following.

(A) Another chain of DNA (B) t - RNA

(C) m - RNA (D) Protein

(68) In a DNA, which region of DNA is diversified ?

(A) Glycerol (B) Nitrogen base (C) Sugar (D) Phosphate

(69) Which base is found only in the RNA ?

(A) Uracil (B) Thymine (C) Guanine (D) Cytosine

(70) m - RNA is a polymer of ..... .

(A) DNA oxyribotide (B) Ribotide

(C) Riboside (D) Deoxyriboside

(71) Structure of ATP is very similar with.

(A) DNA Nucleotide (B) Amino acid (C) Fatty acid (D) RNA Nucleotide

(72) Find out the incorrect sentence for m- RNA.

(A) It is synthesis on the template strand of a DNA (B) Transfers the information for the protein synthesis

(C) It degrades after its functions over (D) Transfer the amino acids

(73) Find out the incorrect pair.

(A) m-RNA - genetic codon (B) t - RNA - anticodon

(C) r - RNA - Ribosome structure (D) DNA - Place for the protein synthesis

(74) Nucleoprotein means...

(A) r - RNA + protein (B) r - RNA + nucleic acid

(C) nucleotide + protein (D) t - RNA + protein

(75) Ribosomes provides the place for which reaction ?

(A) Protein synthesis (B) Respiration (C) DNA synthesis (D) m-RNA synthesis

Answers : (42-A), (43-C), (44-A), (45-C), (46-D), (47-C), (48-C), (49-D), (50-A), (51-B), (52-D), (53-B), (54-C), (55-A), (56-C), (57-A), (58-B), (59-A), (60-C), (61-B), (62-D), (63-A), (64-A), (65-A), (66-D), (67-D), (68-B), (69-A), (70-B), (71-D), (72-D), (73-D), (74-A), (75-A)

- **Enzymes :**
  - All these reactions occur very slowly at low temperature and atmospheric pressure the conditions under which living cells carry on their life processes.
  - In the living cells these reactions proceed at extremely high rates.
  - Specific chemicals which act as biological catalysts are called enzymes which are available in the body of living organisms.
  - Enzymes are proteinaceous.
  - Some enzymes behave of nucleic acids, which is known as ribozymes.

Properties of enzymes :

- Enzymes are amphoteric in nature.
- Each enzymes has an effect on a particular reaction only.
- An enzyme is effective for one reaction is not useful in another reaction.
  E.g., Lipase can digest lipids only while sucrase can digest sucrose only.
- Effect of most enzymes is unidirectional.
- Some enzymes is bidirectional.
- Some enzyme is functional within a specific range.
- At high temperatures enzymes are denatured.
- At very low temperature they become inactive.
- After the complete the reaction enzyme gain in original form.
- Each enzyme is active at a specific pH.

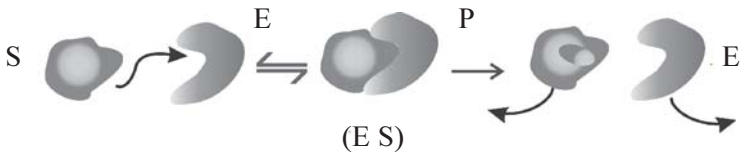
Method :

- It has three dimentional structure.
- It has specific location as lock key linkage.
- For chemical reaction an essential enelgy level is called active energy level.

Equation :

Enzyme + Substrate $\rightarrow$ Enzyme - Substrate complex $\rightarrow$ Enzyme + Product

$E + S \rightarrow E - S \text{ complex} \rightarrow E + P$



(76) Under which physiological factor living organisms perform their activity ?

(A) At low temperature and atmosphere pressure (B) At high temperature and low atmospheric pressure

(C) At high temperature and atmospheric pressure (D) At low temperature and high atmospheic pressure

(77) Enzyme means ..... .

(A) Biological protein (B) Biological hormone (C) Biological catalyst (D) Biological chemical

(78) What is Ribozyme ?

(A) Ribosome + r - RNA (B) Enzymes made up of nucleic acid

(C) r - RNA + protein (D) r - RNA

(79) Which group is correct for the reaction takes place on enzyme at higher and lower temperature sequentially ?

(A) Denaturation and inactive (B) Inactive and denaturation

(C) Inactive and damage (D) Degerate and Inactive

Answers : (76-A), (77-C), (78-B), (79-A)

- **Nomenclature and classification of Enzymes :**

Nomenclature can be done in two ways :

(1) A suffix - ase can be attached to the name of the substrate on which it acts. e.g., The enzyme which acts on sucrose is called sucrase.

(2) It is named after the reaction which it carries. e.g, Hydrolysis are called hydrolases.

Classification of Enzyme/Catagory	Reaction	Example
(1) Oxidoreductases	Oxidation and reduction	- Succinic dehydrogenase - Cytochrom oxidase
(2) Transferases	Except hydrogen from one substrate to another substrate link	- Hexokinese one phosphate group of the ATP to join with glucose
(3) Hydrolases	Complex organic substance by adding a molecule of water	- Maltase (Maltose + H_2O → glucose + glucose)
(4) Lyases	Cleavage of macromolecules into smaller molecules	- Aldose (Fructose 1, 6 biphosphate → 2, triose phosphate)
(5) Isomerases	Changes only in the molecular structure of the structure	- Fructose isomerase (Glucose-6 phosphate → Fructose-6 phosphate)
(6) Ligases or Synthetases	Synthesize a new molecule by joining two molecules with the help of energy obtained from phosphate bond of ATP	- Acetyl co-enzyme synthetase (Acetic acid + ATP + Co.A → Acetyl Co.A + AMP + $H_4P_2O_7$ (Pyrophosphate)

- **Cofactors :**
  - Sometime a non-protein component of enzyme with the protein in enzyme structure is called cofactor.
  - In which protein component is apoenzyme and non-protein component is called cofactor.
  - Cofactor are two types :

(1) Inorganic : eg, = Fe^{+2}, Ca^{+2}, Na^{+}, Zn^{+2}

(2) Organic : eg, NAD, FAD, NADP, FMN

(i) Co enzyme : Loosely bound with apoenzyme.

(ii) Prosthetic group : Tightly bound with apoenzyme.

- For the activation of carbonic anhydrase → Zn
- In azatobacter bacteria - Nitrogen, for the N_2 fixation and activation of Nitrogenase → V
- For the activation of enolase → Mg, Mn, Zn
- Activation of nitric oxide synthetase, protein phosphatase and adenyle kinase → Ca^{+2}

(80) Which structure is responsible for the active site of an enzyme ?

(A) Its colloidal form (B) Its amphoteric property

(C) Its three dimensional structure (D) Activation energy level of a reaction

(81) What is essential for performing every biochemical reaction ?

(A) Three dimensional form (B) Activation energy level (C) Active site (D) All of the given

(82) What is dehydrogenation ?

(A) Joining of hydrogen (B) Donation of a hydrogen

(C) Duplication of hydrogen (D) Absence of hydrogen

(83) Succinic dehydrogenase is example of which type of enzyme ?

(A) Transferases (B) Hydrolysis (C) Oxido-reductases (D) Lyases

(84) What is called that join the phosphate to one substrate to another substrate ?

(A) Transferase (B) Hydrolysis (C) Isomerases (D) Synthetase

(85) Which type of enzyme Hexokinase is ?

(A) Transferases (B) Lyases (C) Isomerases (D) Ligases

(86) Find out incorrect pair :

(A) Maltase - hydrolases (B) Aldolase - lyases

(C) Haxokinase - isomerases (D) Acetyle co-A enzyme synthetase-ligases

(87) Form where Ligase enzyme gets energy in use ?

(A) From the pyrophosphate bond of GTP (B) From the pyrophosphate bond of ATP

(C) From the pyrophosphate bond of AMP (D) From the pyrophosphate bond of GDP

(88) Full form of FMN :

(A) Flavin mononucleotide (B) Flavin mononucleic acid

(C) Ferredoxin mononucleic acid (D) Ferredoxin mononitrite

(89) Find out the correct pair of the following.

(A) Apoenzyme - none protein part of enzyme (B) Prosthetic group - protein part of enzyme

(C) NADP - Co-enzyme (D) Ribozyme - Ribosome + r-RNA

(90) Apoenzyme group is made up of what ?

(A) Carbohydrate (B) Lipid (C) Protein (D) Nucleic acid

(91) Full form of NAD.

(A) Nicotinamide di-nucleotide (B) Nicotinamide Adenine di-nucleotide

(C) Nucleic acid di-nucleotide (D) Nitric acid di-nucleotide

(92) What is cofactor ?

(A) Apoenzyme (B) Inorganic substances

(C) Non protein part of the enzyme structure (D) All of these

(93) Zn is require for the activation of.

(A) Glucose phosphatase (B) Carbonic Anhydrase

(C) Nitrogenase (D) Hydrogenase

(94) Which pair is correct of minerals for the activation of enolase ?

(A) Mg, Ca, V (B) Mg, Zn, B (C) Cu, Zn, Mo (D) Mg, Mn, Zn

Answers : (80-C), (81-B), (82-B), (83-C), (84-A), (85-A), (86-C), (87-B), (88-A), (89-C), (90-C), (91-B), (92-C), (93-B), (94-D)

- **True - False (T - F) types questions :**

Select the true-false statements from the following :

(95) (1) Protein is homopolymer of amino acid.

(2) Protein transports the nutrients through cytoplasmic membrane.

(3) Except 'R' group remaining is same in the amino acid.

(4) Two similar amino acids join to form dipeptide.

(A) T T T F (B) T F F T (C) T F T T (D) F T T F

(96) (1) Isoleucine is having polar and neutral 'R' group.

(2) One terminal of a polynucleotide is known as N - terminal.

(3) Primary structure of protein is decided from the amino acid units.

(4) Skin colour is based on the melanin protein.

(A) T T F T (B) F T T F (C) F F T T (D) T F F T

(97) (1) Dipeptide bond is formed between one carboxyl group and another $-NH_2$ group.

(2) In the structure of haemoglobin there are two α and β polypeptide chains.

(3) Hormones of pituitary gland possesses the ester bonds.

(4) Nuclein were first isolated in human.

(A) T T F T (B) F F T T (C) F T F T (D) F F F T

(98) (1) Actin protein is responsible for the locomotion which is present in our muscles.

(2) There are two polynucleotide chains present in the structure of DNA and RNA.

(3) Pyrimidines are joined together by weak hydrogen bond.

(4) Nucleus is responsible for the heredity in all the organisms.

(A) T T T T (B) T T F T (C) T F F T (D) T F T F

(99) (1) Adenine and guanine joined by the 3 weak hydrogen bond and thymine and cytosine joined by the 2 weak hydrogen bond.

(2) In a every DNA, the proportion of purine and pyrimidine is same.

(3) There is absence of Uracil in Ribose Nucleotide.

(4) A strand of DNA which synthesize the m-RNA is known as Template strand.

(A) F T F T (B) T T F T (C) F T T F (D) F T T F

(100) (1) Enzyme degenerates after the reaction is complete.

(2) Some enzymes show irreversible effect.

(3) Enzyme take part in the intra cellular or extra cellular reaction at a body temperature.

(4) By the lowering of activation energy level, rate of reaction is increased at unexpected speed.

(A) T T F F (B) F T T F (C) F T T T (D) F T F T

(101) (1) Glucose-6 Phosphate, Fructose isomerase, fructose - 6 - phosphate

(2) Nitrogen fixed by the Azatobacter bacteria

(3) A protein which is found in the hair (collagen) is insoluble.

(4) Haemoglobin is the protein with 4 haem molecule.

(A) T T F T (B) T F F T (C) T T F F (D) F F F T

Answers : (95-D), (96-C), (97-C), (98-D), (99-A), (100-C), (101-A)

- **A - Assertion and R - Reason type questions :**

Which option is correct for given question ?

(A) A and R both are true, R is correct explanation of the A.

(B) A and R both true, R is not correct explanation of the A.

(C) A is true and R is false.

(D) A is false and R is true.

(102) Assertion A : For the photosynthesis essential chlorophyll is a conjugated protein.

Reason R : In enzyme if protein and non-protein part is joined then it is called prosthetic group.

(A) (B) (C) (D)

(103) Assertion A : Three dimensional structure of protein is indicated by tertiary structure.

Reason R : Disulphide, Hydrophobic bonds takes part in the quaternary structure of protein.

(A) (B) (C) (D)

(104) Assertion A : Hormones of pituitary gland shows the steroidal nature.

Reason R : Haemoglobin acts as a respiratory pigment.

(A) (B) (C) (D)

(105) Assertion A : Each Nucleotide is made up of 3 subunits.

Reason R : There are two rings in the pyrimidine

(A) (B) (C) (D)

(106) Assertion A : Ribose sugar and Nitrogen base joined to form Riboside

Reason R : Sugar and Nitrogen base joined to form Nucleoside

(A) (B) (C) (D)

(107) Assertion A : Ribonucleoside is phosphorylated to become deoxyribonucleotide.

Reason R : In the structure of DNA two chains of Nucleotide are arranged in a Antiparallel manner.

(A) (B) (C) (D)

(108) Assertion A : Enzymes are colloidal form of a catalyst.

Reason R : Enzymes are secreted in minute concentration by the living cells.

(A) (B) (C) (D)

(109) Assertion A : Enzyme binds to substrate to form enzyme substrate complex.

Reason R : Enzyme is active between fired temperature range.

(A) (B) (C) (D)

(110) Assertion A : Succinic dehydrogenase enzyme is a type of Oxido-reductase.

Reason R : Enzyme plays an important role in the oxidative reaction at Kreb's cycle.

(A) (B) (C) (D)

(111) Assertion A : NADP acts as a co-enzyme.

Reason R : Co-enzymes are small in size compare to Apoenzyme

(A) (B) (C) (D)

Answers : (102-C), (103-D), (104-D), (105-C), (106-A), (107-D), (108-A), (109-B), (110-A), (111-A)

(112) Select the appropriate pair :

Column - I	**Column - II**	
(i) Polar positively charged 'R' group	(p) Glutamine	(A) i - r, ii - p, iii - q, iv - s
(ii) Polar, Neutral 'R' group	(q) Valine	(B) i - q, ii - s, iii - r, iv - p
(iii) Non-polar 'R' group	(r) Histidine	(C) i - q, ii - p, iii - s, iv - q
(iv) Polar, Negatively charged 'R' group	(s) Aspartic Acid	(D) i - r, ii - q, iii - s, iv - p

(113) Select the appropriate pair :

Column - I	**Column - II**	
(i) Haemoglobin	(p) Amino Acid	(A) i - q, ii - r, iii - s, iv - p
(ii) Globular	(q) Immunity	(B) i - q, ii - r, iii - p, iv - s
(iii) Immunoglobulin	(r) Contractbility	(C) i - s, ii - q, iii - p, iv - r
(iv) Glycine	(s) Conjugate protein	(D) i - s, ii - r, iii - q, iv - p

(114) Select the appropriate pair :

Column - I	**Column - II**	
(i) Nucleotide	(p) Peptide bond	(A) i - t, ii - r, iii - p, iv - q
(ii) Lipid	(q) Glycosidic bond	(B) i - r, ii - s, iii - p, iv - q
(iii) Protein	(r) Phospho di-ester bond	(C) i - t, ii - s, iii - p, iv - q
(iv) Carbohydrate	(s) Ester bond	(D) i - r, ii - s, iii - t, iv - q
	(t) Di-sulphide bond	

(115) Select the appropriate pair :

Column - I	**Column - II**
(i) m-RNA	(p) Brings Amino Acid on Ribosome
(ii) DNA	(q) Transfer of genetic code into cytoplasm
(iii) t-RNA	(r) Forms the structure of Ribosome
(iv) r-RNA	(s) Synthesize m-RNA
(A) i - q, ii - s, iii - p, iv - r	(B) i - s, ii - p, iii - q, iv - r
(C) i - p, ii - q, iii - s, iv - r	(D) i - r, ii - s, iii - q, iv - p

(116) Select the appropriate pair :

Column -I	Column -II	Column-III
(i) Lyases	(p) Changes in the molecule structure	(w) Aldolase
(ii) Isomerases	(q) Addition of oxygen	(x) $H_4 P_2 O_7$
(iii) Ligases	(r) Catalyse macro molecule	(y) Fructose Isomerase
(iv) Oxido-reductase	(s) Joining of two molecule	(z) Cytochrome oxidase

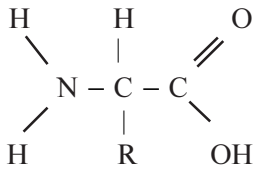
(A) i - q - z, ii - p - y, iii - s-w, iv - r -x (B) i - r - y, ii - s - w, iii - p - x, iv - q -z

(C) i - r - x, ii - p - y, iii - s - w, iv - q - z (D) i - r - w, ii - p - y, iii - s - x, iv - q - z

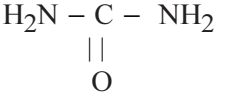
Answers : (112-A), (113-D), (114-B), (115-A), (116-D)

(117) Find out the correct structure of Amino Acid from the following :

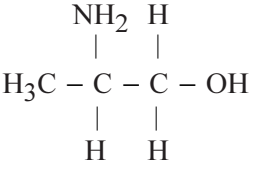
(A)



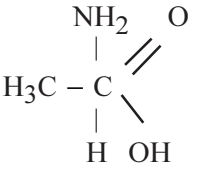
(B)



(C)



(D)



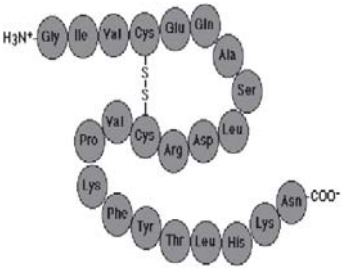
(118) The structure given below is of what ?

(A) Polynucleotide chain

(B) Polypeptide chain

(C) Polysaccharide chain

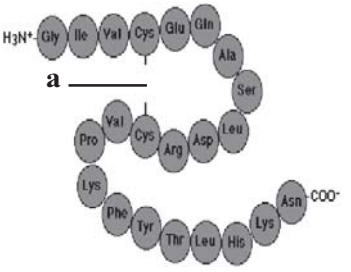
(D) Haemoglobin molecule



(119) Find out what is 'a' in the given figure :

(A) Hydrophobic bond (B) H-H bond

(C) S-S bond (D) Peptide bond



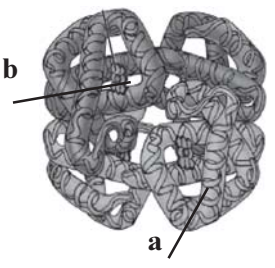
(120) Indicate the a, b regions of the given structure :

(A) Haemoglobin, Haeme group

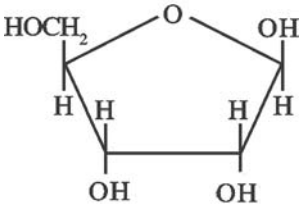
(B) Chlorophyll, Mg group

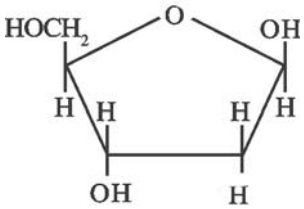
(C) Chlorophyll, Iron group

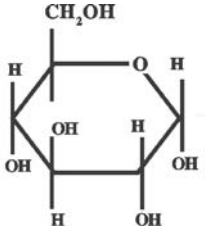
(D) Haemoglobin, Mg group

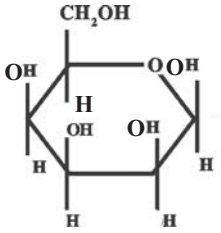


(121) Which structure is correct for the de-oxyribose sugar ?

(A) 

(B) 

(C) 

(D) 

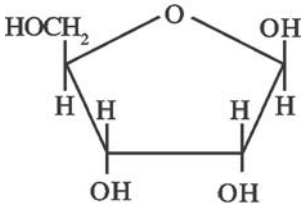
(122) Given structure is of which sugar.

(A) Fructose

(B) Glucose

(C) Galactose

(D) Ribose



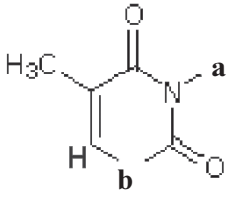
(123) Indicates 'a' and 'b' in given diagram.

(A) a = H, b = N - H

(B) a = H_2, b = N - H

(C) a = H, b = H

(D) a = O, b = H



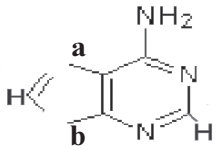
(124) Indicates 'a' and 'b' in given diagram.

(A) a = N - H, b = N - H

(B) a = H, b = N-H

(C) a = N, b = N

(D) a = n, b = N-H



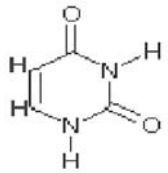
(125) Which Nitrogen base is in the given structure ?

(A) Adenine

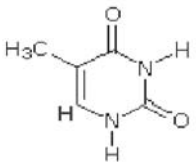
(B) Thymine

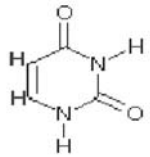
(C) Cytosine

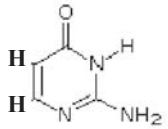
(D) Uracil

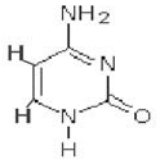


(126) Which is the correct structure of Cytosine ?

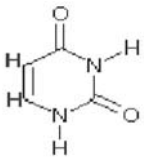
(A) 

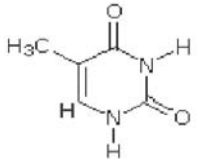
(B) 

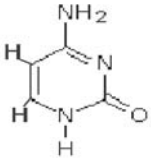
(C) 

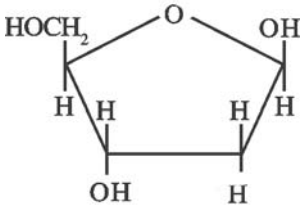
(D) 

(127) Which one of the following structure is absent in the DNA ?

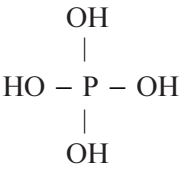
(A) 

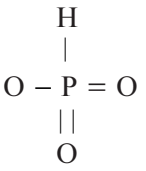
(B) 

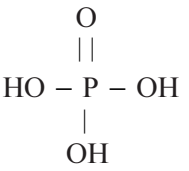
(C) 

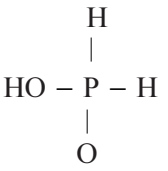
(D) 

(128) Which pair of phosphoric acid is correct for the Nucleotide ?

(A)


(B)


(C)


(D)


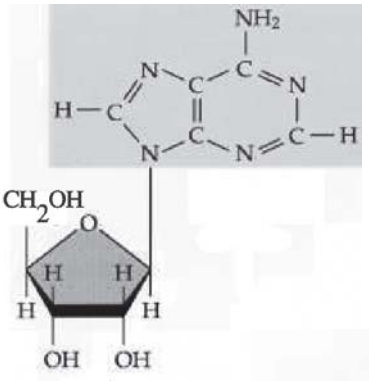
(129) Identify the given structure :

(A) Deoxyriboside

(B) Riboside

(C) Deoxyribotide

(D) Ribotide



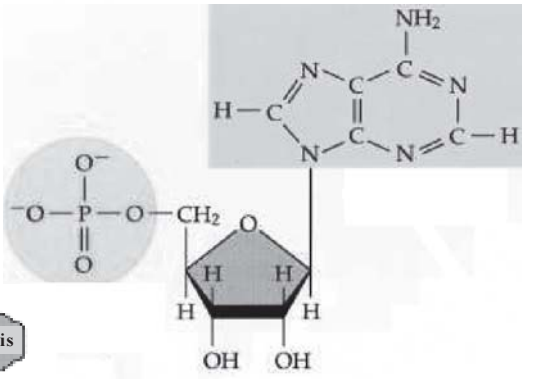
(130) Identify the given structure :

(A) Phosphoribonucleotide

(B) Ribonucleotide

(C) Deoxyribonucleotide

(D) Phospho de-oxy ribonucleotide



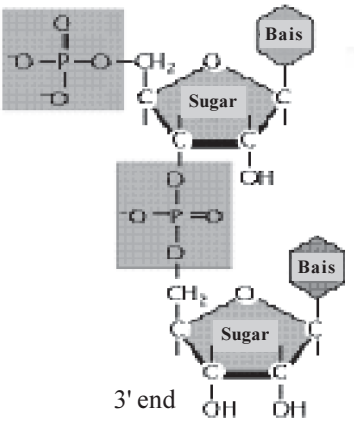
(131) Identify the given structure :

(A) Dipeptide

(B) Disaccharide

(C) Dinucleotide

(D) Polynucleotide



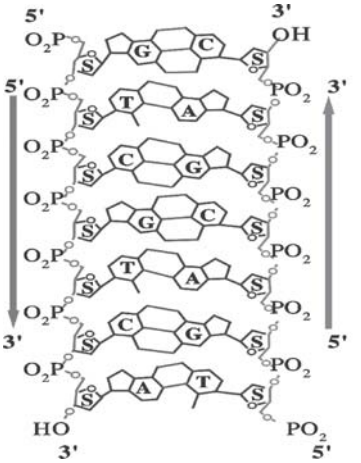
(132) What is given structure ?

(A) Polynucleotide chain

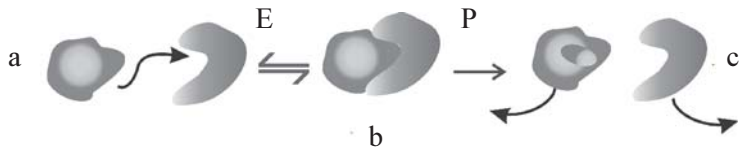
(B) Polypeptide chain

(C) DNA helical structure

(D) DNA molecule structure



(133) Indicates a, b, c in the given diagram :



(A) a = product, b = enzyme, c = substrate

(B) a = product, b = enzyme-substrate complex, c = substrate

(C) a = substrate, b = enzyme-substrate complex, c = enzyme

(D) a = substrate, b = enzyme-substrate complex, c = product

(134) $E + S \rightarrow \underline{\quad a \quad} \rightarrow E + \underline{\quad b \quad}$

(A) a = E - S complex, b = P (B) a = E - P complex, b = S

(C) a = E - S complex, b = S (D) a = P, b = S

Answers : (117-A), (118-B), (119-C), (120-A), (121-B), (122-D), (123-A), (124-D), (125-D), (126-D), (127-A), (128-C), (129-B), (130-B), (131-C), (132-D), (133-D), (134-A)

- **Questions for NEET :**

(135) Pyrenoids are made up of...

(A) Central protein and starch layer (B) Lateral protein and surrounded by lipid layer

(C) Central starch and surrounded by protein layer (D) Central Nucleic acid surrounded by protein layer

(Hint : Pyrenoids are small globular proteins which is surrounded by starch it is found in the chloroplast of Algae as well as bryophytes.)

(136) What enzymes do ?

(A) Decrease in the rate of reaction (B) Increase in the rate of reaction

(C) Increase in the energy level of reaction (D) Decrease in the energy level of reaction

(Hint : It reduces the energy level and enhance the rate of reaction.)

(137) By which bond two individual Ribonucleotide units are bind together ?

(A) Peptide bond (B) Co-valent bond

(C) Hydrogen bond (D) Phospho di-ester bond

(138) Physically and chemically in both the ways which biomolecule is majority differentiated in the cell ?

(A) Protein (B) Carbohydrate (C) Nucleic Acid (D) Lipid

(Hint : Proteins are heteropolymer of Amino Acid. Different 20 types of amino acids form different protein by the arrangement of different sequences and in different numbers.)

(139) By which the gene is arranged in the bacterial chromosome ?

(A) Histones (B) Basic protein (C) Acidic protein (D) Actin

(Hint : Eukaryotic DNA's are attached to the histones. Which is incorporated in the structure of chromatin, when bacterial chromosome is opened and unlist of double helical DNA then by the help of RNA it folded into complex spherical structure. Where Non-histones polyamines carry positive ion ?)

(140) Which Radioactive is required for the measuring of DNA synthesis ?

(A) Uracil (B) Adenine (C) Thymine (D) De-oxy ribose

(Hint : Rate of DNA synthesis is decided by Thymine)

(141) During protein synthesis, in a cytoplasm ribosome gets a appropriate amino acids from the flow of it, which RNA is further get there amino acid ?

(A) m - RNA (B) t - RNA (C) r - RNA (D) DNA

(Hint : The RNA which transfer the Amino Acid into the ribosome in the cytoplasm is known as t - RNA)

(142) Mutations takes place at the gene positions by.

(A) Point Mutation (B) Chromosomal Mutation

(C) Reverse Mutation (D) Forward Mutation

(Hint : Mutation which differes the sequence and numbers of Nucleotide is a kind of genetic mutation known as forward mutation which further convert into wild type reverse mutation. If the numbers and structure of chromosome differs it is known as chromosomal mutation.)

(143) In a human cells, when a new DNA synthesis take place, the surrounding medium is filled with the Radioactive thymine. Now which type of the following chromatid will become radioactive when comes in contact with thymine in the S-phase ?

(A) Hetrochromatin (B) Euchrometin (C) A and B both (D) Non of these

(Hint : Euchromatin is the region which is lightly stained during interphase and in the initial stage of S-phase during DNA replication. Where Heterochromatin is rigid and complex and stained brightly of the chromosome)

(144) Amino Acids structure forms ..... .

(A) Protein (B) Fatty Acid (C) Essential Oil (D) α - Keto Acid

(Hint : Amino Acid contain - NH_2 , - COOH group - H Molecule and R - group.)

(145) A small fragment of DNA consisting 120 Adenine and 120 Cytosine base so the total base pairs are present in this fragment is ..... .

(A) 120 (B) 240 (C) 60 (D) 480

(Hint : A + G = T + C 120 + 120 = 120 + 120 = 480)

(146) Which units are repeated in the structure of DNA ?

(A) Ribonucleoside (B) De-oxy ribonucleoside

(C) De-oxy ribonucleotide (D) Ribonucleotide

(147) What does it called an enzymes which have the different molecule structures but the same functions ?

(A) Holoenzyme (B) Isoenzyme (C) Apoenzyme (D) Co-enzyme

(Hint : Apoenzyme is the protein part of an enzyme. Co-enzyme is the non-protein part of an enzyme. Both binds together form a Holoenzyme, where Isoenzyme is having the same functions but different structures.)

(148) What is wrong for the Nucleic Acid ?

(A) Few viruses are having one strand of DNA (B) Sometimes RNA is double stranded

(C) Length of the one spiral of β-DNA is 45 $\overset{0}{A}$ (D) 12-base are present in the turn of Z-DNA

(Hint : Length of the β-DNA is 34 $\overset{0}{A}$)

(149) Enzymatic reaction is inhibited by.

(A) Final product (B) Substrate (C) Enzyme (D) Increase in temperature

(Hint : Glucose + ATP $\xrightarrow{\text{Hexokinase}}$ Glucose 6 - phosphate + ADP (Final product)

(If inhibitor is present then no final product is formed.)

(150) With the structure of clover leaf which RNA is incorporated ?

(A) r-RNA (B) hn-RNA (C) m-RNA (D) t-RNA

(Hint : hn RNA is a heterogenetic cellular RNA. r-RNA is present in the Ribosome, m-RNA transfer the information. Where it binds as at t-RNA.)

(151) Which sentences is right from the following ?

(A) In general only 20 enzymes are found

(B) In general there are so many proteins

(C) In general four types of DNA molecules are found

(D) In general only 80 types of chromosomes are found

(152) Enzymes are mainly...

(A) Globular protein (B) Fibrous protein (C) Structural protein (D) Nucleo protein

(153) Which is the most abundant organic substance on the earth ?

(A) protein (B) cellulose (C) lipid (D) steroid

(Hint : Most abundantly organic substance found is polysaccharide.)

(154) Match the following pairs :

Column-I	**Column-II**
(i) RNA	(p) Storage product
(ii) Haemoglobin	(q) Protein synthesis
(iii) Steroid	(r) Transport of gases
(iv) Starch	(s) Sexual Hormone

(A) i - q, ii - r, iii - s, iv - p (B) i - s, ii - r, iii - p, iv - q

(C) i - r, ii - s, iii - p, iv - q (D) i - q, ii - s, iii - r, iv - p

(155) In the DNA, Nucleotides are joined by.

(A) Hydrogen bond (B) Co-valent bond (C) Van-der-vals (D) Electro-valent bond

(156) What is collagen ?

(A) Globular protein (B) Carbohydrate (C) Lipid (D) Fibrous protein

(Hint : Collagen is made up of Fibers)

(157) Collagen is full of ..... .

(A) Glycine (B) Serine (C) Glutamic Acid (D) Aspartic Acid

(158) In 1980's RNA term were used as a result of evolution on the basis of...

(A) Genetic material is RNA in the viruses

(B) Because of the importance of m-RNA, t-RNA, r-RNA in the protein synthesis

(C) Because of the enzymatic property of RNA

(D) All the cells does not having RNA

(Hint : RNA is showing the evolution from abiotic substances to biotic system. Active by enzymatically.)

(159) What is the length of one complete spiral of β - DNA ?

(A) 3.4 nm (B) 0.34 nm (C) 34 nm (D) 0.034 nm

(Hint : 1 $\overset{0}{A} = 10^{-1}$ nm, 34 $\overset{0}{A} = 10^{-1} \times 34 = 3.4$ nm)

(160) In the structure of DNA.

(A) Now the chain is Anti parallel where one strand is $3' \rightarrow 5'$ and another is $3' \rightarrow 5'$

(B) Both are in the direction of $5' \rightarrow 3'$

(C) Proportion of Adenine is different then thymin in different organisms.

(D) Total purine Nucleotide and Pyrimidine Necleotide are not same.

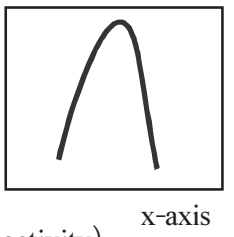
(161) The given graph shows the curve of enzymatic activity with related to three conditions. (pH temperature and substrate concentration) So what does it indicates of x-axis and y-axis ?

(A) x-Activated enzyme, y-pH

(B) x-Temperature, y-Activated enzyme

(C) x-Concentration, y-Activated enzyme

(D) x-Activated enzyme, y-Temperature



(Hint : Gradient indicates the Relation between temperature and enzymatically activity)

(162) Which is the inhibitor of succinate dehydrogenase ?

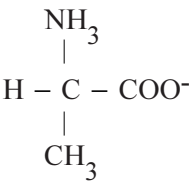
(A) Melonate (B) Malate (C) Succinate (D) Oxaloacetate

(Hint : Melonate is inhibitor for the respiration CH_2 - CH_2 is absense so dehydrogenation does not take place. So the reaction will stops.)

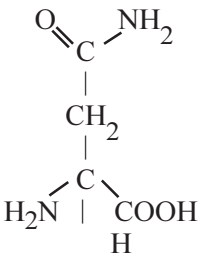
(163) Which Amino Acid is very common ?

(A) Alanine (B) Aspergine (C) Glycine (D) Tyrosine

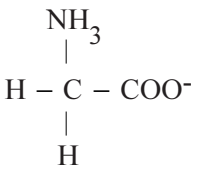
Hint :



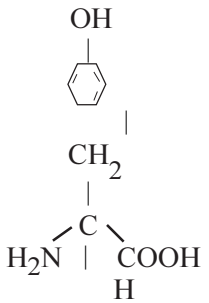
Alanine



Aspergine



Glycine



Tyrosine

(164) How it appears in the 3D structure of transfer - RNA ?

(A) Y - shape (B) S - shape (C) E - shape (D) L - shape

(Hint : By t - RNA model)

(165) In the Nucleus, how many types of RNA polymerase is required for the synthesis of RNA ?

(A) 1 (B) 2 (C) 3 (D) 4

(Hint : For r - RNA synthesis in one nucleus. For synthesis in nucleoplasma for synthesis of m- RNA, For synthesis of t - RNA which is present in Nucleoplasma.)

Answers : (135-A), (136-B and D), (137-D), (138-A), (139-B), (140-C), (141-B), (142-A), (143-B), (144-A), (145-B), (146-C), (147-B), (148-C), (149-D), (150-D), (151-B), (152-A), (153-B), (154-A), (155-B), (156-D), (157-A), (158-C), (159-A), (160-A), (161-B), (162-A), (163-C), (164-D), (165-C)

Biomolecules

Carbohydrates
Hydrates of Carbon, Units are linked by glycosidic bond

- **Monosaccharides**
  Simple sugars,
  All reducing sugars
- **Oligosaccharides**
  Condensation of 2-9 monosaccharides. Disaccharides are the common oligosaccharides
- **Polysaccharides**
  Made by joining of many monosaccharides
  - Homopolysaccharides
    e.g., starch, cellulose, glycogen
  - Heteropolysaccharides
    e.g., chitin, pectin, hemicellulose

Proteins
- Most abundant bio macromolecules
- Synthesized from amino acids linked by peptide bonds
- 20 amino acids

- **Primary Structure**
  Sequence of amino acids linked by peptide bonds
- **Secondary Structure**
  Stable arrangement to form structural patterns α - helix and β - pleated sheets
- **Tertiary Structure**
  3D folding of polypeptide
- **Quaternary Structure**
  3D Molecule of protein formed by two or more polypeptides

Nucleic Acids
- Genetic material of all living organisms
- Deoxy ribonucleic acid (DNA)
- Ribonucleic acid (RAN)

- **Sugar (Pentose)**
  Ribose in RNA, Deoxyribose in DNA
- **Nitrogenous base**
  Purines (Adenine and Guanine)
  Pyrimidines (Cytosine and Thymine / Uracil)
- **Phosphoric acid**
  Forms ester linkage during condensation with sugar

Lipids
Water insoluble carboxyl group attached to R-group

- Simple lipids (homolipids)
- Tryglycerides and waxes
- Compound lipids (heterolipids)
  Phospho lipids, Glycolipids, lipoproteins
- Derived lipids, Steroids, glycezol, fatty acid

Enzymes
- Proteinaceous substances; Holoenzyme is made up of apoenzyme (protein part) and cofactor (non-protein part)
- Cofactors are of 3 types : Inorganic ions (Zn, Mg), Prosthetic group (haem, biotin) and coenzyme (NAD^+, $NADP^+$)

Competitive inhibition
- Reversible
- Due to substrate or enzyme analogue
- K_m is increased but V_{max} remains the same Where, Km = Michaelis - Menten constant V_{max} = Maximum velocity

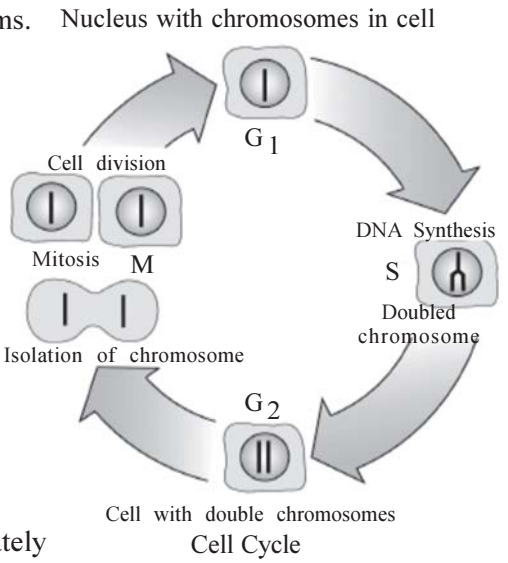
Non-competitive inhibition
- Inhibitor forms a complex with enzyme other than the active site
- V_{max} decreases

Uncompetitive inhibition
- Inhibitor binds to E_S complex
- V_{max} and K_m decrease

Feedback inhibition
- End product or intermediates functions as temporary inhibitor which combines with the regulatory site of the enzyme and thus functions as negative modulator

8 Cell cycle and Cell divisions

- Growth is a fundamental property of all living organisms.
- The body of an adult person contains 10^{14} cells.
- Zygote is formed by the fertilization between male and female gametes.
- Cell division → Cell multification → Growth



Cell Cycle

- **Cell cycle :**
  - The cell cycle is the series of events that takes place inside a cell thus leading to cell division and cell duplication.
  - The period between two successive cell divisions is called cell cycle.
  - The human cell divides in culture once in approximately every 24 hours.
  - Yeast cell can complete one cell cycle in every 90 minutes.
  - Cell cycle divide in two phases :

(1) **Interphase :**

- Materials duplicate. - It is recognized as chromatin material. - Orgenelle, DNA, centriole are duplicated. - It is divided in three subphases.

(i) G_1 Phase (Gap-I phases) :

- Growth phase. - For the replication purpose DNA, essential enzyme protein and RNA are synthesize.

(ii) S Phase (Synthesis or Replication phase) :

- Duplication of DNA. - becomes 4C. - becomes 4C from 2C.

(iii) G_2 Phase (Gap-II phase) :

- Formation of protein. - For the mitosis formation of microtubules are necessary.

---

(1) What is the essential criterium for growth of organisms ?

(A) Increase in cell mass (B) Duplication of genetic material

(C) Cell division (D) Cell degradation

(2) Which of the following statement is incorrect for cell cycle ?

(A) Duplication of cell occurs (B) Enhance the fertilisation of gamete cells

(C) The period between two successive cell divisions (D) The series of events that takes place inside a cell

(3) Which of the following statement is incorrect for Interphase ?

(A) Replication of DNA (B) Duplication of centrioles

(C) Cell size increases (D) Nuclear division occurs

(4) During which phase does chromatin material present ?

(A) Metaphase (B) Interphase (C) Anaphase (D) Telophase

(5) Which subphase of the following is not of Interphase ?

(A) G_1 (B) division (C) G_2 (D) S

(6) Which of the following is a process of G_1 phase ?

(A) DNA synthesis (B) Production of micro tubules

(C) Enzyme, RNA, Protein synthesis (D) All of these

Answers : (1-C), (2-B), (3-D), (4-B), (5-B), (6-C)

- **Mitosis :**
  - It is continuous process.
  - But meant for easy understanding it is divided into four phases.

(1) **Prophase :**

- Condensation of chromosomes along their lengths.
- Nuclear membrane and nucleolus disintegrate.

(2) **Metaphase :**

- Chromosome is clearly observed under the microscope. - Structure of kinetochores are found.
- The line is referred as the equatorial plate or metaphase plate.

(3) **Anaphase :**

- The spindle fibers shorten. - Equal number of chromosomes on both polar ends.
- Division of centromere occur.

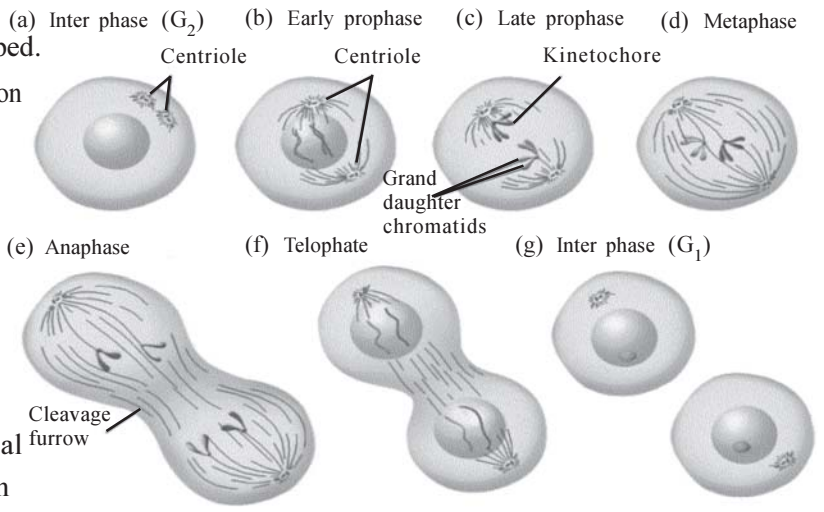
(4) **Telophase :**

- Formation of chromatin material. - Organelles are reestablished.
- Formation of new nucleus.

Cytokinesis :

- It is not a part of mitosis but it is a separate process that completes the entire process of cell division.
- In an animal cell it takes place from the peripheral region.
- In plant cell it take place from the centre to peripheral region.
- In plant cell middle lamella made up of pectin and cell wall develops around it.
- In some organisms karyokinesis does not take place as a result multinucleate condition arises leading to the formation of syncytium.

- **Significance of Mitosis :**
  - Multicellular body is developed.
  - Occurrence of asexual reproduction two daughter organisms arises.
  - The number of chromosomes is maintained in all cells.
  - Cells are made available for growth and development.
  - A very significant contribution of mitosis is cell repair
  - Mitotic divisions in the apical and lateral meristem result in continuous growth in the plants.



(7) Which of the following is the correct sequence for the Mitosis ?
(A) G_1 - S - G_2 - G_2 - M
(B) Prophase - metaphase - anaphase - telophase
(C) anaphase - metaphase - telophase
(D) leptotene - zygotene - pachytene - diplotene

(8) Which of the following process is not involved in prophase ?
(A) the condensation of chromosomes along their lengths (B) bipolar spindle is developed
(C) nuclear membrane and nucleolus disintegrate (D) production of equatorial plate

(9) During which phase condensation of chromosomes is completed and they can be seen clearly ?
(A) Anaphase (B) Prophase (C) Metaphase (D) Telophase

(10) Structures serve as the site of attachment of spindle fibers to the centromere of the chromosomes.
(A) Kinetochores (B) Centromere (C) Metaphase plate (D) All of these

(11) Which of the following process is involved in anaphase ?
(A) Chromosome produces equatorial plate.
(B) Chromosome converts into chromatin.
(C) Centromere splits, will separate and start moving to the opposite poles of the cell.
(D) Chromosomes spread in the entire cell area.

(12) How does the metaphase differ from anaphase ?
(A) Chromatids having centromere (B) For chromatin
(C) Condensation of spindle fibres and the centromere splits and will separate
(D) All of the above.

(13) During which phase of mitosis chromatin network is observed ?
(A) Anaphase (B) Metaphase (C) Prophase (D) Telophase

(14) Cytokinesis means ..... .
(A) Last stage of cell division
(B) Stage of duplication of cellular material
(C) a separate process that completes the entire process of cell division
(D) All of these

(15) Syncytium means ..... .
(A) Cytokinesis (B) Multinucleate condition (C) Metaphase plate (D) Kinetochores

(16) How does cytokinesis begin in higher plants ?
(A) Separation of the cytoplasm from the centre to the periphery
(B) Separation of the cytoplasm from the periphery to centre region
(C) At any plate (D) All of these

(17) Select incorrect pair :
(A) Prophase : Chromosome made up of two charomatids and a centromere holding them together.
(B) Metaphase : Chromosome can be seen clearly.
(C) Anaphase : Chromosome are arranged at the equatorial plate in the middle of the cell.
(D) Telophase : Nuclear membrane, golgi complex are reformed.

(18) Select the correct pair :
(A) Syncytium - nuclear cell division (B) Metaphase plate - telophase
(C) Recombination - anaphase (D) Bipolar spindle - metaphase

(19) A very significant contribution of mitosis is cell repair. Because...
(A) The number of chromosomes is maintained in all cells.
(B) By asexual reproduction produce two new progeny cells.
(C) The cells of lining of the gut and blood cells are constantly replaced.
(D) Cells can maintain their efficient size.

(20) Which tissue shows mitosis in plants ?

(A) Meristematic tissue (B) Complex tissue (C) Permanent tissue (D) All of these

(21) Mitotic divisions in ..... result in continuous growth in the plants.

(A) simple permanent tissue (B) apical meristematic tissue

(C) lateral meristematic tissue (D) B and C

(22) By which process the number of chromosomes is maintained in cells ?

(A) Mitosis (B) Amitosis (C) Meiosis (D) Meiotic division

(23) During cytokinesis middle lamellae is made up of ..... in Sequoia.

(A) cellulose (B) lignin (C) chitin (D) pectin

(24) What is formed on both the sides of middle lamella in angiosperm plants ?

(A) Cytoplasm membrane (B) Tonoplast (C) Cell wall (D) Lipid layer

(25) The formation of syncytium leading because...

(A) Nuclear cytokinesis is not followed. (B) Absence of telophase

(C) Karyokinesis is not followed by cytokinesis (D) Multinucleate condition arises

(26) How many animal cells are reproduced at the end of mitosis ?

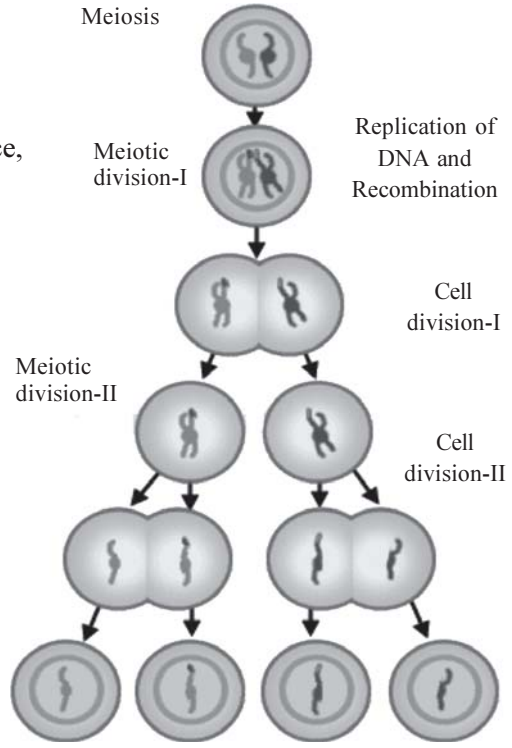
(A) 2 (B) 4 (C) 3 (D) 1

Answers : (7-B), (8-D), (9-C), (10-A), (11-C), (12-C), (13-D), (14-C), (15-B), (16-A), (17-C), (18-A), (19-C), (20-A), (21-D), (22-A), (23-D), (24-C), (25-C), (26-A)

- **Meiosis :**
  - During formation of reproductive cells meiotic cell devision occurs.
  - During meiosis the genetic material is replicated once, whereas the cell divides twice.

  (1) Meiosis-I (Reduction phase) :
  - First division
  - Half the number of chromosomes

  (2) Homotypic division-II (Equational phase) :
  - Second division
  - Maintain the number of chromosome



- **Meiosis-I (Reduction phase) :** It has four phases :

(i) **Prophase-I** It is divided into five substages.



(a) **Leptotene :**
- Initial stage
- Condensed chromosome
- Two chromatids and a centromere

(b) **Zygotene :**



- Synapsis occur
- Process progress in a zipper fashion
- Structure can be understood by electron micrographs

(c) **Pachytene :**



- Appear as tetravalent
- Formation of recombination nodules (Chiasmata)
- Where as exchange of genes occur it is known as crossing over

(d) **Diplotene :**



- Homologous chromosomes start moving away from one another
- The number of chiasmata depends on the length of chromosomes.

(e) **Dikinesis :**



- Condensation complete
- Bipolar spindle fibreforms
- Nucleolus and Nuclear membrane disappears

(ii) **Metaphase-I :** Pairs of homologous chromosomes become arranged at the equatorial plane.

(iii) **Anaphase-I :** Chromosomes of a homologous pair move away towards their respective pole. Thus, at the end of this phase, half the number of total chromosomes are collected at each pole.

(iv) **Telophase-I :** Nuclear membrane and nucleolus are reformed.

- The stage between two meiotic stages is called interkinesis.

- **Meiosis-II (Equational division)**
  - Replication of genetic material does not occur.
  - Divided in four stages.

(1) **Prophase-II :** Reformation of bipolar spindle fibres. The chromosomes again become compact.

(2) **Metaphase-II :** Formation of equatorial plane.

(3) **Anaphase-II :** Centromere splits..

- Centromere of two chromatids of a chromosome migrate towards opposite poles.
- Same as the number of chromosomes in parent cell.

(4) **Telophase-II :** Reformation of nucleolus, nuclear membrane.

- **Significance of meiosis :**
  - Definite number of chromosomes is maintained in organisms from generation to generation.
  - Due to crossing over, meiosis provides an opportunity for the exchange of genes and thus causes the genetic variations among species.
  - It is important process for evolution.

---

(27) How many times cell divides in meiosis ?

(A) 1 (B) 2 (C) 3 (D) 4

(28) Meiosis division means....

(A) the cells must be half in numbers.

(B) the number of chromosomes in the new cells are half at time of cell division.

(C) size of the cells are half. (D) All of these

(29) Mitosis means...

(A) the cells must be half in numbers.

(B) the number of chromosomes is maintained the same in new cells.

(C) the number of nucleus is maintained the same in new cells.

(D) the number of bipolar spindles is maintained the same in new cells.

(30) How many cells are produced at the end of meiosis ?

(A) 2 (B) 1 (C) 8 (D) 4

(31) How many stages are there in Meiosis - I ?

(A) 4 (B) 9 (C) 8 (D) 2

(32) Synapsis means...

(A) appearance of recombination nodules.

(B) chromatids clearly appears as tetrads.

(C) gene exchange occurs at the location of Chiasmata.

(D) chromosomes start pairing together along their length.

(33) At what phase the process progresses in a zipper fashion ?

(A) Pachytene (B) Diplotene (C) Zygotene (D) Dikinesis

(34) During pachytene stage...

(A) chromatids of chromosomes are twined around one another.

(B) appearance of Chiasmata.

(C) exchange of genes, occurs at various places along their length due to crossing over.

(D) All of these

(35) The chronological sequence of stages in prophase-I of meiosis is...

(A) Leptotene → Pachytene → Zygotene → Diplotene → Dikinesis

(B) Leptotene → Zygotene → Pachytene → Diplotene → Dikinesis

(C) Zygotene → Leptotene → Pachytene → Dikinesis

(D) Leptotene → Pachytene → Zygotene → Diplotene → Dikinesis

(36) In which of the following stage chromosome appears filamentous ?

(A) Leptotene (B) Dikinesis (C) Diplotene (D) Pachytene

(37) Interkinesis means ...

(A) The period between two cell cycle.

(B) The stage between interphase and mitosis stage.

(C) The stage between two meiotic stage.

(D) The stage between mitosis and cell cycle.

(38) Intrameiotic interphase means...

(A) Synapsis (B) Crossing over (C) Interkinesis (D) Syncytium

(39) During meiosis chiasmata are observed at ....

(A) Zygotene (B) Pachytene C) Diplotene (D) Dikinesis

(40) The number of chiasmata depends upon....

(A) length of chromosome. (B) numbers of chromosome.

(C) numbers of gene. (D) exchange of gene.

(41) Which is the specific place for gene exchange ?

(A) Zipper (B) Chiasmata (C) Bipolar spindle (D) Equatorial plate

(42) Crossing over means...

(A) duplication of gene. (B) distribution of gene. (C) exchange of gene. (D) division of gene.

(43) During which phase condensation of chromosome is completed ?

(A) Metaphase - I (B) Prophase - I (C) Dikinesis (D) Diplotene

(44) What happened at the end of dikinesis ?

(A) Exchange of gene (B) Complete condensation of chromosome

(C) Chromosome changes place (D) Nucleolus disappears and nuclear membrane

(45) In which direction centromere of chromosome is located at equatorial plate ?

(A) Poles of a cell (B) North pole (C) South pole (D) None of these

(46) At which stage the half the number of chromosomes than parent cell are collected ?

(A) Metaphase-II (B) Metaphase-I

(C) Anaphase-II (D) Anaphase-I

(47) Which of the following stage is not proper for prophase-I ?

(A) Leptotene (B) Diplotene (C) Dikinesis (D) Interkinesis

Answers : (27-B), (28-B), (29-B), (30-D), (31-A), (32-D), (33-C), (34-D), (35-B), (36-A), (37-C), (38-C), (39-B), (40-A), (41-B), (42-C), (43-A), (44-D), (45-D), (46-D), (47-D)

- **Differences between Mitosis and Meiosis :**

Mitosis	Meiosis
(1) It occurs in somatic cell	It occurs in formation of reproductive cells
(2) End of the division two daughter cells are formed	End of the division 4 daughter cells formed
(3) A mitotic mother cell can either be haploid or diploid	A meiotic mother cell is always diploid
(4) Number of chromosomes per nucleus remains the same	Number of chromosomes are half than the original mother cell
(5) No pairing of homalogous chromosomes	Pairing of all homologous chromosomes take place
(6) No crossing over take place	Occurrence of crossing over
(7) Centromere spits during anaphase	The centromeres do separate during anaphase-II, but not during anaphase-I
(8) The genotype of the daughter cells is identical to that of the mother cells	Meiotic products differ in their genotype from the mother cell

(48) Which of the following statement is true for meiosis ?

(A) Formation of two new daughter cells at the end of the process.

(B) Prophase is short and easy.

(C) Once duplication of genetic material and twice cell division.

(D) Cells are made available for growth and development.

(49) Mitosis differs from meiosis in :

(A) based on number of chromosomes (B) based on number of cells

(C) based on stages of divisions (D) All of these

(50) Which type of gametes are produced by meiosis of diploid mother cell ?

(A) 2n (B) n (C) 3n (D) 4n

(51) Find out, incorrect statement for meiosis.

(A) A meiotic mother cell is always haploid.

(B) The genotype of the daughter cells is identical to that of the mother cells.

(C) In a meiotic daughter cell, chromosome is always haploid.

(D) Each daughter cell has exactly same DNA strands.

(52) Which is incorrect sentence for mitosis ?

(A) During prophase homologous chromosomes start pairing.

(B) A mother cell can either be haploid or diploid.

(C) At anaphase stage centromere of each chromosome divides.

(D) The genotype of the daughter cells is identical.

(53) For which of the following mitosis and meiosis are similar ?

(A) For number of chromosomes in daughter cell

(B) For the genotype of daughter cells

(C) For the amount of DNA which becomes double in synthesis phase

(D) For pairing of homologous chromosomes

Answers : (48-C), (49-D), (50-B), (51-A), (52-A), (53-C)

- **True - False (T - F) type questions :**

Choose the correct option for true-false :

(54) (1) The number of chiasmata depends on the number of chromosomes.

(2) In telophase-I, Each nucleus contains half as many chromosomes as were present in the parental cell.

(3) Cytokinesis connects each nucleus from the other.

(4) Similarity of the gene occurs at the location of chiasmata.

(A) F, F, T, T (B) F, T, F, F (C) F, T, T, F (D) T, T, T, F

(55) (1) Electron micrographs of this stage indicate that chromosome synapsis is accompanied by the formation of complex structure called synaptonemal complex.

(2) Dikinesis is the first stage of the homologous pair move away from each other.

(3) Pachytene stage is characterized by apperance of recombination nodules (Chiasmata).

(4) In the period between two meiotic divisions no replication occurs.

(A) T, F, T, T (B) T, T, T, T (C) T, F, F, F (D) F, T, T, F

(56) (1) In meiosis meiotic product is idnetical to that of the mother cells.

(2) After mitosis, each daughter cell has exactly same DNA strands.

(3) Homologous chromosomes are responsible at crossing over.

(4) A meiotic mother cell can either be haploid or diploid.

(A) F, T, F, T (B) T, F, F, T (C) F, T, T, F (D) T, F, T, F

(57) (1) Haploid gemetes are produced by meiosis.

(2) The cells of upper layer of epidermis are replaced in Meiotic division.

(3) Prophase-I of Meiosis is long and complex.

(4) An organism beginning as a single cell develops its multicellular body.

(A) F, T, T, F (B) F, T, T, T (C) T, F, F, T (D) T, F, T, T

(58) (1) Asexual reproduction is indication of Mitosis.

(2) In animal cell constriction of cytoplasm begins from the peripheral region at the cell.

(3) When the centromere splits, the paired chromosomes will separate.

(4) Condensation of chromosomes is completed and it cannot be seen clearly under the microscope.

(A) T, T, F, F (B) F, F, T, T (C) T, T, T, F (D) F, T, F, T

(59) (1) In plant cells, centrosome is not present.

(2) In S-phase proportion of DNA is 4C.

(3) The mitosis process is a continuous one.

(4) In the G_1 phase DNA replication occurs.

(A) F, T, T, F (B) T, T, F, F (C) T, T, T, F (D) T, F, F, T

(60) (1) In interphase chromosomes can not be observed as they are highly dispersed.

(2) Yeast cell can complete one cycle in 60 minutes.

(3) Cell cycle is the time period between Interpase and cell division.

(4) A nucleous develops on the nucleolar organizer region of a specific chromosome.

(A) T, F, F, T (B) F, T, F, T (C) F, T, T, T (D) T, F, T, T

Answers : (54-B), (55-A), (56-C), (57-D), (58-A), (59-C), (60-A)

- **A - Assertion, R - Reason types questions :**

Choose the correct option from following for each questions.

(A) Both A and R are true and R is correct explanation of A.

(B) Both A and R are true and R is not correct explanation of A.

(C) A is true and R is wrong.

(D) A is wrong and R is true.

(61) Assertion A : Meiotic products differ in their genotype from the mother cell.

Reason R : In pachytene stage crossing over that is exchange of genes, occurs.

(A) (B) (C) (D)

(62) Assertion A : In meiosis, only meiosis-I is preceded by a S-phase.

Reason R : During meiosis the DNA is replicated once, whereas the cell divides twice.

(A) (B) (C) (D)

(63) Assertion A : A meiotic mother cell is always haploid.

Reason R : In mitotic, there is no pairing of homologous chromosomes.

(A) (B) (C) (D)

(64) Assertion A : One single division of the mother cell results in two daughter cells.

Reason R : In meiosis, the cell divides twice.

(A) (B) (C) (D)

(65) Assertion A : Meiosis cause the genetic variation among species.

Reason R : Gene exchange occurs at the location of chiasmata.

(A) (B) (C) (D)

(66) Assertion A : Meiosis is important process for evolution.

Reason R : Meiosis maintains the genotype of the daughter cells identical to that of the mother cells

(A) (B) (C) (D)

(67) Assertion A : Diplotene maintains the points of crossing over.
Reason R : The process of synapsis progresses in a zipper fashion.
(A) (B) (C) (D)

(68) Assertion A : Anaphase, anaphase-I, and anaphase-II are similar process.
Reason R : Equatorial plate means the spindle fibres attach themselves to the centromere of the chromosomes and align them along the middle of the cell.
(A) (B) (C) (D)

(69) Assertion A : Meiosis follows interphase.
Reason R : In heterotypic division, the chromosomes are distributed in two cells in half, their number.
(A) (B) (C) (D)

(70) Assertion A : Very significant contribution of mitosis is cell repair.
Reason R : Blood cells must be replaced constantly.
(A) (B) (C) (D)

(71) Assertion A : In animal cells, a plate called middle lamella, gradually develops from centre towards periphery of the cell.
Reason R : Middle lamella is made up of pectin.
(A) (B) (C) (D)

(72) Assertion A : In telophase-I the chromosomes are not seen as chromatin.
Reason R : Chromosomes can be observed as they are highly dispersed during interphase.
(A) (B) (C) (D)

(73) Assertion A : During S stage of interphase centrosome duplicates.
Reason R : Bipolar spindle is made up of cytoplasmic fibres of proteins.
(A) (B) (C) (D)

Answers : (61-A), (62-A), (63-D), (64-B), (65-A), (66-C), (67-B), (68-D), (69-A), (70-A), (71-D), (72-A), (73-B)

(74) Match the following :

Column - I	**Column - II**	
(i) Mitosis	(p) Synapsis	(A) i - r, ii - s, iii - q, iv - p
(ii) Reduction division	(q) Meiosis-II	(B) i - r, ii - q, iii - p, iv - s
(iii) Equational division	(r) asexual reproduction	(C) i - q, ii - s, iii - r, iv - p
(iv) Zygotene	(s) heterotypic division	(D) i - q, ii - r, iii - p, iv - s

(75) Match the following :

Column - I	**Column - II**	
(i) Prophase	(p) the spindle fibres shorten	(A) i - p, ii - s, iii - r, iv - q
(ii) Mitosis	(q) organelles reforms	(B) i - r, ii - s, iii - p, iv - q
(iii) Metaphase	(r) bipolar spindle develops	(C) i - r, ii - p, iii - q, iv - s
(iv) Telophase	(s) formation or equatorial plate	(D) i - p, ii - q, iii - r, iv - s

(76) Match the following :

Column - I	**Column - II**
(i) Zygotene	(p) bivalent chromosomes appears tetravalent
(ii) Pachytene	(q) nucleolus disappears
(iii) Dikinesis	(r) chromosomes start pairing together along their length
(iv) Diplotene	(s) chromosomes start moving away from one another

(A) i - p, ii - r, iii - q, iv - s (B) i - p, ii - q, iii - r, iv - s

(C) i - r, ii - p, iii - q, iv - s (D) i - r, ii - q, iii - s, iv - p

(77) Match the following :

Column - I	**Column - II**
(i) Anaphase - II	(p) chromosomes expands
(ii) Anaphase - I	(q) chromosome is made up of two chromatids held together by a centromere
(iii) Telophase - II	(r) each chromatid has independent centrosome
(iv) Telophase - I	(s) chromosomes of a homologous pair move away towards their respective poles

(A) i - s, ii - r, iii - q, iv - p (B) i - r, ii - s, iii - q, iv - p

(C) i - s, ii - r, iii - p, iv - q (D) i - r, ii - s, iii - p, iv - q

(78) Match the following :

Column - I	**Column - II**
(i) Prophase - I	(p) takes place within somatic cells
(ii) Anaphase - II	(q) takes place within gametic cells
(iii) Meiosis	(r) pairs of homologous chromosomes arranged
(iv) Mitosis	(s) centromere divides

(A) i - r, ii - s, iii - q, iv - p (B) i - s, ii - r, iii - q, iv - p

(C) i - s, ii - q, iii - r, iv - p (D) i - p, ii - r, iii - s, iv - q

(79) Match the following :

Column - I	**Column - II**
(i) G_2 phase	(p) consists two distinct but integrated activities
(ii) S stage	(q) biosynthetic stage
(iii) G_1 stage	(r) Later the cell enters mitosis
(iv) M stage	(s) chromosomes replicated and they all have two sister chromatids

(A) i - q, ii - s, iii - p, iv - r (B) i - r, ii - s, iii - q, iv - p

(C) i - r, ii - p, iii - q, iv - s (D) i - p, ii - r, iii - s, iv - q

(80) Match the following :

Column - I	**Column - II**
(i) Yeast cell division	(p) 10^{14}
(ii) Human cell division	(q) 5
(iii) Human cell number	(r) 90 min
(iv) Substages of prophase	(s) 1440 min

(A) i - r, ii - s, iii - p, iv - q

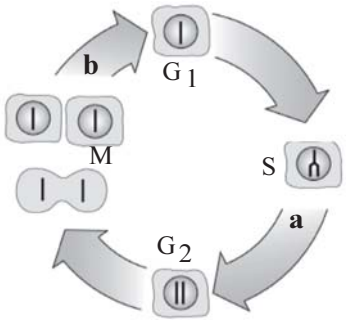
(B) i - r, ii - p, iii - s, iv - q

(C) i - p, ii - r, iii - s, iv - q

(D) i - p, ii - s, iii - r, iv - q

Answers : (74-A), (75-B), (76-C), (77-D), (78-A), (79-B), (80-A)

(81) What indicates 'a' and 'b' ?



(A) cell division, isolation of chromosomes

(B) duplication of chromosomes, isolation of chromosomes

(C) cell division, n

(D) replication of DNA, cell division

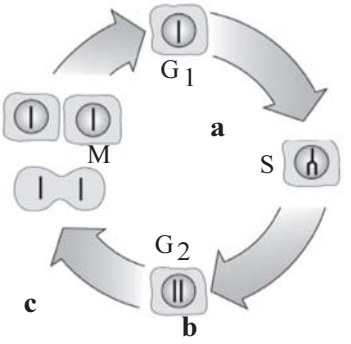
(82) What indicates 'a', 'b' and 'c' ?

(A) DNA synthesis, cell having duplicated chromosome, isolation of chromosome

(B) Duplication of chromosome, cell division, isolation of chromosome

(C) DNA Synthesis, RNA Synthesis, Mitosis

(D) S stage, cell division, isolation of chromosome



(83) Identify the given figure ?



(A) Stages of cell division

(B) Stages of cell cycle

(C) Stages of interphases

(D) Stages of meiosis

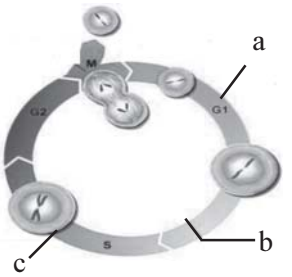
(84) What indicates a, b and c ?

(A) Cycle begins, cell grows, cell devides whether to continue

(B) Resting phase, cell devides whether to continue, cell prepares to devide

(C) Cell grows, cell devides whether to continue, DNA replication

(D) Cycle begins, cell grows, DNA replication



(85) Which stage is shown by given figure ?

(A) Interphase

(B) Prophase

(C) Late Prophase

(D) Early Prophase



(86) Which stage is shown by given figure ?

(A) Metaphase

(B) Interphase

(C) Late prophase

(D) Early prophase



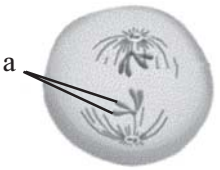
(87) What indicates 'a' in figure ?

(A) Kinetochores

(B) Centrosome

(C) Sister chromatids

(D) Spindle poles



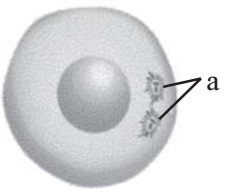
(88) What indicates 'a' in figure ?

(A) Centrosome

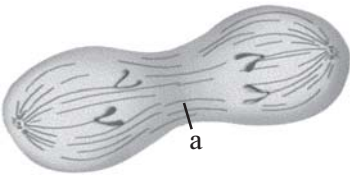
(B) Kinetochores

(C) Sister chromatids

(D) Spindle poles



(89) What indicates 'a' in figure ?



(A) transeverse furrow

(B) longitudinal furrow

(C) cleavage furrow

(D) divisional furrow

(90) Which stage is shown by given figure ?



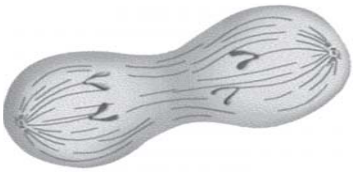
(A) Interphase

(B) Metaphase

(C) Late prophase

(D) Early prophase

(91) Which stage is shown by given figure ?



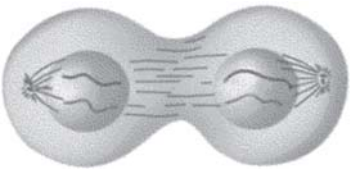
(A) G_1

(B) G_2

(C) Anaphase

(D) Telophase

(92) Which stage is shown by given figure ?



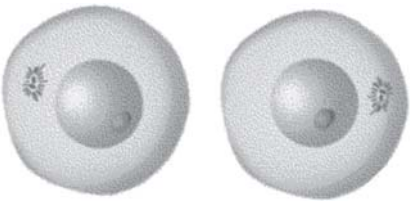
(A) G_2

(B) G_1

(C) Metaphase

(D) Telophase

(93) Which stage is shown by given figure ?



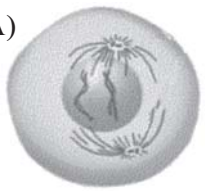
(A) G_1

(B) G_2

(C) Early prophase

(D) Telophase

(94) Which of the following figure indicates late phase of prophase ?

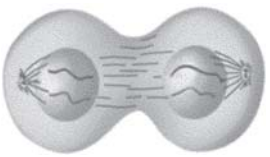
(A) 

(B) 

(C) 

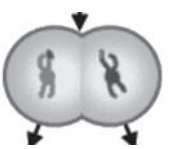
(D) 

(95) Which of the following figure indicates telophase ?

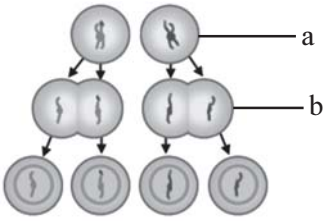
(A) 

(B) 

(C) 

(D) 

(96) What indicates 'a' and 'b' ?



(A) Meiosis-II, Cell division-II

(B) Meiosis-I, Cell division-I

(C) Meiosis, DNA replication

(D) Meiosis-II, Recombination of DNA

(97) Which substage of prophase-I is shown in given diagram ?



(A) Zygotene (B) Pachytene

(C) Diplotene (D) Leptotene

(98) Choose the diagram which correctly shows the pachytene substage ?

(A)  (B)  (C)  (D) 

(99) Choose the diagram which correctly shows the zygotene substage ?

(A)  (B)  (C)  (D) 

(100) Which substage is indicated in given diagram ?



(A) dikinesis (B) diplotene

(C) leptotene (D) pachytene

Answers : (81-D), (82-A), (83-B), (84-C), (85-A), (86-D), (87-C), (88-A), (89-C), (90-B), (91-C), (92-D), (93-A), (94-B), (95-A), (96-A), (97-D), (98-C), (99-A), (100-D)

- **Questions for NEET :**

(101) Meiosis-II Shows...

(A) Isolation of sex chromosomes (B) Synthesis of DNA and centromere

(C) Isolation of homologous chromosome (D) Separation of chromatids

(Hint: To maintain the haploid number chromatids are separated.)

(102) Which is the best stage for observing number of chromosomes, size and shape ?

(A) Interphase (B) Metaphase (C) Prophase (D) Anaphase

(103) Meiosis is important for evolution, what is true for its result ?

(A) The genotype of the daughter cells is identical. (B) Mother cells results in four daughter cells.

(C) Ovules and sperms are produced. (D) Recombination occurs.

(Hint : During meiosis crossing over that is exchange of genes occurs)

(104) At the end of the first meiosis gene changes between chromatids of homologous chromosomes. What is it called ?

(A) Transformation (B) Chiasmata (C) Crossing over (D) Synapsis

(105) In which stage DNA replication occurs in cytoplasm ?

(A) G_1 (B) G_2 (C) Metaphase (D) S Stage

(106) During which stage lempbrush chromosomes are formed ?

(A) Prophase (B) Diplotene (C) Metaphase-I (D) Interphase

(Hint : Lemp brush chromosomes are long dimicrotic chromosomes formed during diplotene stage, which are joined by chiasmata.)

(107) What happens when colchicine is added in haploid cell during diploid condition ?

(A) Inhibits mitosis (B) Inhibits formation of mitotic spindles

(C) DNA duplicates (D) Inhibits formation of centromere

(Hint : Colchicine stops the spindles)

(108) Which of the following organelle is absent in mitotic cells of higher plants ?

(A) Cell plate (B) Centrosome (C) Centromere (D) Spindle fibre

(Hint : Centromere is absent in plant cell.)

(109) Meiosis-I is heterotypic division and Meiosis-II is homotypic division why ?

(A) Pairing of homologous chromosomes (B) Crossing over occurs

(C) Separation of chromatids (D) It is the process of separation of homologous chromosomes

(Hint : Meiosis-II is equational division in which separation of chromatids of cromosome.)

(110) In which stage mendlian factor (Aa) is segregrated ?

(A) Diplotene (B) Anaphase-I

(C) Zygotene (D) Anaphase-II

(Hint : Mendelian segregration means separation of homologous chromosomes and only one chromosome from homologous chromosome enters in zygote.)

(111) How mother cells differ from daughter cells ?

(A) Segregration, independent, dominance, crossing over (B) Segregration, crossing over

(C) Independent, dominance, crossing over (D) Segregration and independent dominance

(Hint : Due to independent assortment and crossing over in meiosis diversity occurs)

(112) What is the number of chromatids in mitosis ?

(A) two in both meiosis and mitosis (B) two in mitosis and one in meiosis

(C) two in mitosis and four in meiosis (D) one in mitosis and two in meiosis

(113) How anaphase differs from methaphase in mitosis ?

(A) Same numbers of chromosomes and chromatids.

(B) Half the numbers of chromosomes and chromatids.

(C) Half the numbers of chromosomes and same the numbers of chromatids.

(D) Same number of chromosomes and half the numbers of chromatids.

(Hint: In anamephase of mitosis chromatids of chromosome are separated to each other and move at opposite polar region.)

(114) How many mitotic divisions must occur in a cell to form 128 cells ?

(A) 7 (B) 14 (C) 28 (D) 64

(Hint : During mitosis from one mother cell results in two daughter cells)

(115) During cell division nucleus of apical meristem tissue are seen ?

(A) Metaphase (B) Anaphase (C) Telophase (D) Cytokinesis

(Hint : During the telophase organelles are formed.)

(116) Role of microtubules...

(A) Cell division (B) Muscle contraction (C) Membrane structure (D) To decide DNA

(Hint : Spindle fibres responsible for cell division are made up of microtubules)

(117) If there are 10^5 cells / ml in a medium in which 175 cells grow in every 35 minutes, How many cells will be there after 175 minutes ?

(A) 5×10^5 cells (B) 35×10^5 cells (C) 32×10^5 cells (D) 175×10^5 cells

(Hint : 175 / 35 minutes = number of cells which divides = $(2)^5 \times 10^5 = 32 \times 10^5$)

(118) Give the name of the structure serve as the sites of attachment of spindle fibers during cell division.

(A) Chromocentre (B) Kinetochore (C) Centriole (D) Chromomere

(Hint : Kinetochore is protein coated layers which is connected with spindle fibre.)

(119) Which is the best material to study mitosis in laboratory ?

(A) Anther (B) Root apex (C) Leaf apex (D) Ovary

(120) What will happen if diploid cell is treated with colcichine ?

(A) Triploid (B) Tetraploid (C) Diploid (D) Haploid

(Hint : Colcichine stops formation of spindles when colcichine is given to diploid cell it duplicates.)

(121) In somatic cell cycle...

(A) DNA duplicates than that of mother cell in G_1 phase.

(B) DNA synthesis occurs in S stage.

(C) Interphase shortened.

(D) Interphase duplicates.

(122) If you have given a root apex of onion and it is said to count chromosomes. Which is the best stage to observe them ?

(A) Metaphase (B) Telophase (C) Anaphase (D) Prophase

(123) At which stage synthesis of histone protein occurs in eukaryotic cell ?

(A) G_2 phase (B) S phase (C) Prophase (D) Telophase

(Hint : During S stage DNA synthesis occurs it is also known as histone protein synthesis.)

(124) Which of the following is related with the formation of nuclear membrane during M-Phase of cell cycle ?

(A) Thickness of chromosomes decreases and formation of nuclear membrane.

(B) Transcription of chromosomes and formation of nuclear membrane.

(C) Formation of contractile ring and nuclear membrane.

(D) Formation of contractile ring and transcription of chromosomes.

(Hint : In telophase nuclear membrane formation occurs through transcription of chromosomes)

(125) What is true for cell cycle ?



(A) C - Nuclear division (B) D - Synthesize stage

(C) A - Cytoplasm division (D) B - Metaphase

(126) Space is present between...

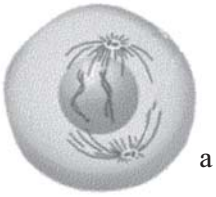
(A) m - RNA and ribosomes (B) Two homologous chromosomes

(C) Spindle fibre and centromere (D) Male and Female gametes

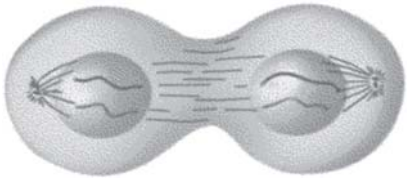
(127) At which stage of mitosis nucleolus and endoplasmic recticulum starts to disappear ?

(A) Late prophase (B) Early metaphase (C) Late metaphase (D) Early prophase

(128) What is indicated by 'a' and 'b' in given figure of cell division ?



a



b

(A) metaphase, telophase (B) telophase, metaphase

(C) Early prophase, Telophase (D) prophase, anaphase

(129) Choose the correct option for mitosis.

(A) In anaphase, chromatids are independent or arranged in the centre of the cell.

(B) In telophase chromatids move towards their respective pole.

(C) At the end of prophase golgibody and endoplasmic reticulum appears.

(D) In metaphase spindle fibres are removed from chromosomes by imaging line.

(Hint : During mitosis chromosomes moves towards equatorial region and during metaphase arranges on equatorial plate.)

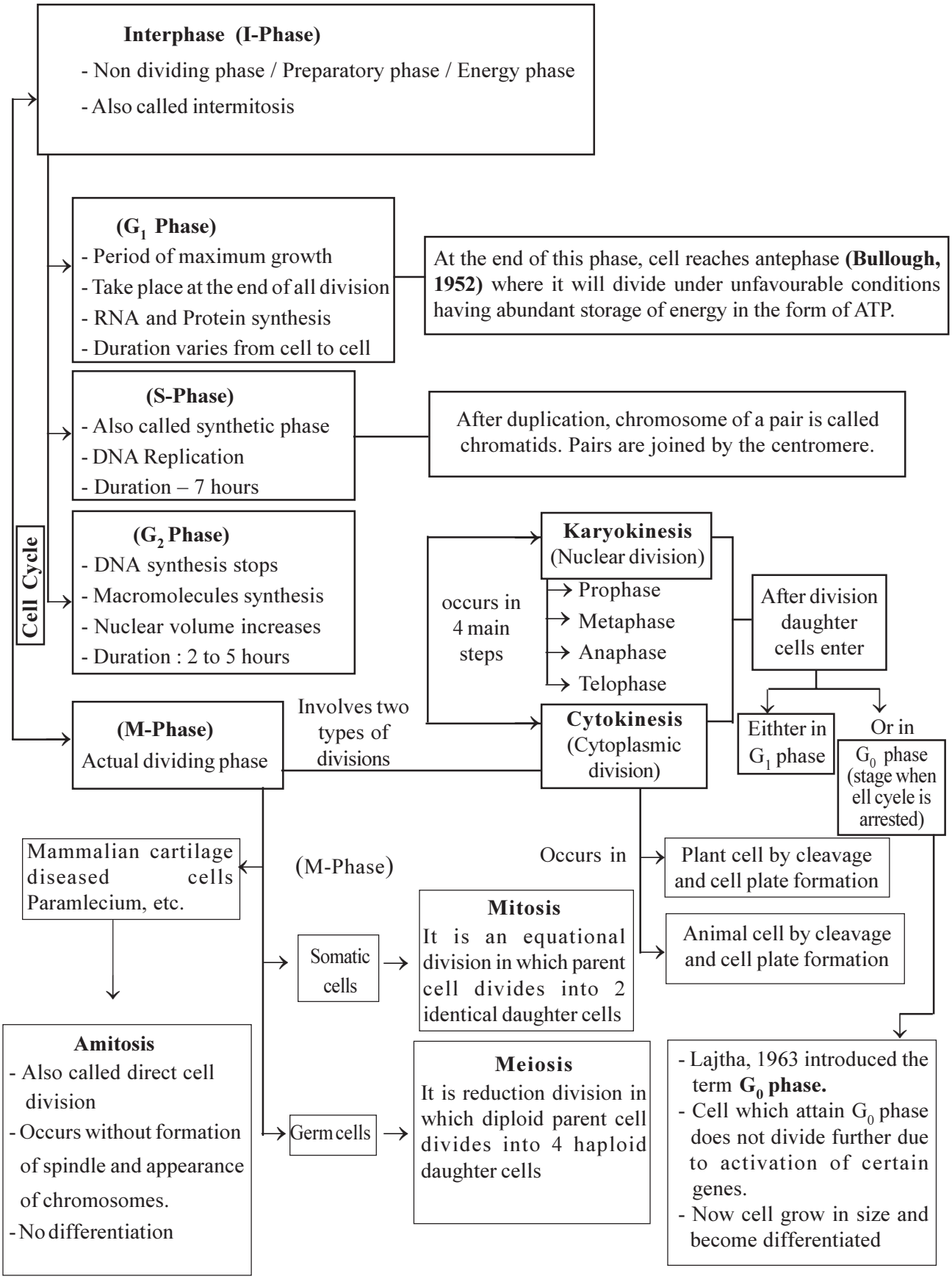
(130) During which stage of meiosis homologous pair of chromosomes separates and chromatids are attached with centromere ?

(A) Metaphase-I (B) Metaphase-II

(C) Anaphase-I (D) Anaphase-II

(Hint : In anaphase-I along with condensation of bipolar spindle chromosomes of homologous pair move towards their respective poles and chromatids are attached with centromere.

Answers : (101-D), (102-B), (103-D), (104-C), (105-D), (106-B), (107-B), (108-B), (109-C), (110-B), (111-A), (112-A), (113-D), (114-A), (115-C), (116-A), (117-C), (118-B), (119-B), (120-B), (121-B), (122-A), (123-B), (124-B), (125-B), (126-C), (127-D), (128-C), (129-D), (130-C)


Interphase (I-Phase)
- Non dividing phase / Preparatory phase / Energy phase
- Also called intermitosis
Cell Cycle
(G1 Phase)
- Period of maximum growth
- Take place at the end of all division
- RNA and Protein synthesis
- Duration varies from cell to cell
At the end of this phase, cell reaches antephase (Bullough, 1952) where it will divide under unfavourable conditions having abundant storage of energy in the form of ATP.
(S-Phase)
- Also called synthetic phase
- DNA Replication
- Duration – 7 hours
After duplication, chromosome of a pair is called chromatids. Pairs are joined by the centromere.
(G2 Phase)
- DNA synthesis stops
- Macromolecules synthesis
- Nuclear volume increases
- Duration : 2 to 5 hours
(M-Phase)
Actual dividing phase
Involves two types of divisions
occurs in 4 main steps
Karyokinesis (Nuclear division)
Prophase
Metaphase
Anaphase
Telophase
Cytokinesis (Cytoplasmic division)
After division daughter cells enter
Eithter in G1 phase
Or in G0 phase (stage when ell cyele is arrested)
(M-Phase)
Mammalian cartilage diseased cells Paramlecium, etc.
Somatic cells
Germ cells
Mitosis
It is an equational division in which parent cell divides into 2 identical daughter cells
Meiosis
It is reduction division in which diploid parent cell divides into 4 haploid daughter cells
Amitosis
- Also called direct cell division
- Occurs without formation of spindle and appearance of chromosomes.
- No differentiation
Occurs in
Plant cell by cleavage and cell plate formation
Animal cell by cleavage and cell plate formation
- Lajtha, 1963 introduced the term G0 phase.
- Cell which attain G0 phase does not divide further due to activation of certain genes.
- Now cell grow in size and become differentiated

9 Animal Husbandry and Plant Breeding

Animals and plants are used for food since evolution of man takes place same it primitive. Thousands of years ago, agriculture began. Today in the animal husbandry; dairy farming, poultry, apiculture, fisheries and in plant breeding. Various methods of breeding and tissue culture are applied for getting more food to fulfill the need of increasing human population.

- **Animal husbandry :**

Animal husbandry is an important from very beginning in development of human beings. It is important for food production, The dairy industry, covers the production, processing and distribution of milk. In Gujarat this industry is well developed.

- Apiculture, is the rearing of honey bee commonly in hives. The chief products of bee keeping industry are (i) Honey and (ii) Bee's wax.
- Fishery deals with catching, processing or selling of fish and other aquatic living organisms. Directly or indirectly the livelihood of over 500 million people in developing countries depend on fisheries.
- In 20th century poultry keeping has become an important small scale industry due to modern need for palatable and nutritive food which it provides in the form of eggs as well as adult animals.

(1) What is apiary ?

(A) honey bee rearing (B) a location where bees are kept

(C) a location for breeding bees (D) Hybridization in honey bees

(2) IARI means,

(A) International Agriculture Research Institute (B) Indian Agriculture Research Institute

(C) Imperial Agriculture Research Institute (D) Indian aeronautical Research Institute

(3) Which of the following group is right for fresh water fishes ?

(A) Catla, Rohu, Mackerel (B) Major carp, Hilsa, Sardines

(C) Mrigal, Mackerel, Promfrets (D) Rohu, Mrigal, Catla

(4) Cattle, Keepers play central role in dairy faming because ..... .

(A) They prepare house hold milk products

(B) They have brought white revolution in dairy farming

(C) Their dairy products are sold at national as well as inernational level

(D) They carries good varieties of cattle.

(5) Honey is a viscous, sugary fluid...

(A) It is thick liquid formed from the nectar within the stomach.

(B) It is thick liquid formed from the nematocytes of honeybee.

(C) It is thick liquid formed from Salivary glands of honeybee.

(D) It is secreted by the abdominal gland of bees.

(6) Numbers of worker bees found in the colony are ..... .

(A) 40,000 to 50,000 (B) 30,000 to 50,000

(C) 40,000 to 60,000 (D) 30,000 to 60,000

(7) Which of the following group is right for edible marine fishes ?

(A) Sardines, Pomfrets, Mackerel (B) Catla, Rohu, Mrigal

(C) Hilsa, Pomfrets, Catla (D) Sardines, Mackerel, Mrigal

(8) What is the full form of IVRI ?

(A) Indian Veterinary Research Institute (B) Imperial Viral Research Institute

(C) Imperial Veterinary Research Institute (D) International Veterinary Research Institute

(9) Which of the following sentence is true for the researches carried out by the IVRI ?

(A) There are several Government poultry farms in India.

(B) They have demonstrated the high biological value of eggs and recommended the consumption of eggs.

(C) Recommended the consumption of meat for proper nutrition.

(D) Little attention has been paid for the proper nutrition.

(10) Who was a pioneer of dairy industry in India ?

(A) Whenson (B) Huber (C) Dr. Varghese Kurien (D) Fahiyan

Answers : (1-B), (2-B), (3-D), (4-D), (5-A), (6-B), (7-A), (8-C), (9-B), (10-C)

- In animal breeding the improved varieties can be more useful to man, various methods are there like, inbreeding, outbreeding, interspecific breeding.
- Through plant breeding methods improved variety can be obtained. The major goals are as per animal breeding. By the various methods species are obtained with genetic variation.
- One of the sources of proteins for animals and human nutrition is single cell protein (SCP) even many people uses mashroom as a food.
- Plant tissue culture means growing and maintaining the cell, tissue and organ in culture medium. In tissue culture (1) Callus culture even suspension culture and (2) Embryo culture are included.

(11) Which type of hybridization gave mule ?

(A) Outbreeding hybridization (B) Inbreeding hybridization

(C) Interspecific hybridization (D) Intraspecific hybridization

(12) How much amount of protein is produced by *Methylophilus methylotrophus* bacteria per day ?

(A) 230 gms (B) 250 gms (C) 200 gms (D) 280 gms

(13) Which of the following statement is true for single cell protein ?

(A) It is the source of proteins for animals and plant nutrition.

(B) It's utilization increases environmental pollution.

(C) It is produced by fermentation.

(D) Fungi cannot be used for single cell protein production.

(14) Select the correct sequence for plant breeding.

(A) Collection of variability → Selection of superior recombinants → Cross hybridization among the selected parents

(B) Collection of variability → Evaluation of selection of parents → Cross hybridization among the selected parents → Selection of testing of superior recombinants → Testing, release and commercialization of new cultivation

(C) Selection and testing of superior recombinants → Collection of variability → Evaluation and selection of parents → Testing, release and commercialization of new cultivation

(D) Selection of parents → Cross hybridization among the selected parents → Collection of variability → Selection of superior recombinants

(15) What is application of the embryo culture method ?

(A) Replication of plantlets (B) Protoplast isolation

(C) Development of plantlets from dormant seeds (D) An increase in the biomass of cells and tissues

(16) Which of the following is not proper for callus and suspension culture ?

(A) Regeneration of saplings (B) Formation of cell biomass

(C) Hybrid plantlet of interspecific plants (D) Protoplast isolation

(17) Which of the following sequence correct for events following during callus culture ?

(A) explant → cell division → callus → cytokinin are added to the medium → cells converted to meristematic tissue

(B) explant → callus → cell division → cytokinin are added to the medium → cells converted to meristematic tissue

(C) callus → cytokinin are added to the medium → cell division → cells converted to meristematic tissue → explant

(D) cell division → cytokinin are added to the medium → callus → explant → cells converted to meristematic tissue

(18) What can be developed in plant cells, tissue or organ by tissue culture ?

(A) totipotency (B) stored food (C) interspecific hybridization (D) seed dormancy

(19) In maize hybrids have twice the amount of ..... amino acid.

(A) tyrosine (B) glycine (C) lysine (D) serine

(20) ..... is the medium for maintaing callus.

(A) Agar Agar gel (B) Agarose gel (C) IBA (D) 2-4 - D

Answers : (11-C), (12-C), (13-C), (14-B), (15-C), (16-C), (17-A), (18-A), (19-C), (20-A)

(21) Select the proper pair :

Column - I	**Column - II**
(P) Milk	(i) Palatable and nutritive food
(Q) Honey	(ii) Callus culture
(R) Agar-agar	(iii) Pasteurization
(S) Rotary shaker	(iv) Medicinal value
(T) Eggs	(v) Suspension culture

(A) P - i, Q - iii, R - ii, S - iv, T - v (B) P - iii, Q - iv, R - v, S - ii, T - i

(C) P - iii, Q - iv, R - ii, S - v, T - i (D) P - i, Q - ii, R - iii, S - iv, T - v

(22) Select the proper pair :

Column - I	**Column - II**
(P) *Santa gertrudis*	(i) Rapid clone distribution
(Q) Orchid	(ii) Interspecific hybridization
(R) *Hybrid maize*	(iii) Reduces environmental pollution
(S) Mule	(iv) Outbreeding
(T) Methylophilus	(v) Biofortification

(A) P - iii, Q - iv, R - i, S - iii, T - v (B) P - iv, Q - i, R - v, S - ii, T - iii

(C) P - v, Q - i, R - ii, S - iii, T - iv (D) P - i, Q - ii, R - iii, S - iv, T - v

(23) Select the proper pair :

Column - I	**Column - II**
(P) Single cell protein	(i) Cytokinin
(Q) Callus calture	(ii) Huber
(R) Hybrid maize	(iii) Twice the around of the lysine and tryptophan
(S) The father of modern bee - science	(iv) Fermentation

(A) P - i, Q - ii, R - iii, S - iv (B) P - ii, Q - i, R - iii, S - iv

(C) P - iv, Q - i, R - iii, S - ii (D) P - iii, Q - iv, R - ii, S - i

(24) Select the proper pair :

Column - I	Column - II
(P) Palanpur	(i) Dudh sagar dairy
(Q) Mehsana	(ii) IARI
(R) Anand	(iii) Banas dairy
(S) New Delhi	(iv) IVRI
(T) Izatnagar	(v) Amul dairy

(A) P - i, Q - v, R - ii, S - iii, T - iv (B) P - ii, Q - i, R - v, S - iv, T - iii

(C) P - ii, Q - v, R - i, S - iv, T - iii (D) P - iii, Q - i, R - v, S - ii, T - iv

(25) Select the proper pair :

Column - I	Column - II
(P) Worker	(i) only breeding female in colony
(Q) Queen	(ii) work as breeding only
(R) Drone	(iii) directly used in inbreeding
(S) *Santa gertrudis*	(iv) unfertile female bees

(A) P - iv, Q - i, R - ii, S - ii (B) P - i, Q - iii, R - ii, S - iv

(C) P - iii, Q - ii, R - iv, S - i (D) P - iii, Q - iv, R - i, S - ii

Answers : (21-C), (22-B), (23-C), (24-D), (25-A)

- **A - Assertion, R - Reason questions :**

Choose the correct option of the following :

(A) A and R both are correct. R is correct explanation of A.

(B) A and R both are correct. R is not correct explanation of A.

(C) A - Correct, R - False

(D) A - False, R - Correct

(26) Assertion A : Single cell protein is produced by fermentation.

Reason R : Genetic engineering is used for production of single cell protein.

(A) (B) (C) (D)

(27) Assertion A : As compared to the callus culture growth is fast in suspension culture.

Reason R : The culture constantly rotates over rotary shaker at 100 to 250 rpm speed in the culture media. So cells easily get aeration and the contents of the medium are throughly mixed.

(A) (B) (C) (D)

(28) Assertion A : Today Mushroom is used as a food.

Reason R : As a food source, heterotrophs mushrooms are being cultivated world.

(A) (B) (C) (D)

(29) Assertion A : Honey is popularly used medicine.

Reason R : Honey is popularly used as cosmetics.

(A) (B) (C) (D)

(30) Assertion A : Continue inbreeding is not desirable.

Reason R : Due to continued inbreeding possibility is increased to collect harmful recessive genes. Therefore, fertility of offsprings gradually increases.

(A) (B) (C) (D)

Answers : (26-B), (27-A), (28-A), (29-C), (30-A)

- **True - False (T - F) types questions :**

Choose the correct option :

(31) (1) In plant breeding nutrient contents in the medium increases.

(2) In this method the biomass of cells and tissues increases.

(3) Decrease in volume of the medium due to evaporation.

(4) In this method, culture is transferred to new glass appratus containing a fresh medium.

(A) T, T, T, F (B) T, T, F, F (C) F, T, T, T (D) F, T, T, F

(32) (1) Milk is important product used as valuable food stuffs.

(2) Milk is the fresh lacteal secretion of animals.

(3) Milk exploited as an commodity of food by human being.

(4) Milk is used for a variety of preparations like curd, butter, cheese, sweet, etc.

(A) F, F, F, T (B) T, T, T, T (C) F, F, F, F (D) F, T, F, T

(33) (1) In embyo culture, a small embryo from a developing seed is kept inside the seed and then cultured on media.

(2) Rapid clone distribution is application of callus culture.

(3) Some seeds stay in dormant condition for longer time. Through embryo culture method plantlets can be developed from such seeds.

(4) Orchid contain stored food.

(A) T, F, T, T (B) T, T, F, F (C) F, T, T, T (D) F, F, T, F

(34) (1) Bee's wax is secreted by the abdominal gland of bees.

(2) Queen is *infertile* female in a colony.

(3) Worker is the only breeding female in the colony.

(4) Drone found in many numbers and work as breeding male drones.

(A) T, F, F, T (B) T, T, F, F (C) F, T, T, F (D) F, F, T, T

(35) (1) Higher yield of milk is the goal of animal breeding.

(2) In the outbreeding approach, animals of two different species are interbred.

(3) *Santa gertrudis* is example of outbreeding approach.

(4) Due to continued inbreeding fertility of offspring gradually increases.

(A) T, F, F, F (B) F, F, T, T (C) T, F, T, F (D) T, F, F, T

Answers : (31-C), (32-B), (33-D), (34-A), (35-C)

- **Questions for NEET :**

(36) Which is the chief method associated with agricultural biotechnology ?

(A) Tissue culture (B) Transformation

(C) Plant breeding (D) Recombinant DNA

(37) Method of breeding crops for higher levels of vitamins and minerals or higher protein.

(A) Biofortification (B) Biomagnification (C) Micro Propagation (D) Somatic hybridization

(38) Which type of breeding product Mule is ?

(A) Hybridization (B) Mutation (C) Interspecific hybridization (D) outbreeding

(39) What is added to the medium for growth of roots in callus culture ?

(A) Auxin (B) Cytokinin (C) Gibberrelins (D) A and B both

(40) Which is generally used auxin in callus and suspension culture ?

(A) Nephthalene Acetic acid (B) Indole butyric acid

(C) Dichlorophenoxyacetic acid (D) ABA

(41) A superior male of any one species is mate with a superior female of other species.

(A) Inbreeding (B) Outbreeding

(C) Test cross (D) Interspecific hybridization

(42) Uses of honey bee.

(A) Honey production (B) Usefulness of larva

(C) Wax production (D) Both A and C

(43) The capacity of producing new organism from one cell.

(A) totipotency (B) pluriopotency (C) heterozygocity (D) cyriendipity

(44) Totipotency of cell means ..... .

(A) Regeneration capacity (B) Producing the whole plant from one cell

(C) Root inducing factor (D) Somatic featus generation

(45) In which method of hybridization, animals of two different species are interbred ?

(A) Inbreeding (B) Interspecific

(C) Inter generic hybridization (D) Outbreeding

(46) Which of the following statement is true for single cell protein ?

(A) Microbes used for single cell protein production include algae, bacteria, yeasts and filamentous fungi.

(B) It does not contain fat and carbohydrates.

(C) It is produced by carboxylation.

(D) None of the above.

(47) Chief product of poultry is ..... .

(A) Eggs (B) Hen (C) Meat (D) Eggs and Meat

(48) Which is the main application of the embryo culture method ?

(A) Production of alkaloids (B) Plantlets can be developed from dormant seeds

(C) Inducing somatic hybridization (D) Protoplast isolation

(49) In which type of bee wax gland is present ?

(A) Worker (B) Queen (C) Drone (D) A and B both

(50) In which part of the body honey bee stores the nectar of flower ?

(A) Stomach (B) Mouth (C) Gizzard (D) Intestine

Answers : (36-A), (37-A), (38-C), (39-D), (40-C), (41-B), (42-D), (43-A), (44-B), (45-B), (46-A), (47-A), (48-B), (49-A), (50-A)

Animal Husbandry

- It is the science of rearing, feeding, caring, breeding and utilization of domesticated animals.
- The ever-increasing demand of food from animals and animal products both in terms of quantity and quality has been met by good animal husbandry practices.
- These practices include (1) management of farm and farm animals and (2) animal breeding.
- In view of the high nutritive value of honey and its medicinal importance, there has been remarkable growth in the practice of bee-keeping or apiculture.
- Fishery is another flourishing industry meeting the ever-increasing demand for fish, fish products and other aquatic foods.

Animal breeding

Natural breeding

Artificial breeding

Inbreeding

- It is mating of more closely related individuals within the same breed for 4-6 generations.

Outbreeding

- It is breeding of the unrelated animals.

Outcrossing

- Mating between unrelated members of the same breed.

Cross breeding

- Mating of superior males of one breed with superior females of another breed.

Interspecific hybridisation

- Mating between members of two different species.

Management of farm animals

- It includes management of cattle, buffaloes, sheep, goat and poultry.

10 Human Health and Diseases
(Immunity Vaccination, Cancer, AIDS)

- **What is a disease ?**
  - Health can expression of physical, mental and social well being.
  - If human-being is healthy that the he/she becomes more inclined towards work and consequently, productivity is boosted leading to economic prosperity.
  - Good health lengthens an individual's life decreases the deaths of children and pregnant woman.
  - Balanced diet, personal hygiene and regular exercises are the most essential for preserving good health.
  - Since years, meditations is done to attain physical and mental health.
  - 21st June is known as Yoga day in the world.
  - For attaining good health, the awarness of the diseases and their influence on body's different functions, vaccination against infections diseases, proper disposal of the wastage, the retraint of organisms and forming of healthy and hygiene food and the arrangement of water resources in method.
  - When the functions of different parts of the body or systems are affected the symptoms of different diseases are seen in the body Hence we will feel uneasy.
- **What is a disease ?**
  - Any physical or actional change form normal condition which produces unhealthiness or weakness or pains the health of living being is called a disease.
  - Alternatively, the bad performance of body or body's part with some particular (Specific) symptoms is called disease. (In French des = away and aise = ease)
  - According to Oxford English Dictionary a disease is such a condition of body or body's parts that obstracts the functioning of them or creats a chaos.
  - The diseases are classified into two different types :
    (1) Infectious diseases (2) Non-infectious diseases
    (1) **Infectious diseases :** Infectious diseases are easily transmitted from one person to another. These diseases are easily carried away by different types of organisms like Virus, Bacteria, Fungi, Protozoa, Helminthes, etc. Some infections diseases like AIDS, are fotal.
    (2) **Non-infectious diseases :** These diseases remain limited to the person in whom it has been developed. They are not transmitted to other persons. Because of the unrestraint growth of specific tissues in the body, non-infectious disease like cancer may occur causing death.

(1) What is expression of physical, mental and social well being by human ?
(A) Health (B) Immunity (C) Diseased condition (D) None of the above

(2) If human-being is healthy, then he/she becomes efficient for ..... .
(A) productivity is boosted leading to economic prosperity.
(B) lengthens individual's life.
(C) decreases death of children and women.
(D) All of the above

(3) What is essential for preserving good health ?
(A) Balanced diet (B) Personal hygiene (C) Regular exercise (D) All of the above

(4) What is necessary for good health ?

(A) Awareness on the diseases and the influence of it on body's different functions.

(B) Vaccination against infectious diseases, proper disposal of wastage, restraint of organisms who carries the pathogens.

(C) Make arrangements for healthy and hygienic food and arrangement of water resources.

(D) All of the above

(5) What is disease ?

(A) Any physical or actional change in normal condition which creates unhealthiness and weakness.

(B) Ruins the health of living being.

(C) Any physical or actional change in normal conditions which create health and energy.

(D) (A) and (B)

(6) If French des = __x__ and aise = __y__ .

(A) x = Ease, y = Away (B) x = Away, y = Ease

(C) x = Near, y = Ease (D) x = Near, y = Away

(7) Which pathogens are responsible for infections diseases ?

(A) Virus, Bacteria, Fungi, Nematodes, Arthropods (B) Virus, Bacteria, Fungi, Protozoa, Coelenterata

(C) Virus, Bacteria, Fungi, Protozoa, Mollusca (D) Virus, Bacteria, Nematode, Protozoa, Echinoderms

Answers : (1-A), (2-D), (3-D), (4-D), (5-D), (6-B), (7-A)

Important Common Diseases

- **Typhoid :**
  - Typhoid is a common disease caused by rod like bacteria *Salmonella typhi.*
  - These bacteria in diseased condition are found in the intestine of human beings. This disease is very common among children belonging to age group 1-15.
  - Every year approximately 2.5 million people suffer from this disease.
  - **Spreading :** This disease is transmitted by the water and food contaminated by patient's feaces.
  - House flies carry these organisms from feaces to food, milk and water.
  - These organisms enter into the body through mouth and reach intestine from where they are carried into other parts of the body through blood circulation.
  - It produces lesion in the wall of intestine.
  - The period of incubation is of one to three weeks, averaging two weeks.
  - **Symptoms :** Common symptoms are fever initially for first and second week that subsides in the third and fourth week.
  - Headache, severe weakness, pain in stomach, constipation and loss of appetite are seen.
  - The liver and spleen get enlarged. Typhoid is diagnosed by widal test.
  - **Control :** Antibiotic Treatment

- **Pneumonia :**
  - Pneumonia is caused in human beings by *Streptococcus pneumoniae* and *Haemophilus influenzae*.
  - Generally streptococcus pneumonia is called *Pneumococcus*. Pneumonia is a disease of respiratory tract.
  - The fluid gets collected in alveoli because of which the lungs do not get enough oxygen.
  - **Spreading :** This disease is spreading by coughing and sputum.
  - Inhaling the pneumococci, it infects the windpipes.
    Burning is felt in alveolal wall which secrets the protein filled fluid. After that it is used as contagious medium by bacteria and obstructs the windpipes.
  - Incubation period is of one to three days. Pneumonia is generally found in aged people.
  - **Symptoms :** The onset of pneumonia is usually sudden chill, followed by fever.
  - Pain in inhalation, cough and headache are seen.
  - In some cases the lips and finger nails turn blue from grey in colour.
  - In pneumonia, sometimes, because of malnutrition, alcohol and side effects of medicines or infections from some other disease like influenza, the resistance power of the body is lost.
  - The sputums are bloody and rusty.
- **Common Cold :**
  - This disease in an important disease among human infections diseases.
  - Which is caused by *Rhino virus*. This virus infect the nose and respiratory passage but not lungs.
  - **Spreading :** The sneezing and coughing of an infectious person or by using, pen, books, cup computer, key-board or mouse used by the infections being, the healthy person may get infected.
  - **Symptoms :** The common symptoms of common cold include nasal congestion and secretion, sore throat, cough, headache, fatigue, etc. It lasts for 3-7 days at least.
- **Malaria :**
  - Microorganism called Plasmodium is responsible for malaria disease.
  - *Plasmodium vivax* and *plasmodium falciparum* are responsible for various kinds of malaria.
  - For severe kind of malaria *plasmodium falciparum* is responsible and some times it proves to be fatal.
  - Its life cycle found in human being in three stages.

  (1) **Pre-erythrocytic cycle :** The Anopheles female mosquito bites, sends small amount of saliva into human being $\rightarrow$ Sporozoites present in saliva are introduced to human blood $\rightarrow$ Sporozoite are spindle shaped, enters in the liver cell $\rightarrow$ Growing over the food from the liver cells it becomes spherical in shape it called cryptosizont $\rightarrow$ Asexual reproduction is found $\rightarrow$ It called schizogony $\rightarrow$ Cryptoschizont is converted in cryptomerozoites.

  (2) **Exo-erythrocytic cycle :** Cryptomerozoites enter a new liver cell and grow to achieve spherical shape $\rightarrow$ It called metacryptoschizont $\rightarrow$ By asexual reproduction it is converted into metacryptomerozoite, breaks the liver cells and comes out enter in blood.

  (3) **Endoerythrocytic cycle :** This cycle is found in RBC. Metacryptomerozoite enter in RBC, becomes spherical in shape. Which is known as trophozoite $\rightarrow$ Trophozoite develops many pseudopodia, this stage is known as amoeboid stage $\rightarrow$ now parasites become rounded in

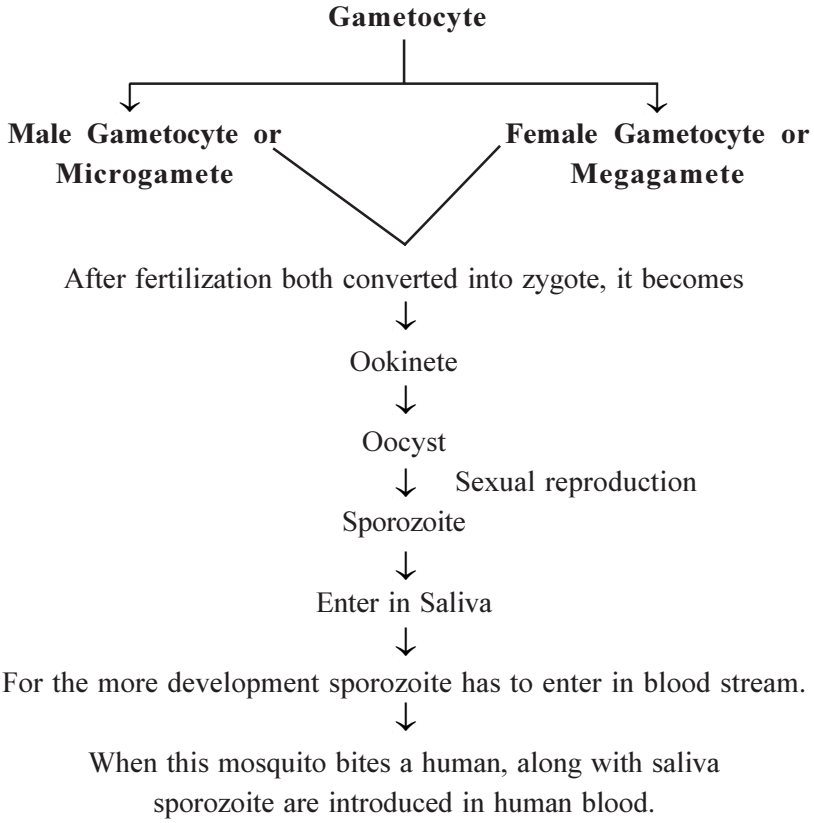
shape, it is called schizont → By asexual reproduction this form is converted into merozoites → it converts into gametocytes. They are two types :

(1) **Male gametocyte :** Small in size and it has large nucleus.

(2) **Female gametocyte :** Large in size and it has small nucleus.

Mosquito and its life cycle :

- When female Anopheles mosquito sucks gametocyte containing blood of human being, it enters in digestive system.



- The life cycle continues in human being.

Filariasis (Elephantiasis) :

- Elephantiasis is caused by the filarial worms.
- Adult male and female are of 40 mm and 80 mm length respectively.
- Filarial worms are long string like, white bodied and pointed at both the ends.
- The female gives birth to the kidworms which are called microfilaria.
- Human being is its first host. This worm is viviparous.
- Then they are transferred to deep blood-venels. Which are sucked by *Culex fatigan* mosquito which is a mediator and carrier. Here within 10 days approximately they developed into larva and are migrated to mosquito's mouth organs.
- When infected mosquito bites the host human; the larva enter in the skin. They enter in new host through the hole in skin made up by the mosquito. Through blood they are transferred to lymphatic vessels and lymphatic glands. Here it gets matured in one year as an adult. An adult worm lives for 5 to 8 years.

- **Effect on the host (Disease agency) :** From filarial infection, there is fever in acute condition.
- In chronic condition, the worm blocks lymphatic vessels. Due to which hands, legs, leg sole, breast and scrotum, etc. get swollen.
- This happens due to thickening of skin and epithelial tissues. The leg gets deformed and swell and thus it is given the name elaphantiasis. This disease is not fatal.
- **Control :** Destroy the mosquito

(6) Ring Worm :

- Fungus is responsible for this disease like *Microsporum, Trichophyton* and *Epidermophyton*.
- **Symptoms :** In common symptoms, there is an appearance of dry scaly lesions on various part of the body like skin, nails and head.
- By contant itching the lesions get expanded. Heat and moisture support fungus to grow in skin folds, thigh corners and between toes.
- **Spreading :** Ring worm disease spreads due to use of towel, cloth and comes from infectious persons.
- For maintaining personal and public hygiene; prevention of the spreading of several infections diseases is very important. To maintain personal health, one should keep the body clean and consume clean water, food, vegetables and fruits.
- For maintaining public health, proper disposal of dirty water to clean the water of dams and barrels periodically and disinfecting them. This is a very important solution for preventing the disease spread by water and food.
- The preventions mentioned above for pneumonia and common cold, one should not come into contact with infected people. Malaria and Filariasis spread by mosquitoes and hence the places where mosquitos lay their eggs should be found and eliminated. The stagnant water pools and poodles surrounding your residence should be cleaned. The water of air-cooller should be changed regularly. Mosquito net should be used. Fishes like *Gambusia* which feed on mosquito larvae can be reared in water reservoirs. Germicides, should be sprayed in drainage canals, ditches.
- The diseases like Dengue and Chikungunia are spread by *Aedes* mosquito. Mosquitoes can be prevented from entering the house by fixing of wire-nets on doors and windows.
- To get protection from polio, diptheria, titanus and pneumonia, vaccination, should be used so that they can properly controlled.

---

(8) Which bacteria causes typhoid ?

(A) *Salmonella typhi* (B) *Pneumococcus*

(C) *Haemophilus influenzae* (D) *Streptococcus*

(9) In which organ, we find bacteria of typhoid ?

(A) Stomach (B) Oesophagus (C) Intestine (D) Respiratory tract

(10) _x_ is very common among children belonging to age group _y_ , Every year approximately _z_ million people suffer from.

(A) x = Pneumonia, y = 1-15, z = 2.5 million (B) x = Typhoid, y = 1-15, z = 2.5 billion

(C) x = Typhoid, y =1 -15, z = 2.5 million (D) x = Pneumonia, y = 1-20, z = 2.5 billion

(11) _x_ disease is transmitted by the water and food contaminated by patient's feaces.

(A) x = Typhoid (B) x = Cold (C) x = Pneumonia (D) x = Cholera

(12) The incubation period of Salmonella typhi is approximately _x_ weeks averaging _y_ weeks.

(A) x = 1 - 3, y = 3 (B) x = 1 - 3, y = 2 (C) x = 1 - 2, y = 2 (D) x = 1 - 4, y = 2

(13) Which of the following is the symptom of typhoid ?

(A) Headache, Stomachache, Constipation, burning sensation in intestine and colon.

(B) Headache, Intestinal pain, Constipation burning sensation in stomach and colon.

(C) Severe weakness, Pain in intestine, burning in kidney and large intestine.

(D) Headache, Severe weakness, Pain in intestine and burning in stomach.

(14) Pneumonia is disease of _x_ and the fluid gets collected in _y_ .

(A) x = Respiratory tract, y = Alveoli and Wind pipe

(B) x = Respiratory tract, y = Alveoli and Lungs

(C) x = Digestive tract, y = Alveoli and Wind pipe

(D) x = Digestive tract, y = Alveoli and Lungs

(15) What is responsible for Pneumonia ?

(A) Patient sputum (B) Patient food (C) Patient faeces (D) Patient urine

(16) What is the incubation period of *Pneumococcus* bacteria ?

(A) 1 - 2 weeks (B) 1 - 3 days (C) 1 - 4 days (D) 1- 3 weeks

(17) Common cold is caused by _x_ which infects _y_ .

(A) x = Rhino virus, y = Nose and Respiratory tract (B) x = Rhino virus, y = Nose and Lungs

(C) x = Retro virus, y = Respiratory tract and Lungs (D) x = Rhino virus, y = Respiratory tract and lungs

(18) In which organ infection does not occur by entry of Rhino virus ?

(A) Nose (B) Respiratory tract (C) Respiratory tube (D) Lungs

(19) _y_ disease occurs by _x_ in human being.

(A) x = Arthropod, y = Typhoid (B) x = Protozoa, y Malaria

(C) x = Protozoa, y = Pneumonia (D) x = Coelenterata, y = Malaria

(20) Which microorganism is responsible for different types of Malaria ?

(A) Plasmodium falciparum (B) Plasmodium vivax

(C) Plasmodium quantum (D) (A) and (B) both

(21) Which host shows lifecycle of Plasmodium ?

(A) Human and Female anopheles mosquito (B) Human being and Male anopheles mosquito

(C) Human being and anopheles mosquito (D) Human being and Female *Culex fatigon* mosquito

(22) Which is the appropriate pair ?

(A) Pneumonia - *Salmonella typhi* (B) Common cold - *Retro virus*

(C) Malaria - *Plasmodium vivax* and *Plasmodium* (D) Filariasis - *Female Anopheles mosquito*

(23) Choose the correct option for Exoerythrocytic cycle of Plasmodium in Human being.

(A) Cryptomerozoites $\rightarrow$ Enters into new liver cell $\rightarrow$ Metacryptoschizont $\rightarrow$ Metacryptomerozoite $\rightarrow$ Blood

(B) Cryptomerozoite $\rightarrow$ Cryptoschizont $\rightarrow$ Metacryptoschizont $\rightarrow$ Metacryptomerozoite $\rightarrow$ Liver

(C) Metacryptomerozoite $\rightarrow$ Blood $\rightarrow$ Saliva $\rightarrow$ Metacryptoschizont $\rightarrow$ Liver

(D) Cryptomerozoite $\rightarrow$ Saliva $\rightarrow$ Blood $\rightarrow$ Liver $\rightarrow$ Metacryptoschizont

(24) Choose the correct option for endoerythrocytic cycle of Plasmodium in Human ?

(A) Metacryptomerozoite → RBCs → Merozoites → Trophozoite → Schizont → Gametocyte

(B) Metacryptomerozoite → Schizont → Trophozoite → RBCs → Gametocyte → Merozoite

(C) Metacryptomerozoite → RBCs → Trophozoite → Schizont → Merozoite → Gametocyte

(D) RBCs → Metacryptomerozoite → Trophozoite → Schizont → Merozoite → Gametocyte

(25) How is Filarial worm which causes Elephantiasis ?

(A) Short thread like, white bodied and sharp at both the ends.

(B) Long thread like, white bodied and sharp at both the ends.

(C) Long thread like, white bodied and sharp at one end.

(D) Long thread like, pale yellow bodied and sharp at both the ends.

(26) Male filarial worm is _x_ mm and female is _y_ mm in length.

(A) x = 40, y = 60 (B) x = 60, y = 40 (C) x = 40, y = 80 (D) x = 80, y = 40

(27) Zygote of malarial parasite becomes spindle shaped, it is known as ..... .

(A) Ookinete (B) Oocyst (C) Trophozoite (D) Sporozoite

(28) Choose the correct sequence following growth :

(A) Cryptomerozoite → Sporozoite → Cryptoschizont

(B) Cryptoschizont → Cryptomerozoite → Sporozoite

(C) Schizogony → Sporozoite → Cryptoschizont

(D) Sporozoite → Cryptoschizont → Cryptomerozoite

(29) Which disease shows the symptoms ?

(i) Severe fever in first and third week.

(ii) Fever decreases sequencially in 3rd and 4th week.

(iii) Headache.

(iv) Enlargement of spleen and liver.

(A) Common cold (B) Malaria (C) Pneumonia (D) Typhoid

(30) Where does filarial worms reside in human ?

(A) Lymphnodes (B) Lymphvessels (C) (A) and (B) both (D) Lymphatic cell

(31) Which disease shows these symptoms :

(i) Thickening of skin and epithelial tissue.

(ii) Swelling in hands, legs, leg sole, breast and scrotum.

(iii) It blocks lymphatic vessels in chronic condition.

(iv) Fever in acute condition.

(A) Malaria (B) Elephantiasis (C) Typhoid (D) Pneumonia

(32) What is responsible for following symptoms ?

(i) Responsible for elephantiasis.

(ii) Adult worm lives for 5 to 8 years.

(iii) They reside in lymphatic vessels and lymphatic nodes in human.

(A) Female anopheles (B) *Salmonella typhi* (C) Filarial worm (D) *Plasmodium vivax*

(33) Which fungi causes Ring worm ?

(A) *Microsporum* (B) *Trichophyton* (C) *Epidermophyton* (D) All of the above

(34) Which fish eats mosquito larvae ?

(A) Gambusia (B) Catla (C) Hilsa (D) Mrigal

(35) Which mosquito is responsible for Dengue and Chikangunia ?

(A) *Aedes* (B) *Culex fatigan* (C) Male anopheles (D) Female anopheles

(36) What should be done to preserve good health ?

(A) Eat fresh vegetables and fruits (B) Take fresh water and food

(C) Keep body hygienic (D) All of the above

Answers : (8-A), (9-C), (10-C), (11-A), (12-B), (13-A), (14-B), (15-A), (16-B), (17-A), (18-D), (19-B), (20-D), (21-A), (22-B), (23-A), (24-C), (25-B), (26-C), (27-A), (28-D), (29-D), (30-C), (31-B), (32-C), (33-D), (34-A), (35-A), (36-D)

Immunity :

- The reason for it is our body is able to protect itself form such foreign agents. This ability of the host to fight against there infections diseases is decided by the system which is called immune system which providing immunity.
- Immunities are of two types : (1) Innate Immunity (2) Acquired Immunity

(1) Innate Immunity :

- Innate immunity is so called because it is acquired by heriditary, since birth of animal.
- Innate immunity includes four barriers.

(A) Physical barrier : The skin of our body is the main physical barrier which prevents the entry of microorganisms.

- Mucus coating of the epithelial lining in the respiratory, gastrointestinal and urogenital tracts also help in trapping microbes entering our body.

(B) Physiological barrier : Acid in the stomach, saliva in the mouth and tears flow from the eyes to prevent the growth of the microorganisms.

(C) Cellular barrier : Certain types of leucocytes like polymorpho Nuclear leucocytes (PMNL); Neutrophils and monocytes leucocytes are natural killer cells found in the blood and macrophage, in tissues can engulf and destroy microbes.

(D) Cytokine barrier : Virus infected cells secrete proteins called interferons which protect uninfected cells from viral infections.

(2) Acquired Immunity (Pathogen Specific) :

- It bears memory characteristics.
- After birth during the lives, animal develop some immunity is their body to fight the diseases which is called Acquired Immunity.
- When our body comes in contact with a pathogen for the first time and then its response which is called 'Primary Response' whose intensity is mild. There after when the same pathogen comes in the contact for the second time, the secondary response is very intense which has the memory of the first encounter by our body.
- By two specific kind of lymphocytes i.e. B-lymphocytes and T-lymphocytes residing in our blood. The primary and secondary immune responses are elicited.

- The B-lymphocytes make an army of proteins to fight with the pathogens into our body. These proteins are called antibodies.
- Each antibody molecule has four peptide chains, two smallest chains are called light chains and two longer chains are called heavy chains. Hence, antibody is represented as H_2L_2.
- Different types of antibodies are created in our body which include I_gA, I_gM, I_gE, I_gD and I_gG.

---

(37) The ability of the host to fight against infections agent is called ..... .

(A) Immunity (B) Digestive system (C) Adversibility (D) Aggressiveness

(38) The disease resistance power obtained by birth in animals is called _x_.

(A) x = Hereditary Immunity (B) x = Innate Immunity

(C) x = Passive Immunity (D) x = Acquired Immunity

(39) _x_ of our body is the main _y_.

(A) x = Skin, y = Cytokine barrier (B) x = Skin, y = Physical barrier

(C) x = PMNL , y = Cytokine barrier (D) x = eye, y = Physical barrier

(40) Which of the following is in physiological barrier ?

(A) Skin, Respiratory tract, Urinary tract, Saliva in mouth

(B) Acid in stomach, Saliva in mouth, Tears flow from eyes

(C) WBCs, Killer cells, Skin, Tears

(D) Acid in stomach, Respiratory tract, Urinary tract, Skin

(41) What is the full form of PMNL ?

(A) Polymorpho Nucleotide Leucocytes (B) Polymorpho Nucleoside Leucocytes

(C) Polymorpho Nuclear Leucocytes (D) Polymorpho Nuclear Lymphocytes

(42) Which of the following protects uninfected cells from viruses ?

(A) Interferons (B) Interside (C) Leucocytes (D) Immunoglobulin

(43) Which is memory based Immunity ?

(A) Innate Immunity (B) Active Immunity

(C) Acquired Immunity (D) Passive Immunity

(44) Which type of response is intense ?

(A) Secondary (B) Innate (C) Primary (D) Tertiary

(45) Acid in stomach : Physiological barrier, Interferon : ..... .

(A) Physical barrier (B) Physiological barrier (C) Cytokine barrier (D) Cellular barrier

(46) Tears of eyes : Physiological barrier Skin : ..... .

(A) Physical barrier (B) Physiological barrier (C) Cytokine barrier (D) Cellular barrier

(47) How is H_2L_2 represented in antibody ?

(A) It has two small light chain and two long heavy chains.

(B) It has two small light chain and two short heavy chains.

(C) It has two small short chains and two small heavy chains.

(D) It has two small long chains and two long heavy chains.

(48) Which of the following is types of antibodies ?

(A) I_gA, I_gB, I_gC, I_gD, I_gE (B) I_gA, I_gM, I_gE, I_gD, I_gG

(C) I_gA, I_gH, I_gK, I_gD, I_gE (D) I_gA, I_gP, I_gQ, I_gR, I_gG

Answers : (37-A), (38-B), (39-B), (40-B), (41-C), (42-A), (43-C), (44-A), (45-C), (46-A), (47-A), (48-B)

Active and Passive Immunity and Vaccination and Immunisation :

- When a host comes into the contact with antigens, the antibodies are produced in his body. Antigen is in the form of dead or alive microorganisms or other proteins. This sort of immunity is called active immunity. Active immunity is slow and takes time to give its full and effective response.
- This immunity develops when the organisms naturally enters or the antigenic beings are introduced in the body in adequate amount so it helps to create antibodies in the body. This is called active immunity.
- When readymade antibodies are directly introduced into body to protect it from foreign agents, it is called passive immunity.
- The yellowish fluid colostrums is secreted by mother during the initial days of lactation which contains plenty of (I_gA) antibody. It gives protection to the infant.
- The foetus also obtains some antibodies from the mother through the placenta during pregnancy. These are the examples of passive immunity only.
- The principle of immunisation or vaccination is based on the property of memory.
- In vaccination, the protein of pathogen or the inactivated or weakened pathogens are introduced in the body. Against this antigen antibody is produced in the body.
- In tetanus, the directly produced antibody and antitoxin is entered.
- In snake biting too the injection which is given to the patients contain performed antibodies against snake venom. This type of immunisation is called passive immunisation.
- By recombinant DNA technology vaccine can be produced at large scale e.g., Heptitis B vaccine from yeasts.

(49) Antiallergens : I_gE, Colostrum : ..... .

(A) I_gG (B) I_gD (C) I_gA (D) I_gM

(50) Which is antibody containing fluid present in mother's milk in initial stages after childbirth ?

(A) Colostrum (B) Cholesterol (C) Clostridium (D) Calcisterol

(51) What is used to prepare Hepatitis B ?

(A) Bacteria (B) Virus (C) Algae (D) Yeast

(52) Vaccination protects individuals from diseases because ..... .

(A) produces antigens. (B) produces antibodies.

(C) increases number of RBCs. (D) helps in digestion process of the body.

(53) On what base principle of immunisation works ?

(A) Memory (B) Self (C) Identification (D) To phage

(54) What enters in the body in Tetanus ?

(A) Antigen (B) Antioxidant (C) Antitoxin (D) Oxytocin

(55) Which microorganisms are used to prepare vaccine ?

(A) Bacteria and Amoeba (B) Bacteria and Yeast

(C) Yeast and Amoeba (D) Yeast and Euglena

(56) Which part helps to obtain antibodies from mother during pregnancy ?

(A) Umbilical cord (B) Uterus (C) Placenta (D) Amniotic fluid

Answers : (49-C), (50-A), (51-D), (52-B), (53-A), (54-C), (55-B), (56-C)

Allergies, Auto Immunity and Lymphoid Organs :

- Some of are sensitive to certain particles of environment, the excess response of the immune system to certain antigens present in the environment is called allergy.
- The substances to which such an immune system responses are called allergens.
- I_gE types of antibodies are produced against such substances.
- Common example of allergen is dust particles.
- In common examples of symptoms include sneezing, watering eyes, running nose and difficulty in breathing.
- Allergies like take place due to the release of chemicals like histamine and seretonin from the mast cells.
- The use of drugs like antihistamine, adrenaline and steroids quickly reduce the symptoms of allergy immediately.
- Due to modern life style the reduction in immunity and more sensitivity are allergens increased.
- In most of the developed cities of India, more and more children suffer from allergies and asthma.
- Sometimes due to genetic and other unknown reasons, the body attacks self-cells resulting into damage to the body. This is called auto-immune disease.
- Rheumatoid arthritis is an auto-immune disease common in our society. Other examples are insuline based diabetes, arthritis, multiple sclerosis, etc.
- The primary lymphoid organs are bone marrow and thymus, in which immature lymphocytes are differentiated into antigen sensitised lymphocytes.
- After getting matured, the lymphocytes are transfered to secondary lymphoid organ like spleen, lymph, tonsils and small intestine. The secondary lymphoid organs provides the space for interaction of lymphocytes with the antigen and produces effective cells after the proliferation.
- The bone marrow is the main lymphoid organ in which all the blood cells along with lymphocytes are produced.
- The thymus is an lobelike organ and is arranged below the breastbone and near the heart.
- The thymus gland is large at the time of birth but decreases in size as the age passes and at adulthoold it becomes very small. Thymus and bone marrow both provide micro environments for the development and maturation of T-lymphocytes.
- The spleen is a large bean shaped organ. Which mostly process lymphocytes and phagocytes.
- Spleen also has large treasure of erythrocytes.

(57) What is the severe response of immune system towards specific antigen of environment known as ?

(A) Allergy (B) Asthma (C) Allergens (D) Primary Allergen

(58) Which diseases shows this symptoms ?

(i) Running nose (ii) Sneezing

(iii) Tears flowing from eyes (iv) Difficulty in breathing

(A) Pneumonia (B) Typhoid (C) Allergy (D) AIDS

(59) Which medicines are used to reduce symptoms of allergy quickly ?

(A) Antihistamine (B) Adrenalin (C) Steroid (D) All of the above

(60) From which diseases more and more children suffer in most of the cities ?

(A) Asthma (B) Bronchitis (C) Pneumonia (D) Malaria

(61) Due to genetic and other unknown reasons the body attacks self cells, it is known as.

(A) Innate Immunity (B) Auto Immunity (C) Acquired Immunity (D) Immunity

(62) Which are the examples of auto immunity ?

(A) Cancer, Rheumatoid, Joundice, Malaria (B) Rheumatoid, Diabetes, Pneumonia

(C) Rheumatoid, Diabetes, Multiple Sclerosis (D) Diabetes, Typhoid, Elephantiasis, Common cold

(63) ..... . produces all types of blood cells.

(A) Thymus (B) Liver (C) Bone marrow (D) Spleen

(64) Spleen is storehouse of ..... .

(A) Monocytes (B) Erythrocytes (C) Lymphocytes (D) Leucocytes

(65) _x_ and _y_ both provide micro environment for development and maturation of T lymphocytes.

(A) x = Thymus, y = Spleen (B) x = Liver, y = Bone marrow

(C) x = Thymus, y = Bone marrow (D) x = Thymus, y = Liver

Answers : (57-A), (58-C), (59-D), (60-A), (61-B), (62-C), (63-C), (64-B), (65-C)

- **AIDS :**
  - AIDS is a serious, incurable, infections and fatal disease.
  - The name AIDS stand for Acquired Immuno Deficiency Syndrome.
  - AIDS was first reported in USA in 1981 and in last 21 years it has spread throughout the world. More that 25 million people have died on account of this disease.
  - In India, the AIDS infection was first seen in Tamilnadu in the year 1986.
  - AIDS is caused by *Human Immuno Deficiency* Virus (HIV).
  - It is belongs to the group of retrovirus. In HIV genetic material is RNA.
  - In macrophages where RNA genome of the virus replicates with help of enzyme reverse transcriptare to form viral DNA.
  - Macrophages continuously produces thus it is acts as a manufacturing unit of HIV.
- **Transmission of AIDS :**
  - Heterosexual or homosexual physical contact with AIDS patient.

- HIV infected blood transfusion.
- Organ transplant from infective donor.
- By resulting the needle or syringe used by an AIDS or containing HIV.
- Through AIDS suffering mother to the foetus during pregnancy as well as through feeding (breast) by a mother to the Child. AIDS is not a contagious disease in common way. It does not spread through contact with clothes utensils, etc.

- **Symptoms of AIDS :**
  - HIV infected patients are classified into three stages :

  (1) The person appears healthy but does not show symptoms of AIDS. But such person as a vector of HIV.

  (2) Some people show a little symptoms of AIDS including fatigue, bodyache lasting over four weeks, fever, blood in stool, loss of appetite, loss of body weights, etc. This condition is called AIDS - Related complex (ARC) from this the full blown of AIDS takes place.

  (3) AIDS at the final stage : That is the last phase of this disease. The stage is marked by symptoms.
    - Loss of body weight by 10 % without any reasons.
    - Continuous fever for one month or more, which is undiognised.
    - Dirrhoea develops.
    - Get affected by cough, cold, pneumonia, etc.
    - Cancer of blood vessels and inflammation of lymph glands.
    - Through Enzyme linked Immunosorbent Assay (ELISA) Test and Western Blot Test (WBT) diagnosis of AIDS can carried on.

- **Prevention of AIDS :**
  - No valid method to discover AIDS has been invented. No vaccine is available AIDS is tottally an incurable disease.
  - In India, the national AIDS Control Organization (NACO) and other Non-Government Organization (NGOs) are doing a lot to educate people for AIDS. WHO has started numerous programmes to prevent the spreading of HIV infection.

- **Preventing Remedies :**
  - Don't cross the limit of single marriage life partner for sexual relationship.
  - At the time of sexual intercourse condom should be used to avoid the danger of AIDS and other sexually transmitted diseases.
  - Do not take blood from profesional blood donor. If it is needed, blood and blood products should be used proper investigation.
  - Injection should be used after steritizing the needle and syringe in boiling water and disinfected. The things like tooth-brush, razor, blades which come in direct contact with blood would not be reused. Once if is used by others.
  - The person suffering from AIDS/HIV should be treated properly and must not be boycotted society.

(66) AIDS was first reported in _x_ in _y_.

(A) x = India, y = 1981 (B) x = U.S.A., y = 1981

(C) x = England, y = 1986 (D) x = U.S.A., y = 1986

(67) In India, the AIDS infection was first seen in _x_ in _y_ year.

(A) x = 1986, y = Gujarat (B) x = 1986, y = Maharastra

(C) x = 1986, y = Tamilnadu (D) x = 1984, y = Tamilnadu

(68) HIV belongs to ..... .

(A) Retro virus (B) Rhino virus (C) (A) and (B) both (D) All of the above

(69) Which enzyme helps to convert viral RNA into viral DNA ?

(A) Reverse transcriptase (B) RNA transcriptase (C) DNA transcriptase (D) None of the above

(70) Which of the following causes Transmission of AIDS ?

(A) Organ transplant from infective donor.

(B) HIV infected blood transfusion.

(C) Heterosexual or Homosexual physical contact with AIDS patient.

(D) All of the above

(71) Which are the symptoms of fully blown AIDS ?

(A) Diarrhoea develops (B) Common cold, Cough, Pneumonia occurs

(C) Inflammation in lymph glands (D) All of the above

Answers : (66-B), (67-C), (68-A), (69-A), (70-D), (71-D)

- **Cancer :**
  - Cancer is a dreaded disease. In the world the most of the deaths occur due to this disease. In India, more than million people suffer from cancer.
  - In our body, process of cell growth and differentiation is regular and in controlled manner. In cancer cell, the process of this control break down and the process of cell divisions becomes uncontrollable. In cancer cell division is continuous, resulting into masses of cells which form as tumors of cancer.
  - Cells are spread to other parts of body through blood or vessels and create the malignant tumors, it called Metastasis.
  - The factor causing the cancer is called carcinogens.
  - Cancer causing viruses are called oncogenic viruses. The gene prevailed in it is called viral oncogenes.
  - Three main types of cancer.
    (1) **Carcinoma :** Abnormal growth in epithelial tissue in the body. This cancer is called carcinoma cancer. Breast cancer, lung cancer, cancer of stomach and pancreas are of this type. Skin melanoma is also of this type of cancer.
    (2) **Sarcoma :** This tissue develops from mesoderm in the body that divides uncontrolably and develops cancer which is called Sarcoma. Bones, cartilage, muscle and lymph glands cancers are of this type.
    (3) **Leukemia :** Leukemia is the cancer of blood. In blood WBC is found, that or; ginates in bone marrow. The number of leucocytes increase abnormally in this type of cancer. The number of immature WBC is also very high.

- **Methods of Diagnoising Cancer :**

  (1) Clinical Examination (2) Histocytological Tests (3) Radiation Technique (4) Laboratory Tests

  - For uterine or uterine mouth cancer a small mass of tissue from that part, is examined. It is called PAP smear.
  - For diagnosing of specific type of cancer antibody is used against cancer specific antigen.
  - The common approches for the treatment of cancer are surgery, radiation therapy and immunotherapy.
  - Some anticancer drugs are also used to destroy the cancerous cells.
  - Immunity too is useful to destroy the tumor cells e.g., Interferon.

---

(72) Which factors causes cancer ?

(A) Carcinogen (B) Metastasis (C) Antigen (D) Differentiation

(73) x = rays, Gamma rays and non-ionizing radiations like UV cause DNA damage leading to ..... transformation.

(A) Oncogenes (B) Oncogenic virus (C) Neoplastic (D) Lipoplastic

(74) Which of the following is inappropriate related to carcinoma cancer ?

(A) Cancer of stomach (B) Cancer of skin (C) Cancer of lymphnodes (D) Breast cancer

(75) The tissue develops from __x__ in the body that divides uncontrollably and develops cancer is called _y_.

(A) x = Mesoderm, y = Leukemia (B) x = Mesoderm, y = Sarcoma

(C) x = Exoderm, y = Carcinoma (D) x = Endoderm, y = Sarcoma

(76) _x_ is blood cancer found in _y_.

(A) x = Leukemia, y = RBCs (B) x = Leukemia, y = Blood plasma

(C) x = Leukemia, y = WBCs (D) x = Sarcoma, y = WBCs

(77) Find out inappropriate pair :

(A) Bone cancer (B) Muscle cancer (C) Cartilage cancer (D) Pancreas cancer

(**Note :** A, B and C are Sarcoma type cancer and D is carcinoma type of cancer.)

(78) Find out inappropriate pair :

(A) Breast cancer (B) Lung cancer (C) Cartilage cancer (D) Pancreas cancer

(79) Which are the diagnostic methods of cancer ?

(A) Clinical examination (B) Histocytological test (C) Radiation technique (D) All of the above

(80) For uterine or uterine mouth cancer a small mass of tissue is examined is called ..... .

(A) PAP smear (B) Tissue culture (C) Tissue transplantation (D) Organ transplantation

(81) In which method of cancer strong magnetic fields and non ionizing radiation are used ?

(A) CT Scan (B) PET (C) MRI (D) UV

(82) What is used to destroy cells of cancer tumor ?

(A) Anticancer drugs (B) Interferon (C) PAP smear (D) (A) and (B) both

(83) Abnormal growth in epithelial tissue of ..... causes cancer of carcinoma type.

(A) bone (B) pancreas (C) muscle (D) lymphnode

(84) Viruses which causes cancer are known as ..... .

(A) Oncogenic virus (B) Retro virus (C) HIV (D) None of the above

(85) ..... protects uninfected cells from viruses.

(A) PMNL (B) Mast cells (C) T cells (D) Interferons

(86) Which rays are used for computed Tomography ?

(A) Y - rays (B) UV - rays (C) α - rays (D) X - rays

Answers : (72-A), (73-C), (74-C), (75-B), (76-C), (77-D), (78-C), (79-D), (80-A), (81-C), (82-D), (83-B), (84-A), (85-D), (86-A)

Drugs and Alcohol Abuse :

- The use of drugs and alcohol is seen in the youth.
- Narcotic drugs like opioids, Cannabinoids, Cocaine, Heroin, Marijuana and LSD are extracted from flowering plants and fungi.
- Opioids are drugs which are bound with specific opioid receptors in our central nervous system and gastro-intestinal tract.
- Opioids is dry latex which is obtained by making fissure in unriped fruit.
- From opioid pain relieving drugs like morphine and codline are obtained.
- Morphine is a well-known pain reliever drug which generally acts on the nerve cells of small brain, obstructs pain and relieve the body from pain which functions as a stimulater. Opium inspires pleasant feelings.
- It draws out anxiety, fear and stress and it leads to the addiction.
- Codeine is another type of alkaloid obtained from opium. That is a pain-reliever medicine. Sometimes it is used in the cough syrup too it cools down the reactions of cough. Relieve the stress felt in stomach and intestine.
- Heroin or diacetylmorphine is white crystalline material which is a pain reliever and is pleasure related drug.
- Smack (brown sugar) is an unclear byproduct of Heroin.
- Cannabinoids are a group of chemical, which gets connected internally to cannabinoid receptors present in the brain.
- Cannabinoids, Bhang, Ganja are obtained from drived levels and flowers of *Cannabis indica* are obtained from the inflorescenes of the plant and dried leaves.
- Marijuana is obtained from canabis sativa. It is obtained from flower tops, leaves and resin of *cannabis* plants. The main active element of it is Della - Tetrahydro cannabinol (THC).
- When this medicine is taken, the eye-pupils get didated. Urine is produced more quantity and the level of sugar in the blood increases.
- Colex alkaloid or cocaine (track) is obtained from dry leaves and branches of coca plants. It has a potent stimulating action on central nervous system. It obstructs hunger, produces sleepness hallucinations which harm the mental performances later on the individual is found to induge in madness.
- The abuse of cocaine leads to severe headache, fit or heart failures or respiratory system leading to death.
- Other well known plants with halucinogenic properties are atro beladona and Datura cannabinoid are also being used by some sports person.

- Hasis is a potent sesin which is obtained from female fruits of *cannabis* plant.
- LSD is basically hallucinogenic chemical helps in changing the mental condition.
- That is produced from lysergic acid found from ergot of mustard seed.
- Amphetamines is a stimulating pill. Mostly use it to keep awake at night.
- Barbiturates is a medicine which makes a person feel comfort and relief. It is generally used as sleeping pills.
- Tobacco contains a large number of chemical substances including nicotine. Nicotine stimulates adrenal gland to release adrenaline and non adrenaline into blood circulation raising blood pressure and increase heart beats.
- Smoking leads to increased incidence of cancers of lungs, urinary bladder and throat, trachea, stomach ulcer, respiratory blockage and emphysema.
- Coronary heart disease can occur.

---

(87) What is known as pain reliever drug ?

(A) Marijuana (B) Morphine (C) Nicotine (D) LSD

(88) Which substance draws out anxiety, fear and strees and inspires pleasant feeling ?

(A) Opium (B) Codeine (C) Morphine (D) Marijuana

(89) Which medicine is used in cough syrup ?

(A) Morphine (B) LSD (C) Codeine (D) Marijuana

(90) _x_ is an unclear byproduct of Heroine.

(A) x = Morphine (B) x = Codeine (C) x = Opium (D) x = Smack

(91) _x_ group of chemicals gets connected to receptors of _y_.

(A) x = Heroine, y = Brain (B) x = Cannabinoids, y = Body

(C) x = Cannabinoid, y = Brain (D) x = Morphine, y = Body

(92) Bhang, Ganja and Charas are obtained from _x_ and Marijuana is obtained from _y_.

(A) x = *Cannabis indica*, y = *Cannabis sativa* (B) x = *Cannabis indica*, y = *Erythroxylum coca*

(C) x = *Cannabis sativa*, y = *Cannabis indica* (D) x = *Erythroxylum coca*, y = *Cannabis indica*

(93) From which part we obtain Bhang, Ganja, Charas and Marijuana ? Choose the sequence.

(A) Dry tip flowers, dry leaves and flowers.

(B) Dry leaves, flowers and green leaves of the apex.

(C) Dry leaves, flowers and dry flowers of the apex.

(D) Dry leaves, branches and dry flowers of the apex.

(94) Choose the correct option for the effect of Marijuana :

(i) Eye pupils get dilated (ii) Failure of respiratory system

(iii) More urine quantity (iv) Obstructs hunger

(v) Blood sugar increases

(A) i, ii and v are true, iii and iv wrong

(B) i, iii and v are true, ii and iv wrong

(C) i, ii and iii are true, iv and v wrong

(D) i, v are true, ii, iii and iv wrong

(95) Which alkaloid obstructs hunger and stimulates central Nervous system ?

(A) Cocaine (B) Cock (C) Crack (D) All of the above

(96) _x_ alkaloid is obtained in West America from dry leaves and branches of plants.

(A) x = Marijuana, y = Cannabis sativa

(B) x = Marijuana, y = Cannabis indica

(C) x = Cocaine or Coke or Crack, y = Erythroxylum coca

(D) x = Heroine, y = Erythroxylum coca

(97) Which medicine is used as sleeping pills ?

(A) Amphetamines (B) Barbiturates (C) Marijuana (D) Cannabinoid

(98) Which stimulating pill people use to keep awake at night ?

(A) Amphetamines (B) Cannabinoid (C) Barbiturates (D) Coke

(99) What is increased in blood by frequent smoking ?

(A) O_2 (B) CO (C) CO_2 (D) H_2O

(100) Nicotine stimulates _x_ gland in the body so _y_ hormone mixes in blood.

(A) x = Pancreas, y = Insulin

(B) x = Pancreas, y = Glucagon

(C) x = Adrenal, y = Epinephrine and Non-Epinephrine

(D) x = Adrenal, y = Insulin and Glucagon

(101) What happens in blood by smoking ?

(A) CO_2 increases and O_2 decreases in Hb (B) CO_2 decreases and O_2 increases in Hb

(C) CO_2 increases and O_2 increases in Hb (D) CO_2 decreases and O_2 increases in Hb

Answers : (87-B), (88-A), (89-C), (90-D), (91-C), (92-A), (93-C), (94-B), (95-D), (96-C), (97-B), (98-A), (99-B), (100-C), (101-A)

Adolescence and Drug / Alcohol Abuse, Addiction and Dependence and Effects of Drugs, Prevention and Control :

- The period between 12-18 years of a child is known as adolescene.
- Adolescene is a link between childhood and adulthood.
- In an adolocent, for excitement and adventure, curiosity is needed and through experiments, common purposes are achieved.
- Some of the adolescents, become addict to it by escaping from facing problems, if fail to problem better in the examination are drawn to stress and pressure they start consuming drugs and alcohol.
- To remove these habits, awareness through newspapers, films, internet, television can be carried out.
- When person taking drugs or alcohol in excess amount, it is called the dependence of addiction.
- Due to dependency mentality is clearly change in any way.
- In certain cases, withdrawal symptoms can be severe and even life threatening and the person may need medical supervision.
- With the abuse of alcoholic substance and drugs, the person is influences intantly and he behaves unthoughtfully, rashly and harms things. Because of excessive intake of alcoholic substances, the respiratory systems fails causing death, unconsciousness and obstructs the heart beats and thus death occur.

- If an addict takes the drugs intravenously, he may get infected to AIDS and Hepatitis - B through of the needle and syringe.
- The excessive and chronic use of drugs and alcohol damages nervous system and it leads to severe liver diseases like cirrhosis.
- If the alcohol is taken during pregnancy, it affects the foetus severely.
- The sports persons abuse the alcohol to increase muscular strength. They misuse anabolis steroids and several; hormones for this.
- The use of anabolic steroids in female causes the following effects, the features like males masaculisication, increased aggressiveness, depression, abnormal menstrual cycles, excessive growth of hair on face and body, etc.
- While in males increasing of acne increased agressiveness depression, reduction in the size of testicles, decreased sperm production, liver dysfunction, increasing haldness, etc. side effect can be seen.
- In adolescent male and female development of acne on face and the growth centres get closed which result into stagnant growth.
- The habits of smoking, taking drugs or alcohol is common in youth. It is there in adolescene as well. So by identifying the situation, it is advisable to keep the adolescent away from using drugs and alcohol.
- The teachers and parents should tackle the situation responsibly.
- Every boy or girl should be grown to his/her personality and choice He/She should not be forced to follow something which they are not interested in and should not be bound within any boundary. The should be allowed to perform the activities like studies, sports and others.
- Alert parents and teachers need to identify dangerous signs and discuss them properly. The friends should also instantly bring the matter to the notice of his parents and teacher.

---

(102) The period between 12-18 years of child.
(A) Adolescence (B) Childhood (C) Infancy (D) Adulthood

(103) Adolescence is link between _x_ and _y_.
(A) x = Adolescence, y = Adulthood (B) x = Adolecence, y = Childhood
(C) x = Infancy, y = Childhood (D) x = Adulthood, y = Childhood

(104) Which are the factors for habits of drugs and alcohol in Adolescence ?
(A) Unsupportive family structure
(B) Not getting proper facilities due to poor financial condition of family
(C) Family pressure
(D) All of the above

(105) Which are the symptoms of withdrawal syndrome ?
(A) Anxiety (B) Nausea
(C) Sweating (D) All of the above

(106) Which system is damaged by excessive use of drug and alcohol as a result of Cirrhosis ?
(A) Respiratory system (B) Digestive system (C) Nervous system (D) Reproductive system

(107) Which of the following effects are seen in women by use of steroids ?
(A) Increasing aggressiveness (B) Depression
(C) Irregularity in menstrual cycle (D) All of the above

(108) Which side effects can be seen by use of steroids in male ?

(A) Development of acne (B) Reduction of sperm production

(C) Decreasing efficiency of liver (D) All of the above

Answers : (102-A), (103-B), (104-D), (105-D), (106-C), (107-D), (108-D)

- **True - False (T - F) type questions :**

(109) Which of the following is correct for Typhoid ?

(1) Typhoid is caused by viruses.

(2) Typhoid is caused by Salmonella typhi.

(3) Typhoid is transmitted by water and food contaminated by patient's feaces.

(4) House flies carry organisms from feaces to food, milk and water.

(A) T, F, T, T (B) T, T, F, F (C) F, T, T, T (D) T, T, T, F

(110) Which of the following is correct option for Typhoid ?

(1) Pathogen of Typhoid are seen in intestine of human being.

(2) Incubation period of Typhoid causing bacteria is 1-3 weeks.

(3) Enlargement of liver and spleen occurs in Typhoid.

(4) Typhoid is diagnosed by Western blots.

(A) T, T, F, F (B) T, T, T, F (C) F, F, T, T (D) F, F, F, T

(111) Choose the correct option for True (T) and False (F) for Pneumonia.

(1) Pneumonia is severe disease of Respiratory tract.

(2) Streptococcus pneumoniae is commonly called as pneumococcus.

(3) Incubation period of *pneumonia* bacteria is 1-3 weeks.

(4) Lips and finger nails turn blue to grey in colour.

(A) T, T, T, T (B) T, T, F, F (C) F, F, T, T (D) T, T, F, T

(112) Which of the following is correct option for True and False ?

(1) A special kind of asexual reproduction is seen in cryptoschizont, known as schizogony.

(2) Endoerythrocytic cycle is seen in blood plasma of human being.

(3) Merozoites produces false legs this stage is known as amoeboid stage.

(4) Size of male gametocyte is small and nucleus is big while size of female gametocyte is big and nucleus is small.

(A) T, T, F, T (B) T, T, F, F (C) F, F, T, T (D) T, F, T, F

(113) Which of the following is correct for True (T) and False (F) for Elephantiasis ?

(1) Elephantiasis is transmitted by Culex fatigon.

(2) Filarial worm is pathogen for Elephantiasis.

(3) Skin and Epithelial tissue becomes thick in Elephantiasis.

(4) Adult filarial worm lives 5 to 10 years.

(A) T, T, F, F (B) T, T, T, F (C) T, F, T, F (D) F, F, T, T

(114) Which of the following is correct as True (T) and False (F) ?

(1) Culex fatigon is the causative organism for Dengue and Chikungunya.

(2) Immunity obtained by birth is called acquired Immunity.

(3) Our skin is the main cellular barrier.

(4) Interferons protect unifected cells from viral infections.

(A) T, T, F, F (B) T, F, T, F (C) F, F, T, T (D) F, T, F, T

(115) Choose the correct option for True (T) and False (F).

(1) Colostrum possess abundant Antibody.

(2) Principle of Vaccination is based on memory.

(3) The excess response of the immune system to certain antigens present in environment is called allergy.

(4) In most of the developed cities of India, more and more children suffer from allergies and asthma.

(A) F, T, F, F (B) F, F, T, T (C) F, F, F, T (D) T, T, F, F

(116) Choose the correct option for True (T) and False (F) :

(1) Insulin based diabetes is example of Autoimmunity.

(2) Spleen is bean shaped organ.

(3) Macrophage behaves like a factory of HIV.

(4) MALT constitutes 50 % of the lymphoid tissue in human body.

(A) T, T, F, F (B) F, F, T, T (C) T, F, T, T (D) T, T, T, F

(117) Choose the correct option for True (T) and False (F) :

(1) Cells are spread to other parts of the body, it is known as Metastasis.

(2) Viruses which causes cancer are called Oncogens.

(3) Leukemia is seen in RBCs of blood.

(4) Strong magnetic field and non-ionizing radiation are used as cancer diagnising methods.

(A) T, T, F, T (B) T, T, F, F (C) F, F, T, T (D) T, F, T, F

Answers : (109-C), (110-B), (111-D), (112-A), (113-B), (114-C), (115-A), (116-C), (117-A)

- **A - Assertion, R - Reason.**

Choose the correct option :

(A) A and R both correct. R is the correct explanation of A.

(B) A and R both correct. R is not the correct explanation of A.

(C) A - True, R - False.

(D) A - False, R - True.

(118) Assertion A : Typhoid is common disease done by bacteria.

Reason R : Typhoid occurs by Salmonella typhi.

(A) (B) (C) (D)

(119) Assertion A : Pneumonia is severe disease of respiratory tract.

Reason R : Fluid accumulates in alveoli and windpipe which causes deficiency of O2 in lungs.

(A) (B) (C) (D)

(120) Assertion A : Retro virus causes common cold.

Reason R : Virus of common cold causes infection in nose and respiratory tract.

(A) (B) (C) (D)

(121) Assertion A : Trophozoite stage is known as Amoeboid stage.

Reason R : Trophozoite produces false legs.

(A) (B) (C) (D)

(122) Assertion A : Schizont is converted into merozoite by asexual reproduction.

Reason R : Then merozoites are converted into gametocyte.

(A) (B) (C) (D)

(123) Assertion A : Filarial worm causes Elephantiasis.

Reason R : Filarial worms are viviparous.

(A) (B) (C) (D)

(124) Assertion A : Ring worm is a common infections disease of humans occurs by Fungi.

Reason R : Ring worm occurs due to use of towel, cloth and comb of infections persons.

(A) (B) (C) (D)

(125) Assertion A : Natural killer cells of blood and Macrophages of tissues behave as phagocytes.

Reason R : It kills microorganisms.

(A) (B) (C) (D)

(126) Assertion A : Skin is the main physical barrier of our body.

Reason R : It prevents entry of microbes in our body.

(A) (B) (C) (D)

(127) Assertion A : The yellowish fluid colostrum is secreted by mother during initial days of lactation which protects infant

Reason R : Colostrum has plenty of IgE.

(A) (B) (C) (D)

(128) Assertion A : When antigen is in contact with host cells, antibodies are formed.

Reason R : Active immunity is low, while it takes time to respond.

(A) (B) (C) (D)

(129) Assertion A : Acquired Immunity has characteristic of memory.

Reason R : Secondary response is intense.

(A) (B) (C) (D)

(130) Assertion A : Things like tooth brush, blades, which come in direct contact with blood should not be reused, once if it is used by others.

Reason R : It can cause infection of HIV.

(A) (B) (C) (D)

(131) Assertion A : Morphine is well known pain relieving drug.

Reason R : Which generally acts on the nerve cells of small brain, obstructs pain and relieve body from pain.

(A) (B) (C) (D)

(132) Assertion A : Marijuana is obtained from *Cannabis sativa*.

Reason R : Bhang and Charas is obtained from *Erythroxylum coca*.

(A) (B) (C) (D)

(133) Assertion A : Amphitamines are stimulating pills which people use to be awake at night.

Reason R : Barbiturates are known as sleeping pills.

(A) (B) (C) (D)

(134) Assertion A : Nicotine stimulates Adrenal gland.

Reason R : Which increases blood pressure and heart beat.

(A) (B) (C) (D)

(135) Assertion A : Adolescence is link between childhood and adulthood.

Reason R : We see many biological and behavioural changes.

(A) (B) (C) (D)

(136) Assertion A : The excessive and chronic use of drugs and alcohol damages nervous system and leads to severe liver disease like cirrhosis.

Reason R : If the alcohol is taken during pregnancy, it affects the foetus severely.

(A) (B) (C) (D)

Answers : (118-A), (119-A), (120-D), (121-A), (122-B), (123-C), (124-B), (125-A), (126-A), (127-A), (128-B), (129-A), (130-A), (131-A), (132-C), (133-B), (134-B), (135-B), (136-B)

(137) Choose the correct option :

Column - A	**Column - B**	**Column - C**
(1) Typhoid	(P) Rhino virus	(i) Sputum of patients
(2) Pneumonia	(Q) *Salmonella typhi*	(ii) Cough, sneezing
(3) Common Cold	(R) Filarial worm	(iii) Culex fatigon
(4) Elephantiasis	(S) *Streptococcus pneumoniae*	(iv) Feaces of patients

(A) (1 - Q - iv), (2 - S - i), (3 - P - ii), (4 - R - iii)

(B) (1 - Q - iv), (2 - S - i), (3 - R - iii), (4 - P - ii)

(C) (1 - Q - iv), (2 - R - iii), (3 - S - i), (4 - P - ii)

(D) (1 - S - i), (2 - Q - iv), (3 - R - iii), (4 - P - ii)

(138) Choose the correct option :

Column - I	**Column - II**
(1) Physical barrier	(P) WBCs
(2) Physiological barrier	(Q) Skin
(3) Cellular barrier	(R) Interferons
(4) Cytokine barrier	(S) Tears flow from eyes

(A) (1 - P), (2 - S), (3 - R), (4 - Q)

(B) (1 - Q), (2 - S), (3 - P), (4 - R)

(C) (1 - Q), (2 - P), (3 - R), (4 - S)

(D) (1 - P), (2 - R), (3 - Q), (4 - S)

(139) Choose the correct option :

Column - I	Column - II
(1) Big bean shaped organ	(P) Bone marrow
(2) Main lymphoid organ	(Q) Spleen
(3) MALT	(R) Thymus
(4) Arranged near to heat	(S) Amount of Lymphoid tissue is 50 %

(A) (1 - Q), (2 - R), (3 - P), (4 - S)
(B) (1 - P), (2 - Q), (3 - R), (4 - S)
(C) (1 - Q), (2 - S), (3 - P), (4 - R)
(D) (1 - Q), (2 - P), (3 - S), (4 - R)

(140) Choose the correct option :

Column - I	Column - II
(1) Metacryptomerozoite	(P) Asexual reproduction
(2) Trophozoite	(Q) Becomes spleen shaped
(3) Schizont	(R) Produces false appandages
(4) Ookinete	(S) Becomes spherical

(A) (1 - Q), (2 - P), (3 - R), (4 - S)
(B) (1 - R), (2 - S), (3 - Q), (4 - P)
(C) (1 - S), (2 - R), (3 - P), (4 - Q)
(D) (1 - Q), (2 - P), (3 - R), (4 - R)

(141) Choose the correct option :

Column - I	Column - II
(1) Dry, scaly lesions on skin, nail and head like parts.	(P) AIDS
(2) Uncontrolled cell division.	(Q) Allergy
(3) Diseases like common cold, cough and pneumonia.	(R) Ring worm
(4) Running nose and difficulty in breathing.	(S) Cancer

(A) (1 - P), (2 - S), (3 - R), (4 - Q)
(B) (1 - Q), (2 - R), (3 - P), (4 - S)
(C) (1 - R), (2 - S), (3 - Q), (4 - P)
(D) (1 - R), (2 - S), (3 - P), (4 - Q)

(142) Choose the correct option :

Column - A	Column - B	Column - C
(1) *Opium Popi*	(P) LSD	(i) Fruit
(2) *Cannabis Indica*	(Q) Marijuana	(ii) Unripened fruits, Latex milk
(3) *Cannabis Sativa*	(R) Ganja	(iii) Obtained from dry leaves and flowers
(4) *Claviceps Purpurea*	(S) Morphine	(iv) Obtained from dry flowers of apex

(A) (1 - S - ii), (2 - R - iii), (3 - Q - iv), (4 - P - i)
(B) (1 - S - ii), (2 - R - iii), (3 - P - i), (4 - Q - iv)
(C) (1 - P - i), (2 - Q - ii), (3 - S - iv), (4 - R - iii)
(D) (1 - Q - ii), (2 - P - i), (3 - R - iii), (4 - S - iv)

Answers : (137-A), (138-B), (139-D), (140-C), (141-D), (142-A)

(143) Choose the correct pair using diagram I :

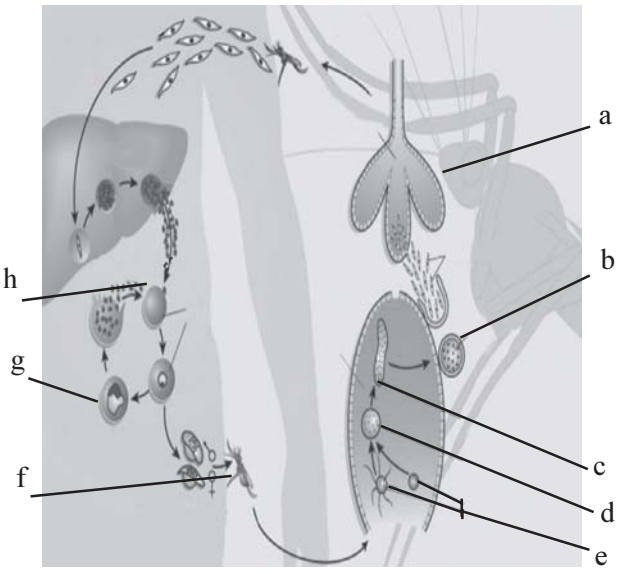
Column - I	Column - II
(a) Sporozoite	(i) g
(b) Trophozoite	(ii) h
(c) Gametocyte	(iii) P
(d) Merozoite	(iv) c
(e) Ookinete	(v) f

(A) (a - v), (b - iii), (c - ii), (d - iv), (e - i)

(B) (a - ii), (b - i), (c - iv), (d - v), (e - iii)

(C) (a - iii), (b - i), (c - v), (d - ii), (e - iv)

(D) (a - i), (b - iii), (c - ii), (d - iv), (e - v)



(144) What indicates b, d and e in diagram ?

(A) b - Gamete, d - Zygote, e - Oocyst

(B) b - Oocyst, d - Zygote, e - Gametophyton

(C) b - Gamete, d - Oocyst, e - Zygote

(D) b - Oocyst, d - Zygote, e - Gamete

(145) What does the given diagram 2 indicate ?

(A) Retro virus (B) E. Coli (C) Microbe cell (D) Eukaryote cell

(146) Which disease occurs in organism showing in diagram 2 ?

(A) AIDS (B) Cancer (C) Malaria (D) Jaundice

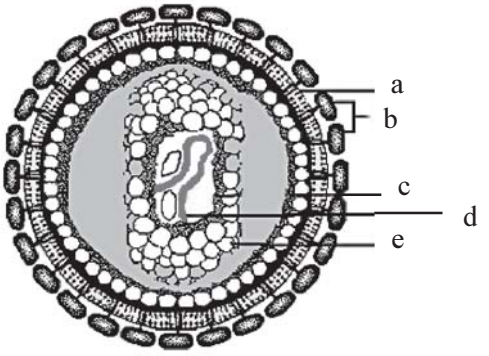
(147) What 'a' and 'b' indicates in the diagram ?

(A) a - Lipid coating, b - RNA core protein

(B) a - Lipid coating, b - Receptor protein

(C) a - Capsid, b - RNA core protein

(D) a - Capsid, b - Receptor binding protein



(148) What indicates 'c', 'd' and 'e' in diagram 2 ?

(A) c - Viral, d - RNA protein, e - Lipid coating

(B) c - Lipid fibre, d - Receptor protein, e - Capsid

(C) c - Flagella, d - Receptor protein, e - Lipid coating

(D) c - Viral, d - RNA core protein, e - Capsid

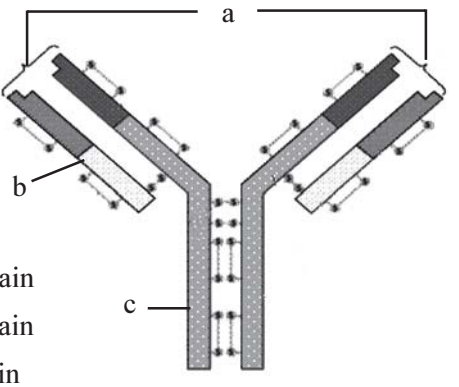
(149) Which is shown in this diagram 3 ?

(A) Molecule of Antibody

(B) Structure of Antigen

(C) Structure of DNA

(D) Structure of RNA



(150) What indicates 'a', 'b' and 'c' in diagram 3 ?

(A) a = Antibody binding site, b = Heavy chain, c = Light chain

(B) a = Antibody binding site, b = Light chain, c = Heavy chain

(C) a = Antigen binding site, b = Light chain, c = Heavy chain

(D) a = Antigen binding site, b = Heavy chain, c = Light chain

Answers : (143-C), (144-D), (145-A), (146-(A), (147-B), (148-D), (149-A), (150-C)

- **Questions for NEET :**

(151) Which of the following is most infectious disease ?

(A) Cold and Cough (B) Malaria (C) AIDS (D) Typhoid

(152) In which of the following barrier saliva in mouth and tears from eyes are included ?

(A) Physical barrier (B) Cellular barrier (C) Physiological barrier (D) Cytokine barrier

(153) Which is the causative agent of Influenza ?

(A) Fungi (B) Bacteria (C) Virus (D) Protista

(154) Which substance is used to prepare medicine for common cold ?

(A) Codeine (B) Cocaine (C) Morphine (D) Heroine

(155) What is the causative agent of typhoid ?

(A) Pseudomonas (B) Salmonella (C) Amoeba (D) Escherichia

(156) What is Interferon ?

(A) Antivirus protein (B) Antibacterial protein

(C) Anticancer protein (D) Complex protein

(157) Which cells are destroyed by HIV in AIDS ?

(A) Leucocytes (B) Thrombocytes

(C) Helper T-lymphocytes (D) B-lymphocytes

(158) Which of following does not cause AIDS infection ?

(A) Breast feeding by infections mother.

(B) Pregnant mother to baby.

(C) Use of clothes of AIDS patients.

(D) Syringe used by AIDS patient.

(159) What is responsible for transmission of AIDS ?

(A) Fungi (B) Virus (C) Bacteria (D) Protozoa

(160) Which plants are used to propare Morphine, Bhang and Tobacco ?

(A) Papaver, Cannabis, Nicotiana (B) Cannabis, Papaver, Nicotiana

(C) Nicotiana, Cannabis, Papaver (D) Papaver, Nicotiana, Cannabis

(161) What is the common name of diacetyl morphine ?

(A) Morphine (B) Cannabis (C) Heroin (D) Cocaine

(162) Which is the infective stage of Plasmodium ?

(A) Sporozoite (B) Trophozoite

(C) Merozoite (D) All of the above

(163) Which organ gets infected by cirrhosis by drinking much alcohol ?

(A) Kidney (B) Liver (C) Heart (D) Lungs

(164) Which hormone causes Allergy ?

(A) Adrenaline (B) Non-adrenaline (C) Glucocorticoids (D) Antihistamine

(165) Which of the following is appropriate for carcinoma ?

(A) Unstable tumor of skin and mucosa. (B) Unstable tumor of colon.

(C) Unstable tumor of connective tissue. (D) Stable tumor of connective tissue.

(166) What is Metastasis ?

(A) Rapid cell division in cancer cells. (B) Regeneration of cancerous cells.

(C) Spreading of cancerous cells. (D) All of the above

(167) What is the treatment known as done by α - rays and γ - rays ?

(A) Radiation treatment (B) Operation

(C) Chemotherapy (D) Anticancer drug treatment

(168) In which organs T-lymphocytes are matured ?

(A) Pancreas (B) Spleen (C) Bone marrow (D) Thymus gland

(169) What produces Antibody ?

(A) RBCs (B) Thrombocytes (C) Lymphocytes (D) Monocytes

(170) Which is an autoimmune disorder ?

(A) Cancer (B) Hepatitis (C) Asthma (D) Rheumatoid

(171) Which of the following substance is CNS reliever ?

(A) Caffeine (B) Morphine (C) Cocaine (D) Amphitamines

(172) Antibodies are group of ..... .

(A) Lipo protein (B) Steroid (C) Glyco proteins (D) Nucleo protein

(173) Each Immunoglobulins possess two heavy chains and two light chains. Where does antigen binding site present ?

(A) Variable region of both heavy and light chains.

(B) Constant regions of both heavy and light chains.

(C) Variable regions of heavy chains.

(D) Variable region of light chain.

(174) Which disease is diagnosed by Western Blot test ?

(A) Cancer (B) AIDS (C) Malaria (D) Pneumonia

(175) Who provides immunity to fight against diseases ?

(A) Immunoglobulin (B) Antigen (C) Histamine (D) HLA protein

(176) Which cells engulf foreign molecules ?

(A) Blood cells (B) Macrophage (C) Mast cells (D) Lymphocytes

Answers : (151-A), (152-C), (153-C), (154-A), (155-B), (156-A), (157-C), (158-C), (159-B), (160-A), (161-C), (162-A), (163-B), (164-A), (165-A), (166-C), (167-A), (168-D), (169-C), (170-D), (171-B), (172-C), (173-A), (174-B), (175-A), (176-B)

Human Health and Disease

- Health defined as a state of complete physical, mental and social well being and not merely the absence of disease or infirmity.
- When the functioning of one or more organs or systems of the body is adversely affected, we say that we are not healthy, i.e., we have a disease.

Congenital diseases

These are inborn diseases present from the birth.

Cancer

- Cancer is a disease of uncontrolled proliferation of cells without any differentiation. It is of two types (i) Benign tumour, (ii) Malignant tumour.
- Cancer is neither hereditary not a contagious disease Chemical and physical agents that can cause cancer are called careinogens e.g. X-rays, UV rays, caffeine, steroid, arsenic air, soot, diethylstilbestrol (DES), vinyl chloride (VC), cadmium oxide.
- Several genes in the host called cellular oncogenes or proto-oncogenes have been identified which under certain conditions on getting activated could lead to oneogenic transformation of cells.
- Tumor suppressor genes normally keep mitosis in check and prevent cancer from occurring.

Acquired diseases

- These are diseases contracted after birth.
- It is of two types :

Communicable diseases

- These can pass from one infected person to a healthy one.

Non-Communicable diseases

- These remain confined to the persons who develop them and do not spread to others.

Bacterial diseases

→ **Typhoid**
Caused by Salmonella typhi

→ **Pneumonia**
Caused by Streptococcus pneumoniae or Diplococcus pneumoniae

→ **Dysentery (Bacillary)**
Caused by bacteria of the genus shigella

→ **Plague**
Caused by Pasteurella/ Yersinea

→ **Diphtheria**
Caused by Corynebacterium diphtheriae

→ **Chlera**
Caused by Vibrio cholerae

→ **Tetanus**
Caused by Clostridium tetani

→ **Tuberculosis**
Caused by Mycobacterium tuberculosis

Viral diseases

- **Common cold**
  Caused by Rhino virus
- **Mumps**
  Caused by Paramyxo virus
- **Measles**
  Caused by rubella virus
- **Dengue fever**
  Caused by arbo virus
- **Chikungunya**
  Caused by alpha virus
- **Chicken pox**
  Caused by Varicellazoster
- **Poliomyelitis**
  Caued by poliovirus (enterovirus)
- **Viral hepatitis**

Protozoan diseases

- **African sleeping sickness or trypanosomiasis**
  Caused by Trypanosoma gambiense
- **Kala-azar**
  Caused by Leishmania donovani
- **Amoebic dysentery or Amoebiasis**
  Caused by Entamoeba histolytica
- **Malaria**
- Caused by Plasmodium
- There are 4 species of Plasmodium with causes 4 types of malaria. These are :

↓

- **Plasmodium**
- Plasmodium vivax or P. vivax (benign tertian malaria)
- P. malariae (quartan malaria)
- P. falciparum (cerebral malaria or malignant tertian malaria)
- P.ovale (mild tertian malaria)

Helminthic disease

- **Ascariasis**
  Caused by Ascaris lumbricoides
- **Filariasis or elephantiasis**
  Caused by Wuchereria (W.bancrofii & W.malayi)

Immunity

It is body's ability to destroy pathogens or other foreign materials and to prevent further cases of certain infectious diseases.

Types of Immunity

Innate Immunity/Inborn

- Non-specific defence mechanisms.

- It is present at the time of birth.
- It is of two types : external defence and internal defence.

External defence first line defense

It comprises physical and chemical barriers to the entry of pathogens into the body.

Internal defense second line defense

It occurs when the first line of defenses fail to prevent access of pathogens to the tissues. The body's internal defenses is carried out by WBC (neutrophils and monocytes) macrophages, natural killer (NK) cells, inflammatory reactions, fever, interferons and complement system.

Acquired immunity / Specific defense mechanism

(Third line of defenses)

- Response of immune system to specific pathogens.

Active immunity

- It is produced due to contact with pathogen or its antigen and subsequent production of antibodies.

Passive immunity

It occurs when readymade antibodies are directly given to protect the body against foreign agents. Colostrum (Contains abundant IgA) and foetus receiving antibodies from mother are some examples of passive immunity.

Antigen Presenting Cells

- These are specialized cells which include macrophages (monocytes as blood macrophages and histocytes as tissue macrophages) B-Lymphocytes and dendrite cells (e.g., Langerhans cells of epidermis of skin)

Lymphocytes

T-Lymphocytes, T-Cells

- T-Cells are involved in cell mediated immune response.

B-Lymphocytes, B-Cells

- B-cells are involved in antibody mediated or humoral immune response.
- They give rise to plasma cells (produce Antibodies) and memory B-cells on activation by antigen specific T-cells.

Immunity

Antigens / Antibodies

- Antigens are substances which when introduced into the body, stimulate the production of antibodies.
- The specific reactivity induced in a host by an antigenic stimulus is known as immune response. It is of two types - (1) Primary and (2) Secondary immune response.
- Antibodies (Abs) are glycoproteins, called immunoglobulins, made up of four peptide chains, two small called light chains and two longer called heavy chains.
- Different types of antibodies such as IgA (heaviest and second most abundant), IgM (Largest and first reach the site of infection) IgG (Most abundant, only Ab that crosses placenta), IgD and IgG produced in our body.

Immune System

- It consists of Lymphoid organs, tissues, cells and soluble molecules such as antibody.
- Lymphoid organs are of two types - (1) Primary lyumphoid organ and (2) Secondary lymphoid organs. Thymus, bone marrow are primary lymphoid organs.
- Lymph nodes, spleen, tonsils, payer's patches, appendix and MALT (Mucosal Associated Lymphoid Tissue) are Secondary Lymphoid Organs.

Vaccination & Immunisation

- Vaccination is the process of development of immunisation against a particular disease by inoculation of harmless antigenic material like attenuated pathogen or its toxoid in the form of vaccine into a healthy person.
- Vaccine is of 3 types - First generation vaccines, Second generation vaccines and Third generation vaccines.

Immunological Disorders

Failure of host defense results into immunological disorders, which falls into 3 different categories.

- Immuno Deficiency
- Auto Immunity
- Hyper Sensitivity

11 Microbes and Human Welfare

- There is diversity in microbes which can not be seen by naked eyes. e.g. Protozoa, Bacteria, Fungi, Virus. Except some microbes many microbes are useful to mankind. These microbes are exploited in different ways for human welfare by the broad knowledge of modern biotechnology and genetic engineering.
- **Microbes in household products :**

  Some food materials which we take in our daily diet are obtained by microbial process. Likely curd from milk which is very old technique; dosa, idli, bread, pickle. Some of the traditional drinks, cheeze and fish, soybean and bamboos plays an important role for the food material products from these by microbes. It is termentation of carbohydrates in green plant tissue.

---

(1) Which organisms are necessary for genetic engineering and biotechnology ?

(A) Animals (B) Microbes (C) Plants (D) None of the above

(2) Which bacteria are useful to produce curd from milk ?

(A) Lacto streptococcus (B) Lacto bacillus (C) Lacto methanogens (D) Lacto coccus

(3) What is produced by LAB during curd formation from milk ?

(A) Base (B) Acids (C) Salt (D) All the above

(4) Which is the substrate digested by LAB in Milk ?

(A) Carbohydrates (B) Lipids (C) Proteins (D) Nucleic acids

(5) Which nutrient's quality is increased by Lactobacillus ?

(A) Vit - K (B) Vit - B_{12} (C) Vit - C (D) Vit - D

(6) Which is the correct option for the bacteria that provides protection from harmful bacteria in human stomach ?

(A) Lactobacillus (B) Azotobactor (C) Methanogens (D) Clostridium

(7) Which are the microbes used in fermentation to prepare Idli and Dosa ?

(A) Virus (B) Protozoa (C) Fungi (D) Bacteria

(8) Choose the correct match for uses of Lactic acid.

(A) It is useful to prepare curd from milk

(B) To increase quality of Vit - B_{12}

(C) In fermentation to prepare pickles

(D) All of the above

(9) Which is the fungi used to prepare bread ?

(A) *Aspergillus niger* (B) *Rhizopus nigricans*

(C) *Propioni bacterium* (D) *Saccharomyces cerevisiae*

(10) Which biomolecule is fermented to prepare ensilage (food for cattle) from green plants ?

(A) Lipid (B) Protein (C) Carbohydrates (D) Vitamins

(11) What is prepared by *Propioni bacterium sharmanii* ?

(A) Roquefort cheese (B) Brewer's Yeast (C) Swiss cheese (D) Toddy drink

Answers : (1-B), (2-B), (3-B), (4-C), (5-B), (6-A), (7-C), (8-D), (9-D), (10-C), (11-C)

- **Microbes in Industrial products :**
  - Many products useful to mankind are synthesized at commercial level through microbes e.g. beverages, antibiotics, carbonic acids, alcohol, enzymes, proteins, industrial chemicals, steroids, vaccines, amino acids, energy fuel etc.
  - Useful microbes are grown in large vessel on industrial scale. Wine, beer, whisky, brandy or rum like beverage are produced. *Saccharomyces cerevisae* yeast helps for the beverages. Ethanol is produced from creals and fruit juices with the help of this yeast. Methane which is used as a energy resource is produced by using methanogenic bacteria.
  - Antibiotics available from market are one type of chemicals. Penicillin is obtained from *Penicillium notatum*. Carbomycin, *bacitracin, fumagillin*, tetracycline are other antibiotics. Carbonic acids like citric acid, acetic acid, *butyric acid*, lactic acid, amino acid, enzymes, vitamins, steroids etc. are produced by bacteria and fungi.

(12) Which organism plays an important role in production of Ethanol ?

(A) LAB (B) *Saccharomyces cerevisiae*

(C) *Monoscus purpureus* (D) *Arabia gossipe*

(13) Which organisms are useful as energy resources ?

(A) Ethanogenic bacteria (B) *Acetobactor aceti*

(C) Methanogenic bacteria (D) *Clostridium butyricum*

(14) Which gas is produced as energy fuel by photosynthetic microbes ?

(A) N_2 (B) H_2 (C) CO_2 (D) O_2

(15) Which organism is used to produce Penicillin ?

(A) *Aspergillus niger* (B) *Acetobactor aceti*

(C) *Penicillium notatum* (D) *Clostridium butyricum*

(16) Which is not the correct option for antibiotics ?

(A) Carbomycin (B) Acetracin

(C) Fumagillin (D) Tetracycline

(17) Which organism is useful to prepare citric Acid ?

(A) *Aspergillus niger* (B) *Clostridium butyricum*

(C) *Acetobactor aceti* (D) *Lacttobacillus aceti*

(18) What is useful as immunosuppressive agent in Organtransplantation ?

(A) Cyclosporin (B) Cyclosporin a

(C) L - malic acid (D) L - Lysine

(19) For what streptokinase is used ?

(A) To remove oily stains from clothes. (B) To decrease cholesterol in blood.

(C) as immunosuppressive agent during organ transplantation in patients.

(D) to prevent blood clotting in blood vessels

(20) What is useful to decrease cholesterol level in blood ?

(A) *Streptococcus* (B) *Rhizopus nigricans*

(C) *Monoscus purpureus* (D) *Trichoderma polysporum*

(21) What is produced by *Arabia Gossipeae* ?

(A) Vit - B_{12} (B) Riboflavin (C) Lipase (D) Glucomylase

Answers : (12-D), (13-C), (14-B), (15-C), (16-B), (17-A), (18-B), (19-D), (20-C), (21-B)

- **Sewage treatment and microbes in production of biogas :**
  - Municipal or city waste water method for cleaning of urban sewage is known as sewage treatment. Such water is treated through heterotrophic bacteria which passes through the clearing process. Such plants are known as sewage treatment plants. Sewage treatment occurs in two phases.

    (1) Primary treatment process : Physical particles present in a sewage are removed through filtration and sedimentation. Which forms primary sludge.

    (2) Secondary treatment method : It is an aerobic process. These bacteria associated with filaments of molds present in the water and form flocks, Microbes, including bacteria consume organic matter present in effluent hence most amount of organic matter present in a water is utilized due to chemical reaction. So BOD of effluent decreases.
  - Biogas is the mixture of gases, which is used as fuel. Due to their anaerobic respiration like metabolic activities, such mixture of gases are produced in biogas plants. Biogas is known as gobar gas by the people also.

(22) Which are the processes used to remove floating debris and grit ?

(A) Filtration and Leaching (B) Filtration and Sedimentation

(C) Sedimentation and Distalization (D) Filtration and Distalization

(23) Choose the correct option for effluent.

(A) bottom water of sludge producd in primary treatment of STPs.

(B) Upper free water of sludge produced by Filtration and sedimentation.

(C) Remnant water of STP.

(D) Remnant water of activated sludge in STPs.

(24) Which is the biological process of STPs ?

(A) Distalization (B) Filtration (C) Secondary (D) Sedimentation

(25) Where does flocks-formation and flocks sedimentation occur ?

(A) In aeration tank, In effluent settling tank

(B) In anaerobic sludge digestor, Primary tank

(C) Primary tank, Secondary tank

(D) anaerobic sludge digestor, settling tank

(26) Which is the correct sequence for purification in STPs ?

(A) Effluent - Primary sludge - flocks - Activated sludge - Biogas

(B) Flocks - Activated sludge - Effluent - Biogas

(C) Primary sludge - Flocks - Activated sludge - Effluent - Biogas

(D) Primary sludge - Effluent - Flocks - Activated sludge - Biogas

(27) Which is the correct sequence to purify dirty water in cities ?

(A) Aerobic process - Physical process - Anaerobic process

(B) Physical process - Anaerobic process - Aerobic process

(C) Physical process - Aerobic process - Anaerobic process

(D) Anaerobic process - Aerobic process - Physical process

(28) Which are useful bacteria for biogas production ?

(A) *Streptococcus* (B) *Methanotrophs* (C) *Eubacteria* (D) *Methanogens*

(29) Which gas is not present in biogas ?

(A) H_2S (B) CH_4 (C) CO_2 (D) NO_2

(30) On which material bacteria act in biogas production ?

(A) Glycogen (B) Cellulose (C) Lignin (D) Suberin

(31) Which institute of India has put efforts to develop biogas technology ?

(A) IVRI (B) KVIC (C) IARI (D) B and C both

Answers : (22-B), (23-B), (24-C), (25-A) (26-D), (27-C), (28-D), (29-D), (30-B), (31-D)

- **Microbes as Biocontrol Agents and Biofertilisers :**
  - The factors which harm crops in the field of agriculture are normally consider as pest. Mainly insects from fungi, bacteria are important pests decreasing crop production. For their control many synthetic insecticides. Pesticides, fungicides chemicals are available in market. But there is pollution in soil, water, air due to heavy use of pesticides. Now, as an alternative to such chemicals, biochemicals are produced through biocontrol agents. By using such pesticides, natural balance (balance of ecosystems is maintained) *Bt*-Cotton by the help of *Bacillus thuringiensis*, *Trichoderma*, *Baculo viruses* etc act as bio controllers.
  - Biofertilizers are developed as alternative to the pollution. Causing chemical fertilizers. Bacteria, Fungi and Cyanobacteria are useful, microorganisms viz, *Rhizobium, Azospirillium* and *Azotobacter* like bacteria, *Glomus fungi, Anabaena*, Nostoc, Oscillatoria like autotrophs algae, etc.

(32) What is used to control nematodes in cereal crops, vegetables and fruits ?

(A) Fungicides (B) Weedicides (C) Bionemedicides (D) Pesticides

(33) Which Bioinsecticide is used to produce disease in insects and some arthropods ?

(A) *Pseudomonas* (B) *Trichoderma* (C) *Baculo virus* (D) *Rhizobium*

(34) Which organism produces Pest Quantum 4000 that is proved effective for cereals and vegetable crops ?

(A) *Pseudomonas* (B) *Cyanobacteria* (C) *Baculo virus* (D) *Trichoderma Fungi*

(35) Which group of organism is used as biofertilizer ?

(A) *Pseudomonas, Rhizobium, Trichoderma* (B) *Baculo virus, Azospirillum, Cyano bacteria*

(C) *Rhizobium, Azotobactor, Anabaena, Nostoc* (D) *Anabaena, Bacillus thurengiensis, Trichoderma*

(36) Mycorrhiza means...

(A) Symbiotic association of bacteria and fungi (B) Symbiotic association of fungi and plant

(C) Symbiotic association of bacteria and plant (D) None of the above

(37) Which group of organism is N_2 fixative ?

(A) *Azotobactor* and *Azospirillum* (B) *Anabaena* and *Nostoc*

(C) *Azotobactor* and *Oscillatoria* (D) *Azospirillum* and *Anabaena*

(38) Which is the correct option for free living and symbiotic fungi ?

(A) *Glomus* and *Azospirillum* (B) *Trichoderma* and *Glomus*

(C) *Glomus* and *Rhizopus* (D) *Trichoderma* and *Azospirillum*

(39) Which is the common organism for Flocks and Mycorihiza ?

(A) Bacteria (B) Fungi (C) Algae (D) Virus

(40) Which is not proper for Mycorrhiza ?

(A) Increases immunity.
(B) Absorbs phosphorus from soil.
(C) Fixes free N_2
(D) Protects against salinity and drought.

(41) Which of the following group is autotrophic ?

(A) *Oscillatoria, Anabaena, Rhizopus*
(B) Nostoc, Aspergillus, Anabaena, Rhizopus
(C) Oscillatoria, Amoeba, Nostoc, Cyanobacteria
(D) Oscillatoria, Anabaena, Nostoc, Cyanobacteria

Answers : (32-C), (33-C), (34-A), (35-C), (36-B), (37-A), (38-B), (39-B), (40-C), (41-D)

- **True - False (T - F) Type questions :**

Choose the correct option for true and false statements.

(42) (1) Lactobacillus produces Lactic acid.
(2) Lactobacillus is useful to produce curd from milk.
(3) Lactobacillus increase quality of Vitamin K.
(4) Lactobacillus partially digests lipids.

(A) T, F, F, T (B) T, T, F, F (C) T, T, F, T (D) T, T, T, F

(43) (1) Organic matter and Microbes are present in sewage.
(2) Water is released into rivers after purification by autotrophic bacteria in sewage.
(3) Aerobic and anaerobic both type of bacteria are used in sewage treatment.
(4) Ethane, CO_2 and H_2S gases are produced in sewage treatment.

(A) T, F, T, T (B) T, F, F, F (C) T, F, T, F (D) F, T, F, T

(44) (1) Statins are obtained by *Trichoderma polysporum*.
(2) Statins are useful to decrease cholesterol level in blood.
(3) Cyclosporin A is used as an immunosuppressive agent in organ transplant.
(4) Riboflavin is obtained by *Rhizopus nigricans.*

(A) F, T, T, F (B) T, F, F, T (C) F, T, F, T (D) T, F, T, F

(45) (1) Biogas is produced by anaerobic bacteria.
(2) Methane is produced in more amount in biogas.
(3) Methanogens are present in lumen of cattle.
(4) Biogas is mixture of Methane, hydrogen sulphide and carbon dioxide.

(A) T, F, F, T (B) F, F, F, F, (C) T, F, T, T (D) T, T, T, T

(46) (1) Mycorrhiza absortbs sulphur from soil.
(2) Mycorrhiza is symbiotic association of *Trichoderma* fungi and plant.
(3) *Azospirillum* and *Azotobactor* fix N_2 during their symbiotic association.
(4) *Trichoderma* is free living fungi.

(A) F, F, F, T (B) T, T, T, F, (C) T, F, F, T (D) F, T, T, F

Answers : (42-B), (43-C), (44-A), (45-D), (46-A)

- **A - Assertion, R - Reason Questions :**

Choose the correct option of the following :

(A) A and R both are correct. R is the correct explanation of A.

(B) A and R both are correct. R is not correct the explanation of A.

(C) A - True, R - False

(D) A - False, R - True

(47) Assertion A : Livorotatory lysin is a type of acid.

Reason R : Pickles are the result of citric acid fermentation of citrus fruits and vegetables.

(A) (B) (C) (D)

(48) Assertion A : Trichoderma helps the plants to obsorb phosphorus.

Reason R : Quantum - 4000 is prepared by Pseudomonas.

(A) (B) (C) (D)

(49) Assertion A : *Bacillus thurengiensis* is useful to control diseases in crops.

Reason R : Some species of fungi is also used to control diseases in plants.

(A) (B) (C) (D)

(50) Assertion A : Bacteria associated with filaments of molds prepare flocks.

Reason R : Upper free water of primary sludge is called effluent.

(A) (B) (C) (D)

(51) Assertion A : Enzyme Proteases are produced by microbes.

Reason R : Proteases are used to remove oily stains in laundry.

(A) (B) (C) (D)

(52) Assertion A : Photosynthetic microbes produces N_2.

Reason R : These microbes are able to convert solar energy into chemical energy.

(A) (B) (C) (D)

(53) Assertion A : As an alternative to chemical pesticides biochemicals are produced through biocontrol agents.

Reason R : By using biopesticides, natural balance is maintained.

(A) (B) (C) (D)

(54) Assertion A : Large amount of methanogens are present in dung of catlle.

Reason R : Cattle dung is useful in biogas

(A) (B) (C) (D)

(55) Assertion A : Ensilage is food for cattle.

Reason R : Ensilage is prepared by fermentation of proteins present in plants.

(A) (B) (C) (D)

(56) Assertion A : CyclosporinA is useful as immuno-suppressive agent in organtransplantation.

Reason R : CyclosporinA is obtained from *Trichoderma polysporum*.

(A) (B) (C) (D)

(57) Assertion A : LAB increases quality of Vit- B12

Reason R : Lactic acid is produced by Lactobacillus.

(A) (B) (C) (D)

Answers : (47-C), (48-D), (49-B), (50-B), (51-C), (52-D), (53-A), (54-A), (55-C), (56-B), (57-B)

(58) Choose the correct pair :

Column - I	**Column - II**	
(p) *Rhizobium*	(i) Leguminous plants	(A) p - i. q - ii, r - iii, s - iv
(q) *Anabaena*	(ii) Autotrophic N_2 fixative	(B) p - ii, q - i, r - iii, s - iv
(r) *Azotobactor*	(iii) Free living N_2 fixative	(C) p - iii, q - i, r - ii, s - iv
(s) *Glomus*	(iv) Absorption of phosphorus	(D) p - i, q - ii, r - iv, s - iii

(59) Choose the correct pair :

Column - I	**Column - II**
(p) BOD	(i) Food for animals
(q) Flocks	(ii) Association of bacteria and filament of molds
(r) Ensilage	(iii) To check pollution level in water
(s) Biogas	(iv) Anaerobic digestion of biological waste
(A) p - iii. q - ii, r - i, s - iv	(B) p - iv, q - i, r - ii, s - iii
(C) p - i, q - iv, r - iii, s - ii	(D) p - ii, q - i, r - ii, s - iii

(60) Choose the correct pair :

Column - I	**Column - II**	
(p) Streptokinase	(i) To prevent blood clotting	(A) p - i, q - ii, r - iii, s - iv
(q) Penicillin	(ii) To decrease cholesterol	(B) p-i, q - iv, r - ii, s - iii
(r) Statins	(iii) To decrease immunity	(C) p - ii, q - i, r - iii, s - iv
(s) Cyclosporin - A	(iv) Antibiotic	(D) p - iii, q - iv, r - ii, s - i

(61) Choose the correct pair :

Column - I	**Column - II**	
(p) Blue green algae	(i) Causes disease in Arthropods	(A) p - iv, q - i, r - ii, s - iii
(q) Baculo virus	(ii) To increase organic matter in soil	(B) p - i, q - ii, r - iii, s - iv
(r) *Bacillus thurengiensis*	(iii) Bio control agent	(C) p - ii, q - i, r - iv, s - iii
(s) *Trichoderma*	(iv) Produces toxins	(D) p - iii, q - iv, r - i, s - ii

(62) Choose the correct pair :

Column - I	**Column - II**	
(p) *Cyanobacteria*	(i) N_2 fixation	(A) p - iv, q - iii, r - i, s - ii
(q) *Pseudomonas*	(ii) Biogas	(B) p - iii, q - ii, r - iv, s - i
(r) *Rhizobium*	(iii) Biofertilizer in Paddy fields	(C) p - ii, q - iv, r - i, s - iii
(s) *Methanogens*	(iv) Quantum - 4000	(D) p - iii, q - iv, r - i, s - ii

(63) Choose the correct pair :

Column - I	Column - II	
(p) *Aspergillus niger*	(i) Butyric Acid	(A) p - ii, q - iii, r - iv, s - i
(q) *Clostridium butyricum*	(ii) Acetic Acid	(B) p - iv, q - i, r - ii, s - iv
(r) *Acetobactor aceti*	(iii) Lactic Acid	(C) p - iii, q - ii, r - i, s - iv
(s) *Lactobacillus*	(iv) Citric Acid	(D) p - i, q - ii, r - iv, s - iv

(64) Choose the correct pair :

Column - I	Column - II	
(p) CyclosporinA	(i) *Rhizopus nigricans*	(A) p - iv, q - ii, r - i, s - iii
(q) Hydroxy progesterone	(ii) *Monoscus purpureus*	(B) p - iii, q - i, r - ii, s - iv
(r) Statins	(iii) *Arabia gossipiae*	(C) p - iv, q - i, r - ii, s - iii
(s) Riboflavin	(iv) *Trichoderma polysporum*	(D) p - iii, q - i, r - iii, s - ii

Answers : (58-A), (59-A), (60-B), (61-C), (62-D), (63-B), (64-C)

- **Questions for NEET :**

(65) Which organism is useful to prepare biofertilizer ?

(A) Azolla (B) E. coli (C) Spirogyra (D) Cassia

(66) Scientist who invented antibiotic ..... .

(A) Louis Pasteur (B) R. Koch (C) W. Fleming (D) A. Fleming

(67) BOD is measurement of what ?

(A) Organic matter present in water

(B) Industrial waste present in waterbodies

(C) amount of CO combined with Hb

(D) amount of O_2 necessary for plants during night

(68) Choose the correct match for BOD.

(A) Microbes (B) Organic matter

(C) Microbes and Organic matter (D) None of the above

(69) The BOD of polluted water is ..... than norml water.

(A) more (B) less (C) normal (D) zero

(70) Which bacteria helps in production of biogas ?

(A) *Methanogens* (B) *Methanotrophs* (C) *Organotrophs* (D) *Eubacteria*

(71) Which is appropriate pair of the following ?

(A) Rhizobium - Parasite living on Leguminous plants

(B) Mycorrhiza - Enables phosphorus to plants

(C) Yeast - Production of biogas

(D) Slime mold - Ringworm disease

(72) Which is inappropriate pair of the following ?

(A) Yeast - Ethanol (B) *Penicillium* - Penicillin

(C) Methanogens - Biogas (D) *Streptococcus* - Statins

(73) Bio-fertilizer means ...

(A) Cow dung and agricultural waste (B) Crop showing rapid growth

(C) Anabaena and Nostoc (D) None of the above

(74) *Bacillus thurengiensis* is useful for production of ..... .

(A) Bio insecticids (B) Biofertilizer

(C) Bio metalogical method (D) Biomineralization

(75) Which is the character of Bt-Cotton ?

(A) Immunity is more (B) Medium production and long fibres

(C) Long fibre and more production (D) Clustaceary and molo-meshy inhibitor

(76) Constituents of Biogas

(A) 50 - 70 % CH_4 (B) Less CH_4

(C) 30 - 40 % CH_4 (D) 80 % C_2 H_6

(77) In which of the following country Alcohol is used as fuel ?

(A) Germany (B) U.S.A. (C) Brazil (D) China

(78) Which of the following provides energy and fertilizer ?

(A) Petrocrop (B) Biogas (C) Energ crop (D) All of the above

(79) By which process products like Cheese and Toddy are obtained ?

(A) Distillation (B) Pasteurization (C) Fermentation (D) Dehydration

(80) *Azotobacter* and *Azospirillum* are examples of what ?

(A) Symbiotic N_2 fixatives (B) Free living N_2 fixative

(C) Decomposers (D) Pathogens

Answers : (65-A), (66-D), (67-A), (68-C), (69-A), (70-A), (71-B), (72-D), (73-C), (74-A), (75-A), (76-A), (77-C), (78-B), (79-C), (80-B)

•

12 Plant Morphology-I (Root, Stem and Leaf)

The flowering plants are dominant plants on the earth with 3 lac species. They exhibit great variety as shown below :

- **Variety by size :** The smallest gymnospermic plant is *Zamia pygmaea* with 20–30 cm length while greatest gymnospermic plant is *Sequoia sempervirens* with 150 meter height. The smallest aquatic angiospermic plant is Lemna which is less than 1 cm and *Wolffia globosa* is about 2–5 mm in size and the tallest angiospermic plant is *Eucalyptus* sp. with 90–100 meter length.
- **Variety by forms :** Some plants are of Herb, Shrub or tree type, while some are of climber type.
- **Variety by life-span :** Some plants are Annual, while some are Biannual or perennials. e.g., Bodhigaya (Pipal) tree is 2500 years old.
- **Variety by Habitat (Life-style) :** Some plants are terrestrial, aquatic, xerophytic, epiphytic, parasitic, Halophytic in reference to their Habitat.

Normally, phanerogames have two parts : (1) Above ground axis (shoot system) and (2) Underground axis (Root System). Root system develops from Radicle and shoot system develops from plumule.

(1) Which option is correct for root which is developed from Radicle ?

(A) Assimilatory root of Tinospora (B) Prop root of Banyan tree

(C) Food storage root of carrot (D) Haustoria of cuscuta

(2) Which option is correct for classification of plants according to their forms ?

(i) Vinca (ii) Banyan (iii) Ocimum (iv) Eucalyptus (v) Papaya (vi) Bittergourd (vii) Pea

	Herb	Shrub	Tree	Climber
(A)	(i), (iii)	(vi), (v), (iv)	(ii)	(vii)
(B)	(iii), (vi)	(iv), (v)	(ii)	(vii), (i)
(C)	(v), (vi)	(i), (ii)	(ii), (iv)	(vii)
(D)	(i), (iii), (vii)	(v)	(ii), (iv)	(vi)

(3) Which option is correct for given Assertions ?

x : Bodhigaya tree (pipal) is 25,000 years old.

y : Zamia pygmaea and Lemna are smallest angiosperm and smallest gymnosperm plant respectively.

z : Sequoia and Eucalyptus are largest gymnosperm and largest angiosperm plant.

(A) All assertions are wrong (B) All assertions are right

(C) x, y are wrong, while z is right (D) x, z are wrong, while y is right

(4) Which option shows uniform life-form (Habitat) of plants ?

(A) Banyan, Passion flower, Maize, Bougainvillea, Lemna

(B) Bittergourd, Passion flower, Gloriosa, Tinospora, Pea

(C) Lemna, Wolffia, Avicennia, Lotus, Orchid

(D) Opuntia, Acacia, Rose, Maize, Orchid

(5) Which pair is matched ?

(A) Plumule - Fruit, seed (B) Radicle - Primary root, Flower

(C) Plumule - Primary root, Stem, Flower (D) Radicle - Flower, Fruit, Seed

(6) In which plant, Root system and shoot system are not clearly seen ?

(A) Avicennia, Muehlenbeckia (B) Marchentia, Moss (C) Cuscuta, Loranthus (D) Tinospora, Orchid

(7) Which care should be taken to cultivate Bean-seed to observe its germination ?

(A) To cultivate it so as to keep its plumule upper side and radicle lower side.

(B) To cultivate it so as to keep its plumule lower side and radicle upper side.

(C) To cultivate it so as to keep both plumule and radicle upper side.

(D) Bean can cultivate by keeping it at any angle.

Answers : (1-C), (2-D), (3-C), (4-B), (5-A), (6-B), (7-D)

- **Root :** Root is the underground part of the plant axis. It develops from the radicle. It is positively geotropic and hydrotropic and negatively phototropic. It lacks chlorophylls. It is of three types.

(a) Tap root system : When primary root develops from radicle and bears secondary and tertiary roots and the main axis develops perpendicular deep into the soil is called Tap-root. E.g., Mustard.

(b) Adventitious root system : When root develops from stem, leaves, that means from other than Radicle, it is called adventitious root. e.g., Prop root of Banyan tree.

(c) Fibrous Root system : In monocot plants, the primary root is short lived. Later on thin fibrous roots develop from the hypocoty and the basal region of stem. e.g, wheat.

- Normal functions of Root are as given below :

(a) Fixation : To anchor the plant to support (Soil)

(b) Absorption : To absorb water-minerals from Soil.

(c) Conduction : To conduct the absorbed water-mineral to the tip of stem.

(d) Balance and protection : In Pistia, root-tip is protected by root-pocket and keeps balance also.

- **Regions of Root :** Morphological root can be divided into four regions.

(a) Root-cap : In plant like Pandanus, the meristematic tissue at root-tip is protected, while in aquatic plant like Pistia it is in the form of loose covering.

(b) Meristematic region : In this region, meristematic cells are present, which are divisible, filled with protoplasm, thin walled and small in size.

(c) Region of Elongation : The cells of this region vigorouly expands in size and increases length-depth and thickness.

(d) Region of Maturation : This region is also called root-hair region, because it develops unicellular, thin and fibrous hair like structure. The cells undergo differentiation to form tissues. From above these regions, small lateral roots are developed.

---

(8) Normally, Root means ..... .

(A) Positively geotropic axis of plants (B) Positively phototropic axis of plants

(C) Negatively hydrotropic axis of plants (D) All of the given

(9) The structure which develops from Radicle is ..... .

(A) Fibrous Root (B) Adventitious Root (C) Tap Root (D) All of the given

(10) Which one is correct for root that is produced from stem or leaf ?

(A) It has originated from Radicle (B) It is a type of Tap Root

(C) It is also called Adventitious Root (D) None of the given

(11) Which is not applicable to cells of Meristematic region of root ?

(A) Continuous cell division (B) Filled with protoplasm

(C) Thin cell wall (D) Cells are large in size

(12) In which region of root, differentiation is seen ?

(A) Maturation region (B) Elongation region (C) Meristematic region (D) Root-cap region

(13) Which region of root is also known as root-hair region ?

(A) Elongation region (B) Maturation region (C) Root-cap region (D) Meristematic region

(14) From which region of root, lateral roots are arised ?

(A) Elongation region (B) Maturation region (C) Root-pocket region (D) Meristematic region

(15) Which pair is not matched ?

(A) Maturation region - Permanent region of root

(B) Meristematic region - Thin cell walled region of root

(C) Root-cap region - Meristematic region of root

(D) Elongation region - Fast growing in length and volume region

(16) Which function is not applicable to root ?

(A) Ascent of sap (B) Fixation

(C) Absorption of water-minerals (D) None of the given

(17) Which is applicable to a structure which is produced from Radicle ?

(A) It is developed perpendicular to soil and deep into soil.

(B) It absorbs water and minerals from soil

(C) It is called normal root or Tap root

(D) All of the given

(18) Which is applicable to root-hair ?

(A) It increases the surface for absorption (B) It conducts the mineral and water

(C) It anchors plant to the soil (D) All of the given

(19) Which is applicable to root which develops from Hypocotyl ?

(A) It is Tap root (B) It is seen always in perennial plants

(C) It will strongly fix the plants (D) None of the given

(20) What will happen if all the root-caps removed from all the lateral branches and main branch of plant root-system ?

(A) No effect on growth and development of plant

(B) There is possibility of increase of height and depth

(C) There is possibility of absorption of water-mineral

(D) In place of root-caps, root-pocket will be developed

(21) In which plant the meristematic region is protected by root-pocket ?

(A) Banyan tree (B) Pandanus (C) Pistia (D) All of the given

Answers : (8-A), (9-C), (10-C), (11-D), (12-A), (13-B), (14-B), (15-C), (16-A), (17-D), (18-A), (19-D), (20-B), (21-C)

- **Adaptations of Root :**

(A) Storage of Food : Some time alongwith fruit, root also stores food to survive during adverse condition and dormat condition.

(A-1) Tap root : • Carrot : Conical shape • Radish : Fusiform shape • Beet : Napiform

(A-2) Adventetious root : • Sweet potato : Simple tuberous root • Asparagus and Dahila : Fasciculated tuberous root

(B) Mechanical support : When main stem of the plant is not capable of enough support to plant, root modifies for support.

(B-1) Stilt root of Maize : In Maize and Pandanus, adventitious roots arise from node of the stem nearer to the ground and these roots grow obliquely downwards.

(B-2) Prop roots of Banyan tree : Thick and pillar like prop roots develop from thick-heavy and horizontal branch of Banyan tree.

(C) Climbing : Due to weak stem in Pothos, long branched or unbranched, brown adventitious roots develops from nodes and internods of the stem. It secretes sticky material.

(D) Photosynthesis : Plants like Tinospora has few and small leaves, So, it develops suspended, smooth, thin and green thread like root from stem which is known as Assimilatory roots.

(E) Breathing : As the soil of creeks near coastal regions are saline and waterlogged, so it is devoid of oxygen. Plants like Rhizophora and Avicennia which are known as Mangroove plants, produces negatively geotropic and positively phototropic aerial roots from their underground roots which bears lenticels (pores) to exchange O_2 and CO_2. This root is called pneumatophores.

(F) Absorption of moisture : Orchids have no contact with soil. So, it grows on host plants for habitat only. It does not harm host, so it is called epiphytic. It produces adventitious long, green, thick and spongy roots, which has specialized velamen tissue to absorb water (or moisture) from surrounding atmosphere. The cells of this tissue are polygonal, thick walled and arranged in many layers.

(G) Parasitism : Plants depends for minerals, water, food on the host.

(G-1) Total parasite Cuscuta : Cuscuta is non green, leafless plant (climbers) which absorbs water, minerals and food from host by its suckers or haustoria.

(G-2) Partial parasite Loranthus : Loranthus absorbs water and minerals only from host by its suckers. It prepares the food in their leaves.

(H) Symbiosis : Leguminous plants like Bean, Groundnuts, etc. gives habitat and food to Rhizobium bacteria in its root. In return, bacteria fixes free atmospheric nitrogen as nitrogenous salts into roots.

(I) Vegetative Propagation : Adventitious buds occur on tuberous roots of sweet potato that takes part in reproduction.

(22) Although, Orchid grows on host, but still it is not called Parasite but Epiphyte.

(A) It acquires water and minerals only from host.

(B) It acquires only food from host.

(C) It utilizes host for its own protection.

(D) In grows on host for habitat only.

(23) Assimilatory roots means...

(A) Photosynthetic roots (B) Epiphytic roots (C) Parasitic roots (D) Breathing roots

(24) In Coastal regions plants have Breathing roots, Because ..... .

(A) They need more oxygen

(B) Oxygen is less due to salinity

(C) They do not get enough oxygen due to saline atmosphere

(D) All of the given

(25) Which option is correct for the purpose of adventitious bud ?

(A) Carissa - Protection (B) Lawsonia - Protection

(C) Sweet potato - Vegetative propagation (D) Passion flower - Climbing

(26) Loranthus is partial parasite while cuscuta is total parasite, because...

(A) The suckers are present in cuscuta, lacks in Loranthus.

(B) Cuscuta absorbs only water - mineral while Loranthus absorbs water, mineral, food from host.

(C) Loranthus have contact to soil, while cuscuta doesn't have.

(D) Cuscuta is leafless, while Loranthus is leafy plants.

(27) What is the function of Velamen tissue ?

(A) Reproduction (B) Photosynthesis (C) Assimilation (D) Absorption of Moisture

(28) Which pair is mismatched ?

(A) Assimilatory root - Tinospora (B) Stilt Root- Pandanus

(C) Breathing root - Rhizophora (D) Parasitic root - Nepenthes

(29) What is main aim of Assimilatory root ?

(A) To get O_2 through respiration (B) Nutrition through parasitism

(C) Nutrition through photosynthesis (D) Mechanical support

(30) Stilt root is present in Maize. Because...

(A) To absorb extra mineral-water (B) For mechanical support

(C) To absorb extra O_2 (D) For extra absorption of moisture

(31) To separate Breathing root from epiphytic root, which option is correct ?

(A) It is present in Rhizophora (B) It is having lenticels

(C) It is negatively geotropic (D) All of the given

(32) Which plants possesses suckers ?

(A) Rhizophora - Avicennia (B) Loranthus - Cuscuta

(C) Banyan tree - Maize (D) Sweet potato - Dahlia

(33) Which option is correct for location and function of sticky root ?

(A) Location : Pothos, Function : Climbing (B) Location : Tinospora, Function : Assimilation

(C) Location : Avicennia, Function : Breathing (D) Location : Orchid, Function : Nutrition

(34) What benefit does Bacteria (Rhizobium) gets from legume plants as rewards for Nitrogen fixation ?

(A) Protection-Respiration (B) Reproduction-Nutrition (C) Habitat-Nutrition (D) Protection-Reproduction

(35) Which characters are matched for negatively geotropic root ?

(A) Thick, Long, greenish

(B) Thin, long, suspended

(C) Spongy, long, porous

(D) Long, unbranched, grey coloured

Answers : (22-D), (23-A), (24-B), (25-C), (26-D), (27-D), (28-D), (29-C), (30-B), (31-D), (32-B), (33-A), (34-C), (35-C)

- **Stem :** It is plant axis which develops from plumule and is negative geotropic, positive phototropic and negative hydrotropic aerial part.
  - Node : It is condensed part of plant stem which produces leaves
  - Internode : Region between two successive nodes
  - Axil : An angle between leaf and stem
  - Apical bud : Meristematic region of Apex or tip of stem
  - Axillary bud : Meristematic region in the axil of leaf
  - Branching : Arrangement of branches on stem. It is of main two types

    (1) Dichotomous : Apical bud divides into two branches. e.g., Hyphaene (palm)

    (2) Lateral : Branches are produced from axillary bud. It is of two subtypes.

    (i) Racemose : Growth of apical bud remains constant and branches develop from axillary bud. e.g., Polyalthea and Cassuarina. All the branches arises from a single main axis. So the axis is called monopodial axis.

    (ii) Cymose : Apical bud becomes inactive after sometime, branches develops from axillary bud. It is of three types.

    (a) Uniparous : One branch develops only from axillary bud. It has two subtypes.

    (i) Helicoid : Branches are formed only on one side, either on left or right. e.g., Ashoka

    (ii) Scorpioid : Branches are formed alternately on both sides. e.g., vitis.

    (b) Biparous : Two branches develops. e.g., Mirabilis, Carissa.

    (c) Multiparous : More than two branches arises. e.g., Croton, Red oleander.

Note : If the axis of the stem is formed by union of many lateral branches, it is called Sympodial axis.

- **Types of stem :**

  (1) Aerial : It is erect, strong, woody, branched or unbranched. Sometimes in climbers or geoprostrated plants it is weak.

  (2) Underground : It is devoid of chlorophyll, posseses scaly leaves and sometime stores food. Sometime they produce aerial leaves.

- **Normal functions of stem :**

  (1) Proper arrangement of leaves, flowers, fruits and branches

  (2) Bidirectional conduction of food and water-minerals

  (3) To support different organs like branches, flower, fruits

(36) Which is not the characteristic of a structure formed from Plumule ?

(A) It is positive phototropic and negative geotropic

(B) It bears node, internode and leaves

(C) It bears apical bud and axillary bud

(D) It is chlorophyll-less throughout life

(37) Which is competible to plant which have sympodial axis ?

(A) All the branches develops from single axis

(B) A single axis is formed by union of many lateral branches

(C) Apical bud continuously divides into two branches

(D) Apical bud will not become inactive

(38) Which is not competible to branching of Vitis ?

(A) It has Dichotomous branching and monopodial axis

(B) It has scorpioid branching and sympodial axis

(C) It has cymose and uniparous branching

(D) All of the given

(39) Which pair is not matched ?

(A) Mirabilis - Multiparous branching

(B) Palm - Dichotomous branching

(C) Carissa - Biparous branching

(D) Ashoka - Helicoid branching

(40) Which is not the characteristic of a underground stem ?

(A) chlorophyll-less

(B) Strong and woody

(C) With scaly leaves

(D) Small size and food storing

(41) Which is not the function of a stem ?

(A) Bidirectional conduction of food

(B) Bidirectional conduction of mineral-water

(C) Provides support to different organs

(D) Arrangement of different organs

(42) In which type of branching plant becomes conical or pyramidal shape ?

(A) Dichotomous

(B) Cymose - Helicoid

(C) Racemose

(D) Cymose - Multiparous

(43) How Dichotomous branching differs from Biparous branching ?

(A) In Dichotomous, apical bud divides in to two branches.

(B) In Dichotomous, two branches are formed from axillary bud.

(C) Dichotomous and biparous branching are of same type.

(D) Dichotomous present in Palm and biparous present in vitis.

Answers : (36-D), (37-B), (38-B), (39-A), (40-B), (41-B), (42-C), (43-A)

- **Adaptations for stem :**

(A) Storage of food :

(1) Ginger and turmeric : Like root, it is pale yellowish and underground but it posseses node, internode, scaly leaves, adventitious roots, buds, so it is food storing adaptation of stem. It is called rhizome or rootstock.

(2) Potato : Apical bud of underground branches becomes round or oval after food storage. It is called tubers. The pits on potato is known as 'eyes' which is useful in vegetative propagation.

(3) Amorphophallus : Only one internode stores food and becomes condensed and is known as corm.

(B) Vegetative propagation :

(1) Grass, oxalis, Hydrocotyl : From axillary bud, a branch has 2–3 internodes and run parallel to ground and produces new plant from node. This modified branch is called Runner.

(2) Pistia - Eichhornia : From axillary bud short, thick, horizontal branches which are parallel to water surface arises. From the node of these branch new plant develops. This branches is called offsets.

(3) Nephrolepis and Strawberry : Branches arising from basal region grow obliquely like arches, come in contact with ground and produce new plants. This branch is called stolons.

(4) Mint : It shows runner; but it remains underground. So it is called suckers.

(C) Storage of food and reproduction :

In Dioscorea and Agave plants, axillary buds and floral buds respectively store food and become fleshy. Later on they separate from parental plant and produce new plants. Such modified bud is called bulbil.

(D) Protection :

(1) Carissa - Apical bud transform into bifid thorns.

(2) Lawsonia and Pomegranate - Axillary bud tranform into thorn which bears leaves - flowers in rainy seasons.

(3) Rose - Pointed, curved Prickles are produced from surface of stem.

- **Note :** Thorns are meant for protection against Herbivory animals.

(E) Climbing : In passion - flower, cucurbita and Bittergourd axillary bud are modified into touch sensitive tendrils, which twines the support and so is useful in climbing.

(F) Photosynthesis : To reduce the transpiration in xerophytic plants like opuntia and Muehlenbeckia, deciduous leaves are short leaved and shed off in short time. Their stem becomes green and flat and carry out photosynthesis. This stem is called phylloclades.

(44) Bulbil means...

(A) Food storing floral bud (B) Food storing Apical bud

(C) Food storing axillary bud (D) All of the given

(45) Why leaves are deciduous in plants having phylloclades ?

(A) To protect against Herbivory animals (B) To reduce rate of transpiration

(C) Reproduction by leaves (D) All of the given

(46) How many plants shows thorn to gain protection against herbivorous animals ?

Carrisa, Muehlenbeckia, Lawsonia, Passion flower, Pomegranate, Acacia, Zizyphus, Dioscorea

(A) 3 (B) 5 (C) 6 (D) 2

(47) Offset means ..... .

(A) Above ground and obliquely develops branch which comes in contact with soil

(B) Above ground branch and develops parallel to soil

(C) Above water surface branch and develops horizontally

(D) Underground branch develops parallel to soil

(48) Which pair is mis-matched ?

(A) Potato-Rhizome (B) Mint-Runner (C) Muehlenbeckia-Flower bud (D) All of the given

(49) Which modification of stem for protection is not considered actually modification for stem ?

(A) Bifid thorn of carrisa (B) Thorn of Pomegranate

(C) Prickles of Rose (D) None of the given

(50) Where does food stored in Amorphophallus ?

(A) Only in one node (B) Only in one internode

(C) Only in one apical bud (D) Through out whole stem

(51) What is true for thorn of Lawsonia ?

(A) It is uniparous and develops from apical bud

(B) It is biparous and develops from apical bud

(C) It is biparous and develops from axillary bud

(D) It is uniparous and develops from axillary bud

(52) Ginger, although underground, but is not root. Because...

(A) It bears scaly leaves and buds on it (B) It stores food and becomes fleshy

(C) It stores food and perform vegetative reproduction (D) It is completely unbranched

(53) Which characteristic is not of Muehlenbeckia ?

(A) Its stem is green and flat

(B) It bears deciduous leaves to reduce transpiration

(C) Both (A) and (B)

(D) None of the given

(54) How many plants shows tendrils ?

Croton, Passion flower, Dioscorea, Cucurbita, Muehlenbeckia, Pandanus, Bittergourd, Nepenthus

(A) 8 (B) 5 (C) 3 (D) 4

(55) 'Rhizome' and 'Eye' are present respectively in which plants ?

(A) Turmeric, Potato (B) Ginger, Amorphophallus (C) Potato, Ginger (D) Amorphophallus, Potato

(56) Which examples are correct as their thorn bears leaves ?

(A) Acacia - Zizyphus (B) Lawsonia-Pomegranate

(C) Opuntia - Agave (D) Carrisa - Lawsonia

(57) 'Stolon' shows which type of modification ?

(A) Food storage and vegetative propagation (B) Climbing and vegetative propagation

(C) Vegetative propagation (D) Protection and vegetative propagation

(58) How many plants shows double modifications ?

Dioscorea, Muehlenbeckia, Croton, Nepenthus, Carrisa, Pandanus

(A) 1 (B) 2 (C) 3 (D) 5

Answers : (44-D), (45-B), (46-B), (47-C), (48-D), (49-C), (50-B), (51-D), (52-A), (53-D), (54-C), (55-A), (56-B), (57-C), (58-A)

- **Leaf :** It develops from node, green, flat, broad and has limited lifespan and determinate growth and bears axillary bud. It has different parts as given below :

(1) Leaf-base : Stipules are developed laterally from leaf-base. In maize, it is sheathing type of leaf-base.

(2) Petiole : It is stalk like and arranges leaf-blade in a proper manner. Leaf can be divided into two types on base of petiole.

(i) Petiolate : Petiolated leaves show petiole. e.g., sunflower, Vinca **(ii) Sessile :** Petiole is absent. e.g., Maize, Gloriosa

(3) Lamina : It bear veins. Veins forms the skeleton of lamina. There are main two types of venation.

(i) Reticulate : (Dicot leaves) : (i) Unicostate and (ii) Multicostate

(ii) Parallel : (Monocot leaves) (i) Unicostate and (ii) Multicostate

- **Normal function of leaf :** (1) Photosynthesis (2) Exchange of air for respiration (3) Transpiration
- **Phyllotaxy :** "Arrangement of leaves"

(1) Alternate : Alternate development of leaves on axis. e.g., Mustard, Sunflower, Hibiscus

(2) Whorled : More than two leaves arises from node. e.g., Red oleander, Alstonia

(3) Opposite : Leaves arises opposite to each other from node.

(i) Opposite superimposed : e.g., Quisqualis, Guava **(ii) Opposite decussate :** e.g., Calotropis

(59) Which is not function of decussate leaf ?

(A) Gaseous exchange (B) Mechanical support (C) Food-production (D) Food-storage

(60) In which plant, leaf base covers the inter-node ?

(A) Lemon (B) Acacia (C) Moringa (D) Maize

(61) Which structure forms skeleton of leaf-blade ?

(A) Phyllotaxy (B) Venation (C) Petiole (D) Leaf-base

(62) Which option is true for venation ?

(A) In maize, parallel venation is present

(B) In sunflower, Reticute venation is present

(C) Multicostate venation can be divided into convergent and divergent type

(D) All of the given

(63) What is main aim of petiole for leaf ?

(A) Skeleton formation for leaf (B) Helps in function of leaf-blade

(C) Arrange leaf-blade (D) Keeps leaf-blade erect

(64) Which pair is not matched ?

(A) Mustard - opposite to superimposed phyllotaxy (B) Alstonia - whorled phyllotaxy

(C) Calotropis - opposite decussate phyllotaxy (D) Hibiscus - Alternate phyllotaxy

(65) Which option is true for phyllotaxy of Guava ?

(A) Alternate (B) Opposite superimposed (C) Opposite deccusate (D) Whorled

(66) Which option is correct for a plants, whose node forms more than two leaves ?

(A) Red-oleander-whorled (B) Mustard-Alternate

(C) Calotropis-opposite deccusate (D) Guava-opposite superimposed

(67) Which character is not applicable to leaf ?

(A) It arises from node (B) It is green, flat and broad

(C) It has infinite life-span and growth (D) None of the given

(68) Whose function is to connect leaf to node ?

(A) Leaf-base (B) Stipule (C) Petiole (D) Lamina

Answers : (59-B), (60-D), (61-B), (62-D), (63-C), (64-A), (65-B), (66-A), (67-C), (68-A)

- **Types of leaf :** It has two types :

(1) Simple leaf : It has axillary bud in its axil and sometime shows dissected lamina but the incisions does not reach to the mid rib.

(2) Compound : When incision of lamina is upto midrib and leaf-blade divides into small leaf-lets. Leaf-lets do not have axillary bud. It has two types :

(i) Pinnate compound leaves : Leaflets arranges lateral side of midrib.

(a) Unipinnate : Leaflets directly arises on midrib. e.g., Cassia

(b) Bipinnate : Leaflets arises on secondary branch. e.g., caesalpinia, Acacia

(c) Multipinnate : Leaflets arises on tertiary branch. e.g., Moringa

(ii) Palmate compound leaves : Leaflets arranges on tip of mid rib.

(a) Unifoliate : Only one leaflet on tip of Mid-rib. e.g., Lemon

(b) Bifoliate : Two leaflets are present on tip of mid rib. e.g., Balanites

(c) Multifoliate : More than two leaflets on tip of midrib. e.g., Bombax, Aegle.

Note : Scaly leaf, Floral leaf, Bract, Cotyledons, etc. are some other types of leaves.

- **Modification of leaves :**

(1) Food storage : Concentrically arrange long leaves stores food at their leaf-base and they are arranged on underground condensed and disc-like stem. This is called tunicated bulb. Outer most leaf-base remains dry and papery. e.g., Onion.

(2) Support and climbing : In smilax stipules, in Gloriosa leaf-opex, in pea apical 3–5 leaflets transforms into tendrils and in Bignonia apical three leaflets transform into hook or claws for climbing.

(3) Protection : To protect themselves from grazing animals in Acacia and Zizyphus stipules, in Agave leaf apex and in opuntia whole leaf modified into spines.

(4) Photosynthesis : In Pea, stipulas becomes leaflike and is known as foliaceous stipulas. While in Australian accacia, petiole becomes green flat and called phyllode.

(5) Insectovory : In Nepenthus leaf is modified into a pitcher, in Utricularia leaf is modified into bladder for capturing insects. Such plants are rich in Protein, Although they are autotrophic plants.

---

(69) Which is the mode of nutrition in Utricularia plant ?

(A) Autotrophic (B) Insectivory (C) Parasitic (D) Saprophytic

(70) Which content is comparative in high amount in insectivory plants ?

(A) Lipid (B) Protein (C) Carbohydrates (D) All of the given

(71) Phyllode is modification for...

(A) Protection (B) Photosynthesis (C) Support and climbing (D) None of the given

(72) Which two modifications are present in Pea plant ?

(A) Insectivory - Reproduction (B) Reproduction - Climbing

(C) Food Storage - Protection (D) Photosynthesis - Climbing

(73) What is correct for Bignonia ?

(A) Petiole becomes flat and performs photosynthesis

(B) Stipules modified into spines for protection

(C) Apical leaflets transform into tendrils for climbing

(D) Stipules transforms into tendrils for climbing

(74) Which option (pair) is correct ?

(A) Pea - Stipules for climbing (B) Gloriosa - Leaf for protection

(C) Bignonia - petiole for climbing (D) Smilax - Stipules for climbing

(75) Which option is incorrect for Agave ?

(A) Leaf apex is transformed into spine to protect against grazing animal

(B) Floral bud is transformed into Bulbil for reproduction

(C) Axillary bud is transformed into tendril for climbing

(D) None of the given

(76) Which is tunicated bulb ?

(A) Sweet potato (B) Dahlia (C) Onion (D) Amorphophallus

(77) In which plant, foliaceous stipules are present ?

(A) Pea (B) Muehlenbeckia (C) Dioscorea (D) Australian acacia

(78) In which plant concentric leaves and food is stored in leafbase ?

(A) Amorphophallus (B) Onion (C) Dahlia (D) Garlic

(79) Which pair is true ?

(A) Lemon - Unipinnate compound leaf (B) Balanites - Bipinnate compound leaf

(C) Aegle - Bifoliate palmate compound leaf (D) Moringa - Multipinnate compound leaf

(80) Which pair is not matched ?

(A) Caesalpinia - Bipinnate compound leaf (B) Acacia - Multipinnate compound leaf

(C) Bombax - Multifoliate palmate compound leaf (D) Cassia - Unipinnate compound leaf

(81) In which plants, more than two leaflets are present on the tip of midrib ?

(A) Bombax (B) Lemon (C) Balanites (D) Moringa

Answers : (69-B), (70-B), (71-B), (72-D), (73-C), (74-D), (75-C), (76-C), (77-A), (78-B), (79-D), (80-B), (81-A)

(82) Match the column I and II :

Column - I	**Column - II**	
(1) Carrot	(p) Root becomes tendrillar	(A) (1-p), (2-q), (3-r), (4-t)
(2) Gloriosa	(q) Leaf apex becomes tendrillar	(B) (1-u), (2-q), (3-r), (4-s)
(3) Passion flower	(r) Axillary bud becomes tendrillar	(C) (1-p), (2-t), (3-q), (4-r)
(4) Bignonia	(s) Leaflets becomes tendrillar	(D) (1-u), (2-q), (3-t), (4-s)
	(t) Stipules becomes tendrillar	
	(u) Root stores food	

(83) Match the column I, II and III :

Column - I	**Column - II**	**Column - III**
(1) Opuntia	(v) Vegetative propagation	(p) Modification of Axillary bud
(2) Pea	(w) Climbing	(q) Modification of Adventitious root
(3) Tinospora	(x) Support	(r) Modification of Stipules
(4) Dioscorea	(y) Protection	(s) Modification of leaflets
	(z) Photosynthesis	(t) Modification of whole leaf

(A) (1-y-p), (2-z-r), (3-x-s), (4-v-t) (B) (1-y-t), (2-w-s), (3-z-q), (4-v-p)

(C) (1-x-p), (2-z-r) (3-v-q), (4-w-t) (D) (1-z-t), (2-w-s), (3-x-q), (4-v-p)

(84) Match the column I and II :

Column - I	**Column - II**	
(1) Pea	(p) Vegetative propagation - protection	(A) (1-t), (2-s), (3-q), (4-p)
(2) Sweet Potato	(q) Photosynthesis - protection	(B) (1-r), (2-p), (3-q), (4-s)
(3) Opuntia	(r) Food storage - protection	(C) (1-t), (2-p), (3-r), (4-q)
(4) Agave	(s) Reproduction - Food storage	(D) (1-q), (2-s), (3-r), (4-t)
	(t) Climbing - Photosynthesis	

(85) Match the column I and II :

Column - I	Column - II	
(1) Caesalpinia	(p) Three leaflets at tip of midrib	(A) (1-q), (2-p), (3-s), (4-t)
(2) Moringa	(q) Leaflets on tertiary branch	(B) (1-p), (2-t), (3-s), (4-r)
(3) Bombax	(r) Leaflets on secondary branch	(C) (1-r), (2-q), (3-p), (4-t)
(4) Lemon	(s) Two leaflets at tip of midrib	(D) (1-r), (2-q), (3-s), (4-p)
	(t) One leaflet at tip of midrib	

(86) Match the column I and II :

Column - I	Column - II
(1) Super imposed two opposite leaves from node	(p) Mustard
(2) More than two leaves from node	(q) Calotropis
(3) Two opposite decussate leaves from node	(r) Hibiscus
(4) Alternate one leaf from node	(s) Red oleander
	(t) Quisqualis

(A) (1-t), (2-s), (3-r), (4-p) (B) (1-q), (2-t), (3-p), (4-r)

(C) (1-s), (2-t), (3-r), (4-p) (D) (1-t), (2-s), (3-q), (4-p)

(87) Match the column I and II :

Column - I	Column - II	
(1) Bulbil	(p) Photosynthesis	(A) (1-t), (2-r), (3-s), (4-p)
(2) Offset	(q) Climbing	(B) (1-r), (2-s), (3-t), (4-p)
(3) Moisture absorbing root	(r) Food storage and Reproduction	(C) (1-p), (2-s), (3-r), (4-q)
(4) Phylloclade	(s) Vegetative propagation	(D) (1-s), (2-r), (3-p), (4-q)
	(t) Epiphytic root	

(88) Match the column I and II :

Column - I	Column - II	
(1) Food storage modification of internode	(p) Dioscorea	(A) (1-r), (2-q), (3-p), (4-t)
(2) Suckers form for Parasitism	(q) Loranthus	(B) (1-r), (2-u), (3-s), (4-t)
(3) Axillary bud modifies for vegetative propagation	(r) Amorphophallus	(C) (1-s), (2-q), (3-p), (4-t)
(4) Climbing modification of stipules	(s) Muehlenbeckia	(D) (1-r), (2-u), (3-t), (4-s)
	(t) Smilax	
	(u) Cuscuta	

(89) Match the column I and II

Column - I	Column - II
(1) Muehlenbeckia	(p) Petiole - Climbing
(2) Strawberry	(q) Stem - Photosynthesis
(3) Groundnut	(r) Stem - Reproduction
(4) Australian acacia	(s) Petiole - Photosynthesis
	(t) Root - Symbiosis

(A) (1-q), (2-r), (3-s), (3-p) (B) (1-r), (2-u), (3-s), (4-t)

(C) (1-q), (2-r), (3-t), (4-s) (D) (1-p), (2-r), (3-q), (4-s)

Answers : (82-B), (83-B), (84-A), (85-C), (86-D), (87-B), (88-A), (89-C)

- **True - False (T - F) type questions :**

(90) Assertion x : In Pistia; vegetative propagation takes place by runner.

Assertion y : For vegetative purpose, bulbil is present in Muehlenbeckia.

Assertion z : Tap root of sweet potato vegetatively reproduces by adventitious bud.

(A) x and y are wrong, z is right (B) x, y and z all are right

(C) x is right, while y and z are wrong (D) x, y and z all are wrong

(91) Assertion x : Bignonia climbes by hook or claws.

Assertion y : Gloriosa is stem tendrillar plant.

Assertion z : Pothos is a plant which climbes by root.

(A) x, y wrong, while z is right (B) x, z wrong, y is right

(C) y is wrong, x and z are right (D) y, z wrong, x is right

(92) Assertion x : In Pea, leaflets modified for climbing and stipules modified for photosynthesis.

Assertion y : In agave, leaf-base modified for protection and flower bud modified for vegetative propagation.

(A) x is right, y is wrong (B) x is wrong, y is right

(C) x and y both are wrong (D) x and y both are right

(93) Assertion x : Pneumatophore of Avicennia, absorbs moisture from atmosphere due to velamen tissue.

Assertion y : Assimilatory root of Tinospora, Secretes sticky material and by photosynthesis, it compensate the requirement of food.

(A) x is right, y is wrong (B) x is wrong, y is right

(C) x and y both are wrong (D) x and y both are right

(94) Assertion x : In maize, prop root gives extra mechanical support to the plant.

Assertion y : In Australian acacia, petiole transforms into phyllode for photosynthesis.

(A) x is right, y is wrong (B) x is wrong, y is right

(C) x is y both are wrong (D) x and y both are right

(95) Assertion x : Although, *Utricularia plant* "eats" insect, still is known as Autotrophic.

Assertion y : For the purpose of vegetative propagation, sometimes bud transform into bulbil by storing food.

Assertion z : In carrisa apical bud bifids, while in pomegranate axillary buds transforms into spines.

(A) x, y and z all are wrong (B) x, y and z all are right

(C) x, y wrong, z is right (D) x, y right, z is wrong

(96) Assertion x : All the branches arises from main axis in vitis.

Assertion y : More than two leaves arises from a single node in whorled type of phyllotaxy.

(A) x is right, y is wrong (B) x is right, y is right

(C) x is wrong, y is right (D) x and y both are wrong

Answers : (90-D), (91-C), (92-A), (93-C), (94-B), (95-B), (96-C)

(97) The phyllotaxy shown in figure can be seen in this plant.



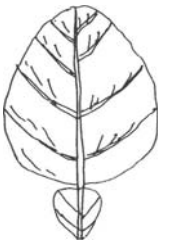
(A) Sunflower, Hibiscus

(B) Quisqualis, Guava

(C) Red oleander, Alstonia

(D) Calotropis, Mustard

(98) Which type of leaf is there in figure ? In which plant is it present ?



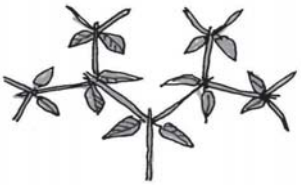
(A) Unipinnate leaf - cassia

(B) Unifoliate palmate leaf - Lemon

(C) Bipinnate leaf - Acacia

(D) Bifoliate palmate leaf - Balanites

(99) Which type of Branching is present in given figure ?



(A) Dichotomous - Palm

(B) Dichotomous - Mirabilis

(C) Biparous - Mirabilis

(D) Biparous - Palm

(100) The branching shown in figure is of ..... ?



(A) Ashoka

(B) Vitis

(C) Balanites

(D) Croton

(101) What is correct for "x" in the given figure ?



(A) Root hair arises from there

(B) New cells constantly added from there

(C) Maximum water-mineral absorbs from there

(D) Maximum size acquired by the cells of this region

Answers : (97-A), (98-B), (99-C), (100-B), (101-B)

- **A - Assestion and R - Reason type questions :**

The answer should be given by following options :

(A) A and R both are correct, R is correct explanation of A.

(B) A and R both are correct, R is not correct explanation of A.

(C) A is right, R is wrong

(D) A is wrong, R is right

(102) Assertion A : Nepenthes is reach in protein.

Reason R : Due to insectivory it is reach in protein.

(A) (B) (C) (D)

(103) Assertion A : Leguminous plants are reach in protein.

Reason R : It gets extra protein by bladder like structure which performs insectivory.

(A) (B) (C) (D)

(104) Assertion A : Loranthus absorbs moisture from host by velamen tissue.

Reason R : Loranthus is partial parasite as it is leafy plant.

(A) (B) (C) (D)

(105) Assertion A : Corm is condensed from of rhizome.

Reason R : In Corm, food is stored only in a single node.

(A) (B) (C) (D)

(106) Assertion A : Palm is an example of biparous branching.

Reason R : In Palm, apical bud continuously divides.

(A) (B) (C) (D)

(107) Assertion A : In opuntia, leaves are modified for protection, while stem is for photosynthesis.

Reason R : As it is xerophytic plants, it shows these type of modifications.

(A) (B) (C) (D)

(108) Assertion A : To reduce the rate of Transpiration, in Australian acacia, compound leaf shed off with petiole.

Reason R : As this is xerophytic plant, so, its show this type of modification.

(A) (B) (C) (D)

(109) Assertion A : Carpels is also called megasporophyll leaf.

Reason R : In course of evolution, leaf transforms into carpels for reproduction purpose.

(A) (B) (C) (D)

Answers : (102-A), (103-C), (104-D), (105-C), (106-D), (107-A), (108-D), (109-A)

- **NEET Questions :**

(110) Ochreate stipules present in which plant family ?

(**Note :** When two stipules joins together to form tubular structure, it is called ochreate stipules.)

(A) Fabaceae (B) Polygonaceae (C) Solanaceae (D) Liliaceae

(111) In which family Nodulated root is present ?

(A) Solanaceae (B) Malvaceae (C) Papilionaceae (D) Liliaceae

(112) What is true for thorn and spine ?

(A) Spine is modified leaf, while thorn is modified stem

(B) Spine is soft while thorn is woody and hard

(C) Both (A) and (B)

(D) None of the given

(113) In which plant, offset is present ?

(A) Mint (B) Eichhornia (C) Cynodon dactylon (D) Pandanus

(114) In which plant, largest leaf-blade is present ?

(A) *Refflesia arnoldii* (B) Nerium (C) Victoria regia (D) Banana

(115) In which plant, Pneumatophores are present ?

(A) Terrestrial plants (B) Xerophyte plants

(C) Halophyte plants (D) Hydrophyte plants

(116) Food storing and beaded root is present in...

(A) Pandanus (B) Ginger (C) Bittergourd (D) Sweet potato

(117) In which plant multicostate convergent reticulate venation is present ?

(A) Cucurbita (B) Cotton (C) Castor (D) All of the given

(118) In plant like Camphor, Hibiscus and Tejpat the venation type is...

(A) Unicostate reticulate (B) Unicostate parallel

(C) Multicostate convergent reticulate (D) Multicostate divergent reticulate

(119) In which plant Radical leaves are present ?

(**Note :** Radical leaf : During favourable season, leaves develop from the nodes of underground stem and seem that they are arising from root.)

(A) Radish (B) Carrot (C) Both (A) and (B) (D) Bryophyllum

(120) How many leaflets are seen on tip of petiole of compound leaf of Marsilia ?

(A) 2 (B) 3 (C) 4 (D) More than four

(121) Which pair is mis-matched for shape of lamina and its related example ?

(A) Grass - Linear (B) Lanceolate - Nerium

(C) Betelnut - Cordate (D) None of the given

(122) Which type of venation is present is Palm and Coconut ?

(A) Multicostate divergent parallel (B) Multicostate convergent parallel

(C) Multicostate divergent reticulate (D) Multicostate convergent reticulate

(123) In which plant more than three multipinnate compound leaf is present ?

(A) Carrot, Coriandrum (B) Sweet potato, Betelnut (C) Grass, Bamboo (D) Radish, Beet

(124) Peripinnate and imperipinnate unipinnate compound leaf respectively present in...

(**Note :** If number of leaflet is odd or even, the unipinnate compound leaf is known as imperipinnate and peripinnate respectively)

(A) Sesbania, Rose (B) Cassia fistula, Neem

(C) Both (A) and (B) (D) Carrot, Coriandrum

(125) In which plant the Intrapetiolar stipules are present ?

(**Note :** When four stipules of two leaves join to form two stipules, are called interpetiolar stipules.)

(A) Gardenia (B) Ixora (C) Hibiscus (D) Desmodium

(126) Which type of stipules are present in Ixora ?

(A) Intrapetiolar (B) Free lateral (C) Foliage (D) Interpetiolar

(127) Ex-stipulated leaves are present in...

(A) Solanaceae (B) Liliaceae (C) Both (A) and (B) (D) Fabaceae

(128) Adnate stipules which are connected to its petiole, is seen in **.....** ?

(A) Zizyphus (B) Smilax (C) Rose (D) Pea

(129) Phylloclade is present in **.....** ?

(A) Australian acacia (B) Pea (C) Passion flower (D) Cactus

(130) Which plant vegetatively propagate by leaf ?

(A) Dioscorea (B) Bryophyllum (C) Muehlenbeckia (D) Agave

Answers : (110-B), (111-C), (112-C), (113-B), (114-C), (115-C), (116-C), (117-D), (118-D), (119-C), (120-C), (121-D), (122-A), (123-A), (124-C), (125-A), (126-D), (127-C), (128-C), (129-D), (130-B)

Adaptation of Root

Adventitious Root

Food stored Fleshy root

This root stores food and swells in different shapes...

- **Fasciculated tuberous root** e.g., Asparagus, Dahlia
- **Tuberous root -** e.g., Sweet potato (*Ipomoea batatus*)
- **Nodulose root -** e.g., Melilotus, Curcuma longa
- **Beaded or Moniliform root** e.g., Portulaca, Momordica
- **Palmate -** e.g. Orchid

Support root

It gives mechanical support

- **Prop root -** e.g., Banyan (Ficus, bengalnensis), *Ficus ementica*
- **Stilt root -** e.g., *Zea mays,* Pandanus
- **Climbing root -** e.g., Pothos, Betel

Tap Root

Food stored Fleshy root

It is also swells up by storing food

- **Fusiform root -** e.g., *Raphanus sativa*
- **Conical Tap root -** e.g., Daucus (Carrot)
- **Tuberous root -** e.g., *Mirabilis Jalapa*
- **Napiform root -** e.g., Brassica rapa, Beta vulgaris

Breathing root

Halophytic plants : Produces negatively geotropic root. They bears lenticels or pneumatophores. e.g., Rhizophora, Avicennia, Jussiaea

Butress root

Root develops from basal part of stem and spreaded parallel to soil.

e.g., Terminalia

Specialized food :

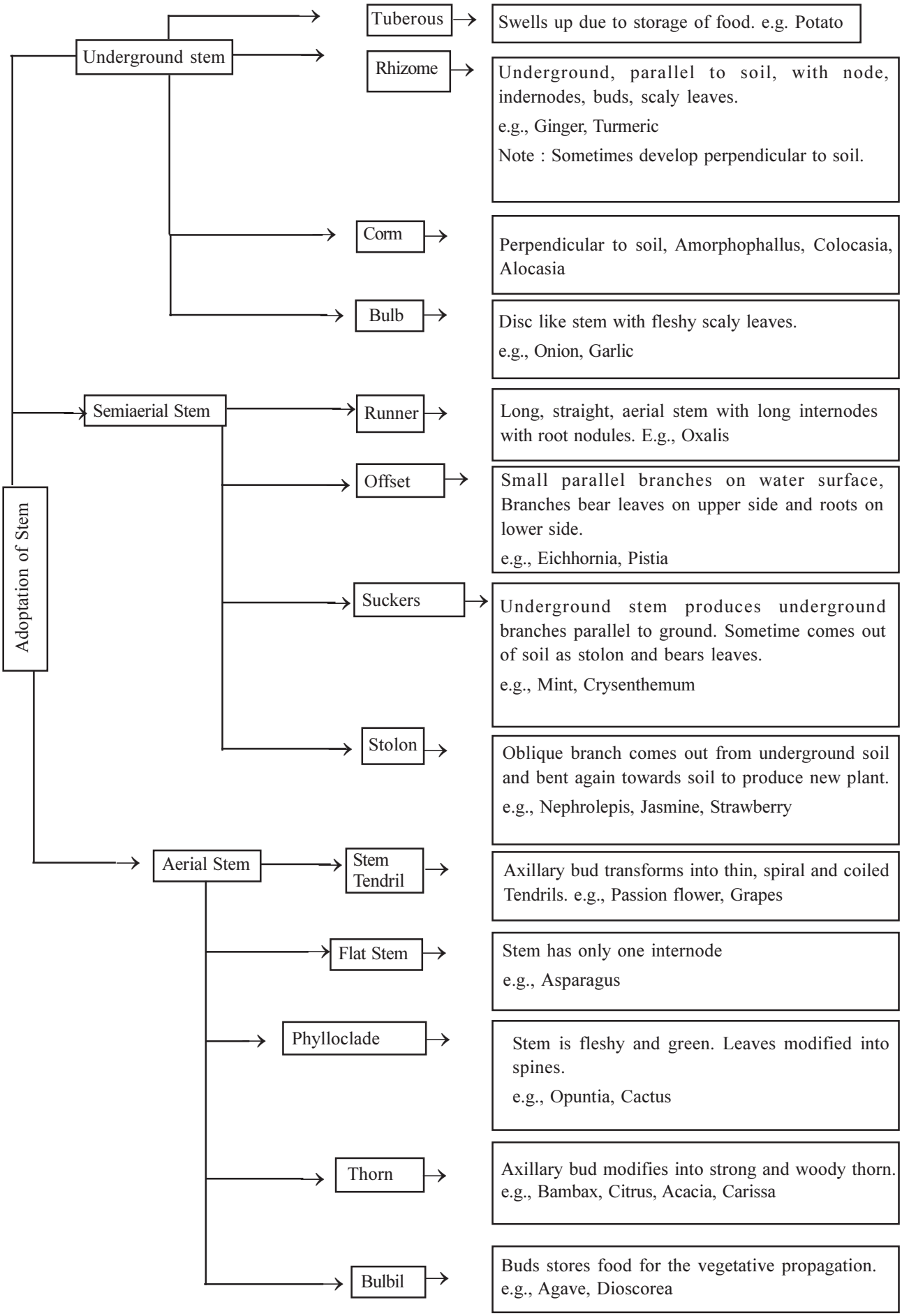
They are performing special functions like photosynthesis, symbiosis etc., according to their functions, they can be classified as under :

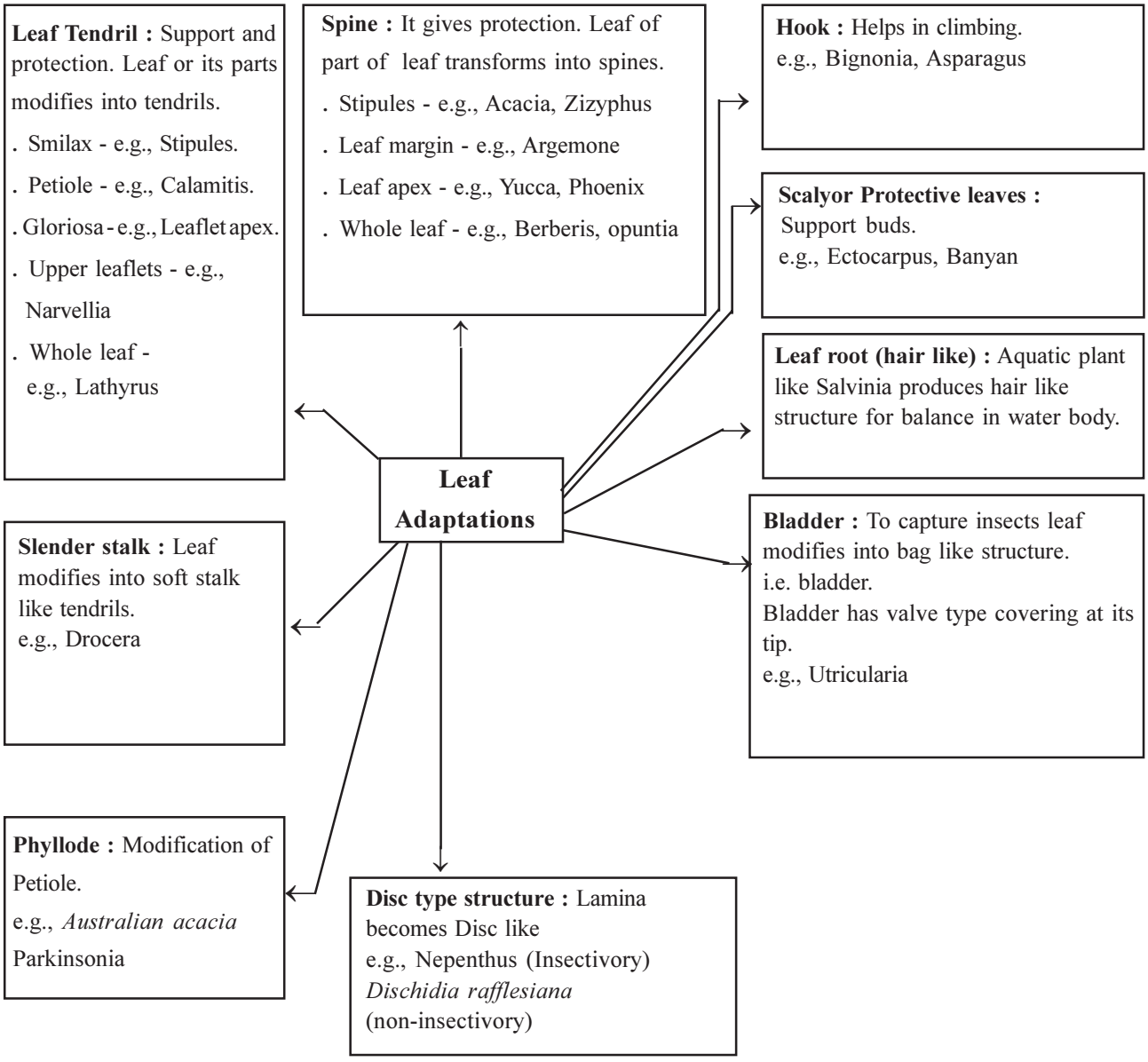
- **Assimilory root -** Tinospora, Trapa
- **Epiphytic root -** Orchid
- **Parasitic root -** Cuscuta, Viscum
- **Root thorn -** Pothos ametum
- **Contractile root -** Crocus, *Alium cepa*
- **Microrhiza root -** Pinus

Root Nonules

Leguminous plantlet let develops Rhizobium into root nodules.

E.g., Pea, Ground nut, *Cicer arietinum*


Adoptation of Stem
Underground stem
Tuberous
Swells up due to storage of food. e.g. Potato
Rhizome
Underground, parallel to soil, with node, indernodes, buds, scaly leaves.
e.g., Ginger, Turmeric
Note : Sometimes develop perpendicular to soil.
Corm
Perpendicular to soil, Amorphophallus, Colocasia, Alocasia
Bulb
Disc like stem with fleshy scaly leaves.
e.g., Onion, Garlic
Semiaerial Stem
Runner
Long, straight, aerial stem with long internodes with root nodules. E.g., Oxalis
Offset
Small parallel branches on water surface, Branches bear leaves on upper side and roots on lower side.
e.g., Eichhornia, Pistia
Suckers
Underground stem produces underground branches parallel to ground. Sometime comes out of soil as stolon and bears leaves.
e.g., Mint, Crysenthemum
Stolon
Oblique branch comes out from underground soil and bent again towards soil to produce new plant.
e.g., Nephrolepis, Jasmine, Strawberry
Aerial Stem
Stem Tendril
Axillary bud transforms into thin, spiral and coiled Tendrils. e.g., Passion flower, Grapes
Flat Stem
Stem has only one internode
e.g., Asparagus
Phylloclade
Stem is fleshy and green. Leaves modified into spines.
e.g., Opuntia, Cactus
Thorn
Axillary bud modifies into strong and woody thorn. e.g., Bambax, Citrus, Acacia, Carissa
Bulbil
Buds stores food for the vegetative propagation. e.g., Agave, Dioscorea


Leaf Tendril : Support and protection. Leaf or its parts modifies into tendrils.
. Smilax - e.g., Stipules.
. Petiole - e.g., Calamitis.
. Gloriosa - e.g., Leaflet apex.
. Upper leaflets - e.g., Narvellia
. Whole leaf - e.g., Lathyrus
Spine : It gives protection. Leaf of part of leaf transforms into spines.
. Stipules - e.g., Acacia, Zizyphus
. Leaf margin - e.g., Argemone
. Leaf apex - e.g., Yucca, Phoenix
. Whole leaf - e.g., Berberis, opuntia
Hook : Helps in climbing.
e.g., Bignonia, Asparagus
Scalyor Protective leaves : Support buds.
e.g., Ectocarpus, Banyan
Leaf root (hair like) : Aquatic plant like Salvinia produces hair like structure for balance in water body.
Leaf Adaptations
Slender stalk : Leaf modifies into soft stalk like tendrils.
e.g., Drocera
Bladder : To capture insects leaf modifies into bag like structure.
i.e. bladder.
Bladder has valve type covering at its tip.
e.g., Utricularia
Phyllode : Modification of Petiole.
e.g., Australian acacia
Parkinsonia
Disc type structure : Lamina becomes Disc like
e.g., Nepenthus (Insectivory)
Dischidia rafflesiana
(non-insectivory)

13 Plant Morphology-II (Flower, Fruit, Seed, Family)

- **Inflorescence :** Arrangement of flowers on rachis (axis) is called Anthotaxy or Inflorescence. It has two types :

(A) **Racemose :** Apical bud is not transformed into flower and flowers formed in acropetal succession. It has 6 types.

(A-i) **Raceme :** Stalked flowers on axis. e.g., Mustard and caesalpinia

(A-ii) **Spike :** Sessile flowers on axis. e.g., Achyranthus

(A-iii) **Spadix :** Flowers bearing axis becomes fleshy. Male flowers on upperside and female flowers remain lowerside. Between male and female flowers, sterile flowers are present. Whole inflorescence in protected by foliage bract, i.e. spathe. e.g., Colocasia, Musa.

(A-iv) **Catkin :** Unisexual flowers on pendulous axis of spike. e.g., Mulberry, Acalypha.

(A-v) **Umbel :** Rachis becomes condensed and stalked flowers are arranged on it like umbrella. The whorl of bracts is called involucre. e.g., Onion, Fennel.

(A-vi) **Capitulum :** Peduncle becomes disc like and flat. It is called receptacle. Flowers are centripetally arranged. Peripheral flowers and central flowers are known as ray florets and disc florests respectively. Like Umbel, it also has involucre. e.g., Sunflowers and Tagetes.

(B) **Cymose :** Apical bud transforms info flowers.

(B-i) **Cymose solitary :** Only one flower is present at the tip of axis. e.g., Hibiscus, Argemone.

(**Note :** There is clear cut demarcation between axis of inflorescence and peduncle.)

(B-ii) **Uniparous (Monochasial) :** The main axis ends in a flower and growth continued by a single lateral branch. It has two types.

(B-ii-a) **Helicoid :** Apical bud gives rise to a single lateral branch before being transformed into flowers. The apical bud of this branch does the same. e.g., Hamelia.

(B-ii-b) **Scorpioid :** In this case lateral branches arise alternately on left and right sides. e.g., Heliotropium.

(B-iii) **Biparous (Dichasial) :** The main axis ends in a flower after producing two daughter axis in flowers. e.g., Jasmine, Nyctanthus.

(B-iv) **Multiparous (Polychasial) :** The main axis ends in a flower after producing more than two branches. e.g., Calotropis, Red oleander.

---

(1) In Rachis of inflorescence bears, male female and sterile flowers. Than this is a type of inflorescence.

(A) Catkin (B) Spadix (C) Spike (D) Raceme

(2) What is correct for inflorescence of Mustard ?

(A) Sessile flowers (B) Stalked flowers

(C) Sterile flowers (D) Small and clustered flowers

(3) In which plant, inflorescence is present with spathe ?

(A) Mulberry (B) Nyctanthus (C) Onion (D) Colocasia

(4) Which type of inflorescence is said if it bears involucre ?

(A) Cymose, Multiparous (B) Racemose, Spike

(C) Racemose, Umbel (D) Racemose, Raceme

(5) What is wrong for inflorescence of Achyranthus ?

(A) It has stalked flowers (B) It is a type of Racemose inflorescence

(C) It has elongated rachis (D) It is a type of spike inflorescence

(6) Which pair is true ?

(A) Nyctanthus - Umbel (B) Heliotropium - Helicoid

(C) Argemone - Scorpioid (D) Jasmine - Biparous, Cymose

(7) "A determinate inflorescence in which the main axis ends in a flower after producing two daughter axis in flowers" – Which option is correct for this sentence ?

(A) Scorpioid - Heliotropium (B) Helicoid - Hamelia

(C) Biparous, Cymose - Nyctanthus (D) Baparous, Racemose - Jasmine

(8) Which statement is true for arrangement of flowers in plant ?

(A) One or more than one flower bearing axis.

(B) Individual axis of every flower is called Pedicel.

(C) Receptacle is modified inflorescence

(D) All of the given.

(9) How many examples are included in cymose inflorescence ?

Red oleander, Hamelia, Mulberry, Musa, Jasmine, Tagetes, Argemone, Mustard

(A) 4 (B) 8 (C) 6 (D) 5

(10) How many examples are included in Racemose type of inflorescence with stalked flowers ?

Nyctanthus, Caesalpinia, Jasmine, Hibiscus, Onion, Musa, Achyranthus, Mulberry

(A) 4 (B) 8 (C) 2 (D) 5

Answers : (1-B), (2-B), (3-D), (4-C), (5-A), (6-D), (7-C), (8-D), (9-A), (10-C)

- **Aestivation :** Arrangement of sepals or petals in the bud codition. It is of five types :

  (i) **Valvate :** Sepals and/or petals without overlaping touch by their margins. e.g., Calotropis, Mustard.

  (ii) **Twisted :** One margin overlapped by other and other margin overlapping. e.g., China Rose, Cotton

  (iii) **Imbricate :** One member is completely outside, one member is completely inside. Rest of the members are arranged like twisted. e.g., Cassia, Delonix.

  (iv) **Quincuncial :** Two members are outside and two members are inside completely. Rest is like twisted. e.g., Cucurbita, Melia.

  (v) **Vexillary :** One largest member is standard, two overlapped members are alae and two smallest members are Keel. e.g., Pea and Bean.

- **Placentation :** The swollen part of ovary from which ovule is produced, is known as Placenta. The arrangement of ovules on placenta is placentation. It has five types.

  (i) **Marginal :** In unilocular ovary, ovules are borne on inner folds of the ovarian wall. e.g., Pea, Bean.

  (ii) **Parietal :** Unilocular ovary becomes bilocular by false septum and ovules are borne on inner periphery wall of ovary. e.g., Mustard and Argemone.

  (iii) **Axile :** Ovules are borne on central axis. Locules of ovary is based on numbers of carples. e.g., Tomato, Hibiscus.

  (iv) **Free central :** There is no partitions in ovary and ovules are arranged on central axis which arises from the base of a ovary. e.g., Dianthus

  (v) **Basal :** Ovule is borne from base of the ovary. e.g., Sunflower

(11) What is applicable to Dianthus for its placentation ?

(A) No partition in ovary (B) False septum is present

(C) Ovule is at base of ovary (D) Ovule is at periphery of ovary

(12) What is true for vexillary aestivation ?

(A) Smallest alae, two, joint (B) Standard are small, joint, two

(C) Standards are largest, two, free (D) Keel are two, joint, and smallest

(13) What is odd for vexillary aestivation ?

(A) Alae are medium of size, two, free, present in pea

(B) Standards are two, largest, free, present in Bean

(C) Keel are smallest, two, joint, present in Pea and Bean

(D) None of the given

(14) Which pair is mis-matched ?

(A) Hibiscus - Axile (B) Dianthus - Basal (C) Bean - Marginal (D) Argemone - Parietal

(15) Which pair is matched ?

(A) Cotton - Valvate (B) Mustard - Twisted

(C) Cucurbita - Quincuncial (D) Cassia - Vexillary

(16) Which plant shows twisted aestivation and axile placentation ?

(A) Sunflower (B) Hibiscus (C) Datura (D) Calotropis

(17) What is true for Mustard plant ?

(A) Valvate aestivation, Parietal placentation

(B) Twisted aestivation, Parietal placentation

(C) Valvate aestivation, Marginal placentation

(D) Twisted aestivation, Marginal placentation

(18) One chambered ovary is becomes two chambered due to formation of false septum-Name the type of placentation and give it location.

(A) Axile - Hibiscus (B) Marginal - Pea

(C) Free central - Dianthus (D) Parietal - Argemone

(19) Which type of placentation is present in a plant having vexillary aestivation ?

(A) Axile (B) Parietal (C) Marginal (D) Basal

(20) What is suitable for a plant having only one ovule ?

(A) Marginal - Pea (B) Basal - Sunflower

(C) Axile - Hibiscus (D) Free central - Dianthus

Answers : (11-A), (12-D), (13-B), (14-B), (15-C), (16-B), (17-A), (18-D), (19-C), (20-B)

- **Flower :** Well developed flower is sexual reproductive organ of angiosperm plants. Its stalk like organ is called Pedicel and its swollen part of tip is called thalamus. From out side to inside calyx and corolla are two accessory whorls and Androecium and gynoecium are two essential whorls.
- **Calyx :** Every member of calyx is sepal. It protects flower in bud condition. It is normally green, fleshy and foliage type. They are either free or united.
- **Corolla :** Every member of corolla is Petal. It is of tubular, bell, funnel, wheel or butterfly in shape. It attracts insects for pollination. As it posses colour, scent and sometime honey gland. It is free or united.

  Note : In plants like crinum. Bougainvillea, Panchratium the petals and sepals are similar in reference to colour and size. They are known as Perianth. (Tepals)
- **Androecium :** Every member of Androecium is stamen. Stamen has Anther at its tip, filament and connective connects filament and Anther. It can be divided in (i) Monodelphous (China rose), (ii) Didelphous (pea) and (iii) Polydelphous (citrus).

Sometime stamens are united with petals, they are epipetalus stamens. e.g., Hibiscus, Cotton, Datura. In beginning the pollen sac is four chambered. When pollen becomes mature and dehiscence from slit or pore, at that time pollen sac becomes two chambered. Pollens are smooth, spiny, edible type.

- **Gynoecium :** Every member of gynoecium is carpel. It is divided into three regions like, stigma which is receptive for pollens, tubular style and basal swollen region ovary bears ovules. It can be divided in Monocarpellary flower (Pea), or Polycarpellary flower. Polycarpellary flowers can be again divided into (a) Apocarpous - e.g., Rose, Lotus (free carpels) or (b) Syncarpous - e.g., Datura, Hibiscus (united carpels).

---

(21) What is true for Hibicus in reference to its aestivation, placentation, Androecium and Gynoecium ?

(A) Twisted, Axile, Monodelphous, Polycarpellary united

(B) Valvate, Axile, Monodelphous, Polycarpellary free

(C) Twisted, Parietal, Diadelphous, United Polycarpellary

(D) Valvate, Marginal, Diadelphous, Free Polycarpellary

(22) Which plant posses syncarpous gynoecium and epipetalous stamens ?

(A) Mustard (B) Pea (C) Datura (D) Lemon

(23) Which family posseses vexillary aestivation, marginal placentation, Diadelphous stamens and monocarpellary flower ?

(A) Solanaceae (B) Liliaceae (C) Papilionoideae (D) Malvaceae

(24) What is Perianth ?

(A) When sepals and petals are dissimilar in shape-size-colour

(B) When sepals and petals are similar in shape-size-colour

(C) When stamens and carpels are dissimilar in shape-size-colour

(D) When stamens and carpels are similar in shape-size-colour

(25) Which pair is matched ?

(A) Syncarpous carpels - Lotus (B) Diadephous stamens - Hibiscus

(C) Monocarpellary flower - Hibiscus (D) Polydelphous stamen - Citrus

(26) Which pair is mis-matched ?

(A) Syncarpous carpels - Hibiscus (B) Diadelphous stamens - Citrus

(C) Monocarpellary flower - Pea (D) Monodelphous stamen - Hibiscus

(27) How many chambers are present in mature anther ?

(A) 1 (B) 2 (C) 3 (D) 4

(28) How many parts are of carpels among given parts ?

Style, Anther, Stigma, Filament, Ovary, Pollen, Connective

(A) 4 (B) 3 (C) 5 (D) 2

(29) These are the parts of a whorl comes next to corolla on thalamus ?

(A) Connective, Stigma (B) Anther, Stigma

(C) Anther, Style (D) Anther, Filament

(30) Calyx and corolla are known as accessory whorls. Because...

(A) They do not take part in reproduction (B) In absence of them, reproduction is possible

(C) They do not produces any reproductive cells (D) They produces helper reproductive cells

Answers : (21-A), (22-C), (23-C), (24-B), (25-D), (26-B), (27-B), (28-B), (29-D), (30-C)

- **Types of flower :**

  (i) According to number of floral whorls : (1) Complete : All whorls are present (2) Incomplete : Any one or more whorls are absent.

  (ii) According to presence of essential whorls : (1) Bisexual : Both essential whorls are present (2) Unisexual : (Male = only Androecium is present) (Female = only Gynoecium is present)

  (iii) According to symmetry : (1) Actinomorphic (Radial symmetry) : Flower can be bisect in two equal halves by any plane. e.g., Hibiscus, Datura. (2) Zygomorphic (Bilateral symmetry) : Flower can be bisect in two equal halves by only one plant. e.g., Bean, Caesulpinia.

  (iv) According to number of floram components : (1) Isomerous : All the components are equal in number. e.g., trimerous, tetramerous, pentamerous.

(2) Heteromerous : All the components are not equal in number.

(v) According to presence of Bract : (1) Bracteate : Bract is present. e.g., Bougainvillea

(2) Ebracteate : Bract is absent. e.g., caesalpinia

(vi) According to location of ovary : (1) Hypogynous flower : Thalamus becomes conical shape on which ovary located at highest position, so, ovary is called superior. e.g., Mustard, China rose, Datura.

(2) Perigynous flower : Thalamus becomes flat and disc like on which ovary located semi-inferiorly. e.g., Rose, caesalpinia.

(3) Epigynous flower : Thalamus envelops ovary and ovary is inferior. e.g., sunflower, cucumber.

---

(31) What type of a flower is called, which does not arise from axil of a leaf ?

(A) Heteromerous (B) Ebracteate (C) Hypogynous (D) Radial

(32) What type of a flower is called, when all the members are equal in number and gynoecium is absent ?

(A) Heteromerous - Unisexual (B) Isomerous - Bisexual

(C) Isomerous - Unisexual (D) Heteromerous - Bisexual

(33) In spadix inflorescence, the types of flowers is ..... .

(A) Unisexual (B) Bisexual (C) Only male (D) Only female

(34) In which plant, a flower is present, which can be divided in equal two halves by any plane ?

(A) Bean (B) Pea (C) Caesalpinia (D) Datura

(35) Zygomorphic flower means ..... .

(A) Unisexual flower (B) All member are equal in number

(C) Bisect by any plane (D) Bisect by only one plane

(36) Which pair is suitable for flower type and its related example ?

(A) Superior ovary - Datura (B) Semi inferior ovary - Rose

(C) Inferior ovary - Cucumber (D) All the given

(37) Which pair is not matched for type of flower and its related example ?

(A) Superior ovary - Caesalpinia (B) Semi inferior ovary - Rose

(C) Inferior ovary - Sunflower (D) All the given

(38) Which option is correct for a flower whose ovary is located at lower position in compare to other members ?

(A) Hypogynous flower, Inferior ovary (B) Epigynous flower, Inferior ovary

(C) Perigynous flower, Inferior ovary (D) Perigynous flower, Superior ovary

(39) What is true for Hibiscus ?

(A) Flower can be divided in equal two halves by any plane.

(B) The aestivation is twisted type.

(C) It has axile placentation and have many carpels but one ovary only.

(D) All of the given.

(40) Which term is improper for Bean flower ?

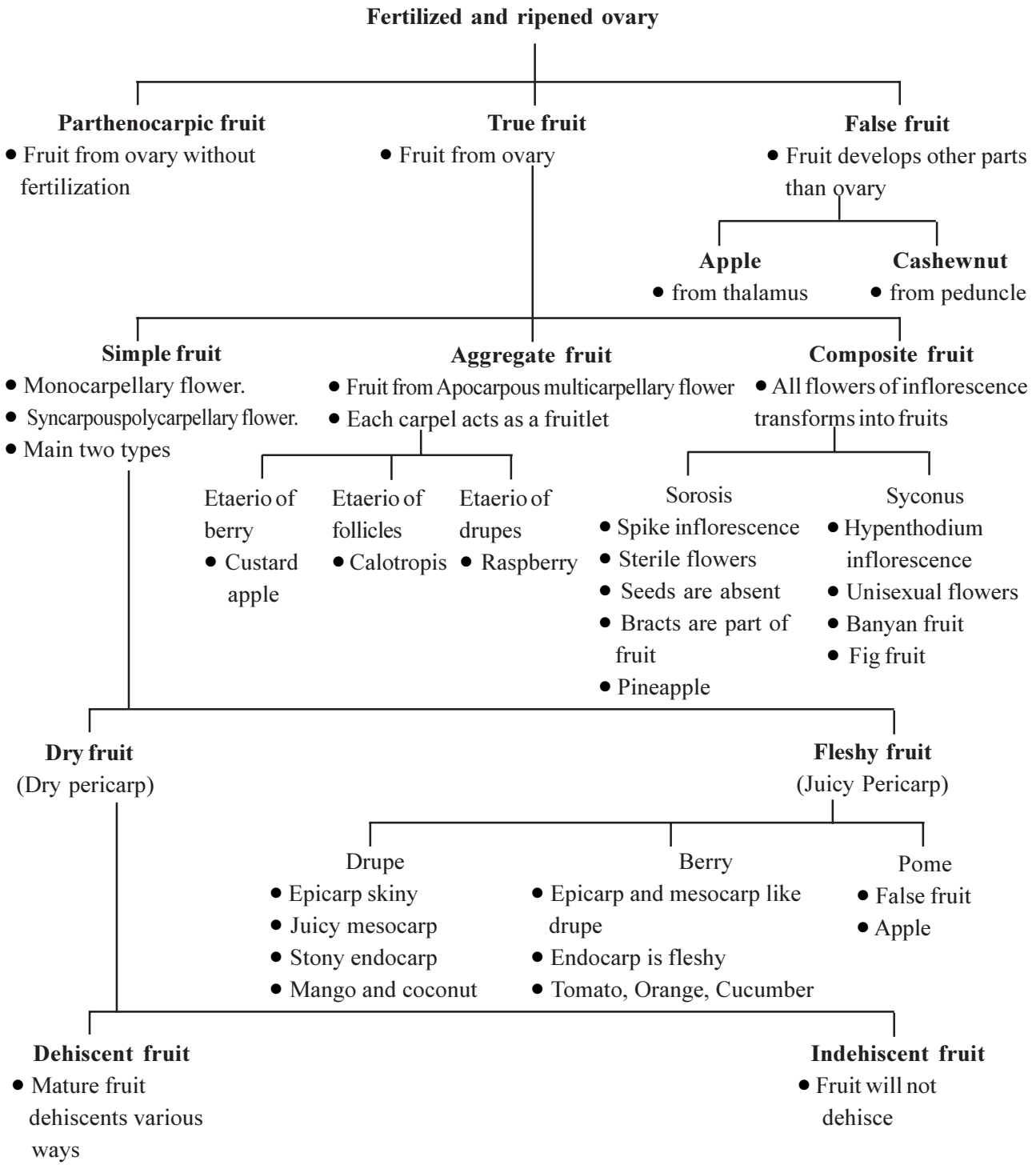
Zygomorphic flower, Vexillary aestivation, Marginal placentation, Isomerous flower, Unisexual flower, Bracteate flower

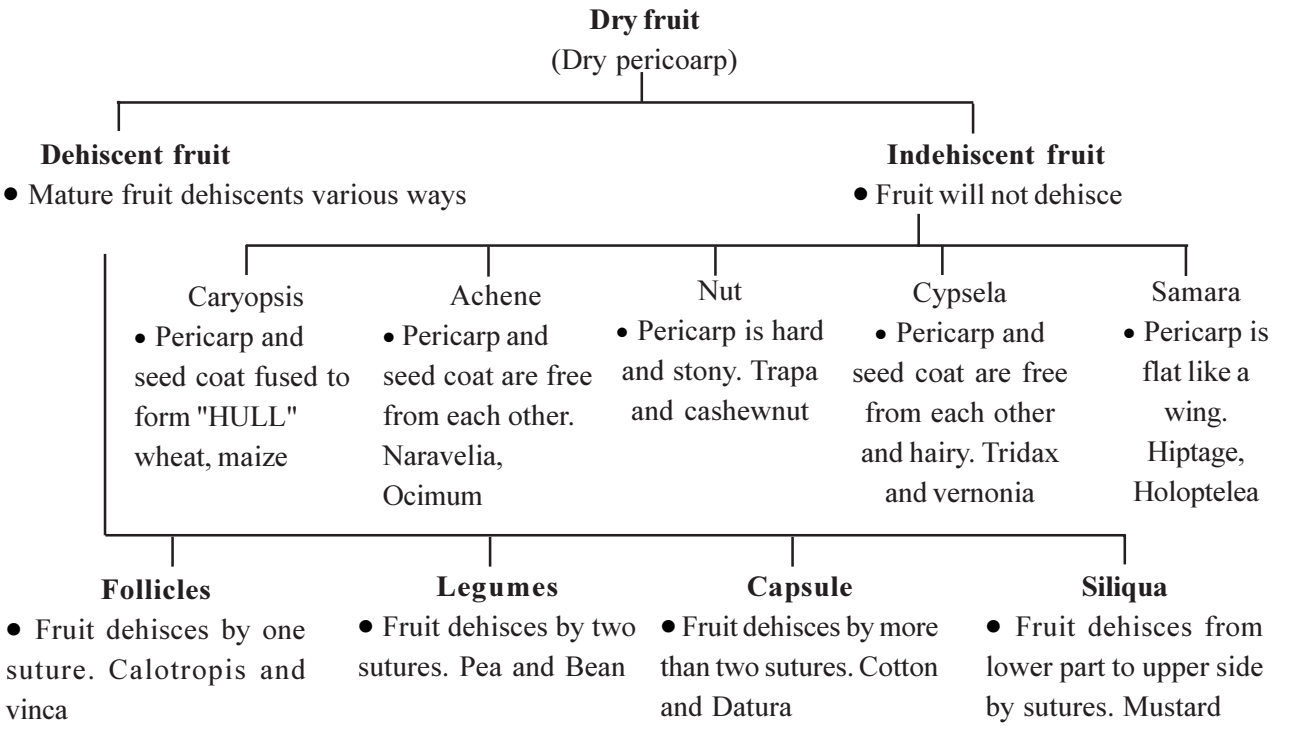
(A) Isomerous flower (B) Unisexual flower

(C) Bracteate flower (D) Marginal placentation

Answers : (31-B), (32-C), (33-A), (34-D), (35-D), (36-D), (37-A), (38-B), (39-D), (40-B)

- **Fruit :**





Note : Calotropis flowers are bicarpellary. Stigma are joint but style and ovary are free, so that it can be included both in simple fruit and aggregate fruit as well.

(41) Which fruit has fleshy mesocarp and endocarp ?

(A) Drupe - Coconut (B) Drupe - Orange (C) Berry - Mango (D) Berry - Tomato

(42) Parthenocarpic fruit means...

(A) Seedless fruit develops after fertilization (B) Seedless fruit develops without fertilization

(C) Seeded fruit develops after fertilization (D) Seeded fruit develops without fertilization

(43) Apple and cashew nut are called false fruit. Because...

(A) Seedless fruit develops without fertilization

(B) Seeded fruit develops without fertilization

(C) Main fruit is not developed from ovary

(D) No role of ovary in formation of fruit

(44) How many dehiscence and indehiscence fruits are given respectively among given list of fruits ?

Mustard, Vinca, Mango, Naravelia, Vernonia, Hiptage, Datura, Coconut, Calotropis, Trapa, Apple

(A) 4, 4 (B) 4, 5 (C) 5, 4 (D) 3, 3

(45) Which pair is mis-matched ?

(A) Hull - maize (B) Wing like pericarp - Holoptelea

(C) Hard pericarp - Nut (D) Hairy pericarp - Naravelia

(46) Which pair is matched ?

(A) Follicles - Pea (B) Legumes - Vinca

(C) Capsule - Datura (D) Siliqua - Cotton

(47) Which is the characteristic of dehiscence in fruit of Mustard ?

(A) Dehisces by more than two sutures

(B) Dehisces by two sutures from lower to upper part

(C) It is a type of fruit which develops without fertilization

(D) Legume fruit with a single seed

(48) What is true for the fruit of Raspberry ?

(A) Mesocarp and endocarp both are fleshy.

(B) Only Mesocarp is fleshy or juicy

(C) It is an example of composite drupe fruit

(D) All the flowers of inflorescence transforms into fruits

(49) Which option is correct for the fruit of Fig ?

(A) It has bisexual flowers (B) It has only male or female flowers

(C) It has male, female and sterile flowers (D) It has only sterile flowers

(50) In development of Banyan tree fruit, which part forms fruit in addition to ovary ?

(A) Peduncle (B) Receptacle (C) Sepals (D) (B) and (C) both

(51) What is main structural difference between fruit of orange and coconut ?

(A) Epicarp (B) Mesocarp

(C) Endocarp (D) None of the given

(52) Composite fruits means...

(A) Many carpels of a single flowers transform into fruit

(B) Every carpels of many flowers transform into fruit

(C) 'A' and 'B' both (D) None of the given

(53) Which statement is not applicable to custard apple ?

(A) Each carpels of a single flower transform into fruitlets.

(B) Endocarp of fruitlet is hard and stony.

(C) It is Etaerio of berry type fruit.

(D) None of the given.

(54) Which option is not applicable to pineapple ?

(A) Fruit develops from cymose infloresence

(B) Flowers are normally Sterile

(C) Seed formation is very rare

(D) Bracts are a part of fruit

(55) Which option is correct for a fruit which forms hull by fusion of pericarp and seed coat ?

(A) Achene - Mirabilis (B) Nut - Cashew nut

(C) Samara - Holoptelea (D) Caryopsis - Maize

Answers : (41-D), (42-B), (43-C), (44-A), (45-D), (46-C), (47-B), (48-B), (49-B), (50-B), (51-C), (52-B), (53-B), (54-A), (55-D)

- **Seed :**

Fertilized Ovule

According to number of cotyledons

Monocot
(Single Cotyledon)
- Maize, Wheat

Dicot
(Two Cotylodons)
- Pea, Bean

According to presence of Endosperm

Endospermic
- (Food Stored in Endosperm)
  Maize, Castor

Non-Endospermic
- (Food Stored in cotyledons)
  Bean, Pea

- **Structure of Bean Seed :** Hilum is on seed which is meant for connection of seed with fruit. It is flat, kindey shape and yellowish white in colour. It is dicot and non-endospermic seed. It has parts are as under :

Seed

Seedcoat

Testa (outer)
- Tough and yellow

Tegman (Inner)
- Thin and Transparent

Embryo

Two large cotyledons
(White, Fleshy)

Tip of plumule
(It is protected by two papery and minute leaves)

Embryonic axis

(Top) Tip region
- Plumule
  (forms shoot system)
- Region between plumule and cotyledons is known as Epicotyl

Lower region
- Radicle
  (forms root system)
- Region between radicle and two cotyledons is known as hypocotyl

- **Structure of Maize Seed :** It is narrow at one end and broad at other end. It is yellow in colour. Longitudinal section of seed stained with iodine shows purple endospermic region due to storage of starch and embryo is stained yellow due to protein. It has parts as mentioned below :

Endosperm Region

Hull or (Seedcoat and pericarp fused)

Aleurone Protein rich layer

Main large region starch is stored

Between these two regious, enzyme rich epithelial layer is present

Embryo Region

Shield shaped Scutellum (Cotyledon)

Embryonic axis

Radicle
- Covered with coleorhiza

Plumule
- Covered with coleoptile

(56) What type of cotyledons are present in non-endospermic seed ?

(A) Thin (B) Fleshy (C) Flesh or thin (D) With endosperm

(57) What is the function of coleoptile and coleorhiza ?

(A) Protection (B) Nutrition (C) Support (D) Photosynthesis

(58) Scutellum means...

(A) Fleshy single cotyledon (B) Thin single cotyledon

(C) Fleshy two cotyledons (D) Thin two cotyledons

(59) What is stored in Aleurone layer ?

(A) Starch (B) Lipid (C) Protein (D) Enzyme

(60) Which one is the largest region in Maize seed ?

(A) Scutellum (B) Endosperm (C) Plumule (D) Radicle

(61) Hilum means...

(A) Food storage region of seed (B) Water absorbing part of seed

(C) Connecting part of seed to fruit (D) Endosperm region of seed

(62) Which is not the characteristic of tegmen of Bean seed ?

(A) Leathery (B) Thin (C) Transparent (D) Soft

(63) What is true for cotyledons of castor seed ?

(A) Endospermic - fleshy (B) Non-endospermic and fleshy

(C) Endospermic - thin (D) Non-endospermic and thin

(64) Why Bean seed is known as non-endospermic ?

(A) It has no nutrition for germination.

(B) It has endospermic tissue in endosperm.

(C) Endosperm is present but has no nutrition.

(D) Nutrition of endosperm is absorbed by cotyledons

(65) Which statement is applicable for a seed which has coleoptile and coleorhiza ?

(A) Largest region of this seed posseses protein.

(B) Aleurone region of this seed posseses starch.

(C) Scutellum is fleshy and purple coloured.

(D) It has hull around it.

Answers : (56-B), (57-A), (58-B), (59-C), (60-B), (61-C), (62-A), (63-C), (64-D), (65-D)

- **Floral formula :** In floral formula, number of each components of flower, connection of each component with each other, location on the thalanus, etc. has been given by symbols as mention below :

Bract :	**Symmetry :**	**Sex :**
Br = Bracteate	$\oplus$ = Regular flower or Actinomorphic	$\bigcirc\!\!\nearrow$ = Male flower (Staminate flower)
Ebr = Ebracteate	θ or % = Irregular flower or Zygomorphic	$\underset{+}{\bigcirc}$ = Female flower (Pistillate flower)
Brl = Bracteolate		$\underset{+}{\bigcirc\!\!\nearrow}$ = Bisexual flower

Calyx :	**Corolla :**	**Perianth :**
K = Sepals	C = Petals	P = Tepals
K_4 = Four free sepals	C_4 = Four free petals	P_6 = Six free tepals
$K_{(4)}$ = Four united sepals (fused)	$C_{(4)}$ = Four fused petals	$P_{(6)}$ = Six fused tepals
		P_{3+3} = Six tepals in two whorls of three each

- **Androecium :**

A = Stamen, A_5 = Five free Stamens, $A_{(5)}$ = Five fused Stamens, K_{5+5} = Ten Stamens in two whorls of five each

A_0 = Stamens absent, A_∞ = Stamens indefinite in number, $\overline{CA}$ = Epipetalous Stamens, $\overline{PA}$ = Epiphyllous Stamens

- **Gynoecium :**

G = Carplels, G_2 = Two free Carpels $G_{(2)}$ = Two fused Carpels, G_0 = Carpels absent, $\underline{G}_{(2)}$ = Bicarpellary, syncarpous superior ovary, $G_{\overline{(2)}}$ = Bicarpellary, syncarpous semi-inferior ovary, $\overline{G}_{(2)}$ = Bicarpellary, syncarpous inferior ovary

- **Floral Diagram :**

Top dot shows position of axis with respect to flower. From outer to inner side Calyx, Corolla Androecium and gynoecium is mentioned. Extrose anthers are to be faced towards the petals and introse authers are to be faced towards the gynoecium. Staminoids are represented by $(\times)$ or $(*)$, i.e. cross or asterisk. Bract is also mentioned and if possible placentation is also mentioned.

(66) What is true for floral diagram with symbol "$(*)$" ?

(A) Only self pollination is possible (B) Only cross pollination is possible

(C) Any type of pollination is possible (D) No pollination takes place

(67) Extrose stamens means; The length of stamens is...

(A) more than petals (B) less than petals

(C) responsible for cross pollination (D) responsible for self pollination

(68) Which is not true for a flower with symbol $\overline{PA}$?

(A) Sepals and petals are similar in colour-shape size.

(B) Stamens are epipetalous type.

(C) Perianth is present instead of calyx and corolla.

(D) Stamens are sterile and in large number.

(69) What is correct for a floral formula with symbol $\bigcirc\!\!\!\!\nearrow$?

(A) G_2 (B) A_0 (C) G_0 (D) All of the given

(70) Which symbols are correct for Datura flower ?

(A) $G_{\overline{(n)}}, \oplus$ (B) $G_{\overline{(n)}}, \Phi$ (C) $G_{\overline{n}}, \oplus$ (D) G_0, Φ

(71) $G_{\overline{\infty}}$ This symbol is applicable to which flower ?

(A) Hibiscus (B) Cucumber (C) Rose (D) Mustard

(72) What can be assumed for a floral formula with "$(\times)$" symbol ?

(A) Sterile Stamens (B) Stamens are absent

(C) Large number of Stamens (D) None of the given

(73) Which pair is mis-matched ?

(A) Bougainvillea - Br (B) Caesalpinia - Ebr (C) Bean - $\oplus$ (D) Datura - %

(74) Which floral formula is not according to rule or is not practical ?

(A) $\oplus$, $\varphi\sigma$, K_{2+2}, C_4 ... (B) Φ, $\varphi\sigma$, K_5, C_5 ...

(C) $\oplus$, $\varphi\sigma$, K_3, C_3, P_{3+3} ... (D) Φ, $\varphi\sigma$, $K_{(5)}$, $C_{(5)}$ ...

(75) Ebr, $\oplus$, $\varphi\sigma$, $K_{(5)}$, $\overline{C_{(5)}, A_5}$, G_0 - What is wrong with this floral formula ?

(A) In regular flower, stamens are not epipetalous (B) Epipetalous stamens are not free

(C) In bisexual flower carpels are not absent (D) All of the given

Answers : (66-D), (67-A), (68-B), (69-C), (70-A), (71-C), (72-A), (73-C), (74-C), (75-C)

- **Family :**

(1) Fabaceae	(2) Solanaceae	(3) Liliaceae
• Class → Dicot	Dicot	Monocot
Subclass → Polypetalae	Gamopetalae	----
Series → Calyciflorae	Bicarpellatae	Coronariae
Order → Rosales	Polymoniales	----
Family → Fabaceae	Solanaceae	Liliaceae
• e.g. *Cajanus cajan* (Pegion pea)	*Datura fastuosa* (Datura)	*Alium cepa* (onion)
Phaseolus mungo (Green gram)	*Solanum melongena* (Brinjal)	*Aloe vera* (Kunwarpathu)
Cicer arietinum (Chana)	*Solanum tuberosum* (Potato)	*Asparagus racemosus* (Satavari)
Derris indica (Karanj)	*Solanum nigrum* (Piludi)	*Gloriosa Superba* (Vachhnag)
Lathyrus aphaca (Pea)		*Asphodelus tenuifolius* (Dungro)
• **Floral formula :**		
Br, Φ, $\varphi\sigma$, $K_{(5)}$, , $C_{1+2+(2)}$, $A_{1+(9)}$, $\underline{G}_1$	Ebr, $\oplus$, $\varphi\sigma$, $K_{(5)}$, $\overline{C_{(5)}\ A_5}$, $\underline{G}_{(2)}$	Br, $\oplus$, $\varphi\sigma$, $\overline{P_{3+3}\ A_{3+3}}$, $\underline{G}_{(3)}$
• **Economic Importance :**		
Pulses : Black gram, green gram, Bean, Pea, Lentil. Edible Oil : Groundnut, Soyabean	Food : Potato, tomato, Brinjal Medicine : Ashwagandha	Medicines : Aloe and Asparagus Ornamental plants : Tulip
Dyes : Indigofera	Spices : Chilli	----
Fibers : Sun hemp	Ornamental plants : Petunia	----
Medicine : Muliathi	----	----
Fodder : Sesbania, crotolaria	----	----
• **Flower :**		
Complete, Zygomorphic, Bisexual, Pedicellate, Bracteate, Perigynous	Complete actinomorphic, Bisexual, hypogyous, pentamerous	Complete, actinomorphic, Bisexual, trimerous, Bracteate, hypogynous

• **Calyx :**		
Five, gamosepalous, Imbricate aestivation	Five, gamosepalous, tubular, persistant, valvate aestivation	----
• **Corolla :**		
Five, polypetalous,	Five, gamopetalous,	----
papillionaceous	Valvate aestivation	----
• **Perianth :**		
----	----	3+3 free, valvate aestivation
• **Androecium :**		
10 diadelphous (9) + 1 dithecous anther	Five epipetalous, introse	3+3, free, periphyllous, introse or extose
• **Gynoecium :**		
Superior ovary, Unilocular, mono-carpellary ovary, marginal placentation	2, syncarpous, ovary superior, axile placentation	3, syncarpous, superior ovary, axile placentation
• **Inflorescence :**		
Racemose	Solitary cyme, Helicoid, apical or axillary	Solitary, axillary, Cymose, Umbellate
• **Leaf :**		
Pinnately compound or simple, alternate, leaf base pulvinate, stipules spiny, reticulate venation	simple, alternate, hairy, exstipulate, pinnately dissected, Reticulate venation	simple, alternate, opposite or whorled, exstipulate, mostly basal, parallel venation
• **Habit :**		
Tree, shrubs, herbs, climbers	Herb, climbers, shrubs, small trees	Herbaceous, climbers, bulb, rhizome
• **Fruit :**		
Legume, one or many seeds. seed non-endospermic	capsule or berry seed endospermic	capsule, berry, seed endospermic
• **Habitat :**		
Distributed all over the world	In tropical and temperate region	Major parts of the world (xerophytes)

(76) Tulip will included in which series ?

(A) Disciflorae (B) Calyciflorae (C) Coronariae (D) Bicarpellatae

(77) Which plant has ebracteate flower ?

(A) *Cajanus cajan* (B) *Aloe vera* (C) *Alium cepa* (D) *Solanum nigrum*

(78) Which symbol is correct for Solanum melongena ?

(A) $\overline{C_{(5)}\ A}_5$ (B) $\overline{P_{3+3}\ A}_{3+3}$ (C) $\overline{C_{(5)}\ A_{(5)}}$ (D) $\overline{C_5\ A_5}$

(79) How many plants are used as medicines ?

Tulip, Petunia, Ashwagandha, Indigofera, Susbenia, Muliathi, Aloe vera, Crotolaria

(A) 3 (B) 4 (C) 2 (D) 5

(80) Which pair is matched ?

(A) Solanaceae - Axile placentation (B) Solanaceae - Inferior ovary

(C) Liliaceae - Marginal placentation (D) Liliaceae - Inferior ovary

(81) Which pair is not matched ?

(A) Fabaceae - seed non-endospermic (B) Solanaceae - seed endospermic

(C) Fabaceae - Imbricate aestivation (D) Solanaceae - Twisted aestivation

(82) How many plants has complete, irregular (zygomorphic), bisexual flowers ?

Datura, Onion, Brinjal, Kunwarpathu, Green gram, Pegion pea, Potato, Satavari, Chana, Piludi, Vachhnag

(A) 3 (B) 4 (C) 2 (D) 1

(83) In which plant epiphyllous stamens are present ?

(A) *Aloe vera* (B) *Solanum tuberosum* (C) *Cajanus cajan* (D) *Datura fastuosa*

(84) In which plant the leaves are exstiputated ?

(A) Fabaceae (B) Liliaceae (C) Solanaceae (D) (B) and (C) both

(85) In which plants calyx is persistant ?

(A) Satavary (B) Brinjal (C) Karanj (D) Pegion pea

(86) Which is subfamily of family Leguminoceae ?

(A) Malvaceae (B) Solanaceae (C) Fabaceae (D) Liliaceae

(87) Ebr, $\oplus$, $\varphi^{\nearrow}$, $K_{(5)}$, , $\overline{C_{(5)}\ A}_{(5)}$, $\underline{G}_{(2)}$ - This floral formula is of which family ?

(A) Malvaceae (B) Solanaceae (C) Fabaceae (D) Liliaceae

(88) Which plant has a heteromerous flower ?

(A) Brinjal (B) Karanj (C) Potato (D) (A) and (C) both

(89) Br, $\oplus$, $\varphi^{\nearrow}$, $\overline{P_{3+3}\ A}_{3+3}$, $\underline{G}_{(3)}$ - which option is correct for leaves of the family which has this floral formula ?

(A) Simple, parallel venation (B) Simple, Reticulate venation

(C) Compound, parallel venation (D) Compound, Reticulate venation

(90) Br, $\oplus$, φ, $K_{(5)}$, C_{3+3}, A_0, $\underline{G}_{(3)}$ - which option is true for given floral formula ?

(A) It has perianth in two whorls of five each.

(B) It is not complete flower.

(C) It is not isomerous flower.

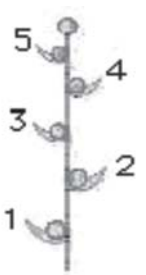
(D) None of the given

Answers : (76-C), (77-D), (78-A), (79-A), (80-A), (81-D), (82-A), (83-A), (84-D), (85-B), (86-C), (87-B), (88-B), (89-A), (90-B)

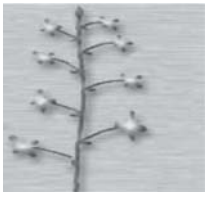
(91) Inflorescence of given figure is present in which plant ?

(A) Mustard (B) Mulberry

(C) Musa (D) Achyranthus

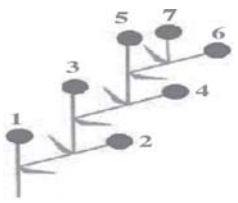


(92) What is correct for inflorescence of given figure ?



(A) Cymose, scorpioid
(B) Racemose, Scorpioid
(C) Racemose, Raceme
(D) Cymose, Uniparous

(93) What is correct for inflorescence of a given figure ?



(A) Cymose, scorpioid
(B) Racemose, Scorpioid
(C) Racemose, Raceme
(D) Cymose, Helicoid

(94) Which option is correct for aestivation and its related plant according to given figure ?



(A) Quincuncial - Cassia
(B) Imbricate - Delonix
(C) Imbricate - Melia
(D) Quincuncial - Cucurbita

(95) Which option is correct for placentation and its related plant according to given figure ?



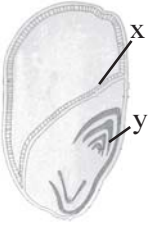
(A) Axile - Hibiscus
(B) Free cental - Dianthus
(C) Free central - Sunflower
(D) Axile - Dianthus

(96) What is type of flower and location of ovary in the give figure ?



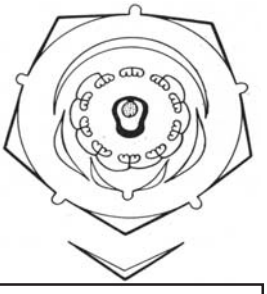
(A) Epigynous, Inferior ovary
(B) Perigynous, Semi-inferior ovary
(C) Hypogynous, Superior ovary
(D) Hypogynous, Inferior ovary

(97) What does X and Y indicates in given figure ?



(A) x = Epithelial layer, y = Plumule
(B) x = Aleurone layer, y = Radicle
(C) x = Epithelial layer, y = Radicle
(D) x = Aleurone layer, y = Plumule

(98) Which option is correct for stamens and petals in the given figure ?



(A) Stamens = Introse, Petals = Vexillary
(B) Stamens = Introse, Petals = Valvate
(C) Stamens = Extrose, Petals = Vexillary
(D) Stamens = Extrose, Petals = Valvate

Answers : (91-D), (92-C), (93-A), (94-D), (95-B), (96-A), (97-A), (98-A)

(99) Match the column :

Column - I	**Column - II**
(1) Umbel	(p) Foliage bract
(2) Spike	(q) Sessile flowers
(3) Spadix	(r) Florets arranged in centripetally
(4) Capitulum	(s) Inflorescence with involucre
	(t) Stalked flowers

(A) (1 - s), (2 - t), (3 - p), (4 - r)
(B) (1 - t), (2 - s), (3 - p), (4 - r)
(C) (1 - s), (2 - p), (3 - r), (4 - t)
(D) (1 - s), (2 - q), (3 - p), (4 - r)

(100) Match the column :

Column - I	Column - II	
(1) Mustard	(p) Hairy fruit	(A) (1 - s), (2 - r), (3 - p), (4 - q)
(2) Tomato	(q) Siliqua	(B) (1 - q), (2 - s), (3 - t), (4 - p)
(3) Ocimum	(r) Berry	(C) (1 - q), (2 - r), (3 - s), (4 - p)
(4) Vernonia	(s) Achene	(D) (1 - p), (2 - q), (3 - s), (4 - t)
	(t) Drupe	

(101) Match the column :

Column - I	Column - II	Column - III
(1) *Cajanus cajan*	(p) Pentamerous	(v) Twisted
(2) *Gloriosa Superba*	(q) Actinomorphic	(w) Imbricate
(3) *Derris indica*	(r) Bisexual	(x) Valvate
(4) *Solanum melongena*	(s) Stalked	(y) Quincuncial
	(t) Trimerous	(z) Vexillary

(A) (1-s-x), (2-t-w), (3-q-z), (4-p-v) (B) (1-r-y), (2-p-x), (3-s-w), (4-q-z)

(C) (1-r-z), (2-t-x), (3-s-z), (4-p-v) (D) (1-s-y), (2-t-y), (3-s-w), (4-q-z)

(102) Match the column :

Column - I	Column - II	
(1) Epithelial layer	(p) Enzymatic layer	(A) (1 - p), (2 - q), (3 - t), (4 - s)
(2) Aleurone layer	(q) Stained yellow by iodine	(B) (1 - p), (2 - q), (3 - r), (4 - s)
(3) Endosperm	(r) Protein rich layer	(C) (1 - p), (2 - q), (3 - s), (4 - t)
(4) Cotyledons	(s) Thin, one, scutellum, shield shape	(D) (1 - t), (2 - q), (3 - r), (4 - s)
	(t) Becomes purple by iodine	

(103) Match the column :

Column - I	Column - II	
(1) Sessile flower	(p) Bean	(A) (1 - s), (2 - q), (3 - t), (4 - p)
(2) Radial symmetry	(q) Pea	(B) (1 - s), (2 - r), (3 - t), (4 - p)
(3) Ebracteate flower	(r) Datura	(C) (1 - s), (2 - r), (3 - p), (4 - q)
(4) Bilateral symmetry	(s) Achyranthus	(D) (1 - q), (2 - r), (3 - t), (4 - p)
	(t) Caesalpinia	

(104) Match the column :

Column - I	Column - II	
(1) $\overline{C_{(5)}\ A}_5$	(p) Potato	(A) (1 - t), (2 - p), (3 - s), (4 - q)
(2) Ebr	(q) *Cicer arietinum*	(B) (1 - p), (2 - s), (3 - r), (4 - q)
(3) $\overline{P_{3+3}\ A}_{3+3}$	(r) Onion	(C) (1 - t), (2 - p), (3 - s), (4 - r)
(4) $C_{1+2+(2)}$	(s) Satavari	(D) (1 - t), (2 - p), (3 - r), (4 - s)
	(t) Piludi	

(105) Match the column :

Column - I	Column - II	
(1) Rose	(p) Apocarpous polycarpellary	(A) (1 - q), (2 - r), (3 - s), (4 - p)
(2) Pea	(q) Diadelphous stamens	(B) (1 - s), (2 - q), (3 - r), (4 - t)
(3) Hibiscus	(r) Monocarpellary	(C) (1 - p), (2 - r), (3 - q), (4 - s)
(4) Lemon	(s) Syncarpous polycarpellary	(D) (1 - p), (2 - q), (3 - s), (4 - t)
	(t) Polydelphous stamens	

Answers : (99-D), (100-C), (101-C), (102-A), (103-B), (104-A), (105-D)

- **A - Assertion and R - Reason type questions :**

Select the proper option for answer of given questions.

(A) A and R are true. R is correct explanation of A.

(B) A and R are true. R is not correct explanation of A.

(C) A is false, R is true.

(D) A is true, R is false.

(106) Assertion A : In onion, inflorescene is umbel type.

Reason R : Rachis becomes condesed and sessile flowers are arranged like spokes of umbrella.

(A) (B) (C) (D)

(107) Assertion A : Pollen germinates only on stigma.

Reason R : Stigma is receptive for pollen or it has adequate chemical for pollen germination.

(A) (B) (C) (D)

(108) Assertion A : In Acalypha, inflorescence is spike type.

Reason R : In Acalypha, the rachis is like pendulum type and bears unisexual flowers.

(A) (B) (C) (D)

(109) Assertion A : Sepals and petals are accessory and stamens and carpels are essential reproductive organs.

Reason R : Sepals and petals are not capable of producing reproductive cells but stamen and carpet can produce pollen and egg cell respectively.

(A) (B) (C) (D)

(110) Assertion A : In Hibiscus, placentation is axile type.

Reason R : Hibiscus, aestivation is twisted type.

(A) (B) (C) (D)

(111) Assertion A : Although Pancratium has not petals, it shows pollination.

Reason R : Instead of petals, in Pancratium, tepals are useful in pollination.

(A) (B) (C) (D)

(112) Assertion A : In Mustard, aestivation is of valvate.

Reason R : In Mustard, the placentation is Parietal type.

(A) (B) (C) (D)

(113) Assertion A : Legumenous fruit is dehiscence fruit.
Reason R : In legumenous fruit, dehiscence takes place by one suture.
(A) (B) (C) (D)

(114) Assertion A : Caesalpinia has perigynous type flowers.
Reason R : In ceasalpinia, thalamus becomes flat like disc, ovary is arranged semi inferiorally into it.
(A) (B) (C) (D)

(115) Assertion A : Apple is known as false fruit.
Reason R : In apple, in addition to ovary, peduncle is also participated in formation of fruit.
(A) (B) (C) (D)

(116) Assertion A : The cotyledons of non-endospermic seeds are always fleshy.
Reason R : Cotyledons comprises food of endosperm, they becomes fleshy.
(A) (B) (C) (D)

(117) Assertion A : Berry fruit is soft in compare to drupe fruit.
Reason R : In Berry fruit, the endocarp is juicy or fleshy.
(A) (B) (C) (D)

(118) Assertion A : Maize is caryopsis type of fruit.
Reason R : In maize, pericarp becomes flat like wings.
(A) (B) (C) (D)

(119) Assertion A : In Fabaceae family, for petals the symbol $C_{1+2+(2)}$ is used in floral formula.
Reason R : In Fabaceae family, aestivation is vexillary type.
(A) (B) (C) (D)

(120) Assertion A : In Vachhnag, the sepals are persistant.
Reason R : Vachhnag is included in Liliaceae family.
(A) (B) (C) (D)

Answers : (106-D), (107-A), (108-C), (109-A), (110-B), (111-A), (112-B), (113-D), (114-A), (115-D), (116-A), (117-A), (118-D), (119-A), (120-C)

- **False and True (T - F) type questions :**

Select correct option for answer of given true and false sentences.

(121) Assertion x : In capitulum inflorescence, the ray florets are in center.
Assertion y : In capitulum inflorescence, the disc florets are in periphary.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(122) Assertion x : In Heliotropium, inflorescence is scorpioid type.
Assertion y : Flowers are arises from left or right side of the rachis.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(123) Assertion x : In crinum, no individual calyx or corolla.
Assertion y : In crinum, perianth attracts insects for pollination.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(124) Assertion x : In Hibiscus, aestivation is twisted type.
Assertion y : In Hibiscus, placentation is like in tomato.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(125) Assertion x : In Bean, stamens are monodelphous.
Assertion y : In Bean, gynoecium is monocarpellary.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(126) Assertion x : Radial symmetry is present in Caesalpinia.
Assertion y : Flowers are bracteate in Caesalpinia.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(127) Assertion x : Fruit of Mustard is called Siliqua.
Assertion y : Ovary is superior in Mustard.
(A) Both are wrong (B) Both are right (C) x = right, y = wrong (D) x= wrong, y = right

(128) Assertion x : *Asparasus racemosus* is useful as Medicine.
Assertion y : Satavary is included in Liliaceae family.
Assertion z : $\oplus$, $\varphi^{\nearrow}$, K_{2+2}, C_4, K_{2+4}, $\underline{G}_{(2)}$ is floral formula of liliaceae family.
(A) x, y = right, z = wrong (B) x, z = right, y = wrong
(C) y, z = right, x = wrong (D) x, y = wrong, z = right

(129) Assertion x : In maize, it has a sheild type scutellum and fleshy endosperm.
Assertion y : Protein is stored in Aleurone layer of maize.
Assertion z : Large amount of starch is stored in epithelial layer of maize.
(A) x, z = right, y = wrong (B) x, z = wrong, y = right
(C) x, y = right, z = wrong (D) y, z = wrong, x = right

(130) Assertion x : Fruit of calotropis is follicle type.
Assertion y : Calotropis posseses multiparous cymose type of inflorescence.
Assertion z : Aestivation is valvate in calotropis.
(A) All are wrong (B) All are right
(C) y, z = right, x = wrong (D) x, y = right, z = wrong

Answers : (121-A), (122-C), (123-B), (124-B), (125-D), (126-A), (127-B), (128-A), (129-C), (130-B)

- **Questions for NEET :**

(131) Which example does not have compound Umbel inflorescence ?
(A) Fennel (B) Onion (C) Cumin (D) Coriander

(132) In which example compound raceme inflorescence is present ?
(A) Delonix (B) Neem (C) (A) and (B) Both (D) Radish

(133) Which plant posses Biparous cymose inflorescence ?
(A) Jasmine (B) Sal
(C) Mirabilis (D) All of the given

(134) Both Racemose and Cymose inflorescence are present in .....
(A) Grapes (B) Palm (C) Hydrocotyl (D) Sunflower

(135) Which example group posseses actionomorphic flower ?

(A) Mustard, Datura, Chilli (B) Pea, Groundnut, Delonix

(C) Datura, Mustard, Delonix (D) Mustard, Delonix, Cassia

(136) What is special feature of Hibiscus flower ?

(A) Have only petals but not sepals

(B) Have sepals in addition to epicalyx

(C) Sepals are useful in pollination

(D) Sepals are persistant or long lived

(137) Water of coconut can be compared to .....

(A) Embryo (B) Endosperm (C) Pericarp (D) Seed coat

(138) Caruncle or stopluole means...

(A) Connection of seed with fruit (B) Spot shows connection of style

(C) Extra seed coat of seed (D) Micropyle of seed

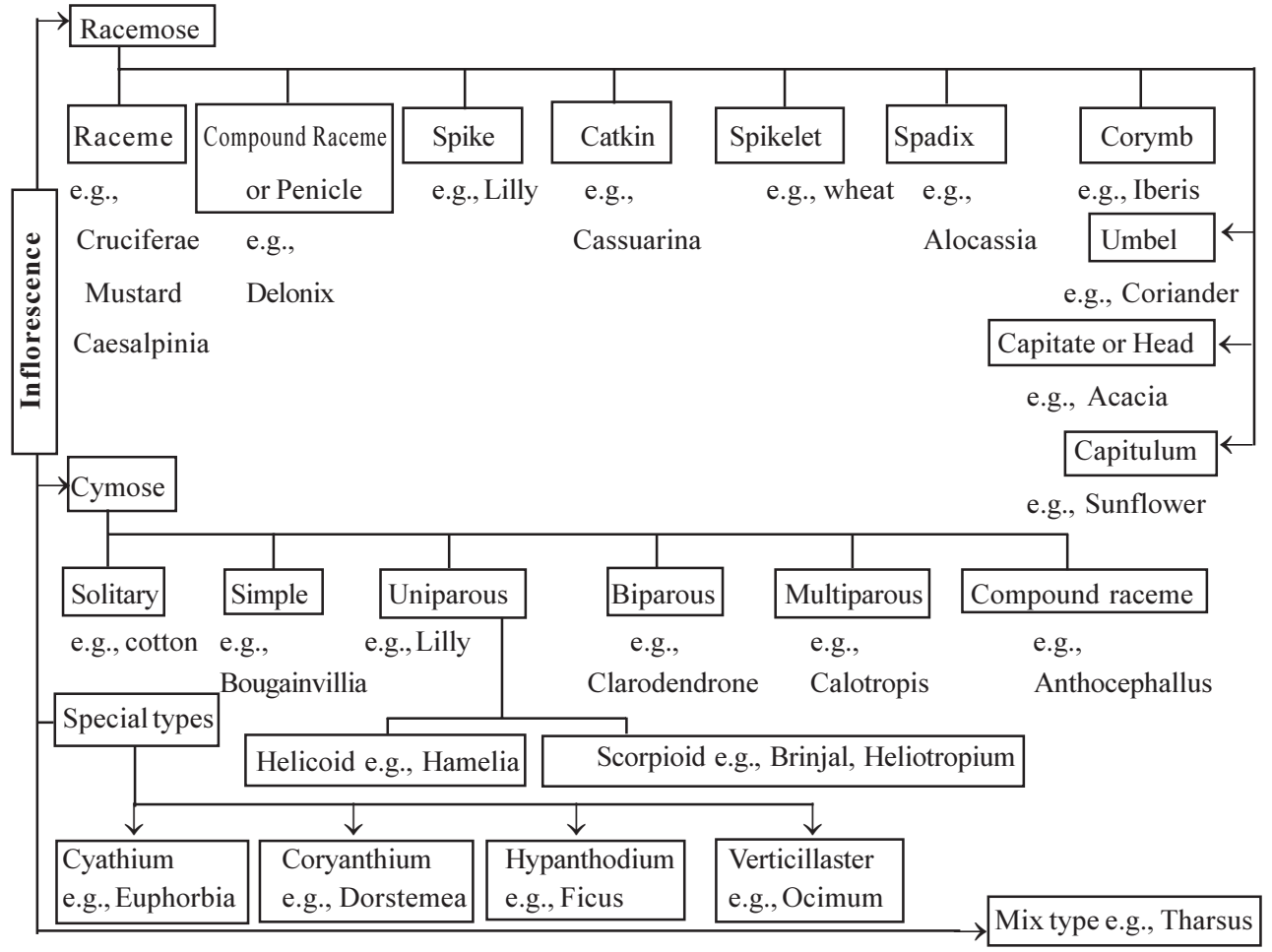
(139) Which example posseses attractive sepals for pollination ? (i.e. Advertisement flag)

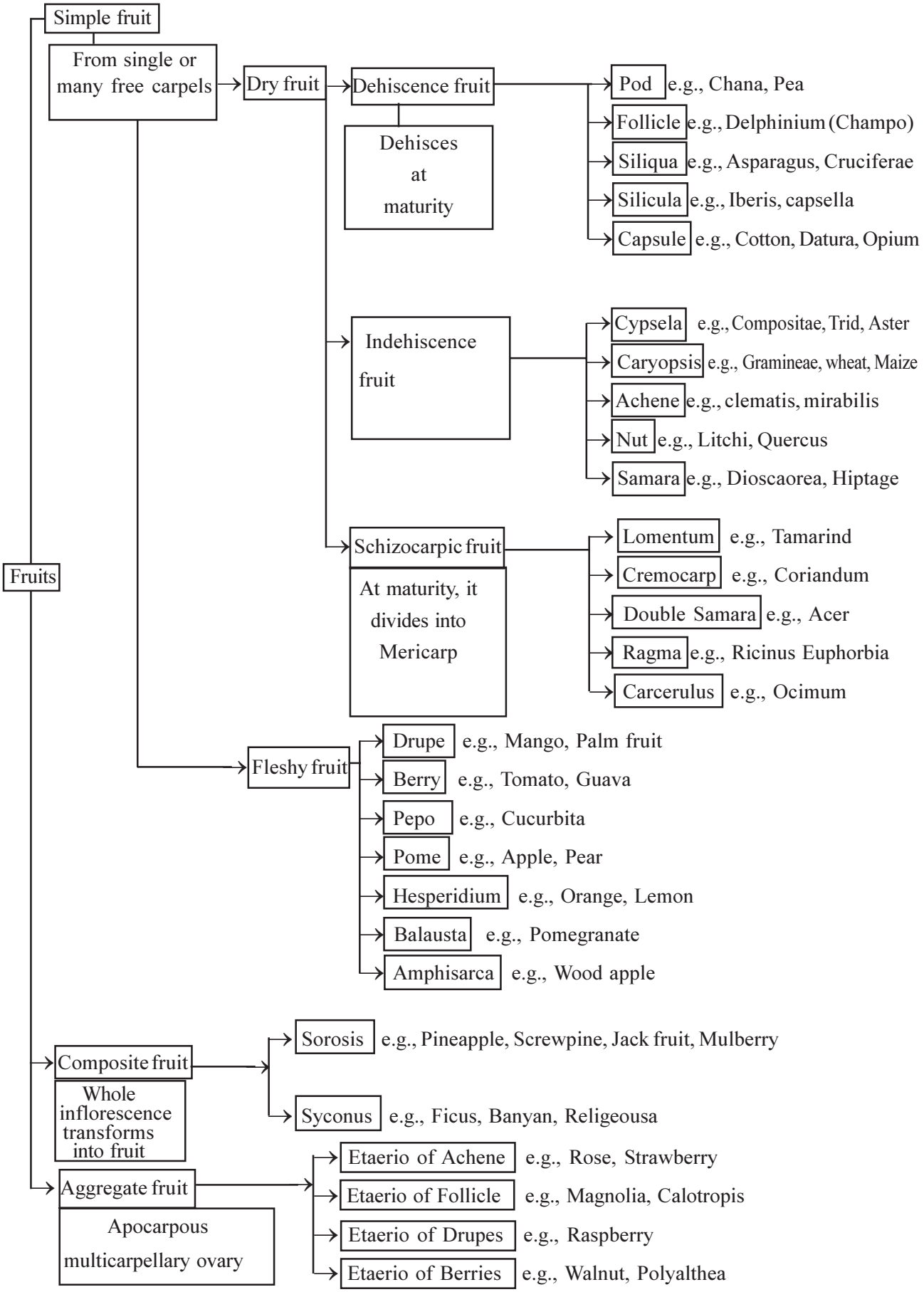
(A) Mussaenda (B) Larkspur (C) Balsum (D) All of the given

(140) In which fruit, edible part is not seen ?

(A) Almond (B) Mango (C) Trapa (D) Pulses

Answers : (131-B), (132-C), (133-D), (134-A), (135-A), (136-B), (137-B), (138-C), (139-A), (140-B)




Fruits
Simple fruit
From single or many free carpels
Dry fruit
Dehiscence fruit
Dehisces at maturity
Pod e.g., Chana, Pea
Follicle e.g., Delphinium (Champo)
Siliqua e.g., Asparagus, Cruciferae
Silicula e.g., Iberis, capsella
Capsule e.g., Cotton, Datura, Opium
Indehiscence fruit
Cypsela e.g., Compositae, Trid, Aster
Caryopsis e.g., Gramineae, wheat, Maize
Achene e.g., clematis, mirabilis
Nut e.g., Litchi, Quercus
Samara e.g., Dioscaorea, Hiptage
Schizocarpic fruit
At maturity, it divides into Mericarp
Lomentum e.g., Tamarind
Cremocarp e.g., Coriandum
Double Samara e.g., Acer
Ragma e.g., Ricinus Euphorbia
Carcerulus e.g., Ocimum
Fleshy fruit
Drupe e.g., Mango, Palm fruit
Berry e.g., Tomato, Guava
Pepo e.g., Cucurbita
Pome e.g., Apple, Pear
Hesperidium e.g., Orange, Lemon
Balausta e.g., Pomegranate
Amphisarca e.g., Wood apple
Composite fruit
Whole inflorescence transforms into fruit
Sorosis e.g., Pineapple, Screwpine, Jack fruit, Mulberry
Syconus e.g., Ficus, Banyan, Religeousa
Aggregate fruit
Apocarpous multicarpellary ovary
Etaerio of Achene e.g., Rose, Strawberry
Etaerio of Follicle e.g., Magnolia, Calotropis
Etaerio of Drupes e.g., Raspberry
Etaerio of Berries e.g., Walnut, Polyalthea

14 Anatomy of Flowering Plants

- **Tissue :** A tissue is a group of cells having common origin and performing specific functions. It has main two types :

(1) Meristematic tissue : Actively dividing cells. It has two sub-types. (a) Primary meristematic tissue : It forms primary plant body : (a-1) Apical meristematic tissue : Apex of shoot and root has apical meristematic tissue. It increases height and depth of shoot and root respectively. (a-2) Intercalary meristematic tissue : It is present in grasses and regenerating parts of plants which is eaten up by Herbivorous animals. (b) Secondary Meristematic tissue : It is also called lateral meristematic tissue and is responsible for secondary growth in plant. Vascular combium is present in between xylem and phloem and beneath the bark of a plant as a cork cambium.

(2) Permanent or ameristematic tissue : Normally cells are not dividing as they are highly differentiated in reference to structure and function. It has two main types : (a) Simple tissue (b) Complex tissue

(a) Simple tissue : Only single type of cells. It has three sub-types.

- **(a-1) Parenchyma :** It is present in organs like root, stem, leaf, flower and fruit of a plant. It is responsible for photosynthesis, storage and secretion. They possess intercellular space with thin cellulose cell-wall and spherical, oval or isodimetric in shape.

Note : It has aerenchyma and chlorenchyma two sub types.

- **(a-2) Collenchyma :** It renders elasticity and flexibility to plants. It is present in tender stem and hypodermis but absent in monocot and underground organs of plant. No intercellular space is found in this tissue as the shape of cells is polygonal. Deposition of pectin is more prominent in the angular walls where number of cells join together. As it does not arrest the growth of a plant, So it is called living mechanical tissue also.

- **(a-3) Sclerenchyma :** Due to deposition of lignin, wall becomes thick. So it is strong, noninelastic and water impermeable. It is dead tissue. No intercellular space is present. It is present in hypodermis, pericycle, xylem and phloem. It provides mechanical support to the plants.

Sclerenchyma is of two types (i) Fibres (ii) Sclereids.

(i) Fibres : They are thick walled, elongated and pointed cells, found in various parts of the plants.

(ii) Sclerieds : They are present in walls of nuts, pulp of fruits like guava, pear, sapota; seed coat of legumes, leaves of tea. As these are the small organs, less strength is required, so they posseses sclereids.

(b) **Complex tissue :** A group of more than one type of cells. As it does conduction, it is also known as conductive tissue. It is of two types.

(b-1) Xylem : In lower plants like pteridophytes and gymnosperms, they posseses tracheids. Tracheids have pointed ends and overlap each other. While in higher plants like in angiosperms, they posseses vessels. Vessels have open end walls and arranged end to end. Both types of xylary components do conduction of water and minerals. They have lignin deposition as secondary wall on the cell wall. Xylem parenchyma is living component and is responsible for storage of starch, lipid, tannin and crystalline substances. While xylem fibers are responsible for mechanical support.

- **(b-2) Phloem :** In gymnosperms, albuminous cells and sieve cells and in angiosperms, sieve tube, companion cells, phloem parenchyma and phloem fibers are present. Sieve tube is made up of many cells arranged longitudinally one above other. The transverse wall separating them become perforated to form sieve plate. Companion cells remain in contact with sieve tube by plasmodesmata.

  Phloem fibers are much elongated, unbranched and provide mechanical support. While phloem parenchyma are elongated and pointed and responsible for storage of mucilage, resin and latex.

  This tissue is responsible for conduction of organic food which is end product of photosynthesis.

---

(1) Which tissue is present in plants having regeneration capacity ?

(A) Intercalary (B) Meristematic (C) Primary meristematic (D) All the given

(2) Which tissue is not responsible in formation Bark ?

(A) Secondary meristematic (B) Lateral meristematic

(C) Cork cambium (D) Intercalary meristematic

(3) Which character is proper for meristematic cells ?

(A) Thin cell wall (B) Spherical cells

(C) Thin cytoplasm (D) Large intercellular space

(4) Which is not character of meristematic tissue ?

(A) Continuous divisibility (B) Thin cell wall

(C) Large intercellular space (D) Dense cytoplasm and conspicuous nucleus

(5) Meristematic tissue is also known as non-permanent. Why ?

(A) Location is not definite (B) Increases depths and Heights

(C) Both (A) and (B) (D) None of the given

(6) Which tissue is responsible for formation of axillary bud of leaf-axil ?

(A) Meristematic (B) Permanent (C) Parenchyma (D) Collenchyma

(7) Which tissue is known as "Packing tissue" which is present in nearly all organs of the plant ?

(A) Collenchyma (B) Parenchyma (C) Chlorenchyma (D) Sclerenchyma

(8) Which tissue is known as living mechanical tissue ?

(A) Collenchyma (B) Chlorenchyma (C) Parenchyma (D) Sclerenchyma

(9) Which is not applicable to the cells of a tissue which is responsible for photosynthesis, secretion and storage ?

(A) Large vacuole (B) Thin cell wall

(C) Pectin deposited wall (D) Intercellular space present

(10) The tissue which gives flexibility to plant is called "Collenchyma". Because...

(A) They have polygonal cells (B) They have pectin deposition

(C) (A) and (B) Both (D) None of the given

(11) Which plant tissue is absent in underground organ ?

(A) Parenchyma (B) Collenchyma (C) Meristematic (D) Conductive tissue

(12) Collenchyma is known as "Living mechanical tissue" Because ..... .

(A) Its cells give mechanical support, and it is living tissue.

(B) The deposition of this tissue will not arrest the growth of a plant.

(C) (A) and (B) Both

(D) None of the given

(13) Collenchyma is absent in underground organ. Because...

(A) Function of Collenchyma is being done by sclerenchyma.

(B) No xylem and phloem in underground organs.

(C) There is no need of storage of any compound in underground organs.

(D) There is no need of flexibility in underground organs.

(14) Sclerenchyma is known as "Dead tissue". Because ..... .

(A) Due to deposition of pectin, it becomes impermeable for water.

(B) Due to deposition of lignin, it becomes impermeable for water.

(C) Due to deposition of pectin, its cytoplasm is destroyed.

(D) None of the given

(15) Which tissue renders mechanical support to plant ?

(A) Meristematic - Collenchyma (B) Ameristematic - Collenchyma

(C) Meristematic - Sclerenchyma (D) Ameristematic - Sclerenchyma

(16) Which tissue is present in small organ like seed, leaves for mechanical support ?

(A) Sclerenchyma fibers (B) Xylem fibers (C) Sclereids (D) All of the given

(17) Which character is proper for sclereids ?

(A) Due to deposition of lignin, they possess narrow lumen

(B) They are spherical, oval or short cylindrical

(C) They possess large vacuole and dense cytoplasm

(D) All the given

(18) Which tissue is concerned for formation of wood ?

(A) Parenchyma (B) Collenchyma (C) Sclerenchyma (D) Meristematic

(19) Which pair is mis-matched ?

(A) Meristematic tissue - Formation of axillary bud

(B) Parenchyma - Function of secretion and storage

(C) Sclerenchyma - Renders mechanical support

(D) Collenchyma - Division and Differentiation

(20) Which option is true for a tissue which is present in hypodermis of aerial parts of a plant ?

(A) Sclerenchyma - deposition of lignin - mechanical support

(B) Collenchyma - deposition of pectin - flexibility and elasticity

(C) Parenchyma - deposition of suberin - division and differentiation

(D) Meristematic - deposition of cellulose - Growth and development

(21) What is not stored by xylem parenchyma ?

(A) Tannin (B) Resin (C) Starch (D) Lipid

(22) What is not stored by phloem parenchyma ?

(A) Mucilage (B) Resin (C) Latex (D) Tannin

(23) Sieve tube of phloem can be considered as living component. Because ..... .

(A) It has only cytoplasm (B) It has only nucleus

(C) It has both cytoplasm and nucleus (D) It possesses companion cells which has nucleus

(24) Instead of companion cell, which cells are present in gymnosperm plants ?

(A) Sclereids (B) Sieve cells (C) Albuminous cells (D) Fat cells

(25) How many components are living and non-living among given list ?

Tracheids, Vessels, Xylem parenchyma, Xylem fibers, Sieve cells, Sieve tube, Phloem fibers, Phloem parenchyma, companion cells.

(A) 4, 5 (B) 3, 6 (C) 6, 3 (D) 5, 4

(26) Which option is not suitable for function phloem component ?

(A) Sieve tube - conduction of the organic food

(B) Companion cells - Regulation of sieve tube by plasmodesmata

(C) Phloem parenchyma - Secretion of Latex and Resin

(D) None of the given

(27) How Tracheids and vessels differs from each other ?

(A) Tracheids shows deposition of pectin, while vessels shows deposition of lignin

(B) Tracheids shows deposition of lignin, while vessels shows deposition of Pectin

(C) Each component of tracheids is open at end, while it is closed in vessels.

(D) Each component of tracheids is closed at end, while it is open in vessels.

(28) Which pair is false for conductive tissue ?

(A) Pteridophyta - Companion cell (B) Gymnosperm - Albuminous cell

(C) Angiosperm - Vessels (D) Gymnosperm - Sieve cells

(29) Which tissue does not have deposition of lignin ?

(A) Collenchyma (B) Sclerenchyma (C) Vessels (D) Phloem fibers

(30) How many components are without deposition ?

Collenchyma, Sclerenchyma, Xylem parenchyma, Phloem parenchyma, Companion cell, Phloem fibers, Xylem fibers

(A) 5 (B) 3 (C) 4 (D) 2

(31) Which is odd of the following ?

(A) Companion cell (B) Albuminous cell (C) Sieve tube (D) None of the given

(32) Deposition of lignin is necessary in xylem. Because...

(A) Xylem wall should be strong against the osmotic pressure which is exerted by soil solution conducted by xylem.

(B) Due to lignin deposition, it will be transformed into living mechanical tissue.

(C) Due to lignin deposition, it will provide flexibility and elasticity to plants.

(D) Due to deposition of lignin, bidirectional conduction of mineral and water becomes possible.

(33) Which option is true for sieve cell and sieve tube ?

(A) Sieve plate is present in sieve tube, but not in sieve cell

(B) Companion cell is present in sieve tube, but not in sieve cell

(C) Conduction efficiency of sieve tube is more than sieve cell

(D) All the given

Answers : (1-D), (2-D), (3-A), (4-C), (5-D), (6-A), (7-B), (8-A), (9-C), (10-C), (11-B), (12-C), (13-D), (14-B), (15-D), (16-C), (17-C), (18-C), (19-D), (20-B), (21-B) (22-D), (23-D), (24-C), (25-D), (26-C), (27-D), (28-A), (29-A), (30-B), (31-B), (32-A), (33-D)

- **The tissue system :** A tissue system is formed by organisation. It is divided into 3.

(A) Epidermal tissue system : It has two type of patterns.

(A-i) Multilayered : It is cutical-less and present in hygroscopic root or orchid.

(A-ii) Unilayered : It is outermost barrel shaped cellular layer, without intercellular space and of parenchymatous tissue. It is cuticularised in stem and leaves to reduce transpiration and for that purpose cells are thick walled. While in root cells are thin walled and Non-cuticularised to absorb the water and mineral from soil. Epidermis of root is known as epiblema which has root-hair.

Root hairs are unicellular while trichomes (hairs of stem and leaves) are multicellular. Epidermis of stem and leaf has numerous pores. These pores open and close by a pair of Guard cell. In dicot plants, these guard cells are bean-shaped and in monocot it is dumbbell-shaped. In monocot, in addition to guard cells, they have one pair of subsidiary cells which are triangular. The stomatal aperture, guard cells and surrounding subsidiary cells are together called stomatal apparatus.

(B) Ground tissue system : All the structures without epidermis and vascular tissue are included in ground tissue system. It has different layers as given below :

(B-i) Hypodermis : For the purpose of protection and mechanical support it is made up of either collenchyma or sclerenchyma.

(B-ii) Cortex : For the purpose of bidirectional conduction and aeration it is made up of loosely packed parenchyma.

(B-iii) Endodermis : Inner most layer of cortex is endodermis. It regulates the exchange of substances between cortex and stele.

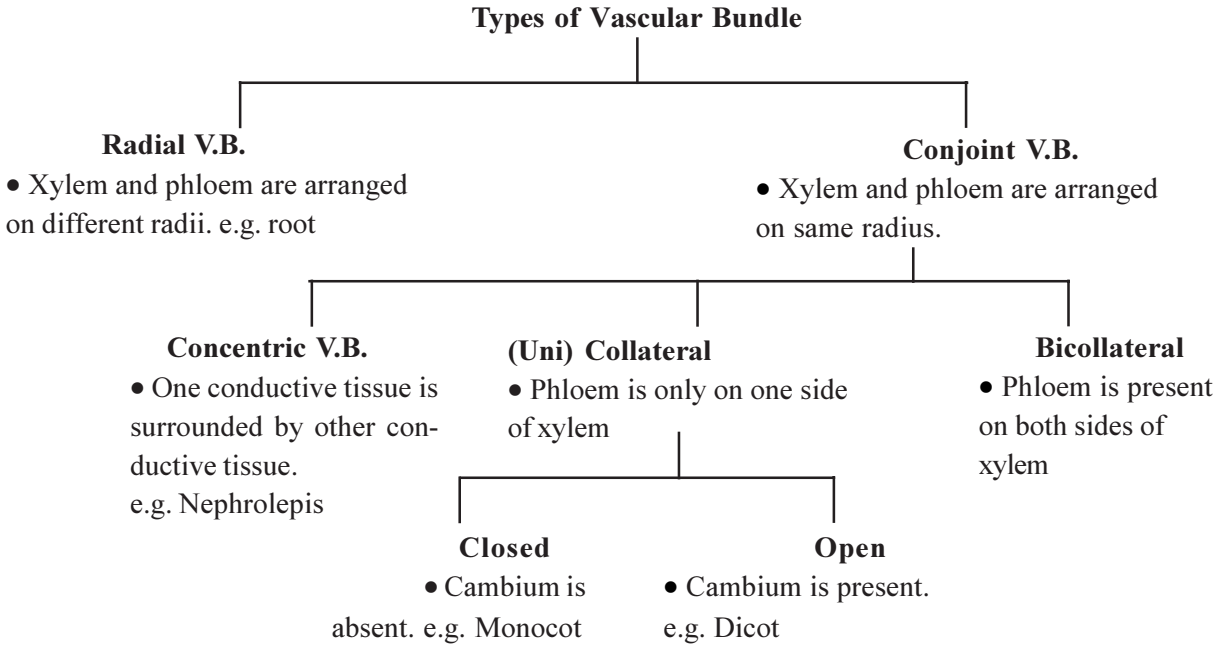
(B-iv) Pericycle : It is responsible for formation of lateral branches. In root it is unicellular parenchyma and in stem it is multilayered and sometime made up to alternatively sclerenchyma and parenchyma.

(B-v) Conjuctive tissue : It is present in between xylem and phloem and made up of parenchyma and sometime of sclerenchyma. In stem it creates medullary rays.

(B-vi) Pith : It is made up of parenchymatous tissue. It is innermost part of stem or Root.

Note : The region between upper and lower epidermis of leaf is known as mesophyll.

(C) Vascular tissue system : Xylem and phloem are collectively called vascular bundle. It is classified as under, according to arrangement of xylem and phloem.



(34) What is the type of vascular bundle which have a phloem and two cambium ?

(A) Bicollateral (B) Concentric (C) Radial (D) None of the given

(35) Which vascular bundle has opposite arrangement to conjoint vascular bundle ?

(A) Radial (B) Unicollateral (C) Open (D) All the given

(36) What is applicable to a vascular bundle of root ?

(A) No cambium between xylem and phloem

(B) Xylem and phloem are on different radii

(C) Both (A) and (B)

(D) None of the given

(37) Which type of vascular bundle is absent in stem ?

(A) Bicollateral (B) Unicollateral (C) Radial (D) Closed

(38) What is special characteristic of Bicollateral V.B. ?

(A) Bicollateral V.B. are same in stem and root as well

(B) Bicollateral V.B. conducts water and mineral bidirectionally

(C) Phloem is only on one side of xylem

(D) Phloem is on both side of xylem

(39) What is not included in stomatal apparatus ?

(A) Subsidiary cell (B) Epidermal cell (C) Guard cell (D) Stomatal aperture

(40) What is wrong for trichomes (stem hairs) ?

(A) It is multicellular (B) It is on aerial organs

(C) Controls transpiration (D) None of the given

(41) What is true for root-hairs ?

(A) It is always multi cellular (B) It is always on aerial organs

(C) Helpful in absorbing water (D) Always covered with cuticle

(42) Which is not observed in hygroscopic (epiphytic) root of orchid ?

(A) Epidermis is multi layered (B) Cuticle is absent on it

(C) Absorbs moisture from air (D) It has lenticels

(43) Which character is not for epidermis of stem ?

(A) Protective in nature (B) Reduced transpiration

(C) No cuticle layer (D) It is unicellular

(44) Which option is true for ground tissue system which has chloroplasts ?

(A) Endodermis is known as Bundle sheath (B) Cortical cells has numerous chloroplasts

(C) It is concerned with photosynthesis (D) It is unilayered

(45) Which sequence is true for ground tissue system from outer to inner side ?

(A) cortex - endodermis - pith - pericycle

(B) cortex - endodermis - pericycle - conjuctive tissue - pith

(C) pith - conjuctive tissue - pericycle - endodermis - cortex

(D) endodermis - pericycle - pith - conjuctive tissue

(46) Which is multilayered region of ground tissue, tissue system and these layers are made up of only Parenchyma ?

(A) Cortex (B) Hypodermis (C) Endodermis (D) Pericycle

(47) Which pair is true for subsidiary cell and companion cell ?

(A) Subsidiary cell - stomatal apparatus, triangular, one pair
Companion cell - phloem, one, Irregular

(B) Subsidiary cell - stomatal apparatus, one, Irregular
Companion cell - phloem, triangular, one pair

(C) Subsidiary cell - phloem, triangular, one pair
Companion cell - stomatal apparatus, triangular, Irregular

(D) Subsidiary cell - phloem, triangular, one pair
Companion cell - stomatal apparatus, one, Irregular

(48) Which is included in Vascular bundle ?

(A) only xylem (B) only phloem (C) xylem and phloem (D) only cambium

Answers : (34-A), (35-A), (36-C), (37-C), (38-D), (39-B), (40-D), (41-C), (42-D), (43-C), (44-A), (45-B), (46-A), (47-A), (48-C)

- **Anatomy of Root :**

(A) Anatomy of sunflower and Maize root : A very thin transverse section of young root of sunflower and maize stained with very dilute solution of safranine, show the structure under the microscope as follows :

	Region	Name of Layer	Maize	Sunflower
(A)	Epidermal tissue system	Epiblema	Structure : Unilayer, barrel shaped cells, parenchymatous tissue, root hairs are present, cuticle is absent Function : Absorption of water and minerals	Like maize root
(B)	Ground tissue system	Exodermis	Structure : Unilayered, parenchymatous tissue, Thick walled. Function : Protection	Absent
		Hypodermis	Structure : Multilayered and sclerenchy matous Function : Mechanical Support	Absent
		Cortex	Structure : Thin walled, parenchymatous, Multilayered, Intercellular space present Function : Conduction and gaseous exchange	Like Maize root
		Endodermis	Structure : Inner most layer of cortex, Unilayesed parenchyma Function : Regulation of conduction	Casperial strips are present on wall of radial endodermis

	Pericycle	Structure : Unilayered, Parenchyma, barrel shaped cells Function : Lateral branch arise from it	Like Maize root
	Conjuctive tissue	Structure : Parenchymatous region between xylem and phloem. It is sometime of sclerenchyma Function : Helps and supports to function of xylem and phloem	Only Parenchyma
	Pith	Structure : Inner most, cells of Parenchyma Function : Support / Storage	Comparatively it is smaller
(C) Vascular tissue system	Stele	Structure : Radial, Alternate, Polyarch, Exarch xylem. Function : Conduction of water - mineral and food	Alternate, Radial, Tetrach, Exarch xylem

Note : Stele includes pericycle, Vascular bundles and pith. In exarch type of xylem, protoxylem remains towards pericycle and metaxylem remain towards pith.

(49) How many layers of ground tissue system are there in sunflower and maize respectively ?
(A) 5, 7 (B) 7, 5 (C) 6, 5 (D) 5, 6

(50) Hypodermis is absent in young root of sunflower. Because...
(A) It has no need of extra-strength, as it is multilayered.
(B) It has no need of extra-protection, as it has sclerenchyma
(C) No need of hypodermis, as it has cuticularised epidermis
(D) No need of hypodermis, as sunflower has tap root system

(51) Sunflower has small pith in compare to maize. Because...
(A) Sunflower is Annual plant
(B) Sunflower has no need of extra-strength
(C) Xylem components are arranged linearly in sunflower
(D) Due to large cortical region, sunflower pith is small

(52) Which layer is made up of only sclerenchymatous tissue in ground tissue system of maize root ?
(A) Hypodermis (B) Pericycle (C) Endodermis (D) Exodermis

(53) Exodermis is present in maize root. Because...
(A) Epiblema is made up of only single layer.
(B) Epiblema is made up of only parenchymous tissue.
(C) Epiblema is easily destroyable due to fibrous root system.
(D) Epiblema is non-cuticularised.

(54) Which layer is absent in stele ?

(A) Pith (B) Pericycle (C) Endodermis (D) All of the given

(55) What is wrong for stele of maize-root ?

(A) Proto and meta-xylem are arranged in V-shape.

(B) Two proto and one meta-xylem are arranged in each and every unit of V.B.

(C) Xylem is endarch.

(D) Protoxylem is arranged towards pericycle.

Answers : (49-A), (50-D), (51-C), (52-A), (53-C), (54-C), (55-C)

- **Anatomy of sunflower and maize stem :**

Region	Layer's name	Maize	Sunflower
(A) Epidermal tissue system	Epidermis	Structure : Unilayered parenchyma, cuticle is present, scattered stomata are present Function: Protection and regulation of transpiration	Multicellular trichomes are present
(B) Ground tissue system	Hypodermis	Structure : 2-4 layers of sclerenchyma Function : Mechanical support	Multilayered collenchyma Flexibility and elasticity
	Cortex	Absent	Structure : Parenchyma with intercellular space, Multilayered Function: Conduction and gaseous exchange
	Endodermis	Absent	Structure : Starch sheath, Inner most layer of cortex Function : control over conduction
	Pericycle	Absent	Structure : Multilayered Alternate parenchyma and sclerenchyma, Hard-bast (Bundle cap) is present above phloem and is schlerenchyma Function: Lateral branches
	Conjuctive tissue	Absent	Structure : Parenchymatous strips from pith to cortex is called medullary rays. Function: Packing tissue or support

	Pith	Absent	Function : support and conduction Structure : Inner most cells of Parenchyma
(C) Vascular tissue system	Vascular bundle	Scatteredly arranged. Small size at periphery and large size toward central part	wavy circular, ring arrangement
		Conjoint, unicollateral and closed - Endarch xylem - Phloem parenchyma absent	Conjoint, unicollateral and open - Endarch xylem - Phloem parenchyma present
		- Lysigenous cavity present	- Lysigenous cavity absent
		- Sclerenchymatous sheath around vascular bundle is present	- Bundle sheath is absent

(56) Which is not participating in formation of medullary rays ?

(A) Parenchymatous pericycle (B) Sclerenchymatous pericycle

(C) Endodermis (D) Conjuctive tissue

(57) Which option is correct for location and function of Hard bast (i.e. Sclerenchymatous pericycle) ?

(A) Location - Sunflower root
Function - Mechanical support

(B) Location - Sunflower stem
Function - Storage of food (starch)

(C) Location - Sunflower stem
Function - Mechanical support

(D) Location - Sunflower root
Function - Storage of food (starch)

(58) Sclerenchymatous bundle sheath is present in and its function is...

(A) Location - Maize root
Function - Storage of water-minerals

(B) Location - Maize stem
Function - Mechanical support

(C) Location - Maize stem
Function - Storage of water-minerals

(D) Location - Maize root
Function - Mechanical support

(59) Ground tissue system of maize stem is not differentiated into cortex, endodermis, pericycle, etc. Because...

(A) It is perennial (B) It shows secondary growth

(C) V.B. are scattered (D) All of the given

(60) What is not applicable for vascular bundle which is devoid of phloem parenchyma ?

(A) Canjoint (B) Unicollateral (C) Open (D) Endarch

(61) Which option is true for location and function of Lysigenous cavity ?

(A) Location - Maize root
Function - Water storage

(B) Location - Sunflower root
Function - Food storage

(C) Location - Maize stem
Function - Water storage

(D) Location - Sunflower stem
Function - Food storage

(62) Which is the similarity between function of trichome (stem hair) and stomata ?

(A) Regulation of transpiration (B) Absorption of water

(C) Protection and gaseous exchange (D) Respiration and photosynthesis

(63) How many sub-layers of ground tissue system are present in a stem which has resin duct ?

(A) 5 (B) 6 (C) 7 (D) 4

Answers : (56-B), (57-C), (58-B), (59-C), (60-C), (61-C), (62-A), (63-B)

- **Anatomy of leaf of sunflower and Maize :**
- **Anatomy of sunflower leaf :**

(A) Epidermal tissue system : Upper and lower epidermis is covered with cuticle. Stomata are guarded by a pair of bean shaped cells and it has multicellular trichomes. Stomata are more in number in lower epidermis compared to upper epidermis.

(B) Ground tissue system (Mesophyll) : Two type of chlorechyma are present between two epidermis. Palisade tissue is made of two layered tightly packed elongated cells just beneath the upper epidermis while loosely packed sponge tissue with large intercellular space is above the lower epidermis. So, it is known as dorsi-ventral leaf or bilateral leaf.

(C) Vascular tissue system : Vascular bundles are scatteredly arranged in mesophyll tissue and are wraped by parenchymatous bundle sheath which are longitudinally, transversaly or obliquely cutin section, as this leaf has reticular venation. Vascular bundle are of conjoint, unicollateral and closed type. Xylem remains towards upper epidermis and phloem is toward lower epidermis.

- **Anatomy of Maize leaf :**

(A) Epidermal tissue system : 5-7 Bulliform cells (Hygroscopic cells) are present in upper epidermis of plant like grass. It is laterally covered with Horn cell. Bulliform cells will shrink in dry weather, to reduce the transpiration by minimizing its open surface due to rolling of leaf blade, when weather becomes adequately moistened, the bulliform cells will absorb water (vapour) from surrounding atmosphere and becomes flaccid to unrol the leaf blade. Like this they are responsible for movement of leaf blade, these cells are known as Motor cells also. During rolling and unrolling of leaf blade, horn cell protects the leaf.

(B) Ground tissue (Mesophyll) system : Only one type of (sponge tissue) chlorenchyma tissue remains in between two epidermis. So, this leaf is known as isobilateral. And that is why the distribution of stomata on both the epidermis is same. Such stomata are guarded by two dumbell shaped cells. The guard cells are also surrounded by triangular subsidiary cells. Stomatal apparatus includes one aperture, two guard cells and two subsidiary cells.

(C) Vascular tissue system : Vascular bundles are transversaly arranged, linearly arranged as this leaf has parallel venation. Vascular bundles are conjoint, unicollateral and closed type, small V.B. are surrounded by parenchymatous bundle sheath and bigger one is by sclerenchymatous bundle sheath.

(64) Which option is correct for location and function of Horn-cells ?

(A) Location - Lower epidermis of maize
Function - Regulation of transpiration

(B) Location - Upper epidermis of maize
Function - Protection of bulliform cells at the time of rolling of leaf

(C) Location - Lower epidermis of maize
Function - Protection of bulliform cells at the time of rolling of leaf

(D) Location - Upper epidermis of maize
Function - Regulation of transpiration

(65) Dicot leaf is known as bilateral. Because ..... .

(A) Distribution of stomata on both the epidermis is unequal.

(B) Distribution of mesophyll tissue between two epidermis is not of same type.

(C) Intensity of green colour of both epidermis is not same.

(D) All of the given.

(66) Monocot leaf is known as isobilateral. Because...

(A) Distribution of stomata on both the epidermis is equal.

(B) Distribution of chlorenchyma in between two epidermis is same.

(C) Intensity of green colour on both epidermis is same.

(D) All the given.

(67) The number of stomata on lower epidermis of dorsiventral leaf is high. Because...

(A) Light impacts directly on upper epidermis

(B) Lower epidermis is non-cuticularized

(C) Sponge tissue remains above the lower epidermis

(D) Distribution of stomata are same on both the epidermis

(68) Bulliform cells are known as Motor cells also. Because...

(A) It stimulate movement of leaf
(B) It renders mechanical support to leaf
(C) It regulates of water
(D) It is responsible for transpiration

(69) Which option is not proper for Motor cells ?

(A) Covered with cuticle
(B) Remain on upper epidermis
(C) It is in 5-7 number
(D) It is also known as bulliform cells

(70) Which option is not proper for vascular bundles of bilateral leaf ?

(A) Scatteredly arranged in mesophyll tissue
(B) Cut in transverse or longitudinal direction
(C) The bundle sheath is sclerenchymatous
(D) Xylem remains towards upper epidermis

(71) Which option is wrong for Vascular bundle of bilateral leaf ?

(A) Linearly arranged in mesophyll tissue

(B) Have parenchymatous or sclerenchymatous bundle sheath

(C) All are transversally cut
(D) Phloem is remain toward lower epidermis

(72) Which option is wrong for Vascular bundle of leaf ?

(A) Radial (B) Conjoint (C) Closed (D) Unicollateral

Answers : (64-B), (65-D), (66-D), (67-A), (68-A), (69-A), (70-C), (71-C), (72-A)

Secondary grwoth : The lengthwise growth of roots and stems is due to apical meristem and this type of growth is known as primary growth. After primary growth, the secondary growth takes place by lateral meristems like vascular cambium and cork cambium. Due to secondary growth the thickness (girth) of plant increases.

Lateral meristematic tissue which remain between xylem and phloem is called intrafascicular cambium and cells of medullary rays between two vascular bundle transformed into meristematic cells is known as interfascicular cells. Both these meristematic tissue joins together to form cambium ring.

Cambium ring forms xylem inside very fast and phloem out side comparative less fast. Newly formed xylem and phloem is known as secondary xylem and secondary phloem. The secondary vascular tissue exerts pressure on primary xylem and phloem, so, these primary tissue becomes dead and forms wool in future.

During winter, cambium is less active, so it will form less wood, it is known as autum wood or late wood. While in spring season, cambium is very active and forms more wood (i.e. xylem elements) which is known as spring wood or early wood.

These wood has two distinct region. Inner blackish part is heart wood or duramen which gives mechanical support to plant. It stores tannin and other Substances. It is hard and durable. It has lost its power of conduction. While outer yellowish part is sap wood or alburnum which gives capacity for conduction to the plants.

Due to high pressure of secondary growth on primary tissue, it will transformed into wood. secondary medullary rays are also formed. Thus, there is increase in girth of plant outer layers like Cortex and epidermis will break down. So, some cells of Cortex will transform into cork cambium (Phallogen). This cork cambium will form new cortex (secondary cortex) inside which is known as phelloderm also, while cork or phellem outside. These phellogen, phelloderm and phellem collectively known as periderm.

By the formation of periderm the stomata are closed. To replace this, some aerating pores are formed in the bark which are called lenticels. Through these pore exchange of gases and evaporation of water can take place.

Note : In dicot root vascular cambium originates in the form of strips from the tissue located just below the phloem bundles. These strips extend both ways in between xylem and phloem and ultimately units with each other to form a continuous ring. Further events are similar to those already describe above for a dicot stem. Initially plant has early or soft bark, at the end of season hard bark is formed.

(73) Some root shows secondary growth, although they have closed vascular bundle. Because...

(A) Pith transforms into cambium (B) Previously they have cambium ring

(C) Medullary rays performs secondary growth (D) All of the given

(74) Meristematic tissue forms initially in which location in secondary growth of dicot root ?

(A) Between Xylem and Phloem (B) Below the Xylem

(C) Below the Phloem (D) Above the Xylem

(75) What is applicable for secondary growth of root ?

(A) No formation of cambium ring (B) No secondary growth takes place

(C) Secondary growth increase length and depth of root (D) Root does not posses cambium

(76) Which is not included in the formation of periderm ?

(A) Phellem (B) Phelloderm (C) Vascular cambium (D) Phellogen

(77) Which compound is deposited on the wall of phloem cells ?

(A) Cellulose (B) Suberin (C) Lignin (D) Pectin

(78) What is called a meristematic tissue formed from cortex ?

(A) Phallogen (B) Cambium ring (C) Vascular cambium (D) Interfascicular cambium

(79) In which region, dead plant cells can be seen ?

(A) Phellem (B) Phelloderm (C) Bark (D) Phellogen

(80) When does formation of phellogen take place ?

(A) After formation of cambium ring (B) Along with formation of cambium ring

(C) Before the formation of cambium ring (D) Phellogen never becomes inactive

(81) Which two tissues are responsible for formation of cambium ring ?

(A) Phellem - Phelloderm (B) Cambium ring - Phellogen

(C) Interfascicular cambium - Intrafascicular cambium

(D) Interfascicular cambium - Phellogen

(82) From where does Interfascicular cambium forms ?

(A) Cortex (B) Endodermis (C) Pericycle (D) Medullary rays

(83) What is proper for sap wood or alburnum ?

(A) It is yellowish (B) It is conducting

(C) It is hard and durable (D) A and B

(84) What is not applicable for duramen ?

(A) It is blackish (B) It is hard and durable

(C) It gives mechanical support (D) It is conducting in nature

(85) Which one exerts pressure on primary xylem ?

(A) Secondary xylem (B) Secondary phloem

(C) (A) and (B) both (D) Phellogen

Answers : (73-A), (74-A), (75-D), (76-C), (77-B), (78-A), (79-C), (80-B), (81-C), (82-D), (83-D), (84-D), (85-C)

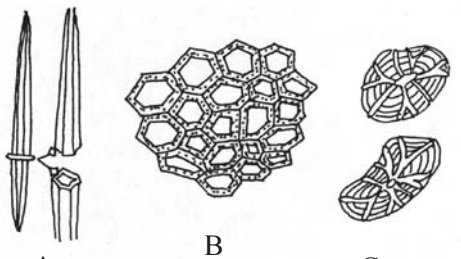
(86) What is correct from A-B-C in the given figure ?

(A) Suberized sclerenchymatous fibers in A

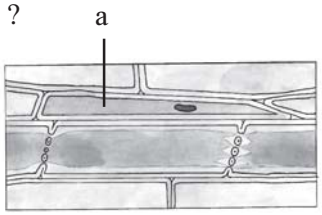
(B) Pectin deposited sclereids in figure B

(C) Lignin deposited sclereids in figure C

(D) All of the given



(87) Which option is correct for name and function of "a" in the given figure ?



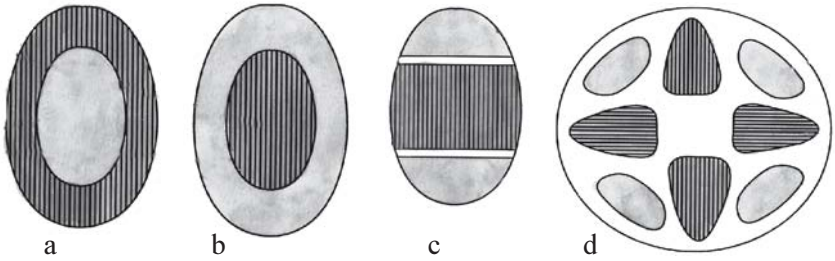
(A) Name - Albumin cell
Function - Helpful in water storage

(B) Name - Companion cell
Function - Helpful in water storage

(C) Name - Albumin cell
Function - Helpful in food storage

(D) Name - Companion cell
Function - Helpful in food storage

(88) Which option for the name of vascular bundle is wrong in reference to its relative figure ?



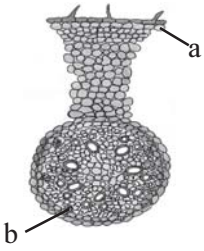
(A) Amphivessel
(B) Amphicribral
(C) Bicollateral
(D) Open

(89) What is wrong for vascular bundle for given figure of transverse section of plant ?



(A) Radial
(B) Closed
(C) Tetrach
(D) Endarch xylem

(90) What is true for "a" and "b" indicated in given figure ?



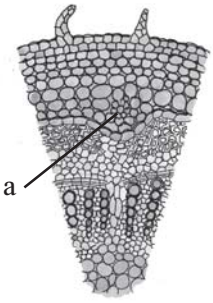
(A) a = Exodermis
b = Protoxylem

(B) a = Hypodermis
b = Protoxylem

(C) a = Exodermis
b = Metaxylem

(D) a = Hypodermis
b = Metaxylem

(91) What is name of "a" in the given figure and its function ?



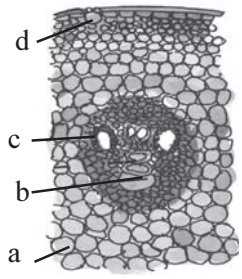
(A) a = Lysigenous cavity
b = Water storage

(B) a = Resin duct
b = Strength and durability

(C) a = Lysigenous cavity
b = Food storage

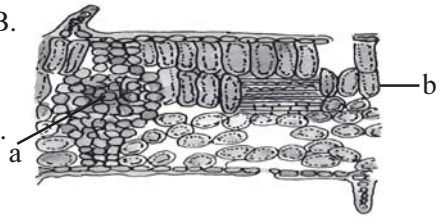
(D) a = Resin duct
b = Photosynthesis

(92) Which symbol is indicated Lysigenous cavity and ground tissue respectively ?



(A) b, a
(B) c, b
(C) a, c
(D) a, d

(93) What is correct labelling for "a" and "b" in the given figure ?



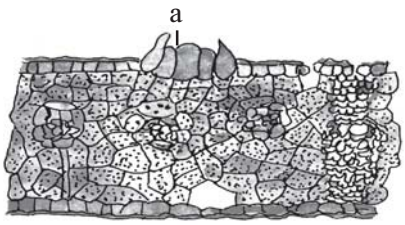
(A) a = Transverse vascular bundle
b = Palisade tissue

(B) a = Longitudinal V.B.
b = Palisade tissue

(C) a = Longitudinal vascular bundle
b = Spongy tissue

(D) a = Transverse V.B.
b = Spongy tissue

(94) What is alternative name of "a" in the given figure ?



(A) Horn cell (B) Bulliform cell

(C) Sclereids (D) Palisade cell

Answers : (86-D), (87-D), (88-D), (89-D), (90-A), (91-B), (92-A), (93-B), (94-B)

(95) Match the column :

Column - I	**Column - II**
(1) Collenchyma	(p) Lignified cells - in all organ
(2) Parenchyma	(q) air-exchange
(3) Sclerenchyma	(r) Albuminous cells are present
(4) Phloem	(s) Only in aerial organ - Pectin deposited cells
	(t) Only in hypogeal organ - Pectin deposited cells

(A) (1 - s), (2 - p), (3 - r), (4 - q) (B) (1 - t), (2 - s), (3 - q), (4 - p)

(C) (1 - s), (2 - q), (3 - p), (4 - r) (D) (1 - r), (2 - p), (3 - q), (4 - s)

(96) Match the column :

Column - I	**Column - II**
(1) Endodermis of sunflower stem	(p) Region contains lysigenous cavity
(2) Cortex of sunflower stem	(q) Region of parenchyma and sclerenchyma
(3) Pericycle of sunflower stem	(r) Unicollateral, Conjoint, open
(4) Vascular bundle of sunflower stem	(s) Starch storing parenchyma
	(t) Region of resin duct
	(u) Conjoint, unicollateral, closed

(A) (1 - t), (2 - s), (3 - p), (4 - q) (B) (1 - s), (2 - t), (3 - q), (4 - r)

(C) (1 - u), (2 - s), (3 - r), (4 - t) (D) (1 - p), (2 - r), (3 - s), (4 - u)

(97) Match the column :

Column - I	**Column - II**	**Column - III**
(1) Leaf of grass	(p) Motor cell	(v) Formation of stomatal apparatus
(2) Stem of Maize	(q) Albumin cell	(w) Necessary for phloem in angiosperm
(3) Stem of Cycus	(r) Sclereids	(x) Necessary for the mechanical support
(4) Seed-coat of seed	(s) Companion cell	(y) Necessary with sieve cell
	(t) Subsidiary cell	(z) Necessary for rolling of leaf in monocot

(A) (1-q-v), (2-p-z), (3-r-w), (4-t-x) (B) (1-r-y), (2-s-w), (3-t-v), (4-q-x)

(C) (1-p-x), (2-r-y), (3-s-v), (4-t-w) (D) (1-p-z), (2-s-w), (3-q-y), (4-r-x)

(98) Match the column :

Column - I	Column - II
(1) Leaf of maize	(p) Radial V.B. with definite number of protoxylem
(2) Stem of sunflower	(q) Vascular bundle in which xylem remains toward upper epidermis
(3) Root of sunflower	(r) Vascular bundle are scattered in transverse section
(4) Stem of maize	(s) Open vascular bundle around the medullary rays
	(t) Radial Vascular bundle around small pith

(A) (1 - q), (2 - s), (3 - t), (4 - r) (B) (1 - p), (2 - s), (3 - r), (4 - t)

(C) (1 - r), (2 - s), (3 - t), (4 - q) (D) (1 - p), (2 - r), (3 - t), (4 - s)

(99) Match the column :

Column - I	Column - II
(1) Hypodermis	(p) Gaseous exchange and conduction
(2) Cortex	(q) Protection and regulation of transpiration
(3) Endodermis	(r) Lateral root or lateral branch arises
(4) Pericycle	(s) Regulation of exchange between cortex and stele
	(t) Flexibility and Mechanical support

(A) (1 - s), (2 - p), (3 - t), (4 - r) (B) (1 - t), (2 - q), (3 - s), (4 - r)

(C) (1 - t), (2 - p), (3 - s), (4 - r) (D) (1 - q), (2 - p), (3 - s), (4 - t)

(100) Match the column :

Column - I	Column - II
(1) Isobilateral leaf	(p) Exodermis is present in ground tissue
(2) Tetrorch root	(q) Resin duct is present in cortex
(3) Dorsi-ventral leaf	(r) Motor cells are present in upper epidermis
(4) Polyarch root	(s) Casperian strips are present in endodermis
	(t) More stomata in lower epidermis

(A) (1 - t), (2 - s), (3 - r), (4 - q) (B) (1 - r), (2 - q), (3 - t), (4 - p)

(C) (1 - r), (2 - p), (3 - t), (4 - s) (D) (1 - r), (2 - s), (3 - t), (4 - p)

Answers : (95-C), (96-B), (97-D), (98-A), (99-C), (100-B)

- **A - Assertion and R - Reason type questions :**

  Choose the correct answer from given options :

  (A) A and R both correct. R is correct explanation of A

  (B) A and R both correct. R is not correct explanation of A

  (C) A - True, R - False

  (D) A - False, R - True

(101) Assertion A : Tissue of Node is known as intercalary meristematic tissue.

Reason R : This tissue arranged between two permanent tissue

(A) (B) (C) (D)

(102) Assertion A : Lateral meristematic tissue is included in primary meristematic tissue.

Reason R : Primary meristematic tissue develops primary plant organs.

(A) (B) (C) (D)

(103) Assertion A : Although more than one type of cells forms xylem and phloem, still it is called tissue.

Reason R : This is actually complex tissue. All types of cells performs same type of function.

(A) (B) (C) (D)

(104) Assertion A : Sclereids are present in unripened fruits.

Reason R : Small organ, i.e. unripe fruits, need less strength, so they posses suberinize components.

(A) (B) (C) (D)

(105) Assertion A : Xylem parenchyma is living and phloem parenchyma is non-living component.

Reason R : Non-deposited cells are present in xylem parenchyma and it is non deposited in phloem parenchyma also.

(A) (B) (C) (D)

(106) Assertion A : Tracheids are present in Nephrolepis and maize.

Reason R : Lower plants like maize and Nephrolepis, can do even by less efficient tracheids.

(A) (B) (C) (D)

(107) Assertion A : Exodermis is present in maize.

Reason R : Their root system is shallow. So there is more possibility of destruction of epiblema. In absence of epiblema, exodermis will give protection to plant.

(A) (B) (C) (D)

(108) Assertion A : Bulliform cells of upper epidermis of dorsiventral leaf is also known as motor cells.

Reason R : Mesophyll of this leaf is differentiated into palisade and sponge.

(A) (B) (C) (D)

(109) Assertion A : Bulliform cells of upper epidermis of dorsiventral leaf is also called as motor cells.

Reason R : As these cells stimulates movement in leaf, they are so called.

(A) (B) (C) (D)

(110) Assertion A : In annual plant, initiation of secondary growth starts by formation of cambium ring with the help of interfascicular cambium and intrafascicular cambium.

Reason R : It is necessary to form cambium ring before secondary growth starts.

(A) (B) (C) (D)

(111) Assertion A : There is no cambium in Vascular bundle of dicot root, still secondary growth takes place.

Reason R : Parenchymatous cells forms cambium ring and makes possible the secondary growth.

(A) (B) (C) (D)

(112) Assertion A : Plants like maize and grass shows rolling of leaf blade by motor cells.

Reason R : As these type of plants belongs to tropical region. They try to reduce rate of transpiration by reducing their open surface by rolling leaf blade.

(A) (B) (C) (D)

Answers : (101-A), (102-D), (103-A), (104-C), (105-B), (106-D), (107-A), (108-D), (109-D), (110-A), (111-A), (112-A)

- **True and False (T - F) type questions :**

Choose correct option for given true-false sentences.

(113) Assertion x : Early wood forms during spring and is not strong.

Assertion y : Late wood forms during autumn and is strong.

(A) Both are true (B) Both are false

(C) x true, y false (D) x false, y true

(114) Assertion x : Mesophyll tissue is uniform in dorsiventral leaf.

Assertion y : Mesophyll tissue is of two types in isobilateral leaf.

(A) Both are true (B) Both are false

(C) x true, y false (D) x false, y true

(115) Assertion x : Medullary rays forms intrafascicular cambium.

Assertion y : Interfascicular cambium joins with intrafascicular cambium.

(A) Both are true (B) Both are false

(C) x true, y false (D) x false, y true

(116) Assertion x : Phellem, Phelloderm and phellogen forms periderm.

Assertion y : Secondary growth takes place by intrafascicular cambium in dicot root.

(A) Both are true (B) Both are false

(C) x true, y false (D) x false, y true

(117) Assertion x : Intercalary meristematic tissue is present in internode.

Assertion y : intrafascicular cambium is known as secondary meristematic tissue.

Assertion z : Permanent or ameristematic tissue never undergo division.

(A) x, z - false, y - true (B) x, y - false, z - true

(C) x, y - true, z - false (D) x, z - true, y - false

(118) Assertion x : Hypodermis of aerial organ has cells with deposition of pectin.

Assertion y : Sieve cells are acompanied by albumin cell in gymnosperm.

Assertion z : Phloem parenchyma is the only living component of phloem.

(A) x, z - false, y - true (B) x, y - false, z - true

(C) x, y - true, z - false (D) x, z - true, y - false

(119) Assertion x : In maize stem, small V.B. remain at periphery and large V.B. towards pith.

Assertion y : In compare to tetrach root, polyarch root has small pith region.

Assertion z : Collenchyma is present out side the cortex in dicot (sunflower) stem.

(A) x, z - false, y - true (B) x, y - false, z - true

(C) x, y - true, z - false (D) x, z - true, y - false

(120) Assertion x : Hypodermis is made up of collenchyma and exodermis is of sclerenchyma in root of maize.

Assertion y : Water storage occurs in phloem parenchyma in a vascular bundle which has Lysigenous cavity.

Assertion z : All the vascular Bundles cut transversaly and are arranged in a raw in isobilateral leaf.

(A) x, z - false, y - true (B) x, y - false, z - true

(C) x, y - true, z - false (D) x, z - true, y - false

Answers : (113-A), (114-B), (115-D), (116-C), (117-A), (118-C), (119-D), (120-B)

- **Questions for NEET :**

(121) Assertion x : N. Grew is known as father of plant anatomy while K. A. Chaudhary is known as father of Indian Plant Anatomy.

Assertion y : The term Tissue has been coined by Nehemiah Grew.

Assertion z : Karl Nageli has divided plant tissue into meristematic and permanent.

(A) All are false (B) All are false

(C) x, y - false, z - true (D) x, z - false, y - true

(122) Which tissue is responsible for enlargement of leaf blade ?

(A) Interfascicular cambium (B) Intercalary meristem

(C) Plate meristem (D) Phallogen

(123) Assertion A: There is no apical meristem in algae and fungi, still they increases in length and volume.

Reason R : Every cell of algae and fungi is divisible, so they do not need meristem.

(A) Both A and R true, R is right explanation of A.

(B) Both A and R true, R is not right explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(124) Which tissue is present in pulp of fruits ?

(A) Parenchyma (B) Collenchyma (C) Sclerenchyma (D) Chlorenchyma

(125) Assertion A : Sometime sieve tube and companion cells are known as "sister cells".

Reason R : Both are evolved from same mother cell.

(A) Both A and R true, R is correct explanation of A.

(B) Both A and R true, R is not correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(126) Xylem fibers are not useful to mankind, but phloem fibers are useful. Because...

(A) Phloem fibers are long, while xylem fibers are inelastic.

(B) Phloem fibers are short, while xylem fibers are elastic.

(C) Xylem fibers can be weaved, while phloem fibers cannot be

(D) All of the given

(127) In which plant, Hydathode gland is absent ?

(A) Tomato (B) Pistia (C) Eichhornia (D) Passion flower

(128) Assertion A : Pericyle of sunflower stem is known as Heterogenous.

Reason R : It is made up of more than one type of tissue.

(A) A and R true, R is correct explanation of A.

(B) A and R true, R is not correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(129) Which two cells take part in formation of Lysigenous cavity ?

(A) Protoxylem - metaxylem (B) Protoxylem - xylem fibers

(C) Protoxylem - xylem parenchyma (D) Metaxylem - xylem fibers

(130) Which option is true for location of passage cell ?

(A) Opposite to metaxylem in endodermis (B) Opposite to protoxylem in endodermis

(C) Opposite to metaxylem in pericycle (D) Opposite to protoxylem in pericycle

(131) Assertion A : Dorsiventral leaf is known as isobilateral leaf.

Reason R : Isobilateral leaf has same number of stomata on both the surface.

(A) A and R true, R is right explanation of A.

(B) A and R true, R is not right explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(132) Which type of cells are present in pith of Root of monocot plant like Canna ?

(A) Parenchyma (B) Collenchyma (C) Sclerenchyma (D) (A) and (B) both

(133) Assertion A : Grafting is not possible in monocot.

Reason R : As monocot plants are annual, they are devoid of cambium.

(A) A and R true, R is correct explanation of A.

(B) A and R true, R is not correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(134) Which tissue is not included in lateral meristematic tissue ?

(A) Interfascicular meristem (B) Intrafascicular meristem

(C) Intercalary meristem (D) Phallogen

(135) What is main function of phallogen ?

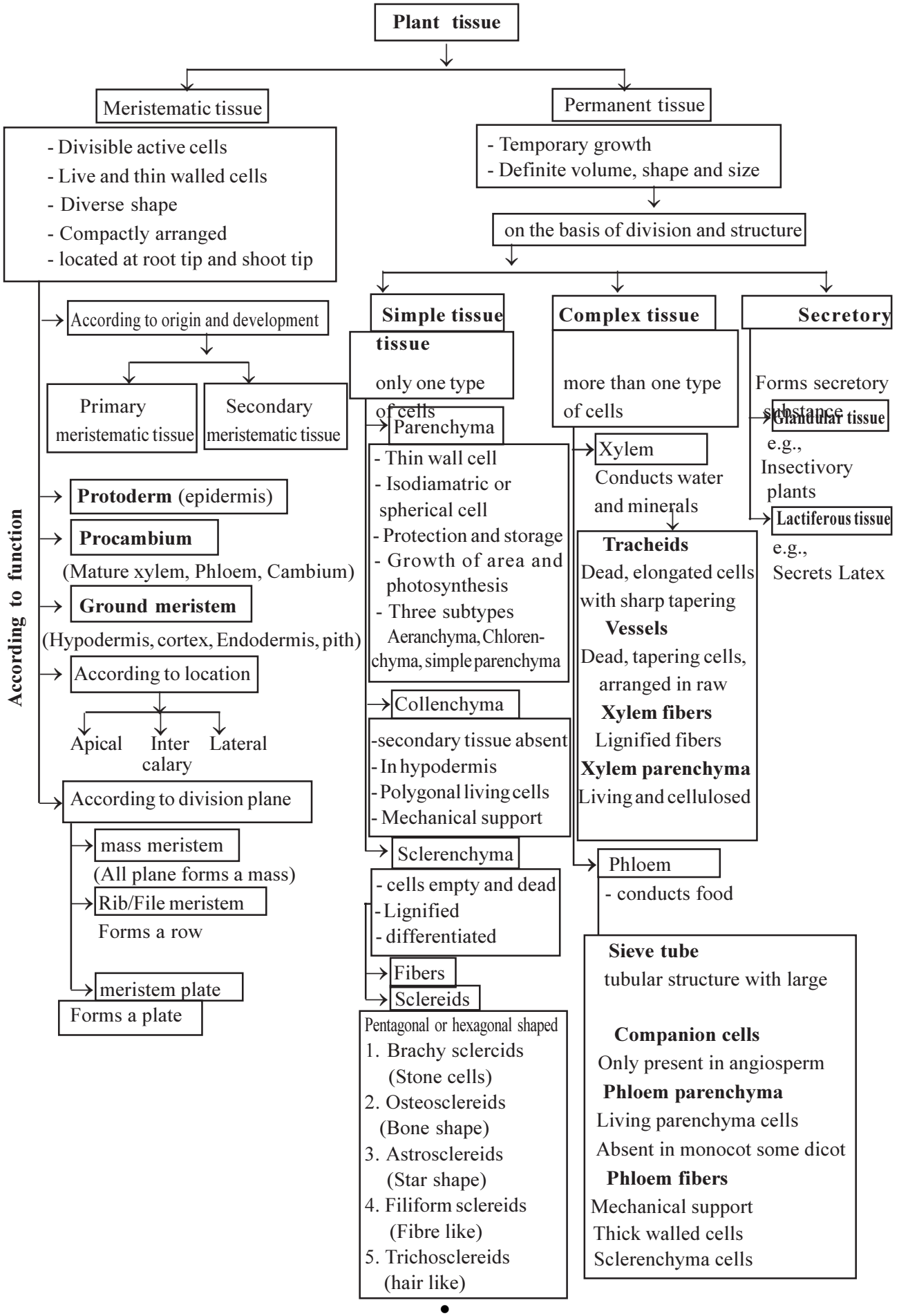
(A) Formation of secondary conducting tissue, for lateral growth

(B) Lateral growth by formation of secondary cortex and phellem.

(C) Secondary growth by formation of secondary cortex and secondary conducting tissue.

(D) Lateral growth by formation of secondary conducting tissue and periderm.

Answers : (121-B), (122-C), (123-A), (124-C), (125-A), (126-A), (127-D), (128-A), (129-C), (130-B), (131-D), (132-C), (133-A), (134-C), (135-B)


Plant tissue
Meristematic tissue
- Divisible active cells
- Live and thin walled cells
- Diverse shape
- Compactly arranged
- located at root tip and shoot tip
Permanent tissue
- Temporary growth
- Definite volume, shape and size
on the basis of division and structure
Simple tissue
tissue
only one type of cells
Complex tissue
more than one type of cells
Secretory
According to origin and development
Primary meristematic tissue
Secondary meristematic tissue
According to function
Protoderm (epidermis)
Procambium
(Mature xylem, Phloem, Cambium)
Ground meristem
(Hypodermis, cortex, Endodermis, pith)
According to location
Apical
Inter calary
Lateral
According to division plane
mass meristem
(All plane forms a mass)
Rib/File meristem
Forms a row
meristem plate
Forms a plate
Parenchyma
- Thin wall cell
- Isodiamatric or spherical cell
- Protection and storage
- Growth of area and photosynthesis
- Three subtypes
Aeranchyma, Chloren-chyma, simple parenchyma
Collenchyma
-secondary tissue absent
- In hypodermis
- Polygonal living cells
- Mechanical support
Sclerenchyma
- cells empty and dead
- Lignified
- differentiated
Fibers
Sclereids
Pentagonal or hexagonal shaped
1. Brachy sclercids (Stone cells)
2. Osteosclereids (Bone shape)
3. Astrosclereids (Star shape)
4. Filiform sclereids (Fibre like)
5. Trichosclereids (hair like)
Xylem
Conducts water and minerals
Tracheids
Dead, elongated cells with sharp tapering
Vessels
Dead, tapering cells, arranged in raw
Xylem fibers
Lignified fibers
Xylem parenchyma
Living and cellulosed
Phloem
- conducts food
Sieve tube
tubular structure with large
Companion cells
Only present in angiosperm
Phloem parenchyma
Living parenchyma cells
Absent in monocot some dicot
Phloem fibers
Mechanical support
Thick walled cells
Sclerenchyma cells
Forms secretory substance
Glandular tissue
e.g., Insectivory plants
Lactiferous tissue
e.g., Secrets Latex

15 Animal Tissue

- **Body Organization :** Body organisation process remains dynamic where multicellularity resulted with complexity. Group of cells structurally and functionally became similar and organised tissue.

  Cells → Tissues → Organs → Organ systems → Complex body.

- There are four basic tissue types :

  (1) Epithelial tissue (2) Connective tissue

  (3) Muscular tissue (4) Nervous tissue

Epithelial tissue :

Definition : A tissue which covers external surface of the body and interal free surface of many organs.

Features :

- It forms external surface and internal surface or many organ and body.
- The cells are arranged very close to each other with very little extracellular material or matrix which is a product of these epithelial cells.
- The cells are thin, polygonal, cuboidal or columnar. They rest on a non-cellular basement membrane.

Location :

- The lining of the skin, alimentary canal, blood vessels, digestiveglands, respiratory organs, etc.

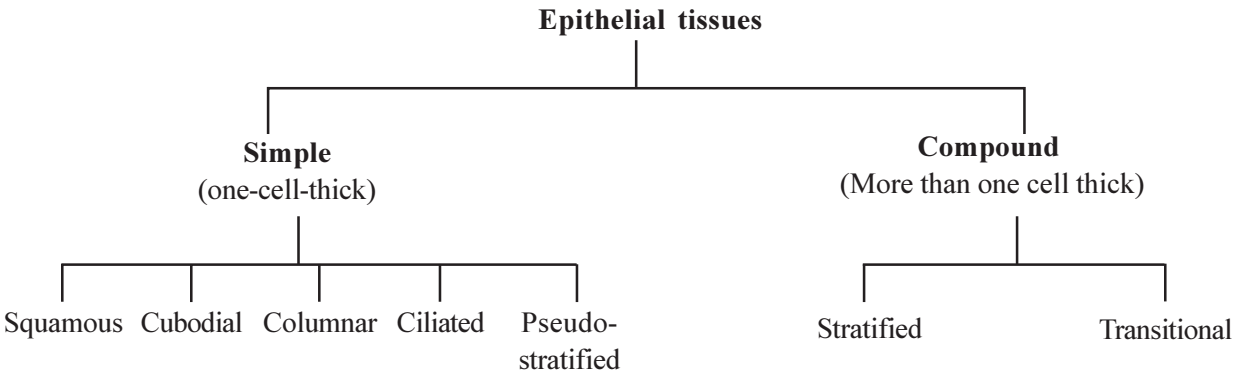
Function :

- It is highly specialised for protection, absorption and secretion.
- The skin and most of the organs are covered by a epithelial membrane for protection.
- Some of the epithelial surfaces are absorptive and secretory.

Group :

- Based on structure and function epithelial tissues are divided into two main groups :

  (1) Covering epithelial (2) Glandular epithelial



- **Squamous epithelial tissue :**

  Location : The outer most layer of skin of frog, inner lining of lung alveoli, lining of blood vessels, Bowman's capsules of kidney, peritoneum of body cavity.

- **Function :** Protection of underlying tissue.

  Features :

  - Cells are joined by cement material.
  - In surface view, this tissue seems to be composed of flat tiles like that of a pavement.

- **Cuboidal epithelial tissue :**

  Location : Proximal tubules of kidney, salivary glands, pancreatic ducts, thyroid gland and ovary.

  Function : In addition to protection, these cells are participate in secretion (gastric juice, hormone, etc.), excretion and absorption.

  Features :

  - Cells are square in vertical sections, polygonal in horizontal section.
  - Cells of absorptive surfaces often bear microvilli on their free ends.

- **Columnar epithelial tissue :**

  Location : It forms the lining of the mucous membrane of the stomach, gall bladder, urinogenital organs and their ducts.

  Function : Secretion and absorption. Reabsorption and secretion in uriniferous tubules and absorption of digested food in alimentary canal.

  Features :

  - Cells of this type of epithelia are elongated and are placed side by side like tall pillars.
  - Their inner ends are generally narrow but free ends are broad having polygonal surface.

- **Ciliated epithelial tissue :**

  Location : Auditory canal, uterine tube, respiratory tract.

  Function : Cilia moves particle, free cells or mucus in a specific direction.

  Feature : It is merely a modification of columnar epithelium.

  - Cilia are developed from basal granules of cytoplasm.
  - Cells bear thin protoplasmic processes on their surfaces called cilia and hence this tissue is called ciliated epithelium.

- **Pseudo - stratified epithelium tissue :**

  Location : Inner lining of trachea, large bronchi.

  Function : Removes mucus and secretions.

  Features : It is the simple columnar epithelium.

  - The regular arrangement of the cells is distorted.
  - Cells are arranged in single layer, but appears multi-layered, which is a false appearance.

- **Compound epithelium tissue :**

Definition : "A tissue which has more than one layer". It is divided in two types. (1) Stratified compound epithelium tissue.

Location : It is present where there is much wear and tear. Such as in the epidermis of skin, the lining of mouth cavity, the tongue, oesophagus and vagina of mammals.

Function : Protection of underlying tissues.

Types : It has two subtypes :

(1) Stratified cuboidal epithelial, the superficial cells are cuboidal in shape. Such tissue is present in the inner surface of larger salivary and pancreatic ducts.

stratified squamous non-keratinised and stratified squamous keratinised are two types of stratified squamous epithelium.

(2) Transitional epithelium tissue : It is present in the passage of the excretory organs.

---

(1) Epithelial tissue means ..... .

(A) A tissue which forms outer and inner surface of many organs of animal

(B) Unilayered tissue and with intercellular fluid

(C) Many layered tissue and no intercellular fluid

(D) (B) and (C) both

(2) Simple epithelial tissue means ..... .

(A) Many layered tissue, more intracellular fluid

(B) Unilayered tissue, no intracellular fluid

(C) Unilayered tissue, more intracellular fluid

(D) Many layered tissue, no intracellular fluid

(3) Which epithelial tissue forms internal wall of blood vessels ?

(A) Cuboidal (B) Squamous

(C) Stratified (D) Columnar

(4) What is the function of intestinal villi ?

(A) Absorption of food (B) Increases absorptive surface

(C) Controls intestinal movement (D) Prevents movement of food

(5) Which pair is mis-matched ?

(A) Squamous - Frog skin (B) Cuboidal stratified - Oesophagus

(C) Columnar - Peritoneum of frog (D) Ciliated - Bronchi

(6) The nucleus of squamous epithelium is having _x_ and _y_ characteristics.

(A) x = Weak, flat, polygonal, y = Spherical or oval

(B) x = Thin, flat, polygonal, y = Spherical or oval

(C) x = Thin, square, uniangular, y = Hexagonal or oval

(D) x = Thick, flat, polygonal, y = Spherical or oval

(7) _x_ is the tissue whose cells use connection by "Cement".

(A) Cuboidal (B) Columnar (C) Cilicated (D) Squamous

(8) The shape of cuboidal tissue _x_ in Longitudinal section and _y_ in transeverse section.

(A) x = Polygonal, y = Square (B) x = Spherical, y = Oval

(C) x = Square, y = Polygonal (D) x = Polygonal, y = Rod shape

(9) Which tissue is present in Thyroid gland ?

(A) Cuboidal (B) Squamous (C) Transitional (D) Columnar

(10) Which epithelial tissue is present in skin which joins eyeball and eyelids ?

(A) Cuboidal stratified (B) Stratified epithelial

(C) Columnar epithelial (D) Transitional epithelial

(11) The epithelial tissue which has brush like free surface is found in .....

(A) Testis (B) Stomach (C) Small intestine (D) Ovary

(12) Which epithelial tissue is present in oesophagus, mouth cavity, cornea, vagina and cervix ?

(A) Transitional (B) Columnar

(C) Keratinized stratified (D) Non-keratinized stratified

(13) Whose cells are arranged side by side and like pillar ?

(A) Cuboidal (B) Ciliated

(C) Columnar (D) Squamous

(14) Which group of organs, have cuboidal epithelium ?

(A) Pancreatic ducts, Thyroid glands, Bowman's cup

(B) Proximal tubules of kidney, Salivary glands, Ovary

(C) Pancreatic ducts, Peritoneum of body-coelom, Ovary

(D) Pancreatic ducts, Thyroid glands, Testes

(15) Pseudostratified tissue remains on internal surface of _x_ and _y_ and removes mucus.

(A) x = Trachea, y = Large bronchioles (B) x = Trachea, y = Small bronchioles

(C) x = Pharynx, y = Fine bronchioles (D) x = Oesophagus, y = Small intestine

(16) _x_ tissue removes mucus in particular direction and _y_ tissue removes mucus.

(A) x = Pseudostratified, y = Ciliated epithelium

(B) x = Ciliated epithelium, y = Columnar epithelium

(C) x = Pseudostratified, y = Cuboidal epithelial

(D) x = Ciliated epithelium, y = Pseudostratified

(17) Which tissue is responsible for division and regeneration ?

(A) Epithelium (B) Muscular (C) Connective (D) Nervous

(18) Which tissue forms glands ?

(A) Muscular (B) Epithelium (C) Nervous (D) Connective

(19) Which tissue is present in Reproductive cells ?

(A) Simple epithelial (B) Nervous (C) Connective (D) None of the given

(20) The outer layer of skin is made up of keratin, Because ..... .

(A) It is secretory and protects against wear and tear

(B) It is widely spreaded overall body

(C) It has more thickness

(D) It prevents entry of pathogens

(21) The cells of squamous epithelial tissue are ..... .

(A) Joins loosely with each other and forms unevenlayer

(B) Constantly divides to support organs

(C) Becomes rigid and supports organs

(D) Makes unilayer by the help of cement material

(22) Nasal cavity, fine bronchioles, oviductal funnel has tissue ?

(A) Ciliated (B) Columnar (C) Cuboidal (D) Germial epithelial

Answers : (1-A), (2-B), (3-B), (4-B), (5-C), (6-B), (7-D), (8-C), (9-A), (10-B), (11-C), (12-D), (13-C), (14-B), (15-A), (16-D), (17-A), (18-B), (19-A), (20-A), (21-D), (22-A)

- **Connective tissue** :

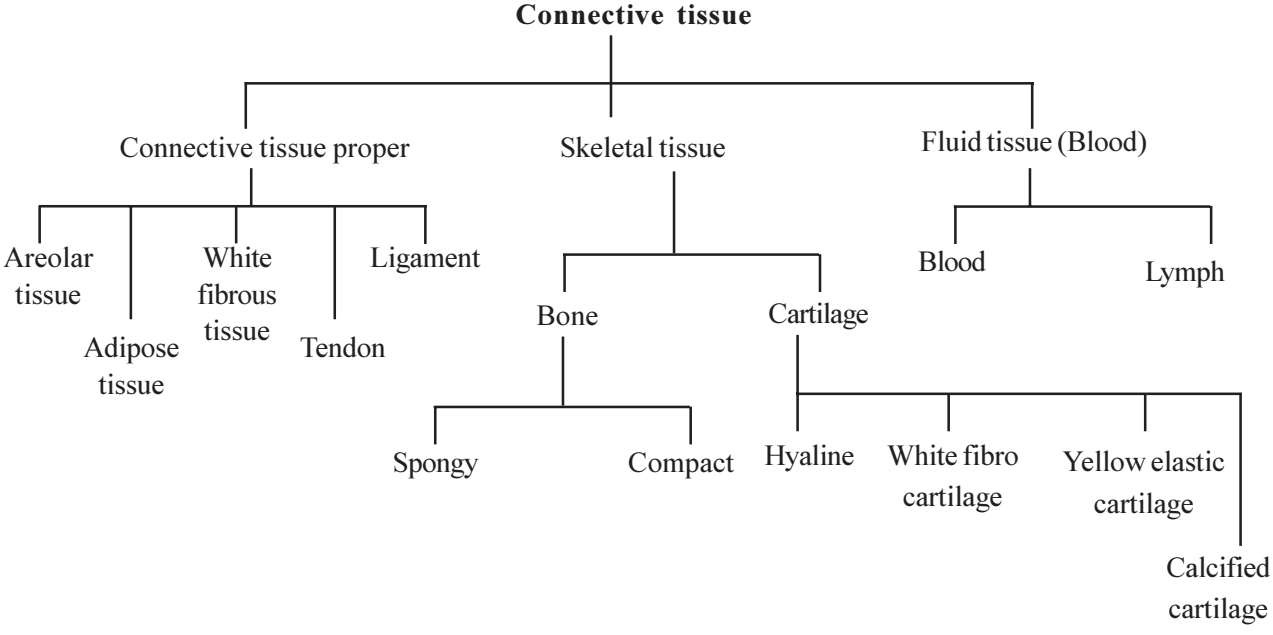
Definition : A tissue which connects together the other tissues of the body.

Features :

- It is the group of cells and matrix containing of intercellular substances secreted by cells themselves.
- Cells are consequently widely spaced.
- They are divisible.
- It forms from mesoderm.

Functions :

- To connect up structures.
- To form packing around organs.
- To replace tissue which have been destroyed by injury.
- To combat foreign toxins.
- To form a supporting framework.



- **Connective tissue proper :**

(1) Areolar tissue :

- This is the simplest and most widely distributed connective tissue. It is also called loose connective tissue.
- It is made up of fibres and cells. Fibres are of two types :
- **White fibers :** It is wavy, unbranched and are arranged in bundles. It is made up of protein collagen. It can bear weight, pressure or tension.
- **Yellow fibers :** It is straight, branched and are not arranged in bundle. It is made up of elastin and is elastic. It forms delicate network. It cannot bear weight, pressure or tension.
- Cells are of three types, i.e. Fibroblasts; Macrophages and Mast cells.
- Mast cells secretes heparin, histamine and serotonin.

(2) Adipose tissue :

Location : Occurs in abundance in the subcuteneous beneanth the skin, around kidney, mesenteries, bone marrow.

Functions : As it is present as subcutenous tissue, it helps to conserve the body heat.

Structure : Besides Adipocytes. It contains fibroblasts, macrophages, collagen fibers and elastic fibers.

(3) White Fibrous tissue :

Location : It is mainly present in the Periosteum of the bones and perichondrium of cartilage. They are also seen at the joint between skull bones and make the immovable.

Function : Connects bones and muscles

- Forms periosteum and perichondrium of bones and cartilage respectively.
- Forms immovable joints (sutures).

Function : White fibers are arranged compactly and parallel in bundles which are made up of collagen white protein. Fibroblast are present in between white fibers.

(4) Tendon and Ligament :

Tendon :

- It is made up of collagen fibers (matrix).
- Skeletal muscles remain connected with tendon.

Ligament :

- Branched yellow elastin fibers are present.
- The ligament connects the joints and holds them in position.

- **Skeletal tissue :**

(i) Cartilage :

- It is partially, strong, soft and elastic.
- Matrix is dense.
- It is special type of skeletal connective tissue, but, it differs from other connective tissue. It has types as follow.

Types :

(1) Hyaline cartilage :

Location : It is present in larynx, trachea, sternum, hyoid and ribs, etc.

Structure : The word hyaline is derived from the correct word "hylos" which means "glass" and the hyaline cartilage was so named because in the gross, it appears as a clear bluish coloured glassy surface.

- Its matrix is clear, homogenous translucent and lacking fibers.
- It is always covered by a tough fibrous membrane.
- It contains blood vessels from which nutritive substances diffuse through the cartilage.
- The cartilage cells are chondroblasts, which secretes chondrin.
- Cells lie in group of two, four or eight in fulfilled space called lacunae.

Function : It protects bones against friction and wear-tear as it has elasticity and limited flexibility.

(2) White fibrous cartilage :

Location : It is typically found in the intervertebral discs, which binds the vertebrae in mammals.

Structure : It consist of dense white fibrous tissue arranged in bundles with cartilage cells between the bundles.

- The cartilage cells are usually ovoid in shape and are surrounded by matrix.

Function : It protects vertebrae from wear and tear during movement.

(3) Yellow elastic cartilage :

Location : This type of cartilage is found in the pinna, tips of nose, epiglottis, in certain other regions.

Function : It contains yellow elastic fibers.

(4) Calcified cartilage :

Location : It occurs in the pubis of old frogs, in supra scapula and at the head of humerus and femur.

Structure :

- It differs from the hyaline type its matrix impregnated with lime salts.
- It is seen in a normal stage in the development of cartilage during early embryonic life but it is also found in a permanent tissue in the external ears of many mammals.

(ii) Bone :

Features :

- It is highly vascular.
- It is mineralized.
- It is constantly changing.
- It is hard and rigid.
- It is resilient.
- It has a regenerating capacity.
- It has a canalicular system.

Mature Bone : It is composed of two tissue.

(1) Compact bone and (2) Spongy bone

- The ground substance or the matrix, composed of protein called ossein.

- Ossein is impregnated with various inorganic salts of lime, namely, calcium phosphate, calcium carbonate, magnesium phosphate and calcium fluorides.

(1) Compact Bone :

- In adult bone flat irregular spaces called lacuna occurs in the solid matrix.
- Each lacuna contains a flat bone cell or osteocyte.
- The lacuna are in communication with one another by fine canalicules.
- In a long dried bone of frog, large number of lamella are present in a ground substance.
- In mammalian bones many column like structures are seen called Haversian system.
- It each Haversian system, several concentric layers (lamellae) of bony matrix encircle a longitudinal central canal (Haversian canal).

(2) Spongy Bone :

- Spongy bones are found in vertebrae, ribs, skull, etc.
- It contains red bone marrow.
- Red bone marrow is a seat for formation of RBCs (erythrocytes) and granulocytes.

- **Liquid connective tissue (Blood) :**

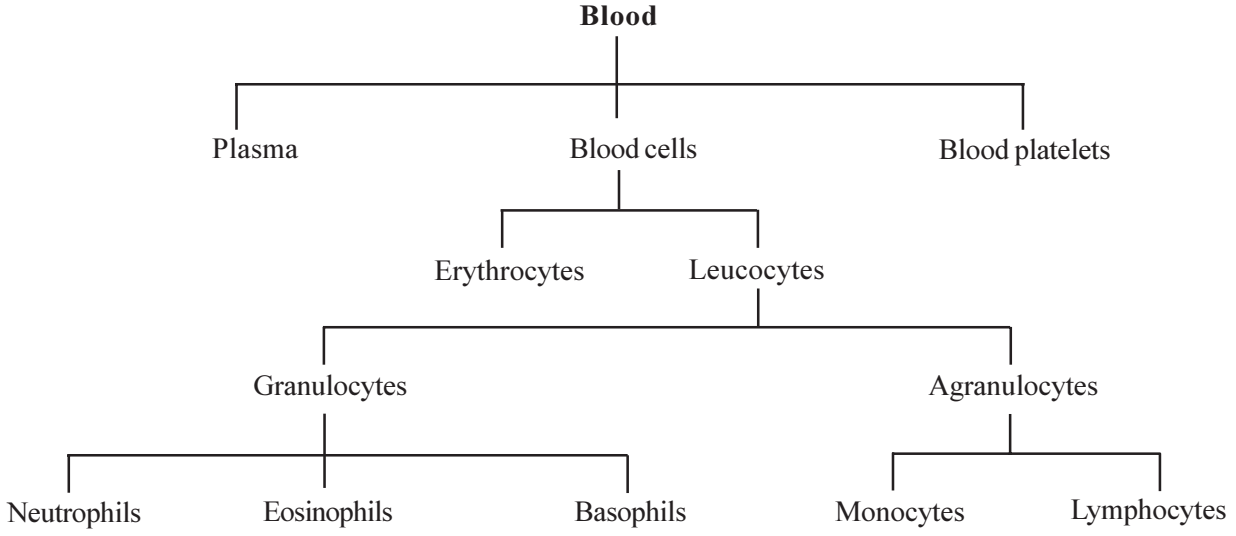
(I) Blood :

- It is an opaque turbid fluid. It is matrix.
- Liquid portion of this tissue is blood plasma.
- Blood cells are of two types : RBCs and WBCs.
- The fragments of cytoplasm that are present in the blood are called platelets.
- Blood differs from other connective tissue in these ways :

  (1) Blood cells are different from cells of other connective tissue.

  (2) Secretion of matrix is not entirely by blood cells.

  (3) Blood cells are not formed by the division of the pre-existing corpuscles.



- **Blood Plasma :**
  - This is the fluid matrix or intercellular substance.
  - 55 % amount occurs in blood by plasma.
  - It is almost colourless having faint yellow tinge.
  - It is alkaline and having pH 7.4.
- **Blood cells :**
  - Erythrocytes (Red Blood Corpuscles - RBCs).
  - The normal human erythrocyte is biconcave disc.
  - The red colour of erythrocyte is due to the presence of haemoglobin, a red coloured pigment which is a conjugated protein made up of globin and Fe^{+2} (Ferrous) containing haeme. It has high affinity for oxygen.
  - **Shape of RBCs :** The shape and size of RBCs vary in different animals. The cells are nucleated in all vertebrates, except mammals. In mammals it is biconcave.
  - **No. of RBCs :** Under normal condition the blood of adult male contains 41,00,000 to 60,00,000 erythrocytes per cubic milliliter, while that of adult female contains 39,00,000 to 55,00,000 per cubic milliliter.
  - **Life span :** Life span is about 120 days.
  - **Origination of RBCs :** From Red bone marrow.
  - **Function :** It participates in transporting carbon dioxide from tissue to lungs, as it contains enzyme carbonic anhydrase.

No. of RBCs

No.	Name of Organism	No. of RBCs
1.	Male (Human)	41,00,000 to 60,00,000 / 1 Cubic milliliter of blood
2.	Female (Human)	39,00,000 to 55,00,000 / 1 Cubic milliliter of blood
3.	Human Embryo	85 lacs / 1 Cubic milliliter of blood
4.	Rabbit	70 lacs / 1 Cubic milliliter of blood
5.	Frog	4 lacs / 1Cubic milliliter of blood

Life Span of RBCs

No.	Name of Organism	No. of RBCs
1.	Mammals and Human	120 days or 4 months
2.	Rabbit	80 days
3.	Frog	100 days
4.	New born baby	100 days

- Haemocytometer is used to calculate the Blood cells.
- **Leucocytes :**

  They are small nucleated semi transparent cells devoid of haemoglobin.

  - The leucocytes are capable of changing their shape and moving independently through the intercellular spaces among tissue. They are originated from bone marrow.
  - The number of these cells in adult humans are $7.5 \pm 3.5 \times 10^3$ per cubic millimeter of blood.
  - The number of cells depends upon the condition of the body. During infection of the body they generally increase in number. They are known as phagocytes because they feed on the bacteria and broken down tissue cells by engulfing particles.
  - **Types of WBCs :**
  - It has two types : Granulocytes and Agranulocytes
  - According to chemical nature of granules, granulocytes are of three kinds :

(i) Neutrophils :

- The neutrophils whose granules stain weakly with both the acidic and basic stain. They have may lobed nucleus.
- It is 60 - 70 % of total WBCs.

(ii) Eosinophils (Acidophils) :

- It can be stain by acidic dyes.
- It has bilobed nucleus and cells are large size.
- It is 2 - 4 % of total WBCs.

(iii) Basophils :

- It can be stained by Methylene blue.
- It has "S' shaped nucleus. It is 0.1 to 1 % of total WBCs.

(2) Agranulocytes :

- It is divided in to two types.
- It has medium alkaline cytoplasm with no lobe in nucleus.

(i) Monocytes

- Largest WBCs
- Nucleus is kidney shape
- 4 - 8 % of total WBCs

(ii) Leucocytes

- Smallest WBCs
- Large spherical nucleus
- 25 % of total WBCs

- **Platelets :**
- It is known as thrombocytes also. They are relatively small non-nucleated.
- They are especially concerned with the clotting of blood.
- They are made in the bone marrow.

- **General function of blood :**
  - Transports oxygen.
  - Clotting of blood.
  - Disposal of cell wreckage.
  - Transport and removal of carbon dioxide.
  - Transport of hormones, antitoxins, etc.
  - Equalization of body temperature.
  - Transport of food materials.
  - To neutralise the toxins.
  - Transport of waste matters.

---

(23) Which option is correct for given statements for function of connective tissue ?

(1) Removes mucus.

(2) Connects structures.

(3) Combat foreign toxins.

(4) Participates in excretion and absorption.

(5) To form a supporting frame work.

(6) To replace tissue which have been destroyed by injury.

(A) 1, 2 Right Statements; 3, 4, 5 and 6 Wrong Statements.

(B) 1 and 4 Wrong Statements; 2, 3, 5 and 6 Right Statements.

(C) 2, 3 and 4 Right Statements; 1, 5 and 6 Wrong Statements.

(D) 3, 4 and 5 Right Statements; 1, 2 and 6 Wrong Statements.

(24) Which tissue is simplest and widely distributed ?

(A) White fibrous (B) Adipose

(C) Areolar (D) Hyaline cartilage

(25) White fibres are x and yellow fibers are y in Areolar tissue.

(A) x = Wavy and branched, y = Thin, less, not in bundle and in bundles

(B) x = Wavy, unbranched, not in bundle.

y = Less, Thin and in bundle.

(C) x = Wavy, unbranched, in bundle.

y = Less, thin, not in bundle.

(D) x = Wavy, branched, in bundle.

y = More, thick and not in bundle.

(26) Which protein is present in white fibers and yellow fibers of Areolar tissue respectively ?

(A) Elastin, Collagen (B) Collagen, Keratin

(C) Collagen, Fibrinogen (D) Collagen, Elastin

(27) Which one is wrong statement ?

(A) Mast cells and Basophils secretes histamine and heparin.

(B) Life is long for mast cell, while it is short for Basophils.

(C) Mast cells are smaller, Basophils are binucleated.

(D) Mast cells are stable while basophils are movable.

(28) Which cartilage has a major role in recovery of nasal septum of Rudransh if it has been injured during cycling ?

(A) Compact cartilage (B) Calcified cartilage (C) Hyaline cartilage (D) Spongy cartilage

(29) Which tissue combat against the toxins which have been entered into the body of Rudransh ?

(A) Epithelial (B) Connective (C) Skeletal muscle (D) Nervous

(30) Which option is proper for secretion of collagen ?

(A) Histocytes (B) Fibroblast (C) Mast cells (D) Phagocytes

(31) Which pair is true for location of Hyaline cartilage ?

(A) Larynx, Trachea, Sternum, Hyoid, Ribs

(B) Larynx, Bonemarrow, Mesenteries, Sternum, Ribs

(C) Larynx, Subcutaneous, Around Kidney, Sternum, Ribs

(D) Larynx, Trachea, Sternum, Hyoid, Bone marrow

(32) Which tissue provides materials for the attachment of the muscles ?

(A) Epithelial (B) Skeletal (C) Nervous (D) Muscular

(33) The intervertebral discs are made up of .....

(A) Elastic cartilage (B) Fibrous cartilage (C) Calcified cartilage (D) Hyaline cartilage

(34) Which tissue is present in Epiglottis ?

(A) White fibrous cartilage (B) Calcified cartilage

(C) Hyaline cartilage (D) Yellow elastic cartilage

(35) Which tissue is present on the head of bone of *Rana tigrina* ?

(A) White fibrous (B) Calcified cartilage

(C) Hyaline cartilage (D) Yellow elastic cartilage

(36) Which option is correct for location of spongy bone ?

(A) Scapula, Ribs, Skull (B) Ileum, Ribs, Skull

(C) Vertebrae, Ribs, Skull (D) Patalla, Vertebrae, Skull

(37) Which option is proper to maintain cells in bone marrow ?

(A) Osteocytes (B) Chondrocytes (C) Osteoclast (D) Bidder canal

(38) What is called a joint between two Haversian canals ?

(A) Semi circular canal (B) Volkman's canal (C) Inguinal canal (D) Bidder canal

(39) Which structure is jointed by Volkman's canal ?

(A) Matrix - Haversian canal (B) Two Haversian canals

(C) Two different bones (D) Matrix - Osteocytes

(40) Which inorganic salts are present in ossein of skeletal matrix ?

(A) Calcium Chloride, Calcium Carbonate, Magnesium Phosphate

(B) Calcium Phosphate, Calcium Carbonate, Magnesium Phosphate, Calcium Flooride

(C) Pottassium Chloride, Magnesium Phosphate, Calcium Carbonate, Calcium Phosphate

(D) None of these

(41) Which option is correct for frog's bone marrow ?

(A) Yellow, Adipose tissue, Blood vessels (B) Green, Adipose tissue, Blood vessels

(C) Yellow, Hyaline cartilage, Blood vessels (D) Red, Adipose tissue, Blood vessels

(42) Which compound of blood is protective ?

(A) Albumin (B) Agglutinin

(C) Hormones (D) Salts of Pottassium

(43) Which one is non-protein group in the respiratory pigment of human RBCs ?

(A) Mg^{+2} (B) Fe^{+2} (C) Ca^{+2} (D) Zn^{+2}

(44) Which are active phagocytes in WBCs ?

(A) Neutrophils, Monocytes (B) Eosinophils, Lymphocytes

(C) Neutrophils, Eosinophils (D) Lymphocytes, Basophils

(45) To which function the globulin of human blood plasma will be responsible ?

(A) Immunity (B) Maintain osmotic pressure

(C) Conduction of oxygen (D) Blood clotting

(46) During resting condition, the process of formation of RBCs will be .....

(A) Increased (B) Decreased

(C) Unchanged (D) None of the given

(47) Arrange the WBCs in increasing order on the base of their quantity in 1 mm^3.
(A) Basophils > Eosinophils > Neutrophils
(B) Eosinophils > Basophils > Neutrophils
(C) Neutrophils > Eosinophils > Basophils
(D) Eosinophils > Neutrophils > Basophils

(48) Which is the correct quantity of Haemoglobin in 100 ml of Human blood ?
(A) 5–11 gram (B) 25–30 gram
(C) 17–20 gram (D) 12–160 gram

(49) In the structure of Haemoglobin .....
(A) 70 % globin + 30 % Haematin (B) 80 % globin + 20 % Haematin
(C) 95 % globin + 05 % Haematin (D) 90 % globin + 10 % Haematin

(50) What is the proportion of RBCs and WBCs in blood ?
(A) 1 : 1000 (B) 1000 : 1 (C) 5 : 3000 (D) 3000 : 5

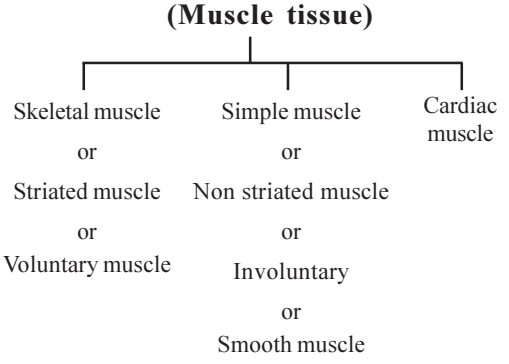
(51) Which blood cells can change their shape ?
(A) WBCs (B) RBCs (C) Platelets (D) All of the given

(52) Which is the waste material in blood ?
(A) Immunoglobulin (B) Thrombin (C) Fibrinogen (D) Carbon dioxide

Answers : (23-B), (24-C), (25-C), (26-D), (27-C), (28-A), (29-B), (30-B), (31-A), (32-B), (33-B), (34-D), (35-B), (36-C), (37-A), (38-B), (39-B), (40-B), (41-A), (42-B), (43-B), (44-A), (45-A), (46-B), (47-A), (48-D), (49-C), (50-C), (51-A), (52-D)

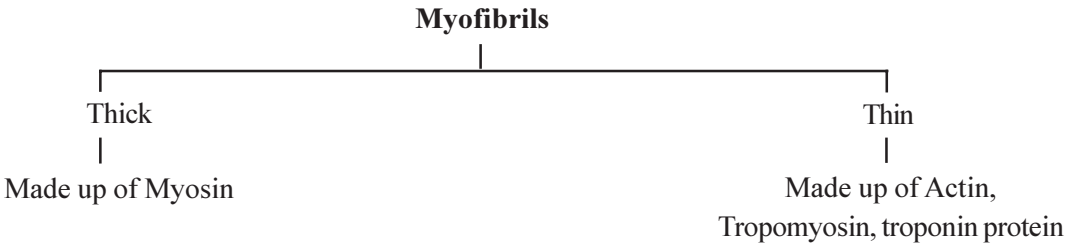
Muscular Tissue

- **Muscular tissue :**
  - It is formd from mesoderm. It consists of cellular elements in the form of fibres of various length.
  - No intercellular liquid is present.
  - All muscular tissue have great contractility.
  - By contraction and relaxation, it make possible the movement of whole body.
  - It has three types.



- **Skeletal muscle or striated muscle :**
  - The muscle fibers are the thin units of skeletal muscle. Each muscle fibre is a single thin and elongated cell provided with many nusclei.
  - The muscle fibers are arranged in bundles (fasciculi).
  - In higher animals, they are found associated with the skeleton.
  - They are attached to bones by tendons.
  - Muscle fibres can be contracted as desired. So, they are also known as voluntary muscles.
  - Because of their appearance they are called striated or striped muscle.
- **Structure of muscle fibers :**
  - The cytoplasm (sarcoplasm) of each fiber contains large number of myofibrils.
  - Each fiber has an extensive sheath, the sarcolemma.

- The muscle fibres show cross or transverse striations of alternative light and dark bands.
- The darker bands are called "A" band.
- The lighter bands are called "I" band.
- Each I band is divided by middle dark line, is known as z-disc or Krause's membrane.
- Two successive z-disc covers the region - "Sarcomere".



- Each sarcomere consists of one full A-band and two halves I-bands.
- In middle of A-band, the actin fibres are absent. So, it looks something lighter. It is known as H-band or Hensen's zone.

- **Smooth muscle or Involuntary muscle :**

  Location : This type of tissue is present in visceral organs - like oesophagus, stomach, intestine, blood vessels and pupil, etc.

  Structure :

  - It is made up of mononucleated spindle shape cells.
  - It has granular Sarcoplasm around its nucleus. These muscles always receive their nerve supply from autonomous nervous system.

- **Cardiac Muscle :**

  Location : In the wall of Heart.

  Structure :

  - It appears as network of branching and anastomosing cylinders.
  - The space between the cardiac muscle cells are occupied by the endomysium. The endomysium contains fibroblasts. Collagen and reticular fibres.
  - Each cell of cardiac muscle shows striation with A, I, Z and H-bands.
  - At the end of the cardiac muscle cells, there are prominent cross striation called intercalated discs.
  - Nucleus is arranged deep into sarcoplasm between two successive intercalated discs.
  - These muscles are capable of contracting rhythmically and are immune to fatigue. There is rich blood supply.
  - It is innervated by autonomous nervous system.

---

(53) What is the location of H-band ?

(A) Middle portion of myosin fibers of A-band

(B) Middle portion of Actin fibers in A-band

(C) Middle portion of myosin fibers of I-hand

(D) Middle portion of Actin fibers in Z-band

(54) Cellular energy is converted into _x_, during muscle contraction.

(A) x = Mechanical (B) x = Heat (C) x = Electric (D) x = Light

(55) What are the symbols for dark band, Krause's membrane, Hensen's zone, light band ?

(A) A, C, H, I (B) I, Z, H, A (C) A, Z, H, I (D) I, C, H, A

(56) What is structural unit for muscle ?

(A) Myosin (B) Actin (C) Sarcomere (D) Tropomyosin

(57) Actin and myosin is connected to .....

(A) Na^+ and K^+ pump (B) Muscle contraction

(C) Nervous system (D) Absorption of water

(58) Where is the strongest muscles are present in Human body ?

(A) Jaw (B) Thigh (C) Neck (D) Hands

(59) Where is the location of biggest muscle in Human ?

(A) Chick (B) Neck (C) Ear (D) Hip muscle

(60) In straited muscle, the thick and thin myofibrils are made up of proteins known as...

(A) Myosin, Collagen (B) Myosin, Troponin

(C) Myosin, Actin-Tropomyosin-Troponin (D) Keratin, Actin-Myosin-Troponin

(61) Which option is correct for smooth muscle ?

(A) Cylindrical, unbranched, straited, multinucleated and voluntary

(B) Spindle, unbranched, non-straited, mononucleated, involuntary

(C) Cylindrical, unbranched, straited, multinucleated, involuntary

(D) Spindle, unbranched, non-straited, multinucleated, involuntary

(62) What is the type of contraction of cardiac muscle ?

(A) Rhythmic, immune to fatigue (B) Rhythmic, not immune to fatigue

(C) Rapidly, not immune to fatigue (D) Rapidly, immune to fatigue

(63) How cardiac muscle defers from Non-straited muscle ?

(1) Intercalated disc (2) Autonomous (3) About nucleus

(A) 1 and 2 (B) 1 (C) 1, 2 and 3 (D) 2 and 3

(64) What is the name of dark and light band respectively ?

(A) I, A (B) A, I

(C) H, Z (D) Z, H

(65) What is Sarcomere ?

(A) Region between two successive Z-band (B) Region between two I-band

(C) Region between two successive A-band (D) The surface of two Z-bands

Answers : (53-B), (54-A), (55-C), (56-C), (57-B), (58-A), (59-D), (60-C), (61-B), (62-A), (63-C), (64-B), (65-A)

- **Nervous tissue :**

Nervous tissue

Neurons	Neuroglia
- Long processes	- Short processes
- It conducts nerve impulse	- It supports neurons

- **Structure of Neurons :**
- Typically each neuron consist of a cell body and branching fibres.
- **Cell-body :** It is called cyton, cytoplasm of cyton contains a large and spherical nucleus. There are large number of dark particles called Nissl's granules.

Processes (or fibres)
(Fibres arises from periphery of cyton)

Axons	Dendrites
- It is longer	- It is shorter
- Conducts impulse away from neuron	- Conducts impulse towards the neuron
- It is called efferent	- It is called afferent

Types of Neurons

Unipolar	**Bipolar**	**Multipolar**
- Only one process	- One axon and one dendrite	- More than two processes
- e.g. In Embryo	- e.g. Retina	- e.g. Brain, Spinal Cord

- The nerve fibres may be surrouned by two concentric sheaths. The inner is known as medulla or myelin sheath.
- It is surrounded by a transparent cellular sheath, the neurilemma.
- The neurilemma sheath is made up of single layer of flat expanded Schwann's cells.
- Each myelinated nerve fiber shows constrictions at regular intervals called nodes of Ranvier.
- The nerve endings of axon are not in direct physical contact with nerve endings of dendrites of another neuron. This physical gap is called synapse.
- Nerve impulses pass between neurons through the synapse with the help of a hormone Acetylecholine called Neurotransmitters.

(66) Which fibre conducts the impulse towards cyton ?

(A) Axon (B) Dendrite (C) White fibres (D) Yellow fibres

(67) Which hormone is responsible for the conduction of impulse in synapse ?

(A) Gastrin (B) Glucagon (C) Collagen (D) Acetylcholine

(68) What is the composition of Nissl's granules ?

(A) Mitochondria (B) Golgi body (C) Lysosome (D) Ribosome

(69) The myelin sheath is present in which tissue ?

(A) Epithelial (B) Connective (C) Muscular (D) Nervous

(70) Which type of Neurons are present in the mammalian brain ?

(A) Monopolar (B) Multipolar (C) Bipolar (D) All of the given

(71) Which type of neuron is present in mammalian retina of eyes ?

(A) Monopolar (B) Multipolar (C) Bipolar (D) All of the given

(72) Which type of neuron is present in Embryo ?

(A) Monopolar (B) Multipolar (C) Bipolar (D) All of the given

Answers : (66-B), (67-D), (68-D), (69-D), (70-B), (71-C), (72-A)

- **True and Flase (T - F) Type Questions :**

(73) Which option is correct for given statement ?

(1) The cells of squamous epithelium are thick, flat and polygonal.

(2) The cells of cuboidal epithelium in longitudinal section are polygonal.

(3) The internal end of the cells of columnar epithelium are broad and narrow at free end.

(4) Ciliated epithelium is modified columnar epithelium.

(A) F, T, F, T (B) T, T, F, F (C) F, T, T, F (D) F, F, T, T

(74) (1) Pseudostratified epithelium is simple columnar epithelium.

(2) Pseudostratified epithelium removes mucus in particular direction.

(3) Transitional epithelium remains in organ of excretory system.

(4) Stratified epithelium provides durable covering to the organs.

(A) F, F, T, T (B) T, F, T, T (C) T, T, F, F (D) T, F, T, F

(75) In reference to connective tissue which option is correct ?

(1) It connects the organs.

(2) Combat foreign toxins.

(3) Makes suture in skull bones.

(4) Adipose tissue remain beneath the skin.

(A) T, T, T, T (B) T, F, T, T (C) F, T, T, T (D) F, F, T, T

(76) (1) Simplest and widely spreaded tissue is areolar.

(2) White fibrous tissue is in tendon and is inelastic.

(3) White fibers are made up of elastic protein and yellow fibres are of collagen protein.

(4) Blood is opaque and turbid liquid.

(A) T, T, F, F (B) F, F, T, T (C) T, T, F, T (D) T, F, T, F

(77) Which option is correct in reference to function of Blood ?

(1) Conducts oxygen (2) Do not conduct waste

(3) Neutralises toxins (4) Do not maintain temperature

(A) T, T, F, F (B) F, F, T, T (C) F, T, F, T (D) T, F, T, F

(78) Which option is correct for skeletal muscle ?

(1) A-band dark and has myosin

(2) I-band is light and has actin

(2) A-band contracts during muscle contraction

(4) Sarcomere is a region between two successive Z-band

(A) T, T, F, T (B) T, T, F, F (C) F, F, T, T (D) T, T, T, T

(79) Which option is correct in reference to Nervous tissue ?

(1) Neurons have long processes which conducts impulses.

(2) Neuroglia has short processes and provide support to Nervous tissue.

(3) Monopolar Neuron has only one process.

(4) Bipolar neuron has one axon and one dendrite.

(A) T, F, T, F (B) T, T, F, F (C) T, T, T, T (D) F, F, F, F

Answers : (73-A), (74-B), (75-A), (76-C), (77-D), (78-A), (79-C)

(80) Match the column :

Column - I	**Column - II**	
(1) Squamous epithelium	(A) Intestinal glands	(A) 1 - A, 2 - B, 3 - D, 4 - C, 5 - E
(2) Cuboidal epithelium	(B) Trachea	(B) 1 - E, 2 - D, 3 - B, 4 - A, 5 - C
(3) Columnar epithelium	(C) Ovary	(C) 1 - D, 2 - E, 3 - A, 4 - B, 5 - C
(4) Ciliated epithelium	(D) Blood vessels	(D) 1 - D, 2 - C, 3 - A, 4 - E, 5 - B
(5) Pseudostratified	(E) Uriniferous tubules	

(81) Match the column :

Column - I	**Column - II**	
(1) Squamous epithelium	(A) Square in longitudinal section	(A) 1 - A, 2 - C, 3 - B, 4 - D
(2) Ciliated epithelium	(B) Very thin and flat cells	(B) 1 - D, 2 - B, 3 - C, 4 - A
(3) Cuboidal epithelium	(C) Polygonal cells and broad at free end	(C) 1 - B, 2 - D, 3 - A, 4 - C
(4) Columnar epithelium	(D) Modified columnar epithelial cells	(D) 1 - C, 2 - A, 3 - D, 4 - B

(82) Match the column :

Column - I	**Column - II**	
(1) Epithelial tissue	(A) Combat foreign toxins	(A) 1 - A, 2 - D, 3 - C, 4 - B
(2) Connective tissue	(B) Capacity for contraction	(B) 1 - B, 2 - A, 3 - C, 4 - D
(3) Muscular tissue	(C) Conduction of nerve impulse	(C) 1 - A, 2 - B, 3 - C, 4 - D
(4) Nervous tissue	(D) Arrange on basal membrane	(D) 1 - D, 2 - A, 3 - B, 4 - C

(83) Match the column :

Column - I	Column - II	
(1) Neutrophils	(A) Spherical large nucleus	(A) 1 - D, 2 - E, 3 - B, 4 - A, 5 - C
(2) Eosinophils	(B) S shaped nucleus	(B) 1 - A, 2 - C, 3 - D, 4 - B, 5 - E
(3) Basophils	(C) Kidney shaped nucleus	(C) 1 - C, 2 - B, 3 - A, 4 - E, 5 - D
(4) Lymphocytes	(D) 2-7 lobed nucleus	(D) 1 - B, 2 - A, 3 - C, 4 - D, 5 - E
(5) Monocytes	(E) Bilobed nucleus	

(84) Match the column :

Column - I	Column - II	
(1) Squamous epithelium	(A) Beneath the skin	(A) 1 - D, 2 - A, 3 - C, 4 - B
(2) Hyaline cartilage	(B) Bowman's capsule	(B) 1 - B, 2 - D, 3 - A, 4 - C
(3) Adipose tissue	(C) Wall of stomach	(C) 1 - A, 2 - B, 3 - D, 4 - C
(4) Simple muscle	(D) Hyoid bone	(D) 1 - D, 2 - C, 3 - B, 4 - A

(85) Match the column :

Column - I	Column - II	
(1) Tendon	(A) Matrix	(A) 1 - C, 2 - D, 3 - A, 4 - B
(2) Ligament	(B) Ossein	(B) 1 - A, 2 - D, 3 - C, 4 - B
(3) Hyaline cartilage	(C) Collagen fibers	(C) 1 - B, 2 - D, 3 - C, 4 - A
(4) Bone	(D) Chondrine	(D) 1 - C, 2 - A, 3 - D, 4 - B

Answers : (80-D), (81-C), (82-D), (83-A), (84-B), (85-D)

- **A - Assertion and R - Reason type questions :**

Which option is correct for given question ?

(A) A and R both correct, R is correct explanation of A.

(B) A and R both correct, R is not correct explanation of A.

(C) A - correct, R - wrong.

(D) A - wrong, R - correct.

(86) Assertion A : Squamous epithelium is like mosaic tiles at a surface view.

Reason R : Cells of this tissue is joined by cement material.

(A) (B) (C) (D)

(87) Assertion A : The cells of squamous epithelia are thin and flat.

Reason R : Main function of this tissue is of protection.

(A) (B) (C) (D)

(88) Assertion A : The cells of cuboidal tissue are polygonal in L.S. and square in T.S.

Reason R : Its function is protection, secretion, excretion and absorption.

(A) (B) (C) (D)

(89) Assertion A : The cells of columnar tissue are elongated and like pillar.

Reason R : Function of this tissue is secretion and absorption.

(A) (B) (C) (D)

(90) Assertion A : Ciliated tissue is modified into columnar tissue.

Reason R : Pseudostratified epitheliam is a kind of simple epithelium tissue.

(A) (B) (C) (D)

(91) Assertion A : White fibres are wavy and branched in Areolar tissue.

Reason R : White fibers are made up of collagen protein.

(A) (B) (C) (D)

(92) Assertion A : Adipose tissue helps in maintaining body temperature.

Reason R : Suture of skull is occupied by white fibrous tissue.

(A) (B) (C) (D)

(93) Assertion A : Cartilage differs from normal connective tissue.

Reason R : It contains dense matrix.

(A) (B) (C) (D)

(94) Assertion A : In bones of mammals, they have large amount of tubular structures in it. It is called haversian system.

Reason R : It is seen in sponge bone, vertebrae, ribs and skull.

(A) (B) (C) (D)

(95) Assertion A : Serum is part of blood other than required components for blood clotting.

Reason R : Human RBCs are biconcave discoid shape.

(A) (B) (C) (D)

(96) Assertion A : Nucleus of Monocyte is kidney shaped.

Reason R : Monocytes are smallest WBCs.

(A) (B) (C) (D)

(97) Assertion A : Platelets are also known as Thrombocytes.

Reason R : The secretes Thromboplastin.

(A) (B) (C) (D)

(98) Assertion A : WBCs are called as phagocytes.

Reason R : They engulf bacteria and broken cells.

(A) (B) (C) (D)

(99) Assertion A : Skeletal muscle is also known as voluntary muscle.

Reason R : They can contract according to desire of an animal.

(A) (B) (C) (D)

(100) Assertion A : Smooth muscle tissue is made up of spindle shaped cells with a single nucleus.

Reason R : It is seen in alimentary canal and pupil.

(A) (B) (C) (D)

(101) Assertion A : Cardiac muscle has cross striation at end, is called intercalated discs.

Reason R : Comparision to Z-disc, it is thinner.

(A) (B) (C) (D)

(102) Assertion A : Blood differes from other connective tissue.

Reason R : Blood cells do not secretes matrix.

(A) (B) (C) (D)

(103) Assertion A : Dendrites are afferent in nature.

Reason R : They conduct impulse towards the cyton.

(A) (B) (C) (D)

(104) Assertion A : Axon are efferent in nature.

Reason R : They conduct impulse away from cyton.

(A) (B) (C) (D)

(105) Assertion A : Monopolar Neuron has only one process.

Reason R : In this Neuron dendrite and axon are same fibers.

(A) (B) (C) (D)

(106) Assertion A : Monopolar Neurons are present in Retina.

Reason R : Monopolar neurons has one fibers only on one side.

(A) (B) (C) (D)

(107) Assertion A : Multipolar neurons has more than two processes.

Reason R : Bipolar neurons has two processes.

(A) (B) (C) (D)

(108) Assertion A : Sarcoplasm of neuron has large and oval shape nucleus.

Reason R : Cyton has some dark granules, which are called as Nissl's granules.

(A) (B) (C) (D)

(109) Assertion A : Axon and dendrite are not in direct physical contact.

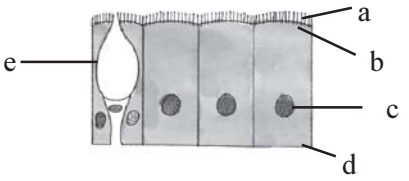
Reason R : Acetylcholine act as a neurotransmitters.

(A) (B) (C) (D)

Answers : (86-B), (87-C), (88-D), (89-B), (90-D), (91-D), (92-B), (93-A), (94-B), (95-B), (96-C), (97-A), (98-A), (99-A), (100-B), (101-C), (102-A), (103-A), (104-A), (105-A), (106-D), (107-B), (108-D), (109-B)

- **Figure based questions :**

(110) Which tissue is mentioned in given figure ?



(A) Columnar (B) Ciliated

(C) Pseudostratified (D) Cuboidal

(111) What is indicated by a, b and c in given figure ?

(A) a = Basal granules, b = pili, c = Nucleus (B) a = Nucleus, b = pili, c = Basal granules

(C) a = Basal membrane, b = pili, c = Nucleus (D) a = Cilia, b = Basal granules, c = Nucleus

(112) What is indicated by d and e in given figure ?

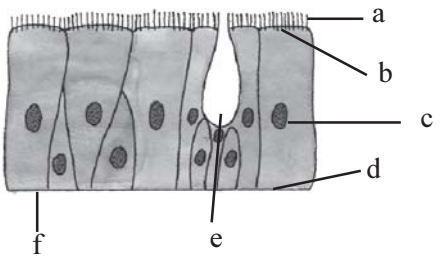
(A) d = Basement membrane, e = mucus secreting cells

(B) d = Nucleus, e = Basal granules

(C) d = Mucus secreting cells, e = Basal membrane

(D) d = pili, e = Mucus secreting cells

(113) Which tissue is mentioned in given figure ?



(A) Columnar (B) Ciliated

(C) Pseudostratified (D) Cuboidal

(114) What is indicated by a, b and c in given figure ?

(A) a = Nucleus, b = pili, c = Basal granules (B) a = Cilia, b = Basal granules, c = Nucleus

(C) a = Basal membrane, b = pili, c = Nucleus (D) a = Basal membrane, b = pili, c = Nucleus

(115) What is indicated by d, e and f in given figure ?

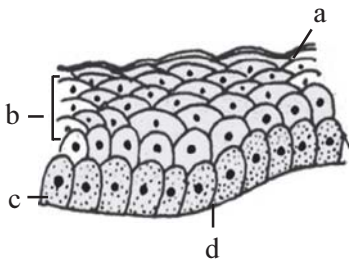
(A) d = Basal cell, e = Mucus secreting cell, f = Basal membrane

(B) d = Basal membrane, e = Basal cell, f = Mucus secreting cell

(C) d = Basal cell, e = Basal membrane, f = Mucus secreting cell

(D) d = pili, e = Basal cell, f = Mucus secreting cell

(116) Which tissue is indicated in given figure ?



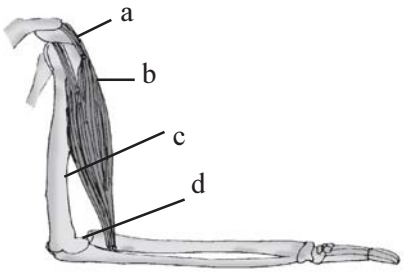
(A) Pseudostratified (B) Ciliated

(C) Cuboidal (D) Stratified epithelium

(117) Match the Column by help of the given figure :

Column - A	**Column - B**	
(p) Basement membrane	(i) b	(A) (p-ii), (q-i), (r-iii), (s-iv)
(q) Epithelial layer	(ii) d	(B) (p-ii), (q-i), (r-iv), (s-iii)
(r) Germinative layer	(iii) a	(C) (p-ii), (q-iv), (r-i), (s-iii)
(s) Dead cells	(iv) c	(D) (p-ii), (q-iii), (r-i), (s-iv)

(118) What is indicated by a, b, c and d in given figure ?



(A) a = Ligament, b = Muscle, c = Bone, d = Tendon

(B) a = Ligament, b = Bone, c = Muscle, d = Tendon

(C) a = Ligament, b = Bone, c = Tendon, d = Muscle

(D) a = Tendon, b = Muscle, c = Muscle, d = Ligament

(119) Which tissue is indicated by given figure ?

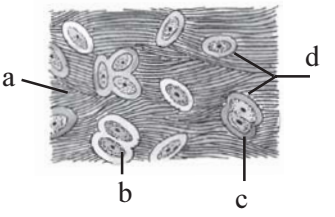
(A) White elastic cells

(B) Yellow elastic cell

(C) Hyaline cartilage

(D) White fibrous tissue

(120) What is indicated by a, b, c and d in the given figure ?



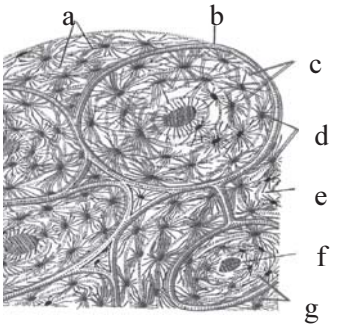
(A) a = Cartilage cell, b = Matrix, c = Collagen fiber, d = Lacuna

(B) a = Collagen fibers, b = Lacuna, c = Matrix, d = Cartilage cells

(C) a = Collagen fibers, b = Matrix, c = Lacuna, d = Chondroblast

(D) a = Matrix, b = Chondroblast, c = Lacuna, d = Collagen fibers

(121) What is indicated by a, b and c in given figure ?



(A) a = Haversian system, b = Interstitial lamellae, c = Concentric lamellae

(B) a = Concentric lamellae, b = Haversian system, c = Interstitial system

(C) a = Interstitial lamellae, b = Haversian system, c = Concentric lamellae

(D) a = Haversian lamellae, b = Canaliculi, c = Haversian system

(122) What is indicated by d, e, f and g in given figure ?

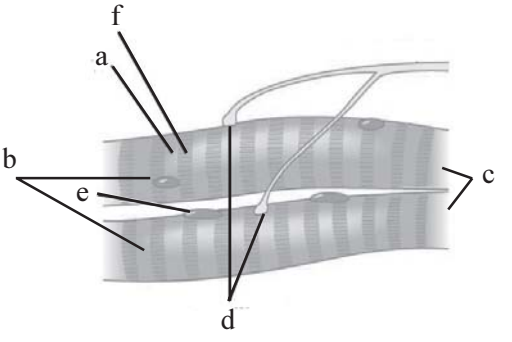
(A) d = Lacunae with bone cell, e = Matrix, f = Haversian system, g = Canaliculi

(B) d = Concentric lamellae, e = Matrix, f = Haversian system, g = Canaliculi

(C) d = Haversian system, e = Canaliculi, f = Matrix, g = Haversian tubules (canal)

(D) d = Canaliculi, e = Matrix, f = Haversian system, g = Haversian tubules (canal)

(123) Identify the parts of striated muscle in given figure ?



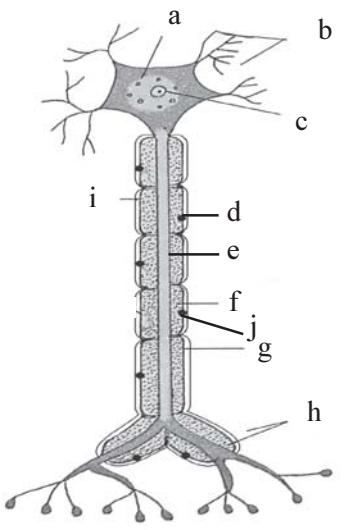
(A) a = Sarcoplasm, b = Nucleus, c = Sarcolemma, d = Muscle fibre, e = Dark band, f = Light band

(B) a = Sarcoplasm, b = Light band, c = Muscle fibers, d = Sarcolemma, e = Nucleus, f = Dark band

(C) a = Light band, b = Sarcoplasm, c = Muscle fibers, d = Sarcolemma, e = Nucleus, f = Dark band

(D) a = Sarcolemma, b = Nucleus, c = Dark band, d = Light band, e = Sarcoplasm, f = Muscle fibers

(124) Match the column by the help of given figure :



Column - A	Column - B
(p) Nucleus	(i) h
(q) Dendrite	(ii) i
(r) Node of Ranveir	(iii) b
(s) Nerve fibers	(iv) c
(t) Schwann's cell	(v) d
(u) Nerve fibers (axis)	(vi) e

(A) (p-iv), (q-iii), (r-v), (s-i), (t-ii), (u-vi)

(B) (p-iv), (q-iii), (r-v), (s-ii), (t-i), (u-vi)

(C) (p-iv), (q-ii), (r-iii), (s-v), (t-i), (u-vi)

(D) (p-iv), (q-vi), (r-ii), (s-iii), (t-i), (u-v)

(125) What is indicated by a, f, g and j by given figure ?

(A) a = dendrite, f = Myelin sheath, g = Nerve fiber, j = Nucleus

(B) a = Nissl's granule, f = Myelin sheath, g = Nerve fiber, j = Nucleus

(C) a = Nissl's granule, f = Nerve fiber, g = Myelin sheath, j = Nucleus

(D) a = Nissl's granule, f = Nucleus, g = Myelin sheath, j = Nerve fiber

Answers : (110-B), (111-D), (112-A), (113-C), (114-B), (115-A), (116-D), (117-B), (118-A), (119-A), (120-B), (121-C), (122-A), (123-C), (124-A), (125-B)

- **Questions for NEET :**

(126) Blood vessels are made up of which cells ?

(A) Columnar epithelial (B) Connective tissue

(C) Simple muscle tissue (D) Squamous epithelium

(127) Where is the location of columnar tissue in Human ?

(A) Intestine and Urethra (B) Auditory canal and Wall of stomach

(C) Bronchus and Fallopian tube (D) Stomach and Oesophagus

(128) Which tissue is present in layer of Alveoli ?

(A) Simple epithelium (B) Cuboidal epithelium

(C) Squamous epithelium (D) Ciliated epithelium

(129) Collagen is made up of ..... .

(A) Lipid (B) Carbohydrate (C) Globular protein (D) Fibrous protein

(130) Ligaments means

(A) Modification of white elastic fibers (B) Modification of yellow elastic fibers

(C) Inelastic white fibers (D) None of the given

(131) Areolar tissue connects ..... .

(A) Bone to bone (B) Lipid to muscle

(C) All the muscles (D) Muscle to bone

(132) What is absent in connective tissue ?

(A) Hyleuronic acid (B) Basement membrane

(C) Collagen fibers (D) Liquid

(133) Which fibers of striated muscles are connected with the activity of ATPase ?

(A) Actin (B) Troponin (C) Tropomyosin (D) Myosin

(134) What is called the cell of Bone ?

(A) Chondroclast (B) Osteoclast (C) Lecuna (D) Osteocytes

(135) To whom the globulin protein of human blood plasma is connected ?

(A) Immunity (B) Maintenance of osmotic pressure

(C) Conduction of oxygen (D) Blood clotting

(136) Which WBCs has highest amount in blood ?

(A) Neutrophils (B) Eosinophils (C) Basophils (D) Lymphocytes

(137) WBCs are true cells, Because ..... .

(A) Nucleus is present (B) Acts as a phagocytes

(C) It is polymorphism (D) None of the given

(138) How much Haemoglobin is present in human RBCs ?

(A) 40 % of biomass (B) 34 % of biomass

(C) 90 % of biomass (D) 50 % of biomass

(139) Which option is correct for content of Nissl's granules ?

(A) Cellular molecular (B) Lipid molecules (C) Ribosomes (D) Mitochondria

(140) What is called the connection of axon of a neuron with dendrite of another neuron ?

(A) Constant bridge (B) Synapse (C) Joint (D) None of the given

Answers : (126-D), (127-C), (128-C), (129-D), (130-B), (131-C), (132-B), (133-D), (134-D), (135-A), (136-A), (137-A), (138-B), (139-C), (140-B)

ANIMAL TISSUES

Epithelial tissues
- One or more layers.
- Outer and inner layer of organ or body.

Simple Epithelium
- **Squamous** mosaic files like flat and single layer, polygonal cells
- **Cuboidal** Small, cube, shapecells, Villi at free end of cells, Brush like Villi
- **Columnar** Elongated cells like pillar
- **Ciliated** Diverse in shape with nucleus with cilia at free end

Compound Epithelium
- Keratinised, non keratinised, Transitional tissue is with 4 - 6 layers
- Transitional epithelial tissue :
  - 4 to 6 layers
  - maintains thickness

  It shows re-organisation

Glandular Epithelium
Secretory in nature. In glands

Connective tissues
- Structural frame, support to body.
- Interstitial liquid, cells and fibers.

Semiliquid/Semi solid
- **Areolar tissue** Connects different body parts
- **Adipose tissue** Eye ball and around kidney for absorption
- **Tendor** Joins muscle to bone
- **Ligament** Joins bone to bone

Supportive skeletal tissue
- **Cartilage tissue** Semi-solid matrix, Hyaline, Fibrous, Caclified and Elastic
- **Bones** Hardest tissue, Solid matrix, Bony and spongy bones

Liquid tissue
- **Blood** Red liquid, RBCs, WBCs platelets
- **Lymph** Pale yellow, Blood plasma and WBCs

Muscular tissues
- Contractive
- Becomes active by electricity
- Helpful in movement

- **Skeletal muscle** Elongated and unbranched muscles. Joined with bones.
- **Smooth muscle** Elongated and spindle shape fibers. Involuntary muscles. Visceral muscles.
- **Cardiac muscle** Elongated and cylindrical fibers. Never get fatigue. Antonomous.

Nervous tissues
- Controls the body according to stimulation and body response.
- It has conduction, co-ordination and response.
- Nerve cells, Neuroglial cells, Neuroganglion, etc.

16 Animal Morphology and Anatomy-I (Earthworm and Cockroach)

General information, morphology and body-wall of Earthworm :

- Earthworm is a typical coelomate animal of annelida phylum.
- In our country, *Pheretima posthuma* is found commonly.
- Habitat : Upper layer of the moist soil.
- During day time it lives in burrow and eat the soil. Whatever the soil it consumes with food is excreted as worm castings and with the help of these castings earthworms can be traced.

Morphology :

- Earthworm is cylindrical, a little bit long and thin.
- Its body is divided into small segments which are approximately 100 to 120 in number.
- The dorsal surface of the body is marked by a dark median dorsal line along the longitudinal axis of the body.
- The ventral surface is distinguished by the presence of genital pores.
- Mouth : Anterior side of the body.
- Prostomium : It is sensitive organ
- Peristomium : It is first segment, in which mouth is found.
- Clitellum : 14 to 16 segments
- Body divided into three regions : (1) Preclitellar area (1 to 13 segments) (2) Clitellar area (14 to 16 segments) and (3) Postclitellar area (17 to last segments)
- Spermathecal apertures : 5/6, 6/7, 7/8, 8/9 intersegmental grooves.
- Genital pore : A single female genital pore is present in the mid ventral line of the 14 th segment.
- Male genital pore : One pair of male genital pore, present on the ventro-lateral side of 18 th segment.
- Nephridia : Number of minute pores called Nephridiopores, get opened on the surface of the body which are found in all segments except first, last and clitellum.
- Setaes : In the midst of each segment a circle of setae is found. These setaes are made up of chitin and help in locomotion.
- Body-wall of earthworm :

  - cuticle - epidermis - circular muscles - longitudinal muscles - parietal layer
- Cuticle : Cuticle is a thin layer secreted from epidermis.
- Epidermis : The epidermis is made up of a single layer of columnar supporting cells and oblong gland cells which secrete albumin and mucus.
- Sensitive cells are found in groups. They bear cilia which takes inspiration from external stimulation.

(1) The earthworm has _y_ genus and _z_ species and _x_ phylum.

(A) x = Coelenterata, y = Rana, z = tigrina (B) x = Annelida, y = Pheritima, z = tigrina

(C) x = Annelida, y = Pheritima, z = posthuma (D) x = Arthropoda, y = Periplanata, z = americana

(2) Which sentence is wrong in reference to earthworm ?

(A) Earthworm can be located upper layer of moist soil

(B) During night, it lives in burrow and eat the soil

(C) Earthworm can be traced by extretory castings

(D) Its colour is reddish brown

(3) The body of Earthworm has _x_ segments and its number is _y_.

(A) x = small, y = 100 to 120 (B) x = small, y = 80 to 100

(C) x = big, y = 100 to 120 (D) x = big, y = 100 to 150

(4) In how many regions the earthworm can be divided on the basis of clitellum ?

(A) One (B) Two (C) Three (D) Four

(5) The Earthworm has preclitellar _x_ segments, clitellar _y_ segments and post clitellar _z_ segments.

(A) x = 4 to 13, y = 14 to 15, z = 16 to last

(B) x = 1 to 13, y = 14 to 16, z = 17 to 20

(C) x = 1 to 10, y = 11 to 13, z = 14 to 16

(D) x = 1 to 13, y = 14 to 16, z = 17 to last

(6) In earthworm, female genital pore are located on _x_ segment and on _y_ line.

(A) x = 18, y = mid-ventral (B) x = 14, y = mid-ventral

(C) x = 14, y = ventro-lateral (D) x = 18, y = ventro-lateral

(7) In earthworm, male genital pore remain on _x_ segment and on _y_ side.

(A) x = 18, y = ventro-lateral (B) x = 18, y = mid-ventral

(C) x = 14, y = ventro-lateral (D) x = 14, y = mid-ventral

(8) On which segments, spermathecal (apertures) are located ?

(A) 4/5, 5/6, 6/7, 7/8 (B) 6/7, 7/8, 8/9, 9/10

(C) 5/6, 6/7, 7/8, 8/9 (D) 14/15, 15/16, 16/17, 17/18

(9) On which segments of earthworm, Nephridiopores are absent ?

(A) First, last, clitellar (B) First, Fourth, Sixth

(C) First, last, 17 to 20 (D) Last and 2 to 4

(10) By whose secretion the cuticle is synthesized in earthworm ?

(A) Circular muscles (B) Longitudinal muscles (C) Epidermis (D) Parietal layer

(11) Which one is secreted by gland cells of epidermis in earthworm ?

(A) Keratin, mucus (B) Fibrinogen, mucus (C) Mucus, Albumin (D) Keratin, Albumin

Answers : (1-C), (2-B), (3-A), (4-C), (5-D), (6-B), (7-A), (8-C), (9-A), (10-C), (11-C)

Digestive canal and circulatory system of Earthworm :

- Alimentary canal of earthworm is simple tubular passage from mouth to anal opening.
- **Terminal mouth :** It opens into the buccal cavity which spreads upto 1-3 segments.
- **Mouth cavity :** Because of the contraction of muscles joining body wall, buccal cavity protrudes out of the body and accepts the food.
- **Pharynx :** It is muscular tube found after buccal cavity and is extended upto fourth segment.
- **Oesophagus :** It is extended from 5 to 7 segments which opens into gizzard.
- **Gizzard :** The wall of Gizzard is thick and there are thick layers of circular muscles. The internal surface of it is made up of columnar cells covered with cuticle. Due to the contraction of circular muscles gizzard grinds the soil particles and decaying leaves like a grinder.
- **Stomach :** It is extended from 9 to 14 segments. Calciferous glands are found in stomach, neutralize the humic acid present in humus.
- **Intestine :** It starts from 15th segments and continues till last segment.
- **Caecum :** A pair of short and conical intestinal caecum projects from the intestine on the 26th segment, which secrets enzyme for carbohydrate digestion.
- **Pretyphlosole region :** From segment 1 to 25, it is called pretyphlosolar region.
- **Typhlosole region :** From 26 to 95 segments internal median fold of dorsal wall called typhlosole is present, which increases the area of absorption.
- **Post typhlosole region :** Last 23 to 25 segments of intestine lack typhlosole, so it is called post-typhlosole region ?
- **Anal aperture :** The alimentary canal opens to the exterior by a small rounded anal aperture.
- The ingested organic rich soil passes through alimentary canal and digestive enzymes breakdown complex food into smaller absorbable units.

- **Circulatory system :**
  - In earthworm, closed type of circulatory system is found.
  - In circulatory system, blood vessels, capillaries and heart is included.
  - Due to closed circulatory system blood is confined to the heart and blood vessels.
  - Due to contraction, blood circulates into one direction.
  - Small blood vessels supply blood to gut, nerve cord and the body wall.
  - Blood glands are present on the 4th, 5th and 6th segments, which produce blood cells and haemoglobin.
  - Blood cells are of phagocytic type.
  - In earthworm there is absence of specialized breathing device.
  - Respiratory exchange occurs through moist body surface and oxygen gets mixed into blood stream.

(12) By which, the internal wall of gizzard is covered in earthworm ?

(A) Keratin (B) Mucus (C) Chitin (D) Cuticle

(13) Which muscle layers are found in gizzard and in which segment it is located ?

(A) Circular muscle, 9th (B) Longitudinal muscle, 8th

(C) Circular muscle, 8th (D) Longitudinal muscle, 9th

(14) In earthworm, stomach is extended upto _x_ segment and it has _y_ glands.

(A) x = 8 to 14, y = Calciferous (B) x = 8 to 13, y = Calciferous

(C) x = 9 to 14, y = Calciferous (D) x = 9 to 14, y = Pancreas

(15) Intestine of earthworm is extended from _x_ to _y_ segments.

(A) x = 14, y = 95 (B) x = 15, y = Last

(C) x = 26, y = Last (D) x = 23, y = Last

(16) In earthworm, small blood vessels supply blood to...

(A) Mouth, Stomach, Intestine (B) Body wall, Nerve cord, Intestine

(C) Body wall, Oesophagus, Stomach (D) Mouth, Nerve cord, Intestine

(17) In which segment of earthworm blood glands are located ?

(A) 4th, 5th and 6th (B) 3rd, 4th and 5th

(C) 1st, 2nd and 4th (D) 7th, 8th and 9th

(18) In earthworm, haemoglobin is dissolved in...

(A) RBCs (B) Platelets (C) Blood plasma (D) WBCs

(19) What type of blood cells are found in earthworm ?

(A) Mast cells (B) Phagocyte cells (C) Fibrous cells (D) All of the given

(20) Which organ is absent in earthworm ?

(A) Sensory (B) Excretory (C) Reproductive (D) Respiratory

(21) From where O_2 enters in to the earthworm ?

(A) Respiratory organs (B) Lungs (C) Body wall (D) Mouth cavity

Answers : (12-D), (13-C), (14-C), (15-B), (16-B), (17-A), (18-C), (19-B), (20-D), (21-C)

Nephridia, Nervous system and reproductive system of Earthworm :

- There are three types of Nephridia in earthworm :

Septal Nephridia	Integumentary Nephridia	Pharyngeal Nephridia
- From 15 to last segment - Found on both side of the inter-segmental septa - Opens into intestine	- From 3rd segment, it is attached to the lining of body woll - Open into external surface of the body through minute pores	- Three pairs in 4th, 5th and 6th segment - Discharge their waste into alimentary canal

- All three types of nephridia basically similar in structure.

Function :

- Regulates the volume and composition of body fluid.
- By the help of nephrostome, they absorb excretory waste from coelome and discharge through a pore to the surface of the body wall and into the digestive tube.

Nervous system of Earthworm :

- Nervous system is ganglionated.
- **Ganglia :** Formed by gathering nerve cells.
- **Ventral nerve cord :** One pair of nerve ganglia is found at the ventral surface of each segment after the fifth segment.
- **Sub pharyngeal ganglion :** One pair of ganglia is found at the posterior part of ventral side of third segment.
- **Suprapharyngeal ganglion :** One pair of ganglia found at the dorsal side of pharynx in the 3rd segment.
- These ganglioa are connected by circumpharyngeal connective.
- These pair of connective and subpharyngeal ganglion connected with suprapharyngeal ganglion and formes nerve ring.
- Suprapharyngeal ganglion and other nerve rings complies the information and reach immediately by making the body muscles to implement it.
- In earthworm, specific sensory organs like eyes are not present.
- Only sensory cells of the body wall perform as a sensory receptor organs.
- These cells are inspired by the intensity of light, vibrations of the land, etc. Some of the sensory cells are inspired by chemical instigator.

Reproductive system of earthworm :

- Earthworm is hermaphrodite (bisexual) i.e. testes and ovary both are present in same animal.

Male reproductive organs :

- **Testes :** Two pairs of testes are found in the 10th and 11th segments.
- **Vasadeferentia :** It runs upto 18th segment where they join with prostate duct.
- **Accessory glands :** Two pairs are present in 17th and 19th segments.
- **Male genital pores :** Common prostate and spermatic duct opens to the exterior by a pair of male genital pores on the ventro–lateral side of the 18th segment.
- **Spermathecae :** It is located in segment 6, 7, 8 and 9. It stores the sperms of companion earthworm during mating.

Female reproductive organs :

- **Ovary :** One pair of ovaries is attached at the inter-segmental septum of 12/13.
- **Oviduct :** Front part of it forms oviductal funnel.
- **Female genital pore :** Both the oviduct joins and opens as a female genital pore of body surface on ventral side of 14th segment.

Mating :

- During mating two earthworms come together and get arranged into juxtaposing directions. Thus the mouth region of one faces the anus of the other.

- Male genital opening of one comes in contact with opening of the spermathecae of another earthworm. In this situation released sperm cells enter into the spermathecae of companion animal. Thus animals get separated after exchanging the sperms.

Formation of cocoon :

- After short time the secretion of clitellum gland forms white cocoon. In this cocoon eggs are released.
- Due to the contraction of body wall the cocoon slips towards anterior end.
- When cocoon slipping towards the anterior end and passes through spermathecal region, sperms are entered into cocoon and fertilizes the eggs of cocoon.
- Both the ends of the cocoon are sealed when it comes out of the body.
- After 3 weeks, two to twenty four baby earthworms are produced from cocoon.
- The development of earthworm is direct. There is no larva formed during development.
- Earthworms are known as friends of farmers because they make burrows in the soil and thus they provide life giving gas for respiration to the developing plant roots.
- To increase fertility of soil earthworms are breeded which is called vermicomposting. In addition to this earthworms as bait are used in fishing.

---

(22) In which segment of earthworms, septal nephridia are located ?

(A) 13 to last (B) 14 to last

(C) 15 to last (D) 23 to last

(23) Which is the nephridia of earthworm which opens on the body surface after 3rd segment ?

(A) Septal (B) Pharyngeal (C) Integumentary (D) All of the given

(24) In which segment, pharyngeal nephridia are located ?

(A) 15 to last (B) 3 to last

(C) 4th, 5th and 6th (D) 15 to 23

(25) Where does the nephridia empty their waste ?

(A) Intestine (B) Pharynx (C) Out of body wall (D) All of the given

(26) Where is the location of subpharyngeal ganglia in earthworm ?

(A) Posterior side of 3rd segment and at dorsal side (B) Posterior side of 4th segment and at ventral side

(C) Posterior side of 3rd segment and at ventral side (D) Posterior side of 5th segment and at ventral side

(27) How many pair of testes and ovaries are present respectively in earthworm ?

(A) 2, 1 (B) 2, 2 (C) 1, 2 (D) 1, 3

(28) Where is the location of two pair of testes in earthworm ?

(A) 9th and 10th segments (B) 9th and 11th segments

(C) 10th and 11th segments (D) 11th and 18th segments

(29) In earthworm, accessory glands are located in x and y segment.

(A) x = 14, y = 17 (B) x = 17, y = 19

(C) x = 18, y = 19 (D) x = 15, y = 16

(30) In _y_ segment, the one pair of ovary is located on posterior side of _x_ septum in earthworm.

(A) x = 11/12, y = 13 (B) x = 10/11, y = 12

(C) x = 12/13, y = 13 (D) x = 12/13, y = 14

(31) How many days it will take to form baby animal from earthworm cocoon ?

(A) 3 weeks (B) 4 weeks (C) 2 weeks (D) 1 week

Answers : (22-C), (23-C), (24-C), (25-D), (26-C), (27-A), (28-C), (29-B), (30-C), (31-A)

General information and morphology of cockroach :

General information :

- It is in class insecta of phylum Arthropoda.
- Generally it is brown or black bodied insect.
- However, in tropical region, they may be yellow, green or red coloured.
- It is omnivore and nocturnal animal.

Morphology :

- **Scientific name :** *Periplaneta americana.*
- **Length and Breadth :** 25-45 mm × 8-12 mm.
- **Exoskeleton :** It is made up of stiff chitin layer.
- **Segmentation :** Only externally segmented not internally.
- In each segment exoskeleton has hardened to thin and flexible articular membrane.
- Body is divided into three regions : (1) Head, (2) Thorax and (3) Abdomen.

(1) Head :

- It is triangular.
- Head lies anteriorly at right angle to the longitudinal body axis. It is formed by fusion of six segments.
- It is attached to thorax through a flexible neck. So that it can move easily in all the directions.
- A pair of sessile, kidney shaped compound eyes occur on head.
- **Mouth parts :** Sensitive mouth parts are associated with mouth. It is useful in collection of food and its chewing.
- It consist of a pair of mandibles, a pair of the first maxillae, labium and labrum.
- **Hypopharynx :** It is located in mouth parts and is muscular, folded structure. Salivary glands open at its basal region.

(2) Thorax : It has 3 parts :

(1) Prothorax, (2) Meso thorax and (3) Meta thorax

- One pair of walking legs arise from ventral side of each segment.
- Each walking leg has 5 segments :
- (1) Coxa, (2) Trochanter, (3) Femur, (4) Tibia and (5) Tarsus.

- The first pair of wings develops from the dorsal side of mesothorax. These wings are horny and protective. The second pair of wings develops from the dorsal side of meta thorax. It is bilobed and transparent and is used for flying.

(3) Abdomen : In male and female, abdomen is made up of 10 segments.

- Each segment has a dorsal tergum, a ventral sternum and lateral pleurae.
- Eight and ninth terga are covered under tergum of seventh segment.
- The 10th tergum a median groove, from where anus open.
- A pair of anal cerci is associated, with a 10th tergum, it is phonoreceptor structures.
- In male cockroach, a pair of anal styles develop from ninth sternum.
- In female cockroach, the 8th and 9th sterna jointly form a genital pouch.
- Seventh sterna is boat shaped.
- The male genital opening occurs in mid-ventral region of ninth segment and female genital opening opens in the eight segment.

---

(32) Cockroach is _x_ colour animal, although in the tropical region it is of _y_ coloured also.

(A) x = Brown, Black, y = Yellow, Red, Green

(B) x = Brown, Black, y = White, Yellow, Red

(C) x = Brown, White, y = Black, Yellow, Green

(D) x = Brown, Yellow, y = Black, Red, Green

(33) Normally, the length of cockroach is _x_ mm and bredth is _y_ mm.

(A) x = 20-40, y = 15-20

(B) x = 15-45, y = 10-15

(C) x = 25-45, y = 8-12

(D) x = 20-25, y = 8-10

(34) Cockroach can move its head easily in all the directions. Because ..... .

(A) Head of cockroach is made up of 6 segments.

(B) Head is connected with throax by a tender neck

(C) Due to structure of mouth parts

(D) Head joins with abdomen, so ..... .

(35) _x_ pair, _y_ and _z_ shaped eyes are located on head of cockroach.

(A) x = 1, y = Stalked, z = compound and round

(B) x = 2, y = Stalked, z = compound and round

(C) x = 1, y = Sessile, z = compound and kidney

(D) x = 1, y = Sessile, z = simple and round

(36) By which structure head of cockroach is connected with thorax ?

(A) neck (B) throat (C) Oesophagus (D) Salivary gland

(37) What remains at anterior of the cockroach mouth ?

(A) Snaut (B) Anal cerci (C) Mouth parts (D) Hypostome

(38) What is correct for mouth parts of cockroach ?

(A) One pair mandible, one pair first maxillae, labium and labrum

(B) One pair first maxillae, one pair labium, mandible and labrum

(C) One pair labium, one pair labrum, first maxillae and mandible

(D) One pair mandible, one pair labrum, first maxillae and labium

(39) From where does wings develops in cockroach ?

(A) Ventral side of thorax (B) Lateral side of thorax

(C) Dorsal side of abdomen (D) None of the given

(40) From where does wings arises in cockroach ?

(A) Ventral side of thorax (B) Lateral side of thorax

(C) Dorsal side of thorax (D) None of the given

(41) Which structure is connected with the 10th segment of cockroach ?

(A) Anal style (B) Anal cerci (C) Tarsus (D) Trochanter

(42) Which type of the organ anal cerci in cockroach ?

(A) Reproduction (B) Excretory (C) Phonoreceptor (D) Respiratory

(43) In which segment, the genital pore opens in male cockroach ?

(A) 8th (B) 7th (C) 9th (D) 10th

(44) In which segment, female genital pore open in cockroach ?

(A) 8th (B) 7th (C) 9th (D) 10th

(45) By which segmental sternum, genital pouch is formed in cockroach ?

(A) 9th and 10th (B) 7th and 8th

(C) 7th and 9th (D) 8th and 9th

Answers : (32-A), (33-C), (34-B), (35-C), (36-A), (37-C), (38-A), (39-A), (40-C), (41-B), (42-C), (43-C), (44-A), (45-D)

Body wall, Alimentary canal and digestion in cockroach :

Body wall :

- It is made up of main three layers.
- Outermost layer is of cuticle, which forms exoskeleton.
- Middle layer is of epidermis, which is single columnar celled.
- Innermost layer is basement membrane.

Alimentary canal :

- It is complete alimentary canal. Mouth is located at the anterior end of head.
- **Mouth parts :** It is adapted for gathering and cutting food and are arranged surrounding the mouth.
- **Pharynx :** The tubular region following mouth is called pharynx. Then onwards, alimentary canal is divided into 3 parts : (1) Foregut, (2) Midgut and (3) Hindgut
- The cavity of foregut and midgut are lined by cuticle.

- On the lateral side of the alimentary canal, a pair of salivary glands are found. Each salivary gland consists of two secreting lobes and two reserviors.
- **Oesophagus :** It is narrow, tubular structure which follows pharynx.
- It enlarges at its posterior end into a crop.
- **Crop :** Crop opens into a muscular gizzard and is spreaded into thorax and abdomen.
- **Gizzard :** It has six chitinous teeth inside.
- At the posterior end of the gizzard, a sieve-like structure made up of chitinous bristles. The region of alimentary canal upto gizzard constitutes the foregut. Gizzard opens into the midgut.
- **Midgut :** Eight blind hepatic caeca occur in the midgut and midgut opens into the hindgut.
- **Malpighian tubules :** It is located at the juncture of mid and hindgut are about 150 and yellowish in colour. It is excretory unit of cockroach.
- **Ileum :** It is the proximal region of hindgut and is narrow. Rest of the part of hindgut is somewhat coiled.
- **Colon :** The distal region of hindgut is colon.
- **Rectum :** The bag like posterior end region of hindgut is called rectum. It is folded inside. Rectum opens to the outside through anus. Anal is located under the tergum of 10th abdominal segment.

Digestion in cockroach :

- It is omnivorous animal. It searches food by its antennae.
- In salivary gland, saliva is prepared by secretory lobes. Mucus present in saliva makes food wet. The enzyme - amylase affect the starch of food and digestion begins.
- Now food enters into crop and digestion proceeds further. Now food enters into gizzard where it cuts down into small pieces with chitinous teeth and food enters into midgut.
- Columnar cells of midgut and hepatic caeca, secrete enzymes.
- Various proteolytic enzymes convert protein into amino acids.
- Lipase digests lipid into fatty acids and glycerol.
- Amylase produces sugar from starch.

---

(46) Which is the correct sequence for the body wall layer from inside to outside ?

(A) Cuticle → Epidermis → Basement membrane (B) Epidermis → Cuticle → Basement membrane

(C) Basement membrane → Epidermis → Cuticle (D) Cuticle → Basement membrane → Epidermis

(47) The cells of epidermis of cockroach body wall are of _x_ type.

(A) x = columnar (B) x = squamous

(C) x = cuboidal (D) x = ciliated

(48) _x_ salivary glands are present in thoracic region of cockroach. Which has _y_ secretory lobes and _z_ reserviors.

(A) x = two pair, y = two, z = one (B) x = one pair, y = two, z = one

(C) x = one pair, y = one, z = two (D) x = one pair, y = two, z = two

(49) Gizzard of cockroach is a _x_ structure. It has _y_ chitinous teeth into its lumen.

(A) x = Muscular, y = Four (B) x = Glandular, y = Six

(C) x = Muscular, y = Eight (D) x = Muscular, y = Six

(50) How many malpighian tubules are found in cockroach ?

(A) 80 (B) 125 (C) 140 (D) 150

(51) Of whose unit, the malpighian tubule is in cockroach ?

(A) Excretory (B) Digestive (C) Respiratory (D) Reproductive

(52) In cockroach, the anal opens at _x_ tergum out of the body.

(A) x = 9th (B) x = 8th (C) x = 10th (D) x = 7th

(53) The rectum of cockroach is _x_ inside and _y_ outside.

(A) x = folded, y = baglike (B) x = Secretory, y = folded

(C) x = Baglike, y = folded (D) x = Baglike, y = processes

(54) Starch $\xrightarrow{x}$ Sugars.

(A) x = Amylase (B) x = Lipase (C) x = Sucrase (D) x = Protease

(55) _x_ enzymes converts protein into _y_ components.

(A) x = Lipase, y = Fatty acids (B) x = Protease, y = Amino acids

(C) x = Amylase, y = Amino acids (D) x = Protease, y = Fatty acids

(56) In which components, lipase will convert the lipid in cockroach ?

(A) Amino acids (B) Sugars (C) Fatty acids, Glycerol (D) All of the given

Answers : (46-C), (47-A), (48-B), (49-D), (50-D), (51-A), (52-C), (53-C), (54-A), (55-B), (56-C)

Circulatory system and Respiratory system in cockroach :

Circulatory system of cockroach :

- Blood vascular system of cockroach is an open type. It means during circulation blood enters into the body cavity instead of blood vessels.
- Thus body cavity acts as a haemocoel and organs of the body and tissues are directly connected with the blood.
- The haemolymph is mostly composed of plasma and uncertain shaped cells.
- The heart is made up of thirteen units. Three units occur in thoracic region and ten occur in the abdominal region.
- Anterior end of the heart is slightly narrow and posterior end is broad. Two valved openings occur between two nearby units. They are called Ostia.
- The cells of haemolymph are of two types. Proleucocytes of smaller size and phagocytes of larger size. Blood from sinuses enters heart through ostia and is pumped anteriorly to sinuses again.

Respiratory system of cockroach :

- Branched tubes known as trachea are main components of respiratory system.
- Terminal branches of tracheal tubes are called tracheoles which carry oxygen to entire body.
- The trachea have direct contact with the air in atmosphere through pores known as spiracles.
- Ten pairs of spiracles occur. Two pairs occur in thoracic region and eight pair on abdominal region.
- The wall of spiracles are framed with chitinous bristles. These bristles works as a filter and prevents water and other waste substances entering into the respiratory system.

- Oxygn gets entered into the trachea through spiracles and from there it comes in contact of tissue fluid and gets dissolved in it.
- The tissues of body use this dissolved oxygen to get energy to work. The carbon dioxide produced consequently gets dissolved in tissue fluid generally which comes out through exhaling.

(57) The blood of cockroach is mainly made up of ..... and ..... .

(A) Haemolymph, RBCs (B) Haemolymph, Haemoglobin

(C) Haemolymph, Uncertain shaped cells (D) Haemolymph, certain shaped cells

(58) The _x_ units of cockroach heart is remain in thoracic region and _y_ units in abdomin region.

(A) x = 10, y = 3 (B) x = 3, y = 10 (C) x = 7, y = 6 (D) x = 4, y = 9

(59) Which sentence is correct for circulatory system of cockroach ?

(A) It has closed type system.

(B) It has complete type system.

(C) Heart has 13 units. Each unit has two ostia.

(D) Blood circulates without help of sinuses.

(60) What is called the end branching of trachea ?

(A) Fine broncheal (B) Fine tracheoles (C) Fine trachea (D) Fine sinuses

(61) The number of spiracles in cockroach are ..... .

(A) 2 = Thorax, 8 = Abdomen (B) 3 = Thorax, 7 = Abdomen

(C) 4 = Thorax, 6 = Abdomen (D) 5 = Thorax, 5 = Abdomen

(62) What is present in the wall of spiracle of cockroach ?

(A) Chitinous bristles (B) Mast cells (C) Areolar tissue (D) Glandular cells

Answers : (57-C), (58-B), (59-C), (60-B), (61-A), (62-A)

Excretion, Excretory process and nervous system of cockroach

Excretion and process of excretion in cockroach :

- At the junction of midgut and hindgut about 150 yellowish thread-like malpighian tubules open. They are main excretory units.
- These blind tubules always float in the haemolymph.
- They absorb nitrogenous waste products and convert them into uric acid which is excreted out through the hindgut. Therefore, cockroach is a uricotelic animal.
- Excretory substances enter into hindgut having large amount of water. This water is absorbed by the wall of hindgut.

Nervous system of Cockroach :

- It is made up of paired ganglia, nerve cords and nerves.
- Supra oesophageal ganglia formed by the fusion of three ganglia or a pair of brain ganglia are located on oesophagus.
- They are connected with suboesophageal ganglia through circum-oesophageal commisures.

- Thus a nerve ring is formed, suboesophageal ganglion innervate the mouth parts.
- Three pairs of ganglia occur in thorax region and six pairs in abdominal region.
- Thus, nervous system of cockroach is spreaded throughout the body.
- Amongst sense organs in cockroach, antennae, eyes, maxillary palps, tarsus of walking legs and anal cercus are included.
- **Mosaic vision :** The compound eyes are situated at the dorsal surface of the head. Each eye consist of about 2000 hexagonal ommatidium. With the help of several ommatidium a cockroach can receive several images of an object. This kind of vision is known as mosaic vision.

(63) Malpighian tubules are ..... .

(A) Excretory organ of insect (B) Excretory organ of frog

(C) Excretory organ of earthworm (D) Respiratory organ of insect

(64) Which one is excreted by cockroach ?

(A) Urea (B) Uric acid (C) Ammonia (D) Amino acid

(65) How many ganglia fuses to form supraoesophageal ganglia in cockroach ?

(A) One (B) Two (C) Three (D) Four

(66) In cockroach, thorax has _x_ ganglia and abdomen has _y_.

(A) x = 2, y = 8 (B) x = 3, y = 5 (C) x = 3, y = 6 (D) x = 2, y = 7

(67) How many ommatidium are present in compound eye of cockroach ?

(A) 20 (B) 2000 (C) 2200 (D) 200

(68) Which one is the phonoreceptor organ of cockroach ?

(A) Anal style (B) Antennae (C) Compound eyes (D) Anal cerci

Answers : (63-A), (64-B), (65-C), (66-C), (67-B), (68-D)

Reproductive system of cockroach :

- It is unisexual animal. Both sexes has well developed reproductive organs.

Male reproductive system of cockroach :

- **Testes :** A pair of testes lying one on each lateral side in the 4 to 6 abdominal segments.
- **Vas deferens :** It arises from each testes and opens into ejaculatory duct through seminal vesicle.
- **Ejaculatory duct :** It opens into male gonopore situated ventral to anus.
- **Mushroom gland :** It occur in segment 6 and 7. It is an accessory reproductive gland.
- **Gonapophysis :** It is located at the end of abdomen and forms external reproductive organs.
- **Seminal vesicle :** Sperms are stored into this vesicle. Sperms are glued together in the form of bundles called spermatophores which are discharged during copulation.

- **Female reproductive system of cockroach :**
  - **Ovary :** Two ovaries lying laterally in 2 to 6 abdominal segments.
  - **Ovarioles :** Each ovary is formed of a group of eight ovarian tubules or ovarioles, containing a chain of developing ova.
  - **Oviduct or Vagina :** Oviduct of each ovary unit into a single median-oviduct called vagina which opens into the genital chamber.

- **Genital chamber :** During copulation ovum come in the genital chamber, where they are fertileized by sperms.
- **Ootheca :** Dark brown coloured Ootheca is formed by the group of fertilized eggs. Each ootheca has 14 to 16 eggs.
- Cockroach is developed through nyphal stage (or young insect). The nymphs look very much like adults.
- The nymph grows by moulting about 6 to 7 times to reach the adult form.

(69) What is the location of testes in male cockroach ?

(A) Lateral side of the 5 to 6 segment of abdomen (B) Lateral side of the 4 to 6 segment of abdomen

(C) Dorsal side of the 5 to 6 segment of abdomen (D) Dorsal side of the 4 to 8 segment of abdomen

(70) What is the location of mushroom gland in cockroach ?

(A) 6 to 7 (B) 4 to 6 (C) 5 to 7 (D) 4 to 8

(71) What is the location of ovary in female cockroach ?

(A) Lateral side of the 2 to 7 abdomenal segment (B) Lateral side of the 3 to 7 abdomenal segment

(C) Lateral side of the 2 to 6 abdomenal segment (D) Lateral side of the 2 to 4 abdomenal segment

(72) Each ovary is made up _x_ in female cockroach.

(A) x = 4 tubular ovarioles (B) x = 6 tubular ovarioles

(C) x = 7 tubular ovarioles (D) x = 4 tubular ovarioles

Answers : (69-B), (70-A), (71-C), (72-D)

- **True - False (T - F) type questions :**

Select the true or false option for given sentences.

(73) (1) Earthworm belongs to annelida and is coelomate animal.

(2) Scientific name of the earthworm is *Pheretima posthuma*.

(3) Earthworm remain lower level of moist soil.

(4) Earthworm is segmented into 100 – 120 segments.

(A) T, T, T, T (B) T, F, T, F (C) T, T, F, T (D) F, F, T, T

(74) (1) First segment of earthworm is called peristomium.

(2) 14 to 16 segment of mature earthworm is called clitellum.

(3) There is only one female genital pore on mid-ventral line in 18th segment of earthworm.

(4) There is one pair of male genital pores on mid-ventral line in 18th segment of earthworm.

(A) T, F, T, F (B) T, T, F, F (C) T, T, F, T (D) T, F, F, T

(75) (1) Earthworm has open circulatory system.

(2) Small blood vessels supply blood to body wall, nerve cord and intestine in earthworm.

(3) Blood glands are present in 4, 5 and 6 segment in earthworm.

(4) Haemoglobin is dissolved in plasma in earthworm.

(A) T, T, F, F (B) T, F, T, F (C) F, T, T, F (D) F, T, T, T

(76) (1) Earthworm has specific respiratory organs.

(2) Gaseous exchange takes place through spiracles in earthworm.

(3) One pair of ganglia posterior side of 3rd segment on ventral side of earthworm. It is called subpharyngeal ganglion.

(4) Earthworm has no sensory organ like eyes.

(A) T, T, F, F (B) F, F, T, T (C) T, F, T, F (D) F, F, F, T

(77) (1) Earthworm is bisexual animal.

(2) Earthworm has two testes in 10th and 11th segment respectively.

(3) Earthworm is useful as a bait in fishing.

(4) After 3 weeks, baby earthworm comes out from cucoon in earthworm.

(A) T, T, T, T (B) T, F, T, F (C) F, T, F, T (D) F, F, T, T

(78) (1) Cockroach has 25-45 mm length and 8-12 mm bredth.

(2) 4 segments are fused to form head of cockroach.

(3) Function of mouth parts is to catch and chew food.

(4) Anterior part of cockroach head, it has mouth.

(A) T, T, F, T (B) T, T, F, F (C) F, F, T, T (D) T, F, T, T

(79) (1) First segment of cockroach walking leg is called coxa.

(2) Third segment of cockroach walking leg is called trochanter.

(3) Fourth segment of cockroach walking leg is called tibia.

(4) Fifth segment of cockroach walking leg is called tarsus.

(A) T, F, T, T (B) T, T, F, F (C) T, T, T, F (D) F, T, T, F

(80) (1) Cockroach anal cerci is phonoreceptor organ.

(2) Cockroach abdomen is made up of 10 segments.

(3) Cockroach alimentory canal can be divided into three region.

(4) 150 malpighian tubuler opens at the juncture of midgut and hindgut in cockroach.

(A) T, F, T, F (B) T, T, F, F (C) T, T, T, T (D) F, F, F, T

(81) (1) Starch give sugar by Amylase in cockroach.

(2) Protein components $\xrightarrow{\text{Proteases}}$ Amino acids.

(3) Lipid components $\xrightarrow{\text{Lipase}}$ Fatty acids and Glycerol

(4) Mucus of salivary gland of Cockroach heats food.

(A) T, T, F, F (B) T, T, T, F (C) F, F, T, T (D) T, F, T, F

(82) (1) Body coelom is called Haemocoelam in cockroach.

(2) Heart is made up of 13 units in cockroach.

(3) Malpighian tubules are 100 in cockroach.

(4) Spiracles are ten pairs in cockroach.

(A) T, T, F, F (B) T, F, T, F (C) T, T, F, T (D) T, F, F, T

(83) (1) Eye of cockroach is made up of 2000 ommatidia.

(2) Mushroom gland of male cockroach is located in 6 to 7 abdominal segment.

(3) Every Ootheca of cockroach posses 14 to 16 eggs.

(4) Cockroach nymph undergoes 4 to 5 time moulting to transform into adult form.

(A) T, F, F, T (B) F, T, T, F (C) F, F, T, T (D) T, T, T, F

(84) (1) In cockroach, gizzard has 6 chitinous teeth and eight tubular hepatic caeca are connected to midgut.

(2) Cockroach has 13 chambered (lobbed) heart.

(3) Cockroach has 10 pair of spiracles and 150 malpighian tubules.

(4) Cockroach is uricotelic animal.

(A) T, F, T, T (B) T, T, F, T (C) T, T, T, T (D) F, T, T, T

Answers : (73-C), (74-B), (75-D), (76-B), (77-A), (78-D), (79-A), (80-C), (81-B), (82-C), (83-D), (84-C)

- **A - Assertion, R - Reason type questions :**

Choose correct option for answer of given questions :

(A) A and R both are correct. R is correct explanation of A.

(B) A and R both are correct. R is not correct explanation of A.

(C) A - correct, R - incorrect.

(D) A - incorrect, R - correct.

(85) Assertion A : Body of earthworm is tubular, slightly elongated and thin.

Reason R : Body of earthworm is divided in small segment which are 100 to 120.

(A) (B) (C) (D)

(86) Assertion A : Genital pores are located on dorsal side in earthworm.

Reason R : Anterior part of earthworm has mouth and prostomium.

(A) (B) (C) (D)

(87) Assertion A : 1 to 13 segment of earthworm is called preclitellar region.

Reason R : Clitellum is present in 14 to 16 segment in earthworm.

(A) (B) (C) (D)

(88) Assertion A : Only one female genital pore is present on mid-ventral line of 14th segment of earthworm.

Reason R : Spermathecal pore is present on 18th segment of earthworm.

(A) (B) (C) (D)

(89) Assertion A : Nephridia are present in first, last and clitellar region.

Reason R : Setae of earthworm are made up of chitin.

(A) (B) (C) (D)

(90) Assertion A : Cuticle is secreted by epidermis of earthworm.

Reason R : In epidermis of earthworm, long, thin support cell and oblong glandular cells are present.

(A) (B) (C) (D)

(91) Assertion A : In earthworm, muscular pharynx is elongated upto 4th segment following mouth cavity.

Reason R : Narrow oesophagus of earthworm elongated upto 5 to 8 segments.

(A) (B) (C) (D)

(92) Assertion A : Stomach of earthworm is spreaded from 9 to 14 segments.

Reason R : Calciferous glands are present in stomach.

(A) (B) (C) (D)

(93) Assertion A : In earthworm, closed type of circulatory system is present.

Reason R : Blood circulates in blood vessels and heart in earthworm.

(A) (B) (C) (D)

(94) Assertion A : Haemoglobin is dissolved in RBC, in earthworm.

Reason R : Blood cells of earthworm are of phagocytic cells.

(A) (B) (C) (D)

(95) Assertion A : In earthworm, after 5th segment one pair of ganglia is present in each segment, which is known as ventral nerve cord.

Reason R : In earthworm, one pair of ganglia is present on ventral side on posterior part of 3rd segment, which is called as subpharyngeal ganglia.

(A) (B) (C) (D)

(96) Assertion A : Earthworm is bisexual Animal.

Reason R : Development in earthworm is direct, that means no larva is formed.

(A) (B) (C) (D)

(97) Assertion A : Earthworm is called friend of farmer.

Reason R : They make burrow in soil and provide life giving gas for respiration to the developing plant root.

(A) (B) (C) (D)

(98) Assertion A : Cockroach has incomplete alimentary canal.

Reason R : Anterior part of head, cockroach has mouth.

(A) (B) (C) (D)

(99) Assertion A : Cockroach is omnivorus animal.

Reason R : Cockroach search food by its antennae.

(A) (B) (C) (D)

(100) Assertion A : Eight blind hepatic caeca joins with midgut of cockroach.

Reason R : 6 Chitinous teeth are present into the gizzard of cockroach.

(A) (B) (C) (D)

(101) Assertion A : Circulatory system of cockroach is open type.

Reason R : Blood circulates only in blood vessels in the cockroach.

(A) (B) (C) (D)

(102) Assertion A : Heart is made up of 13 units in cockroach.
Reason R : Initial three units are located in thorax and rest of the ten in abdomen.
(A) (B) (C) (D)

(103) Assertion A : Ten pair of spiracles are present in cockroach.
Reason R : Four pairs are present in thorax and 6 pairs in abdomen.
(A) (B) (C) (D)

(104) Assertion A : Cockroach is called uricotelic animal.
Reason R : Cockroach tranforms nitrogenous waste into uric acid.
(A) (B) (C) (D)

(105) Assertion A : In cockroach, suboesophageal ganglia innervates mouth parts.
Reason R : Nervous system of cockroach is spreaded through out its body.
(A) (B) (C) (D)

(106) Assertion A : Cockroach is bisexual animal.
Reason R : One pair of testes in arranged posterior part of 4 to 6 abdominal segment in male reproductive system of cockroach.
(A) (B) (C) (D)

(107) Assertion A : Gaseous exchange takes place through moist body wall in earthworm.
Reason R : Certain respiratory organs are absent in earthworm.
(A) (B) (C) (D)

Answers : (85-B), (86-D), (87-A), (88-C), (89-D), (90-B), (91-C), (92-B), (93-A), (94-D), (95-B), (96-B), (97-A), (98-D), (99-B), (100-B), (101-C), (102-B), (103-C), (104-A), (105-B), (106-D), (107-A)

(108) Match the column :

Column - I	**Column - II**
(a) First segment	(i) One pair male genital pore
(b) 14th segment	(ii) Female genital pore
(c) 18th segment	(iii) Clitellum-Glandular tissue band
(d) 14 to 16 segment	(iv) Peristomium

(A) (a – iv), (b – ii), (c – i), (d – iii) (B) (a – i), (b – iii), (c – ii), (d – iv)
(C) (a – ii), (b – i), (c – iii), (d – iv) (D) (a – iv), (b – ii), (c – iii), (d – i)

(109) Match the column :

Column - I	**Column - II**
(a) Setae	(i) Secretes albumin and mucus
(b) Cuticle	(ii) Helps in locomotion
(c) Excretory pores	(iii) Made up of secretion by epidermis
(d) Long-thin support cells, Glandular cells in epidermis,	(iv) Present in all segment, except first, last and clitellar segment

(A) (a – i), (b – ii), (c – iii), (d – iv) (B) (a – iii), (b – i), (c – iv), (d – ii)
(C) (a – ii), (b – iii), (c – iv), (d – i) (D) (a – ii), (b – iii), (c – i), (d – iv)

(110) Match the column :

Column - I	Column - II
(a) Pharynx	(i) 8th segment
(b) Small narrow oesophagus	(ii) 9 to 14 segments
(c) Gizzard	(iii) 5 to 7 segments
(d) Stomach	(iv) 4th segment

(A) (a – ii), (b – iii), (c – iv), (d – i)
(B) (a – iv), (b – iii), (c – i), (d – ii)
(C) (a – iv), (b – iii), (c – ii), (d – i)
(D) (a – i), (b – ii), (c – iii), (d – iv)

(111) Match the column :

Column - I	Column - II
(a) Blood glands	(i) From 15th segment to last segment
(b) Spermatheca	(ii) After 3rd segment, internal surface of body wall
(c) Septal nephridia	(iii) 6, 7, 8 and 9 segments
(d) Integumentary nephridia	(iv) 4, 5 and 6 segments

(A) (a – iv), (b – iii), (c – i), (d – ii)
(B) (a – iv), (b – ii), (c – iii), (d – i)
(C) (a – i), (b – ii), (c – iii), (d – iv)
(D) (a – ii), (b – iii), (c – i), (d – iv)

(112) Match the column :

Column - I	Column - II
(a) First segment	(i) Trochanter
(b) Second segment	(ii) Femur
(c) Third segment	(iii) Coxa
(d) Fourth segment	(iv) Tibia
(e) Fifth segment	(v) Tarsus

(A) (a – ii), (b – i), (c – iv), (d – iii), (e – v)
(B) (a – iii), (b – i), (c – ii), (d – iv), (e – v)
(C) (a – iii), (b – i), (c – iv), (d – ii), (e – v)
(D) (a – v), (b – ii), (c – iii), (d – i), (e – iv)

(113) Match the column :

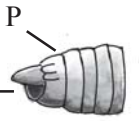
Column - I	Column - II
(a) Testes	(i) 14 to 16 eggs
(b) Ovary	(ii) 6 to 7 abdominal segments
(c) Mushroom gland	(iii) 4 to 6 abdominal segments
(d) Cocoon	(iv) 2 to 6 abdominal segments

(A) (a – ii), (b – i), (c – iv), (d – iii)
(B) (a – ii), (b – i), (c – iii), (d – iv)
(C) (a – i), (b – ii), (c – iii), (d – iv)
(D) (a – iii), (b – iv), (c – ii), (d – i)

Answers : (108-A), (109-C), (110-B), (111-A), (112-B), (113-D)

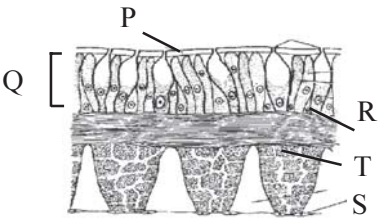
Figure based questions :

(114) What is indicated by P and Q in given figure ?



(A) P = Mouth, Q = Peristomium
(B) P = Peristomium, Q = Mouth
(C) P = Mouth cavity, Q = Peristomium
(D) P = Mouth cavity, Q = Mouth

(115) What is indicated by P and Q in given figure ?

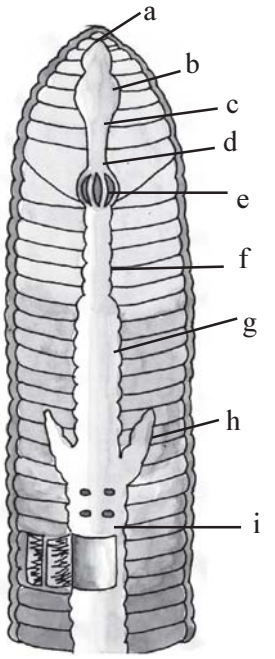


(A) P = Cuticle, Q = Longitudinal muscle layer
(B) P = Parietal layer, Q = Circular muscle layer
(C) P = Cuticle, Q = Epidermis
(D) P = Cuticle, Q = Parietal layer

(116) Which layer is formed by secretion of epidermis in the above figure ?
(A) P (B) Q (C) R (D) S

(117) What is indicated by R, S and T in given above figure ?
(A) R = Circular muscle, S = Parietal layer, T = Cuticle
(B) R = Circular muscle, S = Longitudinal muscle, T = Parietal layer
(C) R = Parietal layer, S = Longitudinal muscle, T = Circular muscle
(D) R = Cuticle, S = Circular muscle, T = Longitudinal muscle

(118) What is indicated by a and b in given figure ?



(A) a = Mouth, b = Pharynx
(B) a = Mouth cavity, b = Pharynx
(C) a = Mouth, b = Mouth cavity
(D) a = Pharynx, b = Oesophagus

(119) What is indicated by c and d in the given figure ?
(A) c = Oesophagus, d = Pharynx
(B) c = Mouth cavity, d = Oesophagus
(C) c = Mouth, d = Pharynx
(D) c = Pharynx, d = Oesophagus

(120) What is indicated by e, f and g in the given figure ?
(A) e = Stomach, f = Gizzard, g = Pre typhlosolar region
(B) e = Gizzard, f = Stomach, g = Post typhlosolar region
(C) e = Gizzard, f = Stomach, g = Pre typhlosolar region
(D) e = Stomach, f = Gizzard, g = Post typhlosolar region

(121) What is indicated by h and i in the given figure ?
(A) h = Typhlosole, i = Caecum (B) h = Caecum, i = Typhlosole
(C) h = Caecum, i = Lateral hearts (D) h = Typhlosole, i = Lateral hearts

(122) Which part indicates neutralization of Humic acid in the given figure ?
(A) f (B) g (C) h (D) e

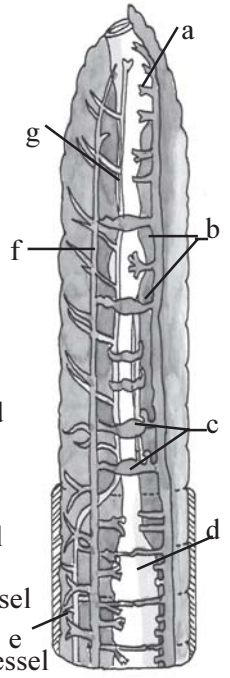
(123) What is indicated by a, c and d in given figure ?

(A) a = Dorsal blood vessel, c = Lateral oesophageal hearts, d = Alimentary canal

(B) a = Alimentary canal, c = Hearts, d = Blood Vessel

(C) a = Nerve vessel, c = Alimentary canal, d = Dorsal blood vessel

(D) a = Lateral hearts, c = Alimentary canal, d = Blood vessel



(124) By which later, lateral hearts are indicated in the given figure ?

(A) a (B) c (C) b (D) d

(125) What is indicated by e, f and g in given figure ?

(A) e = Dorsal blood vessel, f = Subnerve vessel, g = Supranerve vessel

(B) e = Supranerve vessel, f = Ventral blood vessel, g = Sub neural vessel

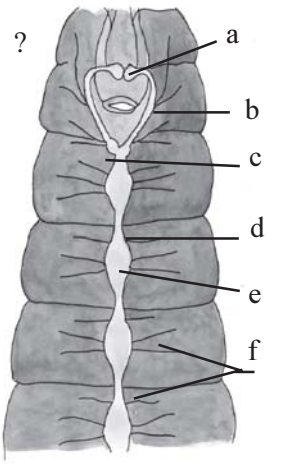
(C) e = Sub neural vessel, f = Dorsal blood vessel, g = Supra neural vessel

(D) e = Sub neural vessel, f = Ventral blood vessel, g = Lateral Oesophgeal Blood vessels

(126) What is the number of segment for given part "a" in the given figure ?

(A) 1-3 segments (B) 2-5 segments

(C) Third segment (D) Second segment



(127) Which parts connected with nerve ring in the given figure ?

(A) b, c, d (B) b, c

(C) a, c (D) a, b, c

(128) What is indicated by a, b and c in the given figure ?

(A) a = Suprapharyngeal ganglia, b = Circumpharyngeal connective, c = Subpharyngeal ganglia

(B) a = Subpharyngeal ganglia, b = Circumpharyngeal connective, c = Suprapharyngeal ganglia

(C) a = Nerve ring, b = Ganglia, c = Connective

(D) a = Circumpharyngeal connective, b = Nerve ring, c = Ganglia

(129) What is indicated by d, e and f in the given figure ?

(A) d = Ventral nerve cord, e = Nerve ring, f = Connective

(B) d = Ventral nerve cord, e = Segmental ganglia, f = Segmental nerve

(C) d = Dorsal nerve cord, e = Segmental nerve, f = Ganglionic cord

(D) d = Neural axis, e = Neural cells, f = Sensory cord

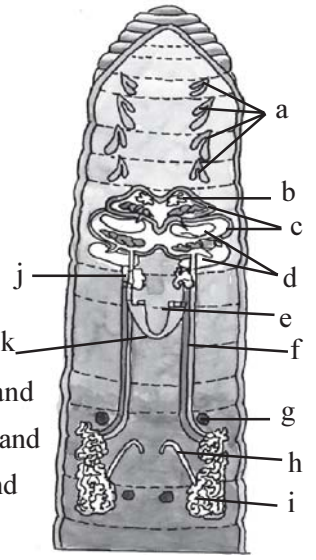
(130) Where does storage of sperm takes place in the given figure ?

(A) a (B) i (C) e (D) j

(131) Which are included in male reproductive organ in the given figure ?

(A) j, k, d (B) a, b, k

(C) a, b, f (D) g, h, i



(132) What is indicated by a, b, c and d in the given figure ?

(A) a = Testes, b = Seminal vesicles, c = Spermiductal funnel, d = Spermatheca

(B) a = Spermatheca, b = Testes, c = Seminal vesicles, d = Spermiductal funnel

(C) a = Seminal vesicles, b = Spermatheca, c = Testes, d = Spermiductal funnel

(D) a = Spermatheca, b = Testes, c = Spermiductal funnel, d = Seminal vesicles

(133) What is indicated by e, f, g and i in the given figure ?

(A) e = Testes sac, f = Vasadeferentia, g = Prostate gland, i = Accessory gland

(B) e = Ovarian funnel, f = Vasadeferentia, g = Accessory gland, i = Prostate gland

(C) e = Spermiductal funnel, f = Oviduct, g = Prostate gland, i = Accessory gland

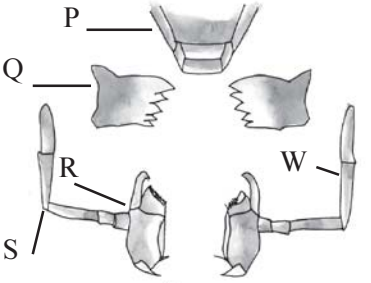
(D) e = Testes sac, f = Oviduct, g = Accessory gland, i = Prostate gland

(134) What is indicated by j and k in the given figure ?

(A) j = Ootheca, k = Accessory gland (B) j = Ovary, k = Female genital pore

(C) j = Testes, k = Male genital pore (D) j = Testes, k = Spermathecal pore

(135) What is indicated by P, Q, R and S in the given figure ?



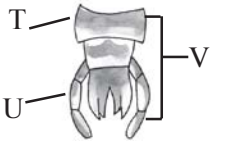
(A) P = Labrum, Q = Mandible, R = Maxillary palp, S = Maxilla

(B) P = Mandible, Q = Labrum, R = Maxilla, S = Maxillary palp

(C) P = Mandible, Q = Labrum, R = Maxillary palp, S = Maxilla

(D) P = Labrum, Q = Mandible, R = Maxilla, S = Maxillary palp

(136) What is indicated by T, U, V and W in the given figure ?



(A) T = Labium, U = Maxillary palp, V = Secondary palp, W = Primary palp

(B) T = Labium, U = Labial palp, V = Primary palp, W = Secondary palp

(C) T = Anterior labium, U = Labial palp, V = Secondary palp, W = Primary palp

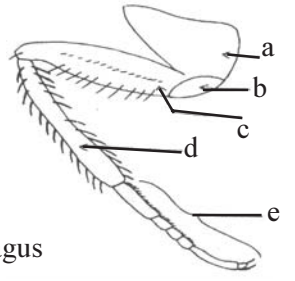
(D) T = Anterior labium, U = Labial palp, V = Primary palp, W = Secondary palp

(137) What is indicated by a, d and e in given figure ?

(A) a = Coxa, d = Trochanter, e = Femur (B) a = Coxa, d = Tebia, e = Tarsus

(C) a = Tebia, d = Femur, e = Trochanter (D) a = Trochanter, d = Coxa, e = Tarsus

(138) What is indicated by b and c in given figure ?



(A) b = Trochanter, c = Femur

(B) b = Tibia, c = Femur

(C) b = Trochanter, c = Tarsus

(D) b = Tibia, c = Tarsus

(139) What is indicated by a, b, c and d in the given figure ?

(A) a = Pharynx, b = Salivary reservior, c = Salivary gland, d = Oesophagus

(B) a = Pharynx, b = Salivary gland, c = Salivary reservior, d = Oesophagus

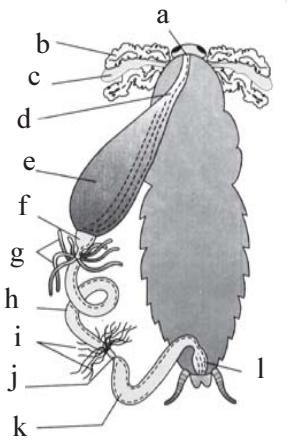
(C) a = Oesophagus, b = Salivary reservior, c = Salivary gland, d = Pharynx

(D) a = Oesophagus, b = Salivary gland, c = Salivary reservior, d = Pharynx

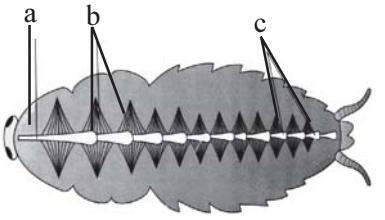
(140) Match the column :

Column-I		Column-II	
(a)	Caeca	(1)	k
(b)	Malpighian tubules	(2)	g
(c)	Ileum	(3)	h
(d)	Colon	(4)	j
(e)	Mesentory	(5)	i

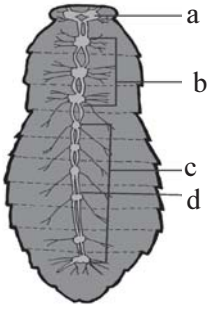
(A) (a – 3), (b – 2), (c – 1), (d – 4), (e – 5)
(B) (a – 2), (b – 5), (c – 4), (d – 1), (e – 3)
(C) (a – 5), (b – 4), (c – 3), (d – 2), (e – 1)
(D) (a – 5), (b – 3), (c – 2), (d – 1), (e – 4)



(141) What is indicated by a, b and c in the given figure ?
(A) a = Alary muscle, b = Posterior aorta, c = Heart chamber
(B) a = Posterior aorta, b = Heart chamber, c = Alary muscle
(C) a = Heart chamber, b = Posterior aorta, c = Alary muscle
(D) a = Anterior aorta , b = Alary muscle, c = Heart chamber



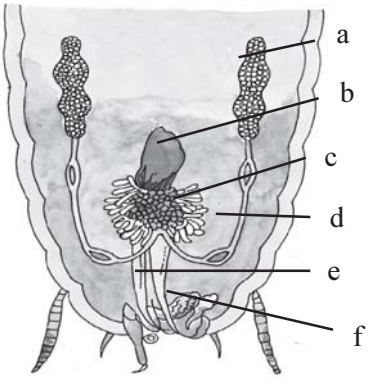
(142) What is indicated by a, b, c and d in the given figure ?
(A) a = Brain, b = Thoracic ganglia, c = Abdominal ganglia, d = Ventral nerve cord
(B) a = Thoracic ganglia, b = Abdominal ganglia, c = Brain, d = Dorsal nerve cord
(C) a = Thoracic ganglia, b = Brain, c = Abdominal ganglia, d = Ventral nerve cord
(D) a = Brain, b = Abdominal ganglia, c = Dorsal nerve cord, d = Thoracic ganglia



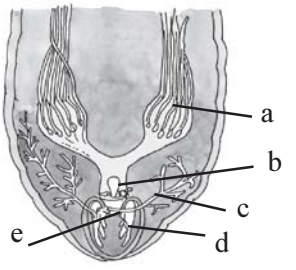
(143) Match the column :

Column-I	Column-II
(i) Ejeculatoryduct	(1) b
(ii) Testes	(2) d
(iii) Conglobate gland	(3) e
(iv) Vasdeference	(4) c
(v) Gonapophyses	(5) f
(vi) Mushroom gland	(6) a

(A) (i – a), (ii – b), (iii – c), (iv – d), (v – e), (vi – f)
(B) (i – a), (ii – f), (iii – c), (iv – e), (v – d), (vi – b)
(C) (i – e), (ii – a), (iii – b), (iv – d), (v – f), (vi – c)
(D) (i – f), (ii – c), (iii – a), (iv – d), (v – b), (vi – e)



(144) What is indicated by a, b, c, d and e in the given figure ?
(A) a = Labrum, b = Mandible, c = Maxillary palp, d = Maxilla, e = Colleterial glands
(B) a = Mandible, b = Labrum, c = Maxilla, d = Maxillary palp, e = Genital chamber
(C) a = Mandible, b = Labrum, c = Maxillary palp, d = Maxilla, e = Genital chamber
(D) a = Labrum, b = Mandible, c = Maxilla, d = Maxillary palp, e = Mushroom gland



Answers : (114-A), (115-C), (116-A), (117-B) , (118-A), (119-D), (120-C), (121-B), (122-A), (123-A), (124-B), (125-D), (126-C), (127-D), (128-A), (129-B), (130-A), (131-C), (132-D), (133-B), (134-C), (135-D), (136-B), (137-B), (138-A), (139-B), (140-B), (141-D), (142-A), (143-C), (144-C)

- **Questions for NEET :**

(145) Which pair is correct in earthworm ?

(A) Mouth cavity = 1 to 5 segment
(B) Stomach = 11 to 12 segment
(C) Typhlosole = 26 to 95 segment
(D) Testes = 10 to 14 segment

(146) Blood of earthworm...

(A) is blue due to presence of haemoglobin in RBCs.
(B) is blue due to presence of haemocyanin in plasma.
(C) is red due to presence of haemoglobin in RBCs.
(D) is red due to presence of haemoglobin in plasma.

(147) From where earthworm gets food ?

(A) Living plants
(B) From decaying leaves and shaded leaves which remains in soil
(C) Insects of soil
(D) Fresh leaves from soil

(148) Where is the correct location of Typhlosole in the earthworm ?

(A) 5 – 9 segments
(B) 9 – 14 segments
(C) 26 – 95 segments
(D) 15 segments onwards

(149) What is correct for earthworm ?

(A) Integumentary and septal nephridia are endo-excretory
(B) Integumentary and pharyngeal nephridia are exo-excretory
(C) Pharyngeal and septal nephridia are exo-excretory
(D) Pharyngeal and septal nephridia are endo-excretory

(150) What is the main function of spermatheca in earthworm ?

(A) Maturation location of sperms
(B) Production of sperms
(C) Storage of sperms of companion earthworm
(D) After mating it will play an important role

(151) What is true for respiration in earthworm ?

(A) It will perform anaerobic respiration
(B) O_2 diffuses from air into haemoglobin of RBCs
(C) No important role of blood in conduction of O_2
(D) O_2 diffuses from air into haemoglobin of plasma

(152) In earthworm....

(A) No eyes
(B) Many eyes
(C) One eye
(D) Two eyes

(153) In which segment, the spermathecal apertures are present in earthworm ?

(A) 9/7, 7/8, 8/9, 9/10 (B) 5/6, 6/7, 7/8, 8/9

(C) 10/11, 11/12, 12/13, 13/14 (D) 4/5, 5/6, 6/7, 7/8

(154) How male and female cockroach can be differentiated from each other ?

(A) In male anal style (B) Anal cerci in female

(C) Anal style and antennae in female (D) (A) and (B) both

(155) In which animal ommatidia is present for photosensitivity ?

(A) Cockroach (B) Frog (C) Human (D) Earthworm

(156) What is main function of salivary reservior in cockroach ?

(A) Storage of saliva (B) Secretion of saliva

(C) Secretion of enzymes (D) Storage of enzymes

(157) In cockroach, Salivary glands opens at ..... .

(A) Labrum (B) Mandible (C) Hypopharynx (D) Secondary palp

(158) How many abdominal segments are present in male and female cockroach ?

(A) 10, 10 (B) 9, 10 (C) 8, 10 (D) 9, 9

(159) In which animal, blood is colourless ?

(A) Leech (B) Earthworm (C) Cockroach (D) Rabbit

(160) What is correct sequence for the parts of walking leg of cockroach ?

(P) Tebia (Q) Trochanter (R) Coxa (S) Tarsus (T) Femur

(A) R, Q, T, P, S (B) Q, T, R, S, P (C) S, T, Q, P, R (D) P, Q, R, S, T

(161) How many ommatidia are present in one eye of cockroach for mosaic vision ?

(A) 100 (B) 2000 (C) 1000 (D) 200

Answers : (145-C), (146-D), (147-B), (148-C), (149-D), (150-C), (151-D), (152-A), (153-B), (154-A), (155-A), (156-A), (157-C), (158-A), (159-C), (160-A), (161-B)

•

17 Animal Morphology and Anatomy - II (Frog)

- Frog is a member of class Amphibia. The class amphibia includes all those animals which live in fresh water as well as on land habitat.
- They are the first tetrapods which evolve from fish-like ancestors.
- They stands between fishes and reptiles.
- The systematic position of frog is as under :

  Phylum → Chordata, Subphylum → Vertebrata, Division → Gnathostomata, Class → Amphibia

  Order → Anura, Genus → Rana, Species → Tigrina

  Binomial nomenclature : *Rana tigrina*
- The common Indian bull frog (*Rana tigrina L*) generally lives in or near the water.
- The dormant life during winter and summer known as hibernation and aestivation respectively.
- It is carnivorous which feeds on small animals.
- There are many natural enemies of frog such as snakes, some birds, man, etc. frog has capacity to change the skin colour according to the environment and this helps frog to escape from enemies.

---

(1) From which type of ancestors, the amphibian has been evolved as first order tetrapoda ?

(A) Reptile (B) Arthropoda (C) Fish (D) Platyhelminthus

(2) Due to special characteristic, amphibians has been placed between x and y phylum.

(A) x = Fish, y = Arthropoda (B) x = Fish, y = Reptile

(C) x = Arthropoda, y = Reptile (D) x = Nemanthelminthes, y = Reptile

(3) Which option for right location of frog in systematics ?

(A) Chordata → Invertebrata → Gnathostomata → Amphibian → Anura → Rana → tigrina

(B) Chordata → Vertebrata → Agnatha → Amphibia → Megascolacedeae → Rana → tigrina

(C) Chordata → Vertebrata → Gnathostomata → Amphibia → Anura → Rana → tigrina

(D) Chordata → Vertebrata → Gnathostomata → Reptilia → Anura → Rana → tigrina

(4) How frog can protect itself from enemies ?

(A) By hibernation and aestivation

(B) As it is carnivorous

(C) It has capacity to change skin colour

(D) It is burrowing animal

Answers : (1-C), (2-B), (3-C), (4-C)

External Morphology and Skin :

- The size of frog varies in the same species depending upon the age of the individual.
- Frog has streamlined body which helps it to swim.
- The colour of the body at the dorsal side is green with black spots but ventrally it is lighter.
- The body is divided into two parts, head and trunk.

- The true neck and tail are absent.
- The anterior most part of the head is snout with two nostrils and laterally proputed eyes. On the mid dorsal line, in between two eyes there is browspot.
- Mid ear is represented by tympanic membrane.
- On the ventral side wide mouth is seen.
- The trunk bears two pairs of limbs; fore limbs are shorter with four digits while hind limbs are larger with five toes.
- Toes of hind limbs are connected by webs, which helps the animal in swimming.
- *Rana tigrina* exhibits external sexual dimorphism as :

Male	Female
- Two vocal sacs are present	- Vocal sacs are absent
- Nuptial pad present in the index finger	- Nuptial pad is absent
- The skin shows dark yellow colour during breeding season	- Skin colour is not changed.
- The abdominal region narrow and flat	- The abdominal region is broad and bulged.

Skin :

- The skin of frog is moist, smooth, slimy and without any exoskeleton. It consist of two layers.

(1) **Outer Epidermis :** The epidermis formed of several layers and it is further divided into two layer, i.e. outer stratum corneum and inner stratum germinativum.

- Stratum corneum formed of a single layer of cells. This layer is dead and sheds periodically.
- Stratum germinativum is formed of columnar cells. New cells are formed from this layer.

(2) **Inner Dermis :** It is differentiated into two layers, outer stratum spongiosum and inner stratum compactum.

- The stratum spongiosum consists of loose network of connective tissue, blood vessels and mucus glands. Superficial part of this layer contains chromatophores.
- The stratum compactum is made by dense connective tissue, smooth muscle fibres, nerves and blood vessels.

Functions of the skin :

- It gives definite shape and texture to the body.
- It protects the body against foreign bodies and fungi.
- It acts as a chief respiratory organ.
- It acts as an important sensory organ.
- The frog never drinks water but absorbs water through skin.

(5) The colour of frog on dorsal side is X and on ventral side is Y.

(A) X = Yellow with black spot, Y = Dark (B) X = Green with black spot, Y = Light

(C) X = Green with black spot, Y = Dark (D) X = Yellow with black spot, Y = Light

(6) What is absent in frog ?

(A) Head (B) Trunk (C) True neck and tail (D) Limb

(7) How many fingers are present in one fore and hind limb of frog ?

(A) 5, 5 (B) 4, 4 (C) 8, 10 (D) 4, 5

(8) How many total limbs and fingers are there in frog ?

(A) Limbs = 4, Fingers = 20 (B) Limbs = 2, Fingers = 20

(C) Limbs = 4, Fingers = 18 (D) Limbs = 2, Fingers = 18

(9) Which finger of forelimb of frog has nuptial pad on its proximal end ?

(A) Second (B) Third (C) Fourth (D) First

(10) Which one is suitable for female frog during breeding season ?

(A) Vocal sac absent (B) Skin colour is not changed

(C) Nuptial pad absent (D) Abdomen is narrow and bulged

(11) Which statement is true in reference to frog skin ?

(A) Skin is moist, smooth, slimy and without exoskeleton

(B) It is moist, smooth, slimy and with exoskeleton

(C) It is not moist, smooth, dry and without exoskeleton

(D) It is moist, smooth, dry and with exoskeleton

(12) The outer layer of frog skin is called _X_ and inner one is called _Y_.

(A) X = Dermis, Y = Epidermis (B) X = Stratum spongiosum, Y = Stratum corneum

(C) X = Epidermis, Y = Dermis (D) X = Stratum compactum, Y = Stratum germinativum

(13) Which layer of frog skin becomes dead and sheds periodically ?

(A) Stratum corneum (B) Stratum spongiosum (C) Stratum germinativum (D) Stratum compactum

(14) Which tissue is present in stratum germinativum of frog ?

(A) Squamous epithelium (B) Cuboidal epithelium

(C) Ciliated epithelium (D) Columnar epithelium

(15) The epidermis of frog is made up of _X_ tissue and dermis is of _Y_ tissue.

(A) X = Connective tissue, Y = Epidermal tissue (B) X = Epidermal tissue, Y = Connective tissue

(C) X = Epidermal tissue, Y = Muscle tissue (D) X = Nerve tissue, Y = Connective tissue

(16) In which layer of frog skin, mucous glands are present ?

(A) Stratum corneum (B) Stratum spongiosum (C) Stratum compactum (D) Stratum germinativum

(17) Which is true for stratum spongiosum of skin of frog ?

(A) Loose network of connective tissue, blood vessels and nerves are present.

(B) Compact connective tissue, blood vessels and mucus glands are present.

(C) Simple muscle fibers, blood vessels and nerves are present.

(D) Loose network of connective tissue, blood vessels and mucus glands are present.

(18) What is true for stratum compactum of frog skin ?

(A) Loose network of connective tissue, nerves and blood vessels are present.

(B) Compact connective tissue, simple muscle fibers, nerves and blood vessels are present.

(C) Compact connective tissue, simple muscle fibers, blood vessels are present.

(D) Compact connective tissue, blood vessels and mucus glands are present.

(19) Which is wrong in reference to function of frog skin ?

(A) It gives definite shape and texture (B) It protects body against foreign bodies and fungi

(C) It acts as a cheif respiratory organ (D) It will not act as a sensory organ

Answers : (5-B), (6-C), (7-D), (8-C), (9-D), (10-D), (11-A), (12-C), (13-A), (14-D), (15-B), (16-B), (17-D), (18-B), (19-D)

Digestive system :

- The digestive system of frog mainly includes the alimentary canal and digestive glands. Alimentary canal starts from mouth and ends at cloaca. The alimentary canal includes buccal cavity, pharynx, oesophagus, stomach, intestine, rectum and cloaca.

Alimentary canal :

- Mouth is wide opening located at the anterior end of the head. It is bounded by the upper and lower jaws. A single raw of teeth is present in the upper jaw.
- Mouth leads into a wide and broad cavity called buccal cavity which lies in between two jaws.
- The buccal cavity contains, maxillary teeth, vomarine teeth, internal nostrils, inbulging of eye-ball, eustachian aperture, opening of vacal sacs (only in male) and bifid tongue.
- The internal nostrils are a pair of openings located near the vomarine teeth. They serve in respiration.
- The two vocal sacs are present only in male frog.
- Behind the vomerine teeth there are two inbulging areas on the roof of the buccal cavity known as eye-balls.
- A pair of eustachian aperture lies on the roof of buccal cavity, almost at the angle of jaws.
- The eustachian aperture connects the pharynx with the middle ear and help in keeping the air pressure equal on both the sides of tympanic membrane.
- The bifid tongue is soft, slimy and fleshy. It is attached by its anterior end and is free at the posterior end. The free end of the tongue is bilobed.
- The posterior end of the buccal cavity is the pharynx due to absence of neck. There is no demarcation between the buccal cavity and the pharynx. Hence, both the structures sometimes are collectively known as a bucco-pharyngeal cavity.
- The oesophagus is short, wide, muscular and highly distensible tube leading into the stomach.
- The stomach lies on the left side in the body cavity. It is large, broad and slightly curved chamber. It is formed of two regions, large broader anterior part is called cardiac stomach and posterior narrow part is called pyloric stomach.
- Its mucus epithelium contains multicellular gastric glands secreting the pepsinogen enzyme, dilute HCl and mucus.
- Posterior end of pyloric stomach contains pyloric sphincter valve.
- The stomach is an organ for temporary food storage churning and partly digestion of protein. The pyloric stomach leads to the small intestine.
- The small intestine divide into anterior duodenum a posterior ileum.
- Duodenum runs ahead parallel to stomach forming a U-shaped structure. It receives the hepatopancreatic duct from liver and pancreas which brings bile and parcreatic juice.
- The coiled ileum is the longest part of alimentary canal.
- Digestion and absorption of digested food occur in the small intestine.
- The ileum leads to the large intestine. Anterior is a short, wide tube, running straight and opens into the cloaca. Its function is the re-absorption of water and the preparation and storage of faeces.
- Cloaca is the small terminal sac like part. The term cloaca is used to denote the end part of the large intestine which opens into the rectum and the urinogenital apertures.
- Cloaca opens outside by the vent or cloacal aperture located at the dorsal end of body.

Digestive glands :

- The organ in which food does not enter but its secretion helps in physiological process of digestion are called digestive gland which includes Liver and Pancreas.

(1) Liver :

- Liver is the largest gland found in frog.
- It is dark brown coloured gland located close to the heart and lungs.
- Liver is divided into two lobes and the left is again subdivided so it appears trilobed.
- A gall bladder lies between the right and left lateral lobes.
- The liver secretes greenish liquid called bile which is stored in gall bladder.
- The bile is transported to gall bladder by small hepatic ducts. Cystic ducts from gall bladder and hepatic ducts from liver combine to form a common bile duct.
- The bile duct passes through the pancreas and receives numerous parncreatic ducts. Now the bile duct is called hepatopancreatic duct. It opens into the duodenum.
- Bile has no digestive enzymes. It only emulsifies fat so that liver is not truly a digestive gland.

(2) Pancreas :

- It is the exocrine and endocrine pale coloured gland.
- It is situated at the junction between the stomach and duodenum. The pancreas produces pancreatic juice which contains various enzymes which helps in digestion of proteins, carbohydrates and fats.
- In between the pancreatic lobule groups of cells are found which are known as islets of Langerhans.
- These cells endocrine in nature which secretes hormones like insulin and glucagon directly into blood vessels.
- Insulin and glucagon maintain level of glucose in blood.

---

(20) The large anterior part of frog in stomach is _X_ and posterior narrow part is called _Y_.

(A) X = Pyloric stomach, Y = Cardiac stomach (B) X = Body, Y = Pyloric stomach

(C) X = Cardiac stomach, Y = Pyloric stomach (D) X = Cardiac stomach, Y = Body

(21) Which is true statement for function of stomach of frog ?

(A) Storage of food, churning, lipid digestion

(B) Storage of food temporarily, churning and partial digestion of protein

(C) Storage of food temporarily, churning and protein digestion

(D) Storage of food temporarily, churning and carbohydrate digestion

(22) Duodenum gets _X_ and _Y_ from hepato-pancreatic duct of liver and pancreas.

(A) X = Saliva, Y = Bile juice (B) X = Saliva, Y = Pancreatic juice

(C) X = Bile juice, Y = Pancreatic juice (D) X = Bile juice, Y = Saliva

(23) Which is the longest and coiled part of alimentary canal of frog ?

(A) Stomach (B) Ileum (C) Oesophagus (D) Large intestine

(24) Cloaca means a part of large intestine in which _X_ and _Y_ opens.

(A) X = Rectum, Y = Ureter (B) X = Anal aperture, Y = Urethra

(C) X = Rectum, Y = Urinary bladder (D) X = Anal aperture, Y = Rectum

(25) Liver is dark brown and remain with _X_ and _Y_.

(A) X = Heart, Y = Lungs (B) X = Oesophagus, Y = Heart

(C) X = Heart, Y = Stomach (D) X = Heart, Y = Small intestine

(26) Where does storage and secretion of Bile juice takes place in frog ?

(A) Liver, gall bladder (B) Pancreas, gall bladder

(C) Gall bladder, liver (D) Gall bladder, pancreas

(27) Pancreas is located at the junction of _X_ and _Y_ and produces _Z_.

(A) X = Ileum, Y = Duodenum, Z = Pancreas

(B) X = Pyloric stomach, Y = Ileum, Z = Bile salt

(C) X = Stomach, Y = Duodenum, Z = Pancreatic juice

(D) X = Stomach, Y = Ileum, Z = Bile salt

(28) Islets of Langerhans of pancreas are _X_ type and it produces _Y_ and _Z_ type of hormones which directly mixes up with blood.

(A) X = Exocrine, Y = Insulin, Z = Glucagon (B) X = Endocrine, Y = Insulin, Z = Secretin

(C) X = Endocrine, Y = Gastrin, Z = Secretin (D) X = Endocrine, Y = Insulin, Z = Glucagon

Answers : (20-C), (21-B), (22-C), (23-B), (24-C), (25-A), (26-C), (27-C), (28-D)

Absortion, adsorption, digestion and egestion :

Digestion :

- Generally frog feeds on insects, worms, crustaceans, molluscs etc.
- The prey is captured by rapid action of its prehensile tongue. If prey is large, it is gripped with jaw, and prevented from escape. The prey is swallowed. The mucus helps in swallowing. The digestion of captured prey takes place in the stomach, duodenum and intestine.

Digestion : Digestion means converting large particals, complex and unabsorbable food into simple, small particals and absorbable by alimentary canal.

- Mucus glands of oesophagus secretes mucin. Which is constantly changes the physical form of the food by peristalsis, so as to send the food into stomach.
- Gastric juice is secreted by gastric glands into the stomach. Gastric juice contains Gastrin and dil. HCl. Dil. HCl destroys micro-organisms of food and stimulates gastric glands. It also provides acidic medium for the activity of pepsinogen enzyme.

Inactive pepsinogen :

Inactive pepsinogen + HCI $\rightarrow$ Active pepsin

Protein + Pepsin $\rightarrow$ Peptones + Proteoses

- Mucus keeps wall of stomach very smooth. Thus, this type of partially digested food in stomach is known as chyme.
- Liver as a digestive gland, produces bile juice which has bile salts. This is greenish basic liquid. It will removes acidity of chyme, emulsification of fats and makes lipase active.

Pancreas as a digestive gland produces pancreatic juice which is basic liquid.

(1) In active trypsinogen + Enterokinase → Active trypsin, this active trypsin makes all the proteolytic enzymes active.

(2) Peptones + Trypsin (E) → Peptide and Amino acid

(3) Polysaccharides + Amylase (E) → Maltose

(4) *Emulsi fied* fats + Lipase (E) → Fatty acids + Glycerol

Various hormones of Duodenum works as .....

(1) Enterogastrone (H) → It reaches to stomach and prevents secretion of Gastric juice.

(2) Cholecystokinin (H) → It reaches to Gall bladder and contracts gall bladder so as to secretes bile juice into duodenum.

(3) Secretin and pancreozymin (H) → Combine effect of both the hormone activates pancreas to secretes pancreatic juice into duodenum.

(4) Enterokinin and duokinin (H) → Stimulates the small intestine to produce intestinal juice.

Secretion of intestinal juice into intestine,

(1) Peptide + Erepsin (E) → Amino acids (2) Maltose + Maltase (E) → Glucose + Glucose

(3) Sucrose + Sucrase (E) → Glucose + Fructose (4) Lipid + Lipase (E) → Fatty acids + Glycerol

Absorption :

- Absorption is a process in which the digested food is taken into blood. Mainly this process take place into the duodenum and ileum.
- The walls of these organs are folded with villi which increase the absorptive surface.
- Water, minerals, salts and other nutrients in solution are directly absorbed through the epithelial lining and then pass into blood capillaries.

Adsorption and egestion :

- As the food passes downward in the alimentary canal the water and the digested products are absorbed whereas the undigested food, debries, epithelial cells, leucocytes, bile pigments and large number of bacteria from faeces are removed time to time trough cloacal opening.

---

(29) Frog takes as a food, like ..... .

(A) Insects, worms, crustaceans and protozoan

(B) Insects, worms, crustaceans and molluscs

(C) Insects, worms, crustaceans and coelenterata

(D) Insects, worms, crustaceans and annelida

(30) Which organs are useful to catch and to engulf prey ?

(A) Jaw (B) Mucus (C) Tongue (D) All the given

(31) Inactive pepsinogen + HCl →

(A) Inactive pepsin (B) Active pepsin (C) Inactive trypsin (D) Active trypsin

(32) Protein + Pepsin →

(A) Amino acid (B) Glycerol (C) Proteoses (D) Fatty acid

(33) X + Y → Active trypsin.

(A) X = Inactive trypsinogen, Y = Enterokinase (B) X = Active trypsinogen, Y = Enterokinase

(C) X = Inactive trypsinogen, Y = Inactive pepsin (D) X = Active pepsinogen, Y = Enterokinase

(34) Peptones or proteoses + Trypsin →

(A) Amino acids (B) Fructose

(C) Peptide (D) Peptide and Amino acids

(35) Polysaccharides + X → Maltose

(A) X = Lipase (B) X = Trypsin (C) X = Amylase (D) X = Pepsin

(36) Emulsified fats + X → Y + Z.

(A) X = Maltase, Y = Fatty acids, Z = Glycerol (B) X = Sucrase, Y = Peptide, Z = Amino acids

(C) X = Lactase, Y = Glucose, Z = Lactose (D) X = Lipase, Y = Fatty acids, Z = Glycerol

(37) Which one is responsible to stimulate the contraction of gall bladder to secrete of bile juice into duodenum ?

(A) Hormone (B) Enzyme (C) Co-enzyme (D) All of the given

(38) Whose combine effects activates pancrease to secrete pancreatic juice into duodenum ?

(A) Secretin and Pancreozymin (B) Cholecystokinin and Secretin

(C) Pancreozymin and Enterocrinin (D) Enterogastrone and Secretin

(39) In which organs, the digested food is absorbed and mixed with blood in frog ?

(A) Stomach and Duodenum (B) Duodenum and Foregut

(C) Duodenum and Ileum (D) Stomach and Ileum

(40) In which organ of frog villi increases the absorption area ?

(A) Ileum (B) Duodenum (C) Stomach (D) Oesophagus

(41) Which is not the composition of frog faeces ?

(A) Dead epithelial cells (B) Leucocytes (C) Bile pigments (D) Sucrose

Answers : (29-B), (30-D), (31-B), (32-C), (33-A), (34-D), (35-C), (36-D), (37-A), (38-A), (39-C), (40-A), (41-D)

- **Respiratory system :**
  - Respiration is a process in which living organisms obtain oxygen for oxidation and CO_2 is removed regularly from the body.
  - Organs involved in respiration are known as respiratory organs.
  - Frog as an amphibian animal shows three types of respiration, i.e. (i) Cutaneous respiration or respi ration through skin, (ii) Bucco-pharyngeal respiration and (iii) Pulmonary respiration or respiration through lungs.
  - Generally frog respires through skin, but when the need of oxygen is greater it respires through bucco-pharyngeal cavity and lungs.

(i) Cutaneous respiration : This type of respiration takes place through the skin, so it is known as cutaneous respiration.

The characteristic of skin : Skin remains moist due to mucus, secreted by the mucous glands.

- Skin is permeable for gases.
- Skin is very thin.

- Skin has rich blood supply through blood capillaries.
- Cuteneous repiration takes place in both medium, i.e. water as well as on land, so, when frog lives in any habitat, either water or land, it respires through skin.

(ii) Bucco-pharyngeal respiration :

- Respiration takes place through the buccal cavity and the pharynx then it is called bucco-pharyngeal respiration.
- It is terrestrial respiration.
- Both the organs are lined with thin mucus, permeable to gases and rich with blood vessels.
- Lowering and raising of the floor of the buccal cavity bring the bucco-pharyngeal respiration.

(iii) Pulmonary respiration :

- Respiration takes place through lungs in terrestrial habitat is called pulmonary respiration.
- The entire process of pulmonary respiration is completed in three steps.

(i) Aspiration : The entry of gases into the bucco-pharyngel cavity is called aspiration.

(ii) Inspiration : The gases pass through the bucco-pharyngeal cavity to lungs is called inspiration. During this process diffusion of oxygen occurs.

(iii) Expiration : The passage of impure air from the lungs to the outside of the body is called expiration.

---

(42) Which respiration is performed by frog in water and land medium ?
(A) Bucco-pharyngel respiration (B) Pulmonary respiration
(C) Cuteneous respiration (D) Nasal respiration

(43) Which type of respiration takes place on terrestrial condition ?
(A) Pulmonary respiration (B) Bucco-pharyngeal respiration
(C) Nasal respiration (D) (A) and (B) both

(44) _X_ respiration is performed by lowering and raising of bottom of buccal cavity. During this process, the air is absorbed continuously by _Y_.
(A) X = Cutaneous respiration, Y = Buccal cavity
(B) X = Pulmonary respiration, Y = Lungs
(C) X = Bucco-pharyngeal respiration, Y = Mouth cavity
(D) X = Cuteneous respiration, Y = Internal nares

(45) In which type of respiration oesophagus remain closed ?
(A) Cuteneous respiration (B) Bucco-pharyngeal respiration
(C) Pulmonary respiration (D) All of the given

(46) Which sequence is correct for pulmonary respiration ?
(A) External nares → Nasal cavity → Bucco-pharyngeal cavity → Glottis → laryngotracheal chamber → Lungs
(B) External nares → Nasal cavity → Epiglottis → Glottis → Lungs
(C) Internal nares → Nasal cavity → Bucco-pharyngeal cavity → Epiglottis → Lungs
(D) External nares → Nasal cavity → Bucco-pharyngeal cavity → Glottis → Lungs

(47) Entry of air into bucco-pharyngeal cavity is called _X_.
(A) X = Inspiration (B) X = Apsiration
(C) X = Expiration (D) All the given

(48) Through bucco-pharyngeal cavity air enters into _X_ and is called _Y_ type of respiration.
(A) X = Lungs, Y = Aspiration (B) X = Lungs, Y = Expiration
(C) X = Nasal cavity, Y = Inspiration (D) X = Lungs, Y = Inspiration

(49) Which is wrong in reference to cutaneous respiration ?

(A) Skin remains slimy due to secretion of mucus from mucus glands

(B) Skin is permeable to air

(C) Skin is very thick

(D) Blood supply is in large amount through skin blood capillaries

Answers : (42-C), (43-D), (44-C), (45-B), (46-A), (47-B), (48-D), (49-C)

Circulatory system :

- Frog as a vertebrate animal has closed type of circulatory system. This system has four main components i.e. blood, heart, arteries and veins.
- Blood is red coloured liquid connective tissue. It is composed of blood cells and blood plasma.

Heart : Heart is protected inside a double walled bag called pericardium. In between this walls pericardium fluid is present.

- It is muscular, conical and three chambered.
- On the dorsal side of heart sinus venosus is located. Which collects deoxygenated blood from various parts of the body and empty up into right auricle through sino-auricular aperture.
- The left auricle receives oxygenated blood from lungs through a common pulmonary vein. The blood from both auricles enters the ventricle through a single opening called auriculo-ventricular aperture.
- Thus, oxygenated blood from left auricle and deoxygenated blood from right auricle is mixed in ventricle. As a result in frog, mixed blood is circulated through arterial system.
- The ventricle contracts and forces the blood into truncus arteriosus from where it enters into arterial system.

Arterial system :

- The arterial system, supplies blood from heart to different parts of the body.
- The system starts from truncus arteriosus and supplies mixed blood through aortic arteries.

(1) Carotid arch : External carotid and internal carotid arteries supply blood to head.

(2) Systemic arch : Both systemic arch unites to form a dorsal aorta. It supplies blood to posterior region through various arteries such as -

- **Coeliaco-mesenteric artery :** Supply blood to alimentary canal.
- **Gonadial artery :** Supply blood to gonads.
- **Renal artery :** Supply blood to kidneys.
- **Iliac artery :** Supply blood to hind limbs.

(3) Pulmo-cutaneous artery : It supply blood to lungs and skin.

Venous system :

- The venous system brings blood from different parts of the body to heart.
- Collected impure blood from all the parts of the body is empty up into sinus venosus through three cavals (two pre-cavals and one post-caval).
- Each pre-cavals receives blood from both sides through external jugular, innominate and sub-clavian veins.
- These veins drains impure blood from different organs like tongue, lower-jaw, head, brain, fore-limb, etc.

- Exceptionally, cutaneous vein and branches of sub-clavian veins carries oxygenated blood.
- Renal veins emerging from two kidney enter right lobe of liver and from this it leads to sinus venosus through post-caval which bring blood into sinus venosus from kidney, gonads and liver through different veins.
- Generally arteries and veins are divided to form capillaries, vertebrate like frog exhibits a special type of vein arrangement called portal system. Frog has two types of portal system.

(1) Renal portal system : In this system blood vessels carrying blood from the hind limbs to the kidneys.

(2) Hepatic portal system : It supplies blood to the liver from alimentary canal.

---

(50) The deoxygenated blood from different parts of body is conducted to P → Q → R → auricle → ventricle valve → S.

(A) P = Right auricle, Q = Veins, R = Sinus venosus, S = Ventricle

(B) P = Veins, Q = Left ventricle, R = Sinus venosus, S = Ventricle

(C) P = Veins, Q = Sinus venosus, R = Right auricle, S = Ventricle

(D) P = Veins, Q = Sinus venosus, R = Left auricle, S = Ventricle

(51) Oxygenated blood from lungs and skin conducted to P → Q → R → ventricle.

(A) P = Pulmonary vein, Q = Left auricle, R = Auriculo-ventricular valve

(B) P = Pulmonary vein, Q = Right auricle, R = Auriculo-ventricular valve

(C) P = Veins, Q = Sinus venosus, R = Right auricle

(D) P = Auriculo-ventricular valve, Q = Left auricle, R = Pulmonary vein

(52) The mixed blood of from ventricle is conducted to various parts of body through P → Q → R → with the help of auriculo-ventricular valve.

(A) P = Artery, Q = Aorta, R = Aortic arch

(B) P = Aorta, Q = Aortic arch, R = Arteries

(C) P = Aortic arch, Q = Aorta, R = Arteries

(D) P = Aorta, Q = Right auricle, R = Arteries

(53) The blood of frog vein, before entering into right auricle, enters into which organ ?

(A) Arterial cavity (B) Lymphatic cavity (C) Sinus venosus (D) None of the given

(54) In which chamber of heart, the blood from sinus venosus will be emptied ?

(A) Right ventricle (B) Left ventricle (C) Right auricle (D) Left auricle

(55) Which vessel is formed by union of two systemic arch ?

(A) Ventral aorta (B) Dorsal aorta (C) Dorsal sinus (D) Ventral sinus

(56) By how many cavals, the impure blood of whole body, is emptied into sinus venosus ?

(A) One (B) Two (C) Three (D) Four

(57) From where blood is collected into pre-caval ?

(A) External jugular (B) Innominate (C) Sub-clavian (D) All the given

(58) To which organs, the coeliaco-mesenteric artery supplies the blood in frog ?

(A) Alimentary canal (B) To the gonads (C) Hind limbs (D) Kindneys

Answers : (50-C), (51-A), (52-B), (53-C), (54-C), (55-B), (56-C), (57-D), (58-A)

Nervous system and sensory organs of frog :

- In the frog the nervous system is located on the dorsal side of the body as found in all vertebrates. It is divided into two sections :
  (i) Voluntary nervous system and (ii) Involuntary nervous system.
- The regulation of voluntary nervous system is under the willingness of animals. Voluntary nervous system divides into central and peripheral nervous system.
- Central nervous system consist of brain and spinal cord. Brain is situated in the head and protected within the cranium. Which is divided into three regions : Fore brain, mid brain and hind brain.
- Fore brain includes a pair of olfactory lobes, a pair of cerebral hemispheres and diencephalons.
- On the ventral side of diencephalons, one hollow, bilobed and pouch like part is located, which is known as infundibulum. Pituitary gland is attached to the broad posterior end of it. It is master endrocrine gland controlling various physiological activities, animal growth and development.
- The mid brain includes of two large oval and obliquely arranged optic lobes.
- While hind brain is composed of cerebellum and medulla oblongata. The medulla oblongata continues as the spinal cord in the vertebral column, spinal cord forms the voluntary central nervous system in the trunk region and terminates in the hollow cavity of urostyle of a vertebral column as a filum terminale.
- Involuntary of autonomous nervous system is associated with the controlling of involuntary activity of animal. It has two sections.
- Sympathetic nervous system and parasympathetic nervous system - both types are complementory to each other.
- For example, the sympathetic nervous system accelerates the heart beat while parasympathetic nervous system deaccelerates the heart beats.

Sensory organs : It is of five kinds.

- Touch sensory, taste sensory, smell sensory, sight sensory and sound sensory.
- Of these, eye and ears has well organized structures.
- Other kinds of sense organs are constituted by specialized groups of cells, connected at near nerve endings.
- Tactile structures occur in skin. The tongue possesses taste receptors. In the lining of olfactory chambers, smell sensory structures are located.

Eye :

- Two eyes of frog are located in the orbits. The wall of eye ball is made up of three layers. The outer layer is sclerotic, the middle layer is choroids and the inner layer is retina.
- Towards the outerside a transparent cornea occurs in the eye ball.
- On the inner side, the choroid coat forms an iris. In the center of iris, an aperture known as pupil occurs.
- Behind iris, the lens is arranged.

Ear :

- The ear consist of an inner ear and a middle ear. Exteral ear is absent.
- The internal ear is also called membranous labyrinth. It is located in a fluid-filled auditory capsule.
- Middle ear is an air-filled chamber which possesses the tympanic membrane at its outer end.
  Over and above the nervous system, frog has endocrine regulatory system made of endocrine glands.
- Which includes pituitary glands located in brain, thyroid gland in throat region, adrenal gland sunk in kidneys and gonads (testis and ovary).
- Islets of Langerhans of Pancreas also secretes hormones. Hormones are chemical regulators. It releases Insulin and Glucagon.

(59) Brain is in _X_ and is protected by _Y_ .

(A) X = Head, Y = Cranium (B) X = Head, Y = Neck

(C) X = Snout, Y = Cranium (D) X = Head, Y = Trunk

(60) Which parts are included in fore brain of frog brain ?

(A) One pair of olfactory lobes, two big eye balls, Diencephalons

(B) One pair of olfactory lobe, one pair of cerebral hemisphere, medulla oblongata

(C) One pair of olfactory lobes, one pair of cerebral hemisphere, Diencephalon

(D) One pair of olfactory lobes, cerebellum, meddula oblongata

(61) The _Z_ is bag like and _Y_ structure, remain _X_ side of Diencephalon.

(A) X = Dorsal, Y = One hollow bilobed, Z = Infundibulum

(B) X = Ventral, Y = Two hollow bilobed, Z = Infundibulum

(C) X = Ventral, Y = One hollow bilobed, Z = Cerebellum

(D) X = Ventral, Y = One hollow bilobed, Z = infundibulum

(62) In frog mid brain _X_ and _Y_ arranged _Z_ are included.

(A) X = Two big spherical, Y = Longitudinally, Z = Optic lobes

(B) X = Two big oval, Y = Oblique, Z = Optic lobes

(C) X = Two big oval, Y = Longitudinally, Z = Olfactory lobes

(D) X = Two big spherical, Y = Oblique, Z = Optic lobes

(63) The hind brain of frog is made up of ..... .

(A) Cerebellum, Diencephalon (B) Cerebellum, Infundibulum

(C) Olfactory lobes, optic lobes, medulla oblongata (D) Cerebellum, medulla oblongata

(64) The frog has _X_ pairs of cranial nerves and _Y_ pairs of spinal cord nerves.

(A) X = 10, Y = 08 (B) X = 09, Y = 10

(C) X = 10, Y = 09 (D) X = 11, Y = 08

(65) Sympathetic nervous sytem _X_ heart beats and parasympathetic nervous system _Y_ heart beats.

(A) X = Accelerates, Y = Deaccelerates (B) X = Deaccelerates, Y = Accelerates

(C) X = Medium, Y = Deaccelerates (D) X = Deaccelerates, Y = Medium

(66) What is called the posterior end of spinal cord ?

(A) Urostyle (B) Filum terminale (C) Plasmo desmata (D) White fibres

(67) Which sensory organs has well organized structure in frog ?

(A) Eye, ears (B) Eyes, skin (C) Tongue, ears (D) Ears, nares

(68) Frog has _X_ touch sensory, _Y_ tastes sensory and _Z_ (olfactory) smell sensory organ.

(A) X= Tongue, Y = Skin, Z = Nares (B) X = Skin, Y = Nares, Z = Tongue

(C) X = Tongue, Y = Skin, Z = Internal ears (D) X = Skin, Y = Tongue, Z = Nares

(69) Which are the chemical regulators in frog ?

(A) Vitamins (B) Protein (C) Hormones (D) Lipid

(70) What is the structure which is made up of cranial nerves and spinal cord nerves ?

(A) Peripheral nervous system (B) Parasympathetic nervous system

(C) Central nervous ystem (D) Sympathetic nervous system

(71) Which structure is made up of brain and spinal cord ?

(A) Peripheral nervous system (B) Parasympathetic nervous system

(C) Central nervous system (D) Parasympathetic nervous system

(72) Filum terminale is the end part of which organ ?

(A) Nerve fibers (B) Muscle fibers (C) Vertebral column (D) Spinal cord

Answers : (59-A), (60-C), (61-D), (62-B), (63-D), (64-C), (65-A), (66-B), (67-A), (68-D), (69-C), (70-A), (71-C), (72-D)

Urinogenital system :

- The urinogenital system comprises the excretory and the reproductive organs.
- The excretory system is closely associated with the genital system in male but they are separate in female.

Excretory system :

- A pair of kidney acts as main excretory system in frog.
- They are arranged on lateral sides of vertebral column in posterior part of the body. Kidneys are flat, oblong and brown in colour.
- A large number of uriniferous tubules occur in each kidney as units of excretion called Nephrons.
- Each uriniferous tubule possesses a double walled bowl like structure at its anterior end called Bowman's capsule.
- Glomerulus is located within the Bowman's capsule which jointly called malpighian body.
- The process of urine formation begins here.
- The liquid urine formed in kidneys, is transported through ureters. Ureters come out from posterior lateral ends of kidneys. They transport urine towards cloaca.
- A bilobed, thin walled urinary bladder is associated with cloaca, urinary bladder stores urine. When it is filled up, it contracts and discards urine through the cloacal aperture.
- The main excretory substance in urine is urea.
- In male frog, transport of sperms also occurs through the ureters, hence, the ureters are called urinogenital ducts.

---

(73) In frog, _X_ pair kidneys, remain _Y_ side of body and lateral side of _Z_.

(A) X = Two, Y = Posterior, Z = Spinal cord (B) X = One, Y = Ventral, Z = Spinal cord

(C) X = One, Y = Posterior, Z = Spinal cord (D) X = Two, Y = Ventral, Z = Spinal cord

(74) The frog kidneys are of _X_ coloured, flat and are _Y_ shaped.

(A) X = Dark red, Y = Oval (B) X = Dark brown, Y = Oval

(C) X = Light brown, Y = Round (D) X = Dark brown, Y = Round

(75) What is called the excretory unit in the frog ?

(A) Uriniferous tubule (B) Malpighian tubule (C) Nephridia (D) Malpighian corpuscle

(76) What is called the cup shaped structure remain at the distal end of the uriniferous tubule in frog ?

(A) Kidney pouch (B) Malpighian tubule (C) Bowman's capsule (D) Malpighian corpuscle

(77) What is called the combine structure of Bowman's capsule and Glomerulus ?

(A) Malpighian corpuscle (B) Malpighian tubule (C) Malpighian cup (D) Malpighian layer

(78) From where does the urine formation starts in frog ?

(A) Uriniferous tubule (B) Malpighian corpuscle (C) Malpighian tubule (D) Malpighian cup

Answers : (73-C), (74-B), (75-A), (76-C), (77-A), (78-B)

Reproductive system :

- Frog is a unisexual animal.
- The male reproductive system includes a pair of testis, vasa efferentia, Bidder's canal and urinogenital ducts.
- Each tests is located at the antero-lateral part of the kidney. It is oval-shaped, small and yellowish in colour. It remains attached to the kidney by mesorchium.
- The sperms produce in testis are transported to vasa efferentia, Bidder's canal and finally, it is transported by urinogenital ducts to cloaca.
- The reproductive organs of frog includes a pair of ovaries, a pair of oviducts and ovisac. Ovaries enlarge during the breeding season and produce ova.
- Each ovary is located at antero-lateral region of kidney.
- It is attached through mesorchium.
- Each oviduct begins as an oviductal funnel.
- It is highly coiled structure which terminates into an ovisac and opens into the cloaca.
- The female frog releases large number of ova in water.
- The fat bodies situated at anterior end of kidney also act as accessory reproductive organs which supply energy during formation of gonadial cells.
- The breeding season of frog is monsoon, when it exhibits external and cross-fertilization. The medium of fertilization is water.
- Embryonic development is incomplete, external and metamorphosis type. So, instead of baby frog, the larval form of tadpole comes out from the egg.
- The tadpole passes through different forms like tadpole with external gills, internal gills, hind limbs, tetrapod tadpole, etc. and it gets converted into baby frog.

(79) Which organs are included in male reproductive system ?

(A) One pair testis, vasa efferentia, Bidder's canal and one urinogenital ducts.

(B) One testis, vasa efferentia, Bidder's canal, one pair of urinogenital ducts.

(C) One pair testis, vasa efferentia, Bidder's canal, one pair of urinogenital ducts.

(D) One pair testis, vasa efferentia, cloaca, urinary bladder.

(80) The testis of male frog is arranged _X_ side of kidney and are _Y_ coloured small organs.

(A) X = Antero-lateral, Y = Round, Yellowish (B) X = Ventro-lateral, Y = Oval, Yellowish

(C) X = Antero-lateral, Y = Oval, Greenish (D) X = Antero-lateral, Y = Oval, Yellowish

(81) What is the right path for conduction of sperms which are produced from testis ?

(A) Sperms → Vasa efferentia → Urinogenital ducts → Bidder's canal → Cloaca

(B) Sperms → Vasa efferentia → Bidder's canal → Urinogenital canal → Cloaca

(C) Sperms → Bidder's canal → Vasa effernita → Cloaca → Urinogenital ducts

(D) Sperms → Vasa effernita → Urinogenital ducts → Cloaca → Bidder's canal

(82) What is included in female reproductive system ?

(A) One ovary, one oviduct, ovisac (B) One ovary, one pair oviduct, ovisac

(C) One pair ovary, one pair oviduct, ovisac (D) One pair ovary, Bidder's canal, ovisac

(83) The location of ovaries is _X_ side of kidney and are connected to _Y_ .

(A) X = Antero-lateral, Y = Mesorchium (B) X = Antero-lateral, Y = Mesorchium

(C) X = Antero-ventral, Y = Mesorchium (D) X = Antero-ventral, Y = Mesorchium

(84) Which accessory reproductive organ is located on anterior of frog kidney ?

(A) Ovary (B) Testes (C) Ovisac (D) Fat body

(85) Which structure provides energy for gametogenesis in frog ?

(A) Fat body (B) Bidder's canal (C) Vasa efferntia (D) Oviduct

(86) The fertilization medium for frog is _X_ and embryo development is _Y_ type.

(A) X = Water, Y = Incomplete, external and metamorphosis

(B) X = Land, Y = Incomplete, external, metamorphosis

(C) X = Water, Y = Complete, external, metamorphosis

(D) X = Land, Y = Complete, internal, metamorphosis

(87) Which sequence is correct for various stage of frog metamorphosis ?

(A) Egg → External gills → Hind limb → Mature frog

(B) Egg → External gills → Internal gills → Hind limb → Tetrapoda → Mature frog

(C) Egg → External gills → Tetrapoda → Mature frog

(D) Egg → Posterior gills → Tetrapoda → External gills → Internal gills → Mature frog.

Answers : (79-C), (80-D), (81-B), (82-C), (83-B), (84-D), (85-A), (86-A), (87-B)

- **True - False (T - F) type questions :**

Choose the correct alternative option for answer :

(88) (1) Frog is descended from the Aves.

(2) Series of frog is Anura.

(3) Frog is omnivorous animal.

(4) Frog can change skin colour according to surrounding environment.

(A) T,T, F, F (B) T, F, T, F (C) F, T, F, T (D) F, F, T, T

(89) (1) True tail and neck is absent in frog.

(2) Anterior part of frog head is called snout.

(3) Frog has eye-brow spot on mid-dorsal line on head.

(4) During breeding season, the abdomen of female frog is narrow and flat.

(A) T, T, F, F (B) F, F, T, T (C) F, T, T, T (D) T, T, T, F

(90) (1) The frog skin is moist, smooth, slimy and with exoskeleton.

(2) The germinative layer is cuboidal in frog.

(3) The skin of frog is main respiratory organ in frog.

(4) The skin of frog protects from external factors and fungi.

(A) F, F, T, T (B) T, T, F, F (C) T, F, T, F (D) F, T, F, T

(91) (1) Anterior part of mouth cavity is called pharynx.

(2) Anterior bigger part of stomach is cardiac stomach and lower narrow portion is pyloric valve.

(3) Frog duodenum runs parallel to stomach to form "U" shape.

(4) In frog, the digested food is absorbed in small intestine.

(A) T, T, F, F (B) F, T, T, T (C) F, F, T, T (D) T, F, T, F

(92) (1) The function of cloaca in frog is reabsorption of water and formation as well as storage of faeces.

(2) Liver is the largest gland of frog.

(3) Pancreas is located at the junction of stomach and ileum.

(4) The aperture of oesophagus is useful to engulf the food as the internal lining of oesophagus is smooth.

(A) T, T, F, T (B) T, T, F, F (C) F, F, T, T (D) T, F, T, F

(93) (1) Peptide + Peptidase $\rightarrow$ Amino acids (2) Maltose + Maltase $\rightarrow$ Glucose + Fructose

(3) Sucrose + Sucrase $\rightarrow$ Glucose + Glucose (4) Lipid + Lipase $\rightarrow$ Fatty acids + Glycerol

(A) F, F, F, T (B) F, F, T, T (C) T, F, F, T (D) F, T, T, F

(94) (1) The skin of frog is impermeable for gas.

(2) The skin of frog is very thin.

(3) Large amount of capillary supplies blood to skin.

(4) Due to mucus, frog skin remains slimy.

(A) F, T, T, T (B) F, F, T, T (C) F, T, F, T (D) T, F, T, F

(95) (1) RBCs of frog are non-nucleated and contains haemoglobin.

(2) WBCs of frog are nucleated and colourless.

(3) Frog platelets are nucleated.

(4) Blood plasma of frog is liquid and contains water and salts.

(A) T, F, T, F (B) T, T, F, F (C) F, F, T, T (D) F, T, T, T

(96) (1) Pituitary gland is located at posterior part of infundibulum.

(2) The wall of eye ball of frog is unilayered.

(3) Behind the iris, the lens is arranged in frog.

(4) Interal ear of frog is arranged in fluid-filled cavity called auditory capsule.

(A) T, F, T, T (B) F, T, T, T (C) F, F, T, T (D) T, T, F, F

(97) (1) Pituitary gland is in throat in frog and Adrenal gland is embaded in kidney.

(2) Hormone of frog is chemical regulators.

(3) Malpighian tubules are excretory units in frog.

(4) Main excretory substance in frog is urea.

(A) T, F, T, F (B) T, T, F, F (C) F, T, F, T (D) F, F, T, T

Answers : (88-C), (89-D), (90-A), (91-B), (92-A), (93-C), (94-A), (95-D), (96-A), (97-C)

- **A - Assertion, R - Reason type questions :**

Choose correct option :

(A) A and R both are correct, R is correct explanation of A.

(B) A and R both are correct, R is not correct explanation of A.

(C) A - correct, R - incorrect.

(D) A - incorrect, R - correct.

(98) Assertion A : Frog belongs to class Amphibia.
Reason R : Frog can live either on land or in water.
(A) (B) (C) (D)

(99) Assertion A : Frog is placed in between Fish and Reptile.
Reason R : Frog is first tetrapod which is evolved from Fish.
(A) (B) (C) (D)

(100) Assertion A : Frog passes dormant life during winter and summer.
Reason R : Frog is carnivorous animal.
(A) (B) (C) (D)

(101) Assertion A : Frog has true neck.
Reason R : Tail is absent in frog.
(A) (B) (C) (D)

(102) Assertion A : Anterior portion of mouth in frog is called snout.
Reason R : Eye-brow spot remains between two eyes in frog.
(A) (B) (C) (D)

(103) Assertion A : Male frog has two vocal sacs.
Reason R : Abdomen of male frog is broad and flat.
(A) (B) (C) (D)

(104) Assertion A : Female frog has naptial pad.
Reason R : Female frog has broad and flat abdomen.
(A) (B) (C) (D)

(105) Assertion A : Frog skin is moist, smooth slimy and without any exoskeleton.
Reason R : The germinative layer of frog is made up of columnar cells.
(A) (B) (C) (D)

(106) Assertion A : Dermis of frog is internal layer of skin.
Reason R : It is divided into two sublayers.
(A) (B) (C) (D)

(107) Assertion A : Stratum spongiosum possesses loose network of connective tissue, smooth muscles and mucus glands.
Reason R : Superficial part of stratum spongiosum contains chromatophores.
(A) (B) (C) (D)

(108) Assertion A : The stratum compactum of frog is made up of dense connective tissue, smooth muscles, nerves and blood vessels.

Reason R : Skin gives definite shape and texture to the body of frog.

(A) (B) (C) (D)

(109) Assertion A : Mouth cavity and pharynx is also known as bucco-pharyngeal cavity.

Reason R : As frog has no neck, there is no demarcation between mouth and pharynx.

(A) (B) (C) (D)

(110) Assertion A : Anterior bigger part of stomach is known as cardiac stomach in frog.

Reason R : Posterior narrow part of stomach is known as pyloric stomach in frog.

(A) (B) (C) (D)

(111) Assertion A : Liver and parcreas are known as accessory digestive glands.

Reason R : Liver is the largest gland of frog.

(A) (B) (C) (D)

(112) Assertion A : Liver is not true digestive gland.

Reason R : It only emulsifies lipid but has no digestive enzymes.

(A) (B) (C) (D)

(113) Assertion A : Pancreas is endocrine as well exocrine gland.

Reason R : Pancreas is located at the juncture of stomach and duodenum.

(A) (B) (C) (D)

(114) Assertion A : Frog is carnivorous animal.

Reason R : Frog generally, takes insects, worms, crustaceans and molluscs as a food.

(A) (B) (C) (D)

(115) Assertion A : As the wall of ileum possesses villi, it increases the surface for absortion.

Reason R : The epithelial layer of frog absorbs water, minerals and nutrient substance and mixes it into blood.

(A) (B) (C) (D)

(116) Assertion A : Frog respires by skin.

Reason R : Due to mucus into the skin of frog, it is moist, permeable to gases and very thin and gets blood supply from capillaries.

(A) (B) (C) (D)

(117) Assertion A : Bucco-pharyngeal respiration is terrestrial type.

Reason R : During this type of respiration aperture of oesophagus remains open.

(A) (B) (C) (D)

(118) Assertion A : Entry of air into bucco-pharyngeal cavity is inspiration.

Reason R : Removal of impure air from lungs is called expiration.

(A) (B) (C) (D)

(119) Assertion A : Frog has closed type circulatory system.

Reason R : In frog, blood circulates in heart, artery and veins.

(A) (B) (C) (D)

(120) Assertion A : In arterial system of frog, mixed blood is circulated.

Reason R : Oxygenated blood of left auricle and deoxygenated blood of right auricle mixes into ventricle.

(A) (B) (C) (D)

(121) Assertion A : Arterial system of frog supplies blood to various organs of the body by heart.

Reason R : Arterial system starts from aorta.

(A) (B) (C) (D)

(122) Assertion A : Carotid arch supply blood to head region in frog.

Reason R : The systemic arch supplies blood to skin in frog.

(A) (B) (C) (D)

(123) Assertion A : Coeliaco-mesentric artery supplies blood to kidney.

Reason R : Sciatic artery supplies blood to hind limbs.

(A) (B) (C) (D)

(124) Assertion A : The venus system of frog collects blood from various parts of body and brings it towards heart.

Reason R : Impure blood from whole body emptied into sinus venosus by three cavals.

(A) (B) (C) (D)

(125) Assertion A : The nervous system is divided into main two parts.

Reason R : The nervous system of frog is located towards ventral side.

(A) (B) (C) (D)

(126) Assertion A : Pituitary gland of frog is located at the posterior side of infundibulum.

Reason R : Pituitary gland is a master gland in frog.

(A) (B) (C) (D)

(127) Assertion A : Pituitory gland is a master gland in frog.

Reason R : In frog, it regulates physiological processes, growth and development.

(A) (B) (C) (D)

(128) Assertion A : In frog, parasympathetic and sympathetic nervous systems are complementary to each other.

Reason R : Sympathetic nervous system accelerates the heart beats while parasympathetic nervous system deaccelerates the heart beats.

(A) (B) (C) (D)

(129) Assertion A : In frog, external ear is developed.

Reason R : In frog, internal ear is called membranous labyrinth.

(A) (B) (C) (D)

(130) Assertion A : In frog, hormones are chemical regulators.

Reason R : Islets of Langerhas of pancreas forms hormones in frog.

(A) (B) (C) (D)

(131) Assertion A : Uriniferous tubules of frog is called excretory unit.

Reason R : Urine formation starts from malpighian tubules in frog.

(A) (B) (C) (D)

(132) Assertion A : Ureter of male frog is called urinogenital duct.

Reason R : It conducts both, sperms as well as urine.

(A) (B) (C) (D)

(133) Assertion A : In male frog, testes are located ventro-lateral side on kidneys.

Reason R : In Female frog, ovaries are located antero-lateral side on kidneys.

(A) (B) (C) (D)

Answers : (98-A), (99-B), (100-B), (101-D), (102-B), (103-C), (104-D), (105-B), (106-B), (107-B), (108-B), (109-A), (110-B), (111-B), (112-A), (113-B), (114-A), (115-B), (116-A), (117-C), (118-D), (119-A), (120-A), (121-B), (122-C), (123-D), (124-B), (125-C), (126-B), (127-A), (128-A), (129-D), (130-B), (131-B), (132-A), (133-D)

- **Match the columns and choose the correct option :**

(134) Match the Column :

Column - I	**Column - II**	
(a) Frog (Amphibian)	(1) Abdomen is narrow and flat	(A) (a - 4), (b - 3), (c - 1), (d - 2)
(b) Eye-brow spot	(2) Abdomen is broad and bulged	(B) (a - 4), (b - 3), (c - 2), (d - 1)
(c) Male frog	(3) Between two eyes, on mid dorsal line	(C) (a - 3), (b - 4), (c - 1), (d - 2)
(d) Female frog	(4) Descended from fish	(D) (a - 2), (b - 1), (c - 3), (d - 4)

(135) Match the Column :

Column - I	**Column - II**	
(a) Stratum corneum	(1) Chromatophores are present.	(A) (a - 3), (b - 1), (c - 4), (d - 2)
(b) Stratum germinativum	(2) It is inner layer of frog skin.	(B) (a - 3), (b - 4), (c - 1), (d - 2)
(c) Stratum spongiosum	(3) It is made up of unicellular thickness.	(C) (a - 3), (b - 4), (c - 2), (d - 1)
(d) Dermis	(4) It is made up of columnar cells.	(D) (a - 1), (b - 2), (c - 3), (d - 4)

(136) Match the Column :

Column - I	**Column - II**
(a) Pharynx	(1) Digetion and absorption of digested food.
(b) Oesophagus	(2) Left side of the body cavity.
(c) Stomach	(3) Posterior part of the mouth cavity.
(d) Small intestine	(4) Short, broad, muscular and special tube.

(A) (a - 2), (b - 1), (c - 3), (d - 4) (B) (a - 3), (b - 4), (c - 1), (d - 2)

(C) (a - 3), (b - 4), (c - 2), (d - 1) (D) (a - 4), (b - 3), (c - 2), (d - 4)

(137) Match the Column :

Column - I	**Column - II**
(a) Ileum	(1) Islets of Langerhans
(b) Cloaca	(2) Emulsifies the fats
(c) Liver	(3) Largest and coiled part of alimentary canal
(d) Pancreas	(4) End part of large intestine

(A) (a - 3), (b - 4), (c - 2), (d - 1)
(B) (a - 3), (b - 4), (c - 1), (d - 2)
(C) (a - 3), (b - 2), (c - 1), (d - 4)
(D) (a - 1), (b - 2), (c - 3), (d - 4)

(138) Match the Column :

Column - I	**Column - II**
(a) Carotid arch	(1) Supply blood to alimentary canal
(b) Coelico-mesentric artery	(2) Supply blood to gonads
(c) Gonadial artery	(3) Supply blood to head region
(d) Pulmocutaneous artery	(4) Supply blood to skin and lungs

(A) (a - 3), (b - 2), (c - 1), (d - 4)
(B) (a - 3), (b - 1), (c - 4), (d - 2)
(C) (a - 3), (b - 1), (c - 2), (d - 4)
(D) (a - 3), (b - 4), (c - 2), (d - 1)

(139) Match the Column :

Column - I	**Column - II**
(a) Fore brain	(1) Medulla oblongata and cerebellum
(b) Mid brain	(2) One pair olfactory lobes, one pair of cerebral hemisphere and diencephalon
(c) Hind brain	(3) Two large, oval and oblique eye balls

(A) (a - 2), (b - 3), (c - 1)
(B) (a - 2), (b - 1), (c - 3)
(C) (a - 1), (b - 2), (c - 3)
(D) (a - 3), (b - 2), (c - 1)

(140) Match the Column :

Column - I	**Column - II**
(a) Eye	(1) Smell sensory structures
(b) Ear	(2) Touch sensory structures
(c) Skin	(3) Vision sensory structures
(d) Tongue	(4) Audio sensory structures
(e) Nares	(5) Taste sensory structures

(A) (a - 3), (b - 1), (c - 2), (d - 5), (e - 4)
(B) (a - 3), (b - 4), (c - 2), (d - 5), (e - 1)
(C) (a - 3), (b - 4), (c - 2), (d - 1), (e - 5)
(D) (a - 3), (b - 4), (c - 1), (d - 2), (e - 5)

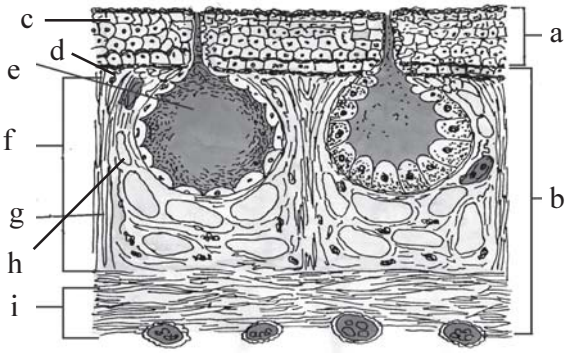
(141) Match the Column :

Column - I	**Column - II**
(a) Cholecystokinin	(1) Activates trypsinogen
(b) Enterokinase	(2) Collects deoxygenated blood
(c) Sinus venosus	(3) Provides energy for formation of gonadial cells
(d) Fat body	(4) Contracts gall blader

(A) (a - 4), (b - 1), (c - 2), (d - 3)
(B) (a - 4), (b - 1), (c - 3), (d - 2)
(C) (a - 4), (b - 3), (c - 2), (d - 1)
(D) (a - 1), (b - 2), (c - 3), (d - 4)

Answers : (134-A), (135-B), (136-C), (137-A), (138-C), (139-A), (140-B), (141-A)

- **Diagram based questions :**

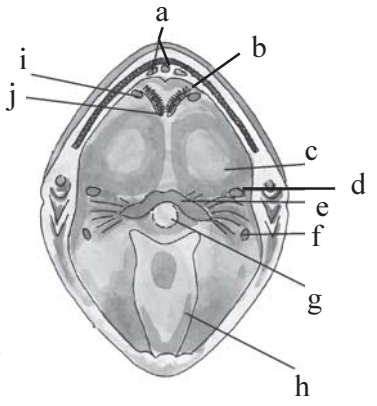


(142) What is indicated by a, c and d in given figure ?
(A) a = Dermis, c = Stratum germinativum, d = stratum corneum
(B) a = Epidermis, c = Stratum corneum, d = Stratum germinativum
(C) a = Epidermis, c = Stratum germinativum, d = Stratum corneum
(D) a = Stratum spongiosum,, c = Epidermis, d = Dermis

(143) What is indicated by b, e and f in given figure ?
(A) b = Dermis, e = Mucus gland, f = Stratum spongiosum
(B) b = Epidermis, e = Mucus gland, f = Stratum spongiosum
(C) b = Mucus gland, e = Dermis, f = Stratum spongiosum
(D) b = Stratum spongiosum, e = mucus gland, f = Dermis

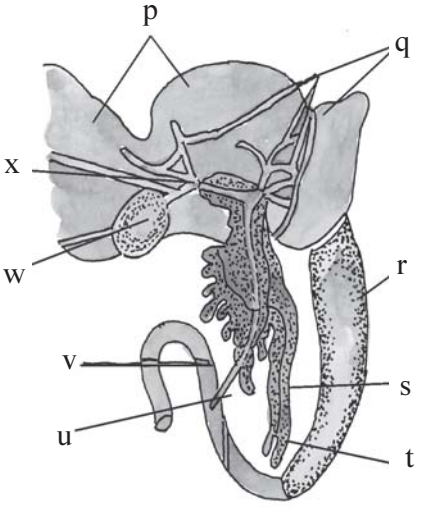
(144) What is indicated by g, h and i in the given figure ?
(A) g = Stratum compactum, h = Longitudinal muscle, i = Blood vessel
(B) g = Longitudinal muscle, h = Stratum compactum, i = Blood vessel.
(C) g = Longitudinal muscle, h = Blood vessel, i = Stratum compactum
(D) g = Stratum spongiosum, h = Blood vessel, i = Longitudinal muscle

(145) What is indicated by a, b, c and d in given figure ?
(A) a = Maxillary teeth, b = Subrostral fossae, c = Inbulging of eye ball, d = Eustachian aperture
(B) a = Subrostral fossae, b = Vomerine teeth, c = Inbulging of eye ball, d = Eustachian aperture
(C) a = Subrostral fossae, b = Maxillary teeth, c = Inbulging of eye ball, d = Vomerine teeth
(D) a = Subrostral fossae, b = Internal nares, c = Maxillary teeth, d = Vomerine teeth



(146) What is indicated by e, f and g in given figure ?
(A) e = Oesophageal aperture, f = Aperture of vocal sacs, g = glottis
(B) e = Oesophageal aperture, f = Maxillary teeth, g = glottis
(C) e = Glottis, f = Maxillary teeth, g = Gullet
(D) e = Gullet, f = Glottis, g = Opening of vocal sacs

(147) What is indicated by h, i and j in given figure ?
(A) h = Vomarine teeth, i = Internal nares, j = Bifid tongue
(B) h = Bifid tongue, i = Internal nares, j = Vomerine teeth
(C) h = Glottis, i = Vomerine teeth, j = Internal nares
(D) h = Bifid tongue, i = Glottis, j = Vomerine teeth



(148) What is indicated by p, q and r in given figure ?
(A) p = Gall bladder, q = Cystic duct, r = Stomach
(B) p = Duodenum, q = Hepatic duct, r = Pancreas
(C) p = Lobes of the liver, q = Hepatic duct, r = Stomach
(D) p = Pancreatic lobes, q = Pancreatic duct, r = Stomach

(149) What is indicated by t, v and w in given figure ?
(A) t = Stomach, v = Duodenum, w = Gall bladder (B) t = Pancreas, v = Duodenum, w = Gall bladder
(C) t = Liver, v = Pancreas, w = Stomach (D) t = Pancreas, v = Gall bladder, w = Duodenum

(150) What is indicated by s, u and x in given figure ?

(A) s = Cystic ducts, u = Pancreatic ducts, x = Hepatic ducts

(B) s = Hepatic ducts, u = Common bile duct, x = Pancreatic ducts

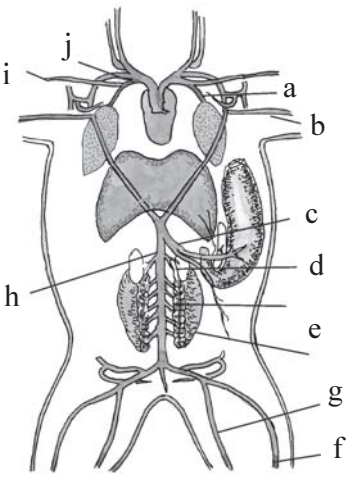
(C) s = Pancreatic ducts, u = Cystic ducts, x = Gastric duct

(D) s = Pancreatic ducts, u = Hepatopancreatic duct, x = Cystic duct

(151) On which organ, cholecystokinin affects ?

(A) p (B) r (C) t (D) w

(152) Which system of frog is indicated in given figure ?



(A) Digestive system (B) Arterial system

(C) Venule system (D) Excretory system

(153) Match the Column :

Column - I	Column - II
(a) Pulmonary artery	(i) f
(b) Renal artery	(ii) a
(c) Dorsal aorta	(iii) e
(d) Pulmocutaneous artery	(iv) j
(e) Femoral artery	(v) h

(A) (a - ii), (b - iii), (c - v), (d - iv), (e - i) (B) (a - iii), (b - i), (c - ii) (d - iv), (e - v)

(C) (a - ii), (b - iii), (c - i), (d - v), (e - iv) (D) (a - iv), (b - v), (c - iii), (d - ii), (e - i)

(154) What is indicated by b, c, d and i in given figure ?

(A) b = Coelico artery, c = Subclavian artery, d = Gonadial artery, i = Cutaneous artery

(B) b = Cutaneous artery, c = Gonadial artery, d = Subclavian artery, i = Coelico mesentric artery

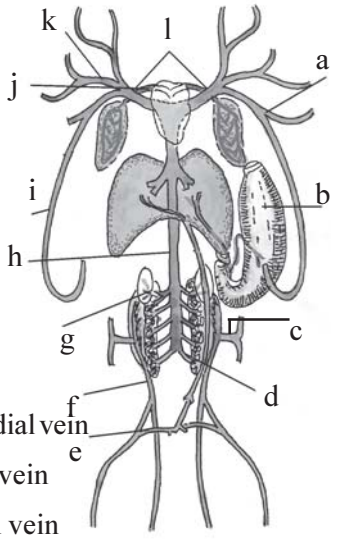
(C) b = Gonadial artery, c = Cutaneous artery, d = Coelico mesentric artery, i = Subclavian artery

(D) b = Subclavian artery, c = Coelico mesentric artery, d = Gonadial artery, i = Cateneous artery

(155) Which system of frog is indicated by given figure ?

(A) Reproductive system (B) Arterial system (C) Venule system (D) Excretory system

(156) What is indicated by a, d, f and j in given figure ?



(A) a = Pulmonary vein, d = Renal portal vein,
f = Renal vein, j = Subclavian vein

(B) a = Gonadial vein, d = Renal vein,
f = Subclavian vein, j = Pulmonary vein

(C) a = Pulmonary vein, d = Subclavian vein,
f = Renal vein, j = Gonadial vein

(D) a = Subclavian vein, d = Renal vein,
f = Renal portal vein, j = Pulmonary vein

(157) What is indicated by b, c, e and g in given figure ?

(A) b = Hepatic portal vein, c = Dorsal lumbar vein, e = Pelvic vein, g = Gonadial vein

(B) b = Hepatic vein, c = Pelvic vein, e = Renal vein, g = Dorsal lumbar vein

(C) b = Pelvic vein, c = Gonadial vein, e = Dorsal lumbar vein, g = Renal vein

(D) b = Renal vein, c = Pelvic vein, e = Gonadial vein, g = Dorsal lumbar vein

(158) What is indicated by h, i, k and l in given figure ?

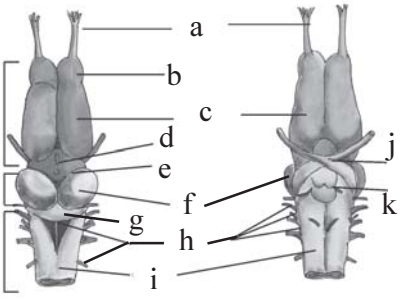
(A) h = Pre caval, i = Cutaneous vein, k = Pelvic vein, l = Musculocuteneous vein

(B) h = Post caval, i = Musculocutaneous vein, k = Innominate vein, l = Pre caval

(C) h = Post caval, i = Innominate vein, k = External jugular vein, l = Pre caval

(D) h = Post caval, i = Pre caval, k = Innominate vein, l = Musculocutaneous vein

(159) What are included in fore brain in given figure ?



(A) b, c, e (B) a, b, c, d

(C) a, b, c, d, e (D) a, b, e

(160) What is indicated by b, c, e and f in given figure ?

(A) b = Olfactory nerve, c = Olfactory lobes, e = Optic lobes, f = Diencephalon

(B) b = Olfactory lobe, c = Cerebral hemisphere, e = Diencephalon, f = Optic lobes

(C) b = Optic lobes, c = Cerebral hemisphere, e = Cerebellum, f = Olfactory lobes

(D) b = Olfactory nerves, c = Optic lobes, e = Olfactory lobes, f = Diencephalon

(161) What is indicated by d, g, j and k in given figure ?

(A) d = Pituitary gland, g = Infundibulum, j = Cerebellum, k = Pineal stalk

(B) d = Pineal stalk, g = Optic lobes, j = Medulla oblongata, k = Pituitary gland

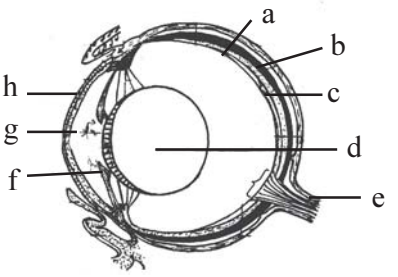
(C) d = Pineal body, g = Pituitary gland, j = Cerebellum, k = Mushroom gland

(D) d = Pineal stalk, g = Cerebellum, j = Optic chiasmata, k = Pituitary gland

(162) Which structure is formed by combining g and i in given figure ?

(A) Fore brain (B) Mid brain (C) Hind brain (D)Vertebral column

(163) What is indicated by a, b, c and h in given figure ?



(A) a = Cornea, b = Sclera, c = Choroid, h = Retina

(B) a = Choroid, b = Sclera, c = Cornea, h = Retina

(C) a = Retina, b = Choroid, c = Sclera, h = Cornea

(D) a = Sclera, b = Choroid, c = Retina, h = Cornea

(164) What is indicated by d, e, f and g in given figure ?

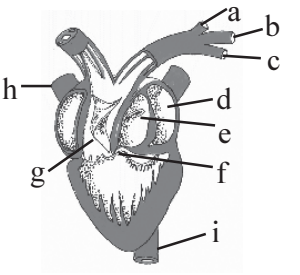
(A) d = Lens, e = Optic nerve, f = Iris, g = Pupil

(B) d = Pupil, e = Optic nerve, f = Iris, g = Lens

(C) d = Eye ball, e = Pupil, f = Lens, g = Musculo-nerve

(D) d = Pupil, e = Iris, f = Optic nerve, g = Lens

(165) Match the column by given figure :



Column - I	Column - II
(p) Systemic arch	(1) i
(q) Dorsal aorta	(2) c
(r) Carotid artery	(3) b
(s) Pre caval	(4) a
(t) Pulmocuteneous artery	(5) h

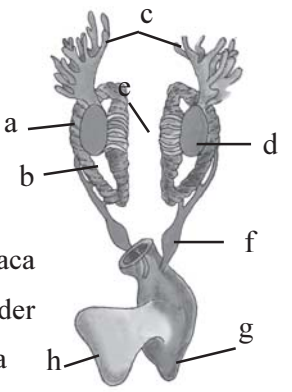
(A) (p - 1), (q - 3), (r - 2), (s - 4), (t - 5)
(B) (p - 3), (q - 1), (r - 4), (s - 5), (t - 2)
(C) (p - 5), (q - 3), (r - 4), (s - 1), (t - 2)
(D) (p - 3), (q - 1), (r - 5), (s - 4), (t - 2)

(166) Match the column :

Column - I	Column - II
(p) Sino-auricular aperture	(i) e
(q) Auriculo ventricular valve	(ii) g
(r) Apperture of pulmonary vein	(iii) f
(s) Pre caval	(iv) h
(t) Aorta	(v) d

(A) (p - iii), (q - ii), (r - i), (s - iv), (t - v)
(B) (p - i), (q - iii), (r - v), (s - iv), (t - ii)
(C) (p - v), (q - iii), (r - i), (s - ii), (t - iv)
(D) (p - ii), (q- iii), (r - iv), (s - v), (t - i)

(167) What is indicated by a, b, c and d in given figure ?



(A) a = Testes, b = Fat body, c = Adrenal gland, d = Kidney
(B) a = Testes, b = Adrenal gland, c = Fat body, d = Kidney
(C) a = Kidney, b = Adrenal gland, c = Fat body, d = Testes
(D) a = Kidney, b = Fat body, c = Adrenal gland, d = Testes

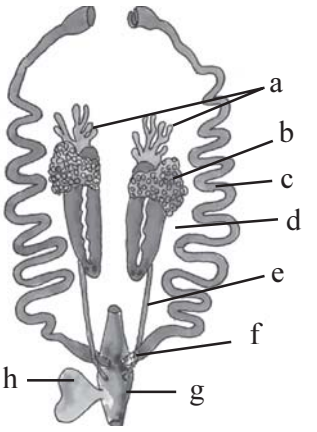
(168) What is indicated by e, f, g and h in given figure ?

(A) e = Urinogenital duct, f = Vasa efferentia, g = Urinary bladder, h = Cloaca
(B) e = Vasa efferentia, f = Urinogenital duct, g = Cloaca, h = Urinary bladder
(C) e = Urinogenital duct, f = Cloaca, g = Urinary bladder, h = Spermatheca
(D) e = Vasa efferentia, f = Urinogenital duct, g = Fat body, h = Urinary bladder

(169) Which is the accessory reproductive organ in given figure ?

(A) a (B) b
(C) c (D) f

(170) What is indicated by b, c, d and e in given figure ?



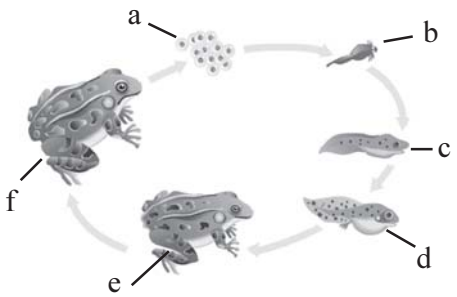
(A) b = Ovary, c = Ureter, d = Kidney, e = Oviduct
(B) b = Kidney, c = Oviduct, d = Ovary, e = Ureter
(C) b = Ovary, c = Oviduct, d = Kidney, e = Ureter
(D) b = Kidney, e = Ureter, d = Ovary, e = Oviduct

(171) What is indicated by f, g and h in given figure ?

(A) f = Ovisac, g = Urinary bladder, h = Cloaca
(B) f = Urinary bladder, g = Cloaca, h = Ovisac
(C) f = Cloaca, g = Ovisac, h = Urinary bladder
(D) f = Ovisac, g = Cloaca, h = Urinary bladder

(172) Match the column in reference to the given figure :

Column - I	Column - II
(i) Internal gill tadpole	(p) a
(ii) Tetrapoda tadpole	(q) c
(iii) Mature frog	(r) b
(iv) External gill tadpole	(s) f
(v) Hind limb tadpole	(t) d
(vi) Egg	(u) e



(A) (i - r), (ii - q), (iii - s), (iv - t), (v - u), (vi - p)

(B) (i - q), (ii - u), (iii - s), (iv - r), (v - t), (vi - p)

(C) (i - t), (ii - r), (iii - s), (iv - u), (v - q), (vi - p)

(D) (i - q), (ii - p), (iii - s), (iv - r), (v - u), (vi - t)

Answers : (142-B), (143-A), (144-C), (145-B), (146-A), (147-B), (148-C), (149-B), (150-D), (151-D), (152-B), (153-A), (154-D), (155-C), (156-D), (157-A), (158-B), (159-A), (160-B), (161-D), (162-C), (163-D), (164-A), (165-B), (166-B), (167-C), (168-B), (169-A), (170-C), (171-D), (172-B)

- **NEET questions :**

(173) Frog is amphibian animal, Because .....

(A) It has no lungs (B) Its tadpole stage is aquatic

(C) It can live on land as well as in water (D) It has no true tail and neck

(174) Which sequence of classification is correct upto genus for *Rana tigrina* ?

(A) Chordata, Vertebrata, Amphibians, Gnathostomata, Rana

(B) Chordata, Vertebrata, Ganthostomata, Amphibians, Anura, Rana

(C) Chordata, Amphibians, Ganthostomata, Vertebrata, Anura, Tigrina

(D) Ganthostomata, Vertebrata, Chordata, Rata, Tigrina

(175) How skin of frog is useful ?

(A) Protection (B) Absorption of water (C) Respiration (D) All the given

(176) Which gland is remain in frog skin ?

(A) Sweat gland (B) Sweat gland and mucous gland

(C) Mucous gland (D) Sebaceous gland

(177) Which animal can respire with skin ?

(A) Cockroach (B) Rat (C) Rabbit (D) Frog

(178) Which characteristic is not of frog ?

(A) Heart is trilobed

(B) Presence of liver and renal portal system

(C) Every fore limb with five fingers which are long

(D) Skin of frog is moist, smooth, slimy and without exoskeleton

(179) Which one is not the function of Gall bladder of frog ?

(A) Destroys micro-organisms (B) Emulsification of fat

(C) Saponification (D) Removes acidity of chyme

Answers : (173-C), (174-B), (175-D), (176-C), (177-D), (178-C), (179-A)

•

18 Transport in Plants

Different types of Transport : Water Relations.

- In plants, the transport of water, mineral, nutrients, organic nutrients and growth, hormones, is important.
- At shorter distances, these components are transported through diffusion and cyclosis (cellular streaming).
- For longer distances, the process of transport is done through vascular tissue system.
- In Xylem the transportation is unidirectional.
- The transport of organic and mineral nutrients is bidirectional.
- The transport of growth hormones is unidirectional.

Types of Transport :

- **Diffusion :** It is only along the concentration gradient, carrier proteins and ATP are not needed. Diffusion rate is affected by, temperature, pressure, gradient of concentration and membrane permeability.
- **Facilitated Diffusion :** Concentration gradient and carrier protein are essential, but metabolic energy (ATP) is not needed.
- **Active Transport :** It is done against the concentration gradient, carrier proteins and ATP are essential.
- **Symport :** When 2 molecules are transported in the same direction.
- **Antiport :** When 2 molecules are transported in the opposite direction.
- **Uniport :** Independant transport of single molecules.
- Transport through semipermeable membrane.
  - From lower concentration to higher concentration.
  - From lower osmotic pressure to higher osmotic pressure.
  - From higher turgor pressure to lower turgor pressure.
  - From higher water potential to lower water potential.

Plant Water Relation :

- Cytoplasm in plants contain 75 % water.
- Herbaceous plants have only 10 - 15 % of its fresh weight as dry matter.
- Hydrophytes have 98 % of water content.
- Xerophytes have 60 % of water content.
- Watermelon has 92 % of water content.
- A mature corn plant absorbs 3 litres of water in a day.
- Mustard plant absorbs water equal to its own weight in 5 hours.

(1) Plants absorb water from soil, through,

(A) Cambial ring (B) Root hair (C) Leaf hair (D) Stem hair

(2) Long distance transport of sap is done by,

(A) Meristematic tissue (B) Cytoplasmic streaming (C) Vascular tissues (D) Diffusion

(3) What is the type of transportation in Xylem ?

(A) Amphibolic (B) Unidirectional (C) Descending (D) Bidirectional

(4) Organic matter synthesized in the leaves is transported to the stem and roots, for the purpose of,

(A) Upward transport (B) Usage (C) Storage (D) (B) & (C) both

(5) How are growth hormones transported in plants ?

(A) Bidirectional (B) Polar (C) Amphibolic (D) Bipolar

(6) What is incorrect for diffusion ?

(A) It is along with the concentration gradient. (B) It is an active transport.

(C) It is passive transport. (D) No energy is utilized.

(7) Facilitated diffusion requires,

(A) Saturation (B) Pressure (C) Energy (D) Gradient

(8) Facilitated Diffusion depends on,

(A) Molecular size (B) Pressure (C) Number of molecules (D) Concentration

(9) Plasma membrane has one of the following as the main component.

(A) Sugar (B) Protein (C) Lipid (D) Complex protein

(10) Which of the following solutes is transported rapidly across the plasma membrane ?

(A) That which is soluble in proteins. (B) That which is soluble both in lipids and proteins.

(C) That which is insoluble in lipids. (D) That which is soluble in lipids.

(11) What are porins made up of ? And what is their function ?

(A) Made up of lipids and passes solutes through the membrane.

(B) Made up of proteins and allows the passage of large sized protein like molecules.

(C) Made up of sugar and checks the passage of micromolecules.

(D) Made up of proteins and checks the passage of large sized protein like molecules.

(12) What helps in the passage of hydrophilic solutes through the plasma membrane ?

(A) Proteins in the membrane (B) Lipids in the membrane

(C) Sugars in the membrane (D) Inhibitors in the membrane

(13) What is helpful in active transport ?

(A) Lipid (B) Inhibitors

(C) Membrane proteins (D) ADP

(14) What is correct for active transport ?

(A) Takes place with the help of membrane proteins.

(B) Energy is not utilised.

(C) It is along the concentration gradient.

(D) None of the above

(15) In active transport what will happen when all the protein transporters are being used.

(A) Transport rates becomes minimum (B) Transport rate is not affected

(C) Transport rate reaches maximum (D) Transport will stop

(16) Carrier proteins in the membrane are sensitive to.....

(A) Catalysts (B) Inhibitors (C) Competitors (D) All of the above

(17) Membrane proteins are not helpful in.....

(A) Active transport (B) Facilitated diffusion (C) Passive transport (D) Simple diffusion

(18) What is correct for carrier proteins ?

(A) They only transport negative ions across the membrane.

(B) They transport all types of ions across the membrane.

(C) They transport only selected solutes across the membrane.

(D) They transport only positive ions across the membranes.

(19) Which part of the plant cell contains 75% of water ?

(A) Mitochondria (B) Nucleus (C) Cytoplasm (D) Nucleolus

(20) In the cytoplasm of the plant cell, what is the percentage of dry matter, excluding water ?

(A) 75 % (B) 25 % (C) 52 % (D) 57 %

(21) Which plant has more than 92% water content ?

(A) Trapa (B) Mustard (C) Maize (D) Water Melon

Answers : (1-B), (2-C), (3-B), (4-D), (5-B), (6-B), (7-D), (8-A), (9-C), (10-D), (11-B), (12-A), (13-C), (14-A), (15-D), (16-B), (17-D), (18-C), (19-C), (20-B), (21-D)

- **Water Potential and Osmosis :**
  - The potential energy of water is called its water potential.
  - It is represented by Greek symbol Psi or ψ.
  - The water potential of any solution is represented by its 3 components.

ψ_s : Solute potential, the amount of water potential.

ψ_p : Pressure potential of water. (as the pressure increases, the water potential increases. Its value is zero in pure water.)

ψ_g : Gravitational component. It depends on water mass. In plant cells, ψ_g is generally ignored. Thus for plants, the equation for water potential is,

$\psi_w = \psi_s + \psi_p$.

- **Osmosis :**
  - When two solutions of unequal concentrations are separated by a semi-permeable membrane, the solvent (water) diffuses from dilute solution, towards concentrated solutions. This process is osmosis.
  - A parchment paper, frog's urinary bladder or an egg membrane from hen's egg are semi permeable membranes.
  - The pressure at which the entry of water across the semi permeable membrane stops is called the osmotic pressure.

(22) The water potential in the leaves of Tradescantia can be measured by

(A) $\psi_w = \psi_s + \psi_g$ (B) $\psi_w = \psi_p + \psi_s + \psi_g$

(C) $\psi_w = \psi_s + \psi_p$ (D) $\psi_w = \psi_p + \psi_g$

(23) Potential energy of water means...

(A) Kinetic energy (B) Water potential (C) Mechanical energy (D) Chemical energy

(24) Which factor does not affect the water potential ?

(A) Pressure (B) Concentration (C) Transpiration (D) Gravitational pull

(25) What is the unit for measuring water potential ?

(A) Density (B) Pascal (C) Meter (D) Hydrostatic pressure

(26) If the ψ_s of solution, in a cell decreases,

(A) Water enters cell. (B) Water goes out of the cell.

(C) Water remains stable. (D) No inward or outward movement of water.

(27) In the chemical context, water potential means...

(A) Physical potential (B) Concentration potential

(C) Biological potential (D) Chemical potential

(28) If the concentration of solutes increases in a cell, then the value of ..... decreases.

(A) ψ_g (B) ψ_s (C) ψ_p (D) ψ_w

(29) Turgor pressure means...

(A) Positive value of ψ_s (B) Negative value of ψ_p (C) Positive value of ψ_p (D) Negative value of ψ_s

(30) With respect to water potential, the component dependant on the bulk of water is,

(A) ψ_g (B) ψ_p (C) ψ_s (D) ψ_w

(31) Water potential of pure water is,

(A) 100 (B) 10 (C) 0 (D) 1

(32) At which hydrostatic pressure does the water potential decreases ?

(A) When the value of ψ_p is positive. (B) When the value of ψ_p is constant.

(C) When the value of ψ_s is positive. (D) When the value of ψ_p is negative.

(33) By which process the root hair of plants absorb water from soil ?

(A) Diffusion (B) Osmosis (C) Imbibition (D) Adsorption

(34) In all living organisms the following acts as a barrier ..... .

(A) Cell wall (B) Nuclear membrane (C) Plasma membrane (D) All the above

(35) Under what circumstances does osmosis stop ?

(A) When the concentration of both the solutes becomes equal.

(B) When concentration of both the solvents becomes equal.

(C) When concentration of both the solutions becomes equal.

(D) None of the above

(36) The pressure at which the entry of water molecules is prevented across a semi-permeable membrane, is

(A) Turgor pressure (B) Root pressure (C) Osmotic pressure (D) Diffusion pressure

Answers : (22-C), (23-B), (24-C), (25-B), (26-A), (27-D), (28-B), (29-C), (30-A), (31-C), (32-D), (33-B), (34-A) (35-B), (36-C)

- **Hypertonic Solution :** If external solution is more concentrated than cytoplasm it is called as hypertonic solution.
- **Hypotonic Solution :** If external solution is less concentrated then it is called as hypotonic solution.
- **Isotonic Solution :** If external solution having equal concentration than that of internal solution is said to be isotonic solution.
- **Plasmolysis :** When living plant cells are kept in salt water solution, water moves out of the cell. The plasmamembrane gets separated from the cell wall and shrinks. Such a cell is called plasmolysed cell and the event is called as plasmolysis.
  - If such a cell is put back in hypotonic solution, there can be a reversal of the above events. This is called as deplasmolysis.
  - This pressure is called turgor pressure and the swollen cell is called its turgidity.

Imbibition :

- Any colloidal system, absorbs a large amount of water from its surroundings. This is imbibition.
- Starch, glucose, proteins and gum do this and swell as a result.
- An imbibition pressure develops due to imbibition. This causes the germination of seeds.
- Energy is released during Imbibition.

Long Distance Transport of Water :

- Water moves through diffusion at shorter distances.
- Long distance transport of water takes place through vascular tissues.

Absorption of Water in Plants :

- Water is absorped by roots in plants. There are two ways of transport in plants :

  (1) Apoplast Pathway (2) Symplast Pathway

(1) **Apoplast Pathway :** The movement of water which occurs only through the intercellular spaces and walls of the cells is called as apoplastic pathway. Such water does not cross any membranous structures. A major part of the water is transported through the apoplastic pathway, by the vascular tissues.

(2) **Symplast Pathway :** When the water moves from one cell to the other through plasmodesmatic, it is called symplast pathway.

Transmembrane Transport : Endodermal cells possess thickenings of casparian strips made up of suberin which is impermeable to water. Water in this layer is pushed forcibly forward through cellular membranes. Such a transport is called transmembrane transport.

Mycorrhiza : In some plants root system has symbiotic association with a fungus. It is known as Mycorrhiza.

- The fungus provides water and minerals to the roots while the roots provide sugar and nitrogen containing compounds to the fungus.
- Pinus seeds cannot germinate and establish without the presense of mycorrhiza.

(37) What will happen if salt is sprinkled on onion which is cut ?

(A) Water comes out of the onion cells. (B) Salt enters the onion cells.

(C) Vacuolar fluid comes out from the onion cells. (D) The cell wall of the onion cells shrinks.

(38) If a square piece of beetroot is cut and put in cold water, the anthocynin pigment does not come out. This indicates the fact that the plasma membrane.

(A) It is dead structure

(B) It is impervious to anthocyanin.

(C) It is pervious to anthocyanin.

(D) The membrane is selectively previous to anthocyanin.

(39) Which phenomenon becomes zero when a cell is fully turgid ?

(A) Turgor pressure (B) Osmotic pressure

(C) Suction pressure (D) Pressure exerted by cell wall

(40) If freshly cut potato chips are kept in salt water.

(A) It becomes turgid (B) It imbibes more starch

(C) It imbibe more sucrose (D) It bends

(41) The turgor pressure of a cell kept in hypotonic solution,

(A) Doesnot undergo a change (B) It decreases

(C) It increases (D) Changes a little bit

(42) No movement of water is seen when a cell is placed in this solution. The solution is

(A) Hypertonic (B) Isotonic (C) Hypotonic (D) Pure water

(43) Which type of solution makes the cell deplasmolysed ?

(A) Isotonic (B) Hypertonic (C) Pure water (D) Hypotonic

(44) Which compounds act like a colloidal system ?

(A) Cellulose (B) Starch (C) Protein (D) All of the above

(45) In the presence of which factors there is a considerable increase in cell size, due to absorption ?

(A) Gaseous form (B) Liquid form (C) Colloidal form (D) All of the above

(46) The force due to imbibition is called...

(A) Osmotic pressure (B) Imbibition pressure (C) Transpiration pull (D) Root pressure

(47) What is the system for long distance transport in plant ?

(A) Ionic channels (B) Transport system (C) Transport paths (D) Transport units

(48) Positive and negative hydrostatic pressure gradients can be obtained by,

(A) Long distance transport (B) Osmosis

(C) Active transport (D) Bulf flow

(49) In which pathway, the water movement doesnot face much barriers.

(A) Apoplastic pathway (B) Symplastic pathway (C) Both (A) and (B) (D) Vacuolar path

(50) Water can cross which of the following membranes in the transmembrane transport ?

(A) Cytoplasmic membrane (B) Tonoplast (vacuolar membrane)

(C) Both (A) and (B) (D) Cell wall

(51) Through which pathway is the least water transported ?

(A) Vacuolar path (B) Apoplastic pathway (C) Symplastic pathway (D) All of the above

Answers : (37-A), (38-B), (39-C), (40-D), (41-C), (42-B), (43-D), (44-D), (45-C), (46-B), (47-B), (48-D), (49-A), (50-C), (51-A)

Ascent of Sap, Transpiration & its types :

- **Ascent of Sap :** The movement of water and minerals dissolved in it, absorbed by the root system of plants, towards the stem and the leaves is called the ascent of sap.

Theories for Ascent of Sap :

Root Pressure Theory : A hydrostatic pressure of water is developed inside roots due to the movement of water from soil into root hairs and from there to cells of xylem, is called root pressure. This pressure pushes the water upwards in the xylem vessels Root pressure is responsible for the ascent of sap in very small plants. However it is not enough to raise water in taller plants.

- Water collected in the form of liquid around special openings of veins near the tips of grass leaves and leaves of many herbaceous plants - these openings are called Hydathodes. Such water loss is called guttation.
- Less temperature at night and lower rate of transpiration results in guttation.
- Root pressure is essential for ascent of sap.

Transpiration Pull Theory :

- The ascent of sap is due to transpiration in leaves which creates a pull. So it is called as the transpiration pull theory.
- Water molecules tend to stick together due to cohesive forces. (Cohesion).
- A strong adhesive force exists between the walls of xylem vessels and water.
- When water moves out of leaves through transpiration as water vapour, the water potential in the cells of the leaves is lowered. As a result, water from the leaf veins (xylem) moves into leaf cells and xylem vessels of the main stem. This is called transpiration pull.

Transpiration : Very little of the total water absorbed from the roots is utilised by the plants. Rest of the absorbed water is lost from the plant in the form of vapour or liquid. This loss of water in the form of vapour is known as transpiration.

There are 3 main types of transpiration :

- **Cuticular Transpiration :** It occurs from the surface of aerial organs of plants.
- **Lenticelar Transpiration :** it occurs through lenticels present on stem of woody plants.
- **Stomatal Transpiration :** It occurs through stoma of the leaves. Maximum transpiration occurs through stomata.
- Each stomata consists of a pore surrounded by two specialised epidermal cells called guard cells.
- The turgidity of guard cells regulate the opening and closing of stomata.
- When guard cells become turgid, the stomata open, when they loose turgidity, they close.
- The process of transpiration is affected by external factors, like humidity, temperature, wind speed and light.
- Plant factors that affect transpiration include number and distribution of stomata, number of stomata, water status in plants, canopy structure, etc.
- Transpiration is important for ascent of sap, absorption of minerals and photosynthesis.

(52) Ascent of sap means,

(A) Transport of water and salts towards the root system and stem.

(B) Transport of food prepared in leaves towards the buds.

(C) Transport of water and mineral salts absorbed by the root system towards the stem and leaves.

(D) Transport of food from the leaves for the roots.

(53) What is correct about ascent of sap ?

(A) Passive transport (B) Utilizes energy

(C) Against the gravitational force (D) Both (B) and (C)

(54) Ascent of sap is the result of,

(A) Root pressure (B) Turgor pressure (C) Diffusion (D) None of the above

(55) Which of the following is involved in the ascent of sap ?

(A) Surface tension (B) Capillary action (C) Cohesive force (D) All the above

(56) How much water is used by plants to maintain its life ?

(A) More (B) Less (C) Very less (D) No water

(57) In which form does the excess water absorbed by the roots is lost ?

(A) Liquid form (B) Solid form (C) Gaseous form (D) Both (A) and (C)

(58) Which organs take part in the process of transpiration ?

(A) Roots (B) Only leaves (C) Stem and leaves (D) Only stem

(59) Which of the following shows the least water potential ?

(A) Root cells (B) Stem cells (C) Leaf cells (D) Cells of cortex

(60) The force present amongst water molecules is

(A) Friction force (B) Cohesive force (C) Adhesive force (D) Gravitational force

(61) In dicots, where are the stomata located in maximum numbers ?

(A) In leaf stalk (B) Dorsal surface of leaf

(C) Leaf margin (D) Ventral surface of leaf

(62) In monocots, which cells are associated with the stomata.

(A) Guard cells and accessory cells (B) Epidermal cells and guard cells

(C) Only guard cells (D) Epidermal cells and accessory cells

(63) The stomata closes under which circumstances ?

(A) When water collects in guard cells. (B) When the turgidity of guard cells decreases.

(C) When water enters guard cells. (D) When guard cells loose turgidity.

(64) What is the shape of the guard cells when they become turgid ?

(A) Crescent shaped (B) Triangular (C) Semicircular (D) Elliptical

(65) As the turgidity of the guard cells increases, the pore of the stomata.

(A) Becomes narrow (B) Becomes broader (C) Gets closed (D) Becomes very narrow

Answers : (52-C), (53-D), (54-A), (55-D), (56-C), (57-D), (58-C), (59-C), (60-B), (61-D), (62-A), (63-D), (64-C), (65-B)

- **Transport of mineral nutrients - Phloem transport :**
  - Plants get CO_2 and O_2 from atmosphere.
  - Other minerals are obtained from soil water.
  - Mineral nutrients are absorbed actively.
  - The carrier proteins persent in the endodermal cells regulate this process.
  - The mineral ions are assimilated into organic compounds.
  - These organic compounds are redistributed to other parts through phloem.
  - Source means the site where the food is synthesized i.e. leaf.
  - Sink means the part that needs or stores food i.e. roots.
  - The direction of the movement of phloem sap in phloem is bidirectional.
  - Mass flow Hypothesis is accepted for the translocation of sugar from source to sink.

(66) Water absorption in roots takes place by which process ?

(A) Active (B) Active absorption

(C) Passive (D) Active and passive

(67) Why can't the mineral nutrients be absorbed passively ?

(A) They can pass from the plasma membrane.

(B) They are in ionic form. Plasma membrane is semi permeable. Their concentration is less in roots.

(C) They are in ionic form Plasma membrane is permeable and their concentration in roots is more.

(D) None of the above

(68) Which cells in the roots behave as regulators ?

(A) Pericycle (B) Cortex (C) Endodermis (D) Epidermis

(69) The mineral nutrients reach the leaves through which path ?

(A) Phloem (B) Symplast pathway (C) Apoplast path (D) Transpiration path

(70) Source means,

(A) Place where food is digested. (B) Where food is synthesized.

(C) Where food is stored. (D) Where food is accepted.

(71) Sink means,

(A) Place where food is required. (B) Place where food is synthesized and stored.

(C) Place where food is stored. (D) None of these

(72) Food is stored in which form in the cells of the roots ?

(A) Protein (B) Sugar (C) Lipids (D) Mineral salts

Answers : (66-C), (67-B), (68-C), (69-D), (70-C), (71-A), (72-B)

A - Assertion, R - Reason.

Answer the following from the options given below :

(A) Both A and R are correct and R is the correct explanation of A.

(B) A and R are both true, But R is not the correct explanation of A.

(C) A is correct, R is wrong.

(D) A is wrong, R is right.

(73) Assertion A : Mass flow hypothesis is responsible for the ascent of sap.

Reason R : The cohesive and adhesive forces play an important role in the ascent of sap.

(A) (B) (C) (D)

(74) Assertion A : Removal of water in liquid form in plants is called as guttation.

Reason R : Hydathodes through which guttation takes place, are present in leaves.

(A) (B) (C) (D)

(75) Assertion A : Some plants have symbiotic association with fungus.

Reason R : The fungus provides N containing compounds and sugar, while the roots provide water and mineral nutrients.

(A) (B) (C) (D)

(76) Assertion A : Facilitated diffusion permits upward and downward movement of solute.

Reason R : Facilitated diffusion, goes against the concentration gradient.

(A) (B) (C) (D)

(77) Assertion A : Water molecules have motion.

Reason R : In gaseous form the motion is random and regular.

(A) (B) (C) (D)

(78) Assertion A : The value of osmotic pressure is dependent on the concentration gradient.

Reason R : When the concentration of the solution is low, the osmotic pressure is high and when solution is concentrated, the osmotic pressure is low.

(A) (B) (C) (D)

(79) Assertion A : For the reversal of plasmolysis, cell must be kept in hypertonic solution.

Reason R : When the cell becomes turgid, this is called turgidity.

(A) (B) (C) (D)

(80) Assertion A : In the cortex, water is mostly transported through apoplastic pathway.

Reason R : Cortical cells are loosely arranged.

(A) (B) (C) (D)

(81) Assertion A : Endodermic cells are impervious to water.

Reason R : They possess thickening of casparian strips made up of suberin.

(A) (B) (C) (D)

(82) Assertion A : There is an adhesive force between the water molecules and the xylem walls.

Reason R : Root pressure is responsible for the ascent of sap.

(A) (B) (C) (D)

(83) Assertion A : Pinus seeds can germinate in the absence of mycorrhiza.

Reason R : Root pressure can be seen in the early morning hours.

(A) (B) (C) (D)

(84) Assertion A : In the leaves of sunflower, the lower epidermis has more transpiration rate.

Reason R : In dicots, the lower epidermis has more number of stomata.

(A) (B) (C) (D)

(85) Assertion A : In the roots, the mineral nutrients are transported passively.

Reason R : Mineral nutrients cannot pass the cell wall.

(A) (B) (C) (D)

Answers : (73-D), (74-A), (75-C), (76-C), (77-B), (78-C), (79-D), (80-A), (81-A), (82-C), (83-D), (84-A), (85-D)

True - False Types of Question :

Of the sentences given below, choose the correct True - False option :

(86) (1) It is because of the cohesive forces between the water molecules that a continuous column of water is formed in xylem.

(2) In plants the cuticular transpiration occurs maximum.

(3) In the symplastic pathway, the water is transported through the cell wall.

(4) The concentration of solutes and the turger pressure affects the process of Osmosis.

(A) T T F F (B) F T T F (C) T F F F (D) T T F T

(87) (1) During monsoon, the doors and windows donot close due to the swelling by imbibition.

(2) For plasmolysis the cells must be placed in hypertonic solution.

(3) In plants, long distance transport of water is executed by the process of diffusion.

(4) The negative value of ψ_p is referred as the turgor pressure.

(A) F T F T (B) T T F F (C) T T T F (D) F T T T

(88) (1) For the ascent of sap, the transpiration pull, cohesive and adhesive forces are responsible.

(2) Osmotic pressure depends on the atmospheric pressure.

(3) The urinary bladder of a frog, is used as a permeable membrane.

(4) The roots provide sugar and N-containing compounds to fungus.

(A) T F F T (B) T F T F (C) F T F T (D) T T F F

(89) (1) The osmotic process depends on the solute concentration of the solution.

(2) In osmosis the molecules move from lower concentration towards higher concentration.

(3) In osmosis selectively permeable membrane takes part.

(4) It is due to osmosis that the soil water enters the root cells.

(A) T F T F (B) T T F T (C) T T F F (D) F T F T

(90) (1) Osmosis is a type of Imbibition.

(2) A semipermeable membrane is involved in the osmotic process.

(3) In the osmotic process, the molecules of solvent move towards the concentrated solution.

(4) Osmosis depends on the concentration of solution and the transpiration pull.

(A) T F T F (B) T F F T (C) F T T F (D) T T F F

(91) (1) Maximum transpiration takes place through cuticular transpiration.

(2) The turgidity of the guard cells decides the opening and closing of stomata.

(3) The underground portions/organs of plants play an important role in transpiration.

(4) The lower surface of the leaves of *Vinca rosea* has more transpiration.

(A) F T F T (B) T F T F (C) F F T T (D) T T F F

(92) (1) Plants obtain carbon and oxygen from soil water.

(2) Transpiration is important for the ascent of sap and absorption of mineral nutrients.

(3) Turgidity is very important for the maintainence of the shape of plant cells.

(4) The mineral nutrients are absorbed by active transport.

(A) T T F T (B) F T T T (C) T F F T (D) F T T F

(93) (1) Porins allow protein sized molecules to pass.

(2) Hydrophilic moieties find it easier to cross the cell-membranes.

(3) The seed germinates due to imbibition pressure.

(4) Root pressure is responsible for the ascent of sap in tall trees.

(A) T F F T (B) T T T F (C) F T F T (D) T F T F

Answers : (86-C), (87-B), (88-A), (89-B), (90-C), (91-A), (92-B), (93-D)

(94) Match Column - I with Column II with the correct option :

Column - I	**Column - II**	
(i) Active transport	(p) Transport of 2 types of molecules in opposite direction.	(A) i - r, ii - s, iii - q, iv - p
(ii) Symplast pathway	(q) Absorbs 3 litres of water.	(B) i - r, ii - p, iii - q, iv - s
(iii) Antiport	(r) Use of metabolic energy.	(C) i - r, ii - s, iii - p, iv - q
(iv) Maize	(s) Transport through plasmodesmata.	(D) i - r, ii - q, iii - p, iv - s

(95) Match Column - I with Column - II and choose the correct option from the answer :

	Column - I	Column - II	
(i)	Cohesive force	(p) Facilitated diffusion	(A) i - r, ii - t, iii - p, iv - q, v - s
(ii)	Imbibition	(q) Bidirectional transport	(B) i - r, ii - t, iii - q, iv - p, v - s
(iii)	Phloem	(r) Ascent of sap	(C) i - r, ii - t, iii - s, iv - p, v - q
(iv)	Membrane protein	(s) Guttation	(D) i - r, ii - t, iii - s, iv - q, v - p,
(v)	hydathodes	(t) Cellulose	

(96) Match Column - I with Column - II and choose the correct answers from the given options :

	Column - I	Column - II	
(i)	More transpiration	(p) 60 %	(A) i - r, ii - p, iii - s, iv - q
(ii)	Xerophytes	(q) Chemical potential	(B) i - r, ii - s, iii - p, iv - q
(iii)	Xylem	(r) Stomata	(C) i - r, ii - p, iii - q, iv - s
(iv)	Water potential	(s) Unidirectional transport	(D) i - p, ii - s, iii - r, iv - q

(97) Match Column - I with Column - II and choose the correct option from the given answers :

	Column - I	Column - II	
(i)	Porins	(p) Transpiration pull	(A) i - r, ii - t, iii - s, iv - q, v - p
(ii)	Transpiration	(q) Guttation	(B) i - r, ii - s, iii - t, iv - q, v - p
(iii)	Sink	(r) Mitochondrial membrane	(C) i - r, ii - s, iii - q, iv - p, v - t
(iv)	Ascent of sap	(s) Turgidity of a cell	(D) i - r, ii - s, iii - t, iv - p, v - q
(v)	Hydathodes	(t) Food storage	

Answers : (94-C), (95-B), (96-A), (97-D)

(98) In the given figure I, what is present in the area marked M in Q.

(A) Concentrated salt solution

(B) Vacuolar sap

(C) Cytoplasm

(D) Dilute salt solution

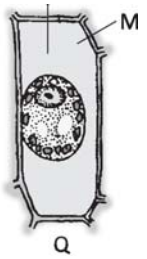


Figure 1

(99) In the given figure I, which is the factor causing the entry of hypertonic solution in the region marked as M.

(A) Plasma membrane (B) Vacuole (C) Cell wall (D) Cytoplasmic fibres

(100) In figure I, which envent does Q indicate.

(A) Incipient plasmolysis (B) Complete plasmolysis

(C) Initial stage of deplasmolysis (D) Intermediate stage of deplasmolysis

(101) In the given figure I, If the cell shown in Q, is kept in a hypotonic solution what is likely to happen ?

(A) Imbibition (B) Plasmolysis (C) Deplasmolysis (D) Exosmosis

(102) What does P indicate in figure 2 ?

(A) Apoplast path (B) Symplast path

(C) Vacuolar path (D) (A) and (C) both

(103) What does Q indicate in figure 2 ?

(A) Vacuolar path (B) Phloem

(C) Symplastic path (D) Apoplastic path

(104) Which process is taking place at 'S' in figure 2 ?

(A) Symplastic path (B) Apoplastic path

(C) Transmembrane transport (D) Vacuolar path

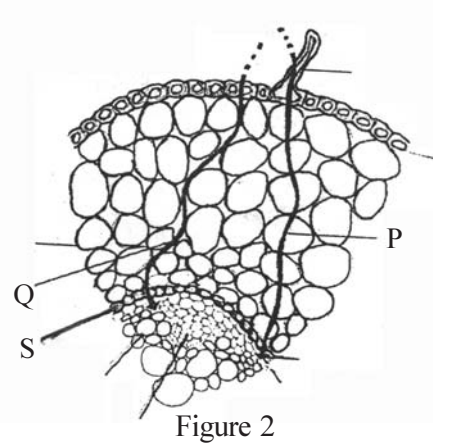


Figure 2

(105) In figure 3, what will happen, if after osmosis, concentrated sugar solution is added in beaker.

(A) Water molecules from the beaker will stop entering the thistle funnel.

(B) Water from beaker enters thistle funnel.

(C) Nothing will occur.

(D) Water will enter the beaker from the thistle funnel.

(106) In the experiment depicted in figure 3, what type of membrane is utilized ?

(A) Permeable

(B) Semipermeable

(C) Selectively permeable

(D) Impermeable

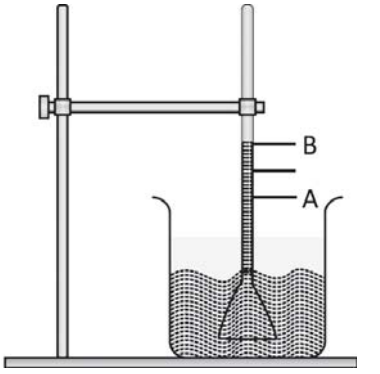


Figure 3

(107) In the given figure 4, what does P indicate ?

(A) Apoplastic path (B) Vacuolar path

(C) Symplastic path (D) Transpiration path

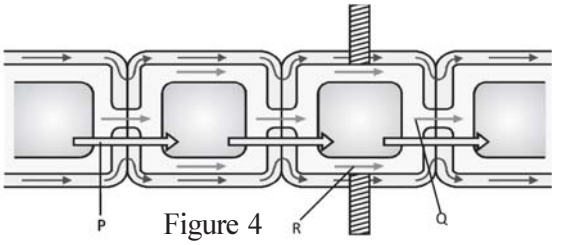


Figure 4

(108) In the given figure 4, of the P, Q and R paths shown, water is transported maximally through which path ?

(A) P (B) Q (C) R (D) None of the above

Answers : (98-A), (99-C), (100-B), (101-C), (102-B), (103-D), (104-C), (105-D), (106-B), (107-B), (108-C)

Points for PMT :

- In imbibition, the forces of capillarity and surface tension are seen.
- Temperature, pressure and concentration gradient are the forces which affect the diffusion rate.

Water Potential :

- **In fully turgid condition :**
- When the cell is fully turgid, water doesn't enter in.
- In a fully turgid condition the ψ_s and ψ_p values are equal and opposite.
- In a fully turgid condition, the OP (Osmotic Pressure) and TP (Turgor Pressure) values are equal.

In General Condition :

- When the cell is not turgid, then TP (Turgor Pressure) = 0, but the ψ_s value is negative.

Diffusion Pressure (DPD) (SP – Suction Pressure) :

- Water Potential Value is equal to DPD, but it is negative in value.
- The DPD indicates the water absorption potential of any solution.
- DPD = (OP - WP) (WP = TP)
- DPD = (OP - TP)

K^+ Ion Theory :

- The solute concentration of the guard cells is due to the presence of the K^+ ions, mainly. These ions are acquired from the neighbouring cells. The guard cells push their H^+ ions outside and in turn, K^+ ions move in (enter).
- This process utilizes energy.
- To maintain the ionic concentration of the guard cells, negatively charged Cl^- ions enter the guard cells.
- The size of the stomatal pore depends on the number of K^+ which enter.
- In the guard cells, due to the hydrolysis of the starch, there is an increase of mallate ion in those guards cells which have open stomatal pores.
- It is due to the splitting of the starch molecule the PEP (Phospho Enol Pyruvic acid) is generated. It is the carboxylation and dehydrogenation of PEP that the mallate ions are formed.
- Plants absorb the capillary water present in the upper layers of the soil by the root hairs.

Questions for NEET :

(109) Transpiration is reduced under which circumstances ?

(A) In humid soil (B) When it is airy/windy

(C) In dry atmosphere (D) In humid atmosphere

(110) An airtight bottle filled with mustard seeds and water, bursts after half an hour. What is it due to ?

(A) Diffusion (B) Imbibition (C) Osmosis (D) DPD

(111) Which water is absorbed by the roots from the soil ?

(A) Gravitational water (B) Capiliary water (C) Bonded water (D) Free water

(112) What causes the movement of water between cells ?

(A) Turgor Pressure (T.P.) (B) Wall Pressure (W.P.)

(C) Diffusion Pressure Deficit (D.P.D.) (D) Plasmolysis

(113) Guttation is mainly due to,

(A) Root pressure (B) Osmosis (C) Transpiration (D) Imbibition

(114) What is created during the water loss which results in stomatal closure ?

(A) Ethylene (B) Abscisic acid (C) Fumaric acid (D) Comairin

(115) What is the difference in the active and passive transport by the plasma-membrane ?

(A) Passive transport is non-selective.

(B) Passive transport follows the concentration gradient, while active transport uses metabolic energy.

(C) Active transport is more rapid.

(D) Passive transport follows anion, while active transport follows cations.

(116) When the water increases near the cell, there is an increase in its turgidity, due to this, the pressure on the cell wall,

(A) Increases (B) Decreases

(C) Increases or decreases (D) No change in pressure

(117) If the plant is covered under water, it dies because ..... .

(A) Mineral nutrients of the soil become dilute.

(B) The roots cannot breathe.

(C) The ascent of sap in the plant cells becomes slow.

(D) The mineral nutrients dissociate / decomposed due to dilution.

(118) In a leaf, during which process the water potential is positive ?

(A) More transpiration (B) Less transpiration (C) More absorption (D) Guttation

(119) What is the water potential and osmotic pressure of pure water ?

(A) 0 and 0 (B) 100 and 100 (C) 0 and 100 (D) 100 and 0

(120) In cells absorbing water, the water potential is,

(A) Always positive (B) Always negative

(C) Always zero (D) Always greater than 1

(121) When water crosses a semipermeable membrane, then which of the following pressures develops ?

(A) O.P (B) S.P (C) T.P (D) W.P

(122) Cell A and Cell B are lying adjacent. The value of DPD is –3 in Cell A and –5 in Cell B. What will be the result ?

(A) No movement of water (B) Both get balanced

(C) Water moves from Cell A to B (D) Water moves from Cell B to A

(123) If there is a damage due to friction or pressure or breakage in the xylem vessels, a continuous column of water is not formed due to,

(A) The presence of lignified thick wall. (B) Cohesive and adhesive forces are opposite.

(C) Weak gravitational force. (D) Transpiration pull is not created.

(124) Guard cells are helpful in,

(A) Transpiration (B) Guttation

(C) Protection against injury (D) Protection against grazing

(125) It is cooler under a tree as compared to a metal shed due to,

(A) Transpiration (B) Guttation (C) Photosynthesis (D) Green leaves

(126) An example of selectively permeable membrane is...

(A) Cell wall (B) Mitochondrial membrane

(C) Chloroplast membrane (D) Plasma membrane

(127) In hypertonic solution, the water potential of the cell is,

(A) Increases (B) Decreases

(C) Initially increases then decreases (D) No change occurs

(128) When we put cell in sugar solution, then which is the correct sequence of processes ?

(A) Endosmosis, Plasmolysis, Mendetory and permanent flaccid

(B) Exosmosis, Plasmolysis, Mendetory and permanent flaccid

(C) Exosmosis, Deplasmolysis, Mendetory and permanent flaccid

(D) Exosmosis, Plasmolysis, Deplasmolysis, Mendetory and permanent flaccid

(129) Osmosis means,

(A) Movement of solute from lower concentration to higher concentration.

(B) Movement of solute from higher concentration to lower concentration.

(C) Movement of solvent from lower to higher concentration.

(D) Movement of solvent from higher to lower concentration.

(130) When a seed is placed in water, imbibition takes place because,

(A) The osmotic pressure inside the seed is less.

(B) The seed coat has lots of salts in it.

(C) The process of absorption takes place.

(D) There is ample amount of vacuolar fluid in the embryonic cells.

(131) Of the following, which pressure becomes zero (0), when the cell is fully turgid ?

(A) Osmotic pressure (B) Turgor pressure

(C) Wall pressure (D) Diffusion pressure deficit

(132) When a small branch is cut, then water oozes out. It indicates that .....

(A) Pressure on the water column. (B) Pull on the water column.

(C) Water column remains continuous. (D) All the above

(133) Of the following, which is the main reason for the increase in stomatal pore size ?

(A) Low temperature at night.

(B) Entry of water molecules in guard cells.

(C) The atmosphere outside the stomatal pores is less humid.

(D) The surrounding guard cells secret salts or salt molecules.

(134) Under which condition the rate of transpiration is maximum ?

(A) When soil is moist and the atmosphere is dry.

(B) When soil is dry and the atmosphere is humid.

(C) Soil is wet and atmosphere is dry.

(D) When soil is dry and atmosphere is dry.

(135) Which force is important in the stomatal opening process ?

(A) Shape of the Guard cell (B) The chlorophyll content of the cell

(C) The amount of hormones in the cell (D) The amount of proteins in the cell

(136) Why do leaves become flaccid in hot atmosphere ?

(A) Too much transpiration (B) More transpiration as compared to absorption

(C) More water absorption by roots (D) Decrease in water absorption

(137) Phloem transport is usually in which direction ?

(A) From source to sink (B) From sink to source

(C) From leaves to xylem and then phloem (D) From leaves to roots

(138) In a solution, the water potential is,

(A) Zero (B) Negative (C) Positive (D) Positive or Negative

(139) Which is the main layer which blocks the entry of a substance into a cell ?

(A) Cell's outermost boundary - the cell wall. (B) Outermost layer of the plasma membrane.

(C) Innermost layer of the plasma membrane. (D) Innermost layer of the cell wall.

(140) What provides the maximum blockage to the process of transpiration ?

(A) Leaf stomata (B) Cuticles

(C) Pectin (D) All provide equal resistance

(141) The guard cells are different from epidermal cells because they possess ..... .

(A) Vacuole (B) Mitochondria (C) Chloroplast (D) Cell wall

(142) Osmotic pressure is expressed as.

(A) ψ_p (B) ψ_s (C) ψ_w (D) ψ_g

(143) In the thistle funnel experiment, the entry of water in the thistle' funnel is blocked due to,

(A) Increase in the hydrostatic pressure in the thistle funnel.

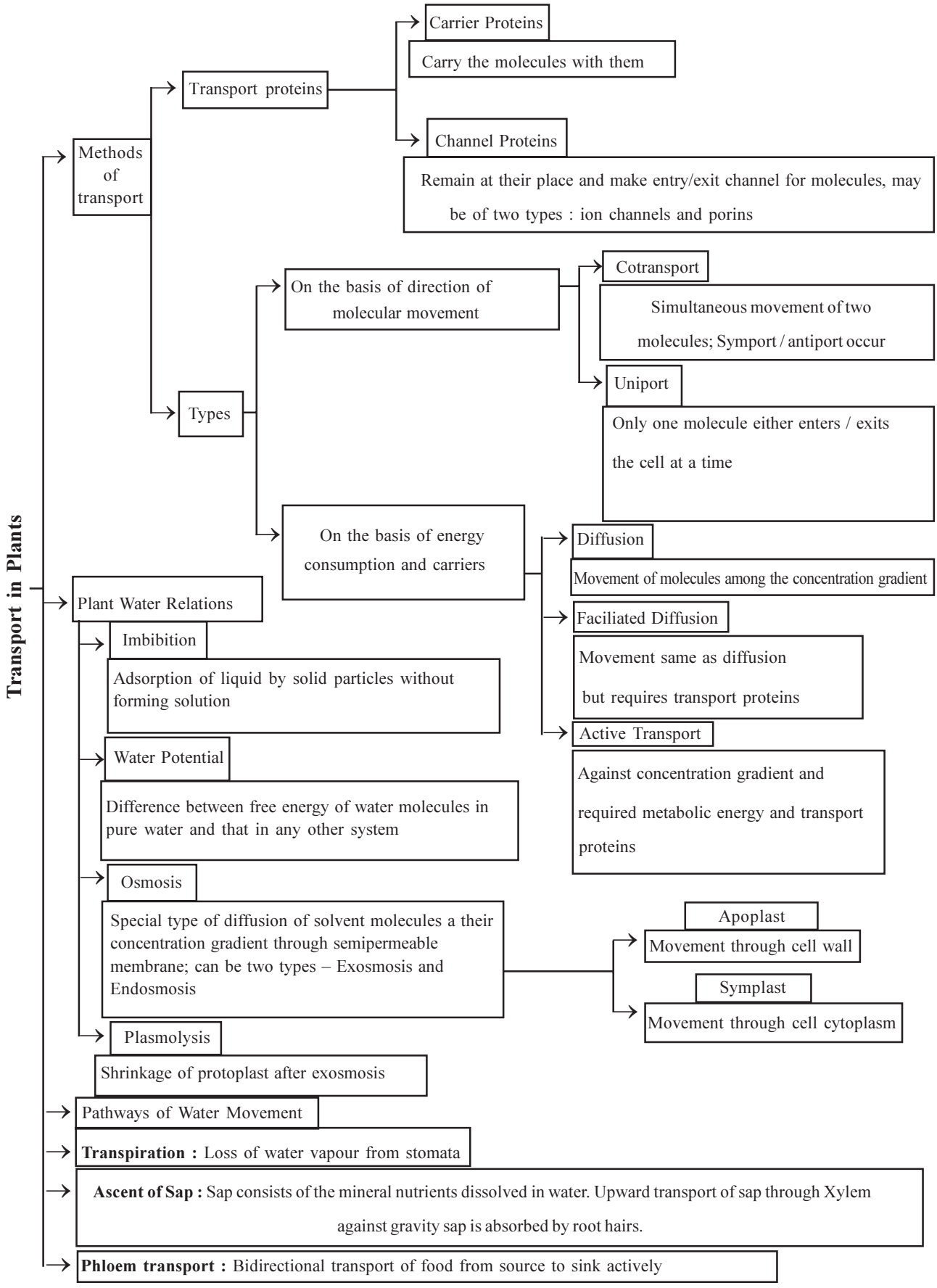
(B) Diffusion of sugar out of the thistle funnel.

(C) The solutions inside and outside become isotonic.

(D) Hydrostatic pressure develops in the beaker.

Answers : (109-D), (110-B), (111-B), (112-C), (113-A), (114-B), (115-B), (116-A), (117-B), (118-D), (119-A), (120-B), (121-A), (122-C), (123-D), (124-A), (125-A), (126-D), (127-B), (128-B), (129-D), (130-C), (131-D), (132-B), (133-B), (134-A), (135-A), (136-B), (137-A), (138-B), (139-B), (140-B), (141-C), (142-B), (143-A)

Transportation in Plants



Hydroponics :

- The absorption, distribution and metabolism of various mineral elements by plants is called **mineral nutrition.**

Importance :

- Important for physical activity of plants.
- Plants derive their inorganic nutrients from soil, water and atmosphere.
- Plants require various essential elements for their survival, growth, proper development and reproduction. These mineral elements occur mainly in their inorganic ionic forms in the soil. Plants absorb them from the soil through their root system.
- If the minerals are not available to plants, specific symptoms appear due to the deficiency of a particular element.

Methods of study :

- Methods to study the requirement of mineral in plants is as under :

Hydroponics :

- To understand the role of an individual mineral element and to understand the effects of its deficiency or absence, 'Hydroponic' method of growing plants is employed.
- The technique of growing plants only in a nutrient medium and in complete absence of soil is known as Hydroponics.
- Demonstrated By : A German Botanis Julius Von Sachs (1980)
- Hydroponics is soil less cultivation of plants. It is a method of growing plants using mineral nutrient solution instead of soil.
- Essential nutrients are introduced to the growth of plants artificially in a water medium.
- A large amount of such 'nutrient solution' is utilized.
- Essential mineral elements are dissolved in their determined amounts.
- Their concentrations and the pH of the medium are periodically checked and maintained at their proper levels.

Types of Hydroponics :

- There are two types of Hydroponic :

(a) Solution culture :

- In solution culture of hydroponics, just the nutrient solution of essential plant nutrients is used for rousing plants.
- There are three types of solution culture.

(I) Static solution culture :

- This is a method of hydroponics, used for raising plants and seedlings in solution filled containers such as glass jars, buckets, tub and water tanks.

(II) Continuous flow solution culture :

- This is a method of hydroponics where continuous flow of nutrient filled solution is automated by using a nutrient film technique-NFT.

(III) Aeroponics :

- This is a method of hydroponics where plants or seelings are raised in an environment saturated with fine drops (aerosol) of nutrient solution.

Advantages of Hydroponics

- Less use of plant nutrients
- Less use of water
- Less utilization of energy and space
- Elimination of soil-borne diseases
- Complete elimination of weeds
- Balanced plant nutrition

---

(1) Why is there a need to add oxygen gas continuously in Hydroponic method ?

(A) Because it is an essential mineral elements (B) Because it is an inessential mineral elements

(C) For the growth of roots of plants (D) For the growth and development

(2) Which scientist coined the term 'essential mineral' ?

(A) Arnon and Stout (B) Khurrana and Mathai (C) Schleiden and Schwan (D) Dixon and Jolly

(3) Continuous flow solution culture : ..... :: Aeroponics : use nutrient solution fine drops (aerosol)

(A) NET (B) NFT (C) NCT (D) NEFT

(4) Hydroponic method means growing plants in ..... .

(A) Water (B) Soil

(C) Mineral nutrient solution (D) in the medium of Tissue culture

(5) Characteristic of essential mineral elements ..... .

(A) in the ash of vegetation (B) Minerals are present in soil

(C) it is essential for growth and development (D) improves the growth of vegetation

(6) Where are the elements, absorbed by the roots present in ..... .

(A) Xylem (B) Phloem (C) Sieve tubes (D) Combium

(7) Hydroponic plant growing method means ..... .

(A) growth of plant in water (B) growth as floating aquatic plants

(C) growth as an aquatic plants (D) growth of plant without soil

(8) From the given options which of the following is not present as nutrition in plants ?

(A) Water (B) Minerals (C) CO_2 (D) N_2

(9) Which minerals are mostly present in the dry mass of plants ...

(A) N, P, K (B) Ca, Mg, S (C) C, N, H (D) C, H, O

(10) What is Hydroponic solvent ?

(A) Liquid (B) Water

(C) Soil culture (D) Saturated solution with contained minerals

(11) Which scientist observed Hydroponic method and in which year ?

(A) Arnon and stout, 1939 (B) Julious wan sach, 1980

(C) Arnon and stout, 1950 (D) Julious wan rech, 1939

(12) What should be added continuously in Hydroponic method ?

(A) O_2 (B) Minerals (C) pH (D) A, B, C

(13) Full form of NFT ...

(A) Nutrient Fine Technique (B) Nutrient Film Technology

(C) Nutrient Film Technique (D) Nutrient Fine Technology

(14) What is used in Medium culture ?

(A) Water tank (B) NFT

(C) agar agar (D) Fine drops of nutrient solution

(15) Without what plants can not complete their life cycle ?

(A) essential land (B) essential solution (C) essential minerals (D) essential gasses

Answers : (1-D), (2-A), (3-B), (4-C), (5-C), (6-A), (7-D), (8-C), (9-A), (10-D), (11-B), (12-A), (13-C), (14-C), (15-C)

Criteria for essentiality of elements :

- Botanist Arnon and Stout (1939) represented criteria for these.
- A plant must be unable to complete its life cycle in the absence of the mineral element.
- The function of the element must not be replaceable by another mineral element.
- The element must be directly involved in plant metabolism.
- The mineral nutrients are divided into two groups.

(A) Macronutrients : Source, importance and Deficiency Symptoms

- The concentration of macronutrients in the dry mass of plants is about 1 to 10 mg per 1 gram.
- **Primary Macronutrients :** Phosphorus (P), Nitrogen (N), Potassium (K)
- **Secondary Macronutrients :** Calcium (Ca), Magnesium (Mg), Sulphur (S)
- **Non mineral Macronutrients :** Carbon (C), Oxygen (O), Hydrogen (H)

(B) Micronutrients : Source, important and Deficiency symptoms

- The concentration of micronutrients is 0.1 mg or less than that micronutrients are also called - Trace elements.
- Micronutrients are those elements which are needed in very small (micro) quantities for plant growth.
- Example : Manganese (Mn), Nickel (Ni), Sodium (Na), Cobalt (Co), Silicon (Si), Vanadium (V), Boron (B), Copper (Cu), Iron (Fe), Chlorine (Cl), Molybdenum (Mo), Zinc (Zn).

Essential plant nutrients : their source, relative amounts, functions and classificaiton :

Element	Chemical Symbol	Source	Absorbed Form	Major functions	Relative % in plant
Macronutrients					
Non mineral elements					
Carbon	(C)	Atmosphere	CO_2	In all organic molecules	–
Oxygen	(O)	Atmosphere	O_2	In most of organic molecules	–
Hydrogen	(H)	Soil	H_2O	In most of organic molecules	–
Nitrogen	(N)	Soil	NH^+_4,NO^-_2 and NO^-_3	In protens, nucleic acids etc.	100
Mineral Nutrients					
Phosphorus	(P)	Soil	$H_2\ PO^-_4$	In nucleic acids, ATP, Phospholipids etc.	6
Potassium	(K)	Soil	K^+	Enzyme, activation, water balance, ion balance	25
Sulphur	(S)	Soil	SO_4^{2-}	In structure of coenzymes ionic balances	3
Calcium	(Ca)	Soil	Ca^{2+}	Affects the cytoskeleton, membranes and many enzymes, second messenger	12.5
Magnesium	(Mg)	Soil	Mg^{2+}	In structure of chlorophyll and many enzymes, stabilizes ribosomes	8
Micronutrients					
Iron	(Fe)	Soil	Fe^{3+}	In active site of many redox enzymes and electron carriers, In chlorophyll synthesis	0.2
Chlorine	(CI)	Soil	CI^-	In photosynthesis, ionbalance	0.3
Manganese	(Mn)	Soil	Mn^{2+}	Activation of many enzymes	0.1
Boron	(B)	Soil	$H_2BO^-_3$, $H_2BO_3^{-2}$	Carbohydrate transport, Cell wall component	0.2
Zinc	(Zn)	Soil	Zn^{2+}	Enzyme activation, auxin synthesis	0.03
Copper	(Cu)	Soil	Cu^{2+}	In active site of many redox enzyme and electron carriers	0.01
Molybdenum	(Mo)	Soil	MoO^{3-}_4	Various process of nitrogen fixation	0.0001
Nickel	(Ni)	Soil	Ni^{2+}	Required for iron absorption	
Sodium	(Na)	Soil	Na^+	Involved in osmotic (water movement) and ionic balance in plants	
Cobalt	(Co)	Soil	CO^{2+}	Required for nitrogen fixation in xeguminous plants	
Silicon	(Si)	Soil	Si^{4+}	As a component of cell walls	
Vanadium	(V)	Soil	$V^{3+,4+,5+}$	Necessary for the activation of nitrogenase in the nitrogen fixing bacteria	

(16) Cause yellowing of leaves means ..... .

(A) Tylosis (B) Nephrosis (C) Florosis (D) Chlorosis

(17) Deficiency kills terminal buds leaving a rosetic effect on the plant ...

(A) Cobalt (B) Boron (C) Calcium (D) Phosphorous

(18) Which of the following group is secondary nutrient ?

(A) N, P, K (B) Ca, Mg, S (C) N, P, Ca (D) B, Cu, Fe, CI

(19) Which mineral is necessary in meristematic tissues and in differentiating tissues ?

(A) P (B) K (C) Ca (D) S

(20) Which mineral is essential for the activity of enzymes involved in photosynthesis and respiration ?

(A) Mg (B) N (C) Cu (D) A and B both

(21) Which substance (minerals) is necessary for process of $H_2O \rightarrow H^+$ and OH^- ?

(A) Fe (B) Mn

(C) Zn (D) None above

(22) Which mineral is essential for synthesis of growth hormone ?

(A) Fe (B) Mn (C) Zn (D) B

(23) Which mineral is essential for thick brown spoted fruits ?

(A) Cu (B) B (C) Ca (D) Mo

(24) What is the quantity of magnesium present per gram in by mass of plant for the deficiancy symptoms observe ?

(A) 10 mg (B) 10^{-1} g (C) 1 g (D) 10^{-4} g

(25) Due to deficiency of which mineral there is a reduction in fruit-size ?

(A) Chlorine (B) Boron (C) Molybdenum (D) Sulphur

(26) Which mineral is not absorbed by the plant in the form of soil solution ?

(A) Mn (B) C (C) Zn (D) CI

(27) Which process is important for Zinc ?

(A) activity of carboxylase type of enzymes (B) Translocation of sugars

(C) Cell differentiating (D) maintenance of ribosomal constitution

(28) Which mineral is necessary for the activity of carboxylase type of enzymes and synthesis of auxin ?

(A) Manganese (B) Zinc (C) Boron (D) Molybdenum

(29) Which mineral deficiency is removal of dominance of apical buds as a result lateral buds develop ?

(A) Phosphorous (B) Sulphar (C) Copper (D) Potassium

(30) Which mineral is present in constitution of vitamin Thiamine ?

(A) Iron (B) Zinc (C) Manganese (D) Sulphar

(31) Which mineral is required for absorption and utilization of calcium ?

(A) Cu (B) S (C) B (D) Fe

(32) Which of the following reaction occurs in presence of phosphorous ?

(A) Cellular differentiation (B) Photosynthesis

(C) Regulation of stomatal pore size (D) Phosphorylation

(33) Which minerals is necessary for constitution and synthesis of nuclic acid ?

(A) Magnesium, Phosphorous (B) Mangese, Phosphorous

(C) Phosphorous, Magnesium (D) Manganese

(34) Which mineral is responsible for bark of tree become rough and get split ?

(A) Copper (B) Zinc (C) Manganese (D) Chlorine

(35) Show the importance of iron element ?

(A) induce photolysis of water during photosynthesis

(B) in the constitution of cytochromes and ferredoxin

(C) essential for germination of pollengrain (D) Synthesis of O_2

(36) The transportation of minerals through Xylem happens ..... with water.

(A) Transpiration (B) Evaporation (C) Guttation (D) Asent of sap

(37) Which of the following is important of sulphar ?

(A) for cambial activity (B) for synthesis of O_2

(C) for constitution of Vitamin B_1 (D) for subcellular skeletone

(38) Which mineral defficiency is responsible for the red and purple spot on leaves ?

(A) Phosphorous, Magnesium (B) Phosphorous, Manganese

(C) Sulphur, Magnesium (D) Sulphur, Manganese

(39) In which form the Molybdenum is absorbed ?

(A) MoO_3^{-3} (B) MoO_3^{-4} (C) MoO_4^{-3} (D) MoO_4^{-4}

(40) In which form the Boron is absorbed from the soil ?

(A) $H_2B_2O_3^-$ (B) $H_2BO_3^+$ (C) $H_2BO_3^{-2}$ (D) HBO_3^{-2}

(41) Which of the following subsieuant mineral is present in constitution and element of cell wall ?

(A) Boron, Silicon (B) Silicon, Boron (C) Calcium, Boron (D) Silicon, Calcium

(42) Which mineral defficiency is responsible for shortening of internodes and reduction in cambial activity ?

(A) N (B) Cu (C) K (D) Fe

(43) Which group is the micronutrients ?

(A) B, Cu, Fe, CI, Mn (B) B, Cu, S, Ca, Mn (C) Mn, Cu, Ca, Zn (D) Zn, Cu, B, N, P

(44) Function of potassium ....

(A) Photosynthesis

(B) increase enzyme activity which helps function of vegetation (plants)

(C) impart red pigment in fruits

(D) produce cambium in vascular bundle

(45) Essential characteristic of macronutrients ?

(A) made during photosynthesis (B) used by growth regulators

(C) absorbed from the soil (D) produced by enzyme

(46) Macronutrient which is essential for synthesis of organic substance but not obtain from soil.

(A) Nitrogen (B) Phosphorus (C) Manganese (D) Carbon

(47) What should be added for premature yellowness of leaves ?

(A) Fe and Mg

(B) give water to crop

(C) Cytokinin and little Nitrogen base fertilizer

(D) Spray 2, 4, 5 - T = (2, 4, 5 - Trichlorophenoxy acetic acid) on green leaves

(48) Between whom the continuous ion exchange take place ?

(A) Root and between the soil particles (B) Root and soil solution

(C) Cell and Cortex (D) Given all

(49) Less quantity of micronutrient is important because ..... .

(A) it is poisonous for vegetation

(B) it is removed by root in plant

(C) physiologist could not find it in plant

(D) it absorb less as directed

(50) In which form the inorganic elements are seen in plants ?

(A) make very dilute solution (B) concentrated solution

(C) very high concentrated solution (D) make dilute solution

(51) What is Trace elements ?

(A) Isotops by Gayger counting method

(B) it is needed in very less quantity

(C) elements which are not present in protoplasm

(D) which is found first in protoplasm

(52) Identify the common defficiency effect of P, K, Ca and Mg in plant ?

(A) proportion of anthocyanine (B) feel curve of leaftip

(C) week development of vascularbundle (D) few dead zone of sight zone

(53) Which elements is present in the middle lamella of plant cell ?

(A) Mg (B) Ca (C) K (D) Fe

(54) Which mineral defficiency is responsible for death of stem tip and root tip ?

(A) Calcium (B) Nitrogen (C) Carbon (D) Phosphorous

(55) What is seen in plant tissue by the defficiency of Mg ?

(A) Hydrolysis (B) Nephrosis (C) Chlorosis (D) Stuntation

(56) Which mineral is present in the constitution of chlorophyll ?

(A) Mg (B) K (C) Mn (D) Fe

(57) Which mineral is removed during the combustion of chlorophyll ?

(A) Na (B) Mg (C) Mn (D) Fe

(58) What will happen by the iron defficiency ?

(A) Decrease protein synthesis

(B) Intervein of young leaf first feel chlorosis

(C) Indicates stemtip curve

(D) Process of less leaf and stunt growth

(59) Which two metalic element is needed for the synthesis of chlorophyll ?

(A) Fe and Mg (B) Cu and Ca

(C) Cu and K (D) Mg and Ca

(60) Which independant mineral is necessary for energy in metabolic reaction ?

(A) P (B) S (C) Ca (D) Na

(61) From the given mineral below which is not the constituent of any enzyme but is necessary for the activeness of enzyme ?

(A) Zn (B) Mg (C) K (D) Mn

(62) Main function of potassium ..... .

(A) it controls many reaction of plant

(B) useful to produce the red pigment in fruit

(C) it helps in photosynthesis

(D) it activates enzyme and therefore plant indicate reaction

(63) Which of the following trace element is useful in plant growth and its radio isotopes which useful in the treatment of cancer ?

(A) Cobalt (B) Calcium (C) Sodium (D) Iron

(64) What happens by the defficiency of Molybdenum ?

(A) it causes leaftip curve. (B) chlorosis in leaves

(C) drying and wilting of leaves (D) weak development of vascular tissue system

(65) Which of the following mineral is responsible for translocation of sugar in plant ?

(A) Boron (B) Molybdenum (C) Manganese (D) Iron

(66) Which micro nutrient is absorbed by foliaceous stipule ?

(A) Zinc (B) Iron (C) Phosphorous (D) None above

(67) Which mineral is usefull for synthesis of IAA ?

(A) Iron (B) Zinc (C) Calcium (D) Nitrogen

(68) Which mineral defficiency causes grayish spot on malt ?

(A) Cu (B) Zn (C) Mn (D) Fe

(69) How many minerals are necessary for healthy growth and development of plant ?

(A) 20 (B) 10 (C) 26 (D) 50

(70) Which mineral is present in sea weed ?

(A) Iodine (B) Iron (C) Chlorine (D) Boron

(71) Which mineral is responsible for death of apical bud and poor growth of plant ?

(A) Mg (B) Mo (C) Zn (D) B

(72) For which process the Manganese is useful in plants ?

(A) Photolysis in Photosynthesis (B) Synthesis of chlorophyl

(C) Synthesis of Nucleolus (D) Production of plant cell wall

Answers : (16-D), (17-B), (18-B), (19-C), (20-A), (21-B), (22-C), (23-B), (24-D), (25-B), (26-B), (27-A), (28-B), (29-D), (30-D), (31-C), (32-D), (33-C), (34-A), (35-B), (36-D), (37-C), (38-A), (39-C), (40-C), (41-C), (42-C), (43-A), (44-B), (45-C), (46-D), (47-A), (48-A), (49-A), (50-A), (51-B), (52-D), (53-B), (54-A), (55-C), (56-A), (57-B), (58-B), (59-A), (60-A), (61-C), (62-D), (63-A), (64-C), (65-A), (66-B), (67-B), (68-C), (69-A), (70-A), (71-D), (72-A)

Absorption of Mineral Nutrients :

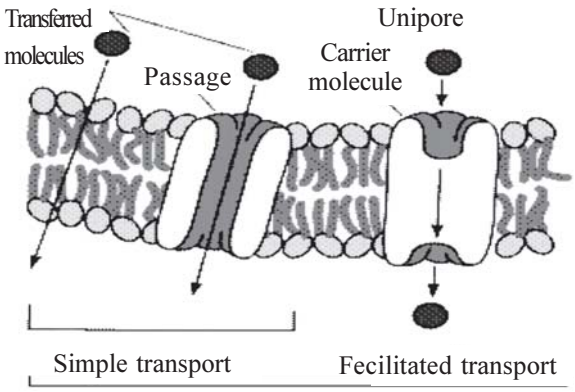
- The process of absorption of mineral nutrients from the soil.
- The nutrients enter the cell walls of root on the outer side of plasma membrane and they also move into the intercellular spaces.
- This process is relatively more rapid and no energy (ATP) is expended in it.
- Then from this extra cellular region, the mineral nutrients enter the cellular region. The cellular region means within the plasma membrane and the region within the vacuole. Plant absorb a large number of minerals from soil.

(A) Simple Absorption : (Passive Absorption)

- Absorption is the initial and rapid phase.
- This type of absorption does not require use of energy (ATP).
- Absorption occurs according to normal physical principles.

(I) Diffusion :

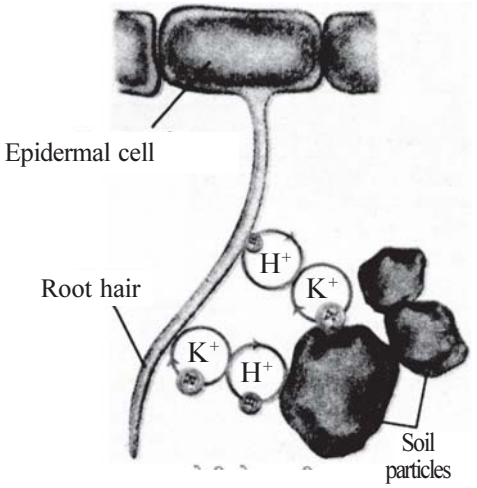
- A substance moves from where it is in a higher concentration to where it is in a lower concentration.
- Thus, ions are absorbed following their concentration gradient.
- For ionic absorption various 'ionic-channels' are located in the plasma membrane.
- Special kinds of proteins which extend along the thickness of the plasma membrane provide pores which act as such 'ion-channels'.



Passive transport

(II) Ion Exchange :

- Both, anions and cations are located on the surface of the cell wall through their absorption.
- The soil solution also contains ions. An ion exchange occurs between them.
- Such an ionic-exchange occurs even against their concentration - gradient.



Ion Exchange

(III) Donnan Equilibrium :

$[Co^+]\ [Ao^-] = [Ci^+]\ [Ai^-]$ જ્યાં, Co^+ = Outer cations

Ao^- = Outer anions

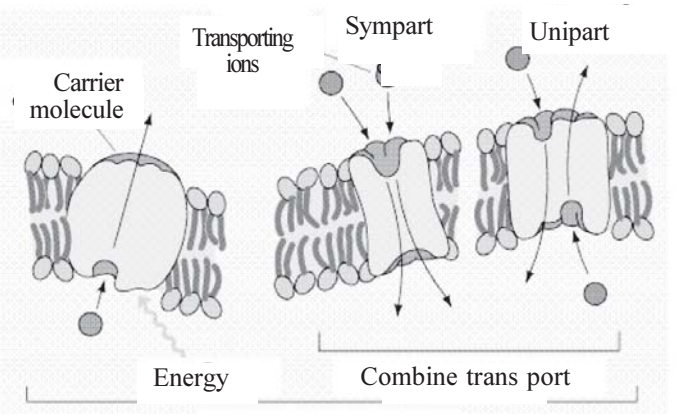
Ci^+ = Inner cutions

Ai^- = Inner anions

- Certain ions are incapable of diffusion against their concentration-gradient.
- The accumulation of such stable and non diffusible ions is explained through this principle.
- It occurs against concentration-gradient.
- Plasma membrane possesses selective permeability it permits exchange of some ions and does not permit that of some others.
- As a result, there occurs an increase in the concentration of positive ions on the inner surface of plasma membrane.
- Due to this the negatively charged ions in the soil solution, become accumulated on the outer surface of Plasma membrane.

(B) Active Absorption :

- This type of absorption needs the expenditure of metabolic energy from the cell in the form of ATP.
- The ions are taken in slowly into the 'inner space'.
- Such absorption of ions occurs against their concentration ingradient. Energy is consumed in it. Based on a variety of evidences.



Active transport

- It seems that specials 'carrier molecules' are involved in this process.
- This carrier molecules are special kind of proteins.
- They combine with ions on the outer side of plasma membrane and form 'ion-carrier complex'.
- This complex migrates across the membrane towards the inside, where the complex releases the ion.

---

(73) Following which way is correct for absorption and transportation or minerals ?

(A) Epiblema → Cortex → Endodermis → Pericycle → Protoxylem → Metaxylem

(B) Epiblema → Endodermis → Cortex → Pericycle → Metaxylem

(C) Epiblema → Cortex → Pericycle → Endodermis → Protoxylem → Metaxylem

(D) Epiblema → Cortex → Endodermis → Pericycle → Metaxylem → Protoxylem

(74) Ionchannels are seen in which types of transportation ?

(A) Diffusion (B) Ion exchange

(C) Donnan equllibrium (D) Principle of mass flow

(75) By which principle the accumulation of such stable and non diffusible ions can be explained ?

(A) Diffusion (B) Ion exchange

(C) Donnan equllibrium (D) Principle of mass flow

(76) If inner surface of plasmamembrane becomes negatively, charged which element is absorbed by the soil solution ?

(A) CI^- (B) $H_2PO_4^-$ (C) MoO_4^- (D) K^+

(77) Donnan equillibrium happens on which surface ?

(A) Cell wall (B) Nuclear membrane (C) Plasma membrane (D) Tonoplast

(78) Simple absorption of elements happens by which process ?

(A) Diffusion (B) Mass flow (C) Donnan equillibrium (D) All above

(79) Ion channels are made up of which element ?

(A) Carbohydrate (B) Lipid (C) Protein (D) Nucleic acid

(80) What is the true equation of Donnan equillibrium ?

(A) $[Ci^+]\ [Ai^-] = [Co^+]\ [Ao^-]$ (B) $[Ci^+]\ [Ai^+] = [Co^+]\ [Ao^+]$

(C) $[Ci^+]\ [Ai^-] = [Co^-]\ [Ao^-]$ (D) $[Co^+]\ [Ci^-] = [Ao^+]\ [Ai^-]$

(81) More suction pressure as ..... .

(A) Less water absorption and more ion absorption (B) More water absorption and less ion absorption

(C) More water absorption and more ion absorption (D) More water absorption and no absorption of ion

(82) Write the correct sentence for ion exchange.

(A) Both the ions are located on the surface of cell wall throught their absorption.

(B) Absorption of ion happens towards their concentration gradient

(C) Ions are absorbed along with the absorption of water in large amount.

(D) This type of absorption needs the expenditure of metabolic energy.

(83) Through which minerals absorb by the root transport to the leaves ?

(A) Xylem (B) Phloem (C) Sieve tube (D) Companion cell

(84) Transportation of minerals in plant root by diffusion is called ...

(A) Exosmosis (B) Osmosis (C) Active absorption (D) Passive absorption

(85) What is indicated in plant ash ?

(A) Minerals which of absorbed by plants.

(B) It shows the organic compounds of the plants.

(C) It shows the organic and inorganic compounds of the plants.

(D) None above.

(87) How does the passive absorption of minerals takes place ?

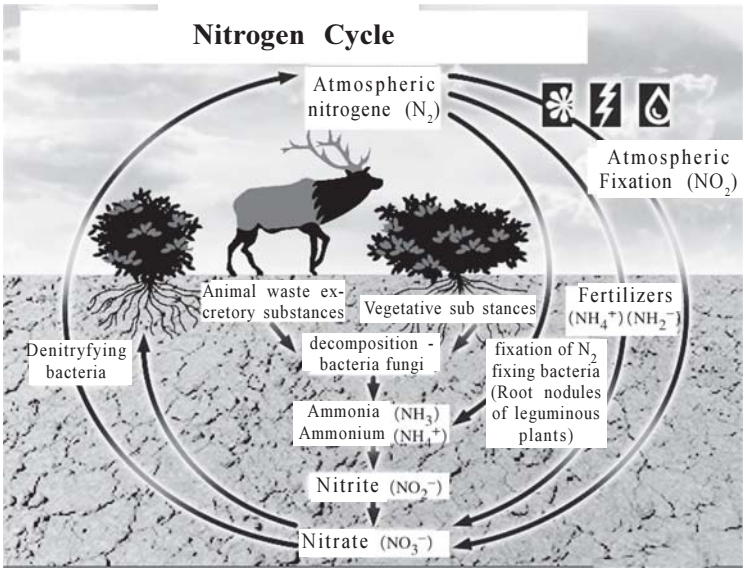
(A) Diffusion and ionexchange (B) Osmosis

(C) Transpiration (D) From the following

Answers : (73-A), (74-A), (75-C), (76-D), (77-C), (78-D), (79-C), (80-A), (81-C), (82-A), (83-A), (84-D), (85-C), (87-A),

Nitrogen Metabolism :

- C, H, O and N are amongst the most important essential elements.
- In living organisms except C, H, and O, N occurs in the largest amount.
- It occurs in the constitution of important protoplasmic substances like amino acids, proteins, nucleotides, vitamins and hormones.
- We also know that the atmosphere contains large amount of N_2 and that it cannot be directly utilized by any organism.
- Plants absorb salts of nitrogen like NH_4^+, NO_2^- and NO_3^- from the soil and incorporate them into their organic constituents.
- Animals obtain these constituents when they consume these plants as food.
- When plants and animals die, their dead bodies are decomposed.
- During such decomposition which is caused by bacteria, NH_3 is released.
- This process is called ammonification.
- This NH_3 is immediately converted to NO_2^- and NO_3^- by a process called – nitrification.
- Specific bacteria are responsible for this. This salts can be absorbed by plants.
- Some bacteria release N_2 into atmosphere by decomposing nitrates.



The Nitrogen Cycle

Nitrogen Cycle :

- Nitrogen cycle is an example of gaseous type of biogeochemical cycle.
- Nitrogen is an essential element of protoplasm, proteins and genetically important nucleic acids such as DNA.
- It is also a major constituent (about 78 %) of the atmosphere.
- Most green plants need nitrogen in the form of nitrate ions (NO_3^-) and ammonium ions (NH_4^+).
- Special N_2^- fixing bacteria (Rhizobium) present in the roots of leguminous plants and certain blue green algae (Nostoc and Anabaena) are capable of fixing nitrogen and converting the same into nitrates (NO_3^-).
- Lightening also converts gaseous N_2 into nitrates.
- But the denitrifying bacteria in the soil are transfer the nitrates into gaseous nitrogen and release the same to the atmosphere.
- The processes involved in the cycle are fixation, ammonification, nitrification and denitrification.

Ammonification :

- Ammonification is the transforming process of complex organic matters into the inorganic matters.
- Dead animals and plant tissues and excretion matters give rise to nitrogenous wastes (complex organic matter).
- This nitrogenous wastes are converted into ammonia (NH_3) by heterotrophic bacteria, fungi and other decomposers in soil and water.
- Part of the ammonia is dissolved in water or taken up by plants. It is one way reaction.

Nitrification :

- Nitrification is a biological process in which ammonia (NH_3) is converted into nitrite (NO_2^-) and nitrate (NO_3^-).
- NH_3 is converted into NO_2^- by Nitrosomonas bacteria.
- NO_2^- is converted into NO_3^- by Nitrobacter.
- Thus, nitrification is a oxidation process in which nitrogen is increased.
- The NO_3^- which is thus formed is once again utilized by green plants for the synthesis of proteins.
- Gaseous N_2 is converted into NO_3^- by lightning N_2 – Fixing bacteria and blue green algae.
- Thus the N_2 cycle goes on continuously in nature again and again.

Denitrification :

- Some bacteria are capable of reducing nitrates to gaseous a process called denitrification.
- NO_3 is reconverted into gaseous N_2 by denitrifying bacteria. Agrobacterium and Pseudomonas.

  E.g., $2NO_3 \rightarrow 2NO_2 \rightarrow 2NO \rightarrow N_2O \rightarrow N_2$

Nitrogen Fixation Process :

- Nitrogen fixation is the conversion process of atmosphere nitrogen to ammonia or nitrate.
- Ammonia is the product of biological fixation.
- In biological fixation molecular nitrogen (N_2) splits into two atoms of free nitrogen.

  $$N_2 \rightarrow N + N$$

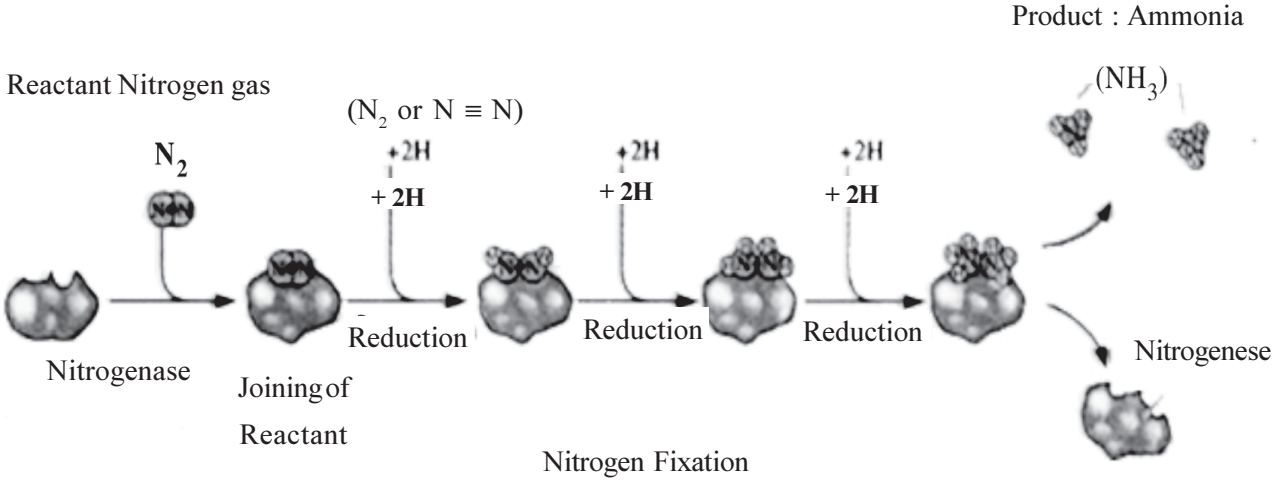
- The free nitrogen atoms can combine with hydrogen to form ammonia.

  $$2N + 3H_2 \rightarrow 2NH_3$$

- This process is mainly carried out by living organisms. i.e. specific bacteria (symbiotic bacteria-Rhizobium), blue green algae (Nostoc and Anabena), Free-living soil bacteria (the aerobic Azotobacter species and the anaerobic clostridium species).

Important Components :

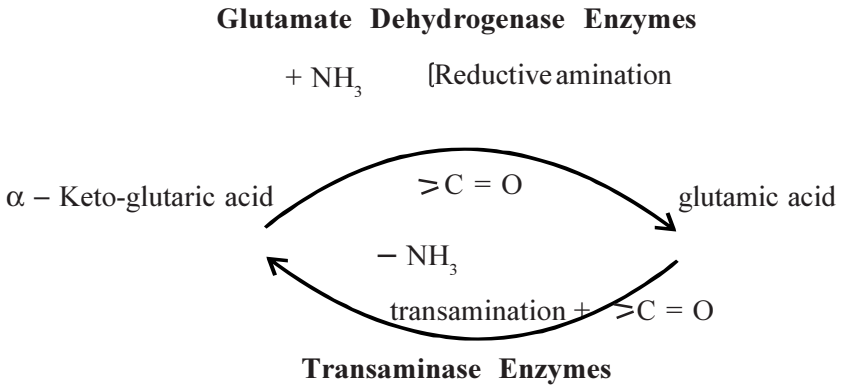
- Leghemoglobin = Red pigment haemeprotein.
- nif gene = nitrogen fixation gene.
- nif gene is responsible for synthesis of nitrogenase enzyme.
- This enzyme is extremely sensitive to the presence of oxygen.
- Leghemoglobin accepts oxygen and thus protects the enzyme from its side effects.
- This process of Biological fixation possess low energy requirement regulated by two enzymes which are nitrogenase (an iron containing protein) and hydrogenase (molybdenum containing protein).



Nitrogen Fixation

Protein synthesis :

- NH_3 is initially utilized in the synthesis of amino acids.
- In the structure of an amino acid, there always occurs at least one ($-NH_2$) amino group and one (COOH) carboxyl group.
- There are two main methods of amino acid synthesis in plants.



(A) Reductive Amination :

$$\alpha\text{–Keto-glutaric acid} + NH_3 \xrightarrow{\text{Glutamate dehydrogenase enzymes}} \text{glutamic acid}$$

(B) Transamination :

- In this method, an amino group is separated from one amino acid and it is transferred to a keto group in another keto kind of acid.
- Glutamic acid acts as a main donor of amino group.
- Other 17 kinds of amino acids are synthesized in this way.
- The enzyme responsible for this are of transaminase types.

Synthesis of Proteins :

- Various kinds of amino acid units become joined with one another in various numbers and various sequences to form polypeptide chains through peptide bonds.
- There can be one or more polypeptide chains in the constitution of proteins.

(88) Which of the process take part in nitrogen cycle ?

(A) Ammonification, Denitrification, Nitrification

(B) Ammonification, Nitrification, Decarboxylation

(C) Nitrification, Denitrification, Decarboxylation

(D) Ammonification, Nitrification, Denitrification

(89) Which bacteria converts ammonia in to nitrate ion ?

(A) Nitrobactor (B) Nitrosomonas (C) Pseudomonas (D) Agrobacterium

(90) Which type of reaction is denitrification ?

(A) Reduction (B) Oxidation (C) Carboxylation (D) Decarboxylation

(91) $2NO_3 \rightarrow 2NO_2 \rightarrow 2NO \rightarrow N_2O \rightarrow N$ it shows which process ?

(A) Nitrification (B) denitrification (C) Ammonification (D) Deammonification

(92) Which of the following is nitrogen fixing anaerobic free living bacteria ?

(A) Azotobacter (B) Clostridium (C) Both A and B (D) Rhizobium

(93) Relation between Leguminous plant and Rhizobium bacteria ?

(A) +, – (B) –, – (C) +, + (D) +, 0

(94) Which protein contain molybdenum ?

(A) Hydrolysis (B) Nitrogenase

(C) Hydrogenase (D) None above

(95) α Ketogtaric acid + ammonia $\rightarrow$ glutamic acid which enzyme responsible for this process ?

(A) Glutamate dehydrogenase (B) Transaminase (C) Glutamate dehydrogenase (D) Transfarase

(96) Select a non similar option.

(A) Rhizobium (B) Azotobector (C) Clostridium (D) Cynobacteria

(97) When animals and plants dies during such decomposition NH_3 is released this process is called ...

(A) Ammonification (B) Nitrification (C) Biological nitrogen fixation (D) Transaminase

(98) Which enzyme is essential for reductive amination ?

(A) Glutamate dehydrogenase (B) Transaminase

(C) Nitrogenase (D) Leg haemoglobin

(99) From the following number which is correct for N_2 reformation ?

(A) $2NO_3 - 2NO_2 - 2NO - N_2O - N_2$ (B) $2NO_3 - 2NO_2 - N_2O - 2NO - N_2$

(C) $2NO_2 - 2NO_3 - 2NO - N_2O - N_2$ (D) $2NO_2 - 2NO - 2NO_3 - N_2O - N_2$

(100) In which process glutamic acids behave as a main donor ?

(A) Reductive amination (B) Trans amination (C) Nitrification (D) Amonification

(101) Which is the similarity for reductive amination ?

(A) $2NH_3$ + α–Succinic acid $\rightarrow$ diglutamic acid (B) $2NH_3$ + α–Keto glutarate $\rightarrow$ glutamic acid

(C) $2N + 3H_2 \rightarrow 2NH_3$ (D) $2NH_3 \rightarrow 2N + 3H_2$

(102) Nitrogen absorption takes place in plant by which form ?

(A) Nitrate (B) Nitrite (C) Ammonium (D) Phosphate

(103) Nonsymbiotic nitrogen fixing bacteria .....

(A) Rhizobium (B) Nostoc (C) Frenchea (D) Azotobacter

(104) In which type of soil carnivorous plants live ?

(A) In wet soil

(B) Sugar defficient soil

(C) The soil which is rich in micro element

(D) Nitrogen compound defficient soil

(105) Which element is important for nitrogen fixation ?

(A) FAD (B) Nitrogenase (C) ATP (D) All above

(106) What is ammonification ?

(A) The formation of NH_3 after decomposition of dead body

(B) The formation of NH_2 after decomposition of dead body

(C) The process of formation of NO_2 from ammonia

(D) The process of formation of NH_3 from N_2

(107) Which compound of N_2 is obtained on a large scale from soil ?

(A) Nitric acid (B) Nitrite (C) Nitrate (D) Gaseous form of N_2

(108) $N_2 + 8e^- + 8H^+ + 16\,ATP \rightarrow 2NH_4 + H_2 + 16\,ADP + 16\,Pi$ which reaction indicate the following equation ?

(A) Ammonification (B) Nitrification (C) Nitrogen fixation (D) Denitrification

(109) What is necessary for fixation of N_2 ?

(A) Wet soil (B) Leghemoglobin (C) Keritinoids (D) Chlorophyll

Answers : (88-D), (89-B), (90-A), (91-B), (92-B), (93-C), (94-C), (95-A), (96-C), (97-A), (98-A), (99-A), (100-B), (101-B), (102-A), (103-D), (104-C), (105-D), (106-D), (107-B), (108-A), (109-B)

A - Assertion, R - Reason :

Choose correct option among the given options for answer.

(A) A and R both are correct, R is correct explanation of A.

(B) A and R both are correct, R is not correct explanation of A.

(C) A - correct, R - incorrect.

(D) A - incorrect, R - correct.

(110) Assertion A : In aeroponic plants are cultured by the fine drop of nutrient solution.

Reason R : In static solution culture film technique is used.

(A) (B) (C) (D)

(111) Assertion A : Potassium is essential for regulation of stomatal pore size.

Reason R : Yellowing of leaves is due to the potassium deficiency.

(A) (B) (C) (D)

(112) Assertion A : Bark of tree becomes rough and gets split and exades gum – like secretion.

Reason R : Mass of copper is less in plants.

(A) (B) (C) (D)

(113) Assertion A : Protoplasm is a permeable membrane.

Reason R : Protoplasm is made up of lipid and protein.

(A) (B) (C) (D)

(114) Assertion A : Nostoc and Anabaena are capable to convert Nitrogen to nitrate.

Reason R : In the cell of nostoc and Anabaena nucleous is non develop.

(A) (B) (C) (D)

(115) Assertion A : N, P, K fertilizer is necessary for the growth of plants.

Reason R : N, P, K fertilizer are usually not enough in the soil.

(A) (B) (C) (D)

(116) Assertion A : They involved in cation exchange.

Reason R : Soil acts as nutrients reservoir.

(A) (B) (C) (D)

(117) Assertion A : Rhizobium bacteria live as a symbiotic relationship in root nodules.

Reason R : A NiF-gene present in Rhizobium is responsible for synthesis of nitrogenase enzyme.

(A) (B) (C) (D)

(118) Assertion A : The concentration of macro nutrients in the dry mass of plants is about 1 to 5 mg per gram.

Reason R : The concentration of micro nutrients in the dry mass of plants is about 0.1 mg or less than that.

(A) (B) (C) (D)

(119) Assertion A : Nitrogen fixation aerobic bacteria – Azotobacter

Reason R : Nitrogen fixation anaerobic bacteria – Clostridium

(A) (B) (C) (D)

(120) Assertion A : Calcium is necessary for cell differentiation

Reason R : Calcium helps for creation of bipolar spindle.

(A) (B) (C) (D)

(121) Assertion A : Ions are transported in massflow with water from root.

Reason R : Metabolic energy is necessary for transportaion of ion from root.

(A) (B) (C) (D)

(122) Assertion A : Carrier molecules are present as a carrier in plasmamembrane.

Reason R : Carrier molecules are carbonic molecules.

(A) (B) (C) (D)

(123) Assertion A : The transportation of minerals towards soil is depended on the pH and the quantity of water.

Reason R : The transportation of minerals through root in the form of ion is responsible pH and high water pressure.

(A) (B) (C) (D)

(124) Assertion A : In epiphytes root system are non developing.

Reason R : In epiphytes maximum water absorbtion take place by the root surface.

(A) (B) (C) (D)

(125) Assertion A : Mineral absorbed by the root dissolve in water and transported in various part of plant through xylem.

Reason R : Phloem transported the inorganic and organic elements.

(A) (B) (C) (D)

(126) Assertion A : In crystaline solution all molecules are in the form of crystal and they transport individually from each other.

Reason R : In true solution the size of molecules are too small which helps to remain as a floating partical in the liquid.

(A) (B) (C) (D)

(127) Assertion A : For plants Fe^{+2} and Fe^{+3} ions are essential.

Reason R : Defficiency of Fe is very common in soil.

(A) (B) (C) (D)

(128) Assertion A : Fe and Cu are main electron transport minerals.

Reason R : Volancy are variable in Fe and Cu.

(A) (B) (C) (D)

Answers : (110-C), (111-B), (112-A), (113-C), (114-B), (115-A), (116-B), (117-B), (118-D), (119-B), (120-B), (121-B), (122-B), (123-A), (124-A), (125-B), (126-B), (127-C), (128-B)

(129) Name of elements, their importance and their defficient effect are given column I, II and III which one is the correct pair identify :

Column – I	**Column – II**	**Column – III**
(P) Potassium	(I) Constitution of co-enzyme	(a) Stemtip become dead
(Q) Calcium	(II) Tugidity of Cell	(b) Less green leaves
(R) Sulphur	(III) In middle lamella	(c) Increase respiration rate

(A) P - II a, Q - I - c, R - II - b (B) P - II c, Q - III - a, R - I - b

(C) P - I c, Q - II - a, R - III - b (D) P - II a, Q - III - b, R - I - c

(130) Method and its concerned option are given column-I and II, which one is the correct pair identify ?

Column – I	**Column – II**	
(P) diffusion	(i) carrier molecules	(A) P - ii, Q - iii, R - i
(Q) mass flow	(ii) ionic channels	(B) P - iii, Q - i, R - ii
(R) active absorbtion	(iii) absorbtion pressure	(C) P - i, Q - ii R - iii
		(D) P - ii Q - i, R - iii

(131) Name of enzymes and their functions are given column-I and II which one is correct pair identify :

	Column – I	**Column – II**	
(P)	Glutamate dehydrogenase	(i) Regulation of nitrogenfixation	(A) P - i, Q - ii, R - iii
(Q)	Hydrogenase	(ii) amino group is separated from amino acid	(B) P - ii, Q - iii, R - i
(R)	Transaminase	(iii) Synthesis of glutamic acid	(C) P - iii, Q - i, R - ii
			(D) P - i, Q - iii, R - ii

(132) Match the column in proper sequence :

	Column – I		**Column – II**	
(1)	defficiency of nitrogen	(P)	stunted growth of root	(A) 1 – R, 2 – P, 3 – S, 4 – Q
(2)	defficiency of potassium	(Q)	induction of dormacy in seeds	(B) 1 – R, 2 – Q, 3 – S, 4 – P
(3)	defficiency of phosphorous	(R)	reduction in cambial activity	(C) 1 – S, 2 – R, 3 – Q, 4 – P
(4)	defficiency of calcium	(S)	Chlorosis	(D) 1 – R, 2 – S, 3 – Q, 4 – P

(133) In the column given below number the proper sequence of defficienty effect of minerals :

	Column – I	**Column – II**	
(1)	Magnesium	(P) black or grayish spots appear	(A) 1 – R, 2 – P, 3 – S, 4 – Q
(2)	Sulphur	(Q) middle area of leaves become yellowish	(B) 1 – R, 2 – Q, 3 – S, 4 – P
(3)	Iron	(R) purple blots	(C) 1 – S, 2 – R, 3 – Q, 4 – P
(4)	Manganese	(S) completely yellowing of leaves	(D) 1 – R, 2 – S, 3 – Q, 4 – P

(134) Match proper pair :

	Column – I		**Column – II**		
(P)	Iron	(i)	activation of many enzymes	(A)	P – i, Q – ii, R – iii, S – iv
(Q)	Chlorine	(ii)	formation of cell wall	(B)	P – iv, Q – iii, R – i, S – iii
(R)	Manganese	(iii)	active space of electron conductor	(C)	P – iii, Q – iv, R – i, S – ii
(S)	Boron	(iv)	photosynthesis and ionic balance	(D)	P – iii, Q – iv, R – ii, S – i

(135) Match proper pair :

	Column – I		**Column – II**
(a)	Sodium	(p)	Osmosis and ionic balance in plants
(b)	Cobalt	(q)	activity of nitrogenase type of enzymes
(c)	Silicon	(r)	for nitrogen fixation
(d)	Vanadium	(s)	the element of cell wall

(A) (a – s), (b – p), (c – r), (d – q) (B) (a – p), (b – r), (c – s), (d – q)

(C) (a – p), (b – q), (c – r), (d – s) (D) (a – q), (b – r), (c – p), (d – s)

(136) Match proper pair :

Column – I	Column – II	
(P) K	(i) Element of ferredoxin	(A) P - ii, Q - i, R - iv, S - iii
(Q) S	(ii) Role in stomatal-pore	(B) P - i, Q - ii, R - iii, S - iv
(R) Mo	(iii) Synthesis of oxygen	(C) P - iv, Q - iii, R - ii, S - i
(S) Mn	(iv) Element of nitrogenase	(D) P - i, Q - iii, R - iv, S - ii

(137) Match proper pair :

Column – I	Column – II	
(P) Mg	(i) are seen in amino acid	(A) P - ii, Q - i, R - iv, S - iii
(Q) S	(ii) not important for plants	(B) P - iii, Q - i, R - ii, S - iv
(R) I	(iii) constituent of chlorophyll	(C) P - iii, Q - i, R - ii, S - v
(S) Mn	(iv) constituent of sugar	(D) P - iv, Q - iii, R - i, S - v
	(v) required in the activation of enzymes	

(138) Match proper pair :

Column – I	Column – II	
(P) N	(i) copper coloured plant body	(A) P - iv, Q - iii, R - i, S - ii
(Q) Mg	(ii) immature leaf fall	(B) P - i, Q - ii, R - iii, S - iv
(R) B	(iii) chlorosis take place in veins	(C) P - i, Q - iii, R - iv, S - ii
(S) P	(iv) chlorosis and Necrosis happen	(D) P - ii, Q - iii, R - iv, S - i

(139) Method and matter are given in column-I and II which one is the correct pair identify :

Column – I	Column – II	
(a) Static solution culture	(i) Automated flow	(A) a - i, b - ii, c - iii
(b) Continuous flow solution culture	(ii) Fine drops (aerosol)	(B) a - ii, b - iii, c - i
(c) Aeroponics	(iii) Plant culture in glass jar	(C) a - iii, b - i, c - ii
		(D) a - iii, b - ii, c - i

(140) Name of elements and their importance are given column I and II which one is correct pair identify :

Column – I	Column – II	
(a) Nitrogen	(i) In ionic balance	(A) a - i, b - ii, c - iii
(b) Potassium	(ii) Constitution of chlorophyll	(B) a - ii, b - i, c - iii
(c) Phosphorous	(iii) Process of phosphorylation	(C) a - iii, b - i, c - ii
		(D) a - iii, b - ii, c - i

(141) Name of elements and their importance are given column I and II which one is the correct pair identify :

	Column – I	Column – II	
(a)	Calcium	(i) In bipolar spindle	(A) a - i, b - ii, c - iii
(b)	Sulphur	(ii) maintenance of ribosomal constituion	(B) a - i, b - iii, c - ii
(c)	Magnesium	(iii) a constituent of amino acid	(C) a - ii, b - i, c - iii
			(D) a - iii, b - ii, c - i

(142) Name of bacteria and their function are given in column I and II which one is the correct pair identify :

	Column – I	Column – II	
(a)	Rhizobium	(i) $NH_3 \rightarrow$ nitrite ion	(A) a - iii, b - iv, c - i, d - ii
(b)	Nitrobactor	(ii) Nitrate $\rightarrow N_2$ gas	(B) a - iii, b - i, c - iv, d - ii
(c)	Nitrosomonas	(iii) Formation of root nodules	(C) a - iii, b - iv, c - ii, d - i
(d)	Pseudomonas	(iv) Nitrite ion $\rightarrow$ Nitrate	(D) a - iii, b - i, c - iv, d - ii

(143) Match proper pair :

	Column – I	Column – II	
(P)	Mg	(i) seen in few amino acid	(A) P - iii, Q - iv, R - ii, S - i
(Q)	S	(ii) not important for plants	(B) P - iii, Q - i, R - ii, S - iv
(R)	I	(iii) constituent of chlorophyll	(C) P - iii, Q - i, R - ii, S - v
(S)	Mn	(iv) constituent of sugar	(D) P - iv, Q - iii, R - i, S - v
		(v) required in the constituent of enzymes	

(144) Match proper pair :

	Column – I	Column – II	
(P)	N	(i) copper coloured plant body	(A) P-iv, Q-iii, R-i, S-ii
(Q)	Mg	(ii) immature leaves fall	(B) P-i, Q-ii, R-iii, S-iv
(R)	B	(iii) chlorosis take place in veins	(C) P-i, Q-iii, R-iv, S-ii
(S)	P	(iv) chlorosis and nephrosis happen	(D) P-ii, Q-iii, R-iv, S-i

Answers : (129-B), (130-A), (131-C), (132-C), (133-A), (134-C), (135-B), (136-A), (137-C), (138-A), (139-C), (140-B), (141-B), (142-C), (143-C), (144-B)

(145) What does 'p' indicate in the given diagram-1 ?

(A) more concentration (B) Transportive molecules

(C) ion channel (D) Uniport

(146) What does 'Q' indicate in given diagram-1 ?

(A) carrier molecule (B) ion channel

(C) protein (D) lipid

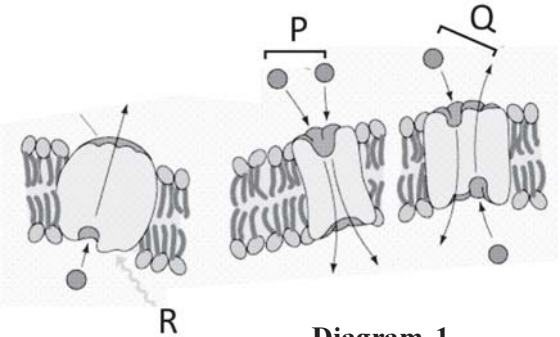


Diagram 1

(147) In the given diagram – 1 the indicated part – R is not used in which process ?

(A) simple diffusion (B) facilitated diffusion (C) active absorbtion (D) ATP

(148) In the given diagram – 1 show the function of 'P'.

(A) carrier molecule (B) inner side of plasma membrane

(C) ion-protein complex (D) Antiport

(149) What does 'R' indicate in given diagram-2 ?

(A) Transpoted ion (B) Energy

(C) Symport (D) Anti port

(150) What does 'S' indicate in given diagram-2 ?

(A) Active transport (B) Coupled transport

(C) Antiport (D) Symport

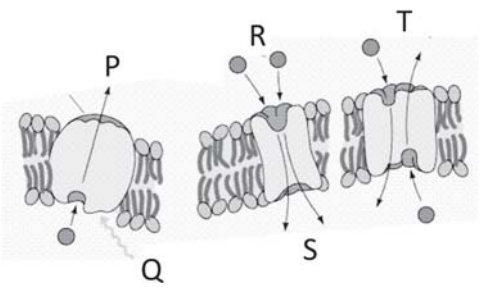


Diagram – 2

(151) In the given diagram-2 which option is not matched with part 'T' ?

(A) Carrier molecule (B) Slowly absorption

(C) Consumption of ATP (D) None above

(152) Which type of passive absorption indicate the given diagram-2 ?

(A) diffusion (B) Donnan equllibrium (C) ion exchange (D) active transport

(153) Which in the given diagram-2 where the absorptive ion take place ?

(A) Outer surface of plasma membrane (B) inner surface of plasma membrane

(C) on surface of cell wall (D) Both A and B

Answers : (145-A), (146-C), (147-A), (148-A), (149-C), (150-D), (151-C), (152-A), (153-C)

True - False (T - F) type questions :

Choose correct alternative option for answer :

(154) (1) Nitrogenase is an iron containing enzyme.

(2) FAD is reduction inducing unit in nitrogen fixation.

(3) The enzyme transaminase 15 responsible for Reductive Amination.

(4) Glutamic acid acts as a main donor of amino group.

(A) T F T F (B) T T F T (C) T T F F (D) F F T T

(155) (1) Sulphur is constituent of some amino acids like Cystiene and Methionine.

(2) Sulphur occurs in constitution of Vitamin Biotin (H_1) and thiamine (B_1).

(3) Iron is in the constitution of feredoxin and cytochrome.

(4) Magnesium is to induce photolysis of water during photosynthesis.

(A) T T T F (B) T F F T (C) T T F F (D) F F T T

Answers : (154-B), (155-A)

NEET questions :

(156) Which element is necessary for the growth of root hair ?

(A) Zn (B) Ca (C) Mo (D) S

(157) The deficiency of micromolecule not only resist the growth in plants but also resist some, living process like photosynthesis, mytochondrial electron transfer from given below which group effect on both photosynthesis and mitochondrial electron transport ?

(A) Cu, Mn, Fe (B) Co, Ni, No

(C) Mn, Co, Ca (D) Ca, K, Na

(158) The process of conversion of NH_3 into NO_3^- by Nitrosomonas bacteria called ...

(A) Nitrogen fixation (B) Nitrification

(C) Denitrification (D) Ammonification

(159) Characteristic of Leghemoglobin ..... .

(i) Source of energy

(ii) pink coloured pigment which shows in rootnodules

(iii) It is active in presence of Mg^{+2}

(iv) It protects the nitrogenase from oxygen side effect

(A) (i), (ii) and (iii) are true (B) (i) and (ii) are true

(C) (ii) and (iv) are true (D) (i) and (iii) are true

(160) 'Little leaf disease' is caused by which mineral deficiency ?

(A) Cu (B) Zn (C) B (D) Fe

(161) Nitrifying Bacteria is : ..... .

(A) Conversion of ammonia in to nitrate.

(B) Free N_2 is converted in to salt nitrogen.

(C) Conversion of protein in to ammonia.

(D) Conversion of nitrate through reduction to free nitrogen.

(162) The function of leghemoglobin.

(A) regulates the mass of O_2 in cell (B) regulates the mass of CO_2 in cell

(C) it produce the phenotic solution in all (D) regulates the mass of Mo in cell

(163) Given below is not the characteristic of essential mineral.

(A) It is necessary for constitution of Biomolecules

(B) It convert the chemical characteristic of soil.

(C) It comes in the constitution of energy giving chemical compound.

(D) It is responsible for activeness and passiveness of enzymes.

(164) Which three group of micro molecule effect on photosynthesis and mitochondrial electron transfer ?

(A) Cu, Mn, Fe (B) Co, Ni, Mo (C) Ca, K, Na (D) Mn, CO, Ca

(165) By whom nitrite is converted in to nitrate ?

(A) Nitrobactor (B) Nitrosomonas (C) Psuedomonas (D) Chlostridium

(166) Suppose in protoplasm of cell there are 3X K^+ ion and 2X K^+ ion outside in active respiration after sometime amount of X of K^+ ion enter in the cell by which reaction K^+ ion transported ?

(A) primary active transport (B) secondary active transport

(C) passive transport (D) diffusion

(167) Which mineral deficiency in leaf is responsible for the dead tissue ?

(A) N, K, Mg, Fe, Mn, Zn and Mo (B) Ca, Mg, Cu and K

(C) N, K and S (D) N, K, Mg and Fe

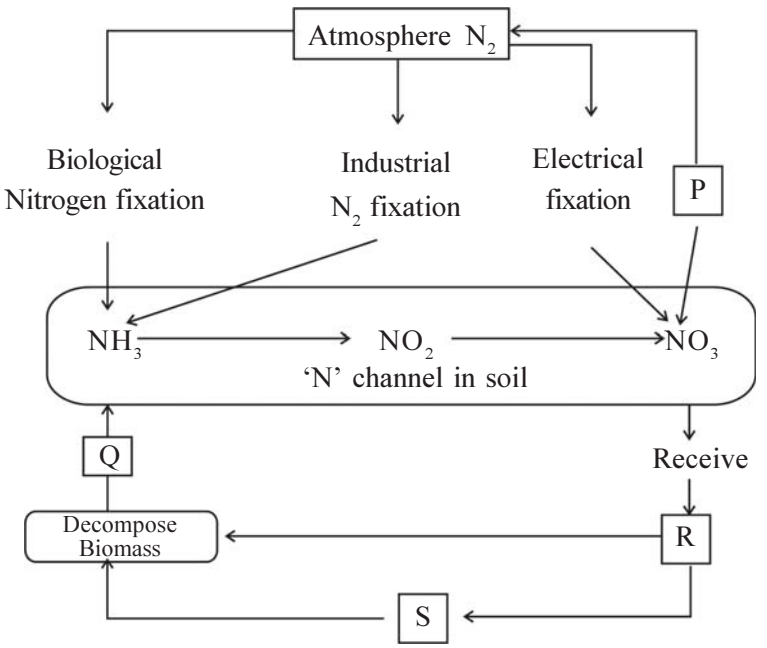
(168) In nitrification who does the first oxidation of ammonia to nitrite ?

(A) Nitrosomonas and Nitrococus (B) Nitrosomonas and Nitrobacter

(C) Pseudomonas and Nitrococus (D) Pseudomonas and Thyobacilus

(169) Choose the proper option which give the real identification of P, Q, R and S :

	P	Q	R	S
(A)	Nitrification	Ammonification	Animals	Plants
(B)	Denitrification	Ammonification	Plants	Animals
(C)	Nitrification	Denitrification	Animals	Plants
(D)	Denitrification	Nitrification	Plants	Animals



(170) Which of the following statement is true ?

(A) Nitrogenfixator who construct root nodules in free living condition are aerobic.

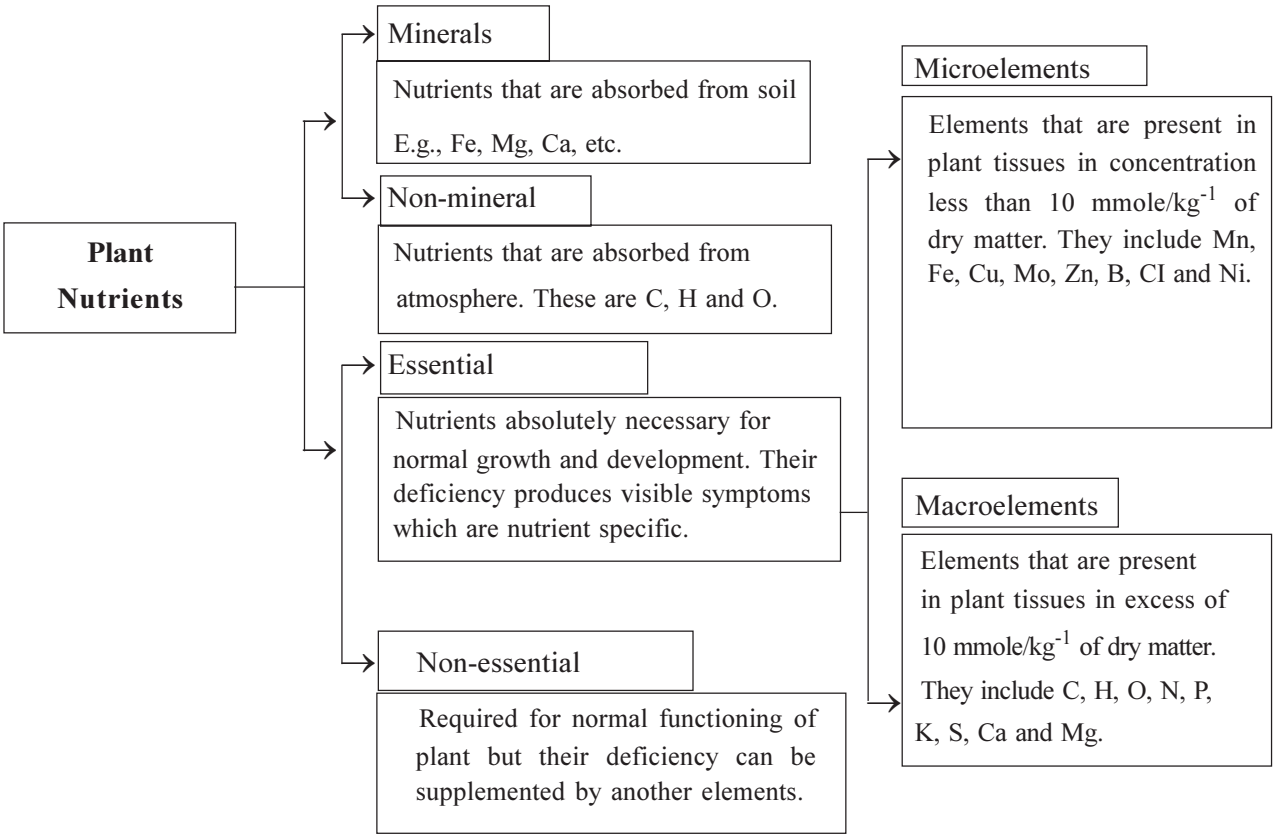
(B) Phosphorous is structural component of plasmamembrane some nucleic acid and some types of proteins.

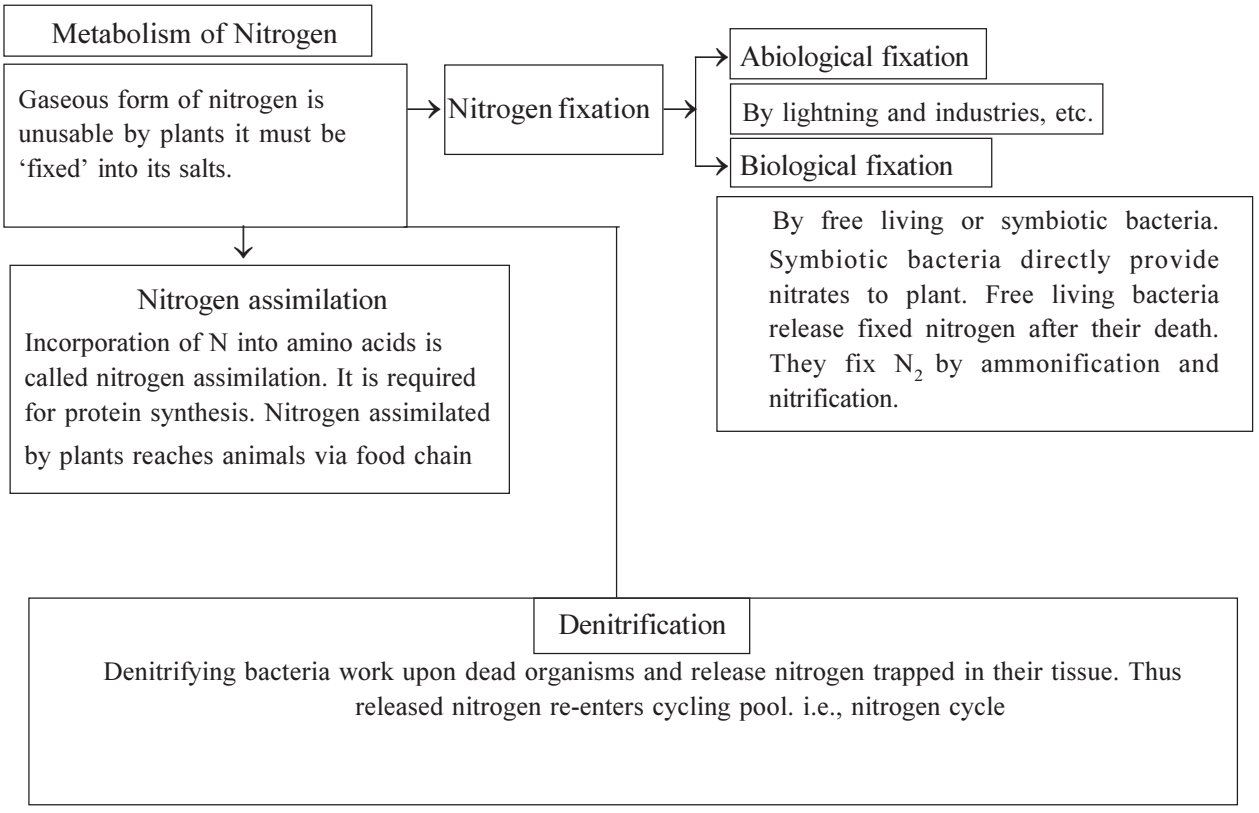
(C) Nitrosomonas and nitrobacter are chemically synthesised free living organisms.

(D) Nostoc and Anabena are capable free living to nitrogenfixation in free living condition.

Answers : (154-B, (155-A) (156-B), (157-A), (158-B), (159-C), (160-C), (161-B), (162-A), (163-C), (164-A), (165-A), (166-B), (167-A), (168-B), (169-A), (170-B)

Concept Map


Plant Nutrients
Minerals
Nutrients that are absorbed from soil E.g., Fe, Mg, Ca, etc.
Non-mineral
Nutrients that are absorbed from atmosphere. These are C, H and O.
Essential
Nutrients absolutely necessary for normal growth and development. Their deficiency produces visible symptoms which are nutrient specific.
Non-essential
Required for normal functioning of plant but their deficiency can be supplemented by another elements.
Microelements
Elements that are present in plant tissues in concentration less than 10 mmole/kg-1 of dry matter. They include Mn, Fe, Cu, Mo, Zn, B, CI and Ni.
Macroelements
Elements that are present in plant tissues in excess of 10 mmole/kg-1 of dry matter. They include C, H, O, N, P, K, S, Ca and Mg.


Metabolism of Nitrogen
Gaseous form of nitrogen is unusable by plants it must be 'fixed' into its salts.
Nitrogen fixation
Abiological fixation
By lightning and industries, etc.
Biological fixation
By free living or symbiotic bacteria. Symbiotic bacteria directly provide nitrates to plant. Free living bacteria release fixed nitrogen after their death. They fix N2 by ammonification and nitrification.
Nitrogen assimilation
Incorporation of N into amino acids is called nitrogen assimilation. It is required for protein synthesis. Nitrogen assimilated by plants reaches animals via food chain
Denitrification
Denitrifying bacteria work upon dead organisms and release nitrogen trapped in their tissue. Thus released nitrogen re-enters cycling pool. i.e., nitrogen cycle

20 Photosynthesis

Plants use sunlight (Photon), (H_2O) and (CO_2) to convert light Energy into chemical Energy by chlorophyll. This process is known as photosynthesis.

Chemical equation :

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Necessary factors for photoynthesis :

- Light
- Chlorophyll
- Water (H_2O)
- Carbon Dioxide (CO_2) (0.03 % in the atmosphere)

Photosynthetic Organisms :

- Different types of algae (Blue green algae, Brown algae, Red algae, Green algae).
- Bryophytes, Pteridophytes, Gymnosperms, Angiosperms.
- Some bacteria.

(**Note :** Fungi is not able to perform photosynthesis due to lack of chlorophyll.)

Advantages of Photosynthesis :

- Main products of photosynthesis are starch and sucrose.
- (O_2) is byproduct which is used by organisms for respiration.

---

(1) What is produced as byproduct of photosynthesis ?

(A) Oxygen (B) Nitrogen (C) Carbon dioxide (D) Sulphur dioxide

(2) Which type of energy can be used by all organisms ?

(A) Light energy (B) Chemical energy (C) Heat energy (D) Water potential

(3) Which of the following type of reaction photosynthesis is ?

(A) Anabolic, Endothermic, Reduction (B) Anabolic, Endothermic, Oxidation

(C) Catabolic, Exothermic, Oxidation (D) Catabolic, Endothermic, Reduction

Answers : (1-A), (2-B), (3-A)

History of Photosynthesis :

No.	Name of Scientist	Contribution
1.	Joseph Priestly (1733-1804)	Plants obtain CO_2 from atmosphere and release O_2.
2.	Ingenhouse (1730-1799)	In bright sunlight, small bubbles were formed around the green parts of the plant.
3.	Julius Von Sachs (1854)	Green substance (chlorophyll) in plants is located in special bodies (chloroplast) of plant cell. This green substances produces glucose which is usually stored in the form of starch.
4.	Robert Hill	O_2 released into atmosphere comes from water during photosynthesis. He gave equation : $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$
5.	Cornelius Van Niel (1897-1985)	Photosynthesis is essentially a light dependent reaction in which hydrogen from a suitable oxidisable compound reduces CO_2 to carbohydrates. He gave equation as follows : $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$

(4) Which scientist showed that plants releases O_2 and obtain CO_2 ?

(A) Jan Ingenhousze (B) Julius Von Sachs (C) Joseph Priestly (D) Van Niel

(5) Which modified equation is given by Hill ?

(A) $6CO_2 + 10H_2O \rightarrow C_6H_{10}O_6 + 5H_2O + 6O_2$

(B) $5CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 5H_2O + 6O_2$

(C) $4CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$

(D) $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$

(6) Which scientist showed that Light is necessary for study of photosynthesis in blue green algae ?

(A) Cornelius Van Niel (B) Jan Ingenhousze (C) Julius Von Sach (D) Robert Hill

(7) Which scientist demonstrated that O_2 is liberated only by chlorophyll containing organs of the plant in presence of O2 ?

(A) Priestly (B) Niel (C) Ingenhousze (D) Helmont

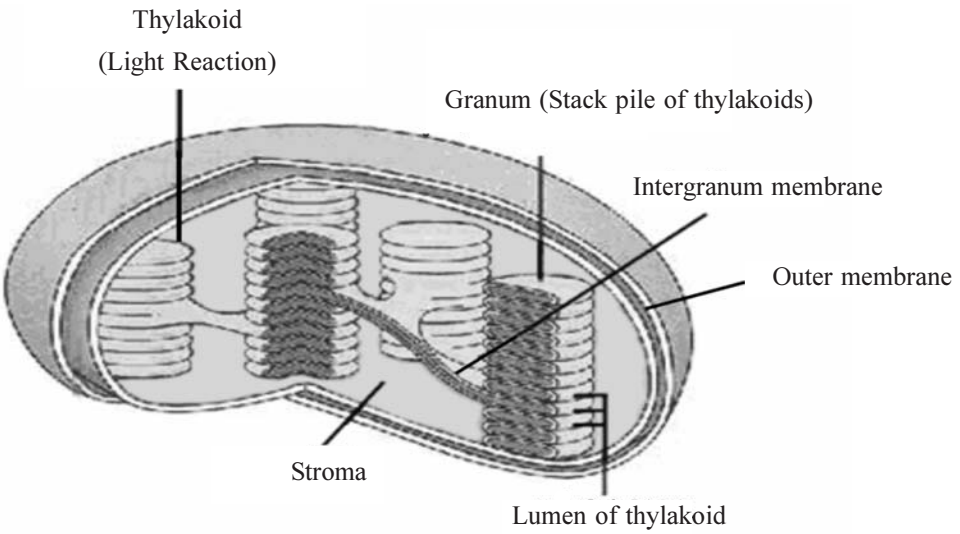
(8) Which type of reaction photosynthesis is ?

(A) Anabolic (B) Catabolic (C) Metabolic (D) All of the above

Answers : (4-C), (5-D), (6-A), (7-C), (8-A)

Structure of Chloroplast : (Site for photosynthesis)

- The proces of photosynthesis takes place in green organs of the plants.



Structure

Chloroplast has main two parts :

(A) Grana (Membranous) (B) Stroma.

Grana :

- Each Granum is made up of flat, lamellar structure called thylakoid.
- This thylakoids contain chlorophyll pigments.

These are the main pigments of chloroplast :

Chlorophyll-a	($C_{55}H_{72}O_5N_4Mg$)	- Green coloured
Chlorophyll-b	($C_{55}H_{70}O_6N_4Mg$)	- Green coloured
Xanthophyll	($C_{40}H_{56}O_2$)	- Yellow coloured
Carotenoids		- Yellow to Orange coloured

- Amount of substances is different in all plants.

Stroma (Chlorenchyma) :

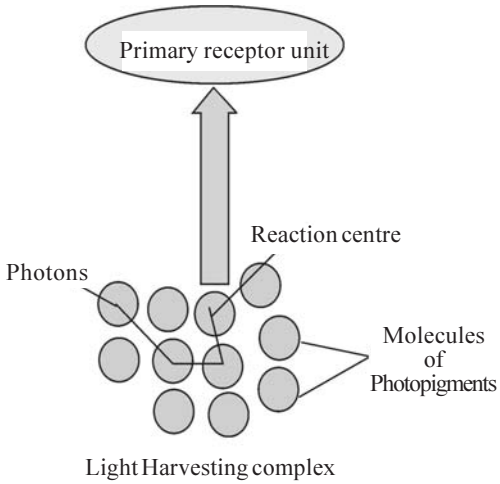
- Enzymes necessary for Calvin cycle are present in Stroma.

Functions :

- Light reaction take place in Grana.
- In Light reaction ATP and NADP are produced.
- Second phase of photosynthesis is dark reaction which occurs in Stroma, Enzymes becomes activated, So CO_2 and H_2O combines to form $C_6H_{12}O_6$.
- Dark reaction is dependent on the product of light reaction.

Photopigment System :

- We can identify different pigments by Chromatography.
- Pigments are classified on the basis of colour.
- These photopigments absorb light at specific wavelength and obtain energy from them.
- Chlorophyll is a main reactant of this process.
- Other pigments works as accessory pigment.
- The photosynthetic pigments at reaction centre are stimulated by specific wavelength. On the basis of is they are divided into two.



- These light harvesting complexes are known as Ps-I and Ps-II. (Ps = Photopigment system)

No.	Features	PS-I	PS-II
(1)	Reaction centre	PS-I is also known as P 700.	PS-II is also known as P 680.
(2)	Process	Participates in cyclic and nonacyclic photo phosphorylation.	Participates only in noncyclic photophosporylation.
(3)	e^- gain	Picks up e^- released by PS-II.	Picks up e^- obtained by photolysis of water.
(4)	Oxygen	Does not show presence of Oxygen.	During photolysis of water O_2 is produced.
(5)	Wavelength	It gets stimulated at 700 nm wavelength.	System is stimulated at 680 nm wavelength.

(9) Number of piles of flat lamellar thylakoids is known as ..... .

(A) Grana (B) Stroma (C) Granum (D) Lamellae

(10) In the process of photosynthesis ..... .

(A) O_2 is liberated and $NADPH_2$ is formed.

(B) Production of ATP and $NADPH_2$ occurs.

(C) Production of ATP and $NADPH_2$ occurs and O_2 is released.

(D) Fixation of CO_2 occurs.

(11) Which of the following is the place for chlorophyll ?

(A) In Stroma (B) On the surface of chloroplast

(C) In Grana (D) Overall chloroplast

(12) In which part of the leaf process of photosynthesis occurs ?

(A) Vascular bundle (B) Mesophyll tissue (C) Upper epidermis (D) Lower epidermis

(13) Where does the chloroplast present in the cell ?

(A) In the centre (B) Near to nucleus (C) Scattered (D) At periphery

(14) The intense light oxidise chlorophyll and causes damage to the plant. It is known as .....

(A) Photorespiration (B) Photooxidation (C) Photolysis (D) Photoreduction

(15) What is LHC ?

(A) Lower Hard Complex (B) Light High Complex

(C) Light Harvesting Complex (D) Light Heavy Complex

(16) Which of the following group is accessory photosynthetic pigment ?

(A) Chlorophyll-b, Xanthophyll, Chlorophylla (B) Chlorophyll-b, Xanthophyll, Carotenoides

(C) Chlorophyll-b, Carotenoids, Chlorophyll-a (D) Chlorophyll-b, Fibrin, Xanthophyll

(17) Which pigments are yellow to Orange ?

(A) Chlorophyll-a (B) Xanthophyll (C) Carotenoids (D) Chlorophyll-b

(18) How many Light Harvesting complexes are formed, by photosynthetic pigments ?

(A) Two (B) Three (C) Four (D) Five

(19) How many chlorophyll pigments are present in each pigment system ?

(A) 300 to 400 (B) 400 to 500 (C) 350 to 800 (D) 250 to 400

(20) Which system is more sensitive to intense wavelength of light in photo pigment system ?

(A) PS-I (B) PS-II

(C) Splitting of water molecule by light (D) Light harvesting system

(21) What is known as PS-II ?

(A) P_{700} (B) P_{680} (C) LHC-I (D) RHC-I

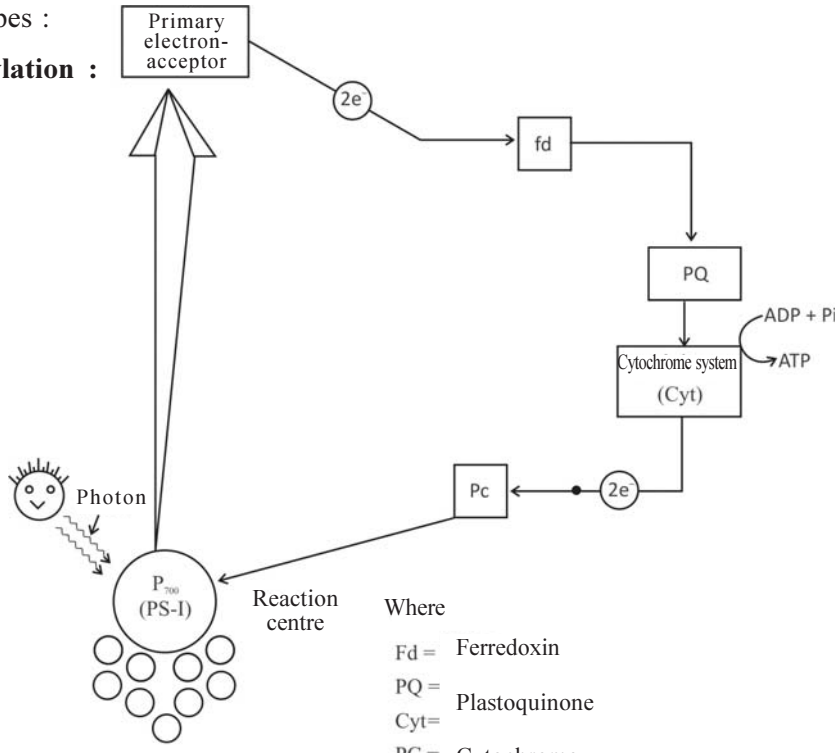
Answers : (9-A), (10-C), (11-C), (12-B), (13-D), (14-B), (15-C), (16-B), (17-C), (18-A), (19-D), (20-A), (21-B)

Phases of Photosynthesis :

(A) Light Reaction :

- There are mainly two types :

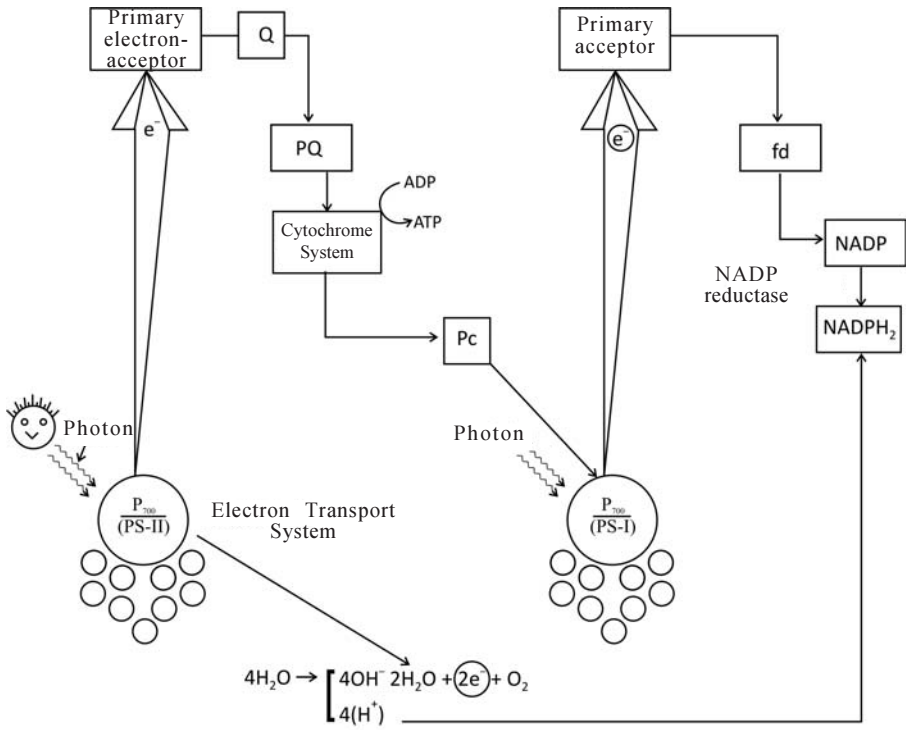
(a) Cyclic Photophosphorylation :



Cyclic Photophosphorylation Reaction :

In cyclic photophosphorylation e^- transporters Ferredoxin (Fd) → Plastoquinone (PQ) → Cytochrome system → Plastocyanin (Pc) sends electron to its main source of PS-I.

(b) Noncyclic Photophosphorylation :



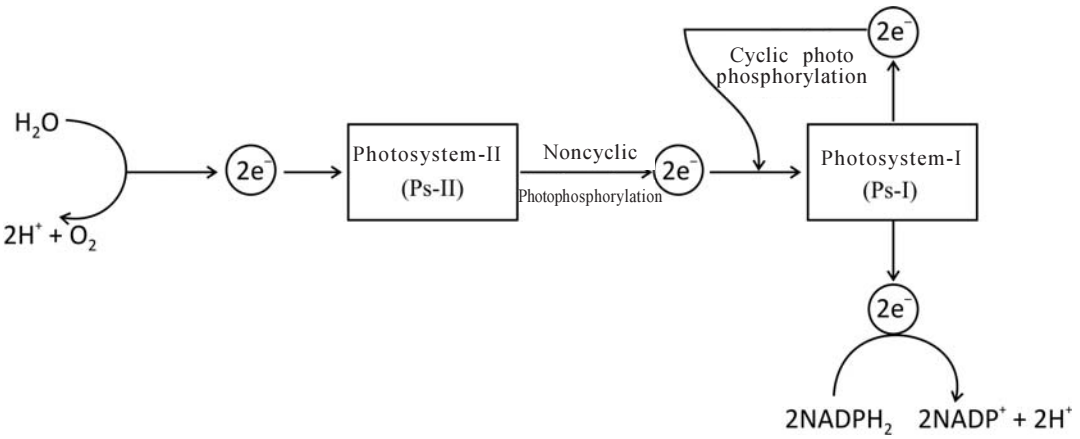
Photolysis of water

From PS-II to PS-I :

Electron passes Quinone → Plastoquinone → Cytochrome system → Plastocyanin → PS-I.

Here in PS-I Ferredoxin accepts the electron and next utilized for reduction of NADP.

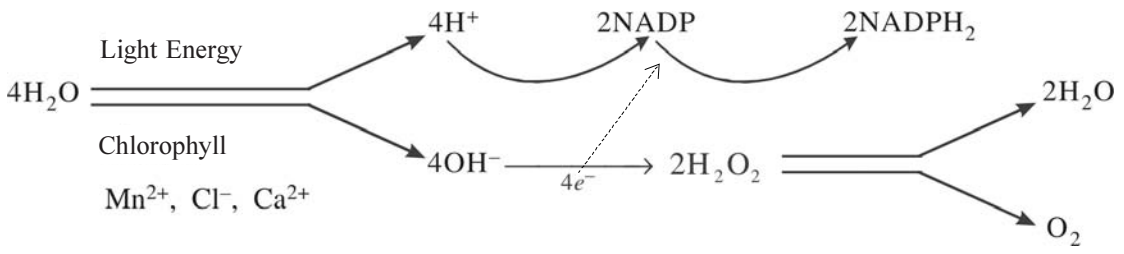
Summary of Cyclic and Noncyclic Photophosphorylation :



No.	Features	Cyclic Photo Phosphorylation	Non Cyclic Photophosphorylation
1.	Production	Produces ATP.	Produces ATP and $NADPH_2$.
2.	Oxygen	Oxygen does not take part.	Oxygen is produced by photolysis of water.
3.	Photopigment system	Only PS-I participates.	Participation of PS-I & PS-II both.
4.	e^- Transportation	Electrons released from PS-I goes back to the same source. No e^- from any other molecules enter in the system.	Electrons released from PS-I and PS-II does not revert to the main source. Outer e^- released from photolysis of water enter in the system.
5.	Efficiency	Usually occurs during anaerobic condition, less light intensity and less CO_2 concentration.	Usually occurs during aerobic condition, high light intensity and high CO_2 concentration.
6.	Process	Usually occurs in bacteria like organisms.	Usually occurs in CO_2 fixative plants and another photosynthetic organisms.

(c) Photolysis of Water :

The splitting of water molecule using energy of light is known as photolysis of water. The entire process can be presented as under :



(22) Which of the following accepts electron at last that released by photolysis of water ?
(A) PS-I (B) PS-II (C) OAA (D) NADP

(23) How many water molecules split at a time in light reaction of photosynthesis ?
(A) 1 (B) 4 (C) 2 (D) 6

(24) Which are the essential ions for photolysis of water ?
(A) Mn^{+2}, Cu^{+2}, $C1^-$ (B) Na^+, Mg^{+2}, $C1^-$ (C) Mn^{+2}, Ca^{+2}, $C1^-$ (D) Mn^{+2}, Fe^{+2}, $C1^-$

(25) Which pigment is associated with oxidation of water ?
(A) PS-I (B) PS-II (C) Phycobilism (D) Carotenoids

(26) What is the product formed by $4OH^-$ at the end of photolysis of water ?
(A) $2H_2O$ (B) $2NADPH_2$ (C) O_2 (D) (A) and (B) both

(27) What is primary acceptor of $4e^-$ released by PS-I in cyclic photophosphorylation ?
(A) Cytochrome (B) Electron transport system
(C) Electron acceptor (D) PS-II

(28) Which of the following process occurs in cyclic photophosphorylation ?
(A) Reduction of NADP (B) Photophosphorylation
(C) Photolysis (D) Production of O_2

(29) How many photons are accepted by PS-I for initiation of cyclic photophosphorylation in light reaction of photosynthesis ?
(A) One (B) Two (C) Three (D) Four

(30) Where does photophosphorylation occur ?
(A) In PS-I (B) In PS-II (C) In (A) and (B) both (D) In thylakoid

(31) What is Phosphorylation ?
(A) Production of ATP from ADP in presence of light
(B) Production of ADP from ATP in presence of light.
(C) Oxidation of $NADPH_2$ in presence of light.
(D) Reduction of $NADPH_2$ in presence of light.

(32) What happens during electron transportation ?
(A) Energy is released (B) Energy is utilized
(C) Reduction (D) Energy is stored

(33) The process associated with cyclic photophosphorylation ..... .

(A) O_2 is produced (B) Phosphorylation occurs

(C) Reduction of $NADPH_2$ (D) Photolysis of water

(34) Sequence of electron transport released from cyclic photophosphorylation ..... .

(A) Fd → Cyt → Pc → PQ (B) Fd → PQ → Cyt → Pc

(C) PQ → Q → Cyt → Pc (D) Q → PQ → Cyt → Pc

(35) Which electron transporter is not present in cyclic photophosphorylation ?

(A) Fd (B) Q (C) PQ (D) Cyt

(36) What is photon ?

(A) Electron transporter (B) Molecules of chlorophyll

(C) Molecule form of light (D) Wave form of light

(37) Which of the following molecule works as a reaction centre in noncyclic photophosphorylation ?

(A) Chlorophyll a of PS-I (B) Chlorophyll a of PS-II

(C) LHC (D) (A) and (B) both

(38) Who accepts $4e^-$ released by PS-I at last in Noncyclic photophosphorylation ?

(A) PS-I (B) PS-II (C) NADP (D) RuBP

(39) Who accepts $4e^-$ released by PS-II at last in Noncyclic photophosphorylation ?

(A) PS-I (B) NADP (C) RuBP (D) H_2O

(40) From which photopigment system e^- is obtained for reduction of NADP during Noncyclic photophosphorylation ?

(A) PS-I (B) PS-II

(C) From PS-I and PS-II both (D) None of the above

(41) How many ATP molecules are formed during noncyclic photophosphorylation ?

(A) 2 ATP (B) 1 ATP (C) 3 ATP (D) 6 ATP

(42) Which of the following process occurs first in photosynthesis ?

(A) Production of O_2 (B) Noncyclic photophosphorylation

(C) Photolysis of water (D) Reduction of NADP

(43) $4H^+ + 2NADP\ 4e^-\ 2NADPH_2$. From where the $4e^-$ are obtained in the given process ?

(A) PS-I (B) PS-II (C) Photon (D) H_2O

(44) Which of the following electron transporters are associated with cyclic and Noncyclic photophosphorylation ?

(A) PQ (B) PC (C) Cyt (D) All of the above

(45) Which of the following accepts e^- released from PS-II ?

(A) PQ (B) Cyt (C) Q (D) Pc

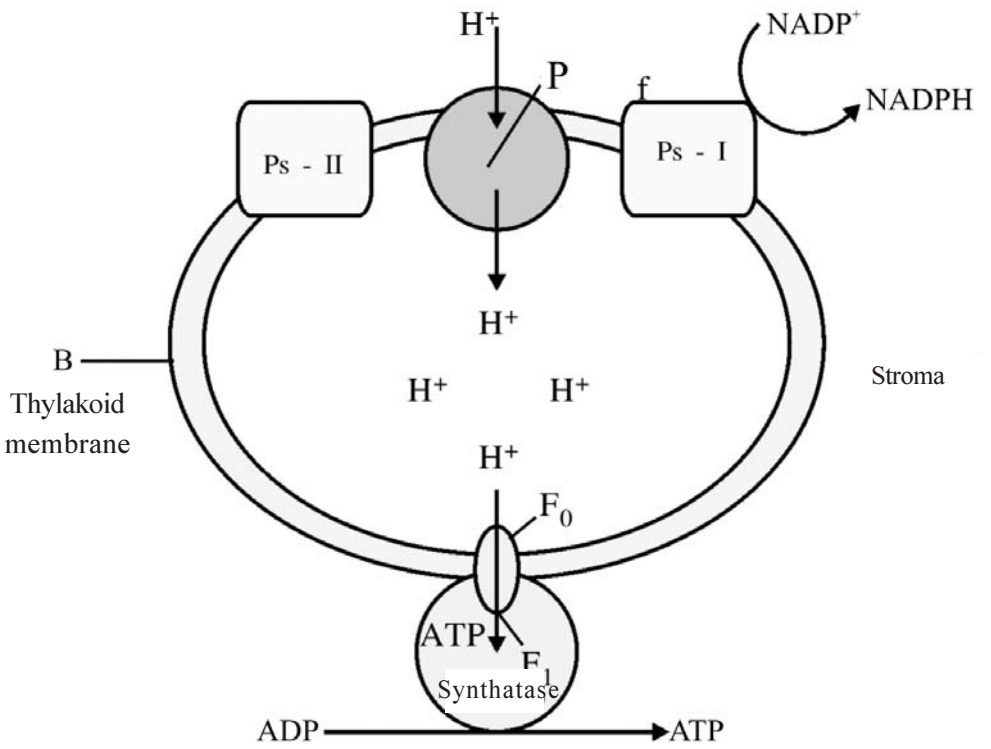
(46) After removal of thylakoid, if we put leaf in CO_2 and H_2O containing medium in presence of sunlight, Hexose is not the Final product, because ..... .

(A) Molecules which obtains light are absent. (B) Photosystem PS-I and PS-II are not present.

(C) Enzymes which fixes CO_2 are absent. (D) CO_2 fixation does not occur in presence of light.

Answers : (22-D), (23-B), (24-C), (25-B), (26-D), (27-C), (28-B), (29-D), (30-D), (31-A), (32-A), (33-B), (34-B), (35-B), (36-C), (37-D), (38-C), (39-A), (40-A), (41-B), (42-C), (43-A), (44-D), (45-C), (46-C)

Chemiosmosis :



ATP synthesis through chemiosmosis

- The chemiosmotic hypothesis has been put forward to understand synthesis of ATP in chloroplast during photosynthesis.
- ATP synthesis is dependent on concentration gradient generated across thylakoid membrane.
- Due to splitting of water molecule protons (hydrogen ions) are liberated, which are accumulated in lumen of thylakoid.
- Outside the thylakoid membrane that is in Stroma - Protons are used to reduce NADP. So number of protons decreases in stroma and number increases in lumen of thylakoid.
- So, concentration gradient is created across the thylakoid membrane.

Function of Proton Gradient :

- Enzyme ATPase has two components : F_0 and F_1.
- F_0 is embeded in the membrane which, works as transmembrane channel.
- F_1 which protrudes on the outer surface of thylakoid membrane facing stroma.
- Abundant energy is produced due to breakage of concentration gradient.
- This conformational change in F_1 particle ATPase makes the enzyme to synthesize several molecules of energy packed ATP. $ADP + Pi \rightarrow ATP$.

Use of ATP :

The ATP produced will be used immediately along with NADPH in the biosynthetic phase taking place in stroma.

---

(47) By splitting of which molecule H^+ formation occurs in thylakoid ?

(A) CO_2 (B) NAD (C) H_2O (D) NADP

(48) Which molecule is reduced in stroma outside the thylakoid membrane according to chemiosmotic hypothesis ?

(A) CO_2 (B) NAD (C) H_2O (D) NADP

(49) What we can understand by chemiosmotic hypothesis ?

(A) Splitting of H_2O (B) Synthesis of ATP in chloroplast

(C) Fixation of CO_2 (D) Reduction of NADP

(50) What is responsible for change in F_1 component of ATPase ?

(A) Presence of proton acceptor $NADP^+$ (B) Noncyclic transportation of electrons

(C) Energy released by proton gradient (D) Enzymes of F_0 component

(51) What is the cause for creation of proton gradient across the Thylakoid membrane ?

(A) Increase in number of protons in thylakoid.

(B) Decrease in number of protons in stroma.

(C) Decrease in pH in cavity of thylakoid.

(D) All of the above

Answers : (47-C), (48-D), (49-B), (50-C), (51-D)

(B) Biosynthetic Phase : CO_2 fixation, Calvin cycle, C_3 path, Dark reaction, Reduction of RuBP.

- This process occurs in stroma of thylakoid.
- Light is not necessary for this process.
- H_2O and CO_2 combines to form sugar.
- Malvin Calvin by using radioactive C^{14} in algal photosynthesis described this process.
- So, this process is known as Calvin cycle.
- They got Nobel Prize in 1961.
- Steps necessary for this process are shown as under :

Main Phases of Biosynthetic Phase :

Carboxylation

6 (Ribulose 1-5 Biphosphate) (RuBP) + $6CO_2$ ⟶ 6 (Unstable product)
(5C) (1C) (6C)

6 (Unstable product) + $6H_2O$ ⟶ 12 (PGA)
(6C) (3C)

Reduction Phase

12 PGA + 12NADPH + 12 ATP ⟶ 12 PGAL + 12NADP + 12 ADP
(3C) (3C)

12 PGAL (3C) ⟶ 10 PGAL (3C) + 2 PGAL (3C)

6 ATP

Regeneration phase ←

6 ADP

3 PGAL (3C) 3-PGAL (3C)

Transportation as per the requirement in different plant organs

3-PGAL (3C) 3-DHAP (3C)

Fructose 1-6 Biphosphate (6C)

Storage in Phloem + Transportation (Sucrose)

ADP
ATP

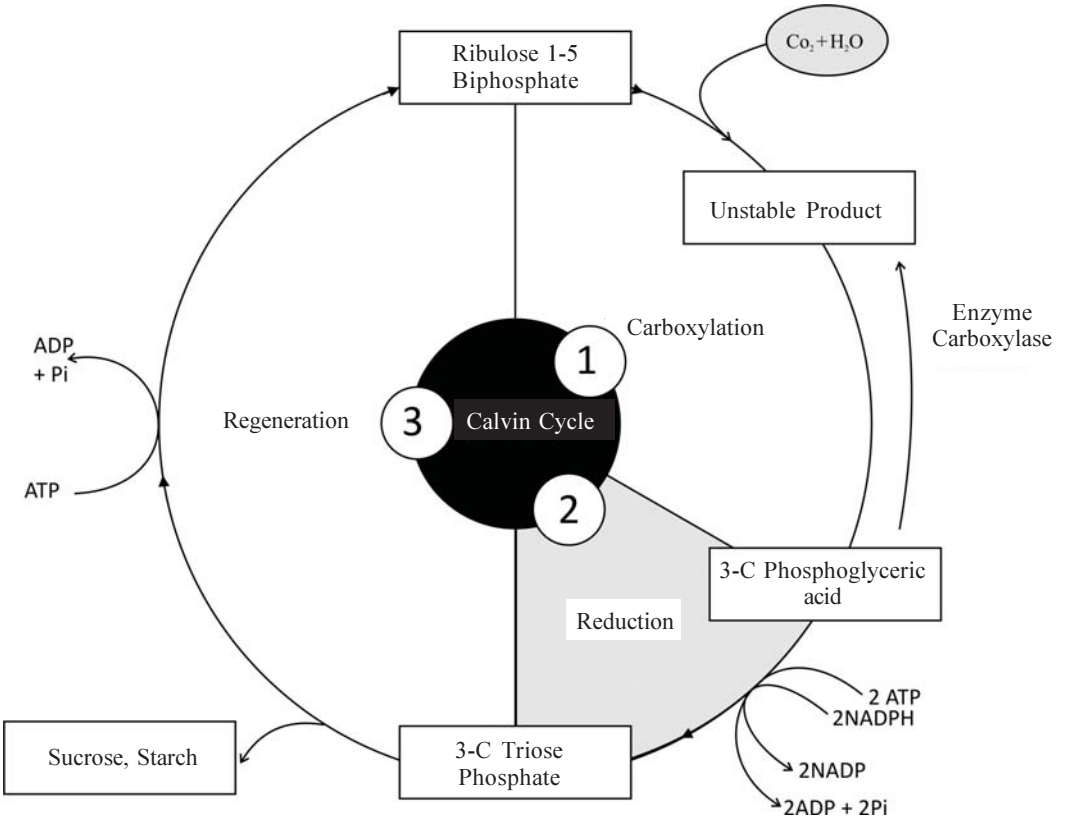
Fructose-6 Phosphate (6C)

Unstable storage in stroma

Glucose-6-Phosphate (6C)

ADP
ATP

Starch ($n \in N$) ← ← Glucose (6C)
($C_6H_{12}O_6$)



(52) In dark reaction of photosynthesis .....

(A) 3C compound is formed by splitting of 6C sugar.

(B) Photolysis of water occurs.

(C) Molecules of chloroplast are stimulated.

(D) Carbonic compounds are formed by reduction of CO_2.

(53) Which radioactive isotope is used to explain C_3 - path ?

(A) C^{12} (B) C^{14} (C) C^{13} (D) C^{16}

(54) Which enzyme is responsible to join CO_2 and RuBP ?

(A) NADP reductase (B) ATPase

(C) RuBP Carboxylase (D) Cytochrome Oxidase

(55) Which sugar is used in first step of Biosynthesis ?

(A) $C_6H_{12}O_6$ (B) RuBP (C) DHAP (D) PGAL

(56) Which type of sugar is used in Calvin cycle ?

(A) Aldotriose (B) Ketotriose (C) Aldopentose (D) Ketopentose

(57) How many ATP and $NADPH_2$ are required for fixation of one CO_2 molecule ?

(A) 3ATP + 1$NADPH_2$ (B) 2ATP + 2$NADPH_2$

(C) 3ATP + 2$NADPH_2$ (D) 3ATP + 3$NADPH_2$

(58) From which molecule 3 - Carbon of phosphoglyceric acid are obtained during carbon fixation ?

(A) PEP + CO_2 (B) CO_2 (C) RuBP (D) RuBP + CO_2

(59) Calvin cycle starts with ..... .

(A) PGA (B) RuBP (C) OAA (D) PGAL

(60) How many ATP and NADPH are necessary for production of one glucose ?

(A) 12, 20 (B) 6, 3 (C) 18, 12 (D) 10, 20

(61) What is the product of carboxylation in stroma of chloroplast ?

(A) PGA (B) PGAL (C) DHAP (D) RuBP

(62) What is the product formed by removal of 'O' from PGA ?

(A) PGAL (B) DHAP (C) RuBP (D) OAA

(63) How many molecules of 3-PGAL are necessary for regeneration of RubP ?

(A) 12 (B) 8 (C) 10 (D) 6

Answers : (52-D), (53-B), (54-C), (55-B), (56-D), (57-C), (58-D), (59-B), (60-C), (61-B), (62-A), (63-C)

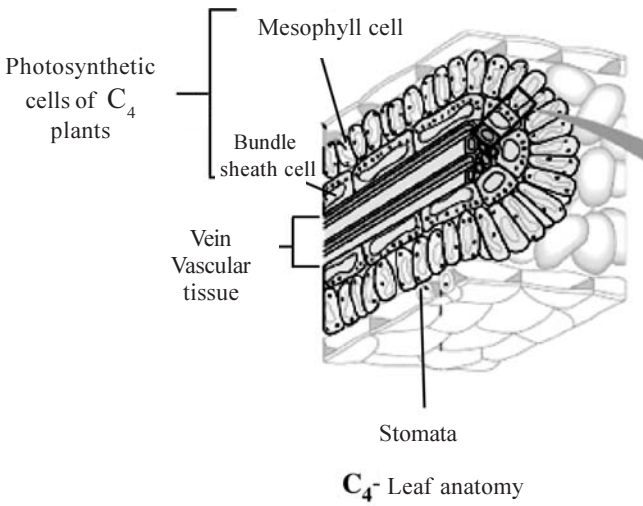
C_4 – Path (Hatch Slack Pathway)

- C_4 pathway is generally observed in the monocot plants which grow in the tropical regions.
- It is very clearly observed in the leaves of sugarcane.
- Special type of Kranz anatomy is seen in C_4 pathway.

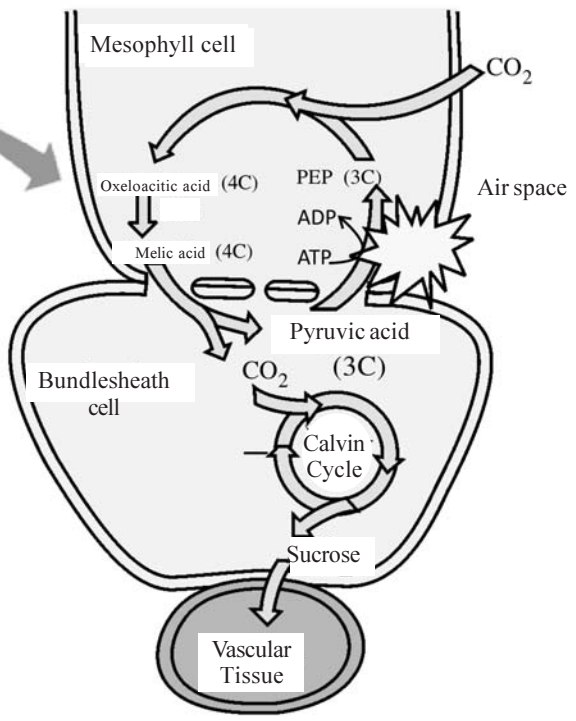
According to Kranz Anatomy :

Bundle Sheath Cell	Mesophyll Cell
• They do not show grana organization. (Only thylakoids are present.)	• They show grana organization. (Thylakoids are arranged in piles.)
• C_3-cycle is obseved for CO_2 fixation.	• Light reaction occurs and OAA is produced.

- This type of arrangement prevents entry of O_2, evolved during light reaction, from entering bundle sheath cells. So, there is no possibility of photorespiration.



C_4- Leaf anatomy



C_4- cycle

Special Characters :

- Due to release of CO_2 in bundle sheath cells, the concentration of CO_2 goes on increasing. As the process of CO_2 fixation occurs in these cells. Also, more food is prepared through biosynthetic fixation.
- As there is no possibility of photorespiration, C_4 is more efficient than C_3.

Differences between C_3 and C_4 path :

No.	Character	C_3 - Path	C_4 - Path
1.	Type of cell	One (Mesophyll)	Two (Mesophyll) and bundle sheath cell
2.	Kranz Anatomy	Does not occur	Occurs
3.	Chloroplast	With grana	With and Without grana
4.	First CO_2 Acceptor	RuBP	PEP
5.	First Product	PGA (C_3)	Oxalo acetic acid (C_4)
6.	Productivity (Food)	Normal	High

(64) Which of the following plant shows C_4 path ?

(A) Sunflower (B) Potato (C) Maize (D) Bignonia

(65) Which of the following plant utilizes maximum light ?

(A) Paddy (B) Sugarcane (C) Wheat (D) Sorghum

(66) How C_4 - cycle is different from C_3 - cycle ?

(A) Photosynthetic product (B) Number of ATP for production of sugar

(C) Molecule of CO_2 fixation (D) Photopigments which play role in photosynthesis

(67) In which part light reaction occurs in C_4 plants ?

(A) cells of Mesophyll (B) In Bundle sheath cells

(C) In epidermal cells (D) In phloem

(68) What is produced by decarboxylation of Malic acid in bundle sheath cells in C_4 plants ?

(A) PGA (B) PEP (C) PEPA (D) Pyruvic Acid

(69) How many Calvin cycles are necessary for production of 1 glucose ?

(A) 2 (B) 3 (C) 4 (D) 6

(70) First and Second CO_2 acceptor in C_4 path are ..... .

(A) RuBP, PEP (B) PEP, RuBP (C) OAA, PEP (D) PEP, OAA

(71) What is the number of ATP for production of 1 glucose in C_4 - pathway ?

(A) 10 (B) 20 (C) 16 (D) 30

(72) How many times biochemical reactions occurs for production of one glucose molecule in C_4 - path ?

(A) 2 (B) 4 (C) 6 (D) 8

(73) By which process Malate is formed from PEP ?

(A) Carboxylation, Dehydrogenation

(B) Carboxylation, Hydrogenation

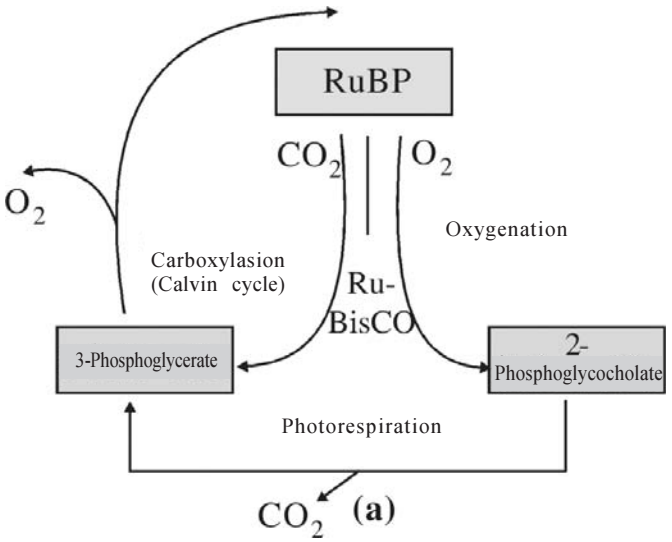
(C) Decarboxylation, Hydrogenation

(D) Decarboxylation, Dehydrogenation

Answers : (64-C) (65-B), (66-C), (67-A), (68-D), (69-D), (70-B), (71-D), (72-C), (73-A)

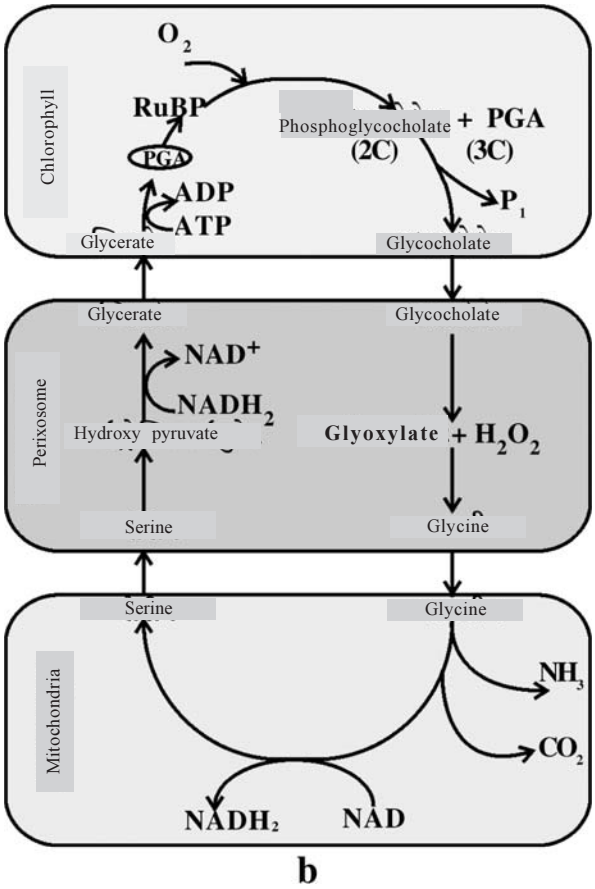
Photorespiration :

- **Definition :** The respiration which occurs in chloroplast in presence of light is known as photorespiration.
- **Location :** Chloroplast, Peroxisome and Mitochondria.
- **Enzyme :** RuBisCo.



- CO_2 and RuBP combines to form 2 PGA in biosynthetic phase which is known as carboxylation.
- Enzyme RuBisCo (Ribulose Biphosphate Carboxylase) is resonsible for this process which is in abundant amount.
- It can bind with O_2 and CO_2.
- In photorespiration it is oxygenase means behaves as enzyme for O_2.

Chart explaining photorespiration



b

Importance :

- RuBP is oxygenated in presence of O_2 forming one molecule of 2C phosphoglycolate and next 1 glycine is formed. Such two glycine × 2 = 4C, 3C serine is formed and 1C is removed as release of CO_2.
- So, Total 4 carbon (100%)

  3 carbon (75%) 1 CO_2 (25%) is wasted
- Under condition of intense light and poor availability of CO_2 photorespiration plays defensive role.
- With insufficient availability of CO_2 and light energy cannot be fully utilized and excess light can cause photooxidation and damage the plant. Photorespiration provides protection against this.
- In C_3 plants some O_2 binds to RuBisco and so CO_2 fixation is decreased.
- In C_4 plants photorespiration does not occur. This is because they have a mechanism that increases the concentration of CO_2 at the enzyme site.
- This ensures that RuBisco functions as Carboxylase minimizing Oxygenase activity.
- So productivity is C_4 is more.

(74) What decreases photorespiration ?

(A) Fe^{+2} (B) Mg^{+2} (C) H_2O (D) CO_2

(75) Photorespiration is harmful to plants ? Why ?

(A) ATP production does not occur. (B) RuBP is decomposed.

(C) CO_2 is synthesized. (D) Amino acid is decomposed.

(76) Photorespiration occurs.....

(A) At low temperature (B) More O_2 and less CO_2

(C) Low intensity of light (D) Less O_2 and more CO_2

(77) What is produced by dephosphorylation of phosphoglycolate during photorespiration ?

(A) Glycerate (B) Glycine (C) Glycolate (D) Glyoxylate

(78) In which orgenelle CO_2 is released in photorespiration ?

(A) Mitochondria (B) Chloroplast (C) Peroxisome (D) Lysosome

(79) Which is the substrate for photorespiration ?

(A) Glycine (B) Serine (C) Glycolate (D) RuBP

(80) By which process Glycolate is formed from phosphoglycolate ?

(A) Reduction (B) Decarboxylation (C) Phosphorylation (D) Dephosphorylation

(81) In photorespiration, Glycolate is produced in ..... .

(A) Chloroplast (B) In mitochondria (C) In cytoplasm (D) In peroxisome

(82) Which product is formed when RuBP is oxygenated ?

(A) Phosphoglycolate, PGA (B) PGA, DHA

(C) Phosphoglycolate, Serine (D) Glyceric Acid, PGA

(83) Which of following is not true for photorespiration ?

(A) It occurs in presence of light (B) It occur in chloroplast

(C) Speciality of C_3 plants (D) Speciality of C_4 plants

Answers : (74-D), (75-B), (76-B), (77-C), (78-A), (79-C), (80-D), (81-A), (82-A), (83-D)

Factors affecting photosynthesis :

The rate of photosynthesis is very important in determining the yields of plants including crop plants. Photosynthesis is under the influence of several factors, both internal and external. When several factors affect any (bio) chemical process.

(1) Chemical process :

- Blackman's (1905) law of limiting factors comes into effect. This states the following :
- "If a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value : it is the factor which directly affects the process it its quantity is changed."
- For example, despite of presence of green leaf, optimal light and CO_2 conditions, photosynthesis may not occur, it temperature is very low. This leaf, if given the optimal, temperature, will start photosynthesizing.

(2) Light :

- Both, the intensity and quality of light influence photosynthesis : Normally as intensity of light increases, the rate of photosynthesis increases. However, at higher intensity it is not so. At high intensity of light, oxidation of chlorophyll, occurs and it is decomposed. This is called photooxidation. Because of this the rate of the process drops.
- The visible spectrum of light ranges from 400 nm to 700 nm plants can absorb light from this range only. The rate increases in orange and red light. It decreases in green light.

(3) CO_2 Concentration :

The concentration of CO_2 is very low in the atmosphere i.e. 0.036%. Increase in the concentration upto 0.05% can cause an increase in CO_2 fixation rates. Beyond this the levels for can become damaging over longer periods.

- At high light intensity both, C_3 and C_4 plants show increase in the rates of photosynthesis.
- C_3 plants responds to increased CO_2 concentration and saturation is seen only beyond 450 $\mu l\ L^{-1}$.
- C_4 plants show saturation at about 360 $\mu l\ L^{-1}$.

(4) Temperature :

(a) There is no notable effect of temperature in the photochemical phase but as dark reaction or biosynthetic phase being enzymatic, temperature has a remarkable effect on it.

(b) The C_4 plants respond to higher temperatures and show higher rate of photosynthesis while C_3 plants have a much lower temperature optimum.

(5) Water :

- When availability of water is reduced, the plant experiences water stress.
- One is that the stomata are closed, as a result the amount of available CO_2 is reduced.
- Secondly, as water potential in leaves is lowered, they curl. Thus available leaf area is reduced for photosynthesis.

---

(84) In which temperature range rate of photosynthesis is maximum ?

(A) 10 ^{0}C to 20 ^{0}C (B) 5 ^{0}C to 10 ^{0}C (C) 10 ^{0}C to 25 ^{0}C (D) 10 ^{0}C to 15 ^{0}C

(85) Which plants shows decreasing rate of photosynthesis in CO_2 concentration ?

(A) C_3 plants (B) C_4 plants (C) Aquatic plants (D) CAM plants

(86) Maximum absorption of light occurs at ..... wavelength.

(A) 400 nm 700 nm (B) 600 nm 1200 nm (C) 600 nm 700 nm (D) 500 nm 900 nm

(87) At high intensity of light ..... .

(A) Rate of photosynthesis increases (B) Rate of photosynthesis decreases

(C) Oxidation of chloroplast occur (D) Reduction of chloroplast occur

(88) What happens when the amount of water decreases in plants ?

(A) Closing of stomata (B) Decrease in CO_2 concentration

(C) Rolling of leaves as water potential decrease (D) All of the above

(89) What happens when the temperature increases in the atmosphere ?
(A) Rate of photosynthesis decreases in C_4 plants.
(B) Rate of photosynthesis decrease in C_3 plants.
(C) Rate of photosynthesis increases in C_4 plants.
(D) Rate of photosynthesis becomes minimum in C_3 plants.

(90) When does C_4 plant show saturation ?
(A) 360 μl^{-1} of CO_2 (B) C 350 μl^{-1} of CO_2 (C) 400 μl^{-1} of CO_2 (D) 436 μl^{-1} of CO_2

(91) ..... is the important for structure of chlorophyll.
(A) Magnesium (B) Nitrogen (C) Phosphorus (D) Iron

(92) Which process of photosynthesis show effect of temperature ?
(A) Light reaction (B) Dark reaction
(C) Cyclic photophosphorylation (D) Photorespiration

Answers : (84-C), (85-B), (86-A), (87-C), (88-D), (89-C), (90-A), (91-B), (92-B)

A - Assertion, R - Reason.

Choose the correct option :

(A) A and R both are true and R is the correct explanation of A.

(B) A and R both are correct, R is not correct explanation of A.

(C) A is true, but R is false.

(D) A is false, but R is true.

(93) Assertion A : ATPase present in thylakoid membrane in chloroplast possess two component.
Reason R : F_0 is embedded in the membrane and F_1 protudes on the outer surface of thylakoid membrane.
(A) (B) (C) (D)

(94) Assertion A : During photorespiration NAD is reduced in peroxisome and NADH is dehydrogenated in mitochondria.
Reason R : During photorespiration ATP is dephosphorylated to form PGA from glycerine.
(A) (B) (C) (D)

(95) Assertion : O_2 is released from $4OH^-$ produced from photolysis of water.
Reason R : NADP are reduced by $4OH^+$ of photolysis of water.
(A) (B) (C) (D)

(96) Assertion A : 6 times kelvin cycle occur for formation of 1 glucose.
Reason R : 18 ATP and 12 $NADPH_2$ are utilised for formation of 1 glucose.
(A) (B) (C) (D)

(97) Assertion A : Rubisco is more active as carboxylase and less active as oxigenase.
Reason R : Productivity is more in C_4 pathway.
(A) (B) (C) (D)

(98) Assertion A : There is no possibility of photosynthesis if termperature is less though amount of CO_2 and light is minimum.
Reason R : Plants follow blackman's law of limiting factor.
(A) (B) (C) (D)

(99) Assertion A : If one molecule of RuBP and oxygen occurs the last product is glycine and PGA.

Reason R : If one molecule of RuBP and O_2 the last product is PGA and phosphoglycolate.

(A) (B) (C) (D)

(100) Assertion A : In C_4 plants PEP is the first CO_2 acceptor and RuBP is last acceptor.

Reason R : In C_3 plants RuBP is first CO_2 acceptor.

(A) (B) (C) (D)

(101) Assertion A : The rate of photosynthesis is less at high intensity of light.

Reason R : At high intensity of light chloroplast oxidation occur.

(A) (B) (C) (D)

(102) Assertion A : In light reaction absorption of light, photolysis, release of oxygen, formation of ATP and NADP occurs.

Reason R : LHC creates PS-I and PS-II

(A) (B) (C) (D)

Answers : (93-A), (94-D), (95-B), (96-A), (97-A), (98-B), (99-A), (100-C), (101-D), (102-B)

True - False (T - F) type questions :

Choose the correct True and False :

(103) (1) Kranz anatomy is seen in C_4-plants.

(2) Melvin-Kelvin found C_4-path.

(3) Photorespiration can be seen in C_3-plants.

(4) 6 ATP and 4 $NADPH_2$ are necessary for fixation of $2CO_2$ in C_3 cycle.

(A) F T T F (B) T F T T (C) T T F T (D) F T F T

(104) (1) Photolysis occur in stroma of chloroplast.

(2) $NADPH_2$ produced by photolysis are used in light reaction.

(3) Electron and NADP are produce from PS-I are utilise in photolysis.

(4) Mn^{+2}, Ca^{+2} and Cl^- are necessary for photolysis of water.

(A) F T F T (B) T T T F (C) F F T T (D) F T T F

(105) (1) The first stable product of bio synthetic is PGA.

(2) Non cyclic photophosphorylation occurs in stroma of chloroplast.

(3) Rate of photosynthesis is more in red light.

(4) Light energy is converted into chemical energy during photosynthesis.

(A) T F T T (B) T F F T (C) F F T F (D) F T F T 111

(106) (1) Light reaction occurs in grana of chloroplast.

(2) Temperature factor affects biosynthetic phase.

(3) Chlorophyll-b works as reaction centre.

(4) Productivity of C_4 is less comparatively to C_3 plants.

(A) F T T F (B) T T F T (C) F T F T (D) T T F F

(107) (1) Hatch-Slack found C_3 plants.

(2) Photorespiration does not occur in C_4 cycle.

(3) Law of limiting factor is found by blackman.

(4) The plants which use PEP during CO_2 fixation are called C_4 plants.

(A) F T T F (B) T F T F (C) F T T T (D) T F F T

(108) Choose Correct pair :

	Column - I	**Column - II**	
P.	Grana region of Chloroplast	(i) Kreb's cycle	(A) P - ii, Q - iii, R - iv, S - i
Q.	Stroma region of Chloroplast	(ii) Light reaction	(B) P - i, Q - ii, R - iii, S - iv
R.	Cytoplasm	(iii) Dark reaction	(C) P - ii, Q - i, R - iv, S - iii
S.	Mitochondrial cytoplasm	(iv) Glycolysis	(D) P - iii, Q - iv, R - i, S - ii

(109) Choose Correct pair :

	Column - I		**Column - II**	
P.	Carboxylation	(i)	Production of O_2	(A) P - ii, Q - ii, R - iii, S - iv, T - v
Q.	Photophosphorylation	(ii)	Photorespiration	(B) P - iii, Q - v, R - i, S - ii, T - iv
R.	Photolysis	(iii)	RuBiSCo	(C) P - i, Q - iii, R - iii, S - iv, T - ii
S.	Phosphoglycolate	(iv)	Chemosynthesis	(D) P - i, Q - iii, R - iv, S - ii, T - v
T.	Nitrosomonas	(v)	ATP	

(110) Choose Correct pair :

	Column - I		**Column - II**	
1.	PS-I	P.	Chemiosmotic hypothesis	(A) 1 - Q, 2 - R, 3 - P, 4 - S
2.	PS-II	Q.	Cyclic photophosphorylation	(B) 1 - Q, 2 - S, 3 - P, 4 - R
3.	Cytochrome-b	R.	Present in thylakoid	(C) 1 - Q, 2 - R, 3 - S, 4 - P
4.	ATPase	S.	Photolysis of water	(D) 1 - R, 2 - S, 3 - Q, 4 - P

(111) Choose Correct pair :

	Column - I		**Column - II**
1.	Cyclic photophosphorylation	P.	O_2 is released and NADP is reduced
2.	Biosynthetic phase	Q.	Depends on concentration of O_2 and CO_2
3.	Non cyclic photophosphorylation	R.	Formation of 2ATP
4.	Photorespiration	S.	Photorespiration phase occurs in stroma

(A) (1-R), (2-P), (3-Q), (4-S) (B) (1-r), (2-S), (3-Q), (4-P)

(C) (1-R), (2-S), (3-P), (4-Q) (D) (1-P), (2-R), (3-Q), (4-S)

(112) Choose Correct pair :

Column - I	**Column - II**
1. $C_6H_{12}O_6$	P. Regenerated during process of C_3 cycle
2. OAA	Q. Amino acid produced in peroxisome
3. RuBP	R. Product of photosynthesis
4. Glycin	S. First stable product of C_4 path

(A) (1-R), (2-S), (3-Q), (4-P) (B) (1-R), (2-S), (3-P), (4-Q)
(C) (1-S), (2-R), (3-P), (4-Q) (D) (1-R), (2-P), (3-DS), (4-Q)

(113) Choose Correct pair :

Column - I	**Column - II**
P. C_4 Chloroplast	p. RuBP
Q. First acceptor of C_3	q. With Grana and without grana
R. First acceptor of C_4	r. Possess Grana
S. C_3 Chloroplast	s.. PEP

(A) P - s, Q - r, R - p, S - q (B) P - r), Q - s, R - p, S - q
(C) P - q), Q - p, R - s, S - r (D) A - q), Q - p, R - r, S - s

(114) Choose Correct pair :

Column - I	**Column - II**
1. 1733-1804	p. Time period of priestly
2. 1897-1985	q. Time period of Ingenhousz
3. 1854	r. Time period of Julius Von Sachs
4. 1730-1799	s. Time period of Cornelius Van Niel
	t. Time period of Robert Hill

(A) (1-p), (2-s), (3-r), (4-q) (B) (1-p), (2-q), (3-r), (4-s)
(C) (1-p), (2-s), (3-t), (4-q) (D) (1-p), (2-r), (3-s), (4-q)

(115) Choose Correct pair :

Column - I	**Column - II**
1. Photolysis of water $\rightarrow$	p. Non cyclic photophosphorylation
2. ATP ase $\rightarrow$	q. Peroxisome
3. RuBisCo $\rightarrow$	r. Mitochondria
4. NAD^+ reduction $\rightarrow$	s. Thylakoid membrane
	t. Stroma

(A) 1 - t, 2 - p, 3 - q, 4 - p (B) 1 - t, 2 - r, 3 - s, 4 - p
(C) 1 - p, 2 - s, 3 - t, 4 - r (D) 1 - p, 2 - s, 3 - t, 4 - q

(A) T. P, Q, R (B) T, R, S, P (C) P, S, T, R (D) P, S, T, Q

Answers : (103-B), (104-C), (105-A), (106-D), (107-C), (108-A), (109-B), (110-B), (111-C), (112-B), (113-C), (114-A), (115-D)

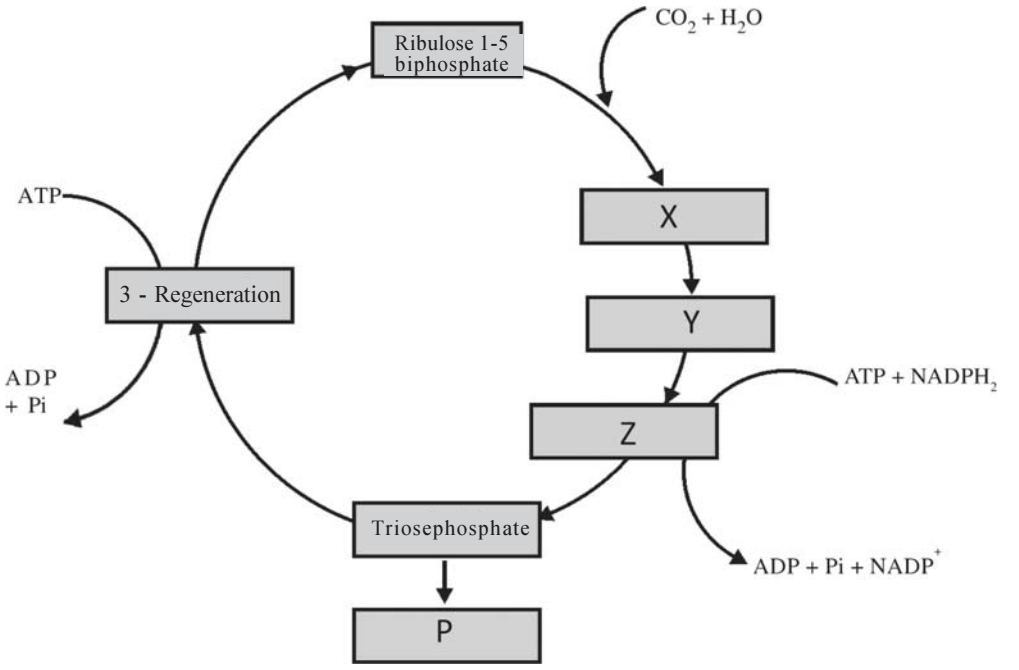


Chart-1

(116) What is shown in the chart-1 ?

(A) Reduction (B) Oxidation (C) Carboxylation (D) Decarboxylation

(117) Which process is indicated by Z in chart-1 ?

(A) Reduction (B) Carboxylation (C) Oxidation (D) Decarboxylation

(118) Which process is indicated by Y in chart-1 ?

(A) 3-Phosphoglycerate (B) 2-Phosphoglycolate (C) Triosphosphate (D) Decarboxylation

(119) Name chart-1

(A) C_4 path (B) Kreb's cycle (C) CAM pathway (D) Calvin cycle

(120) What is indicated as P in chart-1 ?

(A) Sucrose, Starch (B) Fructose, Starch (C) Maltose, Starch (D) Glucose, Starch

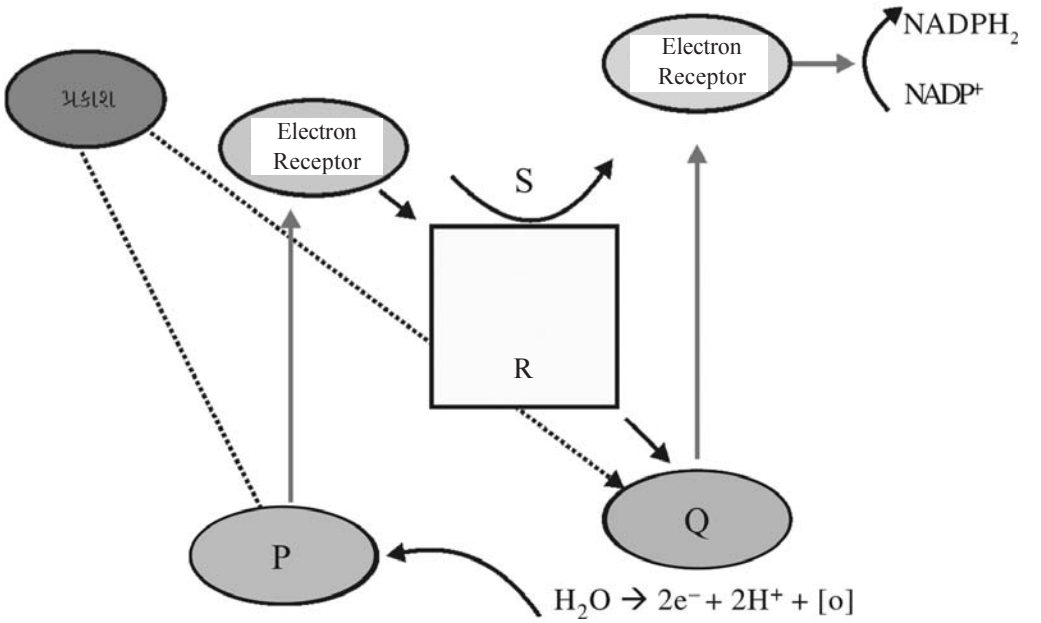


Chart-2

(121) Name chart-2

(A) Cyclic photophosphorylation (B) Photorespiration

(C) Non-cyclic photophosphorylation (D) Photolysis

(122) What is indicated by P in chart-2 ?

(A) P_{700} (B) P_{680} (C) P_{670} (D) P_{750}

(123) What is indicated as Q in chart-2 ?

(A) P_{700} (B) P_{750} (C) P_{680} (D) P_{580}

(124) What is indicated by R in chart-2 ?

(A) Thylakoid membrane (B) Cytoplasmic stroma

(C) Electron transport system (D) Mitochondria

(125) Which process is indicated by S in chart-2 ?

(A) ATP + ADP + Pi (B) ADP + Pi $\rightarrow$ ATP

(C) $2H_2O_2 \rightarrow 2H_2O + O_2$ (D) $CO_2 + H_2O \rightarrow CH_2O + O_2$

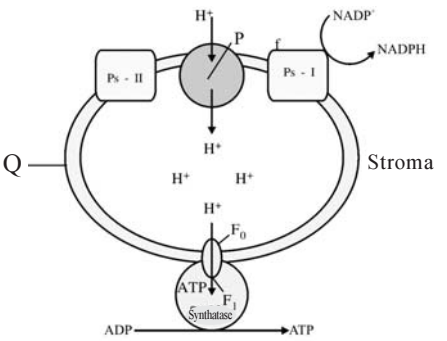
(126) What is indicated by P and Q in the chart ?

(A) P – Cytochrome A & A, Q – Thylakoid membrane

(B) P – Thylakoid membrane, Q – Cytochrome A & A

(C) P – Cytochrome b & f, Q – Cytochrome a & a

(D) P – Stroma, Q – Thylakoid membrane



(127) Which process is indicated by this figure ?

(A) ATP synthesis by chemiosmosis (B) Food synthesis by chemiosmosis

(C) Photolysis by chemiosmosis (D) Electron transport by chemiosmosis

Answers : (116-C), (117-A), (118-A), (119-D), (120-A), (121-C), (122-B), (123-A), (124-C), (125-B), (126-C), (127-A)

Questions for NEET :

(128) Which of the following step is necessary for photosynthesis ?

(A) ATP formation

(B) Photolysis of water

(C) Production of chlorophyll with the help of photon

(D) Production of $C_6H_{12}O_6$

(129) Who found the laws of limiting factors of photorespiration ?

(A) Hill (B) Blackman (C) Niel (D) Calvin

(130) How many H^+ ions are formed by photolysis of $12H_2O$ during Non-cyclic photophosphorylation ?

(A) $24H^+$ (B) $12H^+$ (C) $36H^+$ (D) $32H^+$

(131) In which light the rate of photosynthesis is more ?

(A) Blue (B) Green (C) Orange or Red (D) Blue or Green

(132) What happens due to absorption of light ?

(A) Release of O_2 (B) Absorption of CO_2

(C) Oxidation of chlorophyll (D) Reduction of chlorophyll

(133) What is correct for C_4 plant ?

(A) CO_2 combines with PEP (B) CO_2 combines with PGA

(C) CO_2 combines with PGAL (D) CO_2 combines with RuBP

(134) How many glycine molecules are necessary to release CO_2 during photorespiration ?

(A) 1 (B) 3 (C) 2 (D) 4

(135) Where does the process of Reduction of NAD occurs during photorespiration ?

(A) Chloroplast and Peroxisome (B) Chloroplast and Mitochondria

(C) Mitochondria and Peroxisome (D) Mitochondria

(136) What happens when we put plants in lower intensity of light ?

(A) Photosynthetic unit becomes more active because of light.

(B) More CO_2 fixation occurs in presence of light in plants.

(C) Root system develops more.

(D) Leaves are converted into thorns.

(137) Where we find Hill-reaction ?

(A) Higher level of plants (B) In complete darkness

(C) During waterstress (D) In presence of fatty cyanide

(138) Which of the following is incorrect for photorespiration ?

(A) Associated with chloroplast (B) Occurs during daytime

(C) Speciality of C_4 plants (D) Speciality of C_3 plants

(139) In which organism chlorophyll is absent during photosynthesis ?

(A) Bacteria (B) Cyanobacteria (C) Red algae (D) Brown algae

(140) Which minerals are necessary for the formation of chlorophyll ?

(A) Fe, Mg (B) Ca, K (C) Fe, Ca (D) Ca, Cu

(141) What is true for puruvic acid in CAM plants ?

(A) Increases during day time (B) Decreases during night

(C) Decreases during day time (D) Increases during night

(142) How many e^- are utilise for formation of 1 mole glucose and $6O_2$ molecule ?

(A) 6 (B) 12 (C) 18 (D) 24

(143) What is present in blue algae ?

(A) Chlorophyll (B) Xanthophyll (C) Phycocyanin (D) Fucoxanthnin

(144) Choose the correct sequence for Kelvin cycle :

(1) ATP and NADPH formed during photosynthesis are utilised and forms carbohydrate.

(2) It is the process in which Ribulose 1, 5 biphosphate except CO_2 and RuBP is regenerated.

(3) It is the process in which ribulose 1, 5 biphosphate bind with CO_2 or carboxylation occur.

(A) 3 – 1 – 2 (B) 3 - 2 - 1 (C) 1 - 2 - 3 (D) 2 - 1 - 3

(145) Which of the following statement is appropriate for light and process of photosynthesis ?

(A) Cytochrome a stimulates 680 nm, PS-I stimulates 700 nm, PS-II absorb move photons.

(B) Na^+ and Mg ions are necessary for photolysis of water.

(C) O_2 participates in photophosphorylation.

(D) PS-I and PS-II both participate in iron cyclic photophosphorylation.

(146) Productivity of C_3 plant are more because .....

(A) Photorespiration is less in it (B) Leaf area is more

(C) More amount of chloroplast occur in leaves (D) Thin cuticle layer is present

(147) Which enzyme is responsible for fixation of CO_2 with PEP in mesophyll of C_4 plants ?

(A) Pyruvate dehydrogenase (B) PEP Carboxylase

(C) Pyruvate decarboxylase (D) RuBisco

Answers : (128-C), (129-B), (130-C), (131-C), (132-C), (133-A), (134-C), (135-B), (136-A), (137-A), (138-C), (139-B), (140-A), (141-A), (142-B), (143-A), (144-A), (145-D), (146-C), (147-D)

Photosynthesis

Formation of glucose from Co_2 and water in the presence of sunlight and chlorophyll

Factors affecting Photosynthetisis

- External Factors
  - Light, Co_2 conc, temperature, Water, O_2 etc.
- Internal Factors
  - Chlorophyll conc, leaf age, phytohormones, etc.,

Site of photosynthesis

Chloroplasts that consists of Stroma, Grana stacks, Thylakoids and stroma lamellae.

Thylakoids contain Photosynthetic pigments.

Photosynthetic Pigments

Photosynthetic pigments include Chlorophylls, Carotenes and phycobilins. These pigments are packed in two major pigment system.

- P_{S-I} — Includes P_{700} as photocentre and light harvesting complex (LHC) with electron carriers
- P_{S-II} — Includes P_{680} as photocentre and LHC

Photosynthetic Mechanism

An enzymatic anabolic reaction involving many intermediates

- Photochemical Phase
  Light dependent phase occuring in Thylakoids
  - Photolysis
    Water molecule is broken down into H^+ and OH^- ions by absorbing light energy. It produces e^-
  - Photophosphorylation
    Electons released in photolysis of water are used for production of ATP
    - Cyclic photophosphorylation
      Only P_{S-I} involves in electron transport and producs only ATP
    - Non-cyclic photophos phorylation
      Both P_{S-I} and P_S-II involve and produce both ATP and NADPH

Biosynthetic Phase

Light independent reaction that occurs in Stroma

- C_3 Cycle
  Also called Calvin cycle. Co_2 combines with RuBP and produces glucose. 3C phosphorlyceric acid is first stable product
- C_4 Cycle
  Glucose is formed by Co_2 fixation but initial acceptor is PEP in mesophyll calls and first stable product is 4C Oxaloacetic acid. C_4 plants have Kranz Anatomy and are more efficient than C_3 plants.
- CAM Cycle
  In xerophytes, CO_2 is fixed at night. It helps in reducing transpiration.

Photorespiration

Light dependent oxygenation of RuBP and releases CO_2 in photosynthetic cells of C_3 plants.

21 Respiration

Word understanding and Scientific contribution

If we distinguish the difference between respiration and respiratory system then,

Respiration = Cell respiration (in which cell organell like mitochondria and cytoplasm take part) where as in respiratory system different organs, tissue and cell participates in it.

- In another words, respiration is a complete biochemical process, where as respiratory system process can be physical as well as chemical process.
- Respiration is attached with ATP process of synthesis.
- Respiratory system is attached with transportation of O_2 and (CO_2 + H_2O) that is respiration.
- Respiration is of two types aerobic (with oxygen) anerobic (without oxygen).
- In respiratory system this type of system are not possible.
- Most importantly respiration is an essential process for both plant kingdom and animal kingdom.
- Where as respiratory system is associated with metabolism process in animal kingdom. Phisiological process = Biochemical process.

Metabolism (Physiological process biochemical process)

- In various organisms different Biologial processes occurs that is known as metabolism.

Endergonic process : Energy storing process

Example : Photosynthesis

Exergonic process : Energy releasing process

Example : Cell respiration

Respiratory substrates : The substances which are oxidized during respiration with in a cell are known as respiratory substrates.

(1) What things occur during respiration ?

(A) Energy stored in form of ADP

(B) Free energy stored in form of ATP

(C) Energy released

(D) Energy used

(2) During which process in a cell, the breakdown of C-C bonds of complex compounds, release considerable amount of energy.

(A) Redox (B) Oxidation

(C) Hydrogenation (D) Reduction

(3) What is the respiration rate in animals as compare to plants parts ?

(A) More (B) less

(C) Speedy (D) Very slow

(4) Plants don't have particular respiratory organ because

(A) As compare to animals, in plants respiration rate and exchange of gases is more.

(B) Each plant part is directly involved in exchange of gases.

(C) Availability of CO_2 is not a problem because CO_2 is released during photosynthesis.

(D) A and B both.

(5) Which of the following process is not related to respiration.

(A) Respiration is a catabolic process.

(B) Respiration is a exergonic process.

(C) The breakdown of C-C bonds of complex compounds through oxidation within the cells releases considerable amount of energy.

(D) Energy released during this process is utilized for the synthesis of ATP.

(6) Which of the following reaction is an aerobic respiration ?

(A) $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6CO_2$

(B) $C_6H_{12}O_6 \rightarrow 6O_2 + 6\ CO_2 + 6H_2O$ + Energy

(C) $C_6H_{12}O_6 + 6O_2 \rightarrow 2C_2\ H_5OH + 6O_2$ + Energy

(D) None of these

(7) What is respiration ?

(A) Catabolic process, in which CO_2 is, used O_2 is released and free energy is converted into ATP.

(B) Anabolism process in which O_2 and CO_2 are used to form ATP.

(C) Anabolism process in which O_2 is used, CO_2 is released and free energy is converted into ATP.

(D) Catabolism process in which O_2 is used, CO_2 is released and free energy is converted into ATP.

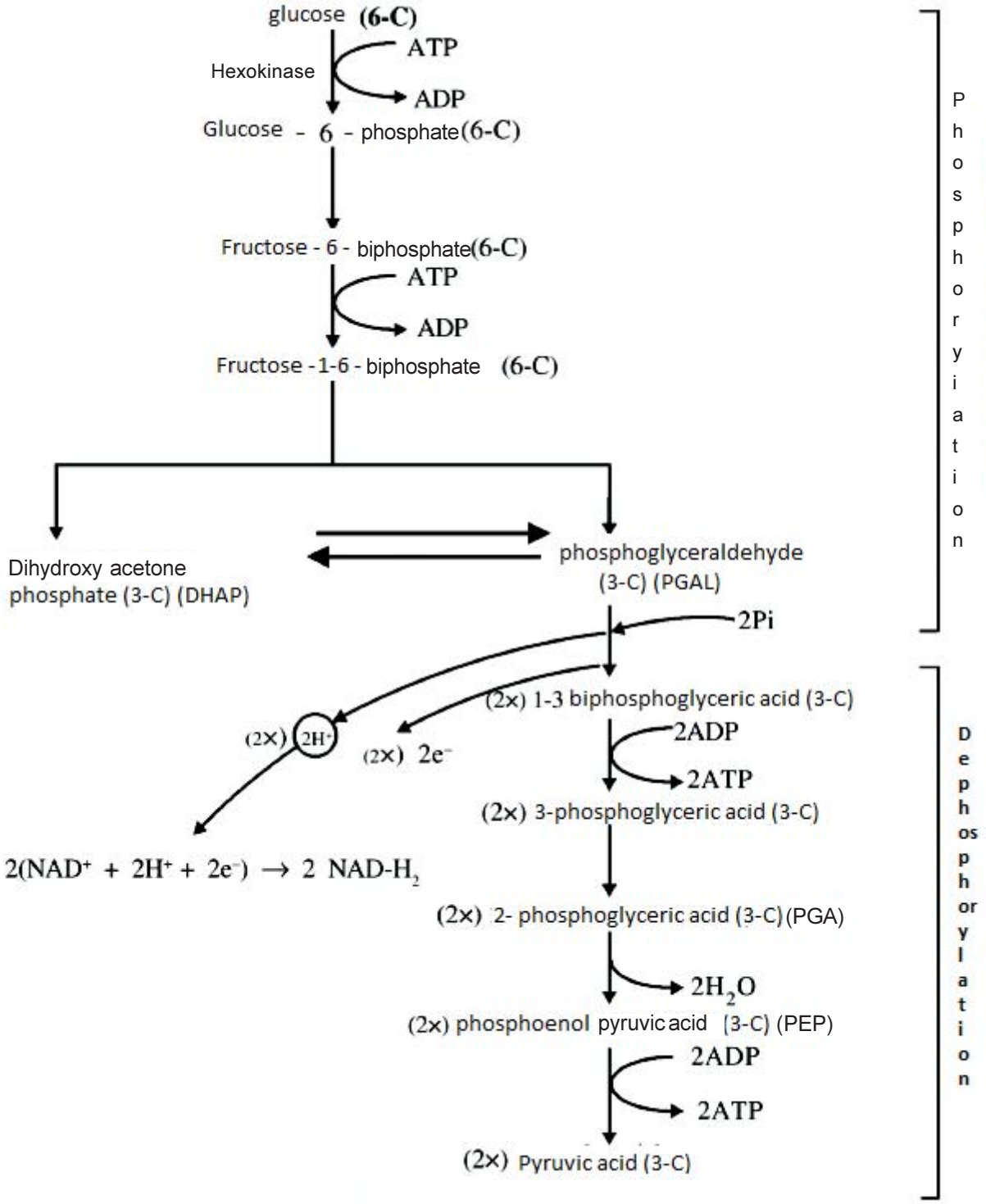
Answers : (1-B), (2-B), (3-D), (4-C), (5-B), (6-B), (7-D)

Glycolysis - EMP - Path

- It is the first step of respiration.
- It is the first step of aerobic and anerobic respiration.
- It is an anerobic phase of aerobic respiration (during glycolysis, presence of O_2 is not required).
- All enzymes are present in cytoplasm which are required during cellular respiration.
- In the end of glycolysis 2 pyruvic acid are formed.
- 2 molecules of ATP are consumed and 4 molecules of ATP are formed. Thus there is a net gain of 2 ATP. (4 ATP - 2ATP = 2ATP)

Formula :

- $C_6H_{12}O_6 + 2NAD^+ + H_3PO_4\ (P_i) + 2\ ATP \rightarrow 2CH_3CO{\cdot}COOH + 2NADH_2 + 4ATP$



Glocolysis

Glycolysis

Sr.	Process	Substrate	Enzyme	Product
1.	Phosphorylation	Glucose + ATP	Hexokinase + Cofactor Mg^{+2}	Glucose-6-phosphate (6 C) + ADP
2.	Isomerization	Glucose 6 phosphate	phosphogluco isomerase	Fructose-phosphate (6 C)
3.	Phosphorylation	Fructose-6-phosphate + ATP	phospho Fructo kinase + Mg^{+2}	Fructose 1, 6 biphosphate + ADP (6 C)
4.	Division	Fructose l, 6, biphosphate phosphoglycera-	Aldolase ⟶	3- phosphoglyceraldehyle + Dihydroxyacetone phosphate (DHAP)

According to above reaction there are 2 molecules of PGAL are fromed. This molecules are used in further reactions.

5.	Phosphorylation	3-PGAL + H_3PO_4 (Pi)	Phosphoglycerokinase	1,3 biphosphoglyce-raldehyde (1,3-PGAL) (3C)
6.	Dihydroginase	1,3 BPGAL+ NAD+	Glyceraldehyde phos-phateDehydrogenase	1,3 disphosphoglycerate (3C) + $NADH_2$
7.	Dephosphorylation	1,3 BPGA + ADP	Dephosphoglyce rodinase	3- PGA + ATP 3- Phosphoglycerate (3C)
8.	Isomerization	3-PGA	Phosphoglycerate Mutase (3-PGA)	2-PGA - (3C) 2- Phosphoglycerate (3C) (3-PGA) - 8
9.	Dehydration (H_2O Remove)	2PGA	Co-factor Enolase+ Mg^{+2}	2 Phosphoenol pyruvate + H_2O
10	Dephosphorylation	2 PEP + ADP	Pyruvat kinase + Mg^{+2}, K^+	Pyruvate + ATP (3C)

- Difference between aerobic and anerobic respiration

	Aerobic Respiration`	Amerobic Respiration
1.	Complete Breakdown of Glucose	– Incomplete breakdown of Glucose
2.	Process in presence of oxygen	– Process in absence of oxygen
3.	Large Amount of energy in form of ATP is released	– Small amount of energy ATP is released
4.	End product CO_2, H_2O and ATP	– End product ethylalcohol, CO_2, lactic acid
5.	This process occurs in higher organisms	– This process occurs in parasite and lower organism
6.	This process occurs in Cytoplasm and mitochondria	– This process occurs in Cytoplasm
7.	Economic value of product is high	– Economic value of product is low
8.	In this process glycolysis and krebs cycle and Oxidative phosphorylation phases are occurs.	– In this process only glycolysis and fermentation occurs

(8) Which bio-chemical reaction is seen in cytoplasm matrix ?

(A) Calvin cycle (B) Amphibolic cycle (C) Glycolysis (D) Krebs cycle

(9) Which enzyme is responsible for the glucose phosphorylation ?

(A) Isomarease (B) Hexokinase (C) Transferase (D) lyasis

(10) In which process of glycolysis the ATP is used ?

(A) Phosphoryslation in cytoplasm matrix (B) Oxidative phosphorylation

(C) Phosphorylation (D) Dephosphorylation

(11) Which enzyme convert sucrose into glucose and froctose ?

(A) Isomerase (B) Invertase (C) Hexokinase (D) Dehydrogenase

(12) Which bio chemial reaction of glycolysis the CO_2 is released ?

(A) 3- Phosphoglyceric acid $\rightarrow$ 2 Phospho glycoric acid

(B) 3-Phosphoglyceraldihyde $\rightarrow$ 3 Phosphoglyceri acid

(C) 2-Phosphoglyceric acid $\rightarrow$ 2 Phosphoenol pyruvic acid

(D) 2-Phosphoenol pyruvic acid $\rightarrow$ Pyruvic acid

(13) How many ATP produce in dephosphorylation of glycolysis ?

(A) 3 ATP (B) 2 ATP (C) 4 ATP (D) 6 ATP

(14) Which is first phase of glucose splitting in eukaryotic organisam ?

(A) Oxidative phosphorylation (B) Krebs cycle

(C) Glycolysis (D) ETS

(15) Glycolysis is connect with

(A) Only in aerobic (B) only in anaerobic

(C) Mitochonotion matrix (D) Cytoplasm matrix

(16) Which type of sugar is formed when fructose 1-6 biphosphate splits in glycolysis ?

(A) Ketotroose (B) aldotriose (C) Pentose sugar (D) A and B both

(17) Which is common path of aerobic and anaerobic respiration.

(A) TCA-Cycle (B) Calvin cycle

(C) Glycolysis (D) Oxidative phosphorylation

(18) In glycolysis process ..... oxidation and ..... reduction process is seen.

(A) 2, 1 (B) 1, 0 (C) 1, 2 (D) 0, 1

(19) Which molecule is produce by phosphorylcation anaerobic phase of erobic respiration ?

(A) DHAP (B) 1,3 PGA

(C) PGAL (D) Fructose 1-6 biphosphate

(20) Released phosphate from ATP is connect with which glucose carbon of glucose ?

(A) 1 (B) 4 (C) 5 (D) 6

(21) How many times the phosphorylation and dephosphorylation occur in glycolysis ?

(A) 2 and 3 (B) 3 and 4 (C) 4 and 4 (D) 3 and 3

(22) What is phosphorylation ?

(A) Removal of H_3PO_4 (B) Change place of phosphate in glucose

(C) Remove phosphate from glucose (D) Joining of H_3PO_4

(23) What need to convert 3 phosphoglyceric acid into 2 phosphoenol pyruvic acid in glycolysis ?

(A) Use of H_3PO_4 (B) decarboxylation (C) dehydration (D) Formation of ATP

(24) How many oxygen are used in glycolysis ?

(A) 6 (B) 0 (C) 12 (D) 3

(25) End product of glycolysis

(A) $2CH_3CO\ COOH + 2NADH_2 + ATP$ (B) $2CH_3CO\ COOH + 2NADH_2 + 2\ ATP$

(C) $CH_3CO\ COOH + NADH + ATP$ (D) $CH_3CO\ COOH + 2H^+ + 2\,e^- + 4ATP$

(26) How many PGAL molecule produce from 3 molecule of glucose and how many ATP molecules produce from PGAL till CO_2 and H_2O produce during respiration.

(A) 6 PGAL – 160 ATP (B) 6 PGAL – 120 ATP

(C) 4 PGAL – 80 ATP (D) 4 PGAL – 40 ATP

(27) How many times isomerisation process seen during glycolysis ?

(A) 1 (B) 2 (C) 3 (D) 4

(28) Net profit of glycolysis ..... .

(A) 3 ATP and 1 NADH + H^+ (B) 2 ATP and 2 NADH + H^+

(C) 6 ATP and 4 NADH + H^+ (D) 10 ATP and 6 NADH + H^+

(29) Mean of dihydroxy acetone phosphate.

(A) 2C component (B) 3C component

(C) 4C component (D) 6C component

(30) When human muscles contracts ..... .

(A) Respiration not occur

(B) In less O_2 anaerobic respiration occur

(C) anaerobic never occur

(D) Always anaerobic respiration seen

(31) Which one is true sentence for 3 PGAL to 1,3 BPGA in glycolysis ?

(A) Use of H_3PO_4 (Pi) and produce of $NADH_2$

(B) Use of ATP and produce of $NADH_2$

(C) Synthesis of ATP and $NADH_2$

(D) Decompose of $NADH_2$ and formation of ATP

(32) How many ATP produce in dephosphorylation in glycolysis ?

(A) 2 ATP (B) 3 ATP (C) 4 ATP (D) 6 ATP

(33) Which sentence is true for glycolysis ?

(A) ATP used in dephosphorylation

(B) Phosphorylation occur in matrix level

(C) $C_6H_{12}O_6 \rightarrow 3CO_2$

(D) ATP produce during phosphorylation

(34) What happened of released H^+ and e^- when oxidation of glyceraldehyde in glycolysis ?

(A) It released in form of CO_2

(B) It convert into pyruvic acid

(C) They oxidise by NAD

(D) It reduce from NAD

(35) Identify me : "I am anaerobic phase of aerobic respiration."

(A) glycolysis

(B) TCA cycle

(C) oxidative phoshphorylation

(D) None of above

(36) Glucose + ATP $\xrightarrow{x}$ glucose : 6 - po_4 + ADP, (here x = Enzyme)

(A) X = carboxylase

(B) X = Hexokinase

(C) X = decarboxylase

(D) TPP

(37) Arrange in sequence of glycolysis process

(i) Glucose $\rightarrow$ glucose - 6-Po_4

(ii) 3PGA $\rightarrow$ 2PGA

(iii) Fructose 6-Po_4 $\rightarrow$ fructose 1-6 biphosphate

(iv) PGAL $\rightarrow$ PEP

(v) PGAL $\rightarrow$ PGAP

(A) (i), (ii), (iii), (v), (iv)

(B) (i), (iii), (v), (ii), (iv)

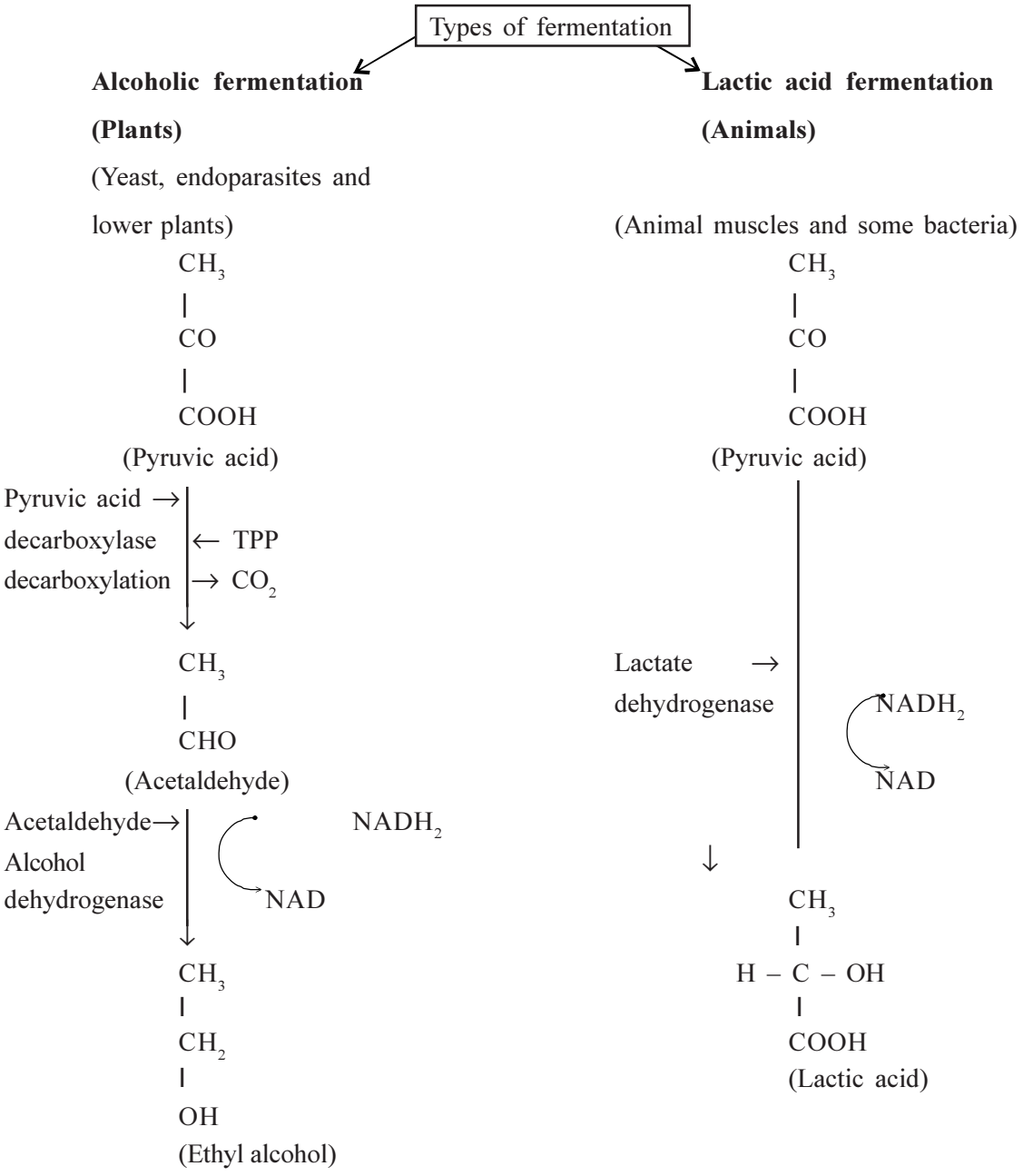
(C) (i), (ii), (iv), (iii), (v)

(D) (i), (iii), (v), (iv), (ii)

Answers : (8-C), (9-B), (10-C), (11-B), (12-C), (13-C), (14-C), (15-D), (16-D), (17-C), (18-B), (19-C), (20-D), (21-C), (22-D), (23-C), (24-B), (25-B), (26-B), (27-C), (28-B), (29-B), (30-B), (31-C), (32-C), (33-B), (34-D), (35-A), (36-B), (37-B)

Anaerobic cellular respiration (fermentation)

- **Takes place without O_2**
- **Glycolysis is the first stage of anaerobic respiration. Alcoholic fermentation is the second stage in plants, while in animals lactic acid fermentation takes place.**
- In this type of respiration comparatively less amount of energy is released.
- In this respiration incomplete breakdown of pyruvic acid occurs.
- During this process compounds like ethanol (ethyl alcohol C_2H_5OH) or lactic acid ($CH_3CHOHCOOH$) are produced from glucose.
- At the end of respiration 54 k cal energy releases.
- Anaerobic respiration occurs in the cytoplasm matrix of cytoplasm.
- Cytoplasm matrix of cytoplasm.



Summary of anaerobic respiration

- Respiration takes place in the absence of O_2.
- During this biochemical process It results in incomplete breakdown of glucose like organic molecule.
- In this type of respiration less amount (54 K cal) energy releases.
- The anaerobic respiration are of two types.

(1) **Alcoholic fermentation**

(yeast and lower class plants, endoparasites)

- equation :

$2CH_3CO\ COOH \xrightarrow[TPP]{decarboxylase} CH_3CHO + 2CO_2$

pyruvic acid — Acetaldehyde

$2CH_3CHO$

$Acetaldehyde + 2NADH_2 \xrightarrow[dehydrogenase]{Alcohol} 2(C_2H_5OH) + 2NAD$

ethanol

(2) Lactic acid fermentation

(Some bacteria and animal muscle)

- equation :

$2CH_3CO\ COOH + 2NADH \xrightarrow[dehydrogenase]{lactate} 2CH_3CHOHCOOH + 2NAD$

pyruvic acid — lactic acid

---

38) Which process is neccessary to convert lactic acid from pyruvic acid ..... .

(A) Carboxylation (B) Reduction

(C) Oxidation (D) Decarboxylation

(39) Which type of respiration occur in human RBC ?

(A) Aerobic respiration (B) Lactic acid enzyme

(C) Alcoholic enzyme (D) None of above

(40) What produce when CO_2 and $2H^+$ Released from pyruvic acid ?

(A) citric acid (B) Lactate (C) Acetate (D) CO–A

(41) Which respiration process is only seen during anaerobic respiration in living organism ?

(A) TCA (B) ETS (C) CAM (D) EMP

(42) $2CH_3CO\ COOH \xrightarrow[TPP]{deccarboxylase}$ ..... + ...

(A) $2(C_2H_5OH) + NAD$ (B) $2CH_3CHO + O_2$

(C) $2CH_3CHO + 2\ CO_2$ (D) $CH_3CHOHCOOH + NAD$

(43) What is the end product of anaerobic respiration in animals ?

(A) $2CH_5CHOHCOOH.\ 2NAD^+$ (B) $2(C_2H_5OH),\ 2CO_2$

(C) $2CO_2,\ 2CH_3CHCOOH$ (D) $2CO_2,\ 2NADH_2$

(44) What is true when lactic acid formed during anaerobic respiration of two pyruvic acid ?

(A) loss of 3 ATP (B) loss of 6 ATP

(C) produce 3 ATP (D) produce 6 ATP

(45) How alcoholic and lactic acid fermentation is different from each other ?

(A) In alcoholic fermentation the end product possess 3 C.

(B) In production of lactic acid reduuction of pyruvic acid occurs.

(C) In alcoholic fermentation oxidation of pyruvic acid occurs.

(D) In alcoholic fermentation 4 ATP and in lactic acid 6 ATP are produced.

(46) During anaerobic respiration which one is first hydrogen receiver and last hydrogen donner ?

(A) PGAL (B) FAD (C) NAD (D) DHAP

(47) How much KJ energy is released when ethyl alcohol produce from glucose ?

(A) 225.93 (B) 686 (C) 2870.22 (D) 54

(Hint : 1 K.Cal = 4.184 KJ)

(48) How many glucose molecule is needed for formation of 38 ATP in anaerobic respiration ?

(A) 2 (B) 4 (C) 38 (D) 19

(49) What we get on reduction of acetaldehyde

(A) Methyl alcohol (B) Glycerol (C) Ethyl alcohol (D) Ethyl aster

(50) Which product is used in alcohol formentation of glycolysis ?

(A) CO_2 (B) $NADH_2$ (C) ATP (D) A and B both

(51) In alcoholic fermentation .....

(A) Triose phosphate is a electron donner when acetaldehyde is electron acceptor.

(B) Triose phosphate is a electron donner when pyruvic acid is electron acceptor.

(C) No electrone donner.

(D) O_2 is electron acceptor.

(52) If 15 CO_2 is produce instead of 18 by complete hydrolysis of 3 molecule of glucose in human muscles then how many CO_2 will produce in anaerobic respiration of glucose as a some living organism ?

(A) 03 (B) 02 (C) 00 (D) 04

(53) How many ATP is produce of oxidative phosphorylation of $NADH_2$ made by 5 molecule of glucose during glycolysis ?

(A) 28 (B) 20 (C) 15 (D) 30

(54) Which process is found in anaerobic respiration ?

(A) Dephosphorylation of PGA

(B) Oxidation of PGAL

(C) Oxidation of glucose

(D) Phosphorylation of PEP

(55) Which product is occur when fermentation of glucose by yeast ?

(A) Water and CO_2 (B) Methanol + O_2

(C) Ethanol + CO_2 (D) Ethanol and water

(56) In which process decarboxilation is seen ?

(A) Melic acid $\rightarrow$ oxalo acetic acid

(B) Pyruvic acid $\rightarrow$ acetaldehyde

(C) 2-phosphoglyceric acid $\rightarrow$ 2-Phospho enol pyruvic acid

(D) Citric acid $\rightarrow$ Isocitric acid

(57) In alcoholic fermentation ..... .

(A) Triose phosphate is electron donner and pyruvidc acid electron acceptors

(B) No electron donner

(C) Oxygen is electron acceptor

(D) Triose phosphate is electron donner and acetaldehyde is electron acceptors

(58) Which enzyme is useful in alcoholic fementation ?

(A) decarboxylase (B) Thimine pyrophosphate

(C) alcohol dehydrogenase (D) all above

(59) How many energy is released during fermentation ?

(A) < 7% (B) > 7% (C) 45% (D) 7%

(60) If 12(CO_2) is produce in alcoholic fermentation then substrate X and product Y

(A) X = 6 ($CH_3COCOOH$); Y = 6 (C_2H_5OH)

(B) X = 6 ($CH_6H_{12}O_2$); Y = 12 (C_2H_5OH)

(C) X = 12 ($CH_3COCOOH$); Y = 6 (C_2H_5OH)

(D) X = 12 ($CH_6CH_{12}O_6$); Y = 12 (C_2H_5OH)

(61) In yeast ..... and in animal muscle cells ..... are used in reduction during anaerobic respiration.

(A) CH_3CHO; $CH_3COCOOH$ (B) CH_3CHO; CH_3CHO

(C) $CH_3COCOOH$; Acetaldehyde (D) acetaldehyde, lactic acid

(62) Identify me " I am insidant of pyruvate reduction

(A) EMP - path (B) Fermentation

(C) TCA cycle (D) ETS

(63) End product of fermentation is ..... .

(A) O_2 and ethanol (B) O_2 and acetaldehyde

(C) CO_2 and ethanol (D) CO_2 and acetaldehyde

(64) Give the name of enzyme which convert pyruvic acid into acetaldehyde.

(A) Alcohol dehydrogenase (B) Pyruvid acid decarboxylase

(C) Hexokinase (D) oxidase

(65) Give the name of enzyme which convert pyruvic acid into lactic acid.

(A) Alcohol dehydrogenase (B) TPP

(C) Pyruvic acid decarboxylase (D) Lactate dehydrogenase

(66) It is a product of fermentation of alcohol ..... .

(A) O_2; CO_2 (B) 2(C_2H_5OH), $2CO_2$

(C) $2CO_2$, $2CH_2CHCOOH$ (D) $2CO_2$, $2NADH_2$

(67) What is true of lactic acid which produce from pyruvic acid during anaerobic respiration ?

(A) Loss of 3 ATP (B) Received 3 ATP

(C) Loss of 6 ATP (D) Received 6 ATP

Answers : (38-B), (39-B), (40-C), (41-D), (42-C), (43-A), (44-B), (45-B), (46-C), (47-A), (48-D), (49-C), (50-B), (51-A), (52-C), (53-D), (54-B), (55-C), (56-B), (57-B), (58-D), (59-A), (60-D), (61-D), (62-B), (63-C), (64-B), (65-C), (66-B), (67-D)

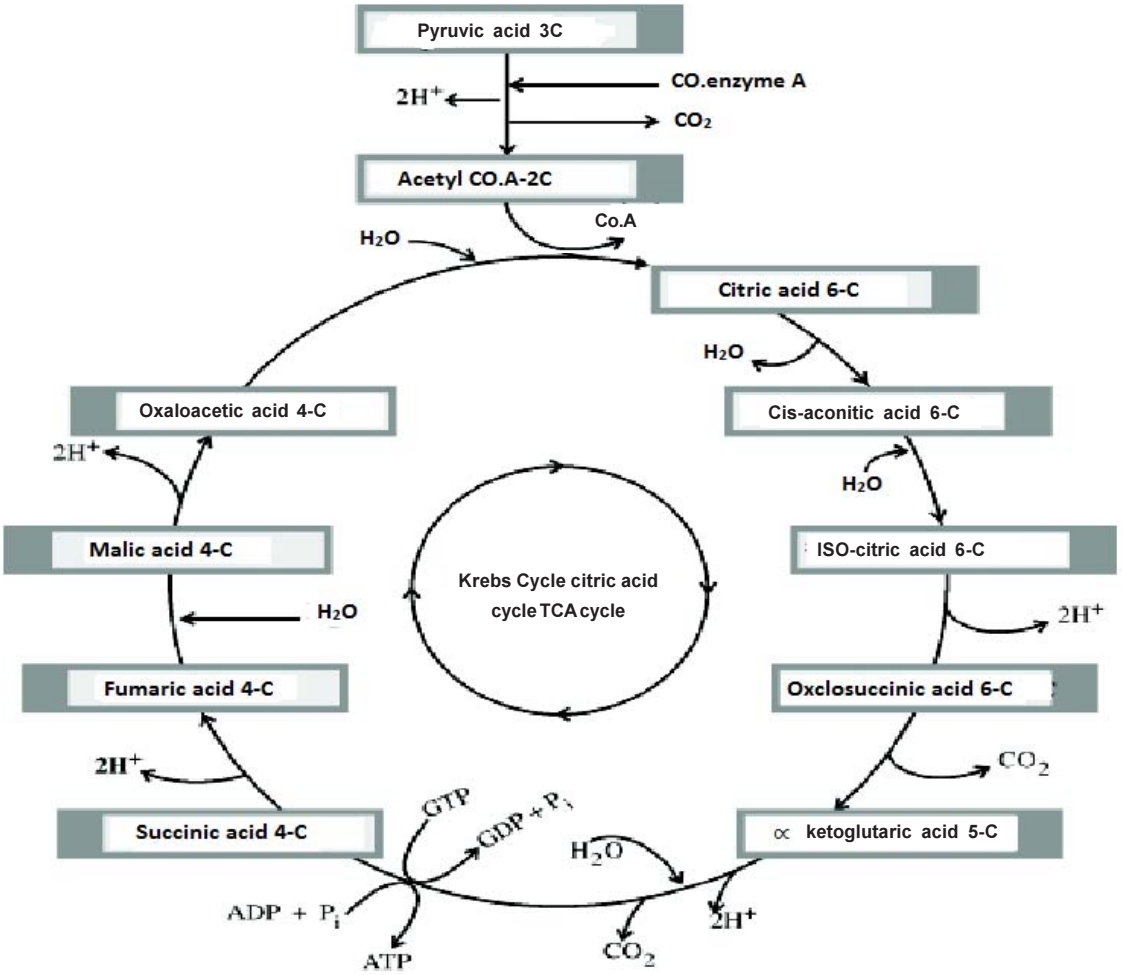
Krebs cycle (TCA cycle)

Discovered by - Hans krebs, Noble prize - 1953

- It is second stage of aerobic respiration followed by glycolysis.
- It provides a pathway for complete breakdown of glucose.
- This respiration is the process occurs in the presence of O_2.
- As the first substance formed is citric acid, this cycle is known as **citric acid cycle**.
- Citric acid possess three carboxylic groups, Hense, this cycle is also known as **TCA cycle** 3(-COOH).
- It provides the main pathway for synthesis of ATP.
- Two ATP are procuced during main pathway, but by the reduction of components of ETS 28 ATP are produced.
- Various carbon-complexes formed during this cycle provide necessary components for growth and maintenance of cell. These components are utilized in the synthesis of substances like amino acids, nucleotides, fats, chlorophyll and cytochromes.

- Stages involved in krebs cycle

No.	Process	Final product
1.	Pyruvic acid (3C) + NAD^+ + Co-A (acetyl group)	acetyl Co-A (2C) + $NADH_2$ + CO_2
2.	Oxaloacetic acid (4C) + acetyl Co-A (2C) + H_2O	Citric acid (6C) + Co-A
3.	Citric acid	Cis-aconitic acid (6C) + H_2O
4.	Cis-aconitic acid (6C) + H_2O	Isocitric acid (6C)
5.	Isocitric acid (6C) + NAD^+	Oxalosuccinic acid (6C) + $NADH_2$
6.	Oxalosuccinic acid (6C)	∝-Ketoglutaric acid (5C) + CO_2
7.	∝-Ketoglutaric acid (5C) + H_2O + NAD^+ + Co-A	Succinyl Co-A (4C) + $NADH_2$ + CO_2
8.	Succinyl Co-A + GDP + H_3PO_4 (Pi) GTP + ADP	Succinic acid (4C) + Co-A + GTP GDP + ATP
9.	Succinic acid (4C) + FAD^+	Fumaric acid (4C) + $FADH_2$
10	Fumaric acid (4C)+ H_2O	Malic acid (4C)
11	malic acid (4C) + NAD^+	Oxaloacetic acid (4C) + $NADH_2$



Krebs cycle

Differentiate between : Glycolysis and krebs cycle.

	Glycolysis	Krebs cycle
1.	It is respiration process occur in cytoplasm.	It is respiration process occurs in mitochondria.
2.	Aerobic as well as anerobic respiration means, either presence or absence of O_2 it occurs.	It is aerobic respiration means presence of O_2 is necessary.
3.	Two molecules of pyruvic acid are formed from one molecule of glucose.	It is continue cyclic process.
4.	At the end of the process net gain of 2 ATP and 2 molecules of $NADH_2$ are formed.	During this process 2 ATP molecules from GTP are formed and 6 $NADH_2$ and are also $2FADH_2$ are also formed.
5.	During this process CO_2 does not release.	In One krebs cycle 3 molecules of CO_2 releases

(68) If two molecule of glucose enter into aerobic respiration and one molecule of glucose enter in anaerobic respiration then how many NAD molecule is formed ?

(A) 20 (B) 10 (C) 14 (D) 08

(69) Which carbonic acid is seen between two decarboxylation in krebs cycle ?

(A) ∝ - Ketoglutaric acid (B) Isocitric Acid

(C) Oxalo succinic acid (D) Succinic Acid

(70) Which product is made if we remove CO_2 and $2H^+$ from pyruvic acid ?

(A) Acetyle Co-A (B) Citric Acid

(C) Acetate (D) Co-A

(71) Which is known as tricarboxylic acid ?

(A) Fumeric acid (B) Pyruvic acid

(C) Citric acid (D) Melic acid

(72) How many carbon molecue is occur in oxalo acitic acid ?

(A) 2 (B) 4 (C) 5 (D) 6

(73) Which of the following process are related in both decarboxylation and dehydrogenation ?

(A) Melate - Oxaloacitate
Succinate - Fumarte

(B) Succinate - fumarate
Fumerate - Melate

(C) C Isocitrate → ∝-Ketoglutarate
∝-Ketoglutarate → Succinate

(D) ∝-ketoglutaric acid → Succinate
Malate → Oxaloacetate

(74) Product of TCA Cycle ?

(A) 2 $FADH_2$, 2 $NADH_2$, 2 GTP (B) 1 $FADH_2$, 2$NADH_2$, 1 GTP

(C) 1 $FADH_2$, 3 $NADH_2$, 1 GTP (D) 1 $FADH_2$, 4 $NADH_2$, 1 GTP

(75) Which is isomer of citric acid ?

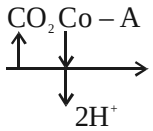
(A) Cis - aconitic acid (B) Malic acid

(C) iso-citric acid cycle (D) ∝ - Ketoglutaric acid

(76) How many ATP formed of two molecule of pyruvic acid in krebs cycle ?

(A) 28 (B) 30

(C) 36 (D) 38

(77) Pyruvic acid 

Acetyl-Co-A where this process occur ?

(A) In matrix of cell (B) In chloroplast

(C) In krebs cycle (D) In matrix of mitochondria

(78) How many carboxylic groups found in citric acid ?

(A) 2 (B) 4

(C) 3 (D) 6

(79) Which is true for dehydrogenation process ?

(A) Iso-citrate → ∝-ketoglutrate → succinyl Co-A → Fumaric acid

(B) Citrate → Iso citrate → succinate → fumaric acid

(C) Acetyle Co - A → citrate → Isocitrate → Malet

(D) Citrate - isocitrate - fumaric acid - Malet

(80) How many CO_2, $2H^+$ and $2e^-$ released when one molecule of pyruvic acid enter into krebs cycle ?

(A) 6, 10, 10 (B) 3, 5, 5 (C) 10, 10, 6 (D) 5, 5, 3

(81) How many ATP formed in one time through substrate in krebs cycle ?

(A) 1 (B) 3 (C) 2 (D) 4

(82) How many $NADH^+$ H^+ ion in one time kreb cycle ?

(A) 2 (B) 6 (C) 4 (D) 5

(83) Where is most less ATP produce in aerobic respiration ?

(A) Pyruvate - acetyl Co - A

(B) Succinyl - Co - A - Succinic acid

(C) Malic acid - oxaloacetic acid

(D) Isocitric acid - oxalosuccinic acid

(84) How many water molecule produce and used respectivly during aerobic respiration in mitochondria matrix ?

(A) 6 and 6 (B) 12 and 6 (C) 14 and 8 (D) 2 and 8

(85) What happened of malic acid in krebs cycle ?

(A) decarboxylation (B) Hydrolysis (C) Oxidation (D) Reduction

(86) By whom FAD as electro acceptor in Krebs cycle ?

(A) succinyl Co-A - Succinic acid (B) Fumaric acid - melic acid

(C) Succinic acid - Fumaric acid (D) Malic acid - Oxaloacetic acid

(87) Which molaule of primary and secondary in aerobic respiration ?

(A) Pyruvic acid (B) Co-A (Co. factor)

(C) Oxaloacetate (D) Acetyle Co-A

(88) If secondery phase is not occur then which product is found in glucose aerobic respiration ?

(A) 4 ATP, 10 $NADH_2$, 2 $FADH_2$ (B) 4 ATP, 6 $NADH_2$, 1 $FADH_2$

(C) 2 ATP, 8 $NADH_2$, 1 $FADH_2$ (D) 2 ATP, 8 $NADH_2$, 2 $FADH_2$

(89) Which primary process is seen of pyruvic acid during both aerobic and anaerobic respiration ?

(A) Decarboxylation (B) Caboxylation (C) dehydrogenation (D) A and C both

(90) Connecting link between glycolysis and krebs cycle.

(A) Oxalo acitate (B) PEP (C) Pyruvate (D) Acetylco-A

(91) How malic acid formed from fumaric acid ?

(A) H_2O (B) H_2O (C) by CO_2 removal (D) CO_2 addition

(92) Pyruvic acid co.A + NAD $\rightarrow$

(A) Acetyl Co-A + $NADH_2$

(B) Acetyl Co. A + $NADH_2$ + CO_2

(C) Acetyl Co-A + $NADPH_2$ + CO_2

(D) Acetyl Co-A + $NADH_2$ + CO_2

(93) In which process dehydrogenation occur but do not decarboxylation in krebs cycle ?

(A) Citric acid - isocitric acid

(B) Succinic acid - fumaric acid

(C) Fumaric acid - Malic acid

(D) Isocitric acid - Oxalo succinic acid

(94) How many CO_2, $2H^+$ $2e^-$ Released when one molecule of pyruvic acid pass on krebs cycle ?

(A) 6, 10, 10 (B) 10, 10, 6 (C) 3, 5, 5 (D) 5, 5, 3

(95) Product of krebs cycle.

(A) $4NAD^+ + FAD^+ + 2H_2O$

(B) $4NAD^+ + FAD^+ + 2H_2O$ + ADP + Pi

(C) $CH_3CO.COOH + 4NAD^+ + FAD^+ + 2H_2O$ + ADP + Pi

(D) $3CO_2$, $4NADH + 4H^+ + FADH_2$ + ATP

(96) Which molecue is contain 4C in krebs cycle ?

(A) Citric acid (B) 2-Kitoglutarate (c) (C) Isocitrate (D) Succinic acid

(97) Importance of krebs cycle.

(A) It provides a pathway for complete breakdown of glucose

(B) It provides the main pathway for synthesis of ATP

(C) Various carbon - complex formed during this cycle provide necessary components for growth and maintenance of cell

(D) All three - A, B, C are important

(98) In which process CO_2 is released during krebs cycle

(A) $\propto$- keto glutaric acid - succinic acid

(B) Citric acid - Iso citric acid

(C) Succinic acid - Fumaric acid

(D) Malic acid - oxalo acetic acid

(99) Matrix related phosphorylation occur when .....

(A) Succinic acid convert into fumaric acid

(B) Succinic acid co-A convert into, succinic acid

(C) Fumaric acid convert into melic acid

(D) Isocitric acid convert into α-keto glutaric acid

(100) Which is true sequence of carbonic acid in krebs cycle ?

(A) Citric acid - oxalo succinic - isocitric acid

(B) Citri acid - isocitric acid - Oxalo succinic acid

(C) Isocitric acid - oxalo succinic acid - citric acid

(D) Oxalo succinic acid - isocitric acid - citric acid

Answers : (68-A), (69-A), (70-A), (71-B), (72-B), (73-C), (74-D), (75-C), (76-A), (77-D), (78-C), (79-A), (80-C), (81-A), (82-C), (83-B), (84-D), (85-C), (86-C), (87-D), (88-A), (89-A), (90-D), (91-B), (92-B), (93-B), (94-C), (95-D), (96-D), (97-D), (98-A), (99-B), (100-B)

Electron transport chain (ETC)

Or

Oxidative Phosphorylation

Or

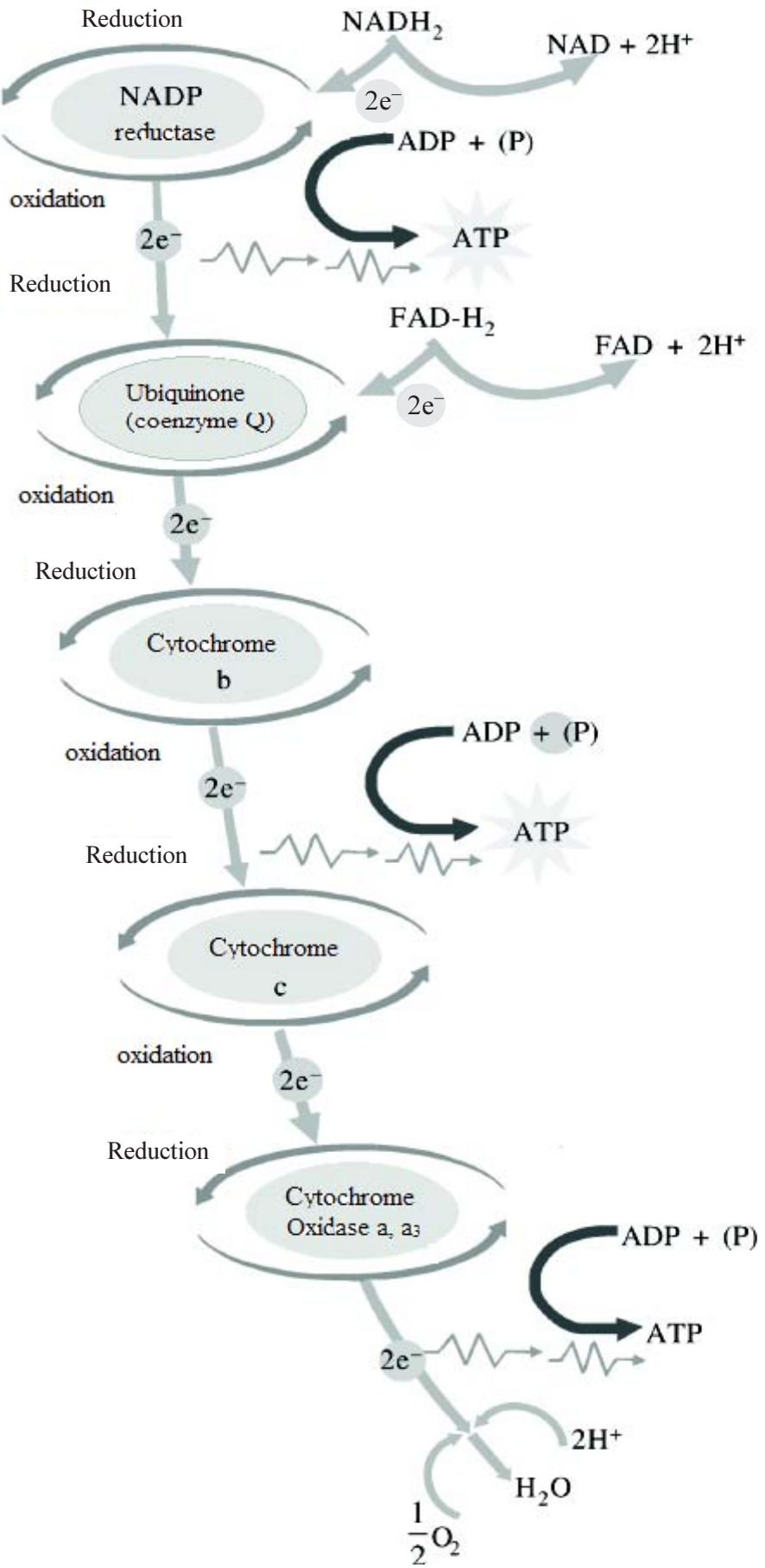
Last (third) stage of cellular respiration

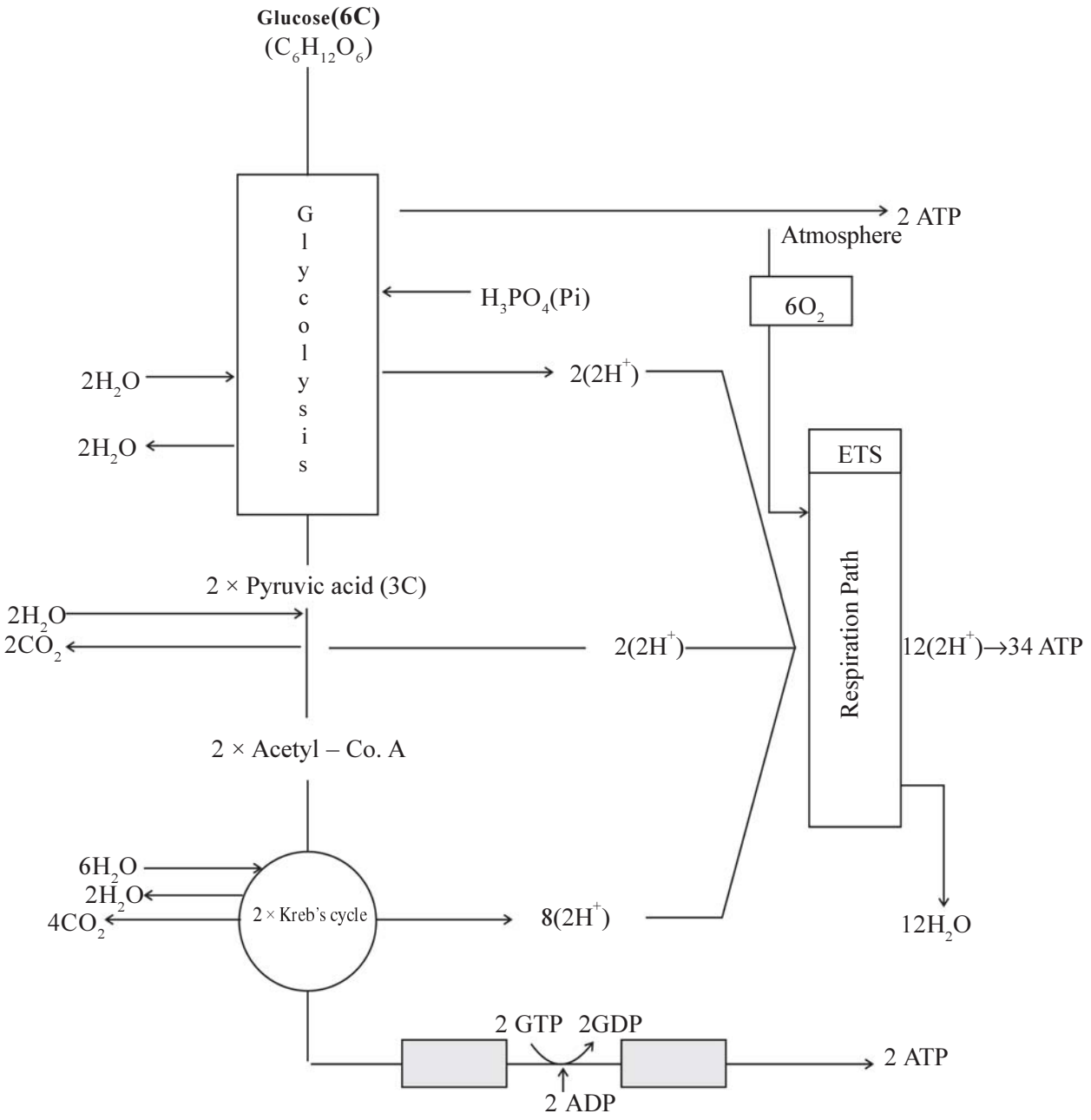
- $NADH_2$ and $FADH_2$ formed by dehydrogenation of $2H^+$ during glycolysis and krebs cycle do not react directly with oxygen (O_2).
- Hydrogen acceptors like NAD and FAD reduce them and they convert into $NADH_2$ and $FADH_2$.
- NAD and FAD work as coenzyme, it reduced in mitochondrial matrix and transports electrons in mitochondrial respiration pathway. So this pathway is known as electron transport system (ETS).
- Where $NADH_2$ and $FADH_2$ formed during this process are transported to atmospheric O_2 and oxidized, ATP is released.
- Transportation of $2H^+$ and $2e^-$ towards O_2 is essential.
- The metabolic pathway through which the electron passes from one carrier to another is called the Electron Transport System (ETS).
- This system is performed of inner membrane of mitochondria.
- By oxidation of one $NADH_2$ and on $FADH_2$ 3 ATP and 2 ATP are formed respectively . Which is as following :

(1) $NAD^+ + H \xrightarrow[\text{Reduction}]{} NADH_2 + \frac{1}{2} O_2 \xrightarrow[\text{oxidation}]{} NAD^+ + 3ATP + H_2O$

(2) $FAD^+ + H \xrightarrow{\text{Reduction}} FADH_2 + \frac{1}{2} O_2 \xrightarrow{\text{oxidation}} FAD^+ + 2ATP + H_2O$

- Mechanism of chemiosmotic generation of ATP.
- Since mitochondrial membrane is impermeable to protons. These can not be diffused back into the matrix across the membrane.


Reduction
NADH2
NAD + 2H+
NADP
reductase
2e−
ADP + (P)
oxidation
2e−
ATP
Reduction
FAD-H2
FAD + 2H+
Ubiquinone
(coenzyme Q)
2e−
oxidation
2e−
Reduction
Cytochrome
b
oxidation
ADP + (P)
2e−
ATP
Reduction
Cytochrome
c
oxidation
2e−
Reduction
Cytochrome
Oxidase a, a3
ADP + (P)
ATP
2e−
2H+
H2O
1/2 O2



Whole equation for aerobic respiration

$$C_6H_{12}O_6 + 6H_2O + 6O_2 \rightarrow 12H_2O + 6CO_2 + 38 \text{ ATP}$$

(101) How many ATP are formed during complete breakdown of glucose from $NADH_2$ and $FADH_2$ oxidation ?

(A) 3 ATP (B) 2 ATP (C) 34 ATP (D) 36 ATP

(102) In which of the following process O_2 is used directly ?

(A) glycolysis (B) ETS (C) Citric acid cycle (D) Alcoholic fermentation

(103) How many ATP are formed during complete breakdown of 3 glucose molecule from $NADH_2$ oxidation ?

(A) 60 (B) 90 (C) 180 (D) 38

(104) Which is true sentence for aerobic respiration ?

(A) NPD is first respiratory substance which accepts hydrogen.

(B) NAD is a last hydrogen donner.

(C) Cy + C_3 is a last electron donner.

(D) Oxygen is a last component for receiving eletron.

(105) How many protons (H^+) are released from $12H_2O$?

(A) $6H^+$ (B) $12H^+$ (C) $24H^+$ (D) $18H^+$

(106) Which process is found during last phase of aerobic respiration ?

(A) Oxidative phosphorylation (B) Electron circulation

(C) A and B both (D) Electron excretion

(107) What is another name of ubiquinone ?

(A) cyt – b (B) NADP Reductase

(C) COA – Q (D) Cyctocrome oxidase

(108) Which is not respiration related enzyme ?

(A) Cytocrome oxidase (B) TPP

(C) Pyruvate dehydrogenase (D) RubisCO

(109) In which process energy released by hydrogen acceptor during respiration ?

(A) Glycolysis (B) ETS (C) TCA-cycle (D) Fermentation

(110) In last process of aerobic respiration, which process is seen after $FADH_2$ oxidation ?

(A) $\frac{1}{2}$ O_2 received $2e^-$ and $2H^+$ during ETS.

(B) $2e^-$ is released in ETS when $2H^+$ remain in the Matrix.

(C) $2e^-$ NADP reductase enter into ETS when $2H^+$ remain in the matrix.

(D) None of the above.

(111) Which is true related to ETS ?

(A) CO. A–Q - $\xrightarrow{2e^-}$ cyt – b + ATP (B) cyt – C $\xrightarrow{2e^-}$ cyt – a, a_3 + ATP

(C) cyt – b - $\xrightarrow{2e^-}$ cyt – c + ATP (D) A and C Both

(112) If aerobic respiration occur in PEP = X how many ETS take place = X.

How many ATP formed in matrix = Y X

How many ATP formed in oxidative phosphorylation = 2 Y

(A) X = 4; Y = 3; Z = 15 (B) X = 5; Y = 2, Z = 14

(C) X = 6; Y = 1; Z = 17 (D) X = 1; Y = 4; Z = None of All

(113) Maltose X $\rightarrow$ Y = NAD reduction

(A) X = glucose Y = 10 NAD (B) X = $C_6H_{12}O_6$; Y = 5 NAD

(C) X = 2($C_6H_{12}O_6$); Y = 10 NAD (D) X = 2($C_6H_{12}O_6$); Y = 20 NAD

(114) Which is true sequence during ATP synthesis of electron receptor ?

(A) Cyt (a_1, a, b, c) (B) Cyt (b, c, a, a_3)

(C) Cyt (b, c, a, a) (D) Cyt (c, b, a, a_3)

(115) In which process energy is released from ETS ?

(A) Ubiquinone $\xrightarrow{2e^-}$ Cytocrome b (B) Cytocrome $\xrightarrow{2e^-}$ Cytocrome c

(C) Cytocrome $\xrightarrow{2e^-}$ Cytocrome a, a_3 (D) above three

(116) All process of krebs cycle : occur in mitochonria matrix : : all process of ETS : ..... .

(A) Matrix of mitochondria (B) Matrix of cell

(C) Cristymembrane of mitochondria (D) Matrix of peroxisome

(117) Which one is first receptor of $2H^+$ and $_{2e^-}$ from $NADH_2$?

(A) NAD Reductase (B) alcolage

(C) Hexokinase (D) None of above

(118) Give the name of enzyme which take place in oxidative phosphorylation.

(A) Aldolase (B) NAD Reductase (C) Hexokinase (D) None of above

(119) How many ETS take place during the one molecule of pyruvic acid breakdown ?

(A) 14 ETS (B) 5 ETS (C) 6 ETS (D) 1 ETS

(120) How many ATP produce during bio-oxidation of $NADH_2$?

(A) 2 ATP (B) 6 ATP (C) 3 ATP (D) 1 ATP

(121) Which is end product of oxidative phosphorylation ?

(A) ATP (B) $FADH_2$ (C) $NADH_2$ (D) Electron

(122) How many ATP produce during oxidative phosphorylation of pyruvic acid ?

(A) 12 (B) 14 (C) 13 (D) 15

Answers : (101-C), (102-B), (103-B), (104-C), (105-C), (106-B), (107-C), (108-D), (109-B), (110-A), (111-D), (112-B), (113-D), (114-B), (115-B), (116-C), (117-A), (118-B), (119-B), (120-C), (121-A), (122-B)

ATP

- However, these can enter the membrane via a proton channel.
- Which is established by the membrane bound adenosine triphosphatase (ATPase).
- ATPase is multienzyne complex containing two parts F_0 and F_1
- F_1 headpiece is a peripheral membrane protein complex.
- This is the site for the synthesis of ATP from ADP.
- For each ATP produced, $2H^+$ passes through F_0 from the intermembrane space to the matrix down the electrochemical proton gradient.
- Thus for each pair of protons flowing back into matrix, one molecule of ATP is synthesized.
- So for three pairs of protons, three molecules of ATP are generated.

  (Note : 3 Molecules of ATP are formed from $NADH_2$ when 2 molecules of ATP are formed from $FADH_2$ because they have only 2 pairs of protons.)

Energy capacity available to glucose by aerobic respiration

- From slow oxidation of 1 mole glucose 180 grams CO_2, H_2O and only

2870 KJ mol^{-1} (686000 cal 686 K cal) energy produce which is stored in form of 38 ATP.

- When ATP is hydrolised then produced energy is useful for biological activity.

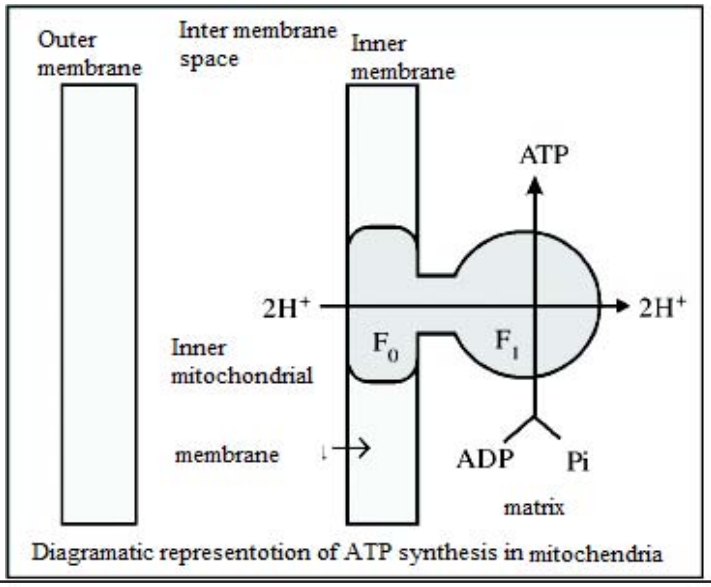
Which is 1 ATP = 34 KJ. (7.3 K cal)

Energy capacity of glucose $= 38 \times \frac{34}{2870}$

$= 1292$ KJ

$= 45$ %

- Thus, the efficiency of the process of transformation of potential chemical energy in glucose into the energy in ATP is about 45%. The remaining 55% energy is dissipated as heat.



Diagramatic representotion of ATP synthesis in mitochendria

(123) Where ATPase F, component is seen in mitochondria ?

(A) Proton channel (B) middle membrane of mitochondria

(C) Peripheral mambrane of mitochondria (D) Matrix of mitochondria

(124) How many pair of protons needed for formation of 3 ATP ?

(A) 3 (B) 6 (C) 9 (D) 12

(125) Chemiosomotic theory is related with ..... .

(A) K^+ ion theory (B) H^+ ion theory

(C) membrane capacity (D) Protongradient

(126) Which characteristic is found in mitochondrial membrane for entry proton ?

(A) Permeable (B) Semi permeable (C) impermeable (D) None of above

(127) By whom proton channel produce ?

(A) ATPase (B) Multi enzyme complex

(C) Adenosine triphosphate (D) above all

(128) How many ATP molecule produce with the help of one pair of proton ?

(A) 1 (B) 2 (C) 3 (D) 4

(129) In which complex ATP format formed ADP ?

(A) F_0 (B) F_1

(C) F_2 (D) F_3

(130) How many protons tranfered by $FADH_2$ towards F_0 and F_1 complex ?

(A) 2 (B) 6

(C) 4 (D) 8

(131) In mitochondria related of chemiosomotic ..... .

(A) On the basis of proton gradient there is movement of $2H^+$ from inter-membrane space to matrix.

(B) ATP produce by passing every pair of protons towards matrix.

(C) $2H^+$ passes from ATPase ATP F_0 to F_1 by proton rediant

(D) above all

Answers : (123-C), (124-B), (125-D), (126-C), (127-D), (128-A), (129-B), (130-C), (131-D)

Respiration balance sheet

	Step of reaction	**Consumed ATP**	**Released ATP**	**Net gain ATP**
●	Glycolysis			
1.	Glucose → glucose 6-P	1 ATP		1 ATP
2.	Fructose 6-p → Fructose 1,6 biphosphate	1 ATP		
3.	3-phosphoglyraldehyde ↓ 1,3-biphosphate glyciric acid		3 ATP	3 × 2 = 6 ATP
4.	1,3-biphosphate glyciric acid ↓ 3-phosphoglyciric acid		1 ATP	1 × 2 = 2 ATP
5.	2-phosphoenol pyruvic acid ↓ Pyruvic acid		1 ATP	1 × 2 = 2 ATP
6.	Pyruvic acid → Acetyl coA.		3 ATP	3 × 2 = 6 ATP
●	Krebs cycle			
7.	Isocitric acid - Oxalo succinic acid		3 ATP	3 × 2 = 6 ATP
8.	α -Ketoglutaric acid → Succnyl co A		3 ATP	3 × 2 = 6 ATP
9.	Succnyl Co A → Succinic acid		1 ATP	1 × 2 = 2 ATP
10.	Succinic acid → Fumaric acid		2 ATP	2 × 2 = 4 ATP
11.	Malic acid → Oxalo acetic acid		3 ATP	3 × 2 = 6 ATP
			Total	**38 ATP**

(132) How many ATP are produce in aerobic respiration without phosphorylation ?

(A) 2 (B) 4 (C) 6 (D) 8

(133) How many ATP are produce after glycolysis and before krebs cycle ?

(A) 1 (B) 3 (C) 12 (D) 8

(134) How many ATP are used when pyruvic acid enter into mitochondria ?

(A) 2 (B) 4 (C) 0 (D) 8

(135) If 162 ATP are phosphorylated during aerobic respiraltion of complet breakdown of glucose then how many NAD reduced in same process ?

(A) 45 (B) 38 (C) 10 (D) 36

(136) Which product is formed during respiration of 360 gm glucose and 384 gm oxygen ?

(A) 528 gm CO_2 + 432 gm H_2O + energy (B) 180 gm CO_2 + 2.64 gm H_2O + energy

(C) 528 gm CO_2 + 180 gm H_2O + energy (D) 264 gm CO_2 + 264 gm H_2O + energy

(137) By complete oxidation of one molecule of glucose, among 38 ATP produce by

(A) 2 ATP in EMP path : 36 ATP in ETS

(B) 2 ATP outside mitochondria : 36 ATP in mitochondria

(C) 8 ATP outside mitochondria : 30 ATP in mitochondria

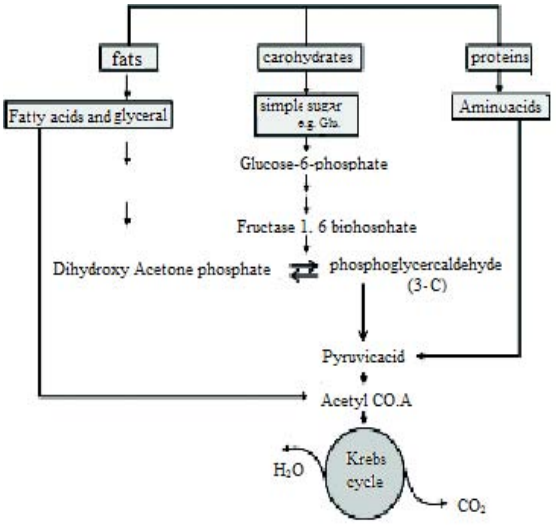
(D) 6 ATP in EMP : 32 ATP in ETS

Answers : (132-A), (133-B), (134-A), (135-A), (136-A), (137-B)

Amphybolic pathway
(bidirectional metabolism pathway)

- Carbohydrate (glucose) oxidation is main respiration pathway.
- Other important organic components or substrates like lipid or protein also can be respired.
- Fats need to be broken down into glycerol and fatty acids first. Now fatty acids degraded into acetyl Co A and glycerol to DHAP → PGAL and then enter into respiratory pathway.
- In case of protein they first broken down into their monomers amino acids after that enter respiratory pathway.
- Both oxidation and reduction process can be seen in respiration pathway.
- It is involved in both anabolic and catabolic processes and hence it is known as an amphibolic pathway.

Respiratory Substrates



(138) Which componant take place in respiration path ?

(A) Fatty acid (B) glycerol

(C) Lipid (D) Acetyl Co-A

(139) Which 2 carbon organic containing substance is elegible to enter into kreb cycle ?

(A) Oxalo acitic acid (B) Acetyle co-A

(C) pyruvic acid (D) glycerol

(140) Which form of glycerol is enter in glycolysis ?

(A) Fructose - 6-phosphate

(B) PGAL

(C) glucose-6-phosphate

(D) Fructose 1-6-biphosphtate

(141) Give the name of 3-c cotaining substance which produce in amphibolic path and enter into respiration path.

(A) Pyruvic acid (B) PGAL

(C) glycerol (D) Acetaldehyde

(142) Which of the following process is occur in both anabolic and catabolic ?

(A) Krebs cycle (B) glycolysis

(C) Amphibolic path (D) Electron transfer system

(143) ..... enter into respiration path by fatty acid breakdown.

(A) Citric acid (B) OAA

(C) PGAL (D) Acetyl Co - A

(144) During the lipid respiration glycerol enter into respiratory path in which sequance ?

(A) Glycerol $\rightarrow$ acetyl co A $\rightarrow$ TCA cycle

(B) Glycerol $\rightarrow$ DHAP $\rightarrow$ PGAL $\rightarrow$ EMP

(C) Glycerol $\rightarrow$ Pyruvic acid $\rightarrow$ Krebs cycle

(D) Glycerol $\rightarrow$ PGAL $\rightarrow$ Krebs cycle

(145) Catabolic path is also called as amphibolic pathway because ..... .

(A) only include anabolic process

(B) it include only respiratory process

(C) it include both anabolic and catabolic process

(D) None of above

(146) Energy released from 1 gram of total breakdown is ..... .

(A) high, when, wheat starch as a reagent

(B) high when potato starch as a reagent

(C) high, when rice starch as a reagent

(D) Energy produce equally in all above

Answers : (138-D), (139-B), (140-C), (141-A), (142-C), (143-D), (144-B), (145-C), (146-D)

Respiratory quatient

Definition : At specific temperature and pressure the ratio of released CO_2 to the O_2 consumed during respiration is called (RQ) respiratory quatient.

Equation

$$RQ = \frac{CO_2 \text{ released during respiration}}{O_2 \text{ consumed during respiration}} = \frac{CO_2}{O_2}$$

- Burning capacity (ignition capacity) of respiratory substance of tissue can be known by RQ.

RQ of Carbohydrates

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$

$$RQ = \frac{6CO_2}{6O_2} = 1$$

- RQ of fats

$2(C_{51}H_{98}O_6) + 145\ O_2 \rightarrow 102\ CO_2 + 98H_2O + \text{energy}$

$$RQ = \frac{102\ CO_2}{145\ O_2} = 0.7$$

- RQ for protein is 0.9
- As O_2 is not consumed during anerobic respiration RQ is infinitive (α).
- When respiratory substances are more than one pure protein or fats never use as respiratory mollecules.

---

(147) How much RQ of carbohydrate ?

(A) 1 (B) less than 1 (C) more than 1 (D) α

(148) What is value of RQ in lipid ?

(A) 1 (B) less than 1 (C) more than 1 (D) α

(149) How much RQ of 2 $(C_{51}H_{98}O_6)$?

(A) 0.8 (B) 0.7 (C) 0.9 (D) 1

(150) $2\ (COOH) + O_2 \rightarrow 4CO_2 + 2H_2O$ + Heat How much RQ of given equation ?

(A) 4 (B) 1 (C) Less then 1 (D) α

(151) If RQ = 0.7 for one respiratory substance. What is indicates ?

(A) For breakdown more O_2 needed (B) For breakdown less O_2 needed

(C) O_2 is not used (D) it contain O_2

(152) $CH_3COCOOH + NADH_2 \rightarrow CH_3CHOHCOOH + NAD$ How much RQ of given equation ?

(A) 0 (B) Not to be conted

(C) 1 (D) α

(153) What will be the RQ for ethenol producing during alcoholic fermentation ?

(A) 0 (B) 1 (C) More than one (D) Infinite

(154) It RQ is α what it indicate ?

(A) more CO_2 produce (B) O_2 is produce (C) Not use of O_2 (D) it contain more O_2

(155) If RQ is maximum then which molecule occur ?

(A) Fat (B) Glucose (C) Proteine (D) carbonic acid

(156) How much RQ of hexosemonosaccharides ?

(A) 4 (B) 3 (C) 2 (D) 1

(157) For which more CO_2 releases than O_2 consumption during respiration ?

(A) Glucose (B) Fat (C) Protein (D) Sucrose

(158) RQ = ...

(A) $\frac{O_2 consumed}{released\ CO_2}$ (B) $\frac{released\ CO_2}{consumed\ O_2}$ (C) $\frac{released\ O_2}{consumed\ CO_2}$ (D) $\frac{released\ CO_2}{released\ O_2}$

(159) ..... change RQ will change.

(A) Light (B) Product (C) Substence (D) Type of respiration

(160) $2(C_{51}H_{98}O_6) + (X) \rightarrow (Y) + 98H_2O$ + Energy ?

(A) X = 145 O_2; Y = 102 CO_2 (B) X = 145 CO_2; Y = 102 O_2

(C) X = 102 O_2; Y = 145 CO_2 (D) X = 102 CO_2; Y = 145 O_2

(161) $2C_6H_{12}O_6 + 3O_2 \rightarrow 3C_4H_6O_5 + 3H_2O$ + 386 K.Cal What is RQ of given equation ?

(A) 0 (B) α (C) 3 (D) 0.3

(162) If RQ is - 0.7 what it indicate ?

(A) More O_2 is needed for respiration (B) less O_2 is needed for respiration

(C) No O_2 is used for respiration (D) it contain more O_2

(163) If RQ is less than one then material will be ..... .

(A) lipid (B) Carbonic Acid

(C) Protein (D) Glucose

(164) What indicates given figure ?

(A) Diagramatic representation of ATP synthesis in mitochondria.

(B) Diagramatic representation of ATP Release in mitochondria.

(C) Diagramatic representation of ATP synthesis in Cytoplasma

(D) Diagramatic representation of ATP release in cytoplasma

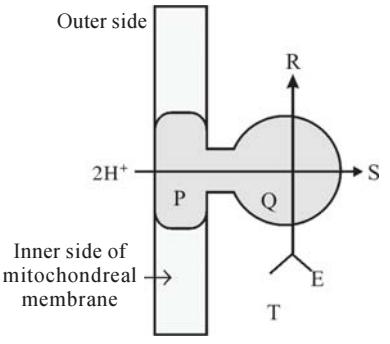


Figure 1

(165) Give the name of P and Q in figure 1.

(A) P–F_1 Q-F_2 (B) P–F_0, Q–F_1 (C) P–F_1 Q–F_0 (D) P–ADP- Q –ATP

(166) What indicate R in given figure 1 ?

(A) F_1 (B) Matrix (C) $2H^+$ (D) ATP

(167) What indicate T in given figure 1 ?

(A) F_1 (B) Matrix (C) $2H^+$ (D) ATP

(168) What indicates in given figure 1 ?

(A) F_1 (B) Matrix (C) $2H^+$ (D) ATP

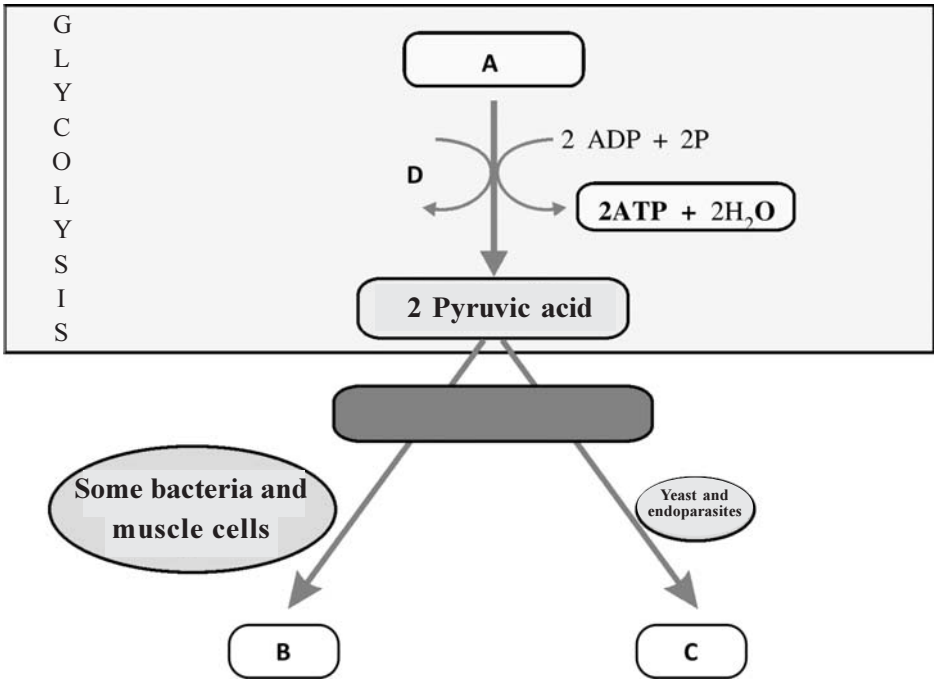


Figure 2

(169) Which biochemical process indicates given figure - 2 ?

(A) Glycolysis (B) Krebs cycle

(C) Fermentation (D) Amphibolic pathway

(170) What indicate P in given figure 2 ?

(A) Lactic Acid (B) Ethanol

(C) Hexose sugar (D) $NADH_2 \rightarrow NAD$

(171) What indicate S in given figure 2 ?

(A) $FADH_2 \rightarrow FAD$ (B) $NADH_2 \rightarrow NAD$

(C) $ADP \rightarrow ATP$ (D) Ethyl alcohol

(172) What indicate Q in given figure 2 ?

(A) Ethyl alcohol (B) Methyl alcohol

(C) Lactate (D) Phospho enol pyruvic acid

(173) What indicate R in given figure 2 ?

(A) Ethyl alcohol (B) Phosphoenol pyruvic

(C) Lactic acid (D) Hexose sugar

Select appropriate answer given below

(A) A and R both are true, R is explanation of A.

(B) A and R both are true, R is not explanation of A.

(C) A is true, R is flase.

(D) A is flase, R is true.

(174) A = ETS performed in cristy

R = Mitochondria matrix - TCA performed

(A) (B) (C) (D)

(175) A : RQ of Fat is less then one.

R : Fat has more oxygen then carbohydrate.

(A) (B) (C) (D)

(176) A : Aerobic and aenaerobic respiration start with glycolycis.

R : Aenaeroblec respiration occur in glycolysis only.

(A) (B) (C) (D)

(177) A : The ratio of released CO_2 to the O_2 consumed during respiration is called RQ

R : Burning capacity of respiratory substance can be known by RQ.

(A) (B) (C) (D)

(178) A : Ethanol produces by reduction of acetaldehyde

R : For that alcohol dehydrogenase enzyme is responsible.

(A) (B) (C) (D)

(179) A : 6 ATP produce during 2 $NADH_2$ oxidative phosphorylation.

R : 6 ATP produce during 3 $NADH_2$ oxidative phosphorylation.

(A) (B) (C) (D)

(180) A : Glycolysis is a first step of animal and plantcell respiration.

R : Other name of glycolysis is EMP.

(A) (B) (C) (D)

(181) A : Citric acid has 4 molecule of carbon.

R : Oxalosuccinic acid has 6 molecule of carbon.

(A) (B) (C) (D)

(182) Choose the correct option from column I and II:

	Coulmn - I	**Coulmn - II**	
a.	Phosphorylation phase of Glycolysis	p. 4 ATP	(A) (a–p), (b–q), (c–r), (d-s)
b.	Dephosphorylation phase of Glycolysis	q. 2ATP	(B) (a–r), (b–s), (c–q), (d–p)
c.	Fermentation of pyruvic acid	r. 30ATP	(C) (a–s), (b–r), (c–q), (d–p)
d.	Oxidation of pyruvic acid	s. 6ATP	(D) (a-s), (b–p), (c-q), (d-r)

(183)

	Coulmn - I		**Coulmn - II**	
(i)	Reduction of NAD	p.	Last stage of aerobic respiration	(A) r, a, s, p
(ii)	Reduction of FAD	q.	Succinate - Fumarate	(B) r, q, t, p
(iii)	Phosporylation of ADP	r.	Pyruvic $\rightarrow$ Acetyl Co. A	(C) a, r, p, s
(iv)	Oxidation of $NADH_2$	s.	2 – PGA $\rightarrow$ PEP	(D) p, t, q, s
		t.	PGAP $\rightarrow$ 3–PGA	

(184)

Coulmn - I	Coulmn - II
(a) Glycolysis	(i) mitochondrial matrix
(b) Krebs cycle	(ii) Cytoplasm
(c) Electron Transpart System (ETS)	(iii) inner membrane of mitochondria
(d) Ps-I and Ps-II	(iv) Thylakoids

	a	b	c	d
(A)	iii	iv	ii	i
(B)	ii	i	iii	iv
(C)	i	ii	iii	iv
(D)	iv	iii	ii	i

(185)

Coulmn - I	Coulmn - II
(a) H^+, OH^-	(i) glycolysis
(b) Pyruvic acid	(ii) Alcoholic fermentation
(c) $C_2H_5OH–CO_2$	(iii) Chemical synthesis
	(iv) Photolysis of water

	a	b	c
(A)	iv	i	iii
(B)	iv	i	ii
(C)	i	iv	ii
(D)	i	ii	iv

(186)

Coulmn - I	Coulmn - II
(a) Citric acid → Cisaconitic acid	(i) Dehydrogenation
(b) Succinic acid → fumaric acid	(ii) hydrolysis
(c) Oxalosuccinic acid → Ketoglutaric acid	(iii) dehydration
(d) cis aconitic acid → citric acid	(iv) decarboxylation

	a	b	c	d
(A)	ii	iv	iii	i
(B)	i	iii	iv	ii
(C)	iii	i	iv	ii
(D)	iv	ii	iii	i

Answers : (147-A), (148-B), (149-B), (150-A), (151-A), (152-A), (153-D), (154-C), (155-D), (156-D), (157-C), (158-B), (159-C), (160-A), (161-B), (162-A), (163-A), (164-A), (165-B), (166-D), (167-B), (168-C), (169-C), (170-C), (171-B), (172-C), (173-A), (174-B), (175-C), (176-C), (177-A), (178-A), (179-B), (180-B), (181-D), (182-D), (183-B), (184-B), (185-B), (186-C)

True - False type questions

(187) (1) The terminal transporter in cytochrome chain are cytochromes a and a_3 in oxidative phosphorylation.

(2) The terminal electron acceptor in last stage of oxidative phosphorylation is $\frac{1}{2}O_2$.

(3) Oxidative phosphorylation is performed on cristae membrane of mitochondria.

(4) Krebs cycle takes place in the matrix of chloroplast.

(5) Glycolysis is also known as bidirectional pathway of metabolism.

(A) T T F T T (B) T F F T F (C) T T F F T (D) T T T F F

(188) (1) Succinic acid is dehydrogenated to form fumaric acid.

(2) Krebs cycle is the second stage of aerobic respiration.

(3) Oxalo acetic acid reacts with acetyl Co-A; this is the first step of krebs cycle. Krebs cycle begins with this.

(4) For fats, respiratory quotient is 0.7.

(A) T T F T (B) T T T F (C) F T T F (D) T T T T

(189) (1) When fats are respiratory substrate, the ratio would be more than 1.

(2) By complete oxidation of one molecule of glucose 34 ATP are produced through oxidative phosphorylation.

(3) EMP pathway means glycolysis.

(4) Lactic acid is formed by the anaerohic respiration takes place in trhe muscles of animals.

(A) F T T F (B) F T T T (C) F T F T (D) T F F F

Answers : (187-D), (188-D), (189-B)

Questions for NEET

(190) Proton is released by proton, but how many protons are required to release 12 H_2O ?

(A) $24H^+$ (B) $48H^+$ (C) $12H^+$ (D) $6H^+$

(191) PGAL is produced from PGA. This process is known as ..... .

(A) Reduction (B) Oxidation (C) Hhydroxilation (D) Isomerisation

(192) What will be R.Q. of stored oil seeds ?

(A) More than one (B) Zero (C) Less than one (D) Equal to one

(193) Injury to plant leads to ..... .

(A) increases in respiration (B) decreases in respiration

(C) no change in respiration (D) first increase and then decrease in respiration

(194) What happens when succinate is converted to fumarate during TCA cycle ?

(A) Hydrogen is liberated (B) Oxygen is added

(C) Proton is removed (D) Electrons is added

(195) Who combinely discovered the scheme of glycolysis ?

(A) Embden, merisan, uts (B) Embden, meyerhof and parnas

(C) immersion, haufman and patterson (D) Avery, macleod mccarty

(196) What is the end product of ETS in mitochondria ?

(A) H_2O (B) H^+ (C) Electrons (D) All of these

(197) Arrange the enzymes increasing order of carbon numbers according their reaction on substrate

(P) Hexokinase (Q) Alcohol dehydrogenase

(R) Pyruvate dehydrogenase (S) Fumarase

(A) Q R S T (B) Q S R P (C) P S R Q (D) S P R Q

(198) Which is the place for ATP Synthesis ?

(A) ion channels (B) Plasma membrane (C) F_0- part (D) F_1- part

(199) Succinate + FAD = ...

(A) Fumarate + $FADH_2$ (B) Malate + $NADH_2$

(C) ISO citrate + $NADH_2$ (D) Citrate + H_2O

(200) During fast exercise glucose is converted into ..... .

(A) starch (B) glycogen (C) lactic acid (D) pyruvic acid

(201) Complete the reaction in a way that RQ = 4 (Choose the correct option)

16 $(COOH)_2$ + 80 = ... + ... + energy

(A) $CO_2 + 2H_2O$ (B) $8CO_2 + 4H_2O$ (C) $CO_2 + 8H_2O$ (D) $32CO_2 + 46H_2O$

(202) In all type of respiration, which important intermediate molecule is produced ?

(A) Acetyl Co. A (B) Oxaloacetate

(C) Pyruvic acid (D) Tricarboxylic acid

(203) What is phosphorylation ?

(A) Replace P in glucose

(B) Combination of some chemical with phosphoric acid

(C) Release of phosphoric acid from any chemical

(D) Removal of P from PGA

(204) Which of the following equation ?

(A) DHAP $\rightarrow$ PGAL (B) PGAL $\rightarrow$ PGAP

(C) Glucose $\rightarrow$ Glucose 6 - phosphate (D) Glucose - 6- phosphate $\rightarrow$ fructose - 6-phosphate

(205) Which of the following equation is true for glycolysis ?

(A) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 38$ ATP

(B) $C_6H_{12}O_6 \rightarrow 2CH_3CH_2OH + 2CO_2 + 2$ ATP

(C) $C_6H_{12}O_6 \rightarrow 2CH_3COCOOH + 2$ ATP + 2 $NADH_2$

(D) $C_6H_{12}O_6 \rightarrow 2CH_3COCOOH + 2$ ATP

Answers : (190-A), (191-B), (192-A), (193-A), (194-C), (195-B), (196-A), (197-A), (198-D), (199-A), (200-C), (201-D), (202-C), (203-B), (204-B), (205-C)

Gaseous Exchange

It occurs between enviroment and organism according to organism's needs through body surface or special respiratory organs. It does not involve any chemical reaction

Respiration

Liberation of energy from food by chemical breakdown

Cellulor Respiration

Chemical breadown of food substrate mediated by several enzymes to release energy stored in chemical bonds

Respiratory Quotient

Ratio of value of CO_2 evolved to the volume of O_2 consumed in respiration per unit time per unit weight at standard temperature and pressure

Anaerobic Respiration

Respiration that occurs in absence of oxygen It is common in microorganism

Aerobic Respiration

It is more efficient in terms of ATP production than anaerobic respiration. It occurs in mitochondria

Glycolysis

Series of biochemical reactions in which glucose is broken down to pyruvate with the release of usable energy in the form of ATP

Fermentation

Conversion of pyruvate to alcohol and organic acids in the absence of oxygen

Depending upon its products

Alcoholic Fermentation

Eathanol and CO_2 are produced

Lactic acid Fermentation

Only lactic acid is produced

Mixed Fermention

Produces lactic acid ethanol, formic acid and succinic acid

Glycolysis

It is same as that in anaerobic respiration and occurs in cytoplasm

Pyruvate Oxidation

It involves conversion of rlycolysis product pyruvate into acetyle coA that enters next step krebs' cycle. It is also a link reaction as it links glycolysis with krebs' cycle.

Kreb's Cycle

Acetyle CoA is converted to oxaloaccetate by enzymes and can be regained by intermediates. Thus, this the stage of cyclic processes

Terminal oxidation

Culminating step in which the energy of oxidation drives the synthesis of ATP

Electron Transport chain

Sequential arrangement of electron carriers in inner mitochondrial membrane that carry electrons to final electron acceptor i.e., oxygen

Oxidative Phosphorylation

Synthesis of energy rich ATP molecules with the help of energy liberated during oxidation of reduced coenzymes (NADH, FADH) produced in respiration.

22 Digestion and Absorption

- The components of the normal diet include carbohydrates, lipids, proteins, vitamins, minerals and water.
- Complex food materials are broken down by digestive enzymes to form simple substances which are digestible and these substances then absorbed in the alimentary canal and this biochemical process is called digestion.
- Digestive tract is muscular, which has mouth, (buccal/oral cavity-teeth), Pharynx, oesophagus, Stomach, duodenum, Small intestine, large intestine, rectum and anus aparture.
- In addition accessory digestive glands are - Liver, Gall bladder and pancreas are included in the digestive system.
- Alimentary tract - Injection of food, Digestion and absorption and ejection like process are associated with this tract.

Mouth (Oral cavity -Teeth)

- The mouth leads to the buccal cavity/oral cavity. It is bounded by the maxillary bones, cheeks, chin and the lips.
- Mouth is connected at postariorly with pharynx.
- Roof of mouth is made up of (1) Hard palate and (2) Soft palate.
- Tongue is present in the basal side of mouth.
- In oral cavity, in jaw tooth is embedded in socket it is called thecodont.
- Dentition of teeth is dichyodont type (Milky teeth and permanent teeth)
- Wjem 20 milky teeth fall, upto 32 number permanent teeth develop.
- In human, all teeth are not identical (Heterodont).
- Four types of teeth are found : (1) Incisor (I), (2) Canine (C), (3) Premolar (PM), (4) Molar (M)
- Each tooth has a crown, root and neck region.
- Crown is made of strong enamel which composed of calcium and phosphate as a dentine metal.
- Dentine / tooth pulp has connective tissue, blood vessels and nerves.
- The tongue is attached to the hyoid bone, at the base of mouth, through frenulum.
- On tongue taste buds are present - on the papillae, which are small projections.

Four types of papillae : (i) Vallate papillae (ii) fungiform papillae (iii) Filiform papillae (iv) Simple papillae

- In humans three pairs of salivary glands open in the oral cavity.
- The parotid glands are the main salivary gland which comes to lie below the ear.
- The sub mandibular glands lie on each side beneath the jaw bone.
- The sublingual glands lie beneath the tongue.

(1) What is responsible for converting food components in digestive tract into simple structural forms ?

(A) Hormones (B) Enzymes (C) Vitamins (D) Mineral elements

(2) Which types of enzymes are found in digestive tract ?

(A) Oxido reductase (B) Transferase (C) Hydrolases (D) Synthetases

(3) Which of the following is not a function of the digestive tract ?

(A) Synthesis of food (B) Churning and swallowing food

(C) Digestion of food (D) Egestion of undigested food

(4) It forms the roof of buccal cavity.

(A) Lips (B) Mandible (C) Hyoid bone (D) Palate

(5) What is the location of uvula ?

(A) Anterior side of hard palate (B) Middle region of hard palate

(C) Middle region of soft palate (D) Anterior region of soft palate

(6) Tongue, is attached to which bone to form floor of oral cavity ?

(A) Mandible (B) Hyoid (C) Maxilla (D) Palatine

(7) What is not correct for oral cavity ?

(A) Muscular tongue (B) Salivary gland

(C) Many teeth (D) Bifid tongue

(8) What is the Dental Formula for adult man ?

(A) $\frac{2123}{2122}$ (B) $\frac{2132}{2132}$ (C) $\frac{2123}{2124}$ (D) $\frac{2123}{2123}$

(9) Number of teeth formed only once in human.

(A) 4 (B) 22 (C) 12 (D) 32

(10) What is the type of human teeth ?

(A) Thecodont (B) Diphydont (C) Heterodont (D) All of above

(11) What is the number of teeth growing twice in human ?

(A) 12 (B) 32 (C) 30 (D) 28

(12) Which is the hardest region of tooth ?

(A) Dentine (B) Bone (C) Enamel (D) Pulp

(13) Select proper pair from the following.

(A) Crown-Bony region (B) Enamel-Calcium Phosphate

(C)Neck-Covered by crown (D) Dentine-Pulp

(14) Which part of the tongue is connected with frenulum in oral cavity ?

(A) Posterior (B) Anterior (C) Dorsal (D) Lateral

(15) Which Structures are found on the tongue besides papillae ?

(A) Mucous glands (B) Lymph glands (C) Lymph nodules (D) A and C both

Answers : (1-B), (2-C), (3-A), (4-D), (5-C), (6-B), (7-D), (8-D), (9-C), (10-D), (11-C), (12-C), (13-B), (14-A), (15-C)

Digestive tract :

Pharynx : It lies behind the nose, mouth and larynx.

- It is a cone shaped Muscular membranous passage.
- It is about 12.5 cm long.
- It is divided into three portions (1) Nasopharynx (2) oral pharynx and (3) Laryngeal pharynx.
- **Oesophagus :** Pharynx opens from distal end in the Oesophagus.

  Which has muscular tube like structure. Its length is 23 to 25 cm.

  It extends posteriorly passing through neck, thorax and diaphragm and leads to stomach.
- **Stomach :** It is the most dilatable portion of alimantary canal. It is divided in three portions.

  Cardiac stomach, fundus and pyloric stomach.

  Pyloric stomach opens in duodenum by pyloric valve.
- **Small intestine :** In man length of small intestine is 6.5 m.

  Duodenum : Initial part of it is 25 cm long and it is U-shaped portion in which common Hepeto-pancreatic duet opens.
- Jejunum - long convoluted area, which have digestive glands.
- Small intestine open in large intestine by ilio - coecal valve.
- **Large intestine :** It is called colon. It is 1.5 m. in length. It starts from close bay like structure is caecum. It has three areas. (1) Ascending colon (2) Transverse colon and (3) Descending colon which opens into the rectum which opens out through anus.

(16) Pharynx is behind which part ?

(A) Mouth (B) Nose (C) Larynx (D) all of above

(17) Pharynx is which type of muscular passage ?

(A) Cone shaped without serosa (B) Cone shaped with serosa

(C) Oblong without serosa (D) Tubular with serosa

(18) What is the length of pharynx ?

(A) 125 mm (B) 125 cm (C) 100 mm (D) 110 mm

(19) Oesophagus is how long ?

(A) 23 to 25 Meter (B) 23 to 25 feet

(C) 2.3 to 2.5 cm (D) 230 to 250 mm

(20) Part of digestive tract, which has sphincter muscle ..... .

(A) Duodenum (B) Pyloric stomach (C) Oesophagus (D) B and C both

(21) Which part of stomach is situated beneath diaphragm in human ?

(A) Cardiac stomach (B) Fundus (C) Pyloric stomach (D) Pyloric mouth

(22) What is the length of small intestine ?

(A) 6.25 cm (B) 6.25 meter (C) 6.25 mm (D) 625 meter

(23) Small intestine ends in which region ?

(A) At the region of pyloric valve (B) At the region of ilio-coecal valve

(C) At the region of 'u' bend of duodenum (D) Region, where Rectum opens

(24) What is the location of small intestine in human body ?

(A) In the Navel region of abdomen surrounded by duodenum.

(B) In the navel region of abdomen surrounded by ribs.

(C) In the Navel region of abdomen surrounded by large intestine.

(D) In the navel region of abdomen surrounded by pancreas and spleen.

(25) What is involved in the formation of common bile duct ?

(A) Cystic duct and hepatic duct. (B) Hepatic duct and salivary duct.

(C) Pancreatic duct and hepatic duct. (D) Bile ducts and pancreatic duct

(26) Caecum opens into which part of digestive tract in human ?

(A) Midgut (B) Colon (C) Fore gut (D) Hind gut

(27) Descending part of colon opens into ..... .

(A) Urinary bladder (B) Appendix (C) Caecum (D) Cloaca

(28) Finger like projection of caecum is called ..... .

(A) Colon (B) Intestinal nodes (C) Caecum tail (D) Intestinal tail

(29) Which organ of digestive tract glands are present in submucosal layer ?

(A) Duodenum (B) Rectum (C) Mid gut (D) Pancreas

(30) What is proper for arrangement of muscles in muscularis layer ?

(A) Circular muscles outside, longnitudinal muscles inside.

(B) Longitudinal muscles outside, circular muscles inside.

(C) Sphincter muscle outside, voluntary muscles inside.

(D) Voluntary muscles outside, Cardiac muscles inside.

(31) What is involved in the formation of submucosal layer ?

(A) cells, nerves, blood vessels, Bones

(B) bones, nerves, blood vessels and digestive glands

(C) tissues, nerves, blood vessels, and lymph vessels

(D) tissues, digestive glands, blood vessels and lymph glands

(32) What is present in papillae ?

(A) Blood vessels or large lymph vessels

(B) Only lymph glands

(C) Network of blood capillaries

(D) Network of blood capillaries and large lymph vessels.

(33) What is the function of papillae in digestive tract ?

(A) Increases absorptive surface area

(B) Storage of food

(C) Absorption of food

(D) A and C both

(34) Which cells secrete mucous in mucosal layer ?

(A) Oxyntic cells (B) Goblet cells (C) Peptic Cells (D) Gall bladder cells

Answers : (16-D), (17-B), (18-A), (19-D), (20-D), (21-B), (22-B), (23-B), (24-C), (25-A), (26-B), (27-D), (28-C), (29-A), (30-B), (31-C), (32-D), (33-D), (34-B)

Accessory digestive glands and functions of Alimentary canal :

- The wall of alimentary canal possesses four layers : from outside (1) Serosa (2) Muscularis (3) Submucosa and (4) Mucosa.
- Glands are present in submucossa of duodenum.
- In stomach, in mucosa small irregular folds are present and small finger like foldings called villi, in small intestine.
- Globlet cells of mucosa secrete mucus.
- Gastric glands are present in mucosa of stomach, which secretes the varuous digestive juices.

Liver :

- It is largest gland in the body. It weighs 1. 2 to 1.5 kg.
- It is wedge - shaped and reddish brown in colour. Bile juice is secreted by hepatic cells. Liver is made up of two lobes.

Gall bladder :

- The duct of gall bladder (cystic duct) along with the hepatic duct forms the common bile duct. The bile duct and the pancreatic duct open together. into the duodenum as the common hepatopancreatic duct.

Pancreas : It is situated between the loop of the 'U' shaped duodenum. It is an exocrine and as well as endocrine gland (endocrine - secretion of Insulin and glucagon, exocrine = secretion of pancreatic juice)

Functions of alimentary canal :

- **Mouth :** Mastication (biting and grinding of food), to soften the food, form the bolus and swallowing.
- **Pharynx :** Passes food to oesophagus.
- **Oesophagus :** Food passes to stomach.
- **Stomach :** Temporary storage of food, partial digestion of food by the gastric juice.
- **Small intestine :** Digestion of chyme, absorption of digested substances.
- **Large intestine :** Absorption of water, salt and glucose.
- **Rectum :** Stores semi-solid stool.
- **Liver :** Largest chemical factory of body prepares waste components for excretion.
- **Pancreas :** Its exocrine part secrets pancreatic juice and endocrine part secrets insulin and glucagon.

Functions of digestive system : Ingestion of food, digestion, absorption and ejection of stool.

---

(35) Which is largest gland in human ?

(A) Pancreas (B) Gall bladder (C) Liver (D) Spleen

(36) What is weight of liver in human ?

(A) 12 to 15 kg (B) 12 to 15 gm (C) 1200 to 1500 gm (D) 1.2 to 1.5 mg.

(37) Pancreas structure is similar to which gland ?

(A) Salivary gland (B) Sebaceous gland (C) Pineal gland (D) Spleen

(38) Which gland is with duct and without duct in digestive tract ?

(A) Gall bladder (B) Liver (C) Pancreas (D) Salivary gland

(39) Which hormones are secreted by pancreas ?

(A) Insulin (B) Glucagon

(C) Insulin or glucagon (D) Insulin and glucagon

(40) Absorption of water, minerals and glucose in large quantity, is by ..... .

(A) Stomach (B) Oesophagus (C) Duodenum (D) Colon

(41) ..... reacts with sugar and digested starch.

(A) Triglyceride (B) Tyrosine (C) Transferases (D) Ptylin

(42) Which organ absorbs water, mineral elements and glucose in less quantity ?

(A) Stomach (B) Large intestine (C) Oral cavity (D) A and B Both

(43) Which organ converts chyme into chyle ?

(A) Stomach (B) Large intestine (C) Small intestine (D) B and C Both

(44) Which organ converts poisonous and useless substances into excretory product ?

(A) Kidney (B) Liver (C) Gall bladder (D) Colon

(45) Which of the following digestive juice is acidic ?

(A) Intestinal juice (B) Salivary juice (C) Gastric juice (D) Pancreatic juice

(46) Which digestive juice contains solid 0.5% substances ?

(A) Gastric juice (B) Saliva (C) Pancreatic juice (D) Bile juice

(47) Which digestive juice contains 0.55 % solid substances ?

(A) Intestinal juice (B) Bile juice (C) Gastric juice (D) Pancreatic juice

(48) Which digestive juice does not contain enzymes ?

(A) Pancreatic juice (B) Bile juice (C) Salivary juice (D) Intestinal juice

(49) Which digestive juice contain lipase ?

(A) Gastric juice (B) Pancreatic juice (C) Intestinal juice (D) All of above

(50) ..... digestive juice, digests all types of components of food.

(A) Salivary juice (B) Pancreatic juice (C) Intestinal juice (D) Gastric juice

(51) Which digestive juice is associated with digestion of milk ?

(A) Salivary juice (B) Gastric juice (C) Pancreatic juice (D) Bile juice

Answers : (35-C), (36-C), (37-A), (38-C), (39-D), (40-D), (41-D), (42-A), (43-C), (44-B), (45-C),(46-B), (47-C), (48-B), (49-D), (50-C), (51-B)

Digestion :

- The process of biochemical transformation of complex food into a simple form suitable for absorption with the help of digestive enzymes is called digestion.

Digestion in Mouth :

- Saliva : Amylase (ptylin), Water (99. 5%) Lysozymes also there.

  Carbohydrates disaccharides

  (Starch) $\longrightarrow$ (maltose)

- Lysozyme acts as a anti bacterial substance.

Digestion in stomach :

- Gastric juice : Water (99.45%) , Dilute HCl, Pepsinogen, Renin, Lipase and mucin also there.
- Cells of pyloric stomach secretes gastrin, which induces the secretion of gastric juice.
- Dilute HCl, converts pepsinogen to pepsin, which digests the protein.

- Protein $\xrightarrow{\text{pepsin}}$ Proteiosis peptones.
- Renin in human babies hydrolyses (a milk protein) into paracasein.
- Lipase ctalyses hydrolysis of fats.
- After the digestion in stomach, the acidic food so formed is called chyme.

Digestion in small intestine :

- By the pyloric valve the partially digested chyme enters the small intestine.
- Chyme is digested by bile juice, pancreatic juice and intestinal juice.

Digestion by pancereatic juice :

Enterokinase of intestinal juice act on trypsinogen and digestion starts here.

Inactive trypsinogen

↓ Enterokinase

Proteiosis + peptones $\xrightarrow{\text{Active trypsin}}$ Polypeptide + Amino acid

Inactive chymotrypsinogen

↓ Trypsin

Proteiosis + peptones $\xrightarrow{\text{Active Chymotrypsin}}$ small polypeptide + few free Amino Acids

procarboxy peptidase

↓ Trypsin

Polypeptides $\xrightarrow{\text{carboxy peptidase}}$ peptide + Amino acid

Nucleic acid $\xrightarrow{\text{Nucleases}}$ Nucleotide + Nucleoside

Polysaccharide $\xrightarrow{\text{Amylase}}$ Disaccharide

Digestion by bile juice :

Fat $\xrightarrow{\text{lipase}}$ Diglycerides $\longrightarrow$ Monoglycerides.

Digestion by intestinal juice :

$$\text{Dipeptides} \xrightarrow{\text{Erepsin}} \text{Amino acid} \qquad \text{Maltose} \xrightarrow{\text{Maltase}} \text{Glucose} + \text{Glucose}$$

$$\text{Lactose} \xrightarrow{\text{Lactase}} \text{Glucose} + \text{Galactose} \qquad \text{Sucrose} \xrightarrow{\text{Sucrase}} \text{Glucose} + \text{Fructose}$$

$$\text{Di and mono glyceride} \xrightarrow{\text{Lipase}} \text{fatty acid} + \text{glycerol}$$

$$\text{Nucleotides} \xrightarrow{\text{Nucleotidase}} \text{Nucleoside} \xrightarrow{\text{Nucleosidase}} \text{Sugar} + \text{Nitrogenous base}$$

- In small intestine complete digestion can occurs. It is alkaline fluid, which is called chyle.
- Huge Mass of water absorption takes place in large intestine.

Role of Hormones in digestion :

No.	Hormone	Secretory part	Function
1.	Gastrin	Wall of stomach	Stimulates gastric gland to release gastric juice.
2.	Pancreozymin (Pz)	Wall of duodenum	Stimulates pancreas to release pancreatic juice and helps in secretion of intestinal juice.
3.	Secretin	Wall of duodenum	Secretion along with pancreozymin stimulates secretion of pancreatic juice and intestinal juice.
4.	Cholecytokinin (cck)	Wall of Duodenum	Stimulates gall bladder to release bile juice.
5.	Enterogastrone Or gastric inhibitory Peptide (GUP)	Wall of duodenum	Inhibits gastric juice secretion and stops stomach churning.

(52) Ptylin enzyme present in salivary juice is responsible for digestion of ..... .

(A) Carbohydrate (B) Lipid (C) Protein (D) Nucleic acid

(53) Ptylin enzyme of salivary juice acts in which medium ?

(A) Neutral medium (B) Alkaline medium (C) Acidic medium (D) None of above

(54) Peristaltic movement is observed in various parts of digestive tract. It is slowest in which organ ?

(A) Duodenum (B) Rectum (C) Stomach (D) Oesophagus

(55) Caesin present in milk is ..... .

(A) Sugar (B) Protein (C) Lipid (D) Vitamin

(56) Protein of milk is converted into paracaesin by ..... .

(A) Renin (B) Pepsin (C) dil. HCl (D) Trypsin

(57) Ptylin in stomach is ..... .

(A) Inactivated by pepsin (B) Inactivated by Renin

(C) Inactivated by HCl (D) Inactivated by trypsin

(58) Stomach is the main location for digestion of ..... .

(A) Lipid (B) Carbohydrate (C) Protein (D) all of above

(59) What is called muscle contraction in digestive tract ?

(A) Contraction (B) Transportation (C) Relaxation (D) Peristalsis

(60) Regulation of digestive juice in stomach is by ..... .

(A) Gastrin (B) Enterogastrone (C) A and B both (D) None

(61) What is function of HCI in Stomach ?

(A) Enzymes become soluble

(B) Pepsinogen is activated into pepsin

(C) Facilitate absorption of food

(D) Protection of stomach wall

(62) Which Digestive enzyme functions only in acidic medium ?

(A) Ptylin (B) Pepsin (C) Trypsin (D) Amylase

(63) Digestion of Carbohydrate initiated in which organ ?

(A) Stomach (B) Intestine (C) Mouth (D) Oesophagus

(64) Which is different, in given below ? Find out the odd one out.

(A) Secretin (B) Renin (C) Calcitonin (D) Oxytonin

(65) Which of the following is stimulated by enterokinase ?

(A) Pepsin (B) Pepsinogen (C) Trypsin (D) Trypsinogen

(66) Digestion of lipid is not possible in absence of ..... .

(A) Pigments (B) Bile salts (C) Cholesterol (D) Calcium

(67) ..... stimulates production of intestinal juice by cells of intestional wall ?

(A) Bile juice (B) Gastrin (C) HCI (D) Secretin

(68) Which enzyme is responsible for the digestion of emulsified lipid ?

(A) Sucrase (B) Lipase (C) Pepsin (D) Trypsin

(69) Find the odd one out.

(A) Glucagon (B) Gastrin (C) Ptylin (D) Secretin

(70) Cells of pancreas are not auto digested by its enzymes ..... .

(A) Enzymes are produced only when required.

(B) Cells are covered by mucous.

(C) Enzymes are produced in inactive form.

(D) Enzymes do not have coenzymes.

(71) Induces contraction of gall bladder : ..... .

(A) Enterogastrone (B) CCK (C) Gastrin (D) Secretin

(72) Enterokinase is associated with transformation of ..... .

(A) Tryspinogen → trypsin (B) Pepsinogen → pepsin

(C) Caesin → paracaesin (D) Protein → polysaccharide

(73) Pyloric valve prevents back flow of food into ..... .

(A) Stomach to oesophagus (B) oesophagus to mouth

(C) Duodenum to stomach (D) Stomach to duodenum

(74) Oxyntic cells are present in which part of digestive tract ?

(A) Ileum (B) Rectum (C) Duodenum (D) Stomach

(75) Fully digested, liquid alkaline food is ..... .

(A) Chyme (B) Chyle (C) Semidigested (D) Bile juice

(76) Which biological catalysts are responsible for the digestion of protein ?

(A) Amylase, pepsin, peptidase

(B) Enterogastrone, peptidase, pepsin

(C) Dipeptides, pepsin, protease

(D) Pepsin, protease, peptidase

(77) What is responsible for converting polysaccharide into disaccharide ?

(A) Protease (B) Amylase (C) Invertase (D) Amylose

Answers : (52-A), (53-B), (54-B), (55-B), (56-A), (57-C), (58-C), (59-D), (60-C), (61-B), (62-B), (63-C), (64-B), (65-D), (66-B), (67-D), (68-B), (69-C), (70-C), (71-B), (72-A), (73-C), (74-D), (75-B), (76-D), (77-B)

Absorption and Disorders of Digestive systems

- Complex molecules of food with very high molecular weight and large molecular size so absorption of those can not be possible.
- Active transport, osmosis and diffusion like processes are included in absorption.
- Glucose, several amino acid are absorbed by simple diffusion.
- Some of the amino acid, fructose absorbed Na^+ carrier.
- Water and dissolved substances absorbed by osmotic gradient.
- Fatty acid and bile salts react and form chylomicrons and then are absorbed by lymphatic vessels ● Maximum absorption takes place in small intestine.

DISORDERS OF ALIMENTARY CANAL :

- Starvation and malnutrition causes kwashiokar.
- Inflammation in gastric portion causes of gastritis.
- In acute gastritis, burning sensation is due due to food poisoning.
- Due to effect of gastric juice peptic ulcer can occur.

Ulcerative colitis : Ulceration and dilation of colons with passing of watery stools containing blood and mucus.

- When the liver is affected, skin and eyes turn yellow due to the deposition of bile pigments.
- It may be due to the production of too much bile pigments and due to failure of the liver cells to modify or exerete the bile or due to obstruction of the common bile duet.

---

(78) Amino acid is assimilated in ?

(A) Wall of rectum (B) Papillae of Ileum

(C) Lymph duct of Ileum (D) Lacteals of Ileum

(79) Completely digested food is mainly absorbed by ..... .

(A) Colon (B) Stomach (C) Small intestine (D) Large intestine

(80) Which components are absorbed by simple diffusion ?

(A) Disaccharides (B) Monosaccharides

(C) Polysaccharides (D) Protein

(81) Which components are absorbed by process of osmosis ?

(A) Lipid (B) Protein (C) Water (D) Minerals

(82) Which components are absorbed by lymph vessels ?

(A) Chyme (B) Chyme granules (C) Pigments (D) Water

(83) What are chyme granules ?

(A) Granules of chyme (B) Alkaline food with gastric juice

(C) Oil droplets of fatty acids (D) Oil droplets of glycerols

(84) What happens in a gastric disorder ?

(A) Reduction in digestion (B) Nausea and Vomitting

(C) Increase in digestion (D) Both A and B

(85) Peptic ulcer is formed due to / Peptic ulcer is formed due to :

(A) Chyme (B) Intestinal juice (C) Gastric juice (D) Saliva

(86) What are symptoms of ulceritis colitis ?

(A) Ulcers of colon (B) Distention of colon and bleeding

(C) Diahorrea with mucous (D) All of the above

(87) What are the reasons for vomitting ?

(A) Motion sickness (B) Mental stress

(C) Poisonous effect in any part of digestive tract (D) all of above

(88) Which organ of digestive tract is affected during jaundice ?

(A) Gall bladder (B) Liver (C) Pancreas (D) Stomach

Answers : (78-B), (79-C), (80-B), (81-C), (82-B), (83-C), (84-D), (85-C), (86-D), (87-D), (88-B)

A - Assertion, R - Reason.

Choose the correct option :

(A) A and R both are true and R is the correct explanation of A.

(B) A and R both are correct, R is not correct explanation of A.

(C) A is true, but R is false.

(D) A is false, but R is true.

(89) Statement A : Complex components are converted into simple structural form by various hydrolysing biocatalyst secreted from wall of digestive tract.

Reason R : Complex components are absorbed in their original form.

(A) (B) (C) (D)

(90) Statement A : Digestion of milk Protein is by pepsin in stomach.

Reason R : Renin converts casein into paracasein in stomach.

(A) (B) (C) (D)

(91) Statement A : Pancreas is ductless gland.

Reason R : Islets of langerhans secretes insulin and glucagon.

(A) (B) (C) (D)

(92) Statement A : Bile juice formed in liver does not contain enzymes.

Reason R : Liver plays important role in chemical digestion.

(A) (B) (C) (D)

(93) Statement A : Enterokinase activates trypsinogen.

Reason R : Enterogastrone inhibits secretion of gastric gland.

(A) (B) (C) (D)

(94) Statement A : Absorption of fructose is by carrier Na^+.

Reason R : Absorption of water and minerals takes place in colon.

(A) (B) (C) (D)

(95) Statement A : Pancreas secretes insulin and glucagon.

Reason R : Protease enzymes are formed in inactive form in pancreas.

(A) (B) (C) (D)

(96) Statement A : In human, digestion of food begins in oral cavity.

Reason R : In oral cavity, salivary juice containing ptylin is secreted.

(A) (B) (C) (D)

(97) Statement A : Liver of human is formed of three lobules.

Reason R : Hepatic cells forms bile salts.

(A) (B) (C) (D)

(98) Statement A : Disaccharides are converted into monosaccharides by erepsin.

Reason R : Lipase converts Di and monoglyceride into fatty acid and glycerol.

(A) (B) (C) (D)

(99) Statement A : Micro organisms and parasites, which have entered through mouth are de stroyed in stomach.

Reason R : Gastric juice contain dil HCL.

(A) (B) (C) (D)

(100) Statement A : Glycerol is totally Soluble in water, hence easily absorbed.

Reason R : Chyme is absorbed by lacteals present in papillae.

(A) (B) (C) (D)

Answers : (89-C), (90-D), (91-A), (92-C), (93-B), (94-B), (95-B), (96-A), (97-D), (98-D), (99-A), (100-C)

True - False (T - F) type questions. Select true or false option in the given statements.

(101) (1) Teeth are embeded in the bones of mandibles.

(2) Oesophagus and trachea opens into larynx.

(3) Human, have diphyodont type of dental arrangement.

(4) Digestive tract starts from anterior cavity, known as oral cavity.

(A) FFTF (B) FFFT (C) TTFF (D) TFTF

(102) (1) Papillae are present in the region of oesophagus.

(2) Colon has ascendings transverse and descending regions.

(3) Carbohydrates are digested in the stomach.

(4) Pancreas has a U shaped structure.

(A) TFTF (B) TTFF (C) FTFT (D) FFTT

(103) (1) Gall bladder secretes bile juice.

(2) Origin of amylase is salivary gland and pancrease.

(3) Valve formed of sphincter muscles are found at the end of pyloric stomach and rectum.

(4) Lysozyme is found in gastric juice.

(A) TFFT (B) FTTF (C) TFTF (D) TTTF

(104) (1) Digestion occurs in large intestine.

(2) Water and mineral ions are absorbed in large intestine.

(3) Faecal formation occurs in large intestine.

(4) Glucose and amino acids are absorbed in large intestine.

(A) TFFT (B) FTFT (C) TTFF (D) FTTF

(105) (1) Dipeptide peptidase amino acid

(2) Pepsinogen HCL pepsin

(3) Trypsinogen enterokinin trypsin

(4) Monoglyceride, protease, fatty acid and glycerol

(1) Dipeptide $\xrightarrow{\text{Peptidase}}$ Amino acid

(2) Pepsinogen $\xrightarrow{\text{HCL}}$ Pepsin

(3) Trypsinogen $\xrightarrow{\text{Enterokinin}}$ Trypsin

(4) Monoglyceride $\xrightarrow{\text{Protease}}$ Fatty acid and glycerol

(A) TFTF (B) TTFF (C) FFTT (D) TTTF

(106) (1) Caecum, produces intestinal juice.

(2) In each jaw, 2 types of teeth present in human.

(3) Malnutrition is reason for Kwashiokar.

(4) Salivary glands are four in human.

(A) TTFT (B) FFTT (C) FFTF (D) TTFF

(107) (1) Lysozyme present in saliva, removes acidity of food.

(2) Digested food is absorbed in small intestine.

(3) Liver is biggest chemical factory of body.

(4) Milk Protein is digested by lipase in stomach.

(A) FTFT (B) TFTF (C) TFFT (D) FTTF

(108) (1) Bile juice contains bile salts and bile pigments.

(2) Glucagon hormone is produced in pancreas.

(3) Pancreatic juice contain enterokinase co-enzyme.

(4) Enterogastrone is produced by the wall of duodenum.

(A) TTFT (B) FFTF (C) TFTF (D) FTTF

Answers : (101-A), (102-C), (103-B), (104-D), (105-B), (106-C), (107-D), (108-A)

(109) Match column - I and column - II :

	Column - I		**Column - II**	
(I)	Ptylin	(p)	Islets of langerhans	(A) (I - r) (II - s) (III - q) (IV - p)
(II)	Renin	(q)	Wall of duodenum	(B) (I - r) (II - q) (III - s) (IV - p)
(III)	Insulin	(r)	Saliva	(C) (I - r) (II - s) (III - p) (IV - q)
(IV)	Enterogastrone	(s)	Gastric juice	(D) (I - r) (II - p) (III - q) (IV - s)

(110) Match column - I and column - II :

	Column - I		**Column - II**	
(I)	Intestinal juice	(p)	Dil. HCl Present	(A) (I - p) (II - s), (III - r) (IV - t)
(II)	Pancreatic juice	(q)	Contain enterokinase enzyme	(B) (I - p) (II - s) (III - t) (IV - r)
(III)	Saliva	(r)	Do not digest any food substance	(C) (I - p) (II - s) (III - r) (IV - q)
(IV)	Bile juice	(s)	Digest all food substances	(D)(I - p) (II - s) (III - q) (IV - t)
		(t)	Digest carbohydrate	

(111) Match column - I and column - II :

	Column - I		Column - II	
(I)	Pyloric valve	(p)	Secretion of HCL	(A) (I - r) (II - p), (III - q) (IV - s)
(II)	Tongue	(q)	Secretion of pancreatic juice	(B) (I - r) (II - s) (III - q) (IV - p)
(III)	Oxyntic cells	(r)	Sphincter muscle	(C) (I - r) (II - q) (III - s) (IV - p)
(IV)	Exocrine cells	(s)	Papillae	(D) (I - r) (II - s) (III - p) (IV - q)

(112) Match column - I and column - II :

	Column - I		Column - II
(I)	Amylase	(p)	Intestinal juice
(II)	Billiverdin	(q)	Saliva
(III)	Enterokinase	(r)	Pancreatic juice
(IV)	Pepsinogen	(s)	Gastric juice
(V)	Lysozyme	(t)	Bile juice

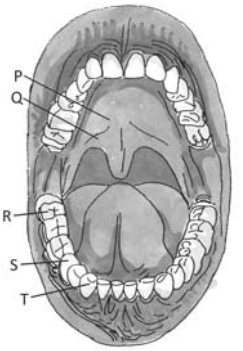


Figure 1

(A) (I - r) (II - t), (III - q) (IV - s) (V - p) (B) (I - r) (II - t) (III - p) (IV - q) (V - s)

(C) (I - r) (II - t) (III - p) (IV - s) (V - q) (D) (I - r) (II - t) (III - s) (IV - p) (V - q)

(113) Match column - I and column - II :

	Column - I		Column - II
(I)	Peptic ulcer	(p)	Ulcers in colon
(II)	Malnutrition	(q)	Ulcers in stomach
(III)	Jaundice	(r)	Kwashiokar
(IV)	Colitis	(s)	Skin and eyes yellow

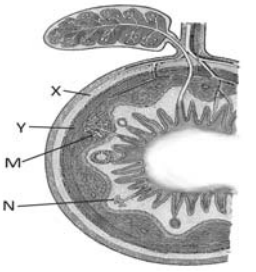


Figure 2

(A) (I - q) (II - r), (III - s) (IV - p) (B) (I - p) (II - q) (III - s) (IV - r)

(C) (I - q) (II - r) (III - p) (IV - s) (D) (I - p) (II - q) (III - r) (IV - s)

Answers : (109-C), (110-B), (111-D), (112-C), (113-A)

(114) P and Q in given diagram is associated with which structure ?

(A) Palate (B) Mandible

(C) Papillae / Frenulum (D) Oral Cavity

(115) What is correct option for R Shown in diagram I ?

(A) Pre molar (B) Grows two times in life

(C) Incisior / Incisor (D) Grows only once in life

(116) What is correct option for S shown in diagram I ?

(A) Its total number is 4 (B) Grows once in life

(C) Its total number is 8 (D) B and C Both

(117) In the given diagram I what is correct option for 'T' ?

(A) Its total number is 4. (B) Its total number is 4 pairs.

(C) It grows only once in life. (D) It is for grinding of food.

(118) In the given diagram 2, X and Y structure are part of ..... .

(A) Serosa (B) Mucosal layer (C) Muscularis (D) Sub mucosa

(119) In the given diagram -2 which type of structure is X and Y ?

(A) Striated muscles (B) Visceral muscles (C) Skeletal muscle (D) Cardiac muscle

(120) In the given diagram 2 which structure is shown by M ?

(A) Sub mucosal gland (B) Sub mucosal layer

(C) Mucosal gland (D) Lymph gland

(121) Which structure is shown by N in the given diagram 2 ?

(A) Mucosal layer (B) Circular muscles

(C) Mucosal gland (D) Sub mucosal gland

(122) Which type of cells are found in N Structure in diagram 2 ?

(A) Oxyntic (B) Goblet cells

(C) Zymogen cells (D) Hepatic cells

Answers : (114-B), (115-D), (116-D), (117-A), (118-C), (119-B), (120-A), (121-C), (122-B)

Points for PMT - NEET :

- 5-6 year old children do not have premolar.
- In man teeth which are formed only once are 8 premolars and 4 molars.
- In man which are formed twice are 20 in number.
- In man third molar is known as wisdom teeth.
- Largest salivary gland - Parotid gland - opens in the oral cavity through stenson duct.
- Smallest salivary gland - sublingual gland
- Lysosyme is present in saliva as well as tears.
- Shortest digestive organ - pharynx
- In humans daily approximately 1 to 1.5 litre saliva is formed.
- Gastrin hormones secreted by gastrin secretory cells are located in pyloric gland.
- Zymogen cells produce pepsinogen.
- Liver daily produces 600 ml bile juice.
- In gall bladder, bilirubin and biliverdin (yellowish) bile pigments are present.
- In pancreas daily 500 - 800 ml pancreatic juice is secreted.
- Neutral sat containing white fluid transported in lymphatic vessel is called chyle (pyolus)
- Energy needed by body is obtained from the energising components of food. It is measured in calories. It is expressed as K. Cal. or J (Joule) units. 1 Kcal = 1000 callories.
- When lgm of fod is metabolised the released energy is called its calorific valve
- Caloric Value of carbohydrates is 4.1 C.
- Caloric value of protein is h- 4.65 C
- Caloric value of lipid is 9.45 C.

● **Questions for NEET :**

(123) Hydrochloric acid is secreted by whom ?

(A) Goblet cells (B) Chief cells (C) Paneth cells (D) Oxyntic cells

(124) What is dental formula of 5 to 6 year child ?

(A) $\frac{2120}{2120}$ (B) $\frac{2012}{2012}$ (C) $\frac{2102}{2102}$ (D) $\frac{2012}{2102}$

(125) What is dental formula for 16 year boy ?

(A) $\frac{2123}{2123}$ (B) $\frac{2122}{2122}$ (C) $\frac{2102}{2102}$ (D) $\frac{2023}{2023}$

(126) Islets of langerhans ..... .

(A) ductless gland found in pancreas (B) modified lymph gland

(C) specific region of pituitary gland (D) ubule found in kidney

(127) Which statement is false for Bile juice ?

(A) It is important only for digestion of sugar. (B) It is essential for digestion of lipid.

(C) It is stored in gall bladder. (D) It has bile pigments.

(128) Where is parotid gland situated ?

(A) Beneath eye ball (B) Beneath jaw

(C) Beneath Ear (D) Beneath tongue

(129) Which specific types of cells present in liver ?

(A) B- Cells (B) Hepatic Cells (C) Cl- Cells (D) r- Cells

(130) Bilirubin and Biliverdin are found in ..... .

(A) Saliva (B) Bile juice (C) Pancreatic juice (D) Blood

(131) What is Hydrolysed by invertase ?

(A) Sucrose from glucose

(B) Maltose from starch

(C) Glucose and fructose from sucrose

(D) Sucrose from starch

(132) Trypsin is secreted by ..... .

(A) Liver (B) Pancreas (C) Stomach (D) Duodenum

(133) In which organ protein is converted into amino acids ?

(A) Stomach (B) Rectum (C) Oral cavity (D) Intestine

(134) What are proteolytic enzymes ?

(A) Amylase, lipase, pepsin

(B) Steapsin, ptylin, urease

(C) Trypsin, erepsin, pepsin

(D) Trypsin, Invertase, ptylin

(135) Chymotrypsin is ..... .

(A) Vitamin (B) Hormone

(C) Enzyme for digestion of lipid (D) Enzyme for digestion of protein

(136) In which organ carbohydrate, protein and lipid is fully digested ?

(A) Small intestine (B) Liver (C) Stomach (D) Large intestine

(137) Pancreatic juice plays important role in the digestion of ..... .

(A) Protein and carbohydrate (B) Protein and lipid

(C) Protein, lipid and carbohydrate (D) Only protein

(138) Which three secretions mix up in small intestine ?

(A) Pancreatic juice, bile juice, saliva

(B) Bile juice, gastric juice, saliva

(C) Bile juice, pancreatic juice, intestinal juice

(D) Bile juice, pancreatic juice, gastric juice

(139) Which of the following is not a digestive enzyme ?

(A) Enterogastrone (B) Amylase (C) Lipase (D) Trypsin

(140) Jaundice is disease linked with ..... .

(A) Circulatory system (B) Excretory system (C) Digestive system (D) Skin and eyes

Answers : (123-D), (124-C), (125-A), (126-A), (127-A), (128-C), (129-B), (130-B), (131-C), (132-B), (133-D), (134-C), (135-D), (136-A), (137-C), (138-C), (139-A), (140-C)

Digestion and Absorption

Mouth

ALIMENTARY CANAL

Salivary Glands
(secrete saliva with pH. 6-7)

- **Parotid Gland**
  - Largest salivary gland
  - Lie on the sides of the face
- **Sub lingual Gland**
  - Lie under front part of the tongue
- **Sub maxillary Gland**
  - Lie at the angles of the lower jaw

Oral/Buccal cavity
- Ingestion occurs; food is tasted by the taste buds, moistened with mucus and saliva, masticated by grinding teeth to smaller smaller particles and swallowed.
- Saliva contains salivary amylase which digests starch into maltose.

Pharynx

Gastric glands

Digestive Glands

- **Fundic (oxyntic glands)**
  - Secrete Hcl (oxyntic /parietal cells)
  - Secrete pepsinogen (peptic / chief cells)
- **Pyloric**
  Secrete Mucin
- **Cardiac glands**
  - Secrete mucin and very little pepsinogen

Oesophagus
- Oral cavity leads through pharynx into tube-like oesophagus.
- Food is pushed through the oesophagus by involuntary muscular movement called peristalsis.
- No digestion occurs here.

Liver
- It secretes bile which contain bile salts, bile pigments (bilirubin and biliverdin, cholesterol and phospholipids.)
- Bile salts and phospholipids help in the digestion of fats in the smal intestine by emulsification.

Stomach
- Widest organ of the alimentary canal; acts as food reservoir.
- The gastric glands of the stomach secrete gastric juice which is thoroughly mixed with food.
- Mostly protein digestion occurs in stomach, carbohydrate digestion gets inhibited due to highly acidic medium.

Pancreas
- Pancreatic juice contains $NaHCO_3$, trypsinogen, chymotrypsinogen, procarboxy peptidase, Amylase, DNAase, RNAase, lipase and helps in digestion of protein, carbohydrate, nucleic acids and fats.

Small intestine
- Largest, narrow, tubular part of the alimentary canal.
- It is divisible into 3 parts - Duodenum, Jejanum and ileum.
- Secretion from intestinal glands called succus entericus acts on all types of food.

Intestinal glands
- They lie in the wall of the small intestine and are of two types : Crypts of Lieber Kuhn and Branner's glands.
- Crypts of Lieberkuhn secrete enzymes and mucus and Brunner's glands secrete alkaline watery fluid, a little enzymes and mucus.
- Intestinal juice contains aminopeptidases, dipeptidases, amylase, maltase, dextrinase, sucrase, lactase, lipase, nucleotidases, nucleosidases.

Large intestine
- It is divisible into 3 parts - caecum, colon and rectum.
- Peristalsis pushes the indigestible materials (faccees) of the small intestine into the large intestine or colon.
- As facces enter the rectum, distension of it induces defaection reflex. It is initiated by peristalsis and forces the facces out towards anus.

23 Breathing and Exchange of Gases

Importance of Respiratory systems :

- Cellular respiration is a catabolic process, so energy can release.
- In cellular respiration oxidation of glucose takes place by glycolysis, Krebs cycle and oxidative phophorylation.
- During the respiration process CO_2 is produced which toxic to cell. Hence it becomes a compultion to cell remove it. Sameway for oxidative phosphorylation cell has necessity to get oxygen.
- In Respiratory system this process is possible. This system has two major paths : (1) Inspiration, by which O_2 is accepted. (2) Expiration by which CO_2 is released.
- In body for the supplies O_2 and removal of CO_2 circulatory system and respiratory system are these.
- The respiratory system consists of organs that exchange gases between atmosphere and blood.
- Blood transports gases between lung and cells.
- The overall exchange of gases between atmosphere, blood and cell is called respiration in general.

(1) It is not proper for complete oxidation of glucose.
(A) TCA Cycle (B) ETS (C) Fermentation (D) Glycolysis

(2) What is not true for human respiratory system ?
(A) CO_2 is removed by it and O_2 is obtained.
(B) It has two passage, Inspiration and expiration.
(C) It makes exchanges of gases possible at pulmonary level.
(D) It makes exchange of gases possible at cellular level.

(3) In human for the exchange of CO_2 and O_2 which system is linked with respiratory system ?
(A) Nervous system (B) Circulatory system (C) Digestive system (D) Excretory system

(4) Sequentially with which exchange of gases takes place, termed as breathing ?
(A) Atmosphere -blood -cells (B) Blood -Atmosphere -cells
(C) Atmosphere -cells- blood (D) Blood - cells - atmosphere

(5) Role of blood in respiration in human ..... .
(A) Connects exchanges of gases at pulmonary and cellular level
(B) Connects exchange of gases at atmospheric and pulmonary level
(C) Connects exchange of gases at atmospheric and cellular level
(D) All of above

Answers : (1-C), (2-D), (3-B), (4-A), (5-A)

Respiratory system of human : The respiratory system of human consists organs like nose, pharynx, larynx, trachea, bronchi and lungs.

(1) Nose : It has external portion jutting out from the face and an internal portion lying hidden inside the skull. It is divided into external noses, nasal chamber and internal nosed.

● **External noses or Nostrils :** Two nostrils are seperated by septums

● **Nosal chamber :** The internal region of nose is a large cavity within the skull. Anteriorly it opens through external noses and posteriorly it communicates with the pharynx through internal noses. Each noses chamber is divided into their regions lower vestibular, middle respiratory and upper olfactory. The two nosal chambers are also separated by the nosal septum.

(2) **Pharynx :** It is a tube about 12.5cm long, serving as a passage way for air and food, Pharynx is divided into three parts.

(i) Nasopharynx : The upper most portion of the pharynx from which air passes.

(ii) Osopharynx : The second portion of the pharynx, lies behind the buccal cavity from which air and food passes.

(iii) Laryngo pharynx : The lowest portion of the pharynx. It extends downward and empties into oesophagus posteriorly and into the larynx (voice - box) anteriorly from which air and food passes.

(3) **Larynx :** It is a passage that connects the pharynx with the trachea. The leaf shaped piece of cartilaege called glottis is always remains open except during swallowing. The mucous membrane of the larynx contains vocal cords. It has ability to vibrate. This ability allows us to speech.

(4) **Trachea :** It is about 12cm in length and 2.5cm in diameter. It extends from the larynx to the middle of the thoracic cavity, where it divides into right and left primary bronchi. It is a tubular passage way for air.

The trachea is divided into right primary bronchus, which enters the right lung and left primary bronchus-which enters the left lung. The right primary bronchus is more vertical, shorter and wider than the left. Along the length in the trachea and bronchi at short distances 'C' shaped incomplete; cartilagenous rings occur. They keep the respiratory passage open and prevent it from blockage.

- When primary bronchi enter into lungs, divide as follows. (In each lobe of lungs)

- Primary bronchi divide in each lung lobe into secondary bronchi $\rightarrow$ Tertiary bronchi or segmental bronchi divides further into bronchioles $\rightarrow$ Bhronchioles further divide into terminal bronchioles ended into alveoli $\rightarrow$ millions of alveoli in each lung, Each alveolus is sac like structure surrounded by pulmonary cell and enveloped by a net work of blorod capillaries.

- Bronchial tree : It is arising by trachea and alveoli between network of tubules are present.

(5) **Lungs :** The lungs are paired, cone shaped organs lying in the thoraeic cavity. Protected by rib-cage, The diaphragm is placed beneath them. Two layers collectively called the plural membrane covers each lung.

- The outer layer is attached with the wall of thoracic cavity.

- The inner layer covers, the lungs themselves.

- The space between two layers contains a lubricating fluid secreted by the membranes. This fluid protects lungs against shock.

Right lung	Left lung
(1) It is thickers, broader, longer and heavier.	(1) It is thinner, narrower, shorter and lighter.
(2) It has three lobes, (upper, lower and middle lobess which are demarkated by two fissures one ablique and one horizontal)	(2) It has two lobes.(upper and lower lobes, which are demarkated by one fissure (oblique)
(3) Cardiac notch in which the heart lies is not present.	(3) Cardiac notch is present.

(6) How many organs included in respiratory system perform functions other than respiration ?

(A) 2 (B) 1 (C) 3 (D) 4

(7) Large part of nose enclosed by bone of cranium ..... .

(A) External nares (B) Nasal chamber (C) Internal nares (D) A and C both

(8) What is connected by internal nares ?

(A) Nasal chamber - pharynx (B) Nasal chamber - larynx

(C) Nasal chamber - trachea (D) Nasal chamber - oesophagus

(9) Which region of nasal chamber is near to internal nares ?

(A) Pre olfactory region (B) Respiratory region (C) A and B (D) olfactory region

(10) Number of vestibular region in nose.

(A) 2 (B) 4 (C) 6 (D) 1

(11) Which region of pharynx,transport air and food ?

(A) Nasopharynx (B) Buccopharynx (C) Larynx (D) B and C

(12) Which part of pharynx opens posteriorly into oesophagus ?

(A) Buccopharynx (B) Laryngo -pharynx (C) Nasopharynx (D) Larynx

(13) Normally epiglottis remains closed in which condition ?

(A) During Yawning (B) While drinking water (C) Speaking (D) A and B

(14) What is length from larynx to the midregion of thoracic cavity ?

(A) 12.5 cm (B) 2.5 cm (C) 12 cm (D) 24.5 cm

(15) Length of respiratory passage, having 'C' shaped cartilagenous ring.

(A) 12 cm (B) Less than 12 cm (C) 2.5 cm (D) More than 12 cm

(16) Total number of secondary bronchioles.

(A) 2 (B) 5 (C) 3 (D) 4

(17) Bronchiole, totally located in pulmonary lobules ..... .

(A) Secondary (B) Primary (C) Tertiary (D) A and B

(18) Having same number as alveoli ..... .

(A) Secondary bronchiole (B) Tertiary bronchiole

(C) Terminal bronchiole (D) Primary bronchiole

(19) It is improper for Bronchial tree.

(A) Primary bronchiole (B) Tertiary bronchiole

(C) Secondary bronchiole (D) Trachea

(20) What is proper for lungs with median fissure ?

(A) Small and light (B) Parallel fissure

(C) Presence of cardiac notch (D) Thin and narrow

(21) What is proper for both lungs ?

(A) cardiac notch (B) Median fissure (C) Oblique fissure (D) All

(22) Cartilagenous tissue in the structure-

(A) Trachea (B) Bronchus (C) Primary bronchiole (D) A and C both

(23) What is proper for left bronchus compared to right bronchus ?

(A) Narrow (B) More straight (C) Short (D) (B) and (C)

Answers : (6-C), (7-B), (8-A), (9-D), (10-A), (11-D), (12-B), (13-B), (14-C), (15-D), (16-B), (17-C), (18-C), (19-D), (20-B), (21-C), (22-D), (23-A)

Mechanism of respiration, respiratory volumes and capacities, transportation of respiratory gases :

Respiration process : It is phenomenon of breathing.

Breathing :

- **Inhalation :** The process of drawn in air from the atmosphere toward lungs is called inhalation.
- **Exhalation :** The process of exporatation of air from lungs into the atmosphere is called exhalation.
- **Inhalation + exhalation = Breathing**

- In this process diaphargm and ribs play important role.
- The diaphragm is dome shaped separates the thoraic cavity and abdominal cavity. It is attached anteriorly with sternum and at its posterior, it remains attached to the vertebral column.
- Inter costal muscles are asssociated with the ribs.

	Inspiration process		**Exhalation process**
–	Contraction of diaphragm muscles.	–	Relaxation of diaphragm muscles.
–	The diaphragm is pulled down wards.	–	The diaphragm is more upwards.
–	The size of thoracic cavity is increase.	–	The size of thoraic cavity is decrease.
–	The partial pressure of air in lungs is reduced.	–	The partial pressure of air in lungs increases by the contraction of lungs.
–	Air from atmosphere move into lungs.	–	Under such pressure the air within the lungs goes out into the atmosphere.
–	Exchanges of O_2-CO_2 in alvadli	–	CO_2 contain air removed from the alveolli.
–	Ribcage moves upper and out side.	–	Ribcage moves lower and interior side.
–	In rapid inhalation process muscles of neck also plays in important role.	–	In rapid exhalation process muscles of abdominal also plays important role.

Respiratory volumes and capacities : Many factors effect respiratory capacity viz : Person's age, size, sex and physical conditions.

(1) Tidal Volume (TV) : Normal quiet breathing moves approximately 500 ml of air into and out of the lungs with each breath. This volume is referred as a tidal volume (TV).

(2) Inspiratory Reserve Volume (IRV) : The amount of air that can be inhaled forcibly over the normal (tidal) volume is the IRV. Normally IRV is between 2500 ml to 3000 ml.

(3) Exparatory Reserve Volume (ERV) : The amount of air that can be forcibly exhaled after a normal (tidal) expiration is known as ERV. The ERV is approximately 1000 ml to 1100 ml.

(4) Residual volume (RV) : It is the amount of air remains in the lungs after a forcible expiration. The approximately RV is 1100 ml to 1200ml.

(5) Inspiratory Capacity (IC) : It is the total volume of air which can be inhaled by a person after normal expiration i. .e IC is a sum of TV and IRV. It is about 3000 ml to 3500 ml.

(6) Expiratory Capacity (EC) : It is the total volume of air which can be exhaled by a person normal inspiration i.e. EC is a sum of TV and ERV. It is about 1500 ml to 1600 ml.

(7) Functional residual capacity (FRC) : It is a volume of air that will remain in the lungs after normal expiration. FRC is the sum of ERV and RV. FRC is about 2100 ml to 2300 ml.

(8) Vital Capacity (VC) : It is the total of air which can be breathe by a person. The VC is the sum of TV, RV and ERV, VC is about 4000 ml to 4600 ml.

(9) Total Lung Capacity (TLC) : It is the amount of air in the lungs and respiratory passage after a maximum inspiration. The TLC is the sum of TV, IRV, ERV or VC + RV. TLC is about 5100 ml. to 5800 ml.

Transport of respiratory gases :

- The interstitial fluid makes possible exhanges between alveoli and blood.
- Here transport of O_2 and CO_2 by simple diffusion.
- The partial pressures (in mm Hg) of O_2 and CO_2 at different parts is compared with atmosphere as under.

Respiratory gas	Atmospheric Air	Alveoli	Blood (deoxygenated)	Blood (oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

(24) Diaphragm is connected with ..... on its dorsal side.

(A) Ribs (B) Sternum (C) Vertebral column (D) Femur

(25) Process associated with breathing.

(A) Contraction of diaphragm muscle (B) Relaxation of diaphragm muscle

(C) Partial pressure of air increases in lungs (D) Decreases size of thoracic cavity

(26) When diaphragm is pushed up, process occurs is ..... .

(A) Contraction of diaphragm muscles (B) Relaxation of diaphragm muscles

(C) Diaphragm is pushed down (D) Increases size of thoracic cavity

(27) Which situation is observed when diaphragm contracts ?

(A) Decreases size of thoracic cavity, partial pressure of air increases in lungs.

(B) Size of thoracic cavity decreases, partial pressure of air decreases in lungs.

(C) Size of thoracic cavity increases, partial pressure of air increases in lungs.

(D) Size of thoracic cavity increases, partial pressure of air decreases in lungs.

(28) Contraction and relaxation of diaphragm is sequentially connected with which process ?

(A) Exhalation - Inhalation (B) Inhalation - Inhalation

(C) Inhalation - Exhalation (D) Exhalation - Exhalation

(29) When breathing process is slow ?

(A) Anger (B) Excersing (C) Excited condition (D) Sleeping

(30) Phenomenon, which do not affect respiratory capacity.

(A) Age (B) Size (C) Caste (D) Physical condition

(31) Total volume of air, normaly inhalation and forcefully taken is ..... .

(A) ERV (B) IRV (C) RV (D) IC

(32) What is obtained when TLC - FRC ?

(A) EC (B) IC (C) VC (D) RV

(33) What is obtained in EC-TV + RV ?

(A) VC (B) TLC (C) FRC (D) IC

(34) What is volume of air by VC - IC ?

(A) 2500 to 3000 ml (B) 4000 to 4600 ml (C) 1000 to 1100 ml (D) 3000 to 3500 ml

(35) Approximate volume of air, coming out during sneezing ?

(A) 1500 to 1600 ml (B) 1000 to 1100 ml (C) 3000 to 3500 ml (D) 2100 to 2300 ml

(36) Total volume of air taken by person during breathing is ..... .

(A) EC (B) VC (C) IC (D) FRC

(37) What has largest volume ?

(A) EC (B) ERV (C) FRC (D) RV

(38) Air entering during yawning in lungs has volume ..... .

(A) EC (B) IC (C) VC (D) TLC

(39) During excessive excercise air taken by person and given out, volume is ..... .

(A) 3000 to 3500 ml (B) 1500 to 1600 ml (C) 5100 to 5800 ml (D) 4000 to 4600 ml

(40) Exchange of O_2 and CO_2 between blood and tissue cells occurs by ..... .

(A) Cerebro spinal fluid (B) Coelomic fluid (C) Inter cellular fluid (D) Lymph

(41) Exchange of gases at respiratory surface occurs by which process ?

(A) Simple diffusion (B) Fascilitated diffusion (C) Active transport (D) All

(42) How less mm Hg is partial pressure of O_2 in alveolus than atmosphere ?

(A) 104 (B) 159 (C) 5 (D) 55

(43) Blood Capillary having similar partial pressure of CO_2 as that of alveolus.

(A) Pulmonary artery (B) Hepatic portal vein (C) Pulmonary vein (D) Renal vein

(44) Partial pressure of O_2 in blood of left auricle ..... .

(A) 40 (B) 95 (C) 45 (D) 104

(45) Difference of partial pressure of CO_2 mm hg between pulmonary artery and renal artery is ..... .

(1) Left ventricle (2) Right ventricle (3) Pulmonary artery (4) Pulmonary vein

(5) Hepatic artery (6) Hepatic portal vein (7) Different vein (8) Renal vein

(A) 1, 3, 5, 7, (B) 1, 3, 5, 6, 8 (C) 2, 3, 5, 6, 8 (D) 1, 4, 5, 7

(46) What is difference of partial pressure of CO_2 mm Hg between pulmonary artery and Renal artery ?

(A) 40 (B) 95 (C) 45 (D) 104

Answers : (24-C), (25-A), (26-B), (27-D), (28-C), (29-D), (30-C), (31-D), (32-B), (33-C), (34-C), (35-A), (36-B), (37-C), (38-B), (39-D), (40-C), (41-A), (42-D), (43-C), (44-B), (45-C), (46-D)

Transport of O_2 in the blood :

- Nearly 97% O_2 is transported through RBCs.
- Haemoglobin, a respiratory pigment, present in RBCs is responsible for transport of O_2; Each RBC transports around one billion molecules of O_2.

Reactions at respiratory surface :

- $H + Hb \rightarrow H{\cdot}HB$ (Haemoglobinic acid)
- $H.Hb + O_2 \rightarrow H{\cdot}Hb\ O_2$ (Oxyhaemoglobinic acid)
- $H.HbO_2 + KHCO_3 \rightarrow KHbO_2 + H_2CO_3$
- Thus, O_2 is transported in a form of potassium oxyhaemoglobin ($KHbO_2$) near tissue surface $KHbO_2$ splits to release Potassium haemoglobin and oxygen.

  $KHbO_2 \rightarrow KHb + O_2$ (Near tissue surface)

(47) Transport of O_2 takes place by ..... .

(A) RBC (B) WBC (C) Blood plasma (D) A and C

(48) How many O_2 molecules are carried by one RBC ?

(A) 10^7 (B) 10^9 (C) 10^8 (D) 10^6

(49) Where is HHb Present ?

(A) At the respiratory surface in RBC of Blood.

(B) Blood Surrounding tissue cells.

(C) Respiratory surface blood plasma.

(D) Blood plasma surrounding tissue.

(50) Which component is formed in the blood by O_2 at respiratory surface ?

(A) HHb (B) $KHCO_3$ (C) $KHbO_2$ (D) $HHbO_2$

(51) Which part of the Hb accepts O_2 ?

(A) R - Group (B) Fe^{+2} (C) NH_2 - Group (D) COOH - Group

(52) By which process potasium oxyhaemoglobin is formed in R.B.C ?

(A) $HHbO_2 + H_2Co_3$ (B) $KHCo_3 + H_2Co_3$ (C) $HHbO_2 + KHCO_3$ (D) $HHbO_2 + KCl$

(53) How much O_2 is transported in the form of $KHbO_2$?

(A) 100 % (B) 3 % (C) 97 % (D) 70 %

(54) Which is correct option for transport of O_2 from respiratory surface to tissue cells ?

(A) $O_2 \rightarrow KHbO_2 \rightarrow HHbO_2 \rightarrow O_2$ (B) $O_2 \rightarrow HHbO_2 \rightarrow KHbO_2 \rightarrow O_2$

(C) $O_2 \rightarrow KHCO_3 \rightarrow HHbO_2 \rightarrow O_2$ (D) $O_2 \rightarrow H_2CO_3 \rightarrow KHbO_2 \rightarrow O_2$

(55) Reactant at respiratory surface for carbonic anhydrase : ..... .

(A) CO_2 (B) H_2O (C) H_2CO_3 (D) A and B

(56) Which is proper option for transport of O_2 ?

(A) $KHbO_2$ (B) NH_2HPO_4 (c) NaH_2PO_4 (D) $HHbO_2$

Answers : (47-D), (48-B), (49-A), (50-D), (51-B), (52-C), (53-C), (54-B), (55-C), (56-A)

Transport of CO_2 in blood :

- CO_2 Produced through cellular respiration in cells diffuses into the blood within the capillaries. It is transported in two forms :
- (1) Physical solution and
- (2) Chemical compound

(1) In a form of physical solution : About 10% of CO_2

- Combines chemically with water of **plasma** forming carbonic acid.

(1) $CO_2 + H_2O \rightleftarrows H_2CO_3$ (Carbonic acid) Any increase in its concentration causes the disassociation of H_2CO_3 into hydrogen ion and bicarbonate ion $H_2CO_3 \rightleftarrows H^+ + HCO_3^-$

If all amout of Co_2 is transported by blood streams. pH of blood would be lowered from its normal level. i.e. 7.4 about 4.5 This would be instantly fatal.

(2) In a form of chemical compound : (In RBCs)

(A) Carbamino compounds : (20% Nearly)

$CO_2 + Hb.NH_2 \rightarrow Hb.NH.COOH$ (carbamino haemoglobin)

(B) As a bicarbonate form (Nearly 70%)

↓

↓	↓
• **In RBCS**	• **In Blood plasma**
• $CO_2 + H_2O \xrightarrow[\text{anhydrase}]{\text{Carbonic}} H_2CO_3$ $H_2CO_3 + KHb \rightarrow KHCO_3 + HHb$ (Thus maintain the pH of RBS)	• By the phosphate buffer $Na_2HPO_4 + H_2CO_3 \rightarrow NaHCO_3 + NaH_2PO_4$ (Bisodium Hydrogen phosphate) sodium carbonate — sodium bihydrogen phosphates
• Wall of RBC is Permeable for negative ions (Cl^-, HCO_3^-) while impearmiable for positive ions. (Na^+, K^+)	• By the blood proteins : $NaPr + H_2CO_3 \rightarrow NaHCO_3 + HPr$ (sodium protein) (proteinic acid) • By the chloride shifting (surrounding to tissue need)
• In RBC, by the $KHCO_3$ increase the alkalinity of RBC so Cl^- shifting is needed for decrease alkalinity.	
• Chloride shifting :(surrounding to tissue cell)	• By the chloride shifting (surrounding to tissue cell)
$Cl^- \leftarrow$	$Cl^- + Na^+ \leftarrow NaCl$
$KHCO_3 \rightarrow K^+ + HCO_3^- \rightarrow$	HCO_3^-
$K^+ + Cl^- \rightarrow KCl$	$Na^+ + HCO_3^- \rightarrow NaHCO_3$
• RBC on respiratory surface has HHb, KCl and Hb.NH COOH.	• Blood plasma on respiratory surface, it has $NaHCO_3$ • Chloride reshifting (From respiratory surphase to lungs)
$HHb + O_2 \rightarrow HHbO_2$	
$KCl \rightarrow K^+ + Cl^- \rightarrow$	Cl^-
$HCO_3^- \leftarrow$	$HCO_3^- + Na^+ \leftarrow NaHCO_3$
$K^+ + HCO_3^- \rightarrow KHCO_3$	$Na^+ + Cl^- \rightarrow NaCl$

Released CO_2 on respiratory surface (lungs)

- $H_2CO_3 \rightarrow H_2O + CO_2 \uparrow$
- $2KHCO_3 \rightarrow K_2CO_3 + H_2O + CO_2 \uparrow$
- $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2 \uparrow$
- $HbNHCOOH \rightarrow HbNH_2 + CO_2 \uparrow$

(57) How much CO_2 is brought at respiratory surface by blood plasma ?

(A) 70 (B) 90 (C) 30 (D) 80

(58) In which form CO_2 is not transported at respiratory surface ?

(A) $NaHCO_3$ (B) $KHCO_3$

(C) Physical Solution (D) Carbamino haemoglobin

(59) Reactant for carbonic anhydrase in blood plasma surrounding tissue cells.

(A) CO_2 (B) H_2CO_3 (C) CO_2 and H_2O (D) None

(60) Which are sequential changes during chloride shift and reshift ?

(A) $NaCl \rightarrow Cl^- \rightarrow KCl \rightarrow Cl^- \rightarrow NaCl$ (B) $NaCl \rightarrow Cl^- \rightarrow NaCl \rightarrow Cl^- \rightarrow KCl$

(C) $KCl \rightarrow Cl^- \rightarrow NaCl \rightarrow Cl^- \rightarrow KCl$ (D) $KCl \rightarrow Cl^- \rightarrow KCl \rightarrow Cl^- \rightarrow NaCl$

(61) Removal of CO_2 by blood plasma at respiratory surface.

(A) $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (B) $Hb{\cdot}NH{\cdot}COOH \rightarrow Hb{\cdot}NH_2 + CO_2$

(C) $H_2CO_3 \rightarrow H_2O + CO_2$ (D) $2\ KHCO_3 \rightarrow K_2CO_3 + H_2O + CO_2$

(62) Which components are present in RBC flowing in blood from pulmonary artery ?

(1) Hb.NH.COOH (2) KCl (3) $NaHCO_3$ (4) $KHCO_3$ (5) HHb (6) $KHbO_2$ (7) H_2CO_3

(A) 1, 2, 3, 6, (B) 1, 2, 5 (C) 1, 2, 4, 5, 7 (D) 1, 2, 4, 5, 6, 7

(63) Buffer component present in blood plasma : ..... .

(A) NaH_2PO_4 (B) $NaHCO_3$ (C) Na_2HPO_4 (D) $KHCO_3$

(64) During transport of CO_2 in blood formation and dissociation of Nacl, Sequentially as -

(A) tissue cell and respiratory surface

(B) respiratory surface and cell

(C) tissue cell and tissue cell

(D) respiratory surface and respiratory surface

(65) During backshift of chloride, Cl^- joins with which component in blood plasma ?

(A) KCl (B) $KHCO_3$ (C) NaCl (D) $NaHCO_3$

(66) Source of necessary Cl^- for chloride shift is ..... .

(A) KCl (B) HCl (C) NaCl (D) NH_4Cl

(67) Transport of CO_2 by blood plasma as chemical composition by ..... .

(A) Na_2HPO_4 (B) Cl^- Shift (C) NaPr (D) $Hb.NH_2$

(68) If all CO_2 is transported in the form of physical solution, what will be pH of blood ?

(A) 7.4 (B) 4.5 (C) 8.5 (D) 11.9

(69) Cl^- from blood plasma, enters RBC, in exchange of which component ?

(A) HCO_3^- (B) H^+ (C) K^+ (D) Na^+

Answers : (57-D), (58-B), (59-D), (60-C), (61-A), (62-B), (63-C), (64-B), (65-D), (66-C), (67-D), (68-B), (69-A)

Regulation of respiration

(1) Nervous regulation

● **First principle**

Respiratory center

↓

(In medulla oblongata)

↓

Order (impulse)

↓

By Vagus nerve

(X^{th} cranial nerve)

↓

Towards the Diaphragm and intercostal muscles

↓

Regulation

● **Second principle**

Respiratory centers

(In Brain stem)

↓

Two folds (Interaction)

↙ ↘

Inhalation Exhalation

↘ ↙

Natural Coordination

↓

Give rhythmical Stimuli to the resptraory muscles

↓

Regulation of Inspiration and expiration

(2) Chemical regulation

↓

Regulation of CO_2 amount in spinal fluid and blood of arteries.

Brain, arota aortic arch coratidsimas

↓

Chemical receptors

↓

detect the CO_2, pH, and O_2 level

↓

Information to the brains rythamicitly maintain centre

↓

Trasmits the appropriate signals to respiratory muscles

↓

Regulation done.

- Respiratory center / (s) gets impulses from substances which lying on lungs.

Disorders of respiratory systems :

(1) Bronchitis :

- **Causes :** infection - smoking.
- **Symptoms :** regular coughing with thick and large amount of ● phlegm - Yellowish / greenish like cough ● burning sensation in trachea. ● **Prevention** - Away from the smoke, chemical and pollutants ● **Treatment :** By antibiotics.

(2) Asthma :

- **Causes :** Allergenes in air.

Symptoms : The muscles wall of tracheal branches remain constantly agitated undergo spasmic contraction - repeated coughing out of phlagm.

Difficulty in breathing - suffocation in trachea.

● **Prevention :** Avoiding the foreign subtance or allergens.

● **Treatment :** With suitable antibiotic, antihistamine drugs.

(3) ● **Emphysema : causes :** Smoking and chronic bronchitis.

● **Symptoms :** lost elasticity of alveoli - after inhalation alveoli are full - difficulty in breathing, suffocatia

● **Side effect :** The heart and brain do not get enough O_2.

● **Prevention :** Avoiding of smoking and pollutants.

● **Teatment :** Use of antibiotics and tracheal dialatory drugs. once it occurs, the obtained, there is no permanent treatment.

(4) **Pneumonia-Causes** : Streptococcus Pneumonia

● **Symptoms** : An acute infection of the alveoli - alveoli of lungs becomes filled with fluid and dead I WBCs, become defanct.

● Children, old individuals and AIDS - patients are susceptible to this disease.

● Treatment : with antibiotics

(5) **Occupational lung disease :**

Causes : exposure of harmful gas and dust

Examples : Silicosis and asbestosis

---

(70) Respiratory centre of medulla oblongata sends message to ..... .

(A) diaphragm muscle (B) Intercostal muscle (C) Pulmonary muscle (D) A and B both

(71) Where is location of respiratory centre with two folds ?

(A) cerebrum (B) Brain stem (C) cerebellum (D) Diencephalon

(72) Chemical regulation of breathing is not regulated by amount of ..... .

(A) CO_2 in artery (B) CO_2 into cerebro spinal fluid

(C) A and B both (D) O_2 in cerebrospinal fluid

(73) Chemical receptors checks amount of ..... in blood

(A) CO_2, O_2 (B) CO_2, O_2 and pH (C) CO_2 (D) O_2

(74) In treatment of which disease antihistamine drugs are given ?

(A) amphisema (B) Asthama (C) Asbestosis (D) Bronchitis

(75) Due to which disease, functions of brain and heart is hampered ?

(A) Asthama (B) Pneumonia (C) Amphysema (D) Bronchitis

(76) Which disease do not occur due to infection of Bacteria or virus ?

(A) Pneumonia (B) Asbestosis (C) Bronchitis (D) All of above

(77) Alveolus loses elasticity and alvelous filled with liquid are symptoms of which disease ?

(A) Amphysema - Asthama (B) Amphysema - Bronchitis

(C) Pneumonia - Amphysema (D) Amphysema - Pneumonia

(78) ..... symptom of bronchitis.

(A) Suffocation in Bronchus (B) Inflammation, irritation in bronchus

(C) Destruction of alveoli (D) alveoli remain filled with air

(79) Probability of which respiratory related disease possibly occur in HIV infected person ?

(A) Bronchitis (B) Asthma (C) Pneumonia (D) Amphysema

(80) Constant contraction of tracheal muscle is symptom of ..... disease.

(A) Asthma (B) Amphysema (C) Bronchitis (D) Asbestosis

(81) It is symptom of Bronchitis.

(A) Alveoli loses elasticity.

(B) Alveoli filled with liquid and WBC.

(C) Expulsion of thick, greenish yellow cough in large quantity.

(D) Mostly difficulty in exhalation.

Answers : (70-D), (71-B), (72-D), (73-B), (74-B), (75-C), (76-B), (77-D), (78-B), (79-C), (80-A), (81-C)

- **Column type questions - Select proper option.**

(82)

Column -I	**Column-II**	
(1) Pre-olfactory region	(P) upper region	(A) 1 - R, 2 - Q, 3 - P
(2) Olfactory region	(Q) middle region	(B) 1 - P, 2 - R, 3 - Q
(3) Respiratory region	(R) lower region	(C) 1 - R, 2 - P, 3 - Q,
		(D) 1 - Q, 2 - P, 3 - R

(83)

Column -I	**Column-II**	
(1) Pharynx	(P) Opens in pharynx	(A) 1 - R, 2 - Q, 3 - S, 4 - P
(2) Larynx	(Q) Connects pharynx with trachea	(B) 1 - R, 2 - Q, 3 - P, 4 - S
(3) Trachea	(R) Opens in larynx and oesophasus	(C) 1 - R, 2 -P, 3 - S, 4 - Q
(4) Internal nares	(S) Gives passage to air towards trachea	(D) 1 - R, 2 - S, 3 - Q, 4 - P

(84)

Column -I	**Column-II**	
(1) Primary bronchiole	(P) Branch entering in each lobules of lungs	(A) 1- Q, 2 - R, 3 - T, 4 - P
(2) Tertiary bronchiole	(Q) Branch entering into lungs	(B) 1 - Q, 2 - S, 3 - P, 4 - T
(3) Secondary bronchiole	(R) Branch forming trachea	(C) 1-Q, 2 - R, 3 - P, 4 - T
(4) Trachea	(S) Branch forming terminal end	(D) 1- Q, 2 - R, 3 - P, 4 - S
	(T) Branch ending in alveolus	

(85)

Column -I	**Column-II**	
(1) Right lung	(P) Contraction of diaphragm muscles	(A) 1 - S, 2 - R, 3 - P, 4 - Q
(2) Left lung	(Q) Relaxation of diaphragm muscles	(B) 1 - R, 2 - S, 3 - P, 4 - Q
(3) Inhalation	(R) Two fissures	(C) 1 - R, 2 - S, 3 - Q, 4 - P
(4) Exhalation	(S) One fissure	(D) 1 - S, 2 - R, 3 - Q, 4 - P

(86)

Column -I	**Column-II**
(1) Volume of air, entering into lungs and coming out, during normal breathing	(P) 500 ml
	(Q) ERV
(2) Residual air after forceful exhalation, in lungs	(R) 2500 to 3500 ml
(3) Amount of air, expelled forcefully from lungs after normal exhalation	(S) RV
(4) Amount of air, taken by lungs after normal inhalation	

(A) (1-P), (2-S), (3-R), (4-Q) (B) (1-P), (2-Q), (3-S), (4-R)

(C) (1-P), (2-S), (3-Q), (4-R) (D) (1-P), (2-R), (3-S), (4-Q)

(87)

Column -I	Column-II	
(1) TLC - RV =	(P) FRC	(A) 1 - R, 2 - S, 3 - Q, 4 - P
(2) TLC - FRC =	(Q) EC	(B) 1 - R, 2 - S, 3 - P, 4 - Q
(3) TLC - IC =	(R) VC	(C) 1 - R, 2 - Q, 3 - P, 4 - S
(4) TLC - (IRV - RV) =	(S) IC	(D) 1 - R, 2 - P, 3 - S, 4 - Q

(88)

Column -I	Column-II	
(1) $HHbO_2$	(P) Haemoglobinic acid	(A) 1 - R, 2 - U, 3 - T, 4 - S
(2) $KHbO_2$	(Q) Disodium hydrogen phosphate	(B) 1 - P, 2 - U, 3 - T, 4 - S
(3) NaH_2PO_4	(R) Oxyhaemoglobinic acid	(C) 1 - R, 2 - U, 3 - Q, 4 - S
(4) H.Pr	(S) Proteonic acid	(D) 1 - S, 2 - U, 3 - T, 4 - R
	(T) Sodium hydrogen phosphate	
	(U) Potassium oxyhaemoglobin	

(89)

Column -I	Column-II	
(1) Silicosis	(P) Irritation, burning sensation in bronchus.	(A) 1 - R, 2 - T, 3 - P, 4 - Q, 5 - S
(2) Asthama	(Q) Recurrence of bronchitis	(B) 1 -R, 2 -T, 3 -P, 4 -S, 5 - Q
(3) Pneumonia	(R) Professional hazards	(C) 1 -R, 2 -T, 3 -S, 4 -Q, 5 - P
(4) Amphisema	(S) Acute infection of aveoli	(D) 1 -T, 2 -R, 3 -S, 4 - Q, 5 - P
(5) Bronchitis	(T) Allergic disease	

Answers : (82-C), (83-A), (84-D), (85-B), (86-C), (87-B), (88-A), (89-C)

A - Statement, R - Reason type questions

Select answers for questions given below from the following options :

(A) A and R true and R is explanation of A.

(B) A and R true, but R is not explanation of A.

(C) A is true, R wrong.

(D) A is wrong, R true.

(90) Statement A : During Asthamatic attack antihistamine drugs give relief.
Reason R : Asthma is alergic disease.
(A) (B) (C) (D)

(91) Satement A : Food does not enter into oesophagus while eating.
Reason R : Oesophagus always remain open.
(A) (B) (C) (D)

(92) Statement A : Alveoli of lungs are its respiratory surface.
Reason R : Exchange of O_2 and CO_2 takes place in lungs by alveoli.
(A) (B) (C) (D)

(93) Statement A : When muscles of diaphragm contracts, Size of thoracic cavity decreases.
Reason R : When diaphragm muscles contracts, diaphragm is pushed downwards.
(A) (B) (C) (D)

(94) Statement A : O_2 is transported only by RBC in blood.
Reason R : Hb is present in RBC.
(A) (B) (C) (D)

(95) Statement A : Respiratory passage remains open and do not suffocate.
Reason R : Trachea possess 'C' shaped cartilagenous rings.
(A) (B) (C) (D)

(96) Statement A : Lungs are prevented from jerks.
Reason R : Lungs are protected by rib-cage.
(A) (B) (C) (D)

(97) Statement A : More O_2 is transported by RBC.
Reason R : $HHbo_2$ is formed at respiratory surface in RBC.
(A) (B) (C) (D)

(98) Statement A : Cl^- Shifting occurs at respiratory surface.
Reason R : $HHbo_2$ is formed in RBC at respiratory surface.
(A) (B) (C) (D)

(99) Statement A : Silica enters during inhalation, causes silicosis.
Reason R : Elasticity of alveolus is destroyed in silicosis.
(A) (B) (C) (D)

(100) Statement A : pH of blood is maintaining 7.4.
Reason R : Na_2HPO_4 is present in blood plasma.
(A) (B) (C) (D)

(101) Statement A : Blood emptying in left auricle has 95 mmHg partial pressure.
Reason R : Blood from lungs emptied into left auricle by pulmonary vein.
(A) (B) (C) (D)

Answer: (90-A), (91-B), (92-A), (93-D), (94-D), (95-C), (96-B), (97-B), (98-D), (99-C), (100-C), (101-A)

True - False type questions (T - F)

Select which option is correct for true or false in following sentences.

(102)(1) Middle region of nasal cavity is called respiratory passage.
(2) Pharynx acts as passage for air in human.
(3) Lungs of human are situated in thoracic cavity.
(4) 'C' Shaped cartilagenous rings present in the trachea.
(A) T T F F (B) T F F F (C) T T T F (D) T F T F

(103) (1) Alveoli tubule is smallest diameter tube in bronchiole tree.
(2) Diaphragm have striated muscles.
(3) Volume of ERV is 1100 to 1200 ml.
(4) Left lung, has two lobes and two fissures.
(A) T F T T (B) T T T T (C) F T T T (D) T F F F

(104) In comparison with exhalation which statments are right or wrong ?

(1) diaphragm is pulled up. (2) Size of thoracic cavity increases.

(3) Partial pressure of air increases in lungs. (4) Diaphragm muscles relaxed.

(A) T F T F (B) T F T T (C) F F T T (D) T T T T

(105) (1) Value of IRV is more than FRC.

(2) Value of IC is less than IRV.

(3) Value of VC is 5100 to 5800 ml.

(4) Value of TV is less than RV.

(A) T F F T (B) T F T T (C) T T F T (D) T T F F

(106) (1) 3% of CO_2 is transported by blood plasma.

(2) 97% of CO_2 is transported by RBC.

(3) 20% of CO_2 is transported as carbamino components.

(4) O_2 in the form of $KHbO_2$ amd CO_2 in the form of $NaHCO_3$, is transported maximum.

(A) F T T T (B) F F F F (C) F F T T (D) F F T F

(107) (1) Message of respiratory centre of medulla oblongata is transported by 10th cranial nerve.

(2) Respiratory centers of medulla oblongata has two folds.

(3) Chemical receptors are present in lungs.

(4) Chemical receptors check percentage of CO_2, pH and O_2 in blood.

(A) T F F T (B) T T F T (C) F F T T (D) F F F T

(108) (1) Antihistamine is useful in the treatment of Asthama.

(2) As a side effect of asthma brain and heart do not receive enough blood.

(3) Alveoli are filled with liquid and dead RBC.

(4) Muscles of trachea remains excited in bronchitis.

(A) T T T F (B) T F F F (C) T T F F (D) T T F T

Answers : (102-B), (103-D), (104-B), (105-A), (106-C), (107-D), (108-B)

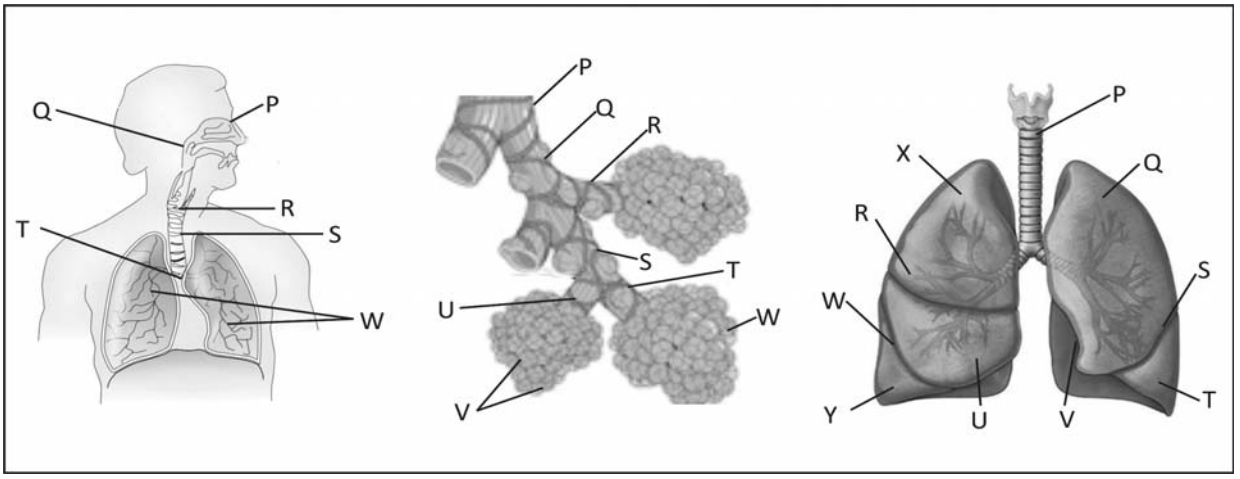


Figure 1 Figure 2 Figure 3

(109) In which structure of Fig. 1 'C' Shaped cartilagenous ring is observed ?

(A) Q and R (B) Q and T (C) R and S (D) R and P

(110) What is indicated by 'R' in Fig. -1 ?

(A) Pharynx (B) Larynx (C) Trachea (D) Bronchus

(111) Which structure of Fig -2 is not included in broncheal tree ?

(A) P (B) V (C) Q (D) P and V

(112) What is indicated by 'S' in Fig. 2 ?

(A) Secondary bronchiole (B) quartnary bronchiole (C) Tertiary bronchiole (D) Primary bronchiole

(113) What is indicated by 'W' in Fig. 3 ?

(A) Oblique fissure of right lung. (B) Oblique fissure of left lung.

(C) Transverse fissure of right lung. (D) Transverse fissure of left lung.

(114) What is indicated by upper lobe in Fig -3 ?

(A) Y and T (B) X and U (C) Q and T (D) X and Q

(115) What is indicated by 'U' in Fig. - 3 ?

(A) upper lobe (B) middle lobe (C) lower lobe (D) lateral lobe

(116) Which process is indicated by Fig. 4 ?

(A) Exhalation (B) Breathing (C) Inhalation (D) Peristalsis

(117) What is not proper for Fig. 5 ?

(A) Relaxation of diaphragm muscle.

(B) Partial pressure increases in lungs.

(C) Size of thoracic cavity reduces.

(D) Contraction of diaphragm muscles.

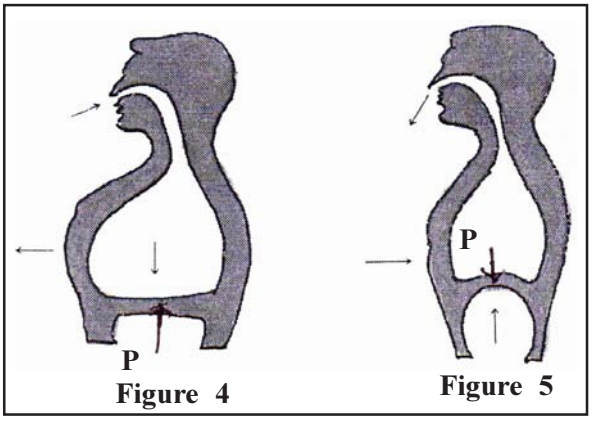


Figure 4 Figure 5

Answers : (109-C), (110-B), (111-A), (112-C), (113-A), (114-D), (115-B), (116-C), (117-D)

Nasal cavity dividing accordingly function

(i) Vastibular - Skin, hair, sebacious gland.

(ii) Respiratory passage - Globlet cells, pseudo columnar epithelial cells, glandular epithelial.

(iii) Oltaetory region : nerve sensory epithelial.

Activity of respiratory center :

- Increases the concentration of CO_2 in blood, decreases the PH so acidity can occur that why respiratory centers become active.
- Respiratory rate in human 12-18 per minute.
- One molecule of haemoglobin attached with four molecules of Co_2.
- Hb as four units : Each unit react with O_2.
- 1gm Hb transport 1.34 ml of O_2
- 100 ml blood transport 20 ml of O_2.

QUESTIONS FOR NEET :

(118) When amount of CO_2 increases in blood, breathing becomes ..... .

(A) Shallow and slow (B) No effect on breathing

(C) Slow and deep (D) Fast and deep

(119) When analysis of patient's blood shows high amount of carboxyhaemoglobin than which of the following summary may mostly correct ?

patient may be breathing more amount of polluted air with high amount of ..... .

(A) carbon disulphide (B) chloroform

(C) carbon dioxide (D) carbon monoxide

(120) Which of the following cells of mammals are not able to metabolizte glucose by aerobic respiration ?

(A) Non - Striated muscle cells (B) Hepati cells

(C) RBC (D) WBC

(121) What is essential capacity of our lungs ?

(A) Reserve volume of air - Tidal volume

(B) Total lung capacity - Resereve volume of exhalation

(C) Reserve volume of air - Reserve volume of exhalation

(D) Total lung capacity - Remaining Volume

(122) Given below is four respiration capacity and four irregular breathing volume in normal human.

Which of the two from the given capacity is correct ?

Capacity of breathing	**Volume of air**
(i) Remaining volume	(p) 2500 ml
(ii) Most essential capacity	(q) 3500 ml
(iii) Reserve volume of air	(r) 1200 ml
(iv) Capacity of inhalation	(s) 4500 ml

(A) (ii-p) and (iii-s) (B) (iii-r) and (iv-p) (C) (iv-q) and (i-r) (D) (i-s) and (ii-q)

(123) Respiratory centre located in the brain is stimulated by ..... .

(A) Concentration of O_2 in venous blood (B) Concentration of CO_2 in arterial blood

(C) Concentration of O_2 in artery (A) Concentration of CO_2 in venous blood

(124) How, is common cold, differ from pneumonia ?

(A) Pneumonia is spreading disease where as common cold is disease of defeciency of nutritive substances.

(B) Pneumonia can be prevented by live, less, weakened bacterial vaccine where as no effective vaccine for common cold.

(C) Pneumonia is caused by virus, where as common cold is due to bacterium haemophils, influenzal.

(D) Microorganisms of bacteria affects alveoli, where as effect of common cold is at nose and respiratory passage, lungs are not affected.

(125) Normal amount of O_2 and CO_2 present in air when inhaled by man, he feels suffocated because ..... .

(A) Haemoglobin unites with CO instead of O_2 to form carboxyhaemoglobin.

(B) CO affects nerves of lungs.

(C) CO and O_2 reacts together to decrease their amount.

(D) CO affects diaphragm and intercostal muscles.

(126) Breathing is regulated by ..... .

(A) Hypothalamus (B) Cerebrum (C) Cerebellum (D) medulla oblongata

(127) Alveolar epidermis of lungs is formed of ..... .

(A) non- ciliated columnar (B) non-ciliated squamous

(C) ciliated columnar (D) ciliated squamous

(128) Transport of CO_2 from tissue to respiratory surface is by ..... .

(A) blood plasma and RBC (B) blood plasma

(C) RBC (D) RBC and WBC

(129) How is air passed into respiratory passage ?

(A) trachea → lungs → larynx → pharynx → alveoli

(B) Nose → larynx → pharynx → trachea → alveoli → micro bronchioles

(C) Nasal aperture → pharynx → larynx → trachea → bronchioles → microbronchiole → alveoli

(D) Nose → mouth → lungs

(130) Normally carbonic anhydrease is active in ..... .

(A) RBC (B) WBC (C) blood plasma (D) blood corpuscles

(131) Presence of more number of alveoli surrounding alveolar tubule, opening in micro bronehioles in lungs indicates ..... .

(A) Non-functional system for, transport of residual air in less quantity

(B) Non-functional system for transport of residual air in more quantity

(C) In absence of residual air, functioning system for transportation

(D) Functional system for transport of residual air in less amount

(132) During transport of CO_2 why blood does not become acidic ?

(A) due to neutralization of H_2CO_3 By Na_2CO_3. (B) Due to absorption of leucocytes.

(C) Due to presence of blood buffer. (D) None of above.

(133) If 1500 ml air increases in air, it is called.....

(A) Tidal volume (B) Inspiratory reserve volume

(C) Residual volume (D) Vital capacity

(134) What protects lungs ?

(A) Ribs (B) Vertebral column

(C) sternum (D) All of above

(135) What is haemoglobin ?

(A) vitamin (B) Pigments of skin (C) Blood carrier (D) Respiratory pigment

(136) CO_2 reacts with water in RBC to form carbonic acid. What is responsible for it ?

(A) Carbonic anhydrase (B) Carboxy peptidase (C) Hydrolase (D) Oxido reductase

(137) What is conditon of diaphragm at the time of exhalation ?

(A) Oblique (B) Normal (C) Flat (D) Dome shape

(138) What is called exchange of bicarbonate and chloride between RBC and blood plasma ?

(A) Chloride shift (B) Effect of Bohr (C) Halden Effect (D) Intracellular respirtion

(139) If CO_2 concentration icreases in blood, then respiration ..... .

(A) increases (B) decreases (C) stops (D) no change occurs

(140) What is vital capacity of our lungs ?

(A) RV +ERV (B) TLC + TV (C) RV +TV (D) TLC - ERV

(141) CO_2 liberated from tissues into blood is in which form ?

(A) Carbamino haemoglobin in RBC

(B) Bicarbonate in blood plasma and RBC

(C) Free CO_2 in blood plasma

(D) 70% carbamino haemoglobin and 30 % bicarbonate

(142) Two friends are taking lunch, one of them has hiccups while swallowing food. It is due to improper movement of ..... .

(A) epiglottis (B) diaphragm (C) neck (D) tongue

(143) Arrange it in ascending order.

(1) TV (2) RV (3) IRV (4) VC

(A) $1 < 2 < 3 < 4$ (B) $1 < 3 < 2 < 4$ (C) $1 < 4 < 3 < 2$ (D) $1 < 4 < 2 < 3$

(144) What is symptoms of amphysema ?

(A) Haemorrhage of bronchieles

(B) Increase in number of alveoli

(C) Infection of microbacterium tubeculi

(D) Inflammation of alveoli

(145) Center for Regulation of rhythemacity in breathing are ..... .

(A) pons (B) aortic arch (C) medulla (D) carotid artery

Answers : (118-D), (119-D), (120-C), (121-D), (122-C), (123-C), (124-D), (125-A), (126-D), (127-B), (128-A), (129-C), (130-A), (131-B), (132-C), (133-C), (134-D), (135-D), (136-A), (137-D), (138-A), (139-A), (140-B), (141-B), (142-A), (143-A), (144-D), (145-B)

Breathing and Respiration

Human Respiratory System

Respiratory Tract

- It is a passage way for the respiratory gases.
- Gaseous exchange does not occur here.
- It consists of nostrils, nasal cavity, pharynx, larynx, trachea, bronchi and alveoli.

Breathing

- It is withdrawing in air and releasing CO_2 rich air into atmosphere.
- It involves inspiration (inhalation) and expiration (exhalation).
- It is brought out with the help of diaphragm and intercostal muscles on rib cage.

Respiratory Organs

- A pair of lungs which comprise of bronchi, bronchioles and alveoli.
- Alveoli are the primary sites of exchange of gases.

Exchange of Gases

- Alveoli are the primary sites of exchange of gases.
- At alveolar surface, the blood has lower PO_2 and higher PCO_2 than the alveolar air. This results in diffusion of CO_2 out of the blood into the alveolus.
- At tissues surface, blood has more PO_2 than that in the tissues. As a result O_2 from blood diffuses in the tissues and CO_2 from tissues diffuses into the blood.

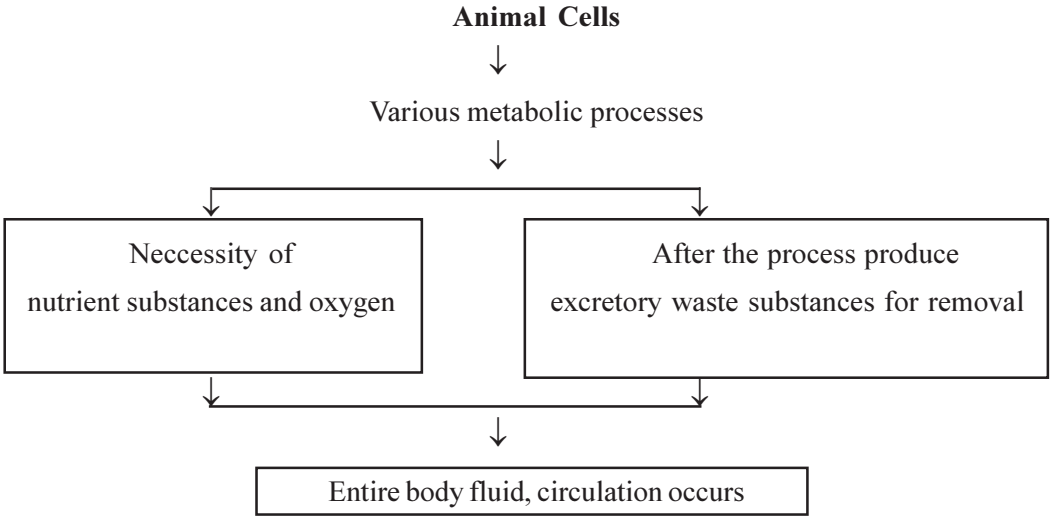
Transport of Gases

- Blood is the medium of transport for O_2 and CO_2. About 97 percent of O_2 is transported by RBC in the blood. The remaining 3 percent of O_2 is carried in a dissolved state through the plasma.
- Nearly 20-25% of CO_2 is transported by RBCs where as 70% of it is carried as bicarbonate.
- About 7% of CO_2 is carried in a dissolved state through plasma.

Regulation of Respiration

- Respiration is controlled by nervous regulation, mechanical regulation and chemical regulation.
- Nervous regulation involves respiratory centre located in the medulla oblongata and pons Varolii. Respiratory centre is very sensitive to CO_2 in the arteries and to the pH level of blood.
- Mechanical control involves stretch receptors located in the walls of bronchi and bronchioles which get stimulated by overstocking of the lungs and inhibit inspiratory centre.
- Chemical regulation is performed by chemo sensitive area near respiratory centre and chemoreceptors present in the carotid and aortic bodies. They get stimulated by rise in CO_2 of H^+ consentration of blood.

24 Body Fluids and Circulation

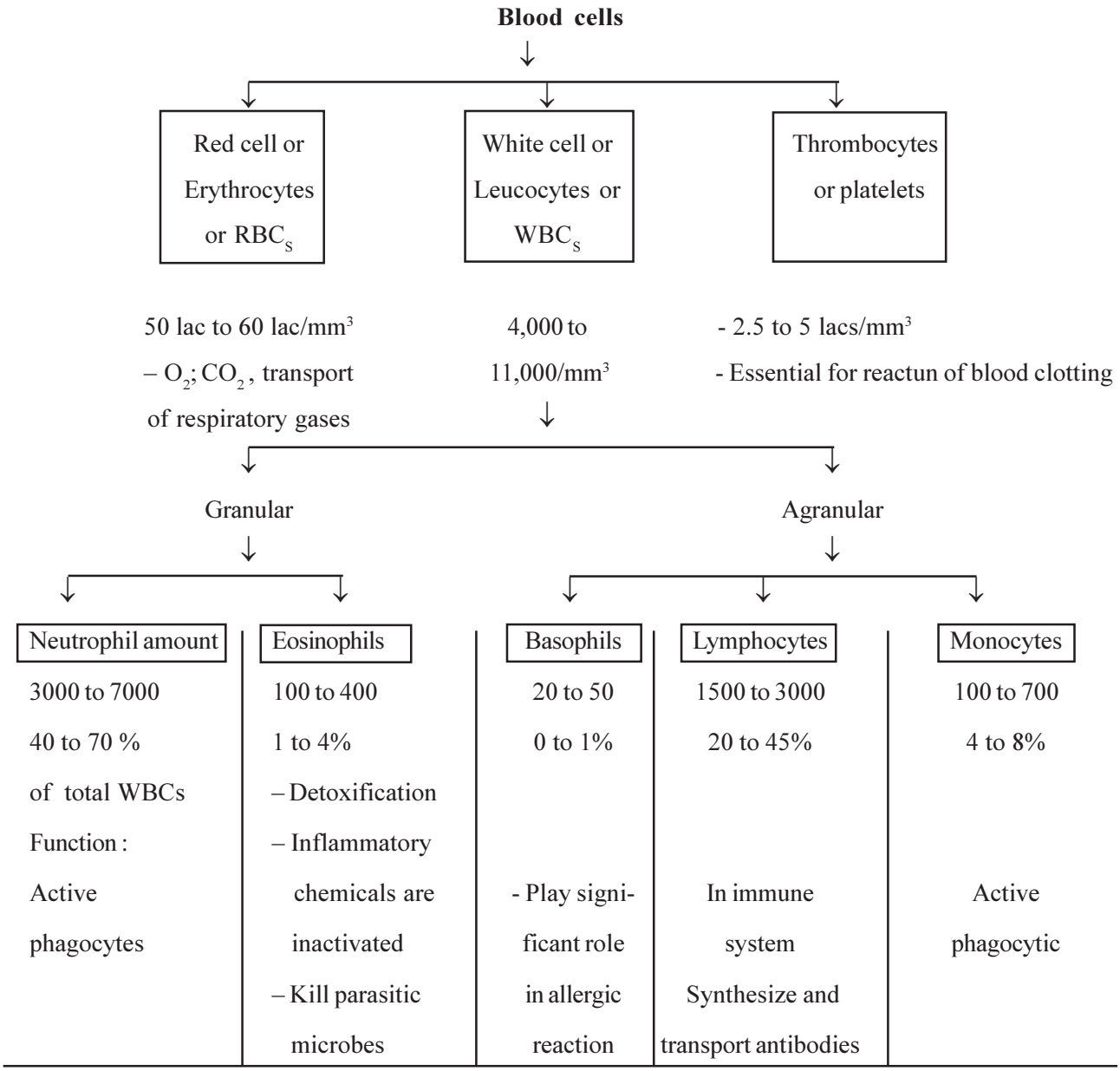


Body fluid - (Blood - Red - Blood plasma 55% , Blood cells - 45%)
Tissue fluid - Colourless - lymph

Blood :

- Liquid connective animal tissue
- Nearly 8 % of total weight of adult body.
- In man 5 to 6 liter blood average in volume.
- In female / woman l to 5 liter blood average in volume.

	Main Substance	**Functions**
Water		As a solvent, which is for the enterance substance in blood plasma.
Salts	[Na^+, Ca^{+2}, Mg^{+2}, K^+ HCO_3, $C1^-$	Osmotic balance pH buffering Regulation of permeability of plasma membrane
Plasma	Fibrinogen	In blood (clotting)
Protein	Globulin	Immunity and transport
	Albumin	Regulation of osmosis and pH buffering.



(1) Blood is liquid connective tissue, because ..... .

(A) It connects WBC and RBC.

(B) Its blood plasma is secrete from blood cells.

(C) It connects all cells, tissues and organs of the body.

(D) Proteins of blood plasma coordinates with each other for functioning.

(2) Find improper substance.

(A) Albumin (B) Haemoglobin (C) Fibrinogen (D) Globulin

(3) State the protein component of blood plasma related with osmoregulatory function.

(A) Globulin (B) Fibrinogen (C) Albumin (D) Thrombin

(4) Which of the following leucocyte participate in allergic reaction?

(A) Erythrocytes (B) Lymphocytes (C) Basophils (D) Eosinophils

(5) If p = neutrophil, q = Eosinophil, r = Basophils, s = lymphocytes, r = monocytes, then find true options for various leucocytes based on their numbers :

(A) $p < s < t < q < r$ (B) $r < q < s < t < p$ (C) $t < r < s < q < p$ (D) $r < q < t < s < p$

(6) State the percentage of blood cells (per mm^3) in the body, which inactivate chemicals, which produces burning sensation in the body.

(A) 1500 - 3000 (B) 100 - 700 (C) 100 - 400 (D) 3000 - 7000

(7) It has 4 to 8% of total WBC.

(A) Monocytes (B) Lymphocytes (C) Eosinophils (D) Basophils

(8) Name leucocyte having bilobed nucleus.

(A) Acidophils (B) Basophils (C) Neutrophils (D) Monocytes

(9) Name blood corpuscle without nucleus in human.

(A) Erythrocytes (B) Leucocytes (C) Monocytes (D) Lymphocytes

(10) State pH of Blood.

(A) 4.7 (B) 6.7 (C) 7.4 (D) 3.7

(11) What is approximate life span of erythrocytes ?

(A) 120 days (B) 85 days (C) 40 days (D) 365 days

(12) Of the following living organism, which lacks RBC in blood ?

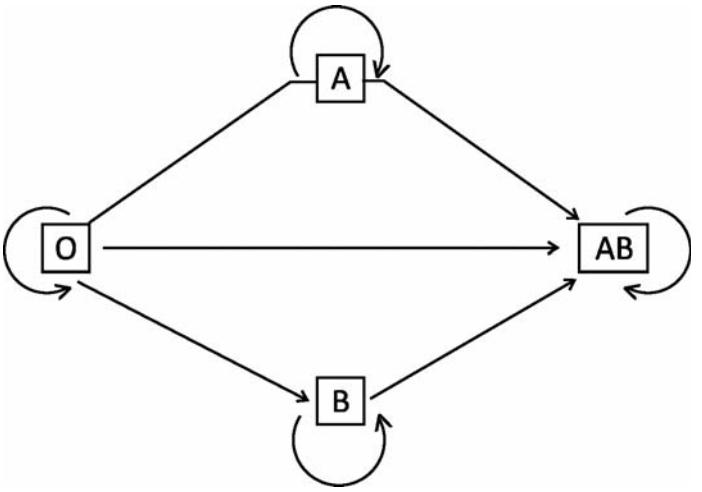
(A) Frog (B) Kangaroo - rat (C) Earthworm (D) Pigeon

Answers : (1-C), (2-B), (3-C), (4-B), (5-D), (6-C), (7-A), (8-B), (9-A), (10-C), (11-A), (12-C)

Blood Group

- Carl Landstiener discovered blood group in human.
- Four types of blood groups A, B, AB and O.
- In blood of person which type of antigen is present and which type of antibody is present, that two point are concentrated for the determined blood group.

Blood Group	Antigen	Antibody
A	A	anti b
B	B	anti a
AB	A and B	None of these
O	None	Anti a, Anti b

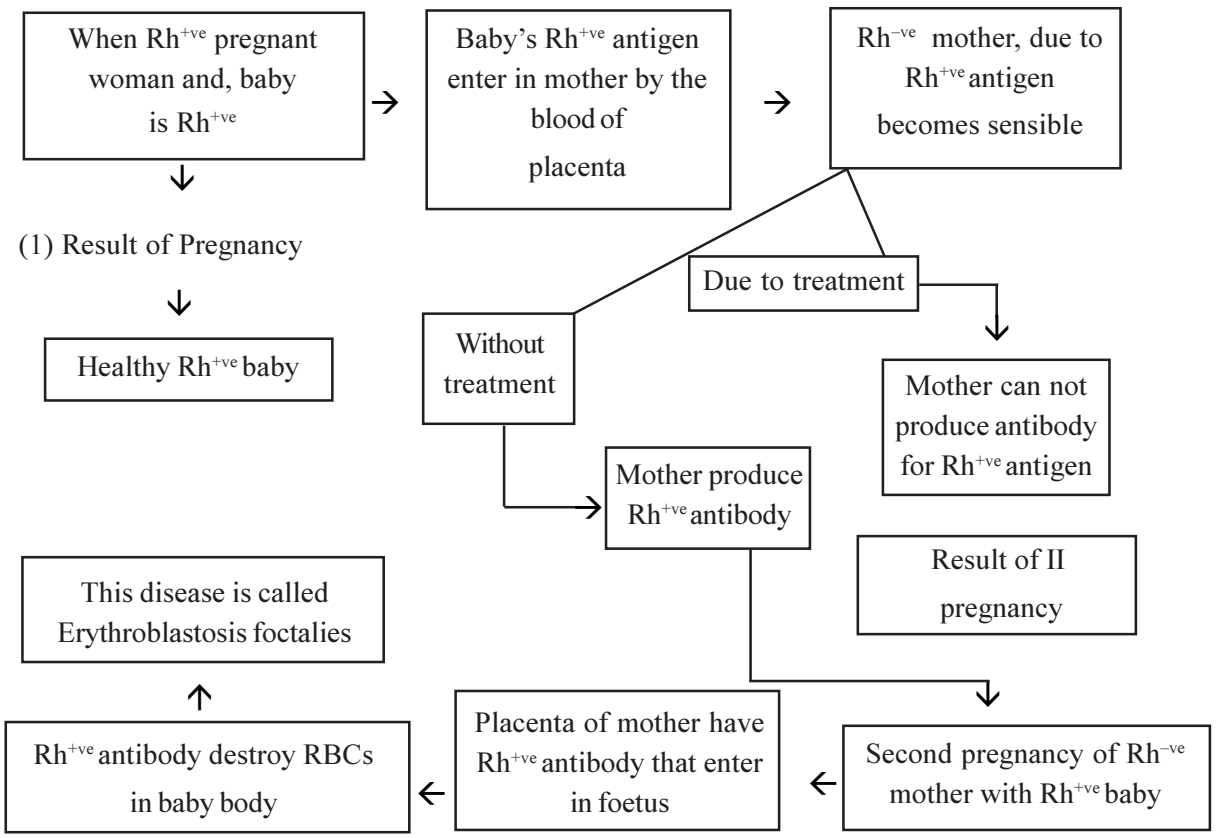


- Universal donor blood group is O.
- Universal acceptor / receptor blood group is AB.

Rh Group :

Human having Rh antigen on membrane of RBCs the

- person is called **Rh^{+ve}.**
- If Rh antigen is absent so it is – **(Rh^{-ve})**
- This antigen was originally discovered in blood of Rhesus monkey.
- In human poputation 85% of people having Rh^{+ve} and 15% of people having Rh^{-ve}
- Rh related problems are seen at the pregnancy time.



(13) Which option is correct for the given statements X, Y and Z ?

X : First child having Rh^{+ve} of Rh^{-ve} women, then Rh^{+ve} anti body should be introduced in woman's blood after the birth of child.

Y : If such woman in second pregnancy carries Rh^{-ve} foetus , such child has possibility of haemolytic disease.

Z : During first pregnancy Rh^{+ve} antibody is produced in her blood.

(A) X and Y correct, Z false. Y is correct explanation of X.

(B) X and Z correct, Y false. Z is correct reason for X.

(C) X and Z correct, Y false. Z is not correct explanation for X.

(D) X and Z false. Y is correct.

(14) Mr Bean is seriously injured in accident, with tremendous loss of blood, there is no time for blood test. So Dr. Amish will advise which blood group for transfusion ?

(A) AB^{+ve} (B) O^{+ve} (C) AB^{-ve} (D) O^{-ve}

(15) When is Rh^{+ve} antibody produced in Rh^{-ve} woman ?

(A) When her second child is Rh^{-ve} (B) When her second child is Rh+ve

(C) Her first child Rh^{+ve} (D) Her first child Rh^{+ve}

(16) Master Masum is suffering from erythroblastosis fits disease. What would be the blood group of mother ?

(A) O^{+ve} (B) A^{-ve} (C) B^{+ve} (D) AB^{+ve}

(17) Which of the following marriage is not proper from biological view point ?

(A) Rh^{+ve} Male, Rh^{+ve} female (B) Rh^{-ve} Male, Rh^{+ve} female

(C) Rh^{-ve} Male, Rh^{-ve} female (D) Rh^{+ve} Male, Rh^{-ve} female

(18) Person, having O blood group is called universal donor, as in his blood, on the surface of RBC ..... .

(A) A antigen present (B) Only B antigen absent

(C) Only O antigen present (D) A and B both antigen absent

(19) If in blood plasma of person with A blood group mixed up with blood of person having B blood group then why blood gets more density?

(A) antibody act as antitoxin. (B) Blood cells become more turgid.

(C) Due to reaction of antibody antigen. (D) Overall increase in amount of antigen.

(20) Which antigen enters through placenta into mother from child with erythroblastosis fits disease ?

(A) Rh^{+ve} (B) Rh^{-ve} (C) Anti – A/a (D) Anti - AB/ab

(21) Which blood corpuscles are destroyed in haemolytic child ?

(A) White corpuscles (B) Red corpuscles (C) Blood platelets (D) Both A and C

(22) Which leucocytes produces serum globulin ?

(A) Monocytes (B) Thrombocytes (C) Lymphocytes (D) Basophils

(23) State improper pair.

(A) Basophils - Allergy (B) Monocytes - Active phagocytes

(C) Lymphocyte - Anti poisonous (D) Neutrophils - Active phagocyte

(24) Number of blood corpuscles per mm^3 associated with blood clotting.

(A) 40 to 60 million (B) 5.5 to 7.5 million (C) 2.5 to 5 million (D) 3000 to 4000

(25) Mention the percentage of blood corpuscles in blood having more lobulated nucleas.

(A) 0 to 1% (B) 20 to 45% (C) 40 to 70% (D) 1 to 4%

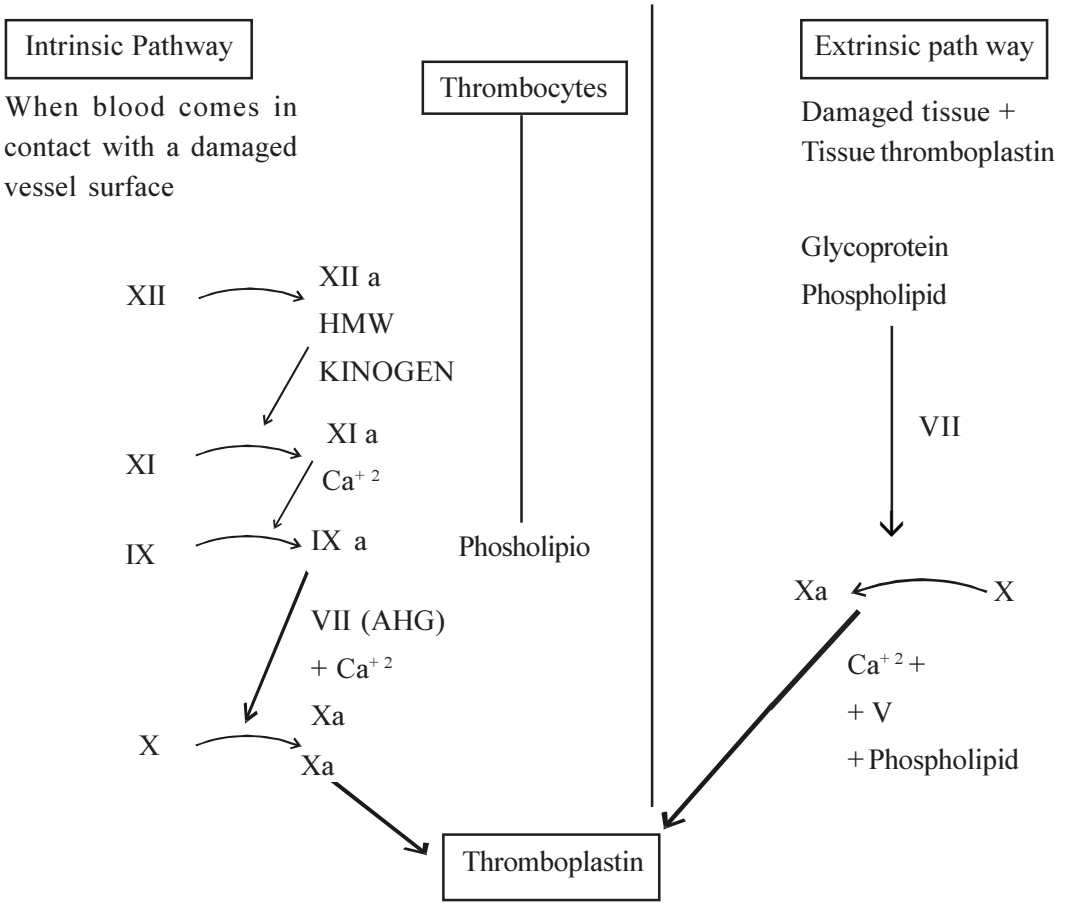
Answers : (13-B), (14-D), (15-C), (16-B), (17-D), (18-D), (19-C), (20-A), (21-B), (22-C), (23-C), (24-C), (25-C)

Coagulation of blood : The major chemical defence against blood loss is the formation of the bloodclot.

- The process in which the conversion of fluid like blood into jelly-like clot is called blood coagulation.
- Complex biological process.
- For this process blood plasma and tissues having 13 factors. Most of the factors play role in formation of thromboplastin by the active inducing principle.
- Numerical system for Nomenclature of blood clotting factors.

Roman Numerical Designation	Common Name	Activation product
I	Fibrinogen	Fibrin
II	Prothrombin	Thrombin
III	Thromboplastin	-
IV	Calcium	-
V	Proaccelerin	Accelerin (VI)
VII	Proconvertin	Convertin
VIII	Anti haemophilic globulin (AHG)	Activated AHG
IX	Christmas factor	Activated Christmas factor
X	Stuart factor	Activated stuart factor
XI	Plasma thromboplastin antecedent (PTA)	Activated (PTA)
XII	Hageman factor	Activate Hageman factor
XIII	Fibrin stabilizing factor (FSF)	Activated FSF

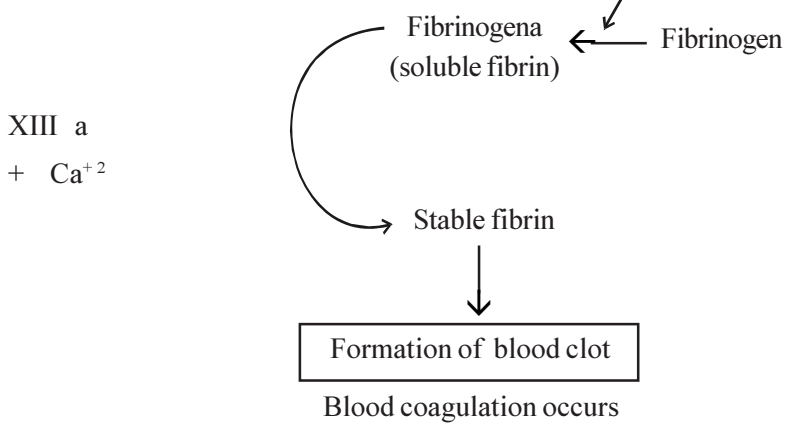
Phase - I



Phase - II



Phase - III



- **External factors effected to blood coagulation :**
- Vitamin K (less amount)
- Heparin
- Defibrination
- Hirudin (Leech - Phylum - Annelida - Secreted from Pharynx of animal)

Intrinsic Pathway :

- Initiator factors which start this system produce from blood plasma.
- This type of Pathway for Coagulation, when blood in contact with surface of blood vessels, that is start.
- Initial factor : Hageman factor XII

Extrinsic Pathway :

- Initiated factors of this system are derived from outside the blood plasma (injured tissue)

Lymph:

- Surrounded fluid to cells ↓ Tissue fluid — Fluid transport by the Lymphatic vessels ↓ Lymph
- The Composition of lymph is very much like that of the blood plasma but the dissolved substances are in different concentrations.
- In lymph protein contents are low and it contains less fibrinogen than blood plasma.
- Lymph in the smallest capillaries have no cells.
- When lymph passes through the lymph nodes; cells are added in it.
- 99% small lymphocytes (agranular lymphocytes)
- 1% rarely erythrocytes; Eosinophils, monocytes (anucleated monocytes)

Function : Responsible for the immunity.

- Important carreer for nutrients, hormones etc.

---

(26) Lymph passed through lymph node is more beneficial from functional view point than lymph present in small lymph vessel ?

(A) It has less amount of nutritive substances.

(B) It has 99% RBC so necessary for respiration.

(C) It has more fibrinogen so readily gets clotted.

(D) It has immunity property.

(27) Of the following group which is essential stuart complex ?

(A) Active X + Active VIII + Active X

(B) Active X + Xa + phospholipid + Ca^{+2}

(C) Active X + Active X + phospholipid + Ca^{+2}

(D) Active X + VIII a + phospholipid + Ca^{+2}

(28) Which factor for blood clotting is less in lymph than blood plasma ?

(A) I (B) IV (C) X (D) XI

(29) Thromboplastin is essential in human for blood clotting, from which cell it is obtained ?

(A) RBCs (B) Platlets (C) lymphocytes (D) Neutrophils

(30) It is function of thromboplastin.

(A) Formation of fibrin from prothrombin (B) Formation of thrombin from prothrombin

(C) Formation of fibrin from thrombin (D) Formation of prothrombin from thrombin

(31) Initiation of formation of thromboplastin through extrinsic passage begins from ..... .

(A) Blood corpuscles (B) Blood plasma (C) Injured tissue (D) A & B Both

(32) Due to formation of stable fibrin in blood ..... .

(A) Inhibits formation of blood nodules (B) Blood nodules dissolved in blood

(C) Hard blood nodules formed (D) Maintains fluidity of blood

(33) In which organism hirudin is normally observed ?

(A) Ascaris (B) Leech (C) Salamander (D) Frog

(34) Who prevents transformation of prothrombin into thrombin in uninjured blood vessel ?

(A) Heparin (B) Soluble fibrin (C) XIII a factor (D) Thromboplastinogenase

(35) Lymph has ..... compare to blood plasma.

(A) less amount of water

(B) some protein factors and collagen in less amount

(C) more amount of erythrocytes

(D) more amount of eosinophils

(36) Which of the following factor is absent in haemophilic patient ?

(A) Factor XII (B) Factor X (C) Factor VIII (D) Factor II

(37) For the activation of stuart factor by active christmas factor, help of which factor is obtained ?

(A) Plasma thromboplastin ancedent (B) Antihaemophilic globulin

(C) Proconvertin (D) Fribin stabilizing factor

(38) Which of the following option is improper for factors participating in blood clotting through intrinsic passage ?

(A) XII (B) VIII (C) VII (D) XI

(39) Which components participate in the formation of thromboplastin ?

(A) Injured tissue (B) Blood corpuscles (C) A and B (D) Thrombin

(40) Thromboplastin

↓ Ca^{+2}

Prothrombin → Thrombin

↓

↓ ↓

[Y] [X]

(A) X - Fibrinogen Y - Insoluble Fibrin

(B) X - Fibrinogen Y - Soluble Fibrinogen

(C) X - Soluble fibrin Y - Fibrinogen

(D) X - Soluble fibrin Y - Stable fibrin

Answers : (26-D), (27-D), (28-A), (29-A), (30-B), (31-C), (32-C), (33-B), (34-A), (35-B), (36-C), (37-B), (38-C), (39-C), (40-C)

- **Circulatory system**
- **Type :**

Open type	**Close type**
• Absence of blood capillaries	• Presence of blood capillaries
• Phylum = Arthropoda	• Phylum annelida
• Mollusca (except cephalopoda)	• Vertebrates Animals

Pathe way :

- Systemic circulation
- Pulmonary circulation

Heart :

- Size : 12 mm length, 9 cm broad, weight nearly 300g.
- Location : Between the two lungs arranged oblique in groove
- Shape : blunt coneshaped, hollow muscular organ
- Pericardium : bilayered - toward outer side : Fibrous coat
- **Toward inner side :** Serosa between the two layer.

  There is a very narrow space which is filled with paricardial fluid which gives protection against - jerks and shocks.
- **Lobes :** Four lobes.

  Right atrium – R A Righ ventricle – R V

  Left atrium – L A Left ventricle – L V

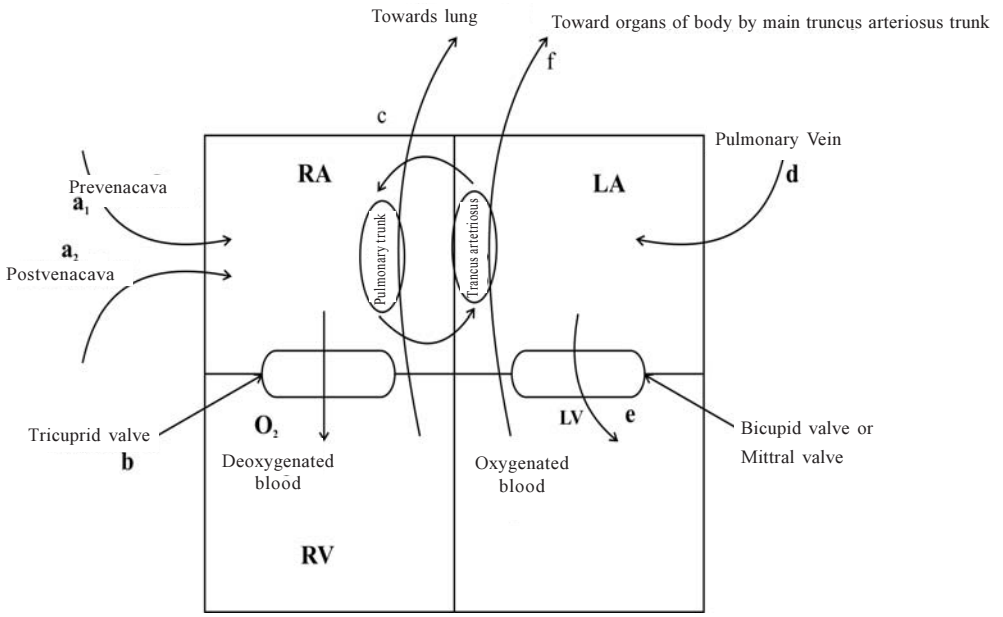
  Atria : Small in size, contain thin wall.

  Ventricles : Large in size, contain thicker wall.
- (Wall of left ventricle is thicker than the wall of right ventricle)
- Two grooves present on ventricles
  - Anterior interventricular sulcus
  - Posterior inter ventricular sulcus.

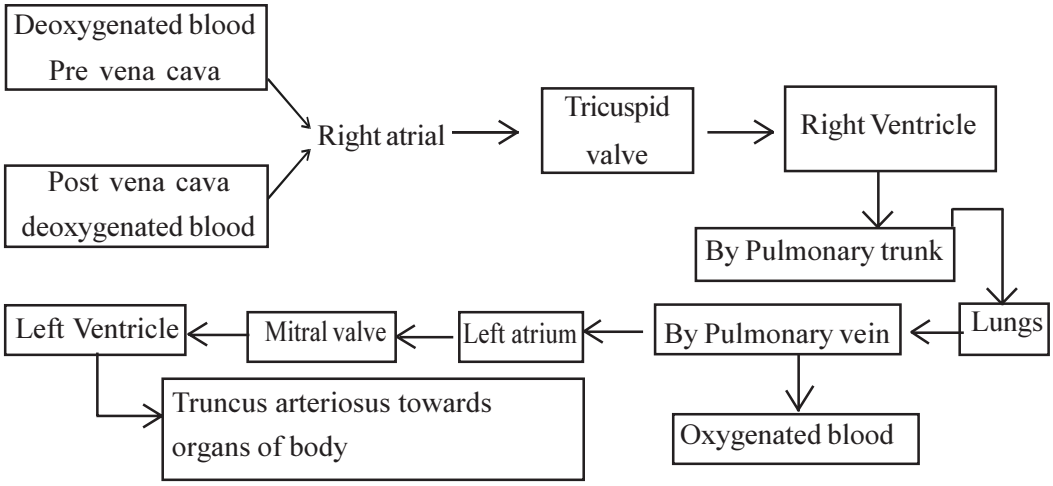
- **Internal structure :**
- Septum between two atrial - Inter atrial septum
- Septum between two ventricles - Inter ventricular septum.
- Septum between atria and ventricles - Atrio ventricular septum.
- Tricupsid valve - Between the right atrial and right ventricle
- Bicupsid / mitral valve - Between the left atrial and left ventricle
- **Semilunar valves :**
- The valve present between right ventricle and pulmonary trunk is called pulmonary semilunar valve.
- A valve present between left ventricle and aorta arteriosis is called aortic semilunar valve.
- Haversian valve : Pre venacava protected by Haversian valve which opens in right atrium.
- The bacien valve : Postvena cava protected by Thebesian valve which opens in right atrium.
- Pulmonary veins are not protected by valves.
- **A Node :** Sinoatrial node : It is present in right upper corner of the right atrium which is like tissue mass.
- **AV Node :** Atrio - Ventricular Node : It is present in the lower left corner of the right atrium close to the atrio - ventricular septum which is like tissue mass.
- Bundles of HIS muscles in interventricular septum.
- Perkinje fibers - branches that emerge from the bundle of HIS.

Double circulation : Discovered by William Harvey.

- Oxygenated and deoxygenated blood circulation is separated, so it is called double circulation.



Circulation : a_1, a_2 → b → c → d → e → f



- **Double pump :** By the heart, blood circulate from the right lobes to left lobes of heart for the entry of blood it has to passes through lungs. Thus heart is called double pump. (Accumulation of blood in atria = suction pump blood circulate from ventricles towards various organs of body pressure pumps because
- Right lobes of heart pushes the blood in lungs.
- Left lobes of heart pushes the blood toward the entire body organs.
- **Systemic heart :** Left part of heart (eg. Left atrium and left ventricle) which supplies blood in various organs of body so it is known as systemic circulation and left part of heart is called systemic heart.
- Main aim of this circulation is circulation of O_2 as well as nutrient substances upto tissues of body and removal of CO_2 and other nitrogenous toxin substances.

Pulmonary heart :

- Right part of heart (eg. Right atrium and right ventricle) has blood which pulmonary circulation for oxygenation, blood absorption or suction. Thus this circulation is known as pulmonary circulation and right part of heart is called pulmonary heart. By pulmonary circulation impure, deoxygenated blood which suction in right atrium, which is responsible for oxygenation.

---

(41) Which option is correct with reference to statement X, Y and Z ?

Statement :

X – wall of left Ventricle is more muscular than right ventricle.

Y – Blood is transported to various body parts from left ventricle.

Z – Blood is transported to lungs from right ventricle.

(A) X and Y correct, Z is wrong.

(B) X is correct, Y and Z wrong.

(C) X is wrong, Y and Z correct.

(D) X , Y, Z all three are correct.

(42) State the correct sequence of systemic transportation.

(A) Right auricle - Right ventricle - Left auricle - Left ventricle

(B) Left auricle - Left ventricle - body tissue system - vena cava - Right auricle

(C) Right auricle - Right ventricle - Left auricle - Left ventricle

(D) Left auricle - Left ventricle - lungs - Right ventricle - right auricle

(43) Which valve is present in the aorta originating from ventricle ?

(A) Semilunar (B) mitral (C) Tricuspid (D) Bicuspid

(44) Which part of heart is more muscular ?

(A) RA (B) LA (C) RV (D) LV

(45) Near which valve in human heart only O_2 blood passes?

(A) Pulmonary semilunar (B) Tricuspid valve (C) Bicuspid valve (D) A and B Both

(46) Which fibres held bicuspid and tricuspid valve to its proper place ?

(A) Bundle of His (B) Purkinje fibres (C) Chordae tendinae (D) A and B both

(47) Which vein is responsible for reentry of blood to left side of heart?

(A) Anterior Venacava (B) Pulmonary vein

(C) Posterior Venacava (D) Renal portal vein

(48) State the function of pulmonary artery.

(A) Transport of oxygenated blood from heart to lungs.

(B) Transport from left ventricle to body organs.

(C) Deoxygenated blood is transported from heart to lungs.

(D) Transport of O_2 blood from lungs to heart.

(49) Exterenal layer of pericardial membrane is formed of ..... .

(A) fibrous envelop (B) serosa (C) muscularis (D) mucosa layer

(50) Blood emptied into heart by venacava Sequentially passes through which valve ?

(A) Bicuspid valve - aortic arch - semilunar valve - mitral valve - pulmonary semilunar valve

(B) Bicuspid valve - pulmonary semilunar valve - mitral valve - aortic arch - semilunar valve

(C) Tricuspid valve - pulmonary semilunar valve - aortic arch - semilunar valve - mitral valve

(D) Tricuspid valve - pulmonary semilunar valve - bicuspid valve - aortic arch - semilunar valve

Answers : (41-D), (42-B), (43-A), (44-D), (45-C), (46-C), (47-B), (48-C), (49-A), (50-D)

Cardiac cycle :

- In healthy man, 72 cardiac cycle occurs in 1 minute (72 time pulsation). 0.80 sec time occur for / cardiac cycle. Systole : Contraction of heart lobes. Diastole : Relaxation of heart lobes.

Time-gap	Atria	Ventricles
0.10 Sec	undergo systole	undergo diastole
0.30 Sec	undergo diastole	undergo systole
0.40 Sec	undergo diastole	undergo diastole

- Cardiac cycle is starts by contraction of atria.
- Ventricles accumulated by blood, it takes 0.10 sec.
- Atria accumulated by blood, it takes o.70 sec.

- During the contraction of ventricles pressure arises due to this.
- From the right ventricle due to blood pressure semilunar valve of pulmonary trunk opens.
- From the left ventricle due to blood pressure semilunar valve of trancus arteriosus opens.
- At the time of contraction of both ventricles due to blood pressure tricuspid valve and bicuspid valve are closed.
- At the time of systole of atria through blood pressure tricuspid valve and mitral valve are opens.
- At the time of diastole of ventricles through blood pressure semilunar valve closed.
- When closing of atrioventricular valve, lubb sound can hear.
- When closing of semilunar valve, dubb sound can hear.

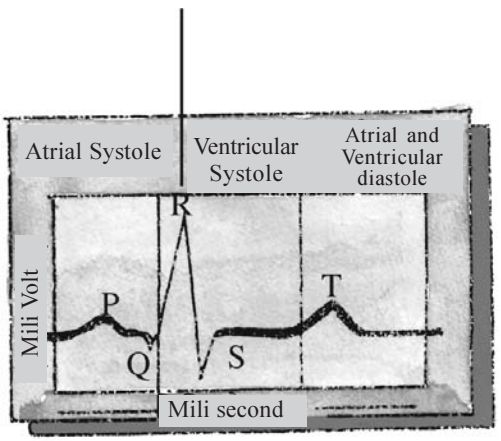
Electrocardiogram (ECG) : Discovered by Inthovan (1906)

The electrical charges induced in cardiac muscles during each cardiac cycle can be registered through electrodes arranged on the body surface. Such a recording is called - Electrocardiogram (ECG)

- Upward and downward directional curves arranged and produce ECG.

Uses : This is important method for checking the health or health related problems of heart.

- Electodes are arranged at different places on the body. Normally they are placed on wrists and ankles. (by clamp)
- The appearance of ECG depends on particular placement of electrodes.



- P wave : Systole of atria.
- Stimulation situation SA node to atria.
- QRS waves : Systole of ventricle.
- T wave : It indicate diastole of ventricles. This time atria also goes for diastole situation.

Regulation of heart : Normally cardiac activity are auto regulated by nodal tissue.

- Heart of man is myogenic because nodal tissue has properties of nerve and muscular both.
- Sympathetic nervous system : Heart activity becomes faster (stimulate situation)
- Parasympathetic nervous system : Prevent heart activity (Increased pulsation becomes normal)
- Nodal tissue :
- Pulsation of heart process starts by it. So it is called “heart” of heart.
- It passes massage of contraction of atria by every 0.80 sec; so it known as pacemaker (SA node).
- It keeps heart pulsation regulate and rhythematic. This stimulation reaches upto AV node by SA node.

AV node :

- From stimuli transport to bundle of His and purkinje fibers of heart by ventricle and by it contraction can occur.
- Failure of S.A. node causes transplants artificial pacemaker in thoracic region and keep heart pulsation control.

---

(51) During one cardiac cycle, total how much time blood can be held by auricles ?

(A) 0.10 Sec (B) 0.30 Sec (C) 0.40 Sec (D) 0.70 Sec

(52) What is indicated by P waves in ECG ?

(A) Contraction of ventricles (B) Relaxation of ventricles

(C) Contraction of auricles (D) Relaxation of all chambers

(53) When Ventricle experiences systole ..... .

(A) Tricuspid and Bicuspid valve closes (B) All valve of right chambers of heart opens

(C) Bicuspid and aortic Semilunar valve opens (D) All four valve opens for separation of blood of body

(54) Heart of heart ..... .

(A) pericardial membrane (B) SAN (C) AVN (D) brain

(55) Improper for SAN ..... .

(A) It is called pacemaker

(B) Responsible for stimulation of auricles

(C) Keeps heartbeat rhythemic and regular

(D) Every 0.10 sec sends messages for contraction of auricles

(56) State number of portal veins in human.

(A) 1 (B) 2 (C) 3 (D) Inumerable

(57) State the passage of conduction of stimulation in heartbeating.

(A) SAN - AVN - Bundle of His - purkinje fibre

(B) SAN – Bundle of His - AVN - Purkinje fibre

(C) AVN – SAN - Purkinje fibre - Bundle of His

(D) SAN – Bundle of His – Purkinje fibre – AVN

(58) State the characteristic of blood in heart at Diastole

(A) Oxygenated (B) De oxygenated (C) Both A and B (D) Arterial blood

(59) During systole of ventricle ..... .

(A) blood enters into ventricle from auricles

(B) O_2 blood emptied in pulmonary artery deoxygenated blood emptied into aortic arch

(C) CO_2 Containing blood into aortic arch and blood without CO_2 emptied in pulmonary vein

(D) Deoxygenated blood into pulmonary arch and O_2 blood into aortic arch

(60) Total diastole period of ventricle during one cardiac cycle

(A) 0.10 Sec (B) 0.40 Sec (C) 0.70 Sec (D) 0.50 Sec

(61) When auricles experience systole ..... .

(A) both Semilumar valve opens

(B) both AV valve opens

(C) only tricuspid and pulmonary semilunar valve opens

(D) only tricuspid and pulmonary semilunar valve opens

Answers : (51-D), (52-C), (53-A), (54-B), (55-D), (56-A), (57-A), (58-C), (59-D), (60-D), (61-B)

Disease Related to Blood Circulation :

(A) Hypertension : Blood pressure measured in redial artery of hand by sphigmomanometer.

- In a normal healthy person systolic pessure is 120 (i.e 120 mm hg) and diastolic pressure is 80; it is recorded as 120/80.
- Minor variations can occur in blood pressure 140/90. Which is considered as normal.
- Situation of Hypertention : If systolic blood pressure > 140 mm Hg. and diastolic blood pressure > 90 mm Hg. Situation is determined as hypertension.

Causes : Smoking, obesity

- Obesity : A person can be considered to be obese, when his weight is higher by 20% from the standard weight determined by age, height and other aspects. In such person more blood is to be supplied to tissues so blood pressure rises.
- Effects of Smoking : Nicotine mixed with blood causes

  (1) Contraction of artery so increase in blood pressure. (2) Decrease the absorption capacity of O_2 in lungs. (3) By CO_2 carring capacity of haemoglobin decreases.
- Atherosclerosis : Also known as hardening of arteries, silent killer.

  Effect : Responsible for the heart attack and strock of brain.

  Treatment : Angioplasty, by-pass surgery.
- Responsible situation for disease : More blood pressure, smoking, diet with more fat etc. By this substances damage occur in lumens of blood vessels's endodermis below that layer lipid and particularly cholesterol like substances are stored and there produce plaque. In the lumens of wall of blood vessel groove form as a node. By the producing of plaque lumens of blood vessel becomes narrower and decreases blood flow. (So that organ can not getting proper amount of blood.)
- Plaque are of two types : (1) Thrombus : It is known as stable mode. (same place as a groove)
- **Embolus :** After formation of plaque which free from that place and if circulate in small blood vessels obstruct occurs in blood flow.

- **Arteriosclerosis :** The walls of artery become thick and inelastic due to deposition of cholesteral and calcium salts, such a brittle artery can break. The blood from it flows out and clots. Such clots may circulate in blood and may cause problem in some organ for blood flow.

(62) Blood pressure is measured by which instrument ?

(A) Sphigmo baro meter (B) Stethoscope (C) Spirometer (D) Sphigmomanometer

(63) Diastolic pressure in healthy normal person is how much mmHg ?

(A) 120 (B) 140 (C) 80 (D) 100

(64) Considering physiological condition of normal healthy person, what will be blood pressure?

(A) 90/140 mmHg (B) 140/90 mmHg (C) 80/120 mmHg (D) 158/98 mmHg

(65) Which of the following option is improper for the effects of smoking ?

(A) O_2 carrying capacity of haemoglobin reduces by CO.

(B) Decrease O_2 accepting capacity of lungs.

(C) Induces contraction of artery increases blood pressure.

(D) Inhibits exchange of gases in excretory unit.

(66) Which method is used for the treatment of atherosclerosis ?

(A) Replacement of AV valve (B) Angiography (C) ECG (D) Angioplasty

(67) Blood nodule formed by plaque start circulating in blood along its flow, it is called ..... .

(A) thrombus (B) angiography (C) embolus (D) stroke

(68) It is most dreadful in human - ..... .

(A) Thrombus (B) Angiography (C) Callus (D) Embolus

(69) Patient of atherosclerosis disease have cholesterol stored in ..... in the form of plaque ?

(A) Endothelium (B) Mesothelium (C) Epithelium (D) B and C Both

(70) Hardening of arterial wall is observed in which disease ?

(A) Atherosclerosis (B) Arteriosclerosis (C) Hypertension (D) Silent killer

(71) How is veins are differentiated from arteries ?

(A) Very narrow lumen (B) Strong and muscular wall

(C) Valves are present to control blood flow (D) Dark coloured wall

Answers : (62-D), (63-C), (64-B), (65-D), (66-D), (67-C), (68-D), (69-B), (70-B), (71-C)

A = Statement, R = Reasoning type questions

Select the answer for questions given below from the options given here.

(A) A and R right, R is explanation of A.

(B) A and R right, but R is not explanation of A.

(C) A is right, R is wrong.

(D) A is wrong, R is right.

(72) Statement A : In each artery of human pure blood is circulated.

Reason R : Left chambers of heart in human receives pure blood.

(A) (B) (C) (D)

(73) Statement A : CO reduces O_2 carrying capacity of haemoglobin.

Reason R : Nicotine induces contraction of arteries and increases blood pressure.

(A) (B) (C) (D)

(74) Statement A : In cockroach, open circulalory system is present.

Reason R : Blood capillaries are absent in cockroach.

(A) (B) (C) (D)

(75) Statement A : Human heart is called double pump.

Reason R : In human heart, blood flows from right auricle to right ventricle and left auricle to left ventricle.

(A) (B) (C) (D)

(76) Statement A : SA node is pacemaker.

Reason R : SA node initiate heart beat.

(A) (B) (C) (D)

(77) Statement A : Heart attack is under the effect of atherosclerosis.

Reason R : Cardiac arteries supplying blood to heart lumen totally or partially obstructed.

(A) (B) (C) (D)

(78) Statement A : Mr. Iyer, call Jethalal obese person.

Reason R : Weight of Jethalal, according to his age, height and other things is 27% more.

(A) (B) (C) (D)

(79) Statement A : Pulmonary veins carry blood containing CO_2.

Reason R : Oxygenated blood is circulated in aortic arch.

(A) (B) (C) (D)

(80) Statement A : In the process of blood clotting, soluble fibrin is converted into stable fibrin.

Reason R : Active Hagmen factor activates PTA.

(A) (B) (C) (D)

(81) Statement A : Normally lymph is responsible for immune reaction.

Reason R : Lymph has 99 % small lymphocytes.

(A) (B) (C) (D)

Answers : (72-D), (73-B), (74-A), (75-B), (76-A), (77-A), (78-A), (79-D), (80-B), (81-C)

True - False (T - F) type questions

Select proper option for true or false in the following sentences.

(82) (1) Blood group is determined based on antigen present in blood serum.

(2) Rh^{-Ve} antigen are present on the surface of RBC.

(3) Child with haemolytic disease has Rh^{-ve} factor's antibody in their blood.

(4) IX + VIII + phospholipid + Ca^{+2} is present as essential factor christmas stimulating complex.

(A) TFTF (B) TTFT (C) FFTF (D) FFTT

(83) (1) Blood is partially acidic fluid.

(2) Hirudin present in human prevents blood clotting.

(3) In right chambers of heart deoxygenated blood flows.

(4) Pericardial fluid maintains concentration of blood in left chambers of heart.

(A) TFFT (B) TFTF (C) TFFF (D) FFFF

(84) Check the truth regarding AV node in following statements.

(1) Stimulation arising in it, spreads into bundle of HIS muscles.

(2) It sends stimulation to SA Node.

(3) It sends message for initiating contraction of auricles.

(4) It initiates heart beating.

(A) TFTF (B) TFFF (C) FFFF (D) TFTT

(85) (1) In aerteriosclerosis disease blood clots, when come out and such clots are circulated in the blood

(2) In obese person more blood plasma is required by tissue so blood pressure increases

(3) SA - node is called heart of heart.

(4) Relaxation of ventricle is possible by purkinje muscle and bundle of His muscles.

(A) TFTF (B) TTTF (C) FTTF (D) FTTT

Answers : (82-C), (83-D), (84-B), (85-B)

(86) In column-I different types of blood cells are given and in column-II functions of blood cells are given. Match the proper pair for column-I from column-II.

	Column-I		**Column-II**		**I**	**II**	**III**	**IV**
I	Lymphocytes	P	Active phagocytes	(A)	R	Q	P	S
II	Monocytes	Q	Transports respiratory gases	(B)	R	P	Q	S
III	Erythrocytes	R	Part of immunity system	(C)	R	S	P	Q
IV	Eosinophils	S	Nullifies effect of poison	(D)	P	Q	R	S

(87) In column-I Roman letters are given for factors of blood plasma. In column-II name of factors are given. Match column-I with column-II.

	Column-I		**Column-II**		**I**	**II**	**III**	**IV**
I	XIII	P	Calcium	(A)	S	P	Q	R
II	IV	Q	Proconvertin	(B)	P	S	Q	R
III	VII	R	Plasma thromboplastin anticedent	(C)	S	P	R	Q
IV	XI	S	Fibrin Stabilizing factor	(D)	S	Q	P	R

(88) Match the proper pair from column-I and column-II

	Column-I		**Column-II**		**I**	**II**	**III**	**IV**
I	O	P	Absence of antibody in blood plasma	(A)	Q	S	P	R
II	A	Q	Anti A and anti b	(B)	P	R	Q	S
III	AB	R	Anti a	(C)	P	S	R	Q
IV	B	S	Anti b	(D)	Q	P	S	R

(89) Match the proper pair for blood vessel and it's functions given below

	Column-I		Column-II
I	Pulmonary vein	P	Regulation of blood flow in right chambers of heart
II	Tricuspid valve	Q	Obtain O_2 blood from left ventricle
III	Anterior venacava	R	Deoxygenated blood emptied in right ventricle
IV	Aortic arch	S	transport of O_2 blood

	I	II	III	IV
(A)	S	R	P	Q
(B)	Q	P	R	S
(C)	S	P	R	Q
(D)	P	Q	R	S

(90) Find proper Sentence for X, Y, P, Q labelled parts of given diagram.

(A) X - First sound responsible for LUB

(B) Y - Prevent flow of blood from left auricle to left ventricle

(C) P - When ventricle show diastole, valve open

(D) Q - de O_2 blood passes from here

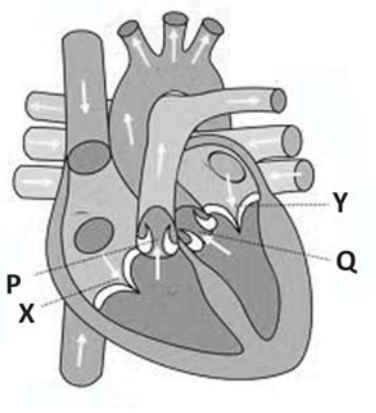


Figure 1

(91) Find the proper option for X and Y labelled part in the given diagram.

	X	Y
(A)	Bile pigments	Mesothelium
(B)	Cholesterol	Endothelium
(C)	Mesothelium	Cholesterol
(D)	Endothelium	Mesothelium

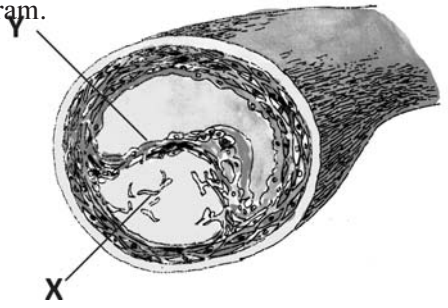


Figure 2

(92) Find the proper option for P, Q, R, S labelled part in the given diagram - 3.

(A) P - obtain blood from LV

(B) Q - give blood to LV

(C) S - give blood to RA

(D) R - obtain blood from LA

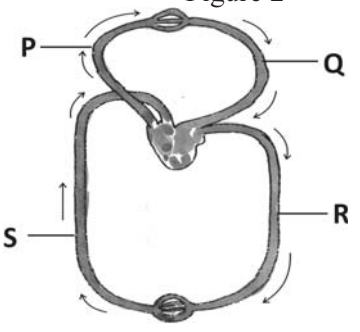


Figure 3

(93) Name the scientist, who has shown blood circulation passage given in diagram - 4.

(A) Carl landsteiner

(B) William parnas

(C) Thomas harvey

(D) William harvey

(94) blood passes through labelled part X- ..... .

(A) Diastole of right auricle

(B) Systole of left auricle

(C) Systole of right ventricle

(D) Systole of left ventricle

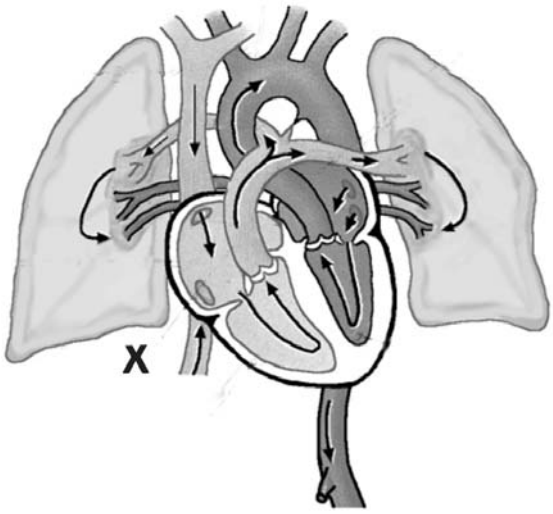


Figure - 4

(95) Identify Atrio ventricular - nodle tissue in given diagram - 5.

(A) p

(B) q

(C) r

(D) s

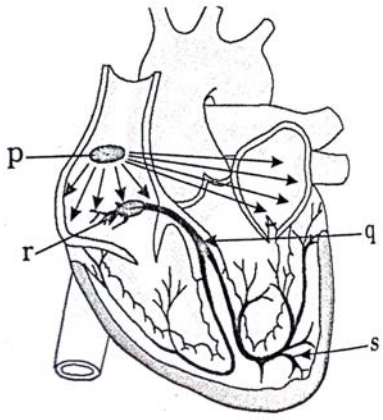


Figure - 5

(96) In the given diagram - 6 which labelled part transports O_2 blood ?

(A) W

(B) X

(C) Y

(D) Z

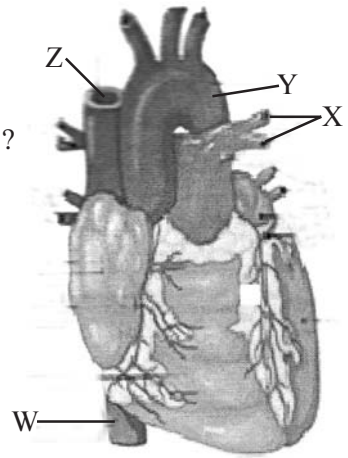
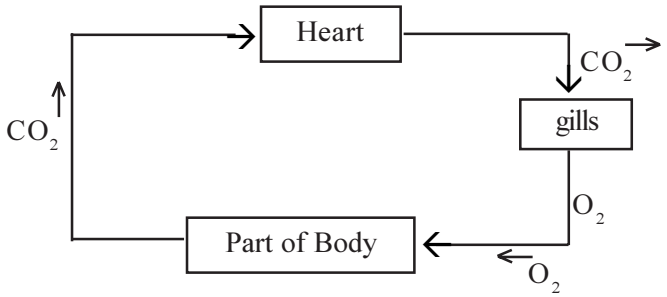


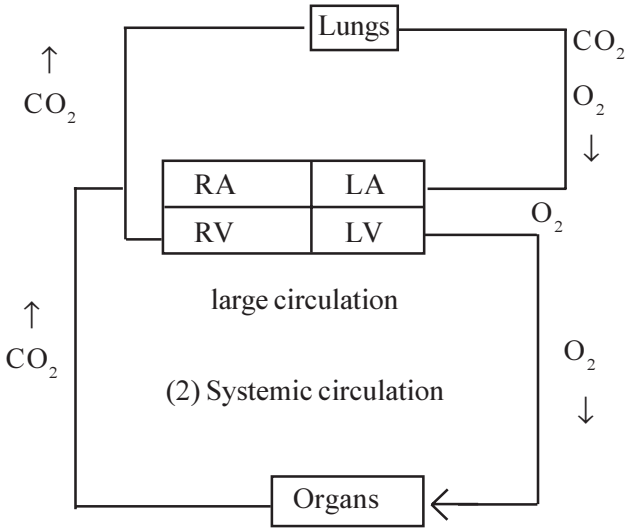
Figure - 6

Answers : (86-B), (87-A), (88-A), (89-C), (90-A), (91-B), (92-C), (93-D), (94-C), (95-C), (96-C)

- Study of circulatory system is known as Angiology.
- Double circulation discovered by William Harvey. He proposed heart as a pumping station of body.
- For unipath flow : eg. fish.



Double Circulation



- Pulmonary artery and systemic arch cross to each other and at the crossing point they connected with each other by ligamentum arteriosum.
- During the embryonic development in inter atrial spetum ovilocur hole is present and after birth it can not be closed so deficiency can occur in baby. Blood of both atrial mixed in each other and decreases the amount of supplies of O_2 and symptoms of cynams's can develop (blue colour of skin can occurs. This type of baby is known as 'bluebabies.'

Pulsation of heart :

Organism	Pulsation per min	Organism	Pulsation per min
Adult man	72	Shrew	600 – 800
Embryo	160	Blue whale	25
New born baby	120 – 140	Elephant	28
Child	100	Sparrow	500
Rabbit	210	Canary bird	1000
Rat	400 – 600		

- Pulsation of heart of animals related.
- In mammals pulsation are more of small animals.
- During the physiological process numbers of pulsation are different. (stressful stage-resting stage)

- Neurogenic heart - eg. Arthropoda, some animals of annelida, phylam.

Myogenic heart - eg. Human being

	Neurogenic heart		Myogenic heart
1.	In heart impulse of pulsation come out from the heart.	–	Impulses of heart pulsation develop only in heart.
2.	Inpulses origin by nervous system.	–	Impulses origin by specific muscular tissue.
3.	Nerve fibers are spread on the heart and contraction and relaxation can occur.	–	In it specific type of muscles spread impulses.
4.	If heart is removed from the body, then pulsation of heart stops.	–	If removal of heart from the body, gives sufficient nutrient, and favourable condition, it give pulses for few time

Control / Regulation of heart beats :

Endocrine regulation : Adrenalin = ↑ rate (Inscrease)

Noradrenalin = ↑ rate (Increase)

Vagus stimulate acetyl coline = ↓ rate (Decrease)

Autonomous nervory system : Sympathetic = ↑ rate (Increase)

Parasympathetic = ↓ rate (Decrease)

Tachycardia : Averagely, in adult man one of the situation can arise where heart pulsation per minute is more than 90. During the fever increase in the temperature of body rate of metabolism can increase and thus spontaneously stimulation and rhythamacity also increase.

• Bradycardia : In adult man where average heart pulsation average below the 60 per minute.

• In common situations heart rate and respiration rate ratio is 4 : 1.

• 0.8 sec time for one cardiac cycle in man.

(1) Atrial systole = 0.1 Sec
(2) Atrial diastole = 0.7 Sec] 0.8 Sec

(3) Ventricle systole = 0.3 Sec
(4) Ventricle diastole = 0.5 Sec] 0.8 Sec

- Volume of blood related to heart :

Volume of blood related to heart :

- 120ml volume of each ventricle is called End Diastolic Volume (EDV)
- During the systole becomes empty while nearly 70 ml volume decrease, it is called stroke volume (SV).
- Remaining volume in ventricle nearly 50 ml, it is called End Systolic Volume (ESV) other systolic volume.
- Ejection fraction (EF) = $\frac{ESV}{EDV} = \frac{70}{120} = 0.583$

$\therefore$ 60 %

Strocke Volume : (SV) = EDV – ESV

= (120 – 50) ml

= nearly 70 ml

During the one pulsation 70 ml blood from ventricle towards the organs.

Cardiac Output (CO) = Stroke Volume × Heart Beats

= 70 × 72

= 5040 ml

= nearly 5000 ml

CO : 5 lit / min

- **Feeling of heart (ventricle)** : Commonly blood flow from venacava towards atria. Approximate 75% of blood directly from the atria to ventricle passes. Thats why contraction of atria can occur before that blood flow can occur. After that contraction of atria can occur and ventricles are full by 25% of more blood.
- Systolic duration of atria - 25 % Ventricles are filled.
- Diastolic duration of ventricles - 75% Ventricles are filled.
- Diastolic of ventricles - First phase - 70% (Rapid transport)

  Second phase - 5% (Open Av.valves)

  Final phase - 25% (Due to contraction of atria)
- Spleen : Biggest lymphnode of the body. Blood bank of body.
- Graveyard of RBCs . Origin from mesoderm of embryo
- One specific connective tissue feel in spleen, which is known as Reticula endo epithelial tissue or spleenic pump.
- Spleenic pump has two parts : (1) White pulp (2) Red pulp.
- Pulp of spleen have large five photocytic cells which is called macrophage cells.

Functions :

(1) Senescenteial blood cell, dead or living microbes or disease infact microbe and waste material of cell are swallowing by marcophage.

(2) RBCs are forms in embryonic stage.

(3) Several anti-bodies synthesis here.

(4) In adult stage spleen act as a blood blank. Its sinuses act as a reservior of blood. When required blood is transported in circulation.

(5) Storage of iron occur in spleen.

(6) At the time of malaria, increase the volume of spleen because increases the number of lymphocytes and dead RBCs.

- Spleen, liver, kidney three are known as instruments of filtration.

Questions for NEET

(97) Which cells produces thrombokinase ?

(A) Erythrocytes (B) leucocytes

(C) Blood corpuscles (D) Endothelium of blood capillaries

(98) State the name of valve, in post-vena cava which opens in right auricle in mammals.

(A) Mitral valve (B) Thabius valve (C) Tricuspid valve (D) Austatian valve

(99) State the reason of murmur sound in heart.

(A) Pulsation of veins (B) Defect in valve

(C) Thromboplastin (D) Less developed auricle

(100) Thromboplastin is produced by ..... .

(A) Injured tissue (B) Blood corpuscles (C) Both A and B (D) Prothrombin

(101) Which situation is responsible for leukaemia ?

(A) Percentage of WBC less than 5000/mm^3. (B) Bone marrow is destroyed.

(C) lymphoocytes reduced by 2 to 5 %. (D) WBC increases by 6000/mm^3.

(102) Which situation is responsible for blood clotting ?

(A) Destruction of RBC (B) Destruction of WBC

(C) Destruction of lymphocytes (D) Change in component of lymph

(103) When auriculo - Ventricular valve closes, which sound is hear ?

(A) Dubb (B) Lubb (C) lubb-dubb (D) Dhakdhak

(104) More than normal amount of leucocytes, indicates which condition ?

(A) Anemic (B) Infection

(C) Protection against bacteria (D) Effect on erythrocyte production

(105) Which of the following organism has red coloured blood without RBC ?

(A) Frog (B) Human (C) Cockroach (D) Earthworm

(106) Pressure of blood vessels is ..... .

(A) Diastolic pressure (B) Systolic pressure

(C) Difference of B and A (D) Pressure in vena cava

(107) Blood entering from lungs into heart by pulmonary vein, has more amounts of ..... .

(A) RBC in 1ml blood (B) Hb in 1ml Blood

(C) O_2 in 1ml blood (D) Nutritive substance in 1ml

(108) State the location of fovea ovalis

(A) Inter auricular septum (B) Intra ventricular septum

(C) Right intra auriculo - ventricular septum (D) Left intra auriculo - ventricular septum

(109) Beginning and end of hepatic portal system

(A) Digestive system to liver (B) Kidney to heart (C) Liver to heart (D) Kidney to liver

(110) What is called blood vessel, shows blood circulation begins from capillary and ends in capillaries ?

(A) Portal system (B) Hepatic circulation (C) Cardiac circulation (D) All of above

(111) Which of the following organ is death-place for R.B.C. ?

(A) Kidney (B) Spleen (C) Bone marrow (D) Pancreas

(112) Which type of leycocytes are linked with secretion of Histamine and heparin ?

(A) Neutrophils (B) Basophils (C) Acidophils (D) Monocytes

(113) What are stages of cardiac cycle ?

(A) Auricle systole - ventricle systole - auricle - ventricle systole

(B) Diastole - auricle systole - ventricle diastole

(C) Auricle systole - ventricle diastole - ventricle systole

(D) Ventricle diastole - auricle diastole - ventricle systole - auricle systole

(114) Billirubin and biliverdin is byproduct of dissociation of ..... .

(A) myoglobin (B) lipid (C) carbohydrate (D) haemoglobin

(115) Increase in percentage of erythrocyte is called ..... .

(A) Poly cythemia (B) leucocytopenia (C) Anaemia (D) leucopenia

(116) Double circulation of blood in heart of normal, healthy person is ..... per minute.

(A) 8 (B) 16 (C) 32 (D) 72

(117) ..... is called father of cardio vascular system.

(A) Wharvey (B) S harvey (C) S thomas (D) Landsteneir

(118) When is difficulty observed in blood circulation ?

(A) Antigen absent in RBC (B) Antibody absent in blood plasma

(C) If antigen is opposite to antibody (D) If antigen - antibody are similar type

(119) When dupp sound of heart is heard ?

(A) When mitral valve opens

(B) When mitral valve closes

(C) When semilunar valve, situated at anterior part of aortic arch closes

(D) When tricuspid valve opens

(120) Output of heart is determined by ..... .

(A) heart - beats (B) size of stroke (C) blood circulation (D) both A and B

(121) Regulation of blood pressure is by ..... .

(A) pituitary gland (B) adrenal gland (C) thymus gland (D) parathyroid gland

(122) Pericardial fluid is secreted from ..... .

(A) myocardium (B) Lateral diaphragm

(C) Visceral membrane (D) pericardial membrane

Answers : (97-C), (98-D), (99-B), (100-C), (101-A), (102-C), (103-B), (104-B), (105-D), (106-C), (107-C), (108-A), (109-A), (110-A), (111-B), (112-B), (113-C), (114-D), (115-A), (116-D), (117-A), (118-D), (119-C), (120-D), (121-B), (122-D)

Blood Vascular System

- It is a circulatory system containing body fluid and pumping organ

Blood Vessels

- They are intricate network of hollow tubes that transport blood throughout the entire body.

Arteries

- They are elastic vessels that transport blood away from the heart

Veins

- They are elastic vessels that transport blood to the heart

Capillaries

- They are extremely small vessels that transport blood from the arteries to the veins. Respiratory gases, nutrients and wastes are exchanged through the capillaries within the tissues

Blood

- It is mobile fluid connective tissue

Plasma

- It is slightly alkaline, non-cellular, pale yellow fluid that constitutes about 60% of the blood

Formed elements

Erythrocytes

- Most abundent cells filled with haemoglobin

Leucocytes

- Colourless cells that show amoeboid movement

Agranulocytes

- Grarnules are not found in cytoplasm and are of two types - Lymphocytes and Monocytes
- They are motile and phagocytic in nature

Granulocytes

- Gradules are present in cytoplasm and are of three types - eosinophils, basophils and neutrophils

Blood Coagulation

- They response to repture of blood vessel or tissue damage, a complex, cascade of chemical reaction occur which results in the formation of blood clot

Platelets

- They release platelet factors that help in the clotting of blood (or thromosplastin) on injury

Heart beat

- Rhythmic contraction and relaxation of the cardiac muscles is known as heart beat

Heart

- It is a mesodermally derived organ that comprises four complete chamers (two auricles and two ventricles)
- Right auricle receives deoxygenated blood from the body and left auricle receives oxygenated blood from lungs

Cardiac Cycle

- It is the sequence of events that occurs in the heart during one full heart beat

25 Excretory Products and their Elimination

- The separation and elimination of toxic nitrogenous waste material from the body is called excretion.
- Excretory substances : Ammonia, Urea, Uric acid, creatin, creatinin, phosphate, Hipuric acid etc.
- First excretory substance is Ammonia.
- Due to deproteinization of amino acid resultant ammonia is formed.
- For the solubility of excretory material water is essential, Ammonia > Urea > Uric acid.
- According to consmption of energy formation of excretory substances are in order Uric acid > Urea > Ammonia.
- For the elimination of excretory substances consumption of water and energy is in receprocal ratio.
- By the types of excretory substances, types of animals.

Ammonotelic animals	Ureotelic animals	Uricotelic animals
• NH_3 dissolve in more water eliminate from the body	In liver $2NH_3 + CO_2 \rightarrow$ Formation of urea Normally elimination in form of urine	Less toxic, semisolid, semiliquid, in round shape of pellet or paste excretion
• Eg $\rightarrow$ Aquatic insects bonyfish, amphibian, tad pole, crocodile	cartilage fish, Adult amphibian, mammals	Insects, Reptiles, Aves, Terrestial snail

(1) Type of nitrogenous poisonous substances and its excretion depends on ..... .

(A) Temperature (B) Availability of water (C) Food components (D) Atmosphere

(2) Poisonous substance ammonia is formed by ..... .

(A) Emulsification of lipid (B) Protein synthesis

(C) Oxidation of carbohydrates (D) Deamination of aminoacids

(3) Find the correct option for formation of poisonous substance urea, location and reactants.

(A) Stomach, $CO_2 + NH_2$ (B) Liver, $O_2 + 2NH_3$

(C) Liver, $CO_2 + 2NH_3$ (D) Kidney, $Co_2 + 2NH_3$

(4) Find correct option for the uric acid excretion group of organisms.

(A) Snake - Calotes - Sparrow (B) Osteichthyes - Crocodile - Aquatic insects

(C) Frog - Man - Reptile (D) Chondrichthyes - Amphibian tadpole - Aquatic insects

(5) Of the following posionous substances which one require more amount of water for its solubility ?

(A) Glycerol (B) Ammonia (C) Urea (D) Uric acid

(6) Find correct option in descending order for utilization of energy for the formation of excretory substances.

(A) Frog, Snake, Tadpole (B) Tadpole, Frog, Snake

(C) Sparrow, Frog, Tadpole (D) Pigeon, Tadpole, Frog

(7) Organism showing two types of excretion during its life span, find the correct option.

(A) Osteichthyes (B) Chondrichthyes (C) Amphibia (D) A and B both

(8) Deamination is ..... .

(A) Reabsorption of amino acids by papillae of intestinal wall from digested food

(B) Release of CO_2 and NH_2 from dissociation of amino acids

(C) Formation of NHCO, for linking amino acid with amino acid

(D) Removal of CO_2 from amino acids by kidney

(9) What is proper statement for amphibian class ?

(A) Urea in tadpole and ammonia in adult amphibian.

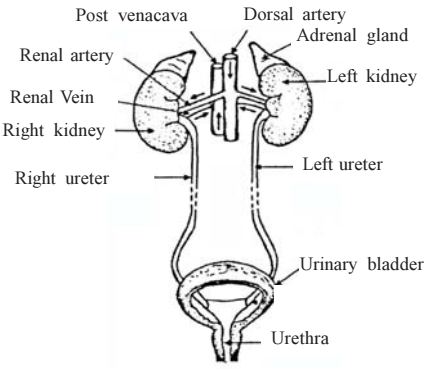
(B) Urea in total lifecycle of amphibian animals.

(C) Ammonia in tadpole and urea in adult frog.

(D) Uric acid in adult amphibia and ammonia in tadpole.

Answers : (1-B), (2-D), (3-C), (4-A), (5-B), (6-C), (7-C), (8-B), (9-C)

Human Excretory system :



Human Excretory System

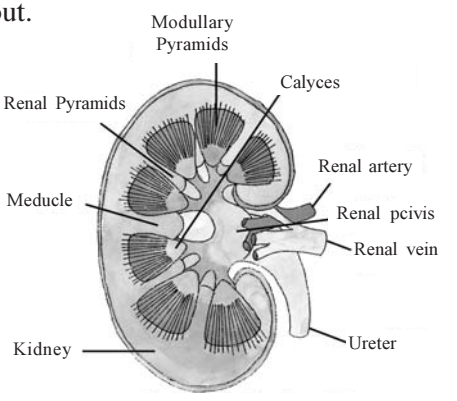
- Main excretory organ-kidney →Colour: Reddish brown
  Shape : Bean shape
- Weight : 125 to 170 gm
- Size : (l : 10 × b : 5 × h : 3) cm
- Location : Both side of vertebral column, inlumber region 12th thoracic vertebrate and first two lumber vertebrate, lateral side ($T_{12}L_1L_2$)
- Arrangement : The two kidney are not placed at the same level. The right kidney is slightly on a lower level than the left. This is due to fact that right side of the abdominal covity is occupied by liver.

External structure :

- Surface : Outer surface convex, inner surface concave.
- Hilum : On concave surface one longitudinal slit like structure is there by which artery, nerve enters inside the hilum, through this slit vein, ureter come out.

Internal structure :

- Cortex : Peripheral area of convex surface dark red and somewhat granular.
- Medulla : Pyrami shaped renal pyramid, area between the cortex and renal pyramid which spread toward the cortex is called column of Bertini.



L.S. of Kidney

(10) Spreading region of cortex ..... .

(A) Renal pyramid (B) Column of bertini

(C) Calyces (D) Renal pelvis

(11) Among pair of kidney in human, right kidney is slightly on lower side, because ..... .

(A) diaphragm is slightly lowered towards right side

(B) right kidney is heavy in weight

(C) right ureter is very short

(D) above right kidney, beneath diaphragm largest accessory digestive gland is situated

(12) Hilum present on the concave surface of kidney ..... .

(A) pure blood by renal artery come out of kidney

(B) pure blood by renal artery enters into kidney

(C) pure blood by renal vein enters into kidney

(D) pure blood by renal vein come out of kidney

(Hint : pure blood - without nitrogenous poisonous substances)

(13) Location of kidney ..... .

(A) 11th and 12th thoracic vertebrae and on either side of 1st Sacral vertebrae.

(B) 12th thoracic vertebrae and either side of first two sacral vertebrae

(C) either side of first three sacral vertebrae

(D) either side of last three thoracic vertebrae

(14) Renal vein transport blood in which direction ?

(A) Inferior venacava to kidney

(B) Kidney to inferior venacava

(C) Inside kidney toward Afferenent renal arteriole

(D) From hind limb to kidney

(15) In which part of kidney glomerulus is situated ?

(A) Renal pelvis (B) Renal pyramid (C) Renal cortex (D) Calyces

(16) In terrestrial organism, excretory substances formed in liver reaches to kidney through which passage ?

(A) Hepatic vein - hepatic artery - post venacava - heart - renal artery

(B) Hepatic artery - heart - dorsal aorta -Renal artery

(C) Hepatic vein - Post venacava - aortic arch - heart - renal artery

(D) Hepatic vein - post venacava - heart - aortic arch - renal artery

---

Neephron : Excretory unit :

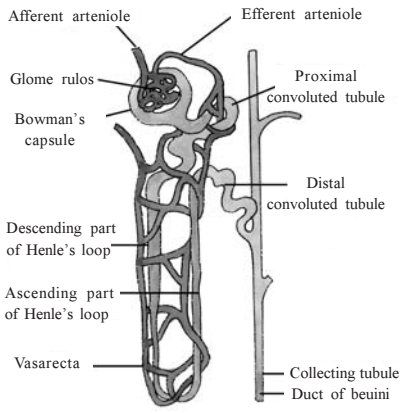
- Structural and functional unit of kidney.
- Each kidney has approximately 10 lacs nephron.
- Lenght 3 cm

  Diameter : 20 to 30 μm (Micrometer)
- Excretory unit has two main part

(a) malpighian body

(b) Uriniferous tubule

(a) Malpighian body : Glomerulus + Bowman's capsule

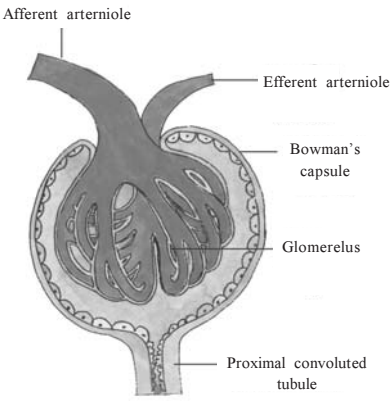


Ultraspructure of uriniforous tubule

Answers : (10-B), (11-D), (12-D), (13-B), (14-B), (15-C), (16-D)

Blood flow :

- Afferent arteriole - Glomerulus - Efferent arteriole
- Diameter of afferent arteriole is more.
- Diameter of efferent arteriole is less.
- Blood capillaries unilayered thickness micro vessel.



Malpighian Body

Bowman Capsule :

- Double walled cup like structure
- Towards outer side wall composed by squmousal epithelial cells
- Inner side wall some. specific type of epithelial cell are situated which is called podocytes.
- Specific arrangement of cells little bit space among them, is called filtration slits.

Uriniferous tubules :

- **PCT - Proximal Convoluted tubule**

  Wall of this region, cells having mitochondria and micro villi are present. Tubular linning made up of cuboidal epithelial cells.

  Location : In cortex region.

- **Loops of Henle :** U-shaped tubule

  Descending loop : Linning made up by flat epithelial cells, permeable for water.

  Asecending loop : Its linning made up by cuboidal epithelial cells impermeable for water.

  Location : In medulla region.

- **DCT : Distal Convoluted tubule**

  It linning made up of cuboidal epithelial cells.

  Location : It is situated in Cortex.

- **Transport pathway :** Bowman's capsule - PCT - Loop of Henle - DCT

Collecting duct :

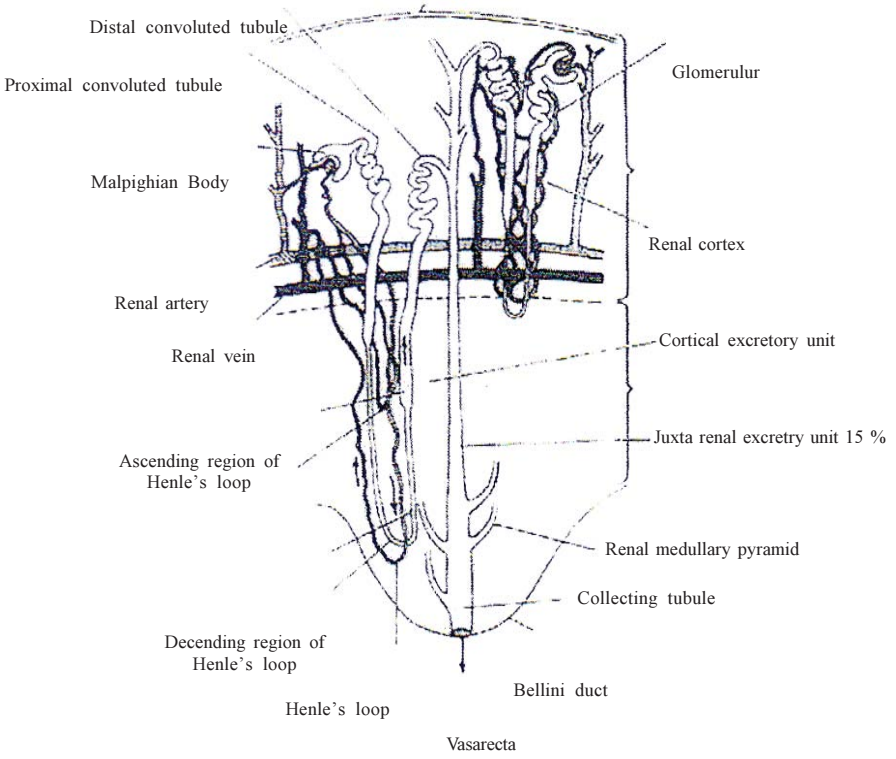
Location : tubule beneath the DCT which is located in pyramid of medulla region

Bellini duct : Many collecting ducts joint and open in large billini duct.

- Bellini duct opens in renal pelvis.
- Urine formation occur only in nephron. While urine transportation occur only in collecting duct.
- Transporation pathway : Collecting duct - Bellini duct - Renal pelvis - uretar - urinary bladder.

Types of nephron :

Juxta medullary nephron	Cortical nephron
– 15% of total nephron	– 85% of total nephron
– length of Henle's loop is more	– Length of Henle's loop is less
– vasa recta is present	– Vasarecta is absent or less developed
– No or less developed peritubular capillaries	– Development of peritubulur capillaries
– Henle's loop situated deep in medulla.	– Limited upto cortex, very less part enlarge in medulla.



Ultrasturcture and types of uriniferous tubule excretory unit

---

(17) State name of cells which by Specific arrangement forms filteration pore in inner wall of Bowman's capsule.

(A) Squamous epithelium (B)Podocytes (C) ciliated cells (D) Stratified epithelium

(18) In healthy person, diameter of afferent and efferent arteriole ..... .

(A) Both equal (B) More and less

(C) Less and more (D) Depending on working capacity of nephron

(19) Improper, accordance with renal pyramid.

(A) Convoluted tubule (B) Henle's loop (C) Vasa recta (D) Collecting tubule

(20) True sequence for transport of excretory substances ..... .

(A) uriniferous tubule - collecting tubule - renal pelvis - duct of bellini

(B) uriniferous tubule - renal pelvis - collecting tubule - bellini's duct

(C) uriniferous tubule - bellini's duct - collecting tubule - renal pelvis

(D) uriniferous tubule - collecting tubule - bellini's duct - renal pelvis

(21) Its layer is formed by columnar epithelial cells ..... .

(A) PCT (B) DCT

(C) Ascending region of Henle's loop (D) All of above

(22) End portion of uriniferous tubule

(A) DCT (B) collecting tubule (C) duct of bellini (D) ureter

(23) Region of renal pelvis is more broad at cortex region because ..... .

(A) Concentration of intercellular fluid is more (B) More amount of Juxta-cortical nephron

(C) More amount of corical excretory unit (D) More amount of convoluted tubule

(24) What is true for external excretory unit ?

(A) It has peritubular capillaries (B) Henle's loop is extended towards medulla

(C) convoluted tubules are less developed (D) collecting tubules are branched

(25) Columnar cells with brush border is characteristic of the following.

(A) PCT (B) Henle's ascending narrow region

(C) Inner wall of Bowman's capsule (D) Henle's descending narrow region

(26) Which structures are more in renal pyramid ?

(A) Malpighian body (B) PCT (C) Henle's loop (D) DCT

Answers : (17-B), (18-C), (19-A), (20-D), (21-D), (22-A), (23-C), (24-A), (25-A), (26-C)

Function of Nephron :

PCT - PCT is lined by cuboidal brush border (having microvilli) epithelium which increases the surface for re-absorption of fitrate.

- The PCT also helps to maintain a constant balance in body fluid by the secretion of H^+ and by reabsorption of the buffer HCO_3^- from the filtrate.
- The filtrate is isotonic blood plasma.
- Absorption of essential nutrients.

Descending Henle's loop : Permeable for water, impermeable for electrolytes, filtrate more concentrated than blood plasma.

Ascending Henle's loop

Impermeable for water, permeable for electrolytes (Diffusion and active transport)

When electrolytes enters from Henle's loop to medulla filtrate becomes dilute.

DCT

Reabsorption of water, HCO_3^-

From efferent arteriole absorption of excretion secretion occur in lumens of DCT.

Na^+ K^+ balance is maintained.

Collecting duct :

- From the DCT urine passes from here and reabsorption of water in huge amount, Urine becomes concentrated.
- pH and Ionic balance by secretion of H^+ and K^+ and elimination occurs.
- End portion of collecting tubule is permeable for urea. So it diffuses in medulla. Therefore concentration of Urine is maintained.

---

(27) Characteristic of endothelial cells of proximal convoluted tubule ..... .

(A) flat cells (B) cuboidal with brush border

(C) many layered (D) columnar

(28) Process occuring in PCT to maintain pH of body fluid

(A) Secretion of H^+ and HCO_3^-

(B) Reabsorption of H^+ and secretion of $HCO3^-$

(C) Reabsorption of H^+ and HCO_3^-

(D) Secretion of H^+ and reabsorption of $HCO3^-$

(29) Descending region of Henle's loop is mainly impermeable for ..... .

(A) electrolytes (B) water (C) both A and B (D) None

(30) Filterate is diluted in ascending region of Henle's loop because ..... .

(A) electrolytes from interstitial medullary fluid enters ascending region

(B) more water from interstitial medullary fluid enters ascending region

(C) from ascending region only electrolytes diffuses or actively transported in interstitial fluid

(D) more secretion of ADH

(31) State correct option, with reference to X, Y , Z

Statements -

X - urine gets more concentrated in distal convoluted tubule

Y - urea come out of DCT

Z : uric acid and ammonia secreted from DCT

(A) X, Y, Z are true, Y and Z are explanation for X.

(B) X and Y are true, Z is False, Y is explanation for X.

(C) X and Z are true, Y is False Z is reason for X.

(D) X true, Y and Z wrong.

Answers : (27-B), (28-D), (29-A), (30-C), (31-C)

Urine formation :

Three phases :

(1) Glomerular filtration (2) Selective reabsorption and (3) Tubular secretion

(1) **Glomerular filtration :**

- For the filtration process, kidney per minute average 1100 to 1200 ml blood gain.
- Pressure filtration process occur in glomerulus.
- Filtration of blood occur by micro membrane.

In filtrate water, essential substances and excretory substances etc. are present.

GFR : (Glomerular filtration rate)

- Every minute from both kidney by all the excretory units forms amount of filtrate which enter in PCT. or The quantity of fitrated formed per minute in all nephrons of both the kidneys is called glomerular filtration rate.
- 125 ml/ minute is the GFR for healthy man.

(2) Selective reabsorption :

99% amount of filtrate reabsorption occur in PCT

= GFR : 1 minute $\rightarrow$ 125 ml, 60 minute $\rightarrow$ 7500 ml or 7.5 lit, 24 hour $\rightarrow$ 180 lit (24 × 7.5) l, per day filtrate $\rightarrow$ 18 li.

Everyday amount of urine release average 1.5 li.

(180 l $\rightarrow$ 1.5 l = 99%)

Reabsorptionable substances : Water, glucose, amino acid, vitamins, $NaHCO_3$ etc.

(3) Tubular Secretion :

- Substances which are not reabsorbed in PCT, passes towards the descending loop.
- H^+, K^+, NH_4^+ like substances are secreted in the PCT.
- Moreover in malpighian body the substances which are not filtered pass through efferent arteriole and such substances are ammonia, urea, uric acid, antibodies like penicillin etc. secreted in DCT.
- In PCT and Henle's loop absorption of water can occur, so urine become concentrated.
- When the filtrate passes through DCT and collecting duct, water is also reabsorbed and urine becomes concentrated.

---

(32) State the volume of blood entering into both kidney for filteration.

(A) 0.125 liter (B) 180 liter (C) 1.5 liter (D) 1.1 to 1.2 liter

(33) In healthy person, volume of filterate per minute passes through neck of malphithian body ?

(A) 125 ml (B) 75 ml (C) 1250 ml (D) 1500 ml

(34) To maintain ionic balance of body fluid, tubular cells secretes ..... .

(A) uric acid (B) ammonia (C) penicillin (D) A and C both

(35) Organisms in which, Henle's loop length is very less in excretory unit urine of such organism is ..... .

(A) isotonic (B) hypotonic (C) hypertonic (D) more concentrated

(36) In normal condition which of the following substance is absorbed totally in PCT ?

(A) Urea (B) Salts (C) Glucose (D) Uric acid

(37) In healthy person components present fiterate formed due to pressure filteration , find true group from the following ..... .

(A) Water, glucose, vitamins, sodium bicarbonate (B)Water, glucose, urea, uric acid, penicilin

(C) Water, salt, uric acid, penicilin (D) Water, vitamins, vasopressin, uric acid

(38) Out going Blood flow is slow in malpighian tubule than blood entered because ..... .

(A) efferent arteriole is unilayered and has very narrow lumen

(B) afferent arteriole diameter is narrower than efferent arteriole

(C) afferent arteriole lumen is broader than efferent arteriole

(D) both afferent and efferent arteriole has same diameter but amount of blood capillaries are more

Answers : (32-D), (33-A), (34-B), (35-B), (36-C), (37-A), (38-C)

Mechanisms of Concentration of the filtrate : (Counter current)

- For the concentration of urine the kindey has a mechanism that is known as concentration mechanism.
- Aves and mammals have ability for urine excretion (More concentrated than blood).
- For this they have evolved a counter current mechanism, for conservation of water.
- Henle's loop and vasa recta (capillary in the form of loop) play a significant role in it.
- The flow of filtrate in the two limbs of Henle's loop is in opposite direction which forms a counter current.
- The flow of blood in the two limbs of vasa recta is also in opposite direction and thus forms a counter current.
- Thus these two counter current systems play an important role in concentrating urine.
- The proximity between the vasa recta and loop of Henle and the counter current in them help in maintaining an increasing Osmolarity towards the inner medullary interstitial fluid. ie from 300 $mOsmoil^{-1}$ the cortex to about 1200 m Osmol/L in the inner medulla.

Participant units of counter currrent :

(1) Henle's loop :

- When the filtrate passes through the ascending limbs of the Henle's loop, it loses Na^+, Cl^- by diffusion in its narrow region to the interstitial fluid in the renal medulla and Na^+ and Cl^- ions by active transport in its wide region of the ascending limbs.
- The water so drawn enters the vasa recta and is transported away.
- The interstitial fluid in the medulla is nearly as concentrated as the urine leaving the collecting duct.

(2) Vasa Recta :

- The walls of the vasa recta is made up of endo thelial cells and are freely permeable to water, urea and ions.
- Blood flows from ascending capillary towards water is drawn out by the process of osmosis from the blood and there will be an increase in the concentration of the interstitial fluid by diffusion of urea and Na^+ and Cl^- enters the blood plasma.
- As the blood enters in the ascending capillary towards the renal cortex, the reverse condition occurs.
- It means water re-enters the blood plasma while urea and Na^+ and Cl^- drawn out due to a decrease in the concentration of interstitial fluid.
- The counter current exchange in the vasa recta :

  (A) Prevent the loss of Na^+ and Cl^- ions, from the renal medulla and

  (B) helps to maintain concentration gradient in the renal medulla.
- Certain urea diffuses from the last part of the collecting duct into interstitial fluid, to increase density which increased water re- absorption from the bottom of Henle's loop as well as from collecting duct.
- The function of counter - current mechanism is to maintain high concentration surrounding nephrons and collecting ducts.

- Thus hypertonic urine is produced in human. Thus urine produced is nearly four times concentracted than the initial filtrate formed.

  (300 mOsmoil^{-1} → 1200 mOsmoil^{-1})

  mOsmoil^{-1} = Osmolarity in milli osmosmol per litre.

Chemical composition of urine :

Water	95%
Urea	2.6%
Electrolytes	2%
Uric acid	0.3%

Creatinine	0.3%
Ammonia	0.3%
Hipuric acid	0.3%

- **Urochrome :** By presence of this pigment Urine is pale yellow in colour.
- **Daily :** 25-30 gm urea is excreted.

---

(39) Counter current mechanism means ..... .

(A) direction of blood flow in glomerulus

(B) in both regions of the loop, flow of filterate is in same direction

(C) flow of urine in collecting tubule

(D) in both regions of the loop, flow of filtrate is in opposite direction

(40) How is Osmosis of interstitial fluid present in renal medulla maintained ?

(A) With the help of counter current (B) Due to proximity of Henle's loop and vasa recta

(C) Due to proximity of PCT and vasa recta (D) A and B Both

(41) Current of osmolarity is formed by which substance ?

(A) Urea (B) NaCl (C) A and B both (D) NH_3

(42) True sequence of blood (mOsmoil^{-1}) in descending capillary ..... .

(A) 900 → 600 → 400 (B) 400 → 600 → 900

(C) 200 → 400 → 600 (D) 600 → 400 → 200

(43) True sequence of urine (mOsmoil^{-1}) flowing in collecting tubule ..... .

(A) 300 → 400 → 600 → 900 → 1200 (B) 1200 → 700 → 400 → 200 → 100

(C) 1200 → 900 → 600 → 400 → 300 (D) None of above

(44) State the true sentence, related with uriniferous tubule

(A) Na^+ and Cl^- flow from interstitial fluid into descending Henle's loop

(B) Na^+ and Cl^- flow from Henle's descending region into interstitial fluid and than into vasa recta

(C) Na^+ and Cl^- flow from Henle's ascending region to interstitial fluid and than enters vasa recta

(D) Na^+ and Cl^- flows from vasa recta to interstitial fluid and from than Henle's ascending region

(45) Function of counter current ..... .

(A) to prepare hypertonic urine

(B) to prepare isotonic ccompare to blood urine

(C) maintain more concentration of nearby excretory unit and collecting tubule

(D) maintain more concentration in adjoining excrectory unit and vasa recta

(46) To increase density of interstitial fluid of renal pyramid which substance diffuse from last portion of collecting tubule ?

(A) NaCl (B) HCO_3^- (C) H_2O (D) Urea

(47) $mOsmoil^{-1}$ of blood entering in vasa recta is ..... comparing interstitial fluid ?

(A) more (B) less (C) isotonic (D) A and B both

Answers : (39-D), (40-D), (41-C), (42-B), (43-A), (44-C), (45-C), (46-D), (47-B)

Regulation of kidney function :

- The function of the kidney is efficiently monitored and regulated by hormonal feed back mechanisms.
- Regulated by hormonal feed back mechanisms mainly by hypothalamus, pituitary, JGA (Juxta Glomerular Appartus) and heart at certain extent.
- When water is abundant in the body tissue kidney excretes large quantity of dilute urine.
- When body tissues are short of water small quantity of concentrated urine is excreted.
- The regulation of water and solute content of the body fluid by kidney is known as osmoregulation.
- Charge in ionic concentration, blood volume and volume of body fluid activate the Osmoreceptor in the body.
- Excessive loss of fluid from the body activates these receptors which stimulate the hypothalamus to release antidiuretic hormone (ADH or vasopressin) from the posterior pituitary.
- ADH facilities water absorption from posterior parts of the tubules and thus prevents diuresis.
- An increase in volume of body fluid can switch off the osmoreceptors and suppress the ADH release to complete the feed back.

Micturition

- Urine formed and drained continuously by the nephrons is ultimately carried cut by the ureters by peristaltic movement and then carried into urinary bladder.
- The function of bladder is to store urine temporarily till a voluntary signal is given by central nervous system (CNS). The signal initiates the stretching of the urinary bladder as it is filled with urine.
- In response, the strech receptors on the walls of the bladder send signals to the CNS.
- The CNS passes messages to indicate the concentration of smooth muscle of the bladder and relaxation of the urethral sphincter surrounding the opening of the bladder with causes release of urine.
- Thus the release of urine from bladder is called micturition or urination and the neural mechanisms causing micturition is known as the micturation reflex.
- contraction and relaxation of the bladder are caused by impulses from the sympathetic and parasympethetic nerve fibres.
- An adult man excretes on an average 1 to 1.5 litres of urine per day.

Properties of urine : The normal urine is a light yellow coloured and slightly characteristic odour; pH : 6.0 (slightly acidic)

- ratio of substances

Urea	:	30 gms	Creatinine	:	1.2 gms
Ammonia	:	1.2 gm	Uric acid	:	1 gm

(48) Which of the following hormone increases permeability of collecting tubule ?

(A) GH (B) LH (C) ACTH (D) ADH

(49) In absence of ADH ..... .

(A) concentrated urine released (B) increase in dilute urine

(C) kidney is inactivated (D) more urea is formed

(50) Urine emptied in renal pelvis, first of all enters ..... .

(A) bellini tubule (B) urinogenital tract (C) ureter (D) urinary bladder

(51) Which of the following organ is not associated with formation or removal of nitrogenous poisonous substance ?

(A) Lungs (B) Liver (C) Skin (D) Kidney

(52) Urine formation is continuous process but removal of urine is not continuous because ..... .

(A) restricition muscles around ureter.

(B) urine is stored temporarily in bellini's duct.

(C) wall of urinary bladder is formed of striated.

(D) aperture of urinary bladder, opening in urinogenital passage is closed by restriction muscle, which relaxes under the effect of message by CNS.

(53) State pH of human urine.

(A) 7.35 (B) 4.5 (C) 5.2 (D) 6.0

(54) Contraction and relaxation of urinary bladder is by nerve impulses of which nerve fibre?

(A) Brain (B) Spinal cord

(C) Both A and B (D) Sympathetic and para sympathetic n. system

(55) Change in which of the following makes osmo - receptor centers of the body active?

(A) Change in the volume of bodyfluid (B) Amount of above

(C) Ionic concentration (D) All of above

(56) State the origin place for vassopressin hormone ..... .

(A) kidney (B) adrenal gland (C) hypothalamus (D) post pituitary gland

(57) When is ADH secretion is inhibited ?

(A) Decrease in volume of body fluid (B) Body fluid volume is normal

(C) More urea is synthesized (D) All of above

(58) State the hormone which make urine hypertonic and blood hypotonic.

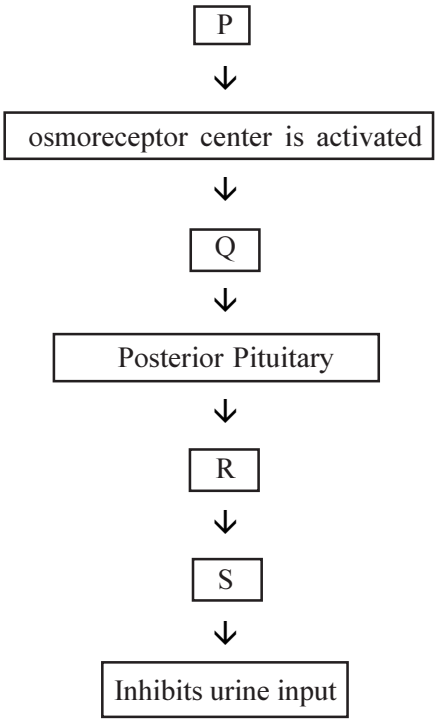
(A) ADH (B) Adrenalin (C) Glucagon (D) Sterols

59. During expulsion of urine, muscles of wall of urinary bladder ..... and passage of urinogenital tract opening into urinary bladder, muscles surrounding it shows ..... .

(A) relaxation, relaxation
(B) relaxation, contraction
(C) contraction, contraction
(D) contraction, relaxation

60. In the chart showing regulation of urine formation what is indicated by P, Q, R, S ?

Kxo :



(A) P = Increases H_2o in the body
Q = Hypothalamus
R = Inhibits secretion of ADH
S = Inhibits reabsorption from post region of tubule

(B) P = Decreases H_2o in body
Q = Hypothalamus
R = ADH secreted
S = H_2o is reabsorbed from post part of tubule

(C) P = Decreases H_2o in body
Q = Hind brain
R = Starts secretion of ADH
S = Reabsorption of H_2o from anterior region of tubule

(D) None of the above

Answers : (48-D), (49-B), (50-C), (51-A), (52-D), (53-D), (54-D), (55-D), (56-D), (57-B), (58-A), (59-D), (60-B)

Role of other organs in excretion :

- Other than kidneys, skin, lungs and liver help in the elimination of excretory wastes.

Skin :

- Many aquatic animals like starfish and Hydra excrete ammonia into surrounding water through skin by diffusion.
- In Mammals :
- Sebaceous glands discharge waxes, sterols, fatty acids and hydro carbon which is collectively known as sebum. It lubricates the hair and preventing drying up of skin and wetting of hair.
- While sweat gland secretion is watery and stimulated by increase in body temperature. The secretion consists of water, salts mainly Nacl, urea, lactic acid, little amino acids, which are discharged on the surface of skin.
- **Lungs :** About 18 litres of CO_2 and about 400 ml of water per day are removed by human lungs.

- **Saliva :** Drugs, heavy metals and probably small amounts of nitrogenous waste could be eliminated.

Disorders of the excretory system : An excessive amount of urea in the blood.

- Due to bacterial infection (nephritis) or some mechanical obstraction excretion of urea by kidney decreases.
- Due to this disease kidney can fail.

Treatment : Haemodialysis

- **Method :** In treatment of haemodialysis or artificial kidney dialyser unit of patient blood and dialysing fluid are in contact to each other.
- Patients' blood and dialyzer fluid passes to both sides through semi permeable membrane.
- By the process of osmosis from the blood of patient urea is filtered in dialiyzer fluid.
- Keep Isotonic solution approximately as dialyzer fluid and blood plasma (which is maintained).
- But percentage of urea in dialyzer fluid necessary to maintain.
- From the patient body by the blood pump from any one artery, blood is pumped out and supplied towards the dialyzer unit, in that added heparin and keep it at 0°C temperature for the cooling.
- When passes through the cellophone duct. Excretion can occur from the blood, urea, uric acid, creatinine and excessive salts come in dialyzer fluid by Osmosis and blood becomes pure.
- Now in this blood added with anti heparine and then pumped back into vein of the patient enter in patient.

A : Unit of thermoregulator,

B : Pump

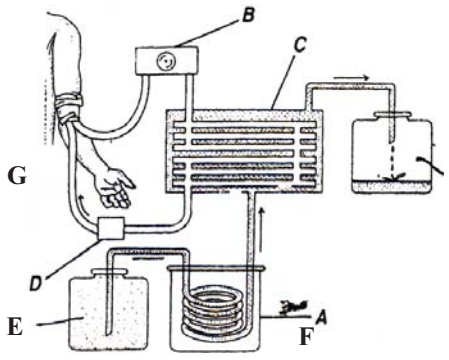
C : Dialyzer unit

D : Bubble strip

E : Dializer fluid

F : Waste fluid

G : vein



Kidney failure :

- Partial or total inability of kidney to carry excretory functions is known as kidney failure.
- **Causes:** Infection tubular injury, bacterial toxin, drug reaction etc.
- **Treatment :** Dialysis (Three times in a week)

Renal calculi :

- It blocks the kidney tubules / nephron.
- **Causes :** Stones or insoluble masses of crystalized salts formed by oxlate or uric acid precipitation formed in the kidney stone.
- **Symptoms :** Severe back pain. The stone may pass into urinary bladder and cause severe pain.
- **Treatment :** Surgery may be done to remove stone.

Nephritis :

- It is an inflammation of renal pelvis interstitial tissue and calyces. It is due to bacterial infection. Bacteria reach kidney via urethra and ureter. Inflammation effects the counter current mechanism disease. Back-pain and painful and frequent urination are its symptoms.
- Glycosurea : Presence of sugar in urine.

Oligourea : less formation of urine

Haematuria : Blood cells presence in urine.

Anuria : Very less production of urine in urine formation, thus kidney failure can occur.

Dysuria: Painful micturition.

---

(61) Substances which required to be removed, what is their amount in Dialyzing fluid ?

(A) More (B) Less

(C) Zero (D) Same as that of blood plasma

(62) Find correct option for the structure of sebum.

(A) Wax, Hydrocarbon, Fatty Acid, Uric acid

(B) Urea, Lactic acid, some Amino acids, Salt

(C) Fatty acids, Sterols, Water, Salt

(D) Wax, Sterols, Fatty acid, Hydrocarbon

(63) State the components of sweat secreted from sweat gland.

(A) Lactic acid, Amino acid, H_2O, Urea, NaCl

(B) H_2O, Urea, Amino acid, Fatty acid

(C) NaCl, Sterols, Wax, Fatty acid, Hydrocarbon

(D) Amino acid, Lactic acid, Sterols, Wax, H_2O

(64) State function of sebum ..... .

(A) hair becomes soft (B) prevents dryness of skin

(C) keeps hair moist (D) all of above

(65) State the amount of daily excreted CO_2 by lungs.

(A) 1.8 litre (B) 18 litre (C) 180 ml (D) 0.18 litre

(66) Of the following, inactivation of which organ requires haemodialysis ?

(A) Kidney (B) Urinary bladder (C) Ureter (D) Renal pelvis

(67) Which of the following component is absent in liquid used in dialyzing unit ?

(A) Glucose (B) Urea (C) Salt (D) Water

(68) More amount of urea is found during laboratory test of blood, so doctor may diagnosed which disease ?

(A) Nephritis (B) Uremia (C) Diahorrea (D) Aneuria

(69) Which crystals are responsible for stone formation in kidney ?

(A) Calcium chloride (B) NaCl (C) Calcium oxalate (D) Na_2CO_3

(70) In the last stage of haemodialysis process, before pumping blood into the body which substance is added ?

(A) Heparin (B) Histamine (C) Antihistamine (D) Antineparin

(71) Burning sensation during which of the following disease affect the process of counter current mechanism ?

(A) Uremia (B) Atherosclerosis (C) Nephritis (D) Hypertension

(72) Find improper pair.

(A) Stone - Acute pain, if enters in urinary bladder

(B) Nephritis - Frequent urine release/explusion

(C) Failure of kidney - Inability for total excretory function

(D) Uremia - Increase amount of urea in urine

(73) Presence of blood in urine is termed as ..... .

(A) ketoneuria (B) oligourea (C) haematuria (D) glycosurea

(74) What is correct for haemodialysis ?

(A) Liquid components of dialyzing fluids are similar to blood plasma but it has more amount of urea

(B) Antiheparin is mixed in pure blood.

(C) Filtered blood is allowed to enter in patient's artery through pump.

(D) Haemodialysis is blessing for the patient of kidney stone.

Answers : (61-C), (62-D), (63-A), (64-D), (65-B), (66-A), (67-B), (68-B), (69-C), (70-D), (71-C), (72-D), (73-C), (74-B)

True - False (T - F) type questions

Of the following sentences which true or false option is correct choose it.

(75) (1) Collecting tubules unite to form renal pyramids.

(2) Collecting tubules transports and synthesizes urine.

(3) Descending region of Henle's loop is formed of cuboidal epithelial cells.

(4) Ascending region of Henle's loop formed of flat epithelial cells.

(A) FTFF (B) FTTT (C) TFTT (D) FFTT

(76) Given Statement ..... .

(1) If length of Henle's loop is less urine is more concenterated.

(2) Secretion in the tubule is in PCT of uriniferous tubule.

(3) Medicinal Substances and H^+ are reabsorbed.

(4) PCT layer is formed of squamous epithelium.

(A) FFFF (B) FFTT (C) TFFT (D) TTTF

(77) Given Statement ..... .

(1) In human body ammonia from urea is formed in the liver.

(2) Aves, snail and osteichthyes are uricotelic organisms.

(3) Left kidney is slightly higher than right kidney.

(4) Layer of PCT is formed of squamous epithelium.

(A) FFFF (B) FFTT (C) TFFT (D) TTTF

(78) Given Statement ..... .

(1) Micturition is reflex action but partially controlled voluntary.

(2) Human can release hypertonic or hypotonic urine as per requirement of body.

(3) Skin helps in removal of nitrogenous excretory substances.

(4) H_2O, Na^+ and HCO_3^- reabsorbed in DCT.

(A) FTFF (B) FTTT (C) TTTT (D) FFTF

(79) Given Statement ..... .

(1) Liquid concentration 400 mosmoil^{-1} in descending capillary's outer side.

(2) Ascending capillary's has liquid concentration 600 mosmoil^{-1}, present in the middle part of renal medulla.

(3) Filterate concentration is 1200 mosmoil^{-1} in Henle's loop, towards curved region of medulla in uriniferous tubule.

(4) Filteration concentration in DCT is 300 mosmoil^{-1}.

(A) FFTT (B) TFTT (C) TFFT (D) TTTT

A - Statement, R - Reason type questions

Select answer of following questions from the options given below.

(A) A and R true, R is explanation of A.

(B) A and R true, but R is not explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

(80) Statement A : NH_3 Synthesized in the body, is to be removed in same or another form

Reason R : NH_3 is not soluble in water.

(A) (B) (C) (D)

(81) Statement A : Vasopressin increases permeability of H_2O in DCT.

Reason R : In absence of Vasopressin there is decrease in reabsorption of H_2O and dilute urine increases.

(A) (B) (C) (D)

(82) Statement A : Mammalian animals living in desert, has concenterated urine.

Reason R : Length of Henle's loop is very less in these animals.

(A) (B) (C) (D)

(83) Statement A : In human of the excretory substances are soluble in H_2O.

Reason R : Water is non-essential reactant.

(A) (B) (C) (D)

(84) Statement A : During physiological process of excretion, deamination process occurs in liver.

Reason R : Ammonia links with O_2 and converted into urea in liver.

(A) (B) (C) (D)

(85) Statement A : Water utilized for the removal of excretory substances and energy, both are in reciprocal proportion.

Reason R : Ammonia is less poisonous, its removal require less energy.

(A) (B) (C) (D)

(86) Statement A : Reabsorption is essential after microfilteration.

Reason R : Filteration through glomerulus has similar component as that of blood plasma.

(A) (B) (C) (D)

(87) Statement A : In descending region of Henle's loop urine become hypertonic, whereas in ascending region urine is hypotonic.

Reason R : Ascending region is impermeable where as descending region is impermeable for Na^+.

(A) (B) (C) (D)

(88) Statement A : Blood pressure is very high in capillaries of glomerulus.

Reason R : Blood capillary is very narrow comparing to afferent arteriole.

(A) (B) (C) (D)

(89) Statement A : Increasing osmolarity of interstitial fluid of medulla is maintained.

Reason R : Both counter current sytem are closely linked.

(A) (B) (C) (D)

(90) Statement A : Presence of urea in urine increases in uremia.

Reason R : Dialysis is essential in the patient of uremia.

(A) (B) (C) (D)

(91) Statement A : Urea is less poisonous comparing uric acid.

Reason R : Synthesis of urea utilizes less energy than uric acid.

(A) (B) (C) (D)

Answers : (75-D), (76-A), (77-B), (78-C), (79-B), (80-C), (81-B), (82-C), (83-C), (84-B), (85-C), (86-A), (87-A), (88-A), (89-A), (90-D), (91-D)

(92) Match a the proper pair.

Column - I	**Column - II**
(I) Aves	(P) uricotelic
(II) Adult amphibian	(Q) ureacotelic
(III) Aquatic insects	(R) Ammonotelic

	I	**II**	**III**
(A)	P	Q	R
(B)	P	R	Q
(C)	R	P	Q
(D)	Q	R	P

(93) Match the proper pair.

Column - I	**Column - II**
(I) PCT	(P) Reabsorption of HCO^-_3 and Na^+
(II) Ascending region of Henle's loop	(Q) Secretion of Penicillin and NH_3
(III) Descending region of Henle's loop	(R) Filtrate, is hypertonic than blood plasma
(IV) DCT	(S) Filtrate, is hypertonic than blood plasma

	I	**II**	**III**	**IV**
(A)	Q	S	R	P
(B)	Q	R	S	P
(C)	R	Q	S	P
(D)	P	Q	R	S

(94) Match the proper pair

Column - I	Column - II		I	II	III	IV	V
(I) Uremia	(P) more protein in urine	(A)	R	T	Q	S	P
(II) Haematuria	(Q) more ketone in urine	(B)	S	R	Q	T	P
(III) Ketonuria	(R) presence of blood cells in urine	(C)	Q	S	P	T	R
(IV) Glycosuria	(S) presence of glucose in urine	(D)	T	R	Q	S	P
(V) Protinuria	(T) presence of urea in blood						

(95) **Match the proper pair.**

Column - I	Column - II		I	II	III	IV	V
(I) Osmoreceptors	(P) microfilter aperture	(A)	S	R	T	P	Q
(II) Podocytes	(Q) counter current structure	(B)	Q	T	S	P	R
(III) Vasarecta	(R) dializer unit	(C)	T	P	Q	S	R
(IV) Peritubular blood capillary	(S) external excretory unit	(D)	R	S	Q	P	T
(V) Cellophane tube	(T) stimulates hypothalamus						

Answers : 92-A), (93-B), (94-D), (95-C)

Give answers, in reference to figure given below Question 96 to 98.

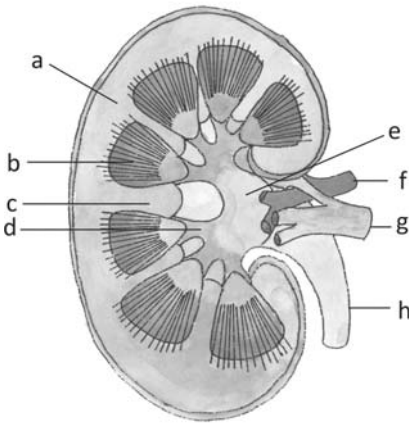


Figure 1

(96) In the given figure in part labelled as mention osmolarity L^{-1} of enetered excretory fluid.

(A) 300 (B) 600 (C) 900 (D) 1200

(97) Mention amount of excretory substance in h region of given diagram.

(A) More (B) Less (C) Nil (D) Not sure

(98) In the given diagram I contraction of which region, stores excretory fluid in temporary storage place ?

(A) e (B) f (C) g (D) h

Answers : (96-D), (97-C), (98-A)

(99) In the given diagram - 2 which labelled region at the time of urine expulsion due to neurostimulation sequentially show contraction and relaxation ?

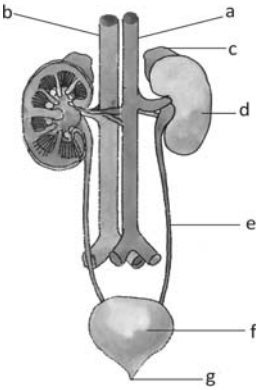


Figure 2

(A) e, f

(B) f, g

(C) g, f

(D) f, e

(100) In which labelled part of diagram-3 pressure filteration takes place ?

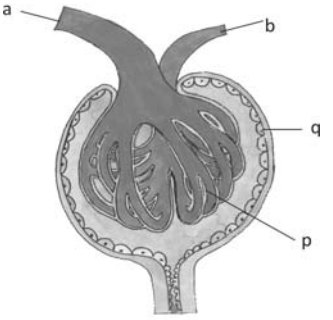


Figure 3

(A) p

(B) q

(C) r

(D) s

For a 101 to 103 select proper option with reference to diagram show below.

(101) In the given diagram-4, d labelled part formed of which of the following cells ?

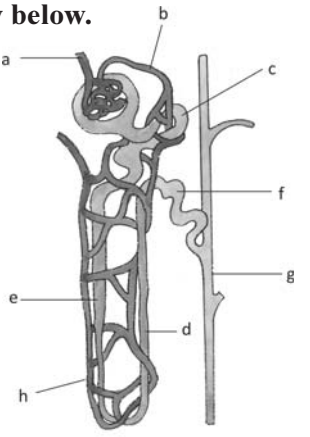


Figure 4

(A) Columnar epithelium

(B) Brush border cells

(C) Squamous epithelium

(D) Cuboidal epithelium

(102) In the given diagram - 4, mention the concentration of filterate mOsmoil^{-1} passing through labelled part f.

(A) 400 (B) 200 (C) 600 (D) 900

(103) In the above shown diagram - 4, find the correct option for process occurring in labelled part 'e'.

(A) Helps in concentration Nacl in filterate.

(B) Electrolytes spread from here into renal medulla.

(C) Filterate become isotonic to blood plasma.

(D) Here urine become hypertonic and removed in ureter.

Answers : (99-C), (100-A), (101-D), (102-B), (103-A)

Summary for NEET PMT

Water regulation:

- **Homeostatic situation :** Maintenance of stable stage.
- Homeostate process is important for normal life. Which maintain metabolic process at specific rate.
- Loss of 12 part of body fluid causes man die. By drinking water and taking water content diet water stress can be reduced or complementry.
- Kangaroo-rat loss very little amount of water because their metabolic fluid covered the 90% of water loss.
- When camel do not get water, it can go for micturition but it stored urea on basis of metabolic fluid in tissue when camel get water, approximately it drinks water in only 10 minute and do rehydrosis.

Homeostasis word proposed by walter canon.

Elimination of excretory substances :

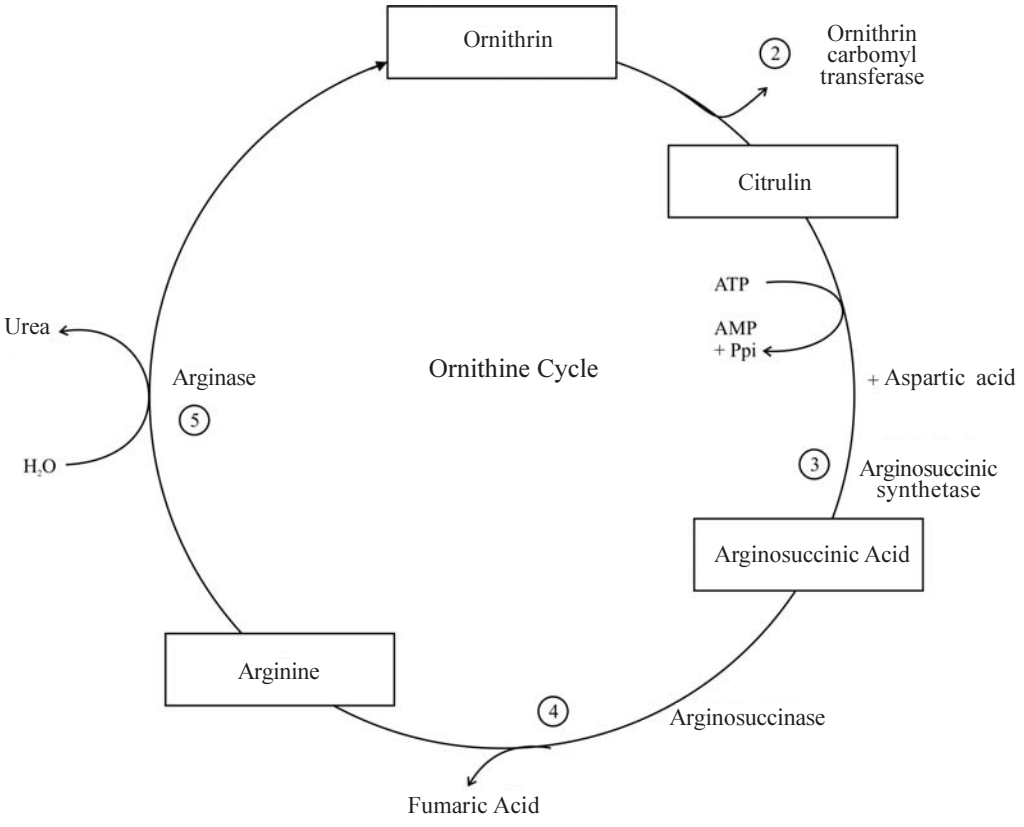
Urea is produce from liver by the ureacycle which take place in liver.

- Shark, for the maintenance of osmolarity of the omolarity equilibrium in surrounding marine water, body fluid, retains some of the urea which produced in it. So here urea act as a osmotic substances.

Ornithine Cycle :

It is also known as kreb -Henslet cycle. In this cycle two molecules of NH_3 act with the 1 molecule of CO_2 as a result urea can form.

$$NH_3 + CO_2 \xleftarrow[2\ ATP \quad 2ADP + 2P]{\text{C.P. Synthetase}} \text{Carbomyl phosphate}$$



Formation of urea in liver

Tri Methylamine Oxide (TMO) :

- In several animals by the methylalous process ammonia is converted into tri-methyl amine and oxidised and form trimethyl amine oxide which is eliminated. It is soluble in water and non-poisonous. e.g. Some marine molluscan. Crustacean, marine animals which have tail.
- **Gaunine :** In spiders ammonia is Converted in guanine. Its structure is like uric acid. It is not soluble in water. Excretion is in the form of crystal.
- **Allantoin :** Mostly mammals, converted purines and pyrimidines to allantoin. In man purines converted into uric acid form for excretion and pyrimidine excreted in form of alaline and isobutyric acid.
- **Hipuric acid :** In mammal Benzoic acid + Glycine = Hipuric acid
- In Aves Benzoic acid + Ornithin - orniturіc acid
- **Creatine :** Absence in normal urine.

  New born body, pregnant woman, feeder woman, in their urine it present. In liver, amino acid - creatine.
- **Regulation of kidney function by feedback pathways :**

(1) Regulation by ADH

(2) Regulation by JGA

Regulation by JGA :

- Multi hormone renin angiotensin alsosterone system is controlled by JGA.
- When decreases in blood volume or blood pressure in arterioles of glomerulus of JGA blood pressure and volume of blood can decrease due to release in blood stream of renin.
- Renin induce chemical process.
- Angiotensinogen is converted into Angiotensin-II protein in blood plasma which act as a hormone.
- By the angiotensin - II increases the blood pressure by the contraction of arterioles. It increases the blood pressure by two way.

(1) By the signals of PCT reabsorbs NaCl and water.

(2) Induce the adrenal grand to releases hormone aldosterone in which more Na^+ and water reabsorbs.

- These processes increases blood pressure and blood volume.
- **ANF (Atrial Natriuretic Factor):**
- When blood volume and blood bressure increases after that wall of artery of heart release ANF which keeps the enlargement of blood vessels and decreases pressure.
- ANF also stop the renin secretion from kidney.

Questions for NEET

(104) If liver is removed from person, which of the following component increases ?

(A) Uric acid (B) Urea (C) Ammonia (D) CO_2

(105) Uric acid is synthesized from where in human ?

(A) Pyrimidine (B) Purine (C) Protein (D) Pentosesugar

(106) Of the following organism, which shows green gland as excretory organ ?

(A) Spider (B) Cockroach (C) Honeybee (D) Lobster

(107) If person observe fasting for prolonged period urine has more amount of ..... .

(A) glucose (B) amino acids (C) lipid (D) katones

(108) State the net value of pressure in malpighian body during pressure filteration.

(A) 30 mm Hg (B) 20 mm Hg (C) 50 mm Hg (D) 75 mm Hg

(109) Urea is transported by ..... .

(A) RBC (B) WBC (C) Thrombocytes (D) Blood plasma

(110) In which direction excretory substances are transported from malpighian body ?

(A) Intestine (B) Rectum (C) PCT (D) DCT

(111) Who regulates secretion of Na^+ by urine ?

(A) Ant. Pituitary (B) Post Pituitary (C) Adrenal cortex (D) Adrenal medulla

(Hint - Adrenal cortex is origin for aldosterone hormone)

(112) Yellow colour of urine in vertebrate animals is due to ..... .

(A) urea (B) uric acid (C) urochrome (D) ammonia

(113) Urea is synthesized by which of the following cycle ?

(A) Kreb's cycle (B) Ornithine cycle (C) Citracin cycle (D) Arginine cycle

(114) In healthy person after transport of blood from kidney, amount of which substances remain unchanged ?

(A) Amino acids and glucose (B) Urea and glucose

(C) Urea and uric acid (D) Uric acid and amino acid

(115) Reason for Diabetes insipidus ..... .

(A) less secretion of vasopressin (B) deficiency of insulin

(C) more secretion of Insulin (D) secretion of ADH

(116) How many NH_3 molecules are required for synthesis of 1 mol. of urea ?

(A) 1 (B) 2 (C) 3 (D) 4

$$[\text{Hint}: CO_2 + 2NH_3 \rightarrow H_2 - \overset{\overset{O}{\|}}{C} - NH_3]$$

(117) Which of the following is included in RAAS ?

(A) Angiotensin (B) Aldosterone (C) Renin (D) All of above

(118) Function of Renin hormone.

(A) Dissociation (B)Decreases blood pressure

(C) Causes relaxation of blood capillaries (D)Stimulates corpus luteum

(119) Where is Angiotensin - 1 is synthesized ?

(A) Pancreas (B) Liver (C) JGA (D) Kidney

(120) For activation of angiotensinogen to angiotensin, responsible ..... .

(A) dil. HCl (B) Caesin (C) Renin (D) Hipuric acid

(121) State the synthesis of main excretory substance and its removal in human.

(A) Synthesized in kidney, removed by liver (B) Synthesized and removed in kidney

(C) Synthesized in liver and removed by kidney (D) Synthesized and removed by liver

(122) If GFR Decreases ..... .

(A) Renin is secreted from JGA (B) Aldosterone is secreted from adrenal cortex

(C) Adrenalin is secreted from adrenal medulla (D) ADH is secreted from post. pituitary

(123) Place of origin of renin hormone ..... .

(A) Renal medulla (B) Collecting tubule (C) JGA (D) Renal pelvis

(124) Which of the following is not observed in external excretory unit ?

(A) Bowman's capsule (B) DCT (C) Vasa recta (D) Renal pelvis

(125) Which of the excretory unit is affected by aldosterone ?

(A) PCT (B) DCT

(C) Bellini's tubule (D) Post part of collecting tubule

(126) Who regulates reabsorption of salts filtered from glomerulus ?

(A) Oxytocin (B) ADH (C) Glucocorticoid (D) Mineralocorticoid

(127) Marine tailed organisms, during degradation produces strong foul smell, it is due to ..... excretory substance ?

(A) ammonia (B) trimethyl amine (C) lactic acid (D) hydrogen sulphide

(128) Osteichthyes of Fresh water, osmoregulates by ..... .

(A) excreting hypotonic urine

(B) chlorine cells present in gills, excrete salts

(C) drinks water frequently

(D) excrete excretory substance in the form of uric acid

(129) Which of the following sentence is true for regulation of kidney function ?

(A) Effect of cold climate stimulates ADH secretion.

(B) When some one drinks more water ADH is not released.

(C) Increase in blood flow in glomerulus stimulates synthesis of angiotensin II.

(D) During summer when due to transpiration body loses more amount of water. ADH is not released.

(130) ..... hormone is secreted by kidney.

(A) Gastrin (B) Secretin (C) Erythropoietin (D) Aldosterone

(131) Blood capillaries entering in Bowman's capsule are called ..... .

(A) renal artery (B) renal vein (C) efferent arteriole (D) afferent arteriole

Answers : (104-C), (105-B), (106-D), (107-D), (108-B), (109-D), (110-D), (111-C), (112-C), (113-B), (114-A), (115-A), (116-B), (117-D), (118-A), (119-B), (120-C), (121-C), (122-A), (123-C), (124-C), (125-B), (126-D), (127-B), (128-A), (129-B), (130-C), (131-D)

Nitrogenous Excretion

- Depending upon the excretory product, animals show the following types of nitrogenous excretion.

Ammonotelism

- Excretion of ammonia. It is readily soluble, excreted by diffusion.
- Bony fishes, aquatic amphibians, aquatic insects.

Ureotelism

- Excretion of Urea. It is less toxic.
- Mammals, Terrestrial amphibians and Marine fishes.

Uricotelism

- Excretion of uric acid, Its crystals are non-toxic and almost insoluble in water, excreted as pellets.
- Reptiles, birds, land snails and insects.

Human Excretory System

Kidney

- Paired, bean-shaped, dark red in colour
- Two distinct Zones- (i) renal cortex, outer and darker. (ii) renal medulla, inner and lighter, it is made up of 8 to 10 conical renal pyramids.

Ureters

- Ureters are paired smooth muscle tubes.
- They open into urinary bladder and carry urine from the kidney.

Urinary Bladder

- Urinary bladder is a chamber of smooth muscles.
- It stores Urine.

Urethra

- It is the duct that conveys Urine from the bladder to be discharged to the outside of the body.
- Muscular urethral sphincters keep the Urethra closed

Regulation of Kidney

- Juxta Glomaruler Appartus (JGA) is special region formed by cellular modifications in the DCT and afferent arteriole supplying Bowman's capsule.
- JG cells contain granules of the proteolytic enzyme renin which activates renin - angiotensin- aldosterone pathway (RAAS) that loads to activation of angiotensin-II.
- Angiotensin-II increases the synthesis and release of aldesterone.
- Aldosterone causes sodium and water reabsorption by the distal part of the tubules.
- Anti Diuretic Hormone (ADH) or vasopressin from posterior pituitary facilities water reabsorption from DCT and CT.

Nephron

- It is the structural and functional unit of kidney. Each nepharon is made up of Bowman's capsules and tubules.

Bowman's Capsule

- Bowman's capsule enclose a tuff of a glomerular capillaries
- Epithalial cells called podocytes from filtration slits through which ultrafiltration is carried out
- Glemerular filtration occurs due to the pressure gradient between glomerular capillaries and Bowman's capsule.

Tubules

Proximal Convoluted Tubules (PCT)

- PCT is lined bycuboidal epithalial cells.
- About 70-80% of electrolytes and water are reabsorbed in PCT.
- It helps in maintaining pH and ionic balance
- Filtrate is isotonic.

Henle's loop

- Descending limb is permeable to water and impermeable to electrolytes. Filtrate becomes hypertonic.
- Ascending limb is impermeable to water but permeable to K^+, $C1^-$ and Na^+ Filtrate becomes hypotonic.

Distal Convoluted Tubule (DCT)

- DCT absorbs sodium chloride and other inorganic salts while retaining water.

Collecting Tubule

- Entire duct is permeable to water. Water is reabsorbed.

26 Locomotion and Movement

Types of movement and Muscular tissue and its types :

- **Simple type of movement** : In amoeba, cytosis
- **Movement in Paramoecium** : By cilia
- **Movement in Hydra** : By the tantacles
- **Movement of locomotion** : in man - By hind limbs
- **Locomotion / Movement** : The act of changing place or position by the entire body or by one part or more of its part is called movement.

Three main basic types of movements :

(1) Amoeboid : Amoeba, WBCs and macrophages in human blood, slime mold

(2) Ciliary : Paramoecium, Human trachea, in oviduct and in vasa efferentia.

(3) Muscular : Appendages of human, jaws, tongue etc.

Muscular tissue and its types :

- Origin of muscle tissue : From mesoderm
- In adult human body 40-50% muscle of total body weight.
- **Typical properties of muscle** : Ability to conduct (impulses), excitability contractibility, extensibility and elasticity.
- Three types of muscles : (1) Skeletal muscle (2) Visceral muscle and (3) Cardiac muscle.
- **Skeleton muscle :** It found as red and white muscle.
- **Location :** Head, trunk and found in appandageal region.
- In it light and dark bands are found, so it is also called striated muscle.
- Their activities are under the voluntary control of the nervous system, they are also known as voluntary muscle.
- **Visceral muscle :** In the inner walls of hollow visceral organs of the body such as alimentary canal, reproductive track and respiratory organs.
- There are involuntary and are innervated by autonomous nervous system.
- Cardiac muscle : It is only found in wall of heart.
- Every cordiac muscles are connected to each other known as intercalated disc by zigzy junctions regulated by ANS is indefatigable fibres.

(1) Living animals differ by very important characteristic from plants, is ..... .

(A) transport (B) locomotion (C) osmosis (D) respiration

(2) Which components of human body show, amoeboid locomotion ?

(A) Macrophage (B) Erythrocytes (C) Leucocytes (D) Both A and C

(3) What is true for amoeboid locomotion ?

(A) It occurs in oviduct and vasa efferentia. (B) It is for obtaining food/change of place.

(C) It occurs with the help of tentacles. (D) It is induced by circular movement.

(4) In human being ciliary movement is observed in which parts ?

(A) Digestive tract (B) Urinogenital passage

(C) Blood vessel (D) Oviduct and vasa deferens.

(5) Which characteristic of muscle helps in locomotion ?

(A) Pulsation (B) Contractility (C) Non-elasticity (D) All

(6) Coordination between which system is essential for muscular movement in higher organisms ?

(A) Skeletal system (B) Nervous system (C) A and B both (D) Digestive system

(7) Which characteristics are not observed in muscles ?

(A) Electricity (B) Durability (C) Contractility (D) Elasticity

(8) Skeletal muscles are responsible for ..... .

(A) movement (B) change in body posture

(C) locomotion of internal organs (D) A and B Both

(9) Which substances are transported in tubule of organs with the help of muscles ?

(A) striated muscle (B) cardiac muscle (C) visceral muscle (D) skeletal muscle

(10) Which muscles obtain large amount of blood supply ?

(A) visceral muscle (B) skeletal muscle (C) cardiac muslce (D) Non-striated muscles

Answers : (1-B), (2-D), (3-B), (4-D), (5-B), (6-C), (7-B), (8-D), (9-C), (10-C)

Structure of skeletal muscle, strucure of contractile proteins and Mechanism of muscle contraction.

- **Skeletal muscle :**
- It is multinucleated, cytoplasm called as sarcoplasma which have mitochondria.
- The light bands are isotropic (having same refractive index in all planes) and are known as isotropic or I-band.
- The dark bands are anisotropic (refract light differently in different plane) and are known as anisotropic or Abands.
- **Sarcomere** : The part of the myofibric between two successive Z-lines in called sarcomere. It is structural and functional unit of skeletal muscle.
- **Controctile protein :** Two types (1) Actin (2) Myosin.
- **Actin :** It is in two form (1) Monomer G-actin and polymeric F actin.
- **Tropomyosin :** Rod shaped fibrous protein. It forms two helical strands, which are wrapped around F actin.
- **Traponin :** It is a complex small globular protein which distributed at regular intervals on the tropomyosin. They are three type : TpC, TpI, TpT
- **Myosin :** Each myosin molecule has two parts.

  (1) A globular head with heavy meromyosin (HMM), which act as a atpase.

  (2) Tail - It consist of light meromysin (LMM).
- Region which is form by myosin only it is called H-region / H-line.
- Region which is form by actin only, it is called I- region / H band.
- A band - having Actin and myosin both present in this region.

Mechanism of muscle contraction :

- Initiation of muscle contraction $\rightarrow$ CNS $\rightarrow$ Motor nerve $\rightarrow$ neuromusculo junction $\rightarrow$ Acetyl choline $\rightarrow$ Potential $\rightarrow$ Ca^{+2} release in sarcoplasma $\rightarrow$ Head of myosin active open location of actin bridge formation between them $\rightarrow$ Attached actin fibers pulled toward the middle part of A band $\rightarrow$ pulled inner side z line which is also attached with the actin so $\rightarrow$ Sarcomere becomes shorten $\rightarrow$ muscle contraction by myosin release the ADP and P_i $\rightarrow$ muscle relaxation a enlargement can occur.
- **Bridge formation** : Only in absence of ATP, Break in presence of ATP.

(11) Which myofibrils are observed in I band ?

(A) Thick (B) Thin (C) A and B both (D) None

(12) Which is structural and functional unit of muscle fibre ?

(A) Myofibrils (B) sarcomere (C) sarcolemma (D) muscle fibre

(13) Thin filaments are of ..... structure.

(A) Monomer G-actin (B) Tropomyosin

(C) polymericfactin (D) A and C both

(14) Which is complex, small and globular protein ?

(A) Tropomyosin (B) Troponin (C) Meromyosin (D) F-actin

(15) Of the following, which one is globular protein ?

(A) Troponin (B) Monomer G-actin (C) A and B both (D) None

(16) Who forms bridge at the open, activated site of actin filament ?

(A) Troponin (B) Tropomyosin (C) Head of myosin (D) Tail of myosin

(17) Where is location of troponin in actin filament of striated muscle ?

(A) On Tropomyosin (B) F-actin (C) G-actin (D) all of above

(18) Which substance is secreted at neuromuscular junction when nerve impulses reach there ?

(A) Acetyl esterase (B) Acetyl choline (C) Acetic acid (D) Oxytocin

(19) Sarcomere shortens when (P) linked wiht actin is also pulled toward ..... inner side.

(A) M-line (B) H-line (C) A-line (D) Z-line

(20) Relaxation of sarcoplasm is due to decrease in concentration of ..... .

(A) Ca^{+2} (B) Mg^{+2} (C) $C1^{-}$ (D) Na^{+}

(21) Accumulation of Lactic acid in muscle fibre is due to ..... .

(A) Less activity of muscles (B) More activity of muscles

(C) Inactivation of muscles (D) Non-elasticity of muscle fibre

(22) Which action is important during muscle contraction ?

(A) Bridge formation (B) Bridge is not formed

(C) Bridge formed and stabilizes (D) Bridge formed and break-down

(23) What is improper for red muscles ?

(A) It stores more amount of CO_2 (B) It has large number of mitochondria

(C) It has more amount of myoglobin (D) Stores more O_2 and ATP formation

Answers : (11-B), (12-B), (13-D), (14-B), (15-C), (16-A), (17-C), (18-B), (19-D), (20-A), (21-B), (22-D), (23-A)

Skeleton system :

- Study of skeletal system is called Osteology.
- In human beings this system is made up of 206 bones and a few cartilages.

Skeleton System of Human being (206)

- Axial skeleton system (80)
  - Skull (29)
    - **Cranial bones (8)**
      1. Frontal bone-1
      2. Parietal bones-2
      3. Temporal bones-2
      4. Occipital condyle bone-1
      5. Ethmoid bone-1
      6. Sphenoid bone-1
    - **Facial bones (14)**
      1. Nasal bones-2
      2. Maxillae bones-2
      3. Palatine bones-2
      4. Zygomatic bones-2
      5. Lacrimal bones-2
      6. Interior nasal conchae-2
      7. Vomer bone-1
      8. Mandible bone-1
    - **Bones of middle ear (6)**
      - Malleus-2
      - Incus- 2
      - Stapes - 2
    - **Floor bone of lower jaw**
      - Hyoid bone - 1
  - Vertebral column (26)
    - Cervical vertebrae(7)
    - Thoracic vertebrae-12
    - Lumbar vertebrae-5
    - Sacral (5 part join-1) vertabrae-1
    - Coccygeal (4 partsjoin-1) vertabrae
  - Ribs (24)
    - 1 to 7 ribs (1 pair) -True ribs
    - 8 to 10 ribs - False ribs
    - 11, 12 ribs - Floating ribs
  - Sternum (1)
- Appendicular skeleton system (126)

(24) Mention the number of bones sequentially of skull, pectoral girdle, one fore limb and verebral column.
(A) 29, 2, 30, 33 (B) 29, 4, 60, 33 (C) 29, 4, 30, 26 (D) 29, 2, 30, 31

(25) Which of the following is function of vertebral column ?
(A) Induces head movement (B) Prevents movement of ribs
(C) Inhibits head movement (D) Connects the femur bone

(26) Bone and cartilage are which type of tissue ?
(A) Liquid connective tissue (B) Connective tissue proper
(C) Skeletal connective tissue (D) Simple epithelial tissue

(27) Which are paired bone in cranium ?

(A) Temporal and Ethemoid (B) Parietal and sphenoid

(C) Frontal and occipital (D) Temporal and parietal

(28) Which bones can not be seen externally on face ?

(A) Vomer (B) Nasal (C) Zygomatic (D) Lacrymal

(29) Which is only bone 'U' shaped ?

(A) Vomer (B) Maxillae (C) Mandible (D) Hyoid bone

(30) Which is pair of false ribs, sequentially in human ?

(A) 7, 8 and 9th pair (B) 8, 9, 10th pair (C) 10, 11, 12th pair (D) 11, 12th pair

(31) In the formation of ribcage which of the following are linked sequentially ?

(A) Clavicle, sternum, scapulla (B) Thoracic vertebral, femur and ribs

(C) Clavicle, sternum, ribs (D) Thoracic vertebral, sternum and ribs

(32) Which vertebrae are located on the post most part of vertebral column ?

(A) Cervical, coccyx (B) Cervical, sacrum

(C) Sacrum, coccyx (D) Sacrum, thoracic

(33) Vertebral column is connected with which part ?

(A) Skull and pelvic girdle (B) Only skull (C) Ribs (D) Both A and C

(34) Sacrum is present in which structure in human ?

(A) Pelvic girdle (B) Vertebral column (C) Pectoral girdle (D) Fore limb

(35) Cumber curve of vertebral column is between ..... vertebrae.

(A) 20 to 24 (B) 8 to 19 (C) 1 to 7 (D) 25 to 29

(36) Thoracic curve of vertebral column is between ..... vertebrae.

(A) 1 to 7 (B) 8 to 19 (C) 20 to 24 (D) 25 to 29

37) Which cells are formed in the bone marrow of hollow bone ?

(A) Muscle cells (B) Stem cells (C) Nerve cells (D) Collar cells

(38) Smallest bone, in human is located at ..... .

(A) Vertebral column (B) Carpals (C) Phalanges (D) Middle ear

(39) Smallest bone in human - ..... .

(A) Incus (B) Malleus (C) Stapes (D) Patella

(40) What is number of bones in axial skeleton of trunk region ?

(A) 50 (B) 80 (C) 29 (D) 51

(41) Which of the following is connected with ribs ?

(A) T_2 (B) S_4 (C) L_5 (D) L_4

(42) Which bone is unpaired in face ?

(A) Nasal (B) Palatine (C) Vomerbone (D) Lacrimal

(43) What is the number of floating ribs in human ?

(A) 2 pair (B) 5 pair (C) 6 pair (D) 3 pair

(44) What is total no. of vertebrae in human ?

(A) 35 (B) 40 (C) 30 (D) 33

(45) Spinal cord is connected with brain by ..... .

(A) Fovea ovalis (B) Foramen of magnum

(C) Large aperture (D) Magendils aperture

(46) Which is helpful in human for neck movement ?

(A) Atlas vertebrae (B) sacral vertebrae (C) lumbar vertebrae (D) Axis

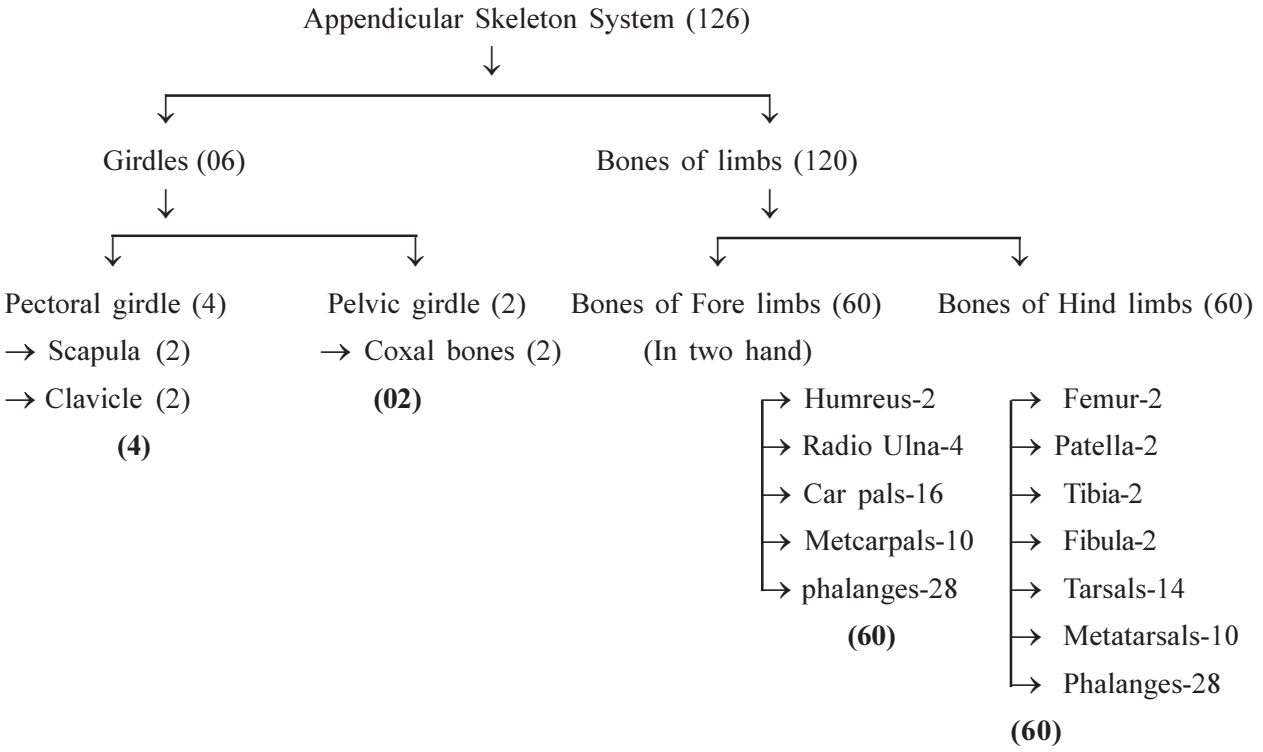
(47) Who possess 7 pairs of true ribs ?

(A) Frog (B) Human (C) Both A and B (D) None

Answers : (24-C), (25-A), (26-C), (27-D), (28-A), (29-D), (30-B), (31-D), (32-C), (33-D), (34-B), (35-A), (36-B), (37-B), (38-D), (39-A), (40-D), (41-A), (42-C), (43-A), (44-D), (45-B), (46-D), (47-B)

Appendicular Skeleton System : Joints and its types :

- Appendicular skeleton system consists of girdles and bones of fore and hind limbs.
- Girdles are attached with axial and appendicular skeleton systems.



All types of joins are essential for locomotion / movement.

(1) Fibrous - Fixed - immovable joints :- Occur between the bones of the cranium.

(2) **Cartilaginous of Slightly movable joints:** Found between the centre of vertebrae, at the pubic symphysis. Joint (cartilage and other) of ribs.

(3) **Freely movable joint / synovial joints :** It is full of fluid in cavity (Synovial).

Disorders of Skeleton system :

(1) **Myasthenia gravis :** It is an auto immune disorder that affects neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

(2) **Tetany :** It is a muscular disorder in which rapid spasms in muscle occur due to lesser Ca^{+2} in the body fluid.

(48) Which is the longest bone in human skeletal system ?

(A) Humerus (B) Femur (C) Tibia (D) Axial bone

(49) Metatarsal of ankle is present in ..... .

(A) Hind limb of Frog (B) Fore limb of rabbit

(C) Fore limb of human (D) Hind limb of human

(50) Scapula is part of ..... .

(A) Pelvic girdle (B) Skull (C) Vertebral column (D) Pectoral girdle

(51) In each hind limb of human, what are the number of tarsals ?

(A) 6 (B) 8 (C) 7 (D) 2

(52) Number of scapula in pectoral girdle ..... .

(A) 1 (B) 2 (C) 3 (D) 4

(53) Wrist joint is of ..... type.

(A) pivotal joint (B) synovide joint (C) hinge joint (D) ball and socket joint

(54) Synovial fluid is secreted by ..... .

(A) bone (B) blood (C) cartilage (D) synovial membrane

(55) Which type of joint is found between numerous and radio-ulna bone ?

(A) Synovial joint (B) Hinge joint (C) Pivotal joint (D) Cartelagenous joint

(56) Which type of joint is found in cranium ?

(A) Movable (B) Partially movable (C) Suture (D) None

(57) What is hinge joint ?

(A) Wrist (B) Between metacarpals (C) Ankle (D) All of above

(58) What is pivotal joint ?

(A) Shoulder joint (B) Atlas - Axis Joint (C) Pelvic joint (D) Thumb joint

(59) Number of Ischium bone in pelvic girdle.

(A) 2 (B) 1 (C) 4 (D) 3

Answers : (48-B), (49-D), (50-D), (51-C), (52-B), (53-C), (54-D), (55-B), (56-C), (57-D), (58-B), (59-A)

A = statement R = Reason, type questions :

Select answers of following questions from options given below.

(A) A and R both correct. R is explanation of A.

(B) A and R both correct but R is not explanation of A.

(C) A is correct, R false.

(D) A is false, R correct.

(60) Statement A : Change of place and obtaining food is through pseudopodia in Amoeba.

Reason R : Pseudopodia is formed due to activation of cilia.

(A) (B) (C) (D)

(61) Statement A : Innervation by nerves in cardiac muscle is by autonomous nervous system.
Reason R : Cardiac muscles are present in the wall of the heart.
(A) (B) (C) (D)

(62) Statement A : H region has only thin filaments.
Reason R : A band has thick and thin Filaments on both side.
(A) (B) (C) (D)

(63) Statement A : Double hellicle chains of tropomyosin is innervated surrounding to G-actin.
Reason R : In myosin filament Head is formed of heavy meromysin and tail is light meromyosin.
(A) (B) (C) (D)

(64) Statement A : Calcium activates inter effect of actin and myosin.
Reason R : Sarcomere, is shortened as Z line is pulled in side, attached with myosin.
(A) (B) (C) (D)

(65) Statement A : Striated muscles get fatigued due to long time activation.
Reason R : Due to anaerobic decomposition of glycogen, adds lactic acid in muscles.
(A) (B) (C) (D)

(66) Statement A : In voluntary muscles, dark and light transverse lines are seen.
Reason R : Voluntary muscles are also known as skeletal muscles.
(A) (B) (C) (D)

(67) Statement A : Active electric potential is generated in sarcomere.
Reason R : Head of myosin, forms bridge with open active place of actin.
(A) (B) (C) (D)

(68) Statement A : Coccyx is formed by 4 tail vertebrae.
Reason R : It is found in vestigeal spinal cord.
(A) (B) (C) (D)

(69) Statement A : Dark band of sarcomere is A disc.
Reason R : A band, reflects light at various angles, in different plane.
(A) (B) (C) (D)

(70) Statement A : Hinge joint is found between first and second vertebrae.
Reason R : Synovial joint is found between vertebrae.
(A) (B) (C) (D)

(71) Statement A : Frontal, parietal and occipital bones are joined by suture joint.
Reason R : Bones of skull are joined by cortilagenous joint.
(A) (B) (C) (D)

(72) Statement A : During muscle contraction I band shortens.
Reason R : During muscle contraction myosin fibre contracts.
(A) (B) (C) (D)

(73) Statement A : In skull only upper jaw is movable.

Reason R : Lower jaw is attached by muscles with head region.

(A) (B) (C) (D)

Answers : (60-C), (61-B), (62-D), (63-D), (64-C), (65-A), (66-B), (67-D), (68-C), (69-A), (70-B), (71-C), (72-C), (73-B)

True / False type questions

In the given sentences, select correct option for true / false.

(74) For human skeletal system select proper option for true/ false sentence.

(1) Human skull is formed of 22 bones.

(2) Human skeletal system is formed of 206 bones.

(3) In adult human vertebral column is formed by 31 vertebrae.

(4) Appendicular skeletal, in human is formed of 120 bones.

(A) TTFF (B) TFTF (C) FFTT (D) FTFT

(75) Select proper option, for the given statement, whether T / F.

(1) All organisms show movement.

(2) All movements are locomotion.

(3) Cardiac muscle are innervated by nerves of voluntary nervous system.

(4) All organisms show locomotion.

(A) TTFF (B) FTFT (C) FTTF (D) TFTF

(76) Select proper option, for the given statement, whether T / F.

(1) In the middle of I disc, line present is called M line.

(2) Myosin fibres are thick muscle fibres.

(3) I disc, shortens during muscle contraction.

(4) Actin, acts as ATpase in muscle contraction.

(A) TFFT (B) FTFT (C) FTTF (D) FFTT

(77) Which option is correct in human skeletal system for number of Axial skeletal system.

(1) Human skull is formed of 29 bones.

(2) Pectoral girdle and pelvic girdle are included in skeletal system.

(3) Sternum, thoracic vertebrae and ribs form ribcage.

(4) Humerus, nasal, vomer and palatine are bones of face.

(A) TFTF (B) TTFF (C) FTFT (D) FFTT

(78) Given statements are (T) or (F), select proper option for it.

(1) In cardiac muscles, specific structure intercalated disc is present.

2) 8, 9, 10 ribs are called false ribs.

(3) Palatine forms the floor of mouth.

(4) 11th and 12th ribs are joined with sternum bone.

(A) TTTT (B) TTFF (C) FTTF (D) TFTF

(79) Given statements are (T) or (F), select proper option for it.

(1) Ball and socket joints are most movable joint.

(2) Bones of skulls are joined by cartilagenous joint.

(3) Fore limbs are connected with axial skeletal system by pelvic girdle.

(4) Muscle contraction is reaction between actin and myosin.

(A) TTFF (B) FFTT (C) FTFT (D) TFFT

80) Select proper option for (T) or (F) for the given statements.

(1) Paramoecium shows amoeboid movement.

2) Clavicle is bone of pelvic girdle, which has two curves.

(3) Sternum is longest bone of hind limb.

(4) Cardiac fluid is present in synovial joint.

(A) FTFT (B) TTTT (C) FFFF (D) TFTF

(81) Given Statements are T or F, select proper option for it.

(1) In skull, cranium has 2 paired and 4 unpaired bones.

(2) In appendicular skeletal system, 126 bones are present.

(3) Between Atlas and Axis, hinge joint is observed.

(4) Between Humerus and glenoid cavity suture type of joint is present.

(A) TTFF (B) TFTF (C) FFTT (D) TFFT

Answers : (74-D), (75-B), (76-C), (77-A), (78-B), (79-D), (80-C), (81-A)

(82) Select proper option for column-I and column-II.

Column - I	**Column - II**	
(I) WBC	(P) Intercalated disc	(A) I-R, II-Q, III- P, IV-S
(II) Visceral muscles	(Q) disc/band	(B) I-R, II-S, III- P, IV- Q
(III) Cardiac muscles	(R) Amoeboid movement	(C) I-R, II-S, III- Q, IV-P
(IV) Striated muscles	(S) Non-striated muscles	(D) I-P, II-S, III- Q, IV- R

(83) Select proper option for column-I and column-II

Column - I	**Column - II**	
(I) Troponin	(P) Rod shaped fibrillar protein	(A) I-Q, II- S, III- P, IV-R
(II) G- actin	(Q) Globular protein	(B) I-Q, II-P, III- S, IV- R
(III) Tropo myosin	(R) Polymeric protein	(C) I-Q, II-S, III- R, IV-P
(IV) F-actin	(S) Monomeric protein	(D) I-P, II-Q, III- S, IV- R

(84) Match column-I with column-II. Select proper options.

Column - I	**Column - II**	
(I) Hyoid bone	(P) Bone of cranium	(A) I-P, II-Q, III- S, IV-R
(II) Ethemoid	(Q) Bone of floor of buccal cavity	(B) I-Q, II-P, III- R, IV- S
(III) Lacrimal	(R) Bone of pectoral girdle	(C) I-R, II-P, III- Q, IV-S
(IV) Clavicle	(S) Bone of Face	(D) I-Q, II-P, III- S, IV- R

(85) Match column-I with column-II. Select proper options.

Column - I	**Column - II**	
(I) Tarsals	(P) 8	(A) I-R, II-P, III- Q, IV-S
(II) Carpals	(Q) 24	(B) I-P, II-Q, III- R, IV- S
(III) Vertebrae	(R) 7	(C) I-R, II-P, III- S, IV-Q
(IV) Ribs	(S) 26	(D) I-R, II-Q, III- S, IV- P

86 Match column-I with column-II. Select proper options.

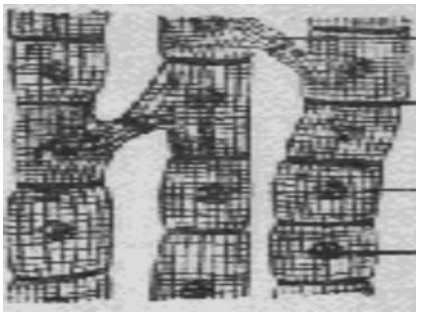
Column - I	**Column - II**	
(I) Pectoral girdle	(P) Coxal bone	(A) I-R, II-S, III- P, IV-Q
(II) Clavicle	(Q) Four curves	(B) I-P, II-S, III- R, IV- Q
(III) Pelvic girdle	(R) Acromion process	(C) I-R, II-S, III- Q, IV-P
(IV) Vertebral column	(S) Two curves	(D) I-Q, II-R, III- S, IV- P

87 Match column-I with column-II. Select proper options.

Column - I	**Column - II**	
(I) Cranium	(P) 37 bones	(A) I-S, II-P, III- R, IV-T, V-Q
(II) Calcium	(Q) 5 metacarpals	(B) I-S, II-T, III- R, IV- P, V-Q
(III) Ribcage	(R) require for muscle contraction	(C) I-S, II-R, III- P, IV-T, V-Q
(IV) Vertebral column	(S) 8 Bones	(D) I-S, II-R, III- T, IV- Q, V-P
(V) Palm	(T) 26 vertebrae	

Answers : (82-B), (83-A), (84-D), (85-C), (86-A), (87-C)

(88) Which is given Figure I ?



(A) Visceral muscle

(B) Unstriated muscle

(C) Cardiac muscle

(D) Striated muscle

(89) What is true for muscle shown in Fig. I

(A) They do not fatigue

(B) They function slowly

(C) They fatigued speedily

(D) None of these

(90) Which is Diagram 2 ?

(A) Ball and socket joint

(B) Pivot joint

(C) Hinge joint

(D) None

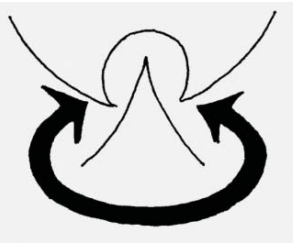


Figure 2

(91) Joint shown in Fig. 2 is seen in ..... .

(A) Forearm and wrist (B) Atals and axis

(C) Sternum and ribs (C) Humerus and glenoid cavity

(92) In the given Fig 3. P indicates ?

(A) Structure of H region

(B) Arrangement of myosin.

(C) Arrangement of actin

(D) B and C Both

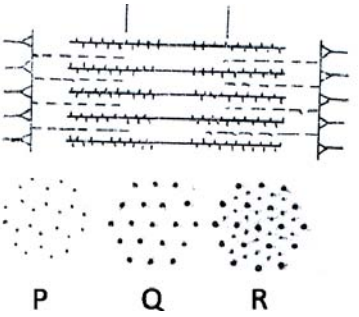


Figure 3

(93) In the given Fig.3 Q indicates which part ?

(A) Arrangement of myosin (B) Arrangement of actin

(C) A and C Both (D) Structure of I disc

(94) What is shown in Fig. 4 ?

(A) Pivot joint (B) Hinge joint (C) Ball and socket joint (D) Cartilagenous joint

(95) Joint shown in Fig. 4 is seen between ..... .

(A) Femur and Acetabulum

(B) Humerus and Radio-ulna

(C) Humerus and glenoid cavity

(D) A and C Both

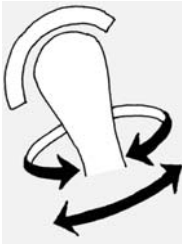


Figure 4

(96) N Part indicated in Fig. 5 suggests ..... .

(A) Linkage of Actin

(B) ATP linkage

(C) Ca^{+} linkage

(D) Mg^{+2} linkage

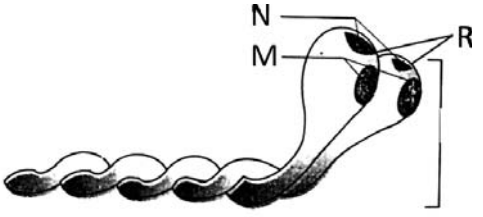


Figure 5

(97) M part shown in Fig.5 indicates ..... .

(A) ADP linkage (B) Ca^{+2} linkage (C) Actin linkage (D) ATP linkage

(98) R part shown in Fig.5 indicates ..... .

(A) LMM (B) HMM (C) ATPase (D) MHH

(99) P indicates what in Fig. 6

(A) TpT

(B) TpL

(C) TpC

(D) All of above

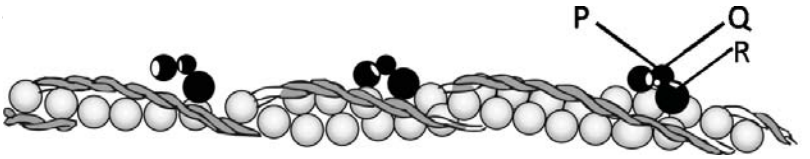


Figure 6

(100) In the given fig. 6 what is indicated by 'Q' ?

(A) TpT (B) Tpl (C) TpM (D) TpC

(101) In the given Fig. 6 what is indicated by 'R' ?

(A) TpT (B) TpC (C) TpI (D) TpM

Answer : (88-C), (89-A), (90-C), (91-B), (92-C), (93-A), (94-C), (95-D), (96-A), (97-D) (98-B), (99-C), (100-B), (101-A)

Questions for NEET :

(102) In muscle fibre Ca^{+2} is stored in ..... .

(A) sarcoplasm (B) sarcosome (C) sarcomere (D) sarcolemma

(103) Cardiac muscle is ..... .

(A) striated, voluntary (B) striated, involuntary

(C) non-striated, voluntary (D) involuntary, non striated

(104) During resting phase, middle of thick filament not innervated by thin filament is ..... .

(A) A - disc (B) M-line (C) H-zone (D) I-disc

(105) Where is ciliary movement observed ?

(A) Macrophage and WBC (B) Oviduct and blood capillary

(C) Oviduct and Trachae (D) Tongue and appandages

(106) Where is amoeboid movement observed ?

(A) Appandages and jaws (B) Macrophage and WBC

(C) Trachae, oviduct (D) RBC, WBC

(107) During muscle contraction ..... .

(A) thin filaments slide over thick filament (B) thick Filaments slide over thin filament

(C) both filaments slide over each other (D) filaments do not slide during contraction

(108) Which character is not linked with white muscle fibre ?

(A) Less amount of myoglobin (B) Less amount of mitochondria

(C) Less amount of sarcoplasmic fibres (D) Depend on anaerobic process for energy

(109) Which point is correct for process of visceral muscle ?

(A) They form the wall of hollow visceral organs. (B) They do not show any bands.

(C) Their contraction is under control of involuntary nervous system. (D) All of above

(110) Which muscle, during activation is coenocytic ?

(A) Cardiac muscle (B) Smooth muscle (C) Voluntary muscle (D) All of above

(111) Elastic fiber in the middle of I disc, divides it into two parts, is ..... .

(A) H-zone (B) M-line (C) Z-line (D) A-band

(112) Motor neuron with nerve fibre forming ..... connects it.
(A) Neuro mucular junction (B) Motor and disc
(C) Motor unit (D) A and B Both

(113) Which type of joints are found in bones of cranium ?
(A) Fibrillar (B) Hinge (C) Synovial (D) None

(114) Coxal bone is formed by ..... .
(A) Ilium (B) Ischium (C) Pubis (D) All of above

(115) Which is contractile protein of muscle ?
(A) Myosin (B) Tropomyosin (C) Tropoin (D) Tubulin

(116) Which of the following pair is proper ?
(A) Hinge joint - Between vertebrae
(B) Sliding joint - Zygapophysis of sequential vertebrae
(C) Cartilagenous joint - Bones of skull
(D) Fibrillar joint - Between phalanges

(117) Which is correct, in human body, anatonically ?
(A) Collar Bone - 3 pair (B) Salivary gland - 1 pair
(C) Cranial nerves - 10 pair (D) Floating ribs - 2 pair

(118) Of the following, one shows its correct number.
(A) Floating ribs - 4 (B) Amino acid obtained from protein - 16
(C) Types of Diabetes - 3 (D) Cervical vertebral in human-8

(119) Wrist joint is example of ..... .
(A) Hinge (B) Sliding (C) Ball and socket (D) Pivotal

(120) For human skeletal system, Find the mismatch.
(A) Sternum and ribs - axial skeleton (B) Clavicle and glenoid cavity - pectoral girdle
(C) Humerus - ulna hind - limb (D) Malleus and stape - ear

(121) During muscle contraction, energy is provided by ..... .
(A) AMP (B) Glucose (C) ATP (D) Acetyl Co-A

(122) Which of the following structure is formed of single bone, in human ?
(A) Upper jaw (B) Zygomatic process (C) Lower jaw (D) Rib cage

(123) What is the form of actin and myosin during muscle contraction ?
(A) Myoplasm (B) Sarcoplasm (C) Plastocine (D) Ectoplasm

(124) What is called part of cyloplasm in striated muscle ?
(A) Sarcomere (B) Sarcoplasm (C) Neuron (D) Large segment

(125) In striated muscle, contractile substance present between two successive z disc is ..... .
(A) sarcomere (B) sarcoplasm (C) myofilament (D) all

(126) Shoudler blade is formed of ..... .
(A) ilium (B) humerus (C) clavicle (D) scapula

(127) Player follows speedily ball in the game of cricket. Which of the following bonegroup participate in it ?
(A) Femur, incus, tibio, metacarpals (B) Tarsals, femur, metatarsals, tibia
(C) Sternum, femur, tibia, fibula (D) Tarsal, pelvic girdle, ulna, tibia

(128) Which is the bone of forelimb ?

(A) Humerus (B) Femur (C) Tibia (D) Fibula

(129) Which is cup shaped cavity on the head of femur bone, for articulation ?

(A) Glenoid cavity (B) Acetabulum (C) Sigmoid notch (D) Femur aperture

(130) What is present in ball and socket joint to reduce friction ?

(A) Coelomic fluid (B) Synovial fluid (C) Pericardial fluid (D) Mucous

(131) Contractile protein of striated muscle, which links with ATPase action process.

(A) Troponin (B) Tropomysin (C) Myosin (D) Actin

(132) Which is largest synovial joint ?

(A) Thigh (B) Knee (C) Shoulder (D) Ankle

(133) Where is neuromotor connection observed ?

(A) Between nerve cell and sarcolemma of muscle (B) Between two nerve cells

(C) Between two muscles (D) B and C Both

(134) Where does synovial fluid is formed ?

(A) Blood (B) Cartilage (C) Bones (D) Synovial membrane

(135) Patella is linked with ..... .

(A) wrist (B) knee (C) neck (D) ankle

(136) Which of the following is associated with appendicular skeleton ?

(A) Bones of cranium (B) Bones of vertebral column

(C) Ribs (D) Bones of fore limb and hind limb

(137) By which vertebral column is linked with pelvic girdle ?

(A) Coccyx (B) Sacrum (C) Lumbar vertebral (D) Cervical vertebrae

(138) Shoulder joint is example of ..... .

(A) Pivotal joint (B) Hinge joint (C) Sliding joint (D) Ball and socket joint

(139) ..... is joint between ribs and sternum.

(A) Cartilagenous joint (B) Curved joint (C) Fibrillar joint (D) Sliding joint

(140) Terminal end of long bones are covered by ..... .

(A) Ligament (B) Bony joint (C) Cartilage (D) Muscle joint

(141) Sarcomere is region between ..... .

(A) two I - disc (B) A and I disc (C) Two Z-disc (D) Z and A disc

(142) Lumbar vertebrae are situated ..... .

(A) neck region (B) abdominal region (C) hip region (D) thoracic region

Answers :(102-A), (103-D), (104-C), (105-C), (106-B), (107-A), (108-C), (109-D), (110-C), (111-C), (112-C), (113-A), (114-D), (115-A), (116-B), (117-D), (118-A), (119-A), (120-C), (121-C), (122-C), (123-B), (124-B), (125-C), (126-D), (127-B), (128-A), (129-B), (130-B), (131-C), (132-B), (133-A), (134-D), (135-B), (136-D), (137-B), (138-D), (139-A), (140-C), (141-C), (142-C)

Locomotion and movement

Skelatal Muscle
- Attached to bones, voluntary, exhibit transverse strips hence called striated.

Sarcomere
- Functional unit of contraction. They are delivered by a very thin and comparitively dense Z-line.

Primary Myofilament
- It is formed of myosin protein
- Myosin (thick) filament is a polymerized protein of many monomers.
- Myosin monomer has head and short arm which project outwords at regular distance and angle from myosin filament.
- Head has ATPase and binding site for actin.

Secondary Myofilament
- It is composed of actin and regulatory proteins tropomyosin and tronin.
- In resting state, subunit of troponin masks the active binding sites for myosin on the actin (thin) filament.

Sliding filament theory of muscle contraction

Neural Signal → Ach released → Action potentinal generated → Ca^{+2} released → Binds to troponin → Change in shape exposes active sites → Myosin - actin cross bridges are formed → Myosin ATPase in presence of Ca^{2+} and Mg^{2+} breaks ATP to ADP → Energized myosin cross bridges bind to actin and move → Thin myofilament slide along the thick myofilament

Skeletal System

Each limb contains 30 bones
- Fore limb : 1 humerus
  1 radius, 1 ulna, 8 carpals
  5 metacarpals, 14 Phalanges
- Hind limb : 1 femur,
  1 tibia, 1 fibula, 1 patella,
  7 tarsals, 5 metatarsals
  and 14 Phalanges

Girdles
- Pectoral girdles (4 bones)
- Pelvic girdle (2 bones)
  articulation between axial and appendicular skeletal components

Joints
- Fibrous / immovable joint
  Eq : Joints of skill bones
- Cartilagenous / partially movable joints
  Eq : Joints between vertabrae
- Synovial / freely movable joints
  Eq : Ball and socket joints hinge joints

Skeletal System
Out of total 206 bones of human, 6 bones are present in ears 3 bones in each ear as ear ossicles and remaining are distributed into axial and appendicular skeleton

Axial Skeleton - 80 bones

Skull
- Skull consists of cranium and facial bones
- 22 total bones ; out of these 8 are bones of cranium and 14 are facial bones

Foramen Magnum
- Skull possesses a posterior aperture called foramen magnum.
- Through which the brain communication with spinal cord.

Ribs
- Ribs joins with strenum and form a rib-cage 12 pairs of ribs 1 to 7 pairs are true ribs; 8, 9, 10 pairs are false ribs 11, 12 pairs are floating ribs.

Sternum
- Single flat bone located at mid- ventral region of theracic cavity

Vertabral column
- formed by union of 26 small bones called vertabrae

7 - Cervical vertabrae
12 - Thoracic vertabrae
5 - Lumbar vertabrae
1 - Sacral vertabrae (5 bones join to from 1 bones)
1 - Coccygeal bone (4 bones join to from 1 bone)

27 Organism and Population

- Ecology word derived from Greek words.

  Oikos (home = habitation = a place to live in) Logos (= study)
- Odum (1969) : ''The Study of the interrelationships between organisms and environment.''

  Branches : Animal Ecology, Plant Ecology, Habitat Ecology, Marine Ecology, Freshwater Ecology, Terrestrial Ecology, Population Ecology, Community Ecology, Applied Ecology, Human Ecology.

  Environment
- Ecology is a part of environment study.
- The science of environment is called environmental studies.

  Defination:

  'The study of basic components of our surroundings and their interactions is called **environmental studies.**''

---

(1) Who gave the defination of Ecology ?

(A) Odum - 1960 (B) Odum - 1969 (C) Tansely - 1935 (D) Tansely - 1965

(2) The study of basic components of our surroundings and their interactions is called ..... .

(A) Ecology (B) Biotic Community (C) Ecosystem (D) Environmental Studies

Answer : (1-B), (2-D)

Habitat : 'Place Where Organisms live'

(1) Four groups of habitat

– Fresh water habitat : **Features:** salinity, osmosis, pH, dissolved gases, temperature

- eg : Ponels, lakes, rivers

(2) Marine Habitat:

– It is the largest habitat in the biosphere.

– It covers an area of 3,62,000,000 sq.kms.

– It is nearly 71% of the earth's surface.

– **Features :** salinity, osmosis, pH, currents, tides, temperature, light

(3) Estuary Habitat

– Water body where river water mixes with sea water.

– Salinity is less here.

(4) Terrestrial Habitat :

– It is land area.

– It is subdivided into a number of sub-unit called biomes.

– Biome is defined as a major terrestrial community characterised by distinctive plants and animals. eg: forest biome, grassland biome, desert biome.

(3) It is the largest habitat of biosphere.

(A) Estuary (B) Terrestrial habitat (C) Marine habitat (D) All given

(4) Marine habitat occupies ..... of area of the earth's surface.

(A) 17 % (B) 71 % (C) 75 % (D) 80 %

(5) Marine habitat covers ..... area of the earth's surface.

(A) 36,20,000 sq.Kms. (B) 3,6,20,000 sq.kms.

(C) 36,20,00, 000 sq.kms. (D) 3,62,00,000 sq.kms.

(6) A waterbody where river water mixes with the sea water.

(A) Estuary (B) Mangroove (C) Creek (D) All given

(7) Osmoregulation is the major problem to living fauna of which habitat?

(A) Fresh water habitat (B) Marine water habitat

(C) Estuary habitat (D) All given

Answer : (3-C), (4-B), (5-C), (6-A), (7-D)

Basic concept of ecology

- **Series of levels of biological organisation:** Macromolecules → cells → tissues → organs → individual organisms → population → communities, ecosystems → biomes
- **Population :** "A group of individual organisms of the same species in a given area."
- **Biotic community :** "A group of populations of different species in a given area."
- **Ecosystem :** "It is a whole biotic community with it's abiotic environment in a given area."

(8) What does an area called that covers population of more than one species?

(A) Biotic community (B) Population (C) Biomes (D) Ecosystem

Answer : (8-A)

Major Abiotic Factors:

- **Environment :** "All the external factors, conditions and influences that affect an organism or a biological community."
- **Environmental Factor :** "Any external force, substance or condition, which surrounds and effects the life of an organism in any way becomes a factor of its environment.
- **Major abiotic factors :** Light, temperature, water, soil.
- **Light :**
- The Sun is the important source of energy.
- It is formed of electromagnetic waves.
- It is a form of energy called radiant energy.

(1) Light in relation to plants :

→ Chlorophyll production, photosynthesis, transpiration, distribution of plants etc. directly or indirectly affected by light.

→ Indirectly light affects transpiration rates through increase in temperature which correspondingly affect water absorption also.

→ Opening and closing of stomata is regulated by light.

→ Total amount of radiation received by the earth's surface differs with latitude.

(2) Light in relation to animals :

→ The intensity of light influences metabolic rate in animals.

→ Increased intensity of light results in increased enzyme activity.

→ Light induces photochemical reactions, because of that cave animals are colourless.

→ Light initiates breeding activities in birds.

→ In some birds gonads become active with increased light intensity during summer.

Temperature :

All metabolic processes and chemical reactions are controlled by temperature.

(1) Temperature in relation to plants : It effects the rate of transpiration, photosynthesis, respiration in plants.

→ It is important factor in phenology of plants and affects distribution of plants.

→ The division of earth's vegetation are dependent on variation in temperature.

→ Altitudinal zonation of vegetation is clearly seen on mountains.

(2) Temperature in relation to animals:

→ In rotifers and daphnids like animals, sex-ratio is effected by temperature.

→ Under normal condition daphnides give parthenogenetic eggs that develop into females.

→ With increase in temperature they give sexual eggs which after fertilization develop either into males or females.

→ Some insects, birds and mammals in warm climatic bear dark pigments than the races of same species present in cool and dry climate.

→ Birds and Animals obtain greater body size in cold regions than in warm areas.

→ Animals are two types with respect to maintenance their body temperature.

(i) Homoeothermic or warm blooded animals - birds, mammals

(ii) Poikilothermic or cold blooded animals - reptiles, fishes, amphibians.

Water :

- It is the universal solvent.
- It is the largest medium of life.
- It exists in three forms : liquid, solid and vapour.
- Liquid state exists in two forms: sea water and fresh water.
- The special properties of water : solvancy, specific heat, latent heat, thermal conductivity, viscosity surface tension, buoyancy, salinity, pressure, transparency, pH etc.
- Temperature variation is much lesser in aquatic habitat than in terrestrial one.
- Temperature variation in fresh water habitat is higher than marine habitat.
- In deep fresh water habitat (lakes, ponds), there is gradual decrease in temperature from the surface to bottom. As a result, different layer of waters with different temperatures are noticed. This is called thermal stratification.

Water balance and adaptations :

(1) In fresh - water :

→ Salinity is very low.

→ Body fluid of fresh water animal is highly concentrated (hypertonic).

→ Fresh water habitat is hypotonic.

→ As a result endosmosis occurs.

→ So water enters into the body constantly.

→ Animal removes excess of water in the form of urine by kidney.

→ Loss of salt is compensated by absorption of salts from fresh water by chloride cells present in the gills.

→ Crustacean Astacus removes excess water in form of urine by green glands.

→ In Amoeba it is maintained by contractile vacuoles.

→ Certain fresh water fishes develop accessory respiratory organs to utilise atmospheric air when fresh water dries up.

→ eg. Ophiocephalus, Clarias, Anabus.

(2) In sea water :

→ Sea water has high salt contents.

→ Sea water is hypertonic, body fluid is hypotonic.

→ As a result exosmosis occurs.

→ Body tends to lose water.

→ To compensate for the loss of water marine animals drink sea water.

→ Excess salt is exuded by chlorine secretory cells present in gills.

→ In marine turtles excess of salt is removed by salt glands present near the eye.

(3) In terrestrial habitat :

→ It is characterized by scarcity of water.

→ The terrestrial animals are faced with two main problems. (a) getting water (b) conserving water. They develop variety of water adaptations -

(1) Evaporation through body surface is prevented by developing spines, scales, scute, shields,...

(2) Certain animals become active during night.

(3) Kangaroo rat and protopterus aestivate durnig summer.

(4) The uromatrix lizard stores water in intestine.

(5) The exess water is stored in water cells of rumen and reticulum of stomach in camel.

(6) Mammals regulate their body temperature by producing sweat.

Soil :

- It is an abiotic ecological factor.
- It acts as suitable substratum and medium for plants and animals.
- It is a bridge between inorganic and organic materials.
- "Soil : Any part of earth's crust in which plants root."
- Treshow (1970) : "Soil is a complex physical - ecological system providing support, water, nutrients and oxygen for plants."
- Soil is formed of five major components : Mineral matter, organic matter, soil water, soil air and biological system.
- All organisms living in soil constitute soil community.

Organisms living in soil are classified into four major groups :

(1) Microflora :

- Bacteria, fungi, blue green algae, algae

(2) Microfauna :

- Protozoans, nematodes, mites (Size 20 to 200 micron)

(3) Mesofauna :

- Mites, nematodes, spiders, insect larvae, molluscs (size between 200 micron to 1 cm)

(4) Macrofauna :

- Animals with size of more than 1 cm large size eg. earthworms, snakes, moles, rates

- Nature and properties of soil in different places vary, dependent on climate.
- Various characteristic of soil such as soil composition, grain size and aggregation determine percolation and water holding capacity of soil.
- Parameter such as pH, mineral composition and topography determine to a large extent vegetation in any area.
- Burrow living animals have special adaptations of digging burrowing, their bodies are spindle shaped and tapering head, short tail, eyes and necks are reduced.

(9) Opening and closing of stomata regulated by ..... .

(A) water (B) soil (C) temperature (D) Light

(10) All chemichal reactions occur in living body are controlled by ..... .

(A) water (B) soil (C) temperature (D) Light

(11) Sexual eggs of daphnids develop as ..... .

(A) male (B) female (C) male or female (D) male and female

(12) ..... develop accessory respiratory organs to utilise atmospheric air when fresh water dries up.

(A) Ophiocephlus (B) Amphioxsus (C) Rohu (D) Katla

(13) The terrestrial animals are faced with which problems ?

(A) Getting water, conserving water. (B) conserving water, loss of salts.

(C) Getting water, loss of water. (D) Getting water, loss of salts.

(14) Its skin is hydroscopic ..... .

(A) Kangaroo rat (B) Spiny lizard (C) Protopterus (D) Uromatrix lizard

(15) Animal with size 200 micron to 1 cm are placed in which group.

(A) Microflora (B) Microfauna (C) Mesofauna (D) Macrofauna

Answers : (9-D), (10-C), (11-C), (12-A), (13-A), (14-B), (15-C)

Response to abiotic factors

Homeostasis : "The organisms try to maintain the constancy of its internal environmental conditions when the external environment changes."

Response of Organisms

(1) To cope with unfavourable condition :

- Maintain homeastasis by physiological means
- eg. birds, mammals

(2) Conformsts :

- Majority of animals and nearly all plants can't maintain a constant internal environment.

(3) Migration :

- Organism can move away temporarily from stressful habitat to a more suitable area and return when stressful period is over.

(4) Suspension :

- In bacteria, fungi, lower plants various kinds of thick walled spores are formed which help them to survive in unfavourable conditions and germinate when suitable environment prevails.
- Animals show hibernation during winter and aestivation during summer.

---

(16) What is the process in which organisms try to maintain the constancy of its internal environment despite varying external environmental conditions that tend to upset their equilibrium.

(A) Management (B) Adaptation (C) Heterostasis (D) Homeostasis

(17) They form thick walled spores in unfavourable condition ..... .

(A) Bacteria (B) Fungi (C) Lower plants (D) All the above

Answers : (16-D), (17-D)

*** Adaptations ***

" The adjustments made by individuals in response to specific, environmental conditions."

- It is in the form of morphological, physiological behavioural.
- Kangaroo rat in desert is capable of meeting all its water requirements through its internal lipid oxidation.
- In most animals, the metabolic reactions proceed optimally in a narrow temperature range (37^0 C).
- Archae bacteria flourish in hot spring where temperature far exceeds 100^0 C.
- Antartic fishes live in temperature below zero.
- A large variety of marine invertebrate fish live at great depths in ocean where pressure could be > 100 times the normal atmospheric pressure.

(18) How does kangaroo rat meet its water requirement ?

(A) Oxidation of internal proteins. (B) Oxidation of internal carbohydrates.

(C) Oxidation of internal Lipids. (D) All the above

(19) Marine fishes are habituated to what pressure as compared to normal atmospheric pressure.

(A) < 100 times (B) > 100 times (C) > 50 times (D) < 50 times

Answers : (18-C), (19-B)

Population : " A group of organisms of the same species living in a particular area of a given time."

Salient features of population :

(1) All individual of a population belong to one species.

(2) The individuals are morphologically and anatomically similar.

(3) The individuals are genetically related.

(4) The individuals are reproductively isolated from other species.

- **Characteristics of a population :**

(1) Population density : It refers to total number of individuals in an unit area or unit volume in a given time.

$$D = \frac{n/a}{t}$$

D = Density , t = time

n = Number of individual

a = Area

(2) Natality or Birth rate : "The average number of new individuals produced by a population in a given time.

$$\text{Birth rate} = \frac{\text{Number of births per unit time}}{\text{Average population}}$$

→ Carrying capacity : "The total number of individuals that can be accomodated in an area at a particular time."

→ There are two aspects of natality. (1) Realised natality (2) Potential natality (Maximum Natality)

(3) Mortality or Death rate : The number of individuals dying in a population at a given time."

Two aspects of mortality : (1) Realised mortality (2) Potential mortality

$$\text{Mortality rate} = \frac{\text{Number of deaths per unit time}}{\text{Average population}}$$

- Vital index : Ratio between birth rate and death rate
- Vital index = Birth rate / death rate × 100
- When Birth rate = Death rate, the population becomes stable and reamains at equilibrium.

- **(4) Age Distribution :** Three groups

(i) Pre-reproductive group → Immature animals

(ii) Reproductive group → Sexually mature animals

(iii) Post → reproductive group → Old animals (reproductive disability)

- On basis of age groups, age, pyramid can be constructed.

Expanding population → Triangular pyramid

Stable population → Bell-shaped pyramid

Declining population → Urn-shaped pyramid

(5) Population growth : The increase in size of population

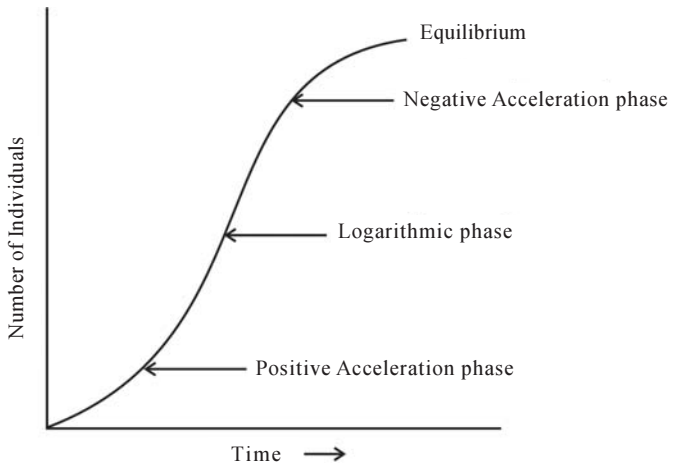
$N_{(t+1)} = Nt + [(B + I) - (D + E)]$

(6) Population equilibrium : Two types.

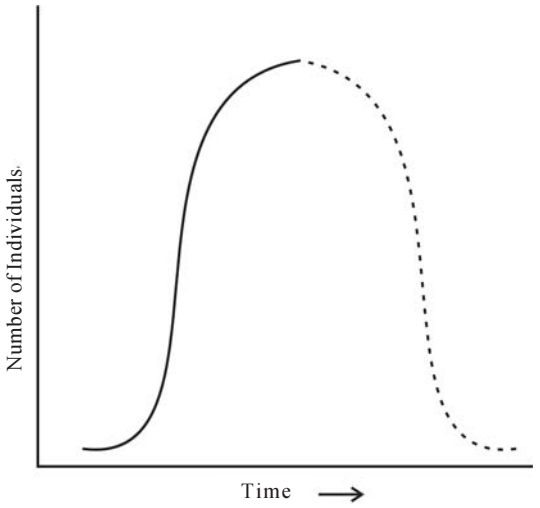
(i) S - Shaped growth curve

(ii) J - Shaped growth curve

(i) S - Shaped growth curve



(ii) J - Shaped growth curve



(7) Population Fluctuations : The increase and decrease in the number of individuals in a population.

- It may be flat, cyclic or irruptive.

• Factors affecting on population growth :

(i) Extrinsic factors : food, space, shelter, weather.

(ii) Intrinsic factors : competition, predation, diseases, emigration, reproductivity.

(8) Population interation :

	Species A	Species B	Name of interaction	General Results
(i)	+	+	Mutualism	Favourable to both
(ii)	-	-	Competition	Harmful to one another
(iii)	+	-	Predation	Favourable to A
(iv)	+	-	Parasitism	Favourable to A
(V)	+	0	Commensalism	Favourable to A
(Vi)	-	0	Amensalism	One species is affected while other is unaffected

(20) The actual number of new individuals added to the population in a given time. It is always ..... than the potential natality.

(A) more (B) lesser (C) equal (D) no relation

(21) Deaths, that occur at all ages is called ..... .

(A) potential mortality (B) realised mortality (C) minimum mortality (D) All the above

(22) The number of organisms increases in compound interest fashion in this growth curve.

(A) J - Growth curve. (B) S-Growth curve. (C) A and B both (D) None of these

(23) When the density increases the reproductive ability ..... .

(A) increases (B) decreases (C) No change (D) None of the above

(24) The interaction of two organism in which one organism get benifitted and another gets harmed, is called ..... .

(A) parasitism (B) predation (C) A and B both (D) None of these

Answers : (20-B), (21-B), (22-A), (23-B), (24-C)

Mutualism : term means ' living together'

Example : (1) Hermit Crab and Sea Anaemone

- Crab lives inside the empty Gastropod shells.
- Sea anemone is inhabited on the shell.
- Stinging cells of sea anemone protect the crab.
- Food acquired / captured by Sea Anaemone is shared by Crab.

Example : (2) Rhizobium bacteria and root nodules of leguminous plants.

- Bacteria fix up atmospheric free N_2 in the nodules in the form of nitrates.
- Plants utilize these nitrates.
- The bacteria, in turn, obtains shelter and nutrients from the plants.

Example : (3) Lichens (algae and fungi)

- Fungi provide moisture and minerals to algae.
- Algae prepare food for both, by photosynthesis.

Competition :

- The two partners may be harmed or only one willl be harmed.
- Two types : **Intraspecific competition :** Occurs between members of same population.

Interspecific competition : Occurs between population of different species.

- **Predation :**
  - It controls the population.
  - All animals and insectivorous plants are predatory.
  - Two types : Herbivores and carnivores.
- **Perasitism :**
  - It is a one sided relationship.
  - The pantner which is benefited is called parasite.
  - The other partner is called host.

- Types of parasites.

(i) Temporary parasites : An animal spends only a part of its life cycle as a parasite. eg. lawal form.

(ii) Permanent pararsites : An animal spends entire life as parasite. eg. Ascaris, Tapeworm.

(iii) Ectoparasites : The parasites which live on the outer surface of the host. eg. Mites, Leeches.

(iv) Endoparasites : The parasites live inside the body of the host. eg. Plasmodium, Ascaris.

Commensalism :

- Partners are called commensals.

It is a symbiotic interspecific relationship.

eg. (i) Sucker fish sucks on the body of shark.

(ii) Flagellates living inside the body of termites.

Amensalism :

- One species is harmed where as the other is unaffected.

- eg. Penicilium fungi, gram positive bacteria.

---

(25) The hermit crab lives inside. the empty ..... shell.

(A) Thoraxopod (B) Abdomenopod (C) Gastropod (D) All the above

(26) Rhizobium fixes free N_2 into which form in root nodule. ?

(A) Nitrous (B) Nitrates (C) Nitrites (D) B and C Both

(27) Which type of competition occurs between Paramoecium caudatum and P. aurelia ?

(A) Intraspecific (B) Interspecific (C) Both (A,B) (D) None of these

(28) In this interaction one partner is harmed where as other is unaffected.

(A) Amensalism (B) Commensalism (C) Parasitism (D) All the above

Answers : (25-C), (26-C), (27-B), (28-A)

Biotic Community : "A group of several species living in a particular area with mutual tolerance (adjustment)".

eg. (i) Pond community : All organisms living in a pond.

(ii) Forest community : trees, shrubs, insects, birds.

Characteristics of a biotic community :

(1) Species diversity : Each community has many varied organisms.

(2) Structure : It is formed of 3 groups of animals.

(i) Producers : The autotrophic green plants are producers – autotrophs.

(ii) Consumers : They are heterotrophs.

- **Two types :** Herbivores (Primary consumers) and carnivores (Secondary consumers)

(iii) Decomposers : They decompose the dead bodies of plants and animals and convert them into micro and macro nutrients. These are utilised again by plants.

eg. Bacteria, fungi

(3) Community dominants : Sometimes one or a few species play a dominant role in a community b y virtue of their number, size etc. These species are called community dominants.

- eg. Trees are community dominants in a forest community.

(4) Ecological Niche : It refers to the functional status of an organisms in its community.

- According to Odum Niche is the profession of an organism in a community.
- An organism's habitat is its residential address and niche is the profession of the organism.

eg. lion and deer live in the same habitat but lion is a carnivore (niche) and deer is a herbivore (niche).

(29) A group of several species living together with mutual adjustment is called ..... .

(A) species (B) population (C) ecosystem (D) biotic community

(30) Biotic community is formed of how many groups of organisms ?

(A) 1 (B) 2 (C) 3 (D) 4

(31) They are community dominants in a forest community ..... .

(A) Trees (B) Shrubs (C) Herbs (D) Climbers.

(32) What is the profession of an organism in biotic community called ?

(A) Structure (B) Species diversity (C) dominants (D) Ecological niche.

Answer : (29-D), (30-C), (31-A), (32-D)

Ecological Succession : The process of development of a new community is called ecological succession.

- It is an orderly and progressive replacement of one community by another till the development of a stable community in that area.
- Succession is directional and predictable.
- Structural complexity increases during succession.

* **Types of succession :**

(1) Primary Succession : It starts from primitive substratum, where there was no living matter previously.

– The first group of organisms establishing themselves is known as the pioneer.

(2) Secondary succession : It starts from previously built up substrates with already existing living matter.

– General process of succession :

– The various developmental stages of a community are called seral stages.

– The first seral stage is called pioneer community.

– The final stable community is called climax community, which is determined by the total environmental factors of the area in which it exists.

Pattern of succession : Three types

(1) Xerosere :

- Begins from a dry place.
- Lichens are the pioneer.
- Mosses, herbs and shrubs develop afterword.

(2) Hydrosere :

- Starts in water.
- The first organisms to inhabit are planktons.
- Then submerged and floating plants enrich the pond.

(3) Mesosere :

- It is intermediate between xerosere and hychosere.
- It occurs in moist places.

---

(33) The process of development of a new community is called ..... .

(A) evolution (B) revolution (C) succession (D) All the above

(34) If the pond is gradually filled with sand and mud then it gets converted into :

(A) Marshy (B) Forest (C) Grassland (D) All the above

(35) The structural complexity ..... during succession.

(A) decrease (B) increase (C) no change (D) fluctuate

(36) What are the stages called in succession ?

(A) Seral stages (B) Serial stages (C) Series stages (D) All the above

(37) The first seral stage is called ..... .

(A) Primary biotic community (B) Initial biotic community

(C) Pioneer community (D) All the above

(38) The final community of succession is called ..... .

(A) seral stage (B) pioneer community (C) A and B both (D) climax community

(39) Which succession starts from rocks ?

(A) Xerosere (B) Hydrosere (C) Mesosere (D) All the above

(40) Which is the pioneer community in xerosere ?

(A) Algae (B) Fungi (C) Lichens (D) All the above

(41) The first organisms to inhabitat in pond succession are ..... .

(A) planktons (B) submerged plants (C) floating plants (D) All the above

Answers : (33-C), (34-A), (35-B), (36-B), (37-C), (38-D), (39-A), (40-C), (41-A)

A = Statement R = Reason, type questions :

Select answers of following questions from options given below.

(A) A and R both correct. R is explanation of A.

(B) A and R both correct but R is not explanation of A.

(C) A is correct, R false.

(D) A is false, R correct.

(42) Statement A : The study of basic components of our surroundings and their interactions is called environmental studies.

Reason : Ecology is a part of environment study.

(A) (B) (C) (D)

(43) Statement A : Marine organisms are adopted to salinity, osmosis, dissolved gases.

Reason R : Fresh water organisms are adapted to salinity, osmosis, pH, temperature.

(A) (B) (C) (D)

(44) Statement A : Estuary water has less salinity.

Reason R : Here river water mixes with the sea water.

(A) (B) (C) (D)

(45) Statement A : Osmoregulation is the major problem to living fauna to aquatic habitat.

Reason R : The dynamic environment and organisms make ways for the development of different kinds of organisms through succession.

(A) (B) (C) (D)

(46) Statement A : Any external force, substance or conditon, which surrounds and effects the life of an organism in any way, becomes a factor of its environment.

Reason R : All external factors, conditions and effects which afffects organisms or biotic community is called environmental study.

(A) (B) (C) (D)

(47) Statement A : There is difference in vegetation at poles and other parts of the earth.

Reason R : Total amount of radiation received by the earth's surface differs with latitude.

(A) (B) (C) (D)

(48) Statement A : Cave animals are colourless.

Reason R : Light induces photochemical reactions resulting in the formation of colour pigment.

(A) (B) (C) (D)

(49) Statement A : In some birds gonads become active during summer.

Reason R : Light initiates the breeding activities in birds.

(A) (B) (C) (D)

(50) Statement A : All metabolic processes are influenced by light in animals.
Reason R : Increased light intensity results into increased enzyme activity.
(A) (B) (C) (D)

(51) Statement A : All metabolic processes are influenced by temperature.
Reason R : Temperature regulates activity of enzymes.
(A) (B) (C) (D)

(52) Statement A : Light affects the plant's life on chlorophyll production, photosynthesis, transpiration, distribution of plants.
Reason R : Temperature effects transpiration, chlorophyll production, photosynthesis in plants.
(A) (B) (C) (D)

(53) Statement A : Altitudinal zonation of vegetation is clearly seen on mountains.
Reason R : Temperature is an important factor in phenology of plants.
(A) (B) (C) (D)

(54) Statement A : Under normal condition daphnids give parthenogenetic eggs that develop into females.
Reason R : In rotifers and daphnids sex - ratio is effected by temperature.
(A) (B) (C) (D)

(55) Statement A : Some insects, birds and mammals in warm humid climatic bear darker pigments than the races of same species present in cool and dry climate.
Reason R : Birds and mammals obtain greater body size in cold regions than in warm areas, but polkilotherms are smaller in cold region.
(A) (B) (C) (D)

(56) Statement A : Temperature variation is much lesser in aquatic habitat than in terrestrial habitat.
Reason R : Temperature variation is higher in fresh water than in marine water.
(A) (B) (C) (D)

(57) Statement A : Thermal stratification occurs in water reserviors.
Reason R : There is a gradual decrease in temperature from the surface to bottom in aquatic habitat.
(A) (B) (C) (D)

(58) Statement A : Body fluid of fresh water animal is hypertonic.
Reason R : The salinity of fresh water is very low.
(A) (B) (C) (D)

(59) Statement A : Clarias is adapted to remove excess of water.
Reason R : It removes excess salt through chlorine secretory cells of gills.
(A) (B) (C) (D)

(60) Statement A : Crustacean Astacus, removes excess of water in the form of urine by the green glands.
Reason R : Amoeba has contractile vacuole for osmoregulation.
(A) (B) (C) (D)

(61) Statement A : In marine turtles the excess of salt is removed by salt glands present near the ear.
Reason R : Its body fluid is hypotonic.
(A) (B) (C) (D)

(62) Statement A : The lizard uromatrix stores water in the intestine.
Reason R : The terrestrial animals are faced with two main problems.
(A) (B) (C) (D)

(63) Statement A : Mites, nematodes, spider belong to mesofauna.

Reason R : Their body size is between 200 micron to 1 cm.

(A) (B) (C) (D)

(64) Statement A : Soil properties and another parameter determine to a large extent the vegetation in any area.

Reason R : The nature and properties of soil in different places vary.

(A) (B) (C) (D)

(65) Statement A : The body of burrowing animals is spindle shaped and tapering head, short tail, the eyes and necks are reduced.

Reason R : The majority of animals living in soil lead either burrowing or cursorial mode of life.

(A) (B) (C) (D)

(66) Statement A : Birds and mammals are able to fight against unfavorable condition.

Reason R : Some organisms maintain their homeostasis physiologically.

(A) (B) (C) (D)

(67) Statement A : The total number of individuals in an unit area or unit volume in a given time is called population density.

Reason R : The total number of individuals that can be accomodated in an area at a particular time is called carrying capacity.

(A) (B) (C) (D)

(68) Statement A : Salman fish produces 2.8×10^8 eggs in a season.

Reason R : The maximum possible rate of reproduction for a population under optimal condition is called potential natality.

(A) (B) (C) (D)

(69) Statement A : The size of a population for any species is not a static parameter.

Reason R : The density of a population in a given habitat during a given period, fluctuates due to changes in four basic processes.

(A) (B) (C) (D)

(70) Statement A : Growth curve of frog in monsoon is J type.

Reason R : In J growth curve number of organisms increases in compound interest fashion.

(A) (B) (C) (D)

(71) Statement A : Bird Population is affected by seasonal effect and migration.

Reason R : Factors are responsible for population fluctuation.

(A) (B) (C) (D)

(72) Statement A : Biological control methods adopted in agricultural pest control are based on ability of the predator.

Reason R : Predator also helps in maintaining species diversity in biotic community.

(A) (B) (C) (D)

Answers : (42-B), (43-D), (44-A), (45-B), (46-C), (47-A), (48-A), (49-A), (50-A), (51-A), (52-C), (53-B), (54-A), (55-B), (56-D), (57-A), (58-A), (59-C), (60-B), (61-D), (62-A), (63-A), (64-B), (65-A), (66-A), (67-B), (68-A), (69-A), (70-A), (71-A), (72-D)

True - False type questions

(73) (1) In xerosere succession mosses, lichens, fungi, herbs gradually develop.

(2) In hydrosere succession planktons. submerged and floating plan to enrich the pond.

(3) Mesosere succession occurs in moist places.

(4) Ecological succession is directional and predictable.

(A) F, T, T, T (B) T, T, T, T (C) F, F, T, T (D) F, T, T, F.

(74) (1) Primary succession starts where there was no living matter, previously.

(2) The various developmental stages of a community are called as seral stages.

(3) Formation of new biotic community is called speciation.

(4) If the pond is filled with sand and mud, grassland biotic community developer.

(A) T, F, F, T (B) F, T, T, F (C) T, F, F, F (D) T, T, F, F

(75) (1) Penicilium fungi, gram positive bacteria occupies Amensalism.

(2) Hermit crab and sea anemone live as commensalism.

(3) Competition between Paramoecium caudatum and P. aurelia is interspecific.

(4) In interaction if one species get benefited and another is not affected, it is called symbiosis.

(A) T, F, F, T (B) F, T, F, T (C) T, F, T, T (D) T, F, T, F.

(76) (1) Food, space, shelter, weather etc are the extrinsic factors on population growth.

(2) The density dependent factors are intrinsic, which arise within the population.

(3) Introducing goats on galapagos island is an example of immigration.

(4) Interaction between animals occurs mainly for food.

(A) T, T, F, F (B) T, T, F, T (C) T, F, T, F (D) T, F, F, T

(77) (1) Realised mortality refers to the number of deaths due to old age.

(2) Realised mortality is higher than potential mortality.

(3) The maximum possible rate of reproduction for a population under optimal codition is called potential natality.

(4) Vital index = Death rate / Birth rate $\times$ 100

(A) F, F, T, T (B) F, T, F, F (C) F, T, T, F (D) F, T, T, T

(78) Select right option for salient features of population.

(1) All individuals of a population belong to one species.

(2) The individuals are genetically isolated.

(3) The individuals are morphologically similar.

(4) The individuals are reproductively isolated from other species.

(A) T, T, F, F (B) T, F, T, F (C) T, F, F, T (D) T, F, F, F

(79) (1) Kangaroo rat in desert is capable of meeting all its energy requirements through its internal lipid oxidation.

(2) Archae bacteria flourish in hot spring where temperature far exceeds 100^0 C.

(3) Homeostasis stabilizes the internal environment.

(4) In most animals, the metabolic reaction proceeds optimally in a narrow temperature range (35^0 C).

(A) T, T, F, F (B) F, T, T, F (C) T, T, T, F (D) F, T, T, T.

Answers : (73-A), (74-D), (75-C), (76- B), (77-C), (78-C), (79-B)

Column Based Questions

(80)

Column - I	**Column - II**	**Column - III**
(I) Microflora	(p) mites	(w) Size less than 20 μ
(ii) Microfauna	(q) spider	(x) Size more than 1 cm
(iii) Mesofauna	(r) bacteria	(y) Size 20 μ to 200 μ
(iv) Macrofauna	(s) earthworms	(z) 200 μ to 1

	I	**II**	**III**	**IV**
(A)	r-w	p-z	q-y	s-x
(B)	q-w	p-y	q-z	s-x
(C)	r-w	q-y	p-z	s-x
(D)	r-w	s-x	p-z	q-y

(81)

Column - I	**Column - II**
(i) Marine turtle	(p) accessory respiratory orgam
(ii) Clarias	(q) green gland
(iii) Crustacean Astacus	(r) Salt gland
(iv) Marine animals	(s) Chlorine secretory cells

	I	**II**	**III**	**IV**
(A)	r	p	s	q
(B)	p	r	s	q
(C)	p	r	q	s
(D)	r	p	q	s

(82)

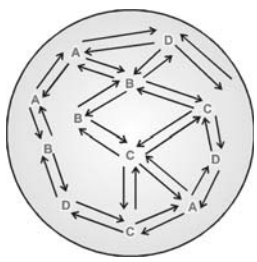
Column - I	**Column - II**	**Column III**
(i) +,+	(p) Predation	(w) flagellates and termites
(ii) +,-	(q) ammensalism	(x) lichen
(iii) +,o	(r) commensalism	(y) gram positive bacteria
(iv) -, o	(s) mutualism	(z) leech and host

	I	II	III	IV
(A)	s-z	p-w	r-y	q-x
(B)	s-x	p-z	r-w	q-y
(C)	q-x	p-w	s-z	r-y
(D)	r-y	q-x	s-w	p-z

Answers : (80-B), (81-D), (82-B)

Diagram based question

(83) Choose correct option for given diagram.



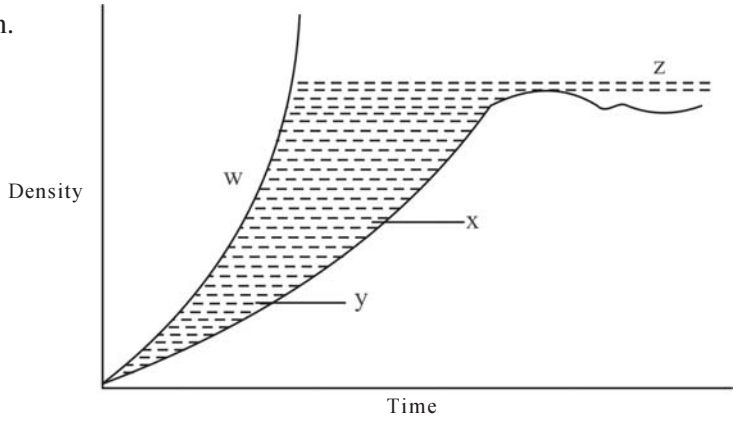
(A) It shows interactions between different species.

(B) It shows interactions between organisms of population.

(C) It shows interactions between ecosystem and biotic factors.

(D) It shows only interaction and interrelation.

(84) What does 'X' indicate in given diagram.



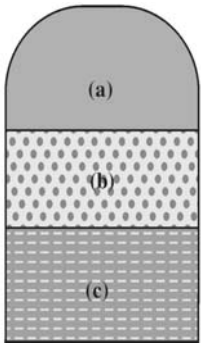
(A) Potantial natality

(B) Environmental resistance

(C) Carrying capacity

(D) Equilibrium

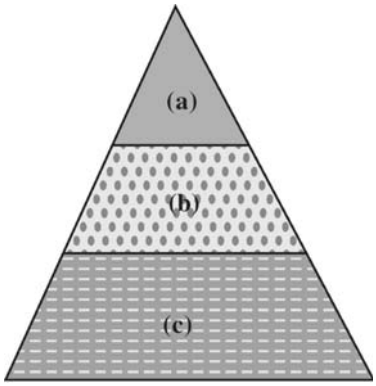
(85) Choose correct option for xgiven pyramid.



(A) $a \cong b \cong c$ (B) $a < b < c$

(C) $a < b \cong c$ (D) $a > b > c$

(86) Choose appropriate option for the given pyramid .



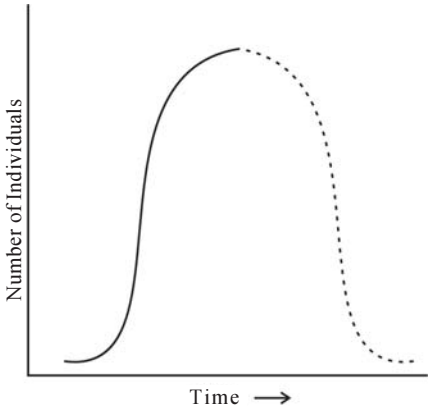
(A) It indicates expanding population.

(B) Pre-reproductive organisms are more in number.

(C) Pyramid is triangle.

(D) All the above

(87) Choose appropriate option for given graph.



(A) Growth occurs rapidly.

(B) Number of organisms increase in compound interest fashion.

(C) Growth suddenly decreases.

(D) All the above.

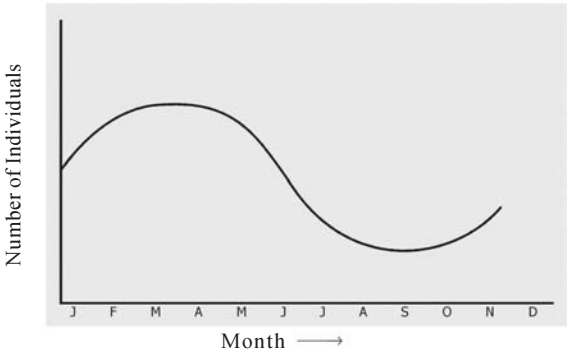
(88) What is true for given diagram ?

(A) Seasonal effect on bird-population

(B) Bird-population affected by migration

(C) Annual effect on bird-population

(D) A and B both



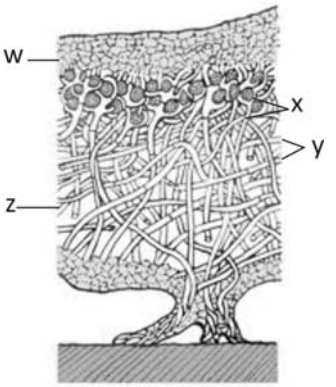
(89) What does 'x' indicates in given diagram.

(A) It provides moisture and minerals to the cells.

(B) It prepares food by photosynthesis.

(C) It absorbs moisture.

(D) B and C both.



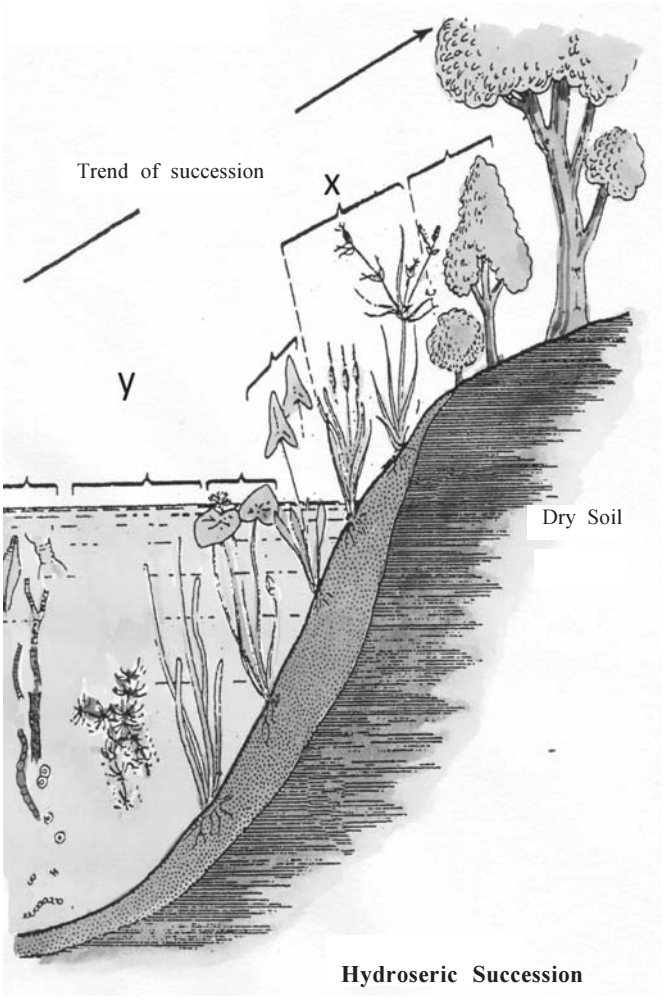
(90) Choose correct option for given diagram.

(A) x = Sedge meadow y = Floating

(B) x = Red swamp
y = Phytoplanktons

(C) x = Sedge-meadow
y = Rooted submerged

(D) x = Red swamp
y = Sedge-meadow



Hydroseric Succession

Answers : (83-A), (84-B), (85-A), (86-D), (87-D), (88-D), (89-B), (90-C)

*** Questions for NEET ***

(91) Intense competition for light, nutrients and habitat occurs between ?
(A) closely related organisms growing in different niches.
(B) closely related organisms growing in the same area / niche
(C) distantly related organisms growing in the same habitat.
(D) distantly related organisms growing in different niches.

(92) A Mutually beneficical association necessary for survival of both partners is ..... .
(A) mutualism/ symbiosis (B) commensalism (C) Amensalism (D) both A and B

(93) Homeostasis is ..... .
(A) tendency of biological systems to change with change in environment
(B) tendency of biological systems to resist change
(C) disturbance of self regulatory system and natural controls
(D) biotic materials used in homeopathic medicines

(94) Which one is true
(A) Commensalism when none of the interacting populations affect each other.
(B) Symbiosis when the interaction is useful to both the populations.
(C) Symbiosis when neither populations affect each other.
(D) Commensalism when the interaction is useful to both the populations.

(95) Soil particles determine its ..... .
(A) texture (B) field capacity (C) water holding capacity (D) soil flora

(96) The sum total of the populations of the same kind of organisms constitute.
(A) colony (B) genus (C) community (D) species.

(97) Association of animals when both partners are benefitted ..... .
(A) colony (B) mutualism (C) commensalism (D) amensalism

(98) xeric environment is charatcterised by ..... .
(A) precipitation (B) low atmospheric humidity
(C) extremes of temperature (D) high rate of vapourisation

(99) In a food chain, the largest population is that of ..... .
(A) producers (B) decomposers
(C) secondary consumers (D) primary consumers

(100) Sunken stomata is the characteristic feature of ..... .
(A) hydrophyte (B) mesophyte
(C) xerophyte (D) halophyte

(101) Which of the following pairs is coreectly matched ?
(A) uricotelism - aquatic habitat
(B) parasitism - intra - specific relationship
(C) excessive perspiration - xeric adaptation
(D) stream lined body - aquatic adaptation

(102) Human population growth in India ..... .
(A) tends to follow a sigmoid curve as in case of many other animal species.
(B) tends to reach a zero population growth as in case of some animal species.
(C) can be reduced by permitting natural calamities and enforcing birth control measures.
(D) can be regulated by following the national programme of family planning.

(103) In increasing order of organizational complexity, which one of the following is the correct sequence ?

(A) Population→species→community→ecosystem (B) Population→variety→species→ecosystem

(C) Population→ecosystem→species→community (D) Species→variety→ecosystem→community

(104) What is the most important factor for the success of animal population ?

(A) Natality (B) Unlimited food

(C) Adaptibility (D) Inter-species activity

(105) Whch part of the world has a high density of organisms ?

(A) Grasslands (B) Savanna (C) Deciduous forests (D) Tropical rain forests

(106) Two opposite forces operate in the growth and development of every population. One of them relates to the ability to reproduce at a given rate. The force opposing it is called ..... .

(A) environmental resistance (B) morbidity

(C) fecundity (D) biotic potential

(107) Certain characteristic demographic features of developing countries are ..... .

(A) high fertility, low or rapidly falling mortality rate, rapid population growth and a very young age distribution

(B) high fertility, high density, rapidly rising mortality rate and a very young age distribution

(C) high infant mortality, low fertility, unever population growth and a very young age distribution

(D) high mortality, high density, uneven population growth and a very old age distribution

(108) Niche overlap indicates ..... .

(A) two different parasites on the same host

(B) sharing of one or more resources between the two species

(C) mutualism between two species

(D) active cooperation between two species

(109) Geometric representation of age structure is a characteristrics of ..... .

(A) population (B) landscape (c) ecosystem (d) biotic community

(110) The population of an insect species shows an explosive increase in numbers during rainy season followed by its disappearance at the end of the season. What does this show /

(A) The food plants mature and die at the end of the rainy season.

(B) Its population growth curve is of J-type.

(C) The population of its predators increases enormously.

(D) S - shaped or sigmoid growth of this insect.

(111) Two plants can be conclusively said to belong to the same speices if they ..... .

(A) have more than 90 % similar genes

(B) look similar and possess identical secondary metabolites

(C) have same number of chormosomes

(D) can reproduce freely with each other and form seeds

(112) A high density of elephant population in an area can result in ..... .

(A) intra specific competition (B) inter specific competition

(C) predation on one another (D) mutualism

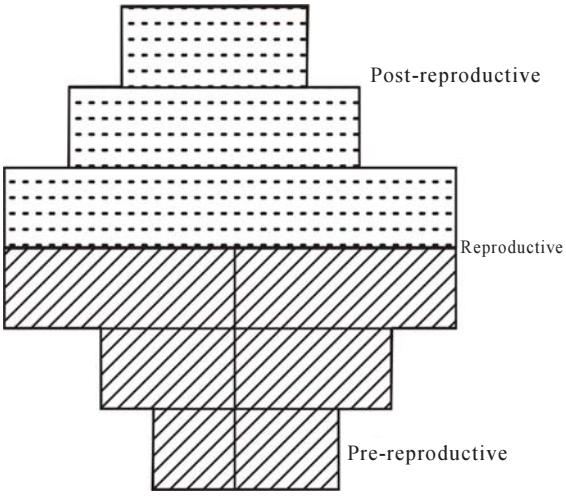
(113) Which one of the following is one of the characteristic of a biological community ?

(A) Stratification

(B) Natality

(C) Mortality

(D) Sex ratio



(114) What type of human population is represented by the following age pyramid ?

(A) Vanishing population (B) Stable population

(C) Declining population (D) Expanding population

(115) Which one of the following statements is correct for secondary succession ?

(A) It begins on a bare rock.

(B) It occures on a deforested site.

(C) It follows primary succession.

(D) It is similar to primary succession exept that it has a relatively fast pace.

(116)

Column - I	**Column - II**		**a**	**b**	**c**	**d**
(a) earthworm	(i) pioneer species	(A)	(i)	(ii)	(iii)	(iv)
(b) succession	(ii) detritus	(B)	(iv)	(i)	(iii)	(ii)
(c) ecological service	(iii) natality	(C)	(iii)	(ii)	(iv)	(i)
(d) population growth	(iv) pollination	(D)	(ii)	(i)	(iv)	(iii)

(117) What does a Niche show in ecosystem ?

(A) Funtion of an organism at its trophic level (B) Its presence

(C) Competitive ability (D) Centre of Origin

Answers : (91-B), (92-A), (93-B), (94-B), (95-A), (96-D), (97-B), (98-B), (99-A), (100-C), (101-D), (102-A), (103-C), (104-D), (105-A), (106-A), (107-B), (108-A), (109-B), (110-D), (111-A), (112-A), (113-C), (114-B), (115-D), (116-A), (117-B)

Ecology

- Study of reciprocal relationship between organisms and their environment.

Organisms

- Basic unit of ecological hierarchy.
- Interaction with physical environment comprised of atmosphere, hydrosphere and lithosphere.
- **Adaptations** enable them to survive and reproduce in their habitats.
- **Environmental factors** (abiotic and biotic) influence the form and functioning of organisms.

Abiotic Factors

- Climatic
  - Light
  - Temperature
  - Wind
  - Rain (Precipitation)
  - Atmospheric humidity
- Edaphic
  - Mineral matter
  - Organic matter
  - Soil
- Topographic
  - Altitude
  - Steepness of slope
  - Direction of slope

Biotic factors

Bacteria, Algae, Fungi, Parasites, Saprophytes, Symbionts, Animals

Population

Organisms of the same kind form population. They are capable of interbreeding among themselves.

Characteristics of Population:
- Density
- Natality
- Mortality
- Dispersal
- Age distribution
  - Emigration
  - Immigration
- Biotic potential
  - J Shaped(Exponential)
  - S-Shaped(Logistic)
- Growth form

Community

Population of different species occurring in a habitat.

Ecosystem

It includes biological community, integrated with its physical environment.

Features:

Species composition

Different kinds of plants and other organisms present in a community.

Dominance

- Only one or few species in a community dominate and influence other species in terms of numbers and biomass production.

Stratification

- Vertical layering of vegetation in which different layers are occupied by different species.

Species interaction

- Positive
  - Mutualism
  - Protocooperation
  - Commensalism
- Negative
  - Competition
  - Predation
  - Parasitism

Biosphere

Includes all ecosystems on the earth.

28 Ecosystem

- Biogeography reveals that living organisms are found practically everywhere on this earth.
- A.G.Tansley (1935) defined ecosystem as the system resulting from the interaction of all the living and non-living factors of the environment.
- The biotic and abiotic components are linked with each other through unidirectional energy flow and nutrient cycling.
- Living organisms are produced through chemical evolution.
- Ecosystem is a structural unit of ecology.
- Each ecosystem is maintained and remained healthy due to energy flow.
- The sun is the main source of energy in ecosystem.

(1) The term 'Ecosystem' given by ..... .

(A) Odum (B) A.G. Tansley

(C) Hatch - Slack (D) Robert Hook

(2) The biotic and abiotic components are linked with each other through ..... energy flow and ..... material flow.

(A) bidirectional, cyclic (B) cyclic, bidirectional

(C) unidirectional, cyclic (D) cyclic, undirectional

(3) Living organisms are produced through ..... .

(A) Biological evolution (B) Physical evolution

(C) Bio-chemical evolution (D) Chemical evolution

(4) What does a biotic community called when it is regulated by abiotic components ?

(A) Biomes (B) Biosphere

(C) Ecosystem (D) Bio geography

(5) What makes an ecosystem healthy and maintained ?

(A) Bio-geochemical cycle (B) Energy-flow

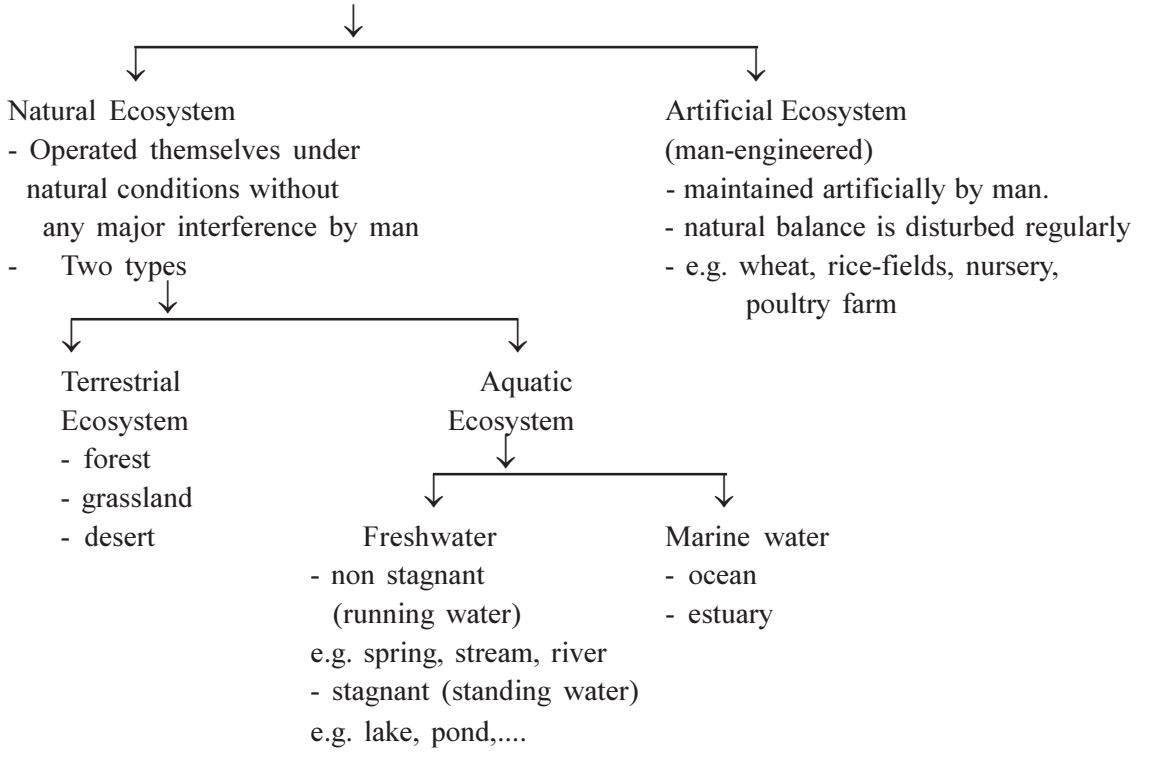
(C) cycling of nutrients (D) All given

Answers : (1-B), (2-C), (3-D), (4-C), (5-B)

Kinds of Ecosystem :

Different types of ecosystem of nature, constitute the giant ecosystem, the biosphere.

→ **Two types of Ecosystem :**



(6) What does an ecosystem call which is operated by itself ?

(A) Artificial ecosystem (B) Physical ecosystem

(C) Natural ecosystem (D) A and C both

(7) Wheat, rice fields includes in what kind of ecosystem ?

(A) Artificial ecosystem (B) Natural ecosystem

(C) Man-engineered ecosystem (D) A and C both

Answers : (6-C), (7-D)

Structure and function of ecosystem

There are major two aspects of an ecosystem

Structure	**functions**
(i) the composition of biological community	(i) the rate of biological energy flow
• species, numbers, biomass, life history, distribution	(ii) nutrient cycle
(ii) the quantity and distribution of the non-living materials - nutriets, water	(iii) biological or ecological regulation
(iii) the range, or gradient of conditions of existence	
• temperature, light	

Thus, in any ecosystem, structure and functions are studied together.

Structure of Ecosystem

Abiotic (non-living) components

(i) inorganic substances :

– Macronutrients :

e.g. C, H, O, P, K, N, S, Mg, Ca

- Micronutrients

e.g. Mn, Cu, Mo, B, Zn, Fe, Cl, Co, Ni, V

(ii) organic materials

- proteins, carbohydrates, lipids

(iii) environmental factors

- climatic factors

e.g. light, temperature, wind precipitation, humidity

- edaphic factors

e.g. soil, water, air

Biotic (living) components

– Two components

Autotrophic components

- green plants, photosynthetic bacteria
- they are autotrophs
- called producers

Heterotrophic components

- animals, micro organisms
- they utilize, rearrange and decompose complex materials
- known as consumers
- categorised as

(a) Macro consumers

- Herbivores (primary consumers)
- Carnivores (secondary consumers)
- Omnivores (tertiary consumers)
- All are depended on herbivores

(b) micro consumers

- known as decomposers or detritivores
- bacteria, actinomycetes, fungi are saprophytes
- they breakdown complex compounds of dead or living protoplasm, absorb them and release inorganic nutrients in environment, making them available again to autotrophs.

(8) How many aspects are there in ecosystem ?

(A) 1 (B) 2 (C) 3 (D) 4

(9) In ecosystem which aspects are studied together ?

(A) Structure and energyflow (B) Functions and types

(C) Kinds and structure (D) Structure and function

(10) Which option belongs to a macronutrients ?

(A) C,H,N,Cu (B) Bo, Zn, P, S (C) Mg, Si, K, S, (D) Mn, C, H, N

(11) Which group belongs to micronutrients ?

(A) Mo, Mn, Mg, C (B) V, Ni, Co, Cu (C) Ca, Si, Mg, Mn (D) Zn, Fe, P, S

(12) How many components has a trophic structure of any ecosystem ?

(A) 2 (B) 4 (C) 6 (D) 3

(13) The members of these components utilise, rearrange and decompose the complex material.

(A) Consumers (B) Producers (C) Decomposers (D) All given

(14) Consumers are further categorised into ..... groups.

(A) 2 (B) 3 (C) 4 (D) 6

(15) They all are depended on herbivores.

(A) macroconsumers (B) microconsumers (C) consumers (D) all given

(16) They decompose complex compounds, release inorganic nutrients, making them available to autotrophs.

(A) macroconsumers (B) microconsumers (C) consumers (D) all given

Answers : (8-B), (9-D), (10-C), (11-B), (12-A), (13-A), (14-A), (15-A), (16-B)

Function of ecosystem :

- Two important functions :

  (1) flow of energy (ii) Biogeochemical cycling of materials

Energy flow :

- It begins when the producer organisms absorb sunlight and convert it into chemical energy by photosynthesis.
- Consumers utilize this chemical energy as food and finally released into atmosphere as heat energy.
- In this way flow of energy is unidirectional.

Biogeochemical cycling of nutrients :

- It begins when inorganic nutrients are absorbed by producers.
- The chemical elements present in these nutrients flow through various organisms are released into the physical environment.
- They are again absorbed by the producers.
- Thus, their transport occur from physical world into living world and from there back into physical world. This process is called Biogeochemical cycle.

(17) The flow of energy begins when ..... organisms absorb sunlight.

(A) producer (B) consumer

(C) decomposers (D) A and C both

(18) Which aspect begins when inorganic nutrients are absorbed by producer ?

(A) Energy flow (B) Nutrient - cycle

(C) Biogeochemical cycle (D) B and C both

Answers : (17-A), (18-D)

*** Structure and Function of Ecosystem ***

- It may be analyzed in terms of :
  (i) food chains, food webs and ecological pyramids (ii) Energy flow (iii) Productivity (iv) Nutrient cycle

Food Chain

- The transfer of food energy from producers, through a series of organisms is known as food chain. (herbivores - carnivores - decomposers)
- food chain starts from primary producers.
- ends with carnivores.

Types of food chain : **Two types**

Grazing food chain	**Detritus food chain**
– begins with chlorophyllus producers	– begins with : decomposers
– extends through herbivores, carnivores, decomposers.	– Primary source of energy is dead organic matter called 'detritus'.
	– eg. fallen leaves, plant parts or dead animal bodies.

(19) Food chain begins with..... and ends in ..... .
(A) carnivores, producers (B) producers, carnivores
(C) decomposers, carnivores (D) producers, decomposers

(20) Grazing food chain begins with ..... .
(A) decomposers (B) carnivores (C) producers (D) consumers

(21) 'Detritus' means ..... .
(A) Primary source of energy is dead organic matter.
(B) Secondary source of energy is dead organic matter.
(C) Primary source of energy is living organic matter.
(D) Primary source of energy is living inorganic matter.

(22) A detritus food chain begins with ..... .
(A) producers (B) consumers (C) carnivores (D) decomposers

Answers : (19-D), (20-C), (21-A), (22-D)

Food web :

- Living organisms, depend on each other for their food requirement and form a chain called food chain.
- The animals are inter - dependent for food and they form a net which is termed as a food web.

(23) How do flow of energy and nutrient - cycling represented in an ecosystem ?
(A) by structure and function
(B) by nutrient pyramid
(C) by composition of ecosystem
(D) food chain

(24) Which term is used for indicating inter-dependent food in net form ?

(A) food chain (B) food web (C) nutrient web (D) all given

Answers : (23-D), (24-B)

Trophic levels

- It is one kind of producer consumer arrangement. Where each food level is known as 'trophic level.'
- "Successive levels of nourishment in the food chains and food web of a community."

(25) In the arrangement of producers and consumers, each food level is known as ..... .

(A) food chain (B) food web (C) trophic level (D) food level

Answers : (25-C)

Ecological Pyramid

- The trophic structure and function at successive trophic levels may be graphically represented by means of ecological pyramid.
- Base of pyramid - first trophic level (producers)
- apex of pyramid - successive levels of trophic level

Three types of ecological pyramid

Pyramid of number

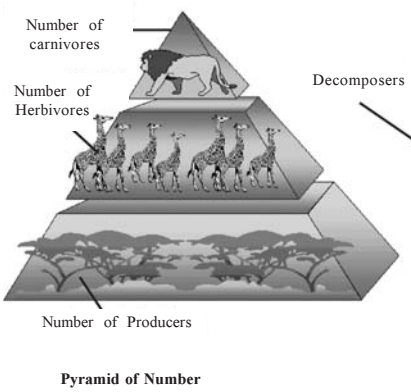
- shows the number of individual organisms
- producers occur in largest number
- top level carnivores are in the smallest numbers
- pyramid is upright

Pyramid of Biomass

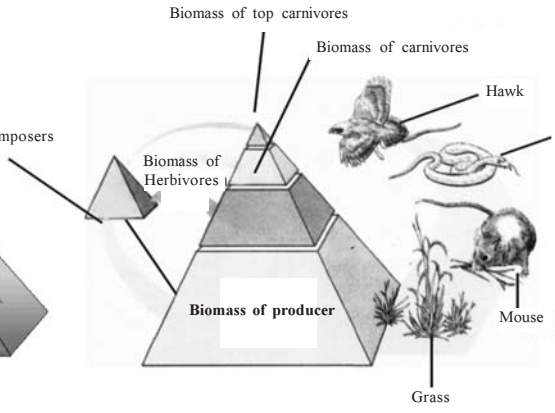
- it shows total dry weight or average biomass of organism at particular
- pyramid is upright

Pyramid of Energy

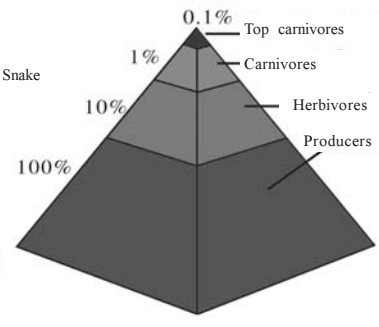
- it shows rate of energy at successive trophic levels
- amount of energy decreases
- pyramid is upright



Pyramid of Number



Pyramid of Biomass



Pyramid of Energy

(26) Base of pyramid is ..... and apex is ..... .

(A) first trophic level and producers (B) first trophic level and successive trophic levels

(C) food chain and food web (D) none of these

(27) Which pyramids are always upright ?

(A) Pyramid of number (B) Pyramid of Biomass

(C) Pyramid of energy (D) All given

(28) In number of pyramid number of producers is ..... and number of carnivores is ..... .

(A) more, less (B) too high, less (C) less, more (D) largest, smallest

(29) Depending upon the nature which pyramids are upright or inverted ?

(A) Biomass (B) Energy (C) Numbers (D) A, C both

Answers : (26-B), (27-C), (28-D), (29-D)

Energy flow :

- The behaviour of energy in ecosystem can be termed "energyflow" due to unidirectional flow of energy.
- Energy is the capacity to do work.
- The flow of energy in ecosystem is regulated by two important laws.

(i) According to first law of Thermodynamics, energy can neither be destroyed nor can it be created.

(ii) According to second law, whenever energy is converted from one form into another, some energy is dissipated in the form of heat.

- Some points to understand energy flow in an ecosystem.

(1) The efficiency of producer in absorption and conversion of solar energy.

(2) This energy used by consumers.

(3) The loss through respiration, heat, excretion.

(4) Total input of energy in form of food and its efficiency of assimilation.

(5) The gross production.

– In a food chain First trophic → producers

Second trophic → herbivores

Third trophic → carnivores

(30) In ecosystem energyflow is regulated by how many laws ?

(A) 1 (B) 2 (C) 3 (D) infinitive

(31) Through which processes heat is lost ?

(A) respiration (B) heat (C) excretion (D) all given

Answers : (30-B), (31-D)

*** Productivity ***

"The amount of organic matter or bio mass produced by an individual organism, population, community or ecosystem, during a given period of time is called productivity."

Types of Productivity :

(1) Primary productivity : In a given region the total production over a given period of time is called primary productivity of that ecosystem."

- measured : annual grams of dry weight (measured as tons) per hectare $g/m^2/yr$.
- It is further two types :

(A) Gross Primary Productivity (GPP) :

- "The total rate of photosynthesis carried out by all producer organisms in an ecosystem."
- It depends on chlorophyll content.
- measured : Chl/g dry weight/unit area, or photosynthetic number
- GPP = NPP + Respiratory loss

(B) Net Primary Productivity (NPP) :

- The rate of storage of organic matter in plant tissues in excess of the respiratory utilization by plants.
- It is referred to as the net primary production (NPP).
- It is located in the plant biomass.
- NPP = GPP - Respiratory loss

(2) Secondary Productivity : "The amount of energy storage at consumer (herbivores) level."

- Consumers obtain energy by consuming producers as food.

(3) Net Productivity : ''The amount of storage of organic matter not used by the heterotrophs (consumers)" .

- meausred as : days, months, season or year basis.

- Different ecosystems have different productivity.
  - high in forest, medium in grasslands and low in deserts.
  - In aquatic habitat decreases with increasing depth.

---

(32) What does formation of organic matter or biomass in an ecosystem call ?
(A) productivity (B) synthesis (C) production (D) all given

(33) It is measured as grams of dry weight (meausred as tons) per hectare $g/m^2/yr$.
(A) GPP (B) NPP (C) primary productivity (D) all given

(34) The total rate of photosynthesis carried out by all producer organisms in an eco-system is called ..... .
(A) GPP (B) NPP (C) primary productivity (D) all given

(35) Its value depends on the chlorophyll content.
(A) Primary productivity (B) GPP
(C) NPP (D) secondary productivity

(36) GPP = ... + Respiratory loss
(A) NPP (B) primary productivity
(C) Secondary productivity (D) NPP

(37) The amount of storage of organic matter not used by heterotrophs (consumers) is known as ..... .
(A) NPP (B) Primary productivity
(C) Secondary productivity (D) GPP

(38) In aquatic habitat productivity ..... with the increasing depth.
(A) increases (B) high (C) low (D) decreases

(39) What is right option for productivity in desert, grasslands and forest.
(A) high, medium, low (B) medium, high, low
(C) low, medium, high (D) high, low, medium

Answers : (32-C), (33-C), (34-A), (35-B), (36-A), (37-C), (38-D), (39-C)

Decomposition

"The process by which dead or complex organic material is broken down into simpler forms of matter which resembles with the soil."

- Decomposers : Bacteria, fungi and some worms break down the body of dead plants, animals and insects. They are called decomposers.
- They need to eat some of the dead things. So they can live and grow.
- Common decomposers are : bacteria, fungi, earthworms.
- The parts of inorganic matters that living plants take to grow are called nutrients.
- Decomposers help to provide these nutrients to plants.

Process of Decomposition : three stages.

(1) Fragmentation :

– First the detritus is fed upon by detritus-feeding animals.

– Changes occur in size and form of detritus when passes through the alimentary canal of these animals.

– The food is chewed, ground and filtered.

– Large surface area is made available for future degradation.

– These materials are added to the soil as faecal material of such animals.

(2) Leaching :

– It is a process in which water leaches deep down into the soil, so dissolved materials in the soil move towards the deeper layers of soil.

(3) Catabolism :

– Extracellular enzymes secreted by various kinds of fungi and bacteria begin the process of catabolic activities.

– First, complex organic matter is converted into simple organic forms and salts.

– Then they undergo to the processes of humification and mineralization.

– When humus is mineralized, these salts and ions are released in the forms which are available to plants.

(40) The process by which dead or complex organic material is broken down into simpler forms of matter which mixes with soil is called ..... .

(A) analysis (B) decomposition (C) leaching (D) fragmentation

(41) What do decompors need to eat to live and grow ?

(A) Organic matter (B) Inorganic matter (C) Dead matter (D) All given

(42) They are common decomposers of an ecosystem.

(A) bacteria, virus, fungi (B) virus, fungi, earthworms

(C) fungi, worms, virus (D) bacteria, fungi, earthworms

(43) In this stage of decomposition size and form of detritus changes.

(A) fragmentation (B) leaching (C) absorption (D) humification

(44) Humification and mineralization occur in this stage.

(A) fragmentation (B) leaching (C) catabolism (D) absorption

(45) When humus is ....., salts and ions are released in the forms which are available to plants

(A) Leaching (B) Ionisation (C) Mineralization (D) All the given

Answers : (40-B), (41-C), (42-D), (43-A), (44-C), (45-C)

*** Biogeochemical Cycles ***

"The cyclic pathways through which chemical elements travel from the environment into the organisms and from the organisms into the environment regularly and continuously are called biogeochemical cycles.

Bio - organisms ; geo - soil, rocks, water;

Chemical - C, N, S etc. chemical elements.

- Two types of cycles :

(1) Gaseous Cycle : Carbon cycle, Nitrogen Cycle, Oxygen Cycle

Carbon Cycle :

- Carbon occurs as CO_2 in atmosphere, as graphite and diamond in nature.
- Carbon is one of the most essential element of organic compounds.
- Green plants obtain carbon as CO_2 from air and by photosynthesis convert it into carbohydrate.
- When animals feed upon plants, thus carbohydrates enter into the bodies of these animals.
- In animals, in mitochondria during final respiratory pathway, carbohydrates are broken down yielding energy, where CO_2 and H_2O are the waste products.
- CO_2 releases in the air or water.
- Plants also liberate CO_2 in air during respiration.
- The burning of coal and fossil fuels (petroleun proudcts) also adds CO_2 in air.

Sedimentary Cycle : Phosphorus cycle, Sulphur cycle.

- Phosphorous Cycle : It is an example of sedimentary cycle.
- It is found in nucleic acids, cell membrane, phospholipids, NADP (Nicotinamide adenine dinucleotide phosphate), bones, teeth and ATP (Adenosine triphosphate) as a universal fuel of living organisms.
- The greatest reservoir of phosphates in the world lies in form of insoluble ferric and phosphates in rocks.
- Erosion caused by rainfall and the runoff of streams remove phosphorous from these rocks.
- This results in phosphorous supply in the soil which is available to plants.
- Animals obtain phosphorous from plants.
- After the death of living organisms, phosphorous returns to the water and soil in ion form and recycle in ecosystem.
- In higher plants, the symbiotic fungus (mycorriza) living on root systems helps in absorption of phosphate.
- A modern source of phosphorous is the common household detergents which now enter waste water system and then released into streams, lakes and estuaries.
- The phosphorous in soil is dissolved in water, in run flows into bodies of water.
- Some of this is used by plankton, which is eaten by fish, which are consumed by sea birds.
- But the majority of phosphorous washed into sea sinks to the ocean floor and is not recycled.
- The loss of phosphorous to the ocean has been greater than the gain to land.

(46) The cyclic pathway through which chemical elements travels from environment into organism and then in reverse mode regularly and continuously is called ..... .

(A) Bio-chemical cycle (B) Bio-geo cycle

(C) Geo-chemical cycle (D) Bio-geo chemical cycle

(47) They are the examples of gasesus cycle and sedimentary cycle respectively.

(A) N-Cycle, S-Cycle (B) P-Cycle, C-Cycle

(C) O_2-Cycle, C-Cycle (D) S-Cycle, P-Cycle

(48) In which form carbon is found in nature ?
(A) CO_2, Graphite (B) Graphite, Diamond (C) CO_2, Diamond (D) CO_2

(49) Which is the greatest reservior of phoshate in the world ?
(A) Soluble ferric and calcium phosphates in rocks.
(B) Insoluble ferric and calcium phosphates in rocks.
(C) Soluble ferric and calcium phosphates in water bodies.
(D) Insoluble ferric and calcium phosphates in water bodies.

(50) Which organisms release bacteria from the dead bodies as phosphate from organic phosphorous compound and makes available to plants ?
(A) Virus (B) Bacteria (C) Fungi (D) All given

(51) Which is the modern source of phosphorous ?
(A) Bathing detergent (B) Utensil detergent (C) Washing detergent (D) All given

(52) Which sequence shows right cycling of phosphorous in water bodies ?
(A) planktons - fishes - sea birds (B) fishes - planktons - large sea creatures
(C) planktons - fishes - large sized fishes (D) planktons - sea birds - fishes

(53) Which organisms living on root system of higher plants helps in absorption of phosphate ?
(A) Symbiotic fungi (B) Mycorriza (C) A , B both (D) None of these

(54) Which phosphorous is not recycled ?
(A) Lies in detergent (B) Sinks to the ocean floor
(C) Lies in dead bodies (D) All given

(55) The loss of phosphorous to the ocean has been ..... than the gain to land.
(A) greater (B) less (C) negligible (D) little

Answers : (46-D), (47-A), (48-B), (49-B), (50-B), (51-C), (52-A), (53-C), (54-B), (55-A)

***Carbon Fixation ***

Carbon fixation is important for both animals and plant life.

- It is conversion CO_2 into organic compounds during photosynthesis, whereby it changed into sugars.
- The calvin cycle is the most common biological method of carbon fixation.
- There are three ways of carbon fixation during photosynthesis.

(1) C_3 plants: During calvin cycle CO_2 incorporates to form a 3-carbon compound as the first stable compound.
eg. majority of terrestrial species.

(2) C_4 plants : Through calvin cycle CO_2 incorporate into 4-carbon compound.
- They have a distinctive internal leaf anatomy.
- e.g. sugar cane, maize plants

(3) CAM (Crassulacean Acid Metabolism) :

- These plants show adaptations for arid conditions.
- At night CO_2 enters through stomata and converted into organic acids, which releases CO_2 for use in calvin cycle during the day, when stomata are closed.

(56) How many ways of carbon fixation in plants during photosynthesis ?
(A) 1 (B) 3 (C) 4 (D) 6

(57) To get adaptations against arid conditions, some plants follow which path for CO_2 - fixation ?
(A) C_2 - path (B) C_3 - path (C) C_4 - path (D) CAM - path

Answers : (56-B), (57-D)

*** Pollution free Oxygen ***

- O_2 is good for healthy body and it is our primary life-support.
- clean air contains 19% - 21% O_2.
- In ancient times people lived longer because the air was composed of 40% O_2.
- The generation and maintenance of all our body processes are supported by four basic life-support components: carbohydrates, water, proteins, energy.
- Most scientists agree that O_2 is actually the over-riding key ingredient in all four of these life-support components.
- 80% of all our metabolic energy production is created by O_2.

Facts about Oxygen :

(1) Lack of oxygen in our universe is due to pollution, burning of fossil fuels and by CFC (Chloro Fluro Carbon) over all destruction of ozone layer.

(2) Everyday we breathe 20,000 times.

(3) Blood is the liquid carrier of O_2 that fuels all systems, stimulates chemical reactions and cleans itself of wastes and toxins.

(4) By mass, oxygen makes up 90% of the water molecule; water makes up 65-75% of human body.

(5) Cancer attacks every organ in our body, except the heart because of its abnormal sypply of oxygen.

(58) Clean air contains ..... oxygen ?

(A) 18-21% (B) 19-22% (C) 18-22% (D) 19-21%

(59) In ancient time air was composed of ..... oxygen.

(A) 20% (B) 40% (C) 36% (D) 41%

(60) How much % of all our metabolic energy production is created by oxygen.

(A) 65-70% (B) 19-21% (C) 40% (D) 80%

(61) They cause destruction in ozone layer.

(A) fossil fuel, CO_2 (B) fossil fuel, CFC

(C) fuels, CO_2 (D) fuels, CFC

(62) How many times we breath everyday ?

(A) 10,000 (B) 20,000 (C) 30,000 (D) 40,000

(63) What mass oxygen makes in water molecule and water in human body ?

(A) 80%, 65-70% (B) 90%, 55-70% (C) 90%, 65-70 (D) 65-70%, 90%

(64) Cancer attacks every organ in our body except this organ ..... .

(A) brain (B) kidney (C) liver (D) heart

Answers : (58-D), (59-B), (60-D), (61-B), (62-B), (63-C), (64-D)

A - Assertion R - Reason type MCQS

Choose the correct option of the following :

(A) A and R both are correct and R is the correct explanation of A.

(B) A and R both are correct but R is not correct explanation of A.

(C) A - True, R - False

(D) A - False, R - True

(65) Statement A : Ecosystem is a structural unit of ecology.

Reason R : Each ecosystem is composed of biotic community and environmental factors :

(A) (B) (C) (D)

(66) Statement A : Ecosystems of ponds and lakes are natural ecosystems.

Reason R : They are operated by themselves under natural conditions without interference by man.

(A) (B) (C) (D)

(67) Statement A : In any ecosystem, structure and functions are studied independently.

Reason R : The characteristic structure of an ecosystem is obtained by the systematic physical organisation of the biotic and abiotic components of that particular ecosystem.

(A) (B) (C) (D)

(68) Statement A : Heterotrophic components are constituted mainly by animals and microorganisms.

Reason R : The utilise, rearrange and decompose the complex materials.

(A) (B) (C) (D)

(69) Statement A : Decomposers are known as detritivores.

Reason R : They breakdown complex compound of dead or living protoplasm, absorb some of the decomposed or breakdown and release organic nutrients in environment.

(A) (B) (C) (D)

(70) Statement A : The energy flow begins when the producer organisms absorb sunlight.

Reason R : The material cycle begins when inorganic nutrients are absorbed by producer organisms.

(A) (B) (C) (D)

(71) Statement A : Detritus food chain begins with decomposers.

Reason R : Primary source of energy is dead organic matter called 'detritus'.

(A) (B) (C) (D)

(72) Statement A : Living organisms depend on each other for their food requirement and form a chain which is termed as food chain.

Reason R : The animals are inter-dependent for food and they form a net is called food chain.

(A) (B) (C) (D)

(73) Statement A : Successive levels of nourishment in the food chains and food webs of a community is called tropic level.

Reason R : Each tropic level contains organisms whose food is obtained from plants by the same number of steps.

(A) (B) (C) (D)

(74) Statement A : In every ecosystem pyramids of energy are always upright.

Reason R : Depending upon the nature of trophic levels pyramids of numbers and biomass may be upright or inverted.

(A) (B) (C) (D)

(75) Statement A : Secondary productivity is measured in dry weight per hectare $g/m^2/yr$.

Reason R : The amount of energy storage at consumer (herbivores) level is called secondary productivity.

(A) (B) (C) (D)

(76) Statement A : Bacteria, fungi and some worms breakdown the body of dead plants, animals and insects.

Reason R : Decomposers need to eat some of the dead things so they can live and grow.

(A) (B) (C) (D)

(77) Statement A : The organic matters that living plants take to grow are called nutrients.
Reason R : Green plants make their own food, but they also need to get nutrients from the soil.
(A) (B) (C) (D)

(78) Statement A : When humus is mineralised, salts and ions are available to plants.
Reason R : Extracellular enzymes secreted by various kinds of fungi and bacteria begin the process of anabolic activities.
(A) (B) (C) (D)

(79) Statement A : The greatest reservior of phosphates in the world lies in soluble ferric and calcium forms in rocks.
Reason R : After the death of living organisms, phosphorous adds to soil from their bodies in ion form.
(A) (B) (C) (D)

(80) Statement A : Cancer attacks every organ in our body except the heart.
Reason R : Heart gets abnormal sypply of oxygen.
(A) (B) (C) (D)

Answers : (65-B), (66-A), (67-D), (68-A), (69-C), (70-B), (71-A), (72-C), (73-A), (74-C), (75-D), (76-A), (77-D), (78-C), (79-D), (80-A)

- **True - False type questions**

(81) (1) The term ecosystem was first proposed by A.G. Tansley (1965).
(2) Individuals of each species can produce population in the same time and space.
(3) Ecosystem is a structural unit of ecology.
(4) The biotic and abiotic components are linked with each other through undirectional energy flow and nutrient cycling.
(A) FFTT (B) FTTT (C) TFFT (D) TTTT

(82) Choose appropriate statements for structure of Ecosystem.
(1) The range or gradient of conditions of existance.
(2) Biological or Ecological regulation.
(3) The quantity and distribution of non -living materials.
(4) The composition of biological community.
(A) TTTF (B) TTFT (C) TFTT (D) FTTT

(83) Choose right option for Abiotic components of an ecosystem.
(1) As macronutrietns C, H, S are present.
(2) Micronutrients are Ca, Mg, Mo, Cu.
(3) Proteins, carbonydrates and lipids are organic materials.
(4) Environmental factors and physical factors act as Climatic factors.
(A) TFTF (B) FTFT (C) TTFT (D) TFFT

(84) (1) Clean air contains 19% - 21% oxygen.

(2) Four basic life components support the generation and maintenance of all our body processes.

(3) All metabolic processes in our body are regulated by oxygen.

(4) Oxygen produces 70% of all our metabolic energy.

(A) FTTF (B) FFTT (C) FTTF (D) FTFT

(85) (1) Specific bacteria release phosphorous from dead bodies as phosphate from organic phosphorous compounds.

(2) After the death of organisms, phosphorous release in salts form in the soil.

(3) Majority of phosphorus washed into the water bodies to the floor.

(4) For biological systems phosphorous is an essential sedimentary cycle.

(A) TTFT (B) TFFT (C) FTTT (D) TTTT

(86) Choose appropriate option for catabolism process of decomposition.

(1) Extracellular enzymes secreted by various kinds of fungi and worms begin the process of catabolic activities.

(2) First of all complex organic matter is converted into simple organic, then to inorganic ions and salts.

(3) Then they undergo mineralization and humification respectively.

(4) When humus is mineralized, salts and ions are released in the forms which are available to plants.

(A) TFTF (B) FTTT (C) TTFT (D) FTFT

(87) (1) Energy remains after used by producers, is available to the consumers is called Net productivity.

(2) GPP depends on the Chlorophyll content.

(3) The amount of stored organic matter not used by heterotrophs is called net productivity.

(4) The productivity increases as increasing depth in aquatic habitat.

(A) TFTT (B) TFTT (C) FTTF (D) FTFF

(88) Choose appropriate option from energetic point of view in an ecosystem.

(1) The efficiency of producers in absorption and conversion of solar energy.

(2) The use of converted chemical energy by producers.

(3) Total input of energy in form of food and its efficiency of cycling.

(4) Gross net productivity.

(A) TFFF (B) TTFF (C) TFFT (D) TFTT

(89) Choose right option for ecological pyramids.

(1) Producers are in the largest number and top level of carnivores are in the smallest number.

(2) Biomass shows the total dry weight of organisms at a particular trophic level.

(3) It shows rate of energy or productivity at successive trophic levels.

(4) According to first law of thermodynamic, during energy conversion some energy is dissipated in the form of heat.

(A) TTTT (B) TTTF (C) TFTT (D) FTTF

Answers : (81-B), (82-C), (83-A), (84-A), (85-B), (86-D), (87-C), (88-A), (89-B)

(90)

Column - I	Column - II
(I) the composition of biological community	(p) water
(II) range or gradient of conditions of existence	(q) life-cycle (life-history)
(III) the quantity and distribution of non-living materials	(r) biological or ecological regulation
(IV) function	(s) temperature

	I	II	III	IV
(A)	s	q	p	r
(B)	s	r	p	q
(C)	q	s	p	r
(D)	r	q	p	s

(91)

Column - I	Column - II
(I) Net productivity	(p) It depends on the chlorophyll content.
(II) secondary productivity	(q) Energy remains after use by producers, is available to con sumers.
(III) GPP	(r) It is consolidated on days, month, season or year basis.
(IV) NPP	(s) The amount of energy storage at consumer (herbivores) level

	I	II	III	IV
(A)	r	p	q	s
(B)	r	s	p	q
(C)	s	r	p	q
(D)	p	r	q	s

(92)

Column - I	Column - II
(I) fragmentation	(p) materials are added to the soil as faecal material of animals
(II) leaching	(q) complex organic matters is converted into inorganic ions and salts
(III) catabolism	(r) dissolved mateirals leach in the soil move towards the deeper layers of soil

	I	II	III
(A)	q	p	r
(B)	r	p	q
(C)	p	r	q
(D)	q	r	p

Answers : (90-C), (91-B), (92-C)

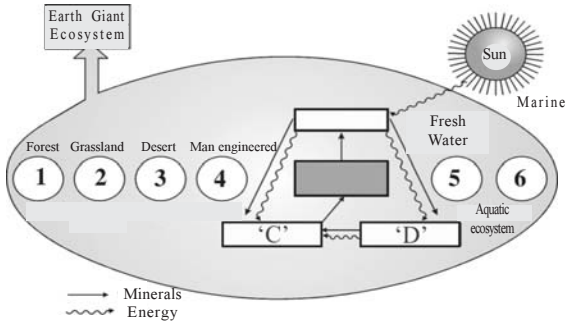
(93) What does 'C' indicates in given diagram.

(A) They are autotrophs.

(B) They belong to second trophic level

(C) They decompose living or dead matters.

(D) B and C both



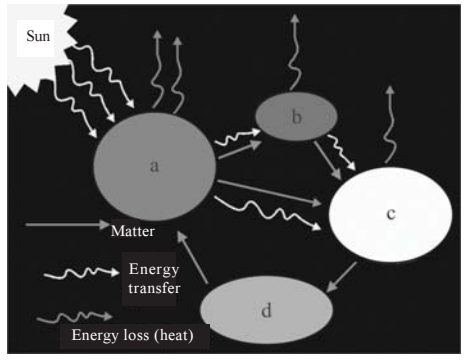
(94) What does 'a' indicate in given diagram.

(A) They are food source to another organisms.

(B) They fix the energy.

(C) They initiate flow of energy.

(D) All given



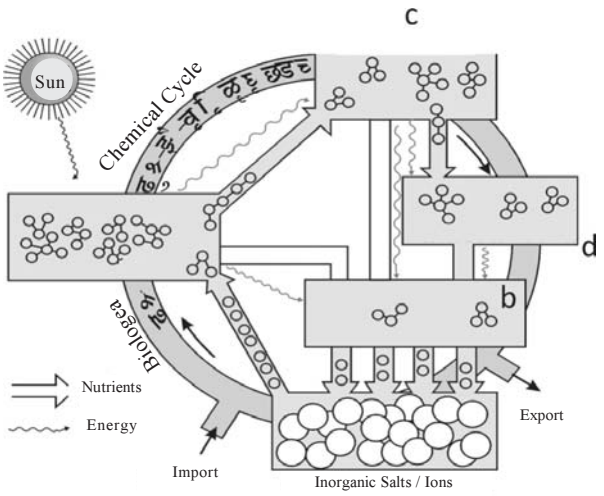
(95) Choose write option for a, b, c, d for diagram.

(A) a = producers b = herbivores
c = carnivores d = decomposers

(B) a = producers b = decomposers
c = carnivores d = herbivores

(C) a = producers b = decomposers
c = herbivores d = carnivores

(D) a = producers b = herbivores
c = decomposers d = carnivores



Ecosystem Functioning

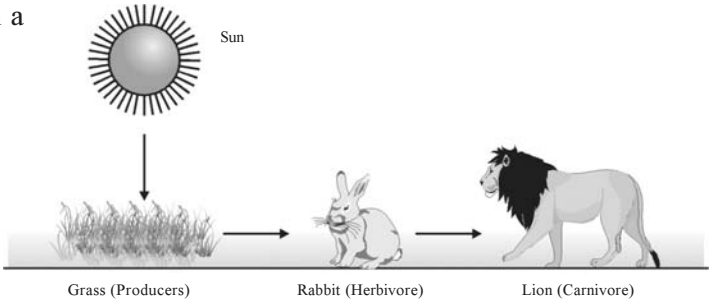
(96) Choose correct option for given diagram.

(A) Living organisms depend on each other for their food requirement and form a chair called food chain.

(B) Interactions of animals for food is called food web.

(C) It is a grazing food chain.

(D) A and C both.



Grassing Food Chain

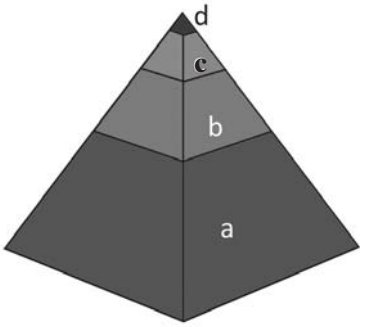
(97) Choose ascending order of energy pyramid.

(A) a → b → c → d

(B) d → c → b → a

(C) c → d → b → a

(D) b → d → c → a



Pyramid of Energy

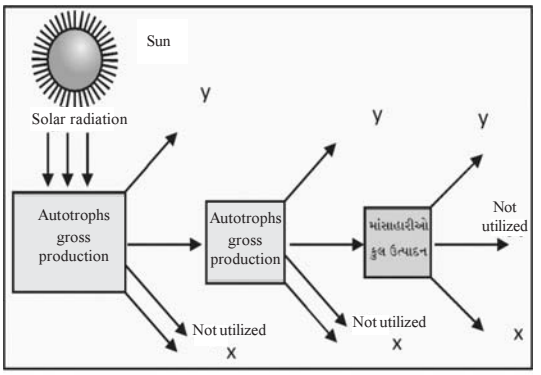
(98) What is 'x", 'y' indicate in given diagram ?

(A) x= decomposition, y = respiration

(B) x= energy y = decomposition

(C) x = respiration y = decomposition

(D) x= decomposition y = energy



Single Channel energy flow

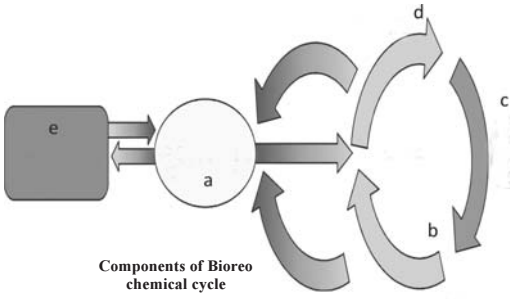
(99) Which is right labeling for 'b' and 'c' in given diagram ?

(A) b = exchange pool, c = consumers

(B) b = exchange pool, c = biotic community

(C) b = reservoir c = exchange pool

(D) b = biotic community c = reservior



Components of Bioreo chemical cycle

Answers : (93-C), (94-A), (95-C), (96-D), (97-B), (98-C), (99-D)

*** NEET Questions ***

(100) Pyramid of numbers in a grassland / true ecosystem is ..... .

(A) always inverted (B) always upright (C) both A and B (D) spindle shaped

(101) Food chain in which micro organims breakdown the food formed by primary producers is ..... .

(A) parasitic food chain (B) detritus food chain

(C) consumer food chain (D) predator food chain

(102) Pick up the correct food chain.

(A) Grass → Chameleon → Insect → Bird (B) Grass → Fox → Rabbit → Bird

(C) Phytoplankton → Zooplankton → Fish (D) Fallen leaves → Bacteria → Insect larval

(103) Study of inter - relationship between organisms and their environment is ..... .

(A) ecology (B) ecosystem (C) phytogeography (D) ethology

(104) Pyramid of numbers deals with numbers of ..... .

(A) species in an area (B) individuals in a community

(C) individuals in a trophic level (C) subspecies in a community

(105) Pyramid of numbers in a pond ecosystem is ..... .

(A) irregular (B) inverted (C) upright (D) spindle shaped

(106) The pyramid which can not be inverted in a stable ecosystem is that of ..... .

(A) Biomass (B) Number (C) Energy (D) All the above

(107) In grass → deer → tiger → food → chain, grass biomass is one ton. The tiger biomass shall be ..... .

(A) 100 kg (B) 10 kg (C) 200 kg (D) 1 kg

(108) In a biotic community the most important factor for survival of an animal is ..... .

(A) day length (B) green food

(C) Soil moisture (D) predators

(109) Second most important trophic level in a lake is ..... .

(A) Zooplankton (B) Phytoplankton (C) Benthos (D) Neuston

(110) Which of the following does not have stomata ?

(A) Hydrophytes (B) Meophytes (C) Xerophytes (D) Submerged hydrophytes

(111) Bulk CO_2 fixation occurs in ..... .

(A) Crop plants (B) oceans

(C) Tropical rain forests (D) Temparature forests

(112) If we completely remove the decomposers from an ecosystem, its fuctioning will be adversely affected because ..... .

(A) energy flow will be blocked (B) herbivores will not receive solar energy

(C) mineral movement will be blocked (D) rate of decomposition will be very high

(113) The primary succession refers to the development of communities on a ..... .

(A) fleshly cleared crop field

(B) forest clearing after devastating fire

(C) pond, freshly filled with water after a dryphase

(D) newly exposed habitat with no record of earlier vegetation

(114) Which of the following is the most stable ecosystem ?

(A) Forest (B) Desert (C) Mountain (D) Ocean

(115) In a biotic community the primary consumers are ..... .

(A) carnivores (B) omnivores (C) detritivores (D) herbivores

(116) Which of the following pairs is a sedimentary type of biogeochemical cycle ?

(A) Oxygen and nitrogen (B) Phosphorus and sulphur

(C) Phosphorus and nitrogen (D) Phosphorus and carbon dioxide

(117) In a food chain the largest population is that of ..... .

(A) decomposers (C) primary consumers

(C) producers (D) tertiary consumers

(118) The transfer of energy from one trophic level to another is governed by the 2nd law of thermodynamics. The average efficiency of energy transfer from herbivores to carnivores is ..... .

(A) 5 % (B) 10 % (C) 25 % (D) 50 %

(119) Which of the following ecosystems has highest rate of gross primary production ?

(A) Grass lands (B) Coral reefs

(C) Mangroves (D) Equatorial rain forest

(120) The rate at which light energy is converted to the chemical energy of organic molecules is the ecosystem's ..... .

(A) net primary productivity (B) gross primary productivity

(C) net secondary productivity (D) gross secondary productivity

(121) What is true of ecosystem ?

(A) Primary consumers are least dependent upon producers.

(B) Primary consumers out - number producers.

(C) Producers are more than primary consumers.

(D) Secondary consumers are the largest and most powerful.

(122) In an ecosystem, which one shows one way passage ?

(A) Free energy (B) Carbon (C) Nitrogen (D) Potassium

(123) Pneumatophores are characteristic of plants growing in ..... .

(A) saline soils (B) sandy soils

(C) marshy places and salt lakes (D) dryland regions

(124) The greatest biomass of autotrophs in the oceans is that of ..... .

(A) sea grasses and slime moulds

(B) free floating microalgae, cyano bacteria and nanoplankton

(C) benthic brown algae, coastal red algae and daphnids

(D) benthic diatoms and marine viruses

(125) Bamboo plant is growing in a forest than what will be trophic level of it ?

(A) First trophic level (B) Second trophic level

(C) Third trophic level (D) Fourth trophic level

(126) Which of the following is expected to have the highest value ($gm/m^2/yr$) in a grassland ecosystem ?

(A) Secondary Production (B) Tertiary Production

(C) Gross Production (GP) (D) Net Production (NP)

(127) If by radiation all nitrogenase enzymes are inactivated, then there will be no ..... .

(A) Fixation of nitrogen in legumes (B) Fixation of atmospheric nitrogen

(C) Conversion from nitrate to nitrite in legumes (D) Conversion from NH_3 to nitrate in soil

(128) An ecosystem which can be easily damaged but can recover after sometime if damaging effect stops will be having ..... .

(A) low stability and high resilience (B) high stability and low resilience

(C) low stability and low resilience (D) high stability and high resilience

(129) Which one of the following is not used for construction of ecological pyramids ?

(A) Number of individuals (B) Rate of energy flow

(C) Fresh weight (D) Dry weight

(130) Which one of following ecosystem types has the highest annual net primary productivity ?

(A) Tropical deciduous forest (B) Temperate evergreen forest

(C) Temperate deciduous forest (D) Tropical rain forest

(131) The slow rate of decomposition of fallen leaves logs in nature is due to their ..... .

(A) low moisture content (B) poor nitrogen content

(C) anaerobic environment around them (C) low cellulose content

(132) Consider the following statements concerning food chains.

(a) Removal of 80% tigers from an area resulted in greatly increase growth of vegetation.

(b) Removal of most of the carnivores resulted in an increased population of deers.

(c) The length of food chain is generally limited to 3-4 trophic levels due to energy loss.

(d) The length of food chain may vary from 2 to 8 trophic levels.

(133) Which one of following types of organisms occupy more than one trophic level in a pond ecosystem?

(A) Fish (B) Zooplankton (C) Frog (D) Phytoplankton

(134) The biomass available for consumption by the herbivores and decomposers is called ..... .

(A) net primary productivity (B) secondary productivity

(C) standing crop (D) gross primary productivity

(135) Which are of following statements for pyramid of energy is incorrect, whereas the remaining three are correct ?

(A) Its base is broad.

(B) It shows energy content of different trophic level organisms.

(C) It is inverted in shape.

(D) It is upright in shape.

(136) Which one of following animals may occupy more than one trophic levels in the same ecosystem at same time ?

(A) Sparrow (B) Lion (C) Goat (D) Frog

(137) Both hydrarch and xerarch successions lead to ..... .

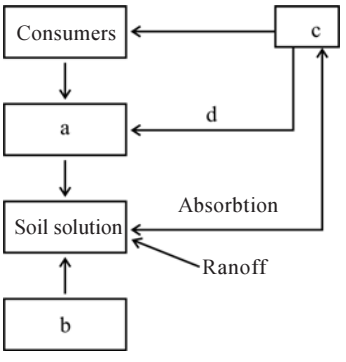
(A) medium water conditions (B) xeric conditions

(C) highly dry conditions (D) excessive net conditions

(138) If storage energy at producer level is 20J, so what mole of energy is available in a food chain ?

(A) 0.02J (B) 0.002J (C) 0.2J (D) 0.0002J

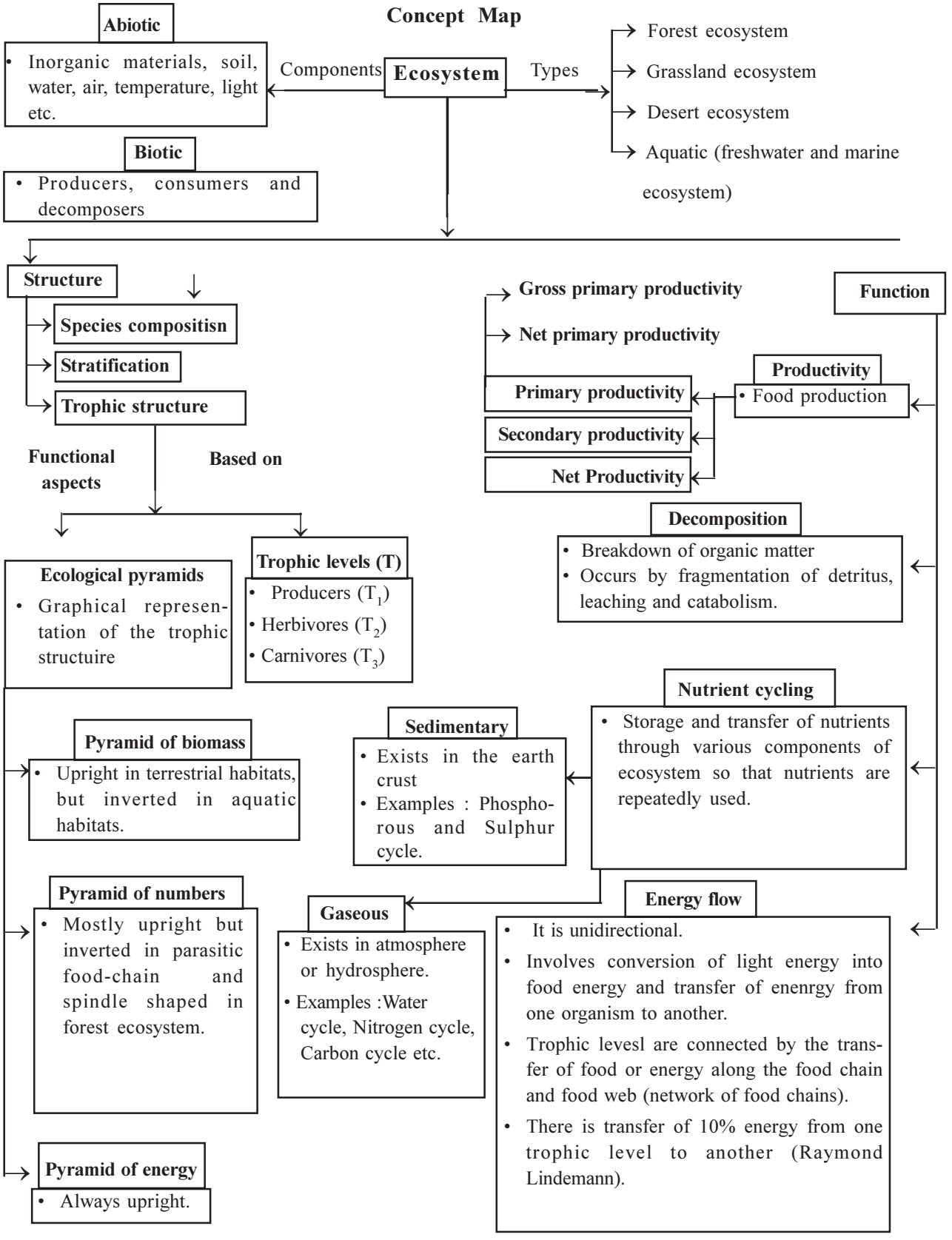
(139) Given chart shows phosphorous cycle. What does a, b, c, d indicates in given diagram ?



	a	b	c	d
(A)	minerals in rocks,	saprophytes,	crop harvest,	producers
(B)	crop harvest,	producers,	mineral rocks,	Saprophytes
(C)	saprophytes,	mineral rocks,	producers,	crop harvest
(D)	producers,	crop harvest,	mineral rocks,	saprophytes

Answers : (100-B), (101-B), (102-C), (103-A), (104-C) (105-C), (106-C), (107-B), (108-C), (109-A), (110-B), (111-D), (112-C), (113-D), (114-D), (115-D), (116-B), (117-B), (118-B (119-D), (120-B), (121-B), (122-A), (123-C), (124-B), (125-A), (126-C), (127-A), (128-A), (129-C), (130-D), (131-D), (132-A), (133-A), (134-A), (135-C), (136-A), (137-A), (138-A), (139-C)

Concept Map


Abiotic
• Inorganic materials, soil, water, air, temperature, light etc.
Components
Ecosystem
Types
Forest ecosystem
Grassland ecosystem
Desert ecosystem
Aquatic (freshwater and marine ecosystem)
Biotic
• Producers, consumers and decomposers
Structure
Species compositisn
Stratification
Trophic structure
Functional aspects
Based on
Ecological pyramids
• Graphical represen-tation of the trophic structuire
Trophic levels (T)
• Producers (T_1)
• Herbivores (T_2)
• Carnivores (T_3)
Pyramid of biomass
• Upright in terrestrial habitats, but inverted in aquatic habitats.
Pyramid of numbers
• Mostly upright but inverted in parasitic food-chain and spindle shaped in forest ecosystem.
Pyramid of energy
• Always upright.
Function
Gross primary productivity
Net primary productivity
Productivity
• Food production
Primary productivity
Secondary productivity
Net Productivity
Decomposition
• Breakdown of organic matter
• Occurs by fragmentation of detritus, leaching and catabolism.
Nutrient cycling
• Storage and transfer of nutrients through various components of ecosystem so that nutrients are repeatedly used.
Sedimentary
• Exists in the earth crust
• Examples : Phospho-rous and Sulphur cycle.
Gaseous
• Exists in atmosphere or hydrosphere.
• Examples :Water cycle, Nitrogen cycle, Carbon cycle etc.
Energy flow
• It is unidirectional.
• Involves conversion of light energy into food energy and transfer of enenrgy from one organism to another.
• Trophic levesl are connected by the trans-fer of food or energy along the food chain and food web (network of food chains).
• There is transfer of 10% energy from one trophic level to another (Raymond Lindemann).

29 Biodiversity and Its Conservation

Introduction :

- Numerous kinds of microbes, plants and animals live in entire biosphere.
- **Tall plants :**
  - sequoia sempervirens (Gymnosperm) 400 feet / 150 mt
  - Eucalyptus sp. (Angiosperm) 100 mt
- **Smallest plants :**
  - Zamia pygmea (Gymnosperm)
  - Wolffia globosa (Angiosperm)
- **Microscopic :**
  - bacteria, yeast (flora)
  - amoeba, euglena (fauna)
- **Large sized mammals :**
  - aquatic mammals - whales, dolphins
  - terrestrial mammal - elephant
- It shows diversity among many aspects like shape, volume, size, colour, life style .....
- 20,000 species of ants
- 3,00,000 species of beetles
- 28,000 species of fishes
- 20,000 species of orchids

---

(1) Sequential form of tallest and smallest angiospermic plants ?

(A) Sequoia, wolfia (B) Eucalyptus, zamia

(C) Eucalyptus, wolffia (D) Sequoia, zamia

(2) Large size mammal ..... .

(A) elephant (B) whale

(C) dolphin (D) all given

Answers : (1-C), (2-D)

Levels of Biodiversity

Three levels : (1) Genetic diversity (2) Species diversity (3) Ecosystem diversity.

(1) **Genetic Diversity :**

- Genes of all organisms are made of DNA.
- The number of genes and individuals of each species are fixed.
- India has more than 50,000 genetically different strains of rice and 1000 varieties of mango.
- "The variation in the constitution of genes is called genetic diversity of that species."
- Any one gene has different alternative. (Allele)
- Genetic diversity refers to totality of all inherited genetic variations within a population.
- All genes of a population are called gene pool.
- Due to genetic diversity the natural selection can operate in evolution.
- Thus, Genetic diversity is basis of formation of new species.

(2) Species Diversity : It is determined by various species which live there.

- The greater the number of species the higher is the species diversity.
- Western Ghats have greater amphibian species diversity than the Eastern Ghats.
- It also increases the genetic diversity of that area.
- It is very important for the normal functioning and sustainability of ecosystems.
- It is measured by number of species live there.
- Larger the area, more the number of species.
- R.H. Whittaker (1965) pointed three types of species diversity :

α-diversity

- It is the relative richness of different species in an habitat.
- The percentage of each and every species within a community.

β-diversity

- It is relative richness of different species along a gradient from one habitat to another habitat within community.
- Birds of pond and birds of agriculture area of same grassland

γ-diversity

- It refers to richness of different species in a range of habitats within a geographical area.
- eg. ocean, lakes, ponds, mountains, forests

(3) Ecological Diversity : "The variation in species richness in different ecosystems of a geographical area."

- It differs from ecosystem to ecosystem.
- More the diversity in ecosystem; more efficiences, productivity and stability is of ecosystem.

---

(3) The variation in the constitution of genes of one species is called ..... .
(A) genetic diversity (B) species diversity
(C) ecological diversity (D) Bio-diversity

(4) What is called the sum total of all genes of a population ?
(A) Gene-frequency (B) Gene-complex (C) Gene pool (D) Gene-mutation

(5) Which diversity is determined by various species of a given area ?
(A) Genetic diversity (B) Species diversity (C) Ecological diversity (D) All given

(6) This diversity is very important for normal functioning and sustainability of ecosystems.
(A) Genetic diversity (B) Species diversity (C) Ecological diversity (D) Bio diversity

(7) Which scientist suggested different types of species diversity ?
(A) Odum (B) Tansley (C) Eichlar (D) R.H.Whittaker

(8) How many types of species-diversity is found ?
(A) 3 (B) 2 (C) 4 (D) 6

(9) The percentage of each and every species within a community belongs to which kind of species diversity ?
(A) α (B) β (C) γ (D) δ

(10) Diversity refers to different species in a range of habitats within a geographical area is called ?
(A) α - diversity (B) β- diversity (C) γ- diversity (D) δ- diversity

Answers : (3-A), (4-C), (5-B), (6-B), (7-D), (8-A), (9-A), (10-C)

* The composition of Biodiversity *

genetic diversity	organization diversity (species)	ecological diversity
↓	↓	↓
populations	kingdom	biomes
↓	↓	↓
species	phyla	ecosystems
↓	↓	↓
chromosomes	families	habitats
↓	↓	↓
genes	genera	niches
↓	↓	↓
nucleotides	species	populations
	↓	
	sub species	
	↓	
	variety	

- Define : "Biodiversity refers to the vareity of microbes, plants and animals of an area."
- It shows totality of ecosystem genes and species of an area.
- It shows hitrarchial levels of diversity of nature.
- It means "variety and variability of life."
- AIMS of biodiversity :
- The knowledge of biodiversity is very important for systematic studies of plants and animals.
- Identification of hot spots
- Ecosystem studies
- Biogeographic studies
- Proposing advance conservation programmes
- Enriching quality of resources
- Conservation of global ethics
- Gandhiji had wrote that, "The nature has capacity to satisfy the need of man, but not greed."

*** Patterns of Biodiversity :**

- **Latitudinal gradients :**
- The diversity of plants and animals is not uniform throughout the world and shows an uneven distribution.
- As we go from polar region to the equator biodiversity increases.

  (trophics latitudinal range of 23.5^{o}N to 23.5^{o} S)
- This localities at lower lattitudes have more species than localities at higher latitudes.
- It is referred as latitutinal gradient in species diversity.

- Colombia located near the equator has nearly 1400 species of birds.
- While New York at 41°N has 105 species and Greenland at 71°N (North) only 56 species.
- India, with much of its land area in the tropical latitudes, has more than 1200 species of birds.
- A forest in tropical region like ecuador has upto 10 times as many species of vascular plants as a forest of equal area in temperate region like the midwest of the USA.
- The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth.
- Where, - 40,000 species of plants
  - 3000 species of fishes
  - 427 species of mammals
  - 427 species of amphibians
  - 378 species of reptiles
  - more than 1,25,000 species of invertebrates.
- Scientists estimated that in this rain forest there might be atleast two million insect species waiting to be discovered and named.

(11) What does biodiverstiy indicate in nature ?

(A) class (B) designation (C) taxon (D) A and B both

(12) As we go from polar region to the equator, the bio-diversity ..... .

(A) decreases (B) increases (C) lower (D) not certain

(13) In India, which region has large land area ?

(A) Temperate region (B) Tropical latitudes

(C) Tundra region (D) All given

(14) The ..... rain forest in ..... , has the greatest biodiversity on earth.

(A) tropical, South America (B) temperate, North America

(C) temperate, South America (D) tropical, North America

Answer : (11-D), (12-B), (13-B), (14-A)

*** High biodiversity in tropics :**

- Ecologists and evolutionary biologists have proposed various hypothesis for greater biodiversity in tropics.

(a) Speciation : It is the formation of new and distinct species in the course of evolution.

- It is time consuming process.
- Tropical latitudes have remained relatively undisturbed for millions of years.
- Where frequent glaciations occured in temperate regions in past.
- It had a long evolutionary time for species diversification.

(b) Effect of environment : Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable.

(c) Species - energy hypothesis : More solar energy available in the tropics, which contributes to higher productivity, which resulted indirectly in high biodiversity.

(d) Species-Area - Relationship : It is a relationship between the area of habitat or part of a habitat and the number of species found in that area.

- According to German naturalist and geographer Alexander Von Humboldt the number of species increases as the size of a geographical area increases but only upto a limit.
- The relation between species richness and area for a wide variety of taxa turns out to be a rectangular hyperbola.

- **Importance of Biodiversity :**

(1) Food : It provides food for man and their pets.

- e.g. Furniture wood, agriculture product, aquatic food, animal husbundary product, medicines, papers, clothes, pesticids.
- Plants are consumed by animals as food.
- It is the consumptive value of biodiversity.

(2) Productive use value : The marketable items of bio-diversity form the productive use value.

- **eg,** Antibiotics are synthesized by microbes.
- **eg,** Penicillin from fungus.

(3) Aesthetic and Cultural Benefits :

- e.g. furniturewood, agriculture products, aquatic food, animal husbundory product, pearls, medicine, papers, cotton, pesticide (neem tree).
- Ecotourism, bird-watching, wildlife, pet keeping, gardening etc. are examples of bio-diversity.
- Throughout human history people have related bio-diversity, to the very existence of human race through cutural and religious believes.
- **eg. :**
  - Oscimum sanctum (Tulsi)
  - Calotropis procera (Akdo)
  - Polyalthia longifolia (Asopalav)
  - Ficus religiosa (Pipal)
  - Prosopis cineraria (Khejdo).....various other trees are planted.
  - Several birds and even snakes have been considered sacred.
  - India : tiger — National animal
    peacock — National bird
  - Symbols of Gujarat
    - Asiatic Lion
    - Flamingo (bird)
    - Indian laburnum (wild flower)
    - Neem (tree)

(4) Ecosystem services :

- Biodiversity is important in the maintenance of biosphere and its ecosystems.

- Forests are the main mechanism for the conversation of CO_2 into carbon and oxygen.
- Green house effect, maintenance of global temperature, oceanic currents and winds for maintenance of climate, regulation of rain fall maintained by forests conservation of soil, pollinations of plants and purification of ground water are made possible due to biodiversity
- The ecosystem services have been valued in the range of 16 to 54 trillion (10^{12}) US dollar per year.

---

(15) Formation of new and distinct species in the course of evolution is called ..... .

(A) evolution (B) succession (C) bio-diversity (D) speciation

(16) The environment of this area relatively more constant and predictable.

(A) Temperate (B) Tropical

(C) Amazon rain forests (D) All given

(17) Which naturalist and geographer has suggested that the number of species increases as the size of geographical area increases but only upto a limit.

(A) Odum (B) Humboldt

(C) Tansley (D) Eichlar

(18) The ecosystem service value per year ..... .

(A) 61 to 74 trillion US dollar

(B) 45 to 61 trillion US dollar

(C) 16 to 54 trillion US dollar

(D) 16 to 45 trillion US dollar

Answer : (15-D), (16-B), (17-B), (18-C)

***Biodiversity at world, National and Gujarat levels :**

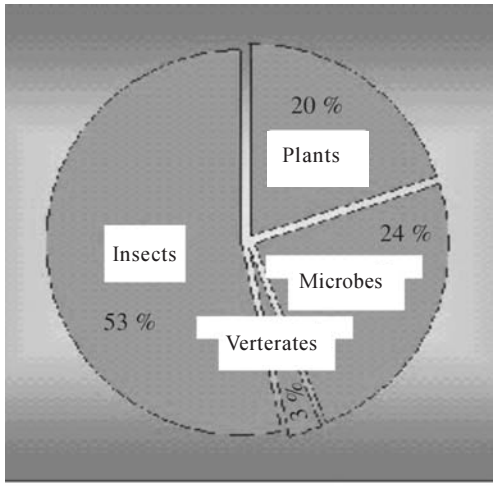
(A) Biodiversity at World Level :

- Estimated species may be approximately 50 lacs to 5 crore.
- Approximately about 1.7 to 1.8 millions species are identified till today.
- That includes - 3,00,000 species of angiosperms.

  - 8,00,000 species of insects

  - 40,000 species of vertebrates

  - 3,60,000 species of microorganisms



Biodiversity at global level

- Largest number of species in the world — Plants - Angiosperms; animal- Arthropods kingdom
- The number of insects alone may be as high as 10 millions.
- More than half of the species on the earth live in moist tropical forests. Which are only 7% of the total land surface.

(B) Biodiversity at National level :

- India is one of the twelve mega biodiversity countries of the world.
- The richness of biodiversity in India is mainly due to wide variety of climatic and altitudinal conditional characteristics of this country.
- The climate varies from humid tropical area (Western Ghats) to icy mountains (Trans-Himalaya) via hot desert (Thar) and plains.
- Ten geographical regions of our country form a wide variety of ecological habitat which give shelter to so many kinds of plants and animals.
- India covers only 2.4% of total area of the world, but it has 11% of the world's total biota, i.e., 45,500 member of species of plants and 86,874 species of animals.
- India has more than 7% of the total animal species in the world.
- Indian flora is the confluence of flora from the surrounding countries like Malaya, China, Tibet, Japan and Europe and even from distant countries like America, Africa and Australia.
- About 5000 species of flowering plants belonging to 141 genera of 47 families had originated in India.
- Hundreds of species of grasses.
- India is a source of traditional crop varieties ranking first amongst the 12 regions of diversity of crop plants and seventh so far in contribution to agricultural species.
- India is origin place of 166 species of crop plants, 320 species of wild relatives of cultivated crops.
- At present 30,000 to 50,000 varieties of cultivated plants are found in India.
- Total number of plant species in India.
  - 4000 species used in medicine
  - 3000 used as food
  - 500 yield fibre
  - 400 species as fodder
  - 300 species yield gum
  - 100 species used to extract essential oils and scents
- **Marine bio-diversity in India :**
  - coastline 7500 km.
  - productive ecosystems : mangroves, coral reefs, estuaries, lagoons.
  - 45 species of mangrove plants
  - 341 species of coral reefs

Endemism : "The occurance of a taxa in a small area and no where else, is called endemism.

- location : North-east, Western ghats, Andaman and Nicobar islands.
- birds representing about 14% of global avian fauna show relatively high endemism.
- endemic birds : Nilgiri pipit, Nilgiri wood pigeon, Malbar parakeet, Rofous babbler.

(C) Biodiversity of Gujarat :

- Gujarat is home of nearly 7000 species of plants and animals.
- It is famous for its unique saline desert called Rann of Kachchh.
- The coral reefs in Gulf of Kachchh as widely admired for their amazing beauty and spectacular diversity.
- Gujarat as seen in the 4320 species of plants, 2198 species of higher plants belonging to 901 genera and 155 families, 2,728 species of animals.

- In Gujarat : 14 % fishes
  9 % amphibians
  19 % reptiles
  37 % birds
  25 % mammals
- Flamingo city : The area between khadir and Paccham islands in the great Rann of Kachchh, is the largest breeding ground of flamingos, a migratory species.
- The great Rann of Kachchh and the little Rann of Kachchh are the worlds only saline desert spread in 25,000 sq.km.
- The Wild Ass Sanctuary is a unique habitat of Indian Wild Ass, the remnant gene pool in the world.

Specialities :

- Bhal region (arid and saline area) known for its indigenous varieties of "Bhalia" and "Rata" Wheat.
- Painted frog (rare and endangered sp) is found in Vansada National Park (Valsad District)
- Safed Musli is available from Vijaynagar forest (Sabarkantha District) and (Danta forest of Banaskantha district)
- Velavedar Black buck National Park (Bhavnagar District), has dense population of Blackbucks.

- Girnar forest (Junagadh district) is a home of Asiatic Lion.
- Narayan Sarovar chinkara sanctuary (Kachchh District) rich in ecosystem in arid region of Gujarat and endangered species such as Guggal among plants and Chinkara, Lesser-Florican, Spring Tailed Lizard among animals.
- Ratanmahal Sloth Bear sanctuary (Panchamahal District) is important habitat for sloth bear.
- Giant flying squirrel is found in shool paneshwar wild life Sanctuary.
- Marine Wild life sanctuary and National Park is famous for its rich coral, marine algae and mangrove diversity.
- Many migratory and local migratory birds take visit to Nalsarovar Bird Sanctuary as well as Thol Bird Sanctuary.

Four categories of forest in Gujarat.

(i) Tropical dry deciduous forests

(ii) Tropical thorn forests

(iii) Tropical moist deciduous forests

(iv) Littoral forests

- Major plants found here are Tectona, Terminalia, Madhuca, Bamboo, Acacia, Zizyphus, Mangroves.

---

(19) How many species estimated on the earth ?

(A) 5 lacs to 50 lacs (B) 50 lacs to 5 crores

(C) 5 lacs to 5 crores (D) 50 thousands to 50 lacs

(20) Which forests are the richest in animal and plant diversity ?

(A) Temperate (B) Tropical (C) Temperate rain forests (D) Tropical decudious

(21) How many biogeographical region of our country form a wide variety of ecological habitat.

(A) 12 (B) 7 (C) 11 (D) 10

(22) India has how many of the total animal speices in the world ?

(A) 7% (B) more than 7% (C) 11% (D) more than 11%

(23) India is a source of traditional crop varieties ranking ..... and ..... so far in the contribution of agricultural species respectively.

(A) 1st, 12th (B) 1st, 7th (C) 10th, 7th (D) 2nd, 7th

(24) There are about ..... species of mangrove plants and over ..... species of coral reefs found in India.

(A) 45, 341 (B) 341, 45 (C) 47, 141 (D) 141, 47

(25) The species which is distributed only in a small restricted area is known as ..... .

(A) migratory species (B) endemic species (C) extinct species (D) dangered species

(26) Gujarat is home of nearly how many species of plants and animals ?

(A) 400 (B) 6000 (C) 7000 (D) 900

(27) It is the only remnant gene pool in the world.

(A) Velavdar Black buck National Park (B) Ratanmahal Sloth Beer Sanctuary

(C) Vasanda National Park (D) Wild Ass Sanctuary

(28) It is a rich ecosystem in arid region of Gujarat.

(A) Narayan Sarovar Chinkara Sanctuary (B) Shool Paneshwar Wild Life Sanctuary

(C) Velavadar Black Buck National Park (D) Ratanmahal Sloth Bear Sanctuary

(29) How many categories of forests in India ?

(A) 6 (B) 4 (C) 2 (D) 8

Answers : (19-C), (20-B), (21-D), (22-B), (23-B), (24-A), (25-B), (26-C), (27-D), (28-A), (29-B)

***Loss of Bio-diversity ***

- The biological wealth has been declining rapidly.
- IUCN (International Union for Conservation of Nature and national resources) Red

List (2004) documents the extinction of 784 species in the last 500 years.

- Examples of recent extinctions
  - Dodo (Mauritius)
  - Guagga (Africa)
  - Steller's sea cow (Russia)
  - Three subspecies of tiger-Bali, Javan, Caspian
- The last twenty years alone have witnessed disappearance of 27 species.
- The current rate of extinction is 1000 to 10,000 times higher than the past rate of extinction.
- Ecologists warn that if the present trends continues by the end of twenty first century there will be a reduction by 50% in the biodiversity of the earth.

*** Causes of Biodiversity Losses ***

(1) Habitat loss and fragmentation :

- It is most important cause driving animals and plants to extinct.
- major reasons : quarrying, laying new railway tracks, constructing dams, felling of trees, cleaning of land for agriculture, deforestation.
- The tropical rain forests was covering more than 14% of the earth's land surface, but now cover more than 6% only.
- The Amazon rain forest, lungs of the planet, is being cut and cleared for cultivating soyabeans.
- Bignonia and Habenaria formed luxuriant growth on rocks of the Western Ghats before quarrying the rocks with dynamite.
- In southern part of Western Ghats, lion tailed macaque found in large numbers has become endangered due to loss of habitat.
- Of the 370 butterfly species available here, upto 70 are on the average of extinction.

(2) Over-exploitation :

- Human uses natural resources for food and other purposes, but when 'need' turns to 'greed', it leads to over-exploitation of natural resources.
- eg. fishes, Molluscs, crustaceans, sea cows and sea turtles.
- victim species : elephants, rhinoceros, tigers, musk deer, crocodiles, spiny tail lizards ..... going to extinct.
- Passenger pigeon of North America was killed for food as result before 500 years this species has been at the stage of extinction.
- All over the world large number of animals are collected for laboratory study, dissections, research point of view.
- Frog is well known example of it.
- Rabbits and monkey are also used.

(3) Alien species invasions :

- New species entering a geographical region are called alien species.
- They cause disappearance of native species.

- **eg.**

(1) The purposely or accidently introduced species like carrot grass (parthenium), Gandhari (Lantana) and Water hyacinth (Eichornia) have lead to extinction of many local species as well as adversely affected human health.

(2) The recent illegal introduction of African catfish Clarias gariepinus for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.

(4) Co-extinctions : When a species become extinct, the plant and animal species associated with it in an obligatory way also become extinct.

- **eg.** When a host fish species become extinct, its unique assemblage of parasites also meets the same fate.

---

(30) According to Red List how many species extincted in last 500 years ?

(A) 784 (B) 748 (C) 874 (D) 478

(31) How many subspecies of tigers disappeared ?

(A) 2 (B) 4 (C) 3 (4) 5

(32) The current rate of extinction is how many times higher than the past rate of extinction ?

(A) 100 to 1000 (B) 100 to 10,000 (C) 10 to 10,000 (D) 1000 to 10,000

(33) These forests are called lungs of the planet.

(A) Tropical (B) Temperate (C) Amazon (D) All given

(34) Where was Lion tailed macaque found ?

(A) Rocks of Western Ghats (B) North of Western Ghats

(C) Southern part of Western Ghats (D) All given

(35) It is not cause for loss of Biodiversity ?

(A) Mass-extinction

(B) Alien species invasions

(C) Habitat loss and fragmentation

(D) Over-exploitation

Answers : (30-A), (31-C), (32-D), (33-C), (34-C), (35-A)

*** Bio-diversity Conservation ***

- Means conversion of gene complexes, species and ecosystems.
- "The maintenance and preservation of biodiversity is called biodiversity conservation."
- In two ways it can be conserved.

(1) In-situ conservation (on site)

- It is the conservation of genetic resources in their natural habitat.
- It includes protected areas and biosphere reserves.
- Wild life sancturies and National parks are examples of protected areas.
- As a September 2002, India has 581 protected areas (89 National parks and 492 wild life sanctuaries.)
- Forests, grasslands and oceanic regions are identified in the concerned state or the country for the purpose of conservation of biodiversity and declared as "Protected areas".

Benefits of Protected Areas :

- Maintain viable populations of all native species and subspecies.
- Maintain the number and distribution of communities and habitats and conserve the genetic diversity of all the present species.
- Prevent man made introduction of alien species.

Hot Spot :

- The area which is rich in plant and animal species, of which many are endemic and endangered is called biodiversity hot spot.
- Initially 25 biodiversity hot spots, late 9 more have been added, so total 34 all over world.

3 hot spots of India are : (1) Western Ghats (2) Indo-Burma (3) Eastern Himalayan region in India.

Western Ghats has a total of 310 endemic vertrebrates among total of 32,678 species in the world.

- **National Parks :**
- Kaziranga National Park (Assam)
- Sundarbans (Tiger Reserves, West Bengal)
- Hazaribagh National Park (Bihar)
- Corbett National Park (Uttar Pradesh)
- Gir National Park (Gujarat dist. Junagadh)
- Kanha Patiala Park (Madhya Pradesh)
- Tandoba National Park (Maharashtra)
- Bandipur National Park (Karnataka)
- Marine National Park (Jamnagar)
- Gujarat has 4 National Parks

Sanctuaries :

- A sanctuary is an area, which is reserved for the conservation of animals only.
- Operations such as harvesting of timber, collection of minor forest products and private ownership rights are allowed with condition that they do not affect the animals adversely.
- **Sancturies of India :**
- Annamali sactuary (Tamilnadu)
- Jaldapara Sanctuary (West Bengal)
- Keoladeo Ghana Bird Sanctuary (Bharatpur-Rajasthan)
- Sultanpur Lake Bird Sanctuary (Hariyana)
- Shikari Devi Sanctuary (Himachal Pradesh)
- Dachigam Sanctuary [Srinagar (Jammu and Kashmir)]
- Mudumalai Sagar Sanctuary (Andhra Pradesh)
- Chilka Lake Bird Sanctuary (Orissa)
- Periyar Sanctuary (Kerala)
- Manas Wild Life Sanctuary (Assam)
- Gujarat : • Nalsarovar Bird Sanctuary • Thol Bird Sanctuary
- Gujarat has 21 Sanctuaries.

Biosphere Reserves

- The man and Biosphere (MAB) programme of UNESCO formulated the concept of biosphere reserves in 1975.
- In this programme people are integral components of the system.
- In India 14 biosphere reserves are found.
- It includes Nanda Devi, Manas, Sundarbans, Great Nicobar, Gulf of Mannar.

A biosphere reserves has 3 zones.

- **Core Zone :**
  - It comprises an undisturbed and legally protected ecosystem.
- **Buffer Zone :**
  - It surrounds the core area and is managed to use for research and educational activities.
- **Transitional Zone :**
  - It is the outermost part of the biosphere reserves.
  - There is active cooperation between reserve management and the local people wherein certain activities are allowed with conservation goals.
  - Some special projects are also established in a special cares.
  - eg. Project tiger, Gir lion project, Crocodile breeding project, Rhinos conservation, Snow-leopard project, Project elephant.

(2) Ex-situ conservation (Offsite conservation) :

- It is the conservation of genetic resources outside their natural habitat.
- They includes : Botanical gardens, Arboreta, Herbal Gardens (for plants), Zoos (for animals), Seed bank, gene bank, Biotechnology use (Tissue culture, genetic engineering), Culture collection (for microorganisms).
- There are 1500 or more botanical gardens in the world.
- About 80,000 species are maintained in these gardens.
- Of these nearly 800 important gardens are documented in the international Association of Botanical Garden (IABG).
- There are more than 800 zoos in the world.
- In our state, the Vaghai Botanical Garden maintains many medicinal plants.

Seed Bank :

- Seed bank can be developed through botanical garden.
- In a seed bank seeds are stored for long duration through cold storage in seed bank.
- Germ plasm of primitive cultivators or land races and other cultivars are kept in seed bank.
- Seeds are maintained here to maintain gene pool.
- The seeds of those plants which have a possibility of extinction are stored in seed bank.
- Germplasm : It is a living genetic source. It can be maintained in the form of seeds or tissues which are used for culture of plants and animals and also used for storage and research.
- Seeds has to be stored for many years, with a minimum loss of viability.
- Many forests and fruit species requires some different ways for the storage involving three fold strategy (1) Dry storage, (2) Cryogenic storage, (3) Storage of fully of practically hydrated seeds at ambient temperature.

Gene Bank : It conserves rare genes.

- Here, crop varieties and wild genetic resources are conserved due to their utility in future crop improvement and a forestation programmes.
- List of gene banks :
  - National Bureau of plant Genetic Resources, Delhi for wild relatives of crop plants.
  - National Bureau of Animal Genetic Resources, Karnal for domesticated animals.
  - National Bureau of Fish Genetic Resources, Allahabad, for economically valuable fish species.
- In India several varieties of rice have been preserved in gene-banks.
- At present gene-bank collections have over 34,000 cereals and 22,000 pulses.
- **Cryopreservation :** The reproductive units of plants and reproductive cells of animals can be deep frozen at a very low temperature and thus conserved for future use.
- This method of conservation at -196°C in liquid nitrogen is called Cryopreservation - Its bank is called cryobank.
- Eggs can be fertilized in and plants can be propagated using tissue culture method.
- The World Conservation Union (WCU) and the World Wildlife Fund (WWF) are active in biodiversity conservatin all over the world.
- Environmental Education (EE) is given first priority to youth.

---

(36) The maintenance and preservation of biodiversity means ..... .

(A) Conservation of genetic-diversity (B) Conservation of ecosystem

(C) Conservation of species-diversity (D) Conservation of bio-diversity

(37) How many ways are there to conserve biodiversity ?

(A) 2 (B) 3 (C) 4 (D) not certain

(38) The conservation of genetic recourses in their natural habitat means ..... .

(A) in-situ conservation (B) ex-situ conservation

(C) in-ex-situ conservation (D) all given

(39) Protected areas includes ..... .

(A) Seed bank, National parks

(B) Gene bank, National parks

(C) Seed bank, Gene bank

(D) National parks, Wild-life sanctuaries

(40) As september 2002, India has ..... national parks and ..... wild-life sanctuaries.

(A) 98, 482 (B) 89, 492 (C) 78, 492 (D) 89, 429

(41) How many hotspots are located in the world ?

(A) 25 (B) 29 (C) 34 (D) 31

(42) How many hot spots are there in India ?

(A) 25 (B) 3 (C) 34 (D) 28

(43) How many National parks has Gujarat ?

(A) 21 (B) 14 (C) 17 (D) 4

(44) Here human activities are allowed with conditions ?

(A) National Parks (B) Sanctuaries

(C) A, B both (D) Reserved areas

(45) How many sanctuaries has Gujarat ?

(A) 13 (B) 4 (C) 17 (D) 21

(46) Which programme of UNESCO formulated the conept of biosphere reserves ?

(A) Man of Biosphere (B) Man of Biography

(C) Man and Biosphere (D) Man and Biography

(47) In India how many biosphere reserves are found ?

(A) 17 (B) 21 (C) 13 (D) 14

(48) How many zones are there in biosphere reserves ?

(A) 6 (B) 2 (C) 3 (D) 4

(49) How many gardens are documented in IBAG ?

(A) 700 (B) More than 700 (C) 800 (D) More than 800

(50) Experimentally seeds are stored at temperature of environment is called ..... .

(A) dry storage (B) cold storage

(C) cryo preservation (D) all given

(51) In gene bank more than ..... cereals and ..... pulses are stored.

(A) 43,000; 22,000 (B) 34,000; 22,000 (C) 22,000; 43,000 (D) 22,000; 34,000

Answers : (36-D), (37-A), (38-A), (39-D), (40-B), (41-C), (42-B), (43-D), (44-B), (45-D), (46-C), (47-D), (48-C), (49-D), (50-A), (51-B)

- **A - Assertion and R - Reason type questions :**

  Choose the correct answer from given options :

  (A) A and R both correct. R is correct explanation of A

  (B) A and R both correct. R is not correct explanation of A

  (C) A is correct but R is wrong

  (D) A is wrong but R is correct

(52) Assertion A : All organisms are not found everywhere.

Reason R : They show diversity in shape, volume, size, colour, life style and many other things.

(A) (B) (C) (D)

(53) Assertion A : The variation in the constitution of genes is called genetic diversity.

Reason R : Species diversity is the basis of formation of new species.

(A) (B) (C) (D)

(54) Assertion A : Western Ghats have greater amphibian species diversity than the Eastern Ghats.

Reason R : The greater the number of species, the higher is the species diversity.

(A) (B) (C) (D)

(55) Assertion A : The percentage of each and every species within a community is called β-diversity.
Reason R : It is the relative richness of different species in a habitat.
(A) (B) (C) (D)

(56) Assertion A : γ- diversity refers to richness of different species in a range of habitats within a geographical area.
Reason R : Bio diversity of ocean, lakes, ponds, mountains, forest etc. is the examples of γ- diversity.
(A) (B) (C) (D)

(57) Assertion A : Species diversity differs from ecosystem to ecosystem.
ReasonR : Each and every ecosystem allows a particular group of plants and animals to grow there in.
(A) (B) (C) (D)

(58) Assertion A : In New York at 41°N has 105 species and Greenland at 71°N only 56 species.
Reason R : Localities at lower latitudes have more species than localities at higher latitudes.
(A) (B) (C) (D)

(59) Assertion A : More than half of the species on the earth live in moist tropical forests.
Reason R : The tropical forests are regarded as the richest in Animal and plant diversity.
(A) (B) (C) (D)

(60) Assertion A : Nilgiri wood pigeon and Malbar parakeet are endemic species.
Reason R : The species which is distributed only in a small restricted area is called endemic species.
(A) (B) (C) (D)

(61) Assertion A : In 1957 UNESCO formulated Biosphere reserve hypothesis under MAB programme.
Reason R : They are special category for protected area of land or coastal environments.
(A) (B) (C) (D)

(62) Assertion A : In a seed bank seeds are stored for long duration.
Reason R : Seed bank has cold storage to store seeds.
(A) (B) (C) (D)

(63) Assertion A : When a species becomes extinct, the plant and animal associated with it in an obligatory way also become extinct.
Reason R : When host fish becomes extinct, parasites lives on it also becomes extinct.
(A) (B) (C) (D)

Answers : (52-B), (53-C), (54-A), (55-D), (56-B), (57-A), (58-A), (59-B), (60-A), (61-D), (62-A), (63-B)

True - False Type Questions

(64) 1. Sequoia sempervirens (Angiosperms) is 400 ft tall.
2. Wolffia globosa (angiosperm) is the smallest plant.
3. Euglena lives as parasite.
4. Shark and Dolphin are large sized mammals.
(A) FTTF (B) TFTF (C) FTFF (D) FTFT

(65) Choose correct option for aims of biodiversity.

1. Ecosystem studies.
2. Proposing advance conservation programme.
3. To identify biosphere reserves.
4. Conservation of global ethics.

(A) TTFF (B) TTFT (C) TFFT (D) FTTF

(66) 1. The largely tropical Amazonian rain forest in North America has the greatest biodiversity on earth.
2. Colombia located near the equator has nearly 1400 species of birds.
3. Localities at higher latitudes have more species than localities at lower latitudes.
4. In rain forest there might be atleast two million insect species waiting to be discovered and named.

(A) TFFT (B) TTFF (C) FFTT (D) FTFT

(67) 1. India covers 2.4% of the total area of the world .
2. India has 11% of the world's total biota.
3. India has more than 7% of the total animal species in the world.
4. 12 biogeographical regions of our country form a wide variety of ecological habitat.

(A) TTTF (B) TTFF (C) TFTF (D) FFFT

(68) 1. The coral reefs in the Gulf of Kachchh is widely admired for their amazing beauty and spectacular diversity.
2. The area of Dhorad and Paccham islands has flamingos in the great Rann of Kachchh.
3. The great Rann of Kachchh and the little Rann of Kachchh are the world only saline desert spread in 2500 sq.km.
4. Chinkara Sanctuary is the remnant gene pool in the world.

(A) TFFT (B) TFFF (C) TTFF (D) TFTF

(69) Choose right option for benefits of protected areas.

1. Prevent man-made introduction of alien species.
2. Maintain viable populations of all native species and sub-species.
3. Maintain the number and distribution of communities and habitats.
4. Biogeographic studies.

(A) TTTF (B) TTFT (C) TFFT (D) FTTF

(70) 1. Gene bank can be developed for conservation of rare genes.
2. In India varieties of wheat are conserved in gene bank.
3. Gametes of threatened species is stored in a gene bank.
4. At present, gene-bank colletions have 34,000 pulses and 22,000 cereals.

(A) TTFF (B) TFTF (C) TTTF (D) TTTT

(71)

	Column-I	Column-II		I	II	III
(I)	Transitional zone	(p) it is managed to use for reaserch and educational activities	(A)	p	q	r
			(B)	q	p	r
(II)	Core zone	(q) certain activities are allowed with conservation goals	(C)	q	r	p
(III)	Buffer zone	(r) legally protected ecosystem	(D)	r	p	q

(72)

	Column-I	Column-II		I	II	III	IV
(I)	West Bengal	(p) Keoladeo Ghana Bird Sanctuary	(A)	q	s	p	r
(II)	Tamilnadu	(q) Jaldapara Sanctuary	(B)	p	s	q	r
(III)	Bharatpur	(r) Sultanpur Lake Bird Sanctuary	(C)	q	s	r	p
(IV)	Hariyana	(s) Annamali Sanctuary	(D)	r	s	q	p

(73)

	Column-I	Column-II		I	II	III	IV
(I)	Shikari Devi Sanctuary	(p) Srinagar	(A)	r	p	q	s
(II)	Dachigam Sanctuary	(q) Orissa	(B)	r	p	s	q
(III)	Mudumalai Sagar Sanctuary	(r) Himachal Pradesh	(C)	s	q	p	r
(IV)	Chilka Lake Bird Sanctuary	(s) Andhra Pradesh	(D)	p	r	q	s

(74)

	Column-I	Column-II		I	II	III	IV
(I)	Kaziranga National Park	(p) Bihar	(A)	p	q	s	r
(II)	Tiger Reserves	(q) Assam	(B)	q	p	s	r
(III)	Hazaribagh National Park	(r) Uttar Pradesh	(C)	s	p	q	r
(IV)	Corbett National Park	(s) West Bengal	(D)	q	s	p	r

(75)

	Column-I	Column-II		I	II	III	IV
(I)	Kanha Patiala Park	(p) Kerala	(A)	q	s	p	r
(II)	Bandipur National Park	(q) Madhya Pradesh	(B)	q	s	r	p
(III)	Manas Wild Life Sanctuary	(r) Assam	(C)	p	q	r	s
(IV)	Periyar Sanctuary	(s) Karnataka	(D)	s	p	q	r

(76)

	Column-I	Column-II		I	II	III	IV
(I)	Gandhari	(p) Lantana	(A)	p	q	s	r
(II)	Catfish	(q) Parthenium	(B)	p	r	q	s
(III)	Grass	(r) Clarias	(C)	q	p	r	s
(IV)	Water hyacinth	(s) Eichornia	(D)	r	p	q	s

(77)

	Column-I	Column-II	Column-III
(I)	Soyabean	(p) rocks of the Western Ghats	(*x*) destroyed due to quarrying
(II)	Bignonia and Habenaria	(q) Amazon rain forests	(y) endangered
(III)	Lion tailed macaque	(r) Southern past of Western Ghats	(z) lungs of planet

	I	II	III
(A)	q→z	r→y	p→*x*
(B)	r→z	p→*x*	q→y
(C)	q→z	p→*x*	r→y
(D)	p→z	r→*x*	q→y

(78)

Column-I	Column-II		I	II	III	IV
(I) painted frog	(p) Vijaynagar forest	(A)	r	q	p	s
(II) Safed Musli	(q) Bhavnagar District	(B)	r	p	q	s
(III) Black Buck National Park	(r) Vasanda National Park	(C)	p	s	r	q
(IV) Chinkara Sanctuary	(s) Kachchh District	(D)	p	r	q	s

(79) Choose correct pairs for Biodiversity of Gujarat.

Column-I	Column-II		I	II	III	IV
(I) Fish	(p) 19%	(A)	s	p	q	r
(II) Amphibians	(q) 14%	(B)	q	r	s	p
(III) Reptiles	(r) 25%	(C)	q	p	s	r
(IV) Mammals	(s) 9%	(D)	q	s	p	r

(80)

Column-I	Column-II		I	II	III	IV
(I) India's coastline ..... km	(p) 45,500	(A)	s	q	p	r
(II) Species used as medicines in india	(q) 7,500	(B)	q	s	p	r
(III) Total biota of plants	(r) 86,874	(C)	r	s	q	p
(IV) Total species of animals	(r) 86,874	(D)	p	q	r	s
	(s) more than 4,000					

(81)

Column-I	Column-II		I	II	III	IV
I Calotropis procera	(p) Asopalav	(A)	r	p	s	q
II Polyalthia longifolia	(q) Khijdo	(B)	s	q	p	r
III Ficus religiosa	(r) Akdo	(C)	r	q	s	p
IV Prosopis Cineraria	(s) Pipal	(D)	p	r	q	s

(82)

Column-I	Column-II	Column-III
(I) α-diversity	(p) percentage of each species in community	(w) richness of different species of two habitats
(II) β-diversity	(q) Lakes, mountains, forests	(x) richness of different species in a habitat
(III) γ-diversity	(r) Birds of pond and agricuture area of same grassland area	(A) richness of different species in a range of habitats

	I	II	II
(A)	p$\rightarrow$y	r$\rightarrow x$	q$\rightarrow$w
(B)	r$\rightarrow x$	p$\rightarrow$w	q$\rightarrow$y
(C)	r$\rightarrow$y	p$\rightarrow x$	q$\rightarrow$w
(D)	p$\rightarrow x$	r$\rightarrow$w	q$\rightarrow$y

(83) Choose Right option for 'b' -

a	b	c
↓	↓	↓
populations	Kingdoms	Biomes
↓	↓	↓
Species	Phyla	Ecosystems
↓	↓	↓
Chromosomes	Families	Habitats
↓	↓	↓
Genes	Genera	Niches
↓	↓	↓
Nucleotides	Species	Population
	↓	
	Sub species	
	↓	
	Variety	

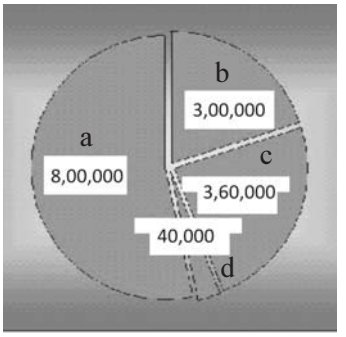
(A) The variation in the constitution of gene is called genetic diversity.

(B) It is important for normal functioning and sustainibility of ecosystems.

(C) Species richness in different ecosystem of a geopgraphical area.

(D) It shows population - diversity.

(84) Choose correct for a,b,c,d in diagram.



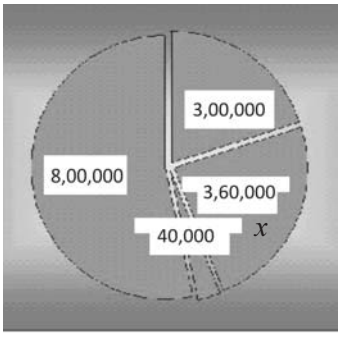
(A) a = plants b = micro organisms
c = insects d = vertebrates

(B) a = vertrebrates b = plants
c = insects d = Micro organisms

(C) a = insects b = plants
c = Micro organisms = vertebrates

(D) a = insects b = vertebrates
c = plants d = Micro organisms

(85) What does '*x*' stand for in given diagram ?



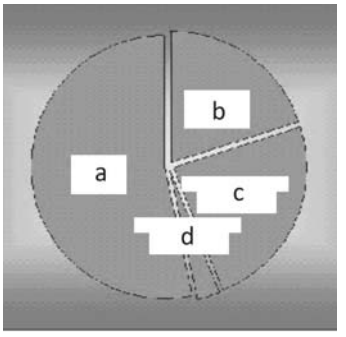
(A) *x* = insects

(B) *x* = micro organisms

(C) *x* = plants

(D) *x* = vertebrates

(86) choose appropriate option for a, b, c, d in given diagram.



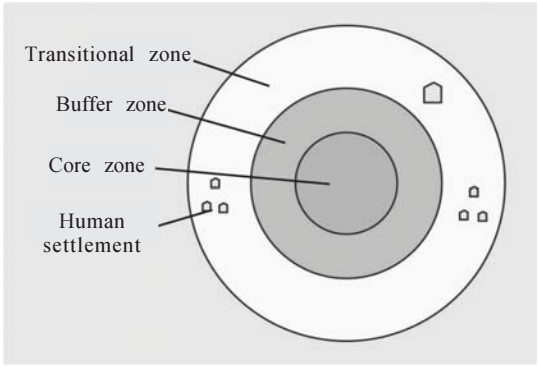
(A) a = 18,00,000 b = 3,00,000
c = 3,60,000 d = 40,000

(B) a = 80,000 b = 30,000
c = 30,600 d = 40,000

(C) a = 8,00,00 b = 3,00,000
c = 3,60,000 d = 40,000

(D) a = 40,00,000 b = 8,00,000
c = 3,00,000 d = 3,60,000

(87) Choose correct option for given diagram.



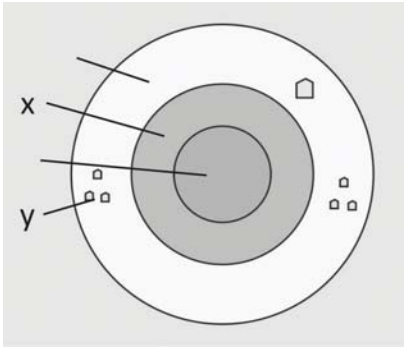
(A) In India 34 in number

(B) It is formulated by MAB

(C) More than 1500 in the world

(D) All given

(88) What does '*x*' and 'y' indicate in diagram ?



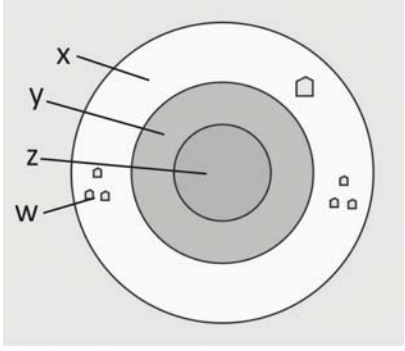
(A) *x* = transitional zone y = human settlement

(B) *x* = buffer zone y = human settlement

(C) *x* = buffer zone y = authorised

(D) *x* = transitional zone y = authorised

(89) What does '*x*' indicate ?



(A) *x* = certain activities are allowed with con servation goals.

(B) *x* = It is managed to use for research and educational activities.

(C) *x* = It is legally protected ecosystem.

(D) *x* = It is undisturbed.

Answers : (64-C), (65-B), (66-D), (67-A), (68-B), (69-A), (70-B), (71-C), (72-A), (73-B), (74-D), (75-B), (76-B), (77-C), (78-B), (79-D), (80-B), (81-A), (82-D), (83-B), (84-A), (85-B), (86-C), (87-B), (88-B), (89-A)

- **NEET Questions :**

(90) Wildlife is destroyed most when ..... .

(A) there is lack of proper care

(B) mass scale hunting for foreign trade

(C) its natural habitat is destroyed

(D) natural calamity

(91) Which animal has become extinct from India ?

(A) Snow leopard (B) Hippopotamus (C) Wolf (D) Cheetah

(92) Ranthambore National park is situated in ..... .

(A) Maharashtra (B) Rajasthan (C) Gujarat (D) U.P.

(93) National park associated with Rhinoceros is ..... .

(A) Kaziranga (B) Ranthambore

(C) Corbett (D) Valley of flowers

(94) Species diversity increases as one proceeds from ..... .
(A) high altitude to low altitude and high latitude to low latitude
(B) high altitude to low altitude and low latitude to high latitude
(C) low altitude to high altitude and high latitude to low latitude
(D) low altitude to high altitude and low latitude to high latitude

(95) Which of the following is the correct matching pair of a sanctuary and its main protected wild animal ?
(A) Gir-Lion (B) Sariska-Tiger
(C) Sunderban-Rhino (D) Kaziranga-Musk deer

(96) A number of natural reserves have been created to conserve specific wild life species. Identify the correct combination from the following.
(A) Gir forest - Tiger (B) Kaziranga-Elephants
(C) Rann of Kachchh-Wild Ass (D) Manas Wild life sanctuary - Musk deer

(97) If the forest cover is reduced to half, what is most likely to happen on a long term basis ?
(A) Tribals living in these areas will starve to death.
(B) Cattle in these and adjoining areas will die due to lack of fodder.
(C) Large areas will become deserts.
(D) Crop breeding programmes will suffer due to a reduced availability of variety of germplasm.

(98) MAB stands for
(A) Man and Biology programme (B) Man and Biosphere programme
(C) Mammals and Biosphere programme (D) Mammals and Biology programme

(99) At present, the most significant cause of dwindling biodiversity is probably ..... .
(A) the deterioration of ozone layer (B) the destruction of habitat
(C) Biological magnification of DDT (D) global warming

(100) An Institution where valuable Plant material likely to become irretrievably lost in the wild or in cultivation is preserved in a viable Condition is known as ..... .
(A) Genome (B) Herbarium (C) Gene library (D) Gene bank

(101) Reason of diversity in living beings is due to ..... .
(A) mutation (B) long term evolutionary change
(C) gradual change (D) short term evolutionary change

(102) Which group of vertebrates comprises the highest number of endangered species ?
(A) Birds (B) Mammals (C) Fishes (D) Reptiles

(103) In your opinion, which is the most effective way to conserve the plant diversity of an area ?
(A) By tissue culture method (B) By creating biosphere reserve
(C) By creating botanical garden (D) By developing seed bank

(104) Diversification in plant life appeared ..... .
(A) due to long periods of evolutionagy change (B) due to abrupt mutation
(C) suddenly on earth (D) by seed dispersal

(105) Biodiversity Act of India was passed by the parliament in the year.
(A) 1992 (B) 1996 (C) 2000 (D) 2002

(106) Which one of the following is the correct matched pair of an endangered animal and National Park ?
(A) Rhinoceros - Kaziranga National Park (B) Wild ass - Dudhwa National Park
(C) Great Indian Bustard - Keoladeo National Park (D) Lion - Corbettt National Park

(107) Which one of the following is not included under in-situ conservation ?
(A) Botanical garden (B) Biosphere reserve (C) National Park (D) Sanctuary

(108) Which of the following is considered a hot-spot of biodiversity in India ?

(A) Indo-Gangetic Plain (B) Eastern Ghats (C) Aravalli Hills (D) Western Ghats

(109) Indentify the odd Combination of the habitat and the particular animal concerned.

(A) Sunderbans- Bengal Tiger (B) Periyar - Elephant

(C) Rann of kachch - Wild Ass (D) Dachigam National Park - Snow leopard

(110) Which one of the following is not observed in biodiversity hotspots ?

(A) Endemism (B) Accelerated species loss

(C) Lesser inter-specific competition (D) Species richness

(111) Which one of the following has maximum genetic diversity in India ?

(A) Mango (B) Wheat (C) Tea (D) Teak

(112) Tiger is not a resident in which one of the following national Park ?

(A) Sunderbans (B) Gir (C) Jim Corbett (D) Ranthambhor

(113) Which one of the following is an example of Exsitu conservation ?

(A) Wildlife Sanctuary (B) Seed bank (C) Sacred groves (D) National Park

(114) A collection of plants and seeds having diverse alleles of all the genes of a crop is called.

(A) herbarium (B) germplasm (C) gene library (D) genome

(115) Which one of the following shows maximum genetic diversity in India ?

(A) Groundnut (B) Rice (C) Maize (D) Mango

(116) Biodiversity of a geographical region represents ..... .

(A) endangered species found in the region

(B) the diversity in the organisms living in the region

(C) genetic diversity present in the dominant species of the region

(D) species endemic to the region

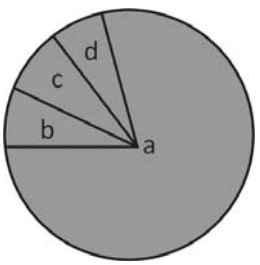
(117) What do species call which will extinct in near future ?

(A) sensitive species (B) endemic species (C) endangerous species (D) extinct species

(118) Which institute makes Red List ?

(A) ICFRE (B) IUCN (C) UNEP (D) WWF

(119) Choose appropriate option for invertebrate animals ?



	a	b	c	d
(A)	insects	curstercean	another animal group	mollusca
(B)	crustacean	insects	mollusca	other animal phyla
(C)	mollusca	another animal phyla	crustacean	insects
(D)	insects	mollusea	crustacean	other animals

Answers : (90-C), (91-D), (92-B), (93-A), (94-A), (95-B), (96-C), (97-D), (98-B), (99-B), (100-D), (101-B), (102-B), (103-C), (104-A), (105-D), (106-A), (107-A), (108-D), (109-D), (110-B), (111-B), (112-B), (113-B), (114-C), (115-B), (116-B), (117-C), (118-B), (119-D)

Biodiversity and Conservation

• Biodiversity is defined as the vast array of species of microorganisms, algae, fungi, plants and animals occuring on the earth and the ecological complexes of which they are a part. This includes diversity within species, between species and of the ecosystem

Alpha diversity
Diversity in the given community of habitat.

Beta diversity
Diversity between community

Gamma diversity
It is regional diversity

Levels of biodiversity

Genetic diversity
It is the diversity in the numbers and types of genes as well as chromosomes present in different species and the variations in the genes and their alleles in the same species.

Species diversity
It is the diversity in the number and richness of species of a region

Community Ecosystem diversity
It is related to the differnt type of ecosystems/habitats eg., terrestrial and aquatic ecosystems

Patterns of Biodiversity

- Species Latitudinal and altiludinal gradient diversity on earth is not uniformly distributed but shows interesting patterns. It is generally highest in the tropics and decreases towards the poles.
- Important explanations for the species richness in tropics: Tropics had more evolutionary time; they provide a relatively constant environment and they receive more solar energy which contributes to greater productivity.

Species area relationships
It is a relationship between the area of habitat or part of a habitat and the number of specis within that area

Importance of Biodiversity

- stability of ecosystem
- sources of food and improved varieties
- sources of fibres
- provide variety of useful products such as gums, resins, dyes, perfumes, lubricants, rubber, latex, tea, dryfruits
- provide drugs and medicines
- scientific value
- ecosystem services

Loss of Biodiversity

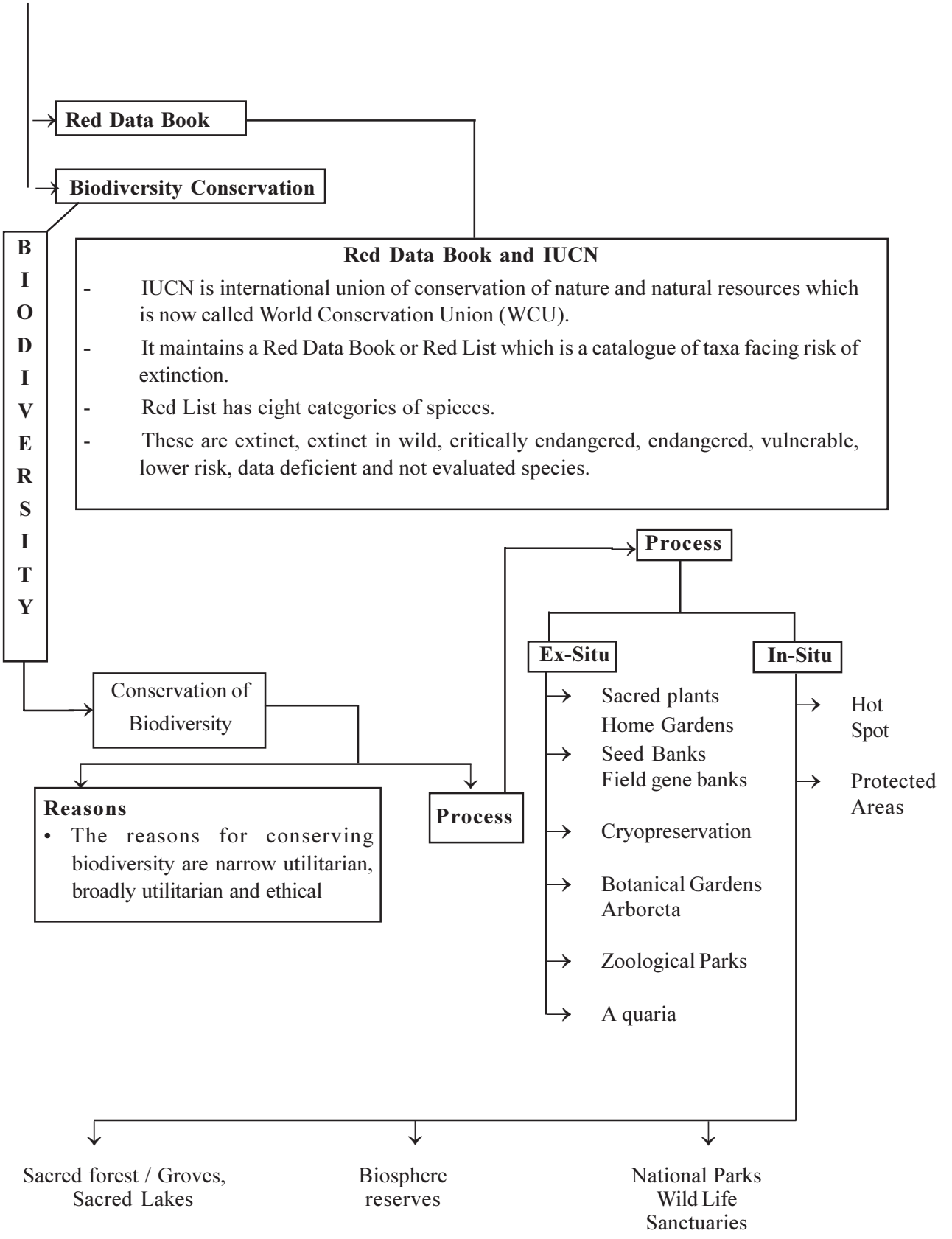
Most serious aspect of the loss of biodiversity is the extinction species

Types of extinction

- Natural extinction
- Mass extinction
- Antropogenic extinction

Factors contributing to extinction

- Habitat loss and fragmentation
- Over exploitation
- Alien species invasions
- Disturbance and Degradation
- Pollution
- Coextinctions
- Forestry
- Intensive Agriculture


Red Data Book
Biodiversity Conservation
BIODIVERSITY
Red Data Book and IUCN
- IUCN is international union of conservation of nature and natural resources which is now called World Conservation Union (WCU).
- It maintains a Red Data Book or Red List which is a catalogue of taxa facing risk of extinction.
- Red List has eight categories of spieces.
- These are extinct, extinct in wild, critically endangered, endangered, vulnerable, lower risk, data deficient and not evaluated species.
Process
Ex-Situ
In-Situ
Conservation of Biodiversity
Reasons
• The reasons for conserving biodiversity are narrow utilitarian, broadly utilitarian and ethical
Process
Sacred plants
Home Gardens
Seed Banks
Field gene banks
Cryopreservation
Botanical Gardens
Arboreta
Zoological Parks
A quaria
Hot Spot
Protected Areas
Sacred forest / Groves, Sacred Lakes
Biosphere reserves
National Parks
Wild Life
Sanctuaries

30 Environmental Issues

Pollutants :

Materials which are harmful to organisms are called pollutants.

Pollution :

Undesirable change in physical, chemical or biological characteristics of air, land, water or soil is called pollution.

- The Government of India has passed the Environment (protection) Act, 1986.

Air pollution and its control :

effects :

- on respiration of organisms
- reduce the growth and yield of crops

main source :

(a) fossil fuel (b) vehicular exhaust

Control :

- Photochemical smog should be eliminated as far as possible.
- Development of electrical and solar-energy controlled vehicles.
- Partiulate matter produced by industries and thermal power plants can be controlled by
  (a) scrubbers (b) electrostatic precipitator (c) filters.

(a) Scrubbers : It can remove gases like SO_2 .

(b) Electrostatic precipitator : Electric charge is generated on the particles in the smoke and they are down towards electrodes and removed. Around 99% particulate matter can be removed.

(c) Filters : Sieves of different sized can remove specific particles of sizes.

- Research by Central Pollution Control Board (CPCB) - particles having size of 2.5 micrometers or less in diameter are responsible for causing greatest harm to lungs and health of human.

- **Act :**

  India, Air (prevention and control of pollution), 1981.
  - In 1987 this act include noise as an air pollution.
  - 80 db (decible) intensity or more sound is harmful to human health, it damages corticells of ears.

- Study of air pollution reveals,
  - Every year about 4,50,000 vehicles are added to the roads of the state.
  - Pollutants added by vehicles : SO_2, NO_2, SPM (Suspended Particulate Matter)
  - 18% increase in CO at evening time.
  - NIOH (National Institute of Occupational Health, Ahmedabad) observed that air pollution at street junctions and people living here suffered from cough and congestions.

Check on transport system :

- State government has established "Commissionerate of Transport." initiated "Pollution Under Control System" in 1995.
- EURO-III norm, newly labeled as BHARAT-III has been implemented in Ahmedabad and Surat.

CNG :

- CNG vehicle increase.
- CNG is not carcinogenic (not cause for cancer).
- Companies provide CNG to cities of Gujarat.
- GGCL - Gujarat Gas Company Ltd; Surat-Ankleshwar
- GAIL - Gas Authority of India Ltd; Vadodara
- GAEL - Gujrat Adani Energy Ltd; Ahmedabad

---

(1) When did India pass the Environment Act ?

(A) 1995 (B) 1981 (C) 1986 (D) 1974

(2) Major sources of NO_2 and SO_2 pollutants.

(A) Vehicular exhaust (B) Fertilized factories

(C) Petroleum refineries (D) B and C both

(3) From which method of following electric charge is generated on particles in the smoke ?

(A) Scrubbers (B) Electrostatic precipitator's

(C) Filters (D) All given

(4) Size of particles to human health, suggested by CPCB

(A) 2.5 micrometer or less in diameter

(B) 2.5 micrometer or more in diameter

(C) 2.5 milileter or more in diameter

(D) 2.5 milimeter or less in diameter

(5) To include noise as an air pollution, Indian government amended the Air act in which year ?

(A) 1981 (B) 1987 (C) 1995 (D) 2002

(6) How much amount of CO increases in evening in city ?

(A) 1.8% (B) 2.0% (C) 18% (D) 20%

(7) Company provides CNG to Ankleshwar - ..... .

(A) GAEL (B) GGCL (C) GAIL (D) GSRTC

(8) Which one is not appropriate for CNG : ..... .

(A) odourless smoke

(B) more efficient

(C) can not be adulterated easily

(D) heavier than air so no chances of fire-catching

(9) Which norm is newly added to register vehicles in Ahmedabad ?

(A) BHARAT - I (B) BHARAT - II (C) BHARAT - III (D) EURO - II

(10) Photochemical smog is formed due to pollution of ..... .

(A) vehicles (B) cement factories (C) plastic factories (D) steel factories

(11) GAIL provides CNG to ..... .

(A) Ahmedabad (B) Anklelshwar

(C) Surat (D) Rajkot

(12) What is 'Smog' ?

(A) Smoke (B) Types of pollen

(C) Radiation (D) Mixture of fog and smoke.

(13) Which of the following is not carcinogenic ?

(A) CNG (B) Smoke by combustion of Nicotin.

(C) CO (D) All given

(14) Recently CPCB researched that ..... .

(A) particles of 2.5 micrometers or less in diameter cause premature death in human

(B) noise of 150 db or high intensity disturb the peaceful life of human

(C) India occupied 24 % area of biodiversity of total area of world

(D) None of the above

(15) Natural air pollutants are ..... .

(A) smoke of vehicles, pollens, SPM

(B) SPM, smoke from chimney, smoke spread during eruption of volcano

(C) pollen of plants, smoke from chimney, vehicular smoke

(D) pollen of plants, smoke spread during volcanic eruption, SPM

(16) Air protection and control act has established in ?

(A) 1990 (B) 1975 (C) 1981 (D) 1985

(17) Scrubbers reduces which pollution ?

(A) water (B) soil (C) air (D) radiative

(18) Which gas can be removed through scrubbers ?

(A) O_2 (B) CO_2 (C) SO_2 (D) NO_2

(19) In Electrostatic precipitator's electric charge is generated on the particles of ..... .

(A) soil (B) smoke

(C) photochemical smog (D) precipitation

(20) According to Gujarat Pollution Control Board, Gandhinagar has declared SPM in 2010-11.

(A) 27.5 (B) 257 (C) 155 (D) 275

Answers : (1-C), (2-D), (3-B), (4-A), (5-B), (6-C), (7-B), (8-D), (9-C), (10-A), (11-A), (12-D), (13-A), (14-A), (15-D), (16-C), (17-C), (18-C), (19-B), (20-D)

Water pollution : When any undesirable substance mixes with water then water becomes contaminated and becomes unfit for use.

Water born desease : typhoid, hepatitis jaundice, cholera, diarrhoea, dysentery.

Law : The Water (Prevention and Control of Pollution) Act, 1974, India.

Domestic sewage as pollutant :

- It has bio-degradable organic matter.
- A mere 0.1% impurities make domestic sewage unfit for human use.
- **Bio-degradable organic matter** : They are readily decomposed by microorganisms from complex organic matter into simple organic and inorganic matter.

BOD (Biological Oxygen Demand)

- The measurement of amount of O_2 required for decomposition of biodegradable substances.
- Higher the bio-degradable matter, more O_2 utilise, more BOD.
- More BOD makes water unfit to drink.
- Drinking water has BOD value less than 1.
- BOD is meausred by Oxymeter.

Eutrophication : Means easy availability of nutrients in an aquatic habitat.

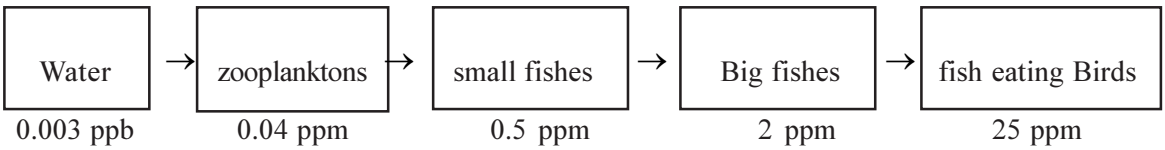
- That cause rapid growth in plants.
- Results in population-blast and pollute the water.
- **Cultural or Accelerated Eutrophications :** "Pollutants from man's activities like effluents from industries and homes can accelerate the aging process called ....."
- **Source of pollutants :** sewage, fertilizers, animal excreta, detergents.
- That increases ammonia, nitrates, nitrites and phosphorous.
  e.g. Water hyacinth Eicchornia crassipes (Terror of Bengal)

  Side effect : Water body becomes shallow and finally converted into land.

Biological magnification :

- Means increasing concentration of some toxic substances at various tropic levels of a food chain of organisms.
- some toxic substance can not degrade (non-biodegradable). They are accumulated by an organism can not be metabolized or excreted and thus passed on to next higher trophic level.

  eg : DDT (Dichloro Diphenyl Trichloro ethane)



Waste water treatment :

- Removes contaminants from waste water.
- It includes physical, chemical and biological process to remove such contaminants.
- Three phases :

Phase I : Large suspended substances are filtered out and made harmless and used as manure.

Phase II : Through bacterial activity organic waste is decomposed with airflow and water is chlorinated.

Phase III : Nitrates, phosphates are removed from this water and pure water is obtained.

'Ecosan' toilets (Ecological Sanitation) are used in Kerala and Srilanka, in which human excreta recycled into natural fertilizer.

(21) To purify water reserviors and maintenance Indian Government has passed which law ?

(A) The water (protection) act, 1986

(B) The water (prevention and control of polution) act, 1974

(C) The water(prevention and control of polution) act, 1981

(D) The water (prevention and purification) act, 1974

(22) In decreases dissolved O_2 in water ..... .

(A) higher concentration of non-biodegradeble materials

(B) lower concentration of biodegradeble materials

(C) high proportion of biodegradable materials

(D) lower concentration of non-biodegradable materials

(23) Dereacing O_2 in water, ... BOD.

(A) increases (B) some what decreases (C) no change (D) negligible

(24) It is known as 'terror of Bengal'.

(A) Clarias genipius (B) Calotropic procera (C) Polyalthia longifolia (D) Eichornia crassipes

(25) Pollutants from man's activities like effluents from industries and homes can accelerate the ageing process is called ..... .

(A) Cultural Eutrophication (B) Accelerated Eutrophication

(C) Inhibitory Eutrophication (D) A and B both

(26) If in water reserviors concentration of bio-non-degadable component is 0.003 ppb, then what will be the concentration in small fishes ?

(A) 25 ppm (B) 0.04 ppm (C) 0.5 ppm (D) 2 ppm

(27) Which is correct option for activity done during second phase of waste water treatment ?

(A) Passes air flow

(B) Bacterial decomposition

(C) Chlorination of water

(D) All given

(28) The higher accumulation of DDT found in ..... .

(A) phytoplanktons (B) eel (C) crab (D) birds

(29) In sewage water, fishes die, because of ..... .

(A) more physical matters

(B) more silt enters the body along with water

(C) specific odour

(D) dissolved O_2 is lower

(30) Drinking water sometimes passes through Chlorination because ..... .

(A) it lowers the salt amount (B) removes suspended particles

(C) kills micro organisms (D) increases O_2 concentration

(31) BOD value for drinking water should be ..... .

(A) >1 (B) =1 (C) <1 (D) not certain

(32) What will affect the BOD value if we add organic materials in water ?

(A) no effect on BOD (B) no effect on COD

(C) decrease in BOD (D) increase in BOD

(33) Arrange them in ascending value of BOD ..... .

(i) high polluted lake water

(ii) non-polluted lake water

(iii) distil water

(A) i → ii → iii (B) ii → iii →i (C) iii → ii → i (D) i → iii → ii

(34) Arrange them in increasing BOD value.

(i) distil water

(ii) tape water

(iii) drainage debris in river water

(A) i → ii → iii (B) ii → iii →i (C) iii → ii → i (D) i → iii → ii

(35) Fish eating birds →large fishes → small fishes → zooplanktons → phytoplanktons

(A) Eutrophication (B) Biological magnification

(C) Aquatic food chain (D) Succession

(36) Herbicides which are accumulate at different trophic levels of an ecosystem as a part of green revolution indicates ..... .

(A) Biological magnification (B) Succession

(C) Eutrophication (D) Organic farming

(37) What is Eutrophication ?

(A) 'Terror of Bengal,' Water Hyacinth destroys

(B) Blue green algae destroys from aquatic habitat

(C) Decrease in NH_3, NO_3, NO_2, PO_4 in aquatic habitat

(D) Easy availability of nutrients

Answers : (21-B), (22-C), (23-A), (24-D), (25-D), (26-C), (27-D), (28-D), (29-D), (30-C), (31-C), (32-D), (33-C), (34-A), (35-B), (26-A), (37-D)

***Solid Waste or Soil Pollution ***

- Commonly known as garbage that goes in trash.

Different types of wastes :

(I) Biodegradable waste : food, and kitchen waste, green waste, paper (Can also be recycled)

(II) Recyclable material : paper, glass, bottles, cans, metals, certain plastics

(III) Inert waste : construction and demolition waste, dirt, rocks, debris

(IV) Composite waste : Waste clothing, tetra packs, waste plastics (toys etc...)

Hazardous waste :

– hospital waste (Bio waste) (BMW-Bio Medical Waste)

– products of paints, chemicals

– e-waste (non-repairable electronics)

Recycling in developing countries involves manual participation thus exposing workers to toxic substances present in e-waste.

Use of plastic waste :

- Waste plastic can be mixed with polymers to build roads.
- Such plastices cleaned, shredded in small pieces, passing through 4.35 mm sieve, heated to 170°C in "Mini hot mix plant".
- Then it gets softened and coated over the aggregate.
- At 160°C Bitumen is added and blend is formed at surface of aggregate.
- Then mixture is transferred to road.
- This technique is extended to "Central Mixing Plant" also.

Benefits of Polymer - Tar :

Stronger roads, no potholes formed, less bleeding during summer, reduces consumption of bitumin, not involve extra machinery.

Agrochemicals and their effects :

As a part of green revolution the use of herbicides, pesticides and fungicides has increased tremendously in order to get more production.

They accumulate and generate harmful effects on various trophic levels.

- causes bio-magnification in terrestrial ecosystem.
- think about organic farming.

Organic farming

- It excludes the use of manufactured fertilizers, pesticides, plant growth hormones etc.
- Ramesh Chandra Dagar, a farmer in Hariyana is making use of organic farming.
- He developed 'Haryana Kisan Welfare Club" and introduced many techniques of bee harvesting, dairy management, water harvesting, composting and agriculture.

Radioactive waste :

- It is by-product of nuclear power generation, nuclear fission or nuclear technology.
- Radiation waste - Radiation - mutation in organism - Cancer.
- The first accidental event is leakage in Three Mile Island and Chernobyl incidents.
- It should be given pretreatment, shielded in containers buried within rocks, about 500 mt deep below the earth's surface.

(38) Which is bio-degradable waste ?

(A) Green waste (B) Cans (C) Clothing waste (D) Metals

(39) What is the diameter of sieve through which plastic waste passed to make tar ?

(A) 1.35 mm (B) 3.54 mm (C) 4.35 mm (D) 5.34 mm

(40) Which in not suitable with polymer-waste-bitumen mix road ?

(A) Less bleeding during summer.

(B) Road strength is twice than normal roads.

(C) No resistants towards water stagnation.

(D) Not involve any extra machinery.

(41) Biomagnification of agrochemical occurs in which ecosystem ?

(A) Forest ecosystem (B) Pond ecosystem

(C) A and B both (D) Terrestrial ecosystem

(42) It is not include in organic farming.

(A) Crop rotation (B) Plant growth hormones

(C) Green manure (D) Bio-pest control

(43) Who established "Haryana Kisan Welfare Club" ?

(A) Ramesh Chandra Duble (B) E. Z. Pillai

(B) Ramesh Chandra Dagar (D) Rajeshchandra Dagar

(44) Which are the victims of incidents of nuclear energy ?

(A) Bhopal (B) Three mile island (C) Chernobyl (D) B and C both

(45) Glohal warming can be controlled by ..... .

(A) increase in forest cutting, decrease in use of energy

(B) decrease in forest cutting, decrease in use of fossil fuel

(C) decrease in re-forestation, increase in use of fossil fuel

(D) increase in forest cutting, decrease in human population

Answers : (38-A), (39-C), (40-C), (41-D), (42-B), (43-D), (44-D), (45-B)

Green house effects and Global warming :

- Sunlight passes through atmosphere and reaches the surface of the earth.
- The CO_2 absorbs long wavelength infrared rays and reflects them back towards the earth.
- That increases temperature of atmosphere is called Green House Effect.
- Another green house gases : Methane, Nitrous, Oxide : Hydroflurocarbon (HFC) Chloro-flurocarbon (CFC)

Global Warming :

- The Earth's average temperature is around 15°C.
- By the year 2050 it can be reach upto 5°C more.
- Results of Global warming :
  - increase in temperature
  - melting of polar ice of Himalayan snow caps, result in rise in sea level that can submerged many costal areas.

Precautions :

- reduce the use of fossil fuel
- alternative source of energy like natural gas, solar-energy, nuclear-energy, should be encouraged.
- reduction in deforestation
- promote plantation

Ozone depletion in stratosphere :

- Ozone column measures in Dobson Unit (DU).
- O_3 absorbs UV-rays and inhabit them to reach on the earth and protect us from its harmful effects.
- Main reason of O_3 depletion is sources of chlorine used as CFC and HFC in Refrigerators and Freon.
- $Cl + O_3 \rightarrow ClO + O_2$

  $Clo + ClO \rightarrow Cl_2 + O_2$

  $Cl_2 \rightarrow Cl + Cl \downarrow$
- Ozone depletion first observed in Antarctic Region
- Formation of a large area of thinned ozone layer is called the ozone hole.
- Depletion of 10% in ozone layer cause 26% increase in skin-cancer and cataract.
- High dose of UV causes inflammation of cornea, called snow -blindness.

(46) Why it is not advisable to use CFC in cooling equipments (Regrigerators and A.C.) ?

(A) Swollen occurs on lungs (B) Deplete ozone layer

(C) More use of energy (D) Combines Cl_2 with CO_2

(47) Chernobyl incident concern with ..... .

(A) Ozone hole (B) Nuclear power Generation

(C) Bio magnification (D) Eutrophication

(48) Dobson Unit indicates measure of ..... .

(A) materials in bio-magnification (B) rate of ozone depletion

(C) thickness of ozone layer (D) to measure length of UV-rays

(49) One atom of Cl_2 can decomposes ..... molecules of O_3 .

(A) 10,000 (B) 1,00,000 (C) 10,00,000 (D) 1,000

(50) Ozone depletion occurs in which layer of atmosphere ?

(A) Trophosphere (B) Stratosphere (C) Mesosphere (D) Ionosphere

(51) Atmospheric CO_2 absorbs this rays : ..... .

(A) IR of short wavelength (B) IR of long wavelength

(C) UV of short wavelength (D) UV of long wavelength

(52) An average temperature of the earth is ..... .

(A) 27°C (B) 28°C (C) 15°C (D) 37°C

(53) Green house effect means : ..... .

(A) atmospheric CO_2 does not allow long wavelength UV-rays so temperature of atmosphere remains hot.

(B) atmospheric CO_2 absorbs long wavelength IR-rays and reflect them on the earth so temperature of the earth increases.

(C) atmospheric CO_2 absorbs short wavelength UV-rays and reflect them on the earth, so temperature of the earth increases.

(D) atmospheric CO_2 doesn't allow short wavelength IR-rays so temperature of the earth remains hot.

54) Which is commonly used in both refrigerators and A.C.

(A) HFC (B) Freon (C) CFC (D) None of these

Answers : (46-B), (47-B), (48-C), (49-B), (50-B), (51-B), (52-C), (53-B), (54-B)

Degradation by improper resource utilization and maintenance :

- It is important that degradation of natural resources not only due to the action of pollutants but also due to improper resource utilization practices.

Soil erosion and desertification :

- Over cultivation, unrestricated grazing, deforestation and poor irrigation practicies are main causes.
- Such large patches extend and meet over a time, a desert is created.

Deforestation :

- At beginning of twentieth centuary, forest covered about 30% of the land of India.
- By end of centuary, it shrunk to 19.4 %.
- National Forest Policy (NFP), 1988 of India has recommended 33% forest cover for plains and 67% for hills.

Reasons :

- rapidly increasing population
- more land demand for agriculture
- use of timbers and fuel wood increases
- increasing industrialization and urbanization

Jhum cultivation : It is slash and burn agriculture.

- Farmers cut down the trees of forest and burn the plant remains.
- The ash is used as fertilizer and land is used for farming or cattle grazing.

Serious effects :

- atmospheric changes occur
- rainfall decreases
- increased soil erosion decreasing soil fertility
- amount of increasing CO_2, causing green house effect
- ecosystem balance is disturbed
- irregularities arise in bio-geochemical cycles.

Reforestation is the process of restoring forest.

*** People's participation in conservation of forest ***

- Seven billion inhitants found in forest all over the world.
- 1.6 billion directly depend on forest for their livelihood.
- Declaration of state forests, national parks or wild-life reserves.
- The states in India have a strategy of "Joint Forest Management" (JFM)
- Forestry departments and communities jointly manage it.
- JFM originated from the management of Sal (Shorea robusta) forests in West Bengal.
- Sal forests increased from 11% to 20% in Midnapore district alone.
- Under JFM, the ownership of land remains with the government.

Study of three beneficial environmental issues :

(I)
- In 1931, Palace of Jodhpur, Rajasthan required timber wood from Forest.
- Bishnois failed them to do so.
- A Bishnoi woman Shri Amrita Devi laid down her life to save the trees.
- Along with her 363 lay dead.
- The Government of India has recently instituted the 'Amrita Devi Wildlife Protection Award."

(II) Chipko Movement :
- Initiated in 1974 in Uttar Pradesh, Garhwal Himalayas.
- It spread to Himachal Pradesh, Karnataka, Rajasthan, Bihar, Vindhyas.
- Mrs. Indira Gandhi (Prime minister) given the leadership to Sunderlal Bahuguna, a Gandhian activist and philosopher.

(III)
- In 1980, the Government of India has introduced the concept of Joint Management Program (JFM), to protect and manage forests.

---

(55) What is responsible for destroy of fertile layer of soil ?

(A) deforestation and weak irrigation

(B) unrestricted grazing

(C) more crop cultivation

(D) all given

(56) According to National Forest Policy (NFP-1988) how many perantage the hills of forest should be expanded ?

(A) 33% (B) 67% (C) 76% (D) 30%

(57) Reason for deforestation : ..... .

(A) increasing agriculture

(B) increasing use of timber and fuel wood

(C) increasing industrialization and urbanization

(D) all given

(58) Slush and burn cultivation means ..... .

(A) harvesting (B) green manure (C) jhum cultivation (D) A and B both

(59) Which idea gave origin of JFM ?

(A) Sal management in Dhaka (B) Cutting of Sal trees in Assam

(C) Sal management in West Bengal (D) Artificial forestation surrounding large cities

(60) Who is owner of land as per JFM ?

(A) Government (B) Village Management Association

(C) Farmer (D) Sahkari Mandli

(61) Which award is declared by Government of India for Wildlife conservation ?

(A) Chipko Movement Wildlife Protection Award

(B) Amritadevi Bishnoi Wildlife Protection Award

(C) Amritadevi Bahuguna Wildlife Protection Award

(D) Sundarlal Bishnoi Wildlife Protection Award

(62) The leadership of Chipko-movement was given to ..... .

(A) Sundarlal Bahuguna (B) Mrs. Indira Gandhi

(C) Amritadevi Bishnoi (D) Shri Vinoba Bhave

(63) Aim of Joint Management Programme is ..... .

(A) involvement of people to conserve forests

(B) forest products are available to people

(C) manage the forests with communities

(D) all given

(64) Chipko- movement initiated in ..... .

(A) 1947 (B) 1980 (C) 1974 (D) 1731

(65) Jhum cultivation has harmful and beneficial effects both because ..... .

(A) decreases food production, increases forest area

(B) deforestation occurs, increases food production

(C) deforestation occurs, only to develop medicinal plants

(D) none of these

Answers : (55-D), (56-B), (57-D), (58-C), (59-C), (60-A), (61-B), (62-A), (63-D), (64-C), (65-B)

*** True - False type questions ***

(66) The measures taken for global warming.

(1) Alternative source of energy Natural gas should be increased.

(2) Solar-energy, geo-thermal energy utilization should be decreased so that Natural resources can be saved.

(3) Inhibit reforestation and encourage plantation

(4) Rapid increase in price of fossil fuel.

(A) TFTF (B) TFFF (C) TTTF (D) TFTT

(67) (1) Sound more than 150 db couse noise pollution.

(2) Pollution can be decreased by use of unleaded petrol and diesel.

(3) 90% gaseous pollutants can remove through electrostatic precipitators.

(4) The Government of India introduced Joint Management programme to control environmental pollution.

(A) TFTF (B) TTTF (C) TTTT (D) TTFF

(68) Choose correct option for water pollution.

(1) to clean water reserviors, Indian Government introduced The water (purification and control of solution) Act, 1974.

(2) Chlorination of water is done in second phase of waste water management.

(3) Non-biodegradable components are bio magnified in food chain of aquatic habitat.

(4) Animal excreta, detergent when add to water causes exponential growth in aquatic plants.

(A) FFTT (B) TFTT (C) TFFT (D) FTTT

(69) Choose appropriate option for improper use of natural resources.

(1) 100 years pass to form fertile layer of soil.

(2) So many issues generate due to marshy soil and high salinity in industrial revolution.

(3) 63% land area should occupy by forests.

(4) Reforestation is not possible in deforest area.

(A) TTFF (B) TTFT (C) TFFF (D) TFTT

Answers : (66-B), (67-D), (68-A), (69-C)

A - Assertion R - Reason type MCQS

Read the assertion and reason carefully to mark the correct option out of the options given below :

(70) Assertion A : Available area for natural ecosystem decreases.

Reason R : Human population increases.

(A) (B) (C) (D)

(71) Assertion A : Falcon and Pelicans are threaten due to bio magnification.

Reason R : To destroy mosquitoes DDT is spreaded in water.

(A) (B) (C) (D)

(72) Assertion A : Forest cutting is one of the reason for global warming.

Reason R : Forest cutting increases green house gas CO_2.

(A) (B) (C) (D)

(73) Assertion A : DDT shows bio magnification.

Reason R : UV rays cause ozone depletion.

(A) (B) (C) (D)

(74) Assertion A : CNG is more beneficial than petrol and diesel.

Reason R : CNG is carcinogenic.

(A) (B) (C) (D)

(75) Assertion A : Smoke is precipitated in electrostatic precipitators.

Reason R : 90% particles remove in electrostatic precipitators.

(A) (B) (C) (D)

(76) Assertion A : BOD measurement decides fitness of water.

Reason R : BOD value of drinking water is always 1.

(A) (B) (C) (D)

(77) Assertion A : Shorea robusta forests expand in Midnapur District.

Reason R : Management of forest with the help of communities under JFM.

(A) (B) (C) (D)

(78) Assertion A : One atom of chlorine degrades 1,00,000 moleules of ozone and it returns back as chloride.

Reason R : Use of freon in refrigerator and AC should be banned

(A) (B) (C) (D)

(79) Assertion A : Main reason for global warming is deforestation.

Reason R : Methane and CFC are also added to list of green house gases along with CO_2.

(A) (B) (C) (D)

Answers : (70-A), (71-A), (72-A), (73-B), (74-C), (75-D), (76-C), (77-A), (78-A), (79-B)

(80)

Column- I	Column- II
(I) paper	(P) toxic waste
(II) plastic waste	(Q) inert waste
(III) spray cans	(R) recyclable material
(IV) construction waste	(S) polymer tar

	I	II	III	IV
(A)	Q	R	P	S
(B)	Q	P	R	S
(C)	R	S	P	Q
(D)	R	P	S	Q

(81)

Column- I	Column- II
(I) Rajasthan-Jodhpur	(P) Chipko-movement
(II) Garhwal-Himalaya	(Q) Ozone depletion
(III) Midnapur	(R) Shorea robusta
(IV) Stratosphere	(S) Bishnoi community

	I	II	III	IV
(A)	S	P	R	Q
(B)	R	S	R	Q
(C)	S	R	P	Q
(D)	P	S	R	Q

(82)

Column- I	Column- II
(I) GAIL	(P) Vadodara
(II) GGCL	(Q) Ankleshwar
(III) GAEL	(R) Ahmedabad
	(S) Rajkot

	I	II	III
(A)	P	S	R
(B)	Q	R	S
(C)	P	Q	R
(D)	Q	P	S

(83)

Column- I	Column- II
(I) Water hyacinth	(P) Kerala
(II) Ecoson	(Q) Three mile island
(III) Shorea robusta	(R) Terror of Bengal
(IV) Radiation incident	(S) Natural resources

	I	II	III	IV
(A)	R	P	Q	S
(B)	P	R	S	Q
(C)	S	Q	R	P
(D)	R	P	S	Q

Answers : (80-C), (81-A), (82-A), (83-D)

Pollutants :

- Non-degradable pollutants : prohibited through laws, use alternative sources.
- Degradable polluntants :
- **Primary pollutants :** They are found in the forms in which form they are added. eg. DDT
- **Secondary pollutants :** They are formed due to chemical reactions between primary pollutants.

  eg. photochemical smog, PAN

  They are more toxic than primary pollutants.

  (PAN : Peroxy Acetyl Nitrate)
- **Quantitative pollutants :** They are present in atmosphere, when their amount increases at maximum level, they act as pollutants.

  eg: CO_2, N_2O
- **Qualitative pollutants :** They are not found in atmosphere. Human activities add them in environment.

  eg. : fungicides, weedicides, DDT, etc.
- **Natural pollutants :** Natural resources like cattles, methane from paddy field, forest-fire etc.

- **Negative pollution :** Soil fertility is lost.

  eg. over grazing, soil erosion, desirable matters are removed or absence makes loss of productivity of soil.

- **Positive pollution :** Undesirable matters are present at wrong places or adding of this matters, decreases soil fertility.

  eg.: biofertilizer compensate the land. (Landfilling)

- **Point source pollution :** external source at definate places.

  eg. exit of garbage of factories, drainage of Nagar-palika.

- **Primary pollutants :**

  CO, 3-4 Benzopyrine, Benzene, Ethylene

  NO, NO_2, SO_2, SO_3

- **Secondary pollutants :** Smog = Smoke + fog

  - Term given by Devoex,
  - Smog is measured by Ringalman technique.

  (a) Losangeles smog or photochemical smog first observed in Los Angeles.

  (PAN + O_3 + Nitrogen Oxide)- Brown Smog

  (b) London Smog - Sulphar Smog

  - Coal, smoke, smog, sulphur oxide and low temperature is favourable.
  - H_2SO_4 is vaporised is called London fog.
  - Along with smog, vapour of H_2SO_4 is breathe in. As a result in 1952 4000 people died in London.

- **Acid-rain :**
  - H_2SO_4 + HNO_3 fall with rain water. (70% H_2SO_4 + 30% HNO_3)
  - Term was given by Robert August.
  - pH < 5.6
- **Green house gases :**

  CO_2, CH_4, CFC, N_2O
  - CO_2 level increases from 280 ppm to 368 ppm during 1956 to 2002.
  - Upto 2020 CO_2 amount will be doubled.
- **Effect of CO_2 fertilization :**
  - Increase in CO_2 concentration also increases rate of photosynthesis (upto some years).
  - Response against increasing CO_2 concentration is called CO_2 fertilization effect.
  - Maximum concentration of Ozone is found in stratosphere (16 km to 25 km from earth's surface).
  - First Ozone hole was found in Antarctica by Satellite Nimbus -7
  - Ozone hole generate after August and before october in Antarctica.
  - During spring season mostly seen during February to April.
  - July-October show less.

To memorised :

5th June – World Environment Day

16th September – Ozone Day

22nd April – Earth Day

2nd December – National Environment Day

3rd December, 1984 – Bhopal gas tragedy (Methyl I_{SO} Cynite)

ODS – Ozone Depleting Substance

Water-blooms – more nutrients in water

Jetropa plants – to make bio-ethanol

Green muphlar – to reduce noise-pollution plant shrubs and trees on two sides of roads

Pyrolysis – burning of solid waste in absense of O_2

Plume – Smoke releases from chimneys

Anthracosis – disease caused by coal particles

Syderosis – disease caused by iron particles

Disease caused by presense of excess metals.

- Arsenic – Black foot disease
- Mercury – Minamata
- Blue-baby syndrome – more amount of Nitrate in drinking water, that reduces O_2 carrying capacity called methamoglobinemia.
- Plumbism – Water polluted by lead
- Itai-Itai or Ouch-Ouch - bone-joint discase caused by cadmium.

Questions for NEET

(84) Increase phospharus and nitrate fertilizer in water causes ..... .

(A) decrease in growth of algae (B) Eutrophication

(C) stability in growth of algae (D) Bio magnification

(85) Air pollutants produce photochemical oxidants ..... .

(A) CO_2, CO, SO_2

(B) Nitrous oxide, Nitric oxide, Nitric Acid

(C) O_2, Cl^-, HNO_3

(D) O_3, Cl^-, SO_2

(86) Presence of E-coli in water indicates pollution of ..... .

(A) pollens of aquatic plants (B) light metals

(C) faecal matter (D) polluted water of industries

(87) In 1984 Bhopal tragedy occured because methyl isocyanide reacted with ..... .

(A) DDT (B) NH_3 (C) CO (D) H_2O

(88) Recently Govt of India has allowed mixing of alcohol in petrol what is the amount of alcohol permitted for mixing in petrol ?

(A) 2.5% (B) 10-15% (C) 10% (D) 5%

(89) Prolonged liberal irrigation of agriculture field is likely to create the problem of ..... .

(A) acidity (B) aridity (C) Salinity (D) metal toxicity

(90) Which of the following is not used for disinfection of drinking water.

(A) chlorine (B) oxygen (C) phenyl (D) florine

(91) Limit of BOD prescribed by Central pollution control board for discharge of industrial and municipal waste emptied into natural surface water is -

(A) < 10 ppm (B) < 100 ppm (C) < 30 ppm (D) < 3.0 ppm

(92) Photochemical smog pollution doesn't contain ..... .

(A) nitrogen dioxide (B) Carbon dioxide

(C) Oxygen (D) PAN (peroxy acetyl nitrate)

(93) In which one of the following BOD (Biochemical Oxygen Demand) of sewage, distillery effluent, paper mill effluent and sugar mill effluent have been arranged in ascending order ?

(A) SE < PE < S < DE (B) PE < S < SE < DE

(C) S < DE < PE < SE (D) SE < S < PE < DE

(94) In a coal fired power plant electrostatic precipitators are installed to control emission of ..... .

(A) NO (B) SPM (C) CO (D) SO_2

(95) Which one of the following is not a bioindicator of water pollution ?

(A) Blood-worms (B) Stone flies (C) Sewage fungus (D) Sludge-worms

(96) Montreal protocol aims at ..... .

(A) biodiversity conservation (B) control of water pollution

(C) control of CO_2 emission (D) reduction of ozone depleting substances

(97) BOD in river water ..... .

(A) not related with amount of O_2 in water (B) indicates presense of salmonela in water

(C) when sewage adds to water, it increases (D) not affected by population-blast of algae

(98) Which of the following plant species you would select for the production of bioethanol ?

(A) Pongamia (B) Brassica (C) Zea mays (D) Jatropha

(99) A renewable exhaustible natural resource is ..... .

(A) coal (B) petroleum (C) minerals (D) forest

(100) Chipko-movement initiated to conserve ..... .

(A) wild animals (B) forests (C) cow (D) grasslands

(101) 'World Environment Day' : ..... .

(A) 5th June (B) 5th August (C) 5th September (D) 5th December

(102) 'Ozone Day' : ..... .

(A) 16th January (B) 16th May (C) 16th August (D) 16th September

(103) "Green Muphlar" is called the pollution of ..... .

(A) Air (B) Sound (C) Soil (D) Water

(104) Gas produces in Paddy field, causes global warming is ..... .

(A) Methane (B) Chlorine (C) H_2S (D) Nitrate

(105) What does 'Water blooms' indicate in lake ?

(A) Presence of more nutrients (B) Deficiency of nutrients

(C) Deficiency of oxygen (D) Absence of aquatic insects

(106) ODS means ..... .

(A) Ozone Developing Substances (B) Ozone Depleting Substances

(C) Ozone Doubling Substances (D) Ozone Deta Sources

(107) In 'Green Muphlar', trees and shrubs can be planted at ..... .

(A) unuseful land (B) on mountains

(C) on road-divider (D) on both side of road

Answers : (84-B), (85-B), (86-C), (87-B), (88-D), (89-C), (90-A), (91-A), (92-B), (93-B), (94-B), (95-B), (96-D), (97-C), (98-D), (99-B), (100-A), (101-D), (102-B), (103-B), (104-A), (105-A), (106-B), (107-D)

Environmental Issues

Pollution

- It is 'undersirable change in physical, chemical and biological properties of air, water and soil, which directly or indirectly affect human beings.'

Air pollution

- Primarily results from burning of fossil in industries and automobiles.
- Effects include reduction in growth and yield of crops and their premature death. Affects respiratory system of humans and of animals.
- Devices such as setting chamber, cyclonic separators, bag filters, electrostatic precipitators etc. are used to control air pollution.
- Noise is also considered air pollutant.

Water pollution

- It is deterioration of water quality that makes it unfit for consumption.
- Causes are domestic sewage, industrial discharge, surface run off containing pesticides, detergents, fertilizers, silt etc., oil spills.
- Consequences of water pollution are :
  - decreased DO
  - increased BOD
  - eutrophication
  - biomagnification
  - thermal pollution
  - health hazards
- Water pollution can be controlled sewage treatment.

Solid waste management

- Solid waste refers to everything that goes out in trash.
- Sources of solid wastes are municipal wastes, industrial wastes, mining wastes, hazardous wastes, defunct ships and electronic wastes.
- Disposal of wastes consists of recovery and recycling, source reduction, burning and dumping (landfills).

Agrochemicals and their effects (Soil pollution)

- Pesticides, fertilizers, chemicals and radioactive wastes are the main source.
- Control measures include :
  - → Organic farming
  - → Solid waste disposal
  - → Soil conservation by terrace farming, growing grazing fields, afforestation, reforestation

Radioactive pollution

- Pollution of air, water and soil with radio-active materials.
- It is caused by atomic weapons and atomic explosions, atomic reactors and nuclear fuel, mining of radioactive materials.
- These waste should be changed into harm-less form and stored in safe place where they may gradually decay in a harmless manner.

Greenhouse effect and global warming

- Greenhouse effect is a naturally occurring phenomenon that is responsible for heating of Earth's surfce and atmosphere.
- Increased level of greenhouse gases (N_2O, CFCs, methane, CO_2) has led to considerable heating of earth leading to global warming. This is resulting in odd climatic changes, melting of ice caps etc.
- Control measures include cutting down use of fossil fuels, improving efficiency of energy usage, reducing deforestation, planting trees and slowing down the growth of human population.

Ozone layer depletion

- Ozone layer (layer which protects life on earth from harmful effects of UV radiations) is being depleted by ozone depleting substances (ODS), such as chlorofluorocarbons, nitrogen oxides, hydrocarbons etc.
- Ozone depletion of thinned ozone layer (ozone hole) is particularly marked over Antarctic region.
- It causes aging of skin, damage to skin cells and various types of cancer, snow blindness, cataract etc.

Deforestation

- It is the conversion of forested areas to non-forested ones.
- It results in enhanced CO_2 concentration, loss of biodiversity, disturbed hydrological cycle, soil erosion, and desertification.

●

31 Neural Control and Coordinations in Animals

- **Nervous System :**

- In Human being all physiological processes control and coordination by Nervous system and Endocrine system.
- The nervous system provides and organised network of nerves for fast coordination.
- The endocrine system provides chemical integration through hormones.
- Nerve cell is the structural and functional unit of the nervous system.
- Three main functions of nervous system : (1) Receiving sensory stimuli from external and internal environment by nerves to the brain. (2) Processing the stimuli informations by brain. (3) Responding to stimuli transrmitting impulses from brain to body parts of cells.
- The nervous system of hydra is composed of network of nerve cells.
- The nervous system is well developed in insects, which consists of brain, ganglia and nervous system.
- Vertebrates have a highly developed nervous system.

- **Structures of Nervecell : (Three parts)**

(1) Cyton : The cyton may be oval, rounded or star shaped neuroplasma, which has a relatively large and spherical nucleus. Neuroplasm has mitochondria, golgibody and small basophilic granules i.e. Nissl's grenules.

(2) Dendrites : Several short branched tapering afferent processes.

(3) Axon : A Single, cylindrical and very long efferent processes with synaptic knob.

- **Synapse :** An ending branch of axon of first neuron and sequincial dendrites of record neuron never directly attached with each other but very small physical space is there so it is called synapse.

- **On basis of processes types of neurons :**

(1) Unipolar neuron : One axon only – e.g. found in embryonic stage.

(2) Bipolar neuron : One axon and one dendrite. eg. found in retina of eye.

(3) Multipolar neuron : One axon and more dendrites – present in the cerebral cortex.

- **Types of Axon or Types of nervefibres :**

Myelinated nerve fibers	Non-myelinated nerve fibers
– Two coat are present.	– Only one coat is present.
– Outer coat is neurilemma or sheath of schwann cell.	– Only one outer neurilemma is present .
– Inner coat is made up of fatty substances.	– Myelin sheath is absent.
– Mylein is absent at a certain points which are known as nodes of Ranvier.	– Nodes of ranvier is absent.
– This type of nerve fiber is in Pariferal nerve.	– This type of nerve fiber include in autonomous nerve.

(1) Control and coordination of all physiological activities in human, is by which system ?
(A) Blood circulatory system (B) Nervous system (C) Endocrine system (D) B and C both

(2) Provides chemical coordination for control and coordination of physiological activities.
(A) Excretory system (B) Endocrine system (C) Nervous system (D) B and C both

(3) For coordination of physiological activities provide quick arrangement.
(A) Nervous system (B) Endocrine system (C) Blood vascular system (D) A and B both

(4) In which animal phylum nerve net type nervous system is present ?
(A) Porifera (B) Coelenterata (C) Protozoa (D) Arthropoda

(5) Cyton is which shape ?
(A) Star (B) Oval (C) Round (D) All

(6) Where is secretory vesicle, located in nerve cell ?
(A) Cyton (B) Synapse (C) Synaptic knob (D) Nerve fibre

(7) Location of neuron, having one axon ..... .
(A) cerebral cortex (B) retina (C) embryonic stage (D) all

(8) Location of neuron having one dendron.
(A) Embryonic (B) Cerebral cortex (C) Retina (D) All of above

(9) Type of neuron having more than two axons.
(A) Multipolar (B) Bipolar (C) Unipolar (D) None

(10) Nerve, which has covering of schwann's cell.
(A) cranial nerve (B) Spinal nerve
(C) Autonomous nerve (D) All of above

(11) Nerve with two membranes surrounding nerve fibre ..... .
(A) cranial nerve (B) autonomous nerve
(C) spinal nerve (D) A and C

Answers : (1-D), (2-B), (3-A), (4-B), (5-D), (6-C), (7-C), (8-C), (9-A), (10-D), (11-D)

- **Structures of nerve fiber resting potential and Formation of nerve impulse :**

- **Structure of nerve fiber :** The neuroplasm possesses a higher negative charge compared to it, there is a higher positive charge on the outerside of plasma membrane. The distribution of positive and negative ions is responsible for this differences in electrical charges.

- This difference in electrical charges between the inside and the outisde plasma membrane is called the 'membrane potential'. In an unexcited state of nerve fibre, it is called 'resting potential' ! It can be measured using electrodes and voltmeter. Plasma membrane of nerve fiber has main two parts.

(1) Bilayer lipids - which is impermeable for ions.

(2) Specific protein present in coat.

Functions :

(1) **Ion channels :** Aqueous and made up of protein - carry out two way transport of ions selectively. - permeable to any one kind of ion only - Na^+, K^+, Cl^- and Ca^{+2} ion channels can occur transport in direction of concentration gradiant- Ion channels can be opened and closed. Chemical and electrical charges are responsible for this - bring differences in electrical potential.

(2) It play key roll to maintain the electrical potential and change.

(3) **Ion pumps** : Transportation of ions opposite to concentration gradient - In transportation energy consume as ATP - Mainly Na^+ and K^+ pump is there. - By the Na^+ pump 2 Na^+ goes outside. While $3K^+$ comes inside by K^+ pump - Ion pump maintain potential.

- **Resting Potential :** The nerve fibre is at rest, a negative electrical charge occurs on its inner side and a positive electrical charge occurs on its outerside.
- Such a nerve fibre is called polarized.
- For this, ions responsible Na^+ are concentrated on the outside of plasma membrane whereas K^+ are concentracted on its inside.
- Moreover protein molecules having negative charges on them also occur. They can not move out of plasma membrane.
- **Active potential or initiation of the Impulse :**
- Nerve fibre can become excited in verious ways. Touch, smell, pressure, chemical changes etc. include this.
- A change in the polarity of nerve fibre is called active potential.
- In the excited region Na^+ channels open up; the electrical excitation generated in plasma membrane in this region is reponsible for this. Na^+ ions are poured inside through its ion channels in plasma membrane. The lower concentration of Na^+ inside is responsible for this. Moreover, negatively charged proteins are also responsible. Thus, due to a sudden influx of a large number of Na^+ towards the inside, the plasma membrane in this region becomes positively charged on its inside. This phenomenon is called- depolarization. It is called active potential.
- It lasts for a very short period i.e a millionth part of a second (nearly 0.5 milli sec). The excited region immediately becomes repolarised.
- For repolarization, the process of closure of Na^+ ion channels is responsible, simultaneously, K^+ ion channels open up and K^+ ions go out of the plasma membrane.
- At the end of this phase, a difference in concentration of positive ions on two sides of plasma membrane is generated. Concentration of Na^+ increases on inside and that of K^+ increases on the outside.
- The activity of Na^+ - K^+ pump removes this imbalance.

---

(12) Conduction of nerve impulse by nerve fibre is which type of process ?
(A) electrical (B) chemical (C) electrochemical (D) physical

(13) Which electrical charge is more towards outerside of neurolemma of inactive nervefibre ?
(A) Negative (B) Positive (C) Negative and positive (D) Neutral

(14) Membrane potential is ..... .
(A) difference of electric charge, innerside of nerve cell
(B) difference of electric charge, on outer surface of nerve cell
(C) difference of outer and inner electric charge of nerve cell
(D) inner and outer electric charge of nerve cell, become similar

(15) Conduction of ions by Na^+ - K^+ pump is of ..... type.
(A) Antiport (B) Symport (C) Uniport (D) biport

(16) Protein which conducts ions without utilization of ATP in neurilemma of nerve cell.
(A) Na^+ - K^+ pump (B) Na^+ passage (C) K^+ passage (D) B and C

(17) ..... is responsible for resting membrane potential of nerve fibre.
(A) Na^+ channel (B) K^+ channel (C) Na^+ - K^+ channel (D) Na^+- K^+ pump

(18) ..... is responsible for changing resting membrane potential of nerve fibre.
(A) Na^+ channel (B) K^+ channel (C) Na^+ and K^+ channel (D) Na^+- K^+ pump

(19) What is duration of active membrane potential on nerve fibre ?
(A) 1 Sec (B) 0.5 Millisec (C) 2 Sec (D) 5 Sec

(20) Which ions are responsible for creating active membrane potential ?
(A) K^+ (B) Ca^{++} (C) Cl^- (D) Na^+

(21) Condition for repolarisation is ..... .

(A) Na^+ - K^+ channels closed (B) Na^+ - K^+ channel open

(C) Na^+ channel closed, K^+ channel open (D) Na^+ channel open, K^+ channel closed

(22) Proper for repolarised nerve fibre is ..... .

(A) K^+ more towards innerside (B) Na^+ more towards innerside

(C) K^+ more towards outerside (D) B and C

(23) Filled with water and formed of protein

(A) Na^+channel (B) K^+channel (C) Ca^{+2} channel (D) All of above

(24) Responsible for opening of Na^+ channel of nervefibre.

(A) Touch (B) Smell (C) Pressure (D) All of above

(25) Important characteristic of active membrane potential ..... .

(A) Intensity of stimulation should constant (B) Intensity of stimulation should not constant

(C) Intensity of stimulation is specific and less (D) All of above

Answers : (12-C), (13-B), (14-C), (15-A), (16-D), (17-D), (18-C), (19-B), (20-D), (21-C), (22-D), (23-D), (24-D), (25-C)

- **Conduction of nerve impulse :**

- The nerve impulse in nerve fibre conducted along one direction.
- This process is self-induced because when the membrane electric potential is reduced in the nearby region, the ion-channels in the region automatically open up.
- Thus sequential depolarization and repolarization progress and the nerve impulse moves along in that direction.
- To protect the impulse weakening through diffusion all arounds nerve fibre is surrounded by a medullary layer. This conduction is very speedy (100 meters per second).
- In these nerve fibres, active potential after it is generated, does not move in cyclic waves gradually. It moves directly from one Ranvier's node to the next node. Such a conductions is known as 'Saltatory Conduction.'

- **Transmission of Impulse by Synapse (Phases) :**

Impuse - The synaptic knob of the axon, it depolarises the presynaptic membrane - Ca^{+2} permeability increases. - Ca^{+2} from the synaptic cleft enter into synaptic knob - Releases acetylcholine in synapse- Acetylcholine receptors complex - Post synaptic membrane allows Na^+ to enter the cell and cause depolarization - Generate new action potential in the post synaptic membrane. Now the nerve impulse passes to next neuron. - Now acetylcholine is hydrolysed in presence of acetylcholine esterase - Resynthesised into acetylcholine in presence of ATP.

- Ca^{+2} and Na^+ ions presence in synapse - postsynaptic membrane having receptive protein.
- Acetyle choline esterase present in synaptic cleft and post synaptic membrane.
- In synapse direction of impulse is not proper because dendrites can not produce neurotransmitters.

(26) Conduction of nerve impulse by nerve fibre is by sequential of ..... .

(A) depolarization and depolarization (B) repolarization and repolarization

(C) depolarization and repolarization (D) repolarization and depolarization

(27) Prevents spreading of nerve impulse intosurrounding medium ..... .

(A) nourilemma (B) myelin layer

(C) sarcolemma (D) both A and B

(28) Shows saltatory conduction of nerve impulse ..... .

(A) cranial nerve (B) spinal nerve

(C) autonomous nerve (D) A and B both

(29) In which nerve, active membrane potential once formed moves ahead cyclic, continuously without interruption ..... .

(A) Sympathetic nerve (B) Parasympathetic nerve

(C) Spinal nerve (D) A and B

(30) Conduction of nerve impulse is possible through ..... .

(A) Acetic acid (B) Choline (C) Acetyl choline (D) Acetyl Co-A

(31) For conduction of nerve impulse through synapse ..... .

(A) Ca^{+2} should come out from presynaptic membrane

(B) Ca^{+2} should enter presynaptic membrane

(C) Ca^{+2} should enter post-synaptic membrane

(D) Ca^{+2} should come out through post-synaptic membrane

(32) When Na^{+} Channel of post synaptic membrane open up ?

(A) When Ca^{+} enters synaptic node.

(B) When acetyl choline released in synapse.

(C) When acetyl choline is hydrolysed.

(D) Acetyl choline receptor unit is formed.

(33) Where does hydrolysis of acetyl choline occur ?

(A) Synaptic node (B) Synaptic groove

(C) Post synaptic membrane (D) Pre-synaptic membrane

(34) True sequence for conduction of nerve impulse through synapse.

(1) Hydrolysis of acetylcholine. (2) Secretion of acetyl choline

(3) Entry of Ca^{++} into synaptic knob. (4) Formation of acetyl choline receptor complex.

(A) 2 - 3 - 4 - 5 - 1 (B) 5 - 3 - 4 - 2 - 1

(C) 3 - 2 - 4 - 5 - 1 (D) 1 - 2 - 3 - 4 - 5

(35) Location of Acetyl choline esterase - ..... .

(A) Synaptic knob (B) Pre synaptic membrane

(C) Post-synaptic membrane (D) All of above

Answers : (26-C), (27-B), (28-D), (29-D), (30-C), (31-B), (32-D), (33-B), (34-C), (35-C)

- **Human Nervous System :**

Human Nervous System (parts)

- Central Nervous System (CNS)
  - Brain
  - Spinal cord
    - Systemic Nervous System
      - Conduction of nerve impulsions toward skeleton muscles
- Peripheral Nervous System(PNS)
  - Cranial Nerves → Sensory, motor or mixed types ← Spinal nerves
    - Auto nomous Nervous system (ANS)
      - Sympathatic Nervous System (SNS) → Involuntary organs and visceral muscles toward coduct of nerve impulse ← parasympathatic Nervous system(PNS)

- **Central Nervous system :** The CNS consists of the brain and the spinal cord. The CNS has two distinct regions (1) White matter- White in colour and consists of myelienated fibres (2) Gray matter- cyton bodies along with their dendrites and axon.

- Meaninges of Central Nervous System (CNS):

(1) The *inner membrane* is pia matter which is thin highly vascular membrane

(2) Middle membrane arachnoid matter which is a thin transparent membrane but non vascular

(3) The outer membrane is dura matter. The thickest membrane which covers the brain. It is also adherent to the inner surface of cranial bones by fibrous and vascular processes.

- **Human brain :**
  - Weight about 1200 to 1400 gm.
  - Number of neurons is about 100 billions.
  - The brain can be divided into three major parts.
    (1) Fore brain (2) Mid brain (3) Hind brain

(1) Fore brain :

A. Olfactory Lobes : Small, club shaped solid appear- only found in ventral view, covered by cerebral, hemisphere.

B. Cerebrum : The largest parts of brain. - They are separated from each other by a longitudinal cerebral fissure. Two hemispheres are connected by a large bundle of myelinated fibres known as corpus callosum. Surface of the ecrebral cortex is highly folded to increase the area for accommodating more nerve cells. The folds are called gyri and the depression between them are Known as Sulci Three wide and deep sulci are termed as fissures. Which divide each hemisphere into four lobes.

(B1) Anterior frontal lobe: The premotor area fronted lobe controls involuntary movements and autonomic nervous system. The associated area of the frontal lobe is concerned with memory, reasoning,learning and creative ability- control voluntery movements.

(B2) Middle parietal lobe : Associated touch, cold, temperature and pain.

(B3) Posterior occipital lobe : Visual and auditory areas which are centres for hearing and sight. Each cerebral hemisphere encloses a cavity called lateral ventricle, whcih is closed in front but open behind into third ventricle of diencephalon by a aperture known as formen of manro.

(B4) Lateral temporal lobe : Concerned with sound, smell, emotion and memory.

C. Diencephalon : Enclosed a cavity of the third ventricle, the roof of the cavity is known as the epithalamus, the right and left sides are known as thalamus and floor as the hypothalamus. Epithalamus is not formed of nervous tissues, but it is made up of blood vessels possessing folds. Which form anterior choroid plexus. Just behind this the epithalamus forms a short stalk; the pineal stalk and at its tip is a rounded body known as pineal body is present. Pineal body recreates hormone melatonin. The hypothalamus is visible in the ventral view of the brain. The optic nerve originating from the eyes forms acrossing, the optic chiarma in front of the hypothalamus. The hypothalamus is small, possessing 4gm weight. The pituitary is attached to the hypothalamus by a stalk called the infundibulum. The diencephalon encloses a cavity termed as third ventricle which communicates arterially with the lateral ventricle of cerebrum by the foramen of monro and posteriosly with the forth ventricles of medulla oblongata by a narrow passage called the iter which is present in mid brain.

(2) Mid brain- Very small and it consists of four small lobes; the coropora quadrigemia.

Quadrigemia

Upper a pair of lobes	Lower a pair of lobes
- Superior colliculi	**- Inferior Colliculi**
Which receives impulses from the eyes and muscles of head and controls visual reflexes.	Which receives impulses from the ears and muscles of the head and controls the auditory reflexes.

(3) Hind brain : It is made up by three structures.

Hind brain

Cerebellum	**Pons Varolli**	**Medulla Oblongeta**
- very large and well developed. - Two large lateral parts,the cerebral hemispheres and small vormis. - Branching tree like structure of white matter surrounded by a sheath of greatly folded grey matter. - Control muscular activities like running, talking and typing.	- An oval mass which consists of mainly nerve fibres. - Bridges the cerebellar hemispheres and medulla oblongata. - It relays impulses between medulla oblongata, cerebellar hemispheres and between the brain and cerebellum.	- It is pyramid shaped. - The posterior most of the part of the brain that connects the spinal cords. - It encloses a cavity, the fourth ventricle. - This ventricle has a very thin non nervous epithelial, folded roof is called postorior choroid plexus. - Centres for cardiac activities, respiration, vasoconstrictor that control heart beat, breathing, blood pressure salivation, swallowing vomiting, sneezing, coughing and other involuntary fuctions.

(36) Included in peripheral nervous system is ..... .

(A) brain and cranial nerves
(B) spinal cord and spinal nerves
(C) brain and spinal cord
(D) cranial nerves and spinal nerves

(37) Function of motor nerves.

(A) Conduction of nerve impulse towards peripheral gland.
(B) Conduction of nerve impulse towards muscles.
(C) From tissue/organs conduction of nerve impulse towards peripheral nervous system.
(D) Both A and B.

(38) Nerve conducting nerve impulse towards striated muscle is of ..... nervous system.
(A) sympathetic (B) parasympathetic (C) visceral nervous system (D) all of above

(39) Menninges cover which structure ?
(A) Brain (B) Central nervous system
(C) Spinal cord (D) Brain and Spinal cord

(40) Menninges without blood capillary ..... .
(A) Arachnoid (B) Middle layer (C) Duramatter (D) Piamatter

(41) Gray matter is formed of ..... .
(A) Medullated nerve fibre (B) Non-medullated nerve fibre
(C) Axon (D) Both B and C

(42) ..... is not part of forebrain.
(A) Diencephalon (B) Cerebrum (C) Cerebellum (D) Olfactory lobe

(43) More broad and deep sulci number is ..... .
(A) Four (B) Three (C) Innumeralbe (D) Two

(44) Making of ship blocks by student, this activity is associated with which lobe of cerebral cortex ?
(A) Parietal lobe (B) Occipital lobe (C) Lateral temporal lobe (D) Frontal lobe

(45) Student become very much happy due to good result, this action is associated with which lobe of cerebral cortex ?
(A) Parietal lobe (B) Occipital lobe (C) Lateral temporal lobe (D) Frontal lobe

(46) Parts of brain, which do not have nervous tissue.
(A) Thalamus - medulla oblongata (B) Hypothalamus - cerebellum
(C) Mid brain - Epithalamus (D) Medulla oblongala - epithalamus

(47) Connected by Iter ..... .
(A) Both lateral ventricles (B) Lateral and third ventricle
(C) Lateral and forth ventricle (D) Third and forth ventricle

(48) What is called left and right wall of third ventricle ?
(A) Hypothalamus (B) Epithalamus (C) Pineal body (D) Thalamus

(49) Iter is present in ..... .
(A) mid brain (B) diencephalon (C) cerebram (D) medulla oblongata

(50) What is proper for inferior coliculi ?
(A) Receives nerve impulses from eyemuscles (B) Pair of lobes in upper region of mid brain
(C) Pair of lobes in lower region of mid brain (D) Controls reflex action of vision

(51) Vermis is part of ..... .
(A) Diencephalon (B) Cerebellum (c) Cerebrum (D) Medulla oblongatal/Hind brain

(52) Oval structure is included in which part of the brain ?
(A) Fore brain (B) Midbrain (C) A and B (D) Hind brain

Answers : (36-D), (37-D), (38-C), (39-B), (40-A), (41-D), (42-C), (43-B), (44-D), (45-C), (46-D), (47-D), (48- (D), (49-A), (50-C), (51-B), (52-D)

- **Reflex Action :** It is a monotonous or unchangeable response to a stimulus. An involuntary response to stimulus given by reflex center of brain /spinal cord without knowledge of the voluntary centres of brain is called reflex action.

- Reflex action is done by brain and spinal cord.

With knowledge ← Reflex action → Without knowledge

↓ ↓ ↓

When see food saliva secretion ocur (conduct by brain) Heart beat, Peristalysis of digestion canal

- **Reflex Arc :** The nerve chain between a receptor and effector organ

Receptor organs - Skin Stimulation → afferent nerve (Sensory impulse) → CNS (Brain, Spinal cord)

↓

Analysis of sensory impulses

↓

Motor nerve (Motor impulse)

Respond ← effector organ (muscle and gland)

- **The animal by reflex action respond quickly to harmful stimuli and serves the animal from harmful effect.**

Types of Reflex Action

Sensory Reception and processing (Interitances)	**Conditioned Acquired Reflexes**
- Away the hand from hot material	- Conditioned reflexes are acquired during life.
- Blinking of eyes	- Gradually developed by training and experience
- Sneezing	- No inheritance
- Yawning	- Animal behaviours
- Knee jerk	- eg. habits, likes and dislikes, prejudices and
- Heart beats	interests. Typing, riding a bicycle, knitting etc.
- Paristalisis are examples	

(53) Where are reflex centers situated ?

(A) Nerves (B) Brain (C) Spinal cord (D) B and C both

(54) Movement of diaphragm during respiration is reflex activity of ..... .

(A) spinal cord (B) brain (C) nerves (D) A and B both

(55) Reflex activity occurring without our knowledge is ..... .

(A) peristalsis of digestive tube (B) shouting for escaping from wild animal

(C) flickering of eye (D) touching of hot substance, hand is removed

(56) Location of unipolar nerve cell in spinal cord.

(A) Ventral root (B) Dorsal root gangtion

(C) Dorsal root (D) Between dorsal and ventral horn

(57) Reflex center for reflex activity occurring in spinal cord.
(A) Sensory neuron (B) Motor neuron (C) Inter connected neuron (D) All of above

(58) How is reflect arch in spinal cord is formed ?
(A) Receptor organ - spinal cord - brain - muscle (B) Receptor organ - brain - muscle - spinal cord
(C) Receptor organ - brain - muscle (D) Receptor organ - spinal cord - muscle

(59) ..... is unconditional reflex action.
(A) Yawning (B) Kneeting (C) Tying (D) Cycling

(60) ..... is acquired reflect action
(A) Sneezing (B) Habits of person
(C) Peristalsis of digestive system (D) Flickering of eyes

Answers : (53-D), (54-B), (55-A), (56-B), (57-C), (58-D), (59-A), (60-B)

- **Eye structure and Mechanism -**

The eyes are located in a deep protective bony cavities, known as orbits. The eye is hollow spherical in shape, about 2.5 cm in diameter and its weight is about 6 to 8 g.

The eye has two parts wall and contents.

↓

Walls

↓

- Made up of three layers.

(1) Sclera :
- The fibrous layer consists of sclera and cornea.
- Sclera froms the posterior 5/6 part of the collagen fibres.
- While cornea forms, 1/6th part made up of connective tissues and lacks blood vessels.
- Conjunctiva is thin, transparent and composed of stratified epithelium which cornea externally covers and exposed part of sclera.

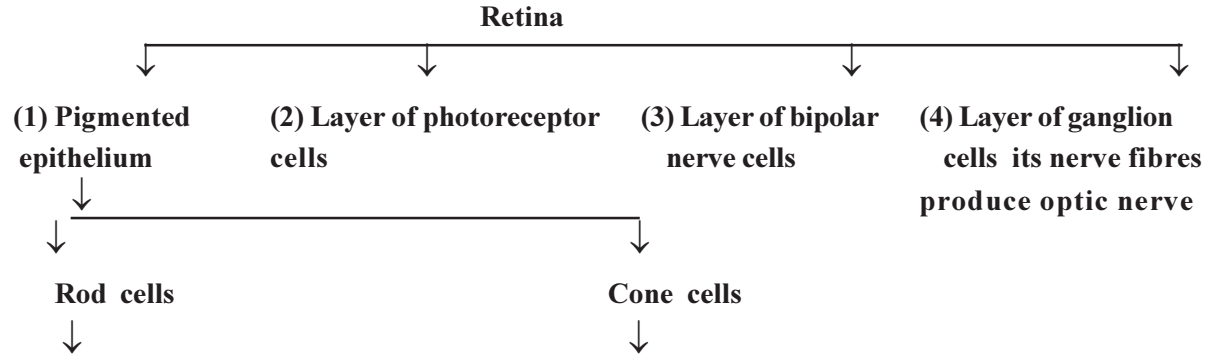
(2) Choroid :
- It is composed of connective tissue and blood vessels. Blood vessels of choroid nourish the retina.
- Iris is a circular shelf like diaphragm and perforated in the centre by an aperture termed as pupil.
- The iris works like diaphragm of a camera. Just behind the iris, the choroid is thickened to form ciliary body. Which contain circular and radial muscle fibers.

Contents

↓

- **Lens is solid, elastic, transparent biconwex** and consists of laminated fibrous tissue which is enclosed in thin elastic membrane known as lens capsule. The lens focuses light on retina.
- The lens divides eyeballs cavity into two chambers.
- The anterior small aqueous chamber and the posterior large vitreous chamber.
- Aqueours chamber is filled with a watery clear fluid, the aqueous humor, which is secreted by ciliary processes of ciliary body.
- The vitreous chamber is full of transparent fluid thick the vitreous humour which is secreted by retina during development of eye.

(3) **Retina :** It is delicate layer of optic part. It compared of four layers.



- Purple pigments termed as rhodopsin
- Very sensible
- Function at night and dimlight.
- In bright light rhodopsin splits in to scotopsin and retinal through a process of bleaching. Spliting of rhodopsin depolarizes the rod cells and that releases a neurotransmitter which sends the nerve impulse to there optic nerve via bipolar and ganglion cells.
- In the dark, rhodopsin is resynthesized from retinal and scotopsin. which becomes functional.
- On the retina there is a small depression known as fovea. Which has only cone cells.
- The retina from where optic nerves starts is known as blind spot. It lacks the receptor cells and is insensitive to light.

- It contains iodopsin pigments.
- It function in day light and produce complete images and give colour vision.
- They are less sensitive as compared to rod cells.
- In human eye, three types of cone cells are found which produce their own photopigment, each absorbing light rays of different wave lengths.
- Erythrolab (red light)
- Chlorolab (green light)
- Cynolab (blue light)
- Lack of one of moe types of conecells cause colour blindness.

---

(61) Cornea of eye is part of which membrane ?
(A) Sclera (B) Fibrous layer (C) Choroid (D) Retina

(62) What is improper for eye membrane ?
(A) It is transparent. (B) It is formed of stratified epithelium.
(C) It innervates eye ball. (D) It completely covers cornea.

(63) Number of layer covering lens from outer side ..... .
(A) Three (B) Two (C) One (D) Four

(64) ..... is layer of eye, made up of collagen.
(A) Retina (B) Choroid (C) Cornea (D) Sclera

(65) Optic nerve is formed of which nervefibre ?
(A) Cone cells (B) Ganglionated cell layer
(C) Bipolar neurons (D) Cone cells and Rod cells

(66) It is second layer, of retina from innerside
(A) photo receptor cell layer (B) ganglionated nerve cells
(C) bipolar neurons (D) pigmented epithelium

(67) Location of blood capillary which providing nutrition to retina.
(A) Retina (B) Choroid (C) Sclera (D) Cornea

(68) What is present in rod cells during intense light ?

(A) Rhodopsin (B) Iodopsin

(C) Scotopsin (D) Scotopsin and retinal

(69) How fovea centralis differ from normal retina ?

(A) Presence of cone cells (B) Presence of cone and rod cells

(C) Absence of Rod cells (D) Absence of cone and rod cells

(70) Cyanolab is sensitive for which colour ?

(A) Blue (B) Red (C) Green (D) All of above

(71) The fluid occurs in the large cavity of eye ball is secreted by whom ?

(A) Choroid (B) Processes of ciliary body

(C) Retina (D) Sclera

(72) During daytime, in visual process, whose position is changed ?

(A) Red cells (B) Cone cells (C) Pigment epithelium (D) A and B both

Answers : (61-B), (62-C), (63-B), (64-D), (65-B), (66-C), (67-B), (68-D), (69-C), (70-A), (71-C), (72-D)

- **Structure of ear, mechanism of hearing and abnormalities of nervous system :**
- **Structure of ear :** It has three parts (1) External ear (2) Middle ear and (3) Inner ear

(1) External ear :

Pinna	External auditory canal
- It is oval, some what funnel shaped - Stiff outer ridge is known as helix. - While its flexible lower lobe is known as lobule.	- It is an S-shaped tube which spreads inwards up to the tympanic membrane. - The outer part of mear bears hair which prevents entry of dust particles. - Its inner parts have wax glands. It secretes a brownish falty substance known as earwax. It protects and lubricates the lining of canal.

(2) Middle ear : Divide in three regions.

Tympanic membrane	Fenestra/Cavity apertures	Inner bony wall
It is a thin, oval membrane made up of connective tissue with fibres, covered with skin outside and muscle membrane inside.	Three small movable articulated bones. The ear ossicles. - Hammer shaped ossicle is called the malleus which attached to tympanic membrane. - Inner ossicle known as stapes is straupshaped. - The middle anvil shaped is known incus. It is externally joined to the malleus and internally to the stapes.The function of ear ossicles is to trasmit the vibration from the tympainic membrane to internal ear and amplify vibration 20 times.	- It has two apertures lining covered. - Upper aperture is known as fenestra ovalis - Lower aperture is known aperture rotunder.

(3) Internal ear : It is an irregular, delicate and complicated organ, it is known as membranous labyrinth, which is enclosed in a bony labyrinth. The space between these two is filled with fluid called perilymph. The membranous labyrinth is also filled by another fluid, the endolymph. The membranous labyrinth consists of 3 parts : (1) Vestibule duct (2) Semi circular ducts (3) Cochlear duct.

(1) **Vestibule** : Vestibule is a sac like part and consists of two chambers, the large utricle and smaller saccule, which leads into cochlear duct. Two Sensory spots are present in macula of utricle and macula of sacrule. They are located in the wall of utricle and sacule. A macula consists of hair cells and supporting cells. The supporting cells contain many minute particles known as earstones or otoliths composed of calcium carbonate and protein.

(2) **Semi circular ducts** : Three semi circular ducts are arranged on outerior porterior and lateral sides. Each duct opens into the utricle on both ends. One end of each duct is swollen to form ampulla. Sensory spot is found on each ampulla and is known as crista its function is to maintain equilibrium.

(3) **Cochlear duct** : It is spirally coiled tube, resembling a shell of snail. It is a part of the bony labyrinth. Which encloses the cochear duct together termed as the cochlea. Cochlea has three longitudinal chambers, known as scalae, which are separated from one another by a thin membrane. The middle chamber is termed as scale media, which consists of organ of corti. The roof of scala media is called as Reissner's membrane and floor is termed as basilar membrane. Organ of corti is an organ of hearing which consists of receptor cells hair cells and supporting cells. The hair cells bear hair at their free surface and at the basal region synaps has contact with dendrites of nerve cells. The tips of hair are embedded in a tectorial membrane, supporting cells are of two types, longer pillar cells and shorter phalangeal cells.

- **Mechanism of Hearing :** Sound waves → way of external ear → vibrations on tympanic membrane → vibratious of ear ossicles → Vibrations through ear ossicles to fenestra ovalis → vibration pass to the basilar membrane of cochlear canal → Travel on the other side of the canal to Reissner's membrane → Vibration cause a movement of endolymph → catch, sensory cells which found in the organ of corti → Nerve impulses develop → Transmitted by nerve fibers via auditory nerves → Auditory area of cerebrum. Where the impulses are analysed and sound is recognised.

- **Disorelers of the Nervous System : -**

Disordes of the Nervous System

Multiple sclerosis	**Parkinson's disease**	**Sciatica**
- **Reason :** Mylin sheath of spinal nerve are damaged, effect occurs on the function of nerve. This	- Associated with deficiency of the neurotransmitted dopamine and ageing.	- Caused by degeneration of intervertebral disc. it is also popular as 'slipped' out of

disease occur in young and middle aged persons. **Symptons :** Shaky movement of limb rapid involuntary movements of the eyes. defects in speech paralysing in greater or less a degree, finally causing death.	Symptoms : Tremor, rigidity and lack of spontaneous movements.	intervertebral disc. - **Symptom :** A continuous pain in back, thigh and leg.

(73) Helix is part of ..... .
(A) middle ear (B) internal ear
(C) tympanic membrane (D) external ear

(74) Location of round window ..... .
(A) external wall of middle ear (B) inner wall of inner ear
(C) inner wall of middle ear (D) inner wall of external ear

(75) Ossicles present in oval window - ..... .
(A) malleus (B) incus (C) stapes (D) all of above

(76) Location of exolymph fluid.
(A) Membranous labyrinth
(B) Bony labyrinth
(C) outside of bony labyrinth
(D) Between bony labyrinth and membranous labyrinth

(77) Otoliths are present in which structure ?
(A) Mecula utricle (B) Crista (C) Mecula sacculi (D) A and C both

(78) What is location of crista ?
(A) Sacculi (B) Utricle (C) Ampulla (D) Organ of corti

(79) Location and function of organ of corti - ..... .
(A) scela, hearing (B) scela, balance (C) saccule, hearing (D) ampulla, balance

(80) Sensory cells of organ of Corti, identify movement of ..... .
(A) Exolymph (B) Ressiner's membrane
(C) Bacillar membrane (D) Endolymph

(81) Tactorial memberane is part of ..... .
(A) Maculla (B) Organ of corti
(C) Crista (D) Ressiner's membrane

(82) It is disease due to irregularity of nervous system due to injury of medulla-membrane ..... .
(A) Multiple sclerosis (B) Parkison's (C) Sciatica (D) Atherosclerosis

(83) Disease due to deficiency of neurotransmitter.
(A) Multiple sclerosis (B) Sciatica (C) Parkinson's (D) Arthritis

(84) Due to dislocation of intervertebral disc ..... discease is observed.
(A) Multiple sclerosis (B) Arthritis (C) Epilepsy (D) Reumatid arthritis

Answers : (73-D), (74-C), (75-B), (76-D), (77-D), (78- C), (79-A), (80-D), (81-B), (82-A), (83-C), (84-C)

Column type questions :

(85)

Column-I	Column-II	
(1) Multipolar neuron	(P) Retina	(A) (1-R) (2-Q) (3-P)
(2) Bipolar neuron	(Q) Embryo	(B) (1-R) (2-P) (3-Q)
(3) Unipolar neuron	(R) Cerbral cortex	(C) (1-P) (2-R) (3-Q)
		(D) (1-Q) (2-P) (3-R)

(86)

Column-I	Column-II	
(1) Cyton	(P) Autonomous nerve	(A) (1-R) (2-S) (3-P) (4-Q)
(2) Dendron	(Q) Secretory vesicles	(B) (1-R) (2-P) (3-Q) (4-S)
(3) Synaptic Knob	(R) Nissi's granules	(C) (1-R) (2-Q) (3-S) (4-P)
(4) Non-medullated	(S) Pointed processes nerve	(D) (1-R) (2-S) (3-Q) (4-P)

(87)

Column-I	Column-II	
(1) Ion pump	(P) Na^+ outside neurilema	(A) (1-R) (2-S) (3-P) (4-Q)
(2) Ion passage	(Q) Na^+ inside neurilema	(B) (1-S) (2-R) (3-P) (4-Q)
(3) Resting potential	(R) Active transport of ions	(C) (1-P) (2-S) (3-R) (4-Q)
(4) Membrane potential	(S) passive transport of ions	(D) (1-Q) (2-S) (3-P) (4-R)

(88)

Column-I (Process)	Column-II (Follow up process)
(1) Depolarilation of presynaptic membrane	(P) Na^+ enters post synaptic membrane.
(2) Active membrane potential in post synaptic membrane	(Q) Acetyl choline released statasco form and further conduction
(3) Ca^{+2} enters synaptick nob	(R) Hydrolysis of Acetyl choline
(4) Receptor complex is formed with acetyl choline	(S) Ca^{+2} enters synaptic knob

(A) (1-Q) (2-S) (3-P) (4-R) (B) (1-S) (2-R) (3-Q) (4-P)
(C) (1-R) (2-S) (3-Q) (4-P) (D) (1-Q) (2-S) (3-P) (4-R)

(89)

Column-I	Column-II	
(1) Visceral nervous system	(P) Thin and capillarized membrane	(A) (1-Q) (2-S) (3-P) (4-R)
(2) Autonomous nervous system	(Q) Conduction of nerve impulse towards striated muscle	(B) (1-Q) (2-S) (3-R) (4-P)
(3) Dura matter	(R) Thick and durable membrane	(C) (1-S) (2-Q) (3-R) (4-P)
(4) Pia matter	(S) Conduction of nervous impulse towards visceral muscles	(D) (1-Q) (2-P) (3-R) (4-S)

(90)

Column-I	Column-II	
(1) Lateral temporal lobe	(P) Control of voluntary nervous system	(A) (1-Q) (2-R) (3-P) (4-S)
(2) parietal lobe	(Q) Associated with sound and smell	(B) (1-Q) (2-R) (3-S) (4-P)
(3) occipital lobe	(R) Sensation of touch	(C) (1-Q) (2-S) (3-R) (4-P)
(4) Frontal lobe	(S) Audio-visual centres	(D) (1-R) (2-Q) (3-S) (4-P)

(91)

Column-I	Column-II	
(1) Iter	(P) Medulla oblongata	(A) (1-S) (2-R) (3-T) (4-Q) (5-P)
(2) central cavity	(Q) Diencephalon	(B) (1-R) (2-S) (3-P) (4-Q) (5-T)
(3) Lateral ventricles	(R) Mid brain	(C) (1-R) (2-S) (3-T) (4-Q) (5-P)
(4) Third ventricle	(S) Spinal cord	(D) (1-S) (2-R) (3-Q) (4-T) (5-P)
(5) Forth ventricle	(T) Cerebrum	

(92)

Column-I	Column-II	
(1) medulla oblongata	(P) Branched structure like tree	(A) (1-Q) (2-R) (3-P) (4-S)
(2) Olfactory lobe	(Q) Pyramid shape	(B) (1-R) (2-Q) (3-P) (4-S)
(3) cerebellum	(R) Dumbel shape	(C) (1-P) (2-R) (3-Q) (4-S)
(4) Bridge	(S) Oval shape	(D) (1-S) (2-R) (3-P) (4-Q)

(93)

Column-I	Column-II	
(1) Retina	(P) Covered fibrous tissue	(A) (1-S) (2-Q) (3-R) (4-P)
(2) Eye membrane	(Q) Delicate layer at the site of vision	(B) (1-P) (2-Q) (3-R) (4-S)
(3) Ciliary body	(R) Thick Choroid	(C) (1-Q) (2-S) (3-P) (4-R)
(4) Lens	(S) Stratified Epithelium	(D) (1-Q) (2-S) (3-R) (4-P)

(94)

Column-I	Column-II	
(1) Rod cells	(P) Absence of rod cells	(A) (1-R) (2-S) (3-Q) (4-P)
(2) Cone cells	(Q) Absence of recepter cells	(B) (1-R) (2-S) (3-P) (4-Q)
(3) Fovea centralis	(R) Rhodopsin	(C) (1-S) (2-R) (3-P) (4-Q)
(4) Blind spot	(S) Iodopsin	(D) (1-R) (2-Q) (3-P) (4-S)

(95)

Column-I	Column-II	
(1) Eustachian tube	(P) $CaCO_3$ + protein	(A) (1-Q) (2-S) (3-R) (4-P)
(2) Stapes	(Q) "S" shaped	(B) (1-S) (2-Q) (3-P) (4-R)
(3) Ossicles	(R) Tympanic membrane	(C) (1-Q) (2-S) (3-P) (4-R)
(4) Oval window	(S) Band shape	(D) (1-P) (2-S) (3-Q) (4-R)

Answers : (85-B), (86-D), (87-A), (88-C), (89-B), (90-B), (91-C), (92-A), (93-D), (94-B), (95-C)

- **A - Statement, R - Reason type questions**

Questions given below, select proper option for it.

(A) A and R both right, R is explanation of A.

(B) A and R are right, but R is not explanation of A.

(C) A is right, R false.

(D) A is false, R right.

(96) Statement A : Conduction of nerve impulse in nerve fibre is autonomous.
Reason R : Ion channels open automatically when active membrane potential increases in nearest region of nerve fibre.
(A) (B) (C) (D)

(97) Statement A : Direction for conduction of nerve impulse is fixed in synapse.
Reason R : Dendrites can not secrete neurotransmitter substances.
(A) (B) (C) (D)

(98) Statement A : In autonomous nerve, saltatory conduction of nerve impulse is not observed.
Reason R : Nerve fibres of autonomous nerves are non-medullary.
(A) (B) (C) (D)

(99) Statement A : Olfactory lobe can be visible only from dorsal view of brain.

Reason R : Olfactory lobes are totally covered by cerebral hemisphere.

(A) (B) (C) (D)

(100) Statement A : Due to repolarisation, on both side of neurilemma, difference occurs in concen tration of positive ions.

Reason R : Na - K pump removes imbalance.

(A) (B) (C) (D)

(101) Statement A : Conduction of nerve impulse takes place through synapse.

Reason R : Acetyl choline esterase is secreted in synapse.

(A) (B) (C) (D)

(102) Statement A : Kneeting is acquired reflex action.

Reason R : Kneeting is inherited by person.

(A) (B) (C) (D)

(103) Statement A : Depolarised rod cells release neurotransmitter conduction of nerve impulse towards optic nerve.

Reason R : In intense light Rhodopsin dissociates into scotopsin and Retinol.

(A) (B) (C) (D)

(104) Statement A : The function of inner ear is heading and maintaining balance.

Reason R : Organ of corti present in the structure of inner ear.

(A) (B) (C) (D)

(105) Statement A : In multiple sclerosis disease rapid, uncontrol movement of eyes is observed.

Reason R : In this disease medulary layer is affected so function of nerves is effected.

(A) (B) (C) (D)

Answers : (96-C), (97-D), (98-A), (99-D), (100-B), (101-C), (102-C), (103-A), (104-B), (105-A)

- **True - False (T - F) Type questions :**

Select proper option for T/F from the questions given below :

(106) (1) Multiple sclerosis is due to dislocation of intervertebral disc.
(2) In Epilepsy deficiency of dopamine is found.
(3) Multiple sclerosis, paralytic attack leads to death.
(4) In parkinson's discease selfinduced movement increases.
(A) FFTT (B) FFTF (C) FFFF (D) FTTT

(107) (1) Phalangeal cells are present in the structure of maculla.
(2) Semicircular canals opens at both the ends of sacculi.
(3) Scale are cavities of cochlea.
(4) Ossicles, amplifies sound waves 20 times.
(A) FFFT (B) FTTF (C) FTTT (D) FFTT

(108) (1) Fovea centralis is active during daylight.
(2) Aqueous humor is secreted from processes of ciliary body.
(3) Cynolab is sensitive for blue colour.
(A) FTT (B) FFF (C) TTT (D) TFF

(109) (1) Ear wax is brown coloured oily substance.
(2) Round window is apeture in the inner wall of mid ear.
(3) Retina is formed of three layers.
(4) Sclera is formed of 6/5 part.
(A) TFFF (B) TFFT (C) TTTF (D) TTFF

(110) (1) Cyton of motor neuron is in dorsal horn of spinal cord.
(2) Third ventricle connects with forth ventricle by iter.
(3) Grooves of cerebral cortex are called sulci.
(4) Upper lobes of mid brain are called superior coliculi.
(A) TTFT (B) TTTT (C) TFFT (D) TTFF

(111) (1) Brain weighs approximate 1200 to 1400 gm.
(2) Feeling of pain is by parietal lobe.
(3) Synapse has approximate 200A^{o} length.
(A) FFF (B) TTT (C) TFF (D) FTT

(112) (1) Speed of conduction of nerve impulse is 100 m/sec.
(2) For Repolarisation Na^{+}channel opens, K^{+}channel closes.
(3) Na^{+} - K^{+} pump, opens and closes.
(4) Floor of Diencephalon is called hypothalamus.
(A) FFTT (B) FTTT (C) FFFT (D) TTTT

(113) (1) Speed of conduction of nerve impulse is physical, chemical, electrical change observed in neurilemma.
(2) Nissl's granules are acidophilic.
(3) Multipolar neuron is found in retina.
(4) Medullary sheath is found on nerves fibres of autonomous nerves.
(A) TFFT (B) TFTT (C) TTFF (D) FFTT

Answer : (106-B), (107-D), (108-C), (109-A), (110-C), (111-B), (112-C), (113-A)

(114) Medullary sheath is shown in Fig. 1.
(A) T (B) U (C) V (D) R

(115) What is indicated by V in Fig. 1 ?
(A) Medullary sheath (B) Neurilemma
(C) Ranvier's node (D) Axon

(116) White, shining structure found in Fig. 1.
(A) T (B) U (C) V (D) R

(117) What is indicated by W in Fig. 1 ?
(A) Node of Ranvier (B) Synaptic Knob
(C) Dendron (D) Nissl's granunule

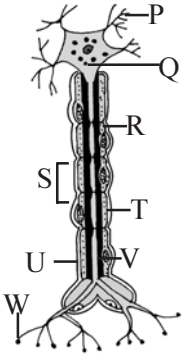


Figure 1

(118) What is indicated by "P" in Fig. 2 ?
(A) Depolarization (B) Polarization
(C) Non polarity (D) Repolarization

(119) In Fig. 2 "T" and "S" indicates sequential change of location.
(A) K^{+} and Na^{+} (B) K^{+} and K^{+}
(C) Na^{+} and K^{+} (D) Na^{+} and Na^{+}

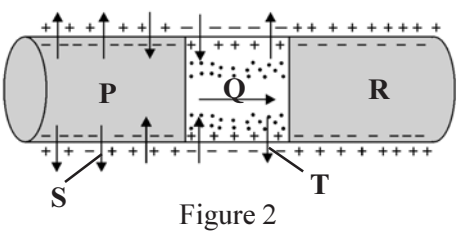


Figure 2

(120) In Fig. 3 where is receptor protein ?

(A) V (B) X

(C) U (D) P

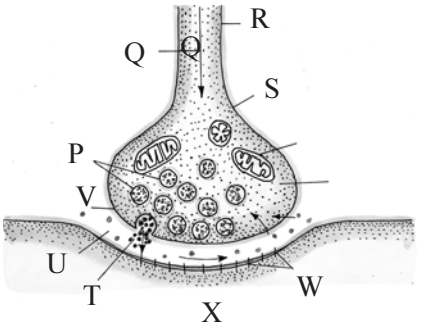


Figure 3

(121) What is indicated by "U" in Fig. 3 ?

(A) Groove of Synapse (B) Presynaptic membrane

(C) Post synaptic membrane (D) Receptor protein

(122) Fig. 4 indicates which view of Brain ?

(A) Dorsal (B) Lateral

(C) Ventral (D) Posterior

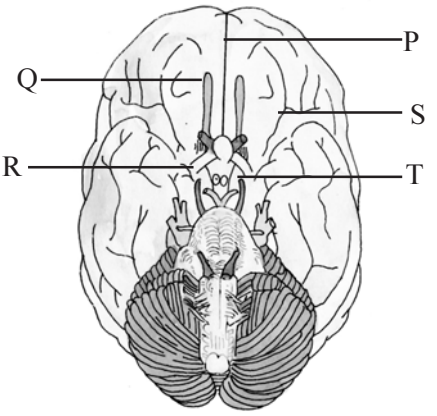


Figure 4

(123)Shows olfactory bulb in Fig. 4.

(A) S (B) T

(C) Q (D) R

(124) What is indicated by "R" in Fig. 4 ?

(A) Infundibulum (B) Pituitary gland

(C) Olfactory bulb (D) Optic chiasmata

(125) What is indicated by "P" in Fig. 4 ?

(A) Sulci (B) Gyri (C) Lateral groove (D) Corpus callosum

(126) In Fig. 5 indicates taste region - ..... .

(A) Q (B) X

(C) W (D) U

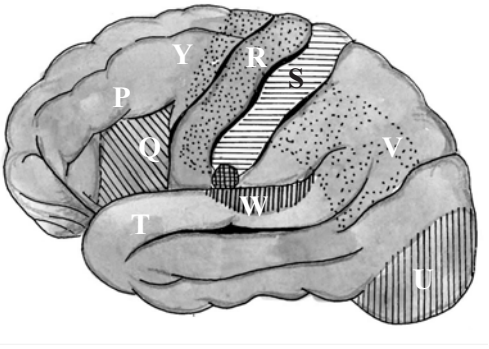


Figure 5

(127) Which region of Fig.5 associated with creative skill ?

(A) Q (B) Y

(C) R (D) P

(128) What is indicated by "Y" in Fig. 5 ?

(A) Premotor area (B) Sensory speech area

(C) Motor speech area (D) Auditory area

(129) Corpus callosum is shown in Fig. 6.

(A) S (B) Q

(C) P (D) W

(130) What is indicated by "S" in Fig. 6 ?

(A) Post choroid plexus (B) Ant. choroid plexus

(C) Coliculi (D) Infundibulum

(131) Which is midbrain part in Fig. 6 ?

(A) Q (B) R

(C) W (D) S

(132) Which structures are hind brain in Fig. 6 ?

(A) W, Y, V (B) Q, X, V

(C) U, W, Y (D) X, W, Y

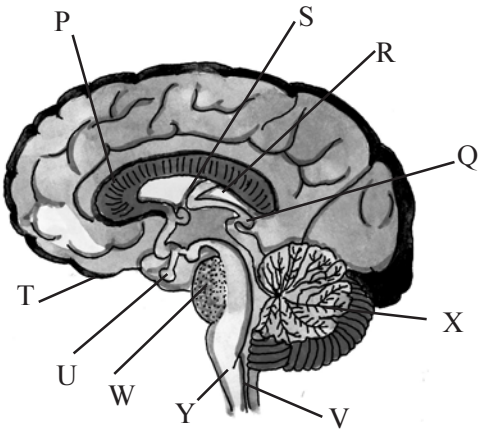


Figure 6

(133) Iter is shown in Fig. 7.

(A) S (B) U

(C) T (D) R

(134) Which part, surrounds "R" in Fig.7 ?

(A) Mid brain (B) Medulla oblongala

(C) Diencephalon (D) cerebram

(135) In Fig. 7, which structure is not for ventricles ?

(A) Q (B) S

(C) R (D) T

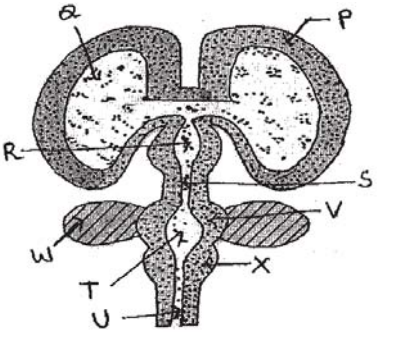


Figure 7

(136) In Fig. 8 indicates dorsal root ganglion.

(A) W (B) Z

(C) U (D) V

(137) What is indicated by "U" in Fig. 8 ?

(A) Sensory nerve cyton (B) Motor nerve cyton

(C) Interglial nerve cells (D) Dorsal root ganglion

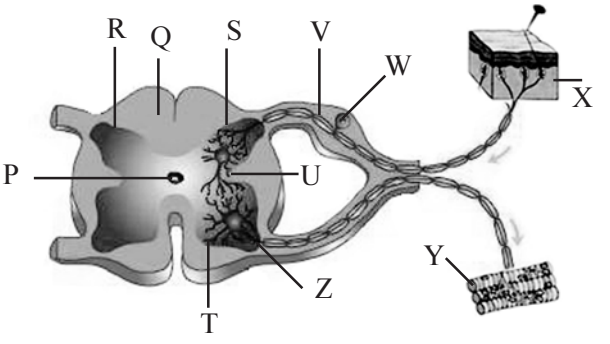


Figure 8

(138) Shows Idopsin in Fig. 9.

(A) Q (B) P

(C) T (D) U

(139) What is indicated by "U" in fig. 9 ?

(A) Photo receptor cells (B) Pigmented epithelium

(C) Bipolar neuron (D) Ganglionated cells

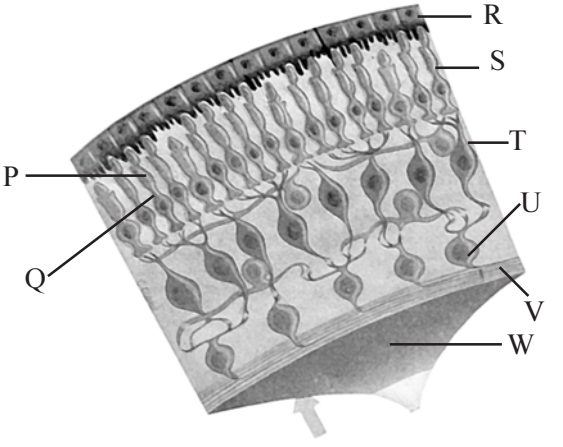


Figure 9

(140) In Fig. 9 Bipolar neurons are shown.
(A) S (B) U (C) T (D) V

(141) What is indicated by "P" in Fig. 10 ?
(A) Post semicircular canal (B) Ant. semicircular canal
(C) Lateral semicircular canal (D) Dorsal semicircular canal

(142) Where are otoliths found in Fig. 10 ?
(A) T (B) V
(C) S (D) W

(143) What is shown by "V" in Fig. 10 ?
(A) Crista (B) Macula
(C) Organ of corti (D) Otolith

(144) In Fig. 10 tissue is indicated.
(A) S (B) T
(C) U (D) V

(145) Which internal structure is shown by Fig. 11 ?
(A) Cristra (B) Macula
(C) Int-ear (D) Organ of corti

(146) Tactorial membrane is shown in Fig. 11 ..... .
(A) P (B) S
(C) R (D) Q

(147) "Q" is cavity of ..... shown in Fig. 11.
(A) Vestibule (B) Cochlea
(C) Ampulla (D) Saccule

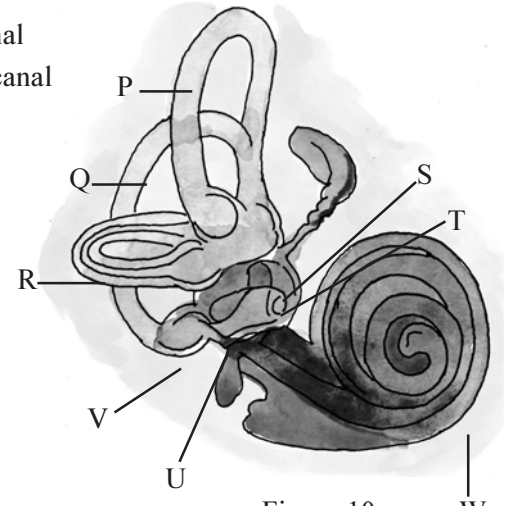


Figure 10

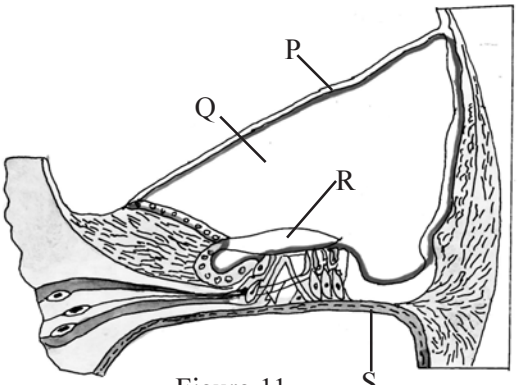


Figure 11

Answers : (114-B), (115-D), (116-A), (117-B), (118-D), (119-C), (120-B), (121-A), (122-C), (123-C), (124-D), (125-C), (126-B), (127-D), (128-B), (129-C), (130-B), (131-A), (132-D), (133-B), (134-C), (135-B), (136-D), (137-C), (138-B), (139-D), (140-C), (141-B), (142-C), (143-A), (144-B), (145-D), (146-C), (147-B)

- **Cerebral Spinal fluid :** It present in middle and inner meanings even in ventricles and cranial, iter also present surrounded to spinal cord. It is structurally identical to blood plasma. Protein and Cholesterole less in amont. It gives protections against jurking. It gives floatting to brain.

- **Systems Occurs in Brain :**

(1) Reticulate System	(2) Limbic system
- It is made up by network of entire nerve fibers which are present in brain item. It has two group centers which is made up by neurons. - Its centers conveys massages to centers of brain accordingly. - It has centers of sleep and wake. Its stimulate situation person get wake cond. less stimuli situation induce sleepness.	- It locates inner side of cerebrum Frontal to be parietal, Thalamus and Hypothamus are connected with it. - This area inspire to person, like Angriness, painfulness, Joyful and emotions type of experience can get. By that way person is learning and express his/her emotions. - Hipocampus is the one part of limbic system.Memory of short duration is remember for the long period.

- **Paripheral Nervous system :** Cranial Nerves + Spinal Nerves.
- **Cranial nerves :** The nerves arising from the brain is called cranial nerves.
  - Human being have 12 pairs fo cranial nerves. even in Reptiles, Birds and Mammal also. Amphibian and fishes having 10 pair of cranial nerves.
  - Sentory nerves : I, II, VIII
  - Motor nerves : III, IV, VI, XI, XII
  - Mixed nerves : V, VII, IX and X
  - In ANS (Autonomous Nervous System) III, IX and X nerve fibers of cranial nerves are found.
  - Vegus nerve (X) - Longest cranial nerve.
  - Trigeminal nerve (V) - Biggest cranial nerve.
  - Abducens nerve (VI) - Smallest cranial nerve
  - Truchear nerve (IV) - Thinest cranial nerve
- **Spinal nerves :** In human 31 pairs, In rabbit 37 pairs of spinal nerves are present. All nerves are mixed type.
  - Types of nerves given on basis of origin.

    cervical nerves - 8 pair, Thoralic nerves - 12 pair, Lumbar nerves - 5 pair, scaral nerves - 5 pair, coccygeal nerve - 1 pair.
  - In rabbit Lumber nerves - 7 pair, and coccygeal nerves - 6 pair.

- **Autonomous Nervous system or visceral Nerous System :**
  - Every impulses are controled / managed by two motor neurons and one gangilon. From this one cyton connected with CNS. and one pre gangionic fibre it have - other which cyton is present is ganglion and it has one post ganglionic fibre.

Parasympathetic Nervous System	Sympathic Nervous system
(1) Act at the resting situation.	(1) Act at the emergency.
(2) Neurotransmitter is Acetyl choline.	(2) Neurotransmitter is Non apinephrine.
(3) Preganglionic fibre is longer than post ganglionic fibre.	(3) Post ganglionic fibre is longer than pre-ganglionic fibre.
(4) Pre ganglionic fibre is origin from lower region of brain and spinal cord.	(4) Pre ganglionic fibres are originated from middle part of spinal cord.
(5) Increases heart beat, quickly breathing, enlargment of pupil, stop digestion.	(5) Normal heart beat, Normal breating contraction of pupil digestion can occur.

- **Deficiencies linked with vision :**

(1) Glucoma- Stopped aqueous humor fluid in vessel so can not spread outside- so pressure increases in anterior cavity - retina can damaged partially and then get partial blindness and after that complet blindness can occur.

(2) Cataract : By the ageing, sunlight or by the other reason catarat can occur. lens get impermeable - vision is totally stop. Transplant synthetic lens and remove this deficiency.

(3) Myopia (Nearsightedness) : Nearby obejcts easily seen but distant object can not seen easily - long eyehalls from so focus. in front of retina - Eyeglasses with concave lenses correct this problem.

(4) Hyper myopia (Hyperopia or Farsightedness) : Distance object is easily seen by person, but can not seen nearby object clearly - Eye ball become shorten. So focus arising behind retina - Eye- glasses with covex lens can correct the problem.

- **Sensory organ : Nose** Olfafactory sensory organ - It has nasal cavity in upper part as olfactory hairs- on those olfactory sensory cells of epithelia is presence for conduction - In this epithelial pillar cells and gland of Bowmann is present - Mucin is serected by bowmnn's gland - Olfactory hair having sensory centres which conected with smell molecule olfactory sersory cells through olfactory related impulses origin - which transmitted in olfactory ganglion of brain. Experience of smell which catch by olfactory sensory centre of celebrum.

- **Questions for NEET**

(148) Location of cerebro-spinal fluid.

(A) Arachnoid and Pia mater (B) Ventricles of brain

(C) Iter (D) All of above

(149) Function of cerebrospinal fluid

(A) Protects brain from friction (B) Gives buoyancy to brain

(C) Both A and B (D) Conduction of impulses

(150) ..... Has centre for wakefulness and sleep.

(A) Reticular system (B) Limbic system (C) Cerebrum (D) Diencepnalon

(151) Function of hippo campus.

(A) Sends messages to related parts of brain.

(B) Experience feeling of pain and anger.

(C) Convert short term memory into long term memory.

(D) Induces sleep and wakefulness.

(152) Number of cranial nerves in human ..... .

(A) 10 pair (B) 8 pair (C) 14 pair (D) 12 pair

(153) Which nerves of brain are motor nerves ?

(A) I, II, VIII (B) III, IV, VI, XI, XII (C) V, VII, IX, X (D) III, IV X

(154) In which cranial nerves, nerve fibres of Parasympathetic nervous system are observed ?

(A) III (B) IX (C) X (D) All of above

(155) Select proper option for pair.

Column - I	**Column - II**	
(1) X	(P) Largest	(A) (1-S) (2-P) (3-R) (4-Q)
(2) V	(Q) Smallest	(B) (1-S) (2-Q) (3-P) (4-R)
(3) VI	(R) Thinnest	(C) (1-S) (2-P) (3-Q) (4-R)
(4) IV	(S) Longest	(D) (1-P) (2-S) (3-Q) (4-R)

(156) Number of spinal nerves in Rabbit ..... .

(A) 31 Pairs (B) 37 Pairs (C) 12 Pairs (D) 10 Pairs

(157) Number of coccygeal nerves in human and rabbit sequentially.
(A) 5 Pairs -7 Pairs (B) 6 Pairs - 1 Pair (C) 8 Pairs - 6 Pairs (D) 1 Pair - 6 Pairs

(158) Which sentence is improper for parasympathetic nervous system.
(A) Functions during resting phase.
(B) Nor-epinephrin is neurotransmitter substance.
(C) Pre-ganglionic fibre is longer than post ganglionic fibre.
(D) Enlarges lens.

(159) From where pre-ganglionic Fibre of para-sympathetic nervous system arises ?
(A) Brain (B) Lower region of spinal cord
(C) Mid region of spinal cord (D) Upper region of spinal cord

(160) Vision related disease due to increase in pressure of aqueous humor ?
(A) Cataract (B) Glaucoma (C) Short sight (myopia) (D) Long sight far

(161) Which defect of spectacle can be removed by concave lens ?
(A) Glucoma (B) Myopia (C) Cataract (D) Far sight

(162) In which defect person can not see nearby things clearly ?
(A) Glucoma (B) Cataract (C) Myopia (D) Far sight

(163) Reason for cataract.
(A) Eye ball shortens (B) Eye ball lengthens.
(C) Vessels blocked (D) Lens becomes nontransparent.

(164) Organ having Bowman's gland - ..... .
(A) Eye (B) Ear (C) Nose (D) Kidney

(165) How many pair of cranial nerves are sensory in human ?
(A) 5 (B) 4 (C) 6 (D) 3

(166) How many pair of peripheral nerves are of mixed type of human ?
(A) 31 (B) 35 (C) 4 (D) 70

(167) Neurotransmitter substance released sympathetic nervous system.
(A) Epinephrin (B) seratonin (C) Nor epinephrin (D) Acetylcholine

(168) Which option is correct for spinal nerves in column -I and column-II.

Column- I	**Column - II**	
(1) Cervical nerves	(P) 1 - Pair	(A) (1-S) (2-R) (3-Q) (4-P)
(2) Thoracic nerves	(Q) 5 - Pairs	(B) (1-Q) (2-S) (3-R) (4-P)
(3) Sacral nerves	(R) 8 - Pairs	(C) (1-P) (2-S) (3-Q) (4-R)
(4) Coccygeal nerves	(S) 12 - Pairs	(D) (1-R) (2-S) (3-Q) (4-P)

(169) Function of parasympathetic nervous system ..... .
(A) increases heart beat (B) increases breathing
(C) increases digestion (D) increase size of eye ball

Answers : (148-D), (149-C), (150-A), (151-C), (152-D), (153-B), (154-D), (155-C), (156-B), (157-D), (158-B), (159-C), (160-B), (161-B), (162-D), (163-D), (164-C), (165-D), (166-B), (167-C), (168-D), (169-C)

Human Neural System

• Neural system performs three basic functions : receiving, processing and responding

Central Nervous System

Brain

• Central information processing organ

Central Nervous system (CNS)

Spinal Cord

• It conducts impulses to the brain and controls most of the reflex activities.

Brain

Central information processing organ

Fore brain

• Fore brain consists of olfactory lobes, cerebrum and diencephalon.
• The largest and complex cerebrum is divided into two cerebral hemisphere, Each one contain frontal, parietal, temporal and occipital lobes.
• Diencephalon contains epithalamus, thalamus and hypothalamus.

Mid brain

• Dorsal portion contains four lobes of corpora quadrigemina which is equivalent to optic lobes of lower animals.

Hind brain

• Cerebellum maintains posture and equilibrium and controls muscular activities.
• Medulla oblongata is cardiac centre, respiratory centre, vasomotor centre and reflex centre
• Pons varolii has respiratory centre pneumotaxic and apneustic centre.

Preripheral Nervous System

• Nerves outside the brain and spinal cord

Cranial nerves

• 12 pairs, 10 pairs from brain stem

Peripheral Nervous System

Nerves outside the brain and spinal cord

Carnial Nerves - 12 pairs, 10 pairs from brain stem

Spinal nerves

• 31 pairs of spinal nevers; (1) cervical nerves-8 pairs, (2) thoracio nerves - 12 pairs, (3) lumbar nerves - 5 pairs, (4) sacral nerves - 5 pairs, (5) coccygeal - 1 pair

32 Chemical Coordination and Control

- **Endrocrine glands and its hormones :**

- ductless gland - with introducing chemical secretion - hormone - transportation by blood - target which is located nearby or at the distance - Inducing or preventing the physiologicaly process which specifically take place in body.

- **Hypo thalamus :** Made up by nerve tissue- form the base of diencephalon, a part of forebrain.

Hypothalamus	
It is joined with anterior pituitary gland by hypophysial portal vessels.	It is linked with pituitary gland by nerve fibre (axis).

Hormones of Hypothalamus	
(1) RH Releasing hormone inducing e.g. GH - RH Inducing the secretion of GH from anterior pituitary gland.	**(2)** IH Inhibiting hormone-Inhibiting GH - RIH Prevents the growth hormone from anterior pituitary gland.

(1) Proper control and coordination of cellular activities are done.
(A) Nervous system (B) Endocrine system (C) Respiratory system (D) A and B Both

(2) Endocrine gland, made up of nervous tissue.
(A) Pituitary gland (B) Pineal gland (C) Thymus gland (D) Hypothalamus

(3) Hypophyseal portal connects ..... .
(A) Hypothalamus - mid pituitary (B) Hypothalamus - Post pituitary
(C) Hypothalamus - ant. pituitary (D) Ant. pituitary - Post pituitary

(4) It is connected with hypothalamus by axon.
(A) Post pituitary (B) Mid pituitary (C) Ant. Pituitary (D) Adenohypophysis

(5) Hormone stimulating growth hormone ..... .
(A) GH - RIH (B) STH - RIH (C) STH (D) STH - RH

(6) It is function of GH - RH.
(A) Stimulates secretion of GH - RH (B) Stimulates secretion of GH
(C) Stimulates secretion of STH (D) Inhibits secretion of STH

(7) Which hormones are obtained by pituitary gland from hypophyseal portal ?
(A) GH - RH (B) GH - RIH (C) PH (D) A and B

(8) Pituitary gland obtains from axon which induces synthesis of ..... .
(A) GH (B) oxytocin (C) ACTH (D) MSH

(9) Hypothalamus is which portion of Diencephalon ?

(A) Roof (B) Right side (C) Floor (D) Left side

Answers : (1-B), (2-D), (3-C), (4-A), (5-D), (6-D), (7-D), (8-B), (9-C)

- **Pituitary gland :**

Location :- A depression of the sphenoid bone of the skull- In sella turcica.

Adenohypophysis : Anteria pituitary gland - Intermidiate pituitary gland.

Neurohypophysis : Posterior pituitary gland.

(A) Anterior Pituitary gland :

Controlled secreation of GH (Growth Hormone) : Stimulate the growth, development - Rapid cell division - Protein synthesis.

Less secreation : Dwarfism

Excess Secretion : In Childhood - Abnormal height (Gaigantism) At mature age - Upper- lower jaws and appandageales get more length (Acromegaly)

- PRL/PH - Prolactin;
  After partuirition secretion of milk is produce from the mammary gland.
- TSH/TTH - Thyroid Stimulating Hormone
  It stimulates the thyroid gland and the hormones which are originated from it eg.Thyroxine.
- ACTH - Adreno Cortico Trophic Hormone; Stimulates the secretion of adrenal cortex. Glucocorticoid and Minarelo corticoid
- GTH - LH ♂ - Stimulate the secretion of testosterone, Induce the complete development of male reproductive system.
  - ♀ - Development of graffian follicle- induce ovulation - Formation of corpus leutium.

  FSH - ♂ - Stimulate the spermatogenesis-
  - ♀ - Stimulate process of oogenesis - Development of follicles occurs in ovary.

(B) Intermediate Pituitary gland :

MSH - Melanocyte Stimulating Hormone, Control of pigments in skin.

(C) Posterior Pituitary gland :

Oxytocin- Stimulates the contraction of smooth muscles of body.- During the Parturition widening of uterus at the time of child birth- milk secretion from the mammary glands.

- Vasoprasin : Stimulates the reabsorption of water and electrolytes by the distal tubules of kidney.
- Reduces loss of water through urine (diuresis dilute excess urine stopped).

- Deficiency of ADH : Reduces reabsorption of water and increases urine output. This disorder is known as diabetes incipidus.

- **Pineal Gland :**

Location : Under the corpus callosum between the two cerebral hemispheres of the brain.

Structure : A very small, solid, vascular, reddish grey and conical body.

Hormone : Melatonin - A biological Clock.

Function : Helps in maintaining sleep - wake cycle regulation.

- Regulates metabolism, menstrual cycle, pigmentation and self defense capacity.
- In dark secretion of melatonin more, due to that more percentrage of sleepness. (More in child
- In adulthood it decreases).

(10) It is not part of adenohypophysis.
(A) Ant Pituitary (B) Mid Pituitary (C) A and B both (D) Post Pituitary

(11) Disease, due to less secretion of somatostatin, which causes irregularities : ..... .
(A) Dwarfism (B) Critinism (C) Giantism (D) Acromegaly

(12) In adult stage, excess secretion of STH leads to irregularities of ..... disease.
(A) Giantism (B) Acromagely (C) Critinism (D) Dwarfism

(13) Stimulates synthesis of thyroxine.
(A) TSH - RH (B) TSH - RIH (C) TSH (D) TCT

(14) It is target tissue for ACTH.
(A) Anterior Pituitary gland (B) Adrenal Cortex
(C) Adrenal Medulla (D) Post pituitary

(15) Stimulates mammary gland for secretion of milk.
(A) PH (B) PIH (C) Oxytocin (D) A and C

(16) LH stimulates secretion of which hormone ?
(A) Tastosterone (B) progesterone (C) A and B (D) oxytocin

(17) Origin of secretion, which stimulates synthesis of FSH.
(A) Ant. Pituitary (B) Ovule (C) Hypothalamus (D) Ovarian follicle

(18) Reason for Diabetes incipedus.
(A) More secretion of insulin (B) More secretion of vasopressin
(C) Less secretion of vasopressin (D) Less secretion of insulin

(19) Number of hormones originating from adenohypophysis.
(A) 6 (B) 2 (C) 9 (D) 7

(20) Origin of hormone, which helps in facilitating normal delivary.
(A) Ant. Pituitary (B) Mid Pituitary (C) Neuro hypophysis (D) Adenohypophysis

(21) Secretion of pituitary gland regulates the process of spermatogenesis.
(A) LH (B) FSH (C) Androgen (D) B and C

(22) Gland present beneath corpus callosum.
(A) Pituitary gland (B) Hypothalamus
(C) Pineal gland (D) Thymus gland

Answers : (10-D), (11-A), (12-B), (13-C), (14-B), (15-D), (16-C), (17-C), (18-C), (19-D), (20-C), (21- B), (22- C)

- **Thyroid Gland :**

Location :- Bilobed and located on either side of the upper part of the trachea. The two lobes connected by a narrow connective tissue band called isthmus.

Mineral elements : For the formation of hormone I_2 is necessary. Deficiency of I_2 goiter can occur.

Hormones : (1) T_3 - Triiodothyronin, (2) T_4 - Tetra iodothyronin and TCT - Thyrocalcitonin.

(A) Functions of Thyroxine :

- During the metabolism of carbohydrates control the oxidation an ATP synthesis. Maintains BMR
- Stimulates the process of RBC formation.
- Maintains water and electrolyte balance.

Hypothyrodism (less secretion) → In child - Critinism - mental retardation - Dumb and Deaf occur.
→ In adult female- Myxodema Accumulation of fact in subcutaneous tissue- puffy face - inflammentary face.

Hyperthyrodism (Excess selection) → Irregular in menstrual cycle
→ Exothalmic Goitre (Grave's disease)

(B) TCT - Thyrocalcitonin :

- Acts upon osteoblast of bones - Decreases the calcium level in blood.

- **Parathyroid Gland :**

Location : Four lobe, the ventral surface of thyroid gland

Horomone : Parathormone (PTH)

Functions : Increases the Ca^{+2} level in blood.

Acts on bone and activates the process of bone resorption.

- Increases Ca^{+2} absorption from the digested food and also activates reabsorption of Ca^{+2} by renal tubuler.
- PTH and TCT are complimantry to each other.

- **Thymus Gland :**

Location : In upper dorsal side of the heart and aorta.

Structure : Bilobed, mass of lymphoid tissue.

Activation : In children it is a prominent gland but it gradually degenerate in the adult.

Hormone : Thymosin

Functions : Maintains the immunity power. Thymosin stimulates the development and differentiation of T-lymphocytes which provide cell-mediated immunity. Besides this thymosin stimulates the production of antibodies to provide hormonal immunity.

(23) Endocrine gland with bilobed structure : ..... .
(A) Thymus (B) Thyroid (C) A and B (D) Parathyroid

(24) Complete colloids found in cells of thyroid glands are formed of ..... .
(A) Zeatin (B) Gelatin (C) Melanin (D) Carotene

(25) What is reason for increase in the size of thyroid gland ?
(A) Less secretion of thyroxine (B) More secretion of thyroxine
(C) Defeciency of iodine in food (D) All of above

(26) ..... is observed due to hypothyroidism.
(A) Goiter (B) Myxodema (C) Exothalmic goiter (D) A and B both

(27) Condition rensponsible for irregularity in menstrual cycle in female
(A) Hyper Thyroidism in adult female (B) Hypothyroidism in adult female
(C) Hyper thyroidism during embryonic development (D) Hypo thyroidism during embryonic development

(28) Reduces Ca^{+2} level in blood
(A) Thyroxine (B) Tri idothyronine (C) PTH (D) TCT

(29) Combine effect of which maintains Ca^{+2} level in body ?
(A) PH - TCT (B) PTH - TCT (C) PIH - TCT (D) PIH - PTH

(30) Activates absorption of Ca^{+2} in uriniferous tubule ..... .
(A) ADH (B) Aldosterone (C) PTH (D) TCT

(31) Endocrine lymph gland ..... .
(A) Thymus (B) Thyroid (C) Parathyroid (D) All

(32) It is function of thymosin.

(A) Stimulates cellular immunity (B) Stimulates production of antibody

(C) Differentiation of B - lymph cells (D) Decrease in cellular immunity

(33) ..... Endocrine gland situated in thoracic region.

(A) Thyroid (B) Pancreas (C) Thymus (D) Adrenal

(34) Endocrine gland with four lobes.

(A) Thyroid (B) Parathyroid (C) Thymus (D) Pituitary gland

Answers : (23-C), (24-B), (25-C), (26-B), (27-A), (28-D), (29-B), (30-C), (31-A), (32-A), (33-C), (34-B)

- **Adrenal Gland :**

Discovered by Eustessious **Location :** One on the anterior part of each Kidney

Origin From : Mesoderm **Weight :** 4 to 6 g.

Two main regions of gland → Adrenal Medulla - 10 to20% region → Adrenal cortex- 80 to 90 %

Adrenal Medulla

↓

Secretion of group of cotecholamine

For the reason of origin - Fear, Mental pressure, it secrete at the time of accident or emergency. These horomones are secreted

↓

Adrenaline/ Epinephirine	**Non-adrenaline/non epinephrine**
- Effects : Warm red face	- Conctraction of blood vessels, but it can not do conctration of coronary artery.
- pupilary dilutions	
- Raising of hairs	
- Faster heart beats	- (Other effects are likely to adrenaline only)
- More sweating	
- Conversion of glucagon to glucose and added in blood	

↓

Target tissues : Straited, smooth and cardiac muscular tissues, blood vessels and adipose tissue.

- **Hormones of Adrenal cortex (These are steroid types)**

(A) Cortex Region (Zona Glomerulosa)

Secretion : Mineralocorticoid → (Main) aldosterone, Deoxy corticosterone

Regulate mineral metabolism and balance of water and Na^+

Target Organ : Kidney

(B) Middle region (Zona Fasciculata)

Secretion : Glucocorticoid - **Main secretion :** Cortisol

Regulate carbohydrate, protein and fat metabolism.

Antiallergic and anti-inflamatory effects-suppress the immune responses.

Target organ- Liver

(C) Inner Region : (Zona Reticularis)

- Very less amount of hormones secretion occurs which are sexual.

In male testosterone - in female - estrogen and progesterone.

Target organ: Genetial organs

- **Abnormal secretion:**

(1) **Less secretion :** Addision Disease (Hypo adrenalism)

Due to deficiency of Mineralo corticoid.

Effects : Weekness, weight loss, nausea and diarrhoea

(2) **Excess Sercetion :** Cushing's Syndrome **Effects :** More secretion of cortisol

Symptoms : High blood sugar, obesity and rise in blood pressure and blood volume, more amount of $Na^{+,}$ H_2O also.

---

(35) Amino acids produced derivatives type of secretions.

(A) Adrenal cortex (B) ovary (C) Testis (D) Adrenal medulla

(36) ..... secretions are not in the control of secretion of pituitary gland.

(A) Adrenal medulla (B) Adrenal cortex (C) Both A and B (D) Testis

(37) It is epinephric endocrine gland.

(A) Testis (B) Ovary (C) Adrenal (D) Pancreas

(38) Pair of hormones regulating metabolism of carbohydrate.

(A) ANF - Relaxin (B) Catechocalamines-cortisol

(C) Tastosterone-Estrogen (D) PTH - PH

(39) Corticoids are which type of hormones ?

(A) Catecholamines (B) Iodothyronins (C) Peptide (D) Steroid

(40) From which layer of adrenal cortex mineralo corticoids secreted ?

(A) Outer (B) Middle (C) Inner (D) All

(41) ..... hormone are secreted in male by zona reticularis.

(A) Cortisol (B) Tastosterone (C) Progesterone (D) A and B both

(42) It has anti inflamatory effect.

(A) Aldosterone (B) Progesterone (C) Cortisol (D) Testosterone

(43) Due to inactivity of which layer of adrenal cortex Addison's disease occur ?

(A) Inner (B) Middle (C) Outer (D) All of above

(44) ..... is disease due to more activity of zona fasiculata of adrenal cortex.

(A) Addison (B) Myxodoma (C) Critinism (D) Cushing syndrome

Answers : (35-D), (36-A), (37-C), (38-B), (39-D), (40-A), (41-D), (42-C), (43-C), (44-D)

- **Pancreas :**

Exocrine and endocrine gland

Endocrine region

- Islets of Langer hans - about 1 to 2 millions

- It occupies 1 to 2% area of pancreas.

(A) α - cells :

secretion : Glucagon

Type : peplide

Hyper glycemic hormone

(I) In liver cells, glycogenolysis takes place.

Glycogen $\xrightarrow{Glucagon}$ Glucose (Added in blood)

(II) Gluconeogenesis :

Lactic acid $\rightarrow$ Pyruvic acid $\xrightarrow{Glucagon}$ Glucose (added in blood)

(B) β - Cells:

Decreases the glucose level in blood - Hypoglysemic

Glycogenesis

Glucose $\xrightarrow{Insulin}$ Glycogen (Hepatic cells - Adiposetissue)

Less secretion of Insulin - Diabetes melitus

(C) δ (Delta) - cells : Somatostatin - Prevents the GH and regulate the growth.

- It controls insulin and glucagon secretion.

- **Sexual endocrine gland :**

(A) Testis : Androgens hormone - Steroid type - Main hormone - Testosterone.

Function : Stimulates male accessory reproductive organs and get maturity.

- Develops secendory sexual characteristics.
- Stimulates the spermatogenesis process.

(B) Ovary :

Hormones

	Estrogen	Progesterone	Relaxin
Origin →	Developed from ovarian follicle	- Corpus luteum	- Corpus lateum
Functions →	- Development of female reproductive system - Develops the secondary sexual characters - Developments of ovarian follicles occurs	- Stimulates embryo development and foetus development - Stop the ovulation inducing implantation of embryo - After parturition Lactation can induced.	- End of the pregnancy it is secrete - Widening of cervix of - uterus and simple child birth can occur.

(45) Which cells of pancreas are exocrine ?

(A) α - Cells (B) cells of acini (C) β - Cells (D) Delta cells

(46) ..... is function of insulin.

(A) Glycogenolysis (B) Gluconeogenesis (C) Glycogenosis (D) Glycolysis

(47) _____ increases blood glucose level.

(A) Adrenaline (B) Cortisol (C) Glucagon (D) All

(48) It is hypoglycemic hormone.

(A) Cortisol (B) Adrenaline (C) Insulin (D) Glucagon

(49) Hormone which inhibits secretion of GH.

(A) GH - RTH (B) Somatostatin (C) A and B (D) STH - RH

(50) Due to inactivation of which cells, Diabetes malitus occur ?

(A) α - Cells (B) β - Cells (C) Delta cells (D) Leydig's cells

(51) Origin of androgen type hormone.

(A) Interstitial cells (B) Langerhans cells

(C) Neuro secretory cells (D) germinal epithelium

(52) Hormone initially secreted from corpus luteum : ..... .

(A) Estrogen (B) Progesterone (C) Relaxin (D) All of above

(53) Hormone secreted from ovarian follicle ..... .

(A) Relaxin (B) FSH (C) Estrogen (D) Progesterone

(54) From where Relaxin is originated ?

(A) Ovarian follicle (B) Leydig's cells

(C) Corpus luteum (D) Germinal epithelium

(55) Secondary sexual character, growth of hair in armpit is induced by ..... .

(A) Androgen (B) Progesterone (C) Estrogen (D) A and C both

(56) Gluco genesis process induces ..... .

(A) Insulin (B) Testosterone (C) Somatostatin (D) glucagon

Answers : (45-B), (46-C), (47-D), (48-C), (49-C), (50-B), (51-A), (52-B), (53-C), (54-C), (55-D), (56-D)

- **Some Hormones of Non-endocrinal glands:**

(certain organs produce some hormones)
eg. Heart, Kidney, Gastro instinal lract

(A) Heart:

Secretion : Atrial Natriuretic Factor (ANF)

Origin : Wall of atrial of heart

Type : Peptide

Function : Relaxation of blood vessels and decreases blood pressure

(B) Kidney :

Secretion : Erythropoietin

Origin : J. G. Cells Jaxta (Glomerular cells)

Function : Stimulates formation of RBCs.

(C) Gastro intestinal tract :

Secretion :

Gastrin, secretin, cholecystokinin (CCK), Gastric Inhibitory peptide (GIP)

(I) **Gastrin :** Stimulate gastric glands to secrete pepsinogen and HCl.

(II) **Secretin:** Acts on the exocrine glands of pancreas and stimulates the secretion of bicarbonate ions and water.

(III) **CCK :** Acts on pancreas and gall bladder and stimulates them to secrete pancreatic enzyme and bile respectively.

(IV) **GIP :** Inhibits gastric secretion.

(D) Growth Factors :

For normal growth of tissues, their repairing and regeneration.

- **Basis on chemical types :**

Types of Hormones :

(A) Peptide, Polypeptide, Proteinous hormone- Insulin, Glucagon, Hormones of pituitary and Hypo thalamus etc.

(B) Steroidal hormone : Cortisol, Testosterone, Progesterone

(C) Iodothyronins : Hormone of thyroid

(D) Derivatives of amino acides : Epinephrine, Norepinephrine

- **Mechanism of Hormone action :**

(A) Endocrinal mechanism of steroidal type hormones

Hormone directly enter in the nucleus of cell

↓

Joint with receptive unit

↓

By the help of specific molecule hormone receptive complex is formed

↓

Link with DNA transcript the specific gene

↓

↓

Formation of m-RNA

Translocation ↓ Trtanslation

Protein synthesis (Enzyme)

(B) Endocrinal mechanism of peptide type hormones

Peptide Hormone

↓

Joined with membrane bound receptor

↓

G type of protein of plasma membrane get active

↓

Activates

↓

ATP → C - AMP

↓

any one specific enzyme gets activation

↓

Other enzyme occur activation

Catch in Memory :

First messanger → Hormone

Second messanger → C - AMP, Ca^{+2}, IP_3

Peptide hormone do not enter in molecule.

Effect of peptide hormone is quick.

---

(57) Non-endocrine tissue (cells), which produces hormone is ..... .

(A) Leydig's cells (B) Langerhans cells (C) Juxta glomerular cells (D) Delta cells

(58) Secretes ANF : ..... .

(A) Wall of ventricle (B) Wall of urinary bladder

(C) Wall of auricles (D) Wall of stomach

(59) Function of erythropoetin ..... .

(A) Widens blood capillaries (B) Induces formation of erythrocytes

(C) Inhibits secretion of gastric juice (D) Stimulate secretion of bile juice

(60) Hormone which interacts with membrane bound receptor.

(A) Epinephrine (B) Tri iodothyronin

(C) Follicle stimulating hormone (D) All of above

(61) Mostly controlling gene expression, hormones : ..... .

(A) Estrogen (B) Testosterone (C) cortisol (D) All

(62) During mechinsm of action forms secondary messenger.

(A) ANF (B) Aldosterone (C) Progesterone (D) Cortisol

(63) In which mechanism IP_3 is formed ?

(A) ADH (B) Testosterone (C) Estrogen (D) Progesterone

(64) Chemically thyroxine is which type of hormone ?

(A) Derivetives of Amino acids (B) Peptide

(C) Iodothyronins (D) Protein

(65) Forms Receptor complex in the nucleus.

(A) GH - RH (B) Mineralo - corticoids

(C) GTH (D) ACTH

Answers: (57-C), (58-C), (59-B), (60-D), (61-D), (62-A), (63-A), (64-C), (65-B)

(66)

Column- I	Column - II	
(1) Secondary messenger	(P) ANF	(A) (1-P) (2-Q) (3-S) (4-R)
(2) Peptide hormone	(Q) IP_3	(B) (1-Q) (2-P) (3-S) (4-R)
(3) Intra cellular receptor	(R) Nor-epinephrin	(C) (1-Q) (2-S) (3-P) (4-R)
(4) Catecholamine	(S) Relaxin	(D) (1-Q) (2-S) (3-R) (4-P)

(67)

Column- I	Column - II	
(1) Androgens	(P) Eythropoesis	(A) (1-Q) (2-P) (3-R) (4-S) (5-T)
(2) Erythropoetin	(Q) Intersticial cells	(B) (1-P) (2-Q) (3-R) (4-S) (5-T)
(3) Progesterone	(R) Increase of mammary glands	(C) (1-Q) (2-R) (3-P) (4-T) (5-S)
(4) Relaxin	(S) Widens fallopian tube	(D) (1-Q) (2-P) (3-T) (4-S) (5-R)
(5) Estrogen	(T) Induces implantation of embryo	

(68)

Column- I	Column - II	
(1) Delta cells	(P) cellular immunity	(A) (1-Q) (2-R) (3-S) (4-P)
(2) Alpha cells	(Q) STH - inhibition	(B) (1-Q) (2-S) (3-R) (4-P)
(3) Beta cells	(R) Hyper glycomia	(C) (1-P) (2-R) (3-S) (4-Q)
(4) T-cells	(S) Hypo glycemia	(D) (1-Q) (2-S) (3-R) (4-P)

(69)

Column- I	Column - II	
(1) Addison disease	(P) More secretion of ADH	(A) (1-S) (2-T) (3-R) (4-U)
(2) Mexydoma	(Q) Deficiency of mineralocorh coids	(B) (1-Q) (2-T) (3-R) (4-P)
(3) Exopthalmic goiter	(R) Hyper thyroidism	(C) (1-Q) (2-T) (3-R) (4-U)
(4) Diabetes incipedus	(S) more secretion of mineralo corticoids	(D) (1-Q) (2-R) (3-T) (4-U)
	(T) Hypothyroidism	
	(U) Less secretion of ADH	

(70)

Column- I	Column - II	
(1) Sex corticoids	(P) Thymus	(A) (1-U) (2-S) (3-T) (4-Q)
(2) Catecholamine	(Q) pineal	(B) (1-R) (2-S) (3-P) (4-Q)
(3) Thymosine	(R) External layer of Adrenal cortex	(C) (1-U) (2-P) (3-T) (4-Q)
(4) Melatonin	(S) Adrenal medulla	(D) (1-U) (2-S) (3-P) (4-Q)
	(T) Thyroid	
	(U) Inner layer of Adrenal cortex	

(71)

Column- I	Column - II	
(1) Oxytocin	(P) Hypothalamus	(A) (1-R) (2-S) (3-P) (4-Q)
(2) MSH	(Q) Ant. pituitary	(B) (1-P) (2-R) (3-S) (4-Q)
(3) GH - RIH	(R) Mid. pituitary	(C) (1-S) (2-R) (3-P) (4-Q)
(4) TSH	(S) Post. pituitary	(D) (1-P) (2-S) (3-R) (4-Q)

Answers: (66-B), (67-D), (68-A), (69-C), (70-D), (71-C)

- **A - Statemen, R - Reasoning type questions.**

Select answers of following questions from the options given below.

(A) A and R both correct, R is explanation of A.

(B) A and R both correct, R is not explanation of A.

(C) A is correct, R is wrong.

(D) A is wrong, R is correct.

(72) Statement A : Immunity in old persons decreases.

Reason R : With the aging thyroid gland degenerates.

(A) (B) (C) (D)

(73) Statement A : STH-RH is neurohormone.

Reason R : STH-RH is synthesized by hypothalamus.

(A) (B) (C) (D)

(74) Statement A : Diabetes malitus caused due to deficiency of ADH.

Reason R : Amount of urine, expelled out increases due to deficiency of ADH.

(A) (B) (C) (D)

(75) Statement A : Due to deficiency of iodine in food goiter occur.

Reason R : Iodine is essential component for thyroxine hormone, in thyroid gland.

(A) (B) (C) (D)

(76) Statement A : FSH hormone, regulates cellular metabolism without entering into cell.

Reason R : FSH shows its effect by reacting with receptor complex.

(A) (B) (C) (D)

(77) Statement A : Ovulation ceases in pregnant woman.

Reason R : At the time of pregnancy, progesterone is proudced by corpus luteum.

(A) (B) (C) (D)

(78) Statement A : Insulin decreases glucose level in blood.

Reason R : Insulin stimulates gluconeogenesis.

(A) (B) (C) (D)

(79) Statement A : At the time of fear and emergency catecholamine level increases in blood.

Reason R : ACTH of pituitary induces secretion of corticoids.

(A) (B) (C) (D)

(80) Statement A : Melatonin functions as biological clock.

Reason R : Melatonin functions to maintain rhythamicity during 24 hours.

(A) (B) (C) (D)

(81) Statement A : Ca^{+2} level is maintained in blood.

Reason R : PTH increases Ca^{+2} level in blood.

(A) (B) (C) (D)

Answers: (72-C), (73-A), (74-D), (75-B), (76-C), (77-A), (78-C), (79-B), (80-A), (81-B)

- **True - False (T - F) type questions**

Select proper option for True / false for the following sentences.

(82) (1) Ca IP_3 is secondary messenger.

(2) Cortisol forms membrane connecting receptor complex.

(3) Steroid hormones normally control genetic expression.

(A) TFF (B) FFT (C) TFT (D) FTF

(83) (1) CCK prevents secretion of bile juice.

(2) GIP induces secretion of gastric juice.

(3) ANF relates blood capillaries.

(4) Erythropoietin induces formation of R.B.C.

(A) FFTT (B) TFTT (C) TTTT (D) FFFT

(84) (1) Progesterone is secreted from ovarian follicle.

(2) Relaxin, broadens the neck of fallopian tube.

(3) Leydig's cells are present in female.

(4) Androgen type secretion occurs from ovary.

(A) TTFF (B) FTFF (C) FTTT (D) TTTF

(85) (1) somatostatin stimulates secretion of GH.

(2) In islets of langerhans 10 to 20 million cells are present.

(3) More secretion of insulin leads to diabetes malitus.

(A) FTT (B) TTT (C) TTF (D) FTF

(86) (1) Blood sugar increases in Addison's disease.

(2) Glucocorticoids and sexcorticoids, secreted from zone fasciculata.

(3) Adrenaline increases glucose level in blood.

(A) FFF (B) FTT (C) TFF (D) TTT

(87) (1) Thymus is lymph tissue.

(2) Parathyroid is bilobed gland.

(3) Secretions of thyroid stimulates R.B.C. formation.

(4) Due to myxodema, irregularities in menstrual cycle occur.

(A) TTFF (B) FFTT (C) TTFT (D) TFTF

(88) (1) Pineal gland is beneath diencephalon.

(2) MSH is secretion from adenohypophysis.

(3) During adult stage more secretion of GH leads to acromegaly.

(A) FTT (B) FTF (C) TTT (D) FFF

(89) (1) STH-RH is secretion of pituitary gland.

(2) Target tissue for TSH is thyroid.

(3) Hormone, stimulating secretion from leydig's cells derives from pituitary gland.

(4) Thymus gland is situated in thoracic region.

(A) FFTT (B) FFFT (C) FTTT (D) FTTF

Answers: (82-C), (83-A), (84-B), (85-D), (86-B), (87-D), (88-B), (89-C)

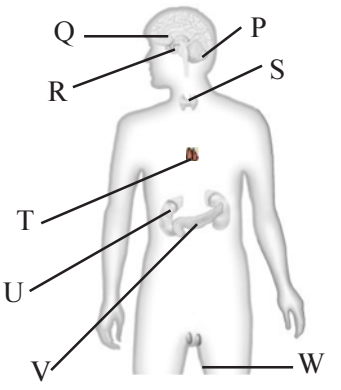


Figure 1

(90) In Fig. 1 labelled P as gland ..... hormone is produced.

(A) oxytocin (B) Melatonin (C) GH-RH (D) TSH

(91) Q is which gland in Fig.1 ?

(A) Pituitary gland (B) Pineal gland

(C) Thyroid gland (D) Hypothalamus gland

(92) Which gland is exocrine and endocrine, shown in Fig.1 ?

(A) V (B) U (C) S (D) Q

(93) Which is parathyroid gland, labelled in Fig.1 ?

(A) R (B) T (C) S (D) Q

Answer Question number 94 - 96 from Fig.2.

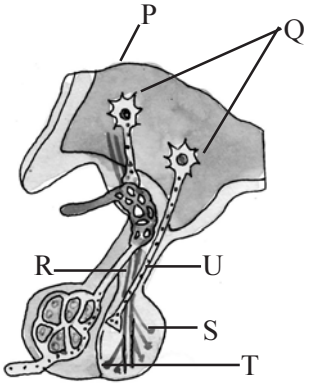


Figure 2

(94) What is indicated by R in fig. 2 ?

(A) Adenohypophysis (B) Infundibulum

(C) Axon (D) Hypophyseal portal

(95) In Fig. 2 vassopressin is secreted by whom ?

(A) T (B) S (C) P (D) R

(96) From which part GH-RH is secreted in Fig.2 ?

(A) T (B) S (C) U (D) Q

(97) In Fig-3, ... hormone secreted by P

(A) PTH (B) TCT (C) PH (D) PIH

(98) ..... Shows bridge in Fig. 3.

(A) S (B) R (C) Q (D) P

(99) In Fig. 3, parts of respiratory system : ..... .

(A) R and S (B) Q and R (C) P and S (D) P and R

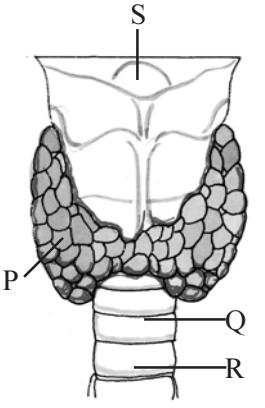


Figure 3

(100) What is indicated by 'P' in Fig. 4 ?

(A) Adrenal cortex (B) Renal medulla

(C) Adrenal medulla (D) Renal cortex

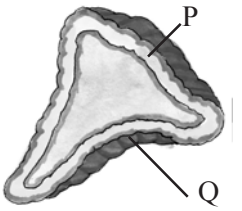


Figure 4

(101) ..... is not hormone secreted from Q in Fig. 4.

(A) Cortisol (B) Estrogen (C) Testosterone (D) Epinephrin

(102) In Fig.5 internal structure of ..... is shown.

(A) ovary (B) pancreas (C) testis (D) thyroid

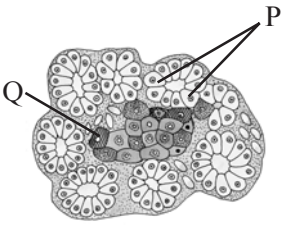


Figure 5

(103) What is indicated by 'P' in Fig. 5 ?

(A) Leydig's cell (B) Hepatic lobules

(C) Langerhan's cells (D) Pancreatic acini

(104) In Fig.6, mode of action of which hormone is represented ?

(A) cortisol (B) progesterone (C) MSH (D) None

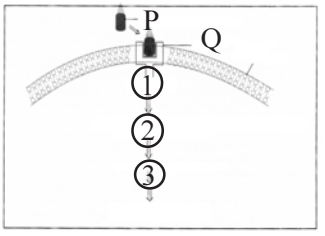


Figure 6

(105) What is suggested by origin of secondary messenger in Fig.6 ?

(A) P (B) 2 (C) Q (D) 3

(106) Fig.7 indicates mode of action of which hormone ?

(A) Peptide (B) Iodothyronins

(C) Catecholamines (D) Steroids

(107) In Fig.7 indicates genome.

(A) Q (B) P (C) R (D) T

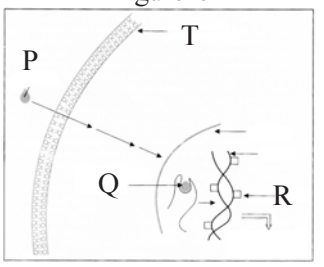


Figure 7

Answers : (90-B), (91-D), (92-A), (93-C), (94-C), (95-B), (96-D), (97-B), (98-C), (99-A), (100-C), (101-D), (102-B), (103-D), (104-C), (105-B), (106-D), (107-C)

- **Summery of PMT-NEET**
  - Endocrinology - is the branch which is connected with a physiology as a branch of biology.
  - Father of Endocrinology - Thomas Adison - Hormone = To Exite.
  - 'HORMONE' word given by Starling.

    In 1902 Secretin first hormone was discovered.
  - **Callons :** A secretion of one endocrine gland which stimulates the target organ for secrete the hormones only, it is called callons.
  - **Autocoids :** Hormones produce from the one endocrinal gland which act on specific target oragan, it is called Autocoid.

Diabetogenic: Hormone which effect on specifically on carbohydrates metabolism.

Ketogenic : Hormone which mainly act on metabolism of fats.

Callorigenic: Mainly BMR is effected.

- Tuber Cinerium

Neurohypophysis → Infudibulum

- Pars Nervosa
- pars distalis
- Adenohypophysis - Pars tuburalis
- pars Intermedia

By the deficiency of pituitary dwarfism can occur, it is called **Ateliosis.**

- Dwarf person shown in circus which is called **midgets.** They are physically and mentally healthy but they have slow maturity in sexually.
- Excess secretion of STH/GH in adult human being - Acromagely.
- FSH also known as Gameto Kinetic Factor. It secrete in male and female both.
- LH is also known as Gamete Releasing Factor.
- **Hashimoto Disease :**
- Acute stress by the lack of thyroxine in this disease. So for the treatment of this disease any medicine or thyroxine hormone also act as a poison. For that reflextion formation of antibody occur in body and destroy the thyroid gland.
- **Plumar Disease :**
- In this disease abnormal growth of thyroid gland occur, but in this gland small buds like nodes are formed which can be visiualised. Which is called Toxic adenoma. During the excess secretion thyroid gland get swollen and enlarged.
- Parathyroid gland was discovered by Reynard. Deep study of it observed by Sandrom.
- Resultant of excess secretion of PTH, Osteolytic cells in more amount unnecessarily phagocytics So that bones becomes rigid and weak. This situation is called an Osteoporosis.
- When increases the amount of Ca^{+2}, decreases the amount of PO_4^{-3}, which is known as Hypercakimia and Hypophospheremia respectively.
- In kidney or Gall bladder accumulation of Ca^{+2} so formation of stone can occur.
- **Benting and Best** - First time insulin discovered.
- Molecular structure of insulin is given by **A.F.Sanger**.
- Human insulin was synthesis by Tsan.
- **14th Novemeber- World diabetes day**
- **Glycosuria :** Amount of glucose in blood > 180 mg/dl.
- **Polydipsia :**. In excess amount diurisis can occur, possibility of dehydration is more. This type of petient thrust more and continuously, electrolytes are remove from the body. More hungry can get.
- Secretion of male hormone by adrenal gland. Mainly is Dihydroapy androsteron.

Keto acidosis :

- Activation of fat and incomplete of ketonebody in adipose tissue is formed.
- As a ketonebody like acetone aceto acetic acid, Beta hydroxy butrate.
- Ketone body is poisonous - Increasing the amount of Ketone in body so ketoacidosis can form in body.

Diabetes Coma :

- Ketoacidosis, Polydipsia, Hyperglycemia with these complex, patient get effect and becomes unconscious.

Heterodrine gland :

- This type of endocrine gland, which secretes the hormone, more over some other functions also they do. .

  e.g. Pancreas, Gonads, Mucaso of digestive system, Kidney, placenta

Simond's disease :

- This disease occur by degradation of anterior part of the pituitory gland. So it is called simond's disease.

Metachrosis :

- Pigmentation change of skin with the help of MSH.

Insulin shock :

- If diabetic patient passes through the physical stress or fasting during that he takes injection of insulin, so decreasing repidly glucose level in blood 40- mg/160 mg. Pateint becomes unconscious and sometimes dies also.

- **Questions for NEET**

(108) Which of the following situation is responsible for acromegaly ?

(A) More secretion of STH (B) More secretion of thyroxine

(C) Defeciency of thyroxine (D) More secretion of adrenaline

(109) FSH is ..... .

(A) glyco protein (B) metallo protein (C) glyco lipid (D) Phospholipid

(110) Posta glandin effects ..... .

(A) Egestion (B) Blood pressure (C) osmoregulation (D) Oxidation reaction

(111) Hormone known as FFF Or 3F ..... .

(A) CCK (B) Glucagon (C) Thyroxine (D) Adrenaline

(Hint : Fight, Fear, Fraction of the second-3F)

(112) Of the following which acts as secondary messenger ?

(A) Phospholipid (B) Glycoprotein (C) carrier protein (D) Immuno globulin

(113) Of the following which is not secondary messanger in hormone action ?

(A) C-AMP (B) Sodium

(C) Calcium (D) IP_3 (Inositol triphosphate)

(114) Deficiency of which hormones from the following reduces Ca^{+2} level ?

(A) Calcitonin (B) Parathormone (C) Thyrotine (D) A and B both

(115) If adrenal cortex is injured, it will affect secretion of following or not ?

(A) Adrenaline (B) Cortisol (C) Aldosterone (D) Androstidione

(116) Which hormone induces relaxation of blood capillaries and increasing O_2 use and induce gluco neogenesis ?

(A) Glucagon (B) Insulin (C) Adrenaline (D) ACTH

(117) Find the improper pair from the following.

(A) Corpus luteum - Relation (secretion) (B) Insulin- Diabetes malitus (disease)

(C) Glucagon - Beta cells (source) (D) Somatostatin - Delta cells (source)

(118) ..... is hormone of adenhypophysis, controlled by feed back mechanism.

(A) Oxytocin (B) TSH (C) Vasopressin (D) TCT

(119) Identify, father of endocrinology.

(A) Land Steiner (B) Renard (C) Harrigton (D) Thomas Adison

(120) Of the following secretion of which is controlled, axon of neurosecretory neuron ?

(A) Adrenal cortex (B) Pineal gland

(C) Ant pituitary gland (D) Post pituitary gland

(121) One person has more urination frequency requires more drinking water but has normal blood glucose level. State the reason for such situation.
(A) More secretion of glucagon.
(B) Decrease in insulin secretion from pancreas.
(C) Imbalance of glucose in urine.
(D) Decrease in secretion of vaso pressin from post pituitary.

(122) Patient of Diabetes malitus, though eating food without carbohydrate, excretet glucose in urine, because ..... .
(A) Glycogen from muscles released in blood.
(B) Amino acids from liver released in blood.
(C) Amino acid dissociates in the kidney and converted into glucose.
(D) In adipose tissue lipid dissociates and transforms into glucose.

(123) Diuretinis is due to which situation ?
(A) Less secretion of vassopressin.
(B) More secretion of ADH.
(C) Less secretion of oxytocin.
(D) Effect of erythropoeitin removed from kidney.

(124) Metachrosis phenomenon ..... .
(A) Change of skin color (B) Failure of kidney
(C) Acromegaly (D) Heart rate become faster

(125) It is known as milk secreting hormone.
(A) STH (B) Oxytocin
(C) PIF (PTH) (D) Thyro calcitonin

(126) Hormone induces metamorphosis in tadpole of amphibia.
(A) Relaxin (B) Melatonin (C) Thyroxine (D) Cortisol

(127) Disease due to acute deficiency of thyroxine.
(A) Tetany (B) Thyrotoxicosis
(C) Hypophosphastamia (D) Hashimoto disease

(128) Which of the following is responsible for Diabetes coma ?
(A) Ketoacidosis (B) Polydipsia
(C) Hyper glysomia (D) All of above

(129) Which scientist explained molecular structure of insulin ?
(A) A. F. Sanger (B) E. C. Kandle (C) Raynard (D) Barger

(130) Hormone discovered for the first time ..... .
(A) Thyroxine (B) Adrenaline (C) Secretin (D) Insulin

(131) Who has coined the term hormone?
(A) Harris (B) E. H. Schally (C) W. M. Baylis (D) E. H. Starling

(132) Size of which gland decreases with increase of age ?
(A) Adrenal (B) Thymus (C) Pituitary (D) Thyroid

(133) Of the following which gland secretes life saving hormone ?
(A) Thyroid (B) Thymus (C) Adrenal (D) Pancreas

(134) Which of the following gland is not considered as Heterofine gland ?
(A) Pituitary (B) Heart
(C) Pancreas (D) Kidney

(135) Less secretion of mineralocorticoid do not depend on RAAS, induces which of the following disease ?
(A) Cushing's disease (B) Cone disease
(C) Adison's disease (D) Grave's disease

Answers : (108-A), (109-A), (110-B), (111-D), (112-C), (113-B), (114-B), (115-A), (116-C), (117-C), (118-B), (119-D), (120-D), (121-D), (122-D), (123-A), (124-A), (125-B), (126-C), (127-D), (128-D), (129-A), (130-C), (131-D), (132-B), (133-A), (134-A), (135-B)

Human Endocrine System

Hypothalamus

- Supreme commander of endocrine system.
- Hypothalamic releasing hormones reach the pituitary gland through a portal circulatory system and regulate its secretion.

Pituitary

- Orchestra of endocrine system.
- Regulate the growth and activity of several other endocrine glands.
- It has two main lobes : anterior labe (or) adenohypophysis (or) parsdistalis and posterior lobe (or) neurohypophysis (or) pars nervosa and an intermediate lobe (or) pars intermedia.

Anterior Pituitary

- Master endocrine gland regulates the activity of several other glands.

Posterior pituitary

- Stores and releases oxytocin and vasopression which are actually synthesized by the hypothalamus.

Pineal gland

- Secretes melatonin which helps in regulating sleep wake rhythmic cycle.

Thyroid

- It is the largest endocrine gland.
- It secretes two iodine - containing thyroid hormones that control the rate of all metabolic processes.
- It also contains Cells which secrete calcitonin that lowers the blood calcium level.

Para thyroid

- Two pairs of endocrine glands located behind the thyroid gland.
- They secrete parathormone which acts opposite to calcitonin.

Adrenal

- Located above kidneys.
- It has two parts-externel adrenal cortex and internal adrenal medulla.
- Adrenal cortex secretes mineralo corticoids, gluco-rticoids and ganado corticoids.
- Adrenal Medulla is an extension of the sympathetic system. It secretes norcepinephrine and epinephrine.

Pancreas

- Islets of langerhans is the endocrine region.
- α-cells secrete glucagon, β-cells produce insulin, δ cells secretes somatostatin.

Goanads

- Ovaries secrete estrogen, progesterone, relaxin and inhibin.
- Testes secrete androgens by leydig's cells.
- Pars qntermedia Secretes melanocyte stimulating hormone which imparts skin colouration.

33 Reproduction in Organisms

All organisms do other physiological processes like they do reproduction as important physiological process, in which an organism gives birth to offspring similar to itself. The offspring grow, mature and then produce new offspring. Thus there is a cycle of birth, growth and death. Reprodution enables the retention continuity of the species, from generation to generation. The genetic variation is created and inherited during reproduction.

There is a large diversity in the biological world and each organism has evolved its own mechanism to multiply and produce offspring. The organism's habitat, its internal physiology and several other factors are collectively responsible for how it reproduces. There are two types of reproduction in organisms. (1) Asexual reproduction and (2) Sexual reproduction.

Asexual reproduction: From a single parents, without formation of gametes offspring are formed so it is called asexual reproduction. Mitosis and amitosis like cellualr reproduction involved in it.

Sexual Reproduction: When two opposite sex parents. involved and form male and female gametes so it is called sexual reproduction. Meiosis and mitosis like cellular divisions take place in it.

(1) Physiological activities, means ..... .

(A) Physiological process in living organisms.
(B) Chemical reaction in living organism
(C) Physical reaction in living organism
(D) complex of physical, chemical and biological processes in living organisms.

(2) Which physiological process is important for the formation of offsprings ?

(A) Copulation
(B) Digestion
(C) Nutrition
(D) Assimilation

(3) When young ones can reproduce ?

(A) by obtaining nutrition
(B) by growth
(C) by attaining maturity
(D) by nutrition, growth and maturity

(4) Which is correct sequence for living ?

(A) Birth, death, growth, maturity
(B) Birth, growth, maturity. death
(C) Death, maturity, Birth, Growth
(D) Growth, maturity, Birth, growth

(5) What is the definition of continuity of life ?

(A) observing variation from one generation to next.
(B) continuous similarity and some variation from one generation to next.
(C) Similarities from one generation to next
(D) None

(6) What is reason for genetic variation ?

(A) Preservation of discontinuous characters
(B) Preservation of continuous characters
(C) Some characters preseved by reproduction
(D) Preservation of inherieance.

(7) On the basis of which factors reproduction method is determined

(A) Habitat, other factors
(B) Internal physiological processes of livings.
(C) Habitat, physiological interactions, other factors
(D) only effective factors.

(8) Which is proper sentence for the Asexual reproduction ?

(A) offsprings obtained from one parent. (B) gamete formation occurs

(C) offsprings obtained from single parent without gamete formation. (D) Meiosis

(9) What is proper for sexual reproduction ?

(A) Two different parents are essential

(B) Forms two similar gametes

(C) Offsprings are exact copy.

(D) Two different parents forms two different gametes, mostly similar offspring obtained.

(10) When is meiosis occur in sexual reproduction ?

(A) At the time of parent cells (B) At the time of gamete formation

(C) At the time of offspring formation (D) At the time of somatic cells formation

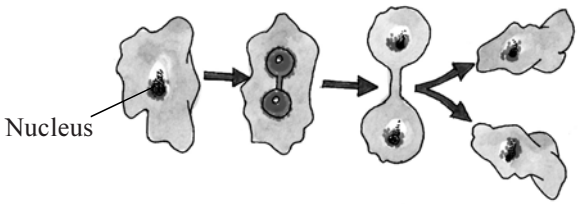
Answers : (1-D), (2-A), (3-D), (4-B), (5-D), (6-C), (7-C), (8-C), (9-D), (10-B)

Asexual reproduction : In asexual reproduction a single, parent is involved and is capable of producing offspring. As a result the offsprings that are produced are not only identical but also exact copies of their parent. Asexual reproduction is common among single-celled organisms and in plants and animals with relatively simple organizations. It is also seen in multicellular organisms.

Asexual reproduction in Animals : In animals asexual reproduction methods have seen as simply Fission, sporulation, Budding, Fragmentation.

(1) Fission : This method is observed commonly in protists and monerans types of organism groups. Three phases of fission. In it first of all division of nucleus occurs, than division or distribution of cytoplasm and finally two identical size of individual daughter cells are formed. This type of fission process is like mitosis. From the single maternal or paternal cell two identical daughter cells are formed so that process is called binaryfision. It occurs in three ways : (1) Simple binaryfision (2) Transverse binaryfission and (3) Longitudinal binary fission.

Simple binary fission : When the cytoplasmic division passes through any directions the fission is called simple binary fission. e.g. Amoeba. In that first nucleus become elongated or cylindrical, groove formed in cell membrane then division of cytoplasm occur in any direction.



Transverse Binary Fission : It is the plane of ctoplasmic division coincides with the transverse axis of the individual, the fission is termed transverse binary fission. eg. Paramoecium and planaria. Their phases are : (1) macronucleus becomes cylindrical (2) Groove formation occur in macronucleus (3) Oral groove disintegrate (4) Amitotic division occurs of macronucleus (5) Formation of new oral groove (6) Formation of new vacuoles (7) Origin of internal groove in plasmamembrane (8) Division occurs in cytoplasm (9) Formation of two individual daughter cells.

Longitudinal binary fission : In euglena and vorticella, the plane of cytoploymic division coincides with the longitudinal axis of the individual. This kind of fission is designated as longitudinal binary fission. Phases of that type of binary fission are as. follows : (1) Elongation of cylindrical nucleus takes place (2) organisation occur in nucleus, ie nucleus groove (3) Formation of longitudinal groove in plasmamembrane. (4) Division occurs in cytoplasm (5) Formation of two individual daughter animals.

Longitudinal binary fission in Euglena : In binary fission only mitotic division take place. (cell division = cellular reproduction for the unicelluler organism.) Resultant progenies are genetically identical with the parental generation, ie from the single parent originated offsprings are identical genetically, which are considered as genetical clones of parents.

Multiple Fission : In this type of fission, dominancy of cell division is amitotic division. The nucleus divides several times by amitotic nuclear division. Thus large numbers of nucleus are formed. Cytoplosm does

not divide during this period. Then cytoplasm collects around each nucleus. Thus, within one maternal cell, innumerable unicellular and uninucleate offspring are formed. In course of time they live as independent unicellular organisms. This method of reproduction is called multiple fission. Multiple fission is observed in Amoeba and paramoecium.

(11) What is sequence of development for asexual reproduction in animals ?

(A) Fragmentation, Budding, sporulation, fission

(B) Buddingm sporulation fission fragmentation

(C) Sporulation, Budding, fragmentation, fission

(D) Fission, Sporulation, Budding, fragmentation,

(12) What is the sequence of development for fission stages ?

(A) Division of nucleus - Division of cytoplasm - 2 daughter cells

(B) Division of cytoplasm - Division of nucleus - 2 daughter cells

(C) Division of nucleus - 2 daughter cells - Division of cytoplasm

(D) Division of nucleus - 2 daughter nuclei - 2 daughter cells

(13) Simple cell division, in which two daughter cells formed is called ....., it is which type of cell division ?

(A) Binary fission, mitosis (B) Binary fission, Amitosis

(C) Binary fission, meiosis (D) Binary fission, cellulardivision

(14) Which sentence is proper for eukaryotic unicellular irregular shaped protozoans ?

(A) Division of cytoplasm in mid-longitudinal axis

(B) Division of cytoplasm transversely

(C) Division of cytoplasm, longitudinally

(D) Division is irregular.

(15) Select proper option for development stage of cell division for eukaryotic, unicellular irregularly shaped protozoans.

(A) Nucleus tubular enlarges - nuclear ridge formed - nucleus divided into two - Internal projection in plasmamembrane - cytoplasm divides.

(B) Internal projection formed in plasma membrane - cytoplasm divides - nucleus tubular - Nuclear ridge formed - nucleus divided into two.

(C) Internal projections formed in plasmamembrane - nucleus divided - nuclear ridge forms - nucleus tubular - nucleus divided into two

(D) Nuclear ridge formed - nucleus tubular - nucleus division - Internal projections in plasma membrane - cytoplasm divided.

(16) In lower form of eukaryotic regular shaped protozoan and platyhelminthes at the time of binary fission, division of cytoplasm is at ..... axis

(A) Mid longitudinal (B) Transverse (C) longitudinal (D) A and C

(17) Which option is proper for nuclear divisional stages in transverse binary fission ?

(A) micronuclei become tubular $\rightarrow$ internal groove in micronuclei $\rightarrow$ two daughter nuclei formed.

(B) Macro nuclei become tubular → internal groove in macro nuclei → two daughter nuclei formed.

(C) Internal groove in micronuclei → micronuclei tubular → two daughter nuclei formed.

(D) Internal groove in micronuclei → micronuclei tubular → two daughter nuclei formed.

(18) Which option is proper for lower form, regular shape eukaryotic protozoans and platyhelminthes for developmental sequence in division of cellular matrix ?

(A) Disappearance of oral groove → New oral groove formed → Internal groove develop in plasma membrane → New contractile vacuoles formed - division of cytoplasm.

(B) New oral groves formed → internal groove develop in plasma membrane → oral groove disappears → new contractive vacuole formed → division of cytoplasm.

(C) Disappearance of oral groove → new oral grooves formed → new contractile vacuoles formed - internal groove develop in plasma membrane → division of cytoplasm

(D) Disappearance of oral groove → new oral groove develops → internal groove develop in plasma membrane → Division of cytoplasm - new contractile vacuoles formed.

(19) Which two earlier stages are found in paramoecium and planaria, in which transverse division give rise to two daughter cells

(A) Division of nucleus → Divison of cytoplasm

(B) Internal groove in meganucleus → internal groove in plasma membrane

(C) Internal groove in plasma membrane division of cytoplasm.

(D) New oral grooves formed, new contractile vacuoles formed.

(20) Select option in which Euglena divides longitudinally to form two daughter cells

(A) Nucleus become tubular → nuclear groove formed longitudinarly groove in plasma membrane - division of cytoplasm

(B) Nuclear groove formed → Nucleus become tubular - longitudinal groove in plasma membrane division of cytoplasm.

(C) Division of cytoplasm → longitadinal groove in plasma membrane - nuclear groove formed -nucleus become tubular.

(D) Longitadinal groove in plasmd membrane - division of cytoplasm - nucleus tubular - nuclear groove formed

(21) Which option is proper for genetic similarity ?

(A) Similarity in number of genes (B) Similarity in shape

(C) Similarity in physiology (D) genetic, morphological, physiological similarities.

(22) In which division processes, Amitosis is main ?

(A) Simple division, transverse binary division (B) Simple division, longitudinal binary division

(C) Simple division, multiple fission (D) Transverse binary fission, mltiple fission

(23) In multiple fission which process occurs frequently and which do not occur ?

(A) Amitosis, division of cytoplasm (B) Mitosis Division of cytoplasm

(C) Mitosis, Division of cytoplasm (D) Amitosis, division of nucleus.

(24) In multiple fission, during amitosis which substage observed frequently ?

(A) G_1 (Gap - 1) (B) G_2 (Gap - 2)

(C) S = DNA replication (D) all of above

(25) offsprings formed as a result of multiple fission from one maternal cell is .....

(A) innumerable, coenocytic (B) innumerable, uninucleated

(C) innumerable, unicellular, uninucleated (D) innumerable, coenocytic.

(26) Which organisms show multiple fission ?

(A) plasmodium, planaria, Amoeba

(B) Plasmodium, Amoeba, paramoecium

(C) Plasmodium, Euglena, paramoecium

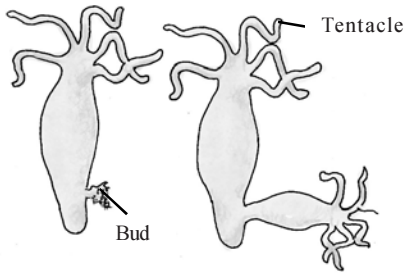
(D) Plasmodium, Euglena, Vorticella.

Answers : (11-D), (12-A), (13-A), (14-D), (15-A), (16-B), (17-B), (18-C), (19-C), (20-A), (21-D), (22D), (23-A), (24-C), (25-C), (26-B)

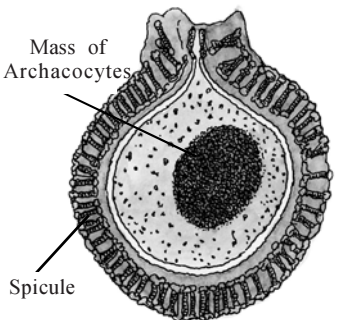
(1) Sporulation : Sporutation occurs during unfavorable conditions. In lower animals to maintain for continuation of life. Amoeba is irregular in shape, uninucleated protist organisms Amoeba withdraw their pseudopodia and become round in shape and form spore like structure, They create a hard protective three layered cyst around themselves, this process is called encystations. When Conditions become favorable, the nucleus in amoeba undergoes multiple divisions and a large number of amoeba are formed. These are called pseudopodiospores. This process is called Sporulation. When the cyst ruptures all new Amoebae are released. In plasmodium this proces occurs at a specific stage in its life cycle.

(2) Budding : In this method, first of all, cells of some part of the body of the animal repeatedly undergo mitotic divisions and the raised regions of cell masses, called bud are formed. From such a bud a young animal develops. It separates from the parent body and lives as an independent animal.

If such a bud is produced on the outer surface of the body it is called exogenous budding. In **Hydra**, exogenous budding is observed.



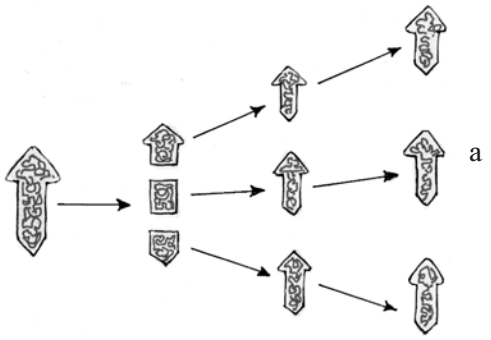
Exogenous Bud In Hydra



Internal bud in Spongilla

In Fresh water sponge eg. **spongilla** and marine sponge eg. **Sycon** specialized cell masses are produced surrounds the inside of the body. An envelope surrounds this cell mass. Such structures are called internal buds or gemmules. Each gemmule gives rise to a new animal. This is called endogenous budding.

(3) Fragmentation : In this method of asexual reproduction, the body becomes fragmented into several distinct parts. Each part develops the remaining body parts and becomes complete animal. This capacity is known as regeneration Fragmentation is observed as regenetation capacity. E.g. lower platyhelminthus - Planaria, coelenterata - Hydra, Animal of ehinodermata - Starfish.


Regeneration in Planaria

(27) Which option is proper for Amoeba ?

(A) It is irregular, unicellular, coenocytic pseudopodial, protozoa.

(B) It is irregular, unicellular, uninucleated pseudopodial protozoa.

(C) It is irregular coenocytic, uninucleated pseudopodial protozoa.

(D) It is irregular, unicellular, uninucleated, protozoa without pseudopodia.

(28) Irregular, unicellular, uninucleated, protozoa shows sporulation in which condition ?

(A) Favourable (B) Normal

(C) Unfavourable (D) specific.

(29) In unfavourable situation, how is round structure formed in amoeba like organisms ?

(A) Showing morphological changes

(B) Spreading of plasmamembrane in periphery

(C) Due to atmospheric pressure - effect.

(D) Pseudopodia contracts and spore formation occurs.

(30) How is cellular capsule ?

(A) Three layered, strong. (B) Three layered, strong, permeable membrane

(C) Three layered (D) Three layered, strong, impermeable

(31) What is encapsulation ?

(A) Three layered, strong, impermeable capsule formation surrounding round structure of spores

(B) Formation of two layered, strong, impermeable capsule surrounding round structure of spores

(C) Formation of three layered smooth, impermeable capsule surrounding round structure of spores.

(D) Formation of capsule, which is three layered smooth, semipermeable surrounding round spores.

(32) Pseudopodiospores possess which structures ?

(A) Nucleus (B) Nucleus + cytoplasm + Plasma membrane

(C) Cytoplasm + nucleus (D) Nucleus + cytoplasmic organelles

(33) Which option is proper for formation of exogenous bud ?

(A) Any organ of parent body divides frequently mitotically to form bud.

(B) All organs of parent body frequently divides mitotically to form bud.

(C) Any organ of parent body frequently divides mitotically as aresult bud forms as an outgrowth

(D) Any organ of parent body frequently divides amitotically, bud froms as an outgrowth.

(34) In which animal group and which type of animal exogenous bud formation is common ?

(A) Coelenterata polygonal shape (B) Coelenterata, tubular

(C) Coelenterata, round (D) Coelenterata, Multicellular

(35) What are gemmules or endogenous buds ?

(A) Bud which is formed outside the parent body

(B) Bud which is developed longitudinally on parent body.

(C) Bud which is developed transversally on parent body.

(D) Specific cell groups encircle to form bud on parent body

(36) What is fragmentation ?

(A) Parent body breaks into smaller, larger parts, and required parts regenerate to form complete organism.

(B) Parent body longitudinally divides in equal parts, required parts regenerate and complete organism is formed.

(C) Parent body tranasversely divides into equal parts, required parts regenerate to form complete animal.

(D) Parent body alternately divides longitudinally and tranasversely, remaining parts regenerated to form complete organism.

(37) What is regeneration ?

(A) Complete organism develops from fragmented segment or organ

(B) Complete organism develops from any cell.

(C) Complete organism develops from any tissue

(D) Complete organism develops from any system.

(38) What is called totipotency ?

(A) Complete organism develops from any broken part or organ

(B) Complete organism develops from any differentiated cell.

(C) Complete organism develops from any tissue.

(D) Complete organism develops from any system.

(39) Which organisms shows asexual reproduction by fragmentation ?

(A) Lizard, crocodile, bird

(B) Anaconda, crocodile, calotes

(C) Cockroach, Locust, housefly

(D) Hydra, planaria, starfish

(40) Complete totipotency is observed in which animal group ?

(A) Platyhelminthes, housefly

(B) Annelida, Arthropoda

(C) Coelenterata, Mollusca

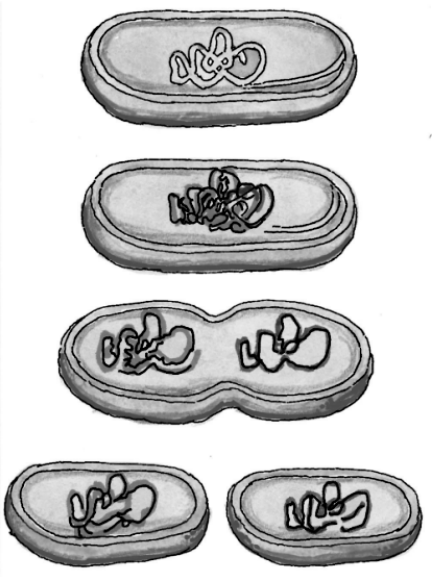
(D) Porifera, coelenterata

Answers : (27-B), (28-C), (29-D), (30-D), (31-A), (32-C), (33-C), (34-B), (35-D), (36-A), (37-A), (38-B), (39-D), (40-D)

-Asexual Reproduction in plants :

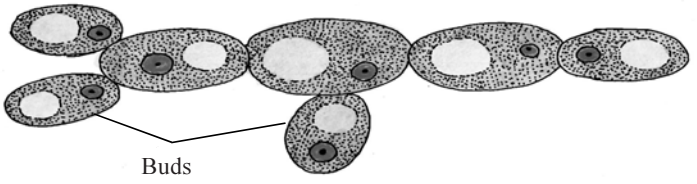
The common mode of asexual reproduction in plants is like fission, Budding, fragmentation, spore formation.

(1) Fission : This method is simplest method of asexual reproduction. In which unicellular maternal cell divides by the mitosis Two daughter cells are formed. Each eventually grow into an independent mature organism. This simple fission process has stages as follows. (1) Enlargment area of nucleus (cylindrical) (2) formation of nucleus groove (3) Formation of two daughter nuclei (4) Inversion of plasmamembrane by the formation of internal groove. (5) Distribution of cytoplasm. (6) Two independent daughter cells are formed. This simple fission method commonly found in lower level plant groups eg. monera (Bacteria = Schizophyta), Algae Fungi, Bacteria/schizophyta are prokaryotic, while Algae and fungi are eukaryotic.



Fission in Bacteria

(2) Budding : In plants amitotic divisions are responsible for asexual reproduction by this method. In that adventitious branches or buds like structures are formed which are attached to parent body. Finally bud forms new organism

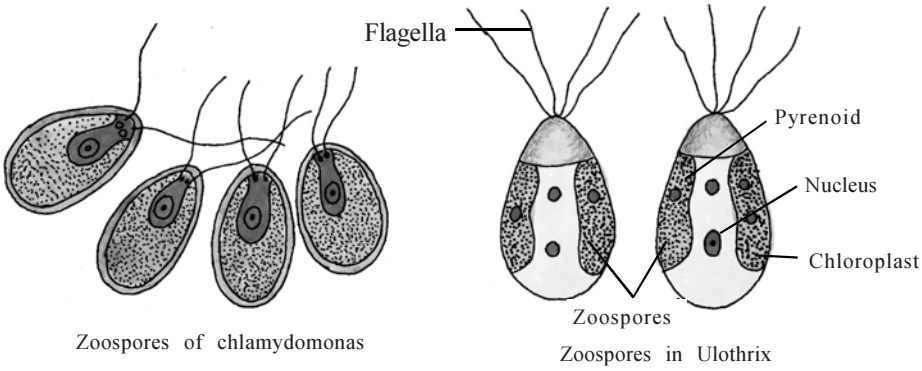


Buds in Yeast

independently. Some algae produce adventitious branches eg. **Dictyota, Fucus** by the amitotic divisions. **Protosiphon** algae produce bud. Even yeast like fungi by amitotic division produce bud.

(3) **Fragmentation :** In plant this type of asexual reproduction break up into small segments due to mechanical pressure and each segment is capable of growing into a new mycelium or orgranism. In some algae e.g **ulothrix, oedogonium, spirogyra** and **zygnema** froms smaller or bigger segments of the vegetative thallus due to mechanical pressure. While **mucor, Rhizophus, saprolegnia** like fungus hyphae break up into small segment due to mechanical pressure. Each segment is capable of growing into a new mycelium.

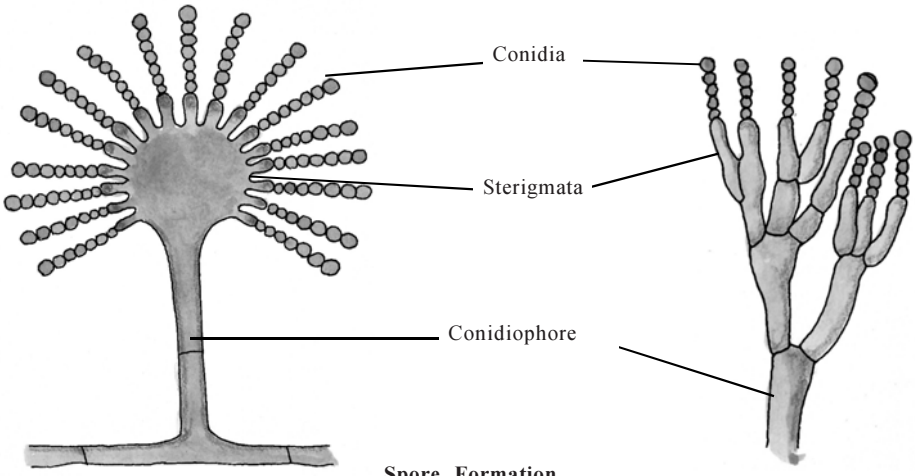
(4) **Spore formation :** In plants in this method of asexual reporduction various types of spores are formed. Spores mainly are two types (1) Zoospores (2) Conidiospores.



Zoospores of chlamydomonas

Zoospores in Ulothrix

Ciliated motile spores called zoospores are produced by algae and fungi which swim in water for some time with the help of their flagella and then directly develop into new independent individuals. e.g **Ulothrix, chlamydomonas, oedogonium.**



Conidia in Aspergillus

Spore Formation

Conidia in Penicillium

Non flegellated and non-motile spores/conidia of various kinds are most common among terrestrial fungi. Such spores are light, dry and provided with a tough coat, and are well adapted for dispersal by wind e.g. Penicillium, Aspergillus.

True spores are always borne by sporophyte. Thus the sporophyte mass reproduces asexually by spores. Similarly ferns (Nephrolepis) bear spores and reproduce asexually by them. These plants are

homosporous (bear only one kind of spores) while in selaginella (a pteridophyte) and gymnosperms they are heterosporous (bears two types of spores).

(41) Select proper option to understand evolutinary sequence for asexual reproduction method in plants
(A) Fission, Bud formation, fragmentation, spore formation
(B) Fission, fragmentation, sporeformation, Bud formation
(C) Bud formation, spore formation, fragmentation, fission
(D) Fission, spore formation, Bud formation, fragmentation

(42) In which plants fission process takes place ?
(A) Unicellular, prokaryotic cell
(B) Multicellular, Eukaryotic cell
(C) Unicellular / Multicellular, prokaryotic cell
(D) Unicellular / multicellular, prokaryotic / Eukaryotic cell

(43) Which option is proper in plants for development sequence of nuclear division during fission ?
(A) Nuclear region increases - nuclear ridge formed - two daughter nuclei formed.
(B) Nuclear region increases - two daughter nuclei formed - nuclear groove is formed.
(C) Two daughter nuclei formed - nuclear groove formed - nuclear region enlarges
(D) Two daughter nuclei formed - nuclear region enlarges - nuclear groove formed.

(44) Which option is proper for development sequence of cytoploasmic division ?
(A) Invagination of plasmamembrane forms internal ridge - Equal distribution of cytoplosm - Two daughter cells are formed
(B) Invagination of plasma membrane forms internal ridge - two daughter cells formed - equal distribution of cytoplasm.
(C) Equal distribution of cytoplasm - internal groove formed by invagination of plasma membrane - Two daughter cells are formed
(D) Two daughter cells formed - equal distribution of cytoplasm - Internal groove formed by invagination of plasma membrane.

(45) In plants during fission, invagination of plasma membrane and internal groove formation occur during which stage ?
(A) Increase in nuclear region
(B) Formation of nuclear region
(C) Two daughter nuclei formed
(D) Equal distribution of cytoplasm occurs.

(46) In which plant group fission process occurs ?
(A) Algae lichen, Fungi
(B) Monera Bryophyta, Pteriophyta
(C) Monera, algae, fungi
(D) Pteridophyta, Gymnosperm, Angiosperm

(47) Which type of cellular structure shown by lower plants, shows fission process ?
(A) Unicellular
(B) Multicellular
(C) Acellular
(D) Unicellular / multi cellalar

(48) Which structure forms by repeated amitosis in parent thallus ?
(A) Adventitions branches
(B) Buds
(C) Adventitions branches or buds
(D) Adventitious branches and buds.

(49) Which algae develops adventitious branches during bud formation ?
(A) Dictyota, Fucus
(B) Dictyota, Protosiphon
(C) Fucus, Yeast
(D) Fucus, Protosiphon

(50) Which algae and fungi sequentially forms bud by fission ?

(A) Dictyota and fucus (B) Dictyota and protosiphon

(C) Fucus and yeast (D) Protosiphon and yeast.

(51) Which plant organ experiences mechanical stress during fragmentation ?

(A) Vegetative thallus (B) Vegetative cells

(C) Spores (D) Reproductive cells

(52) Which group of algae shows asexual reproduction by fragmentation ?

(A) Ulothrix, oedogonium, spirogyra, zygnema

(B) Ulothrix, oedogonium penicillium, Aspergillus

(C) Ulothrix, oedogonium, spirogyra, dictyota

(D) Ulothrix, oedogonium, spirogyra, fucus.

(53) Which fungi show asexual reproduction by fragmentation ?

(A) Mucor, Rhizopus, saprolegnia (B) Mucor, yeast, penicillium

(C) Mucor, yeast, Aspergillus (D) Mucor, penicillium, Aspergillus

(54) In spore formation which are main type of spores ?

(A) Round spores, oval spores (B) Motile spores, non motile spores

(C) Motile spores, Round spores (D) Non-motile spores, oval spores

(55) What is pecularity of motile spore ?

(A) Flagellated (B) Without flagella

(C) Anuclear (D) Flagellated with locomotory capacity

(56) In which algae motile spores are formed ?

(A) Ulothrix, Chlamydomonas, oedogonium (B) Chlamydomonds, Ulothrix, Dictyota.

(C) Ulothrix, Chlamydomonas, Fucus (D) Ulothrix, Chlamydomonas, Protosiphon

(57) Non motile spore, conidia have which pecularity based on structure ?

(A) Light and dry

(B) Without flagellum with strong envelop

(C) Pollination by wind

(D) Inumerable, light, dry, non-flagellated, wind pollinated

(58) Which are known examples of terrestrial fungi which produces conidia ?

(A) Penicillium, Aspergillus (B) Penicillium, oodogonium

(C) Penicillium, Aspergillus (D) Penicillium, ulothrix

(59) When are true spores formed in plants which have alternate gametophytic and sporophytic life cycle?

(A) Sporophytic stage (B) Gametophytic stage

(C) Megasporangiam. (D) Gametes.

(60) Which are isogametes and heterogametes sequentially ?

(A) Same type of gametes - similar morphological physiological and genetically are isogametes. Disimilar morphological physiological and genetically are heterogametes.

(B) Isogametes - genetically similar but physiological and morphologically different. Heterogametes - genetically different but morphological and physiological similar

(C) Isogametes - morphological, physiological similarity, genetic difference.

(D) None of above.

(61) Select proper option for classifying nephrolepis, selaginella, cycus, maize and bean for isogametic or Heterogametic

(A) Isogametic - Nephrolepis, selaginella Heterogametic - cycus, maize, Bean

(B) Isogametic - Nephrolepis, cycus Heterogametic - selaginella, maize, Bean

(C) Isogametic - Nephrolepis Heterogamctic - selaginella, cycus, maize Bean.

(D) Isogametic - Nephrolepis, cycus, maize, Bean. Heterogametic - Selaginella.

(62) Which is proper option for development sequence for conidia ?

(A) Conidiophore - Sterigmata - Conidia

(B) Sterigmata - Conidia - Conidiophore

(C) Conidia - Sterigmata - Conidiophore

(D) Conidia - Conidiophore - Sterigmata

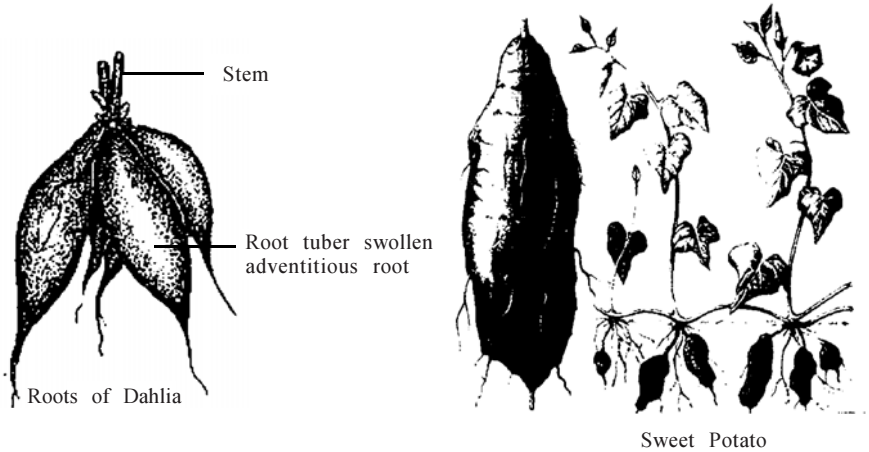
Answers : (41-A), (42-D), (43-A), (44-A), (45-B), (46-C), (47-D), (48-C), (49-A), (50-D), (51-A), (52-A), (53- A), (54-B), (55-D), (56-A), (57-D), (58-A), (59-A), (60-A), (61-C), (62-A)

(1) Vegetative propagation / vegetative reproduction :

Vegetative organs of plant asexual reproduction occur so it is vegetative propagation. It is one type of asexual reproduction for Amimals and lower level organisms. Vegetative propagation - vegetative reproduction takes place by various vegetative organs by natural ways. Even vegetative propagation by artificial means can be induced. i.e. vegetative reproduction is divided in two main methods (1) Natural method and (2) Artificial method.

(2) Natural methods of vegetative propagation : In natural methods of propagation, the development of a new plant from some organ of the mother plant under suitable environmental conditions is very common. In this method root, stem, leaf, floral bud are such reproductive organs.

Vegetative Reproduction by root : Sweet potato like plant vegetative reproduction by adventitious root for food storage occurs. Thus adventitious root becomes fleshy. This type of food storage occurs in adventitious root of root base middle part and rootapex. Food storage is less in root apex and root base but middle part of the root food storage is more. This type of root store food in the form of starch and sucrose molecules. So it called simple tuberous root. This type of adventitious root in Asparagus and Dahlia also store food in cluster form. Such roots are called fasciculated tuberous root. They too involve in vegetative propagation.



Vegetative Reproduction by Root

Vegetative reproduction by stem : Ginger, Turmeric like plants store food in underground stem. Such prostrate underground stem possesses nodes, internodes, scaly leaves, axillary buds and terminal bud. This type of food storing underground stem is called rhizome; by which new daughter plants are formed. while in Amorphophallus food storage occur in only one internode of underground stem so it is called Corm by which vegetative reproduction occur. In patato food storage occur in scaly leaf of apical bud of underground stem which is called tuber stem. Which produce meristem vegetative bud, it is called 'eye'. by it vegetative propagation can occur. In an onion food storage occur in leaf base of scaly leaf stem is underground, disc like and pointwise, flatten so apical and axillary buds growth would be stunted, it is called simple tunicated bulb. By this simple tunicatedbulb vegetative reproduction can occur.

Vegetative propagation by leaf : In Bryophyllum like plants vegetative propagation by leaf. Leaf margins of leaf of this plant vegetative meristem buds can occur. which make the contact with ground /soil and form daughter plant by vegetative reproduction. This types of buds are known as adventitious buds. Thus in bryophyllum vegetative propagation occured by adventitous buds.



Vegetative Reproduction by Leaf in Bryophyllum

Vegetative Reproduction by Buds : Agave has longest inflorescence in the plantkingdom, even in oxalis food storage occurs in floral buds. Such food storing floral buds becomes fleshy and weight increses, which fall in the ground and in favourable condition new daughter plant develops by vegetative propagation. Thus, in Agave, oxalis vegetative propagation occur by floral bud and converted into floral bulbil. Same in Dioscorea like plant food storage occur in axillary bud. so it becomes fleshy, heavy, which falls on ground and produce new daughter plant in favourable conditions. So, in Dioscorea vegetative reproduction occur in axillary bud and converted into axillary bulbil.

Other specific methods like natural methods are also available for vegetative propagation. These modifications are of subaerial stems. A part of the stem is underground and the remaining part is above the ground. In Grass, oxalis and Hydrocotyl, thin long, branches develops. They possess internodes and run parallel to the ground. The nodes which come in contact with ground develop new plants. This modification is called runner. In aquatic plants like pistia, marsilea from the node of stem branch is formed for vegetative propagation which is called offset.

while in mint branch of stem partially underground and partially aerial which do vegetative propagation by suckers. While in fern or nephrolepis and strawberry branch of underground -stem grows away from the ground level and take curve and then enter into soil and develops new plant, such branch is called stolon. Hence in fern (nephrolepis) and strawberry vegetatives propoagation occur by stolon.

(63) What is vegetative reproduction ?

(A) Asexual reproduction by vegetative organs of plants

(B) Asexual reproduction by reproductive organs of plants

(C) Asexual Reproduction by plant organs such as fruits, seeds

(D) Asexual reproduction by flower, microspores megaspores of plants.

(64) In which organisms asexual reproduction is very prominent ?

(A) Lower organism and animals.

(B) Lower category plants and animals

(C) Lower category prokaryotic organism and animals.

(D) Lower category Eukaryota and animals.

(65) Which plant organ participate in vegatative reproduction ?

(A) Root, stem, leaf

(B) Root, stem, leaf, floral bud

(C) Root, stem, leaf, flower

(D) Root, stem, leaf, flower.

(66) Which vegetative organ of sweet potato transforms into ..... for vegetative reproduction

(A) Adventitions root, simple tuberous root.

(B) Primdry root - simple tuberous root

(C) Adventitious root - simple repeated structure

(D) Primary root - composite, covered root.

(67) What is called modified structure of underground stem for storage of food in ginger or turmeric and Amorphophallus ?

(A) Rhizome, tuberous stem

(B) Rhizome, simple tunicated bulb.

(C) Rhizome, simple tuberous

(D) Rhizome, corm.

(68) Which modified structure formed by food storage in underground part of potato and onion ?

(A) Rhizome, Tuberous stem

(B) Tuberous stem, Rhizome

(C) Tuberous stem, simple, tunicated bulb

(D) Tuberous stem, compound tunicated bulb.

(69) Ginger, Amorphophallus potato and onion , food is stored in which part of the plant sequentially ?

(A) Rhizome, Nodes and internodes of rhizome, scaly leaf of apicalbud of under ground stem, base of scaly leaf

(B) Rhizome, Nodes of rhizome, scaly leaf of axillary bud of rhizome, Base of scaly leaf

(C) Rhizome, Nodes and internodes of rhizome, scaly leaf of apical bud of rhizome, scaly leaf apex.

(D) Base of rhizome, nodes of rhizome, scaly leaf of apical bud of rhizome, scaly leaf base.

(70) Which option is proper for vegetative roproduction in Bryophyllum ?

(A) Axillary buds at the leaf margin - vegetative reproduction - adventitious buds.

(B) At the leaf margin vegetative, meristematic buds - food storage - vegetative reproduction - adventitious buds.

(C) Buds on lamina - vegetative reproduction - Adventitious buds.

(D) Buds at leafbase - vegetative reproduction - adventitious buds/

(71) Bulbils are .....

(A) Only do vegetative reproduction

(B) Climbing and vegetative reproduction

(C) Food storage and vegetative reproduction

(D) Support and vegetative reproduction

(72) Which is proper modified structure for vegetative reproduction for hydrocotyl, Eicchornia, Mint and Nephrolepis respectively ?

(A) Runner, offset, sucker, stolon

(B) Offset, Sucker, Stolon, Runner

(C) Stolon, offset, Runner

(D) Stolon, Offset, Runner

(73) What is main difference in Runner and stolon for vegetative reproduction ?

(A) Branch develops parallel to soilsurface, branch develops in the soil

(B) Branch of underground stem develop parallel to soil surface, Branch of underground stem develops above soil-surface again come in contact with soil.

(C) Branch of underground stem remains in semi soil sub aerial and parallel to underground stem

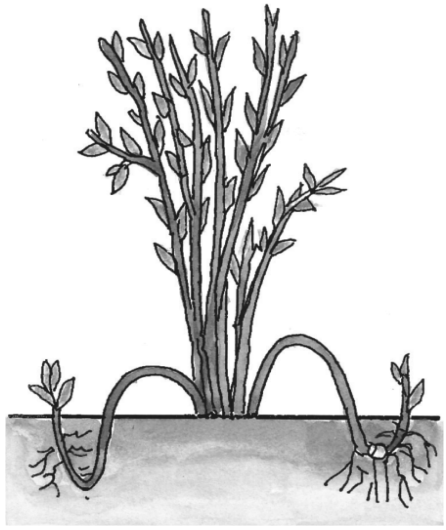
(D) No Difference.

Answers : (63-A), (64-A), (65-C), (66-A), (67-D), (68-C), (69-A), (70-B), (71-C) , (72-A), (73-B)

Artificial methods of vegetative Reproduction : Methods are developed for artificial vegetative propagation in which some part of the plant organ is utilized for obtaining a new complete plant. Amongst them the most common methods are. Cutting, Layering and Grafting.

(1) Cutting Method : By the root and stem this type of artificial method induce as a vegetative reproduction. Cut the pieces of root and are planted in moist soil, development of adventitions roots, is artificially indcued. New plants are developed in this way in **lemon** and **tamarind**.

Proper sizes of stem pieces are obtained and are planted in moist soil to develop into new plants. (xylem in contact with the soil surface). From the underground parts of stem, adventitions roots develop and buds on the aerial parts of stems sprout. The plant, so developed is called 'cutting'. Later, there cutting are transplanted in proper places. eg. **Rose, sugarcone, croton, china-rose** and **chrysanthemum** plants.



Layering

(2) Layering Method : The lower branches of the plant are bent and pressed under the soil in such a way that the tip of the branch remains outside the soil and middle portion is buried inside the soil. When adventitions roots develop from this buried region of plant stem; this branch is cut and separated from the parent plant. Thus, a new plant is obtained. This method is employed in the cultivation of **Rose, lemon, Grape Hibiscus** and **Jasmine.**

(3) Grafting method : Grafting is practiced in plants which do not root easily, or have a weak root system. In this method a union is established between two plant of the same or different kinds. Such a union is estabhished between tissues of the two plants. This process can be induced more successfully amongst those plants which possess meristematic tissue (Dicots)

The main supporting plant or which plant cannot develope root system easily or which plant has weak root system, is called stock plant. The plant which is being grafter on it is called scion. A plant possessing higher and desirable characters is selected as 'scion' various methods of grafting scion are practiced.

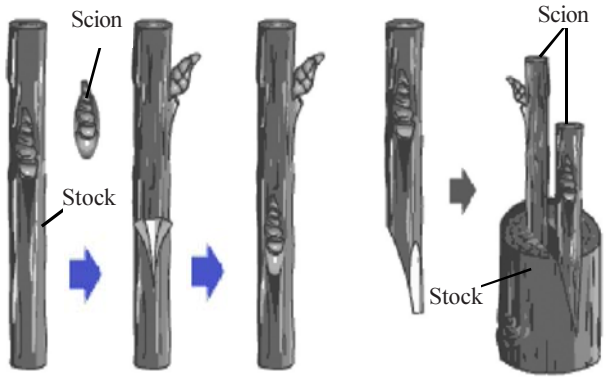
Grafting may be of different types, named budgrafting, side grafting, tongue grafling wedge grafting / clift grafting and crown grafting depending on the methods of uniting the two parts. **mango, Apple, pear, citrus, Guava, litchi** and many other fruit - yielding plants are thus obtained and maintained.

Significance of vegetative Reproduction :

(1) Vegetative reproduction is an ideal method of reproduction in plants in which it is desirable to maintain the same characteristice in the offspring which are present in the parents.

(2) Plants showing reduced power of sexual reproduction, long dormant period of seed or poor viability can also be multiplied easily through this method.

(3) Vegetative reproduction also helps in removing common infections from the parent plant.



Bud-Grafting **Clift-Grafting**

(4) In the plants raised through grafting, it is even possible to bring together the desired characters from plants. even best species are cultured by this method.

---

(74) Which plant tissue is kept in contact with soil during cutting ?

(A) Parenchyma (B) Phloem (C) Xylem (D) Conductive tissue.

(75) When is cutting become ready ?

(A) When Root system develops (B) Adventition root system, Buds and leaf lets develop.

(C) When Buds develop (D) When leaf-lets develop.

(76) In which plants cutting is done by root parts ?

(A) Lemon and Tamarind (B) Rose and sugarcane

(C) Chrysanthemum and china rose (D) Croton, Rose

(77) Which plant group do not show vegetative reproduction by stem cutting ?

(A) Lemon, Tamarind (B) Rose and sugarcane

(C) Chrysanthemum, China Rose (D) Croton, Rose

(78) Why during cutting, xylem tissue, is kept in contact with soil, which type of force developed ?

(A) Capillary force for ascent of sap (B) Capillary force for transpiration

(C) Osmosis, Passive transport (D) Active transport, suction pressure

(79) In which plant group layering is done ?

(A) Rose, Lemon, grapes (B) Hibiscus, Jasmine, Grape

(C) Sugarcane, Chrysanthemum, Croton (D) Lemon, Tamarind sugar-cane

(80) Which option is proper for grafting ?

(A) Between two different or same species of plants, artificial vegetative reproduction for obtaining desired characters.

(B) Artificial vegetative reproduction between two similar species of plants to obtain undesired characters.

(C) Artificial vegetative reproduction between two different species to remove disease causing factors.

(D) Artificial vegetative reproduction between two similar species type plants to maintain hormonal balance.

(81) Which pecularities considered in plants for grafting method ?

(A) Stock possess developed root system, scion possess buds and leaflets and stock and scion has meristematic tissue

(B) Stock possess meristematic tissue scion possess buds and leaflets. Both stock and scion has developed rootsystem

(C) Stock possess developed root system, scion has only meristematic tissue and stock and scion both has Buds and leaflets.

(D) Stock and scion undifferentiated, both possess meristematic tissue.

(82) Grafting method is beneficial for whom ?

(A) Fruit production (B) Garden

(C) Horticulture (D) Decoration

(83) Which sentence is improper for importance of vegetative reproduction ?

(A) Decreasing work efficiency of sexual reproduction is avoided by vegetative reproduction.

(B) To remove prolonged dormancy of seed vegetative reproduction is beneficial.

(C) Disease causing factors do not enter reproduction is beneficial.

(D) To retain unwanted characters of parent plant, vegetative reproduction is useful.

Answers : (74-C), (75-B), (76-A), (77-A), (78-A), (79-A), (80-A), (81-A), (82-A), (83-D)

- **Sexual Reproduction :** Sexual reproduction involves formation of the male and female gametes, either by the same individual or by different individuals of the opposite sex. These gamete fuse to form the zygote which develops to form the new organism. It is complex and slow process as compared to asexual reproduction. Because of the fusion of male and female gametes, sexual reproduction results in offspring that are not identical to the parents or amongst themselves.

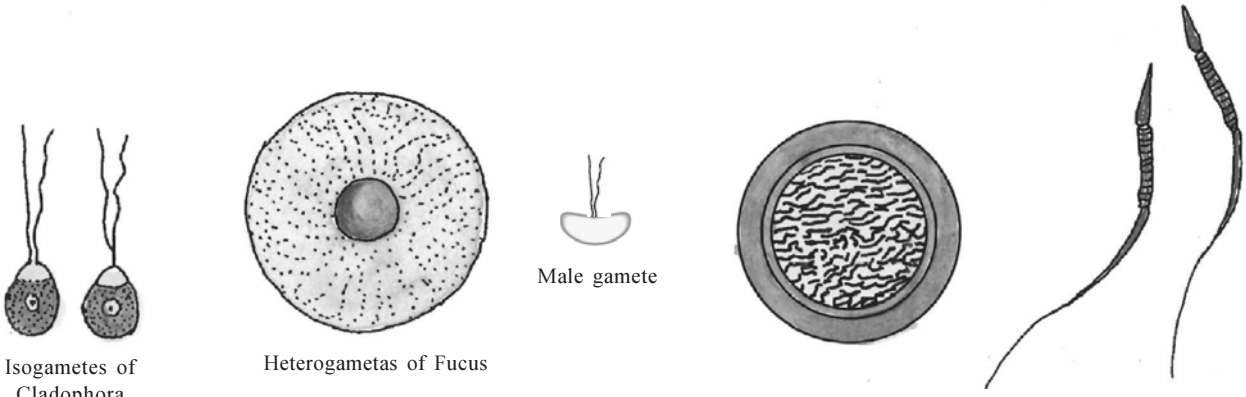
Though the plants, animals or fungi differ in external morphology, anatomy and physiology, yet their sexual mode of reproduction is similar in pattern. All organism reach a certain rate of growth and maturity in their life before they can reproduce sexually. This period is called the juvenile phase and in plants it is known as vegetative phase.

After attaining maturity, all asexually reproducting organisms show events and processes which have fundamental similarity, but the Structures associated with sexual reproduction are quite different. In all cases, the sexual reproduction is characterized by the fusion of the male and female gametes of the species. For convenience there sequential events may be grouped into three distinct stages namely, the (1) Pre-fertilization (2) Fertilization and the (3) post - fertilization events.

- Pre-fertilization Events : The pre-fertilization events of sexual reproduction are found prior to the fusion of gametes. The two main pre-fertilization events are (1) gametogenesis and (2) gamete transfer.

(1) Gametogenesis : Gametogenesis is the process of formation of gametes. Generally gameter are of two types. i.e. male and female gametes. Gametes are haploid (n) cells. In some algae where two gametes are similar in appearance they are called isogametes or homogametes. It is morphologically and physiologically similar and usually motile and has flagellat e.g. clodophora, Ulothrix. However in a majority of sexually reproducting organisms the gametes produced are of two types. Morphologically and phyiologically distinct types which are known as heterogametes or anisogametes. The male gametes are smaller and active whereas the female gametes are large and sluggish. In such cases the male gamete is called anthrozoid or sperm and the female gamete is called the egg or ovum.

(1) Isogametes of cladophora (2) Heterogametes of fucus (3) Gametes of Human beings.



Gametes

Gametes are always haploids, but the parent body from which they arise may be either haploid or diploid A haploid parent produces gametes by mitotic division. Several organisms belonging to **monera, fungi, Algae** and **Bryophyta** have haploid plant body but in majority of organisms belonging to **pteridophyta, Gymnosperms, Angiosperms** and **most of the animals,** the parental body is diploid. Here meiosis takes place to produce haploid gametes.

In diploid organisms the meiocytes (gamete mother cells) undergo meiosis. At the end of meiosis, only one set of chromospmes (n) gets incorporated in each gamete. Table showing diploid and haploid chromosome numbers of organisms.

Sr. No.	Name of Organisms	Chromosome in Meiocyte (2n)	Chromosome Number in Gamete (n)
1.	Apple	34	17
2.	Maize	20	10
3.	Onion	32	16
4.	Potato	48	24
5.	Rice	24	12
6.	Cat	38	19
7.	Dog	78	39
8.	Human beings	46	23
9.	House fly	12	06

(1) Gamete Transfer : After formation, the male and female gametes are brought together to facilitate fertilization. In a majority of organisms, male gamete is motile and the female gamete is stationary. There is a need for a medium through which the male gametes move. In Algae, Bryophytes and preridophytes, water is the medium through which this gamete transfer takes place. A large numger of the male gametes, however, fail to reach the female gametes. To compensate this loss of male gametes during transport, the number of male gametes produced is several thousand times the number of female gametes produced.

In Angiosperms pollen grains are the carrier of male gametes and ovule has the egg cell. Pollen grains are produced in anthers and are transferred to stigma, a phenomenon which is known as pollination. This

phenomenon requires the involvement of external agents such as insects, animals, wind and water. Pollen grains germinate on the stigma and the pollen tubes carrying the male gametes reach the ovule and discharge two male gametes near the egg cell.

- Fertilization : The fusion of two similar or dissimilar gametes is called syngamy and in its result diploid zygote is formed. This process in known as fertilization.

In majority of **algae, fishes** and **amphibians,** syngamy occurs in the external medium i.e. water (outside the body of the organism).This type of gametic fusion is called external fertilization. This happens in the bonyfishes and frogs where a large number of offspring are produced. A major disadvantage is that the offspring are extremely vulnerable to predators threatening their survival upto adult hood.

In plant groups (i.e. Fungi, Bryophytes, Pteridophytes) as well as Reptiles, Birds and Mammals, syngamy occurs inside the body of the organism, hence the process is called internal fertilization. In this process, male gametes are motile and have to reach and fuse with egg. this takes place inside the female body.

- **Post-fertilization Events :** The formation of zygote and the process of development of embryo (embryogenesis) are called post-fertilization events.

(1) Zygote : Formation of zygote (2n) is common in all sexually reproducing organisms. In organism with external fertilization, zygote is formed in the external medium (water), whereas in those exhibiting internal fertiliation, zygote is formed inside the body of organism. Further development of zygote depends on the type of life cycle the organism possesses and the environment to which it is exposed. In organisms, such as algae and Fungi, zygote develops a thick wall that is resistant to desiccation and damage commonly it indergoes a period of rest prior to germination.

Some unicellular protist animals (e.g. Paramoecium) exhibits sexual reproduction by forming male and female gamete nuclei. Which they exchange through temporary cytoplasmic bridge, later the cytoplasmic bridge disappears and the gametes nuclears of one individual fuses with that of the other to form zygote nucleus. This mode of sexual reproduction is known as conjugation.

Zygote is the vital link that ensures continuity of species between organisms of one generation and the next.

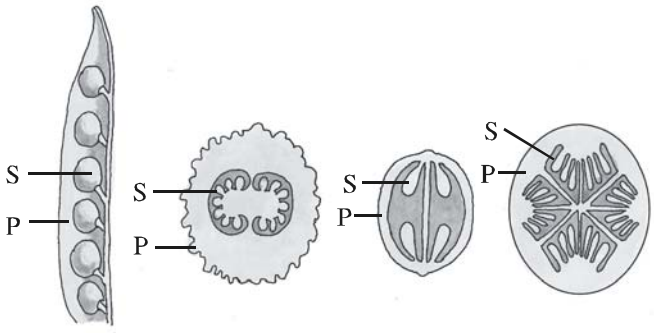
(2) **Embryogenesis :** Embryogenesis is the process of development of embryo from the zygote. During embryogenesis. Zygote undergoes cell division (mitosis) and cell differentiation.

Cell division increases the number of cells in the devleoping embroys while cell differentiation helps group of cells to under go certain modifications to form specialized tissues and organs to form an organism.

In animals, when the devleopment of zygote takes place outside the body of the female parent, it is called oviparous, while when it develops inside then it is called viviparous.

In oviparous animals like Reptile and Birds, the fertilized eggs are covered by hard calcareous shell are laid in a safe place in the environment, after a period of incubation, young ones hatch out. On the other hand, in viviparous animals like Mammals including human beings the zygote develops into a young one inside the body of the female organism. After attaining a certain stage of growth, the young ones are delivered out of the body of the female parent. Because of proper embryonic care and protection, the chances of survival of young ones is greater in viviparous organisms.

In Angiosperms, the zygote is formed inside the ovule. After fertilization the sepals, petals and stamens of the flower fall off. The pistil however, remains attached to the plant, The zygote develops into the embryo and the ovules develops into the seed. The ovary develops into the fruit which develops a thick wall called pericarp that is protective in function. After dispersal seeds germinates under favourable condition to produce new plants.



P = Pericarp S = Seed

(84) In between which organisms sexual reproduction occur ?

(A) Organisms of same species

(B) Organisms of two different species.

(C) Same species / different species organism

(D) Two independent organisms.

(85) How is zygote formed in sexual reproduction ?

(A) After formation of male and female gametes and their union.

(B) Due to formation of male and female gametes

(C) Distraction of male and female gametes

(D) Union of male, female and somatic cells.

(86) How is sexual reproduction compares to asexual reproduction ?

(A) Simple and speedy (B) Simple and slow

(C) Complex and speedy (D) complete and slow

(87) Which type of offspring obtained at the end of sexual reproduction ?

(A) Similar to parent generation (B) Possess characteristics like parent generation.

(C) Totally different than parent generation (D) None of above.

(88) Plants, animals and fungi have difference in morphology, physiology but have similarity in .....

(A) Asexual reproduction (B) Embryology

(C) Sexual reproduction (D) Structure

(89) What is called juvenile phase ?

(A) After birth of organism upto growth, development and death time birth of organism

(B) Time period, after birth of organism showing growth development and maturity

(C) Time period, after birth of organism growth, development and obtaining aging.

(D) Time period after birth of organism growth, development and production of offspring

(90) Organism, who has obtain maturity, shows similarities and differences in .....

(A) Mature organisms show structurally similarity in sexual reproduction and difference in the structure produced during sexual reproduction.

(B) Mature organisms, physiologically similar in asexual reproduction and similar in structures produced during sexual reproduction.

(C) After attaining maturity, organisms do not show similar events and processes during sexual reproduction.

(D) Organisms who has not obtained maturity, structurally show similarity in sexual reproduction and disimilarity in structure formed during sexual reproduction.

(91) How is the structure of gametes ?

(A) Haploid, unicelleular, uninucleated
(B) Haploid, unicellular, coenocytic
(C) Diploid, unicellular, uninucleated.
(D) Haploid, multicellular, uninucleated.

(92) What are isogametes ?

(A) Gametes which show difference in morphology, physiology, Flagellated and motile
(B) Gametes which show similarity in physiology and morphology, flagellated and motile.
(C) Gametes which show difference in physiology, morphology, unflagellated and nonmotile.
(D) Gametes whaich show similarity in morphology and physiology, unflagellated and non-motile.

(93) Which of the following algae is isogametic ?

(A) Chlamydomonas, chlorella
(B) Volvox chalamydomonas
(C) Cladophora, ulothrix
(D) Spirogyra, Anabaena

(94) In 'aplanospore (Heterogametes), which type of difference is observed ?

(A) Morphology and physiology
(B) Working pattern and physiology
(C) Embryology and anatomy
(D) All of above

(95) What is characteristics of male gametes ?

(A) Large active, unicellular uninucleated.
(B) Small active, unicellular, haploid, uninucleated.
(C) Small, active, haploid, unicellular, uninucleatd.
(D) Small, active, diploid, unicellular, coenocytic

(96) What is pecularity of female gamete ?

(A) Haploid, unicellular, uninucleated, small, nonmotile.
(B) Haploid, uninucleated, unicellular, large non-motile.
(C) Diploid, uninucleated, unicellular, large, non motile
(D) Haploid, unicellular, uninucleated, large, motile.

(97) In pteridophyta, Gymnosperm and angiosperm, generally parent plant body is - type

(A) n (B) 2n (C) 3n (D) 4n

(98) Select true option based on number of chromosomes in gametes of cat, Dog, Man and housefly and in meiotic cell of Apple, maize, onion, potato and rice.

(A) 17, 10, 24, 16, 12, 19, 39, 23, 06
(B) 34, 20, 32, 48, 24, 19, 39, 23, 06
(C) 34, 20, 32, 48, 24, 38, 78, 46, 12
(D) 17, 10, 16, 24, 12, 38, 76, 46, 12

(99) Why male gametes are more in number and female gametes less in number ?

(A) To show fertilization process speedy
(B) To show speedy transport of male gametes.
(C) Gametes are destroyed fertilization to prove it.
(D) Male gametes are wasted in transport of male gamete to suppot it.

(100) By whom sequentially male gametes and female gametes are formed ?

(A) Anther megasporophyll
(B) Anther and carpel.
(C) Pollen and ovule
(D) Pollen sac and megasporophyll.

(101) At the time of pollination, pollen grain is released and transferred to which place ?

(A) It is released from Anther and transferred to carpel

(B) Released from pollen and transfered to carpel.

(C) Released from pollen sac and transfered to ovule

(D) Released from pollen sac and transfered to stigma

(102) Which option show proper sequence of living and non-living carriers for pollen grain ?

(A) Wind, water, insects, animals

(B) Insects, wind, Animals, water

(C) Animals, wind, Insects, water

(D) Animals, water, wind, insects.

(103) By which gamete conjugation occur ?

(A) Due to conjugation of two similar gamete

(B) Conjugation of two disimilar gametes

(C) Two similar and disimilar conjugation of gametes

(D) None.

(104) Which animal group show External fertilizaion ?

(A) Algae, Fungi, arthropoda

(B) Algae, Fish, Birds

(C) Algae, Fish, Amphibia

(D) Fungi, Fish, amphibias

(105) Which animal group show External fertilization ?

(A) Algae, Fungi, lichen

(B) Algae, Fungi, Bryophyte

(C) Algae, Bryophyta, pteridophyta

(D) Pteridophyta, Gymnosperm, Angiosperm

(106) Which animal group show internal fertilization ?

(A) Fish, Amphibia, Reptilia,

(B) Amphibia, Reptiliam Birds

(C) Reptilia, Birds, mammals

(D) Mollusca, Echinodermata, Fish

(107) In seed bearing plants which type of male gametes are seen ? How are they transported ?

(A) non motile, pollen tube

(B) Motile, Pollentube

(C) Flagellated, pollen tube

(D) Ciliated, pollentube

(108) Which are proper sequential stages of sexual reproduction ?

(A) Pre Fertilization, Fertilization, post Fertilization.

(B) Fertilization, pre fertilization, post Fertilization

(C) Post fertilization, fertilization, pre fertilization

(D) Pre fertilization, post. fertilization, post fertilization

(109) Which are stages of post fertilization ?

(A) Gamatogenesis, Embryogenesis

(B) Transport of gametes, zygote formation

(C) Zygote formation, Embryogenesis

(D) Embryogenesis, Transport of gametes

(110) Which sentence is improper for sexual reproduction by conjugation ?

(A) Micronuclei is responsible for sexual reproduction in paramoecium.

(B) During conjuction, temporarily conjugation tube is formed, male and female gamete transfer take place through cytoplasmic bridge.

(C) Eukaryota - porifera

(D) Akaryota - virus

(111) Which sentence is improper for sexual reproduction in paramoecium.

(A) Micronuclei is responsible for sexual reproduction in paramoecium.

(B) During conjuction, temporarily conjugation tube is formed, male and female gamete transfer take place through cytoplasmic bridge.

(C) After exchange of nucleus, cytoplasmic bridge disappear and conjugation tube disintegrates.

(D) In paramoecium, always transfer of two similar type nucleus through cytoplasmic bridge and conjugation tube occurs.

(112) During embryogenesis, zygote passes through which processes and converted into multicellular form from unicellular form ?

(A) Cell formation, cell enlargement (B) Mitosis, cell enlargement

(C) Meiosis, cell differentiation (D) Mitosis, cell differentiation

(113) What is called when zygotic development occurs inside female body and outside female body sequentially ?

(A) Gamopary, oviparous (B) Oviparous, viviparous

(C) Viviparous, oviparous (D) None.

(114) Due to development of zygote, embryosac and wall of embryosac which structures formed sequentially ?

(A) Seed, Fruit, seed coat (B) Seed, fruit, Fruit coat (Epicarp)

(C) Fruit, seed, seed coat (D) Fruit seed, epicarp

(115) Teacher wants to explain various stages of sexual reproduction in higher organism so which are points included in which sequence ?

(A) Gametogenesis, Transfer of gametes, fertilization zygote formation, Embryogenesis

(B) Gametogenesis, Transfer of gametes zygote formation, Embryogenesis, Fertilization

(C) Gametogenesis, Embryogenesis, Transfer of gametes, zygote formation, fertilization

(D) Transfer of gamete, zygote formation, gametogenesis, fertilization.

Answers : (84-C), (85-A), (86-D), (87-D), (88-C), (89-B), (90-A), (91-A), (92-B), (93-C), (94-D), (95-C), (96- B), (97-B), (98-B), (99-D), (100-C), (101-D), (102-A), (103-D), (104-C), (105-C), (106-C), (107-A), (108-A), (109-C), (110-B), (111-D), (112-D), (113-B), (114-B), (115-A)

- **A - Statement R - Reason A - R type questions Answer of following question, select from following option.**

(A) A and R both true, R is explanation of A

(B) A and R both true, R is not explanation of A

(C) A is true R is false

(D) A is false R is true.

(116) Statement A : Gamete formation occur from independent, diploid or haploid parent, Gametes are always haploid.

Reason R : Gamete formation occur from diploid by mitosis and meiosis process, where as in haploid parent show only meiosis

(A) (B) (C) (D)

(117) Statement A : Grafting method is important in plants where roots are not formed easily

Reason R : In Jasmine and Hibiscus grafting method is important.

(A) (B) (C) (D)

(118) Statement A : Gametogenesis and transfer of gametes are main Prefertilization events.

Reason R : Isogametes and heterogametes formed during gametogenesis

(A) (B) (C) (D)

(119) Statement A : Reptiles and Aves are oviparous

Reason R : Zygote develops in the body of reptiles

(A) (B) (C) (D)

(120) Statement A : Vegetative reproduction is best for maintainance of desirable characters of parents into offspring.

Reason R : Vegetative reproduction removes infection of parent plant into offsprings.

(A) (B) (C) (D)

(121) Statement A : True spores are formed by sporulation

Reason R : Nephrolepis plants have isogamete

(A) (B) (C) (D)

(122) Statement A : Offsprings produced by sexual reproduction is copy of parent.

Reason R : Offsprings formed by asexual reproduction differs from parent.

(A) (B) (C) (D)

(123) Statement A : Motile gametes fromed from ulothrix, develops as independent organism

Reason R : Bud formation by protosiphon is due to mitosis

(A) (B) (C) (D)

(124) Statement A : In sexual reproduction of paramoecium exchage of gametes take place by cyto plasmic bridge.

Reason R : Majority of organism in sexual reproduction produces heterogametes.

(A) (B) (C) (D)

(125) Statement A : Gametes of cladophora are similar, motile by morphology and physiology

Reason R : Majority of organism in sexual reproduction produces heterogametes.

(A) (B) (C) (D)

(126) Statement A : In vorticella body of individual, cytoplasm divides longitudinally.

Reason R : New organism formed by longitudinal division in Euglena.

(A) (B) (C) (D)

(127) Statement A : When condition become favourable in encysted hydra nucleus show multiple fission

Reason R : In spongilla cytoplasm divides transversely

(A) (B) (C) (D)

(128) Statement A : Gamete formation take place by individual haploid or diploid parent, Gamete are always haploid.

Reason R : Gametes formation occurs in diploid parent by mitosis and meiosis, whereas in haploid parent only by meiosis

(A) (B) (C) (D)

(129) Statement A : Gametes are always haploid.

Reason R : Gametes are always unicellular

(A) (B) (C) (D)

Answers : (116-C), (117-C), (118-A), (119-C), (120-A), (121-B), (122-D), (123-C), (124-C), (125-B), (126-A), (127-D), (128-C), (129-B)

(130)

Column I	**ColumnII**	**Column III**	
(1) Maize	(p) 48	(a) 10	(A) : (1 - s - a), (2 - q - c), (3 - p - d), (4 - r - b)
(2) onion	(q) 32	(b) 12	(B) : (1 - p - a), (2 - q - b), (3 - r - c), (4 - s - d)
(3) Potate	(r) 24	(c) 16	(C) : (1 - r - a), (2 - q - b), (3 - s - c), (4 - p - d)
(4) Peddy	(s) 20	(d) 24	(D) : (1 - r - d), (2 - S - c), (3 - p - b), (4 - q - a)

(131) **Column I** **Column II**

(1) Protosiphon	(p) non-flagellated spores	(A) : (1 - q), (2 - r), (3 - s), (4 -p)
(2) Sapro legnia	(q) Bud	(B) : (1 - r), (2 - s), (3 - p), (4 - q)
(3) Chlamydomonas	(r) Fragmentation	(C) : (1 - q), (2 - r), (3 - p), (4 - s)
(4) Aspergillus	(s) Motile spores.	(D) : (1 - s), (2 - p), (3 - q), (4 - r)

(132) **Column I (Stage)** **Column II (Phenomenon)**

(1) Juvenile	(p) Formation of unicellular fertile cells	(A) : (1 - q), (2 - p), (3 - r), (4 - s)
(2) Gametogenesis	(q) period to reach reproductive stage	(B) : (1 - q), (2 - p), (3 - s), (4 - r)
(3) Transfer of gametes	(r) By fusion of haploid cells period of obtaining diploid stage	(C) : (1 - r), (2 - s), (3 - p), (4 - q)
(4) Fertilization	(s) Motile Foreign cells reach non-motile stage	(D) : (1 - s), (2 - r), (3 - q), (4 - p)

(133) **Column I (Stage)** **Column - II Phenomenon**

(a) True spores	(w) Through spores	(A) : (a - y), (b - z), (c - x), (d - w)
(b) Nephrolepis	(x) Heterosporic plants	(B) : (a - x), (b - w), (c - z), (d - y)
(c) Selaginella	(y) Produced by sporogenesis	(C) : (a - x), (b - z), (c - y), (d - w)
(d) Asexual reproduction	(z) Isosporic plant	(D) : (a - w), (b - z), (c - x), (d - y)

(134) **Column - I** **Column - II**

(a) Mint	(w) Vegetative reproduction by sucker	(A) : (a - x), (b - w), (c - y), (d - z)
(b) Pistia	(x) Vegetative reproduction by runner	(B) : (a - y, (b - w), (c - z), (d - x)
(c) Hydrocotyl	(y) Vegetative reproduction by stolon	(C) : (a - w), (b - z), (c - x), (d - y)
(d) Nephrolepis	(z) Vegetative reproduction by offset	(D) : (a - x), (b - y), (c - w), (d - z)

(135) **Column - I** **Column - II**

(a) Reptiles and mammals	(w) Post fertilization	(A) : (a - z), (b - x), (c - y), (d - w)
(b) Embryogenesis	(x) Development of body out side	(B) : (a - y, (b - w), (c - x), (d - z)
(c) Paramoecium	(y) Internal fertilization	(C) : (a - y), (b - w), (c - z), (d - x)
(d) Oviparous	(z) Sexual reproduction by conjugation	(D) : (a - w), (b - x), (c - z), (d - y)

(136) **Column - I** **Column - II**

(1) Jasmine	(p) Suckers	(A) : (1 - u), (2 - t), (3 - s), (4 - q) (5 - p), (6 - r)
(2) Chrysanthemum	(q) Stolon	(B) : (1 - t), (2 - u), (3 - s), (4 - q) (5 - p), (6 - r)
(3) Bryophyllum	(r) Floral bud	(C) : (1 - t), (2 - u), (3 - s), (4 - p) (5 - q), (6 - r)
(4) Nephrolepis	(s) Adventitious bud	(D) : (1 - t), (2 - u), (3 - s), (4 - r) (5 - q), (6 - p)
(5) Mint	(t) Cutting	
(6) Oxalis	(u) Layering	

(137) **Column - I** **Column - II**

(P) Asparagus	(i) Vegetative propogation by floral bud	(A) : (P - i), (Q - ii), (R - iii), (S - iv)
(Q) Turmeric	(ii) Vegetative reproduction by leaf	(B) : (P - ii), (Q - iii), (R - iv), (S - i)
(R) Bryophyllum	(iii) Vegetative reproduction by rhizome.	(C) : ((P - iv), (Q - iii), (R - ii), (S - i)
(S) Agave	(iv) Vegetative reproduction by root	(D) : (P - iii), (Q - iv), (R - ii), (S - i)

(138) **Column - I** **Column - II**

(a) Algae and Bryophyta	(w) Plant body is diploids	(A) : (a - z), (b - y), (c - x), (d - w)
(b) Pollengrain	(x) Transports ovule	(B) : (a - x), (b - z), (c - w), (d - y)
(c) Ovule	(y) Transports gametes	(C) : (a - y), (b - z), (c - x), (d - w)
(d) Pteridophytes	(z) Plant body is haploid angiosperms	(D) : (a - z), (b - w), (c - y), (d - x)

Answers : (130-A), (131-A), (132-B), (133-A), (134-C), (135-C), (136-A), (137-C), (138-A)

(139) What is indicated by S and P in the Fig 1

(A) Pericarp and fruit (B) Epicarp and seed

(C) Fruit and seed (D) Seed and pericarp

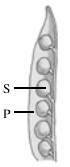


Fig 1

(140) What is indicated by 'a' in the given Fig 2

(A) Nucleus (B) Conidia

(C) Zoospores (D) Chlorophyll

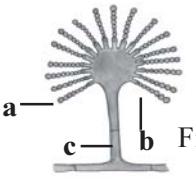


Fig 2

(141) What is indicated by 'a' in the Fig 3 ?

(A) Chlorophyll (B) Pyrenoid

(C) Tentacles (D) Nucleus

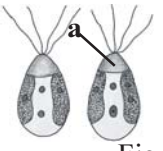


Fig 3

(142) What is indicated by 'a' part surrounded by spicules in Fig 4

(A) Exogenous Bud (B) Nucleus

(C) Endogenous Bud (D) Cell body

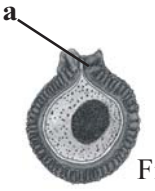


Fig 4

(143) In the given Fig 5 which is correct option for X and Y.

(A) X - Conidiophore Y - Conidia (B) X - Sterigmata Y - Conidia

(C) X - Conidia, Y - Sterigmata (D) X - mycellium, Y - Hyphae

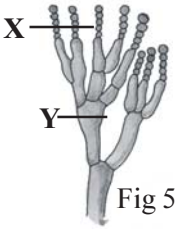


Fig 5

(144) Which is proper option for given Fig 6 ?

(A) Tongue grafting (B) Cleft grafting

(C) Bud grafting (D) Whip grafting


Fig 6

(145) What is function of a in the given Fig 7 ?

(A) Buds forms new organisms

(B) Shows fission in Bacteria

(C) Spore Formation

(D) adventitious branch is formed.

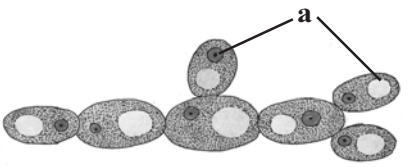


Fig 7

(146) Explain imp of b in the Fig 8.

A Photo synthesis B Vegetative reproduction

C Food storage D Climbing


Fig 8

(147) What is indicated in the given Fig 9 ?

A Vegetative reproduction by root

B Vegetative reproduction by stem

C Vegetative reproduction by leaf

D Vegetative reproduction by flower


Fig 9

Answers : (139-D), (140-D), (141-B), (142-C), (143-C), (144-B), (145-A), (146-B), (147-C)

- **Questions For NEET**

(148) Chara is monoecious plant.

(A) Anther and carpel on the same plant

(B) On the same plant male sporophyll on upper part and female sporophyll on lower part

(C) On the same plant female sporophyll on upper part and male sporophyll on lower part

(D) Both male and female sporophyll on the same plant.

(149) **Select false sentence from the following**

(A) Heterogametes are different structurally functionally or behaviourly.

(B) In Heterogametes female gamete is small and motile, female gamete is large and non-motile

(C) Chlamydomonas shows isogametism and heterogametism. Fucus hetero gametism.

(D) Iso gametes similar, structurally, functionally and behaviour point.

(150) Presence of which is required in grafting between stock and scion ?

(A) Xylem (B) Phloem (C) Meristematic tissue (D) tissue.

(151) In monocot plant grafting is impossible because .....

(A) Scattered vascular bundle (B) Lack of meristematic tissue

(C) Colateral, open vascular bundle (D) Radial vascular bundle

(152) Before it is separated from maternal plant, root formation induced on the plant

(A) Grafting (B) layering (C) Stolon (D) Rhizome

(153) If branch of one plant is taken as scion, which is sweet and stock on which it is grafted is juicy. So such plants by grafting gives which types of fruits ?

(A) Sweet and fibrous (B) Sweet and juicy

(C) Sour and juicy (D) Sour and fibrous

(154) It stock has 58 chromosome and scion possess 30 chromosome, plant produced as a result has how many chromosome in root and egg cell ?

(A) 30 and 29 (B) 15 and 58 (C) 58 and 15 (D) 29 and 30

(155) Isogametic, non-flagellated gametes are seen ?

(A) Spirogyra (B) Volvox (C) Fucus (D) Chlamydomonas

(156) What is called menthod by which plants prepared in large number ?

(A) Cutting (B) Organ formation (C) Micro propogation (D) Macro propogation.

(157) What is called study of anther ?

(A) Ethamology (B) Paleontology (C) Paleo-Botany (D) Omega - tatonomy

(158) In sporophytic stage of plants, where maturation is seen ?

(A) Gemma (B) Primary structure (C) Sporophyll (D) Egg cell

(159) During regeneration formation of one organ from another is called .....

(A) Morphogenesis (B) Epimorphosis

(C) Morpholaxis (D) Specialized cell growth

(160) In mint, vegetative reproduction is by .....

(A) Runner (B) Rhizome (C) Suckers (D) Stolon

(161) What is similarity in vegetative ropreduction and apomixis ?

(A) Both can be included in dicot plants.

(B) Both passes through flowering stage

(C) Both occur any time in the year

(D) Both produces offspring similar to parent plant

(162) If 2n female plant and 4n male plants are fertilized, what is poidy of endosperm ?

(A) 4n (B) 3n (C) 2n (D) 5n

(163) For 100 zygote / 100 seed formation of wheat, necessary meiosis number is .....

(A) 100 (B) 75 (C) 125 (D) 50

Answers : (148-C), (149-B), (150-C), (151-B), (152-B), (153-B), (154-C), (155-A), (156-C), (157-B), (158-C), (159-B), (160-C), (161-D), (162-A), (163-C)

Reproduction in Organisms

Asexual reproduction

- It is the production of offspring by and single parent wihtout the formation and fusion of gametes

Types

- Binary fission
- Budding
- Strobilisation
- Fragmentation
- Gemmae
- Regeneration
- Spore formation

Natural vegetative reproduction

- Examples are root tubers, suckers, rhizomes, corm, bulbs, runners, stolons, offsets, etc.

Vegetative reproduction

- Formation of new plants from vegetative units such as buds, tubers, rhizomes, etc.

Artificial vegetatitive reproduction

- Examples are cutting, layering, grafting, micropropagation, etc.

Sexual reproduction

- it is the process of development of new individual through formation and fusion male and female gametes.
- It is grouped into three distinct stages

Pre-fertilization

- This includes gametogenesis (process of formation of gametes) and gamete transfer.

Fertilization

- It is complete and permanent fusion of two gametes to form a diploid zygote.

Internal fertilizations

Enternal fertilizations

Post-fertilization

- It is described under 2 headings :

Zygote

- It is formed after fertilization in all sexually reproducing organisms.

Embryorgenesis

- It is the process of development of embryo from zygote.

34 Sexual Reproduction in Flowering Plants

Reproduction is the most important characteristic of all living beings. The mode of reproduction varies but is mainly of two types : (1) Sexual reproduction and (2) Asexual reproduction. In higher plants flower is sexual reproductive organ. Thus the reproduction takes place by various parts of flower.

In angiosperm plants the main sexual reproductive organs are stamen and pistil respectively. moreover accessory organs like sepals and petals are also present.

These types of main and accessory reproductive organs arranged in circular form so suffix word they are suffixed with the word 'whorl' i.e. Gynoecium, androecium, calyx and corolla etc.

(1) Which are the reproductiove methods in lower level organisms ?

(A) Budding (B) Celldivision

(C) Developed reproductive - organs (D) A and B both

(2) Which are the reproductive methods in higher level organisms ?

(A) Budding (B) Cell Division

(C) Developed reproductive organs (D) A and B both

(3) In angiosperms which type of reproduction is take place sometimes ?

(A) Multiple fission (B) Polyembryony (C) Apomixis (D) B and C both

(4) Which are the Lateral organs in stunted shoots ?

(A) Carpel, stamen (B) Calyx and corolla (C) Thalamus (D) Petiole

(5) Which are the sterile appendages of typical flower ?

(A) Calyx (B) Corolla (C) Gynoecium (D) A and B both

(6) In typical flower megasporophyll and microsporophyll together and form respectively .....

(A) Gynoecium, Androecium (B) Androecium, Gynoecium

(C) Corolla, Calyx (D) Calyx, Corolla

(7) Which part of stamen is sterile ?

(A) Filament (B) Connective (C) Anther (D) A and B both

Answers : (1-D), (2-C), (3-D), (4-D), (5-D), (6-A), (7-D)

- Pre Fertilization : The differentiation and further development of the floral primordium to a flower is due to several hormonal and structural changes. In the flower, the androecium (representing the male reproductive organs) and the gynoccium (representing the female reproductive organ) differentiate and develop.

- **Stamen, microsprorangium and pollen grain :** A stamen is known as microsporophyll. It is regarded as the male reproductive part of flower. A stamen consists of a filament, connective and an anther. The proximal end of the filament is attached to the thalamus of the flower. The numbers and length of stamens are variable in flowers of different species.

An anther is bilobed and tetrathecoused which are known as microsporangia, which is filled with the pollen grains, which consisting different stages in it. Microsporangia arranged in the distiguished layers which are made up by cells.

From these layers pollen grain (microspores) are produced by the process of microsporogenesis and bydehiscence anther are released from the pollen grains.

Pollen grains are produced in different sizes and clusters, which have two layers. Exine and Intine There is a germ pore in the exine through which the germ tube comes out - which is collect the pollen tube. It carries 2 make gametes upto the ovary.

(8) Variety of stamens, in different species of plants, is seen in

(A) Length (B) Number (C) Sterility (D) A and B both

(9) Microsporagia when developed get converted to ..... .

(A) Pollengrain (B) Stamen (C) Pollensac (D) A and B both

(10) Name the fibrous layer of microsporania ?

(A) Endothecium (B) Endomyocium (C) Myocium (D) Myothecium

(11) For the dehiscence of anther which layer is helpful ?

(A) Middle layer (B) Fibrous layer (C) Epidermis (D) Tapetum

(12) Function of the Tapetum of anther is, ...

(A) Gives nutrition to develop anther (B) Gives nutrition to develop pollen grain.

(C) Gives nutrition to develop epidermis (D) Gives nutrition to develop pollen in other cells.

(13) The sporogenous tissue undergoes meiotic divisions to form microspore tetrad. Each one has .....

(A) Potential to convert into active pollen grain.

(B) Only one has potential to convert into PMC (Pollen mother cell).

(C) First meiotic division occurs then it is converted into pollen mother cells.

(D) First mitotic divison occurs then it is converted into pollen mother cell.

(14) What is pollen tetrad ?

(A) In microsporogenesis a group of four cells arise by the meiotic divisions occur in pollen mother cell.

(B) In microsporogenesis a group of four cell arising by mitotic division from pollen mother cell.

(C) Arrangement of microspores

(D) Four cells group in anther.

(15) Which of the following represents the male gametophyte stage ?

(A) Pollen grain (B) Ovule (C) Pollen tetrad (D) Pollen mother cell

(16) What is the function of exine of pollen ?

(A) Give protection against temperature.

(B) Give protection against concentrated acid and base.

(C) Made of resistant organic substance.

(D) All the above

(17) Name the spot in exime where sporopollelin is absent ?

(A) Embryonic, micropylar (B) Micropyle. (C) Germpore (D) Seed micropyle

(18) Pollen grains are stored as a fossils, because ..... .

(A) Its intine are thickened by pectin and cellulose (B) Its wall are made up of two layers

(C) Sporopollnin is present. (D) It has Germpore.

(19) Intine of pollen is thickened by which substance ?

(A) Pectin (B) Cellulose (C) Sporopollinin (D) A and B both

(20) From which stage development of male gametophyte starts ?

(A) When pollen is presented with in anther. (B) When pollen grain becomes bilayered.

(C) When anther dehisces can occur. (D) When anther get matures.

(21) When initine of pollen come out form the germ pore it is called ..... .

(A) Style (B) Pollen tube (C) Pollengrain tube (D) A and B both

(22) Formation of male gametes occurs in pollen tube by ..... .

(A) Mitotic division of vegetative cell

(B) By the mitotic divison of tube cell.

(C) By the mitotic division of germinative cell.

(D) By the process of degradation of tube cell.

(23) When pollen tube develops in style, its terminal region will contain ?

(A) Vegetative cell (B) Germ cell (C) Pollenmother cell (D) None of these

(24) Which part of the pollen grain possesses huge storage nucleus.

(A) Vegetative cell (B) Germinative cell

(C) Pollen mother cell (D) None of these

Answers : (8-D), (9-C), (10-A), (11-B), (12-B), (13-A), (14-A), (15-A), (16-D), (17-C), (18-C), (19-D), (20-A), (21-B), (22-C), (23-A), (24-A)

- A megasporophyll is the femal reproductive part containing stigma, style and ovary. Inside the ovary is the ovarian cavity or locule. The placenta is located inside the ovarian cavity. From the placenta arise megasporangia which are called ovules. The ovule is attached to the placenta by funicle. Funicle and ovule meet at hilum. Ovule has abandant reserve food material. The nucellus is enclosed in integuements. A small opening at the apex of integuements is the micropyle. Opposite the micropyle is the chalazal end. A large oval cell in the nucellus towards the micropylar end is the embryosac or female gametophyte.

  The formation of megaspores from the megaspore motehr cell (MMC) is megasporogenesis. It divides meiotically to form 4 haploid megaspores. Of these, 3 degenerate and only 1 becomes functional to form the female gametophyte. This is called monosporic development.

(25) Which plant is monocarpellary ?

(A) Papaya (B) Maize (C) Hibiscus (D) Pea

(26) Identify parts of carpel

(A) Stigme, style (B) ovary (C) A and B both (D) Pollen tube

(27) Which part of the carpel accepts pollen ?

(A) Style (B) Pollen tube (C) Stigma (D) Ovary

(28) Name the plant in which number of ovule is one in ovary ..... .

(A) Wheat (B) Mango (C) Papaya (D) A and B both

(29) By which is structure ovule is attached with placenta ?

(A) Chorionic duct (B) Funicle (C) Oviduct (D) Embryonic duct.

(30) Region where ovule is attached with funicle ..... .

(A) Egg nucleus (B) Funicle nucleus

(C) Hilum (D) Embryonic nucleus

(31) Point of attachment between ovule and funicle ?

(A) Egg nucleus (B) Funicle nucleus (C) Hilum (D) Embryonic nucleus

(32) Opposite to micropylar end chalazal end is, it is also known as ..... .

(A) Hilum (B) Planetal part (C) Antipolar (D) None of these.

(33) In nucelus, at the micropylar end, oval shaped cell is present it is known as ..... .

(A) Embryossac (B) Female gametophyta (C) Storage cells (D) A and B both

(34) Megasporogenesis process starts from which cell ?

(A) Megaspore mother cell (B) Egg cell (C) Egg - apparatus (D) Synergid cell

(35) How is differentiation occur in ovule ?

(A) From the synergid cell which lies at micropylar end.

(B) From the megaspore mother cell which lies at micropylar end.

(C) From the eggs which lies at micropylar end.

(D) From the antipolar cell which lies at micropylar end

(36) What is monosporic development ?

(A) Process of formation of embryo sac from only one megaspore.

(B) Formation of embryo sac from four megaspores.

(C) Four single cell megaspore development

(D) Three antipodar cell and one egg cell

(37) Which cells are present at micropylar end ?

(A) One egg cell and two synergid cells (B) Two egg cells and one syner gid cell

(C) Three antipodal cells (D) One antipodal cell and one egg cell.

(38) The nuclei situated at chalazal end it known as .....

(A) Egg cell (B) Synergid cell (C) Antipodal cell (D) Secondary nucleus

(39) Where is the embryosac present in the secondary nucleus ?

(A) Chalazal End (B) Micropylar end (C) Middle area (D) None of there

(40) What is discovery of Panchanan and Maheshwariji ?

(A) How many numbers of megaspore nuclci participate in female gametophyte ?

(B) How many cells are found in female gametophyte ?

(C) How many numbers of cells are present in female gametophyte ?

(D) Given the development of female gametophyte.

Answers : (25-D),(26-C), (27-C), (28-D), (29-B), (30-C), (31-C), (32-A), (33-D), (34-A), (35-B), (36-A), (37-A), (38-C), (39-C), (40-A)

- **Pollination and its types :**

  The process of transfer of pollen released from the anther to the stigma of a carpel is called pollination.

- There are two types / kinds (1) Self pollination and (2) Cross pollination
- Self pollination is of two types : Autogamy and Geitonogamy

  Thus, for the self pollination, plants have certain types of adaptations like. Homogamy, cleistogamy etc.

- Cross pollination also occurs in certain plants and for the cross pollination Dichogamy, self sterility, Heterostyly and herkogamy like adaptation contrivances can be seen.

(41) What it is called when both male and female flowers are borne on single plant ?

(A) Dioecious (B) Monoecious (C) Apiaceae (D) Lamiaceae

(42) In which members of the given family self fertilization occur ?

(A) Apiaceae (B) Lami aceae (C) Cactaceae (D) All above

(43) In the process of self fertilization How Lamiaceae does stigma come near the anther ?

(A) Rapid growth occurs in style (B) Visible movement of style

(C) Style develop upto the stigma (D) Stigma of style large and flat.

(44) In the process of pollination pollen grain is deposited on other flower that phemomenon is known as a .....

(A) Allogamy (B) Xenogamy (C) Geitonogamy (D) Cleistogamy.

(45) When anther and stigma mature at the same time it is known as .....

(A) Cleistogamy (B) Dichogamy (C) Self sterility (D) Homogamy

(46) In which plant Homogamy is found ?

(A) Viola (B) Catharanthus roseus (C) Oxalis (D) Commelina.

(47) What is cleistogamy ?

(A) Flowers never open (B) Flowers open after long time of generation

(C) Flowers open immediaely (D) Flowers open after time of some generation

(48) In which type of flower self fertilization occurs in closed buds ?

(A) Frequent flower (B) Cleistogamous flower (C) Allogamy (D) Xenogamy

(49) In which plant chasmogmous flowers are bright coloured and attractive ?

(A) Violat (B) Oxalis (C) Commelina (D) Catharantus

(50) What is the synonym for cross pollination ?

(A) Geitonogamy (B) Allogamy (C) Cleistogamy (D) Xenogamy

(51) In bisexual flowers, maturation occurs in stamen and carpel at different time it is known as ..... .

(A) Cleistogamy (B) Self sterility (C) Heterostyle (D) Herkogamy

(52) When Pollen reaches upto stigma even though fertilization does not take place, it called ..... .

(A) Cleistogamy (B) Self sterility (C) Heterostyly (D) Herkogamy

(53) What type of adaptation is it when styles have different length in flowers ?

(A) Cleistogamous (B) Self sterility (C) Heterostyly (D) Herkogamy

(54) In bisexual flowers physical barrier between anther and stigma is present it is known as ..... .

(A) Cleistogamy (B) Self sterility (C) Heterostyly (D) Herkogamy

Answers : (41-B), (42-D), (43-B), (44-C), (45-D), (46-B), (47-A), (48-B), (49-C), (50-B), (51-A), (52-B), (53- C), (54-D)

- Agents for pollonation : Various transporting agents are required for carrying pollen from one flower to stigma of a flower to another plant. Plants use abiotic (wind and water) and biotic (animals) agents to achieve pollination. Majority of plants use biotic agents for pollination. Only a small proportion of plants use abiotic agent. Each type of plants possess some distinct characters related to their pollinating agent. Anemophily - pollination by wind, Hydrophily - pollination by water,

zoophily - pollination by animals; Entomophily - Pollination by insects etc. According to these changes in characters in plants.

(55) Which type of vectors / pollinatory are more useful in most of plants ?
(A) Biotic (B) Abiotic (C) Wind (D) Water

(56) In anemophily pollen is ..... .
(A) Less amount of pollen is produced, no wastage is there.
(B) Pollen is produced in less amount if pollinated by insects
(C) Pollen is produced in large amount, more possibility of wastage
(D) Pollen is produced in large amount, but immature pollen are more innumber ..... .

(57) Which type of pollen are seen in anemophilous plant ?
(A) Small, dry (B) Smooth, light (C) Spiny, sticky (D) A and B both.

(58) In anemophilous plants arrangement of flowers is ..... .
(A) Male flowers are at a height and female flowers are dwarf.
(B) Male flowers are dwarf and female flower is at a height.
(C) Male and female flowers both are at the same height.
(D) Male and female flowers both are dwarf.

(59) How does stigma occur in anemophily ?
(A) Branched, feathery, (B) Hairy, sticky
(C) Towrds the top (D) A and B both

(60) Identify the characteristic of flower in which plant is pollen grains are produced in large amount ?
(A) Contain honey (B) Attractive (C) Smellless (D) All three

(61) What is hydrophily ?
(A) Water pollination (B) Insect pollination (C) Wind pollination (D) Animal pollination

(62) How may genera are pollinationted by water ?
(A) 10 (B) 20 (C) 30 (D) 40

(63) Which type of plants are mostly water pollinated ?
(A) Monocots (B) Dicots (C) Gymnosperms (D) Angiosperms.

(64) Zostera is a ..... .
(A) Vallisneria (B) Hydrilla (C) Marine weed (D) Algae

(65) In seaweed / Marine weed, which type of the female flowers are produced ?
(A) Deep in water (B) Free in water
(C) Floating on the water sucface (D) A and C both

(66) Which type of the pollen grains are present in hydrophilous plant ?
(A) Covered by cellulose (B) Covered by lignin
(C) Covered by mucilage (D) Covered by pectin.

(67) Which of the following are associated with zoophily ?
(A) Sunbird and humming bird (B) Bat, squirrel
(C) A and B both (D) None of these.

(68) Who pollinates Aloe vera and Carrisa like plants ?
(A) Sunbird (B) Humming bird (C) Snails (D) A and B both.

(69) Who pollinates kigelia plant ?
(A) Bat (B) Lemur (C) Gecko lizard (D) Garden lizard.

(70) Which is the most effective pollinator, from the following.

(A) Honey bees (B) Butterfly

(C) Wasp (D) Ants

(71) Which characters are attractive for insects involved in entemophily flowers ?

(A) Various arrangement (B) Specific shape, colour and odour

(C) Attractive inflorescese (D) All the above

(72) Amorphophalus and Yucca plants are respectively pollinated by which insects ?

(A) Fly, Honeybee (B) Honey bee, Butter fly

(C) Fly, moth (D) Moth, Fly.

Answers : (55-A), (56-C), (57-D), (58-A), (59-D), (60-C), (61-A), (62-C), (63-A), (64-C), (65-B), (66-C), (67-C), (68-D), (69-A), (70-A), (71-D), (72-C)

Outbreeding Devices : Many flowering plants produce bisexual flowers and pollen grains are likely to come in contact with the stigma of the same flowers. Continued self-pollination result in inbreeding depression. Flowering plants have developed many devices to discourage self-pollination and to encourage cross-pollination. Maturity of pollens and stigma at different times and dichogamy like devices can seen.

(73) Flowering plants have developed which devices to discourage self - pollination and encourage cross pollination ?

(A) Outbreeding devices (B) Autogamy

(C) Geitonogamy (D) A and B both

(74) In which plant, the stigma becomes receptive much before pollens are released ?

(A) Palms (B) Sunflower (C) Primula (D) Malva

(75) Which plant releases the pollen long time before the stigma becomes receptive ?

(A) Palms (B) Sunflower (C) Primula (D) Malva

(76) In which plant anthers and stigma located at different positions ?

(A) Palms (B) Sunflower (C) Primula (D) Malwa

(77) Prevention of pollen germination and pollen tube development in a carpel by the pollen of the same flower or other flowers of the same plant, is called as,

(A) Inbreeding (B) Geitonogamy (C) Self incompatibility (D) A and B both

(78) In which type of plants, self fertilization can be prevented, but not in geitanogamy ?

(A) Monoecious (B) Dioecious (C) Triocious (D) Tetraocious.

(79) Which types of plants are inhibition selffertilization and geitonogomy both ?

(A) Monoecious (B) Dioecious

(C) Triocioms (D) Tatraocious.

Answers : (73-D), (74-B), (75-A), (76-C), (77-C), (78-A), (79-B)

Pollen-pistil Interaction : Nature does not guarantee the transfer of the pollen of the same species on the stigma (compatible pollen.) Often pollen from other species or a self incompatible pollen from the same plant, can also land on the stigma.

- **Artificial Hybridization :** In artificial hybridization desired pollen grains are used for pollination and the stigma is protected from unwanted pollen. This is achieved by emasculation and bagging techniques for which different types of methods are available and by the using that we get the plant with desirable characters.

(80) Which method is useful for artificial hybridization ?

(A) Emasculation (B) Bagging

(C) Fumigation (D) A and B both

(81) From the bisexual flowers, with the help of forceps anthers are removed from the floral bud method is .....

(A) Bagging (B) Emasculation

(C) Fumigation (D) A and B both

Answers : (80-D), (81-B)

Double fertilization : When pollen becomes deposited on stigma of carpel, there is development of pollen tube on the stigma. The pollen tube grows through the style enters the ovary and reaches an ovule. Two male gametes are included within the pollen tube. Two male gametes are released in the ovary which fertilizes egg cell and secondary nucleus respectively, This is called double fertilization.

(82) When two male gametes are released in the cytoplasm of the Z synergids, the structures found in the embryo sac are,

(A) Three antipolar cells (B) One egg cell

(C) One secondary nucleus (D) All the above.

(83) Male gametes which enter the embryo sac are fused with .....

(A) Secondary nucleus (B) Nucleus of egg cell.

(C) A and B both (D) Antipodal cells

(84) Where do double fertilization and zygote formation occur in embryo sac ?

(A) Towards the antipodal cell end (B) Toward chalazal end

(C) Towards the pollen tube end (D) Towards the micropylar end

(85) During double fertilization, in embryosac, where is the endosperm nucleus formed ?

(A) Towards the antipodal end (B) Towards the chalazal end

(C) Middle of embryo sac (D) Towards the micropylar end.

(86) Which structure is developed from the endosperm nucleus ?

(A) Ovary (B) Zygote

(C) Endosperm (D) Eggcell

Answers : (80-A), (81-D), (82-D), (83-C), (84-D), (85-C), (86-C)

Post - Fertilization : Structure and Events : Following double fertilization, events of endosperm and embryo development, maturation of ovules(s) and ovary into fruit are collectively termed post-fertilization events.

- **Formation of endosperm :** As stated earlier, the endosperm develops from the primary

endosperm nucleus (3n) by its repeated mitotic division and forms a triploid endosperm tissue. Lateron, this merges with the cotyledons or independent – depending on the type of plants.

(87) In a common endospermic, development what is PEN ?

(A) Primary endosperm nucleus. (B) Primary embryo and nucleus

(C) Primary embryo and cellular nucleus (D) Formation of primary endosperm tissue

(88) In a fresh coconut, the fluid is ..... .

(A) Embryonic fluid (B) Free endospermic nucleus

(C) Cellular endosperm (D) cytoplasm of degradable cells

(89) In a fresh coconut, the white kernel present inside is ..... .

(A) Embryonic fluid (B) Frce endospermic nucleus

(C) Cellular endosperm (D) Cytoplosm of degradable cells

(90) In which plant endosperm is consumed by the developing embryo ?

(A) Dicot (B) Castor

(C) Coconut (D) B and C both

(91) In which plants endosperm persists in mature seeds ?

(A) Dicot (B) Castor

(C) Coconut (D) B and C both.

Answers : (87-A), (88-B), (89-C), (90-A), (91-D)

Embryo :

Embryo develops from the zygote which is located near the micropyle. Zygotes divides after formation of a certain amount of endosperm. A four celled pro-embryo stage is followed by the formation of suspensor and an octant. At last the embryo is formed. This process is seen in Capsella (dicotyledenous) plants, and so it is called as the capsella type embryo development.

(92) In embryo development division of basal cell is transverse and apical cell divides longitudinal and produce four celled structure it is known as ..... .

(A) Proembryo (B) Primary embryo

(C) Tetrad cellular (D) A and B both

(93) The developing embryo is pushed towards the middle of embryo sac by what ?

(A) Apical octant (B) Basal cell (C) Suspensor (D) Hypophysis cell

(94) Name the large apical cell of the suspensor.

(A) Basal cell (B) Apical cell (C) Hyphysis cell (D) Micropylar cell.

(95) Where is apical octant located ?

(A) Towards hilum (B) Towards microrpyle.

(C) Towards upper side of hypophysis (D) Towards lower side of hypophysis.

(96) Where basal octant is located ?

(A) Towards hilum (B) Towards micropyle

(C) Towards upper side of hypophysis (D) Towards lower side of hypophysis.

(97) From the chalazal octant which region of embryo develop ?

(A) Shoot apex (B) Epicotyl (C) Cotyledons (D) All the above

(98) Which regions of embryo develop from the micropylar octant ?

(A) Central part of radicle (B) Hypocotyl

(C) A and B both (D) Peripheral region of radicle root.

(99) Which region is formed from the hypophysis ?

(A) Central region of radicle (B) Peripheral region of radicle

(C) Root cap (D) B and C both.

(100) Which regions develop from chalazal octunt and micropylor octant respectively ?

(A) Epicotyl, Hypocotyl (B) Hypocotyl, epicotyl

(C) Peripheral region of radicle, central region of radicle

(D) Central region of radicle, peripheral region of radicle.

(101) Monocot embryo has only one cotyledon. This type of cotyledon is known as ..... .

(A) Coleoptile (B) Coleorhiza (C) Scutelum (D) A and B both

Answers : (92-A), (93-C), (94-C), (95-A), (96-B), (97-D), (98-C), (99-D), (100-A), (101-C)

Apomixis : Apomixis is a modified form of reproduction in which seeds are produced without fertilization. This special mechanism is seen in species of Asteraceae and grasses. In other words, apomixis is a form of asexual reproduction that mimics sexual reproduction. Development of embryo from unfertilized cell or from a diploid cell is also Apomixis.

In Apomixis, since chismata formation or conjugation of chromosomes does not occur, no variations are introduced. The desirable characters of the species continue, indefinitely. Thus this phenomenon is very important in Apomictic species.

(102) In which species apomixis is found ?

(A) Asteraceae (B) Grass (C) Apomictic (D) A and B both.

(103) Which reproductive method is in the form of asexual reproduction in which it mimics sexual reproduction ?

(A) Polyembryony (B) Apomixis (C) Simple embryonic (D) A and C both

(104) Plants shows apomixis these plants are known as ..... .

(A) Apomictic (B) Astereceae (C) Sporophytic (D) Adventive.

(105) If seed develops from nucellus or integuments and other cell, it is known as ..... .

(A) Generative apossory (B) Somatic apospoly

(C) Haploid paltheno genesis (D) Sporophytic budding.

(106) Apomixis development of embryo from unfertilized egg is called ..... .

(A) Haploid parthenogenesis (B) Haploid apogamy

(C) Somatic apogamy (D) Somatic apospory

(107) From any cell of embryosac apart from the egg is called ..... .

(A) Haploid parthenogenesis (B) Haploid apogamy

(C) Somatic apogamy (D) Somatic - apospory.

(108) If embryo is developed from unfertilized egg, then the embryo will be ..... .

(A) Triploid (B) Haploid (C) Diploid (D) Without chromosomes.

(109) Cell of ovule lying outside the embryo sac is referred to as ..... .

(A) Apomictic (B) Adventive embryonic cells

(C) Sporophytic budding (D) Sporophytic

(110) In which species sexual and asexual reproduction occur simaltaneously ?

(A) Apomictic (B) Adventive (C) Sporophytic (D) B and C both.

Answers : (102-D), (103-B), (104-A), (105-B), (106-A), (107-B), (108-B), (109-B), (110-A)

Polyembryony : The phenomenon of development of more than one embryo in the seed is called polyembryony. Polyembryony is commonly observed in conifers (gymnosperms). It is also recorded in angiosperm plants. It may be due to the presence of more than one egg cell in the embryo sac or more than one embryo sac in the ovule. It plays an important role in plant breeding and horticulture.

(111) Which plants commonly shows Polyembryony ?

(A) Monocot (B) Dicot (C) Gymnosperms (D) Agniosperms

(112) What is the reason for polyembryony ?

(A) In embryo sac more than one egg cells are present.

(B) More than one embryo sac in the ovule.

(C) In embryo sac number of ovule is more than one.

(D) In embryosac more than two ovules and numerous endospermic nucleus are present.

Answers : (111-C), (112-A)

- **Fruit Formation and Development of seed :**

In the angiosperms fertilization produces changes not only in embryo - sac but also in the ovary and at times, in other parts of the flower. Ovules mature in to seeds. The wall of ovary develops into the wall of fruit called pericarp. The fruits may be fleshy or dried and spread with mechanism of seed dispersal. In several plants thalamus also participates in the formation of fruit formation. Fruit is a result of fertilization but in some species do fruit formation occurs without fertilization (Parthenocarpic fruits.)

In angiosperm, the seed is the final product of sexual reproduction. It is often described as a fertilized ovule. Seeds are formed inside the fruits. A seed typically consists of seed coats, cotyledon (s), and an embryo axis. The cotyledons of the embryo are a simple stucture, generally thick and swollen due to stored food reserve. Seeds can be non endospermic or endospermic seed.

(113) If thalamus participates in the formation, the fruit is known as ..... .

(A) True fruit (B) False fruit (C) Unfertilized fruit (D) Seedless fruit.

(114) In fruit formation is from the ovary the fruits are known as ..... .

(A) True fruit (B) False fruit (C) Unfertilized fruit (D) Seedless fruit.

(115) Fertilised ovule means ..... .

(A) Fruit (B) True fruit (C) seed (D) Unfertilized fruit.

(116) Why cotyledons of seed are thick and swollen ?

(A) Cotyledons plays important role in embryonic development.

(B) Reserve food is stored in it.

(C) Formation of cotyledons by suspensor

(D) Numerous free nuclei are here.

(117) In non endospermic seed why stable endosperm is not present ?

(A) Amount of water is more in seed. (B) Amount of water is less in seed.

(C) Metabolic activities are quick in seed.

(D) During the embryonic development embryo is completely consumed.

(118) Which plant possesses persistant residual nucellus ?

(A) Pepper (B) Clove (C) Beet (D) A and C both

(119) What happens due to seed dormancy ?

(A) It can be store easily (B) Can be sowing in next few years.

(C) Problem occurs in seed germination (D) All the above

(120) How old were the date palm seed discovered from the palace of King Herod.

(A) 200 (B) 2000 (C) 1000 (D) 10,000

Answers : (113-B), (114-A), (115-C), (116-B), (117-D), (118-D), (119-D), (120-B)

(121) Identify mis match from the following.

(A) Perisperm - pepper (B) Non endospermic seed - Ground nut

(C) Endospermic seed - Pea (D) Fertilized fruits - Banana

(122) Identify mis match from the following.

(A) Vernonia - papus (B) Martynia -Hook like structure

(C) Ruellia - mechanism dispersal (D) Cotton - Hairy outgrowth

(123) Identify mis match from the following ?

(A) Anemophily - Grass (B) Entomophily - wasps

(C) Hydrophily - Grass (D) Kigelia - Bat.

(124) Identify proper matching pair from the following ?

(A) Unfertilized fruits - Ground nut (B) False fruit - Apple

(C) Dry pericarp - Guava (D) Flesly pericap - mustard

(125) Identify proper matching pair from the following ?

(A) Proembryo - four cellular structure (B) Suspensor - sixteen cellular structure

(C) Hypophysis cell - six cellular structure. (D) Basal cell - unicellular structure.

(126) Identify proper matching pair from the following ?

(A) Acarpellary - flower of pea (B) Many numbers of ovule - orchid

(C) Monoecious-papaya (D) polyembryony -onion.

(127) Which sentance is related to funicle ?

(A) It attaches embryosac with micropyle (B) It attaches pericarp with seed.

(C) It attaches ovary with anther. (D) It attaches placenta with ovule.

(128) Find the odd one out.

(A) Antipodal cell (B) Synergid cell (C) Vegetative cell (D) Egg cell.

(129) Which is not related to anemophily ?

(A) Hairy and sticky stigma

(B) Pollen grain produced in large amounts

(C) Attractive inflorescence

(D) Male flowers are located higher than female flowers

(130) Which option is not related ?

(A) Aloe - moth - Entomophily (B) Hemp - winds - anemophily

(C) Zostera - Water - Hydrophily (D) Kigelia - Bat - zoophily.

Answers:(121-C),(122-D),(123-C), (124-B), (125-A), (126-D), (127-D), (128-C), (129-C), (130-A)

- **A - Assertion R - Reason type Questions :**

Choose the propare option given following question for answer :

(A) A and R both are true and R is correct explanation of A.

(B) A and R both are correct but R is not the correct explanation of A.

(C) A is correct but R is wrong.

(D) A is wrong but R is correct.

(131) Assertion A : During dormancy, the seed is without water.
Reason R : In dormancy metabolic activities of embryo is slow.
(A) (B) (C) (D)

(132) Assertion A : Flowering plants develop methods for cross pollination ?
Reason R : Continuous self pollination creates in breeding depression.
(A) (B) (C) (D)

(133) Assestion A : All flowers have four whorls.
Reason R : Flowers are organs of sexual reproduction.
(A) (B) (C) (D)

(134) Assertion A : Stamen consists of anthers connective and filament.
Reason R : Stamen is male reproductive organ which produce pollen grains.
(A) (B) (C) (D)

(135) Assertion A : Male and female gametes conjugate to produce zygote.
Reason R : Zygote produces new plant
(A) (B) (C) (D)

(136) Assertion A : Mitosis occurs on MMC (Megaspore mother cell) and produce four haploid megaspore.
Reason R : Formation of embryo sac is from megaspore
(A) (B) (C) (D)

(137) Assertion A : Development of pollen tube occurs from the pollen grain.
Reason R : Germination of pollen tube occurs where there is no sporopollenin.
(A) (B) (C) (D)

(138) Assertion A : The phenomenon of development of more than one embryo in the seed is called polyembryony.
Reason R : Polyembryony may be due to the presence of more than one egg cell in the embryosac.
(A) (B) (C) (D)

(139) Assertion A : Asteraceae is apomitic plant.
Reason : Reproduction occur by apomixis.
(A) (B) (C) (D)

Answers : (131-B), (132-A), (133-D), (134-A), (135-B), (136-D), (137-A), (138-A), (139-A)

(140)

Column - I	Column - II	
(1) Hollow bag like structure	(p) Ovule	(A) : (1 - t), (2 - u), (3 - q), (4 - s) (5 - r)
(2) End of the mitosis creation of small nucleus	(q) Megaspre - mother cell	(B) : (1 - r), (2 - t), (3 - p), (4 - s) (5 - q)
(3) Structure made up by integuments and megasporangia	(r) Anther	(C) : (1 - t), (2 - u), (3 - p), (4 - v) (5 - q)
(4) Structure made up by three nuclei at micropylar end	(s) Antipodal cell	(D) : (1 - r), (2 - u), (3 - p), (4 - v) (5 - t)
(5) Bilayered uninucleate structure	(t) Pollen grains	
	(u) Generative cell	
	(v) Egg apparatus	

(141)

Column - I	Column - II	
(1) Non endospermic seed	(p) Banana	(A) : (1 - s), (2 - r), (3 - q), (4 - p)
(2) Endospermic seed	(q) Beet	(B) : (1 - r), (2 - s), (3 - q), (4 - p)
(3) Perisperm	(r) Castor seed	(C) : (1 - q), (2 - r), (3 - s), (4 - p)
(4) Parthono carpic	(s) Pea.	(D) : (1 - p), (2 - s), (3 - q), (4 - r)

(142)

Column - I	Column - II	
(1) Homogamy	(p) Species which exhibit crosspollination	(A) : (1 - s), (2 - r), (3 - p), (4 - q)
(2) Geitonogamy	(q) Self pollination in closed floral bud	(B) : (1 - s), (2 - p), (3 - q), (4 - r)
(3) Xenogamy	(r) Pollination between the two flowers of one plant	(C) : (1 - s), (2 - q), (3 - r), (4 - p)
(4) Cleistogamy	(s) More possibility of self pollination	(D) : (1 - r), (2 - s), (3 - q), (4 - p)

(143)

Column - I	Column - II	
(1) Embryo from unfertilized egg	(p) Sporophytic	(A) : (1 - q), (2 - r), (3 - t), (4 - p)
(2) Embryo from integuement cell	(q) Haploid apomixis / Parthenogenesis	(B) : (1 - q), (2 - s), (3 - t), (4 - r)
(3) Embryo from any other cell of embryosac except egg	(r) Somatic sporophytic	(C) : (1 - q), (2 - t), (3 - s), (4 - r)
(4) Embryo from any cell of Chalazal region	(s) genetic apospory (f) Haploid apogamy.	(D) : (1 - r), (2 - q), (3 - p), (4 - t)

(144)

Column - I		Column - II	
(1) Moths	(a)	Grass	(A) : (1 - d), (2 - c), (3 - a), (4 - b)
(2) Anemophily	(b)	Amorphophallus	(B) : (1 - c), (2 - a), (3 - d), (4 - b)
(3) Hydrophily	(c)	Yucca	(C) : (1 - b), (2 - a), (3 - d), (4 - c)
(4) Flies	(d)	Zostera	(D) : (1 - c), (2 - d), (3 - b), (4 - a)

(145)

Column - I		Column - II	
(1) Anemophily	(p)	Mucilage	(A) : (1 - r), (2 - s), (3 - p), (4 - q)
(2) Hydrophily	(q)	Dried, smooth light	(B) : (1 - s), (2 - r), (3 - q), (4 - p)
(3) Zoophily	(r)	Hairy spiny	(C) : (1 - q), (2 - p), (3 - s), (4 - r)
(4) Entomophily	(s)	Edible	(D) : (1 - p), (2 - r), (3 - s), (4 - q)

(146) **Column - I** **Column - II**

(1) Cactaceae	(p)	Geitonogamy	(A) : (1 - q), (2 - r), (3 - s), (4 - p)
(2) Catharanthus	(q)	Self fertilization	(B) : (1 - r), (2 - q), (3 - s), (4 - p)
(3) Commilina	(r)	Homogamy	(C): (1 - q), (2 - p), (3 - s), (4 - r)
(4) Maize	(s)	Cleistogamy	(D) : (1 - r), (2 - s), (3 - q), (4 - p)

(147) **Column - I** **Column - II**

(1) Apomixis	(p)	Apiaceae	(A) : (1 - q), (2 - s), (3 - p), (4 - r)
(2) Polyembryony	(q)	Astaraceac	(B) : (1 - r), (2 - s), (3 - p), (4 - q)
(3) Self fertilization	(r)	Conifers	(C) : (1 - q), (2 - p), (3 - r), (4 - s)
(4) Cleistogamous	(s)	Conifers	(D) : (1 - r), (2 - p), (3 - s), (4 - q)

(148) **Column - I** **Column - II (Mechamism of seed dissersal)**

(1) Xanthium	(p)	Papus	(A) : (1- q), (2-s), (3-t), (4- p) (5- r)
(2) Cotton	(q)	Stiff hairs	(B) : (1-q), (2-t), (3 - s), (4 - r) (5 - p)
(3) Vernonia	(r)	Hook like structure	(C) : (1-s), (2- t), (3- p), (4- q) (5 - r)
(4) Calotropis	(s)	Hairy out growth	(D) : (1-t), (2- p), (3- q), (4- r) (5 - s)
(5) Martynia	(t)	Hairy sepals	

(149) **Column - I** **Column - II**

(a) One ovule in ovary	(i)	Viola	(A) : (a - iv), (b - ii), (c - i), (d - iii)
(b) Many ovules in ovary	(ii)	Commelina	(B) : (a - v), (b - iv), (c - ii), (d - i)
(c) Homogomy	(iii)	Catharanthus	(C) : (a - iv), (b - v), (c - iii), (d - ii)
(d) Cliestogamy	(iv)	Papaya	(D) : (a - v), (b - iv), (c - iii), (d - ii)
	(v)	Mango	

(150) **Column - I** **Column - II**

(a) Self parthenogenesis	(i)	Lemon	(A) : (a - v), (b - iv), (c - iii), (d - i)
(b) Parthenogenesis	(ii)	Mango	(B) : (a - i), (b - ii), (c - iii), (d - iv)
(c) Adventive embryony	(iii)	Poaceae	(C) : (a - iv), (b - iii), (c - ii), (d - i)
(d) Polyembryony	(iv)	Malva	(D) : (a - i), (b - iii), (c - ii), (d - v)
	(v)	Mango	

Answers : (140-D), (141-A), (142-A), (143-A), (144-B), (145-C), (146-A), (147-A), (148-A), (149-D), (150-C)

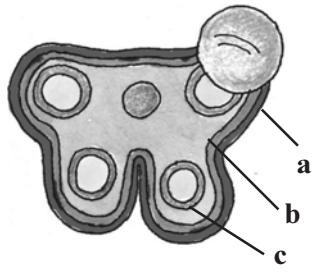
(151) Inlicates parts a, s, c, in given diagrom

(A) a - Connective b - Epidermis, c - Endothecium

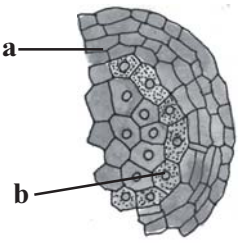
(B) a - Connective, b - Endothccium, c - Tapetum

(C) a - Epidermis, b - Endothecium, c - Sporophytic tissue

(D) a - Epidermis, b - Endothecium, c - Tapetum



(152) Indicate part a, b in given diagram.



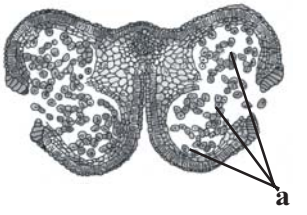
(A) a = middle layer, b = Tapetum

(B) a = Endothecium, b = Tapetum

(C) a = Middle layer, b = Microspore mother cell

(D) a = Endothecium, b = Microspore mother cell

(153) What does 'a' represent in the given figure ?



(A) It is representative of gametophyte staye.

(B) It has two layer other than plasma membrane.

(C) A and B both

(D) By the meiosis diviing production of male gametes occurs

(154) Identify the diagram.



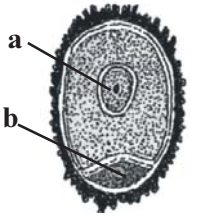
(A) Pollen tetrad

(B) Pollen mother cell

(C) Mature pollengrain

(D) Mega spore mother cell

(155) Identify part a, b in given diagram respectively ?



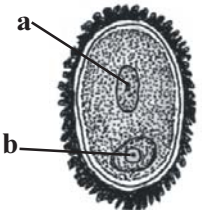
(A) a - Cytoplams b - Intine (Inner coat)

(B) a - Nucleus, b - Inner coat

(C) a - Cytoplasm, b - Outer coat

(D) a - Nucleus, b - Outer coat

(156) Identify part a, b in given diagram respectively ?



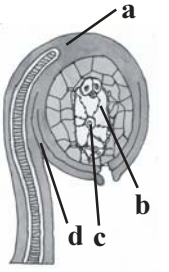
(A) a - Tube nucleus, b - Generative cell

(B) a - Generative cell, b - tube nucleus

(C) a - cytoplasm b - tube cell

(D) a - Nucleus b - Germpore

(157) Identify correct name of parts a,b,c,d. in given diagram.



(A) a - Chalaza, b - Secondary nucleus, c - Egg cell d - himum diagrum

(B) a - Chalaza, b - Secondary nucleus, c - Synergid cell d - Hilum

(C) a - Chalaza, b - Embryo sac, c - Egg cell, d - hilum

(D) a - Chalaza, b - Embryo sac, c - Synergid cell, d - Funicle.

(158) Identify processes which are indicated by a, b in diagram



(A)a - Self pollination, b - Cross pollination

(B) a - Cross pollination, b - Self pollination

(C) a - Self fertilization, b - Geitonogony

(D) a - Self fertilization, b - Cross pollination

(159) Identify part a,b in given diagram



(A) Primula, a - Stigma, b - Anther

(B) Primula, a - Anther, b - Stigma

(C) Malva, a - stigma, b - Anther

(D) Malva, a - Anther, B - Stigma

(160) Identify given figure



(A) Carpel

(B) L.S of flower which showing growth of pollen tube.

(C) Female gametophyte

(D) Germination of pollen grain

Answers : (151-C), (152-B), (153-C), (154-C), (155-D), (156-A), (157-C), (158-C), (159-B), (160-B)

- **Questions for NEET Specially :**

(161) How many whorls of are present in typical flower ?

(A) Two (B) Three (C) Four (D) Five

(162) Which is the sterile part of carpel ?

(A) Ovary (B) Stigma (C) Style (D) B and C both

(163) Intine of pollen at the time of germination comes out and is convert into .....

(A) Pollen tube (B) Ovary (C) Style (D) Stigma

(164) Integuments while covering the embryosac leave an aperture called ..... .

(A) Micropyle (B) Ovule pore (C) Chalaza (D) A and B both.

(165) At the maturity, embryo sac of typical angiospern has how many cells ?

(A) 8 (B) 7 (C) 4 (D) 3

(166) Which type of flower open at the time of dehiscence of anther ?

(A) Chasmogamous flowers (B) Cleistogamous flower

(C) Aerial flowers (D) None of these

(167) Species exhibiting cross pollination is known as .....

(A) Xenogamy (B) Recombmation (C) Allogamy (D) Homogamy

(168) What is anemophily ?

(A) Water pollination (B) Insect pollination (C) Wind pollination (D) Animal pollination

(169) What is zoophily ?

(A) Water pollination (B) Insect pollination (C) Wind pollination (D) Animal pollination

(170) Modified form of reproduction in which seed formation occurs without fertilization is known as .....

(A) Polyembryony (B) Simple embryony (C) Apomixis (D) A and C both

Answers : (161-C), (162-C), (163-C), (164-D), (165-B), (166-C), (167-A), (168-C), (169-D), (170-C)

Sexual Repreduction in floweing plants

Flower

Gynoecium (carpels)

- **Stigma** (pollen receiving structure)
- **Style** (tube connecting ovary and stigma)
- **Ovary**
  - ovule present inside the ovary serves as integumented megasporangium.
  - It develops into seed after fertilization

1 Gametogenesis

- Male gametes and female gametes formed inside the male and female gametophyte respectively

Female gametophyte

- The megaspore mother cell divide meiotically to form 4 haploid megaspores. Out of which one remains functional and others degenarate
- Functional megaspore grows in size and divides by mitosis of form 2-nucleated cell forms embryosac, the female gametophyte.
- Each nucleus moves to other poles and divides there twice to form 8 nucleate embryo sac.

Androecium (stamens)

- **Anther**
  - It is called as microsporangium.
  - It contains sporogenous tissue
  - Cells of sporogenous tissue transform into microspore mother cells or pollen mother cells (PMC)
  - PMC produce poller grains by meiosis
- **Filament**

Male gemetophyte

- inside the microspore (pollengrain) mitosis occur and 2-celled male gametophyte is formed, large one is tube cell and the small one is generative cell

2 Pollination

- Transfer of 2-celled male gametophyte from anther to stigma of a flower

Self Pollination

- Transfer of pollen from anther to stigma of the same flower or different flower of the same plant

Cross pollination

- Transfer of pollen from anther to stigma of different flower belonging to different plant of same species

3 Fertilization

- Pollen tube along with generative cell and tube cell enters the embryo sac through micropyle (porogamy) chalaza (chalazogamy) or sides (mesogamy),
- Generative cell divides into two by mitosis. Tube cell degenerates, Double fertilization occurs, ie - one male gamete fuses with egg to form 2n-zygote and another male gamete fuses with secondary nucleus to form 3n-endosperm.

35 Growth and Development in Plants

- **Growth** : Irreversible increase in size and weight in organisms is called Growth.
  Growth is outcome of metabolism.
- **Anabolism** : All Synthetic biochemical processes which take place in living organisms is combinly called anabolism
- **Catabolism :** All degrading biochemical processes which take place in livning organisms is combinly called catabolism.
- **Note :** Resultant of Growth increasing in weight

Anabolism	**>**	**Catabolism**	**⇒**	**Growth**
Catabolism	**>**	**Anabolism**	**⇒**	**Ageing/ Senescence**
Anabolism	**=**	**Catabolism**	**=**	**Stationary Growth**

---

(1) When anabolic processes occur more than catabolic processes, what possibility occur in living ?
(A) Development (B) Growth (C) Differentiation (D) Senescence

(2) When catabolic processes occur more than anabolic processes, so what possibility occur in living ?
(A) Growth (B) Division (C) Differentiation (D) Senescence

(3) Growth is limited upto which tissues in the plant ?
(A) Meristematic tissues (B) Simple tissues (C) Complex tissues (D) Secondary tissues

Answers : (1-B), (2-D), (3-A)

Types of Growth :

→ Primary Growth :

- It is due to activities of apical meristematic tissues which located at the apex or vegetative organs. so length of vegetative organs increase. This growth is known as primary growth.
- Due to activation of apical meristem Primary growth can take place.
- Root apex and shoot apex are responsible for primary growth.

Note : Intercalary meristem present in nodes of monocots, is the primary tissue.

→ Secondry growth :

- The formation of secondary tissues after the elongation occurs in length of root, stem, due to these tisues increase in grith, so it is called secondary growth.
- Due to activation of lateral meristem secondary growth take place and devloped as a cambium
- Cork cambium is resposlible for the secondary growth of bark of tree.

---

(4) It is responsible for the primary growth in plant :
(A) Apical meristem (B) Intercalary meristem
(C) Lateral meristem (D) A and B both

(5) It is responsible for the secondary growth in plant.
(A) Apical meristom (B) Intercalary meristem
(C) Lateral meristem (D) All above

(6) Which tissue is present in nodal region of monocot plant ?

(A) Apical meristem (B) Intercalary meristem

(C) Lateral meristem (D) All above

(7) It is responsible for secondary growth which located at .....

(A) Shoot apex (B) Root apex

(C) Internode (D) In form of cork cambium below the bark of tree.

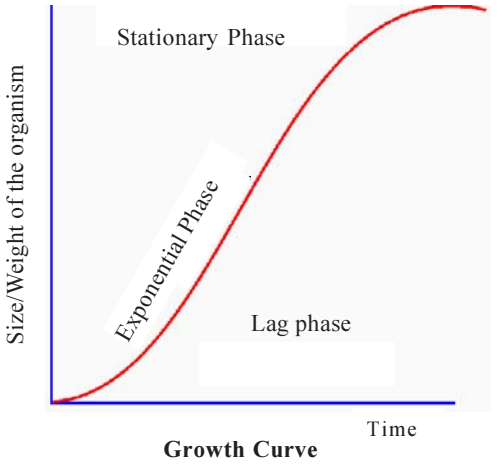
Answers : (4-D), (5-C), (6-B), (7-D)

Growth rate: The increared growth per unit time is termed as growth rate

S - Shaped Growth curue:

- Initially, the rate of growth in plant is slow. Then it increases very rapidly. In course of time, it again slows down.
- Suppose we draw a graph of growth rate raised on the increase in number of cells against time taken. such a graph will be a typical s-shaped graph. (**CS** = Sigmoid curve).
- After an initial period of slow growth rate,

  An exponential period of growth follows and finally a stable state of growth occurs.
- **Note :** such a graph of growth is obtained for growth of an organism and that of a population also parameters other than an increase in the number of cells can also be used in measurement of growth



Growth Curve

(8) S-Shaped growth curve and grand period of growth is changed by which factor ?

(A) Suddenly change in intensity of light

(B) Change occur in temperature

(C) increase - decrease in relation to light and temperature

(D) No effects of given factors on it.

(9) It is correct order of stages of s-shaped growth curve ?

(A) Initial growth stage, stationary growth stage, Exponential growth stage.

(B) Stationary growth stage, Initial growth stage, Exponential growth stage.

(C) Initial growth stage, Exponential growth stage, stationary growth stage.

(D) Exponential growth stage, stationary growth stage, initial growth stage.

(10) What is the parameter for sigmaid curve graph ?

(A) Glowth of population (B) Numberof cell (C) Size of organism (D) All above

Answers : (8-D), (9-C), (10-D)

Phases of Growth :

→ **Phase of cell division (Formation of Meristematic) :** The meristematic cells located in shoot apex and rootapex divide repeatedly and continuously and add new cells.

- The meristematic cells possess dense protoplasm, a large nucleus and thin cell wall made up of cellulose
- Faster rate of metabolism occurs in them.

→ **Phase of cell enlargement (Elongation) :** In this phase, the new cells formed through cell divisions increase in size.

- The volume of cells increases. The growth in cell wall is mainly responsible for such enlargement. The size of vacuole in the cell also increases.

→ **Phases of cell differentiation (Maturation) :** The cells assume forms based on their functions. Their size and form become permanent

- They become associated with the constitution of various tissues
- The phase of becoming differentiated is called differentiation phase.

Grand period of growth: The entire period covering the period from cell division to cell differentiation is called grand period of growth.

(11) What is characteristics of cells which is situated in divisional region ?
(A) Dense cytoplasm (B) Large nucleus (C) Rapid metabolic rate (D) All above

(12) In which stage of growth size of vacuole increases ?
(A) In stage of cell differentiation (B) In stage of cell division
(C) In stage of cell expantion (D) None of the above

(13) It is characteristic of cell which is situated in cell differentiation stage ?
(A) Growth of cell wall (B) Cells assume forms based on their functions
(C) Cells posses dense protoplasm (D) Cells possess large nucleus

(14) In which stage of growth cells obtain various tissue structure ?
(A) In stage of cell division (B) In stage of cell enlargement
(C) In stage of cell differentiation (D) All above

Answers : (11-D), (12-C), (13-B), (14-C)

Factors affecting Growth :

Water : It is essential for turgidity of cells undergoing growth. Water is also required as a medium for various biochemical processes

Oxygen: Oxygen is inevitable for respiration

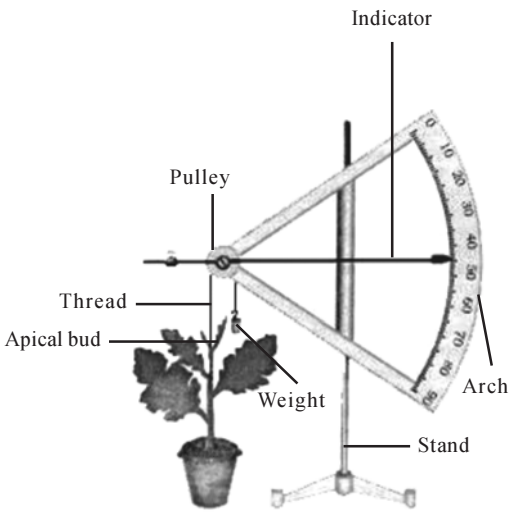
Temperature : proper temperature is required for germination. Normally the optimum range is between 28° C and 30° C.

Light : Light is required for photosynthesis. Food is prepared in this way.

Nutrients : Availability of proper amount of nutrients is required for plant growth. The materials and energy required for synthesis of Protoplasm are obtained from the nutrients. Deficienty of various nutrients and various kinds of stresses hinder the process of growth.

Measurement of Growth: Measurement by scale or tape, number of branches, number of leaves, measured by dry weight, measured by arial expansions.

- **Specific Growth-measurement Method :**
- **Method of ArC - Auxanometer :**
- One end of a thread is tied to the apical bud area of the plant. The thread is then passed over a pulley which is attached to the apparatus and suspended by a weight attached to the other end of the thread



- The position of the indicator on the arch is noted.
- As growth occurs, the weighed end of the thread is lowered.
- As this happens, the indicator also moves along the arch.
- The distance is measured at proper time-intervals and growth is calculated.

(15) What is an optimum temperature of germination ?

(A) 30^0C to 35^0C (B) 28^0C to 30^0C

(C) 25^0C to 30^0C (D) 20^0C to 25^0C

(16) Which of the following supplies material and energy for synthesis of cytoplasm ?

(A) Light (B) Oxygen (C) Nutrients (D) Water

(17) It is not correct option for Arc-Growth measurement.

(A) One end of thread tied with apical bud.

(B) Second end of thread passes on the indicator.

(C) Second end of thread hanging weights

(D) When growth occurs weight end moves downward.

Answers : (15-B), (16-C), (17-B)

Development: Development is a term that includes all changes that an orgamism goes through during its life cycle from germination of seed to senescence.

- Growth, differentiation and development are very closely related events in the life of a plant.
- Development in plants is under the control of two factors
- (i) Internal Factors (ii) Externel cellular factors

(i) **Internal cellular Factors** (Genetical control) or Intercellular plant growth regulators

(ii) **External cellular factors"-** Light, temperature, water, oxygen, nutrients etc.

(18) In plant genetic regulation of growth is which type of regulation ?

(A) Intercellular (B) Intracellular (C) External factor (D) A and B both.

(19) Regulation by growth regultors is known as .....

(A) Intercellular (B) Intracellular (C) External factor (D) None of above.

(20) It is affected to development of growth by means of external factor. ?

(A) Nutrients (B) Gene (C) Growth regulatory (D) All above

Answers : (18-B), (19-A), (20-A)

Plant - Growth Regulators :

- They are conducted to some specific regions and influence growth occurring there or influence some activities from specific place such influence may be either stimulatory or inhibitory such chemicals are called plant growth regulators
- **Properties :**
- They are specific types of organic chemicals which produce from specific regions.
- Origin and activation loci of such chemicals differ.
- Effects are stimulatory or inhibitory
- Their amount decreases after the functions are over.

Note : PGRs are divided in five groups. Auxin, Gibberrelin, cytokinin, Ethylene and Abscisic acid

- **Growth promotor group:** Auxin, Gibberrelin, cytokinin
- **Growth inhibitors group:** Ethylene, Abscisic acid.

(21) PGRs is ..... .
(A) Small molecules of diverse chemical composition.
(B) Simple molecules of diverse chemical composition.
(C) Special Organic Chemicals
(D) All of the above

(22) It is not a characteristic of PGRs.
(A) They are formed in specific regions.
(B) Origination and funcations differ.
(C) They are stimulatory or inhibitory.
(D) They are complex molecules of derivatives of chemical substances.

(23) It is a group of growth regulators.
(A) ABA, Ethylene, Cytokinin
(B) Auxin, Gibberrelin, Cytokinin
(C) ABA, Ethylene,
(D) Ethylene, Cytokinin, Gibberrelins.

(24) It is group of growth inhibitiors.
(A) ABA, Ethylene, Cytokinin
(B) Auxin, Gibberrelin, Cytokinin
(C) ABA, Ethylene,
(D) Ethylene, Cytokinin, Gibberrlins.

Answers : (21-D), (22-D), (23-B), (24-C)

Auxins: It was first isolated from human urine.

Naturel Auxins: IAA (Indole - Acetic Acid), IBA (Indole- Butyric acid)

Synthetic Auxins: 2-4-D (2-4 Dichloro phenoxy acetic acid)
NAA (Napthalene Acetic Acid)

Effects of Auxins: Dominancy of adventitious roots and apical bud.

- Induce the formation of seedless fruit
- Stimulate the process of floweling
- Stimulate the respiration process
- Stops premature fall of leaves and fruits
- Regulates phototropic movement of plant organs.
- Acts as weed-controllers and weed - eradicators
- Are useful in stimulating cell division in tissue-culture also.

(25) It is natural auxins.
(A) IAA, NAA (B) IAA, IBA (C) IBA, 2-4-D (D) 2-4-D, NAA

(26) It is synthetic auxins.
(A) IAA, NAA (B) IAA, IBA (C) IBA, 2-4-D (D) 2-4-D, NAA

(27) Which is the use of synthetic auxin ?
(A) Regulation of cell enlargment
(B) Weed - regulators
(C) Decrease the dominancy of axillary bud
(D) Maintain chrophyll in leaves.

(28) It is used for stimulation for cell divison in tissue culture.
(A) 2-4-D (B) ABA (C) GA (D) IBA

(29) Which auxin is commonly used in callus and suspension culture both ?

(A) NAA (B) IBA (C) 2-4-D (D) ABA

(30) Which growth regulator is used commonly for tea crop ?

(A) Ethylene (B) Gibberelin (C) Cytokinin (D) Indol-3-Acetic acid.

Answers : (25-B), (26-D), (27-B), (28-A), (29-C), (30-D)

Gibberrelins : This hormone was discovered during the investigation of disease in paddy for which gibberrela fungus is responsible (Discovered In Japan)

- The disease was named 'bakane' which means 'foolish plant'

Characters of infected plant : Abnormally long, yellow, thin and normally sterile.

- This happens due to the disease induced by secretion of gibberrela fungus. Their secretion is called gibberrelin.

Plant's Gibberrelin : From the Fungi and other higher plants more than hundred (100) different kinds of gibberrelin have been discovered which are known as GA_1, GA_2, GA_3....etc.

- All kinds of gibberrelins are acidic in nature.
- Their synthesis is higher in darkness.

Effects of Gibberrelins:

- Remove the expression of genetic dwarfism
- Induce elongation of stem. The internodes develop longer
- They also increase leaf area.
- Mobilization of storage compounds during germination that stimulates activation of embryo.
- Responsible for removal of dormancy of buds and seeds.
- They stimulate synthesis of various enzymes which activate the embryo.
- Induce flowering in some plants.

(31) Which of the following is responsible for cell Enlargement in inter nodes ?

(A) Gibberrelin (B) Ethylenc (C) Cytokinin (D) IAA.

(32) When gibberrelic acid gives stimulation for flowering ?

(A) In situation of long day in angisperms

(B) In situation of short day in long day plants

(C) In situation of long day in short day plants

(D) During night in dayneutral plant.

(33) It is discoverd from foolish plant of paddy.

(A) Indole-3-acetic acid (B) Ethylene (C) Gibberrelin (D) Kinetin.

(34) Which hormone is essential for change the genetic expression ?

(A) Cytokinin (B) Auxin (C) Gibberrelin (D) Ethylene

(35) It is an effect of gibberrelins.

(A) Induce germination by conduction of storage compound.

(B) Maintainance of chlorophyll in leaves.

(C) Induce respiration process.

(D) Produce adventitious root.

Answers : (31-A), (32-B), (33-C), (34-C), (35-A)

Cytokinins Discovery :

- It was first discovered as kinetin (a modified form of adenine- purine) from the sperm of herring fish.
- A substance called zeatin, having effects similar to that of cytokinin was obtained from maize grain as well as from coconut milk.

Effect of cytokinin :

- Stimulate the processes of cell division cell enlarglement and cell differentiation.
- Reduce apical dominance.
- Retain chlorophyll in leaves.
- Translocate nutrients and organic substances.

Note: Kinetin hormone gives remarkable effect of cell division.

---

(36) It is gained from water of coconut ?

(A) Kinetin (B) Zeatin (C) Auxin (D) Gibberrelin.

(37) Which hormone treatment is neccessary to keep picked flowers and cut vegetables for long time ?

(A) Gibberrelin (B) Auxin (C) Ethylene (D) Cytokinin.

(38) From which kinetin is first discovered ?

(A) From human urine (B) From herring fish

(C) From fruit of cotton (D) From plant of paddy

(39) What is Kinetin ?

(A) Active form of guanine Nitrogen base (B) Derivative form of cytosine Nitrogen base

(C) Derivative form of adenine Nitrogen base (D) Active form of thimine Nitrogen base

Answers : (36-B), (37-D), (38-B), (39-C)

Abscisic Acid (ABA) : Discovery: It was first discovered as a substance inducing fall of cotton fruits.

Effects of Abscisic Acid: Preventing plant metabolism.

- Stimulate abcission process
- Inducing buds dormancy
- Under the condition of water stress, it stimulates the process of closing of stomata.
- It inhibits seed germination and the development of isolated embryo
- Resistance to stress conditions

Note : Abscisic acid act as a vegetative growth inhibitor hormone.

---

(40) If ABA is more given to plant, ...

(A) Increase the area of leaves (B) Increase in length of stem

(C) Closing of stomata (D) Increase in length of root.

(41) It stimulate the closure of stomata of leaf in water stress condition.

(A) ABA (B) IAA (C) GA (D) Ethylene.

(42) It is known as water stress increased hormone.

(A) IAA (B) 2-4-D (C) ABA (D) GA

Answers : (40-C), (41-A), (42-C)

Ethylene (Volatile hormone):

- It is a simple gaseous plant growth regulator, It is volatile in nature.
- Its concentration remains high in tissues undergoing senescence and in the ripening fruits.

Effects Ethylene:

- It inhibits the length wise growth in root, stems and leaves.
- It induces senescence in plants
- It induces leaf fall and fall of flowers.
- It stimulate the process of ripening of fruits.
- It induces drooping of leaves and flowers.

(43) In which amount / ratio of ethylene is higher ?

(A) In green banana (B) In green apple

(C) In ripen mango (D) In healthy patato.

(44) Plant growth regalator which gives triple response ?

(A) IAA (B) ABA (C) C_2H_4 (D) GA_3

(45) In pea plant which growth regulator is responsible for stunted growth ?

(A) Gubberrelic acid (B) Auxin (C) cytokinin (D) Ethylene

Answers : (43-C), (44-C), (45-D)

Seed Dormancy:

- Seed dormancy is defined as a state in which seeds are prevented from germinating even under favorable environmental condilions. For this various internal factors are responsible.
- During the period of dormancy, the growth of seed is arrested. Some seeds remain dormant for days, while others remain dormant over month or even years.

Types of Seed dormancy : There are mainly four types :

(1) Exogenous dormancy (2) Endogenous dormancy

(3) Combinational dormancy (4) Secondary dormancy

(1) Exogenous Dormancy:

- It is caused by conditions outside the embryo and is often classified into three subgroups.

(a) **Physical Dormancy:** Which occurs when seeds are impermeable to water or to the exchange of gases.

(b) **Mechanical Dormancy:** Mechanical dormancy occurs when seedcoats or other coverings are too hard to allow the emtryo to expand during germination.

(c) **Chemical Dormancy:** Includes growth regulators that are present in the coverings around the embryo.

(2) Endogenous Dormancy: Endogenous dormancy is caused by conditions within the embryo itself and it is also often divided into the following subgroups.

(a) **Physiological Dormancy:** Physiological dormancy presents embryo growth and seed germination until chemical changes occur.

(b) **Morpholgical Dormancy:** It Occurs when the embryos are not differentiated into different tissues at the time of fruit ripening it means embryo is undeveloped or undifferentiated

(c) **Combined Dormancy:** Seeds have both morphological physiological and dormancy (morpho physilogical dormancy)

(3) Combinatonal Dormancy: This is both exogenous (Physical) and endogenous (Physiological) conditions in some seeds

(4) Secondary Dormancy: The conditions that are not favorable for seed germination like high temperatures causes secondary dormancy

---

(46) With the internal situation of embryo which type of dormancy occur ?

(A) Physical dormancy (B) Chemical dormancy

(C) Endogenous dormancy (D) Combinational Dormancy

(47) Which dormancy prevent the germination and embryo growth until the chemical changes ?

(A) Physiogical dormancy (B) Mechanical dormancy

(C) Chemical dormancy (D) Exogenous dormancy

(48) Which is responsible for the exogenous dormancy ?

(A) Presence of growth inhibitors surrounding coat of embryo.

(B) At time of fruit ripening. In various tissue differentiation does not take place

(C) Seed is impermeable for water or gaseous exchange.

(D) Internal situation of embryo is favourable.

(49) Mixed dormancy means .....

(A) Physiological, chemical (B) Chemical, combinational

(C) Physical, physiological (D) Physiological, Exogenous (physical)

Answers : (46-C), (47-A), (48-B), (49-D)

Various causes for seed dormancy:

- undeveloped embryo
- Seed coats impermeable to water
- Mechanically hard and strong seed coats which do not permit germination
- Physiologically immature embryo
- Presence of some germination inhibiting chemicals amongst them the main one is ABA

Note: GA is absent in dormant seeds. At this time ABA remains active, inhibits the transcription of genes. As a result proper enzymes are not synthesized.

Removal of seed dormancy:

(1) Natural Method : When concentration of GA increases more tha ABA, effects of ABA removed and embryo becomes active which results into removal of dormancy.

(2) Seed can scraped lightly with sandpaper. Thus, their seed coats become permeable to water and germination is induced.

- A similar result can be achieved by using chemicals
- Soaked seeds in an O_2 containing environment, can be provided with higher or lower temperature for a definite period. Dormancy is removed is this way.

---

(50) It removes the effect of ABA.

(A) GA (B) IAA (C) CKN (D) 2 - 4 - D

(51) It is absent in dormant seed.

(A) C_2H_4 (B) ABA (C) GA (D) IAA

(52) What is the main role of ABA in dormant seed ?

(A) It inhibits translation (B) It obstract transcription

(C) It continues translation (D) It continues transcription.

(53) It is phenomenon which removes the effect of ABA.

(A) Continues process of trascription (B) Formation of specific enzynes occur

(C) Embryo can be activated (D) All of the above

Answers: (50-A), (51-C), (52-B), (53-D)

Seed Germination: The entire process from the sowing of the seed in the soil up to the emergence of a young sapling from it is called seed germination.

Precondition: Available water, sufficent O_2 and favourable temperature are preconditions for seed germination

Sprouting: The seed, first of all, imbibes water. As the seed coats become loose, rapid absorption of water occurs through the micropyle. Embryo becomes active. Digestion of food, either stored in endosperm or with the help of proper enzymes. The developing embryo is nourished in this way. First or all, the radicle of the embryo develops into a primary root which comes out from the micropyle This is known as sprouting.

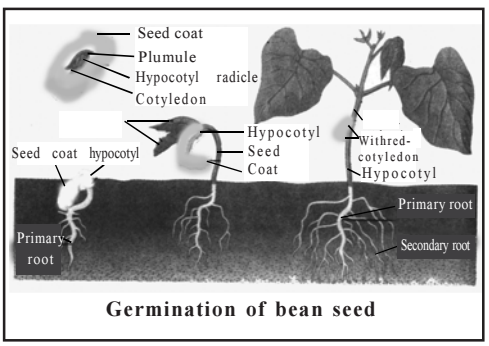
Stucture of rootsystems: This indicates initiation or germination of the primary root developed and forms a root system.

Development of shootsystem: In the mean time, development of plumule also begins, shoot that is stem and leaves, is formed through development of plumule.

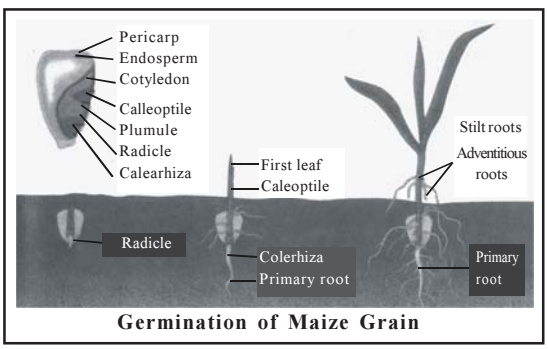
- Growth related processes like cell division, cell enlargement and cell differntiation take place in the formation of root system and shoot systems during germination.

- During the entire process a high respiratory rate is maintained.

- Some hormones are secreted and some enzymes become active.



Germination of bean seed

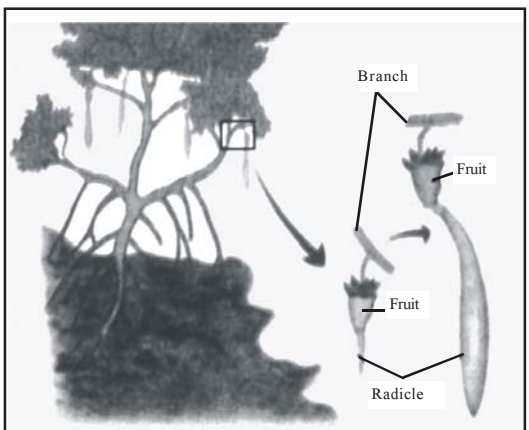


Germination of Maize Grain

Viviparous Germination: Mangrooves are a special type of vegetation live in the bassein (creek) region around sea shore. They exihibit a different kind of germination.

- The seed germinates while it is still in the fruit attaches to the parent plant.

- Later as the weight of sapling increases it drops vertically straight into the mud and becomes anchored. lateral roots. develop and proper establishment is achieved. Such a germination is called - viviparous Germination. eg. Rhizophora and Avicennia are the examples.



Viviparous Germination

(54) What is pre condition for seed germination ?

(A) Avaibility of water (B) Sufficient O_2

(C) Favourable temperature (D) All of the above

(55) By which process seed absorb water ?

(A) Osmosis (B) Imbibition (C) Diffusion (D) Absorption

(56) Sprouting means .....

(A) Root formation from radicle

(B) Formation of primary root from radicle and it comes out from the micropyle.

(C) Formation / development of shoot system from plumule.

(D) Dormant seed becomes active.

(57) During the embryo development first leaf and coleoplile produce one sturucture is called .....

(A) Primary structure (B) Coleoptile (C) Prop root (D) Plumule axis

(58) (i) With this initiation of plant growth occurs

(ii) After the completion of this stage essential factors for growth are avalibable (iii) Removal of seed coat first out growth / development of shoot apex can take place.

(A) (i) Seed coat, (ii) Seed germination (iii) First leaf

(B) (i) Dormancy, (ii) Seed germination (iii) Seed coat

(C) (i) Seed germination (ii) Dormancy period (iii) Seed coat

(D) (i) Seed germination (ii) Dormancy period (iii) First leaf

(59) In rhizophora which type of seed germination is seen ?

(A) Epigeal seed germination (B) Hypogeal seed germination

(C) Viviparous germination (D) Vernalization

Answers : (54-D), (55-B), (56-B), (57-B), (58-D), (59-C)

Senescence:

It is a period between complete maturation of an individual and the death of that individual and it is also knownes 'aging'.

- **Changes ocours during senescence:**
- High rate of catabolism -There is depreciation in the organ and in the body.
- Efficiency decreases -It seems that hormones influences this proces

(i) **Types of Senescence :**

Senescence of entire body : It affects the individuals organs as well as the entire body. All organ are involved eg. Annual plants like wheat and other cereals.

(ii) **Sencescence of aerial Shoots:** In this biennial plants, the aerial shoot experiences this. External portion from above ground like flower and fruit, after formation they die and underground portion devolops new shoot at next year. e.g. Banana, Ginger etc.

(iii) **Sequencial senesence:** This type of sencescence occurs in perennial plants, in which branches of stem becomes meristematic and devolop and formation of new organs like flower, fruit, leaf occurs when old organs such as the branches and leaves fall down. eg. mango, eucalyptus, Cassia etc.

Abcission: The phenomenon of dropping of leaf, flower and fruit is called abscisssion

- The amount of auxin is high in healthy leaf
- During the sencescence decreases the amount of auxin and ABA, ethylene like growth inhibitors are increases.
- Under their influence, the middle lamella which is made up of pectin and which connects th cellulose cell wall, become degraded with other accompanying degradatory activities, the abcission layer develops.
- The cells in this layer undergo degradatory process and become weak. As a result, at proper time the leaf, flower or the fruit breaks.

(60) It is a type of senescence which takes place in annual plant.

(A) Senescence of entire plant (B) Senescence of aerial shoot

(C) Sequential senescence (D) All of the above

(61) In which plant senescene of aerial shoot can be seen ?

(A) In annual plants (B) In biannula plants

(C) In perennial plant (D) A and B both

(62) In healthy leaf which hormone is present in more amount ?

(A) Cytokinin (B) Gibberrelin (C) Auxin (D) Ethylene

(63) In yellowing of leaf which hormone is in more amount ?

(A)ABA (B) C_2H_4 (C) GA (D) A and B both

Answers : (60-A), (61-B), (62-C), (63-D)

Photoperiodism: In the development of plants process of flowering response to the stimulus of period of available light by plants is called photoperiodism.

- The period for which light is available has a remarkable influene on flowering.
- **Classtication of plant on basis of sensention of light availbility.**

(i) **Long day plants:** Some plants required a longer period of light to come into flowering, such plants are called-long day plants eg : **wheat, poppy, Oat, Beet** are long day plants.

(ii) **Short Day plants:** Some plants require a shorter period of light to flower. Such plants are called short day plants. **Soyobean, xanthium** etc are short day plants.

(iii) **Day- Neutral plants:** In some other plants the period of available light does not have any influence on their flowering. Such plants are called - day neutral plants. **Tomato, cucumber** and **Maize** etc are any day neutral plants.

Note: For a photoperiodic response, the continuity of the period of available light and its following period of dark is inevitable. When mendatary period of dark breaks the continuity of light period in the long day plants or such a period of light breaks the continuity of dark period in the short day plants. The response of flowering to this stimulus is not observed.

- It is believed that some specific pigment (Phytochromes) as well as some specific hormone is responsible for this process.

(64) Phytochrome pigment is related to ..... photonastism

(A) Photoperiodism (B) Photorespiration

(C) Photonastism (D) Geotropism

(65) In which plants phytochrome is found ?

(A) Algae (B) Fungi (C) Flowering plants (D) A and B both

(66) When a mendatary period of light is given in dark duration to short day plant ?

(A) Flowering response is not seen

(B) Spontaneous flowering take place

(C) In that season flowering is more

(D) It is converted from short day to long day plant.

(67) What is responsible for responsse of photoperiodism ?

(A) Leaf (B) Bud (C) Shoot apex (D) Flower

(68) Most of the plant produce flowering in season which situation is responsible for this ?

(A) Photonastism (B) Photoperiodism (C) Photosynthesis (D) Photorespiration.

Answers : (64-A), (65-C), (66-A), (67-D), (68-B)

Vernalization:

- Better and earlier germination is induced when seeds are provided with specific low temperature for a definite period of time. Flowering is also earlier in the plants which develop from them. This artificial treatment is called Vernalization.
- Seeds of Wheat, Paddy, Millet and Cotton are provided low temperature between 1^0C to 10^0C and earlier end higher yield of crop is obtained.
- Plants growing under natural conditions must get low temperature for a specific period.
- Only then, they come into flowering. This natural uncertainty can be avoided by vernalization and timely yield can be obtained.

---

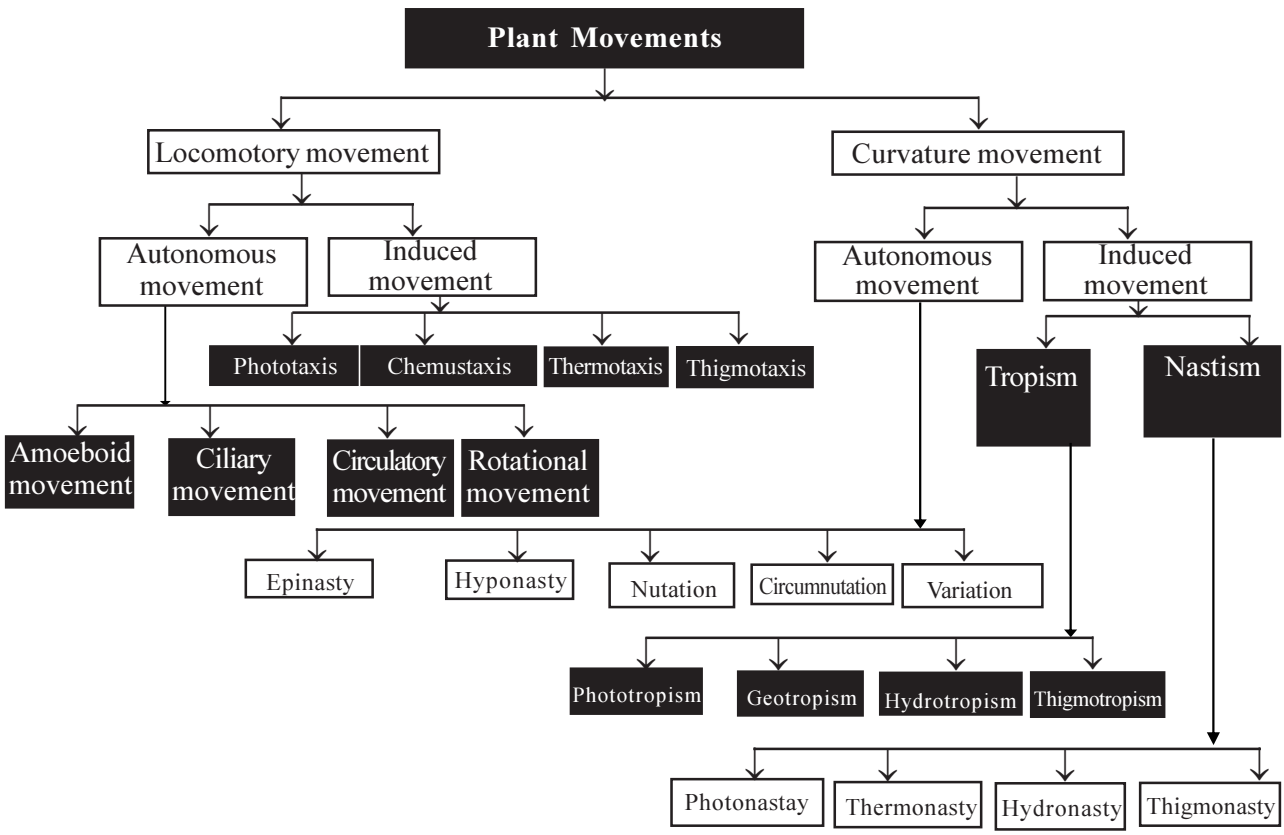
(69) Low temperature treatment given to seed and get rapid germination that method known as .....
(A) Photoperiodism (B) Vernalization (C) Thermonasty D) Phototropisrm.

(70) How much temperature is given to cotton seed so it give yield in proper time ?
(A) 10^0C to 15^0C (B) 15^0C to 20^0C
(C) 1^0C to 10^0C (D) 25^0C to 30^0C

(71) By which process the natural uncertainity for flowering is removed ?
(A) Photoperiodism (B) Photorespiration
(C) Nutation D) Vernalization.

(72) Which hormone is responsible for vernalization ?
(A) Auxin (B) Gibberrelin
(C) Vernalin (D) ABA

Answers : (69-B), (70-C), (71-D), (72-C)



(A) Locomotory Movement:

- Such a movement can be that of the protoplasm, or an organ or the entire organism
- There are two main kinds of movements of locomotion

(1) Autonomous and (2) Induced locomotion

(1) **Autonomous movement:**

- No external factor is responsible for causing this type of movements.

(i) **Amoeboid movement :** Plosmodia of slimemold

(ii) **Cillary movement:** Chlamydomonas algae

(iii) **Circulatory movement:** Cytoplasm in Tradescantia anther

(iv) **Rotational movement:** Cytoplasm in Hydrilla leaves

(2) **Induced Movement:**

- The induced movements of locomotion are caused as a response to the external stimulus
- Such a locomotory movement inducing a change of place called Taxis.

(i) Phototaxis : Zoospore of Volvox.

(ii) Chemotuxis : Antherozoids of bryophytes and pteriodophytes.

(iii) Thermotaxis : Diatoms

(iv) Thigmotaxis : Zoospore in Oedogonium.

(B) Curvatuve Movements: Higher plants show the movements of curvatuve which help them to orient their organ for their work efficiency. Uneven or unbalanced growth causes such a curvature. These are two main kinds of movement of curvature

(1) Autonomous Movement:

- No external factor is responsible for causing such uneven growth.

(i) Epinasty: Growth ratio of upper surface is more then upper susface in leaves -open leaf blade.

(ii) Hyponasty: Growth ratio of lower surface is more then upper surface in leave closing of leaves.

(iii) Nutation: Zigzag movement in apicalbud of stem.

(iv) Circumnutation: Spiral and helical growth of shoot in climbers and tendrilar plants.

(v) Variation: Pulsation in leaflets of Indian telegraph plant (*Desmodium gyrans*)

(2) Induced Movement:

- The induced movement of curvature are caused as a response to the external stimulus. There are two types (i) Tropism and (II) Nastism

(i) Tropism or Tropic movements: If the movement of curvatuve in a plant organ is induced by an external and directional stimulus, it is called tropism The curuature induced by tropism is having a directional relationship with the direction of the stimulus. Kinds of tropism are derived on the basis of the directional stimulus.

(a) Phototropism : (Bylight)

- Stem shows positive phototropism and root shows negative phototropism. Oat coleoptiles.

(b) Geotropism : (By gravitation)

- Stem shows negative geotropism and root shows positive geotropism- Radicle of maize seedling.

(c) Hydrotropism (By Water):

- Roots of higher plants.

(d) Thigmotropism: (By Touch) - Coccinia

(ii) Nastism or nastic Movements:

- This kind of movement depends on the presence and intensity of external stimulus It is not neccessary that it should affect from any definite direction. Based on the external stimulus these movements are called - Nastism
- Based on the external stimulus these movements are called

(a) Photonasty: The flowers in lotus and suflowers open in the morning.

(b) Thermonasty: Flowers of crocus and Tulip open at higher temperature.

(c) Hydronasty: Plants show nastism through heavy rain and water flow.

(d) Thigmonasty: The leaves of Drosera and Mimosa close and droop when they are touched.

---

(73) In which amoebiod movement is found ?

(A) Plasmodia of slime mold (B) Chlamydomonas algae

(C) Stamen of tradescantia (D) Leaf protoplasm of Hydrilla

(74) Which type of movement is found in cytoplasm of hydrilla leaf ?

(A) Amoebiod (B) Thigmotropism

(C) ciliary (D) Circumnutation.

(75) It is correct movemt for zoospore of oedogonium ?

(A) Photonasty (B) Thigmotropissm (C) Chemotaxis (D) Thigmonasty

(76) Where thermotaxis is found ?

(A) Diatoms (B) Slime mold

(C) Male gamete of bryophyte. (D) Coccinia

(77) Spiral and helical growth of shoot of climbers and tendrilar plants is the example for .....

(A) Epinasty (B) Hyponasty

(C) Nutation (D) Circumnutation

(78) Variation movement is shown by

(A) *Desmodium gyrans* (B) Coccinia

(C) Coleoptile of oat (D) Radicle of maize seed germination

(79) Which plant leaflet show pulsation ?

(A) *Desmodium gyrans* (B) Tulip and crocus

(C) Sunflower and lotus (D) Coccinia

(80) Which type of tropism is found in coleoptile of oat.

(A) Phototropism (B) Hydrotropism

(C) Thigmotropism (D) Mimosa

(81) Which type of tropism occurs in radicle in maize of seed germination ?

(A) Phototropism (B) Hydrotropism (C) Thigmotropism (D) Geotropism

(82) It shows thigmotropism.

(A) Coleoptile of oat (B) Coleorhiza of barley

(C) Coccinia (D) Mimosa

(83) It shows photonasty.

(A) Flowers of crocus and tulip

(B) Flowers of lotus and sun flower

(C) Drosera and mimosa

(D) Leaves of *Desmodium gyrans*

(84) Its flowers open at high temperature.

(A) Sunflower (B) Lotus (C) Tulip (D) Coccinia

(85) A phenomenon due to turgidity opening of stomata is .....

(A) Photonasty (B) Hydronasty (C) Thigmonasty (D) Thermonasty

(86) Flowers of sunflower and lotus opens at morning.

(A) Photonasty (B) Thermonasty (C) Hydronasty (D) Phototaxis

Answers : (73-A), (74-D), (75-B), (76-A), (77-D), (78-A), (79-A), (80-A), (81-D), (82-C), (83-B), (84-C), (85-B), (86-A)

- **A - Assertion R - Reason types of questions**

Choose the correct option which given below for following questions.

(A) A and R both are correct while R is a correct explanation of A.

(B) A and R both are correct but R is not correct explanation of A.

(C) A is correct but R is wrong

(D) A is wrong but R is correct.

(87) Assertion A : Gibberrelin is neccessary for flowering in long day plants.

Reason R : If GA treatment is given to tall plant, they can be converted into dwarf plant.

(A) (B) (C) (D)

(88) Assertion A : In presense of ABA seed gemination of seed can be prevented.

Reason R : ABA is inhibitor for transcription process.

(A) (B) (C) (D)

(89) Assertion A : Ethylene is sprinkled on green fruits.

Reason R : Ethylene is neccessary for fruit ripening.

(A) (B) (C) (D)

(90) Assertion A : 2-4-D is useful for tissue culture method.

Reason R : During gemination 2-4-D conducts the storage compounds.

(A) (B) (C) (D)

(91) Assertion A : Cytokinin slows down the senescence.

Reason R : Ethylene is stimulates senescence

(A) (B) (C) (D)

(92) Assertion A : In S-Shaped growth curve three / stages have been found.

Reason R : Initial growth stage is called grand period of growth.

(A) (B) (C) (D)

(93) Assertion A : For the response of photo periodism, its following period of dark is the period of available light and inevitable continuation of dark period is essential.

Reason R : In photoperiod of long day plant for mendatary duration of darkness, flowering respone found on high rate.

(A) (B) (C) (D)

(94) Assertion A : Vernalization is artificial process.

Reason R : Vernalizaion is a process in which entire plant is affected.

(A) (B) (C) (D)

(95) Assertion A : Inshort day plants need less light than the specific time period for flowering.

Reason R : In dark period of shortday plant, introducing of light for moment, response of flowering is not found.

(A) (B) (C) (D)

(96) Assertion A : Cells of short apex and root apex is continuously divides.

Reason R : At the tip of the root apex and shoot apex apical meristematic tissue is present.

(A) (B) (C) (D)

(97) Assertion A : Wheat and other cereals are annual plant

Reason R : In them senescence found in all organs of the plant.

(A) (B) (C) (D)

Answers : (87-C), (88-A), (89-A), (90-C), (91-B), (92-C), (93-C), (94-B), (95-B), (96-A), (97-A)

- **True (T) - False (F) Types Questions**

Given statements are true or false choose the correct option.

(98) (1) In reference of physiology growth is resultant of metabolism.

(2) Anabolic is process of degradation.

(3) Catabolic is process of synthesis.

(4) Dry weight inereases as a result of growth.

(A) TFFT (B) FTTF (C) FTTT (D) TFTF

(99) (1) Rapid metabolic rate of cells is found in meristamatic regions.

(2) Nucleus size of cell increase in enlargment region.

(3) In maturation region, shape and size of the cell become permanent.

(4) Meristematic cells divides repidly which are arranged in shoot apex.

(A) FTTF (B) FFTT (C) TFTT (D) TFTF

(100) (1) Gibberrelin removes genetical dwarfness.

(2) Cytokinin maintain chlorophyll of leaves.

(3) Ethylene prevents the situation of stress.

(A) TTT (B) TTF (C) FTT (D) FFT

(101) (1) Dominancy of apical decreases bud under the effect of cytokinin.

(2) Auxin stimulates respiration.

(3) ABA inhibits stress situation.

(A) FFF (B) TTF (C) TTT (D) TFT

(102) (1) Seed absorb water by imbibition firstly.

(2) Primary root develops from the radicle and it comes out from micropyle.

(3) Sprouting is indication of initiation of seed germination.

(A) TTF (B) TTT (C) FTT (D) FFF

(103) (1) Phyto chrome pigment is responsible for the photoperiodism.

(2) Florigen hormone is responsible for the photo periodism.

(A) FF (B) TF (C) FT (D) TT

Answers : (98-A), (99-C), (100-B), (101-C), (102-B), (103-D)

(104)

Column - I	**Column - 2**	
(I) Auxin	(p) It act as a weedicide and weed regulator.	(A) : (I - p), (II - q), (III - r)
(II) Gibberrlin	(q) Maintain the chlorophyll in leaves	(B) : (I - r), (II - q), (III - p)
(III) Cytokinin	(r) It is more synthesized in absence of light	(C) : (I - p), (II - r), (III - q
		(D) : (I - r), (II - p), (III - q)

(105)

Column - I	**Column - 2**	
(I) IAA	(p) Synthetic auxin act as a weedicide.	(A) : (I - r), (II - q), (III - p)
(II) ABA	(q) Inhibit plant physiology.	(B) : (I - r), (II - p), (III - q)
(III) NAA	(r) Stimalate process of flowering	(C) : (I - p), (II - q), (III - r)
		(D) : (I - q), (II - p), (III - r)

(106)

Column - I	**Column - 2**	
(I) Primary growth	(p) Growth to increase work efficienty in higher plants of organs.	(A) : (I - r), (II - p), (III - q) (B): (I - r), (II - q), (III - p)
(II) Secondary growth	(q) Activation of lateral meristem that growth.	(C) : (I - q), (II - p), (III - r)
(III) Annual growth	(r) Activation of apical meristem that growth.	(D) : (I - p), (II - q), (III - r)

(107)

Column - I	Column - 2	
(I) Phase of cell enlargement	(p) Cells having a dense cytoplasm, large nucleus	(A) : (I - p), (II - q), (III - r)
(II) Phase of call differentiation	(q) Cells included in structure of various tissues	(B) : (I - r), (II - p), (III - q)
(III) Phase of cell division	(r) Increases in size of cell and increase in size of vacuole	(C) : (I - q), (II - r), (III - p)
		(D) : (I - r), (II - q), (III - p)

(108)

Column - I	Column - 2	
(I) More synthesis of Auxin	(p) In helthy leaf	(A) : (I - q), (II - p), (III - r)
(II) Less synthesis of Auxin	(q) In senescence leaf	(B) : (I - r), (II - p), (III - q)
(III) More amount of gibberrelin	(r) During seed germination	(C) : (I - q), (II - r), (III - p)
		(D) : (I - p), (II - q), (III - r)

(109)

Column - I	Column - 2	
(I) Short days plant	(p) Rice, soyabean, Xanthium	(A) : (I - p), (II - q), (III - r)
(II) Long day plant	(q) Wheat, Barley, poppy	(B) : (I - p), (II - r), (III - q)
(III) Day neutral plant	(r) Tomato, cucumber, maize	(C) : (I - r), (II - p), (III - q)
		(D) : (I - q), (II - p), (III - r)

(110)

Column - I	Column - 2	
(I) Epigeal seed germination	(p) Avicennia, Rhizophora	(A) : (I - p), (II - q), (III - r)
(II) Hypogeal seed germination	(q) Bean pea	(B) : (I - r), (II - p), (III - q)
(III) Viviparous germination	(r) Wheat, Maize	(C) : (I - q), (II - r), (III - p)
		(D) : (I - p), (II - r), (III - q)

(111)

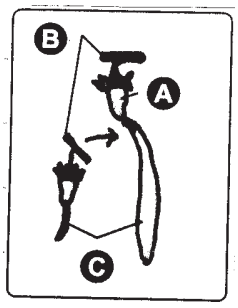
Column - I	Column - 2	
(I) Thigmotaxis	(p) Drosera	(A) : (I - p), (II - q), (III - r)
(II) Thigmotropism	(q) Coccinia	(B) : (I - r), (II - p), (III - q)
(III) Thigmonasty	(r) Zoospore of oodogonium	(C) : (I - q), (II - p), (III - r)
		(D) : (I - r), (II - q), (III - p)

Answers : (104-C), (105-A), (106-B), (107-D), (108-D), (109-A), (110-C), (111-D)

(112) Diagram shows germination of which plant group ?

(A) Mangroove (B) Gymnosperm

(C) Pteridophyta (D) Bryophyta.



(113) True habitat of plant which shows germination in given diagram ..... .

(A) Grassland (B) Creek area of sea shore

(C) Mountain area (D) Wetland area

(114) Indicate parts A,B, C in given diagram.

(A) A = Radicle, B = Branch, C = Fruit (B) A = Fruit, B = Radicle, C = Branch,

(C) A = Fruit, B = Branch, C = Radicle, (D) A = Branch, B = Fruit, C = Radicle.

(115) Which is true plant group for given germination ?

(A) Bean, pea (B) Rhizophora, Bean (C) Wheat, maize (D) Rhizophora, Avicennia

Answers : (112-A), (113-B), (114-C), (115-D)

- **Questions for PMT - NEET**

(116) 2-4-D agent means.

(A) Insecticide (B) Weedicide (C) Pneumatocide (D) Pesticide

(117) 6-furfuryle aminopurine, 2-4- dichlorophenoxy acetic acid and Indole-3 acetic acid are examples of ..... respectively.

(A) Synthetic auxin, kinetin and natural auxin (B) Gibberrelin, natural auxin and kinetin

(C) Kinetin, synthetic auxin and natural auxin (D) Natural auxin, Gibberrelin and kinetin.

(118) Which process is not affected when amount of auxin decreases ?

(A) Hypopulsatory of leaves

(B) Entrance of root in ground

(C) Movement of sunflower in direction of sunlight

(D) Movment of shoot in direction of sun.

(119) Hormone is responsible for phototropism.

(A) Auxin (B) Gibberrelin (C) Cytokinin (D) ABA

(120) Treatment of which hormone induce parthehogenesis ?

(A) Ethylene (B) Auxin (C) Kinetin (D) Cytokinin

(121) Which hormone is not found in plants ?

(A) 2-4-D (B) GA_2 (C) Gibberrelin (D) IAA

(122) Who discovered auxin ?

(A) Thiamann (B) Went (C) Butlar (D) Skoog.

(123) Mostly dormant seeds are found in Leguminous plant, because .....

(A) less developed embryo (B) hard seed coat

(C) Absence of cytokinin (D) Absence of GA_3

(124) Rhizokalin is excess hormone, which is synthesied in .....

(A) Seed coat (B) Root (C) Leaf (D) Stem

(125) Climachteric effect means .....

(A) Ripening process of fruit (B) effect of environment of growth.

(C) Yellowing of leaves mostly in plant (D) Process of dropping of flowers of plant.

(126) A substane which regulate the other organs growth rate and originate from shoot apex and root apex of plant, what is it ?

(A) Enzyme (B) Hormone (C) Vitamin (D) Nutritional substance.

(127) During the whole year for getting the fruit of pineapple which treatment is given to plant ?

(A) NAA, 2, 4-D (B) Phynyl acetic acid (C) Cytokinin (D) IAA, IBA

(128) Evergreen trees are covergreen during whole year what is the reason of it ?
(A) No leaf fall is there.
(B) Leaf fall occur at regular time interval.
(C) During year proper amount of atmospheric moisture is available
(D) effect of cold atmosphere.

(129) Where is active cell division found ?
(A) In medulla region
(B) In cortex region
(C) In internode region
(D) In shoot apex and root apex.

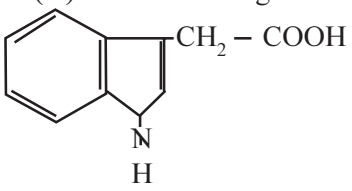
(130) It is a function of phytotron.
(A) Bombardment of electron
(B) release protons.
(C) Plants grows in restrictive atmosphere
(D) Mutation occur in plants.

(131) When itiolation is found in plants ?
(A) When plant grow in dark
(B) When deficiency occur due to mineralnutrition
(C) When more intensity of light
(D) When plant grow in green light

(132) Sowing of seed in ground at depth can not grow well because .....
(A) Lack of light
(B) lack of water
(C) lack of oxygen
(D) absence of nutrients.

(133) Dendrocronology means ..... .
(A) Secondary growth
(B) Development of shoot
(C) Biological Phenology
(D) To measure age of tree.

(134) Identity substance from given structure ?



(A) IAA (B) ABA
(C) GA_3 (D) CKN

(135) Which pair is not correct ?
(A) ABA - closistomata
(B) GA - Leaf fall
(C) CKN - cell division
(D) IAA - cell wall growth.

(136) How seedless tomato can be produce ?
(A) Sowing seed which is vernalized
(B) Seed is treated by phenyl mercuric acetate and then sowing
(C) Removal of another before pollen releases
(D) Plant is treated by auxin and gibberrelin.

(137) Gibberrelin stimulate seed germination because ..... .
(A) It stimulates celldivision
(B) It produces the hydrolase enzymes of.
(C) It produces ABA
(D) From the hard seed coat water is also absorbed.

(138) During seed germination amylase enzyme is produced in response to .....
(A) Auxin (B) Gibberrelin (C) Kinetin (D) Ethylene

(139) By the treatment of which hormone rosette pattern of cabbage is changed.
(A) Gibberrelin (B) 2-4-D (C) ABA (D) IAA.

(140) Who proposed the word cytokinin ?
(A) Yabuta (B) Brown (C) Latham (D) Fitting.

(141) Choose correct sentances from following.
(1) Senescence is removed by cytokinin.
(2) Auxin is responsible for the apical dominance.
(3) Ethylene is neccessary for seed germination.
(4) Gibberrelin induce immature leaf fall.
(A) 1 and 2 are correct
(B) 1 and 3 are correct
(C) 1 and 4 are correct
(D) 2 and 4 are correct

(142) A farmer having farm and he grew paddy crop but most of the leaves of paddy become yellow that decreases the yield. Thus, for the increase of crop yield which correct advice should be given to farmer ?

(A) Remove all yellow leaves and spread 2-4-D on green leaves.

(B) Fe and Mg are given to plant thus synthesis of chlorophyll is increased.

(C) Frequently irrigation given to the farm

(D) Give treatment of N_2 fertilizer with cytokinin.

(143) Who is proposed zeatin word ?

(A) Skoog (B) Letham (C) Miller (D) Yabota

(144) It is neccessary for flowering in short day plant.

(A) Gibberrelin (B) Cytokinin (C) Auxin (D) Ethylene

(145) For the ripening of fruits what is the suitable ratio of gases.

(A) 80% C_2H_4 and 20% CO_2 (B) 80% CO_2 and 20% CH_2

(C) 80% CO_2 and 20% O_2 (D) 80% NO_2 and 20% CO_2

(146) Which acid is gained from carotenoids ?

(A) Indole butyric acid (B) Indole - acetic acid

(C) Abscisic acid (D) Gibberrelic acid

(147) What is morfectin ?

(A) Synthetic growth regulator (B) Synthetic auxin

(C) Synthetic gibberrelin. (D) Synthetic drugs.

(148) Which hormone is opposite for gibberrelin action ?

(A) Zeatin (B) Ethylene (C) IAA (D) ABA

(149) Which is mismatch pair ?

(A) Auxin : growth (B) Gibberrelin : Gibberrela fungi

(C) Cytokinin : DNA of sperm of herring fish (D) ABA : flowering.

(150) In plants biological clock is regulated by which ?

(A) Phytochrome (B) Cytochrome (C) Gibberrelin (D) A and B both.

(151) Some of plants are getting 12 hours light and 12 hours darkness so flowering can occur. Now in these plants 12 hours of dark period is discontinued by introducing when light for mendetary period that breaks flowering response. Thus these types of plants are called .....

(A) Day neutral (B) Long day (C) Short day (D) Neutral darkness

(152) Some of the seedlings of tomato plants are kept in darkness for few days so most of the seedlings show white colour this phenomenon is known as .....

(A) Etiolation (B) Defoliation (C) Mutation (D) Dehydration

(153) Soyabean gives flowering in winter season, because .....

(A) It is long day plant (B) It is short day plant

(C) It is day neutral plant (D) It is terresterial plant.

(154) One plant gives flowering three times in a year october, January, and July, so what you tell about this situation ?

(A) It is photo sensible but not thermo sensible

(B) It is thermo sensible but not photo sensible

(C) It is not sensible for light and heat both.

(D) It is sensible for light and heat.

(155) When plant get light from only one side, plant gets curvature movement in direction of light. Which reason is responsible for this phenemenon ?

(A) Buds of stem is attractive towards light.

(B) For the phototynthesis light is essential.

(C) Auxin is present in shoot apex due to that more cell elongation occur.

(D) More neccessity of photorespiration.

(156) In which plant, treatment is given by gibberelin so male flowers are converted into female flowers?

(A) Papaya (B) Banana (C) Cucumber (D) Coccinia

(157) One hormone is induced early seed germination in barley plant, second hormone respossible for flowering in pineapple. and third hormone is inhibiting leaf senescence, give the name of hormones respectively.

(A) Auxin, Gibberrelin and cytokinin (B) Gibberrelin, cytokinin and Auxin.

(C) Gibberrelin, Auxin and cytokinin (D) Auxin, cytokinin and Gibberrelin

(158) How fruit ripening process induced ?

(A) When fruit become ripen, water level should decrease in ground

(B) Surrounding the fruits increase the ratio of N_2

(C) Surrounding atmosphere become warmly

(D) Artificial ethylene spread in atmosphere.

(159) Which pair is not related

(A) Adenine - Kinetin (B) Carotenoids - ABA

(C) Turpin - IAA (D) Gas - Ethylene.

(160) Which sentance is correct about phytochrome from following ?

(A) Phytochrome is plant growth regulator.

(B) Phytochrome is useful pigment in photosynthesis process.

(C) Phytochrome pigment stimulate growth process light based physiological. and development.

(D) Phytochrome is restricted protein, which control dark-reaction.

(161) What is useful to check the viabillity capacity of seed ?

(A) 2, 6, dichloro phenol indo phenol (B) 2,3,5, triphenyl tetra sodium chloride

(C) DMSO (D) Safranin

(162) In internode of stem for the cell enlargement which hormone is responsible ?

(A) IAA (B) CKN (C) GA (D) Ethylene

(163) Process in which floral bud is converted into flower.

(A) Induced locomotion (B) Palatonic locomotion

(C) Autonomous movement (D) Induced movement.

(164) Which hormone is responsible for differntiation and organogenesis ?

(A) Auxin (B) Gibberrelin (C) Cytokinin (D) ABA

(165) Why coconut water is useful for tissue culture method ?

(A) Auxin (B) Ethylene (C) cytokinin (D) Gibberrelins.

(166) By which physical method, seed coat becomes permeable for the growth of embryo ?

(A) Bending (B) Degradation (C) Vernalization (D) through scrapping

(167) IAA is produced from derivatives of which amino acid ?

(A) Tryptophane (B) Tyrosine (C) Phenyl alanine (D) Serine

(168) Which hormonal treatment is given to plast to stop immature leat fall ?

(A) IAA (B) GA_3 (C) CKN = Cytokinin (D) ABA

(169) The process of rolling of leaves after sunset is known as ..... .

(A) Phototropism (B) Photonasty (C) Phototaxis (D) Chemotaxis

(170) In which plant polyembryony of nucellus is found ?

(A) Citrus (B) Brassica (C) Malva (D) Yucca

(171) Which sentances are correct ?

(1) Gibbrerrelins is responsible for foolish plant.

(2) ABA is growth promotor hormone.

(3) Auxins and cytokinin are regulators of cell differentiation.

(4) Dormancy of axillary bud in cabbage is due to auxin.

(A) 1, 2 and 3 (B) 1 and 2 (C) 2 and 4 (D) 1 and 3

(172) Do mathes coulmns

Coulmn - 1		**Coulmn - 2**	
(I) C-Darwin and F-Darwin	(p)	Cytokinin	(A) : (I - s), (II - p), (III - r), (IV - t)
(II) Miller and Skoog	(q)	ABA	(B) : (I - p), (II - r), (III - s), (IV - q)
(III) Letham	(r)	Zeatin	(C) : (I - q), (II - p), (III - s), (IV - t)
(IV) Kurosuruva	(s)	Auxin	(D) : (I - t), (II - r), (III - p), (IV - s)
	(t)	GA	

(173) Mostly wax coating is done on apple, so that ..... .

(A) Colour can save against light

(B) Stop the aerobic respiration and O_2 cannot enter in fruit

(C) Save from injury and stop the entry of ethylene

(D) Apple is seen bright and beautiful.

(174) In flowering plant senscence is active developmental process which indicate growth and functions it is seen in ..... .

(A) Leaffall (B) Xylem vessel and vessel differntiated

(C) Annual plant (D) Floral parts

(175) In garden tendril of pea climb with support, this process is related to ..... .

(A) Thigmontasty (B) Thigmotaxis (C) Thermotaxis (D) Thigmotropism

(176) Flowering depending on cold, what is called ?

(A) Cryotherapy (B) Cryogenix (C) Cryoscopy (D) Vernalization

(177) What is the uses of clinostat instrument ?

(A) Osmosis measurement (B) Measure Velocity of growth

(C) Measure the rate of Photosynthesis (D) Measure the respiration rate

(178) What is useful for spinch vegetable to keep green for long period ?

(A) GA_3 (B) CKN = Cytokinin (C) IAA (D) ABA

(179) Which hormone is used in place of vernalization ?

(A) Auxin (B) Cytokinin (C) Gibbrerelin (D) Ethylene

Answers : (116-B), (117-C), (118-A), (119-A), (120-B), (121-A), (122-B), (123-B), (124-C), (125-A), (126-B), (127-A), (128-B), (129-D), (130-C), (131-A), (132-C), (133-D), (134-A), (135-B), (136-D), (137-B), (138-B), (139-A), (140-C), (141-A), (142-D), (143-B), (144-A), (145-A), (146-C), (147-A), (148-D), (149-D), (150-D), (151-C), (152-A), (153-B), (154-B), (155-C), (156-B), (157-C), (158-D), (159-C), (160-C), (161-B), (162-A), (163-C), (164-D), (165-C), (166-D), (167-A), (168-C), (169-B), (170-A), (171-D), (172-A), (173-B), (174-A), (175-D), (176-D), (177-B), (178-B), (179-C)

Plant growth and development

Growth
- Irreversible permanent increase in size of an organ or its parts or even of an individual call.
- Growth includes 3 phases division, elongation and maturation.

Development
- Includes all changes that an organism goes through during its life cycle from seed germination to senescence. It involves differentitation, dedifferentitation and redifferentitation of cells.

Photoperiodism
It is the flowering response of plants to periods of day/night phytochrome is the receptor pigment.

Vernalisation
Low temperature treatment of plants to decrease the time period of vegetative phase and to promote early flowering.

Growth Rate
Increased growth per unit time is termed as growth rate. It is of two types.

→ **Arithemetic Growth**
After mitotic cell division only one daughter cell continues to divide while other differentates and matures.

→ **Geometric Growth**
Both of the daughter cells of mitotic cell division keep on dividing regularly

Growth Curve
Graphical representation of growth rate of an organism or population.

→ Arithmetic growth curve linear graph obtained by plotting arithmetic growth.

→ Exponential growth curve curved graph showing initial slow growth and later rapid increase in growth which is obtained by plotting a geometrical growh

→ Sigmoid growth curve
S-shaped curve obtained in natural condition where exponential growth is limited or even ceased at later stages. It occurs so because in nature nutrient supply is limited.

Growth Regulators
Chemical substances also called as phytohormones, produced naturally in plants which regulate one or more physiological conditions.

Growth Promoters
Chemicals that increase growth of cells tissues or organs

→ Auxins : Weakly acidic hormone which induces apical dominance, photoropism and geotropism etc., Sequence of amino acids linked by peptide bonds.

→ Cytokinins : Basic hormone which promotes cell division and differentitation along with many other physiological effects.

→ Gibberellins : Weakly acidic hormones with gibbane ring structure. They mainly enhance seed germination, overcome seed dormancy and induce, bolting in rosette plants.

Growth inhibitors
Phytohormones which perform growth inhibiting activities like inducing dormancy and abscission.

→ Abscisic acid - Mildly acidic stress hormone that induces abscision and dormancy.

→ Ethylene - Gaseous hormone which enhances abscision and senesence induces fruit ripening.

36 Human Reproduction

Reproduction is one of the essential process in organisms. The reproduction is the mechanism by which continuity of generation is sustained and single cell duplicates its genetic material. In the process, genetic material is passed from generation to generation and reproduction maintains the life of the species.

Human is higher level vertebrate social, unisexual animal. So it exhibits external as well as internal sexual dimorphism.

Male (man)	External characters	Female (Woman)
- A pair of mammary gland is namesake only or undeveloped at ventral side of thoracic region.		A pair of mammary gland is well developed at the maturation at ventral side of thoracic region.
- Generally at the maturation time beard and moustache develops.		- At the maturation time beard and mustache is not seen.
- Its muscles are strong.		- Its muscles are comparatively weak.
- Age of musculine voice is deep		- Feminine voice is shrill.
	Internal characters	
- As male gonad a pair of testes occur.		- As female gonad, pair of ovaries occur.
- Testes are located in scrotum.		- Ovaries are located in the abdominal cavity.
- At the age of maturation by the process of spermatogenesis testes produce sperms.		- At the age of maturation by the process of oogenesis ovaries produce ova.
- From the testes, testosterone maleness hormone is released.		- From the ovaries, Estrogen and progesterone hormones, which has characteristic femaleness are released.

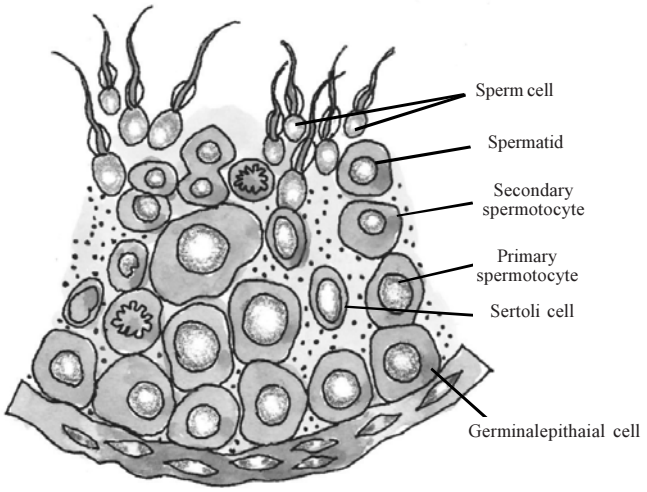
- **Reproductive system :** The male and female reproductive systems are formed by several types of organs. Let us discuss.
- **Male reproductive system :** The organs of the male reproductive system are a pair of testes, a pair of epididymis a pair of vas deferens, a pair of seminal vesicles, a prostate gland, a pair of bulbourethral gland (couper) urethra and penis.
- **Testes :** The testes are situated in the scrotal sac; which are located outside the body. The development of testes starts when they are within the abdominal cavity. Later they descend into the scrotal sac. The scrotal sac helps to maintain the low temperature of the testes. The temperature of scrotal sac is almost 3°C lower than the normal body temperature, which is essential for spermatogenesis.

- **Internal stucture of testes :** The testes are paired oval shaped glands measuring about 5cm in length and 2.5 cm in diameter. A fibrous connective tissue, the tunica albuginea surround each testis - It extends inward and divides each testis into several lobules. Each lobule contains one to four tightly coiled seminiferous tubules, which produces sperms.

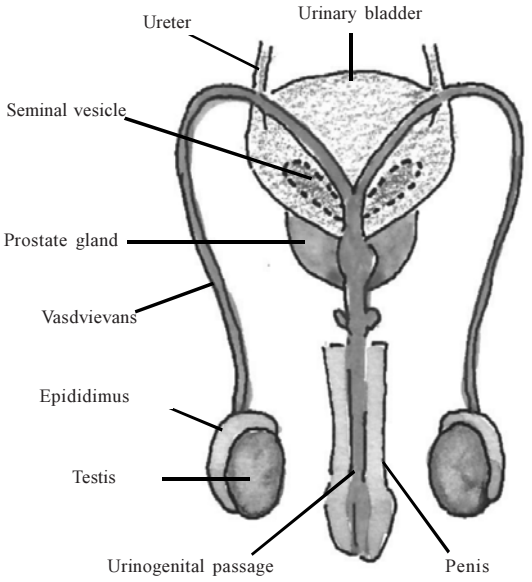
Each seminiferous tubule is lined by two types of cells. (1) Germinal cells : They are arranged in 4 to.8 layers. These cells divides many times and differentiate into sperm (2) Sertoli cells : They are placed in between the developing sperm cells and provides nutrition to the sperm cells.

The cells present in the interstitial space between seminiferous tubules are known as interstitial cells or leydig's cells. They secrete the male sex hormone testosterone.

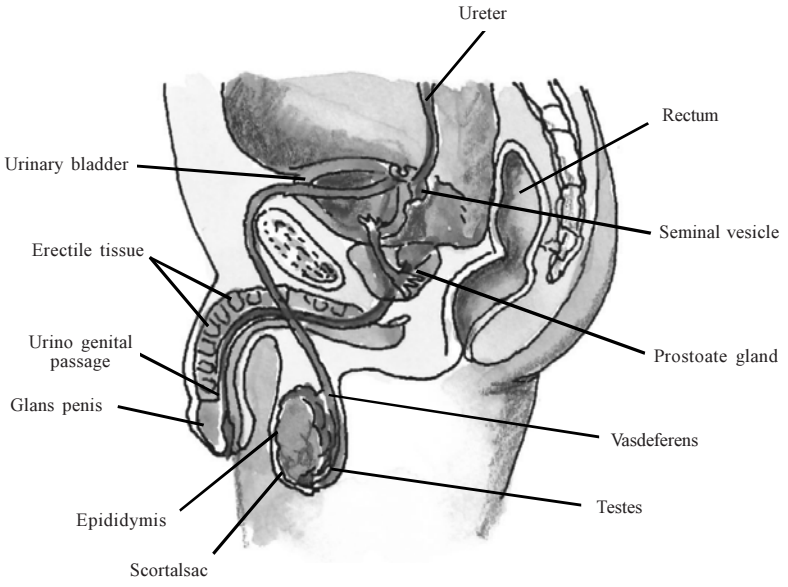
Seminiferous tubules of each lobe empty sperms into the vas efferentia. Sperms travel through vasa efferentia to the epididymis; which surrounds the external surface of the testis.



T.S. of Semini ferous tubule



Male Reproductive System (Front View)



Male Reproductive System (Lateral View)

- **Epididymis :** The epididymis is highly coiled tube, about 6 meters long. It provides temporary storage site for the immature sperms, in which sperms complete their maturation process and gain the ability to swim. When a male is sexually stimulated, the walls of the epididymis contract and sperms are transported into vas deferens.
- **Vas deferens :** The vas deferens is a tube which is about 45 cm long. It turns upward from the epididymis through the inguinal canal and enters the abdomen where it loops over the urinary bladder. This tube is connected with blood vessels and nerves. The distal end of vas deferens is expanded and in this region the seminal vesicle opens. Afterwards it is known as an ejaculatory duct. The duct of urinary bladder joins with ejaculatory duct. Now it is known as urethra.
- **Urethra :** In the structure of uretra ejaculatory duct is formed from the vas deferens and ureter linked, even seminal vesicle (accessory reproductive gland) also secrete in it that path or passage is known as urethra. It called as urinogental passage because ureter passes urine and vas deferens passes the male gametes in this path way. So it called urinogenetal path way. Now urethra passes through the penis. Before that secretions of prostate and bulbourethral of glands poured into urethra, which opens at the tip of the penis.
- **Penis :** The penis is a cylindrical organ located at the frontal region of scrotal sacs. It deposite sperms into the vagina of female. The distal end of the penis is slightly enlarged, called glans - penis. Glans - penis is covered by loosly fitting skin known as foreskin.
- **Internal structure of penis :** Internal structure of the penis is composed of three cylindrical masses of tissue bound together by fibrous tissue. Out of these two are located dorsally and one is located ventrally which contains the urethra. All three masses of tissue are sponge like and contain blood sinuses. It is filled with blood during sexual arousal. This causes the penis to enlarge and become rigid. This event is called erection.
- **Accessory reproductive glands of male / man :** The accessory reproductive glands of man includes paired seminal vesicles, a prostate gland and a pair of bulbourethral glands. Secretion these glands produce semen.
- **Seminal vesicle :** The seminal vesicle are located at the base of the urinary bladder. They produce 60% fluid volume of semen. Their thick and yellowish secretion is rich in sugar, vitamin-C and other substances; which nourishes the sperms. The duct of each seminal vesicle joins with vas deferens and form the ejaculatory duct. Thus sperms and seminal fluid enter in the urethra together.
- **Prostate gland :** The prostate gland is located at the posterior region of the urinary bladder. The secretion of prostate gland is milky. It plays a major role in activating sperms. It enters the urethra through several small ducs.
- **Bulbourethral (cauper)gland :** The paired bulbourethral glands are located beneath the prostate gland on lateral side of urethara. Like prostate gland, they secrete alkaline fluid which serves as a lubricant during sexual intercourse.

- **Semen :** Semen is milky white and sticky mixture of sperms and secretion of accessory glands. The relative alkalinity of semen as a whole (pH 7.2 - 7.6) helps to neutralize the acidic environment (pH 3.5-4.0) of the vaginal fluid thus protecting the delicate sperms and enhancing their motility. The average volume of semen for each ejaculation is 3 to 4 ml.

---

(1) Which option is improper with reference to reproduction ?

(A) It is essential process for livings.

(B) Continuity of life is maintained generation after generation due to reproduction.

(C) During reproduction haploid genetic material is maintained.

(D) Reproduction maintains characteristic of species, genetic continuity by genetic material generation after generation.

(2) Which type of animal is Human ?

(A) Highest level, quardiplage

(B) Highest level, vertebrate, quardiplage, social

(C) Highest level, vertebrate, quardiplage, unisexual, social

(D) Highest level, Bisexual, vertebrate

(3) Which statement is improper for external dimorphism in male ?

(A) Pair of developed mammary gland present on ventral side of thoracic region.

(B) In adult stage well developed beard and moustache appear on face.

(C) They have strong muscles.

(D) Voice is deep and harsh in adult

(4) Which of the following is improper for external dimorphism in female ?

(A) They have well developed pair of mammary gland on ventral side in thoracic region.

(B) In adult stage beard and moustache on face.

(C) Their muscles are weak.

(D) Their voice become shrill, faminine at adult stage.

(5) Which of the following character is not for internal sexual dimorphism of man.

(A) One pair of testis present as gonads.

(B) Testis are protected outside the body in scortal sac.

(C) It releases aldosterone in the form of male hormone from testis.

(D) In adult stage by the process of spermatogenesis forms sperm from testis.

(6) Which of the following is not character of female internal sexual dimorphism ?

(A) Pair of ovary is present as gonad.

(B) One pair of ovary present in abdominal cavity.

(C) Female sex hormone estrogen and testosterone secreted from ovary.

(D) In adult form, by the process of oogenesis, ovules are formed by ovary.

(7) Which is correct sequence for main reproductive organs in male reproductive system?

(A) Pair of testis - Pair of epididymis - pair of vas deferens - urinogenital passage - Penis

(B) Pair of testis - Pair of vas deferens - Pair of epididymis - urinogenital passage - Penis

(C) Pair of epididymis - Pair of vas deferens - Pair of testis - urinogenital passage - Penis

(D) Pair of vas deferens - pair of epididymis - pair of testis - urinogenital passage - Penis

(8) Which arrangement is correct for accessory reproductive organs in male reproductive system ?

(A) Pair of seminal vesicle - one prostate gland - Pair of bulbourethral gland

(B) Pair of seminal vesicle - one prostate gland one bulbourethral gland

(C) Pair of seminal vesicle - Pair of bulbourethral gland - one prostate gland

(D) One pair of bulbourethral gland - one pair seminal vesicle - One prostate gland

(9) Which sentence is proper for scrortal sac with reference of testis ?

(A) Scortal sac is not situated outside body, below abdominal region.

(B) Testis are not protected in scortal sac.

(C) In scortal sac, temperature of testis is less than body temperature, sperms are produced by testis.

(D) Scortal sac inhibits activeness of testis.

(10) What are main functions of scortum ?

(A) Protection of sperms and maintaining shape

(B) Protection of testis, maintaining shape of testis

(C) Protect the testis, lowering its temperature

(D) Protection of testis, maintain amount of water

(11) Which process occurs in testis and for what purpose ?

(A) Process of development of testis, for maturity

(B) Process of growth of testis, for maturity

(C) Testis show spermatogenesis, for formation of sperms

(D) Testis show spermatogenesis, for maturity of sperms

(12) Which is proper statement for the size of testis ?

(A) Oval, pink shaped, 5 cm diameter - 2.5 length

(B) Oval, Pink coloured, 5 cm length 2.5 cm diameter

(C) Round, Red coloured, 5 cm diameter, 2.5 cm length

(D) Round, Red coloured, 5 cm length, 2.5 diameter

(13) Which structure is present outside the testis ?

(A) Structure of tunica albugenia, formed of white fibrous connective tissue.

(B) Structure of tunica albugenia, formed of fibrous connective tissue.

(C) Tunica albugenia, formed by yellow fibrous connective tissue.

(D) Structure of tunica albugenia, formed by white and yellow connective tissue.

(14) Spermatic cord is formed by which structure ?

(A) Fibrous (B) Hollow structure (C) Segmentated (D) Cellular structure

(15) Which vessels are found in the structure of spermatic cord ?

(A) Epididymis tubule (B) Vas deferens (C) Seminiferous tubules (D) Seminal vesicle

(16) Which type of cells are present in seminiferous tubules ?

(A) Germinal epithelium, somatic cells (B) Somatic cells, germinal epithelium

(C) Spermatogonia, sertoli cells (D) Somatic cells, sertoli cells

(17) The cells present in the interstitial space between seminiferous tubule are known as ..... .

(A) sertoli cells (B) gap cells (C) interstitial cells (D) internal cells

(18) What is function of sertoli cells ?

(A) Nutrition for testis (B) Nutrition for vas deferens

(C) Nutrition for sperms (D) Nutrition for developing sperms

(19) What is main function of interstitial cells / Leydig's cell ?

(A) Secretion of male sex hormone - testosterone.

(B) Secretion of male sex hormone androgens.

(C) Development of male secondary sex characters.

(D) Develop sexual characters.

(20) What is specific location of vas deferens ?

(A) Seminiferous tubules, in the vas deferens.

(B) Between seminiferous tubule and ejaculatory duct.

(C) Between seminiferous tubule and epididymis.

(D) Between seminiferous tubule and testis.

(21) What is proper sentence with reference of epididymis ?

(A) One pair of straight tubule, 12 mt. long (B) One pair of straight tubule, 6 mt. long

(C) One pair of convoluted tubule, 12 mt. long (D) One pair of convoluted tubule 6 mt. long

(22) Which tubules are present at both ends of epididymis ?

(A) Vas efferentia, Inguial tubule (B) Vasa efferentia, vas deferens

(C) Vasa efferentia, urinogenital tubule (D) None of above

(23) Epididymis has storage function for ..... .

(A) Storage of immature sperm cells (B) Storage of mature sperm cells

(C) Temporary storage of immature sperms (D) Permenant storage for immature sperms

(24) Which ability is gained by sperm cells in epididymis ?

(A) To obtain nutritive substances (B) Swimming

(C) Ciliary movement (D) Amoeboid movement

(25) When is sperm cells transported from epididymis to vasa efferentia ?

(A) Due to contraction of wall of epididymis, sperm cells gain buoyancy and transported to vasa efferentia.

(B) Secretion of epididymis transports sperm cells to vasa efferentia.

(C) Hormones of epididymis, gives motility to sperm cells into vasa efferentia.

(D) Enzymes of epididymis gives motility to sperm cells to vasa efferentia.

(26) Which tubule connects scrotum and abdominal cavity in male ?

(A) Epididymis (B) Vasa efferentia (C) Inguinal tubule (D) Vas deferens

(27) Inguinal canal is present between, which two tubules ?

(A) Epididymis, vasa efferentia (B) Epididymis, vas deferens

(C) Epididymis, collecting tubule (D) Epididymis, Ejaculatory duct

(28) Which sentence is proper for vasa efferentia ?

(A) Connected with inguinal, 45 cm length, folds around urinary bladder, results in ejaculatory duct.

(B) Connected with ejaculatory duct, 25 cm in length form loop around urinary bladder, results in ejaculatory duct.

(C) Connected with epididymis, 45 cm length, loop around urinary bladder and form ejaculatory duct.

(D) Connected with vasa efferentia, 45 cm length, loops around urinary bladder and form ejaculatory duct.

(29) Which structures are connected with spermatic cord ?

(A) Blood capillaries + lymph capillaries

(B) Blood capillaries + nerves + vasa efferentia

(C) Blood capillaries + lymph capillaries + vasa efferentia

(D) Blood capillaries + nerves + lymph capillaries

(30) When is ejaculatory duct formed from vasa efferentia ?

(A) When secretion of prostate mixes (B) When secretion of Bulbourethral mixes

(C) When secretion of seminal vesicle mixes (D) When secretion of cowper's gland mixes

(31) Which structures are connected with urinogenital passage ?

(A) Vas deferens + seminal vesicle

(B) Vas deferens + seminal vesicle + prostate gland.

(C) Vas deferens + seminal vesicle + prostate gland + ureter

(D) Vas deferens + seminal vesicle + prostate gland + ureter + bulbourethral gland

(32) Urinogenital passage is connected with structure at both ends ..... .

(A) epididymis, penis (B) vasa efferentia, penis

(C) vasa efferentia, penis (D) ejaculatory duct, penis

(33) Structure of penis is formed of ..... .

(A) fibrous tissue (B) erectile tissue

(C) erectile tissue and internal cavities (D) eractile tissue, internal cavities and ureter passes through it

(34) Which sentence is proper for seminal vesicle ?

(A) One pair, accessory organ for storage, at the base of urinary bladder.

(B) One pair, accessory organ for secretion, at the base of urinary bladder.

(C) One pair, accessory organ for secretion, upper side of urinary bladder.

(D) One pair, accessory organ for secretion, lateral side of urinary bladder.

(35) Which are secretory components of seminal vesicle ?

(A) Thick, jelly like fluid + yellowish sugar

(B) Thick, jelly like fluid + yellowish sugar + vitamin C

(C) Thick, jelly like fluid + yellowish sugar + vitamin C + other substances

(D) Thick, jelly like fluid + yellowish sugar + vitamin C + other substances + vitamin B_1 , B_2, B_{12}

(36) Which secretion is essential for entry of sperm cells into urinogenital passage ?

(A) Secretion of seminal vesicle (B) Secretion of prostate gland

(C) Secretion of bulbourethral gland (D) Secretion of ejaculatory duct

(37) Secretion of prostate gland is of which type ?

(A) Colourless, without viscousity (B) Red, without viscousity

(C) Milky, viscous (D) Pink, viscous

(38) What is effect of secretion of prostate gland on sperm cells ?

(A) Reduces activeness (B) Increases activeness

(C) Increase inactivity (D) Maintains inactivity

(39) What is exact location of pair of bulbourethral gland ?
(A) Ventral side of urinary bladder (B) Lateral side of urinary bladder
(C) Ventral side of urinogenital passage (D) Lateral side of urinogenital passage

(40) Secretion of which gland increases motility of sperm cells ?
(A) Seminal vesicle (B) Prostate (C) Bulbo urethral (D) Cowper's gland

(41) Secretion of which gland serves as lubricant during copulation ?
(A) Seminal vesicle (B) Prostate (C) Bulbourethral (D) Ejaculatory duct

(42) Secretion of which gland is useful in transport of sperms into vagina without friction ?
(A) Seminal vesicle (B) Prostate (C) Bulbourethral (D) Vasa efferentia

(43) Which type of fluid is semen ?
(A) Colourless, acidic, white (B) White , oily, alkaline fluid
(C) White, sticky, alkaline, with sperm cells (D) White, sticky, acidic with sperm cells

(44) What is approximate PH value of semen ?
(A) 7.0 to 7.2 (B) 7.2 to 7.6 (C) 7.4 to 7.8 (D) 7.6 to 7.8

(45) Approximate PH in vaginal passage is ..... .
(A) 1.5 to 2 (B) 2.6 to 2.5 (C) 2.5 to 3.0 (D) 3.5 to 4.00

(46) Due to which process PH value become neutral in vaginal passage ?
(A) Growth (B) Development
(C) Differentiation (D) Ejaculation of semen during copulation

(47) Which processes become easy with removal of acidity in vaginal passage ?
(A) Transport of delicate sperm cells and maintain its shape.
(B) Prctection of delicate sperm cells and increage in its motility.
(C) To keep sperms alive, increases motility of sperm cell.
(D) To protect and increase motility of sperm cell.

(48) What is approximate volume of semen ?
(A) 1 to 2 cm (B) 1 to 2 ml (C) 3 to 4 cm (D) 3 to 4 ml

(49) Which option is proper for proper passage for sperm cells in male reproductive system ?
(A) Testis - Epididymis - Vasa efferentia - urinogenital passage - Penis - Vaginal passage
(B) Seminiferous tubule - Vasa efferentia - Epididymis - Inguinal canal - Vasa deferens - ejaculatory duct - urinogenital passage - Penis - Vaginal passage
(C) Seminifercus tubule - Vasa efferentia - epididymis - Inguinal canal - urinogenital passage - Penis - Vagina
(D) Testis - Epididymis - Vasa efferentia - Vasa deferens ejaculatory duct - urinogenital passage - Penis - Vagina

Answers : (1-C), (2-C), (3-A), (4-B), (5-C), (6-C), (7-A), (8-A), (9-C), (10-C), (11-C), (12-B), (13-B), (14-C), (15-C), (16-C), (17-C), (18-D), (19-A), (20-C), (21-D), (22-B), (23-C), (24-B), (25-A), (26-C), (27-B), (28-A), (29-B), (30-C), (31-D), (32-D) (33-D), (34-B), (35-C), (36-A), (37-C), (38-B), (39-D), (40-B), (41-C), (42-C), (43-C), (44-B), (45-D), (46-D), (47-D), (48-D), (49-B)

- **Female Reproductive System :**

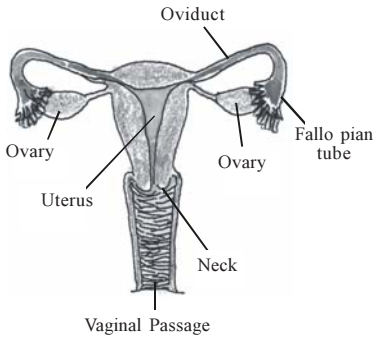
The female reproductive system consists of a pair of ovaries, fallopian tubes or oviducts or uterine tubes, uterus, vagina and external genitalia or vulva or pudendum. The mammary glands are also included in female reproductive system as an accessory organs.

- **External and internal structure of ovary :**

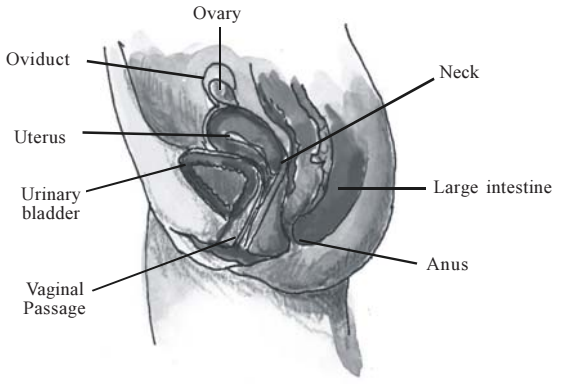
The ovaries are paired glands with the size and shape of almonds. It is about 3 cm long, 2 cm wide and 1 cm thick. They are situated in the upper pelvic cavity, one on each side of the uterus. The ovaries

maintain their position by a series of ligaments. Each ovary contains a hilus, the entry point for blood vessels and nerves. In sectional view, ovary consists of the following parts.

- **Germinal Epithelium :** It is a layer of simple cuboidal epithelium which covers ovary.
- **Tunica Albuginea :** It is a capsule of collagenous connective tissue immediately after the germinal epithelium.
- **Stroma :** This is a region of connective tissue deep to the tumica albuginea. It is composed by outer cortex and an inner medulla. The cortex contains ovarian follicles.
- **Ovarian follicles :** It consist of ova and their surrounding tissues in various stages of development. After that ovarian follicle gets maturation stage.
- **Graffion fallicle :** It consists of mature ovum and its surrounding tissues.
- **Corpus Luteum :** Graffion follicle after ovulation produces glandular body. It secrete hormone progesterone. Thus, ova are produced by ovaries and female sexualy hormones estrogen and progesterone are secreted also.



Female reproductive system (Front view)



Female reproductive system (Lateral view)

- **Oviduct :** A pair of oviduct or fallopian tube, are included in female reproductive system. Uterine or fallopian tube or oviduct transports ova from ovaries to uterus. It is 10cm long and situated between the folds of the ligaments of the uteras. It has vannel-shaped open end, called infundibulum (Oviducal funnel) which lies very close to the ovary but not attached to it. About once a month ovum is released from ovary near the infundibulum of uterine tube; this process is called ovulation. The collected ovum run forward in uterine tube by ciliary action. The uterine tube from side runs forwards and becomes associated with the uterus. If ovum is fertilized by sperm. Phenomenon is observed in the parts after oviducal funnel.
- **Uterus :** The uterine tubes from both side join and form the uterus. The uterus is situated in between the urinary bladder and rectum. It is inverted pear-shaped and thick walled muscular structure, where menstruation, implantation of a fertilized ovum and development of the embryo occurs. The wall of the uterus is made up of three layers.

(i) **Endometrium :** It is an inner most layer. If fertilization occurs, the fertilized egg is implanted here and resides there for further development. If the woman does not conceive, the endometrial lining slough off periodically, usally after every 28 days.

(ii) **Myometriums :** It is bulky middle layer of the uteras. It is composed of bundles of smooth muscle. This layer plays an active role during the delivery of a baby.

(iii) **Epimetrium :** It is the outermost layer of the uterus.

- **Cervix :** The distal narrow end of the uterus is called cervix, which connect uterus to the vagina.
- **Vagina :** Vagina is a thin walled tube, it lies between urinary bladder and rectum and it extends from cervix to the outside of the body. Vagina provides a passageway for the delivery and for the menstrual flow to leave the body. The distal end of the vagina is partially closed by a thin fold of the mucosal membrane called the hymen. It can break at anytime either due to vigorous exercise or due to other reasons.
- **External genitalia of female/Accessory genital organs of female :**

  Female have external genitilia. They are mons pubis, labia majora, labia minora and clitoris.
- **Mons pubis :** It is a cushion of fatty tissues which is covered by skin and pubic hair.
- The labia majora are folds of tissue, which are located below the mons pubis and surrounds the vulva. The labia minora are also a fold of tissue under the labia majora. The clitoris is a tiny finger like structure which lies at the upper junction of two labia minora. It contains erectile tissue and is considered equivalent to the male penis. The clitoris differs from the penis in that it lacks a reproductive duct.

  Mammary glands are present in both sexes, but normally they are functional in females only. The biological role of the mammary glands is to produce milk and nourish a newborn baby. In the puberty stage, the female mammary glands increase in size, this is stimulated by sex hormone estrogen. They are also considered as accessory reproductive glands.

---

(50) Which are main argans in female reproductive system from the given below ?

(A) Pair of ovary - pair of oviduct - uterus - pair of vaginal passage

(B) Pair of ovary - pair of oviduct - pair of uterus - pair of vaginal passage

(C) Pair of ovary - pair of oviduct - uterus - vaginal passage - external genital organs - mammary gland

(D) Pair of ovary - pair of oviduct - uterus - vaginal passage

(51) Which of the following sentence is not proper for female gonads ?

(A) Pair of ovary, almond shaped, glandular 3 cm length, 2 cm width, 1 cm thickness

(B) Pair of testis, almond shaped, glandular 3 cm length, 2 cm width, 1 cm thickness

(C) Pair of ovaries on lateral sides of pelvic cavities attached with one one - muscle cord

(D) Ovary has hilum, which is believed to entry place for blood capillaries and nerves

(52) What is exact location of Ovary ?

(A) Pelvic region, behind abdominal cavity one - one on lateral side

(B) Pelvic region, behind abdominal cavity one- one, each in the middle.

(C) Pelvic region, behind abdominal cavity one - one each on ventral side.

(D) Pelvic region, behind abdominal cavity one - one each on both lateral side

(53) Which is outer most layer in medulla of ovary ?

(A) germinal epithelium (B) Tunica albuginea

(C) Squamous epithelium (D) Nutritive layer

(54) Which tissue layer is found in tunica albuginea ?

(A) Fibrous connective tissue (B) Yellow fibrous connective tissue

(C) Connective collagen tissue (D) liquid connective tissue

(55) Which cells are included in ovarian follicle?

(A) Ovules

(B) Ovules + Follicular epithelial cells

(C) Ovules + Follicular epithelial cells + squamous epithelium

(D) Ovules + Follicular epithelial cells + columnar cells

(56) Developmental stages of ovarian follicle are shown by which hormone and gland ?

(A) LTH, Anterior Pituitary (B) GTH, Ant. Pituitary

(C) LH, Anterior Pituitary (D) Estrogen , Ant. Pituitary

(57) Which are developmental stages of ovarian follicle, which at the end shows ovulation ?

(A) Primary ovarian follicle - primary follicle - secondary follicle - graffian follicle

(B) Primary ovarian follicle - secondary follicle - primary follicle - graffian follicle

(C) Primary follicle - graffian follicle - primary ovarian follicle - secondary follicle

(D) Primary follicle - graffian follicle - secondary follicle - primary ovarian follicle

(58) Which ovarian follicle are found in secondary oocute ?

(A) Secondary follicle, primary follicle (B) Primary ovaridn folicle, primary follicle

(C) Primary follicle, secondary follicle (D) Graffian follicle, secondary follicle

(59) Where arrangement of graffian follicles occurs in human ?

(A) Ovary (B) Oviduct

(C) Uterus (D) Bulges out on ovary

(60) Which structure develop due to rupturing of graffian follicle ?

(A) Corpus albunics (B) Corpus metsuosa (C) Corpus luteum (D) Corpus cortex

(61) Which hormone is released due to rupturing of graffian follicle ? Which type of hormone it is ?

(A) Estrogen, steroid (B) Progesterone, catecholamine

(C) Progesterone, steroid (D) Estrogen, catecholamine

(62) Which hormone level increases due to development of corpus luteum ?

(A) GTH - LH (B) GTH - estrogen

(C) Estrogen - progesterone (D) LH - Progesterone

(63) Which sentence is not proper for fallopian tube ?

(A) Pair of fallopian tube is present in female reproductive system.

(B) It transports ovules from ovary to uterus.

(C) It is arranged between 10 cm - long and folds of uterus muscle joints.

(D) Its anterior portion is narrow, from it ovules are transported.

(64) What is oviducal funnel ?

(A) It is filamentous, funnel shaped structure on anterior. region of oviduct.

(B) Oviducal funnel is structure connected with oviduct.

(C) Funnel shaped structure between oviduct and uterus.

(D) Filamentous structure between oviduct and uterus.

(65) How are eggs transported by oviduct ?

(A) Amoeboid movement (B) Cilliary movement

(C) muscular movement (D) Oscillatory movement

(66) How are of oviduct and uterus connected ?

(A) From Anterior region (B) From Ventral region

(C) From Posterior region (D) From Lateral side

(67) Where is fertilization of ovulm takes place by sperms ?

(A) Posterior region of oviduct (B) Oviducal funnel

(C) Oviduct region after oviducal funnel (D) Middle of oviduct

(68) Which option is proper for location and shape of uterus ?

(A) Above rectum and urinary bladder, pear shaped.

(B) Below urindary bladder and rectum, pear shaped.

(C) Between urinary bladder and rectum, inverted pear shaped.

(D) Lateral side of urinary bladder and rectum, inverted pear shaped.

(69) Which sentence is not proper for uterus ?

(A) Between two oviduct, in between urinary bladder and rectum, inverted pear shaped organ.

(B) It is organ with thick muscular wall, posterior opens in cervix region which is narrow structure.

(C) It forms three layered wall, and structure connected with menstrual cycle, embryo implantation.

(D) It is concerned with fertilization and abortion.

(70) Which layer of uterus, normally breaks by 25 to 30 days ? Which is hormone deficiency controlls it ?

(A) Endometrium (B) Myometrium (C) Ectometrium (D) Mesometrium

(71) Which layer is associated with implantation of embryo in uterus ?

(A) Endometrium (B) Myometrium (C) Epimetrium (D) Ectometrium

(72) Which is most developed layer of uterus for embryo implantation ?

(A) Endometrium (B) Myometrium (C) Epimetrium (D) Ectometrium

(73) Which layer induces contraction and relaxation of uterus during child birth ?

(A) Endometrium (B) Myometrium (C) Epimetrium (D) Ectometrium

(74) Which organs are related with menstrual cycle and delivary ?

(A) Uterus, vaginal passage (B) Oviduct, oviducal funnel

(C) Oviducal neck (D) Neck, oviducal funnel

(75) Which is proper sentence for hymen membrane ?

(A) Partially covers mucosal part of vagina present at the far end of vaginal passage.

(B) Thick membrane connected at the far end of vaginal passage.

(C) Membrane present at the far end of vaginal passage, conctractile in nature.

(D) Posterior membrane formed by angular membrane.

(76) Which option is proper for external genital organs of female ?

(A) Labia majora monspubis, Labiaminora, clitoris

(B) Monspubis, labia majora, Labia minora, clitoris

(C) Monspubis, 1 pair labia majora, labia minora, clitoris

(D) Monspubis, one pair labia majora, labia minora, pair clitoris

(77) Which sentence is proper for monspubis ?

(A) It is accessory organ for vaginal passage, pillow like structure with pubic hair and adipose tissue.

(B) It is accessory organ for vagina, pillow like, white connective tissue containing structure.

(C) Main organ for vaginal passage, structure covered with pubic hair, pillow like formed of adipose tissue.

(D) Main organ for vaginal passage, white fibrous tissue, pillow like structure.

(78) Which is proper sentence for clitoris ?

(A) It is small finger like, attached at upper part of labia minora, having contractile tissue, comparative structure with penis.

(B) It is small finger like, structure present at upper region of labia minora, having non-striated muscles, not comparative to penis.

(C) It is spongy, fibrous structure, attached with labia majora, having striated muscles comparative with penis.

(D) It is spongy fibrous structure, independent of labia majora, having striated muscles comparative with penis.

(79) Which sentence is proper for function of mammary gland ?

(A) It is accessory reproductive organ, for breast feeding of baby.

(B) Main reproductive organ, storage of adipose tissue.

(C) Main reproductive organ, contraction by striated muscle.

(D) Breast feeding, production of antibiotics

Answers : (50-D), (51-B), (52-A), (53-B), (54-C) , (55-B), (56-B), (57-A), (58-D), (59-D), (60-C), (61-C), (62 -C), (63-D), (64-A), (65-B), (66-D), (67-C), (68-C), (69-D), (70-A), (71-A), (72-B), (73-B), (74- A), (75-A), (76-A), (77-A), (78-A), (79-A)

Gametogenesis : The gametogenesis is the process of gamete formation in the sexually reproducing animals. The animals have two types of cells in their body : Somatic cells and Germinal cells. Somatic cells form various organs of the body; and divide by mitotic division. The germinal cells produce gametes by successive mitotic and meiotic divisions. The male gamete is known as spermatozon or sperm and female gamete is known as ovum or egg. The process of sperm production is known as the the spermatogenesis and the process of production of ovum is known as oogenesis.

Spermatogenesis : The process of spermatogenesis occurs in male gonads or testes; continues process of miltiplication, growth and maturation all the end of this process sperms are formed so it is known as spermatogenesis, for the sake of convenience this process can be studied in two different stages.

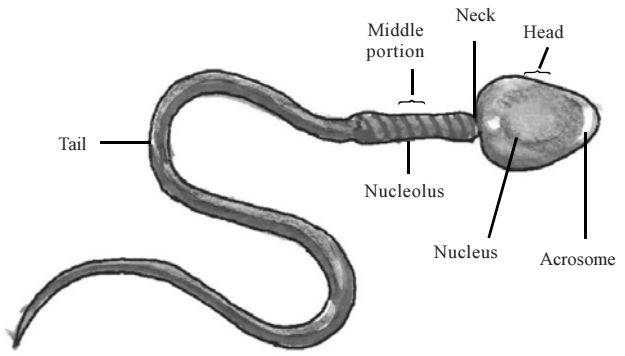
(1) Formation of spermatids (2) Spermiogenesis.

(1) Formation of spermatids : The male germinal cells, which produce the sperms are known as primary germinal cells. These primary germinal cells pass through the following three phases for the formation of spermatids.

(i) Multiplication phase : The undifferentiated germ cells or primary germinal cells, contain large size and chromatin rich nuclei. These cells multiply by mitotic division and produce spermatogonia. Each spermatogonium is diploid.

(ii) Growth phase : In the growth phase the spermatogonia accumulate large amount of nutrient and chromatin material. Now each spermatogonium is known as the primary spermatocytes.

(iii) Maturation phase : Now primary spermatocytes are ready for first meiotic or maturation division. By this, two secondary spermatocytes are formed. Each secondary spermatocyte is haploid. Each secondary spermatocyte passes through the second meiotic or maturation division and produces two spermatids. Thus, by a meiotic or maturation division a diploid spermatogonium produces four haploid spermatids. These spermatids can not act directly as the gametes, so they have to pass through spermiogenesis.



Mature sperm cell

(2) Spermiogenesis : The metamorphosis of differentiation of the spermatids into the sperms is known as spermiogenesis. Three changes occurs in the spermatids.

(i) Changes in the nucleus : The nucleus loses water, shrinks and assumes ovoid and laterally flattened shape. The RNA and the nucleolus are greatly reduced. The DNA become more concentrated.

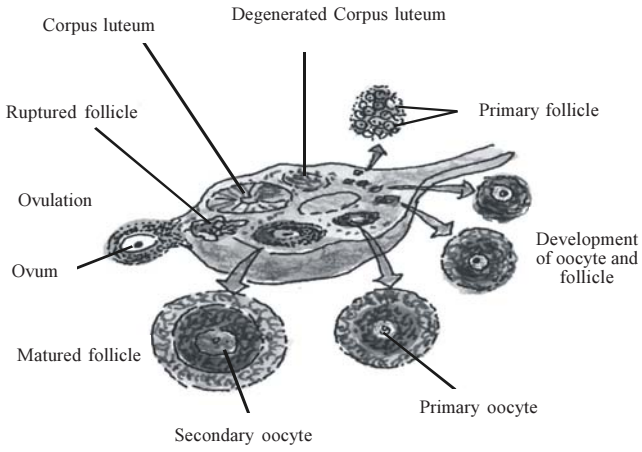
(ii) Acrosome formation : The acrosome occurs at the anterior side of the sperm and contains the protease enzyme hyaluronidase which helps it to penetrate into in the ovum. The acrosome is formed by the Golgi complex. It is concentrated near the anterior end of the sperm nucleus. One or two vacuoles of the Golgi complex become large and occupy the place between the tubes of Golgi complex. Soon after, a dense granule known as proacrosomal granule develops in the vacuole. The proacrosomal attaches with the anterior end of the nucleus and enlarges, which is now known as acrosome.

(iii) **Centrioles :** The two centrioles of the spermatids become arranged one after the other behind the nucleus. The anterior one is known as the proximal centriole and the posterior one is known as the distal centriole. The distal centriole changes into the basal bodies and form axial filament of the sperm. The mitochondria fuse together and twist spirally around the axial filament. These form a middle piece of the sperm.

- **Oogenesis :** Multiplication phase accurs by the germined epithelial cells of ovary due to mitosis and nutritive material include in cells in growth phase. Then by the meiosis division in maturation phase take place so unicellular ovum can be formed. The oogenesis is completed in three successive phases (1) Multiplication (2) Growth (3) Maturation division.

(1) **Multiplication phase :** The germinal cells of ovary divide repeatedly to form oogonia. Germinal cells divide repeatedly to form the oogonia or eggmother cells. By the repeated division, there is increase in the number of cells so this phase is called multiplication. In this phase germinal cells and oogonia both are diploids (2n).

(2) **Growth phase :** This phase is comparatively longer than the spermatogenesis. In this phase, the size of the primary oocyte increases enormously. In the primary oocyte, fats and proteins present in the form of yolk. The cytoplasm becomes rich in RNA, DNA, ATP and enzymes, moreover mitochondria, golgi complex, ribosomes etc. are also concentrated in it. During this phase, changes also occurs in the nucleus of primary oocyte and it becomes large due to the increased amount of nucleoplasm. When the growth of the cytoplasm and nucleus of primary oocyte is completed, it becomes ready for the maturation phase.

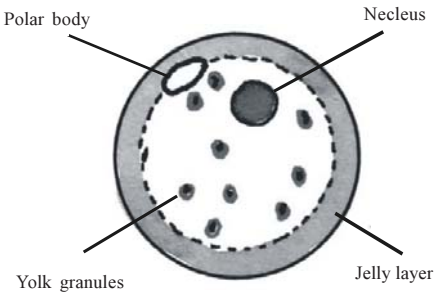


T.S. of Ovule

(3) **Maturation phase :** The maturation phase is accompanied by the maturation division or meiotic division (meiosis). This division is quite different from the meiotic division of spermatocyte. Here, after the first division primary oocyte divides unequally to form one large sized haploid secondary oocyte and one small sized haploid first polar body.

Ovulation takes place at the secondary oocyte stage only and enters into oviduct. When sperm

penetrates into secondary oocyte it undergoes unequal second meiotic division and produces second polar body and an ovum. In the same way the first polar body undergoes equal second meiotic division and produces two second polar bodies. However, if the sperm does not penetrate into the secondary oocytes, it simply deteriorates without completing meiosis to form the ovum. The mature ovum has a cell like structure.



Secondary oocyte

(80) Which type of cells are present in animal body ?

(A) Epithelial cells, connective tissue, blood cells, bone cells, reproductive cells

(B) Epithelial cells, conenective tissue , blood cells, bone cells, reproductive cells, nerve cells

(C) Epithelial cells, connective cells, blood cells, bone cells, reproductive cells, nerve cells, nutritive cells

(D) Somatic cells, reproductive cells

(81) Which sentence is improper for somatic cells ?

(A) Cell division of somatic cells is of mitosis type.

(B) Various organs of animal body is formed due to cell division of somatic cells.

(C) Cell division of somatic cells , doubles number of cells in animal body.

(D) Somatic cells shows structure of reproductive organs and reproductive cells.

(82) Which cells are formed by cell division of germinal epithelial cells in seminiferous tubule.

(A) Somatic cells (B) Spermatogonia (C) Oocytes (D) Male reproductive cells

(83) Arrange given sentences in sequence for spermatogenesis . Select proper option.

(1) Germinal epithelial cells produces male gametes by cell division in seminiferous tubules.

(2) Spermatids are always haploid, without morphogenesis.

(3) Germinal epithelial cells passes through multiplication, growth and maturation division.

(4) Spermatids results in mature sperm cells by morphogenesis .

(5) Germinal epithelial cells first divides mitotically and than meiotically.

(6) Male reproductive cells have various enzymes at their anterior region.

(A) 1, 2, 3, 4, 5, 6 (B) 1, 2, 3, 5, 4, 6 (C) 1, 3, 5, 2, 4, 6 (D) 2, 4, 6, 1, 2, 3

(84) What is known as formation of spermatids ?

(A) Formation of spermatids from spermatogonia.

(B) Development of spermatid from primary spermatocyte.

(C) Spermatids formed from primary germinal cells.

(D) Formation of spermatids from secondary spermatocyte.

(85) What is very specific matter about spermatocyte cells ?

(A) Undifferentiated, large size, nucleus having chromatin .

(B) Differentiated, large size, nucleus having chromatin.

(C) Undifferentiated, small size, nucleus having chromatin.

(D) Differentiated, small size, nucleus with chromatin.

(86) Which type of spermatocyte cells are there during spermatogenesis, diploid (2n) ?

(A) Sperm cells, spermatids

(B) Spermatocyte, primary spermatocyte, spermatogonia.

(C) Spermatocyte, primary spermatocyte, secondary spermatocyte.

(D) Spermatocyte, secondary spermatocyte, primary spermatocyte.

(87) What are sequential stages in metamorphosis of sperm cells ?

(A) Development of centriole, formation of acrosome, change in nucleus

(B) Development of centriole, changes in nucleus, formation of acrosome

(C) Changes of nucleus, acrosome formation, centriole form

(D) Formation of acrosome - formation of centriole - changes in nucleus

(88) Which sentence is proper for changes in nucleus ?

(A) Nucleus of spermatid enlarges, become turgid, Decrease in RNA and nucleolus, DNA concentration increases

(B) Nucleus of spermatid enlarges, become turgid, RNA and nucleolus amount reduces as a result DNA concentration increases.

(C) Nucleus of spermatid constricted, loss of turgidity, decrease of more amount of RNA and nucleolus, DNA concentration increases.

(D) Nucleus of spermatid constricted, loss of turgidity, increases of amount of RNA and nucleolus, DNA concentration decreases.

(89) Which sentence is not proper for formation of Acrosome ?

(A) At the anterior portion of sperm cells enzymes present, hyaluronidase dissolves ovary

(B) Golgi body form acrosome, it is accumulated at anterior part of sperm cells, one or two vacuoles of golgi body enlarges, situated in golgi body.

(C) Spermatid granule connects with anterior region of nucleus enlarges and acrosome is formed.

(D) Spermatid granules form long nuclear structure, connects and form constricted acrosome.

(90) Which sentence is not proper with the function of centriole ?

(A) Two centriole are one by one arranged behind nucleus of spermatid.

(B) Distal centriole forms basal granule to form axial filament.

(C) One centriole attached to anterior part of nucleus of spermatid, another to posterior part of nucleus. (D) Centriole form mid piece and tail of sperm cells.

(91) Which are sequential true name for spermatogenesis, according to development of sperm cells ?

(A) Spermatogonia - primary spermatocyte - secondary spermatocyte - spermatids - sperm cells.

(B) Spermatogenic cells - primary spermatocyte - spermatogonia - secondary spermatocyte - spermatids - sperm cells.

(C) Spermatogenic cells - secondary spermatocyte-primary spermatocyte - spermatogonia - spermatid sperm cells.

(D) Spermatogenic cells - spermatid - spermcell spematogonia - primary spermatocyte - secondary spermatocyte.

(92) Which type of stages found during of oogenesis and cell division ?

(A) Multiplication - Mitosis, growth - mitosis maturation division - meiosis

(B) Multiplication - meiosis, growth - mitosis maturation division - mitosis

(C) Multiplication - mitosis, growth - no cell division maturation division - meiosis

(D) Multiplication - mitosis, growth - no cell division maturation division - A mitosis

(93) Which are main difference in growth phase of spermatogenesis and oogenesis ?

(A) It has long phase than spermatogenesis. It increases in size by increase of cytoplasmic fat, protein and yolkmatter.

(B) It is short phase than spermatogenesis. In its cytoplasm fat, protein and yolk matter decreased, size decreased.

(C) It is long phase than spermatogenesis. In its cytoplasm fat, protein and carbohydrate increase, as a result size increases.

(D) It is long phase than spermatogenesis. In its cytoplasm fat, protein, DNA, RNA increased as a result size increases.

(94) In growth phase of oogenesis which region enlarges ?

(A) Nuclear region increases

(B) Cytoplasmic region increases

(C) Peripheral region increases

(D) Nuclear, cytoplasmic region increase peripheral cell region increases.

(95) In oogenesis amitotic division occur in which stage ? Cells produced by it are called ..... .

(A) Mitosis oocyte, primary polar body

(B) Maturation division (meiosis) - secondary oocyte, primary polar body

(C) Meiosis primary oocyte, primary polar body

(D) Mitosis, secondary oocyte, secondary polar body

(96) Which cells have (2n) diploid chromatin structure during oogenesis ?

(A) Germinal epithelial cell, secondary oocyte, mature oocyte.

(B) Germinal epitherial cell, primary oocyte, first primary oocyte.

(C) Germinal epithelial cell, oogonia, secondary oocyte.

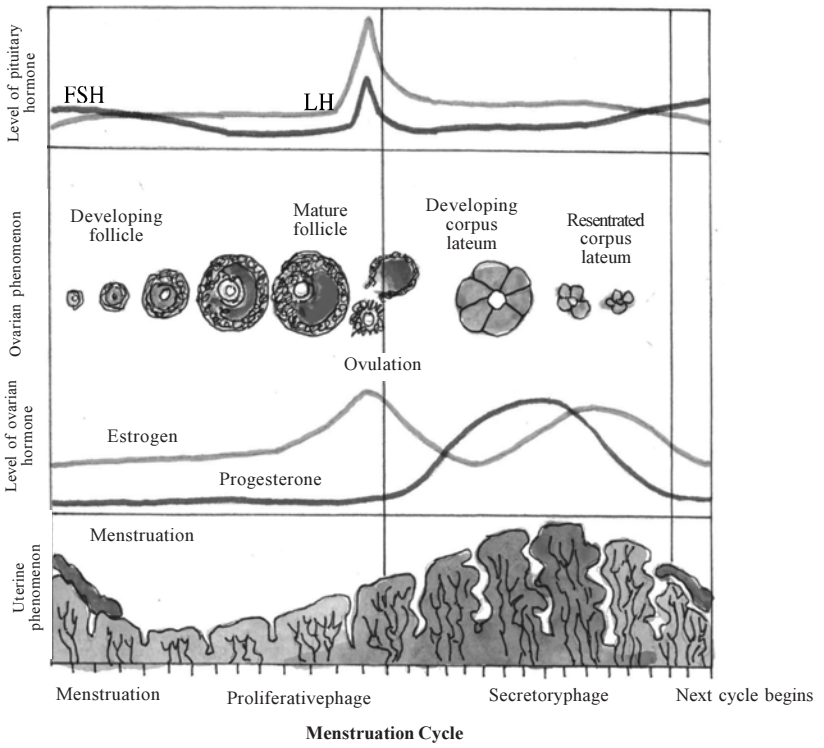
(D) Germinal epithelial cell, mature egg cell, secondary oocyte.

Answers : (80-D), (81-D), (82-D), (83-C), (84-A), (85-A), (86-C), (87-C), (88-C), (89-D), (90-C), (91-A), (92-D), (93-A), (94-D), (95-B), (96-C)

(4) Menstrual cycle : The events of the menstrual or uterine cycle are the cyclic changes in the endometrium; which occur in cyclic form every month. This is due to the change in the levels of female sex hormones like estrogen and progesterone, in the blood. The events in this cycle can be divided into 28 days.

- **Day1-5 (Menstruation phase) :** Due to lower concentration of female sex hormones in the blood, the endometrium disintegrates and blood vessels within it breaks up. Due to this secretion of blood through vagina takes place. It lasts for 3 to 5 days. During this period about 50 ml to 150 ml blood is lost. This phase is known as a menstrual phase.

- **Days 6-14 (proliferative phase) :** This phase of the menstrual cycle is known as a proliferative phase. This phase in stimulated by rising estrogen levels which is produced by the growing follicles. The endometrium becomes glandular, vascularized and thick. At the end of this phase on 14th day ovulation occurs.



Menstruation Cycle

Ended day of 14th in this phase releave the secondary oocyte from the graffian follicle. So this process is known as ovulation. So 14th day is known as the ovulation phase.

- **Days 15-28 (Secretory phase) :** Rising levels of progesterone produced by the corpus luteum induces development of endometrium and increases its blood supply. Now endometrium is ready for implantation of embryo. If fertilization does not occur, the corpus luteum degenerates. This phase is known as secretory phase. At this juncture, menstruation begins.

---

(97) In which layer of uterus menstrual cycle is shown ?

(A) Endometrium (B) Myometrium (C) Epimetrium (D) Exometrium

(98) Menstrual stage is shown for which purpose ?

(A) Estrogen level increases in blood, progesterone level decreases.

(B) Estrogen level increases in blood, progesterone level increases.

(C) Decrease in estrogen, increase in progesterone.

(D) Decrease in estrogen, Decrease in progesterone.

(99) Which sentence is improper for menstrual cycle ?

(A) It is for 1-5 day, decrease in female sex hormone shown.

(B) Endometrium of uterus is degenerated, secreted along blood capillaries.

(C) In this stage 50-100 ml blood is lost which is released out side the body through vaginal passage.

(D) Endometrium of uterus dissociate due to estrogen, progesterone concentration.

(100) Which hormonal change sound in proliferative stage ?

(A) Increase GTH, estrogen increase (B) Decrease GTH, Estrogen increase

(C) LH increase, Estrogen increase (D) LH reduces, Estrogen increases

(101) Which sentence is not proper for changes occurring in proliferative stage ?

(A) Sex hormone estrogen secreted from developing ovarian follicle, so this stage is stimulated.

(B) In this endometrium of uterus develops to become thick for embryo implantation.

(C) It is observed in 6-13 day period on 14th day release the ova is called ovulation.

(D) In this endometrium gradually develops and readily results in embryo implantation.

(102) In which stage graffian follicle develops and ruptures on which day ?

(A) Menstrual stage 6-13 days (B) Proliferative stage 1-5 days

(C) Proliferative stage 15-28 days (D) Proliferative stage 14th day

(103) Which stage corpus luteum is formed ?

(A) Menstrual stage (B) Proliferative stage (C) Ovulation (D) Secretory phase

(104) What changes are found in uterus due to progesterone ?

(A) Endometrium develops in uterus (B) Myometrium develops in uterus

(C) Epimetrium develops in uterus (D) Endometrium degenerates in uterus.

(105) When is uterus become prepared for embryo implantation ?

(A) Endometrium develops by progesterone and blood supply increases with blood vessels.

(B) Myometrium develops by estrogen and blood supply increases with blood vessels

(C) Myometrium develops by progesterone blood supply increases with blood vessels.

(D) Endometrium develops by progesterone and blood supply increases with blood vessel.

(106) When is corpus luteum dissociates ?

(A) Fertilization of ovule by sperm cells.

(B) Ovule is not fertilized by sperm cells.

(C) Fertilization of ovule by sperm cell in the uterus.

(D) Fertilization of ovule by sperm cell in oviducal funnel.

(107) Corpus luteum dissociates by the effect of which hormone ?

(A) Decrease in progesterone level (B) Increase in progesterone level

(C) Decrease in estrogen level (D) Increase in estrogen level

(108) Which are sequential stages of uterine cycle ?

(A) Secretory phase, proliferative stage, ovulation, menstrual stage

(B) Menstrual stage, ovulation, proliferative stage, secretory stage

(C) Menstrual stage, proliferative stage, ovulation, secretory stage

(D) Menstrual stage, secretory stage, ovulation, proliferative stage

(109) Menstruation occurs with the joint effects of which hormones ?

(A) GTH, LH

(B) Estrogen level decrease progesterone level increases

(C) Estrogen, progesterone both level decreases

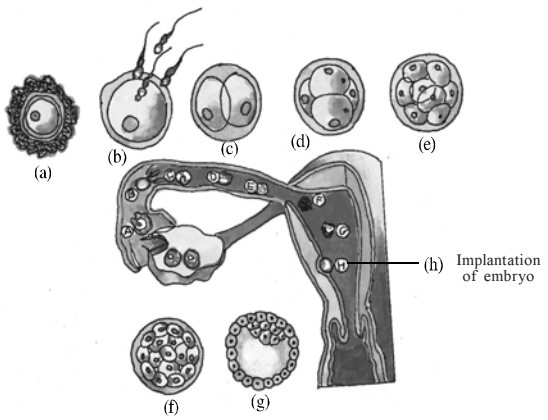
(D) GTH increases, LH increases

Answers : (97-A), (98-D), (99-D), (100-B), (101-D), (102- D), (103-D), (104-A), (105-A), (106-B), (107-A), (108-C), (109-C)

(5) Fertilization and Implantation : At the time of intercourse penis enlarge and become rigid at the climax stage so man ejacts semen in vagina of female. The average volume of semen for each ejaculation is 3 to 4 ml in which billions of sperms are present.

Ejaculated semen with sperms, in vagina, sperms travelled from vagina to oviduct via uterus. For the movements of sperms which get the velocity by contraction of wall of uterus and vagina. Even sticky secretion of oviduct also help to the sperm movement. This process take approximately 5 to 6 hours i.e. sperm reach upto the ovum for fertilization they take 5 to 6 hours.

The secondary oocyte is surrounded by numerous sperms. Secondary oocyte is surrounded by egg membrane and layer of jelly. Various enzymes are located in the arosome of sperms, one of which hyluronidase, which makes the entry of sperm, into secondary oocyte possible. The head and middle part of sperms enters the secondary occyte. Then head and middle piece of sperms become spherical in shape and form male pronucleus.



Transport of ovum in oviduct and Implantation of embryo

(a) secondary oocyte
(b) fertilization
(c) and (d) cleavage
(e) morulla
(f) 16th cell stage
(g) blastocyst
(h) implantation of embryo

The entry of sperm occur in secondary oocyte so certain physical and chemical changes occur in it. The egg menbrane of secondary oocyte becomes slightly separated from the protoplasm. Now egg membrane and jelly layer becomes dense and impermeable which separates from protoplasm. Now it is known as fertilization membrane. This membrane prevents entry of other sperms in secondary oocyte. Entry of sperm induces completion of maturation of secondary oocyte forms a female pronucleus. Thus one sperm and one ovum become involved in fertilization. A diploid zygote nucleus (2n) is formed through fusion of male and female pronucleus. This phenomenon is known as fertilization.

After the fertilization zygote move anterior part of oviduct and then zygote move to uterus through fallopian tube. During the movement of zygote from oviduct to uterus division of zygote will start. Division or cleavages inreducing form. Cleavages occur in zygote resultant cells are known as blastomeres. First five cleavages are regular. By the first cleavage occur in verticle plane zygote becomes two cells. By the second cleavage perpendicular to first cleavage in verticle four cells are formed. Third cleavage occur transveresly and eight cells occur in embryo. Afterwards forth cleavage occurs complete vertically and embryo becomes sixteen celled. This sixteen cells stage is called morula stage. At the time of cleavages hollow or space occur among the cells. It is called Blastocoel. Bag like structure at this time is known as blastocyst. Blostomeres produce fluid it is known as a blastula fluid. Fifth cleavage occur in transversely, it becomes 32 celled stage. Now the irragular cleavages occur in

embryo so multi cellular embryo can occur. Thus by the regular cleavages forms 2,4,8, 16 and 32 daughter cells formed. All these changes take place in period of one week.

Now, the process of implantation of embryo in the uterine wall takes place. The jelly like layer around the embryo is removed. The enzymes secreted from trophoblast cells digest some tissues and blood vessels of uterine wall and make implantation possible. the inner wall of uterus / endometriums develops and partially envelops the embryo. This process is called implantation of embryo.

(110) What is passage of sperm cells which are entered due to semen discharge in vaginal passage who helps in conduction of sperm cells ?

(A) Vaginal passage to uterus muscles of wall of uterus.

(B) Vaginal passage - uterus - towards oviduct contraction of vaginal passage and contraction of uterine wall muscles.

(C) Vaginal passage - uterus - oviduct contractions of vaginal passage and muscle of uterine wall.

(D) Vaginal passage - uterus - oviduct - oviducal funnel - towards ovule, vaginal passage, uterine wall, oviducal wall muscle contraction and slimy secretion helps in transport.

(111) Which statement is not proper for fertilization ?

(A) Innumerable sperm cells surrounds secondary oocyte, one of such sperm which is near to oocyte physically and chemically secretes protease type of enzyme hyaluronidase from Acrosome.

(B) Ovarian membrane and jelly layer disintegrate by hyaluronidase, Head and middle part of sperm enters egg cell.

(C) Head and middle region of sperm enters in egg cell, it is called secretion.

(D) Head and middle region of sperm for physical, chemical balance become round - as male pronucleus, during that period oogenesis is completed, mature egg nucleus become female pronucleus unite and reproductive nucleus and diploid cell formed. It is called fertilization.

(112) Sperm, ovule, spermatogenic cell, spermatid primary spermatocyte, zygote which of the following option shown based on chromosome number ?

(A) n, n, 2n, n, 2n, 2n (B) 2n, 2n, 2n, n, 2n, 2n

(C) n, n, n, n, n, n (D) 2n, 2n, 2n, 2n, 2n, 2n

(113) When is formation of fertilization membrane become possible ?

(A) With the formation of male pronucleus

(B) With the formation of female pronucleus

(C) Zygotic nucleus and zygotic cell formation and constriction of cytoplasm

(D) Enlargement of egg membrane and jelly layer

(114) Which layers are found in fertilization membrane ?

(A) External jelly layer + egg membrane (B) Egg membrane + layer without albumin

(C) Sperm cell membrane + egg membrane (D) None

(115) Morula stage and formation of blastocyst start at which cleave period ?

(A) 5th cleavage and 4th cleavage (B) 4th cleavage and 3rd cleavage

(C) 3rd cleavage and 2nd cleavage (D) 2nd cleavage and 4th cleavage

(116) What is similarity in multicellular embryo and zygote ?

(A) Chromatin matter, total weight and size of cell are similar

(B) Shape of cells similar

(C) No. of cells similar

(D) Cytoplasmic and nuclear component similar

(117) After which cleavage Blastocyst is formed and how many cells are found in it ?

(A) After 4th cleavage, 16 cells (B) After 3rd cleavage, 8 cells

(C) After 5th cleavage, 32 cells (D) After 2nd cleavage, 4 cells

(118) Which type of enzymes are produced by trophoblast ?

(A) Transferases type (B) Lyases type (C) Hydrolyzing type (D) Oxido-reductase type

(119) When is hydrolyzing enzymes released ?

(A) When embryo is at anterior part of oviduct.

(B) When embryo is at middle part of oviduct.

(C) When embryo is at posterior part of oviduct.

(D) When embryo reaches at the basal region of uterus.

(120) After degeneration of which layer by hydrolyzing enzymes, developed trophoblast cells completes ..... process.

(A) Epimetrium, embryo implantation (B) Myometrium, embryo implantation

(C) Endometrium, embryo implantation (D) Trophoblast layer, embryo implantation

Answers :(110-C), (111-C), (112-A), (113-C), (114-A), (115-C), (116-A), (117-A), (118-C), (119-D), (120-C)

(6) Pregnancy and Embryonic Development : The period of development of young one in the female reproductive system is known as pregnancy. In humans the normal period of pregnancy is approximately 266 days to 280 days (40 weeks) from last menstruation from ovulation. However, many babies are born 1 or 2 weeks earlier or later. The fertilized ovum during the first 12 weeks is called embryo and thereafter it known as foetus.

After implantation the trophoblast part of the blastocyst develops elaborate projections, called chorionic villi, which cooperate with the tissues of the mother's uterus to form placenta. The placenta functions to deliver nutrients and oxygen to embryo and remove wastes from the embryonic blood. The placenta is connected to the embryo through an umbilical cord; placenta also acts as an endocrine tissue and produces many hormones like human chorionic gonadotropin (HCG), human placental lactogen (HCL), estrogens and progesterone. In the later stage of pregnancy, hormone relaxin is produced from ovary. The hormones like HCG, HPL and relaxin are produced only during pregnancy.

These hormones help in fetal growth, metabolic changes in the mother and maintenance of pregnancy.

The embryonic development is a continuous process which is summarized, as under.

Stages of embryo development :

	Period	**Changes**
-	First week	- Zygote undergoes cleavage.
		- The blastocyst implants in the uterus.
		- Begins to receive nutrients from the mother.
-	Second week	- Implantation of blastocyst becomes deep in the endometrium.
		- The embryonic disc and amniotic cavity develop.
		- The mesoderm is spreading between ectoderm and endoderm.
-	Third week	- The embrayonic disc broadnens.
		- The primitive heart is formed but it is not yet beating.
-	Fourth week	- The embryo is protected and suspended in amniotic fluid.
		- The primary brain, eyes, stomach, kidneys and heart develop.

- Heart starts beating, approximately 60 times per minute.
- The primitive umbilical cord is formed.
- In this stage, the embryo length is less than 4 cm.

- Second month
  - The embryo is now looks like a human.
  - The main organs of the body develop and begin their function.
  - In this stage, head is larger compared to body.
  - It is 2.5 cm long.

Foetus Development

- Third month
  - Now embryo called foetus means 'young one'.
  - Foetus reaches 7.5 cm height and about 14g weight.
  - Body has grown, but head is larger then body.
  - The limbs become longer.
  - The external genitalia appear; but it is difficult to identify sex.
  - Some movement of the body and limbs occurs.
- Fourth Month
  - Body is bright red in colour because the blood vessels grow through its transparent skin.
  - The muscles become active.
- Fifth month
  - Skin is now less transparent and covered with hair.
  - From this stage onwards the growth of the placenta slows down.
- Sixth month
  - The skin is wrinkled because it lacks fat.
  - Two eyelids are separated but a mambrane covers the pupils.
  - The foetus measures about 32 cm and weight about 650g.
- Seventh month
  - The foetus moves round vigourously within the uterus.
  - It can open its eyes.
  - If born at this stage it can now breath but with difficulty.
- Eighth month
  - In this stage, foetus is about 42 cm long and its weight is about 1800g.
  - Development of lungs can now support life.
  - At this stage, if baby is born, it is necessary to provide expert care.
- Ninth month
  - At the end of this month, foetus measures about 46cm.
- Tenth month
  - In this stage mother awaits the birth of her child.
- Generally the child is about 50 cm long and weighs 3300g. There are wide variations in the weight of the child at birth.

Toward the later part of pregnancy, the human foetus normally assumes a position with its head directed downwards i.e at the time of birth.

- Parturition and lactation : Parturition is also called childbirth. It is the culmination of pregnancy. It usually occurs within 15 days of the calculated due date. The series of events the expel the infant from the uterus are collectively referred to as labour.

Parturition is induced by a complex neuroendocrine mechanism. The signals for parturition originate from the fully developed foetus and the placenta which induce mild uterine contractions called foetus ejection reflex.

At the time of birth, two chemical signal co-operate to creat real labour pain. Some cells of the foetus begin to produce oxytocin, which stimulates the placenta to release prostuglandins. Both hormones stimulate more frequent and powerful contractions of the uterus. At this point, signals for the release of oxytocin is sent by the posterior pituitary. The combined effects of rising levels of oxytocin and prostaglandins initiate true labour. Stronger contractions cause the release of more oxytocin which causes even more vigorous contractions, forcing the baby even deeper into the mother's pelvis. This leads the baby out of the uterus. Soon after the infant is delivered.

The mammary glands of the female undergo differentiation during pregnancy and start producing milk after delivery; this process is called lactation. Mother feeds this to her new born baby. Milk secreted during initial days of lactation is known as colostrum, which contains antibodies.

(121) Gestation period, in human after ovulation and menstrual period, is how many days ?
(A) 266 days, 250 days (B) 280 days, 266 days
(C) 240 days, 380 days (D) 380 days, 240 days

(122) After embryo formation, what is 12th week fertilized egg cell is called ? And after fertilized egg cell is known as ..... .
(A) Foetus, Embryo (B) Embryo, Foetus
(C) Foetus, new born baby (D) New born baby, foetus

(123) What are chorionic villi ?
(A) At multicellular embryonic stage trophoblastic cells irregularly bulge out.
(B) Group of cells encircling embryo at morula stage.
(C) Trophoblast layer, irregularly bulge out during blastocyst stage.
(D) Trophoblast layer irregularly bulge out during gastrulation stage.

(124) What is placenta ? What is its function ?
(A) Structure connected with uterine tissue provides nutritive components to embryo.
(B) Structure connected with uterine tissue provides O_2 to embryo.
(C) Structure connected with uterine tissue removes excretory substances from embryo
(D) Structure connected with uterine tissue provides O_2 and nutritive substance to embryo, removes excretory substances.

(125) Which hormones are produced by placenta ?
(A) HCG, HPL, estrogen, progesterone (B) HCG, estrogen, progesterone
(C) HCG, HPL, progesterone (D) HPL, estrogen, progesterone

(126) During Gestation period what is the function of hormones ?
(A) Maintains Growth of foetus
(B) Metabolic changes in mother
(C) Maintains growth of foetus, metabolic changes in mother
(D) Parturition

(127) Which change is not proper for changes during first week of embryo development ?
(A) Zygote formation, cleavage process (B) Implantation of Blastocoel sac in embryo
(C) Starts obtaining nutrition from mother (D) Blastocyte sac enters deep in endo metrium and implanted

(128) Which change is not possible during 2nd week of pregnancy ?

(A) Blastocoel sac implantation and penetrate deeper in endometrium

(B) Development of blastodisc and Amniotic sac

(C) Enlargement of blastodisc

(D) Mesoderm spreads between ectoderm and endoderm

(129) Which changes are observed in 3rd week of embryo development ?

(A) Blastodisc enlarges and primary heart is formed.

(B) Heart is developed, starts beating.

(C) Primary brain, eyes, stomach, kidney and heart is developed.

(D) Heart starts beating, approximate 60 times per minute.

(130) Which is not proper for changes in 4th week embryo development ?

(A) At this stage embryo is less than 4cm in length.

(B) Primary umbilical cord develops.

(C) Compare to trunk region of body, size of head is more, approtimate 2.5 cm. in length.

(D) Now embryo is called foetus, which has 7.5 cm height and 14gm weight.

(131) Which is not possible in 2nd month of embryo development ?

(A) Embroy looks like human (B) Main organs are developed and starts functioning

(C) Head is larger than body (D) Height is 7.5 and weight is 14 g.

(132) Statement 1 - At this stage, embroy is called foetus which has 7.5 cm height and 14 gm weight.

Statement 2 - Body grows, but head remains enlarged.

Statement 3 - Appandages become larger, body and appandages show slight movement.

Statement 4 - External reproductive organs seen, but sex is difficult to identify.

Which of the above sentences show changes in third week ?

(A) Statements 1, 2 (B) Statements 1, 2, 3, 4

(C) Statements 1, 3 (D) Statements 2, 4

(133) Body appears shining red, blood capillaries seen from transparent skin, activity of muscle, indicates embryo development of which month ?

(A) Second month (B) Third month (C) Fourth month (D) Fifth month

(134) Statement - (1) Skin become less transparent and covered by hair.

Statement (2) At this stage growth of placenta becomes slow.

Statement (3) Its eye lashes separated but lens is covered by membrane.

Statement (4) Skin become wrinkled due to lack of lipid.

Statement (5) Foetus has approximate 32cm in size and 650 gm weight.

Statement (6) Foetus vigorously moves round in uterus.

Statement (7) Foetus opens its eyes.

Statement (8) If born at this stage, breath with difficulty.

Above which statements in fifth, sixth and seventh month which embryological changes are found. Select by option.

(A) Fifth month - statement 1,2,3; 6th month - statement 4,5,6; 7th month - statement 7,8

(B) Fifth month - statement 1,2; 6th month - statement 3,4,5; 7th month - statement ,2,8

(C) Fifth month - statement 6,7,8; 6th month - statement - 3,4,5; 7th month statement 1,2

(D) Fifth month - statement 1,2,3,4; 6th month - statement 4, 6; 7th month - statement 7,8

(135) Which of the following statements related with 8th, 9th and 10th month embryological changes arrange sequentially.

Statement - (1) Foetus is having approximate 42 cm long and 1800 gm weight.

Statement - (2) -Lungs developed which supports life.

Statement - (3) - At this stage if child is born, it should care according specialist advise.

Statement - (4) - At the end of month size of embryo is approximate 46 cm in length and having 1800 gm weight.

Statement - (5) At the end of the month, weight of foetus is 330 gm and size approximate 50cm.

Statement - (6) Mother waits for child birth.

Above statements shows which changes of third week?

(A) Statement 1, 2, 3, 4, 5, 6 (B) Statement 1, 2, 5, 3, 4, 6

(C) Statement 1, 2, 4, 3, 5, 6 (C) Statement 1, 2, 6, 4, 5, 3

(136) What is delivery ?

(A) At the climax of pregnancy in calculated days sequential processes in uterus depending on hormones, together result in removal of child from mothers body.

(B) Discharge of child from vaginal passage.

(C) Discharge of child from uterus.

(D) Based on hormones, expulsion of child from maternal body.

(137) What is improper for embryo / foetus discharge reflex action ?

(A) For foetal removal neuro - hormonal action induced.

(B) Signals for foetus removal originated from foetus and placenta.

(C) By neuro - endocrinal activities contraction of smooth uterus begins.

(D) Contraction of vaginal passage is also begins.

(138) Arrange the sentences sequentially for child birth.

Statement - 1 when child birth is possible two chemical signals unite and gives actual labour pain.

Statement - 2 Some of foetal cells starts synthesizing hormones.

Statement - 3 Oxytocin induces placenta for release of prostaglandins.

Statement - 4 Oxytocin and prostoglandin induces, continuous and powerful contractions.

Statement - 5 Message for oxytocin releases from post pituitary.

Statement - 6 Increase level of oxytocin and prostaglandin together induces real parturition

Statement - 7 Increase of oxytocin, induces powerful contraction of uterus.

Which draws child out from uterus and parturition take place.

(A) Statements 1, 2, 4, 5, 6, 7, 3 (B) Statements 1, 2, 3, 4, 5, 6, 7

(C) Statements 1, 3, 5, 7, 2, 4, 6 (D) Statements 2, 4, 6, 1, 3, 5, 7

(139) What is milk secretion ?

(A) Mammary gland of female develops during gestation period and after delivery starts producing milk is called lactation or milk secretion.

(B) Milk is secreted from mammary glands of female is called lactation.

(C) Sticky fluid secreted from mammary glands of mother is called lactation.

(D) Breast - feeding is called lactation.

(140) Which hormone is milk inhibiting and milk inducing ?

(A) PIF, LTH (B) LTH, PIF (C) PIF, LH (D) LH, PIF

Answers : (121-A), (122-B), (123-C), (124-D), (125-A), (126-C), (127-D), (128-C), (129-A), (130-D), (131-D), (132-B), (133-C), (134-B), (135-A), (136-A), (137-D), (138-B), (139-A), (140-A)

- **A = Statement R = Reason A/R type questions : answers of questions are given below, select the correct answer given option :**

(A) A and R both correct. R is explanation of A.

(B) A and R both correct but R is not explanation of A.

(C) A is right, R is false.

(D) A is false, R is right.

(141) Statement A : Epididymis is very long tubular and 6 mt. long tubule.

Reason R : Provides temporary storage sight for immature sperm cells.

(A) (B) (C) (D)

(142) Statement A : Internal structure of penis formed of two tissue groups, which possess blood sinuses.

Reason R : During sexual intercourse these blood sinuses filled with semen, require for erection.

(A) (B) (C) (D)

(143) Statement A : Third week of foetal development primary heart is produced.

Reason R : At the end of 4th week heart beat starts.

(A) (B) (C) (D)

(144) Statement A : Seminal vesicle produces 80% fluid of semen.

Reason R : Semen is thick and whitish substance.

(A) (B) (C) (D)

(145) Statement A : Mature oocyte moves in oviduct, do not get fertilized by sperm cell.

Reason R : Endometrium layer disintegrates and menstruation type phenomenon occur.

(A) (B) (C) (D)

(146) Statement A : Increasing level of oxytocin and prostaglandin, induces true delivery.

Reason R : These hormones induces powerful contractions in uterus, which takes child out from uterus.

(A) (B) (C) (D)

(147) Statement A : Myometrium is middle layer of uterus.

Reason R : It is formed by group of striated muscles.

(A) (B) (C) (D)

(148) Statement A : Somatic cells form various organs of body.

Reason R : Reproductive cells by reproduction carries hereditary characters.

(A) (B) (C) (D)

(149) Statement A : Primary spermatocyte meiotically divides to form secondary spermatocyte.
Reason R : Germinal epithelial cells by meiotic division produces spermatogonia.
(A) (B) (C) (D)

(150) Statement A : Spermatids formed due to maturation division do not behave as reproductive cells.
Reason R : Spermatids by metamorphosis becomes mature sperms.
(A) (B) (C) (D)

(151) Statement A : Clitoris is external genital organ of female.
Reason R : It is at the connecting place of labia majora.
(A) (B) (C) (D)

(152) Statement A : In 1- 5 days of menstrual cycle endometrium degenerates.
Reason R : Less concentration of female sex hormones in blood.
(A) (B) (C) (D)

(153) Statement A : Clitoris is small finger like projection, which is similar to male penis.
Reason R : Clitoris differs from penis as reproductive tubule is absent.
(A) (B) (C) (D)

(154) Statement A : During spermiogenesis, of the two centrioles of spermatids, distal centriole forms axial filament.
Reason R : Mitochondria together innervates axial filament.
(A) (B) (C) (D)

(155) Statement A : Zygote by the process of cleavage, converted into blastocyst sac.
Reason R : These changes occur within period of one week.
(A) (B) (C) (D)

(156) Statement A : Sperm when enters secondary oocyte, it is called zygote.
Reason R : Due to entry of sperm, nucleus in head region is called male pronucleus.
(A) (B) (C) (D)

(157) Statement A : Secondary spermatocyte by mitosis transforns into spermatids.
Reason R : Spermatids by spermiogenesis are converted into sperms.
(A) (B) (C) (D)

(158) Statement A : HCG, HPL and relaxin produced at the time of conception.
Reason R : Placenta as endocrine gland produces many hormones.
(A) (B) (C) (D)

(159) Statement A : Messages for oxytocin released from anterior pituitary gland.
Reason R : Due to effect of prostaglandins powerful contraction occur.
(A) (B) (C) (D)

Answers : (141-A), (142-E), (143-B), (144-D), (145-A), (146-A), (147-B), (148-A), (149-C), (150-A), (151- C), (152-A), (153-B), (154-B), (155-D), (156-A), (157-D), (158-A), (159-E)

(160)

(1) Fourth month	(p) Skin become less transpatent and covered by hair
(2) Sixth month	(q) Due to lack of fat skin is wrinkled
(3) Seventh month	(r) Foetus turns vigorously in uterus
(4) Fifth month	(s) Blood capillaries developed through transparent skin

(A) : (1 - s), (2 - r), (3 - p) (4 - q)
(B) : (1 - s), (2 - q), (3 - p) (4 - r)
(C) : (1 - s), (2 - r), (3 - q) (4 - p)
(D) : (1 - s), (2 - q), (3 - r) (4 - p)

(161)

Column - I	**Column - II**
(1) Bulbourethral gland	(i) Nutrition to seminal vesicle
(2) Seminal vesicle	(ii) Activates sperm cells
(3) Prostate gland	(iii) Act as lubricant during copulation
(4) Urinogenital passage	(iv) Temporary storage of sperm cells
(5) Epididymis	(v) Opens at the anterior region of penis

	1	2	3	4	5
(A) :	(ii)	(i)	(iii)	(v)	(iv)
(B) :	(i)	(iii)	(ii)	(v)	(iv)
(C) :	(iii)	(ii)	(i)	(v)	(iv)
(D) :	(iii)	(i)	(ii)	(v)	(iv)

(162)

Column - I	**Column - II**	
(P) Tunica albuginea	(i) Provide nutrition	(A) (P - ii) (Q - iv) (R - iii) (S - i)
(Q) Seminiferous tubule	(ii) Fibrous connective tissue	(B) (P - ii) (Q - iv) (R - i) (S - iii)
(R) Sertoli cells	(iii) Secrete testosterone	(C) (P - iii) (Q - i) (R - ii) (S - iv)
(S) Leydig's cells	(iv) Produces sperm cells	(D) (P - iv) (Q - iii) (R - ii) (S - i)

(163)

Column - I	**Column - II**	
(P) Endometrium	(i) Outermost layer of uterus	(A) (P - ii) (Q - iv) (R - i) (S - ii)
(Q) Myometrium	(ii) Innermost layer of uterus	(B) (P - iii) (Q - i) (R - ii) (S - iv)
(R) Epimetrium	(iii) Fold at the far region of vaginal passage	(C) (P - i) (Q - iii) (R - iv) (S - ii)
		(D) (P - iv) (Q - iii) (R - i) (S - ii)
(S) Hymen	(iv) Middle layer of uterus	

(164)

Column - I	**Column - II**		1	2	3	4
(1) Golgibody	(i)	Mid region of sperm cell	(A) :	(iii)	(i)	(ii) (iv)
(2) DNA concentrated	(ii)	Nucleus of sperm cell	(B) :	(iii)	(ii)	(i) (iv)
(3) Mitochondria	(iii)	Acrosome formation	(C) :	(iii)	(i)	(iv) (ii)
(4) Meiosis	(iv)	Primary spermatocyte	(D) :	(i)	(iii)	(ii) (iv)

(165)

Column - I	Column - II		1	2	3	4
(1) First week	(p)	Primary heart is formed, which do not beat.	(A) :	p	q	r	s
(2) Second week	(q)	Development of Amniotic sac	(B) :	r	q	p	s
(3) Third week	(r)	Implantation of blastocyst in to embryo	(C) :	s	r	q	p
			(D) :	r	q	s	p
(4) Fourth week	(s)	Primary umbilical cord is formed.					

(166)

Column - I	Column - II		
(P) Mons pubis	(i)	Small finger like structure	(A) (P - iv) (Q - iii) (R - ii) (S - i)
(Q) Labia majora	(ii)	Folded tissue	(B) (P - iii) (Q - iv) (R - ii) (S - i)
(R) Labia minora	(iii)	Pillow of adipose tissue	(C) (P - ii) (Q - iii) (R - iv) (S - i)
(S) Clitoris	(iv)	Surrounding reproduction organs	(D) (P - ii) (Q - iv) (R - iii) (S - i)

(167)

Column - I	Column - II		
(P) Multiplication stage	(i)	Sperm cell	(A) (P - iv) (Q - iii) (R - ii) (S - i)
(Q) Growth phase	(ii)	Primary sperm cell	(B) (P - iii) (Q - iv) (R - ii) (S - i)
(R) Maturation phase	(iii)	Primary spermatocyte	(C) (P - ii) (Q - iii) (R - iv) (S - i)
(S) Spermiogenesis	(iv)	Spermatid	(D) (P - ii) (Q - i) (R - iv) (S - iii)

Answers : (160-D), (161-D), (162-B), (163-A), (164-B), (165-B), (166-B), (167-C)

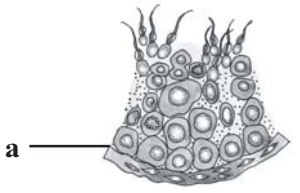
(168) Which is 'a' part in the given diagram 1 ?

(A) Sertoli cells

(B) Primary spermatocyte

(C) Spermatid

(D) Sperm cell



(169) What is indicated by part a in given diagram ? Which organelles are more in it ?

(A) Middle piece - golgibody (B) Middle piece - centriole

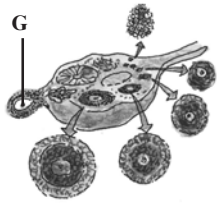
(C) Tail part - Nucleus (D) Middlepart - mitochondria



(170) In given diagram what is indicated by 'G' labelled part ?

(A) Primary oocyte (B) Ovarian follicle

(C) Oocyte (D) Graffian follicle



(171) What is indicated by part labelled as 'a' ?

(A) Trophoblast layer (B) Blastocyst

(C) Internal cell mass (D) Thick covering



Answers : (168-B), (169-D), (170-C), (171-A)

- **Questions for NEET**

(172) ..... region is of grey crescent.

(A) Shows place for entry of sperm into egg cell

(B) It is on the opposite part of the place for entry of sperm into egg cell

(C) In the region of animal pole

(D) In the region of vegetative pole

(173) ..... is key for begining of reaction in acrosome of sperm cell.

(A) Receiving (B) Development of lysine (C) Yolk sac (D) Amniotic sac

(174) Which extra embryonal membrane prevents drying of foetus in uterus.

(A) Trophoblast (B) Allantois (C) Yolk sac (D) Amniotic sac

(175) Human egg cell are ..... .

(A) alecithal (B) micro lecithal (C) mesocecithal (D) poly lecithal

(176) ..... Semen is saturated in human.

(A) Fructose and calcium, no enzyme

(B) Glucose and some enzymes, no calcium

(C) Fructose and some enzymes, calcium is very less

(D) Fructose, calcium and some enzymes present

(177) During secrectory phase of menstrual cycle ..... .

(A) luteal phase for approximate six days

(B) follicular phase for approximate six days

(C) luteal phase for thirteen days.

(D) follicular phase for thirteen days.

(178) Secretory source of ..... is leydig's cell of human

(A) progesterone (B) mucosa in intestinal region

(C) glucagon (D) androgens

(179) Which of the following sentence is false in reference to vitality of sperm cells in mammals ?

(A) Sperm cells are living for 24 hours.

(B) Vitality of spermcell depends on PH of medium more active in alkaline medium.

(C) Vitality of sperm cells are determined by its motility.

(D) It is necessary that sperm cells concenterated in thick medium.

(180) Micro tubule is vasa efferentia coming out from ..... .

(A) festicular lobules to testis (B) testis to vasa efferentia

(C) testis to epididymis (D) epididymis to urinogenital passage

(181) Correct chain, in testis of adult human for formation of sperm cells by spermato genesis is .....

(A) Spermato gonia - primary spermatocyte - spermatid - sperm

(B) Spermatid - primary spermatocyte - spermatogonia - sperm

(C) Spermatogonia - spermatid - primary spermatocyte - sperm cell

(D) Spermatogonia - primary spermatocyte - spermatid - sperm cell

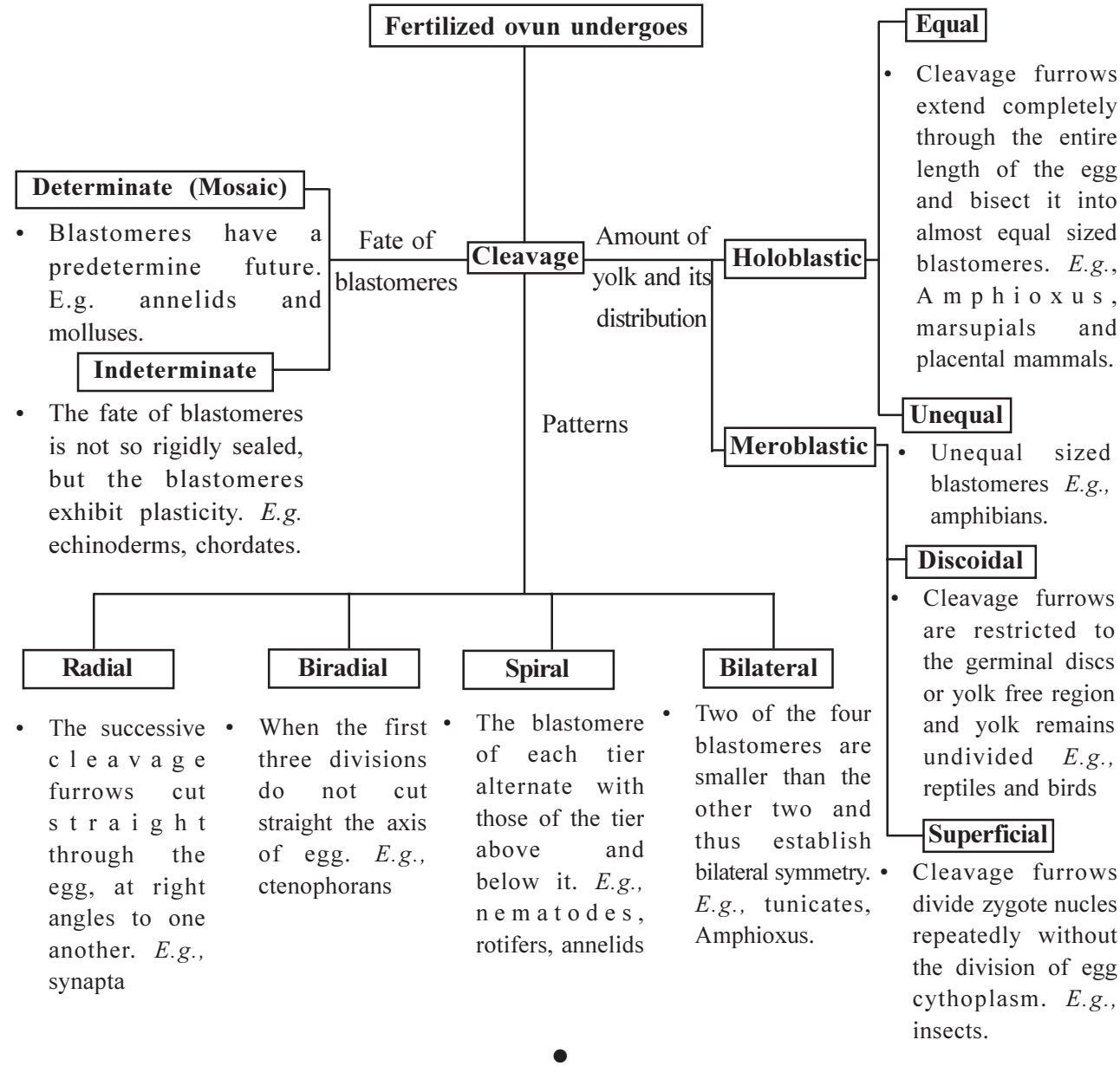
(182) Stimulation for child birth spreads in human female through ..... .
(A) release of oxytocin from pituitary gland (B) completely developed child and placenta
(C) separation of mammary glands (D) pressure by amniotic fluid

(183) At which month of conception, appearance of hair on head and movement of foetus experienced for the first time ?
(A) Fourth month (B) Fifth month (C) Sixth month (D) Third month

(184) ..... amniocenecsis, amniotic fluid analysis technic is used with permission.
(A) To determine sex of unborn child (B) Artificial conception
(C) Transfer foetus in uterus of surrogate mother (D) To study genetic deformitios

(185) In normal pregnant female no. of total stimulating gonadial hormones can determined, result may be .....
(A) High level of transpoting FSH and LH in uterus stimulates implantation of embryo
(B) High level of transpoting HCG is required to increase thickness of endothelium
(C) High level of FSH and LH of uterus, stimulates thickness of endothrlium
(D) High level of transpoting HCG stimulates synthesis of estrogen and progesterone

Answers : (172-B), (173-(D), (174-D), (175-A), (176-C), (177-C), (178-D), (179-D), (180-B), (181-A), (182-B), (183-B), (184-D), (185-D)



Human Reproduction

Primary Sex Organs
- Produce gametes called gonads male gonad testes and female gonad ovaries.

Secondary Sex organs
- Do not produce gametes
- Essential for transport of gemetes fertilizaion etc.
- In males, prostate, seminal vesicles, vasa deferentia, penis.
- In females, fallopian tubes, uterus, vagina and mammary glands.

Male reproductive system

Testes
- Paired structures lying in scrotum.
- Helps in spermatogenesis.

Accessory glands
- Include paired seminal vesicles, a prostate and paired bulbourethral glands.

Female Reproductive System
- Consists of a pair of ovaries along with a pair of oviducts, uterus, cervix, vagina and external genitalia.
- Ovaries help in Oogenesis.
- A cyclic change in the reproductive tract of female occur, which is called menstrual cycle.

Fertilization
- It is the process of fussion of sparn and egg to produce zygote.
- Amphimixis occurs.
- The presence of X and Y chromosomes daternime the sex of the embryo.

Significance
- Restores diploid number of chromosomes
- Initiates cleavage
- Results in determination of sex
- Introduces variation

Pregnancy
- Zygote undergoes repoated mitotic division to form a blastocyst, which is implanted in the uteras resulting in pregnancy.

Embryo Development
- Embryo after implantation differentiates into 3 germinal layered embryo.
- Three germinal layers of the embryo undergo organogenesis.
- By the end of 9th month of the embryo the foetus is fully developed.

Parturition and Lactation
- Parturition is the process of child birth.
- Parturition is induced by neuro-endocrine mechanism.
- Mammary glands differentiates and secrete after child birth.
- New born baby is fed with mother's milk is called lactation.
- Milk produced during initial few days of lactation is called colostrum which is rich with antibodies.

37 Reproductive Health

- The normal functions of reproductive system is referred to as reproductive health. According to the world Health Organization (WHO), reprodutive health means a total well-being in all aspects of reproduction i.e. Physical, emotional, behavioural and social. Therefore, a society with people having physically and functionally normal reproductive organs and normal emotional and behavioural interactions among them in all sex-related aspects might be called reproductively healthy.

Reproductive health - Problems and strategies :

India was the first country in the world to kick off action plans and programmes at a national level to get total reproductive health as a social goal. These programmes are called 'Family Planning' and were initiated in 1952 and were periodically assesed over the part decades. Improved programmes covering wider reproduction-related areas are currently in operation under the popular name- 'Reproductive and Child Health Care (RCH) programmes'. Creating awareness among people about various reproduction, related aspects and providing facilities and support for building up a reproductively healthy society are the major tasks under these programmes. With the help of audio-visual and print media; govermental and non-govermental agencies have taken various steps to create awareness among the people about reproductive health. Parents, other close relatives, teachers and friends, also have major role in the dissemination of the above information. Introduction of sex education in schools should also be encouraged to provide right information to the young so as to discourage children from believing in myths and having misconceptions about sex related aspects. Proper information about reproductive organs adolescence and related changes, safe and hygenic sexual practices, sexually transmitted diseases (STD, AIDS etc would help people, especially those of adolescence age. Educating people, especially couples and those in marriageable age group, about available birth control options, care of pregnant mothers, post-natal care of mother and child; importance of breast feeding, equal opportunities for the male and the female child etc. will make society healthy. Awareness of problems like uncontrolled population growth and of social evils like sex-abuse and sex-related crimes etc needs to be created to take up necessary steps to prevent them and there by build up a socially responsible and healthy society.

The successful implementation of various action plans for reproductive health requires strong infrastructural facilities, professional expertise and material support. It is essential to provide medical assistance and care to people in reproduction related problems like pregnancy; delivery, AIDS, abortions, contraception, menstrual problems, infertility etc. Implementation of better techniques and new strategies from time to time is required to provide more efficient care and assitance to people. Research on various reproduction related areas is encouraged and supported by govermental and non-governmental agencies.

Population Explosion and Necessity and Patterns of Birth control :

In the developed European countries like spain and Italy where the population is decreasing , this might be considered as an advantage, however, for the developing countries like India, population explosion is a nuisance and causes damage to the development of the country and its society. The developing countries are already facing a lack of resources, and with the rapidly increasing population the resources available per person are reduced further, leading to increased poverty,

malnutrition and other large population related problems.

India is the second most populous country in the world after China. India supports 16.87 percent of the world's population on its meager 2.4 percent world surface area. At the time of independence, the country's population was 342 million. The country's population size had grown from 361 million in 1951 to around 846 million in 1991 and 1027 million in 2001. The population of India almost triples during the period of 1951-2001. The phenomenal increase in the population during the last fifty years has led to rapid industrialization and high rate of urbanization which have created tremendous pressure on natural resources like land, air and water.

Decline in death rate and increase in birth rate are the main factors affecting population growth. Increased health facilities along with better living conditions also had impact on the growth of population. Such alarming growth rate could lead us to a scarcity of the basic requirements, i.e. food, shelter and clothing.

The government of India has been organizing several programmes for controlling the population increase and has been spending millions of rupees on controlling the birth rate. One of the programmes has been successful; and the rate of increace has also reduced, but has still to reach the sustainable rate. Our main goal to control population growth is decreasing the birth rate. Several government - funded agencies like the Family Planning Association of India spend billion of rupees on promoting family planning. These organizations aim to promote family planning as a basic human right. The family planning methods provided by the family planning programme are vasectomy, tubectomy, IUD (Intra Uterine Devices), copper - T, conventional contraceptives (Condoms, Diaphragms etc) and oral pills.

Barrier method : A variety of barrier methods suitable for both men and women are available. The aim of these methods is to prevent live sperm from meeting the ovum. Barriar methods have increased in popularity due to the absence of side effects and some protection from sexually transmitted diseases.

(I) **Physical methods :** In this method condoms are used . It is made of thin rubber and used to cover the penis in the male or vagina and cervix in the female just before coitus so that the ejaculated semen does not enter into the female reproductive tract.

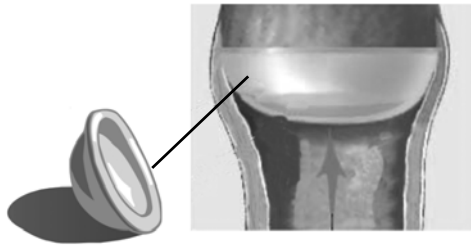
Diaphragm is a vaginal barrier made of synthelic rubber or plastic material and inserted into the female reproductive tract to cover the cervix during coitus. Variations of diaphragm include the conical cap; vault cap and the vinule cap.



Condom for male



Condom for female



Diaphragm

(II) **Chemical Methods :** Spermicides are in the form of creams which attach themselves to sperm and inhibit oxygen uptake and kill sperms.

(1) Commonly what is reproductive health ?

(A) Health of reproductive system (B) Maintenance of reproductive organs

(C) Common functions by reproductive system (D) Arrangement of reproductive organs

(2) What is reproductive health with reference to WHO ?

(A) A total well - being in all aspects of reproduction, i.e. physical, emotional behavioural and social.

(B) A reference of reproduction aspects of physical and maintenance of emotional aspects.

(C) Maintenance of mental and social aspects in reference of reproduction.

(D) Maintenance of effects of behaviour and emotional in reference of reproduction.

(3) India was first country in the world to kick off action plans and programmes at a national level of reproductive health, How ?

(A) With all acceptance, according to social aspects, legally improved.

(B) With all acceptance, according to individual voluntary improved.

(C) Partially, according to individual target voluntary improve.

(D) Partially, according to social target, legally improved.

(4) Reproductive health programme are popular by which name ?

(A) Reproductive (B) Child health care

(C) Reproductive and child Health care (D) Reproductive or child health care

(5) In adolescence which are the changes occurs in youth ?

(A) Neurochemical (B) Mental (C) Physical (D) Neurochemical, mental and physical changes

(6) By which name programmes of population control is popular ?

(A) Child health care (B) Reproductive

(C) Family planning programme (D) Sterilization

(7) In which countries population growth decreases ?

(A) In Britain (B) In entire Europe

(C) Spain and Italy like countries (D) In India and china

(8) In which countries population explosion is a nuisance ?

(A) Spain and Italy (B) Britain and America

(C) Thailand and Greenland (D) China and India

(9) By the population explosure which issues are arising in developing countries ?

(A) Lacks of resources

(B) increased in poverty

(C) Malnutrition

(D) Economically, socially as well as reduces in available resources, incresed poverty and malnutrition like issues.

(10) How much population has of the world's population and How much land area coveres it respectively ?

(A) 16.87%, 2.4% (B) 18.67%, 4.2%

(C) 67.16%, 4.2% (D) 76.16%, 4.2%

(11) How much polulation at the time of independent in 1951 in India respectively ?

(A) 361 million, 342 million (B) 342 million, 361 million

(C) 432 million, 613 million (D) 613 million, 432 million

(12) What is the population of India at 1991 and 2001 respectively ?

(A) 486 million, 2027 million (B) 846 million, 2027 million

(C) 646 million, 1027 million (D) 486 million, 1027 million

(13) Which are the main causes for tripled population growth in 1951-2001 ?

(A) Rapid industrialisation, urbanisation.

(B) Rapid industrialisation, urbanisation increase in birth rate.

(C) Rapid industrialisation, urbanisation, decrease in death rate.

(D) Rapid industrialisation, urbanisation, decrease in death rate, revolution in clinical facilities.

(14) Which option is correct from following for physical barrier method divices ?

(A) Condom, female Condom, Copper - T (B) Condom, Diaphragm, loop

(C) Condom, female Condom, Diaphragm (D) Loop, Copper - T

(15) Which types of chemical choosen in chemical barrier method ?

(A) Which induce production of sperm have rapid locomotory capacity of sperms.

(B) Which induce production of sperm, rapid locomotory capacity of sperms by the production of foam and survival of sperm by availability of O_2.

(C) Creans which produce foam for destroyed sperm, which join with sperm and decrease availability of O_2 and kill-the sperm.

(D) Foam which killes sperm, which digest sperm.

Answers : (1-C), (2-A), (3-A), (4-C), (5-D), (6-C), (7-C), (8-D), (9-D), (10-A), (11-B), (12-C), (13-D), (14- C), (15-C)

Intrauterine devices (IUDs**) :** Intrauterine devices contraception could also be achieved by introducing a foreign body into the uterus of the female. Such devices are known as Intra uterine Devices (IUDs).

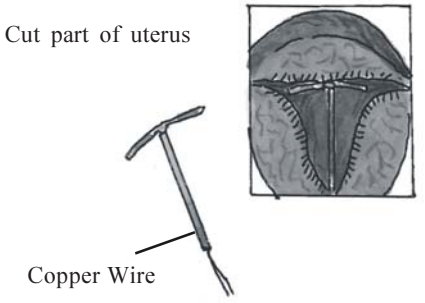
The non-medicated or inert IUDs are often referred to as first generation IUDs. The copper IUDs comprise the second generation IUDs which release metal ions which have strong anti fertility effect. The third generation IUDs release hormones (progesterone) which have a direct local effect on the uterine lining.

Hormonal Methods : Hormonal contraceptives when properly used are the most effective spacing methods of contraception. Oral administration of small doses of progesterones or progestogen estrogen combination in the form of tablets which are known as the pills. The pill is given orally for 21 consecutive days beginning on the 5th day of the menstrual cycle followed by a break of 7 days during which period mentruation occurs. They are used to prevent the release of ovum from the ovary and also render the cervical mucus thick and scanty and thereby, inhibit sperm penetration.

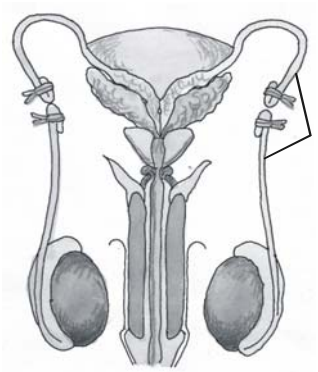
'Saheli' The new oral contraceptive for female contains a nonsteroidal preparation. It was developed at 'Central Drug Research Institute (CDRI)' in lucknow, India. It is a 'once a week' pill with few side effect and high contraceptive value.

Sterilization : Sterilization is generally advised for the male-female partner as a terminal method to prevent any more pregnancies. Surgical intervention blocks gamete transport and thereby

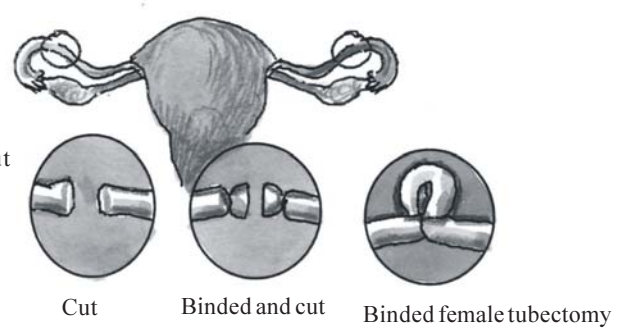
prevents conception. Sterilization procedure in the male (men) is called 'vasectomy' and that in the female (women) tubectomy'. In vasectomy a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through vagina. This type of micro surgery is known as laproscopy. These techniques are highly effective but their reversibility is very poor. It needs to be emphasised that the selection of a suitable contraceptive method and its use should always be undertaken in consultation with qualified medical practitioner.



Copper-T (IUDs)



Vasectomy



Tubectomy

Natural Methods of contraception : Other than above mentioned methods, natural methods are also used which work on the principle of avoiding chances of the meeting of ovum and sperms. Periodic abstinence is one such method in which the couples avoid or abstain from coitus from 10 to 17 days of menstrual cycle when ovulation could be expected, because chances of fertilization are very high during this period. Therfore, by avoiding intercourse during this period, conception can be prevented. Withdrawal or interruption - coitus interruptus is another method in which the male partner withdraws his penis from the vagina just before ejaculation. In the lactational amenorrhea (absence of menstruation) method; menstruation cycle does not occur during the period of intense lactation following parturition. Therefore, as long as the mother breast-feeds the child fully; chances of conception are almost nil. However, this method has been reported to be effective only upto a maximum period of six month following parturition. As no medicines or devices are used in these methods side effects are almost nil.

(16) What is full form of IUDs ?

(A) Intra Uterine Dividing Syndrome (B) Internal Ureters Dividing Samples

(C) Intra Uterine Devices (D) Intra Uretine Devices

(17) Hormonal release devices, loop, metal ion release device, copper - T, These devices classified in first, second and third generation devices and choose correct option for these.

(A) First generation IUDs = Loop, Copper - T, Second generation IUDs = metal ion releases device, Third generation IUDs = Hormal release devices

(B) First generation IUDs = Loop, Second generation IUDs = Copper - T, Third generation IUDs = Hormonal released devices

(C) First generation IUDs : Hormonal released devices, Second generation devices = copper - T, Third generation IUDs = Loop.

(D) First generation IUDs = Loop, Second generation IUDs = Hormonal released devices, Third generation IUDs = Copper - T.

(18) Second generation IUDs is working on which principle ?

(A) On metal ion

(B) On the acute response against fertility of free metal ion.

(C) On metal Fertility of free metal ion.

(D) On response fertility.

(19) Which hormones are useful for hormonal method which have combined effect as oral pill which is useful for contraceptives ?

(A) GTH, LH (B) GTH, Estrogen (C) Estrogen, progesterone (D) LH, progestrone

(20) From which time starts to take pills are taken for contraceptive purpose ?

(A) Minus the 7 days of the menstrual, cycle time.

(B) 7 Minus the 7 days of the menstrual cycle time, star on the 5th day of the menstrual cycle.

(C) As soon as menstrual cycle starts

(D) Any time take

(21) What is the function of contraceptive pills ?

(A) To inhibits the releasing of eggcell from ovary.

(B) Create the inactive and mucilage thick layer of cervix.

(C) To prevent the releasing of eggcell from the ovary, create the inactive and mucilage thick layer of cervix so no transport of eggcell.

(D) Present the entry of spern in cervix.

(22) 'Saheli' contraceptive pill is which type of the chemical pill ?

(A) Non steroidal (B) Catecolamine type (C) Steroidal (D) Oxidental

(23) What is full form of CDRI ?

(A) Central Drugs and Reprohuetive Institute (B) Central Drugs and Reproductice Institute

(C) Central Drugs and Research Institute (D) Central Divices and Research Institute

(24) What is sterilization ?

(A) Surgery in male to avoid pregnancy.

(B) Surgery in female to avoid pregnancy.

(C) Surgery in male and female for pregnancy.

(D) Surgery in male and female to avoid pregnancy.

(25) In vasectomy and tubectomy surgery of which parts can be done ?

(A) Vasa deferentia and fallopian tube (B) Epididymis and fallopian tube

(C) Inguinal canal and oviduct (D) Vas efferentia and oviduct

(26) In which periodic time the couples shall avoid coitus ?

(A) During 10 to 17 days of the menstrual cycle (B) During 10 to 17 days of the menstrual cycle

(C) During 570 to days of uterine cycle. (D) During 17 to 27 days of menstrual cycle.

(27) What is meaning of lactational amenorrhea ?

(A) Menstruation cycle does not occur during the period of intense lactation following parturition.

(B) Menstrual is absent after parturition.

(C) Menstruation cycle is absent during lactation.

(D) Mentruation is absent during to feed baby.

(28) Which method is more favourable in recent era ?

(A) Physical barrier method (B) Chemical barrier method

(C) Hormonal method (D) Natural barrier method

(29) After which method no chances of embryo implantation ?

(A) Physical barrier method (B) Chemical barrier method

(C) Sterilization (D) Natural barrier method

Answers : (16-C), (17-B), (18-B), (19-C), (20-B), (21-C), (22-A), (23-C), (24-D), (25-A), (26-B), (27-A), (28-A), (29-C)

Medical Termination of pragnancy (MTP) : Voluntary termination of pregnancy before foetus becomes viable is called medical termination of pregnnancy (MTP) or induced a abortion. Nearly 45 to 50 million MTPs are performed in a year all over the world. Whether to accept / leaglize MTP or not is a subject debated in many countries due to emotional, ethical, religious and social issues involved in it.

The MTP Act was passed by Indian Government in 1971 and it comes into force from April 1, 1972. It is a health care measure, which helps to reduce maternal mortality resulting from illegal abortion. MTP is required to terminate pregnancy in following cases:

- Continuation of pregnancy might endanger the mother's life.
- There is substantial risk of the child being born with serious handicaps.
- Where the pregnancy is the result of rape.
- Where the pregnancy is unwanted resulting from failure of any contraceptive device.

MTP can be performed only by registered doctors and it can be done only upto 20 weeks of pregnancy only.

- **Sexually Transmitted Diseases (STDs)** : Diseases or infection which are transmitted through sexual intercourse are collectively called sexually trassmitted diseases (STDs) . It is caused by baeteria, virus, protozoan and fungi. STDs are becoming a major health problem in India. More than 20 pathogens have been found to be spread by sexual contact. The highest cases are observed in the 20-24 age groups it is followed by the 25-29 age group and the 15-19 age group. Some of these diseases are summarized below.

Summary of STDs

Name of Disease	Name of pathogens	Major Symptoms
(1) Gonorrhoea	*Neisseria gonorroheae* (Bacteria)	- Pain during passing urine. - Pain in lower abdomen.
(2) Syphilis	*Treponema pallidium* (Bacteria)	- A painless rush. - Flu like illness, tiredness. - White patches on the tounge or roof of the buccal cavity - Patchse hair loss

(3) Genital herpes	Herpes simplex virus	- Many people will not have any visible signs and symptoms. - If person do get signs and symptoms they feel uneasi-ness. - Symptoms such as fever, tiredness, headache, itching in the genital or anal area etc are seen. - Pain while passing urine. - Small, fluid -filled blirsters anywhere in the genital or anal area.
(4) Hepatitis - B	Hepatites B virus	- Symptoms like fever, joint pain, fatigue, loss of appetite, jaundice, pain in upper light abdomen etc. are seen.
(5) AIDS	Human immunodeficiency virus (HIV)	- Person looses immunity so all diseases are dominated. - No specific symptoms are seen, but some symptoms like, fever for over months, diarrhoea, repid weight loss, a cough that won't go away short term memory loss etc. are seen.
(6) Trichomoniasis	Trichomonas vaginalis (Protozoan)	- Soreness, inflammation and itching in and around vagina. - Pain or burning sensation while passing urine.

The diagnosis of these diseases is based on causative organism and the symptoms. The medical examination and symptoms reveal their STD nature. Certain diagnostic tests for these diseases includes culture of the pathogenic organism. Through culturing, microorganisms can be isolated, observed and identified, by microscopic examination using special stains ELISA (Enzyme Linked Immune Absorbent Assay) test is used for identification of antigen-antibody. In ELISA method, antibodies against HIV antigen are search out from the patient's blood. This helps in establishing the identification of the genetic material of pathogenic organism is utilized. Polymerase Chain Reaction (PCR) : In the PCR method the specific section of a gene of the pathogenic organism is multiplied with the help of a suitable primer.

Someone has said "prevention is better than cure". Thus prevention of STDs is in your hands. You could be free of these infections if you follow the simple principles given below.

- Avoid sex with unknown partners.
- Always use condoms during coitus.
- In case of doubt, consult a doctor and get treatment if the disease is diagnised.

(30) What induces abortion ?

(A) Voluntary type abortion

(B) To prevent of embryo implantation voluntary

(C) A method of voluntary termination of pregnancy before foetus viability

(D) Abortion induced by pills

(31) How many people shows MTP. (Medical Termination of Prengnancy) in a year all over the world ?

(A) nearly 35 to 40 million people (B) nearly 45 to 50 million people

(C) nearly 45 to 50 million females (D) nearly 50 to 55 million people

(32) Which types of question are arising against - MTP ?

(A) Emotional and, ethical (B) Religious and ethical

(C) Emotionad, ethical religious and social (D) None of above

(33) The MTP Act was passed in 1971 and it came into force from ?

(A) 1st April, 1972 (B) 1st April, 1971 (C) 1st April,1973 (D) 1st April, 1970

(34) By MTP Act which two programme will succeed ?

(A) Control on population growth, Birth rate

(B) Control on population growth, Death rate

(C) Control of population growth, Mother and child health care

(D) Family planning, Control of population growth

(35) Which programme is inducable by the given statements ?

Statement : 1 Continous pregnancy is harmful for mother health.

Statement : 2 There is substantial risk of the child being born with serious handicaps.

Statement : 3 Where the pregnancy is the result of rape.

Statement : 4 Where the pregnancy is unwanted resulting from failure of any contraceptive device.

(A) Family planning (B) Child health care (C) Control of population growth (D) MTP

(36) After How much period MTP is advicable ?

(A) upto 20 weeks or 140 days or four months

(B) Upto 10 weeks or 70 days or three months

(C) Upto 30 weeks or 210 days or seven and half months.

(D) Upto 25 weeks or 175 days or five months.

(37) What is Sexual Transmitted Diseases ?

(A) Diseases occurs due to lack of sexual knowledge.

(B) Diseases occurs cause the sexual development.

(C) Diseases occurs due to incomplete sexual knowledge.

(D) Disease occurs by sexual intercourse.

(38) What is full form of STDs ?

(A) Standard Terminational Diseases (B) Standard Trasmitted Diseases

(C) Sexual Terminational Diseases (D) Sexual Transmitted Diseases

(39) By which microbes STDs are tramsmitted ?

(A) Virus, bacteria (B) Fungi, protozoan

(C) Virus, bacteria, fungi (D) Virus, bacteria, fungi, protozoan

(40) Choose the correct option according to evolutionary ascending order classification of STDs vectors.

(A) Virus, Bacteria, Fungi, protozoa, (B) Viras, Bacteria, protozoa, fungi

(C) Virus, protozoa fungi, bacteria (D) Protozoa, fungi, bacteria, virus

(41) How many STDs vectors are approximately present ?

(A) Less than 20 (B) More than 20 (C) Only 20 (D) Only 10

(42) Which age group pateints are highest suffering by STDs ?

(A) 15-19 (B) 20-24 (C) 25-29 (D) 30 - 35

(43) Which microbe is responsible for the spread gonorrhoea and which are the major symptoms of it ?

(A) *Neisseria gonorrhoea*, pain during passing urine, pain in lower abdomen.

(B) *Siphilis gonorrhoeae*, pain during passing urine, pain in lower abdomen.

(C) *Neisseria gonorrhoeae*, burning in urethra, headache

(D) *Streptococeus gonorhoeae*, pain during passing urine, pain in lower abdomen.

(44) Main symptoms of syphilis and which is the correct name of disease vector ?

(A) A painless rush, flu like illness, tiredness, white patches on the tongue or roof of the buccal cavity, patchy hairloss, *treponema pallidium*.

(B) A painless rush, tiredness, Flu like illness; *Treponema pallidium*.

(C) White patches on the tongue or roof of the buccal cavity, *treponema valgaris*.

(D) Patches hairloss, *Treponema streptococcus*.

(45) Give the name of disease vector /pathogen of genital herpes and identify major symptoms of it ?

(A) *Herpes zoster,* person do get signs and symptoms they are feeling of uneasiness, symptoms such as fever, tiredness, headache, itching in the genital or anal area etc. are seen. Pain while passing urine.

(B) *Herpes zoster,* cysts in the genital or anal area.

(C) *Herpes simplex* virus, person do get signs and symptoms they are feeling of uneasiness, symptoms such a fever, tiredness, headache, itching in the genital or anal area etc are seen, pain while passing urine, small fluid - filled blisters any where in the genital or anal area.

(D) None of the above.

(46) Name the pathogen of Hepatitis B and identify its major symptoms.

(A) Hepatitis B virus, fever, joint pain, fatigue loss of appetite, Jaundice, pain in upper right abdomen etc are seen.

(B) Hepatitis B - Virus, uneasiness

(C) Hepatitis B - Virus, only pain in joint.

(D) Hepatitis - B Virus, decreases immunity.

(47) Which symptom is not include in symptoms of AIDS ?

(A) Occurance of various diseases when person looses immunity.

(B) Fever for over months, accidently diarrhoea for long duration.

(C) Rapid weight loss, short term memory loss

(D) Loss of immunity in body, uncertain changes in nature, shivering experienced in body.

(48) What is full form of AIDs ?

(A) Acquire Immuno Defensive Syndrome (B) Acquire Immuno Defending Syndrome

(C) Acquire Immuno Deficiency Syndrome (D) Aquire Immuno Deficiency System

(49) Name the pathogen of Trichomoniasis and identity major symptoms of it ?

(A) *Trichomonas vaginalis,* soreness, inflammation, itching in and around vagina; pain or a burning senasation while passing urine.

(B) *Trichomonas derminalis*, soreness, inflammation itching in and around vagina.

(C) *Trichomonas varginalis*, soreness, inflammation, itching in and around vagina.

(D) *Trichomonas varginalis*, soreness, burning in urine passing.

(50) For the diagnosis, which methods are used sequencially ?

(A) PCR, DNA, hybridization, ELISA, Specific staining method, Isolation of micro organisms, culture pathogenis.

(B) Culturioaof patheges, Isolation of microbes, specific stain method, DNA hybridization, PCR method.

(C) Culture of pathogenic organisms, Isolation of microbes specific stain method, ELISA method, DNA hybridization, PLR method.

(D) Culture of pathogenic organisms, Isolation of microbes, stain method, DNA hybridization, ELISA method, PCR method.

(51) What occurs in ELISA (Enzyme Linked Immuno Absorbant Assay) method ?

(A) Emzymatic mechanism can occur (B) Synthesis of antibody

(C) Search of antigen (D) Find out the HIV antigen and antibody from the blood

(52) What is in DNA hybridisation and PCR method respectively ?

(A) A short polynucleotide chain of the genetic material of the pathogenic organism is utilized, the specific pieces of gene of the pathogenic organism is multiplied with the help of a suitable primer.

(B) A long polynucleotide chain of the genetic material of the pathogenic organism is utilized, the undefinated pieces of gene of the pathogenic organisms is multiplied with the help of a suitable primer.

(C) The pieces section of gene of the pathogenic organism is multiplied by three times.

(D) DNA replication and culturing of gene by primer can be done.

(53) Which is not statement of preventing of STDs ?

(A) Avoid sex with unknown partners.

(B) Always use condoms during coitus.

(C) In case of doubt, consult a doctor and get treatment.

(D) Always avoid condoms during coitus.

(54) Rapid weight loss; coughing upto long time, influenza, fever long time, Which is not diagnostic. Whooping, memoryloss of short duration; Dominancy occurs on main immunity and complimentary of Human body system, Failure of immunity. These symptoms of which disease and what is full name of this disease ?

(A) AIDS, HIV, Human Immuno Deficiency Virus

(B) AIDS, HIV- Huminity Immuno Deficiency Virus

(C) AIDS, HIV- Human Immuno Deficieney vaccine

(D) AIDS, HIV- Human Immuno Aquire, Deficiency Vaccine

Answers : (30-C), (31-C), (32-C), (33-A), (34-C), (35-D), (36-A), (37-D), (38-D), (39-D), (40-A), (41-B), (42-B), (43-A), (44-A), (45-C), (46-A), (47-D), (48-C), (49-A), (50-C), (51-D), (52-A), (53-D), (54-B)

(4) Infertility : When female loss ability of embryo implantation and egg formation and ability to transfer of egg cell. Even though loss or lesser ability of spermatogenesis in male which very less formation of sperm then couple can not found reproductive fertility. So it known as infertility. Examination and diagnostic methods are available for the infertility.

Amniocentesis : Amniocentensis is also known as amniotic fluid test or AFT. It is a medical procedure used in prenatal diagnosis of chromosomal abnormalities, in which a small amount of amniotic fluid is sampled from the amnion surrounding a developing foetus, and its DNA is examined for genetic abnormalities. Using this process the gender of the foetus can also be determined and hence this procedure has been legally restricted for sex determination in India.

A number of couples all over the world including India are facing infertility. i.e. they are anable to produce children. In India, mostly female is blamed for this, but it is not always true; the problem can be in the male or female partner. Infertility clinic could help in diagnosing and curing these disorders and helping these types of couples to have children. The couples could be assisted to have children through certain special techniqnes commonly known as Assisted Reproductive Technologies (ART). Assisted Reproductive Technology (ART) is general term referring to methods used to achieve pregnancy by artificial or partially artificial means. Common methods of ART are In Vitro Fertilization (IVF), Zygote Intra Fallopian Transfer (ZIFT) and Gamete Intafallopian Transfer (GIFT).

In Vitro Fertilization (IVF) : It means fertilization outside of the body. IVF is the most effective ART. It is often used when a woman's fallopian tubes are blocked or when a man produces very few sperms. Doctors treat the woman with a drug that causes the ovaries to produce multiple eggs. Once mature, the eggs are removed from the woman. They are kept in a dish in the lab along with the man's sperm for ferlilization, After 3 to 5 days, healthy embryos are implanted in the woman's uterus.

Zygote Intra Fallopian Transfer (ZIFT) or Total Embryo Transfer : It is similar to IVF. Fertilization occurs in the laboratory. Then the very young embryo is transferred to the fallopian tube instead of uterus.

Gamete Intra Fallopian Transfer (GIFT) : It involves transferring eggs and sperm into the woman's fallopian tube. So fertilization occurs in the woman's body.

ART procedures sometimes involve the use of donor eggs (eggs from another woman), donor sperm or previously frozen embryos. Donor eggs are somtimes used for women who can not produce eggs. Also, donor eggs or donor sperm are sometimes used when the woman or man has a genetic disease that can be passed on to the baby. An infertile woman or couple may also use donor embryos. These are embryos that were either created by couples in infertility treatment or were created from donor sperm and donor eggs. The donated embryo is transferred to the uterus. The child will not be genetically related to either parent.

(55) What is an infertility ?

(A) Loss the capacity of embryo implantation in female and pregnancy.

(B) Loss of egg formation in female and transport capacity.

(C) Loss of sperm capacity by male and very less production of sperms.

(D) All of the above.

(56) What is Amniosentensis ?

(A) Examining of blastomeres

(B) Examinng blato coel

(C) Examining of amniotil fluid blastula

(D) Examine of nucleas of blastomeres

(57) When AFT (Amniotic Fluid Techniques) or Amniosentensis can apply ?

(A) After birth of baby (B) Before birth of baby

(C) At the time of baby birth (D) At the time of implantation

(58) For the clinical diagnosis from where amiotic fluid is taken ?

(A) From amniotic sac (B) From cytoplasm of cells

(C) From cells of chorinary layer (D) From the somatic cells

(59) Why AFT is applied ?

(A) For the examine of disorders of chromosomal study

(B) For the examine sex

(C) To examine abnormality of chromosomes and sex

(D) To examine abnormality of chromosomes or sex

(60) What is full form of ART ?

(A) Aquerate Reproductive Technique (B) Acalte Reproductive Technique

(C) Assesory Reproductive Technique (D) Artificial Reproductive Technique

(61) What is full form of IVF ?

(A) Internal Vaccine Fertilization (B) Internal Veins Fertilization

(C) Intra Veins Fertilization (D) In Vitro Fertilization

(62) When IVF is apply ?

(A) Failure of oviduct (B) No formation of egg cell

(C) less counting of sperms (D) Absence of gonad

(63) What is full form of ZIFT ?

(A) Zygote Intra Fallopian Transfer (B) Zoospore Intra Fallopian Transfer

(C) Zoosperm Intra Fallopian Transfer (D) Zygote Internal Fallopian Tube Transfer

(64) Where fertilization occurs in ZIFT and where embryo is implanted ?

(A) In uterus, in oviduct (B) In laboratory, in uterus

(C) In laboratory, in oviduct (D) In oviduct, in laboratory

(65) What is GIFT ?

(A) Gamete Internal Fallopian Transfer (B) Gamete Intra Fallopian Transfer

(C) Gonad Internal Fallopian Transfer (D) Gonad Intra Fallopian Transfer

(66) In GIFT where fertilization occurs ?

(A) In female body (B) In oviduct (C) in uterus (D) In cervix

(67) Which animal group is having first amniotic sac ?

(A) Amphibian (B) Reptiles (C) Aves (D) Mammals

● **A - Assertion, R - Reason type questions**

Choose the correct option from the following for given questions.

(A) A and R both are correct, while R is correct explaination of A.

(B) A and R both are correct but R is not correct explaination of A.

(C) A is correct, but R is wrong.

(D) A is wrong but R is correct.

(68) Assertion A : Highly pressure occurs in India on land, air and water.

Reason R : India supports 16.87% of the worlds population.

(A) (B) (C) (D)

(69) Assertion A : Population growth is an essential devil.

Reason R : By the growth occur in population social crime, sexual crime and available resources, decreases.

(A) (B) (C) (D)

(70) Assertion A : Various programmes can not applied for reproductive health by non-governmental organizations

Reason R : Professionals and equipments are necessary for the programmes.

(A) (B) (C) (D)

(71) Assertion A : Copper - T is included in second generation IUDS.

Reason R : Copper has acute antifertility effects.

(A) (B) (C) (D)

(72) Assertion A : The couples avoid from coitus from day 10 to 17 days of the menstrual cycle.

Reason R : During the menstrual cycle 10 to 17 days possibility of fertilization is more.

(A) (B) (C) (D)

(73) Assertion A : Pills are taken as a contraceptives in oraly.

Reason R : Which has progestero and estrogen combines form.

(A) (B) (C) (D)

(74) Assertion A : Bacteria, Virus, Protozoa and fungi are cause of STDs.

Reason R : Highest cases of STDs in age group of 25 to 29.

(A) (B) (C) (D)

(75) Assertion A : Main aim of chemical method is that conjugation occur of vaible spern and egg cell.

Reason R : Sperm killer foam attached with sperm and decrease O_2 consumption capacity and sperms die.

(A) (B) (C) (D)

(76) Assertion A : In Gonorrhorea, painful urine passing occurs.

Reason R : In syphilis white patches on the tongue or roof of the buccal cavity occurs.

(A) (B) (C) (D)

(77) Assertion A : Amniosentensis is also known as AFT.

Reason R : By this method sex determination can be done.

(A) (B) (C) (D)

(78) Assertion A : In GIFT method fertilization can occur in female body.

Reason R : In this method egg cells and sperms can transfer in oviduct / fallopian tube of female.

(A) (B) (C) (D)

(79) Assertion A : In vasectomy A small part of the vasadeferens is removed through a small incision in the scrotum.

Reason R : In tubectomy, a small part of the fallopian tube is removed by small incision in the abdomen or vaginal tract.

(A) (B) (C) (D)

(80) Assertion A : In our country abortion is illegal.

Reason R : When abortion is result of rape.

(A) (B) (C) (D)

(81) Assertion A : In IVF fertilizatied, occurs outer side of the body.
Reason R : In ZIFF fertilization occurs in laboratory, after that foetus is transfer in the fallopian tube.
(A) (B) (C) (D)

(82) Assertion A : Saheli is taken once in a week.
Reason R : Use of saheli sometime it produce side effects also.
(A) (B) (C) (D)

(83) Assertion A : AFT test is used for to determine genetic disorder and sex determination of foetus.
Reason R : Small amount of amiotic fluid is taken from surrounding embryo.
(A) (B) (C) (D)

(84) Assertion A : After the pregnancy MTP can be applied 24 to 30 weeks.
Reason R : MTP is neccessary while it is result of rape.
(A) (B) (C) (D)

Answers :(55-D), (56-C), (57-B), (58-A), (59-D), (60-D), (61-D), (62-C), (63-A), (64-C), (65-B), (66-A), (67-B), (68-A), (69-B), (70-D), (71-A), (72-B), (73-A), (74-C), (75-D), (76-B), (77-B), (78-A), (79-B), (80-B), (81-B), (82-B), (83-B), (84-D)

(85)

	Column : I	**Column : II**	
(P)	Male condom	(i) Uterus	(A) (P-ii) (Q-iv) (R-i) (S-iii)
(Q)	Female condom	(ii) Penis	(B) (P-ii) (Q-iii) (R-iv) (S-i)
(R)	Diaphragm	(iii) Vaginal tract	(C) (P-ii) (Q-i) (R-iv) (S-iii)
(S)	Copper - T	(iv) Cervix	(D) (P-ii) (Q-iii) (R-iv) (S-i)

(86)

	Column : I	**Column : II**
(P)	ZIFT	(i) External fertilized cell by sperm and ovun which transfer in fallopian tube.
(Q)	GIFT	(ii) External fertilized embryo is implanted in uterus.
(R)	IVF	(iii) Donated embryo is implanted in uterus.
(S)	ART	(iv) Gametes are transferred in oviduct /fallopian tube.

(A) (P-i) (Q-iv) (R-ii) (S-iii)
(B) (P-i) (Q-ii) (R-iv) (S-iii)
(C) (P-iv) (Q-i) (R-iii) (S-ii)
(D) (P-ii) (Q-iii) (R-i) (S-iv)

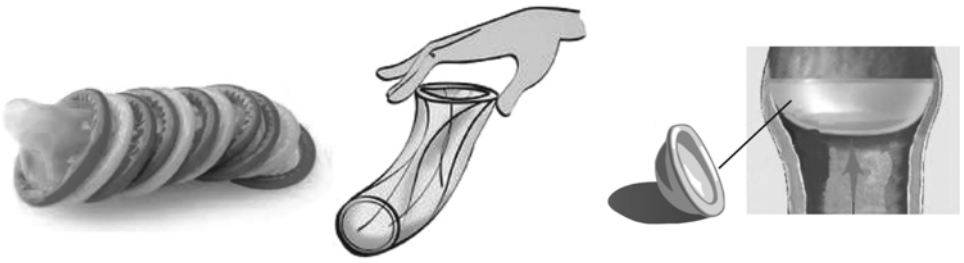
(87)

	Column : I	**Column : II**	
(P)	Gonorrhoea	(i) Herpes simplex	(A) (P-ii) (Q-i) (R-iii) (S-iv)
(Q)	Syphilis	(ii) *Neisseria gonorrhoeae*	(B) (P-iii) (Q-iv) (R-ii) (S-i)
(R)	Genetical Herpes	(iii) *Trichomonus veginalis*	(C) (P-ii) (Q-iii) (R-i) (S-iv)
(S)	Trichomoniasis	(iv) *Treponema pallidium*	(D) (P-ii) (Q-iv) (R-i) (S-iii)

(88)

	Column : I	**Column : II**	
(P)	ELISA - Test	(i) Ag. Ab. test.	(A) (P-i) (Q-ii) (R-iii)
(Q)	DNA hybridization	(ii) With help of primer doubled genetic segment of microbe is obtained	(B) (P-iii) (Q-i) (R-ii) (C) (P-i) (Q-iii) (R-ii)
(R)	PCR	(iii) Short chain of polynucleotide chain consume of genetic material of microbes	(D) (P-ii) (Q-i) (R-iii)

(89) What is indication about fignre B from given diagrams ?



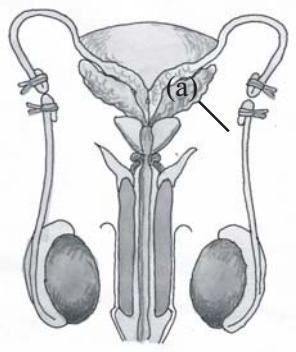
(A) Female condom (B) Male condom (C) Diaphragm (D) Sterilization

(90) What is variation included in given figure - c ?

(A) Cone shaped cap (B) Dome shaped cap (C) Traditional cap (D) A and B both

(91) What is the principle to present the conjugation of viable - spern with egg cell from which given figure above ?

(A) A (B) B (C) C



(92) What is indicating part A in given diagram ?

(A) Female condom (B) Sterilization

(C) Diaphragn (D) Male condom

(93) Given diagram represents .....

(A) Tubectomy (B) Vasectomy

(C) Sterilization in male (D) A and B both

Answers : (85-B), (86-A), (87-D), (88-C), (89-A), (90-D), (91-D), (92-D), (93-D)

- **Question for NEET :**

(94) Testtube baby means when baby is born .....

(A) develops from the unfertilized egg/ovum.

(B) it is developed in test tube

(C) it is developed by tissue culture method.

(D) egg cell is fertilized externally than trassplant in uterus.

(95) Which females are advised for GIFT method ?

(A) Who can not produce egg cell / ovum.

(B) Who can not maintain foetus / embryo in uterus.

(C) Whose cervix portion in narrower, thus sperms can not passed through it.

(D) Who can not produce specific atmosphere for fertilization.

(96) How many weeks after pregnancy is safe for MTP ?

(A) 8 Weeks (B) 18 Weeks (C) 12 Weeks (D) 6 Weeks

(97) Which method of contraception is more choosen in india recently ?

(A) Cap of cervix (B) Tubectomy

(C) Diaphragms (D) IUDs (Intra uterine devices)

(98) Copper containg IUDs put in uterus, copper ions are released and act as a ..... .

(A) uterus misbehave for embryo implantation

(B) rises in the activity of sperms

(C) locomotory capacity of sperm is stopped

(D) stop the process of ovulation..

(99) Do matches according to method (p-s) and its uses (a -d) which is given below and choose correct option for these.

Column - I (Methods)	**Column - II (Types of uses)**
(P) Pills	(a) Prevents the sperms to reach cervix
(Q) Condom	(b) To inhibit the embryo implanlation
(R) Vasectomy	(c) Prevents the production of ova.
(S) Copper - T	(d) No sperms are found in semen.

(A) (P-c), (Q-a), (R-d), (S-b) (B) (P-d), (Q-a), (R-b), (S-c)

(C) (P-c), (Q-d), (R-a), (S-b) (D) (P-b), (Q-c), (R-a), (S-d)

(100) Read given statement carefully and identify that which are two statements are correct for contraceptive methods ?

Statement (i) During the first trimster MTP is safe.

Statement (ii) Mother feeds child upto two years, commonly in that period no probability of pregnancy.

Statement (iii) Copper - T like devices which is IUDs resultant contraceptive.

Statement (iv) For prevention of pregnancy contraceptive pills should be taken for a week after coitus.

Above given statements which two statements are correct ?

(A) (ii) and (iii) (B) (iii) and (iv) (C) (i) and (iii) (D) (i) and (ii)

Answers : (94-D), (95-A), (96-B), (97-D), (98-C), (99-A) , (100-C)

CONCEPT MAP

Reproductive Health

Strategies of reproductive health

- It is counselling and creating awareness among people about :
  - → Reproductive organs
  - → Adolescence and associated changes
  - → Safe and hygienic sexual practices
  - → Sexually transmitted diseases
  - AIDS, syphilis, genital herpes, genital warts, gonorrhoea, hepatities B etc are some of the diseases transmitted through sexual intercourses with infected person.
  - It can be prevented by :
    - Avoiding sex with unknown/ multiple partners.
    - Using condoms during coitus.
    - Early detection and complete treatment if diagnosed with disease.
- Programmes undertaken regarding this are :
  - → Child immunisation
  - → Ban on amniocentesis to legally check female foeticides
  - → Maternity and child health and family planning.

Polulation explosion and birth control

- All around develoment, increased health facilities, better living conditions, these all has resulted in population explosion.
- This can be overcome by motivating smaller families by using contraceptive methods-measures.

Types of contraceptive measures

Temporary methods

- Coitus interruptus
- Lactational amenorrhea
- Condoms, cervical caps etc.
- Foam tablets, jellies, creams etc.
- IUCD
- Pills, norplant etc.

→ **Permanent methods**

- Vasectomy
- Tubectomy

→ **Medical termination of pregnancy**

Infertility

- It is inability to conceive or produce children inspite of umprotected sexual cohabitation.
- It can be caused by physical, congenital, immunological or even psychological disorders.
- Specialized **infertility clinics** can help in the diagnosis and proper treatment of some of these disorders.
- However, where such diagnosis and treatment are not possible, the couples can be assisted to have children through certain special techniques called Assisted Reproductive Technologies (ART).
- Assisted Reproductive Technologies (ART) assist infertile couple to have children.
- Some important techniques of ART are :
  - Test tube baby programme
  - Artificial Insemination Techniques (AIT)
  - Gamete Intra Fallopian Transfer (GIFT)
  - Intra Cytoplasmic Sperm Injection (ICSI)

38 Heredity and Variation

- **Genetics :**

- The term genetics was first coined by William Bateson in 1906.
- The word genetics is derived from the Greek term gen which means 'to become'.
- "Genetics is the branch of Biology which deals with heredity and variation among related Organisms."

- **Heredity :**

- "Transmission of the characters from parents to offsprings."

- **Variation :**

- "The occurrence of differences among the individuals of same species is known as variation."

- **<u>Historical Background of Heredity :</u>**

- **Babylon and Assyria :** Selective breeding of horses, donkeys and date palm was also done during the ancient civilization of Babylone and Assyria nearly 6000 years ago.
- **Ancient Chinese Writing :** Ancient chinese writing mentions creating better varieties of paddy nearly 5000 years ago.
- **Hippocrates (400 B.C.) :** Hippocrates believed that characteristic are inherited from parents because reproductive material is handed over from all parts of the body of an individual.
- **Gregor Johann Medel :** The science of heredity and Variation, the scientific principle of the science of genetics originated in 1900 with re-discovery of a scientific article published in 1866 by Gregor Johann Medel.

* **Johansen :** Mendel's 'factors', the carriers of heredity information are known as 'genes' a term coined by Johansen in 1909.

---

(1) The term genetics was first coined by ..... .

(A) Bateson (B) Mendel

(C) Morgan (D) Correns

(2) In which civilization selective breeding of horses, donkeys and date palm was done?

(A) Babylon (B) Assyria

(C) Harappa (D) A and B Both

(3) Who believed that characteristic are inherited from parents because reproductive material is handed over from all parts of the body of an individual ?

(A) Mendel (B) Bateson

(C) Hippocrates (D) Johansen

(4) Who believed that 'factors' responsible for hereditary information ?

(A) Mendel (B) Johansen

(C) Bateson and Punnet (D) Morgan

(5) Mendel's factors the carriers of heredity information are known as 'genes' a term coined by ..... .

(A) Bateson (B) Johansen

(C) Morgan (D) Tschermak

(6) Mendelism is related with ……

(A) Heredity is living beings. (B) Meiosis during sexual reproduction

(C) Mutation in living organisms (D) Crossing over and Linkage

(7) The first great "geneticist" was ..... .

(A) Engler (B) Mendel (C) Bateson (D) Boveri

(8) Nearly 5000 years ago ancient Chinese writing mentions for creating better varieties of ..... .

(A) Horses (B) Donkeys (C) Paddy (D) Date Palm

Answers : (1-A), (2-D), (3-C), (4-A), (5-B), (6-A), (7-B), (8-C)

- **Gregor Mendel – The Father of Genetics :**

The contribution of Mendel to Genetics is called Mendelism. Mendel is called the father of Genetics.

Christien Johann Mendel was born in 1822. He came to the monastery at Brunn and was appointed as priest in 1848. In 1856, he began to collect and observe the numerous varieties of the garden pea. These varieties differed in seed, pod, flower and a number of other characteristics. He grew each variety in different plots, so that any variation from the listed characteristics could be easily spotted. He carried out experiments for seven years (1856 – 1863) in the monastery gardens. He presented the result of his study of hybridizations together with the generalizations at the Natural History Society of Brunn in 1865. No one at that time read Mendel's research papers. They lay neglected until 1900 when they were discovered almost simultaneously and independently by Karl currens, Hugo de varies and Von Tschermak. He died in 1884. When Mendel's work was recognized and appreciated, he was no more.



Gregor Johann Mendel
(1822 - 1884)

(9) Mendel Presented the results of his study of hybridization together with generalizations at ..... .

(A) British Government in 1865 (B) American Government in 1865

(C) Natural History Society of Brunn in 1865 (D) De. Varies, Correns and Tschermak in 1865

(10) Which scientist had rediscovered Mendel's work ?

(A) De Varies (B) Tschermak (C) Correns (D) All above

(11) Mendel was the native of ..... .

(A) France (B) Austria (C) Sweden (D) Italy

Answers : (9-C), (10-D), (11-B)

- **Mendel's work :**
- Mendel did his work on Garden pea (*Pisum satium L.*)
- It is very easy to cultivate the pea plant in open ground.
- The flowers of pea plants are normally self- fertilized.
- The pea plant shows a number of contrasting characters.
- The hybrid of garden pea are perfectly fertile.

- Cross pollination is not very difficult in pea plant.
- Artificial fertilization was almost Successful.
- It produced large number of OFFSpring.

Speciality of Mendels work :

- Many scientists earlier to the period of Mendel (1856) had attempted Inheritance experiments. How ever, most of them failed to arrive at any specific conclusion or principles to explain the pattern of inheritance.
- During Mendels investigations into inheritance patterns it was for the first time that a statistical analysis and mathematical logic were applied.
- He studied the inheritance of only one character at a time in most of the experiments.
- He maintained statistical records of his results. It helped Mendel to drive numerical ratios of significances.

---

(12) Mendel did his work on…..

(A) *Pisum sativum* (B) *Drosophila moelanogaster*

(C) *Mirabilis jalapa* (D) *Lathyrus odoratus*

(13) The hybrid of garden Pea are…

(A) Intraspecific sterile (B) Perfectly fertile

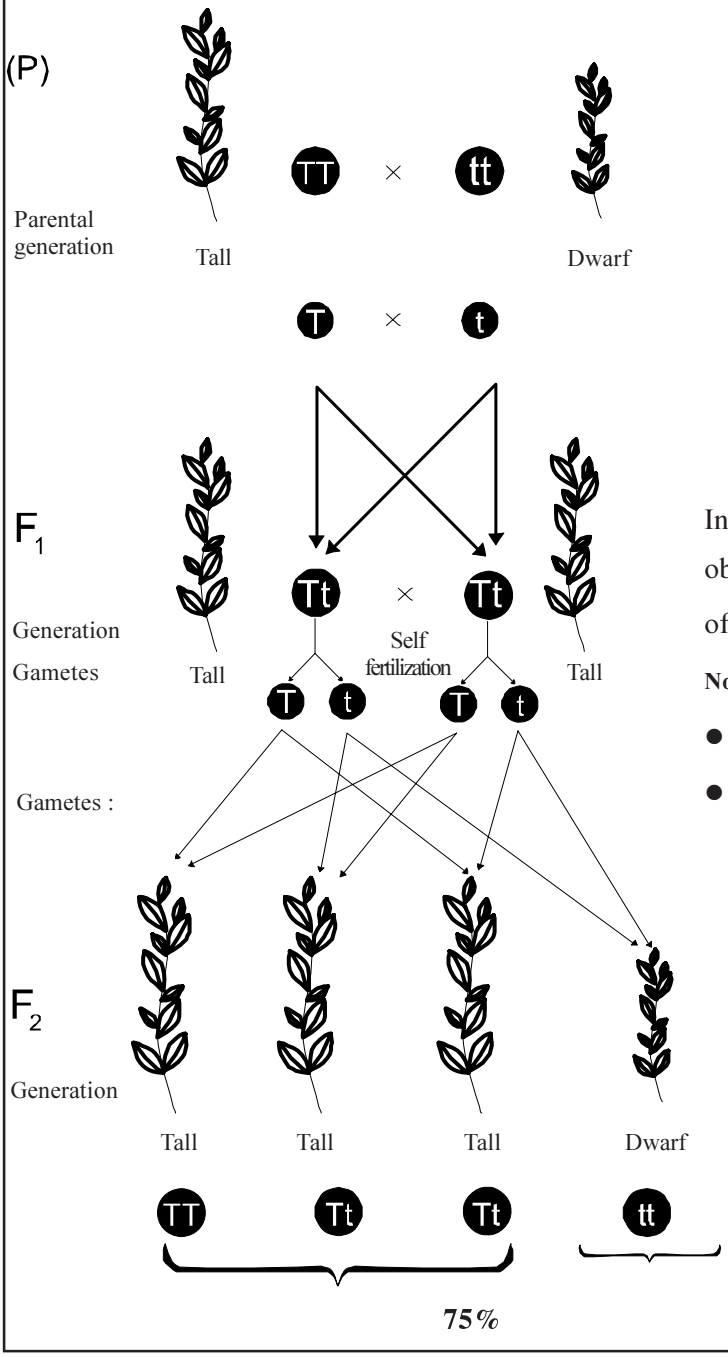
(C) Interspecific sterile (D) Dwarf

Answers : (12-A), (13-B)

- **Monohybrid Cross Experiments or Single Gene Inheritance**
- The experiments Considering the inheritance of any one character are called Monohybridization experiment.
- Mendel selected two pea plants one with a tall plant and other with a dwarf or short plant.
- These plants were considered as parental Plants (P) and were pure breeding.
- Pure means it maintain some characters generation to generation out of two plants any one can be taken either as a male or female.
- This was done by first removing the anther of an immature flower of all plants. This flower was than covered with a small plastic bag when this flower matured, the pistil was dusted with pollen received from the dwarf plant. Seeds were collected from this plant.
- These seeds were sown and a group of plants were raised. These plants represents F_1 generation.
- In Mendel's above referred experiment, all plants in F_1 generation were tall.
- They were as tall as their parents in P generation.
- The F_1 plants were inbreed. The seeds were collected and the next generation F_2 was raised.
- In the F_2 generation two type of plants were found.
- They were tall and dwarf.

- Mendel counted the number of tall and dwarf plants of the 1064 plants of F_2 generation, 787 plants were tall and 277 plants were dwarf, this ratio is approximately 3:1.
- Mendel carried out similar experiments, involving seven different characters. Each time he obtained similar results.
- Mendel's differentiating characters have been variously called 'factors' or 'genes'.
- Bateson Proposed the name 'Alleomorph' or 'Allele' for them.

Experimetns with garden pea for single pair of contrasting characters.



In F_2 generation 3/4 of the total OFFsprings obtained exhibit dominant expression and 1/4 of them exhibit recessive expression.

Note :

- Monohybride experiment phenotypic ratio = 3:1
- Monohybride experiment segnotypic ratio = 1:2:1

• Phenotypic ratio : 3:1 • Genotypic ratio : 1:2:1

Seven Pairs of contrasting traits in pea plant studied by Mendel

	Character	Dominant Trait	Recessive Trait
(1)	Seed Shape	Round	Wrinkled
(2)	Seed Colour	Yellow	Green
(3)	Flower Colour	Violet	White
(4)	Pod Shape	Full	Constricted
(5)	Pod Colour	Green	Yellow

	Character	Dominant	Recessive
(6)	Flower position	Axial	Terminal
(7)	Stem Height	Tall	Dwarf

(14) In Mendel's monohybrid experiment how many different characters selected to carry out similar experiments?

(A) 2 (B) 4 (C) 7 (D) 1

(15) Which of the parental plants Mendel has selected for his experiments ?

(A) Heterozygous and pure (B) Homozygous and mixed

(C) Homozygous and pure (D) Heterozygous and mixed

(16) In a Monohybrid cross of Mendel's the different phenotypes available in F_1 generation are …… .

(A) pure, tall (B) mixed, dwarf

(C) heterozygous, tall (D) heterozygous, dwarf

(17) How many different types of plants are formed in F_2 progeny obtained from self-pollination of a F_1 ?

(A) 1 (B) 2 (C) 4 (D) 16

(18) How many different type of plants counted by Mendel in the F_2 generation ?

(A) 1199 tall and 787 dwarf (B) 1064 tall and 787 dwarf

(C) 1064 tall and 277 dwarf (D) 787 tall and 277 dwarf

(19) What is the Phenotype ratio of Mendel's monohybrid cross ?

(A) 2:2 (B) 3:1 (C) 1:1 (D) 1:2:1

(20) What is the Genotype ratio of Mendel's monohybrid cross ?

(A) 1:2:1 (B) 3:1 (C) 1:1 (D) 2:2

(21) The term 'Alleomorph' was coined by ..... .

(A) Morgan (B) Johansen (C) Bateson (D) Tschermak

(22) In a monohybrid cross between two heterozygous individuals, the number of pure homozygous individuals obtained in F_1 generation is ..... .

(A) 2 (B) 4 (C) 6 (D) 8

(23) Two pea plants were subjected cross pollination of the 183 plants produced in the next generation, 94 plants were found to be tall and 89 plants were found to be dwarf. The genotypes of the two parental plants are likely to be ..... .

(A) TT and tt (B) Tt and Tt (C) Tt and tt (D) TT and TT

(24) What will be the genotypes of parental generation ? If all progeny obtained in F_1 were recesive and Dwarf.…

(A) TT and tt (B) tt and tt (C) tt and Tt (D) Tt and Tt

Answers : (14-D), (15-C), (16-C), (17-B), (18-D), (19-B), (20-A), (21-C), (22-A), (23-C), (24-B)

Laws of Inheritance :

– Based on his observation on monohybrid crosses, Mendel proposed two general rules to consolidate his understanding of inheritance in monohybrid crosses. Today these rules are called the principles or laws of inheritance.

– The first law or law of dominance and the second law or law of segregation.

Law of Dominance :

– When two different alleles for a character occur in an organism, only one of the two alleles expresses itself. The other allele remains unexpressed. The allele which is expressed is called dominant gene and the allele which is not expressed is called recessive gene.

– Let us now examine results obtained from self-fertilization among F_1 individuals. All F_1 plants are tall and have Tt genotype.

– As a male parent, they will produced two types of gamets (T and t gamets).

– As a female parent also they will produce two types of gamets (T and t).

– Two types of gamets can fertilize another two types of gametes in four possible ways (TT, Tt, Tt, tt). Of these, three kinds will be tall (TT, Tt, Tt) and one kind will be dwarf (tt).

– Thus in F_2 generation $\frac{3}{4}$ of the total offsprings obtained exhibit dominant expression and $\frac{1}{4}$ of them exhibit recessive expression. Thus, the ratio of 3:1 is obtained.

Law of Segregation :

– When a pair of contrasting traits are brought together in a hybrid, the two factors (alleles) remain together without mixing.

– When the gametes are formed from each other, only one enters each gamete. Thus any gamete contains only one gene for an expression of a character, this is also called Law of purity of gametes.

– An organism can be homozygous or heterozygous for a character, but its gametes will always be pure for a particular expression of that character.

(25) What is dominat gene ?

(A) Both the genes expresses their expression together.

(B) The allele which is expressed their character.

(C) The allele which is remains unexpressed.

(D) Multiple effect of a single gene.

(26) The allele which is not expressed is called ..... .

(A) Recessive gene (B) Dominant gene

(C) Homozygous gene (D) Co-dominant gene

(27) Both the genes of a character are identical is…..

(A) Homozygous (B) Heterozygous

(C) Dominant (D) Co-dominant

(28) Both the genes of character are unlike is…

(A) Homozygous (B) Incomplete dominant

(C) Heterozygous (D) Co-dominant

(29) The gametes will always be…… for a particular expression of that character.

(A) n, mixed (B) 2n, pure (C) n, pure (D) none of this

(30) The results obtained from self- fertilization amongst F_1 individuals in Mendel's Monohybrid cross is ..... .

(A) $\frac{3}{4}$ Dominant and $\frac{1}{4}$ Recessive (B) $\frac{1}{4}$ Dominant and $\frac{3}{4}$ Recessive

(C) $\frac{1}{4}$ Dominant and $\frac{2}{4}$ Recessive (D) $\frac{9}{16}$ Dominant and $\frac{4}{16}$ Recessive

Answers : (25-B), (26-A), (27-A), (28-C), (29-C), (30-A)

Characters	**Explanation**	**Eample**
Characters	Mental and moral qualities distinctive to an individual	Plant height
Expression	Phenomena of representing character	Tall or Dwarf height
Gene	Unit of heredity	T or t
Phenotype	It is the expression of a character	Tall or Dwarf plant
Genotype	Genotype is the gene complement of an organism	TT or Tt or tt
Homozygous	An individual with two indentical alleles	TT or tt
Heterozygous	An individual with two dissimilar alleles	Tt
Dominant factor	It express itself even in the presence of recessive allele.	T
Recessive factor	Recessive allele fails to express it effect in presence of dominant allele	t

* **Test Cross :**

– A cross arranged for deciding whether an organism is homozygous or Heterozygous is called test cross.

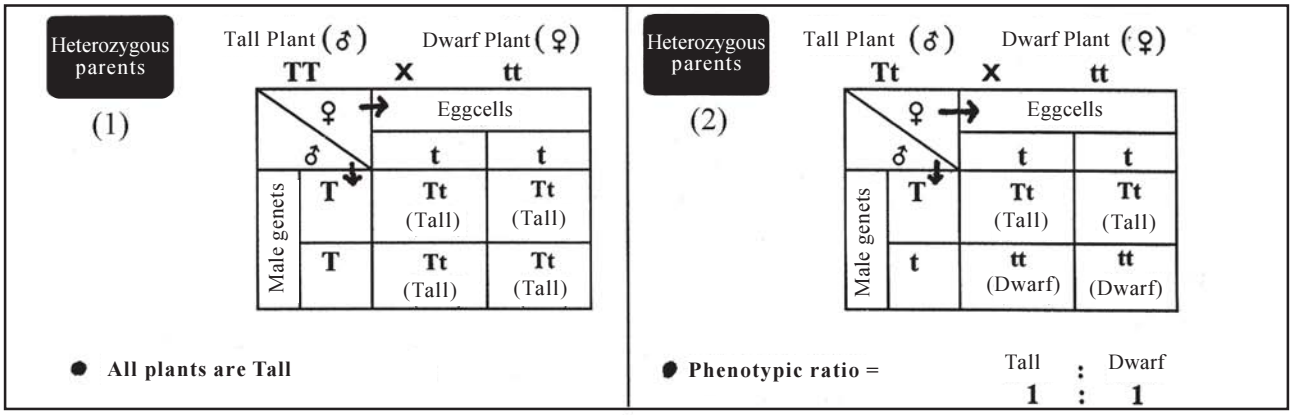
– Selected cross breed of pea may be homozygous tall (TT) or heterozygous tall (Tt). The genotype of both (TT and Tt) are different but phenotype are same.

– If we cross a tall plant with a dwarf plant, two outcomes are possible. This will depend on the genotype of tall plant.

– If homozygous tall (TT) is cross with dwarf plant (tt) all offspring will be tall.

– If it is heterozygous tall (Tt) 50% will be tall and 50% will be dwarf)

– Thus, through such a cross, genotype of the plant can be determined. Hence, it is called test cross. The ratio obtained is 1: 1.



---

(31) Which option is true for test – cross?

(A) Tt ×Tt (B) TT × TT (C) Tt × tt (D) tt × tt

(32) A cross, arranged for deciding whether an organism is homozygous or Heterozygous is ..... .

(A) Epistasis (B) Test-cross (C) Back cross (D) Monohybrid-cross

(33) The phenotype and Genotypes of test-cross is ..... .

(A) 3:1 and 1:1 (B) 1:1 and 3:1 (C) 1:1 and 1:1 (D) 9:3 and 3:1

(34) If Homozygous recessive parent cross with homozygous dominant parent. Then what will be the result ?

(A) 100% heterozygous dominant progeny

(B) 50% dominant and 50% recessive progeny

(C) 70% dominant and 25% recessive progeny

(D) 70% dominant and 30% recessive progeny

(35) Heterozygote tall plant (Tt) is crossed with homozygous dwarf (tt) plant. Then what will be the result ?

(A) All progeny will be (Tt) tall

(B) 50% TT and 50% tt progeny

(C) 70% TT and 30% tt progeny

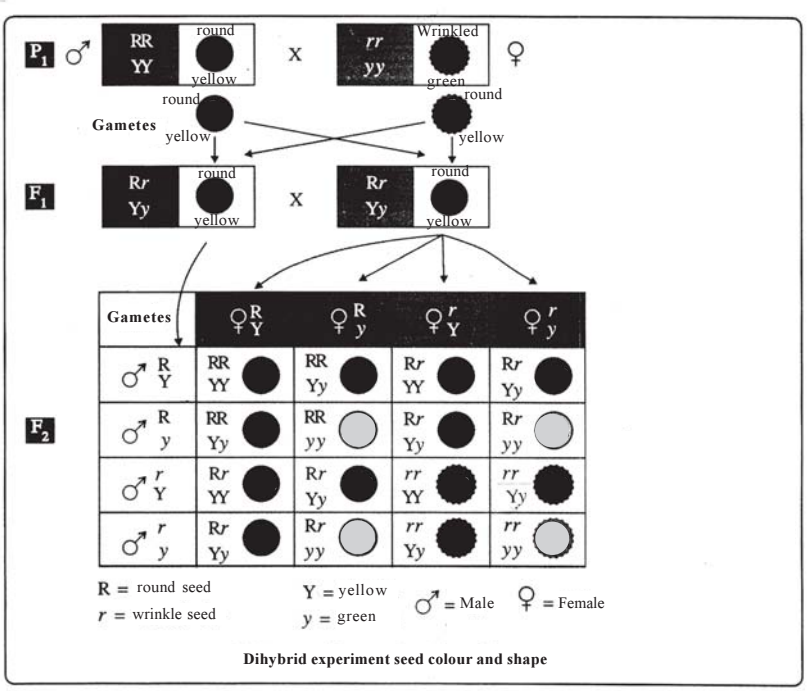
(D) 75% Tt and 25% tt progeny

Answers : (31-C), (32-B), (33-C), (34-A), (35-B)

Inheritance of two Genes/Dihybrid Experiment :

- The crossing of two plants differing in two characters is called dihybrid experiment.
- Mendel arranged experiments to follow simulataneous inheritance of two characters in pea plant e.g. shape of seed and colour of seed.
- Mendel selected pure- breeding yellow, round seed producing plant and another pure breeding green, wrinkle seed producing plant.
- These two plants were treated as parents and were crossed.
- Here, the gene for round shape of seeds 'R' is dominant over the gene for wrinkled shape of seed 'r'.
- Gene for yellow colour of seed 'Y is dominant over the gene for green colour seed 'y'.
- In his experiment, the F_1 generation plants produced only yellow round seed.
- Results in F_2 offsprings are as follow.

Round yellow seeded plant	Wrinkled yellow seeded plant	Round green seeded plant	Wrinkle green seeded plant
9/16	3/16	3/16	1/16



Dihybrid experiment seed colour and shape

R = smooth seed Y= Yellow colour

R = Rough seed y = Green colour

Note : Phenotypic ratio of Dihybrid Experiment } 9:3 : 3:1

Genotypic ratio of Dihybrid Experiment } 1:2:2:4:4:1:2:1:2:1

Law of Independent Assortment : (x × y = xy)

- The segregation of genes controlling one character is independent of the segregation of genes controlling another character. This law is based on dihybrid experiment.
- During gametes formation of a dihybrid cross, the factors (genes) for yellow colour assort out independently of the factors for round shape.

- The gene Y may combine with the dominant gene R or the recessive gene r of the other character and enter a gamete.
- In the same way the gene y may combine with the dominant gene R or the recessive gene r and enter a gamete.

So, the F_1 dihybrid plants produce four type of gametes and they are YR, Yr, yR and yr.

Hence, [x × y = xy]

Shape	Round	:	Wrinkled	
(X)	3	:	1	
Colour	Yellow	:	Green	
(Y)	3	:	1	

9	:	3	:	3	:	1
Round and yellow seeded plant		Wrinkled yellow seeded plant		Round green seeded plant		Wrinkled green seeded plant

This law also has only limited expression. It is true only in those case where the two pairs of genes, independently controlling two different characters are located on two different pairs of hamologous chromosomes. Gene on the same pairs of chromosomes are not assorted.

(36) How many different kind of phenotypic form will be obtain if we arrange Mendel's dihybrid experiment result in Punnett 16 square boxes ?
(A) 8 (B) 4 (C) 2 (D) 16

(37) What was different genotypic form of four different kinds of phenotypic plants obtained in F_2 generation of Mendel's dihybrid experiment?
(A) 16 (B) 4 (C) 8 (D) 9

(38) How many Genotypes of RrYy in F_2 generation of dihybrid experiment ?
(A) 3 (B) 2 (C) 4 (D) 9

(39) How many Genotypes of rryy in F_2 generation of dihybrid experiment ?
(A) 1 (B) 4 (C) 2 (D) 3

(40) What type of gametes will form by genotype RrYy ?
(A) RY, Ry, rY, ry (B) RY, Ry, ry, ry (C) Ry, Ry, ry, ry (D) Rr, RR, Yy, YY

(41) How many different kinds of gametes will be produced by a plant having the Genotype AABbCC ?
(A) 9 (B) 2 (C) 3 (D) 4

(42) How many different (Re combinant) progeny obtained at the end of dihybrid experiment ?
(A) 4 (B) 2 (C) 9 (D) 16

(43) In dihybrid cross, the factors for yellow colour assort out independently of the factors for ..... .
(A) Green colour (B) Round shape (C) Wrinkle shape (D) Long shape

(44) What is the mathematical formula for independent Assortment ?
(A) x + y = w + z (B) $(a + b)^2$ (C) x × y = xy (D) $(a + b + c)^2$

Answers : (36-B), (37-D), (38-C), (39-A), (40-A), (41-B), (42-B), (43-B), (44-C))

Evaluation of Mendel's work :

– All conclusions and deductions by Mendel have not been found true in all cases.

– His belief that of two alleles of a gene, one is dominant and the other is recessive is not true in all cases.

– Many exceptions are found where both genes express their effects jointly.

– It is also not true that there are only two alleles of a gene.

– It is also not true that one character is controlled by one pair of genes.

(45) What was Mendel's belief for two alleles of a gene ?

(A) dominant, dominant (B) dominant, co- dominant

(C) dominant, incomplete Dominant (D) dominant, recessive

(46) According to Mendel's belief, How many gene require for controlling one character ?

(A) 2 pairs (B) 4 pairs (C) 1 pair (D) One

(47) One character is controlled by one pair of genes. This statement is incorrect for ..... .

(A) Co-dominance (B) Polygenic Inheritance

(C) Sex linked inheritance (D) Incomplete dominance

Answers : (45-D), (46-C), (47-B)

Incomplete Dominance : [1 : 2 : 1]

– Incomplete Dominance can be studied though experiments on Mirabilis Jalapa plant. Three colours occur in the flowers of this plant: red, white and pink.

– When homozygous red flowered and homozygous white flowered plants are crossed, all offsprings in F_1 generation are pink flowered.

– When these F_1 generation are self- fertilized in F_2 generation. Three kinds of plants are obtained. 25% plants are red flowered, 25% are white flowered and 50% are pink flowered plants.

– If we represent gene for red colour by 'R' and gene for white colour by 'W' the details of experiment results can be displayed as under :

– Homozygous red flowred plants have both genes of 'R' type. Their genotype is RR.

– All gametes which they produce will contain gene 'R'.

– Homozygous white flowered plants have both genes of 'W' type. Their genotype is WW. All gametes produced by them will contain gene 'W'.

– F_1 plants formed through their cross will contain one 'R' gene and one 'W' gene. Their genotype is RW. All plants will produce pink coloured flowers.

– If 'R' gene were dominant over 'W' gene, the plants should have been red flowered.

– If 'W' gene were dominant over 'R' gene the plants should have been white flowered. However, all plants are obtained pink flowered.

– This indicates that a mixed effect of both genes is observed. No allele is dominant over the other.

– F_2 plants produced by self cross of F_1 plants yields 1 red flowered : 2 pink flowered : 1 white flowered ratio. It means RR produces red colour flowers, WW produces white flowers and RW produces pink flowers.

– Incomplete dominance (1:2:1).

– The phenotypic and Genotypic ratio of incomplete Dominance is 1:2:1.

(P) ♀ Red Flower RR × White Flower WW ♂

Gamet : [R] [W]

(F_1) : RW

♀ Pink Flower RW × RW ♂

Self -Cross

(F_2) :

♀ ↓ / Male ♂ → Gametes	R	W
R	RR Red	WW Pink
R	RR Pink	WW White

Female Gametes

Where :

RR = Red= 1/4

RW = pink = 1/2

WW = White= 1/4

The phonotypic and Genotypic ratio of Incomplete Dominance is 1:2:1.

---

(48) Incomplete dominance can be studied through experiments on ..... .

(A) Sweet pea (B) Mirabilis (C) Drosophilla (D) E.Coli

(49) What was Phenotypic and genotypic ratio of incomplete dominance in F_2 ?

(A) 3:1 and 1:2:1 (B) 1:2:1 and 1:2:1

(C) 3:1 and 3:1 (D) 1:2:2 and 1:1

(50) When homozygous red flowered and homozygous white flowered are crossed all offsprings in F_1 generation are ..... .

(A) Pink (B) White (C) Red (D) All of above

(51) In Mirabilis a plant with RW crossed with RW then the expected percentage value of Red, Pink and White is ..... .

(A) 25%, 50%, 25% (B) 25%, 25%, 50%

(C) 10%, 20%, 70% (D) 50%, 25%, 25%

(52) In Mirabilis a hybrid for pink (RW) flower is crossed with white flower (WW) the expected phenotypic and genotypic ratio is ..... .

(A) Phenotype : Red, White, Genotype : RR, WW

(B) Phenotype : Pink, Pink, Genotype : RW, RW

(C) Phenotype : Pink, White, Genotype : RW, WW

(D) Phenotype : White, White, Genotype : WW, WW

Answers : (48-B), (49-B), (50-A), (51-A), (52-C)

Co-dominance :

– In co-dominance both dominant and recessive alleles lack their dominant and recessive relationships and both the genes expresses their expression independently.

– In these cases the dominant character is not mixed with recessive character.

– In short horn cattle, there are two pure varieties, both red and white for coat colour. A cross between the two varieties (Red RR × white rr) leads to the formation of a new variety (Rr) with reddish grey colour coat. Such contains both red hair and white hair.

(P) Red White

♂ RR × r r ♀

↓ ↓

R r

$[F_1]$: Rr

Reddish grey

♂ Rr × Rr ♀

Self- cross

$[F_2]$

♂ → / ↓ ♀	R	r
R	RR Red coat	kr Reddish grey coat
r	Rr Reddish coat grey	rr white coat

$[F_2]$ RR (Red coat) = $\frac{1}{4}$ [25%]

Rr(Reddish grey coat) = $\frac{1}{2}$ [50%]

rr(White coat) = $\frac{1}{4}$ [25%]

The phenotype and Genotype of co-dominance is 1:2:1.

(53) In which case dominant and recessive alleles lack their dominant and recessive relationships ?

(A) Incomplete dominance (B) Polygenic inheritance

(C) Co-dominance (D) Dominance

(54) When Reddish grey (Rr) short horn cattle crossed with white (rr) short horn cattle then what will be result?

(A) Reddish grey : white (1 : 1) (B) Reddish grey : white (3 : 1)

(C) Reddish grey : Red (1 : 1) (D) White : Red (3 : 1)

(55) What is genotypic ratio of co-dominance ?

(A) 3 : 1 (B) 1 : 2 : 2 (C) 1 : 2 : 1 (D) 1 : 1

(56) Short horn cattle variety (Rr) having …….. colour coat.

(A) Red (B) Black (C) Red, White (D) White, Red, Black

(57) What is genotype of Reddish grey variety ?

(A) RR (B) rr (C) Br (D) Rr

Answers : 53 (C), 54 (A), 55 (C), 56 (C), 57 (D)

Polygenic Inheritance :

- Two or more independent pairs of factors or genes which affect the same characteristic but in an additive manner are known as multiple genes or cumulative genes.
- These affect the degree of development of a character quantitatively.
- Here the effect is dependent upon the number of doses of genes present in the individual.
- According to Dervenport, skin colour in man is determined by multiple genes.
- Human skin colour is generally controlled by three separate genes.
- Each gene contributes to a unit of darkness due to incomplete dominance.
- These three genes can be designated as A, B and C, and thus the skin shade has to vary from a very dark in a AABBCC individual to very light in aabbcc individual.

---

(58) Which gene effect is depend upon the number of doses of genes present in the individual?

(A) Cumulative gene (B) Co-dominance gene

(C) Incomplete dominance gene (D) Multiple alleles gene

(59) Human skin colour gene contributes to a unit of darkness due to ..... .

(A) Complete dominance (B) Incomplete dominance

(C) Co-dominance (D) Multiple allele

(60) How many different skin colours available in human?

(A) 16 (B) 4 (C) 30 (D) 64

(61) That is a genotype of very dark skin : ..... .

(A) AA BB CC (B) aa bb cc (C) AA bb cc (D) aa BB cc

(62) That is a genotype of very light skin ..... .

(A) AA bb CC (B) AA BB cc (C) AA Bb cc (D) aa bb cc

Answers : (58-A), (59-B), (60-D), (61-A), (62-D)

Multiple alleles :

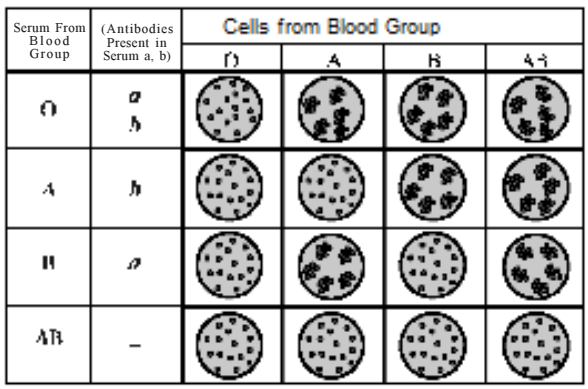
– As per Mendel, every character is controlled by one pair of alleles. There are only two optional forms, one of which is dominant and the other one is recessive.

– Cases are observed where there are more than two optional forms of a gene for one character. Thus when three or more alleles are responsible for a single characteristic, they known as multiple alleles.

- All these alleles occupy the same specific locus on the chromosomes.
- In a diploid cell, only two alleles can be present at a time on the homologous chromosomes.
- A wellknown example is the ABO blood type in humans. Here the inheritance is based on three alleles i.e. I^A , I^B, i
- I^A and I^B are dominant and i is recessive.
- The gene for producing antigen is I and its allele for non-producing antigen is i.
- I^A is responsible for producing antigen - A.
- I^B is responsible for producing antigen - B.
- These two alleles are codominant which means that both can express themselves in presence of each other.
- Thus three alternatives are possible.
- Various persons in different blood groups can have following genotypes.

Blood group	Possible Genotype
A	I^AI^A or I^Ai^o
B	I^AI^B or I^Bi^o
AB	I^AI^B
O	i *i*

- If blood group of parents are known, probable blood groups in their children can be known. Conversely, if blood group of a child is known, the blood group of its parents can be known.
- Landsteiner describe human blood groups. Four blood groups occur. These are A, B, AB and O.
- Two aspects are to be considered in deciding these blood groups- which kind of antigen occurs on RBCs, and which kind of antibody occurs in blood plasma in a person.
- **A blood group :** Person belonging to A blood group has A antigen on RBCs and b antibody in blood plasma.
- **B blood group :** Person belonging to B blood group has B antigen on RBCs and a antibody in blood plasma.
- **AB blood group :** Person belonging to AB blood group has A antigen and B antigen in other RBCs where as blood plasma does not have a or b antibodies.
- **O blood group :** Person belonging to O blood group does not have any antigen in RBCs. Its blood plasma contain 'a' as well as 'b' antibodies.
- An antigen is indicated by capital letter and the antibody effective against it is indicated by the same letter in small script.
- Antibody reacts against the antigen and causes agglutination of RBCs which possess that antigen. Thus clotting takes place.
- **Serum test :** Compability of blood groups is checked for blood transfusion. Serum test is carried out to determine the blood group of a person.



Human Blood Groups - Serum Test

- Following table shows which person can accept blood from whom and can donate blood to whom.

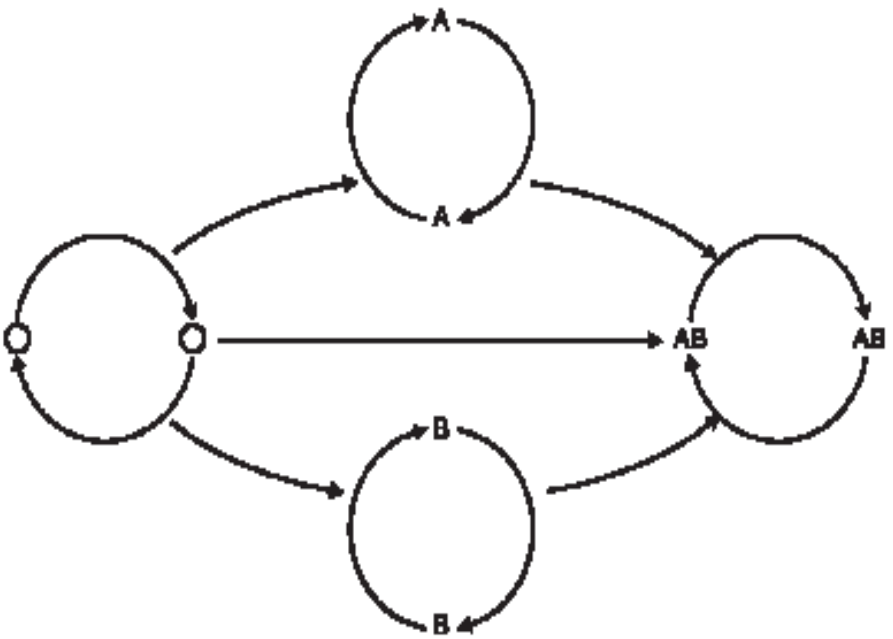
Blood Group	Antigen on RBC	Antibody in plasma	To whom can he/she Donate	From whom he/she receive
A	A	b	A, AB	A,O
B	B	a	B, AB	B ,O
AB	A , B	--	AB	A, B, O , AB (universal recipient)
O	--	a, b	A ,B,AB,O (universal donor)	O

- The following table indicates the effects that would be produced during transfusing blood between antigen and antibody of different types of donor and recipient.

Blood group based on corpuscles of Donor	Blood Group of Receipient			
	A	B	AB	O
A	-	+	-	+
B	+	-	-	+
AB	+	+	-	+
O	-	-	-	-

+ Agglutination takes place.

- Agglutination does not take place.



Simple explanation

Exclusion of paternity based on ABO Blood Groups.

Child blood group	Mother blood group	Father must be of blood group	Father can not be of blood group
A	O	A or AB	O or B
A	A	A ,B, AB, or O	-
A	B	A or AB	O or B
B	B	A , B, AB, or O	-
B	A	B or AB	O or A
B	O	B or AB	O or A
AB	A	B or AB	O or A
AB	B	A or AB	O or B
AB	AB	A, B or AB	O
O	O	O, A or B	AB
O	A	O, A or B	AB
O	B	O, A or B	AB

* **Blood groups of parents and probable blood groups of child :**

No	Blood group of parents	Probable blood group of child
1	O × O	O
2	O × A	O, A
3	O × B	O, B
4	O × AB	A, B
5	A × A	A, O
6	A × B	A ,B, AB ,O
7	A × AB	A, B, AB
8	B × B	B ,O
9	B × AB	A ,B ,AB
10	AB × AB	A, B, AB

(63) Inheritance of blood type in humans is ..... .

(A) multiple Alleles (B) co-dominance

(C) polygenic inheritance (D) incomplete dominance

(64) Which gene produce antigen for blood group ?

(A) I (B) i

(C) A (D) B

(65) Which genes are co-dominance for producing antigen?

(A) I^AI^A or I^Ai (B) I^BI^B or I^Bi (C) I^AI^B (D) i i

(66) Who describe human blood groups?

(A) Morgan (B) Landsteiner (C) Dervenport (D) Mendel

(67) Which condition decided blood group in human beings?

(A) Antigen types on RBCs (B) Antibody in blood plasma

(C) Antigen types in blood plasma (D) A and B both

(68) Which blood group having antibody 'a' as well as antibody 'b'?

(A) A (B) AB (C) B (D) O

(69) If mother and father both having 'O' blood group, then what will be probable blood group of child?

(A) O (B) O, A (C) O, B (D) A, B

(70) If mother having O blood group and father having A blood group then what will be probable blood group of child ?

(A) A, B, AB (B) O, A (C) B, O (D) AB, O

(71) If mother having O blood group and father having B blood group then what will be probable blood group of child ?

(A) A, B, O (B) A, O (C) O, B (D) AB, AB

(72) What will be a probable blood group of parents having child has sequencely A and O blood group?

(A) A, A (B) A, B (C) B, AB (D) A, AB

(73) What will be a probable blood group of parents having children has sequencely A, B, AB and O blood groups ?

(A) A,B (B) A,AB (C) B ,B (D) AB, AB

(74) What will be a propable blood group of couple having three children blood group sequencely are A, B and AB ?

(A) A, B (B) B ,B (C) A ,AB (D) AB, O

(75) A couple having two sons out of one son having B blood group and another having O blood group, then what is probable blood group of couple?

(A) AB, O (B) B ,B (C) AB, AB (D) A, B

(76) A woman with blood group O has a child with blood group O. She claims that a man with blood group 'A' is the father of her child. What would be the genotype of the father, if her claim is right?

(A) I^oI^o (B) I^AI^B (C) I^Ai (D) I^AI^A

(77) For a child having blood group B, if father has blood group A, what may be the blood group of the mother?

(A) B or AB (B) O or A (C) A or B (D) AB or A

(78) A person with unknown blood group under ABO system, has suffered much blood loss in an accident and needs immediate blood transfusion. His one friend who has a valid certificate of his own blood type, offers for blood donation without delay. What would have been the type of blood group of the donor friend?

(A) Type A (B) Type B (C) Type AB (D) Type O

Answers : (63-A), (64-A), (65-C), (66-B), (67-D), (68-D), (69-A), (70-B), (71-C), (72-A), (73-A), (74-C), (75- B), (76-C), (77-A), (78-D)

Pleiotropism :

– It is the effect of a single gene upon two or more characters which are not related.

– Let us see some of their examples in Drosophila, the recessive gene for Vestigial wings produces

vestigial wings in homozygous condition. In addition to wing length it is also responsible for the production of ..... .

(i) the tiny wing like balancer behind the wing

(ii) certain bristles

(iii) the structure of the spermatheca and

(iv) low number of eggs

– This phenomenon of multiple effect of a single gene is called pleiotropism.

– Genes which have multiple effects are called pleiotropic genes.

– The ability of a gene to have many effects is known as pleotropism.

– Its important example is sickle cell anemia.

(79) The effect of a single gene upon two or more characters is ..... .

(A) pleiotropism (B) multiple alleles

(C) polygenic inheritance (D) incomplete dominance

(80) In Drosophila, the recessive gene for vestigial wings produces vestigial wings in homozygous condition. In addition to wing length it is also responsible for ..... .

(A) low number of eggs (B) certain bristles

(C) the structure of spermatheca (D) all of above

(81) Pleiotropic genes mean ..... .

(A) gene which is most dominant (B) gene which have multiple effects

(C) gene responsible for polygenic inheritance (D) gene which have incomplete dominance

(82) The example of pleiotropism is ..... .

(A) red eyes of drosophila (B) sickle cell-anemia

(C) klinefelter's syndrome (D) phenyl ketonuria (PKU)

Answers : (79-A), (80-D), (81-B), (82-B)

Chromosomal basis of inheritance :

– Mendel published his work on inheritance of characters in 1866 but for following reasons, it remained unrecognized till 1900.

– Communication was not easy.

– His thoughts on factors that controlled the expression of traits were not accepted by his contemporaries.

– Mendel's approach of using statistical analysis is to explain biological phenomena was totally new in those days.

– He could not provide any physical proof for the location of these factors (now genes) in the cell was unknown to him.

– In those days neither the role of nucleus in reproduction nor the existence of chromosomes in the nucleus was known.

– **De varies, Correns and Van Tschermak :**

In 1900, De varies, Correns and Van Tschermak independently rediscovered Mendel's results on the inheritance of characters. Thus Mendel's work was rediscovered. Also, by this time, due to advancements of microscopy that were taking place, scientists were able to carefully observe meiotic cell division.

– **Sutton and Boveri :**

In 1902 by Sutton and Boveri they put forward the theory that chromosomes from the physical basis of

factors or genes which determine the heredity of living organisms. This is known as the "chromosome theory of heredity".

– It was also established that chromosomes were separated during formation of Gametes. So haploid gametes are formed through meiosis. When ovum(n) fuses with a sperm (n) during fertilization the diploid status is reestablished in the zygote (2n).

– **Sutton :**

– Sutton demonstrated similarities in the behaviour of chromosomes located with the nucleus and the behaviours of Mendel's hypothetical 'factors'. For example genes (= Factors) occur in pairs. Chromosomes also occur in pairs.

– Each gamete possesses any one gene from a pair each gamete possesses any one chromosome from a pair of homologous chromosomes.

– Mendel's law of independent assortment can also be explained on chromosomal basis.

– Organisms of each species have a fixed number of chromosomes. But Mendel's hypothetical units are not chromosomes. They are genes.

– Genes are located on chromosomes in various numbers of each chromosome.

– Chromosomes as well as genes occur in pairs.

– The two alleles of a gene pair are located at homologous sites on homologous chromosomes.

– Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the chromosomal theory of inheritance.

– **Thomas Hunt Morgan :**

– He worked on fruit fly, *Drosophila melanogaster*. He suggested that genes are arranged in a linear fashion on chromosomes.

– All such aspects establish that there is a chromosomal basis for laws of inheritance proposed by Mendel.

---

(83) Which is correct reason for unrecognized Mendel work on inheritance of characters?

(A) Communication was not easy.

(B) No physical proof for the existence of the factors.

(C) Mendel's approach of using statistical analysis to explain biological phenomena was totally new in those days.

(D) All of above.

(84) The scientists who rediscovered Mendel's results on the inheritance of characters are ..... .

(A) Varies, Correns, Tschermak (B) Bateson and Punnet

(C) Sutton and Boveri (D) Thomas Hunt Morgan and Sutton

(85) Which scientist has proposed the theory of "chromosome theory of heredity"?

(A) Boveri, Tschermak (B) Bateson, Punnet

(C) Sutton, Boveri (D) Thomas Hunt Morgan

(86) Who demonstrated similarities in the behaviour of chromosomes located with the nucleus and the behaviour of Mendel's hypothetical 'factors'?

(A) Boveri (B) Sutton (C) Tschermak (D) Morgan

(87) Mendel's laws of inheritance was based on ..... .

(A) Gene (B) DNA (C) RNA (D) Chromosomes

(88) Which statement given by Thomas Hunt Morgan?

(A) Similarities in the behaviour of chromosomes located with the nucleus and the behaviour of Mendel's hypothetical factors.

(B) Genes are arranged in a linear fashion.

(C) The free assortment behaviours of chromosomes depend on linkage and crossing over.

(D) Gamete contains only one gene for an expression of a character.

Answers : (83-D), (84-A), (85-C), (86-B), (87-D), (88-B)

* **Linkage and Recombination :**

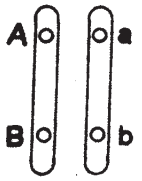
- However in 1903, Sutton and later T.H. Morgan in 1911 found that genes do not assort freely as envisaged by Mendel.
- Basteson and Punnet in their study in the same pea plant found that when they crossed red flowers and spherical pollen plant with a plant having purple flower and cylindrical pollen the test cross yielded a ratio of 7 : 1 : 1 : 7 instead of the expected 1 : 1 :1 :1 ratio.
- Thus we have noted that Mendel's law of independent assortment is not true in all cases.
- If the two pairs of genes controlling two different characters are located in the same pair of homologous chromosomes they can not be segregated separately. Such gene are called linked genes and their inheritance is called linkage.

* **Linkage in Sweet pea plant :**

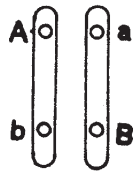
- Experiments indicative of linkage were first performed by Bateson and Punnet on sweet pea plant (*Lathyrus odoratus L.*).
- In those plants purple flower colour is dominant over red flower colour. The respective genes are R and r.
- Long shape of pollen grains is dominant over round shape of pollen. The respective genes are L and *l*. They used the same method as Mendel followed.
- In F_1 generation, all plants had purple flowers and long pollen. This result was as per expectation. But the expected F_2 result, if assortment of these characters was independent, was as under :

Result	**Purple flower Long pollen**	**Purple flower round pollen**	**Red flower Long pollen**	**red flower round pollen**
Expected	9 :	3 :	3 :	1
Actual	11 :	1 :	1 :	3

Cis and Trans arrangement of genes :



Cis (Coupling)



TRANS (Repulsion)

- **Cis arrangement :**

Linkage has coupling phase and repulsion phase. In coupling phase both the linked genes have their dominant alleles in one chromosome and recessive alleles in other chromosomes. The heterozygotes with such constitution is called cis heterozygote. Cis arrangement is an original arrangement which form two types of gametes as (AB) and (ab)

- **Trans arrangement :**

In repulsion phase the normal alleles as well as mutant alleles lie in opposite chromosomes of the homologous pair, such heterozygote is called trans heterozygote. It is not original arrangement, caused due to crossing over, which form two type of gametes (Ab) and (aB).

---

(89) When AaBb parent and aabb parent are crossed. What will be the genotypes of offsprings?

(A) AaBb, Aabb, aaBb, aabb (B) AbBb, Aabb, aaAB, ABab

(C) AABB, BBaa, AbAB, ABaa (D) AABB, BBaa, abAB, aabb

(90) When red flowers and spherical pollen plant crossed with a plant having purple flower and cylindrical pollen, the test cross yielded a ratio of ..... .

(A) 1:1:1:1 (B) 9:3:3:1 (C) 7:1:1:7 (D) 11:1:1:3

(91) If the two pairs of genes controlling two different characters are located in the same pair of homologous chromosomes, they can not be segregated separately, such gene are called ..... .

(A) Homologous genes (B) Dominant genes

(C) Recessive genes (D) Linked genes

(92) When pp L*l* plant crossed with pp L*l* Bateson found the ratio of progeny as ..... .

(A) 1:7:7:1 (B) 1:1:1:1 (C) 9:3:3:1 (D) 11:1:1:3

(93) The percentage of recombination progeny obtained in F_2 generation compairision to expected progeny because of linkage is ..... .

(A) 12.5 % (B) 30 % (C) 87.5 % (D) 70 %

(94) The Actual results of purple flower long pollen in F_2 generation is ..... .

(A) 3 (B) 11 (C) 9 (D) 1

(95) This is true for *Lathyrus odoratus L.*

(A) Complementry genes (B) Mutant genes

(C) Recombination genes (D) Linked genes

(96) Coupling and repulsion associated with ..... .

(A) crossing over node (B) linkage

(C) assortment (D) mutation

(97) Couple gene is separated by which event ? ..... .

(A) pleiotropism (B) epistasis

(C) mutation (D) crossing over

(98) Linkage reduces possibility of ..... .

(A) recombinant progeny (B) dominant gene

(C) recessive gene (D) B and C both

(99) The condition in which both linked genes have their dominant alleles in one chromosome and recessive alleles in other chromosomes is ..... .

(A) TRANS (B) PERI CENTRIC (C) CIS (D) PARA CENTRIC

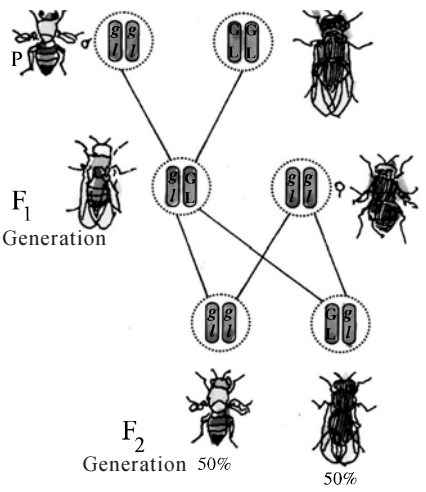
Answers : (89-A), (90-C), (91-D), (92-D), (93-A), (94-B), (95-D), (96-B), (97-D), (98-A), (99-C)

* **Morgan use Drosophila as the experimental material :**

- Drosophila can be easily grown in laboratory.
- Its lifespan is also of about fifteen days.
- It produces a large number of offsprings.
- More over, male and female flies are separate.
- Thus, chances of self-fertilization are not there.

* Linkage in Drosophila :

- Morgan had collected data regarding various characters through monohybrid experiments.
- Body colour grey is dominant over black body colour.
- Long wing is dominant over vestigial wing.
- G represents gene for grey body colour. Its allele g is recessive for black colour.
- Similarly, gene L is for long wing and its allele l for vestigial wings is recessive.
- He took flies with grey body colour and long wings as one parent and flies with black body colour and vestigial wings as another parent.
- All F_1 generation flies were grey and long winged. This was expected result.
- Now, Morgan test – crossed F_1 flies with a parent which is recessive for both the characters of the flies he had obtained, 50% were grey and long winged and 50% were black and vestigial winged.
- No flies were obtained with new combinations of characters. Such results represent complete linkage.
- This is because no crossing over occurs in male drosophila.



Drosophila : Complete linkage

Crossing over :

- In the second experiment like the one above, when he used F_1 female flies to cross with double recessive male flies, he obtained the following results :

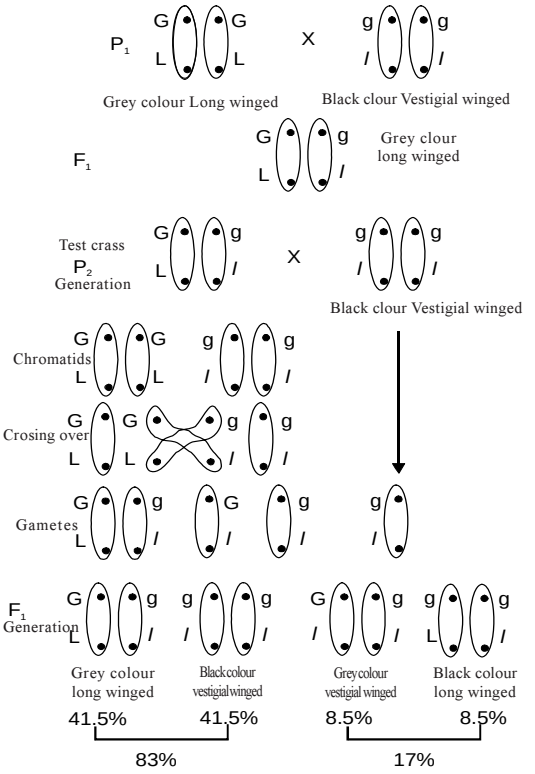
Parental combinations 83% → Grey long winged = 41.5 %

Parental combinations 83% → Black vestigial winged = 41.5 %

Recombinations 17% → Grey vestigialwinged = 8.5%

Recombinations 17% → Black long winged = 8.5 %

- Morgan explained that this unusual ratio is because of the presence of the genes for black and vestigial on the same chromosomes.
- Here the genes are linked together. The two pairs of gene Gg and Ll are not assorted independently.
- Morgan explained crossing over as given in the following diagram :
- Morgan attributed this due to the physical association of genes on a chromosome. The term recombinations is used to describe the generation of non-parental gene combination.



(100) The result of test-crossed between GgLl (F_1 male flies) with a parent which is recessive for both the characters (ggll) are ..... .
(A) Grey colour long winged (B) Black colour vestigial winged
(C) Gery colour vestigial winged (D) A and B both

(101) Odd flies for crossing over is ..... .
(A) male Drosophila (B) female Drosophila
(C) drone bees (D) worker bees

(102) Recombination means ..... .
(A) as like parents (B) progeny without evolution
(C) progeny unlike parents (D) progeny like dominant parent

(103) In second experiment when morgan used F_1 female flies to cross with double recessive male flies, he obtained…….result.
(A) Parental combination = 83%, Recombinations = 17%
(B) Parental combination = 17%, Recombinations = 83%
(C) Parental combination = 75%, Recombinations = 25%
(D) Parental combination = 50%, Recombinations = 50%

(104) The four daughter cells derived from a single meiosis differ from each other due to ..... .
(A) difference in chromosome number
(B) crossing over only
(C) independent assortment of chromosomes only
(D) crossing over as well as independent assortment of chromosomes

(105) Crossing over takes place at a stage of ..... .
(A) Leptotene (B) Pachytene (C) Zygotene (D) Diakinesis

(106) Mendel observed that some characters did not assort independently. Later researchers found it to be due to ..... .
(A) Crossing-over (B) Linkage (C) Dominance (D) Amitosis

(107) Unit of crossing over is ..... .

(A) Cistrone (B) Centimorgan (C) Recon (D) Mutone

Answers : (100-D), (101-A), (102-C), (103-A), (104-D), (105-B), (106-B), (107-C)

* **Chomosomal Theory of Sex Determination :**

* **Henking :**

- Henking observed that in insects, two kinds of sperms are produced.
- The difference was in the presence or absence of one chromosome.
- He had described this chromosome as X-body. He was unable to indentify it as chromosome.

* **Mc Lung :**

- He had indentified X-body as a chromosome.
- Mc Lung also noted that in insects like grasshopper, male has an odd number of chromosomes.

* **Miss. Stevens (1905) and Bridges (1922) :**

- The choromosomal theory of sex determination was proposed by miss Stevens and Bridges.

* **Gold Schmidt (1938) :**

- Gold Schmidt supported chromosomal theory.
- According to this there are two types of chromosomes in an organisms.
- They are the autosomes and allosomes (sex chromosomes).
- The autosomes contain genes which determine the somatic characters of the organisms.
- The sex chromosomes determine the sex of an organism.
- There are two types of sex chromosomes : They are x chromosomes and y chromosomes.
- These two chromosomes differ not only in appearance but also in genetic composition.
- The X chromosome is larger than Y. X is straight while y has a bend at one end.
- In normal animal, there are two sex chromosomes. The two sex chromosomes may be XX or XY.
- In man, insect etc. Female has two X chromosomes.
- But in birds the male has two ZZ chromosomes and the female has one Z chromosome and one W chromosomes.

(108) Which scientist had identified X chromosome as a X body?

(A) Henking (B) Mc Lung (C) Bridges (D) Morgan

(109) Which scientist had identified X body as a chromosome?

(A) Mc Lung (B) Mendel (C) Bridges (D) Dervenport

(110) Which theory is supported by Gold Schmidt ?

(A) Sex Determination (B) Heredity

(C) Chromosomal theory of sex determination (D) Mutation theory

(111) According to which theory there are two types of sex chromosomes in an organism?

(A) Inheritance theory (B) Natural selection

(C) Mutation (D) Chromosomal theory of sex determination

(112) The sex chromosomes determine ..... .

(A) Sex characters (B) Somatic characters

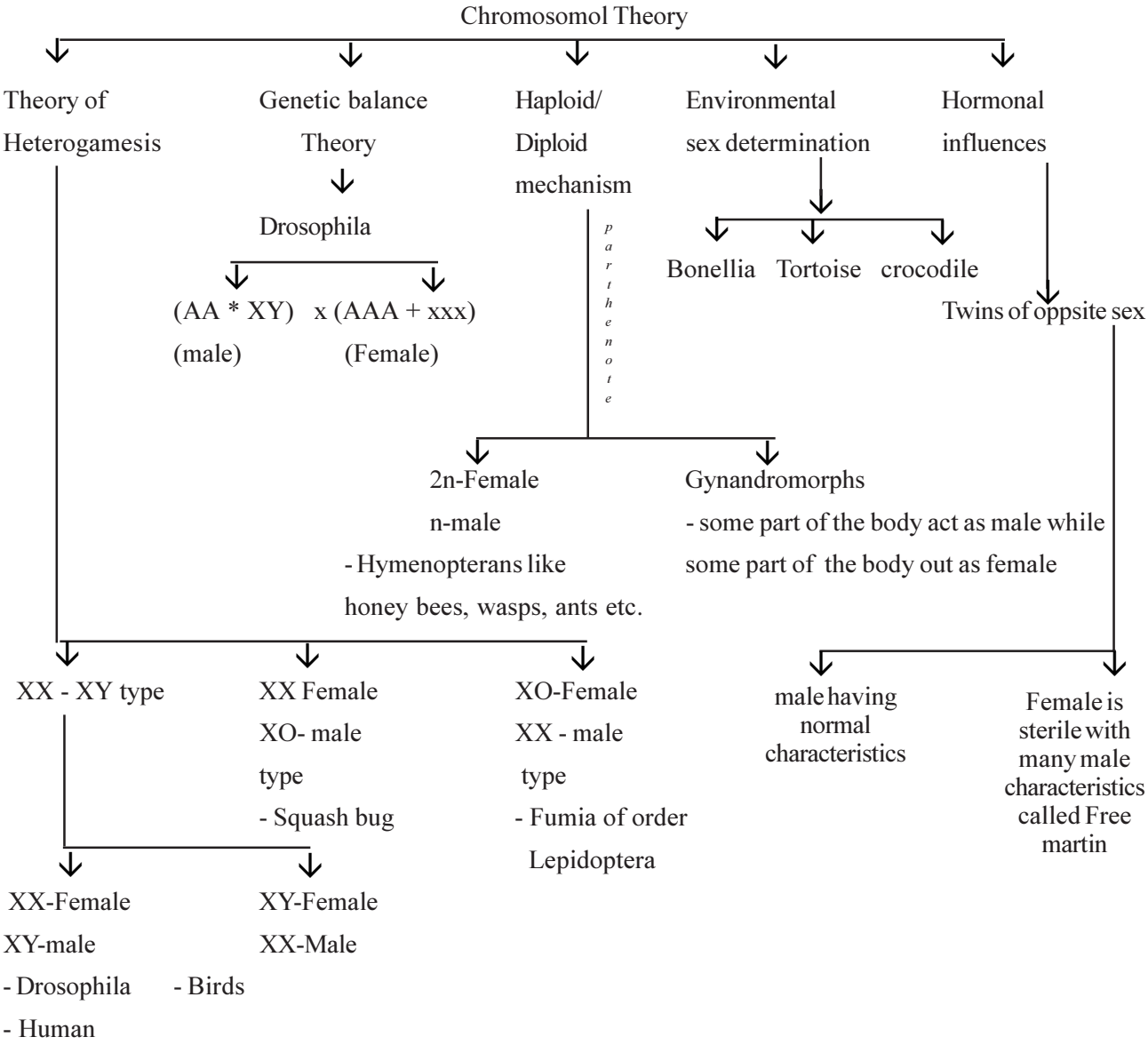
(C) Shape, growth (D) Volume, length

Answers : (108-A), (109-A), (110-C), (111-D), (112-A)

Sub division of chromosomal theory :

(1) Theory of heterogamesis (2) Genetic balance theory (3) Haploid and Diploid mechanism

(4) Environmental effect on determination of sex (5) Hormonal influences



(1) **Theory of Heterogamesis :**

– This theory was proposed by correns in 1906. according to this theory, one sex produce two type of gamete and each type of gamete determine a different sex on fertilization.

– It can be (A) XX - XY type or (B) XX- XO type

(A) XX - XY type :

– There are two different patterns of sex determination (a) through XX - female, XY - male and (b) through XY - female, XX - male type.

(a) **XX - female, XY - male type :**

It was studied in Drosophila and man.

* **Sex determination in Drosophila :**

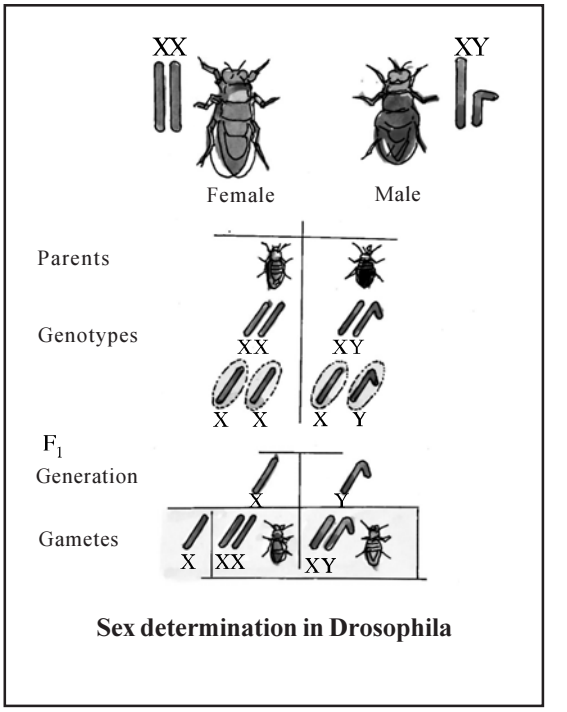
– Morgan had discovered sex chromosomes in

– Drosophila, Morgan studied chromosomal constitution in male and female.

– Drosophila has four pairs of chromosomes, out of it there are three pairs of autosomes (3AA) and one pair of sex chromosome.

– In female this is represented by 3AA + XX and in male 3AA + XY.

– The female produces only one type of Ova which carry 3A + X. But the male produces two types of sperms and they are 3A + X and 3A + Y type of chromosomes.



Sex determination in Drosophila

– Sex is determined by the type of sperm fertilizing an egg.

– If an egg is fertilized by the X type sperm, the resulting individual is a female and by Y type sperm, resulting individual is male.

– This method of sex determination is also called XY male method.

Sex determination in Human :

– In Humans, 23 pairs of chromosomes occur. Of these, twenty – two pairs are of autosomes. They are similar in man and woman.

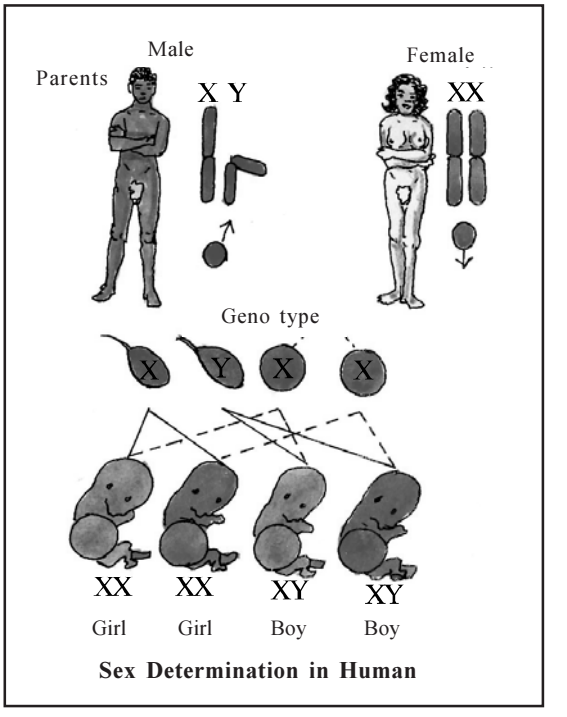
– In woman, twenty third pair consist of two similar X sex chromosomes.

– In man, one chromosome in twenty third pair is X – chromosome, and its homologous chromosome is smaller in size and is called Y chromosome.

– All egg of woman are similar. Each egg contains 22 autosomes and one X sex chromosome.

– In man sperms are of two type.

– Half the number of sperms have 22 autosomes and X sex chromosome, while the other half contains 22 autosomes and one Y sex chromosome.



Sex Determination in Human

– Whether the child will be a boy or a girl depends on the kind of sperm that fertilizes the egg.

(b) **XY – female, XX – male type :**

– In this type of sex determination, the female is sexually heterozygous having X and Y chromosomes.

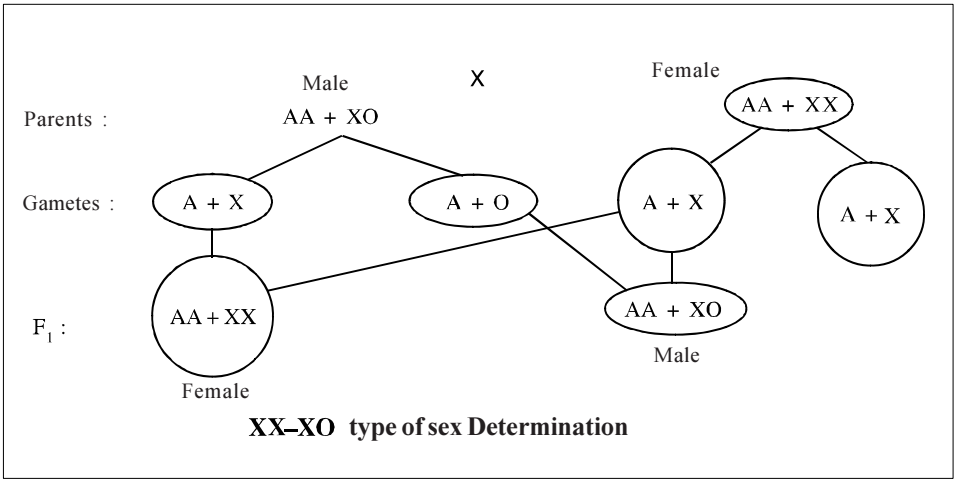
– The male is sexually homozygous having two x chromosomes.

– Males produce only one type of sperms while the females produce two types of eggs.

– In birds the X and Y chromosomes are designated as Z and W chromosomes so the chromosomal structure of the female is AA + ZW and the male is AA + ZZ.

(B) XX –XO type :

- This type of sex determination was first studied in squash bug (protenor).
- Here the sex of the animal is determined by the number of X chromosomes present in the cell.
- In one sex, XX chromosomes are present and in the other sex only one X chromosome is present e.g. bugs and grasshopper.



XX–XO type of sex Determination

* **Female XO and Male XX :**

In some insects like Fumia of order Lepidoptera, the female has one X – chromosome and the male has two X chromosomes.

(113) In which organisms female is sexually heterozygous and male is sexually homozygous?

(A) Drosophila (B) Human (C) Birds (D) All of above

(114) According to which theory, one sex produces two types of gametes and each type of gamete determine a different sex on fertilization?

(A) Genetic balance theory (B) Inheritance theory

(C) Mutation (D) Theory of heterogamesis

(115) The type of sex determination in Drosophila is ..... .

(A) XX – male, XY – female (B) XX – female, XY – male

(C) XX – male, XO – female (D) XO – male, XX – female

(116) The theory of heterogamesis proposed by ..... .

(A) Derven Port (B) Mc Lung (C) Morgan (D) Correns

(117) 3A + X and 3A + Y means ..... .

(A) Types of sperms in male drosophila (B) Types of sperms in birds

(C) Types of sperms in Humans (D) Types of female eggs in drosophila

(118) Which is responsible for sex-determination in drosophila ?

(A) Type of egg cells (B) Type of sperm cells

(C) Type of somatic cells (D) Type of dominant genes

(119) It is a type of sperm cells in man.

(A) 22A +X, 22A + Y (B) 23A +X, 23A + Y

(C) 44A +X, 44A + Y (D) 46A +X, 46A + Y

(120) XX-XO sex determination mean ..... .

(A) Sex- determined by number of x chromosomes present in the cell.

(B) Sex- determined by autosome present in the cell.

(C) Sex-determined by number of sex chromosomes present in the cell.

(D) Sex – determined by XX female.

(121) They follow XX- XO sex-determination ..... .

(A) drosophila, human (B) bug, grasshopper

(C) birds, fumia (D) honey bees, drosophila

(122) What is the order of Fumia?

(A) Hymenoptera (B) Coleoptera (C) Lepidoptera (D) Orthoptera

(123) It is true for XO- Female and XX - male ..... .

(A) bug (B) fumia (C) grasshopper (D) beetles

Answers : (113-C), (114-D), (115-B), (116-D), (117-A), (118-B), (119-A), (120-A), (122-B), (123-C)

* **Genetic Balance Theory :**

- This theory was formulated by bridges student of Morgan.
- According to this theory, sex is determined by the relative number of X – chromosomes and autosomes.
- It is actually the ratio between the X chromosomes and autosomes that determines the sex.
- Drosophila flies having XO- chromosomes were male. However they were sterile.
- They had only one X- sex chromosome it means, Y – sex chromosome is not essential for maleness.
- Bridges during his experiments found triploid female flies.
- These were fertile, they had three sets of autosomes and three X-sex chromosomes.
- He arranged cross breeding amongst such triploid female flies and normal diploid male flies.
- The probabilities are indicated in the table given below :

Female (AAA + XXX)

Kinds of eggs

Kinds of sperms	A + X	AA + XX	A + XX	AA + X
A+X	AA+XX Normal female $\frac{XX}{AA}=\frac{2}{2}=1$	AAA+XXX Normal female $\frac{XXX}{AAA}=\frac{3}{3}=1$	AA+XXX Super sterile female $\frac{XXX}{AA}=\frac{3}{2}=1.5$	AAA+XX Inter sex Sterile $\frac{XX}{AAA}=\frac{2}{3}=0.67$
A+Y	AA+XY Normal male $\frac{X}{AA}=\frac{1}{2}=0.5$	AAA+XXY Inter sex sterile $\frac{XX}{AAA}=\frac{2}{3}=0.67$	AA+XXY Normal female $\frac{XX}{AA}=\frac{2}{2}=1$	AAA+XY super male sterile $\frac{X}{AAA}=\frac{1}{3}=0.33$

- Bridges noted that the offsprings contained normal males, normal females, sterile males, sterile females and sterile intersex flies.
- It seems that the ratio of X-sex chromosomes to autosomal chromosomes (X/A) is responsible for these results.

If $\frac{X}{A}$ = 0.5 than flies will be Normal male.

If $\frac{X}{A}$ = 1 than flies will be Normal female.

If $\frac{X}{A}$ = 1.5 than flies will be Super sterile female.

If $\frac{X}{A}$ = 0.33 than flies will be Super male sterile.

If $\frac{X}{A}$ = between 0.5 to 1.0 than flies will be Intersex sterile.

- Bridges suggested on the basis of results that in Drosophila, the genes for maleness are distributed over autosomes and those for femaleness are located on X-sex chromosomes. Sex depends on their balance.

---

(124) On the basis of Genetic balance theory sex-determination conducted by ..... .

(A) X- sex chromosome

(B) Balance of autosomal chromosomes and X-sex chromosomes.

(C) Number of X and Y chromosomes

(D) Autosomes and hormones balance.

(125) Drosophila flies having XO chromosome were ..... .

(A) Sterile male (B) Fertile male (C) Fertile female (D) Sterile female

(126) According to the genetic balance theory the ratio of the $\frac{X}{A}$ chromosomes obtained 1, than what will be the nature of flies?

(A) Normal female (B) Inter sex sterile (C) Sterile female (D) Supermale

(127) The ratio of $\frac{X}{A}$ chromosomes obtained 0.5, than what will be the nature of flies develop?

(A) Inter sex sterile (B) Super male(C) Normal female (D) Normal male

(128) The ratio of $\frac{X}{A}$ chromosomes is 1.5. It means ..... .

(A) Normal female (B) Super male(C) Normal male (D) Super female

(129) The ratio of $\frac{X}{A}$ chromosome is 0.33. It means ..... .

(A) Intersex sterile (B) Super female (C) Super male(D) Normal female

(130) According to the genetic balance theory the ratio of the $\frac{X}{A}$ chromosomes obtained between 0.5 to 1, it means ..... .

(A) Super female (B) Inter sex sterile (C) Normal female (D) Super male

(131) It is a conclusion of genetic balance theory…

(A) In drosophila the genes for maleness are distributed over autosomes and those for femaleness are located on X- sex chromosomes.

(B) In drosophila the genes for maleness are distributed over Y chromosomes and those for femaleness are located on X- chromosomes.

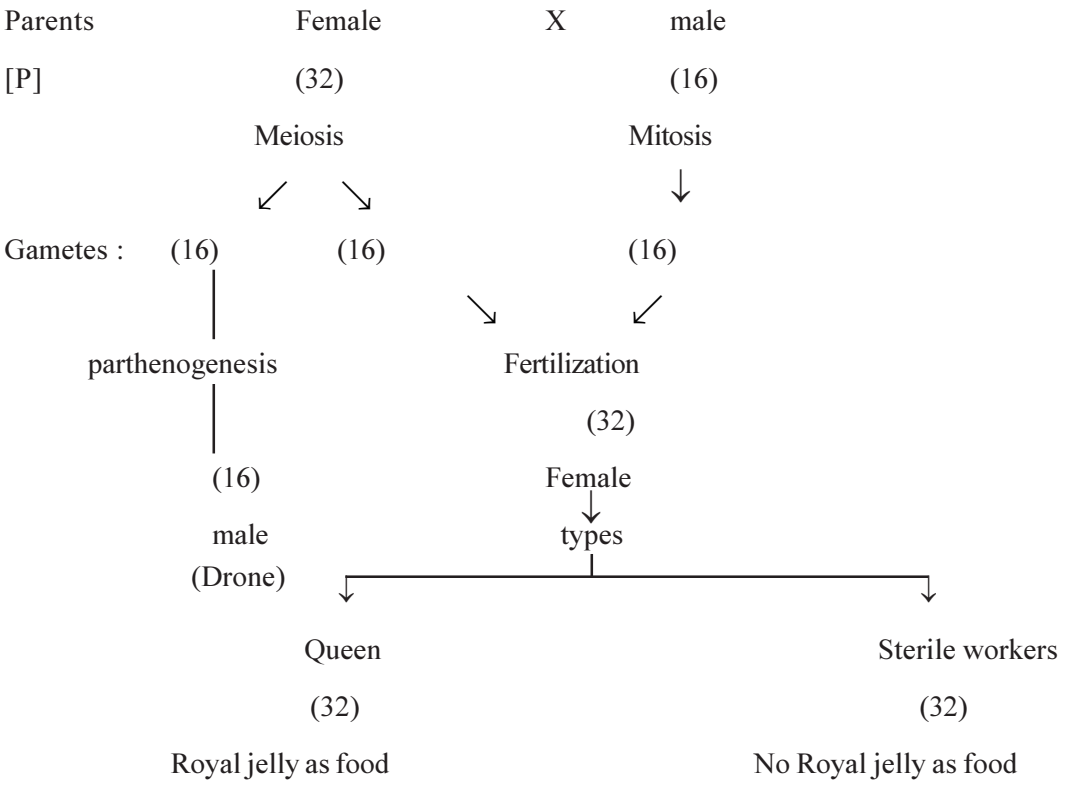
(C) In drosophila Y- chromosome responsible for length of wings.

(D) In Drosophila autosomes are responsible for sex-determination.

Answers : 124 (B), 125 (A), 126 (A), 127 (D), 128 (D), 129 (C), 130 (B), 131 (A)

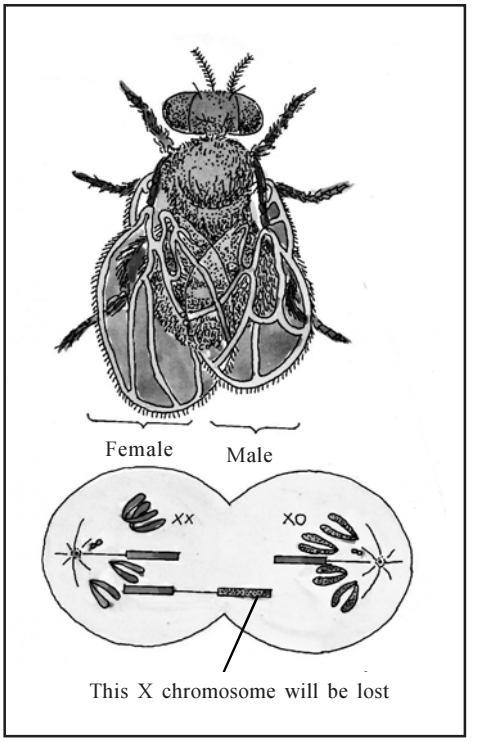
* **Haploidy and diploidy Mechanism :**

- Development of an ovum into a young one without fertilization is known as parthenogenesis.
- An individual produced parthenogenetically is called parthenote.
- In some hymenopterans like honey bees, wasps and ants, the females are diploid and the males are haploid.
- The female lays the normal egg which carries the haploid sets of chromosomes.
- The unfertilized haploid eggs develop parthenagenetically into functional males or drones.
- If the eggs are fertilized; the zygote develops into a diploid female.
- In these two types of females are produced from the fertilized eggs and they are (1) fertile normal diploid queen and (2) Sterile non-functional diploid female workers.
- The diploid larva which gets the Royal jelly as the food material develops into Queen and the other develop into workers.



* **Gynandromorphs :**

- Gynandromorphs are individuals who show male characters on some part of the body and female characters on other parts of the body. They are sterile.
- It happens in rare cases.
- They occur in Drosophila, butterflies beetles, wasps, bees, silkworm etc.
- It happens due to loss of X- chromosomes or due to binucleated eggs.
- The loss of an X- chromosome during mitosis in a 2A+XX cell leads to the derivation of two daughter cells one having 2A+XX and the other having 2A+X.



* **Barr-body Test :**

- The mammalian cells of certain sexes contain a darkly stained body in the nucleus. It is called sex-chromatin or barr-body.
- It was discovered by Barr and Bertram in 1949.
- It helps to identify the sex of the animal.
- The number of Barr bodies is always one less than the number of X chromosomes (X-1).

Chromosomes	No. of Barr bodies	Sex
22AA+ XY	No Barr body	Male
22AA+ XX	One Barr body	Female
22AA+ X	No Barr body	(Turner's syndrome) Female
22AA+ XXY	One Barr body	Male (Kline felter's syndrome)

(132) Pathenote mean ..... .
(A) development of an ovum in to young one without fertilization.
(B) best hybrid species
(C) inter sex sterile species
(D) super male and super female

(133) The order of honey bees, wasps and ants is ….
(A) Lepidoptera (B) Coleoptera (C) Hymenoptera (D) Orthoptera

(134) Which members are having 2n female and n male?
(A) Ant, beetles, honey bees, drosophila (B) Ant, grasshopper, wasps, drosophila
(C) Ant, beetles, honey bees (D) Drosophila, bonellia, honey bees

(135) How many chromosomes are there in drone ?
(A) 32 (B) 16 (C) 6 (D) 8

(136) The diploid larva which gets the Royal jelly as the food material develops into ..... .
(A) Drone (B) Sterile workers (C) Queen (D) Super female

(137) If diploid larva does not get Royal jelly as food material it will develop in to ..... .
(A) Drone (B) Sterile workers (C) Queen (D) Super female

(138) Gynandromorphs mean ..... .
(A) individuals who show male and female characters.
(B) Parthenote
(C) fertile female
(D) Sterile species having only male characters.

(139) The loss of X- chromosome in beetles during mitosis it leads to form ..... .
(A) 2A + XX, 2A +Y (B) 2A + X, 2A +X
(C) 2A + XX, 2A +X (D) 2A + X, 2A +Y

(140) The Darkly stained body in the nucleus of mammalian cells is ..... .
(A) Nucleolus (B) Barr – body (C) Ribosomes (D) DNA

(141) Who discovered sex chromatin ?
(A) Barr and Bertram (B) Derven port and Barr
(C) Bateson and Punnet (D) Bertram and Morgan

(142) It is true for Barr-body.
(A) (2x-1) (B) (x-1) (C) (xy-2) (D) (xxx-1)

(143) How many barr-body in person having Turner's syndrome?
(A) 0 (B) 2 (C) 1 (D) 3

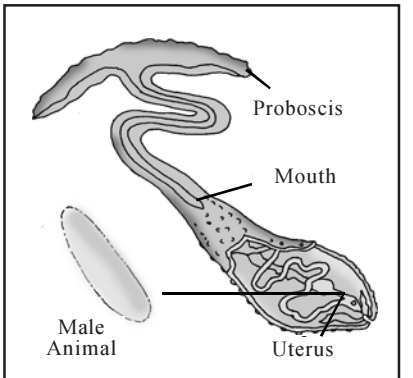
(144) How many barr-body in male having XXY chromosomes?
(A) 2 (B) 0 (C) 3 (D) 1

(145) How many barr –body in female having super female syndrome?
(A) 0 (B) 2 (C) 1 (D) 3

Answers : (132-A), (133-C), (134-C), (135-B), (136-C), (137-B), (138-A), (139-C), (140-B), (141-A), (142-B), (143-A), (144-D), (145-D)

* **Environmental effect on Determination of sex :**

- Baltzar (1935) stated that in Bonellia, sex is determined by environmental factor.
- In Bonellia (marine animal) all zygote are genetically identical-whether the embryo will develop into a male or a female depends on where it develops.
- If it develop near proboscis of a female it enters the body of the female and develops into a male animal.
- If it develops away from female, it develops into a female animal.
- Bonellia exhibits sexual dimorphism.
- It is believed that the proboscis secret a hormone like substance which prevents the development of femaleness.



Sea Determination in Bonellia

* **Sex determination in Tortoise :**

- In tortoise if the water temperature where it lives is higher than 30^0C zygote develops as a female, at a lower temperature, male development occur.

* **Sex determination in Crocodile :**

- In Crocodile the reverse is observed. A higher temperature induces male development and a lower temperature induces female development.

(146) Who stated that in Bonellia sex is determined by environmental factor?

(A) Baltzar (B) Barr (C) Bertram (D) Morgan

(147) In Bonellia all zygotes are genetically ..... .

(A) Dominant (B) Heterozygous (C) Recessive (D) Identical

(148) In Bonellia embryo will develop into a male or a female depends on ..... .

(A) temperature (B) position of zygotes in body

(C) light (D) royal jelly

(149) When male development occur in tortoise?

(A) Water temperature 25°C (B) Water temperature 40°C

(C) Water temperature 50°C (D) Water temperature 35°C

(150) Sex-determination in crocodile is ..... .

(A) higher temperature male lower temperature female

(B) higher temperature female lower temperature male

(C) higher temperature super female, lower temperature male

(D) higher temperature super male, lower temperature female

Answers : (146-A), (147-D), (148-B), (149-A), (150-A)

* **Hormonal Theory of Sex Determination :**

- Generally in higher animals secondary sexual characters are under the influence of their related sex hormones.
- But Lillie found that when twins of opposite sex (one female and other male) are born, the male is normal but the female is sterile with many male characteristics.
- Such sterile females are called free martin.
- In cattle, twins occurs frequently.
- During development both the twins are connected by a common umbilical cord.
- The gonads of the male develop earlier than those of the female at that time, the male hormones reach the female embryo and influence the development of male sex in the female embryo.

(151) Who found that when twins of opposite sex are born, the male is normal but the female is sterile with many male characteristics ?

(A) Baltzar (B) Bridges (C) Dervenport (D) Lillie

(152) In hormonal theory of sex determination sterile female are called as ..... .

(A) free martin (B) super female (C) parthenote (D) twins baby

(153) Which type of gonads development in twins of opposite sex?

(A) Male and female gonads develop at similar rate.

(B) The gonads of the male develop earlier than those of the female.

(C) The gonads of the female develop earlier than those of the male.

(D) The gonads of the male develop poor than those of the female.

Answers : (151-D), (152-A), (153-B)

* **Sex-determination in plants :**

- The mechanism of sex determination has been studied in a large number of plants.
- In most of the plants the sex is controlled by the Y chromosomes as in the case of man.

- If Y - chromosomes are absent, the plant will be female.

* **Liver worts :**

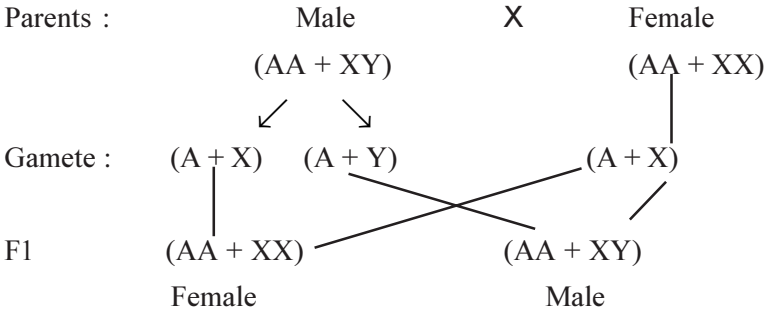
- In plants chromosomes were first studied by Allen in Liver worts.
- In Liver worts the sex organ are located on haploid gametophyte.

* **Sphaerocarpes :**

- In sphaerocarpes the male gametophyte has seven autosomes and one Y chromosomes (7A +Y), the female has (7A +X).

* **Angiosperm :**

- In Angiosperm female is homogametic and male heterogametic type.



* **Flagellaria :**

- In Flagellaria plant, female is heterogametic and male homogametic.

* **Dioscoria sinulate :**

- In Discoria sinulate female is homogametic (AA + XX) and male is heterogametic (AA+XO)

* **Humulus japanic :**

- In Humulus japanic, the female has two X chromosomes, but the male has one X chromosome and two Y chromosomes.

* **Sex-differentiation in Monoecious plants :**

* **Maize :**

- Maize is monoecious, having both staminate flowers and pistilate flower on the same plant.
- The female flowers normally develop along the sides of the stalk, and the male flower at the top of the plant.
- But in some cases the grains of corn may actually be produced at the tip of the plant because of some mutation. There is a mutant gene called ta.
- In homozygous condition (ta ta) it converts the male flowers into female flowers.
- There is another mutant gene (bs); it suppresses the development of the female flowers.

* **Spinach :**

- In spinach sex is controlled by single gene ‘m’ which is located in the X-chromosome.

---

(154) In most of the plants the sex is controlled by the ..... .

(A) X-chromosome (B) Y-chromosome

(C) Somatic A chromosome (D) By ratio of autosomal and sex chromosomes.

(155) In plant chromosomes were first studied by Allen.

(A) Sphaerocarpes (B) Liver worts (C) Flagellaria (D) Dioscoria sinulate

(156) It is true for Sphaerocarpes …..

(A) (7A +Y) male gametophyte and (7A + X) female gameto phyte.

(B) Female is homogametic (AA +XX) and male is heterogametic (AA+XO).

(C) Male (8A+X) and female (8A+Y).

(D) Male has one X chromosome and female has three X chromosomes.

(157) In which plant female heterogametic and male homogametic ?

(A) Liver worts (B) Flagellaria (C) Dioscoria sinulate (D) Humulus japanic

(158) In which plant female has XX and male has XYY chromosomes?

(A) Liver Worts (B) Dioscoria sinulate (C) Flagellaria (D) Humulus japanic

(159) It is a mutant gene in maize.

(A) ta (B) bs (C) m (D) A and B both

(160) What is a role of ta gene in Maize ?

(A) It converts the male flowers into female flowers.

(B) It converts the female flowers into male flowers.

(C) It suppresses the development of the female flowers.

(D) All of above.

(161) What is function of 'bs' mutant gene in Maize?

(A) It converts the male flowers into female flowers.

(B) It converts the female flowers into male flowers.

(C) It suppresses the development of the female flowers.

(D) It suppresses the development of maize.

(162) In spinach sex is controlled by a ..... .

(A) Single gene m (B) Mutant gene ta

(C) Mutant gene bs (D) XX and XY chromosomes.

Answers : (154-B), (155-B), (156-A), (157-B), (158-D), (159-D), (160-A), (161-C), (162-A)

* **Mutation :**

- The term 'Mutation' was first utilized by De vries.
- Mutation is a phenomenon which result in alteration of DNA sequence and consequently results in changes in the genotype and the phenotype of an organism.
- Mutation is a sudden change of a gene or chromosome from one form to another.
- It produces an alteration in the character under its control.
- Dobzhansky stated that mutation is a mistake or misprint in cell division.

* **Types of mutations** :

No	Type	Explanation
1	Somatic mutation	Occurring in the somatic cells. It is not inherited.
2	Germinal mutation	Occurring in the germ cells. It is a inherited.
3	Gametic mutation	Takes place in gametes.
4	Zygotic mutation	It occurs in zygote.
5	Dominant mutation	Mutation produces a dominant gene.
6	Recessive mutation	Mutation produces recessive gene, it does not express immediately
7	Back mutation	This is the reversal of mutation rarely does it happen.
8	Lethal mutation	As a result the mutant dies.

9	Spontaneous mutation	It occurs in the absence of any obvious cause. Most of the mutations occurring in nature are of this type.
10	Induced mutation	Mutations caused by external factors. The factors are called mutagens. E.g. ionizing radiation, mustard gas, peroxides, colchicines, formaldehyde, dimethyl sulphate, nitrous acid etc….
11	Biochemical mutation	Mutation causing change in the metabolites or their end products. Mostly it happens in enzymes. These are metabolic errors.

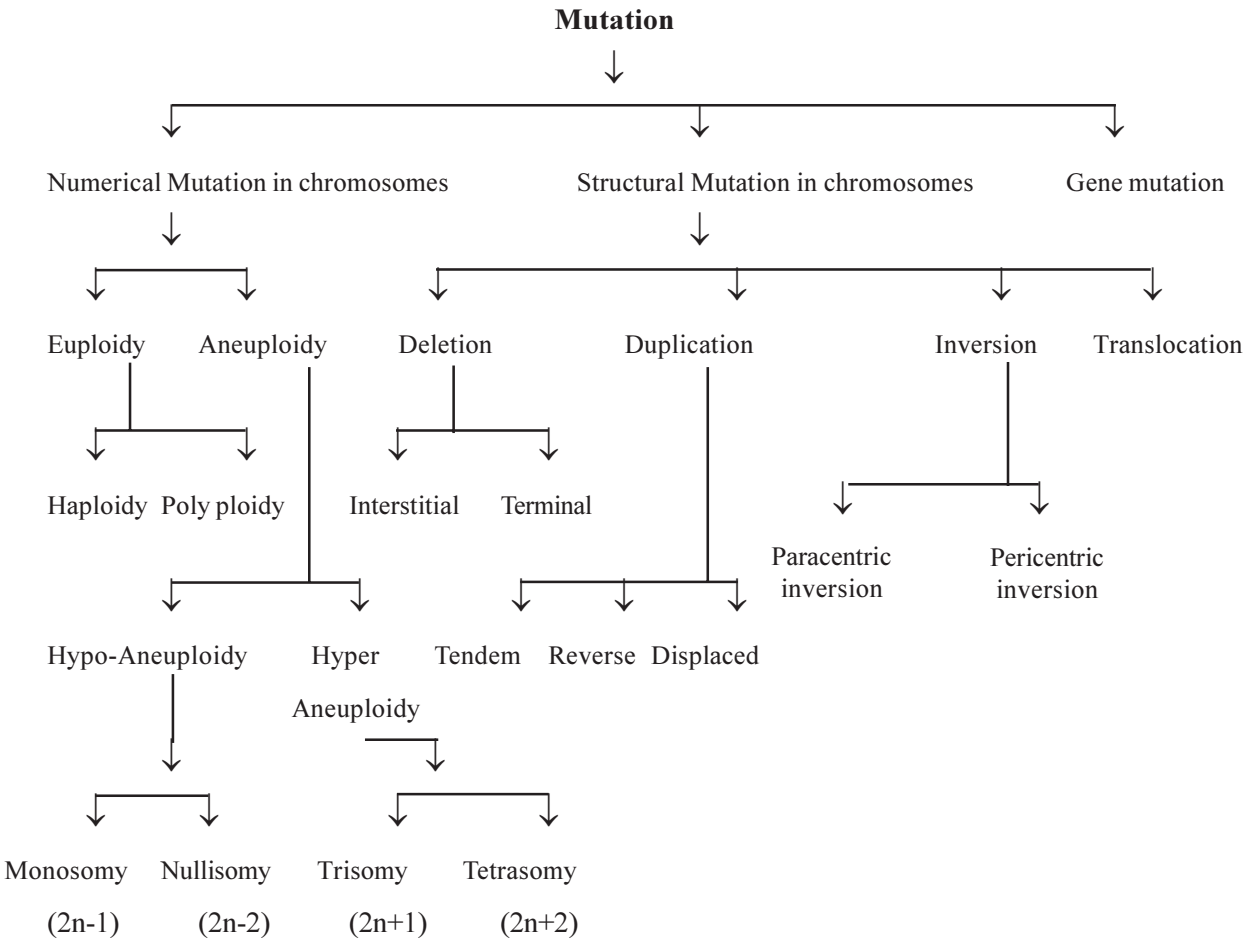
* **Types of mutation :**

• **Mainly three types of mutation :**

(1)Numerical mutation in chromosomes

(2) Structural mutation in chromosomes

(3) Gene mutations



(1) Numerical mutation in chromosomes ploidy :

- We know that the number of chromosomes is fixed for every species.
- Change in the number of chromosomes can be either in the number of sets of chromosomes or in the number of chromosomes in one set.
- This changes cause loss of one chromosome from one set or an addition of chromosomes.
- Chromosomal abberration are broadly classified into two namely euploidy and aneuploidy.

(A) Euploidy

It refers to the change in the number of chromosomes sets. It is further classified into two namely (i) Haploidy (ii) Polyploidy

(i) Haploidy or monoploidy :

Sometimes a set of chromosomes is lost and leads to haploidy.

(ii) Polyploidy :

- If there is an increase in the number of chromosomes which is in a multiple of the basic number n, the change is called polyploidy.
- The number can be 3n, 4n, 5n etc…
- Many of our cultivated crops are developed in this way.
- In plants, generally polyploidy produces larger leaves and flowers and heavier fruits and seeds.

(B) Aneuploidy :

- Aneuploidy refers to the loss or gain of one or more chromosomes in a set.
- Normally here are two members in a homologous pair of chromosomes.
- Instead there may be only one or nil member or three or four members.

* Monosomy (2n-1) :

If there is only one member instead of two in a pair, the condition is called monosomy.

* Nullisomy (2n-2) :

If a homologous pair of chromosomes is totally missing, the condition is called nullisomy. Normally it is lethal.

* Trisomy (2n+1) :

If there are three members instead of two in a given pair of chromosomes, the condition is called trisomy.

* Tetrasomy(2n+2) :

If there are four members in a pair of chromosomes instead of two, the condition is called tetrasomy.

Depending on the kind of chromosomes involved, aneuploidy can be of two kinds:

(1) Autosomal aneuploidy : It involves autosomal pairs of chromosomes.

(2) Sex chromosomal aneuploidy : It involves sex chromosomes.

---

(163) Mutation means ..... .

(A) Change in genes present on DNA (B) Change in structure of chromosomes

(C) Increase in number of cells. (D) None of this.

(164) Which mutation is not inherited?

(A) Somatic (B) Genetic (C) Gametic (D) Zygotic

(165) Which is rarely happen?

(A) Spontaneous (B) Induced (C) Back (D) Biochemical

(166) Which option is true for mutagens?

(A) Mustard gas, padiaton, CO_2, nitrous oxide (B) Colchicine, mustard gas, formaldehyde

(C) Mustard gas, benzene, toluene, formaldehyde (D) Ethanol, methanol, nitrous oxide

(167) Which mutation causing changes in the metabolites ?

(A) Induced (B) Genetic (C) Biochemical (D) Back

(168) Haploidy mean ..... .

(A) 2n-1-1 (B) 2n-2

(C) Deletion of set of chromosome (D) 2n+1

(169) Polyploidy mean ..... .

(A) Increase in the number of chromosomes which is in a multiple of the basic number n.

(B) Decrease in the number of chromosomes

(C) 2n+1

(D) 2n+2

(170) Which mutation responsible for production of larger leaves, flowers and heavier fruits?
(A) Haploidy (B) Aneuploidy (C) Polyploidy (D) Back mutation

(171) It is true for polyploidy ..... .
(A) 3n (B) 4n (C) 5n (D) All above

(172) The loss or gain of one or more chromosomes in a set is ..... .
(A) Aneuploidy (B) Haploidy (C) Polyploidy (D) All above

(173) Nullisomy mean ..... .
(A) 2n-2 (B) 2n-1 (C) 2n+1 (D) 2n+2

(174) It is true for double monosomy.
(A) 2n-2 (B) 2n-1-1 (C) 2n-2 (D) 2n-2-2

(175) Tetrasomy mean..
(A) 2n-2 (B) 2n-1-1 (C) 2n+2 (D) 2n-4

(176) Which mutation is normally lethal?
(A) Monosomoy (B) Nullisomy (C) Trisomy (D) Tetrasomy

Answers : (163-A), (164-A), (165-C), (166-B), (167-C), (168-C), (169-A), (170-C), (171-D), (172-A), (173-A), (174-B), (175-C), (176-B)

* **Structural abnormalities in chromosomes :**

- Each species is characterized by the presence of a specific number of chromosomes and each chromosome is arranged at definite location and in a definite sequence.
- Sometimes, changes occur in the number and arrangement of genes and in the number of chromosomes. These changes are called chromosomal aberrations or chromosomal nutation.
- Four kinds of chromosomal aberrations can occur. These are deletion, duplication, inversion and translocation.

* **Deletion :**

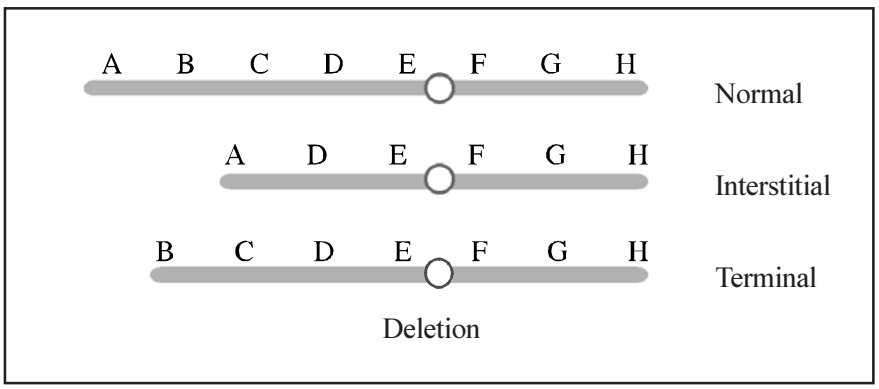
- Deletion is the loss of one or more genes from a chromosome and it is due to the loss of a chromosome segment.
- During deletion some genes are lost, so the organism shows some defects.
- Such a loss can be from the end region or from the inner region of a chromosome i.e. deletion may be 'terminal' or 'intercalary'.

* **Cri-du-chat :**

- In human a disease called cri-du-chat result due to deletion from the short arm of chromosome number five.

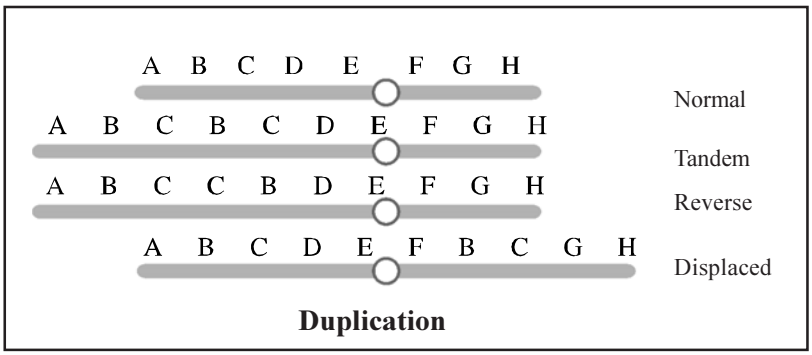
* **Philadelphia syndrome :**

• Similarly a deletion from the long arm of chromosome number twenty two is responsible for the disease Philadelphia syndrome.



* **Duplication :**

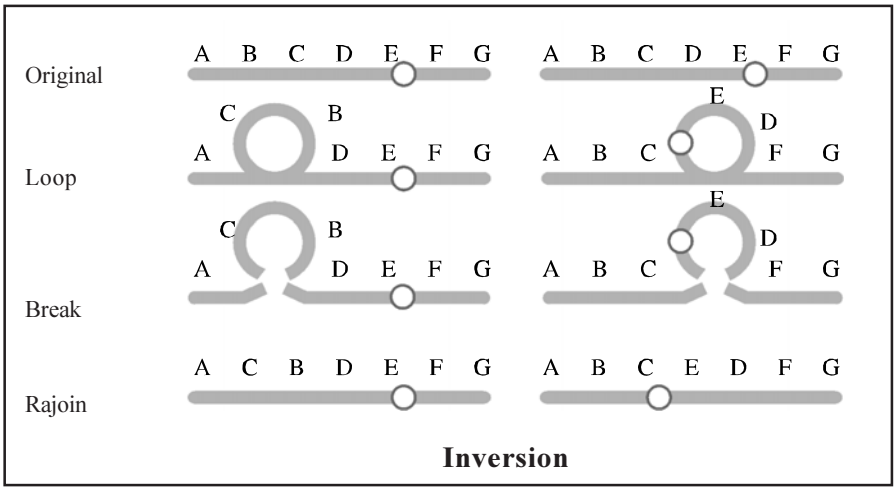
- In such an abnormality a part of a chromosome occure twice
- This can occur during replication of DNA.
- In such cases, the sequence of genes is either maintained or inverted.
- For example, say the sequence of gene is A B C D E F G H and genes BC are duplicated. The new sequences can either ABCBCDEFGH (Tendem) or A B C C B D E F G H (reverse) or ABCDEFBCGH(displaced).
- Abnormally in size of eyes in Drosophila appears to be due to this reason.
- Duplication play a role in evolution.



Duplication

* **Inversion :**

- In inversion, there is no loss or gain of genes.
- But a particular segment of a chromosome is broken and is attached to the same chromosome in an inverted position. So there is rearrangement of original genes.
- For example : chromosome ABCDEFG is cut between A and B and between C and D the separated piece B C undergoes inversion and then rejoins the original chromosome. The new chromosome will have a sequence A B C D E F G. The sequence of gene is changed. Due to this, sometimes phenotypic effects may changed.
- Two types of inversion : (i) paracentric inversion and (ii) pericentric inversion.



Inversion

Translocation :

- Sometimes a part of a chromosome become separated.
- This separated piece joins with another chromosome which is not its homologus chromosome this is called translocation.
- It plays a significant role in evolution.
- Chromosomal aberrations are commonly observed in cancer cells.

(177) The name of disease cause by due to deletion of from the short arm of chromosome number five is ..... .

(A) Philadelphia syndrome (B) sickle cell anemia

(C) cri-du-chat (D) thalassaemia

(178) The name of disease cause by due to deletion of long arm of chromosome number twenty two is ..... .

(A) philadelphia syndrome (B) cri-du-chat

(C) thalassaemia (D) PKU

(179) The name of disease in which child suffer from blood cancer is…..

(A) Philadelphia syndrome (B) Sickle cell anemia

(C) Albinism (D) Alkaptonuria

(180) In the sequence of gene A B C D E F G the C D E gene become tendem then the new sequence can be ..... .

(A) A B C D E F G C D E H (B) A B C D E C D E G F H

(C) A B C D E E D C F G H (D) A B C D E D C E F G H

(181) Abnormality in size of eye in Drosophila appears to be due to ..... .

(A) Deletion (B) Aneuploidy (C) Duplication (D)Inversion

(182) Inversion mean ..... .

(A) a particular segment of a chromosome is broken and is attached to same chromosome in an inverted position.

(B) loss of one or more gene from a chromosome and it is due to the loss of a chromosome segment.

(C) a particular segment of a chromosome is broken and it attached to homologous chromosomes.

(D) duplication of some part of chromosomes.

(183) Chromosome A B C D E F G H is cut between B and C and between E and F the separated piece undergoes inversion and then rejoins the original chromosomes. Then what will be the new sequence will have ..... .

(A) A BC BC D E E F G H (B) A B E D C F G H

(C) A B C D E C D E F G H (D) A B C E D F G H

(184) Which chromosomal aberrations play a significant role in evolution?

(A) Duplication (B) Translocation

(C) Deletion (D) A and B both

Answers : (177-C), (178-A), (179-A), (180-A), (181-C), (182-A), (183-B), (184-B)

* **Gene Mutation :**

- Any mutation induced by a change in the constitution of a gene is called gene mutation.
- DNA is the genetic material.
- A definite length of a DNA molecule acts as one gene.
- Mutation also arises due to change in a single base pair of DNA such mutation may alter the sequence of the nucleotides within a part of the DNA molecules.
- This alteration change the information on the DNA and results in differences in the proteins being produced this is known as point mutation.
- A classical example of such mutation is sickle-cell anemia, where haemoglobin defective and RBCs take a sickle shape.

* Points are note worthy regarding gene mutation :

- Generally a mutated gene is harmful to the individual.
- Any gene can undergo mutations.
- Mutations may be spontaneous or they may be induced.
- Mutation is an evolutionary agent and mutability is a property of the genetic material.
- Such genetic variations are useful in natural selection and evolution of a species.

(185) It is a best example of point mutation.

(A) Thalassaemia (B) Colour blindness

(C) Sickle-cell-anemia (D) Alkaptonuria

(186) It is true for gene mutation ..... .

(A) variation are useful in natural selection and evolution of a species.

(B) any gene can undergo mutations.

(C) mutated gene is harmful to the individual.

(D) all of above

(187) In which diseases haemoglobin becomes defective?

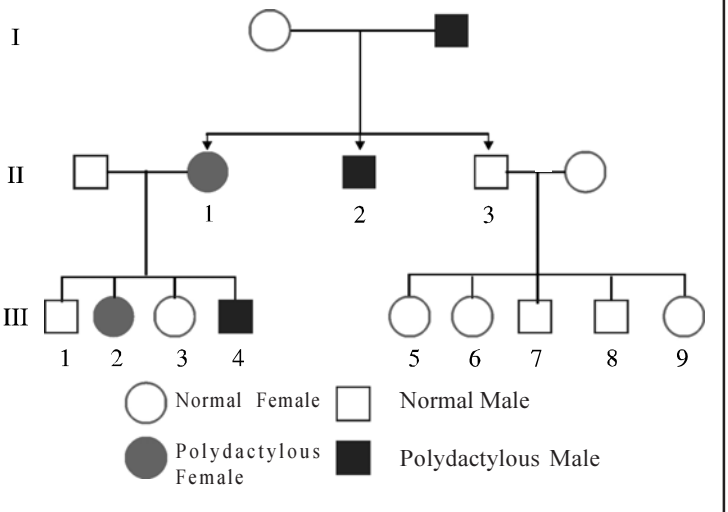
(A) Thalassaemia (B) Sickle cell anemia

(C) Philadelphia (D) Colour blindness

Answers : (185-C), (186-D), (187-B)

Pedigree Analysis :

- A record of the occurrence of trairs in several generations of a human family is known as pedigree analysis.
- For such type of analysis, information about the family history for particular traits is first collected. Then the expressions of the traits are assembled in a chart.
- The pedigree of a family is represented in the form of a chart.
- The females are represented by circles (○) and the males are represented by square (□).
- The marriage is indicated by horizontal bar connecting a circle and a square.
- The offspring are suspended from the marriage bar by vertical lines.
- In individuals in one horizontal line belong to the same generation.
- Each generation is numbered by roman numbers (I, II, III etc) and the individuals in each generation are numbered by 1, 2, 3, 4 and so on.
- Normal individuals are represented by open circle or square and affected individuals are represented by closed circle or squares.



Pe digree of Polydactyl in Man

- Heterozygous individuals are represented by closing half of the circle or squares.
- For example, in the following illustration, a pedigree analysis has been made for polydacty (the occurrence of extra fingers).
- The chart show a marriage between a poplydactylous man and a normal woman (generation I). They produce three children, a polydactylous daughter, a polydactylous son and a normal son (generation II).
- The first and the third individuals of second generation each marry normal persons, their children are shown in third generation from the results.
- We can conclude that a polydactyl offspring appears only when at least one person is polydactylous. Pedigree of polydactyl in man

---

(188) A record of the occurance of traits in several generations of a human family is known as a ..... .
(A) record (B) pedigree analysis (C) family record (D) family tree

(189) The expressions of the traits are assembled in a ..... .
(A) Chart (B) Graph (C) Line (D)Table

(190) Which sign use for females and males in pedigree analysis?
(A) Line, squares (B) Squares, circle (C) Circle, squares (D) Circle, circle

(191) In pedigree analysis marriage is indicated by ..... .
(A) square (B) horizontal square
(C) horizontal bar connecting a circle and a square (D) line

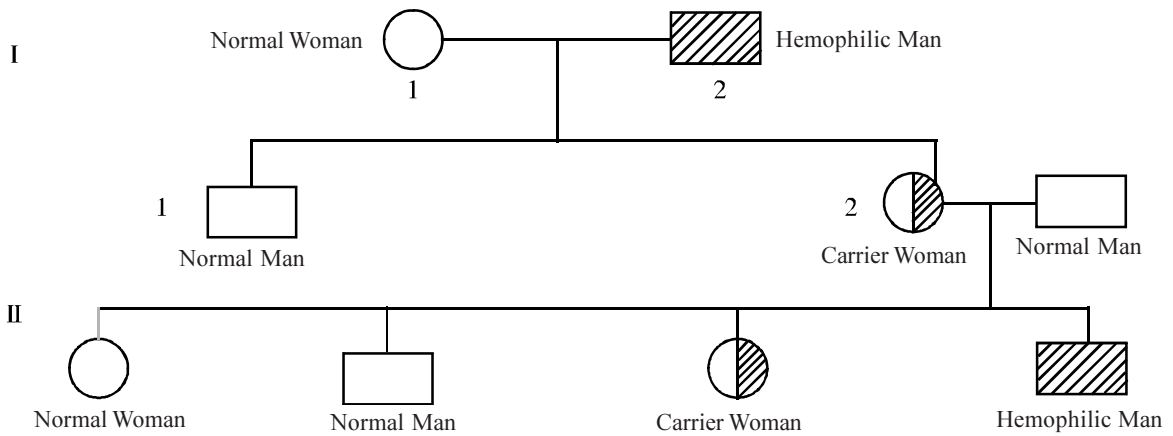
(192) In pedigree analysis normal individuals are represented by ..... .
(A) open circle or squares (B) closed circle and square
(C) closing half circle and square (D) roman numbers

(193) In pedigree analysis, heterozygous individuals are represented by…
(A) open circles and squares (B) closed circle and square
(C) closing half circle and square (D) roman numbers

Answers : (188-B), (189-A), (190-C), (191-C), (192-A), (193-C)

Haemophilia :

- It is a hereditary blood disease by delayed blood clotting.
- This is because of the absence of a factor in the blood-antihaemophilic globulin.
- Haemophilia is a sex linked recessive character and the genes are located on x-chromosome.
- It is caused by recessive genes represented by 'hh' and the normal condition is due to dominat gene H.
- The family pedigree of Queen Victoria show a number of haemophilic descendants as she was a carrier of the diseases.



(194) It is a hereditary blood disease caused by delayed blood clotting.
(A) Haemophilia (B) Sickle cell anemia (C) Thalassaemia (D) PKU

(195) Which reason is correct for haemophilia ?
(A) Absence of a factor in the blood antihaemophilic globulin.
(B) Sickle shaped RBCS.
(C) Required RBCs not generated in blood.
(D) Homogenetic acid oxidase is not produced.

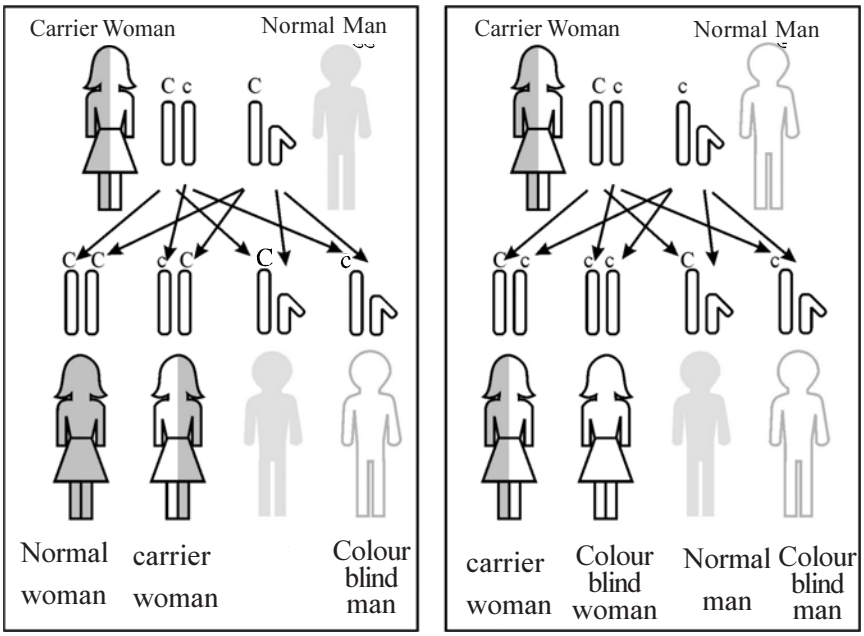
(196) Type of gene which responsible for causing haemophilia is ..... .
(A) co-dominant (B) incomplete dominant (C) recessive (D) epistatic

(197) Haemophilia is represented by gene ..... .
(A) HH (B) hh (C) Hh (D) Hb^s, Hb^s

Answers : (194-A), (195-A), (196-C), (197-B)

* **Colour – Blindness :**

- In it, the affected persons can not distinguish red colour and green colour.
- It is a recessive character. It is caused by recessive genes represented by 'cc'.
- The normal person contain the genes CC or Cc or C alone (in man).
- The genes for colour blindness are located on the x-chromosomes.
- Their alleles are absent from y-chromosomes.
- This character is common in man but rare in woman.
- The daughter carring one recessive gene for colour blindness is called carrier.
- The carriers are normal in their vision.



Inheritance of Colour Blindness

(198) What are all the chances of colour blind child being born in a marriage of normal female marriying a colour blind man?
(A) 0% (B) 25% (C) 75% (D) 50%

(199) What are all the chances of normal daughter being born in a marriage of normal female. Marrying a colour blind man?
(A) 0% (B) 50% (C) 100% (D) 25%

(200) What are the chances of colour blind daughter being born in a marriage of normal male marrying a carrier female?

(A) 100% (B) 50% (C) 0% (D) 75%

(201) Out of two, one son is normal and one son is colour blind of a couple, what are all the chances of colour blind daughters?

(A) 0% and 50% (B)25% and 0% (C)50% and 75% (D) 100% and 0%

(202) Why colour blindness is common in man but rare in woman?

(A) The genes for colour blindness are located on the X-chromosomes.

(B)The genes for colour blindness are located on the Y-chromosome.

(C)The dominant gene for colour-blindness are located on Y-chromosome.

(D) The genes for colour-blindness are located on X and Y-chromosomes.

Answers : (198-A), (199-C), (200-C), (201-A), (202-A)

* **Thalassaemia :**

- In this case, the required haemoglobin is not generated in the blood of a person who suffers from this disease through inheritance.
- It has different types. Out of it test of β thalassaemia is essential before arranging marriage.
- Out of male or female, one or both can be thalassaemic minor or thalassaemia major is possible.
- If out of father or mother, this effective gene is inherited from any one or the two, such offspring does not have the disease, but become a carrier for this effective gene.
- Effective gene from both parents passed to the next generation offspring can be the source of Thalassaemia.
- In such a case both parents are Thalassaemic minor. Their child is known as Thalasaemina major.
- Out of husband-wife, if one is Thalassaemic minor, they live their family life without any trouble, their children have no fear for Thalassaemia major but any one child can be a thalassaemic carrier.

---

(203) Which test of Thalassaemia is essential before arranging marriage?

(A) α (B) β (C) γ (D) δ

(204) If a Thalassamic minor man marries a thalassamic minor woman, their child will be ..... .

(A) Normal (B) Thalassamic minor

(C) Carrier of Thalassaemia (D) Thalassamic major

(205) A man and woman among two one is thalassamic minor. Then the child would be ..... .

(A) Thalassamic minor (B) Thalassamic major

(C) Normal but carrier (D) None of above

(206) The pattern of thalassaemia follows ..... .

(A) Mendelian Principles (B) Morgan Principles

(C) Bridges Principles (D) Derven Port Principles

Answers : (203-B), (204-D), (205-C), (206-A)

* **Sickle Cell Anaemia :**

- It is a hereditary disease, this disease first pointed out in 1952 by Linus Pauling.
- This disease is characterized by the presence of sickle-shaped RBCs under low oxygen pressure.
- It is due to the presence of defective haemoglobin called haemoglobin s(**Hb^S**).

- Sickle cell anaemia is a recessive character caused by the recessive genes **Hb^S, Hb^S**.
- The normal adult haemoglobin is produced by dominant genes **Hb^A, Hb^A**.
- The heterozygous **(Hb^A, Hb^S)** person are normal and are the carriers of sickle cell genes.
- The defect is caused by the substitution of Glutamic acid by Valine at the sixth position of the beta chain of the haemoglobin molecule.

Note : The genetic codes for glutamic are : GAA, GAG
The genetic codes for Valine are : GUA, GUG

---

(207) It is true for sickle cell anemia.
(A) Sickle shaped RBCs. (B) Less haemoglobin production.
(C) Poorly developed reproductive organs. (D) Absence of anti-hemophilic globulin.

(208) Which genes are responsible for sickle cell anemia ?
(A) Hb^A, Hb^A (B) Hb^A, Hb^S (C) Hb^S, Hb^S (D) All of Above

(209) It is a carrier gene for SCA
(A) Hb^A, Hb^S (B) Hb^A, Hb^C (C) Hb^H, Hb^A (D) Hb^S, Hb^H

(210) It is a main reason for causing SCA…
(A) The defect is caused at the sixth position of α-chain of a haemoglobin molecule.
(B) The defect is caused at the sixth position of β-chain of a haemoglobin molecule.
(C) Change at fifth position of β-chain of a haemoglobin.
(D) Change at forth position of α-chain of a haemoglobin molecule.

(211) Which amino acid occupied sixth position of the beta chain of haemoglobin molecule?
(A) Serine (B)Valine (C) Glutamic acid (D) Aespartic acid

(212) The SCA is caused by the substitution of Glutamic acid by…..at the sixth position of the beta chain of haemoglobin molecule.
(A) Glutamic acid (B) Valine (C) Serine (D) Aespartic acid

Answers : (207-A), (208-C), (209-A), (210-B), (211-C), (212-B)

* **Inborn Errors of Metabolism** :

(1) Phenyl Ketonuria (PKU) :

- It is an inborn error in metabolism.
- It is a recessive character caused by recessive genes represented by pp.
- When these recessive genes are present, the enzyme phenylalanine hydroxylase is not produced.
- In the absence of this enzyme, phenyl alanine can not be converted into tyrosine.
- Phenyl alanine and its derivatives accumulate in the blood and cerebrospinal fluid.
- The excess of phenyl alanine is ercreted in the urine.

(2) Alkaptonuria :

- It is an inborn error in metabolism.
- It is a recessive character caused by recessive genes represented by ‘aa’.
- When these genes are present, the enzyme homogentisic acid oxidase is not produced.

- In absence of enzyme, homogentisic acid can not be converted to acetoacetic acid as a result homogentisic acid accumulates in the blood.
- The urine of such person turns black when exposed to air.

(3) Albinism :

- It is a hereditary defect where the melanin pigment are absent from the skin, hair, eye etc..
- It is also an inborn error in metabolism caused by recessive genes. Represented by 'cc'.
- When 'cc' are present, the enzyme tyrosinase can not be produced.
- Hence tyrosine can not be converted into melanin pigments.

---

(213) Which enzyme convert phenylalanine into tyrosine ?

(A) Tyrosinase (B) Homogentisic acid oxidase

(C) Phenyl alanine hydroxylase (D) Uriase

(214) In PKU phenyl alanine and its derivatives accumulate in ..... .

(A) blood (B) cerebro spinal fluid

(C) adipose tissue (D) A and B both.

(215) Alkaptonuria caused by absence of enzyme ..... .

(A) Homogentisic acid oxidase (B) Phenyl alanine hydroxylase

(C) Tyrosinase (D) Urease

(216) In which disease the urine of person turns black when exposed to air ?

(A) PKU (B) Alkaptonuria (C) SCA (D) Albinism

(217) Albinism caused by in the absenece of ..... .

(A) Uriase (B) Maltase (C) Tyrosinase (D) Phenyl alanine hydroxylase

(218) Name of disease in which excess of homogenetisic acid accumulates in the blood?

(A) Sickle cell anemia (B) Thalassaemia

(C) Haemophilia (D) Alkapto nuria

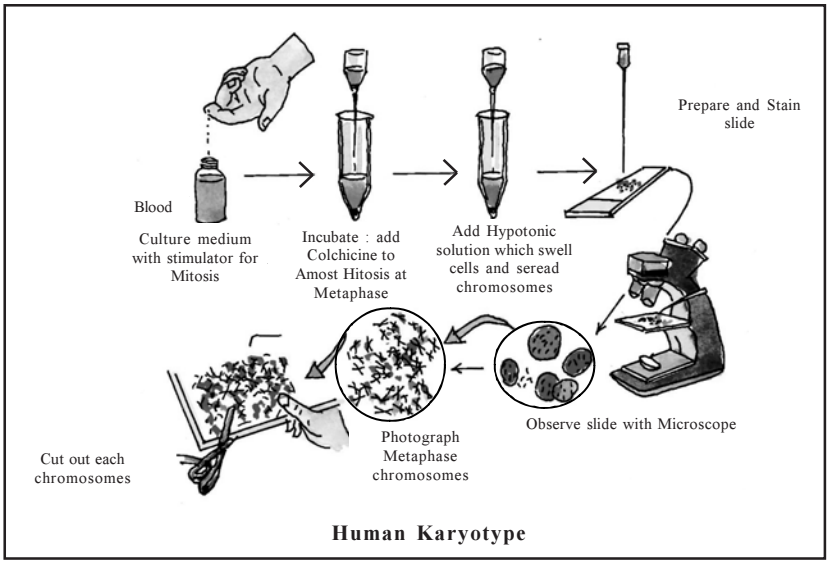
Answers : (213-C), (214-B), (215-A), (216-B), (217-C), (218-D)

* **Human karyotype : Blood culture method :**

Karyotyping of human chromosomes : Chromosomes are clearly visible only in rapidly dividing cells. Human chromosomes are studied in blood cells (WBCs), cells in bone marrow, amniotic fluid and cancerous tissues. The WBCs divide when added with phytohaemaglutinin(PHA).

The division stops when colchicine is added at metaphase stage. These dividing WBCs are then treated with hypotonic saline solution. Chromosomes are now stained with stains like orcein, Giemsa dye or recent quinacrine dye.

When viewed with special microscope in ultraviolet light the stain produces fluorescent bands on chromosomes. The chromosomes are then arranged on photographic plate for making diagram and their study. The pictorial representation of a person's chromosomes is called Karyotype.



Human Karyotype

(219) It is usefull for understanding of chromosomal disorders.
(A) Human pedigree analysis (B) Human family chart
(C) Human Karyotype (D) Inheritance

(220) In blood culture method which growth medium is use for cultivation of WBCs ?
(A) Phyto haemogglutinin (B) Agar–agar (C) Glucose (D) Lac-agar

(221) Which chemical is added at metaphase stage to arrest division of WBCs?
(A) Colchicine (B) Phytohaemogglutinin (C) Phenylalanine (D) Glucose

(222) In blood culture method harvested WBCs are treated with ..... .
(A) Isotonic solution (B) Hypertonic solution
(C) Hypotonic solution (D) None of above

(223) For preparing Human Karyotype, cells are arrested mitosis at ..... .
(A) Prophase (B) Metaphase (C) Anaphase (D) Telophase

(224) If individual chromosomes are cut out, paired as per size and shape and then arranged in a descending order of size, we can construct a ..... .
(A) Chromosomal chart (B) Pedigree analysis
(C) Karyotype of chromosomes (D) Parthenote chart

Answers : (219-C), (220-A), (221-A), (222-C), (223-B), (224-C)

* **Chomosomal Disorders :**

(1) Down's syndrome :

- This disorder is caused by trisomy of 21st pair of chromosomes.
- There are three members in 21st pair instead of two.
- The total number of chromosomes become 47.

Symptoms :

- Short structure, large head, short neck, flat, round face.
- Folded eyelids as are commonly observed in mongoloid race.
- Large thick and swollen tongue and drooping lips.
- Mental retardation lower sensitivity.
- Short, stubby finger, Flat palm
- Poorly developed reproductive organs.
- Sterile

(2) Turner's syndrome :

- This is a sex-linked disorder.
- When a woman has only one x-sex chromosome, instead of the normal two, this disease occurs. Thus this disorders is caused by monosomy of sex-chromosomes.



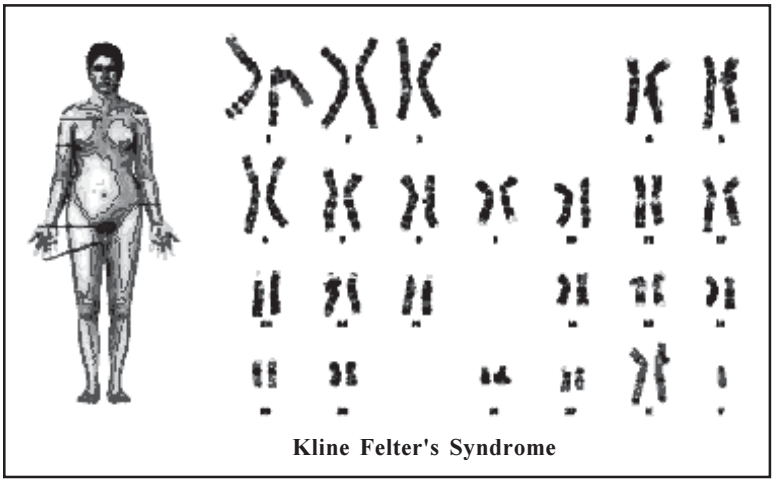
Turner's Syndrome

Symptoms :

- Short structure, short webbed neck
- Phenotypically a woman but reproductive organs are poorly developed
- Almost flat chest
- Uterus underdeveloped
- Sterile

(3) Kline felter's syndrome :

- This is a sex-linked disorder.
- When there are two or more x-sex chromosomes in a man instead of one this disorder occurs.
- Thus, this disorder is caused by trisomy of sex-chromosomes.



Kline Felter's Syndrome

Symptoms :

- Phenotypically a man, but sterile.
- Testes under developed.
- Tall in size, legs much longer, trunk shorter.
- Sparse hairs on body.
- Amount of facial hairs low.
- Breast development as in woman.
- Broad and flat pelvic girdle and shrill, feminine voice.
- Mentally retarded.

(225) Down's syndrome mean ..... .

(A) Trisomy of 21st pair of chromosomes (B) Trisomy of 22nd pair of chromosomes

(C) Monosomy of 5th pair of chromosomes (D) Monosomy of 23rd pair of chromosomes

(226) It is true for Down's syndrome ..... .

(A) flat round face, flat palm (B) folded eyelids, flat chest

(C) flat chest, poorly developed uterus (D) webbed neck, spare hairs on body.

(227) Type of female having only one X-sex chromosome is ..... .

(A) Down's syndrome (B) Super female

(C) Klinefelter's syndrome (D) Turner's syndrom

(228) Turner's syndrome ..... .

(A) Monosomy of autosomes (B) Monosomy of sex-chromosomes

(C) Trisomy of sex-chromosomes (D) Tetrasomy of autosomes

(229) Types of man having extra x-sex chromosome is ..... .

(A) Down's syndrome (B) Turner'syndrome

(C) Klinefelter's syndrome (D) Super male

(230) In which disorder folded eyelids as are commonly observed in mongoloid race?

(A) Turner's syndrome (B) Klinefelter's syndrome

(C) Down's syndrome (D) Super male

(231) It is true for Turner's syndrome.

(A) Uterus underdeveloped (B) Sterile female

(C) Almost flat chest (D) All of above

Answers : (225-A), (226-A), (227-D), (228-B), (229-C), (230-C), (231-D)

- **A = Assertion, R = Reason type questions**

Choose the correct option from the following for given questions.

(A) A and R both are correct, while R is correct explaination of A.

(B) A and R both are correct but R is not correct explaination of A.

(C) A is correct, but R is wrong.

(D) A is wrong but R is correct.

(232) Assertion A : Person suffering from haemophillia fail to produce blood clotting factor (VIII).

Reason R : Prothrombin producing platelets in such person are found in very low concentration.

(A) (B) (C) (D)

(233) Assertion A : Mustard gas act as a mutagen.

Reason R : It transfers alkyl groups to the bases in DNA.

(A) (B) (C) (D)

(234) Assertion A : The principle of segregation given by Mendel is the principle of purity of gametes.

Reason R : Gametes are pure of character.

(A) (B) (C) (D)

(235) Assertion A : Among the primates, chimpanzee is the closest relative of the present day humans.

Reason R : The banding pattern in the autosome numbers 3 and 6 of man and chimpanzee is remarkably similar.

(A) (B) (C) (D)

(236) Assertion A : Holandric genes are found on y chromosome.

Reason R : Inheritance of Holandric genes are always from father to son.

(A) (B) (C) (D)

(237) Assertion A : Haemophillia never occurs in woman.

Reason R : Gene for haemophillia is located on X-chromosome.

(A) (B) (C) (D)

(238) Assertion A : In human most sex-linked genes are present on the xchromosome.

Reason R : x-chromosome contain a large number of gene with major effect on phenotype.

(A) (B) (C) (D)

(239) Assertion A : Haploids are used to study mutation.

Reason R : Most of the mutation are recessive.

(A) (B) (C) (D)

(240) Assertion : Gene mutation is true mutation.

Reason : Genetic mutation are useful in natural selection and evolution of a species.

(A) (B) (C) (D)

(241) Assertion A : If $\frac{X}{A}$ ratio obtained 1, than flies will develop into normal female.

Reason R : In drosophila the gene for femaleness are distributed over x-chromosome and those for maleness are distributed on y-sex chromosome.

(A) (B) (C) (D)

(242) Assertion A : In twins of opposite sex sterile female called free martin.

Reason : During development both the twins are connected by separate umbilical cord.

(A) (B) (C) (D)

Answers : (232-C), (233-A), (234-A), (235-A), (236-A), (237-D), (238-A), (239-A), (240-A), (241-C), (242-C)

- **True - False (T - F) types MCQs :**

(243) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) 22AA + XY $\rightarrow$ 0 Barr Body

(ii) 22AA + XX $\rightarrow$ 1 Barr Body

(iii) 22AA + XXY $\rightarrow$ 1 Barr Body

(A) FTT (B)FFF (C)TTF (D)TTT

(244) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them :

(i) During gametogenesis there is no meiosis in Drone.

(ii) In Flagellaria plant male is heterogametic.

(iii) In Humulus japanic female has XXX.

(A) TFF (B) TFT (C) FFT (D)TTF

(245) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) □ = Normal Female

(ii) ○ = Normal male

(iii) □—○ = Marriage bar

(A) TTT (B) FFT (C) FTT (D) TFT

(246) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) Aneuploidy leads to Down's syndrome.

(ii) PKU is caused by dominant genes of autosomes.

(iii) SCA is a recessive character caused by the recessive gene located on X-sex chromosome.

(A) FFT (B) FFF (C) TFF (D) TTF

(247) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) The allele which is expressed is called dominant gene and the allele which is not expressed is called recessive gene.

(ii) The genes having single effects are called pleiotropic genes.

(iii) Mirabilis jalapa is best example of pleiotropism.

(A) TFF (B) FTF (C) FFT (D) FFF

(248) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) A blood group : Genotype I^AI^A and I^Ai (ii) AB blood group : Genotype I^AI^B and I^Bi

(iii) O blood group : Genotype ii

(A) TTT (B) TFT (C) FTT (D) FFT

(249) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) For child having O blood group, mother having O blood group then the blood group of father may be O, A and B

(ii) If mother having A blood group, father having A blood group then the blood group of child will be A, B, AB or O.

(iii)If mother and father both having B blood group then the blood group of child will be B, O or AB.

(A) FFF (B) FFT (C) TFF (D) TTT

(250) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them:

(i) X/A =1 means normal female (ii) X/A =0.5 means sterile female (iii)X/A = 0.66 means intersex sterile

(A) FFT (B) TFT (C) FTF (D) TTF

(251) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them :

(i) Gynandromorphs are individuals who show male characters on some part of the body and female characters on other parts of body.

(ii) Examples of gynandromorphs are Drosophila, butterflies, beetles, wasps, bees etc.

(iii) The loss of Y-chromosome cause Gynandromorphs.

(A) FFT (B)TTT (C)TTF (D)TFF

(252) Read each of the following statement if statement is true select (T) and if statement is false select (F) for them :

(i) In the case of thalassaemia the required RBCs is not generated in the blood of person.

(ii) In SCA RBCs converted in shape of sickle.

(iii) The person suffering with colour blindness can not distinguish red colour and green colour.

(A) FTT (B) FFT (C) TTT (D) FFF

Answers : (243-D), (244-A), (245-B), (246-C), (247-A), (248-B), (249-C), (250-B), (251-C), (252-A)

* **Column type MCQs :**

(253) Match column-I with column-II and find the correct answer :

Column – I	**Column – II**	
(I) $\frac{X}{A}=1$	(p)Super female	(A) (I-p) (II-q) (III-r) (IV-s)
(II) $\frac{X}{A}=0.5$	(q) Normal male	(B) (I-r) (II-p) (III-q) (IV-s)
(III) $\frac{X}{A}=1.5$	(r) Normal female	(C) (I-s) (II-r) (III-p) (IV-q)
(IV) $\frac{X}{A}=0.66$	(s) Intersex sterile	(D) (I-r) (II-q) (III-p) (IV-s)

(254) Match column-I with column-II and find the correct answer :

Column – I	**Column – II**	
(I) xx-xo sex determination	(p)Klinefelter's syndrome	(A) (I-s) (II-r) (III-p) (IV-q)
(II)xo- type female	(q) Birds	(B) (I-s) (II-p) (III-q) (IV-r)
(III)xxy- type male	(r) Turner's syndrome	(C) (I-q) (II-p) (III-r) (IV-s)
(IV) zz-zw sex determination	(s) Hymenopterans	(D) (I-r) (II-q) (III-p) (IV-s)

(255) Match column-I with column-II and find the correct answer :

Column – I	**Column – II**	
(I) Sickle cell anemia	(p)7th chromosome	(A) (I-r) (II-s) (III-p) (IV-q)
(II)PKU	(q) 4th chromosome	(B) (I-s) (II-p) (III-r) (IV-q)
(III)Cystic fibrosis	(r) 11th chromosome	(C) (I-p) (II-q) (III-r) (IV-s)
(IV) Huntington	(s) 12th chromosome	(D) (I-q) (II-r) (III-p) (IV-s)

(256) Match column-I with column-II and find the correct answer :

Column – I	**Column – II**	
(I) Bateson and Punnet	(p) Observed that in insects, two kinds of sperms are produced.	(A) (I-p) (II-q) (III-r) (IV-s)
(II) Henking	(q) Identified X-body as a chromosome	(B) (I-s) (II-r) (III-q) (IV-p)
(III) Mc Lung	(r) The chromosomal theory of sex-determination was proposed.	(C) (I-s) (II-p) (III-q) (IV-r)
(IV) Miss stevens	(s) Experiments indicative of Linkage were first performed on sweet pea plant.	(D) (I-q) (II-r) (III-p) (IV-s)

(257) Match column-I with column-II and find the correct answer :

Column – I	**Column – II**	
(I) Correns	(p) Genetic balance theory	(A) (I-r) (II-q) (III-p) (IV-s)
(II) Bridges	(q) Theory of heterogamesis	(B) (I-p) (II-q) (III-s) (IV-r)
(III) Barr and Bertram	(r) Environmental effect on determination of sex	(C) (I-q) (II-p) (III-r) (IV-s)
		(D) (I-q) (II-p) (III-s) (IV-r)
(IV) Baltzar	(s) Sex chromatin	

(258) Match column-I with column-II and find the correct answer :

Column – I	**Column – II**	
(I) Liver Worts	(p) The sex organs are located on a haploid gametophyte.	(A) (I-p) (II-q) (III-r) (IV-s)
		(B) (I-q) (II-p) (III-s) (IV-r)
(II) Angiosperm	(q) Male heterogametic and female homogametic	(C) (I-s) (II-r) (III-p) (IV-q)
(III) Flagellaria	(r) Female is heterogametic and male is homogametic	(D) (I-r) (II-q) (III-s) (IV-p)
(IV) Dioscoria sinulate	(s) Female (AA+XX), male (AA+XO)	

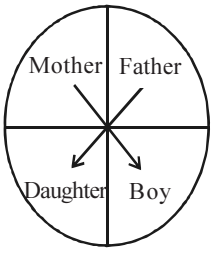
(259) Match column-I with column-II and find the correct answer :

Column – I	Column – II	
(I) Phenyl ketonuria	(p) Phenylalanine can not be converted into tyrosine.	(A) (A) (I-q) (II-p) (III-r) (IV-s)
(II) Alkaptonuria	(q) Homogentisic acid can not be converted into acetoacetic acid.	(B) (I-p) (II-r) (III-q) (IV-s)
(III) Albinism	(r) Tyrosine can not be converted into melanin pigment.	(C) (I-s) (II-r) (III-p) (IV-q)
(IV) Thalassaemia	(s) Required haemoglobin is not generated in the blood.	(D) (I-p) (II-q) (III-r) (IV-s)

Answers : (253-D), (254-A), (255-A), (256-C), (257-D), (258-A), (259-D)

* **Figure based MCQs :**

(260) Represented below is the inheritance pattern of a certain type of traits in humans. Which one of the following conditions could be an example of this pattern.



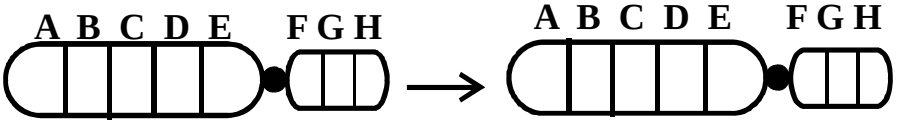
(A) Phenyl Ketonuria

(B) Sickle cell anaemia

(C) Haemophilia

(D) Thalassemia

(261) Given below is the representation of a kind of chromosomal mutation. What is the kind of mutation represented.



(A) Deletion (B) Duplication (C) Inversion (D)Reciprocal translocation

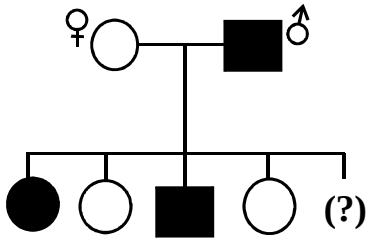
(262) In the given human hand pedigree which character is represented and what is the probability of disease occurrence in fifth child.

(A) Polydactyly (x-linked recessive disorder), 50%

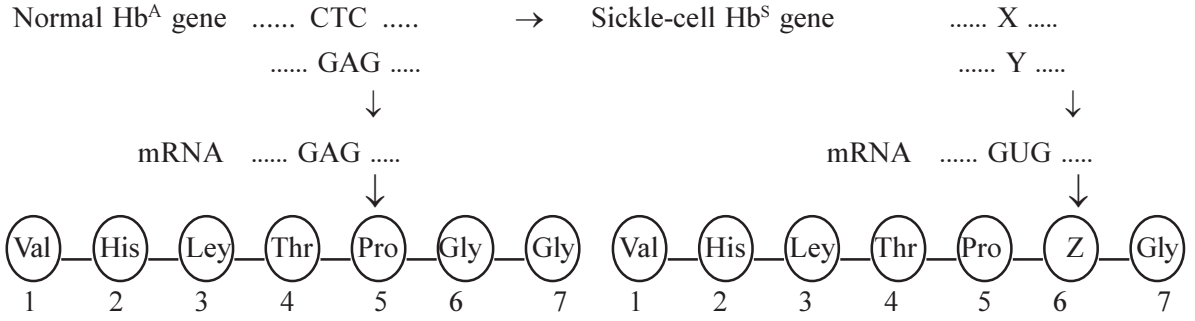
(B) Polydactyly (x-linked dominant disorder), 50%

(C) Polydactyly (autosomal recessive disorder), 50%

(D) Polydactyly (autosomal dominant disorder), 50%



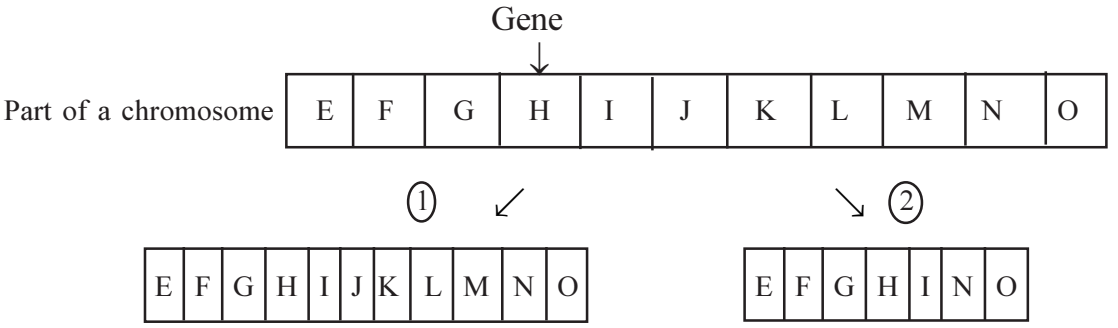
(263) Sickle cell anemia is an autosome linked recessive trait that can be transmitted from parents to the offspring when both the partners are carrier for all the gene (or heterozygous). The disease is controlled by a single pair of allele Hb^A and Hb^S. Out of the three possible genotypes only homozygous individuals for Hb^S(Hb^S, Hb^S) are lethal. Select the right option in which x, y and z are correctly identified.



(A) X - CAC, Y - GTC, Z - His (B) X - GTG, Y - CAC, Z - Val

(C) X - CAC, Y - GTG, Z - phe (D) X - CAC, Y - GTG, Z - Val

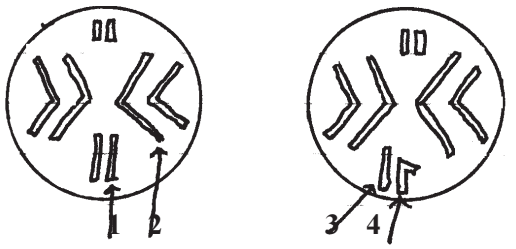
(264) The given figure shows two types of chromosome mutation. These are called ..... .



(A) 1- Inversion, 2 – substitution (B) 1- Inversion, 2 - Deletion

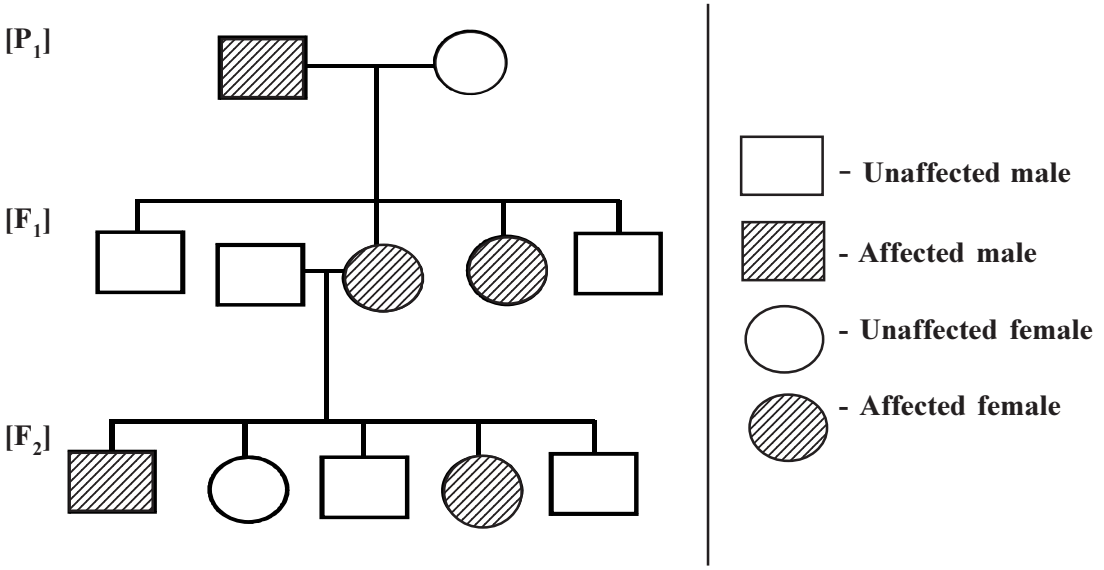
(C)1- Duplication, 2- Substitution (D) 1- Duplication, 2- Deletion

(265) The following figure refer to the chromosome complement of each sex of fruit bfly by which number is a Y chromosome labelled.



(A) 4 (B) 3 (C) 2 (D) 1

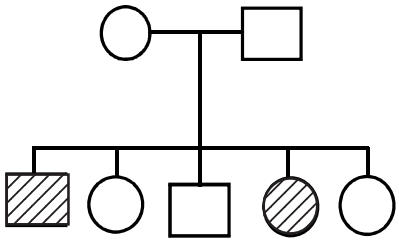
(266) Given below is a pedigree chart showing the inheritance of a certain sex-linked trait in humans. The trait traced in the given pedigree chart is ..... .



(A) X-linked dominant inheritance (B) X-linked recessive inheritance

(C) Y-linked dominant inheritance (D) Y-linked recessive inheritance

(267) The following is a pedigree chart of a family with five children. It shows the inheritance of attached ear-lobes as opposed to the free ones. The square represent the male and circle the female individuals. Which one of the following conclusions drawn is correct.



(A)The parents are homozygous dominant (B) The parents are homozygous recessive

(C) The parents are heterozygous (D) The traits is y – linked

Answers : (260-C), (261-B), (262-D), (263-D), (264-D), (265-A), (266-A), (267-D)

* **MCQs for NEET :**

(268) A man having the genotype EEFf and GgHH can produce P number of genetically different sperms, and a woman of genotype IiLLMmNn can generate Q number of genetically different egg. Determine the value of p and Q.

(A) P =4, Q =4 (B)P = 4, Q = 8 (C) P = 8, Q = 4 (D) P = 8, Q = 8

(269) Ratio of progeny when a red coloured heterozygote is crossed with a white coloured plant in which red colour is dominant in white colour.

(A) 3 : 1 (B) 1:1 (C)1:2:1 (D) 9:3:3:1

(Hint): The cross of heterozygous dominant with its recessive parent is called test cross. The test cross gives 1:1 ratio in monohybrid condition where as 1:1:1:1 in dihybrid condition.

(270) How many types of gametes may be produced by genotype D/d:E/e:F/f.

(A) 27 (B)8 (C)3 (D)6

(Hint) : Kinds of gametes may be calculated by following formula : Number of gametes = $(2)^n$ n is number of alleles

e.g. D/d: E/e : F/f have trihybrid cross

i.e. n =3 than kind of gametes = $(2)^3 = 2f2f2 = \boxed{8}$

(271) The term 'genotype' was coined by ..... .

(A) H.J. Muller (B)T.Boveri (C) W.S. Sutton (D)W.L.Johanssen

(272) When a tall plant with round seeds(TTRR) crossed with a dwarf plant with wrinkle seeds(ttrr), the F_1 generation consists of tall plants with round seeds. What would be the proportion of dwarf plant with wrinkle seeds in F_1 generation ?

(A)1/4 (B)1/16 (C)0 (D)1/2

(Hint) : (P) : Tall palnt with round seeds TTRR × Dwarf plants with wrinkled seeds tt rr

gametes : TR tr

(F_1) generation : Tt Rr Tall plant with round seeds

∴ Therefore, the proportion of dwarf plants with wrinkle seeds is zero.

(273) The dwarfness in plants of F_2 generation is ..... .

(A) recessive gene (B) dominant gene

(C) co-dominant gene (D) incomplete dominant gene

(274) In pea plants, yellow seeds are dominant to green. If a heterozygous yellow seeded plant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in F_1 generation.

(A) 50:50 (B) 9:1 (C) 1:3 (D) 3:1

(275) When yellow round heterozygous pea plants are self fertilized, the frequency of occurrence of RrYY genotype among the offspring is ..... .

(A) 9/16 (B) 3/16 (C) 2/16 (D) 1/16

(276) Test cross is used to ..... .

(A) check heterozygosity in F_1 generation (B) check heterozygosity in F_2 genration

(C) check independent assortment (D) check Dominance

(277) In Mendelism, linkage was not observed due to ..... .

(A) mutation (B) independent assortment

(C) synapsis (D) crossingover

(278) Among the seven pairs of contrasting traits in pea plants as studied by Mendel, the number of traits related to flower, pod and seed respectively were ..... .

(A) 2,2,2 (B) 2,2,1 (C) 1,2,2 (D) 1,1,2

(279) A cross in which an organism showing a dominant phenotype is crossed with the recessive parent in order to know its genotype is called ..... .

(A) monohybrid cross (B) back cross (C) test cross (D) dihybrid cross

(Hint): Test cross include cross of F_1 the recessive parents i.e.(Tt×tt).

(280) In a dihybrid cross where two parents differ in two pairs of contrasting traits like seed colour yellow (YY) and seed colour green (yy) with seed shape round (RR) and seed shape wrinkled (rr), the number of green coloured seeds(yy) among sixteen products of F_2 generation will be.

(A) 2 (B) 4 (C) 6 (D) 8

(Hint): Dihybrid cross

(281) In man, the blue eye colour is recessive to the brown eye colour. If the boy has brown eye and his mother is blue eyed, what would be the phenotype of his father ?

(A) Black eye (B) Brown eye (C) Green eye (D)Blue eye

(282) Pure homozygous offsprings in a dihybrid cross in the F_2 generation will be.

(A) 1/2 (B) 1/4 (C) 1/8 (D) 1/16

(283) Mendel crossed a pure white flowered recessive pea plant with a dominant pure and red-flowered plant. The first generation of hybrids from the cross should show.

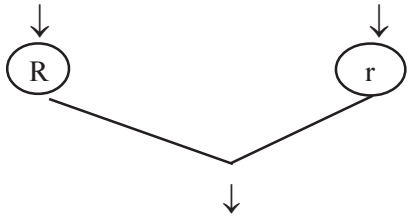
(A) 50% white flowered and 50% red flowered plants. (B) All red flowered plants.

(C) 75% red flowered and 25% white flowered plants. (D) All white flowered plants.

(Hint): All red flowered plants; according to Mendel's law of dominance.

(P) : RR x rr

Red dominant Recessive white



F_1 Rr

All Red

(284) If in a dihybrid cross Mendel had used two such characters which have, linked, he would have faced difficulty in explaining the results on the basis of his ..... .

(A) law of segeregation (B) law of multiple factor hypothesis

(C) law of independent assortment (D) law of dominance

(Hint) : In this experiment, Mendel stated that each character is governed by a single gene and there is no linkage and gene interaction. He failed to explain his law of independent assortment in the presence of linkage.

(285) From a cross AaBB × aaBB, following genotype ratio will be obtained in F_1 generaion.

(A) 1 Aa BB : 1 aa BB (B) 1 Aa BB : 3 aa BB

(C) 3 Aa BB : 1 aa BB (D) All Aa BB : No aa BB

(Hint):

AaBB × aaBB Gametes for F_1 = AB, aB and aB, aB

After crossing = AaBB, aaBB Ratio 1:1

(286) From a single ear of corn, a farmer planted 200 Kernels which produced 140 tall and 40 dwarf plants. The genotype of these offspring are most likely ..... .

(A) TT, Tt and tt (B) TT and tt only (C) TT and Tt only (D) Tt and tt only

(Hint):

TT is homozygous tall plant; (48 plants)

Tt is heterozygous tall plant (90 plants)

tt is homozygous dwarf plant (40 plants)

So, on the basis of Mendel's Monohybid cross raio will be 1 : 2 : 1.

(287) When AABB and aabb are crossed, in F_2 generation the ratio of AaBb will be

(A) 1/16 (B) 2/16 (C) 8/16 (D) 4/16

(288) If a cross is made between AA and aa, the nature of F_1 progeny will be ..... .

(A)Genotypically AA, Phenotypically a (B) Genotypically Aa, Phenotypically a

(C) Genotypically Aa, Phenotypically A (D) Genotypically aa, Phenotypically A

(Hint): AA × aa

↓ ↓

(F_1) : A a

(289) A self fertilizing trihybrid plant forms ..... .

(A)8 different gametes and 16 different zygotes.

(B) 8 different gametes and 32 different zygotes.

(C) 8 different gametes and 64 different zygotes.

(D) 4 different gametes and 16 different zygotes.

(Hint) : The offsprings show 27:9:9:9:3:3:3:1 ratio is found in trihybrid cross.

(290) When heterozygous red (Dominant) flower is crossed with white flower the progeny would be.

(A) 350 red : 350 White (B) 450 red : 250 White

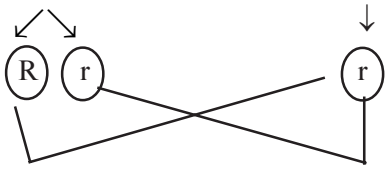
(C) 380 red : 320 White (D) None of these

(Hint) : When heterozygous red (dominant) flower (Rr) is crossed with white flower (rr), red and white flowered plants will be produced in equal ratios. The cross can be given as ..... .

(Hint): Rr × rr

Red White

Gametes



(F_1) Rr rr

generation 1 : 1

(291) In a population of 1000 individuals 360 belong to genotype AA, 480 to Aa and the remaining 160 to aa. Based on this data, the frequency of allele A in the population is ..... .

(A) 0.6 (B) 0.7 (C)0.4 (D)0.5

(Hint): According to hardy Weinberg principle

$P^2 + 2pq + q^2 = 1$;

$(p+q)^2 = 1$

(AA) p^2 = 360 out of 1000 individual or p^2 = 36 out of 100

q^2 = 160 out of 1000 or q^2 = 16 out of 100

so, $9 = \sqrt{16} = 4$

As p + q = 1

So , p is 0.6

(292) Which event responsible in Drosophila for no independent assortment of gene A and B ?

(A) Repulsion (B) Linkage (C) Crossing over (D) Recombination

(293) Which condition is responsible for PKU ?

(A) Trisomy (B) Monosomy

(C) Autosomal Dominant gene (D) Autosomal Recessive gene

(294) The loss of one member of chromosome instead of two in pair, the condition is called ..... .

(A) Monosomy (B)Nullisomy (C)Trisomy (D)Tetrasomy

(295) A diseased man marries a normal woman. They get three daughter and five sons. All the daughters were diseased and sons were normal. The gene of this disease is ..... .

(A) Autosomal Dominant (B) Sex linked dominant

(C) Sex limited character (D) Sex linked recessive

(296) Genes present in the cytoplasm of eukaryotic cells, are found in.

(A) Mitochondria and ribosomes (B) Plastids and mitochondria

(C) Plastids and lysosomes (D) Mitochondria and golgibody

(297) In Drosophila, sex is determined by ..... .

(A) whether the egg is fertilized or develops parthenogonetically

(B) X and Y chromosomes

(C) The ratio of number of X-chromosomes to the sets of autosomes

(D) The ratio of pairs of X-chromosomes to the pairs of autosomes

(298) In plants the cytoplasmic male sterility found in ..... .

(A) cytoplasmic genes (B) plastid gene complex

(C) mitochondrial gene complex (D) nuclear gene complex

(Hint) : The pollen sterility which is controlled by cytoplasmic genes is known as cytoplasmic male sterility. Usually the cytoplasm of zygote comes primarily from the egg cell and due to this progeny of such male sterile plants would always be male sterile.

In plants male sterility is of five types : (1) Genetic male sterility

(2) Cytoplasmic male sterility

(3) Cytoplasmic genetic male sterility

(4) Chemical induced male sterility and

(5) Transgenic male sterility

(299) Recessive traits studied by Mendel in pea is ..... .

(A) Round shape (B) Axial position of the flower

(C) Green colour of seed cotyledon (D) Green colour of the pod

(300) The linkage map of x-chromosome of fruit fly has 66 units with yellow body gene (y) at one end and bobbed hair (b) gene at the other end. The recombination frequency between these two genes (y and b) should be.

(A) 100% (B) 66% (C) 40% (D) 50%

(Hint) : The actual distance between two genes is said to be equivalent to the percentage of crossing over between these genes i.e. 66 %.

Crossing over chances between y and b genes suggest that these are to be placed on the chromosome at a distance of 66 untis.

(301) It is also known as Christmas disease ..... .

(A) Sickle cell anemia (B) Thalassaemia (C) Hemophilia (D) Colour blindness

(Hint) : Christmas disease also called Hemophilia B or factor IX (Christmas factor) Hemophilla, is a rare genetic disorder in which your body doesn't clot properly.

(302) Down's syndrome is caused by an extra copy of chromosome number 21. What percentage of offspring produced by an affeceted mother and normal father would be affected by this disorder ..... .

(A) 25% (B) 75% (C) 50% (D) 100%

(303) In which of the following colour blindness is inherited ?

(A) In male and female both (B) In female only

(C) In males only (D) In none of the above

(304) A male human is heterozygous for autosomal genes A and B and is also hemizygous for hemophilic gene h. what proportion of his sperms will be abh ?

(A) $\frac{1}{8}$ (B) $\frac{1}{32}$ (C) $\frac{1}{16}$ (D) $\frac{1}{4}$

(Hint) : Chance for getting A will = $\frac{1}{2}$ Chance for getting B will = $\frac{1}{2}$

Chance for getting h will = $\frac{1}{2}$ Hence, ABh = $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

(305) A normal woman, whose father was colourblind is married to a normal man. The offsprings would be.

(A) 70% colour blind (B) 25 % colour blind

(C) Normal (D) 100% colour blind

(Hint) : Colour blind father's daughter always carrier.

(306) Kappa particles indicate ..... .

(A) nuclear inheritance (B) nucleo cytoplasmic inheritance

(C) mutation (D) cytoplasmic inheritance

(307) At a particular locus frequency of A allele is 0.6 and that of a 0.4. What would be the frequency of heterozygotes in a random mating population of equilibrium?

(A) 0.36 (B) 0.48 (C) 0.16 (D) 0.24

(Hint) : According to Hardy Weinherg

$(p+q)^2 = 1$

$P^2 + 2pq + q^2 = 1$

Heterozygotic frequency is (2pq).

$\therefore$ $2pq = 2 \times 0.6 \times 0.4 = 0.48$

Where,

p = frequency of dominent gene

q = frequency of dominent gene

pq = Heterozygous dominant

$\therefore$ p = A = 0.6

p = a = 0.4

(308) It is not a hereditary disease.

(A) Thalassaemia (B) Hemophilia (C) Cystic fibrosis (D) Cretinism

(309) Lampbrush chromosomes are visible ..... .

(A) in diplotene of meiosis (B) in prophase of meiosis

(C) in interphase (D) in metaphase of meiosis

(310) It is an example of polygenic inheritance.

(A) Production of drone (B) The shape of pod in pea

(C) Skin colour in man (D) Colour of flowers in mirabilis

(311) Why SCA is a very common in African race ?

(A) It is not a lethal disease. (B) It protect against Malaria.

(C) It controlled by dominant genes. (D) It controlled recessive genes.

(312) Character choosen by Mendel are located on how many chromosome ?

(A)4 (B) 6 (C) 5 (D) 7

(313) Which is correct for chromosomal disorders?

(A) Klinefelter's syndrome – 44 autosomes + XXY

(B) Colour blindness – Y-linked disease

(C) Erythroblastosis - X-linked disease

(D) Down's syndrom – 44 Autosomes + XO

(314) It is true for ADA (Adenosine De Aminase) Deficiency.

(A) By activaters of ADA

(B) Bone marrow transplantation from matched sibling donor.

(C) By enzyme replacement therapy

(D) By introducing lymphocytes at short interval which produced through genetic engineering having active ADA

(315) In pea plants, yellow seeds are dominant to green. If a Homozygous yellow seeded is crossed with a green plant. What ratio of green seeded plants would you expect in F_2 generation ?

(A) 50% (B) 25% (C) 75% (D) 100%

(Hint) : Mendels Monohybrid cross.

(316) It is a mutagen ..... .

(A) IR- rays (B) IAA (C) Ehylene (D) Gama-rays

(Hint) : IR short wavelength rays which having capacity to penetrate in deep of body and change the base sequences of DNA. So it called mutagen.

(317) Mating between two different pure-bred lines that lead to best offsprings because ..... .

(A) Heterosis (B) Transformation (C) Splicing (D) Metamorphosis

(Hint) : Heterosis also called hybrid vigour, the increase in such characteristics as size, growth rate, fertility and yield of a hybrid organism over those of its parents.

Plant and animal breeders exploit heterosis by mating two different pure-bred lines that have certain desirable traits.

(318) The zygotic cell will develop into female child due to presence of ..... .

(A) XX (B) Y (C) X (D) XY

(Hint) : Sex-determination in woman, twenty third pair consists of two similar x- sex chromosomes.

(319) F_2 generation in a Mendelian cross showed that both genotypic and phenotypic ratios are same as 1:2:1. It represents a case of ..... .

(A) Co-dominance

(B) Dihybrid cross

(C) Monohybrid cross with complete dominance

(D) Monohybrid cross with incomplete dominance

(320) It is true for hemophilia.

(A) Hemoglobin is not produced. (B) Melanin accumulate in skin.

(C) Blood clotting is not induced. (D) Delayed blood clotting.

(321) A dihybrid for qualitative trait is crossed with homozygous recessive individual of its type, the phenotypic ratio is : ..... .

(A) 1:2:1 (B) 3:1 (C) 1:1:1:1 (D) 9:3:3:1

(322) In a medico-legal case of accidental interchange between two babies in a hospital; the baby of blood group A could not be rightly given to a people.

(A) Husband of group A and wife of group O (B) Husband of group O and wife of group A

(C) With both husband and wife of group O (D) Both husband and wife of group A

(323) If Mother is A blood group and father is AB blood group. What will be the possible blood group in their progeny.

(A) O, A (B) A, B, AB (C) O, A, B (D) O, A, B, AB

(324) Which chromosome determine sex in human ?

(A) X - chromosome (B) Y - chromosome

(C) A/X - chromosome (D) A and B both

(325) If dwarf pea plant was treated with Gibberellic acid, it grew as tall as the pure tall pea plant. If this treated plant is crossed with pure tall plant then the phenotypic ratio of is likely to be ..... .

(A) all dwarf (B) 50% dwarf 50% tall

(C) 75 % tall 25% dwarf (D) All tall

(326) It is true for Turner's syndrome ..... .

(A) XO (B) XXY (C) XXX (D) XYY

(Hint) : This is a sex-linked disorder. When woman has only one X-sex chromosome, instead of the normal two, this Turner's syndrome occur. This disorder is caused by manosomy of sex chromosomes.

(327) Complete linkage : ..... .

(A) Male drosophila (B) Female drosophila

(C) Female butterfiles (D) Nome of above

(Hint): No crossing over occurs in male drosophila such results represent complete linkage.

(328) Which chemical induced artificial polyploidy in plant ?

(A) colchicine (B) Benzene (C) Acetocarmine (D) None of above

(329) The loss of a chromosomal segment is due to ..... .

(A) inversion (B) duplication (C) deletion (D) transversion

(330) It shows genotypic ratio and phenotypic ratio similar.

(A) Mirabilis (B) Drosophila (C) Pisum sativum (d) Silkworm

(331) A woman is married for second time. Her first husband was ABO blood type A, and her child by that marriage was Type O. Her new husband is Type B and their child is type AB. What is the woman's ABO genotype and blood type ?

(A) I^AI^O : Blood type A (B) I^AI^B : Blood type B

(C) I^BI^O : Blood type B (D) I^OI^O : Blood type O

(332) Name the sex linked disorder when a woman has extra two x-sex chromosome, instead of the normal two.

(A) Turner's syndrome (B) Kline felter's syndrome

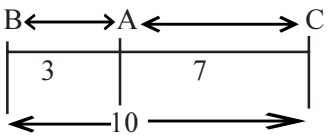
(C) Super female (D) Down's syndrome

(Hint) : Non disjunction of chromosomes lead to change the member of chromosomes in gametes. Such gametes take part in the process of fertilization and causes abnormality.

(333) The map distance between genes A and B is 3 units between B and C 10 units and between C and A 7 units; the order of the genes in a linkage map constructed on the above data would perhaps be ..... .

(A) A, B, C (B) A, C, B (C) B, C, A (D) B, A, C

(Hint):



(334) How many linked gene present in bacteria ?

(A) 4 (B) 2 (C) 1 (D) 5

(Hint) : Number of linked gene is always same then the number of chromosome.

(335) How may linked gene group present in pisum sativum ?

(A) 2 (B) 5 (C) 7 (D) 9

(Hint) : Prokaryotic bacteria E.coli having only one circular DNA hence no. of linked gene in bacteria will be 1.

(336) If Maize having 10 pairs of chromosomes : then the number of linked genes in maize is ....

(A) 5 (B) 10 (C) O (D) 20

(Hint) : In maize n = 10, hence number of linked gene is also 10.

(337) It is true for descending order :

(A) Gene → Cistrone → muton → Recon (B) Gene → muton → Cistrone →Recon

(C) Gene → Recon → Cistrone → muton (D) Gene → Cistrone → Recon → muton

(338) Jumping genes are found in ..... .

(A) Eukaryotes (B) Bacteriophage

(C) Bacteria (D) Eukaryotes and prokaryotes

(Hint) : Jumping gene found both in eukaryota and prokaryota it is discovered by mac.clintock in maize.

(339) Smallest structure having the power of replicating itself is ..... .

(A) Chloroplast (B) Gene

(C) Mitochondria (D) Ribosome

(340) Nucleosome consists of : ..... .

(A) Nucleolus (B) Genes (C) Micro filaments (D) Histones

(Hints) : Histones are main structural Protein found in eukaryotic cells.

(341) Plant A is having chromosome no. 2n = 12 and B having zn = 16 Both are crossed to form allotetraploid C what is the Chromosome number of C.

(A) 32 (B) 14 (C) 28 (D) 7

(342) From a cross AABb × aaBb, the genotypes AaBB : AaBb : Aabb: aabb will be obtained in the following ratio.

(A) 1:1:1:1 (B) 1:2:1:0 (C) 0:3:1:0 (D) 1:1:1:0

(Hint): ♂ AABb × aaBb ♀

Male →

	Gametes →	AB	Ab	AB	Ab
↓	aB	AaBB	AaBb	AaBB	AaBb
FeMale	ab	AaBb	Aabb	AaBb	Aabb
	aB	AaBB	AaBb	AaBB	AaBb
	ab	AaBb	Aabb	AaBb	Aabb

∴ AaBB : AaBb : Aabb : aabb

4 : 8 : 4 : 0

∴ AaBB = 1, AaBb = 2, Aabb = 1, aabb = 0

(343) How many different types of gametes can be formed by F_1 progeny resulting from the following cross ?

AA BB CC X aa bb cc

(A) 3 (B) 8 (C) 27 (D) 64

(344) The cause of cat-cry syndrome is due to : ..... .

(A) loss of a segment of X-chromosome (B) loss of a segment of 5th chromosome

(C) loss of segment of Y-chromosome (D) none of the above

(345) Transition type of gene mutation is caused when, ...

(A) GC is replaced by TA (B) CG is replaced by a GC

(C) AT is replaced by CG (D) AT is replaced by GC

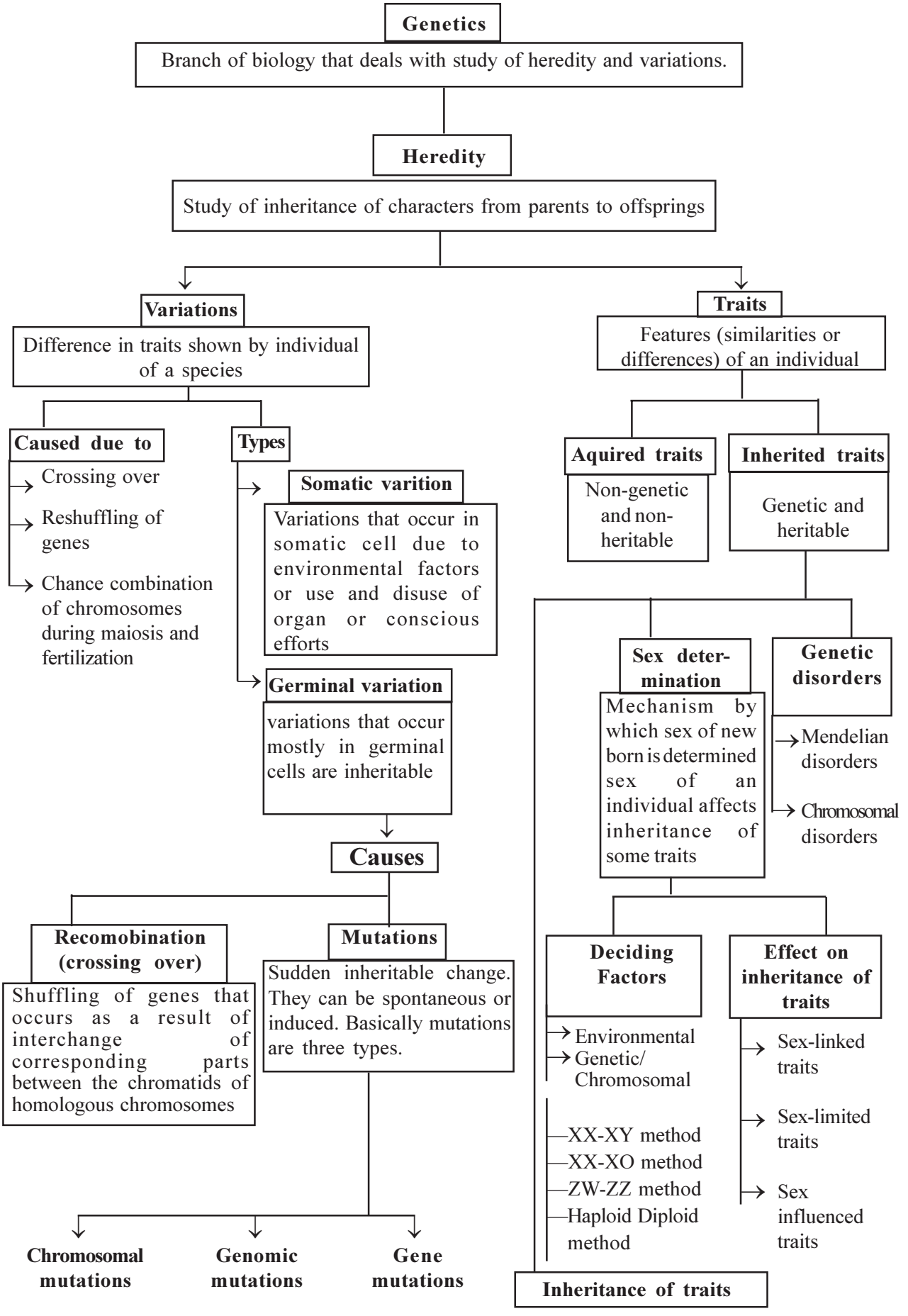
(Hint) :In transition a nitrogen base is replaced by another of its type. i.e. one purine is replaced by another purine (A=G) while one pyrimidine by another pyrimidine (C=T or U)

(346) The term 'mutation' was first utilized by ..... .

(A) Gregor Johann Mendel (B) De Vries

(C) Hardy Weinberg (D) Charles Darwin

Answers : (268-B), (269-B), (270-B), (271-D), (272-C), (273-A), (274-A), (275-C), (276-A), (277-B), (278- A), (279-C), (280-B), (281-B), (282-C), (283-B), (284-C), (285-A), (286-A), (287-D), (288-C), (289-C), (290-A), (291-A), (292-B), (293-D), (294-A), (295-B), (296-B), (297-C), (298-A), (299- C), (300-B), (301-C), (302-C), (303-A), (304-A), (305-B), (306-D), (307-B), (308-C), (309-A), (310-C), (311-B), (312-A), (313-A), (314-B), (315-B), (316-D), (317-A), (318-A), (319-D), (320- D), (321-C), (322-C), (323-A), (324-B), (325-D), (326-A), (327-A), (328-A), (329-C), (330-A), (331-A), (332-C), (333-D), (334-C), (335-C), (336-B), (337-D), (338-D), (339-B), (340-D), (341-C), (342-B), (343-B), (344-B), (345-D), (346-B)


Genetics
Branch of biology that deals with study of heredity and variations.
Heredity
Study of inheritance of characters from parents to offsprings
Variations
Difference in traits shown by individual of a species
Caused due to
Crossing over
Reshuffling of genes
Chance combination of chromosomes during maiosis and fertilization
Types
Somatic varition
Variations that occur in somatic cell due to environmental factors or use and disuse of organ or conscious efforts
Germinal variation
variations that occur mostly in germinal cells are inheritable
Causes
Recomobination (crossing over)
Shuffling of genes that occurs as a result of interchange of corresponding parts between the chromatids of homologous chromosomes
Mutations
Sudden inheritable change. They can be spontaneous or induced. Basically mutations are three types.
Chromosomal mutations
Genomic mutations
Gene mutations
Traits
Features (similarities or differences) of an individual
Aquired traits
Non-genetic and non-heritable
Inherited traits
Genetic and heritable
Sex determination
Mechanism by which sex of new born is determined sex of an individual affects inheritance of some traits
Genetic disorders
Mendelian disorders
Chromosomal disorders
Deciding Factors
Environmental
Genetic/ Chromosomal
XX-XY method
XX-XO method
ZW-ZZ method
Haploid Diploid method
Effect on inheritance of traits
Sex-linked traits
Sex-limited traits
Sex influenced traits
Inheritance of traits

Inheritance of Traits

Pedigree Analysis

Analysis of a family pedigree to find the movement and distribution of certain genetic traits. It helps to find out dominance or recessive etc., nature of certain genes

Mendelism

Factors = genes are the carriers of traits. Their inheritance follow 3 laws.

→ **Low of Dominance**

The allele of a gene shows dominance over other

→ **Law of segregation (or) purity of gametes**

Alleles present in a generation retain their purity (do not mix up) and segregate in next generation

→ **Law of Independent Assortment**

Alleles of genes for two different characters can assort independent of each other

Chromosomal theory of inheritance

Sutton and Boveri (1902) proposed this theory which states that mendelion factors or genes are located at specific loci on chromosomes and it is the chromosome that segregates and assort indepenently during meiosis

Linkage

Phenomenon of certain genes staying together and their enblock inheritance from one generation to another. This affects in heritance of traits and interferes with independent assortment

→ Complete linkage

→ Incomplete linkage

39 Molecular basis of Inheritance

DNA is the Genetic Material :

- Recent developments in molecular biology revealed that the nucleic acids of the chromosomes are responsible for the transmission of heritary character.
- DNA is found to be the genetic material in almost all the living beings except plant viruses where DNA is not found and RNA act as a genetic material.

(1) Which branch revealed that the Nucleic acid of the chromosomes are responsible for the transmission of hereditary character ?

(A) Bio-chemistry (B) Molecular biology (C) Genetics (D) Biology

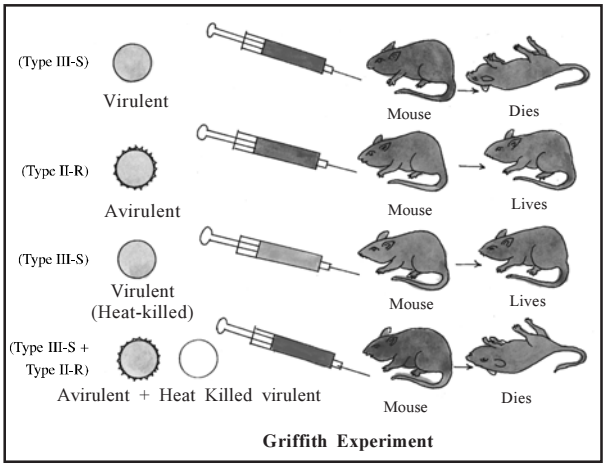
(2) Which one of the following is responsible for the transmission of heriditary character ?

(A) Cytoplasm (B) RNA (C) Chromosomes (D) None of above

Answers : (1-B), (2-C)

Bacterial Transformation or Griffith Effect :

- **Griffith** in 1928 carried out a series of experiments with **Pneumococcus bacteria** namely Virulent (S-III) and avirulent (R-II).
- Virulent strains have smooth polysaccharides capsules and give smooth (S) colonies.
- Avirulent strains have no capsules and give rough colonies (R).
- These two strains also differ in their antigenic properties and virulence for the disease pneumonia.
- Virulence is determined by genetic factors.
- When virulent strain were injected into mice; they killed them by causing pneumonia fever. When mice were inoculated with a virulent bacterium, there was no ill effect when mice were injected with virulent bacteria. After killing the bacteria with heat, no ill effect is produced and the mice survive.
- When mice were injected with a mixture of avirulent and heat-killed virulent bacterium, the mice were found dead due to pneumococcus infection.
- The analysis of dead mice shows that they contain virulent strain responsible for the transformation of avirulent bacteria into virulent smooth bacteria.
- Something from the heat-killed (dead) virulent bacteria transformed the live avirulent bacteria to virulent bacteria.
- As a result the avirulent bacteria are transformed into the virulent bacteria. This phenomenon is known as Griffith effect or bacterial transformation.
- In 1944, **Avery, Mccarty and Macleod** supported the Griffith experiment by molecular explanations.
- They found that the DNA isolated from the heat killed smooth (S) bacterial cells, when added to rough (R) bacterial cells have changed their surface character from rough (R) to smooth (S), and also made them virulent.



Griffith Experiment

- By this experiment it was shown that DNA was the genetic material responsible for inducing the smooth (S) character of the cells and their property of virulence in mice.
- Their experiment proved that bacterial transformation involves transfer of a part of DNA from the dead bacterium to the living bacterium that expresses the character of dead cells and so it is known as recombinant.

Dr. Hargovind Khorana has been successful in synthesizing one gene of yeast containing 77 nucleotides.

- American biochemists in Harward University have synthesized a more complex gene in rabbit which contains 650 nucleotides and controls the synthesis of haemoglobin.

These synthetic activities leave no doubt that genes are molecules.

---

(3) Who among the following did experiment of bacterial transformation ?

(A) Avery, McCarty, Macleod (B) Griffith (C) Hershey, chase (D) Watson, crick

(4) The organism used in Griffith's experiment was ..... .

(A) Spirulina (B) Bacillus (C) Pneumococcus (D) Salmonella

(5) Which two strains used by Griffith on mouse to performing experiment ?

(A) S-III, R-II (B) T-III, S-III (C) R-III, R-II (D) S-III, S-II

(6) Pneumococcus R-II is ..... .

(A) virulent (B) Avirulent (C) contagious (D) none of above

(7) Griffith's experiment famous with ..... .

(A) viral transformation (B) pneumococcus transduction

(C) bacterial transformation (D) bacterial recombination

(8) How many nucleotide having the gene which synthesizing from yeast ?

(A) 77 (B) 650 (C) 87 (D) 230

(9) What is function of a gene which synthesized from rabbit having 650 nucleotides ?

(A) Controls the synthesis of haemoglobin. (B) Controls the height of rat.

(C) Controls the synthesis of red blood cells. (D) It is responsible for growth in mouse.

Answers : (3-B), (4-C), (5-A), (6-B), (7-C), (8-A), (9-A)

Biochemical Evidences for DNA as a genetic material :

(1) The amount of DNA in any given species of cell or organism is remarkably constant and can not be altered by environmental circumstances or by changes in the nutrition etc.

(2) The amount of DNA per cell appears to be in proportion to the complexity of the cell and the amount of genetic information it contains.

- **Erwin Chargaff's rules :** Erwin Chargaff in 1949 proposed Chargaff's rules regarding the composition of bases in DNA as follows:

(i) Total amount of Purine nucleotides always equals the total amount of pyrimidine nucleotides.
i.e. (A) + (G) = (T) + (C)

(ii) The proportion of A is equal to T and also proportion of G is equal to C, but amount of (A) + (T) is not necessarily equal to (G) + (C).
Therefore (A) = (T) : (G) = (C) But, $\frac{A+T}{G+C}$ Varies with the organism.

- **Properties of Genetic Material (DNA) Versus RNA :**

- We have seen that by various experiment it became an established fact that it is DNA that acts as genetic material.
- However, it subsequently became clear that in some viruses; RNA is the genetic material (For example, Tobacco Mosaic Virus).
- A molecule that can act as a genetic material must fulfil the following criteria :

(i) It should be able to generate its replica (Replication).

(ii) It should be chemically and structurally stable (see Griffith's experiment).

(iii) It should be able to express itself in the form of Mendalian characters.

- Both DNA and RNA are able to mutate.
- RNA can directly code for the synthesis of proteins.
  DNA, however is dependent on RNA for synthesis of proteins.
- The above discussion indicates that both RNA and DNA can function as genetic material, but DNA being more stable, is preferred for storage of genetic information.
- For the transmission of genetic information; RNA is better.

---

(10) Chargaff's rules regarding ..... .
(A) Composition of bases in RNA (B) Number of chromosomes
(C) Composition of bases in DNA (D) complexity of genetic material

(11) Who among the following proposed rules regarding the composition of bases in DNA.
(A) Watson Crick (B) Erwin Chargaff (C) Dr. Hargovind Khorana (D) Griffith

(12) Which one is not true ?
(A) A/T = 1 (B) A + T = G + C (C) A + G = T + C (D) G/C = 1

(13) The first rule of Chargaff :
(A) A + G = T + C (B) A + T = G + C (C) A + C = T + C (D) A = G

(14) It varies with the organism.
(A) A + G = T + C (B) A = G (C) A = T and G = C (D) $\frac{A+T}{G+C}$

(15) Who can directly code for the synthesis of proteins ?
(A) DNA (B) RNA (C) Gene (D) Nucleotide

(16) If the total amount of Adenine (A) and guanine (G) is 100 % then the total amount of thymine (T) and cytosine (C) will be ..... .
(A) 50 % (B) 75 % (C) 25 % (D) 100 %

(17) If the total number (T) is 77 than the total number of (C) will be : ..... .

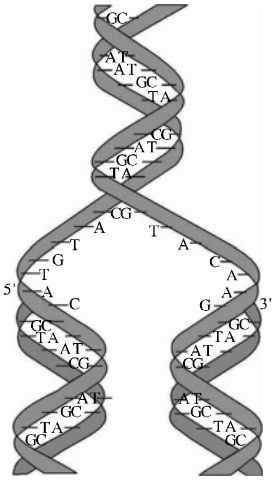
(A) 77 (B) 100 (C) 154 (D) Can't say

Answers : (10-C), (11-B), (12-B), (13-A), (14-D), (15-B), (16-D), (17-D)

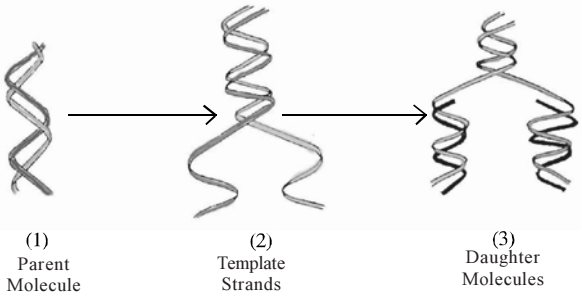
- **Replication of DNA :**

- "The synthesis of new DNA molecule from pre-existing DNA is called replication."
- By replication, each DNA molecule produces exact copies of its own structure.
- Watson and Crick suggested that there is a self duplication of DNA molecule.
- The constituents of DNA viz; nitrogen bases, sugar molecules and phosphates are synthesized within the nucleus, these substances pair upto form the nucleotides and form new DNA molecule during replication.
- They have concluded that the two strands would separate and act as a template for the synthesis of new complementary strands.

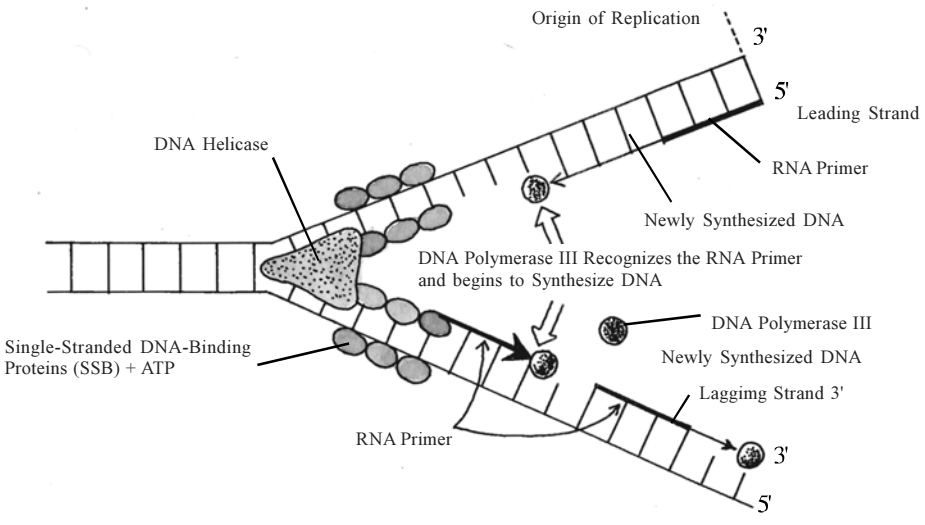
After the completion of replication, each DNA molecule would have one parental and one newly synthesized strand, this method was termed as semi conservative DNA replication.



Watson-Crick Model for Semiconservative DNA Replication



Semiconservative Method of Replication



The process of DNA replication with the help of RNA primer and DNA polymerase III

- **Mechanism of DNA Replication :**

- In the process of DNA replication, the hydrogen bonds between two polynucleotides chains are sequentially broken with the help of proper enzymes.
- The two chains move away from each other. New complementary chains are synthesized along these two separated chains.
- In the end, in the molecules of DNA, one chain is of parental DNA and the other chain is a new one.
- The process begins from a specific site. Then it progresses towards both directions. The enzymes responsible for this are gyrase and helicase.
- The two separated poly nucleotide chains appear like a fork. This is called replication fork. As the process of replication progresses in two directions, it is considered bidirectional.
- As each separated polynucleotide chain provides information for synthesis of the new chain, it is called template chain. The sequence of nucleotides or the template chain determines the sequence of the new chain. The synthesis of new chain is carried out by enzyme. DNA polymerase-III. Synthesis of new chain always begins from 5^1 end and progresses towards 3^1 end.

  First of all RNA – polymerase forms a short RNA chain, complementary to the template DNA as its initial region. This is called primer. Only after the primer is formed DNA, polymerase III become active.

- Based on the direction of the replication, one chain is synthesized continuously in 5^1-3^1 direction. Such a chain is called leading chain.
- The new chain synthesized on the other template is synthesized discontinuously and in the 3^1-5^1 direction. Such a chain is called lagging chain.
- On this template, short pieces of DNA along with RNA- primer are joined. Such fragments are called OKAZAKI fragments.
- At the end of DNA replication the enzyme ligase joins pieces of nucleotides with phospho-diester bonds.
- Thus, two new molecules are formed which are exact replicates of parent-DNA molecule.

---

(18) Choose the correct statement about the direction of DNA strand.

(A) $3^1 \rightarrow 5^1$ (B) $2^1 \rightarrow 5^1$ (C) $5^1 \rightarrow 3^1$ (D) $5^1 \rightarrow 5^1$ and $3^1 \rightarrow 3^1$

(19) It stimulate DNA replication.

(A) DNA polymarese (B) DNA Ligase (C) Transcriptase (D) A and B both

(20) The direction and type of DNA replication is ..... .

(A) Bidirectional and conservative (B) Unidirectional and conservative

(C) Bidirectional and semi-conservative (D) Unidirectional and Semi-conservative

(21) Choose correct option for formation of okazaki ..... .

(A) Replication (B) Transcription (C) Translation (D) Transduction

(22) When DNA replication will start ..... .

(A) While leading strand produce okazaki fragments.

(B) While hydrogen bonds between two polynucleotides chains are sequentially broken with the help of proper enzymes.

(C) While phospho di ester bonds between sequentially two nucleotides are broken.

(D) While bonds between pentose sugar and Nitrogen base are broken.

((23) Enzyme that breaks the bond between DNA helix is known as ..... .

(A) Helicase (B) Gyrase (C) Ligase (D) DNA polymerase

(24) Enzyme that does not allow unwind DNA helix is known as ..... .

(A) Helicase (B) Gyrase (C) RNA polymerase (D) DNA polymerase

(25) What is primer ?

(A) long strand of DNA (B) short strand of DNA (C) long chain of RNA (D) short RNA chain

(26) Ligase joins of nucleotides with ..... .

(A) glycosidic bond (B) ester bond (C) peptide bond (D) phospho-di-ester bond

Answers : (18-C), (19-D), (20-C), (21-A), (22-B), (23-A), (24-B), (25-D), (26-D)

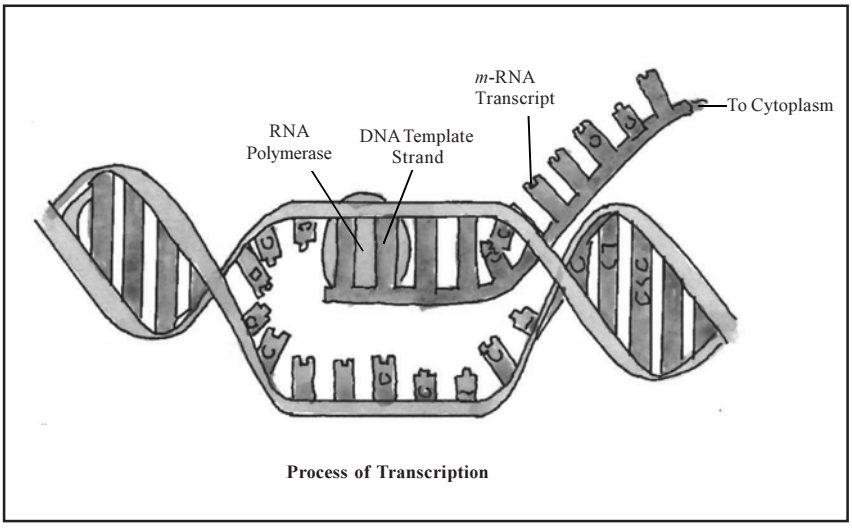
Transcription :

- " The process where DNA gives its information to m-RNA is called transcription."
- Which involves the formation of an RNA strand on the DNA template.

Transcription occurs in three steps.

- **First step :**
- The enzyme transcription becomes attached at the initial region of the DNA (gene) which is to be transcribed.
- As a result the two polynucleotide chains in that region move away from one another.
- **Second step :**
- One of these two separated chains acts as a template chain. This chain provides information for synthesis of m-RNA.
- Based on the sequence of nucleotides on the template chain, enzyme transcriptase arranges complementary RNA – nucleotide. Units one after the other and phospho-di-ester bonds are formed between them. Thus m-RNA chain is synthesized.
- **Third step / Last phase :**
- The third and Last phase are of completion. A specific protein attaches with m-RNA and separates it from template - DNA and transcriptase enzyme.

Sometimes m-RNA gets separated by itself. Then the m-RNA is translocated to cytoplasm where it associates with ribosomes.



Process of Transcription

(27) Which enzyme responsible for transcription ?

(A) DNA polymerase (B) RNA polymerase (C) Transcriptase (D) All of above

(28) Which RNA synthesized by transcription?

(A) m-RNA (B) r-RNA (C) t-RNA (D) All of above

(29) Enzyme which constrict phospho-di-ester bond in the process of transcription is ..... .

(A) DNA polymerase – III (B) DNA Ligase (C) DNA helicase (D) None of above

(30) At the end of transcription m-RNA is translocated to ..... .

(A) Nucleus (B) Cytoplasm (C) Mitochondria (D) One cell to another cell

(31) m-RNA is translocated to cytoplasm where it associates with …

(A) Chloroplast (B) Mitochondria (C) Ribosomes (D) Golgibody

Answers : (27-C), (28-A), (29-D), (30-B), (31-C)

Genetic code :

- Each gene is represented in a small sector of a DNA molecule and may have its unique sequence of nucleotides in that length of DNA.
- The information present in the DNA is transcribed and carried by the m-DNA to the cytoplasm for protein synthesis. This information is designed as genetic codes.
- Thus genetic code is defined as the sequence of nitrogen bases (nucleotides) in m-RNA molecule which contains the information for the synthesis of protein molecules.
- The main problem of genetic code was to determine the exact number of nucleotides in a codon which codes for one amino acid.

  Since there are only four nitrogenous bases in m-RNA for 20 amino acids, combination of only one or two nitrogenous bases can not provide sufficient codons of 20 amino acids.

- **Singlet code :**

Singlet code

A
C
G
U

Doublet code

	A	C	G	U
A	AA	AC	AG	AU
C	CA	CC	CG	CU
G	GA	GC	GG	GU
U	UA	UC	UG	UU

- A singlet code consisting of only one nucleotide provides just four codons A,C,G and U. These are insufficient to code 20 amino acids.
- **Doublet code :**
- Combination of two nitrogenous bases (doublet code) provides 4×4 = 16 codons which are still insufficient for 20 amino acids.
- **Triplet code :**
- Gamov (1954) pointed out the possibility of three letter code, i.e. each codon consists of three nitrogen bases.

- Note: Nirenberg and Matthaei with Khorana (1961) experimentally proved that a single amino acid is determined by a sequence of three nitrogen bases. Khorana has got nobel prize on genetic code.

  This will give $4 \times 4 \times 4 = 64$ codon words of codons, which are more than enough to code twenty amino acids.

Main properties of genetic code are :

- Genetic code is universal type of code which determines the same amino acid in all organisms including virus, bacteria, plants and animals.
- Genetic code is specific. Any one codon specifies the position of one kind of amino acid only.
- The codon is triplet. 61 codons code for amino acid and 3 codon do not code any amino acid. A single amino acid may be specified by many codons: such codons are called degenerate codons .
- The synthesis of a poly peptide chain is initiated by a codon called initiation codon.
- **UAA, UGA and UAG** codons are termination codons. They indicate termination at synthesis of protein. These codons do not code for any amino acid, hence these are called non-sense codons.
- AUG has dual functions. It is the codons for methionine and it also acts as initiator codon.
- The sequence of genetic codes and sequence of amino acids in protein molecule shows linear parallelism.

Triplet code

First Letter	U	C	A	G	Third Letter
U	UUU, UUC } Phe UUA, UUG } Leu	UCU, UCC, UCA, UCG } Ser	UAU, UAC } Tyr UAA Non sense (STOP) UAG Non sense (STOP)	UGU, UGC } Cys UGA Non sense (STOP) UGG Try	U C A G
C	CUU, CUC, CUA, CUG } Leu	CCU, CCC, CCA, CCG } Pro	CAU, CAC } His CAA, CAG } Gln	CGU, CGC, CGA, CGG } Arg	U C A G
A	AUU, AUC, AUA } Ileu AUG Met (START)	ACU, ACC, ACA, ACG } Thr	AAU, AAC } Asn AAA, AAG } Lys	AGU, AGC } Ser AGA, AGG } Arg	U C A G
G	GUU, GUC, GUA, GUG } Val	GCU, GCC, GCA, GCG } Ala	GAU, GAC } Asp GAA, GAG } Glu	GGU, GGC, GGA, GGG } Gly	U C A G

Triplet codons of m-RNA for Amino acids Represented in Tabular form

(32) The secret of the blue print of genetic information lies in the arrangement of ..... .

(A) definite linear sequence of nitrogen bases of a RNA

(B) definite linear sequence of nitrogen bases of a DNA

(C) on the quality of nitrogen bases

(D) number of purine and pyrimidine bases

(33) What is genetic code ?

(A) The information present in the m –RNA is transcribed and carried by the DNA to the cytoplasm for protein synthesis.

(B) The sequence of nitrogen base in t-RNA for synthesis of protein.

(C) The sequence of nitrogen base in r-RNA molecule which contains information for Ribosomes.

(D) The sequence of nitrogen bases in m-RNA molecule which contains the information for the synthesis of protein molecule.

(34) Whose experiments cracked the DNA and discovered unequivocally that a genetic code is a "triplet ?"

(A) Griffith (B) Watson, crick (C)Nirenberg, Matthaei and Khorana (D) Tschermak

(35) Any one codon specifies the position of ond kind of amino-acid only is ..... .

(A) specific (B) universal (C) degenerate (D) non- sense

(36) In triplet codons how many codons specify for amino acids ?

(A) 4 (B) 61 (C) 64 (D) 3

(37) How many non-sense codon in triplet codon ?

(A) 20 (B) 61 (C) 3 (D) 1

(38) What is degenerate codons ?

(A) One code which determine the same amino acid in all organisms.

(B) Any one codon specifies the position of one kind of amino acid only.

(C) A single amino acid may be specified by many codons.

(D) Codons do not code for any amino-acid.

(39) Which of the following in codon code for methionine ?

(A) AUG (B) UAG (C) CCC (D) AGU

(40) Which codons causing chain termination of protein synthesis ?

(A) AUG, UAG, UGA (B) UAA, UGA, UAG (C) UAA, GAU, GUC (D) AUG, UAA, UCA

(41) It is a site for termination codon.

(A) At 3^1 Initial region of m-RNA (B) At 5^1 Initial region of m-RNA

(C) At 3^1 end of m-RNA (D) At 5^1 end of m-RNA

(42) Which one of the following codons code for serine ?

(A) UCU, UCC, UCA, UCG (B) UUU, UUC, UUA, UUG

(C) CUU, CUC, CUA, CUG (D) CCU, CCC, CCA, CCG

(43) Which one of the following codons code for lysine ?

(A) AAA, AAG (B) AGU, AGC (C) GGU, GGC (D) UUA, UUG

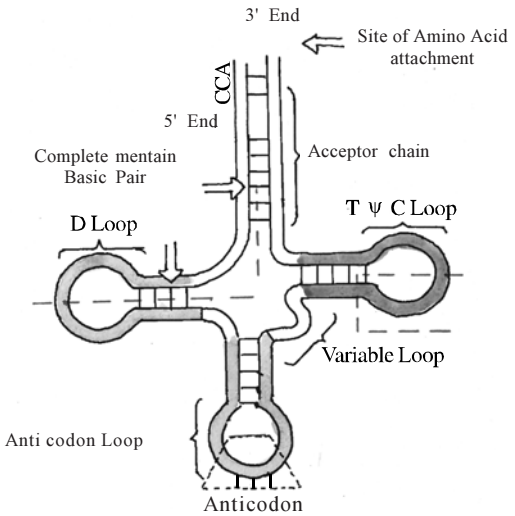
(44) How many codons are code for serine amino acid ?

(A) 4 (B) 2 (C) 8 (D) 6

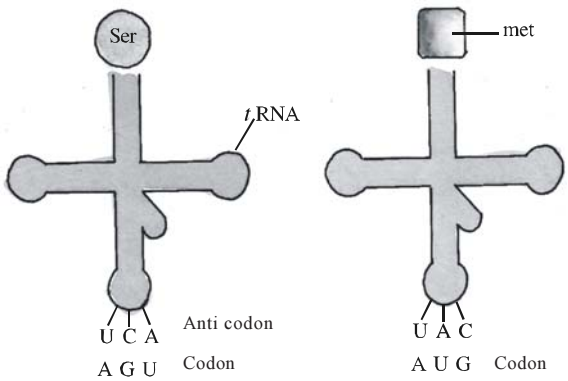
Answers : (32-B), (33-D) (34-C), (35-A), (36-B), (37-C), (38-C), (39-A), (40-B), (41-C), (42 A), (43-A), (44-D)

Transfer – RNA (t-RNA) :

- Amino acid molecules of various kinds located in the cytoplasm are transported by t-RNA
- Each t-RNA specific for the each amino acid molecule.
- There are 61 kinds of t-RNA present in cytoplasm for transportation of 20 amino acids.
- t-RNA molecules are smallest, containing 75 to 95 nucleotides.
- t-RNA has an anticodon loop that has bases complementary to the code, and it also has an amino acid acceptor by which it binds to amino acid.



Clover leaf model of t-RNA structure



Structure of t-RNA

(45) The Function of t-RNA is

(A) It provides information for synthesis of protein.

(B) Amino acid molecules of various kinds located in cytoplasm are transported.

(C) Amino acid transported from cytoplasm to nucleus.

(D) It act as template strand in the process of m-RNA synthesis

(46) How many effective nucleotides present in t-RNA ?

(A) 25 to 30 (B) 80 to 90 (C) 75 to 95 (D) More than 95

(47) How many types of t-RNA present in cytoplasm ?

(A) 61 (B) 20 (C) 64 (D) 16

(48) A specific nucleotide sequence which attached to 3^1 end of t-RNA is ..... .

(A) AAC (B) CCA (C) AUG (D) UUA

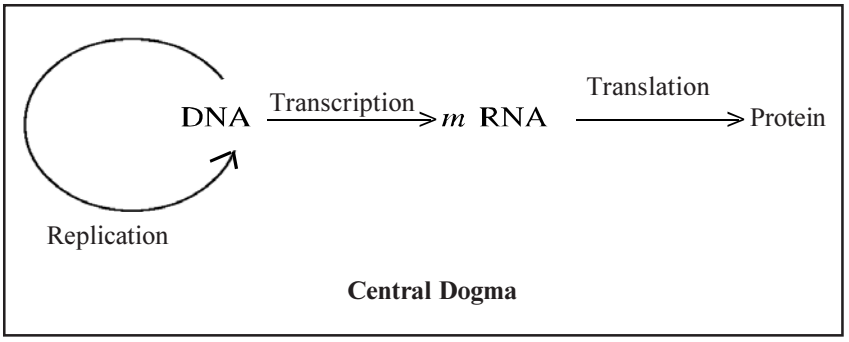
(49) The site for amino acid attachment on t-RNA is ..... .

(A) Anticodon region (B) on TQC 100P (C) On D LOOP (D) At 3^1 - OH end

Answers : (45-B), (46-C), (47-A), (48-B), (49-D)

* **Central – Dogma :**

- With the help of m-RNA and RNA protein synthesis takes place. This unidirectional flow of information was first described by F.H.C crick in 1958 as the central dogma of molecular biology.
- The expression of the genetic material (DNA) which occurs generally through the production of protein synthesis involves two consecutive steps: (1) Transcription and (2) Translation.
- It states that the genetic information flows from DNA → m-RNA → protein replication.



Central Dogma

-- In some viruses the flow of information is in reverse direction.

- It was brought to light by H.M Temin and D. Baltimore. According to them, many tumour viruses contain RNA as a genetic material and replicate by first synthesizing a complementary DNA. This process is called reverse transcription.
- It is done with the help of RNA - dependant DNA polymerase called reverse transcriptase. It is observed in HIV.

---

(50) The unidirectional flow of genetic information from DNA - m-RNA - protein replication is known as ..... .

(A) Transcription (B) Replication (C) Central dogma (D) Translation

(51) Reverse transcription means .....

(A) Formation of m- RNA from DNA (B) Formation of DNA from RNA

(C) Formation of DNA from DNA (D) Formation of t-RNA from m-RNA

(52) Which enzyme responsible for reverse transcription ?

(A) Reverse polymerase (B) Reverse transcriptase (C) Reverse ligase (D) Reverse polymerase – III

(53) The expression of the genetic material which occurs generally through ..... .

(A) production of carbohydrates (B) production of protein

(C) accumulation of genetic material (D) production of lipids

(54) The phase of gene expression is ..... .

(A) Transcription (B) Translation (C) Transduction (D) A and B both

(55) Reverse transcription observed in ..... .

(A) TMV (B) Tumor viruses (C) HIV (D) All of above

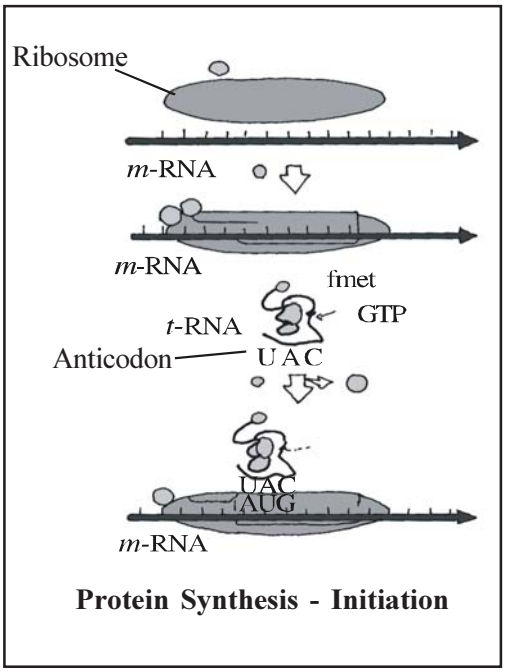
Answers : (50-C), (51-B), (52-B), (53-B), (54-D), (55-D)

Translation / Biosynthesis of Proteins :

- Translation is process in which the genetic message carried by messanger RNA from the DNA is converted into the form of a polypeptide chain having a specific sequence of amino acids.
- Translation occurs in three steps : Initiation, elongation and termination.
- Three other key components which take part in these three steps. They are (i) ribosomes (ii) t-RNA and (iii) amino acids.

(i) Initiation :

- Synthesis is always initiated with amino acid metheionine.
- Amino acid molecules of various kinds located in the cytoplasm are transported by t-RNA molecules.
- A t-RNA synthesis enzyme, specific for the amino – acid molecule is essential in this.
- The amino acyl t-RNA with its specific amino acid now moves towards m-RNA.
- In eukaryotic cell methionine is arranged on m-RNA at its proper location. To decide this, a sequence of three nucleotides occurs on the t-RNA which is complementary to the genetic codon on m-RNA molecule. This unit of three nucleotides on t-RNA is called anticodon.
- The genetic code for methionine on m-RNA is AUG. Hence, anticodon on t-RNA is UAC.

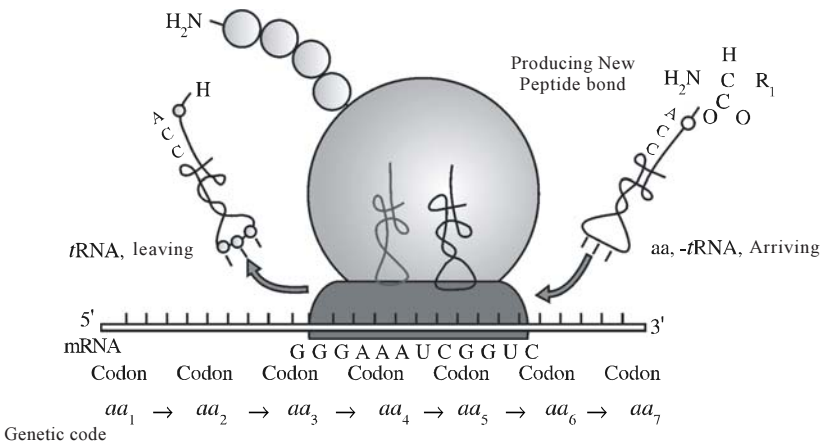


Protein Synthesis - Initiation

- Formyl –methionine acyl t-RNA with its amino acid gets attached here. Thus, synthesis of protein is initiated.

(ii) Elongation :

- Now, other t-RNA molecules carrying their specific amino acid molecules and possessing anti codon specific to the genetic codes on m-RNA become sequentially arranged.



Protein Synthesis Elongation

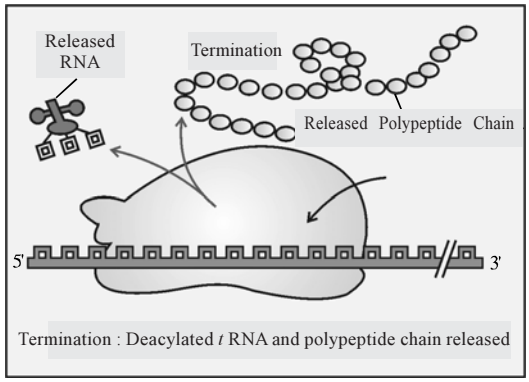
- Proteins known as elongation factors and GTP as energy source are helpful in this.

  Methionine at the first position is joined to the second position amino acid by a peptide bond.
- Now the t-RNA carrying formyl – methionine is released. Thus a dipeptide is formed.
- Now, ribosome moves along m-RNA in 3^1 direction by distance of one codon. As a result, t-RNA which is responsible for methionine transfer is released from the surface of the ribosome.
- The t-RNA caring dipeptide now occupies the vacant region.
- Thus one by one, t-RNA molecules carrying their amino acid get arranged, and the peptide chain elongates. This phase is elongation phase.

(iii) Termination :

- The genetic code located at 3^1 end of m-RNA does not indicate the position or any amino acid. Such codon is called nonsense codon or termination codon.
- The function of such codons is to release the synthesized polypeptide chain from ribosomes.
- From the first amino acid of the released polypeptide chain, formylmethionine, the formyl group is removed.

  Sometimes, even methionine is removed. Then, the polypeptide gets ready to perform its function.



Protein Synthesis Termination

(56) Which RNA contains information for synthesis of protein ?

(A) m-RNA (B) r-RNA (C) t-RNA (D) All of above

(57) A unit of three successive nucleotides indicates the position of particular ..... in the constitution of protein.

(A) Monomer of carbohydrates (B) Fatty acids

(C) Nucleotides (D) Amino acids

(58) The Sequence of nitrogen bases in a particular region of DNA molecule was found to be CAG, CCC, GAT. What would be the sequence of nitrogen bases in the m-RNA ?

(A) CAG, CCC, CAU (B) GUC, GGG, CUA (C) GUC, CCC, GAT (D) GAC, TAG, CTA

(59) Main components for translation process are ..... .

(A) ribosome, t-RNA, amino acids (B) ribosome, m-RNA, DNA

(C) DNA, m-RNA, t-RNA (D) ribosome, cytoplasm, t-RNA, r-RNA

(60) Which one of the following amino acid is always initiate the synthesis of translation ?

(A) Serine (B) Valine (C) Methionine (D) Tryptophan

(61) Which enzyme essential for transportation of amino acids ?

(A) m-RNA synthetase (B) DNA polymerase - III (C) Transcriptase (D)t-RNA synthetase

(62) The code of nitrogen bases in a particular region of m-RNA is AUG, then the anticodon will be on t-RNA is ..... .

(A) AUC (B) TAC (C) UAC (D) TAG

(63) Out of the following which having region of anticodon ?

(A) m-RNA (B) DNA (C) r-RNA (D) t-RNA

(64) Which source are helpful in process of elongation ?

(A) GTP as energy source, t-RNA synthetase. (B) GTP as energy source, elongation factors.

(C) ATP as energy source, elongation factors. (D) ATP as energy source, t-RNA synthetase.

(65) The ribosome moves along m-RNA in 3^1 direction by distance of ..... .

(A) Two codons (B) Three codons (C) Four codons (D) One codon

(66) The position of non- sense codon is ..... .

(A) On 3^1 end of t-RNA (B) On 3^1 end of m-RNA

(C) On 5^1 end of t-RNA (D) On 5^1 end of m-RNA

(67) What is the function of non-sence codon ?

(A) It translocate m-RNA to cytoplasm from DNA.

(B) Release the synthesized polypeptide chain from ribosomes.

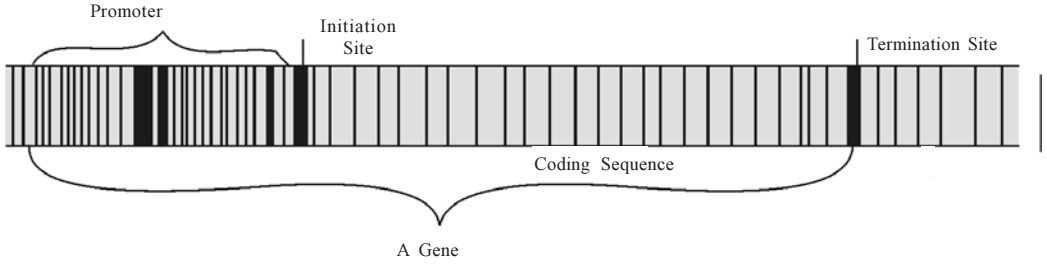
(C) It carrying specific amino acid molecule on m-RNA .

(D) It release polypeptide chain from methionine.

Answers : (56-A), (57-D), (58-B), (59-A), (60-C), (61-D), (62-C), (63-D), (64-B), (65-D), (66-B), (67-B)

*** Gene :**

- It is actually a "Factor" as per Mendelian theory. It is a segment of DNA present in the chromosome.
- Thus gene is the hereditary unit. It controls characters. It transmits characters from parents to offspring.
- The gene consists of four main regions namely promoter, initiation site, coding sequence and termination site.
- **Promoter :** The promoter is at one end of the gene. The RNA polymerase is attached to this region during RNA synthesis.



DNA

- **Initiation site :** Initiation site is located next to promoter. The m-RNA synthesis begins from this site.
- **Coding sequence :** The coding sequence is the middle segment of the gene. All these nucleotides are copied into the m-RNA.
- **Termination site :** The termination is at the other end of the gene, where m-RNA synthesis is stopped.

(68) It controls characters ..... .

A) gene (B) chromosomes (C) RNA (D) DNA

(69) How many regions consisted by gene ?

(A) 1 (B) 4 (C) 3 (D) 5

(70) Which enzyme attached to the region of promoter ?

(A) DNA polymerase (B) DNA Ligase (C) RNA polymerase (D) t-RNA synthetase

(71) The m-RNA synthesis begins from ..... .

(A) promoter (B) termination site (C) initiation site (D) coding sequence

(72) Name the region where all nucleotides are copied into the m-RNA.

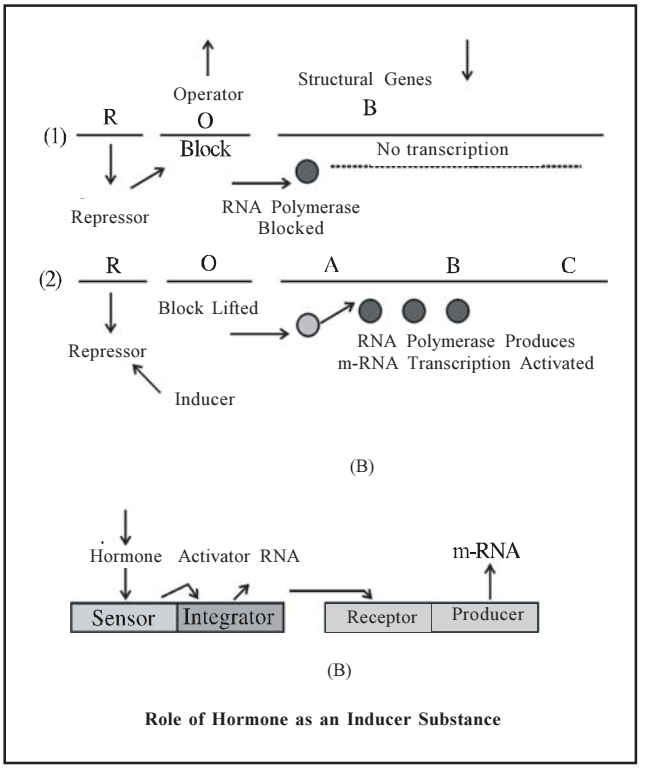
(A) Coding sequence (B) Promoter site (C) Operator site (D) Termination site

Answers : (68-A), (69-B), (70-C), (71-C), (72-A)

Regulation of Gene Expression :

- Different genes in an organism are meant for the synthesis of different proteins. All these proteins are not needed at the same time.
- The mechanism involved in this process is difficult to work out particularly in higher organisms. While in bacteria (prokaryotic cells) detailed information has been obtained regarding regulation of gene expression. Mechanisms in prokaryotes and eukaryotes differ due to their different levels of complexity of organisms.
- French scientists Jacob and Monad discovered a mechanism in E-coli bacteria and won the Noble Prize for the same.
- They found that for an expression of a gene there may be a few other genes are involved.
- The gene whose product is required may be expressed; such a gene is called a structural gene.
- The repressor gene can produce a product which will block operator gene so that the RNA polymerase will not form m-RNA. That means it prevents transcription of one or many structural genes.
- If an inducer which could be a substrate; precursor or a hormone when provided will bind the repressor proteins and thus open up operator gene.
- Now RNA polymerase gets bound to promotersite and starts transcription process to produce m-RNA.
- **Jacob** and **Monad** proposed that metabolic path ways are regulated as a unit.
- They studied the change in the nature of enzymes of E-coli; when provided with different kinds of sugars.
- They found that when sugar lactose is added to the cultures of E.coli, it induces three enzymes necessary to break down the lactose into galactose. These newly synthesized enzymes are:

(i) β -galactosidase **(ii)** permease and **(iii)** transacetylase.



Role of Hormone as an Inducer Substance

- According to them, the synthesis of these three enzymes are controlled by a long DNA segment known as operon which is divisible into an operator site (o) and three structural genes; z, y and a.
- The action of structural genes is regulated by operator site with the help of a repressor protein produced by the action of gene 'i' known as the regulator gene.
- The genes are expressed or not expressed depends on whether the operator switch is on or off.
- When the switch is on, the three genes are transcribed by RNA polymerase into a single stretch of m-RNA covering all the three genes.
- The switching on or off of the operator switch is achieved with the help of a protein known as repressor.
- When this protein ties together to the operator (O) and blocks, it the switch is turned off and the three genes (z, y and a) are not expressed.
- Thus an operon consists of :

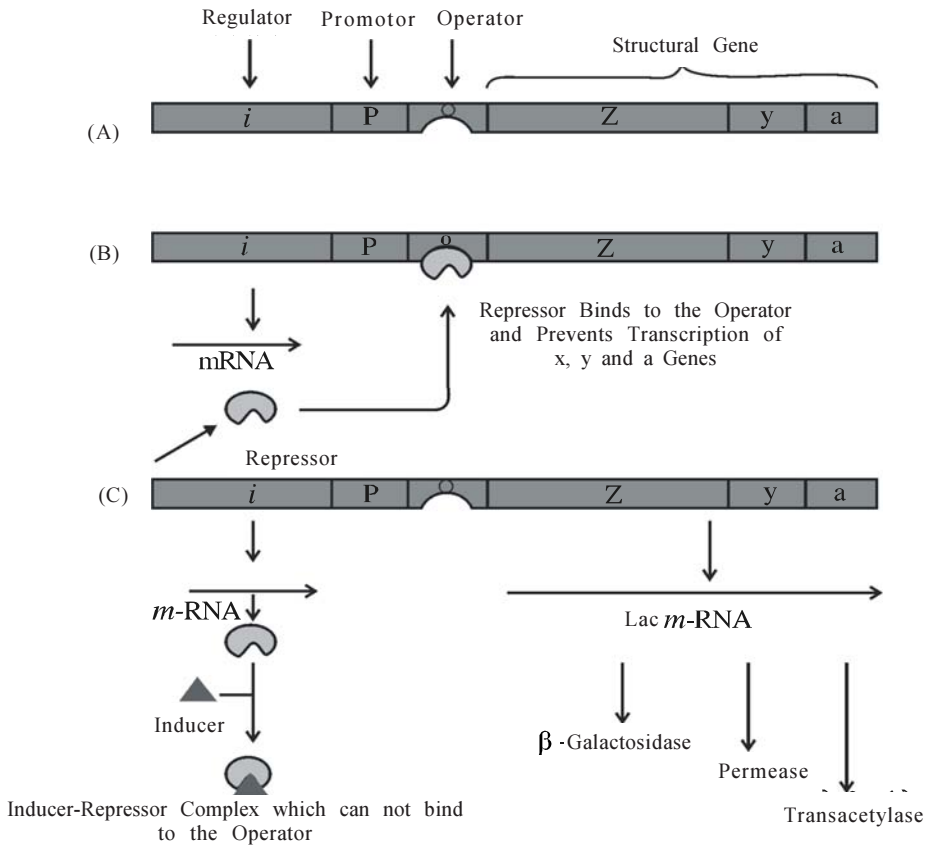
  Operon = Regulator gene + promoter gene + Operator gene + Structural gene
- The genes composing operon are :

(1) **Structural gene** : They are segment of DNA which carry codes for the synthesis of proteins.

(2) **Control genes** : These control the activity of structural genes either by induction or suppression.

These genes are as follow :

(i) **Regulator gene** : The regulator gene produces some specific enzyme which RNA polymerase binds. It initially acts as a repressor substance.

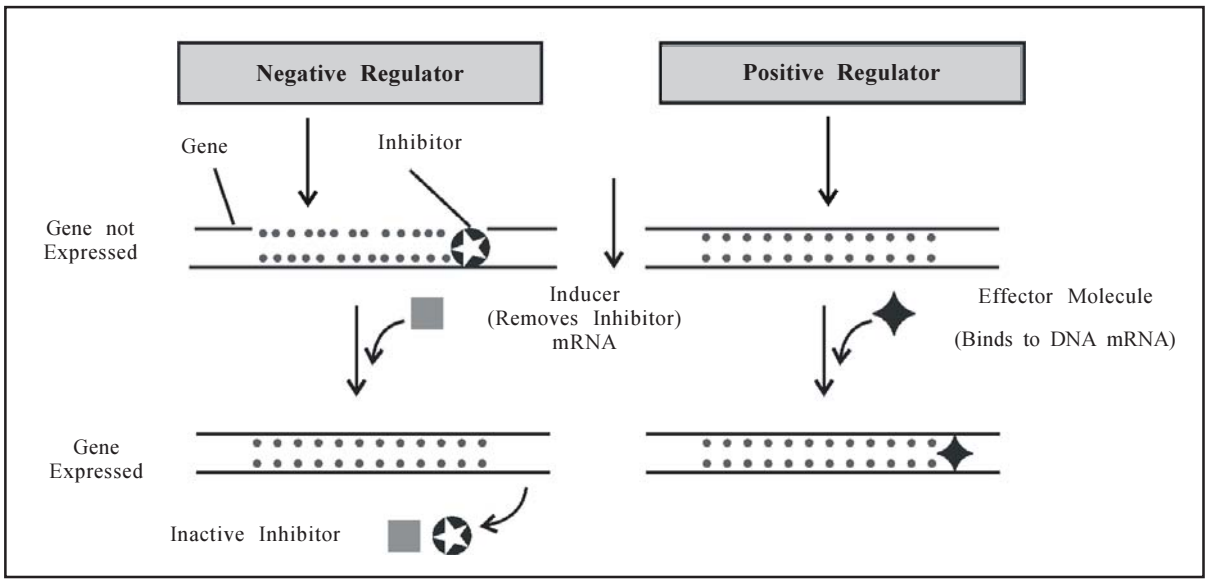


(A) Lac operon structure (B) Repressed state of Lac operon (C) Active state of Lac operon

(ii) **Promoter gene** : The promoter gene (p) is the DNA segment at which the transcription of the structure genes takes place. It controls the rate of m-RNA synthesis.

(iii) **Operator gene** : The operator gene (O) is the segment of DNA which exercises a control over transcription.

Negative and Positive Control :



- The regulator gene exercises a negative control on the functioning of lac operon, because the enzyme produced by the regulator gene switches off the activity of operon gene. It is not allowed to function the protein is not required.
- Certain substances exercise a positive control. These substances when added to the medium initiate the regulator gene to produce the stimulant which stimulates the production of enzyme by the operator gene.

(73) Who discovered a lac-operon ?

(A) Jacob and Monad (B) Watson and Crick (C) Gamov and Khorana (D) Avery and Mccarty

(74) Name the gene which can produce a product which will block operator.

(A) Promoter gene (B) Structural gene (C) Repressor gene (D) Control gene

(75) The Repressor protein is a product of ..... .

(A) promoter gene (B) repressor gene (C) operator gene (D) structural gene

(76) The sugar which used by Jacob and Monad for explanation of regulation of gene expression is …

(A) glucose (B) lactose (C) surcose (D) A and B both

(77) Which Bacteria has been used regarding regulation of gene expression ?

(A) Streptococcus (B) Bacillus (C) E-coli (D) Salmonella

(78) Which enzyme necessary to break down the lactose into galactose ?

(A) β-glycosidase (B) Permease (C) Transacetylase (D) All of above

(79) The segment of DNA which exercises a control of an enzyme which responsible for digestion of lactose is …

(A) Operator gene (B) Promoter gene (C) Structural gene (D) Regulator gene

(80) It is responsible for switching on the lac-operon in bacteria is ..... .

(A) lactose (B) numbers of bacteria (C) structural gene (D) RNA polymerase

(81) Operon Contains :

(A) Structural gene + promoter gene + operator gene

(B) Regulator gene + operator gene

(C) Regulator gene + promoter gene + operator gene

(D) Regulator gene + promoter gene + operator gene + structural gene

(82) It controls the rate of m-RNA synthesis.

(A) Operator gene (B) Regulator gene (C) Promoter gene (D) Structural gene

(83) It exercises a control over transcription.

(A) Operator gene (B) Promoter gene (C) Termination gene (D) Regulator gene

(84) In lac – operon if mutation occurs in the y gene of the 'Structural gene' then ..... .

(A) Permease will not be synthesized. (B) β - Galactosidase will not be synthesized.

(C) Transacetylase will not be synthesized (D) Lactose digestion will be rapid.

Answers : (73-A), (74-C), (75-B), (76-D), (77-C), (78-D), (79-A), (80-A), (81-D), (82-C), (83-A), (84-A)

Human Genome Project (HGP) :

- The Human Genome project (HGP) is an international scientific research project.
- The primary goal of the project is of determining the sequence chemical base pairs which make up DNA, and identifying and mapping the approximately 20,000 – 25,000 genes of the human genome.
- In 1990, the project was initiated by the joint efforts of the of U.S. Department of energy and the National Institute of Health. A working draft of the genome was announced in 2000. In February 2001, the analysis of the working draft was published.

* **Following set of goals was there :**

- To prepare a "genetic map" of all genes in human beings in full detail.
- The information of nucleotide sequence of each gene and its mode of expression should be determined and stored.
- To store information in data bases and improve the project tools for data analysis and transfer related technologies to the private sector.
- To understand social, ethical and legal complications related in the project and find out proper solution for them.
- In April 2003 the HGP sequencing was completed.
- The human genome contains 3 billion chemical nucleotide bases (A, C, T and G). The average gene consists of 3000 bases.
- The total number of genes is estimated at around 35,000 – lower than previous estimated of 80,000 to 1,40,000.
- The deciphering of nucleotide sequence and its store is referred to as "data base". It is a molecular data base.
- While the objective of HGP is to understand the genetic makeup of the species, the project has also focused on several other non human organisms such as E-coli, the fruit fly and the laboratory mouse.
- Scientists have found the following information :

Organism	Genomesize	Estimated genes
Human (Homo sapians sapians)	>3 billion	30,000
Laboratory mouse (M. musculus)	2.6 billion	30,000
Fruit fly (D. melanogaster)	137 million	13,000
Bacteria (E. coli)	4.6 million	3200
HIV	9700	9

Future Challenges of HGP :

- Gene number, exact locations and functions
- Gene regulation - DNA sequence organization - Coordination of gene expression.
- To understand disease susceptibility prediction based on gene sequence variation.

Application of HGP :

(1) **Molecular Medicine :** It improves diagnosis of disease and is also used in gene therapy. Improve understanding of several diseases like Alzheimers, Parkinsons diseases etc.

(2) **Microbial Genomics :** It rapidly detects and treats pathogens in clinical practice; develops new energy sources and monitors environment to detect pollutants.

(3) **Risk Assessment :** It is used to evaluate the health risks faced by individuals who may be exposed to radiation and to cancer – causing chemicals and toxins.

(4) **DNA Identification (Forensics) :** It is used to identify potential suspects whose DNA may match the evidence left at crime scenes, to establish paternity and other family relationships.

(5) **Agriculture, livestock breeding :** It is used to develop biopesticides in corporate edible vaccines incorporated into food products.

(6) **ELSI :** The project's goal included not only identifying all the genes in the human genome, but also addressing the ethical, legal and social issues (ELSI).

Thus HGP will provide a healthy and disease free life.

Salient features of Human Genome : The human genome contains 3 billion nucleotide bases.

- The average gene consists of **3000** bases.
- The total number of genes is estimated at **30,000.** Almost all (**99.9 %**) nucleotide bases are **exactly the same** in all people.
- The functions are unknown for over **50 %** of the discovered genes.
- Less than **2 %** of the genome codes for proteins.
- Repeated sequences make up a very large portion of the human genome.
- **Chromosome 1** has most genes (**2968**) and **Y** has the fewest (**231**).
- It is expected that we will soon have Snapshots of more than 1200 genes that responsible for common cardiovascular aliments, endocrine diseases like **diabetes**, neurological disorders like **Alzheimer's** diseases**, cancer** etc.

- The human genome sequencing not only holds promise for a healthier living, It also holds the prospects of a vast database of knowledge about designer drugs, genetically modified diets and finally our genetic identity.

(85) It is not for HGP goal.
(A) To prepare a genetic map.
(B) The deciphering of nucleotide sequence and store is referred to data base.
(C) To understand ELSI.
(D) To understand genetic mutation

(86) When HGP sequencing was completed ?
(A) April - 2001 (B) July - 2003 (C) April - 2000 (D) April - 2003

(87) Human Genome size is ..... .
(A) >3 billion (B) >5 billion (C) > 10 billion (D) >1 billion

(88) The deciphering of nucleotide sequence and its store is referred to as ..... .
(A) genome (B) data size (C) data base (D) genetic map

(89) Genome size of M. musculus D. melanogaster and E-coli are as ..... .
(A) 2.6 billion, 137 million, 4.36 million (B) 3 billion, 2.6 million, 137 million
(C) 4.6 billion, 137 million, 4.6 million (D) 9700 billion, 137 million, 2.6 million

(90) How many genes present in HIV ?
(A) 4000 (B) 9 (C) 1300 (D) 30

(91) What is a use of Microbial genomics?
(A) Improve understanding of diseases.
(B) Develop new energy sources.
(C) To evaluate the health risks faced by individuals who may be exposed to radiation.
(D) To establish paternity and other family relationships.

(92) It improves diagnosis of disease and is also used in gene therapy.
(A) Forensics (B) Microbial genomics (C) Molecular medicine (D) Risk assessment

(93) It is a function of Forensics
(A) To establish paternity
(B) To establish family relationships
(C) To identify potential suspects whose DNA may match the evidence left at crime scenes
(D) All of above

(94) How many genome are codes for proteins ?
(A) 10% (B) 2% (C) 99% (D) 50%

(95) In Human most genes present on chromosome number is ..... .
(A) first chromosome (B) fifth chromosome (C) X chromosome (D) Y chromosome

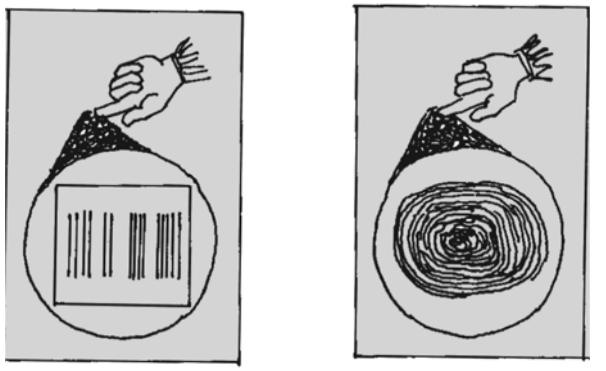
(96) In Human fewest genes present on chromosome is ..... .
(A) X (B) Y (C) First (D) Both X and Y

(97) How many gene present on Y chromosome ?
(A) 231 (B) 2968 (C) 3000 (D) 900

Answers : (85-D), (86-D), (87-A), (88-C), (89-A), (90-B), (91-B), (92-C), (93-D), (94-B), (95-A), (96-B), (97- A)

DNA fingerprinting:

- Like the fingerprinting that came into use by detectives and police labs during the 1930s, due to its specification, each person has a unique DNA fingerprint.
- Unlike a conventional fingerprint that occurs only on the fingertips, a DNA fingerprint is the same for every cell, tissue and organ of a person.
- DNA fingerprint can not be altered by any known treatment but fingerprint can be changed by surgery.
- Consequently, DNA fingerprint is rapidly becoming the primary method for identifying and distinguishing individual human beings.
- An additional benefit of DNA fingerprint technology is the diagnosis of inherited disorders in adults, children and unborn babies.
- Even bloodstained clothing from **Abraham Lincoln** has been analyzed for evidence of a genetic disorder called **Marfan's syndrome**.



DNA Fingerprint Conventional Fingerprint

Two types of Fingerprints

- There are **23 pairs** of human chromosomes, with >3 billion genome size.
- You known that not all segments of DNA code for proteins; some DNA segments have a regulatory function, while others are **intervening sequences (introns)**, and still others are repeated DNA sequence.
- For DNA fingerprinting short repetitive nucleotide sequence which is specific for a person is important. These nucleotide sequence is known as **variable number tandem repeats (VNTR).**
- It was initially developed by Alecjeffreys. He used a satellite DNA as probe that shows a very high degree of polymorphism.

Making DNA Fingerprints :

- As every cell contains DNA, extremely small amount of blood, semen, hair bulb or any other cell from the body of a person are sufficient to detect the individual.
- Thus it is a laboratory procedure that requires the following steps :

Application of DNA Fingerprints: DNA fingerprints are useful in several areas of society.

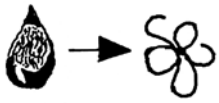
- They are used by professionals in human health and justice system. It can be used to help identify

individuals, who may have carried out crimes such as rape, settling paternity disputes, deterring relationships in inherited diseases and **monitoring** bone marrow transplants.

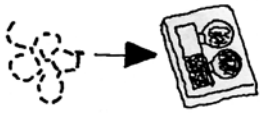
- It is used to diagnose inherited disorders like **cystic fibrosis, haemophilia, Huntington's diseases, Alzheimer's sickle cell anemia thalassemia** and many others.

---

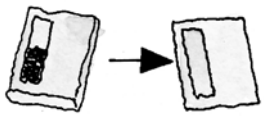
(1) The process begins with a blood or cell sample from which the DNA is extracted



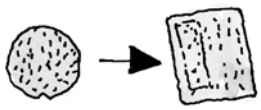
(2) The DNA is out into fragments using a restriction enzyme. The fragments are then separated into bands by electrophoresis through an agarose gel.



(3) The DNA band is transferred to a nylon menbrane.



(4) A radioactive DNA probe are introduced. The DNA probe binds to specific DNA sequences on the nylon membrane.



(5) The excess probe materials is washed away leaving the unique DNA band pattern.



(6) The radioactive DNA pattern is transferred in X-ray film by direct exposure. When developed, the resultant visible pattern is the DNA finger print.



---

(98) Bloodstained clothing from Abraham Lincoln has been analyzed for evidence of a genetic disorder called ..... .

(A) Alzheimer's (B) Marfan's syndrome (C) Huntington's disease (D) Haemophilia.

(99) Which of the following is correct about DNA segment ?

(A) Code for proteins (B) Intervening sequences (C) VNTR (D) All of above

(100) In DNA fingerprinting separation of DNA Fragments by ..... .

(A) Gas – chromatography (B) TLC (C) Electrophoresis (D) PCR

(101) It is known as DNA scissors.

(A) REN (B) Ligase (C) DNA polymerase (D) RNA polymerase

(102) What is the use of radioactive DNA probe ?

(A) The DNA probe binds to specific DNA sequence on the nylon membrane.

(B) Separation of DNA fragments.

(C) It makes x-ray film.

(D) Extraction of DNA fragments.

(103) When radioactive DNA pattern becomes visible ?

(A) While bombarding with DNA probes.

(B) While radioactive DNA pattern is transferred to x-ray film by direct exposure.

(C) While washing DNA fragments.

(D) While DNA fragments are separated by electrophoresis.

Answers : 98 (B), 99 (D), 100 (C), 101 (A), 102 (A), 103 (B)

- **A - Assertion, R - Reason type questions**

Choose the correct option from the following for given questions.

(A) A and R both are correct, while R is correct explaination of A.

(B) A and R both are correct but R is not correct explaination of A.

(C) A is correct, but R is wrong.

(D) A is wrong but R is correct.

(104) Assertion A : Polytene chromosomes are consisting of thousands of DNA strands.

Reason R : Specialized cells undergo repeated rounds of DNA replication without cell division to increase cell volume forming giant polytene chromosome.

(A) (B) (C) (D)

(105) Assertion A : The termination codon is present on the end of m-RNA.

Reason R : End of translation process occurs with or without termination codon.

(A) (B) (C) (D)

(106) Assertion A : Semi conservative DNA replication proposed by Avery and Mccarty.

Reason R : At the end of DNA replication the enzyme ligase joins pieces of nucleotides with phosphodiester bond.

(A) (B) (C) (D)

(107) Assertion A : (A) + (G) = (T) + (C)

Reason R : (A) = (T) : (G) = (C)

(A) (B) (C) (D)

(108) Assertion A : The synthesis of new chain in DNA replication is carried out by enzyme RNA polymerase.

Reason R : RNA polymerase forms a short RNA chain complementary to the template DNA as its initial region, this is called a primer.

(A) (B) (C) (D)

(109) Assertion A : Genetic code is defined as the sequence of nitrogen bases in m-RNA molecule.

Reason R : Genetic code contains the information for the synthesis of protein molecules.

(A) (B) (C) (D)

(110) Assertion A : Genetic code is specific.

Reason R : Any one codon specifies the position of one kind of amino acid only.

(A) (B) (C) (D)

(111) Assertion A : Transcription observed in HIV.

Reason R : Tumour viruses follow reverse transcription.

(A) (B) (C) (D)

(112) Assertion A : The RNA polymerase is attached to promoter region.
Reason R : The operator gene is the segment of DNA which produces repressor substances.
(A) (B) (C) (D)

(113) Assertion A : DNA is cut into section by using restriction endonucleases.
Reason R : Separation of DNA fragments by electrophoresis.
(A) (B) (C) (D)

Answers : (104-A), (105-B), (106-D), (107-B), (108-D), (109-A), (110-A), (111-D), (112-C), (113-B)

- **True – False type MCQs :**

(114) **Choose correct option : True for (T) and False for (F)**
(1) (A) + (G) = 500 then (T) + (C) = 500
(2) (A) = 600 then (G) = 600
(3) (A) = 300 then (T) = 300, (G) = 200 then (C) = 200
(A) TFT (B) TTF (C) FFT (D) TTT

(115) **Choose correct option : True for (T) and False for (F)**
(1) DNA - DNA = Replication
(2) DNA - m-RNA = Transcription
(3) DNA - Protein = Translation
(A) TTF (B) TTT (C) FTT (D) FFT

(116) **Choose correct option : True for (T) and False for (F)**
(1) Helicase : Breaks down hydrogen bonds between two polynucleotide chain.
(2) DNA – polymerase – III : The synthesis of new chain is carried out by enzyme DNA polymerase -III.
(3) RNA polymerase : It removes RNA primers from logging strand.
(A) FTT (B) TTF (C) TTT (D) FFT

(117) **Choose correct option : True for (T) and False for (F)**
(1) In transcription process two separated chains of DNA act as template chain.
(2) In transcription the hydrogen bonds between two polynucleotides chains are sequentially broken by gyrase and helicase.
(3) m-RNA is translocated to nucleus where it associates with DNA.
(A) FFF (B) TTF (C) TFT (D) FFT

(118) **Choose correct option : True for (T) and False for (F)**
(1) AUG = Methionine
(2) CCC = proline
(3) UAA = Leucine
(A) FFF (B) TFF (C) TTF (D) TFT

(119) **Choose correct option : True for (T) and False for (F)**
(1) AUG = START
(2) AGA = STOP
(3) UAA = STOP
(A) TTF (B) TFT (C) FTF (D) TTT

(120) **Choose correct option : True for (T) and False for (F)**

(1) In t-RNA 3^1 end is a site for the attachment of activated amino acid.

(2) In t-RNA definite sequence of three nitrogenous bases, which constitute the anti codon.

(3) In t-RNA the region of D LOOP responsible for releasing amino acids.

(A) TFF (B) FFT (C) TTF (D) TFT

(121) **Choose correct option : True for (T) and False for (F)**

(1) Genetic information flows form DNA $\rightarrow$ *m*-RNA $\rightarrow$ protein.

(2) Unidirectional flow of information was first described by F.H.C. Crick.

(3) Tumour viruses contain RNA as a genetic material. It follows transcription.

(A) TTT (B) FFT (C) TTF (D) TFT

(122) **Choose correct option : True for (T) and False for (F)**

(1) Genome size in Human is > 3 billion.

(2) Genome size in Drosophila is 137 million.

(3) Genome size in E-coli is 4.6 million.

(A) TTF (B) FFT (C) FTF (D) TTT

(123) **Choose correct option : True for (T) and False for (F)**

(1) **Structural gene :** They are segments of DNA which carry codes for the synthesis of protein.

(2) **Promoter gene :** RNA polymerase is attached to this region during DNA synthesis.

(3) **Operator gene :** The segment of DNA which exercises a control over transcription.

(A) FTF (B) TFT (C) TFT (D) FFT

(124) **Choose correct option : True for (T) and False for (F)**

(1) DNA is cut into sections by using restriction ligase.

(2) DNA band pattern is transferred to a nylon membrane or nitro cellulose.

(3) A radioactive DNA Probe is introduced. The DNA Probe binds a specific DNA sequence on the nylon membrane.

(A) FTT (B) FTF (C) TFT (D) FFT

Column Type MCQs :

(125) **Match the enzyme in column - I with its function in column - II and select correct option:**

Column - 1	**Column - 2**	
(I) β- galactosidase	(p) Joining of DNA fragments	(A) (I- r) (II-q) (III-p) (IV-s)
		(B) (I-r) (II-s) (III-p) (IV-q)
(II) Permease	(q) Peptide bond formation	(C) (I-s) (II-r) (III-p) (IV-q)
(III) Ligase	(r) Hydrolysis of lactose	(D) (I-q) (II-r) (III-p) (IV-s)
(IV) Ribozyme	(s) Increase permeability to β - galactosidase	

(126) **Match column - I with column - II and select the correct option:**

Column - 1	**Column - 2**	
(I) AUG	(p) Phenyl alanine	(A) (I- p) (II-q) (III-r) (IV-s)
(II) UAA	(q) Methionine	(B) (I-q) (II-s) (III-r) (IV-p)
(III) UUU	(r) Tryptophan	(C) (I-s) (II-q) (III-r) (IV-p)
(IV) UGG	(s) Termination	(D) (I-q) (II-s) (III-p) (IV-r)

(127) **Match column - I with column - II and select the correct option:**

Column - 1	**Column - 2**	
(I) t-RNA	(p) Amino acid attachment	(A) (I- s) (II-r) (III-p) (IV-q)
(II) m- RNA	(q) It brings genetic information into cytoplasm	(B) (I-q) (II-p) (III-r) (IV-s)
(III) r-RNA	(r) Nucleolus organization region	(C) (I-s) (II-q) (III-r) (IV-p)
(IV) Peptidyl transfrase	(s) Transport aminoacids from cytoplasm towards ribosomes	(D) (I-p) (II-r) (III-q) (IV-s)

(128) **Match column - I with column - II and select the correct option:**

Column - 1	**Column - 2**	
(I) Griffith	(p) Lac- operon	(A) (I- s) (II-q) (III-r) (IV-p)
(II) Erwin Chargaff	(q) Rules regarding the composition	(B) (I-s) (II-q) (III-p) (IV-r)
(III) Temin and Baltimore	of bases in DNA	(C) (I-p) (II-q) (III-r) (IV-s)
(IV) Jacob and Monad	(r) Reverse Transcription	(D) (I-q) (II-r) (III-s) (IV-p)
	(s) Bacterial Transformation	

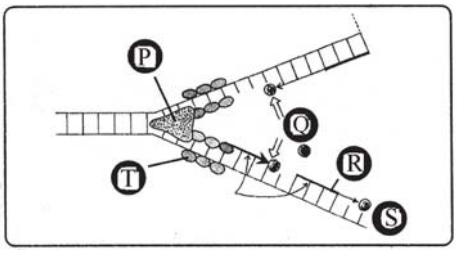
(129) **Match column - I with column - II and select the correct option:**

Column - 1	**Column - 2**	
(I) Molecular Medicine	(p) Improve understanding of diseases	(A) (I- r) (II-q) (III-p)
(II) Forensics Science	like Alzheimers, Parkinsons etc	(B) (I-r) (II-p) (III-q)
	(q) To establish family relationships	(C) (I-p) (II-q) (III-r)
(III) Microbial Genomics	(r) develop new energy sources and monitor environment to detect pollutants	(D) (I-q) (II-r) (III-p)

Answers : (114-A), (115-A), (116-B), (117-A), (118-C), (119-B), (120-C), (121-C), (122-D), (123-B), (124-A), (125-B), (126-D), (127-C), (128-A), (129-C)

Figure based MCQs :

Note : Use given figure for question number 130 to 133.



(130) what is Name and Function of 'P' indicated region in given diagram?

(A) Helicase : It Starts Synthesis of new strand.

(B) Helicase : It breaks down H_2 bond between two strands.

(C) RNA polymerase : It forms primer.

(D) DNA polymerase - III : It start synthesis of new DNA strand.

(131) Which Process illustrated in given diagram?

(A) RNA replication (B) Translation (C) DNA replication (D) Transcription

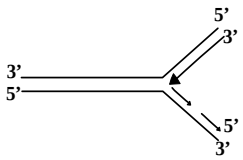
(132) What is indicated by 'T' in given diagram?

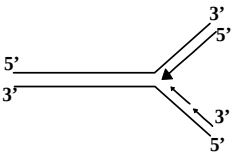
(A) DNA helicase (B) RNA polymerase (C) DNA polymerase - III (D) SSB protein

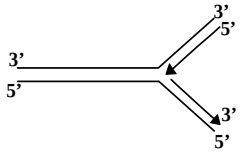
(133) Give the name of cell- division phase where given process occurred :

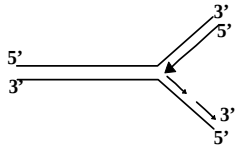
(A) G_1- phase (B) S-phase (C) G_2- phase (D) M- phase

(134) which one of the following correctly represents the manner of replication of DNA ?

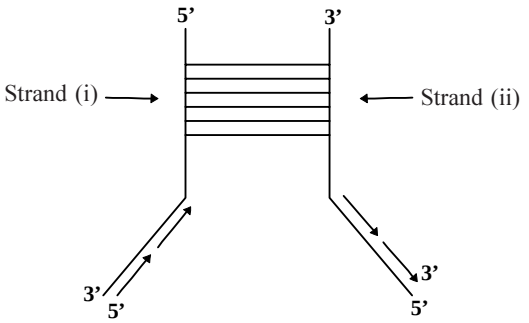
(A) 

(B) 

(C) 

(D) 

(135) Choose the correct option by using given diagram.



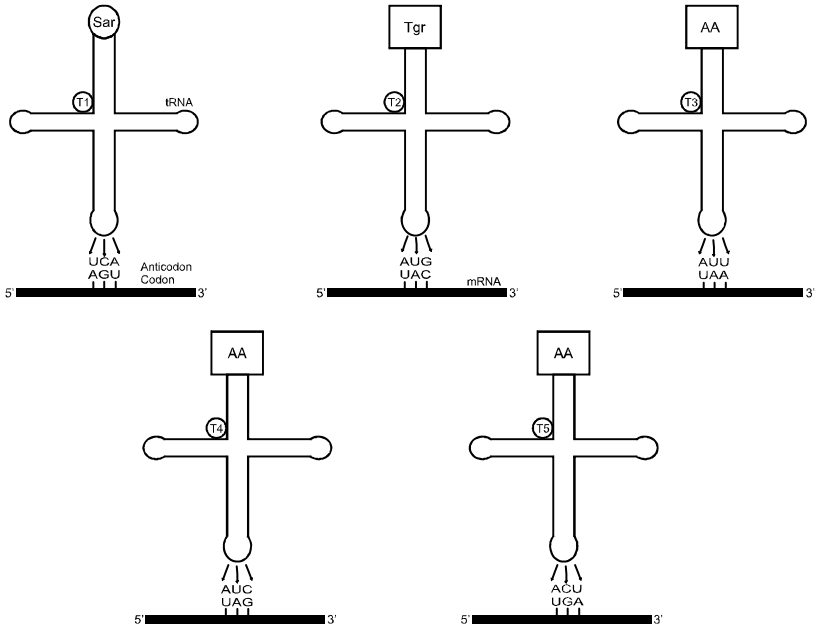
(A) Strand (i) shows direction of DNA replication

(B) Strand (ii) shows direction of DNA replication

(C) Strand (i) shows direction of discontinuous replication

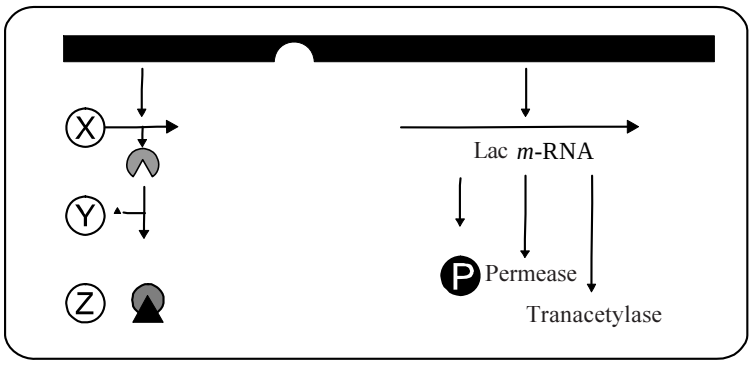
(D) Strand (ii) shows direction of discontinuous replication

(136) Study the following t-RNA molecules related with their anti-codon pairing with respective codon. Which type of t-RNA is impossible ?



(A) T_1 and T_2 (B) T_1 T_2 and T_3 (C) T_1 and T_4 (D) T_3, T_4 and T_5

Note: Use Given diagram for question number 137 to question number 140.



(137) What is indicated by 'X' in given diagram?

(A) Protein (B) m-RNA (C) Inducer (D) t-RNA

(138) What is indicated by Y and Z in given diagram?

(A) y - Inducer, z - Inducer-repressor complex (B) y - Repressor, z - Inducer-repressor complex

(C) y - Protein, z - Promoter complex (D) y - Promoter, z - Operator

(139) What is function of 'Y' labelled region in given diagram?

(A) Inducer which could be a Substrate, Precursor, when provided will bind the repressor protein and thus open up operator gene.

(B) It prevents transcription of one or many Structural genes.

(C) Help in formation of â-galactosidase.

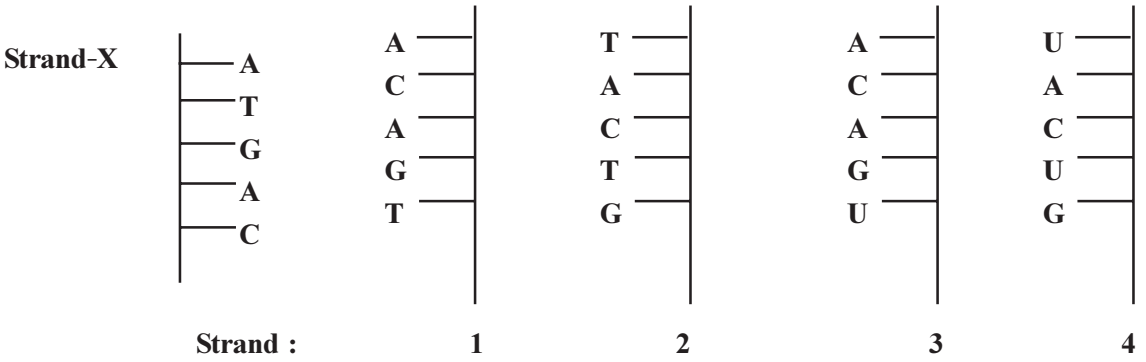
(D) It removes repressor from structural gene region and starts transcription of Z, Y, a genes.

(140) what is the function of 'P' labelled region in given diagram?

(A) Digestion of glucose (B) Hydrolysis of lipids

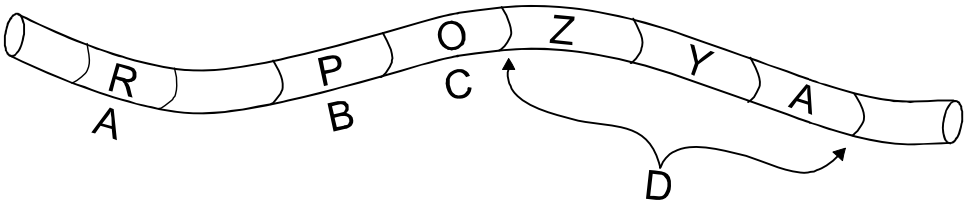
(C) Digestion of galactose (D) Hydrolysis of lactose.

(141) Strand 'X' in the figure shows a small part of a nucleic acid molecule. Which pair of the following strands are complementary to strand X.



(A) 1 and 3 (B) 2 and 4 (C) 1 and 2 (D) 3 and 4

(142) The figure of the Lac-operon from E.coli is shown below. Each alphabet indicates its components may be used more than once, then which of the following option is correct in which all the alphabets are matched with their functions.



DNA Strand

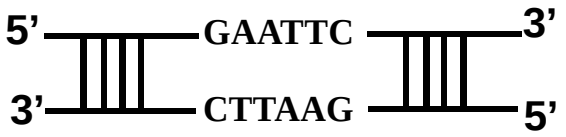
(A) D - The binding site for repressor protein.
B - Structural gene.
C - The binding site for RNA polymerase.
A - The gene that codes for the repressor protein.

(B) A - The binding site for repressor protein.
B - Structural gene.
D - The binding site for RNA polymerase.
C - The gene that codes for the repressor protein.

(C) A - The binding site for repressor protein.
C - The structural gene.
B - The binding site for RNA polymerase.
D – The gene that codes for repressor protein.

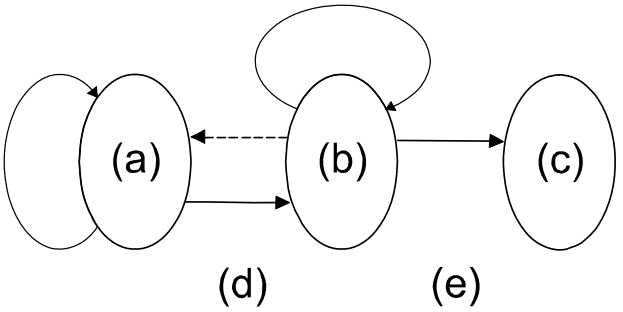
(D) C- The binding site for repressor protein.
D- The structural gene.
B- The binding site for RNA polymerase.
A-The gene that codes for the repressor protein.

(143) Given below is a sample of a portion of DNA strand giving the base sequence on the opposite strands. What is so special shown in it ?



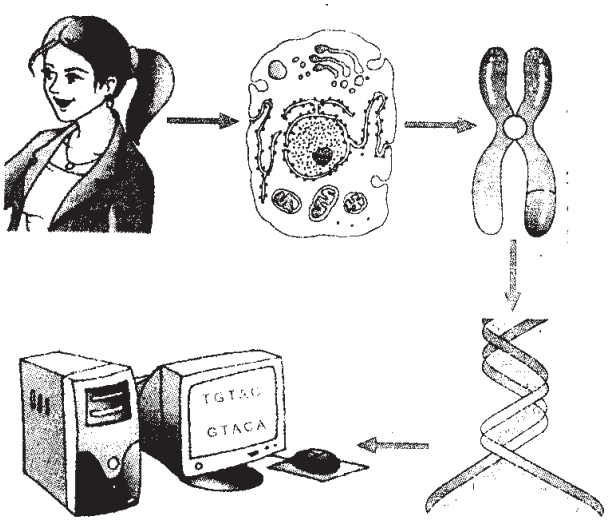
(A) Palindromic sequence of base pairs.
(B) Replication completed
(C) Deletion mutation
(D) Start codon at the .5[1] end

(144) The diagram represents the "central dogma" of molecular biology. Choose the correct combination of labelling.



(A) a - protein, b - RNA, c - DNA, d - translation, e - transcription
(B) a - RNA, b - DNA, c - protein, d - transcription e - translation
(C) a - transcription, b - translation, c - protein, d - DNA, e - RNA
(D) a - DNA, b - RNA, c - protein, d - transcription, e - translation

(145) The given diagram illustrates:



(A) Chromosomal walking
(B) Humoral Analysis
(C) Human genome project
(D) Method of DNA finger printing

Answers : (130-B), (131-C), (132-D), (133-B), (134-D), (135-C), (136-D), (137-B), (138-A), (139-A), (140- D), (141-B), (142-D), (143-A), (144-D), (145-C)

● **MCQs for NEET :**

(146) Mode of DNA replication is ..... .

(A) Semi conservative and bidirectional (B) Conservative and complete

(C) Semi conservative and unidirectional (D) Only conservative

(Hint) : After the completion of replication, each DNA molecule would have one parental and one newly synthesized strand; this method was termed as semi-conservative DNA replication.

- As the process of replication progresses in two directions, it is called bidirectional.

(147) If DNA strand have CTGATGC sequences then what would be the correct base sequences on m RNA ..... .

(A) GUCTUTCG (B) GACUAUCG (C) GACTATGC (D) UACTATCU

(Hint) : The process where DNA gives its information to RNA is called transcription.

- In transcription one of these two DNA separated clains acts as a template chain. This chain provides complementory information for synthesis of m-RNA.
- RNA possesses 'U' instead of 'T'.

(148) The polypeptide synthesis in eukaryotes is initiated by ..... .

(A) CCA (B) GCA (C) AUG (D) GUU

(Hint) : Translation is always initiated with amino acid methionine.

AUG is starting codon as well as code for methionine amino acid.

(149) It having anticodon region ..... .

(A) m-RNA (B) s-RNA (C) t-RNA (D) r-RNA

(Hint): - t-RNA has an anticodon loop that has bases complementary to the code and it also has an amino acid acceptor by which it binds to amino acid.

(150) What is the meaning of Lac in "Lac-operon" ?

(A) Digit of 1 Lac (B) Lactose (C) Lactase (D) Lactic acid

(Hint) : Jacob and Monad proposed that metabolic pathways are regulated as a unit. They studied the change in the nature of enzymes of E.coli when provided with different kinds of sugars. They found that when sugar lactose is added to the culture of E.coli it induces three enzymes necessary to break down the lactose into galactose.

(151) What would happen if there occurs mutation in 25th codon (ie-UAA to UAG) of a gene encoding a polypeptide chain of 50 amino acids ?

(A) enhancer (B) promoter (C) regulator (D) receptor

(Hint) : The promoter gene is the DNA segment at which the transcription of the structure genes take place. It controls the rate of m-RNA synthesis.

(152) During translation initiation in Prokaryotes, a GTP molecules is needed in ..... .

(A) association of 50s subunit of ribosome with initiation complex

(B) formation of formyl-met- t-RNA

(C) association of 30s subunit of ribosome with m-RNA

(D) association of 30s m-RNA with formyl-met-tRNA

(Hint) : For creating complicated complex the elongation process required energy, so GTP used as source of energy.

(153) What would happen if in a gene encoding a polypeptide of 50 amino acids, a mutation occurs in 25th codon is mutated and it change from UAA to UAG ?

(A) A polypeptide of 24 amino acids will be formed.

(B) A polypeptide of 50 amino acids will be formed.

(C) Two polypeptide of 24 and 25 amino acids will be formed.

(D) A polypeptide of 49 amino acids will be formed.

(Hint) : The codon present on 25th position of a gene is a termination codon and due to mutation new codon is formed that also termination codon, so this newly formed codon UAG cause termination of process, hence A polypeptide of 24 amino acids will be formed.

(154) How many effective codons are there for the synthesis of twenty amino acids?

(A) 60 (B) 20 (C) 64 (D) 61

(Hint) : According to possibility of three letter code four nitrogen bases A,U, G, C provide 64 codons. Out of this three codon do not code for any amino acids such codons are called non sense codons hence 61 codon code for 20 amino acids.

(155) A naturally occurring coding strand composed of alternating C and U residues would result in the formation of ..... .

(A) a polypeptide containing alternating leu and ser. residues

(B) a polypeptide containing either leu. or ser. residues

(C) a polypeptide containing only leu. residues

(D) a polypeptide containing only ser. residues

(Hint) : Because CUC and UCC code for respectively Leucine and Serine.

(156) Which one of the following is correct for gene map?

(A) It shows different species distribution at different geological regions.

(B) It helps describe the spatial arrangement of genes on a chromosomes.

(C) It trigers cell-division.

(D) It gives proper information regarding genetic evolution.

(157) DNA fingerprinting method is very useful for ..... .

(A) Molecular analysis of DNA samples (B) Forensic studies and gene map

(C) Polytene chromosome (D) Combination of different-different samples of DNA

(Hint) : For DNA fingerprinting, short repetitive nucleotide sequence which are specific for a person are important. These nucleotide sequences are known as variable number tandem repeats (VNTR).

(158) A mutation which substitutes one purine (A) base with another purine (G) base is called ..... .

(A) transversion (B) transduction (C) transition (D) transfection

(159) Which proportion of Nitrogen bases varies with the organism:

(A) $\frac{A+G}{T+C}$ (B) $\frac{T+C}{G+A}$ (C) $\frac{A+T}{G+C}$ (D) $\frac{A+C}{T+G}$

(Hint) : According to Erwin Chargaff: The proportion of A is equal to T and also proportion of G is equal to C, but amount of (A)+(T) is not necessarily equal to (G)+(C).

Therefore (A)=(T) : (G)=(C).

But $\frac{A+T}{G+C}$ = varies with the organism.

(160) During transcription holoenzyme RNA polymerase binds to a DNA sequence and the DNA assumes a saddle like structure at the point. What is that sequence called ?

(A) AAAT Box (B) TATA Box (C) GGTT Box (D) LAAT Box

(Hint) : TATA box is present in eukaryotic promoter region. It has a resemblance with pribnow box of prokaryotes. TATA box was identified by Dr. Hogness and so it is also called as Hogness box. During the process of transcription the RNA polymerase binds to TATA box due to which DNA assumes a saddle like structure at this place.

(161) Which of following is famous as a RNA dependent DNA polymerase ?

(A) DNA polymerase (B) RNA polymerase (C) Reverse Transcriptase (D) Transcriptase

(Hint) : According to Temin and Baltimore, many tumour virsues contain RNA as a genetic material and replicate by first synthesizing a complementary DNA. This process is called reverse transcription and it is done with the help of reverse transcriptase. It is observed in HIV.

(162) What is telomerase?

(A) Protein (B) RNA (C) Ribonucleoprotein (D) Palindromic sequence

(Hint) : It is a ribonucleo protein that adds a species-dependent telomere repeat sequence to the 3^1 end of telomeres. A telomere is a region of repetitive sequence at each end of a eukaryotic chromosomes in most eukaryotes. Telomeres protect the end of the chromosome from DNA damage or from fusion with neighbouring chromosomes.

(163) One gene one enzyme hypothesis was proposed by ..... .

(A) Hershey and chase (B) Watson and Crick (C) Beadle and Tatum (D) R. Franklin

(164) The okazaki fragments in DNA chain growth : ..... .

(A) Polymerize in the 3' to 5' direction and forms replication fork.

(B) Prove semi-conservative nature of DNA replication.

(C) Polymerize in 5' to 3' direction and explain 3' to 5' DNA replication.

(D) Result in transcription.

(Hint) : Okazaki segment is a short fragment of DNA having RNA primer at 5' terminus during the process of DNA replication. During the synthesis of complementary DNA strands the leading strand always reads - 5' to 3'. Anti parallel complementary strand i.e lagging strand reads from 3' to 5' direction. As original strands of DNA are anti-parallel the only one new continuous strand can be formed at 3' end of leading strand. This is due to intrinsic 5' to 3' polarity of DNA polymerase. The remaining other strand of DNA grows discontinuously in opposite side.

(165) The length of DNA molecule greatly exceeds the dimensions of the nucleus in eukaryotic

- Cells, How is this DNA accomodated ?

(A) Super coiling in nucleosomes. (B) DNAase digestion.

(C) Through elimination of repetative DNA. (D) Deletion of non-essential genes.

(Hint) : The solenoid defines the packing of DNA as a 30 nm fiber of chromatin and results from the helical winding of at least five nucleosome strands.

(166) Which orgenelles provide platform for translation process?

(A) Mitochondria (B) Ribosome (C) Lysosome (D) Golgibody

(167) Which one of the following statement is correct ?

(A) GUU, GCU - Alanine (B) UAA, UAG - Terminator

(C) AUG, ACG - Methionine (START) (D) UUA, UCA - Leucine

(Hint) : GUU – vaccine, GCU – Alanine, UAA, UAG, and UGA – Termination codones. AUG. starting codon as well as code for methionine amino acid, UUA – Leucine while UCA code for serine.

(168) Haploids are more suitable for mutation studies than the diploids. This is because ..... .

(A) all mutations, whether dominant or recessive are expressed in haploids

(B) haploids are reproductively more stable than diploids

(C) mutagen penetrate in haploids more effective than is diploids

(D) haploids are more abudant in nature than diploids.

(169) Removal of introns and joining the exons in a defined order in a transcription unit is Called ..... .

(A) Tailing (B) Transformation (C) Capping (D) Splicing

(Hint): In molecular biology splicing is the editing of the nascent pre-mRNA transcript. After splicing introns are removed and exons are joined together (ligated).

(170) Semi-conservative replication of DNA was first demonstrated in ..... .

(A) *Streptococcus pneumoniae* (B) *E.coli* (C) *Salmonella typhimurinum* (D) *Drosophila melanogaster*

(Hint) : Genome size in E.coli is 4.6 million having estimated genes are 3200. Beside E.coli can easily grow in culture medium and multiply very fast so it is best for making more copies of DNA, hence semi-conservative replication of DNA was first demonstrated in E.coli.

(171) Select the correct statements out of the four (i-iv) given below about lac operon.

(i) Glucose or galactose may bind with the repressor and inactivate it.

(ii) In the absence of lactose the repressor binds with the operator region.

(iii) The Z- gene code for permease.

(iv) This was elucidated by Francis Jacob and Monod.

(A) i and iii (B) i and ii (C) ii and iv (D) i and iii

(172) Whose experiments cracked the DNA and discovered unequivocally that a genetic code is a triplet ?

(A) Nirenberg, Matthaei and Khorana (B) Hershey and Chase

(C) Beadle and Tatum (D) Watson and Crick

(Hint) : Nirenberg and Matthaei (1961) experimentally proved that a single amino acid is determined by a sequence of three nitrogen bases which is known as triplet code. Khorana has got Nobel Prize on genetic code.

(173) Out of following which organism is odd for central dogma?

(A) Bean (B) Yeast (C) Sunflower (D) HIV

(Hint) : According to crick genetic information flows from DNA $\rightarrow$ m-RNA $\rightarrow$ protein replication. But in some viruses the flow of information is in reverse direction, some HIV like viruses contain RNA as genetic material and replicated by first synthesizing a complementary DNA. This process is called reverse transcription.

(174) Removal of RNA polymerase- III from nucleoplasm will affect the synthesis of : ..... .

(A) t-RNA (B) r-RNA (C) Ribosomes (D) m-RNA

(Hint) : RNA Polymerase-I : It is responsible for synthesis of r-RNA.

RNA Polymerase-II : It is responsible for synthesis of m-RNA.

RNA Polymerase-III : It is responsible for t-RNA synthesis, hence nucleoplasm will affect the synthesis of t-RNA.

(175) During transcription the nucleotide sequence of the DNA strand that is being coded is AACCGG, then the nucleotide sequence in m RNA would be : ..... .

(A) GGUUCC (B) UUGCGC (C) UUGGCC (D) TTGGCC

(176) Melting of DNA at an elevated temperature (90^0 C) is primarily due to the result of : ..... .

(A) DNA converted into RNA (B) It breakdown in numbers of pieces

(C) Nothing will happen. (D) Breakdown of Hydrogen bonds and two chain become separated

(177)) It is a vector of genetic information ..... .

(A) RNA (B) DNA (C) Enzyme (D) Protein

(Hint): Recent developments in molecular biology revealed that nucleic acids of the chromosomes are responsible for the transmission of hereditary character. DNA is found to be the genetic material in almost all the living beings except plant viruses and in some retro viruses.

(178) If the DNA codons are ATG ATG ATG and a cytosine (c) base is inserted at the beginning, which of the following will result ..... .

(A) A non-sense mutation (B) CATGATGATG

(C) CATGATGATG (D) CATGATGATG

(179) Non-genetic RNA is of ..... .

(A) Two types (B) Three types (C) Only one type (D) None of these

(Hint) : Non-genetic RNA having three Types (i) m-RNA, (ii) t-RNA and (iii) r-RNA. It helps in Translation process.

(180) It is known as a chemical scissors :

(A) Ligase (B) Polymerase (C) Endonuclease (D) Transcriptase

(Hint) : A endonuclease is an enzyme that cuts DNA at or near specific recognition nucleotide sequences known as restriction site.

(181) which form of RNA has a structure resembling clover leaf?

(A) m-RNA (B) t-RNA (C) r-RNA (D) hn-RNA

(182) Unit of nucleic acid is ..... .

(A) Nucleotide (B) Nucleoside (C) Nucleic Acid (D) All of Above

(Hint) : Nucleic acid is made up of units of nucleotide. In nucleotide, purine or pyrimidine nitrogenous base joined by deoxyribose pentose sugar (D), which is further linked with phosphate (P) group to form nucleotides.

(183) The double helix model of Watson and Crick is known as ..... .

(A) C-DNA (B) B-DNA

(C) Z-DNA (D) D-DNA

(Hint) : Five different morphological forms of DNA double helix have been described. These are A,B,C,D and Z forms. Most of these forms (except B and Z) occur in rigidly controlled experimental condition. Watson and Crick model represents commonest form Biotic-form (B-form or B- DNA) of DNA. It is a handedness of the double helix is right handed.

(184) Which of the following statements are correct ?

(A) Introns are present in m RNA and exons are present in t RNA.

(B) Codons are present in m RNA and anticodons are in t RNA.

(C) Every intron is a set of three terminator codons.

(D) Exons are present in eukaryotes while introns are present in prokaryotes.

(185) If a segment of an m-RNA molecule has the sequence 5' GUACCGAUCG 3', which of the following could have been template DNA molecule ?

(A) 5' GCU AGC CAU G 3' (B) 5' GUA CCG AUC G 3'

(C) 5' CAT GGC TAG C 3' (D) 5' CGA TCG GTA C 3'

(186) If percentage of cytosine is 18% then percentage of adenine will be : ..... .

(A) 32 % (B) 64 % (C) 36 % (D) 23 %

(Hint): If 100% of Nitrogen bases A,T,G and C, then C having 18%. So G also has 18%

$\therefore$ G = C

So, 100 - G - C

100-18-18 = 64% Nitrogen bases

So, $\frac{64}{2}$ = 32% (A = T)

(187) Select the correct Option :

Sr.	Direction of RNA Synthesis	Direction of reading of the template DNA Strand
(A)	5' $\rightarrow$ 3'	5' $\rightarrow$ 3'
(B)	3' $\rightarrow$ 5'	3' $\rightarrow$ 5'
(C)	5' $\rightarrow$ 3'	3' $\rightarrow$ 5'
(D)	3' $\rightarrow$ 5'	5' $\rightarrow$ 3'

(Hint) : RNA polymers catalyse polymerization only in one direction, that is 5' $\rightarrow$ 3' and the strand that has the polarity 3' $\rightarrow$ 5' act as a template.

(188) Which one of the following Nitrogen bases is not correctly matched with termination codons ?

(A) Cytosine (c) (B) U (C) A (D) G

(189) Which one of the following has dual functions?

(A) UGA (B) UUU (C) AUG (D) AAA

(190) Which of the following codons has no t RNA ?

(A) UAA (B) UAU (C) UGU (D) UGG

(Hint): UAA is a termination codon. It does not indicate the position of any amino acid. So such a codon is to release the synthesized polypeptide chain from ribosomes.

(191) Genes that are involved in turning on or off the transcription of a set of structural genes are called

(A) Polymorphic genes (B) Promoter genes (C) Regulatory genes (D) Operator genes

(192) Which one of the following codons codes for the same information as UGC ?

(A) UGU (B) UGA (C) UAG (D) UGG

(193) Which one of the following gene is not matched with name of cartoon characters?

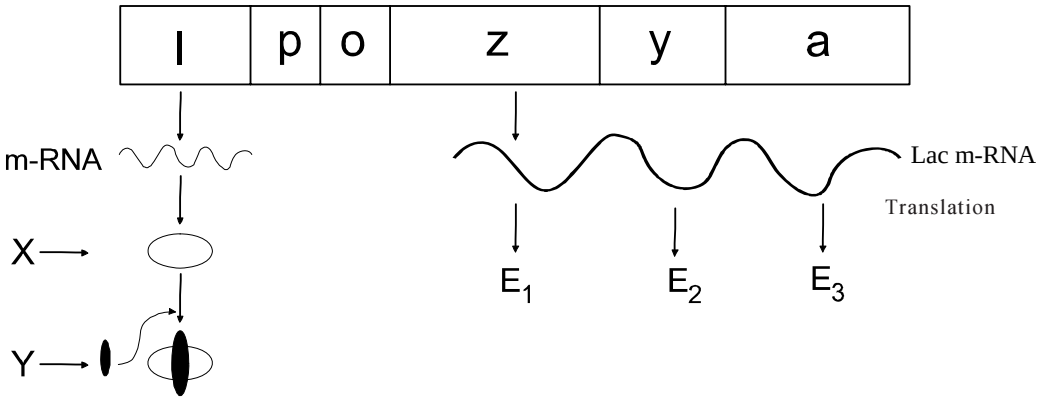
(A) TinTin (B) Popeye (C) Asterix (D) Obelix

(Hint): Popaye gene : Protein is responsible for development of stroted muscels

Asterix gene : Mutant gene is found in rabbit.

Obelix gene : It produce Nuclear binding protein.

(194) In the given figure of the lac operon, an operon for inducible enzymes, Identify Components and enzymes.



Sr.	X	Y	E_1	E_2	E_3
(a)	Repressor protein	Inducer (lactose)	Permease	Transacetylase	β-galactosidase
(b)	Repressor protein	Inducer (lactose)	β-galactosidase	Transacetylase	Permease
(c)	Inducer (lactose)	Repressor protein	β-galactosidase	Permease	Transacetylase
(d)	Repressor protein	Inducer (lactose)	β-galactosidase	Permease	Transacetylase

(195) Which option is correct for the amino acid and the total number of their genetic code ?

(A) Arg = 6, His = 6

(B) Val = 6, Pro = 6

(C) Pro = 4, Thr = 4

(D) Thr = 4, Arg = 4

(196) Which region of gene below become transcription but does not translate?

(A) Exon (B) Intron (C) Cistron (D) Codon

(197) Balbiani rings are sites of ..... .

(A) DNA replication (B) RNA and protein synthesis

(C) synthesis of lipids (D) synthesis of polysaccharides

Answers : (146-A), (147-B), (148-C), (149-C), (150-B), (151-B), (152-D), (153-A), (154-D), (155-A), (156- B), (157-A), (158-C), (159-C), (160-B), (161-C), (162-C), (163-C), (164-C), (165-A), (166-B), (167-B), (168-A), (169-D), (170-B), (171-C), (172-A), (173-D), (174-A), (175-C), (176-D), (177- B), (178-C), (179-B), (180-C), (181-B), (182-A), (183-B), (184-B), (185-C), (186-A), (187-C), (188-A), (189-C), (190-A), (191-D), (192-A), (193-A), (194-D), (195-C), (196-B), (197-B)

Chromosomes

'Carries of heredity'. They are thread like structures composed of chromatin that carry the genes in linear sequence. They determine the individual characteristics of an organism.

Gene

Unit of heredity that is composed of DNA. Gene is visulalised as s discrete particle that determines particular characteristics.

Expression

Gene expression is the mechanism at the molecular level by which a gene is able to express itself in the phenotype of an organism. In the process the information in DNA is transformed into proteins via following sequential process.

Regulation

The mechanism of switching off and switching on of the genes depending upon the requirement of the cells and the state of development. In prokaryotes, operon system is responsible for gene regulation.

DNA

DNA is a long polymer of deoxyribonucleotides that consists of deoxyribose sugar, phosphoric acid and nitrogenous bases (A,T,C,G). DNA contains hereditary information in the form of sequence of nitrogenous bases. Three bases together constitute a codon and the whole may of these codons is teremed genetic code.

Replication

Formation of new exact copy of the DNA is a semiconservative and semidiscontinuous process. DNA replication is essential for maintaining quantity and quality of genetic material in every new cell generation.

Types

Positive regulation
Positive regulation is the one in which the genes remain unexpressed unless they are induced to do it. It is therefore inducible regulation.

Negative regulation
Negative regulation is the one in which the genes continue to express till their activity is repressed. It is also called repressible regulation.

Exerted at five levels

- Transcriptional level when primary transcript is formed.
- Processing level of primary transcript
- Translational level
- Post translational level
- During transport of *m*RNA

Transcription

Genetic information of DNA is transferred to RNA (*m*RNA) via transcription. RNA polymerase enzyme in presence of several other factors prepares single stranded RNA that is complementary to DNA with the substitution of thymine by uracil.

RNA

Single stranded, unbranched nucleic acid molecule consisting of ribose sugar, phosphoric acid and nitrogenous bases (A,U,C,G). They perform various functions in cells.

hnRNA

The orimary transcript that consists of both introns (interrupting sequence) and exons (expressing sequences) is called heteronuclear RNA . It undergoes procesing to produce *m*RNA.

tRNA

*t*RNA or transfer RNA works as the adapter molecule during protein synthesis (gene expression). It brings amino acids present in cytoplasm to the *m*RNA strand corresponding to the sequence of codons on the mRNA and facilities protein synthesis.

rRNA

*r*RNA (ribosomal RNA) are the component of ribosomes. Ribosomes are essential for protein synthesis.

Processing

hnRNA splicing occurs to remove introns. Besides, poly A tail and a methyl guanosine triphosphate cap is added in order to increase its stability and to facilitate transportation of resulting *m*RNA.

***m*RNA**

*m*RNA or messanger RNA carries the genetic information present in DNA inside nucleus to the cytoplasm where proteins are synthesized in accordance to the genetic code.

Translation

Translation is the process during which the genetic information stored in the sequence of nucleotide in an *m*RNA molecule are translated into proteins.

Proteins

Proteins are polymers of amino acids that are essential for body organization. They also act as enzymes for various biochemical processes.

Phenotypic expression

If proteins that catalyze a particular process are produced more, than that process will occur more efficiently causing changes in phenotype.

40 Evolution

- Origin of life and its theoryes :

- Origin of life of protobiogenesis means, the origin of first life from non-livings on the earth about 3000 million years ago. Different religions have also advocated different ideas regarding the origin of life.

- **Abiogenesis** : Different living organisms evolved from mud, soil, meat, organic manure.
- **Biogenesis :** F. Reddy had proposed this principle in 17th century. According to him biogenesis means formation of new life from pre existing organisms through reproduction.
- **Meteorite Theory :** This theory believed in cosmozin, as per it for life formation cosmos came from other planets and on getting favourable conditions they develped into organisms.
- **Theory of Eternity :** As per this theory life was there when abiotic components were also present on earth in its earlier stage. This theory was not accepted by any one.
- **Theory of catatrophism :** Georges Cuvier proposed this theory. He thought that the universe has been subjected to catatropes or sudden revolution at different intervals.
- **Theory of special creation :** There were no evidences for this theory. However, Father Suarez believed that all elements were created within six days.
- **Theory of organic Evolution :** As per this theory, the world has been evolved and not been created. Non living substances have reacted to form organic compounds which developed into colloidal systems. Simple life was developed from them.
- **Oparin-Haldane Hypothesis :** Modern biologists strongly believe that life is originated in the re remote past from interaction of chemical substances. It was formulated by Haeckel, on this basis theory was develpoed by oparin and Haldane. This theory suggests that life is originated from simple inorganic substances. These substances might be transformed into a colloidal system to produce life.

(A) Chemical evolution

(1) **Origin of Earth :** It is now scientifically established that the earth was formed from the sun some 5000 millions years ago. When it was fragmented from the sun, it was a glowing fire containing a molten mass of gases and vapour of various elements. The temperature was very high about 5000°C to 6000°C. As the earth was moving away from the sun, it was getting cooled. The heavy elements like iron, nickel etc. occupied the core of the earth where as the lightest elements like helium, hydrogen, Oxygen, nitrogen carbon etc. occupied the atmosphere of the earth. The chemical changes leading to the formation of life is called chemical evolution or molecular evolution of origin of life.

(2) **Formation of water, Ammonia and Methane :** In lighest elements hydrogen was very active, It combined with nitrogen to form ammonia (NH_3) with Oxygen to form water (H_2O), and with carbon form methane (CH_4). As the temperature of the earth was high, ammonia and methane remained as gases and water as superheated streams.

$N_2 + 3H_2 \rightarrow 2NH_3$

$C+2H_2 \rightarrow CH_4$

$1/_2O_2+H_2 \rightarrow H_2O$

As years passed, the temperature of the earth came down, stream condensed into water and resulted in rain, but the rain drops were evaporated immediately, this procers was repeated for millions of years, the surface of the earth became cooled. Water gradually accumulated and this led to the formation of rivers, streams, lakes, seas and oceans. Compounds like ammonia, methane etc. were

dissolved in rain water and were accumulated in the sea. Mineral rocks also dissolved leading to the accumulation of mineral and salts in sea water. The first chemicals formed on the earth were water, ammonia and methane.

(3) **Formation of Micromolecule :** In molecular evolution the way of formation of micromolecules. amino acids, fatty acids, monosaccharides, purines, pyrimidines, adenosine monophosphate (AMP) and Adenosine diphosphate (ADP) appeared as colloidal droplets in the sea. These were distinct bodies which did not mix with the surrounding sea water. They were contents of proteins, nucleoproteins and other organic and inorganic molecules in various ratio. The surface layer of the droplet had the ability for selective absorption of susbstances from the medium.

(4) **Formation of macromolecules :** When the micromolecules were formed, they combined in various ways to form large molecules called macromolecules. This process is called polymerization.

(5) **Formation of Nucleic acid / Nucleoproteins :** The nucleic acid and proteins combined together to from nucleoproteins which are very important macromolecules to create a life.

(B) Biological evolution :

(1) Formation of coacervates : When macromolecules were formed they undergo aggregation and precipitation in the sea . Which led to the formation of organised structures called coacervates. They were distinct bodies. They contained proteins, nucleopreteins and other organic and inorganic molecules in various ratios.

(2) Formation of precells or protocells : The protocells were spherical in shape and a double - layered membrane was present around them. The protocells were heterotrophs; they obtained the energy formed by the fermentation of organic substances which were dissolved in the sea. Thus the protocells were anaerobic.

(3) Formation of precells to cells : When DNA-RNA system developed within protocells; they looked like a bacteria or virus. The DNA acquired the ability for self - duplication and protein synthesis. Gradually time passed the protocells differentiated into cells.

---

(1) From how many years ago the origin of first life from non-livings on the earth ?
(A) 3000 million years (B) 3000 Trilion years (C) 300 million years (D) 30,000 million years

(2) What was the characteristic of first living organisan ?
(A) replication (B) nutrition
(C) Adaptition and Biosynthesis (D) all the above

(3) By Whom proposed the theory of biogenesis in 17th centuary ?
(A) Aristotle (B) Halden (C) Cuvier (D) F. Reddy

(4) Who proposed the theory of catarophism ?
(A) Christian suarez (B) Cuvier (C) F. Reddy (D) Millor

(5) Who proposed the theory of special Creation ?
(A) Prayer (B) Aristotle (C) Christian Suarez (D) F. Reddy

(6) "The world has been evolved and not been created; Non living substances have reacted to form organic compounds which developed into colloidal systems," Find out the theory propose the above statements.
(A) Theory of organic Evolution (B) Substrate base (C) Meteority theory (D) Abiogenesis

(7) Who cleared that interaction of chemical substances initiated origin of life in the past ?
(A) Haldene (B) Operin (C) Haeckel (D) Father suarez

(8) In the beginning how much was the tempreture of the earth ?
(A) 5000^0C to 6000^0C (B) 6000^0C to 7000^0C (C) 7000^0C to 8000^0C (D) 8000^0C to 9000^0C

(9) Which elements are foumd in the core of the Earth ?
(A) Iron (B) Helium (C) Hydrogen (D) Carbon

(10) Which substance first take place in the earth envionment ?
(A) Helium (B) Hydrogen (C) Carbon (D) above three

(11) Which substance found in huge mass in primitive earth ?
(A) Carbon (B) Iron (C) Nickel (D) Phosphrous

(12) Which element is most active in primitive earth ?
(A) Iron (B) Hydrogen (C) Oxygen (D) Carbon

(13) Which compound is first formed in the earth ?
(A) H_2O (B) HCN (C) NH_2 (D) A and B

(14) Which chemical is first formed in the earth ?
(A) Ammonia (B) Methane (C) A and B both (D) None of the above

(15) Which is the step before molecular evolution ?
(A) Formation of Micromolecule (B) Origin of life
(C) Formation of Microsubstance (D) Formation of chemicals

(16) Which molecules found free in sea ?
(A) AMP (B) ADP
(C) A and B both (D) None of the above

(17) Which is macromolecule ?
(A) Polysaccharide (B) Nucleo protein (C) Proteins - lipids (D) Above three

(18) Name the phenomenon of micromolecules to macromolecules
(A) Conjugation (B) Polymerization (C) Multiple fission (D) Synthesize

(19) Which substances are found in nucleotide ?
(A) Purine (B) Sugar (C) A and B both (D) None of the above

(20) Which molecule is important to make a life ?
(A) Protein (B) Nucleoprotein (C) carbohydrates (D) Lipid

(21) By the macromolecules with the process of precipitation which structure is formed ?
(A) Microspheres (B) Micromolecule (C) Colloiddrop (D) Coacervates.

(22) Which substances have coacervates ?
(A) Proteins (B) Nucleoproteins
(C) Organic and inorgamic molecules (D) All above

(23) When DNA -RNA system developed in protocells, In which form it was ?
(A) Bacteria (B) Spherical (C) Bilayered (D) None of these

Answers : (1-A), (2-D), (3-D), (4-B), (5-C), (6-A), (7-C), (8-A), (9-A), (10-D), (11-A), (12-B), (13-A), (14- C), (15-A), (16-C), (17-D), (18-B), (19-C), (20-B), (21-D), (22-D), (23-A)

- **Urey'sand Miller's Experiment** : The origin of life by molecular evolution first proposed by Oparin and Haldane. Urey and miller in their experiment created a condition which was similar of the primitive earth. The steam was allowed to mix with vapours of ammonia, methane and hydrogen in a reaction chamber. The mixture was treated with an electric shock by electrodes, mixture was then allowed to cool in a condensor and liquified; then liquid was collected in another flask. After weeks of treatment the liquid was analysed by chromatography. The liquid contained simple organic compounds such as amino acids, hydroxy acids and alophatic acid.

- Evidences for evolution :

(1) Morphology : The sources of evidenes of evolution from morphology and comparitive anatomy are the following.

(i) **Homologous organs :** The organs which are similar in their morphology, anatomy and embryology but dissimilar in their functions are called homologous organs. The relationship between homologeus organs is called homology.

eg: The forelimbs of higher vertebrates, The thorn and tendrils of Bougainvillea and cucurbita respectively.

(ii) **Analogous organs :** The organs which are superficially similar but anatomically dissimilar doing similar functions are called analogous organs. Similarity of structures connects with similarity of function is termed analogy.

eg: wings of insects, birds and bats, fins of fishes and whales.

(iii) **Vestigial organs :** Vestigial organs are the useless and functionless degenerate structures which are functional in some other animals or in ancestors.

eg: Vermiform Appendix, Nictitating membrane, Ear muscles, wisdom teeth.

(iv) **Connecting Links** : The animals exhibiting characters of two adjacent taxonomic groups are called connecting links.

eg: Peripatus : Connecting link between Annelida and Arthro poda.

Balanoglossus : Connecting link between invertebrates and chordates.

Lung fish : Connecting link between fishes and amphibians.

Archeopterix : Connecting link between reptiles and Aves.

(2) **Embryology :** The early embryos of all vertebrates are very similar in their appearance. This will be very clear when embryos of a fish, an amphibian, a reptile, an opposum, a monkey and a man are arranged side by side.

(3) **Physiological Evidence** : The studies on physiology and biochemislry prove that the process of evolution has occured biochemically. The protoplasm which is the basic unit of life has the same quantitative and qualitative attributes in cell of living organism.

(4) **Palaeontological Evidence** : The direct evidences of the Process of evolution can be obtained by comparing fossils of the organisms that lived in the part with the organisms living today. The fossils are the written documents of evolution.

e.g. Available fossils of mesozoic era of different dinasaurs, were large-sized Lizards, that can be proved.

(24) Who proposed molecular evolution of life ?

(A) Oparin (B) Haldane (C) A and B both (D) None of the above

(25) In which year Urey and Miller and his student proved molecular evolution ?

(A) 1947 (B) 1948 (C) 1951 (D) 1953

(26) Which method of anylysis of experiment by Urey and Miller ?

(A) Chromotography (B) Staining Method (C) Isolation method (D) None of the above

(27) In plant kingdom, sequence of evolution is according to which plants division ?

(A) Thallophyta (B) Bryophyta (C) Pteridophyta (D) All above

(28) Which group is first developed with well developed nucleus in animal kingdom ?

(A) Archeae (B) Protista (C) Monera (D) Eukaryota

(29) Which type of animals are evolved after protista ?

(A) Multicellular (B) Without tissue (C) A and B both (D) None of the above

(30) In which class monkey, ape and man include ?

(A) Aquatic (B) Mammals (C) Aves (D) Reptiles

(31) Which living organisms evolved first during evolutionary order ?

(A) Amphibians (B) Marine´ (C) reptiles (D) Vertebrates

(32) Which word is proper for the similarity in their morphology anatomy and embryology ?

(A) Analogous (B) Vertigeal (C) Homologous organs (D) None of above

(33) Example of Homologous organs.

(A) Wings of insects, birds and bats (B) The thorn and tendrili of bougainvillea and cucurbita

(C) The forelimbs of higher vertebrates (D) B and C both

(34) The organs which are superficially similar but anatomically dissimilar doing similar functions are called ..... .

(A) Analogous organs (B) Homologous organs (C) Vestigial organs (D) none of above

(35) Which of the following example of analogous organs ?

(A) Vermiform Appendix, (B) Nictitating membrane and ear muscles

(C) The thorn and tendrils of Bougainvillea and cucurbita (D) A and B both

Answers : (24-C), (25-D), (26-A), (27-D), (19-C), (20-B), (21-D), (22-D), (23-A), (24-C), (25-D), (26-A), (27-D), (28-B), (29-C), (30-B), (31-B), (32-C), (33-D), (34-A), (35-B)

(36) The useless and functionless degenerate structures which were functional in some other animals or in ancestors are called ..... .

(A) Homologous organ (B) Analogous organ (C) Vestigial organ (D) None of the above

(37) Which example is vestigial organ ?

(A) Vemiform appendix (B) Nictitating membare (C) Ear muscles (D) above all three

(38) The animals exhibiting characters of two adjacent taxonomic groups are called .....

(A) Vestigial organs (B) Analogous organs (C) Connecting link (D) Homologous organs

(39) Connecting link of peripatus X ..... and Y .....

(A) X = Annelida Y = Arthropoda (B) X = Amphibians Y = Annelida

(C) X = Fishes Y = Amphibians (D) X = Chordates Y = Fishes

(40) Which two groups are connected by lungfish.

(A) Annelida and Arthropoda (B) Invertebrate and chordates

(C) Reptiles and fishes (D) Fishes and amphibian

(41) Connecting link of Balanoglossus X = ... and Y = ...

(A) X = Fishes Y = Amphibians (B) X = Annelida Y = Arthropoda

(C) X = Invertebrates Y = Chordates (D) X = Reptiles Y = Chordates

(42) Which animal is conncting between Reptiles and Aves

(A) Balanoglossus (B) Archeopterix (C) Peripatus (D) Lung fish

(43) Available fossils of Mesozoic era of different dinosaurs were large sized Lizards, That can be proved by which evidence ?

(A) Embryology (B) Physiologial evidence (C) Palaeontological Evidence (D) None of the above

Answer : (36-C), (37-D), (38-C), (39-A), (40-D), (41-C), (42-B), (43-C)

- **Adaptive Radiation :**

Organisms depend on the environment for their survival. Environmental factors are not stable. They are changing time to time and place to place. Hence organisms also adjust themselves to the changes of the environment in which they live. This adjustment of the organisms to their environment is called adaptation. When organisms of the same group occupy different environ ments or habitats, They develop different Kinds of adaptations. Hence organisms of the same group show different adaptation in different habitats and this is called adaptive radiation. The law of adaptive radiation was proposed by osborn.

- Adaptive radiation is nothing but divergence or divergent evolution. It is caused by following.
- (1) Need for food (ii) Need for safety (iii) Need for better breeding grounds (iv) Migration to a new habitat (v) Absence of enemies (vi) Isolation etc. Adaptive radiation is broadly classified into three types:
- **Local Adaptive Branching** : It refers to the development of diversification in a largely distributed population. eg: In Africa, two distinct types of Rhinoceres are found.

(i) One is grazing which lives in openland areas and (ii) second is a browsing type which lives in wooded areas.

(ii) **Continental Adaptive Radiation** : It refers to the evolutionary lines occurring in organisms of particular classes living in a continent.

eg: Australian marsupials, Each marsupial differs from other

(iii) **Contemporareous Radiation** : It refers to the adaptive radiations in the great zoo-geo graphic division of the world.

Darwin during his journey went to Galapagos islands. There he observed diversity in small black birds (finches), later this bird is known as Darwin finches. He observed that these were many varieties of finches in the same island. He conjectured that the varieties evolved on the island itself from the original seed - eating finch. Modifications in the beaks of finches according to the habitat and the avaibility of food. Modifications in the beaks of finches developed according to the habitat and the availability of food.

- **Biological Evolution :** The Sequential evolution of an organism is a result of adaptation as per the requirement. With respect to this different theories were proposed. It abstracts are as follows.
- **Lamarckism :** Lamarckism is also referred as "Inheritance of acquired characters'. The main aspects of the principle were as under.

(1) Organisms and their organs are constantly increasing in size

(2) Development of organs is induced and maintained under environmental pressure.

(3) Those organs which are constantly used they also develop constantly. Unused organs become defunct gradually or degenerate.

(4) Those characters which organism acquires in this way are inherited to their offsprings. Thus, the development of a charaeters increases or decreases generation to generation

- **Drawbacks of Lamarckism** :
- The first point is in cases of horse and elephant it is true. But in case of flowering plants it is not true.
- The second point is also not true. eg: The neck of an Giraffe cannot be made longer as per its wish, more over organs cannot be moulded as per of organism wish
- The third point oppears partially true. Muscles can be developed by constant use loss of limbs, due to disuse in snakes or weak eyesight in cave-dwellers can be quoted as other examples. But eyes of a voraceous reader do not become larger or brighter. The heart which beats through out life does not become bigger.

It is also difficult ot accept that acquired characters get inherited. Hence, Lamarkirm is not accepted.

- **Darwinism :** The theory of 'Natural selection' Proposed by Darwin is the prime one to explain evolution of living world. It was proposed in 1859 in his book title 'Origin of species by means of Natural selection'. New species originate only from a pre existing one. Darwinism is based on the following main aspects.
- **Large Population** : The reproduetive rate in all organism is high, if all the offsprings survive the Population growth will be unimaginable. If all organisms reproduce and produce a large number of offsprinsg the world will be flooded by them. This does not happen. The population of all species, generally remains the same from generation to generation. Diseases, lack of food and other natural factors are responsible for this.
- **Struggle for Existense or for survival** : If the resources are limited and the number of organisms dependent on them is unlimited. it is obvious that all organisms can not get sufficient resources. Thus a competition for resources amongst organisms of one species becomes inevitable. Such competition can also be with organisms of another species, if their, requirements are identical. Thus organisms have to compete for survival, growth and reproduetion.
- **Natural selection or survival of the fittest** : It is quite obvious that when organisms compete for specific resource, some may get it and others may not get it or may get in enough quantity. some may fail or success which depends on the nature. Those who succeed will live longer and reproduce Thus, their hereditary characters will survive in the next generation.
- **Variations** : All individual of a species are not alike. These differences in characters amongst individuals of a species are called variations. Those organisms which can use their environment in the best way succeed and those cannot do so fail. This phenomenon is called survival of the fittest. In due course of time, the degree of variations increases so that the organisms become totally isolated from their parents and constitute a new species.
- **Limitatious of Darwinism :** The greatest draw back of Darwinism is that it does not explain the origin of variations; it characters are hereditary; for which genes are responsible If we can explain the mechanism of inheritance, we can explain the variations in offsprings and their conservation over generations.
- **De Veries theory** : De vries theory, is also called ' Theory of mutation.'
- New characters, ie new variations in characters appear spontaneously. Such a sudden variation is called mutation.
- Once formed, a mutation becomes permanent. This would mean that it will appear constantly once it is formed.
- Mutation occurs simultaneously in a large number of organisms. This occurs frequently. Thus, Chances of natural setection improve.
- Mutation does not occur in a specific direction. Mutation creates many variations.
- De veries does not believe that evolution is a gradual process.
- **Modern concept of Evolution** : According to the modern concept, there are five basic factors involved in the process of biological evolution. These are (i) Gene mutations (ii) Changes in chromosome structures and number (iii) Genetic recombination (iv) Natural selection and (v) Reproductive isolation.

Besides these, there two accessory processes (i) migration of individuals from one population to another and (ii) Hybrization between races, species and related genera, are responsible for biological evolution.

Some important worker like Dobzhansky; R.A Fisher, J.B.S. Haldane, sewall wright, mayr and G.L. Stebbins gave their contribution to the modern synthetic theory.

- **Mechanism of Evolution** : Natural selection, adaptation to habitat, and mutation are the major and sequencial factors for mechaninsm of evolution.

(44) For the survival, living organisms depend on ..... .

(A) water (B) environment (C) gene (D) all above

(45) Adaptive Radiation is nothing but divergence or divergent evolution is caused by ..... .

(A) Need for food (B) Isolation (C) Absence of enemies (D) all above three

(46) How many types of adaptive radiation ?

(A) Three (B) Five (C) Four (D) Six

(47) In which bird Darwin observed diversity

(A) Black bird (B) White bird (C) Redbird (D) Yellow bird

(48) De vries theory is also called

(A) Isolation (B) Survival of fittest (C) Theory of mutation (D) Variation

(49) How many factors involved in the process of modern concept of biological evolution ?

(A) 4 (B) 3 (C) 5 (D) 6

(50) The differences in characters amongst individuals of a species are called .....

(A) Natural selection (B) Variation (C) Struggle for existence (D) Recombinant of gene

Answers : (44-B), (45-D), (46-A), (47-A), (48-C), (49-C), (50-B)

- **Law of Hardy-Weinberg** : They laid the foundation for the development of population genetics. A clear understanding of Hardy - Weinbery law gains through knowledge of gene pool gene frequencey.

  (i) **Genepool** : The sum total of genes present in a Mendelian population means gene pool.

  (ii) **Gene frequency** : The ratio of a gene in a gene pool or in a population is called gene frequency.

  When the gene frequency of one allele is known, the frequency of ther allele in the population can be calculated by applying a simple formula. If the gene frequency of M-allele is p and that of m-allele is q. Then p+q = 1. When q is known, p can be calculated, p = 1-q; similarly, when p is known q can be calculated; q = 1 - p; for example p is 0.6. Then q = 1 - p = 1 – 0.6 = 0.4.

  Practical Application of the law : This formula can be appried to any population to find out the frequency of genes, Let us apply this formula to a hamster population containing equal number of M and m genes.

  The Freqvency of m gene in population is = 50 % = 1/2

  So. $q = 1/2 = 0.5$

  $(p + q)^2 = p^2 + 2pq + q^2$

  $= (0.5)^2 + 2(0.5)(0.5) + (0.5)^2$

  $= 0.25 + 0.5 + 0.25$

  = 25 %, MM + 5.0 % Mm + 25 % mm

  Importante : This law states that the gene frequencies in a large population, remain constant generation after generation when there is no selection and mutation. In small population this equilibrium cannot be maintained.

- When the population is in equilibrium there is no possibility of change and hence the rate of evolution is zero. Evolution occurs only when this equilibrium is upset.

- **Gene flow** : Animals are not static. They have the tendency to migrate. When an animal migrate and comes in contact with another population, it mates with the immate of the population. Thus the genes of one population or transferred in to another population. This is called geneflow. If genes are carried to a population, where these genes previously did not exist, this event of geneflow can be very important source of genetic variation.
- **Genetic Drift**: Genetic drift is an evolutionary force in small population. This matter was described by Sewall wright 1931. Hence it is called Sewall wright effect. According to Hardyweinberg law in a large population the gene frequency remains constant from generation to generation when there is no selection and mutation. But in small populartion, the gene frequerncies are found to fluctuate purely by chance This change in the frequency of gene purely by chance is called geneti drift. The effect of genetic drift is insignificant in large population. But in small population it has a significant effect. As result of this, in small population, some gene may be reduced in frequency or even lost by chance and other may be increased in frequency by chance.
- **Brief account of Evolution :** Earth originated about 500 millions years ago.
- Life in the first form of cell, originated about 3000 millions years ago, some of these cells had the ability to release O_2.
- About 500 million years ago invertebrate were originated.
- sea weeds and few plants existed probably about 320 millions years ago. Fishes originated 360 millions, amphibians 325 million, Reptiles 250 million, Mammals 165 million and First man originated before 2 million years ago.
- The account of biological evolution and age of different animals will be still more clear if the time of the origin of life is placed to a 12 hours scale. If it is imagined that life originated at 12.00 midnight, then fishes appeared at 8.00 pm. amphibians at 8.30 p.m, reptiles appeared at 9.00 pm, mammals appeared at 10.00 p.m. and man appeared at 1159 pm

  Scale :

  1 second = 52, 000 yrs, / minute = 3,25,000 yrs 1 hour = 187,5000 yrs.

---

(51) How many million year ago life have originated ?

(A) 5000 (B) 3000 (C) 4000 (D) 4500

(52) How many millions year ago sea weeds are few plants existed ?

(A) 120 (B) 400 (C) 220 (D) 320

(53) The origin of life is compared to a ..... scale

(A) 13 hours (B) 10 hours(C) 12 hours (D) 14 hours

(54) At what time fishes are appeared

(A) 9:00 PM (B) 8:00 PM (C) 8:00 AM (D) 8:05 PM

(55) In geological clock, 1 miniute is equal to how many years ?

(A) 3,25,000 (B) 52,000 (C) 1,87,000 (D) None of all above

(56) In which year Handy-weinberg law was proposed ?

(A) 1908 (B) 1907 (C) 1808 (D) 1900

(57) What is the importance of Hardy-weinberg law ?

(A) Evolution (B) Gene pool (C) genetics (D) genetic variation

(58) The sum total of genes present in a mendelian popolation is called ..... .

(A) gene pool (B) gene frequency (C) genetic variation (D) Genetics.

(59) How many years ago the earth was originted ?

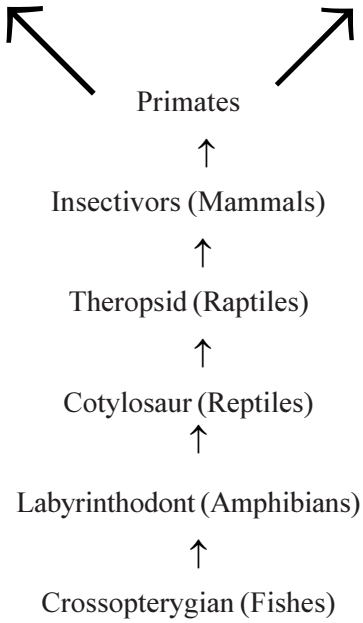
(A) 6000 (B) 5500 (C) 5000 (D) 6500

Answers : 51 (B), 52 (D) 53 (C), 54 (B), 55 (A), 56 (A), 57 (A), 58 (A), 59 (C)

- Origin and Evolution of Man : About 480 million years ago our distant ancestors of our order lived in fresh water ponds in the form of crossopterygian fishes. The other ancestors inthe path of primates evolution are as under.

Prosimians **Anthropoids**

[Tree shrew, Lemur, Lorise etc] [Gibbon, Orangutan, Chimpanzee, Gorilla] Monkeys and man]



- **Human Euolutionary Trends:** Like other animals, human evolution exhibits directional Progressive changes, which are summurised below:

  The change of arboreal life into terrestrial life is the most important evolutionary trend, Loss of opposite toes from hind limbs, Development of erect posture, Development of bipedal locomotion, The development of chin, Increase in the size of brain development of intelligence, Disappearance of simian shelf, Use of fore limbs for non-locomotory purposes, etc.

- **Fossil Record of Man's Evolution:** The gradual evolution of man from ape is fully supported by available fossils.

(1) **Propliopithecus** : It was an ape-like primate. But in the possession of short arms, it resembled man. It lived, about 30 million years ago.

(2) **Aegytopithecus** : It is similar to proplipithecus; but it is more identical to apes than propliopithecus. It is believed that it was ancestral of the Dryopithecus. Another view did not support this.

(3) **Dryopithecus** : It is a group of apes that lived in miocene period, about 20 million years ago. It decended from Aegytopithecus or propliopithecus. It's forelimbs were shorter than hindlimbs. In this respect, it is belived that it is the distant ancestor of man. It is also ancestor of modern apes like chimpanzee and Gorilla.

(4) **Oreopithecus** : In the structure of teeth, small face of and erect walking, it resembles to man, but having long forelimbs it resembles to apes. Straus (1963) and simpson (1963) Suggest that man and oreopithecus have parallel evolution and hence it is not ancestral to man.

(5) **Ramapithecus** : It lived 12 to 14 millons years ago Late miocone and early pliocene. The denitition is more identical to dentition of man. Fossils of Ramapithecus was collected from India and Africa.

(6) **Kenyapithecus** : It is closely related to Ramapithecus.Its fossil is collected from East Africa.

(7) **Australopithecus** : It was 2 to 5 millions years old. The characters of Australopithecus like man and ape.

Man like charactes : Erect posture, Bipedal locomotion, Dentition is like that of man.

Apelike characters : The teeth were larger than modernman, Absence of chin, the eye brow ridges projected over the eyes.

(8) **Homoerectus** : Their fossils were collected from java so it is called Java man. Other fossils were collected from peking in china so, it was commonly called peking man. The Java man and peking man are similar. They lived about 5,00,000 years ago. It forms connecting link between Aurtralopithecus and Homo sapiens (Modern man).

Characters : They had upright bipedal locomotion, They were slightly taller than Austrolopithecus. The face was chinless. They used fire and variety of tools, They inhabited caves. They were hunters.

(9) **Homo sapiens** : Homo sapiens were descended from Homoerectus. Number of fossils are known as below.

(i) **Neanderthal man** : The fossils were collected form the Neanderthal valley of Germany. They existed about 75,000 years ago and became extinct by 25,000 years ago.

Characters : Their eyebrow ridges were heavy and protruding. They have no chin.Their teeth were large. They prepared their tools much more skillfully than the pre-previous men. Their cranial capacity was about 1400cc.

(ii) **Rhodesian man**: The fossils were collected from Rhodesia the cranial capacity was about 1300 CC.

(iii) **Cro-Magnon man** : These were the men who lived in Europe during the last 30,000 years. They possessed almost all characteristics of modern man.

(iv) **Modernman** : The successors of cro-Magnon man, which originated about 8000 years ago attempted for food production from plants and domestication of animals led them towards present day civilization. The cranial capacity of mordern man is 1450 CC.

Man classified different races according to appearance, characters and zoo-geographical distribution.

Main Man subspecies : Caucasiod race, Negroid race Mongoloid race, American race, Indian race. and Australian race.

---

(60) How many million years ago our distant ancestors of our order lived in fresh water pond in the form of crossopterygian fishes ?

(A) 470 million years (B) 500 million years (C) 480 million years (D) 400 million years

(61) How many years ago propliopithecus was lived ?

(A) 30 million (B) 32 million (C) 35 million (D) 40 million

(62) Aegyptopithecus is similar to ..... .

(A) Dryopithecus (B) Oreopithecus (C) Propilopithecus (D) Ramapithecus

(63) From where the fossils of Ramapithecus found ?

(A) India and srilanka (B) India and Africa (C) Pakistan and srilanka

(D) America and Africa.

(64) Whos closely related with kenyapithecus ?

(A) Australopithecus (B) Oreopithecus (C) Ramapithecus (D) Dryopithecus

(65) Which characters of Australopithecus is like man

(A) Erect posture (B) inhabited in cave (C) Absence of chin (D) Hunter

(66) How many years ago the java man and peking man was lived ?

(A) 5,00,000 years (B) 3,00,000 years (C) 5,50,000 years (D) 4,00,000 years

(67) From whom Homo sapiens were descended ?

(A) Ramapithecus (B) Kenyapithecus (C) Australopithecus (D) Homo erectus

(68) How many years ago the fossils of Neanderthal man found ?

(A) 75,000 years (B) 70,000 years (C) 76,000 years (D) 60,000 years

(69) From where the fossils of Rhodesion man collected ?

(A) Valley of Germany (B) Rhodesia (C) Australia (D) Iran

(70) When the successors of cro-magnon man originated ?

(A) 7000 year (B) 9000 year (C) 8000 year (D) 8800 year

(71) What is the cranial capacity of modern man ?

(A) 1450CC (B) 1400CC (C) 1500CC (D) 1550CC

(72) How many races of human are found ?

(A) Five (B) Four (C) Six (D) seven

(73) Which is main races of human ?

(A) Erectus (B) Caucasoid (C) Homo sapien (D) Australopithecus

(74) When Neanderthalman extinct ?

(A) 30,000 years ago (B) 35,000 years ago (C) 25,500 years ago (D)25,000 years ago

(75) Which group was decended from aegyptopithecus or propliopithecus ?

(A) Ramapithecus (B) Australopithecus (C) Dryopithecus (D) Oreopithecus

(76) Straus and simpson suggest that X and Y have parallel evolution ?

(A) X = man, Y = Oreopithecus (B) X = Kenyapithecus, Y = Oreopithecus

(C) X = man, Y = Australopithecus (D) X = man, Y = dryopithecus.

(77) What is the cranial capacity of Neanderthalman Rhodisian man and modern man respectively ?

(A) 1450, 1300CC, 1400CC (B) 1400CC, 1300CC, 1450CC

(C) 1300CC, 1400CC, 1450CC (D) 1450CC, 1400CC, 1300CC

Answers : (60-C), (61-A), (62-B), (63-C), (64-C), (65-A), (66-A), (67-D), (68-A), (69-A), (70-C), (71-A), (72- A), (73-B), (74-D), (75-C), (76-A), (77-B)

- **A - Assertion R - Reason**

Answer the following questions, by choosing the appropriate option.

(A) A and R both are true and R is a explanation of A.

(B) A and R both are true and R is not a explanation of A.

(C) A is true and R is false

(D) A is false and R is true

(78) Assertion A : After the formation of macromolecules they undergo aggregation and precipitation in the sea

Reason R : The formation of organ structures called coacervates.

(A) (B) (C) (D)

(79) Assertion A : At the end of longtime complicated living organism developed

Reasin R : According to requirement of living organism they change their body organisation.

(A) (B) (C) (D)

(80) Assertion A : Primitive earth have huge mass of hydrogen, Nitrogen, carbon, and oxygen

Reason R : Among them Nitrogen is most active

(A) (B) (C) (D)

(81) Assertion A : Marine invertebrate animals originated

Reason R : Protista originated having developed nucleous.

(A) (B) (C) (D)

(82) Assertion A : If population is in equilibrium

Reason R : The rate of evolution is zero

(A) (B) (C) (D)

(83) Assertion A : Lamarkism is acceptable

Reason R : It is difficult to accept that acquired characters get inherited.

(A) (B) (C) (D)

(84) Assertion A : Mutations occurs simultaneously in a large number of organisms

Reason R : Thus chances of natural setection improve

(A) (B) (C) (D)

(85) Assertion A : The protocells obtained the energy formed by fermentation of organic substance

Reason R : In primitive atamosphere lack of free oxygen.

(A) (B) (C) (D)

(86) Assertion A : Protocell were heterotrophic

Reason R : Protocell were anerobic

(A) (B) (C) (D)

(87) Assertion A : DDT nearly uneffective against moquitoes

Reason R : Continuous consumption of DDT, DDT sensible gene resistance gene become mutant.

(A) (B) (C) (D)

Answers : (78-B), (79-A), (80-C), (81-D), (82-A), (83-D), (84-A), (85-A), (86-B), (87-A)

● Match the column and find out the correct option trom following.

(88)

Collumn I	**Collumn II**
(i) Genefic drift	(p) The gene frequencies are found to fluctuate purely by chance
(ii) Gene flow	
(iii) Gene frequency	(q) The genes of one population are transferred in to another population
(iv) Gene pool	(r) The ratio of a gene in a gene pool
	(s) The sum total of genes present in a mendelian population.

(A) (i-q) (ii-r) (iii-p) (iv-s)
(B) (i-r) (ii-s) (iii-q) (iv-p)
(C) (i-p) (ii-q) (iii-r) (iv-s)
(D) (i-s) (ii-p) (iii-r) (iv-q)

(89) **Collumn I** **Collumn II** (Connecting link)

(i) Peripatus	(p) Invertebrate and chordates	(A) (i-r) (ii-s) (iii-q) (iv-p)
(ii) Balanoglossus	(q) Fishes and Amphibians	(B) (i-p) (ii-r) (iii-q) (iv-s)
(iii) Lung fish	(r) Annelida and Arthropoda	(C) (i-r) (ii-q) (iii-p) (iv-s)
(iv) Archeopterix	(s) Reptiles and Aves	(D) (i-r) (ii-p) (iii-q) (iv-s)

(90)

Collumn I	**Collumn II**	
(i) F.Reddy	(p) Theory of catatrophism	(A) (i-p) (ii-s) (iii-r) (iv-q)
(ii) Qvier	(q) Theory of mutation	(B) (i-r) (ii-p) (iii-s) (iv-q)
(iii) S wright	(r) Biogenesis	(C) (i-s) (ii-r) (iii-p) (iv-q)
(iv) De vries	(s) Modern concept of Evolution	(D) (i-q) (ii-r) (iii-p) (iv-s)

(91)

Collumn I	**Collumn II**	
(i) Ramapithecus	(p) Java	(A) (i-p) (ii-s) (iii-q) (iv-r)
(ii) Kenya pithecus	(q) Germany	(B) (i-r) (ii-p) (iii-q) (iv-s)
(iii) Homo erectus	(r) India and Africa	(C) (i-r) (ii-s) (iii-p) (iv-q)
(iv) Neanderthal man	(s) East Africa.	(D) (i-q) (ii-r) (iii-s) (iv-p)

Answers : (88-C), (89-D), (90-B), (91-C)

- **True-Flase (T - F)**

Given sentence are true or flase choose appropriate answer.

(92) (i) Coacervates is a free molecules.

(ii) Protocells were heterotrophs.

(iii) Oparin considered the coacervates as the sole living molecules.

(iv) The protocells in course of time differentiated in to cells.

(A) TFTT (B) TTTT (C) FTFT (D) FTTT

(93) According to Lamarkism

(i) Organisms and their organs are constantly increasing in size

(ii) Development of organs is supress under enviornment pressure

(iii) Unused organs become defunct gradually

(iv) These characters which organism acquires in this way are inherited to their of spring.

(A) TFTT (B) TTFT (C) TFTF (D) TFTT

(94) (i) The earth was formed from the sun some 5000 billions year ago.

(ii) The temperature was 3000°C to 4000°C when earth was formed.

(iii) The heavy elements occupied the core of the earth.

(iv) Life originted only from lightest elements

(A) TFTT (B) FTTF (C) FTFF (D) FFTT

(95) (i) Analogous organs : insects, birds

(ii) Vestigial organs : Vermiform Appendix wisdom teeth

(iii) Homologous organs : Man, bird, whale

(iv) Connecting Link : Fishes and Reptiles

(A) TFFT (B) TTFF (C) TTTF (D) TTTT

(96) According to gene flow

(i) Animal have the tendency to migrate

(ii) Animal not mating with the inmate of the population.

(iii) The gene of one population are not transferred in to another population.

(iv) The event of gene flow can be very important source of genetic variation

(A) TFTT (B) TFFT (C) FFFT (D) TTFT

(97) According to Genetic Drift

(i) In a large population the gene frequency remains constant from generation to generation when there is selection.

(ii) In small population the gene frequencies are found to stable purely by chance.

(iii) The effect of genetic drift is significant in large population.

(iv) In small population some gene may be increased in frequency

(A) FTFT (B) TFTF (C) FFFT (D) FFFF

(98) (i) The earth was formed some 5000 millon years ago.

(ii) Life originatel about 3000 millions years ago.

(iii) Invertebrate originated about 500 millions years ago

(iv) Sea weed originated about 320 million years ago.

(A) TTFF (B) TTFF (C) TFTT (D) TTTT

(99) Texom and originated in million years.

(i) Fishes - 500 Millon years ago (ii) Amphibian 400 Millon years ago

(iii) Reptiles - 250 Millon years ago (iv) Mammals 165 million years ago

(A) TTFF (B) FFTT (C) TFTF (D) FTFT

(100) According to characters of Homo erectus

(i) They were slightly taller than australopithecus

(ii) The face was chinless

(iii) They prepared their tools much more skillfully than the preprevious men

(iv) They inhabited in caves.

(A) TTFT (B) TFFT (C) FFTT (D) TFTF

(101) According to characters of Neanderthalman

(i) They had chin

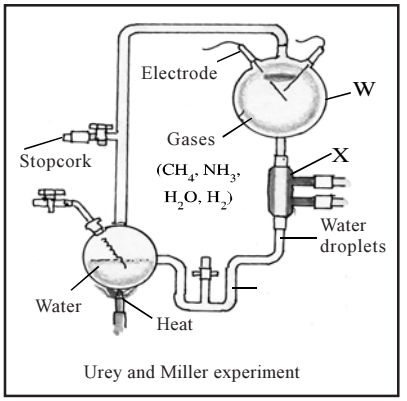
(ii) Their teeth were large

(iii) The cranial capacity was about 1300 C.C.

(iv) Their eyebrow ridges were heavy and protruding

(A) FTFF (B) TTFT (C) TFTF (D) FTFT

Answers : (92-B), (93-A), (94-D), (95-C), (96-B), (97-D), (98-D), (99-B), (100-A), (101-D)



Urey and Miller experiment

(102) Identify the nane of W and X

(A) W - Reaction chamber X - Stopcork (B) W - Condenser X : - Reaction Chamber

(C) W Reaction chamber X : Condensor (D) W condensor X : Stopcork

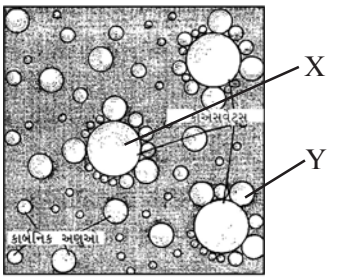
(103) Give the function of W

(A) To Separate the substanee of mixture

(B) To help the mixture cold

(C) Path for addition of mixture

(D) The mixture was treated with an electric shock.



Coacervates

(104) Indicate part x in given diagram, which type of absorbing capacity, occur for substances of surface medium ?

(A) impermeable (B) Permeable (C) Selectivly permeable (D) Non selectively

(105) What is the chemical nature of 'X' organic molecule

(A) Organic molecule (B) Nucleo protein (C) Proteins (D) alll above three

(106) Give the name indicate y

(A) Organic molecule (B) Protocells (C) inorganic molecule (D) Molecular particles


Y


X

(107) Give the name of X and Y

(A) X - Archeopterix Y - Balanoglssus (B) X - Balanoglossus Y = Peripatus

(C) X - Peripatus Y -Balanoglossus (D) X - Balanoglossus Y = Archeopterix

(108) Give, the name of connecting link of X and Y

(A) X - fishes and Amphibians Y - invertebratesand chordates

(B) X - Annelida and Arthropoda Y - invertebrates and chordates

(C) X - Invertibrates and chordeaes Y - Annelida and Arthropoda

(D) X - Annelida and Arthropoda Y = fishes and Amphibiam

Answers : 102 (C), 103 (D), 104 (C), 105 (D), 106 (A), 107 (B), 108 (B)

● **NEET Questions :**

(109) Random reproduction remains balanced Which of the following undirectional method brings genetic fluctuation ?

(A) Migration (B) Genetic drift (C) Mutation (D) Selection

(110) Random genetic drifts in a population are a result of

(A) Small size of population (B) Major genetic differences in individuals

(C) Interspecific hybridization (D) Slower rate of mutation

(111) Which type of evidences shows that Human being is more related to chimpanzee than man like apes ?

(A) Evidences of DNA of only sex chromosomes

(B) Comparison of chromosomes. - according to Morphology.

(C) Fossils, only mitochondria and DNA and evidences of fossils.

(D) Sex chromosomes, autosomes and DNA taken up from mitochondria

(Hint): Sex chromosomes, Autosomes and mitochondrial DNA shows genomic limitations. Chromosomal morphology is onesided.

(112) Which of the following is accurate method to decide fossil dating ?

(A) Carbon - dating method (B) Potassium - Argon method

(C) Electron - spin resonence method. (D) Uranium - Lead method.

(Hint): For Fossil dating electron spin resonence method is very appropriate method.

(113) There are two opposing opinions regarding the evolotion of modern man. According to one Homo erectus is the ancestor of the modern man, Based on genetic differences, The origin of modern man is African. Based on the DNA studies, what do the differences indicate ?

(A) More differences in Asia as compared to Africa

(B) More differences in Africa as compared to Asia

(C) Similar differences in Africa and Asia.

(D) No differences in Africa, only differences are seen in Asia.

(Hint): According to Neo-Darwinism, Variation is the main cause of evolution.

(114) De- Vris Organic evolution theory of mutation was given after research on ..... ?

(A) Pisum sativum (B) Drosophila melanogaster (C)Oenothera lamarckiana (D) Alicia rosa

(Hint): Pisum sativum - Mendel

Drosophila melanogaster - T.H. Morgan)

(115) Evolutionary History of living being is known as

(A) History of ancestors (B) Fossil study (C) Ontology (D) Phylogeny

(Hint): Study of fossils is known as palaentology.

Ontology is study of embryonic development.

Phylogeny is history of living being evolution.

(116) Which of the following is not living fossil ?

(A) Sphenodon. (B) Archeopterix (C) Peripatus (D) King crab

(Hint): Archeopterix fossil are obtained from meoszoic basins. They were connecting links between reptiles and Aves.

(117) When species of two different geneology found together as a result of the event known as ..... adaptation.

(A) Homology (B) Mutation (C) Analogy (D) Variation

(118) Which of the following is a correct statement ?

(A) There is no evidence of gills during the embryonic development of mammals.

(B) All plant and animal cells are totipotent.

(C) Ontogeny recapitulates phylogeny.

(D) Stem cells are special cells.

(Hint): Ontogeny recapitulates phylogeny.

(119) What is common amongst parrot, platypus and knagaroo ?

(A) Teethless jaws (B) Active tail at the posterior end.

(C) Similar egg cell (ovum) (D) Homeothermy.

(Hint) :Only birds and mammals are homeotherms.

(120) Which of the following is the result of geographic isolation ?

(A) Checks speciation (B) Speciation due to reproductive isolation

(C) Random formation of new species (D) No differences in the separated organisms.

(121) Which of the following birds has reptilian ancestors ?

(A) Scales on forelimbs. (B) Four chambered heart

(C) Presence of crop and gizzard in the digestive canal (D) Calcareous Eggs.

(Hint): Calcareous eggs and scales on the hind limbs indicate reptilian ancestors

(122) The thorns on Bougainvilea stem and the tendrils on cucurbita are examples of ..... .

(A) Analogous organ (B) Homologous (C) Vestigeal organ (D) Regression

(Hint): Thorns of Bougainvillea. and tendrils of cucurbita are homologous organs.

(123) Which is the most important factor for the evolution of modern man from his ancestors ?

(A) Erect posture (B) Narrowing of jaws (C) Bifocal three dimensional vision.

(D) Increase in cranial eapacity.

(Hint): Increase in brain capacity is the main factor for the evolution of modren man.

(124) In a given region; the evolution of species from a point to different geographic regions, is called as

(A) Contemporareous Radiation (B) Natural selection (C) Radiation (D) Divergent evolution.

(Hint): The process of speciation is a given region from a point to different biogeographic regions is contemporaneous radiation process. For Example : Darwin's Finches, Australian Marsupials.

Answers : (109-B), (110-A), (111-D), (112-A), (113-A), (114-C), (115-D), (116-B), (117-C), (118-C), (119-D), (120-B), (121-D), (122-B), (123-D), (124-A)

Origin of Life

Theories of origin of unitvers

Nebular Hypotbesis

It was propsed by Kant-Leplace according to which earth originated about 4.5-5 billion years ago from a gaseous cloud solar nebula.

Big bang Theory

According to this theory of Abbe Lemaitre univers formed by a big bang (thermonuclear expansion) of a dense entilty.

Chemical Theory of orogin of lice

- Oparin- Haldane's of chemical origin theory is most widely accepted theory of origin of life. It involves some basic stpes that are :
- **Formation of inorganic molecules**
- From the gases present in early earth's atmosphere.
- **Formation of simple organic compounds**
- From the inorganic compounds formed in previous step.
- Spontaneous formation of complex organic molecules
- Simple organic molecules combined to from large organic complex biomolecules like starch, proteins, fatty acids.
- **Spontaenous formation of molecular aggregates, concervates, eobionts and first living cell**
- Complex organic compounds synthesized on primise earth grouned togeths spontaneously and due to intermolecular attraction formed large colloidal aggregates called coacervates or Micropheres which tuned into eobionts (controlled by nucleic acids). These developed cel membrane and formed first living cells or prokayotes.

Eukaryotes (Unicellular)

Evolution

(Latin evolvere-to unroll) Multicellular organisms evolved from tiny eukayryotic unicellular ones.

Evidence of evolution

Morphological and anatomical

Comparaive studies of morphology and anatomy, homology of organisms provide evolutionary proofs, e.g. vestigial organs, atavism etc.

Embryological

Comparative study of embryonic development of various organisms shows various similarities and resemblance to embryos of lower organisms indicating evolution (Biogenetic law of Haeckel)

Pateontological

Paleontology is the study of fossils (preserved remains of dead organisms). It provides most direst and reliable evidence of evolution.

Connecting Back

Connecting links are living organisms with characteristics intermediate between two groups, e.g., Euglena (link between plants and animals)

Molecular and phystological

various physiological processes, cellular structure, biochemistry, genetic composition, etc. also provide evolutioonary evidence e.g., blood plasma proteins.

Theories of evolutions

Lamark's Theory

It says that "The changes in structure or function of any organ are acquired during the lifetime of an individual by use of disuse of that organ and are inherited" causing evolution.

Darwain's Theory

It says that variation occur in organisms and useful amoung them are selected by nature (i.e., natural selection) and get accumulated in the organism. This leads to evolution.

Mutation Theory

This theory put forward by Hugo de Vries says that evolution is a discontinuous, saltatory process that occurs due to sudden inheritable variations.

Modern Synthetic Theory

According to it five factor : genetic variations, heredity, natural selection, reproductive isolation and speciation lead to evolution

Mechanism of evolution

Evolution starts with generation of variations, Variations in a population may occur by mutations, genetic drift, gene migration, hybridization, etc. Out of these variations inheritable variations undergo natural selection and the individual with highest survival value in the present environmental conditions evolve.

Human evolution

All human beings present today belong to a simple species Homo sapines which has evolved bipedal locomotion high cranial capacity, opposable thumbs and etc. dominates in today's life forms.

41 Biotechnology

- **Principles and processes :**

- General introduction about Biotechnology :
- Biotechnology may be defined as the use of micro organisms, animals or plant cells or their components to generate products and services useful to human being.
- Biotechnology does not mean hunting and gathering animals and plants for food; however domesticating animals such as sheep and cattle for use as livestock is classical example of biotechnology.
- Our early ancestors also took advantage of micro organisms and used fermentation to make breads, cheeses, yogurts and alcoholic beverages such as beer and wine. During fermentation some strains of yeast decompose sugars to derive energy; and in the process they produce ethanol as a waste product.
- Biotechnelogy is also associated with medical therapies to treat diabetes, Hepatitis B, Hepatitis C, Cancer, Arthritis, Haemophilia, Bone fracture etc.

Principles of Biotechnology :

Mainly two methods

Genetic Engineering :

- It is a broad term referring to manipulation of nucleic acids (DNA and RNA) of an organism.

↓

Selectable culturation :

- Organisms whose genes have been artificially altered for a desired affect is often called genetically modified organism. (GMO)

Maintenance of Sterile condition :

- Condition in chemical engineering processes in order to enable the growth of only the desired microorgamisms or eukaryotic cell in large quantities for the production of biotechnological products like antibiotics, enzymes, hormones, vaccines etc.

↓

Method of genetic engineering :

- In the technique of genetic engineering, the desired DNA Fragment would not be able to multiply unless and until it gets integrated into the genome of recipient. This is because the DNA of recipient cell has a specific sequence called origin of replication, which initiates DNA replication. Thus integrated DNA fragment replicates and multiplies itself in the recipient cell.

- The basic principles of recombinant DNA Technology are very simple.

(1) Advantage of what has been taken up by our ancestors to produce bread, cheeses, Beer and alcoholic beverages by fermentation ?

(A) Micro organisms (B) Plants

(C) Sheeps and cattle like animals (D) Tissue culture

(2) What is decomposed by yeast to derive energy during fermentation ?

(A) Polysaccharide (B) Lipids (C) Sugar (D) Disaccharide

(3) What is produced when sugar is decomposed ?

(A) Alcohol (B) Ethane (C) Methanol (D) Ethanol

(4) Which information is preserved by Asexual reproduction ?

(A) Undesirable genes (B) Exact gene information

(C) Desirable hybridization (D) Sexual Reproduction diversity

(5) Which fragment would not be able to multiply unless and until it gets integrated into the genome of recipient ?

(A) mRNA (B) DNA (C) RNA (D) Plasmid

(6) How can we introduce Bt - cotton ?

(A) Transgenic animals (B) Selective breeding (C) Domestic animals (D) Micro organisms

(7) What is the full form of EFB.

(A) Environmental Federation of Britan

(B) European Federation of Biotechnology

(C) Ecological Federation Board

(D) European Foundation of biotechnology

Answers : (1-A), (2-C), (3-D), (4-B), (5-B), (6-A), (7-B)

- **Useful tools of Recombinat DNA technology :**

(1) Restriction enzymes (2) Cloning vectors and (3) Competent host

- **Restriction Enzymes :** The technique involved in recombinant DNA technology is to cut the desired DNA segment and introduce it into a vector (e.g. plasmid)
- This is achieved using specific bacterial enzymes called restriction emzymes or restroction endonucleases.
- We known more than 900 restriction enzymes that have been isolated from over 230 strains of bacteria. These enzymes are named according to their bacteria which is gained.
- **Nucleases are of two types :** (1) Exonucleases and (2) Endonucleases.
- Exonucleases remove nucleotides from the ends of DNA where as, endonucleases make cuts at specific site within DNA.
- Restriction endonucleases recognize short palindromic sequences and cut at specific sites.
- Palindromic sequence is a sequence of base pairs in double stranded DNA that reads the same backwards and forward across the double strand.
- **Cloning vectors** : Vectors are the DNA molecules which can carry a foreign DNA Fragment to be cloned.
- The most important vectors are plasmids and bacteriophages which can replicate in the bacterial cells, indepndent of the central of chromosomal DNA.
- The following features are required to facilitate cloning into a vector.

(1) **Origin of replication** : The origin of replication (also called the replication origin) is a particular sequence in a genome at which replication is initiated. Any fragment of DNA when linked to this sequence can be made to replicate within the host cell.

(2) **Selectable Marker** : Beside origin of replication, the vector should have a selectable marker. Selectable markers are the genes which help in identifying and eliminating nontransformants and selectively permitting growth of the transformants.

(iii) **Cloning sites** : In order to link the desired DNA, the vector should have very few, preferably single recognition site for the commonly used restriction enzymes. The plasmid P^{BR322} Carries within its sequences an origin of replication and two antibiotic resistance genes.

(1) amp^R (ampicillin resistance) (2) tet^R (Tetracycline resistance)

(iv) **Vectors for cloning Genes in plants and Animals** : A cloning vector is a carrier DNA molecule in which a desired DNA fragment can be integrated in such a way that the carrier molecule does not loose its capacity for self replication. It is used to introduce desired (donor) DNA into host cells.

(3) **Competent Host :** DNA usually can get across cell membrane since it a hydrophilic molecule. In order to force bacteria to take up the recombinant DNA, the bacterial cells must be made competent take up DNA.

- **Electroporation :** Cells are exposed to rapid pulse of high voltages current to render the plasma membrane permeable to recombinant DNA.
- **Micro-injection :** Recombinant DNA is directly injected to the nucleus of animal cell though micro - injection.
- **Lipofection** : Recombinant DNA is coated with lipid which allows it to pass though the plasma membrane.
- **Porticle Bombardments :** The high velocity particles of tungston or gold are coated with recombinant DNA and then-shot into cells. This method is also known as biolistics or gene gun.

---

(8) Which enzyme helps to join cut pieces of DNA ?

(A) RNA - Ligase (B) RNA - Protease (C) DNA - Ligase (D) DNA - Polymerase

(9) ..... has capacity to transform normal cells into cancer cell.

(A) TMV (B) Retrovirus (C) DNA - ligase (D) DNA - Polymerase

(10) In which of the following process, to make easy entry of DNA and permeability of cell membrane cells are given high voltage current ?

(A) Micro - injection (B) Lipofection (C) Particle bombardment (D) Electroporation

(11) Which coating on DNA helps to enter in cells membrane ?

(A) Protein (B) Lipid (C) Carbohydrate (D) Cholesterol

(12) When the process of cloning is possible ?

(A) DNA of one species gets fixed into another cell.

(B) Recipient cell DNA is destroyed.

(C) Integrated DNA in recipient cell replicates itself.

(D) Need for cloning vector.

Answers : (8-C), (9-B), (10-D), (11-B), (12-C)

- **Process of recombinant DNA Technology :**
- Recombinant DNA is DNA created artificially by combining the DNA from two or more organisms into a single recominant molecule.
- The process involves a sequence of steps as set below.

(1) **Isolation of the Genetic Material (DNA) :**

- Recombinant DNA technology begins with the isolation of a gene of interest i.e the donor DNA. This can be achieved by cutting the DNA with the enzymes. In order to cut DNA with restriction enzymes, it needs to be in pure form, free from other macromolecules.

- As in eukargotes, DNA is enclosed within the membranes, we have to break membranes to release DNA along with other molecules like RNA, proteins, carbohydrates and also lipids.

(2) **Cutting of DNA at specific Locations :**

- To do this, a known DNA sequence from the donor cell is identified and removed with a restriction enzyme.
- When cut, each fragment of DNA has an overhanging piece of single stranded DNA on its ends called sticky ends.

(3) **Amplification of Gene of Interest using PCR** :

(1) Denaturation

(2) Annealing

(3) Extension

(4) **Insertion of Recombinant DNA into the Host cell / organism :**

The recombinant DNA is introduced into the host cell in a process called 'transformation.' Recipient cells after making them competent to receive, take up DNA present in its surrounding. If a recombinant DNA possessing antibiotic resistant gene e.g ampicillin is transferred into ampicillin resistant cells.

(5) **Obtaining the foreign Gene product :** As the ultimate aim of recombinant technology is to produce a desirable protein, the gene introduced in a recipient cell should be allowed to express.

(6) **Down stream processing :**

The processes like separation and purification of the products are collectively referred to as downstream processing.

---

(13) What is produced by combination of DNA from one or more living organisms in recombinant DNA ?

(A) Artificial DNA (B) Natural RNA (C) Artificial RNA (D) Natural DNA

(14) What can be removed by treatment of Ribonuclease enzyme ?

(A) DNA (B) RNA (C) mRNA (D) rRNA

(15) What is precipitated by chilled ethanol ?

(A) RNA (B) r RNA (C) mRNA (D) DNA

(16) Which enzyme enables sticky ends to bind with complementory DNA ?

(A) Ligase (B) Protease (C) Lipase (D) Restriction

(17) Which pieces can be separated by agarose gel electrophoresis ?

(A) RNA (B) mRNA (C) DNA (D) rRNA

(18) Which group is responsible negative charge of DNA ?

(A) Carbon (B) Phosphate (C) Sulphur (D) Iodine

(19) What is used to create New copies of DNA ?

(A) GMO (B) EFB (C) REE (D) PCR

Answers : (13-A), (14-B), (15-D), (16-A), (17-C), (18-B), (19-D)

- **A - Assertion R - Reason**

Answer the following questions, by choosing the appropriate option.

(A) A and R both are true and R is a explanation of A.

(B) A and R both are true and R is not a explanation of A.

(C) A is true and R is false

(D) A is false and R is true

(20) Assertion A : DNA fragment would not be able to multiply unless and until it gets integrated into the genome of recipient.

Reason R : Recipient cells possess specific sequence.

(A) (B) (C) (D)

(21) Assertion A : During fermentation some strains of yeast decompose sugars to derive energy.

Reason R : In this process ethanol is produced as waste product.

(A) (B) (C) (D)

(22) Assertion A : When both vector (plasmid) and desired DNA cut by the same DNA.

Reason R : The resultant DNA fragments have the same kind of sticky ends.

(A) (B) (C) (D)

(23) Assertion A : Bread is sweet in taste.

Reason R : When bread is produced by yeast, if produces carbohydrates.

(A) (B) (C) (D)

(24) Assertion A : Exonucleases remove nucleosides from the ends of DNA.

Reason R : Endonucleases make cuts at specific site within DNA.

(A) (B) (C) (D)

(25) Assertion A : Treating cells and the DNA with high Ca^{+2} which allows DNA uptake by cells.

Reason R : High Ca^{+2} causes membrane changes that reduce barriers to DNA movement.

(A) (B) (C) (D)

(26) Assertion A : The plasmid pBR322 Carries within its sequence an origin of replication and two antibiotic resistance genes.

Reason R : The tet^R gene has Pst - I recognition site.

(A) (B) (C) (D)

(27) Assertion A : Resistance to tetracyclin becomes inactivated.

Reason R : Bam H I cuts the plasmid in the gene that codes for resistance to tetracyclin.

(A) (B) (C) (D)

Answers : (20-A), (21-B), (22-A), (23-C), (24-B), (25-A), (26-C), (27-A)

· **T - F type Questions :**

Choose the correct option for given statements are true or false.

(28) Choose the correct option for important vector :

(i) Plasmid (ii) Endonuclease (iii) Bacteriophage (iv) Ampicillin

(A) TTTF (B) TFTF (C) FTFT (D) FFTT

(29) Which of the following option is correct for restriction enzyme ?

(i) More than 9000 restriction enzymes have been identified.

(ii) Nomenclature is done from the bacteria they were isolated.

(iii) Restriction enzymes are included in major class of enzymes.

(iv) Restriction endonucleases works on palindromic sequences and breaks both DNA strands.

(A) FTFT (B) TTTT (C) TFFT (D) FTTT

(30) Choose the correct option in the context of gene isolation ?

(i) Donor DNA enzyme can break RNA.

(ii) In order to cut DNA with restriction enzymes, it needs to be in pure form.

(iii) RNA can be removed by treating with ribonuclease.

(iv) Pure DNA can be precipitated by adding hot ethanol.

(A) FTFT (B) FTTF (C) TTTF (D) TFTF

(31) Choose correct option in reference to produce identical copies of short DNA sequences by PCR.

(i) A DNA molecule of interest is denatured by heating at 90 - 95°C.

(ii) The primers are complementary to either end of the target sequence but lie on opposite strand.

(iii) As the mixture cools at lower temperature 50-65°C each sequence of DNA molecule becomes annealed with an oligonucleotide primer.

(iv) DNA polymerase is then added and complementary strands are synthesized.

(A) TTTF (B) FTTT (C) TTTT (D) TFTT

(32) Choose the correct option for competent Host.

(i) High Ca^{+2} causes membrane changes.

(ii) Barriers are reduced for DNA movement.

(iii) Recombinant DNA can be forced into bacterial cells by incubating the cells with recombinant DNA on ice followed by placing them at high temp.

(iv) Treated cells are put at 100°C and then again placed on ice.

(A) TTFF (B) TFTT (C) TTTT (D) TTTF

Answers : (28-B), (29-D), (30-A), (31-C), (32-D)

(33)

Column - I	**Column - II**
(i) Electroporation	(p) Recombinant DNA is coated with lipid.
(ii) Micro - injection	(q) Tiny high velocity particles of tungsten or gold are coated with recombinant DNA.
(iii) Lipofection	(r) Cells are exposed to rapid pulse of high voltage current.
(iv) Particle Bombardments	(s) Directly injected to the nucleus of animal cell through micro injection.

(A) (i-r) (ii-q) (iii-p) (iv-s) (B) (i-r) (ii-s) (iii-p) (iv-q)

(C) (i-r) (ii-q) (iii-p) (iv-s) (D) (i-p) (ii-s) (iii-q) (iv-r)

(34)

Column - I	Column - II
(i) P^{BR322}	(p) Cuts plasmid
(ii) amp^R	(q) Recognition sites for Hind - III, Bam H I and Sal I
(iii) tet^R	(r) Carries two antibiotic resistance genes
(iv) Bam H I	(s) Recognition site for Pst - I

(A) (i-r) (ii-s) (iii-q) (iv-p)
(B) (i-p) (ii-q) (iii-r) (iv-s)
(C) (i-q) (ii-p) (iii-s) (iv-r)
(D) (i-s) (ii-r) (iii-p) (iv-q)

(35)

Column - I	Column - II
(i) Exonuclease	(p) Make cuts at specific site within DNA
(ii) Endonuclease	(q) Can separate DNA pieces
(iii) Agarose gel electrophoresis	(r) Synthesizes complementary chains
(iv) DNA polymerase	(s) Remove nucleotides from ends of DNA

(A) (i-p) (ii-r) (iii-q) (iv-s)
(B) (i-q) (ii-r) (iii-p) (iv-s)
(C) (i-s) (ii-p) (iii-r) (iv-q)
(D) (i-s) (ii-p) (iii-q) (iv-r)

(36)

Column - I	Column - II
(i) Restriction enzyms	(p) Eco RI
(ii) Restriction	(q) Selectable marker
(iii) Plasmid	(r) Important vectors
(iv) Ampicillin	(s) Works on palindrome

(A) (i-q) (ii-s) (iii-r) (iv-p)
(B) (i-s) (ii-p) (iii-r) (iv-q)
(C) (i-r) (ii-q) (iii-p) (iv-s)
(D) (i-p) (ii-s) (iii-r) (iv-q)

Answers : (33-B), (34-A), (35-D), (36-D)

(37) What is indicated as 'X' and 'Y' in the following diagram 1 ?

(A) X-amp^R, Y-tet^R

(B) X-amp^R, Y-Pvu II

(C) X-tet^R, Y-amp^R

(D) X-tet^R, Y-Pvu II

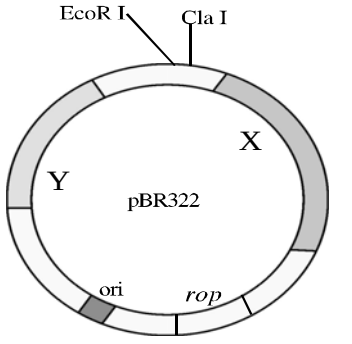


Figure 1

(38) Which recognition sites are indicated as : 'X' and 'Y' in diagram 1 ?

(A) X-Sal I, Y-Pst I

(B) X-Bam HI, Y-Sal I

(C) X-Pst I, Y-Sal I

(D) X-Hind III, Y-Sal I

(39) What 'X' and 'Y' indicateds in given diagram 2 ?

(A) X - Foreign DNA Y - Sticky ends

(B) X - Plasmid, Y - Foreign DNA

(C) X - Sticky ends, Y - foreign DNA

(D) X - Plasmid and Y - Sticky ends

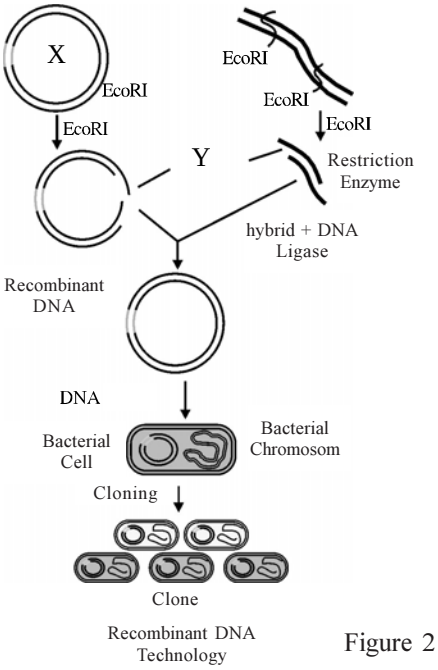


Figure 2

(40) What is indicated as 'X' and 'Y' in given diagram 3 ?

(A) X - Monitor, Y - Submerged aerator

(B) X - Submerged aerator, Y - Sensor probes

(C) X - Sensor probes, Y - Submerged aerator

(D) X - Monitor, Y - Sensor probes

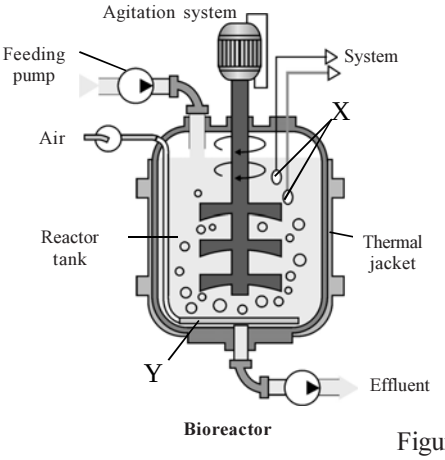


Figure 3

(41) What is used to join antibiotic resistance gene to plasmid ?

(A) DNA ligase (B) Endonuclease (C) DNA polymerase (D) Exonuclease

(42) For what polythene glycol method is used ?

(A) To prepare biodiesel (B) To create seedless fruits

(C) To produce energy from sewage (D) All of above

(43) Find the correct statement :

(A) Biogas can be prepared using aerobic bacteria and animal waste.

(B) Methanogens are anaerobic bacteria.

(C) Biogas known as gobargas is pure methane.

(D) Activated sludge in settling tank is the rich source of aerobic bacteria in STP.

(44) What is used as cloning vector in higher living organisms ?

(A) Baculo virus (B) *Salmonella typhimureum*

(C) *Rhizopus nigricans* (D) Retrovirus

(45) Which nitrogen base sequence can be broken from the middle by restriction enzymes in DNA molecule ?

(A) 5'..... CGTTCG ..... 3'
3'..... ATGGTA ..... 5'

(B) 5'..... GATATG ..... 3'
3'..... CTACTA..... 5'

(C) 5'..... GAATTC ..... 3'
3'..... CTTAAG ..... 5'

(D) 5'..... CACGTA ..... 3'
3'..... CTCAGT ..... 5'

(46) The part of DNA or RNA connected with Radio active molecule is known as ..... .

(A) Vector (B) Probe (C) Clone (D) Plasmid

(47) Enzyme restriction endonucleases ..... .

(A) Cuts DNA molecule at specific sites

(B) Identify specific nucleotide sequence to connect DNA ligase

(C) Prevents the process of enzyme DNA polymerase

(D) Remove nucleotides from ends of DNA molecule

(48) What is appropriate for corestriction endonucleases in eco RI ?

(A) Colon (B) Coelom (C) Coenzyme (D) Coli

(49) What is the use of agarose extracted out from sea weeds ?

(A) Spectrometry (B) PCR (C) Tissue culture (D) Gel electrophoresis

(50) Crystalline insecticidal protein prepared by *Bacillus thurengiensis*.

(A) Binds to the surface of midgut epithelial cells and destroys the pest at the end.

(B) Some genes possessing cay gene are transcribed.

(C) Protein is activated by acidic pH of foregut.

(D) It does not destroy the vector bacteria which possess resistance to toxin.

Answers : (37-C), (38-A), (39-D), (40-C), (41-A), (42-D), (43-B and D), (44-D), (45-C), (46-B), (47-A), (48-D), (49-D), (50-A)

Traditional views

- Brotechnology (Erey' 1917) is the technological employment of microganisms, plant cells, animal cells, their components or biological process to generate products and services useful to human beings.

Modern view

- Modern biotechnological is the use of genetically modified organisms (GMOs) or generally engineered plant and animal cells in developing protection technologies that not only enhance their productivity but also give rise to new products.

Biotechnology

Principles of Biotechnology → **Chemical engineering**

Genetic engineering

- Techniques to alter the chemistry of genetic material (DNA and RNA), to introduce these into host organisms and thus change the phenotype of the host organism.
- The techniques of genetic engineering include creation of recombinant DNA (rRNA).

Tools recombinant DNA technologies

Restruction enzymes

- Restruction eazymes belong in a larger class of enzymes called nucleases. These are of three kinds. exonucleases. endonucleases and restriction endonucleases. Exaleases remove nucleotides from the ends of the DNA. whreas endonucleases make cuts at specific positions within the DNA.
- Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA.
- For R1, Bam !!, 1, Hin d III are some of the example of restriction endonucleases.

Gerl electrophoresis

- It is a technique to seperate and isolate DNA fragments.

Cloning vectors

- They are carriers or vehicles of desired DNA fragments passenger DNA) which can undergo independent replication to increase copies of desired genes.
- Cleaning vectors are of four type - plasmids, viruses, cosmids and artificial chromosomes.

Competent host

(For transformation with recombinant DNA)

- It involves vector transfer and direct or vectorless transfer.
- Vector transfer involves transformation with cloning vectors.
- Vectories transfer involves.

(i) Microinjection (ii) Particle gene gun or biolistic
(iii) Electroporation (iv) Direct DNA injections

Processes of rDNA technologies

Isolation of genetic material

- To isolate the DNA from the bacterial cells plant or animal tissue it is treated with enzymes such as lysozyme (bacteria), cellulase and pectinase (plant cells), chitinase (fungus).
- The treated cells are homogenised and centrifuged.
- Hormogenised product is then treated with proteases (for digesting histones and other proteins). ribonucleases (digestion of RNAs). amylases and lipases.
- DNA remains intact. It is precipitated by the addition of chalked ethylalcohol.

Amplification of gene of interest using PCR

- PCR or polymicrase chain reaction is the in vitro synthesis of multiple copies of a gene or DNA segment.
- Amplified DNA segment can be used to ligate with the vector for formation of rDNA and its cloning.

Otbatining the desirable gene product

- When rDNA is transferred into a bacterial, plant or animal cell, the foreign DNA gets multiplited.
- The foreign gene gets espressed under appropriate conditions.
- If any protein encolding gene is expressed in a heterologous host, it is called a recombinant protein. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory. The cultures may be used for extracting the desired protein and then purifying it by using different separation techniques.
- Small scheme culture can not yield appreciate quantities of products. To produce in large quantities, the development of bioreactors where large volumes of (100-1000 litres) of culture can be processed is used.

Cutting of DNA at specific locations

- Restruction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzyme.
- The process is repeated with the vector DNA also.
- After having cut the source DNA as well as the vector DNA with a specific restriction enzyme. The cut our gene of interest from the source DNA and the cut vector with space are mixed and ligase is added.
- This results in the preparation of recombinant DNA.

Insertion of recombinant DNA into the host cell / organism

- The rDNA is inserted into host bacterium by transformation using cold CaCl solution.
- The bacterial cell containing the desired rDNA is selected uing selective antibiotics in the culture medium.

Downstream processing

- Down stream processing is the recovery of product from the fully grown genetically modified cells, its purification and preservation.
- The procedure and vigour or downstream processing and quality control varies from product to product.

42 Biotechnology and its Applications

- **Biotechnology and its application** : Biotechnology is the most exciting and revolutionary science of this century this is one of the applied biological field.
- **Application of Biotechnology :** In general biotechnology has applications in four major industrial areas including (1) Health care (medical) (2) Crop production and agriculture (3) Non-food (industrial) uses of crops and other products (4) Environmental uses.
- **Biotechnological Applications in Agricutture :** There are following three options in order to increase the food production.

  (1) Agro - chemical based agriculture (2) Organic agricalture (3) Genetically engineered Crop based agricalture.
- The world population has topped 6 billion people and is predicted to double in the next 50 years. Ensuring an adequate food supply for this becoming population is going to be a major challenge in the years to come.
- The green revolution succeeded in tripling the food supply but yet it is not enough to feed the growing human population.
- The increased production have not only been due to the use of improved crop varieties but also due to use of agrochemicals like fertilizers and pesticides and pollution can occur.
- Features exibits in genetically modified plants :

  (1) **Pest resistance :** Growing pest resistance plant can help to eliminate the application of chemical pesticides and reduce the cost of bringing a crop to market.

  (2) **Herbicide Tolerance** : Crop plants genetically - engineered to be resistant to very powerful herbicide, could help to prevent environmental damage by reducing the amount of herbicides needed.

  (3) **Disease Resistance** : These are many viruses, fungi and bacteria that cause plant diseases. GM Plants are resistance to these diseases.

  (4) Cold, drought, salt and heat tolerance.

  (5) With enhanced nutritional value of food.
- **Genetically Modified Organisms** : A Genetically modified organism is an organism whose genetic material has been altered using genetic engineering technique.
- **Advantage of GMO method** : The main advantage of utilizing this technique in agriculture is possibility to increase productivity through the use of newer varieties that possess property as resistance to pest.
- **Introduction of specific charatars for pest resistance :**
- Imparting the property of pest resistance through the transfer of gene from *Bacillus thuringiensis* (Bt) into target plant through modern biotech method is presently considered to be one of the most advanced application of biotechnology.
- Bt cotton is a genetically modified crop that contains a foreign gene isolated from *Bacillus thuringinesis*.
- This bacterial gene produces toxic insecticidal crystalline protein destroys boll worms.
- Actually, *Bacillus thuringiensis* produces this toxic protein in an inactive form, but when an insects ingets this inactive protein, it is converted into an active form of toxin due to the alkaline pH of gut which solubilises the crystals.

- This activated toxin binds to the surface of midgut epithelial cells and creates pores that causes death of the insect. The toxin is coded by a gene named cry and hence, toxin is termed an cry protein.
- **Biotechnological Applications in Medicine :**
- Most development in biotechnology originated for their potential applications in health care.
- In medicinal field, biotechnology techniques are used frequently in diagnosing and treating different diseases.
- The field of biotechnology has introduced techniques like gene theropy, recombinant DNA technology and polymerase chain reaction which use genes and DNA molecules to diagnose diseases and insert new and healthy genes in the body which replace the damaged gene or DNA.

(1) Which type of science is biotechnology from centuries ?

(A) Exciting (B) Revolutionary

(C) A and B both (D) Healthy

(2) The world population has topped ..... .

(A) more than 12 billions (B) more than 8 billions

(C) more than 7 billions (D) more than 6 billions

(3) How many times food supply has increased due to green revolution ?

(A) Two times (B) Three times (C) Many times (D) None of the above

(4) The use of which thing is very expensive in developing countries ?

(A) Agriculture based on agrochemicals (B) Organic agriculture

(C) Genetic engineered crop based agriculture (D) None of the above

(5) What is the main advantage of utilizing genetically modified organisms technique ?

(A) Irrigation (B) Pest control (C) In agriculture (D) None of the above

(6) Bt stands for ..... .

(A) Biotechnology (B) *Bacillus thuringiensis*

(C) Biologically trained plant (D) None of the above

(7) Inactive toxin, is converted into toxin form due to

(A) Alkaline pH (B) Enzymes (C) Neutral pH (D) Acidic pH

(8) By which type of gene toxin protein is produced ?

(A) Cry gene (B) Bt gene (C) Cyr gene (D) Toxic gene

(9) Which hormone, regulates sugar metabolism in human ?

(A) Insulin (B) Crystal (C) Albumin (D) Testosterone

Answers : 1 (C), 2 (D), 3 (B), 4 (A), 5 (C), 6 (B), 7 (A), 8 (A), 9 (A)

- **Genetically Engineered Insulin :**
- Insulin, a hormone, regulates sugar metabolism in haman and it is of immense value to diabetics.
- Insulin is produced by the B-cells of islets of langerhans of pancreas.
- **Human Insulins :**
- Human insulin contains 51 amino acids, arranged in two poly peptide chains. The chain A has 21 amino acids while B has 30 amino acids.
- Both are held together by disulfide bonds.

- In 1980 recombinant DNA technology was used to produce human insulin in bacteria (E.Coli) which is called Humulin. Normally insulin is synthesized as proinsulin which has an extra stretch called the C peptide.
- This C peptide is not found in mature insulin.
- In 1983, Eli Lilly company of United states has produced two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of E.Coli to produce insulin chains.
- **Gene-therapy :**
- Gene-therapy is a technique of biotechnology which is to treat and diagnise diseases like cancer, parkinson's disease etc.
- Gene-therapy may be defined in broad general term as introdution of a normal functional gene into cells, in order to replace defective or mutated gene.''
- **Significance of gene-therapy :**

(1) Identification of the gene that play the keyrole in the development of a genetic disorder.

(2) Determination of the role of its product in health and disease.

(3) Isolation and Cloning of gene.

(4) Development of an approach for gene theropy.

- **Types of Gene-therapy**

(1) Germ line gene-therapy :

- In germ line gene theropy, germ cells, i.e sperms or eggs are modified by introduction of functional gene, which are ordinarily integrated into their genomes. Therefore, change due to therapy would be heritable.

(2) Somatic cell gene theropy :

- In somatic cell gene therapy the gene is introduced only in somatic cells, especially of those tissue in which expression of the concerned gene is critical for health. Expression of the introduced gene relieves or eliminate symptoms of the disorder.

Gene therapy treatment :

(1) Exvivo : Exvivo means 'outside the body'. Cells from the patients' blood or bone marrow are removed and grown in the laboratory.

(2) Invivo : Invivo, which means ' inside the body'. No cells are removed from the patient's body.

- **Transgenic Animals :** The dependence of man on animals such as cattle, sheep, poultry, pig and fish for various purposes (milk, meat, eggs, wool etc.) is well known.
- With the advent of modern biotechnology, it is now possible to carry out manipulation at the genetic level to get the desired characteristics in animals.
- **Tranrgenesis / Transgenic animals :** Transgenesis refers to be the phenomenon of introduction of exogenous DNA into the genome of an animal to create and maintain a stable heritable character. The foreign DNA that is introduced is called transgene. And the animal whose genome is altered by adding one or more transgenes is said to be transgenic animals.

---

(10) From where insulin is produced ?

(A) α - cells of islets of langerhans of pancreas (B) From liver

(C) β - cells of islets of langerhans of pancreas (D) None of the above

(11) How many amino acids are present in human insulin ?

(A) 51 (B) 49 (C) 90 (D) 61

(12) How many amino acids are present in the polypeptide chain A ?

(A) 25 (B) 22 (C) 29 (D) 21

(13) How many amino acids are present in the polypeptide chain B ?

(A) 50 (B) 19 (C) 30 (D) 25

(14) Change due to which therapy would be heritable ?

(A) Somatic cell gene therapy (B) Exvivo gene therapy

(C) Germ line gene therapy (D) Invivo gene therapy

(16) Transgene means ..... .

(A) Foreign DNA (B) External DNA

(C) Internal DNA (D) Any Gene

(17) Which animal is being developed for use in testing the safety of polio vaccine ?

(A) Mice (B) Rabbit (C) Pig (D) Chick

(18) In which year the first transgenic cow was developed ?

(A) 1997 (B) 1979 (C) 1799 (D) 1977

(19) Which animal is associated with the production of milk for human babies ?

(A) Buffalo (B) Cow (C) Goat (D) Sheep

(20) Which product is useful for human being produced by transgenic cow ?

(A) Milk for human babies (B) Trapa natans (C) Meat (D) Leather

(21) What was the amount of protein per litre in the milk of transgenic cow ?

(A) 2.4 gm (B) 0.25 gm (C) 250 gm (D) 24 gm

Answers : (10-C), (11-A), (12-D), (13-C), (14-C),(15-C), (16-A), (17-A), (18-A), (19-B), (20-A), (21-A)

- **Ethical Issues :**
- Ethics includes 'moral principle' that control or influence a person's behavior. It includes a set of standards by which a community regulates its behaviour and decides as to which activity is legitimate and which is not.
- Bioethics may be viewed as a set of standards that may be used to regulate our activities in relation to the biological world. Now-a-days, biotechnology, particularly recombinant DNA technology, is used for exploitation of the biological world by various ways.
- **Biopatent :**
- Patents for bioscientific researches are called biopatents.
- A patent is granted for :

(1) An invention , including a product

(2) An improvement in an earlier invention

(3) The process of generating product

(4) A concept or design

- **Biopiracy :** When big organization and multinational companies exploit patent biological resources or bioresources of other nations without proper authorization from the countries concern; such exploitation is called biopiracy.
- While developing nations are poor in technology and finacial resources, but quite rich in biodiversity and traditional knowledge related to bioresources.
- By modernizing the methods, they become more prosperous. Thus it is necessary to bring awarness regarding biopiracy and to stop it.
- A plant *Pentadiplandra brazzeana* of west Africa, produces a protein called brazzein.
- **Biosafety :**
- Biosafety is the prevention of large scale loss of biological integrity; focusing both on ecology and human health.

  Recombinant DNA technology enables man to combine DNA sequences from different sources to create functional DNA molecules with novel properties.
- These molecules are expressed in genetically modified organisms (GMO), Which are then used in biomedical, agricultural and environmental areas.

---

(22) Biopatents are awarded for ..... .

(A) Specific DNA sequences (B) Strains of micro organisms

(C) The proteins encoded by DNA sequence (D) A, B and C

(23) According to which act biopatent is awarded in India ?

(A) Indian Patent Act (B) Indian Institute of Pharma

(C) Indian Biotech Company (D) None of the above

(24) The awarded duration of biopatent is of ..... .

(A) 5 years (B) 1 year (C) Lifetime (D) 7 years

(25) In which of the folowing plant a brazzein protein is produced ?

(A) *Pentadiplandra brazzeana* (B) *Zamia pygmea*

(C) *Cucurbita pepo* (D) *Datura fastuosa*

(26) Which ethics includes in ethical issues ?

(A) Independancy (B) moral issues (C) A and B both (D) None of the above

(27) When human beings use animals in biotechnology ..... .

(A) It is biothical issue (B) Only exploits animals for the benifit of selfish nature of human

(C) A and B both (D) None of above

(28) A biopatent is granted for ?

(A) An invention ot chemistry (B) Patents granted to a bioresearcher

(C) An astrologist for research in space (D) All of the above

(29) In which year Indian patent Act was introduced in India ?

(A) 1979 (B) 1989 (C) 1970 (D) 1999

(30) The brazzein protein is approximately ..... as sweet as sugar ?

(A) 250 times (B) 2000 times (C) 400 times (D) 300 times

(31) Which disease treatment can be done by brazzein protein ?

(A) Plague (B) Cancer (C) Parkinson's disease (D) Diabetes

(32) From the following country which has obtained a patent for brazzein and fused it in maize ?

(A) India (B) America (C) Canada (D) Australia

(33) Who can have supreme control over, if a single biopatent covers all patents of a plant like Brassica?

(A) Some individual (B) A country (C) institute (D) All of the above

(34) Biopiracy means ..... .

(A) stealing of bioresource without proper authorization.

(B) exploitation of bioresources without authentic permission from country

(C) organization and multinational companies exploit those organisms which can be used to derive commercial benefits

(D) all of the above

Answer : (22- D), (23-A), (24-A), (25-A), (26-B), (27-C), (28-B), (29-C), (30-B), (31-D), (32-B), (33-D), (34-D)

- **A - Statement, R - Reason**

Read the given statements and select the correct option.

(A) Both A and B is correct, R is description of A.

(B) Both A and R is correct, but R ios not description of A.

(C) A is correct, R is wrong.

(D) A is wrong, R is correct.

(35) Statement A : To eliminate the application of chemical pesticides.

Reason B : Pest resistance plant can be grown.

(A) (B) (C) (D)

(36) Statement A : Causes death of insects.

Reason R : It is converted into an active form of toxin due to the alkaline pH of gut which solubilises the crystal protein.

(A) (B) (C) (D)

(37) Statement A : Bacterial gene produces a toxic insecticidal crystalline protein.

Reason R : It provides nutrition to bollworms.

(A) (B) (C) (D)

(38) Statement A : Insulin is one type of enzyme.

Reason R : Insulin regulates sugar metabolism in human.

(A) (B) (C) (D)

(39) Statement A : Human insulin contarins 51 amino acids.

Reason R : The chain A has 30 amino acids, while B has 21 amino acids.

(A) (B) (C) (D)

(40) Statement A : C peptide is not found in mature insulin.

Reason R : Insulin is synthesized as proinsulin.

(A) (B) (C) (D)

(41) Statement A : The foreign DNA is introduced in animals.
Reason R : One can get desired characteristics in animals.
(A) (B) (C) (D)

(42) Statement A : Cells from the patient's blood are removed in Exvivo.
Reason R : The desired gene is delivered to cells in the patient's body.
(A) (B) (C) (D)

(43) Statement A : A plant of west Africa, produces a protein called brazzein. This protein is approximately two thousand times as sweet as sugar.
Reason R : By consuming this protein diabetes increases.
(A) (B) (C) (D)

Answers : (35-A), (36-A), (37-C), (38-D), (39-C), (40-B), (41-A), (42-B), (43-C)

- **True-False (T - F) type questions**

(44) Select the correct option for Bt cotton.
(i) Bt cotton is a gentically modified crop.
(ii) That contains a foreign gene isolated from *Bacillus thuringiensis*.
(iii) *Bacillus thuringiensis* produces this toxic protein in an active form.
(iv) This toxic protein is known as cry protein.
(A) TTTT (B) TTFT (C) TTTF (D) TTFF

(45) Select the correct option for human insulin.
(i) Human insulin contains 51 amino acids.
(ii) The chain A has 30 amino acids.
(iii) The chain B has 21 amino acids.
(iv) Both chains combined by creating peptide bonds to form human insulin.
(A) TTTT (B) TFFT (C) TFFF (D) TFTF

(46) Select the correct option for Ethical issues.
(i) Use of animals in biotechnology causes great suffering to them.
(ii) Introduction of a transgene from one species into another species violates the integrity of species.
(iii) Biotechnology exploits animals only for the benefit of human indicator.
(iv) Biotechnology may pose unforeseen risks to the environment, including risk to biodiversity.
(A) TFTT (B) TFFT (C) TTTF (D) TTTT

(47) Select the correct option for biopiracy.
(i) Africa obtained a patent for germplasm of our basmati rice.
(ii) Biomolecules of many plants are patented in other countries.
(iii) Isolate useful genes and then obtain patent for them.
(iv) Pirate or steal the traditional knowledge and publish the same as a new finding.
(A) TTTT (B) FTTT (C) FTFT (D) FTTF

(48) Select the correct option for risks due to genetically modified plants.
(i) Production of toxic or allergic metabolites.
(ii) Unexpected new susceptibilities to pathogens.

(iii) Transmission of new traits to related sexually compatible weed species.

(iv) The ecosystem may be disturbed by altered reactions.

(A) TTFT (B) TTFF (C) FTTF (D) TTTT

Answers : (44-B), (45-B), (46-D), (47-B), (48-D)

(49)

Column - I	**Column - II**	
(i) Germline gene therapy	(p) Introduction of exogenous DNA into the genome of animals	(A) (i-q) (ii-s) (iii-p) (iv-r)
(ii) Somatic cell gene	(q) The functional gene is incorporated into Genome of germ cells	(B) (i-q) (ii-s) (iii-r) (iv-p)
(iii) Transgenesis	(r) First human insulin was produced	(C) (i-r) (ii-p) (iii-s) (iv-q)
(iv) Eli Lilly	(s) The gene is introduced only in somatic cells, especially of those tissues in which expression of the concerned gene is critical for health.	(D) (i-s) (ii-q) (iii-p) (iv-r)

(50)

Column - I	**Column - II**	
(i) Lepidoptera	(p) Beetles	(A) (i-p) (ii-q) (iii-r) (iv-s)
(ii) Coleoptera	(q) Aphids	(B) (i-q) (ii-s) (iii-p) (iv-r)
(iii) Orthoptera	(r) Bollworms	(C) (i-r) (ii-p) (iii-s) (iv-q)
(iv) Homoptera	(s) Grasshoppers	(D) (i-s) (ii-r) (iii-p) (iv-q)

(51)

Column - I	**Column - II**	
(i) Gene therapy	(p) Pirate the traditional knowledge and publish the same as a new finding	(A) (i-q) (ii-s) (iii-p) (iv-r)
(ii) Bio patent	(q) To extract gene and cloning	(B) (i-r) (ii-q) (iii-s) (iv-p)
(iii) Transgenic animals	(r) Brazzein	(C) (i-q) (ii-r) (iii-p) (iv-s)
(iv) Biopiracy	(s) It was used to test the safety of the polio vaccine	(D) (i-q) (ii-r) (iii-s) (iv-p)

Answers : (49-A), (50-C), (51-D)

(52) Which option is correct for X and Y in this diagram ?

(A) X = C - Peptide Y - disulfied bond

(B) X = B chain of insulin

Y = A chain of insulin

(C) X - A chain of insulin

Y = B chain of insulin

(D) X = disulfied bond

Y = C - peptide

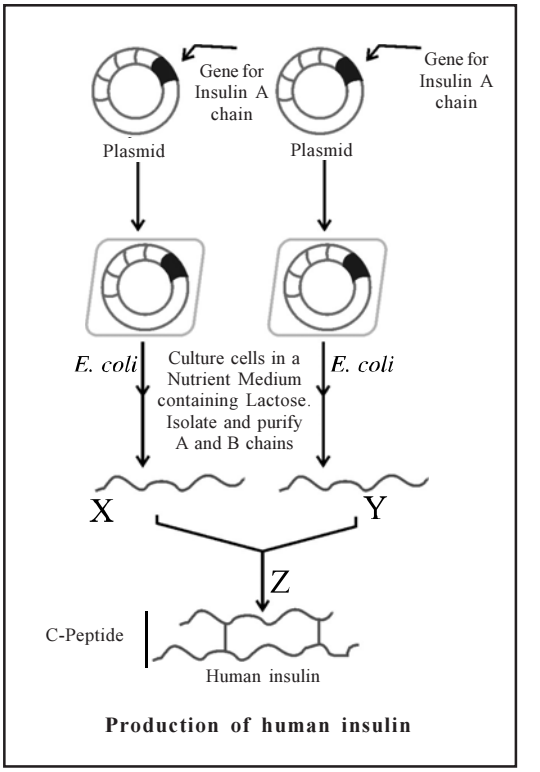
(53) What is the function of Z in given diagram ?

(A) It regulates sugar metabolism in human.

(B) Ioining of chain A and B of insulin.

(C) Synthesis of proinsulin.

(D) Breaking of chain A and B of insulin.



Production of human insulin

Answers : 52 (C), 53 (B)

- **Questions for NEET**

(54) Animal cell culture technique used maximally used these days ..... .

(A) to prepare edible proteins (B) to prepare insulin

(C) for interferon production (D) for vaccine production

(55) Which is the main reactant in elisa test for detection of virus ?

(A) RNAase (B) Alkaline phosphatase (C) Catalase (D) DNA probe

(56) In future 'Golden rice' developed through transgene approach is enriched with ..... .

(A) pest resistance (B) high lysine content

(C) high protein content (D) high vitamin A content

(57) Norman Borlaug is credited for which revolution ?

(A) white revolution (B) Green revolution (C) Yellow revolution (D) Blue revolution

(58) For what Bacillus thuringiensis is used ?

(A) To prepare biofertilizer (B) Biometal technique

(C) Bio mineralization (D) Bio pest - resistance plant

(59) In developing countries for treatment of night blindness useful genetically modified plant ..... .

(A) Tomato (B) Starlink maize (C) Bt soyabean (D) Golden rice

(60) ..... microorganism is useful to transfer foreign DNA into target plant.

(A) Meloidogyne incognita (B) Agrobacterium tumefaciens

(C) Penicillium notatum (D) Trichoderma harzianum

(61) In India genetically modified brinjal is produced for what ?

(A) To make pest resistance (B) To increase life resistance

(C) To increase mineral level (D) Drought resistance

(62) What can be used for production of ethanol in brewery ?

(A) Cornmill (B) Ground gram (C) Soyabean (D) Molasses

(63) The highest numbers of transgenic animals ?

(A) Fish (B) Cow (C) Rat (D) Pig

(64) Why continuous feeding of sugar in medium is needed in fed batch fermentation ?

(A) For production of methane (B) For production of antibiotics

(C) To activate enzymes (D) For dissolution of sewage

(65) By introducing fragments of m-RNA in genetically modified plant, it can be made resistance to ..... .

(A) bollworms (B) aschelminthes

(C) white rust (D) bacterial blight

(66) Find true statements for organic farming ?

(a) Use of genetically modified plant like Bt cotton.

(b) Use of only natural elements like compost.

(c) No use of chemical pesticides and urea.

(d) Production of vegetables having vitamins and minerals.

(A) b, c and d (B) b and c (C) c and d (D) a and b

(67) For development of What Human Genome project was introduce in biology ?

(A) Biotechnology (B) Biomonitoring (C) Bioinformatics (D) Biosystematics

(68) In genetic engineering by which method organisms can be developed ?

(A) Recombinant DNA (B) X - ray diffraction

(C) Heavier isotope labeling (D) Hybridization

Answers : (54-D), (55-B), (56-D), (57-B), (58-D), (59-D), (60-B), (61-A), (62-D), (63-C), (64-C), (65-C), (66- C), (67-C), (68-A)

Applications of Biotechnology

Ethical issues

Biopiracy

- It is exploitation of bioresources of a country by organisations and multinationals for commercial exploitation with or without patent but without any Access and Benefit Sharing Agreement (ABA).
- Some nations are developing laws to prevent such unauthorised exploitation of their bioresources and traditional knowledge.

Biopatent

- It is an official licence to use a particular biological material for commercial exploitation like (i) A strain of microorganism (ii) GM plant animal (iii) DNA sequence (iv) Biotechnological procedure (v) Production process. (vi) Product and (vii) Product application.
- Biopatents are going to give not only increased economic benefits to holders, but will also provide a spurt in research.

Biotechnlogical application in agriculture

- Genetically modified crops (GM crops) or Transgenic plants : GM crops contain and express one or more useful foreign genes or transgenes (e.g. Bt cotton, golden rice, Flaver Savr tomato)
- GM plants have been useful in many ways :
  - Growing GM crops can help to reduce the use of chemical pesticides.
  - GM crops are more tolerant to abiotic stresses (cold, drought, salt, heat etc.).
  - They have helped to reduce post-harvestlosses.
  - Increased efficiency of mineral usage by plants prevents early exhaustion of fertility of soil.
  - GM plants enhance nutritional value of food. e.g., vitamin A enriched rice.
  - Herbicides (weed killers) do not harm the GM crops.
  - Plant biologists are working to create genetically engineered plants having resistance to these deseases.
  - GM plants have been used to create alternative resources to industries in the form of starches, fuels and pharmaceuticals (molecular farming.)

Biotechnological application in medicine

- Recombinant DNA technology is engaged in production of newer and safer therapeutic drugs, called recombinant drugs. These drugs do not induce immunological reactions as some other drugs obtained from nonhuman sources (e.g., insulin from animals).
- A large number of human genes encoding pharmaceutically valuable proteins have been cloned and expressed in microorganisms.
- Gene theraphy is a collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo. Here genes are inserted into a person's cells and tissues to treat a disease.

Production of transgenic animals

- These are animals which have been genetically modified by incorporating foreign and other specific genes through recombinant DNA technology. A number of animals have been genetically modified,e.g., rats, rabbits, pigs, sheep, cows, fish. 95% of them are transgenic mice.
- Some of the important uses of transgenic animals are as follows :
  - Transgenic animals enable scientists to study the working of genes, their regulation, effect on normal body functions and development.
  - Transgenic animals are used to study how genes take part in development of diseases.
  - The important bilogical products being harvested by this method are protein α-1 antitrypsin for treating emphysema, tissue, plasminogen activator (goat), blood clotting factors VIII and IX (sheep) and loctoferrin (cow).
  - Transgenic animals are the best for checking the safety of vaccines before their use in humans.
  - Trasgenic animals are used in studying the effect of toxic chemicals.

Biotechnological application in industry

Mostly microorganisms are used in many industries to produce alcohol, enzymes, vaccines, vitamins, antibiotics, organic acids etc.